Computing the Riemann Hypothesis

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Thorin Measures, Pólya Densities and Hitting Times

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Cover: the Thorin density of $\xi(\frac{1}{2})/\xi(\frac{1}{2}+\sqrt{u})$ at eight resolutions, computed by the authors.

Source, code and data: `https://github.com/VadimSokolov/computing_rh`

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Preface

The Riemann Hypothesis can be stated as the positivity of a measure. The reciprocal Xi clock $u \mapsto \xi(\frac{1}{2})/\xi(\frac{1}{2}+\sqrt{u})$ is the Laplace transform of a generalised gamma convolution exactly when every nontrivial zero lies on the critical line, and then its Thorin measure is the discrete measure $\sum_{\gamma>0}\delta_{\gamma^2}$ on the squared ordinates. This book asks what computation can say about that positivity condition.

Its answer has three parts. First, the Thorin measure is computable at every positive resolution, with explicit error bounds, from the Pólya kernel alone, with no primes and no list of zeros. Second, the distance between the computed measure and the measure demanded by RH can be measured exactly, and the signature an off line zero would leave can be calibrated. Third, a finite computation could refute RH by certifying a zero off the line, but neither a computation of the zeros up to a finite height nor a finite set of positivity tests on the heat trace can prove it: for every sufficiently large height R such that RH has been verified up to $R+50$, there is a perturbation of ξ, with the same order, type and symmetry and the same zeros below R, that has zeros off the line between R and $R+50$ and still passes the tests. The book makes this precise and locates the remaining difficulty. A last question is how much of the positivity can be proved. Moved from the centre of the critical strip to the basepoint one, the clock has a positive Thorin measure with no hypothesis. Chapters 9 to 11 follow each construction of the book back across the strip as far as it holds unconditionally; below the basepoint one, zero density estimates still show that the zeros to the right of the basepoint, the only possible source of negativity, carry a vanishing fraction of the mass of the density.

The book grew out of the first author's programme on generalised gamma convolutions and the Pólya density, collected in the monograph *On Hilbert's 8th Problem* [200], and out of a joint paper computing the Thorin measure directly [210]. The first author's arXiv paper of the same title is a separate work, which we cite by version: [198] is its version 24, of 2026, and [199] its version 5, of 2017. Chapters 6 and 7 incorporate Part IX of that monograph [200], the first author's work on the reciprocal zeta function, written as the manuscript [205], and Chapters 9 to 11 a paper of the authors on the reciprocal Xi function at the basepoint one and across the critical strip, together with notes on Riesz's criterion and on almost generalised gamma convolutions. Sections 8.5 and 11.3 draw on a manuscript of the first author on the Thorin measures of the Dedekind

eta function [203], and Chapter 13 on a note of the first author on ferromagnetic random variables. The core evaluator runs in ball arithmetic, and Appendix A describes the scripts and data files behind the tables and figures and what the archive lacks. No proof of RH is claimed.

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Contents

Preface v

1 Introduction and foundations 1
1.1 The Thorin condition and what this book shows 1
1.2 The Xi function, the Pólya density and the clock 10
1.3 Generalised gamma convolutions and the Thorin condition 20
1.4 Exercises 28

2 Evaluating the Xi function 31
2.1 The Pólya route 31
2.2 Classical evaluators, Turing's method and the Pólya route 40
2.3 A worked computation from first principles 45
2.4 Exercises 48

3 The Thorin measure at finite resolution 51
3.1 What computation can reach, and what the sine squared lemma adds 51
3.2 Distances to the discrete measure implied by RH 56
3.3 What an off line zero would do 61
3.4 Counting zeros with the Thorin measure 64
3.5 Further readings of the sine squared lemma 69
3.6 Exercises 73

4 Weak limits, hyperbolic complete monotonicity and a Bosch type theorem 77
4.1 The limit of vanishing resolution 77
4.2 Bosch's half Cauchy theorem and HCM of the reciprocal Xi function 81
4.3 A Bosch type theorem for the reciprocal Xi function 90
4.4 Exercises 112

5 The heat trace, the arithmetic and the limits of finite information 115

5.1 The heat trace and the hierarchy of positivity 115
5.2 What finite information cannot decide 137
5.3 The Thorin functional from the arithmetic 144
5.4 Spacings, irregularity and the cell distance 148
5.5 Exercises 152

6 The reciprocal zeta function 155
6.1 Thorin type forms of the reciprocal zeta function 156
6.2 The gamma factor, the Meixner law and the sine squared lemma . 169

7 Infinite divisibility and truncations of the reciprocal zeta function 191
7.1 Infinite divisibility, necklaces and solved models 191
7.2 Truncations, finite conditions and the road to RH 217

8 Probability, hitting times and other criteria 243
8.1 The clock as a random variable 243
8.2 Brownian boundaries and Williams' structure 251
8.3 Keiper Li coefficients from the Pólya density 259
8.4 The de Bruijn Newman flow from the Pólya density 263
8.5 Beyond zeta: the Dedekind eta kernel, $L(s, \chi_{12})$ and Ramanujan's Δ 287
8.6 The inverse problem: recovering the atoms by nonnegative deconvolution 337
8.7 Pólya's Bessel function: hitting times, a clock and the flow 341
8.8 Exercises 370

9 The basepoint one 375
9.1 The Thorin measure and the Lévy form at the basepoint one . . . 376
9.2 Hitting times at the basepoints 406

10 The critical strip and Riesz's criterion 421
10.1 Normal scale mixtures and completeness across the strip 421
10.2 Riesz's criterion and the Thorin path 446
10.3 The squarefree series and the clock 461

11 Infinite divisibility, the eta analogue and almost generalised gamma convolutions 487
11.1 Infinite divisibility and a map of processes 488
11.2 Free probability and the Upsilon transform 508
11.3 The Dedekind eta analogue at the basepoint one 530
11.4 Almost generalised gamma convolutions 539
11.5 Pakes, the Lambert law and the Thorin condition 557

12 Conclusions and open problems 595
12.1 The Thorin approach in summary, and a path to RH 595

12.2 Open problems and lessons for the Thorin programme 604

13 Ferromagnetic random variables 615
13.1 Ferromagnetic laws and their clocks 616
13.2 Laws on an interval 622
13.3 The Pólya law 627
13.4 Ferromagnetic approximations of the Pólya law 632
13.5 The exponential of the Pólya variable 640
13.6 Open problems 650

A Code and reproducibility 653
A.1 The Pólya evaluators 653
A.2 Scripts, data files and reference constants 656
A.3 The scripts for the basepoint one and the critical strip 660
A.4 The scripts of the ferromagnetic chapter 666
A.5 Reruns and gaps 667

B A short history of computations on the zeros 671

C Notation 673

D Verification code for the Bosch type theorem 683

List of Figures

1.1 The Thorin density $\rho_\varepsilon(\theta)$ on $10 \le \theta \le 60$, computed from the Pólya density at eight resolutions from $\varepsilon = 1$ (bottom) to $\varepsilon = 0.005$ (top), on a logarithmic colour scale. As the resolution sharpens, the density concentrates on bright vertical lines at the ordinates of the zeros, all of which lie on the critical line in this window: the Cauchy flow converging to the RH measure $\sum \delta_{\gamma^2}$. 4

1.2 Left: the Pólya kernel $\Phi(u)$ on the real line. Right: $|\Phi(x+i\eta)|$, on a logarithmic scale, on the lines $\eta = 0.1000$, 0.3927, 0.6854 and 0.7654 inside the strip of analyticity, that is at distances 0.6854, 0.3927, 0.1 and 0.02 below its edge $\eta = \pi/4$ (the tilts of Section 2.1.3). As the line approaches the natural boundary $\eta = \pi/4$ the kernel grows and spreads (compare Table 2.1), and it still decays doubly exponentially in $|x|$. 13

1.3 Cell distance divided by the number of atoms in the window, for the sinh clock (50 atoms, cells $(k-\frac{1}{2}, k+\frac{1}{2}]$) and for the 79 zeros below height 200 (Voronoi cells of Section 3.2). Both vanish linearly in ε. For the sinh clock the undivided distance stays below $4\varepsilon/\pi$ whatever the number of atoms, so the gap between the curves depends on the window, and the factor of three in the title of the figure holds for these two windows only. 28

2.1 The Thorin density $\rho_\varepsilon(\theta)$ at basepoint $\frac{1}{2}+\varepsilon$ computed from the Pólya density, for $\varepsilon = 0.25, 0.1, 0.05$ and $10 < \theta < 56$, on a logarithmic scale; dashed lines mark the ordinates. Each ordinate carries a Cauchy kernel of height $1/(\pi\varepsilon)$ and mass one (Proposition 1.2.3). . 35

2.2 The Thorin measure at $\varepsilon = 0.5, 0.1, 0.02$ in the variable $z = \theta^2$, for $81 < z < 3600$, with density $u_\varepsilon(z) = v_{1/2+\varepsilon}(\sqrt{z})/(2\pi\sqrt{z})$ as in (1.2.6). Dotted lines mark the atoms γ^2 of U_{RH}. 35

2.3 Left: decimal digits lost to cancellation as a function of height, untilted and tilted ($\delta = 0.1, 0.02, 0.01$), with the runs $(\theta, \delta) = (1000, 0.1)$, $(5000, 0.02)$ and $(7005, 0.01)$ of this book marked. Right: the kernel constant K_η of Table 2.1 (labelled M_η in the plot) against δ, following $\delta^{-5/2}$ for small δ. 39

2.4 The Lehmer pair near 7005 computed by the tilted Pólya route: $\Xi(\theta)e^{\pi\theta/4}$ (left) and the Thorin density at three resolutions (right), drawn through values on a grid of step 0.005, with the two ordinates dashed. 44

2.5 The Thorin density $\rho_{0.1}$ near the first two zeros, split into the kernel of γ_1, the kernel of γ_2 and the rest, that is all other zeros and the reflected kernels at $-\gamma_j$, computed from the first 1700 zeros without a tail correction. 47

3.1 Left: the four pieces of the sine squared decomposition at $\theta = 2$ for three basepoints, with the bracket $v_\alpha(2)$. Right: the pole piece and the condition number grow like $1/(\alpha-1)$ as the basepoint approaches one. 55

3.2 Pointwise difference $|\rho_\varepsilon - P_\varepsilon * \widetilde{U}|$ between the Pólya computed density and the Poisson smoothing of the model $\widetilde{U}$ of the RH atoms, at $\varepsilon = 0.5$, 0.05 and 0.005. 58

3.3 Cumulative Thorin mass $\mu_\varepsilon((0,\theta^2]) = \pi^{-1}\arg\xi(\frac{1}{2}+\varepsilon+i\theta)$ at $\varepsilon = 1$, 0.25 and 0.05 against the counting function of U_{RH}, on the whole window (left) and on $28 \le \theta \le 52$ (right). 60

3.4 The distances of Table 3.2 as functions of ε: the cell distance, the mass defect and the heat trace error at $t = 0.01$ vanish linearly, and the cumulative L^1 distance like $\varepsilon \log(1/\varepsilon)$. The matched total variation $\tilde{d}_{\mathrm{match}}(\varepsilon)$ is the floor $3.66 \times 10^{-9}\,\varepsilon$ of the tail model for the zero list. 61
3.5 The three signatures of Proposition 3.3.1 as functions of δ/ε: $\|g\|_{\mathrm{TV}}$ per displaced zero (part (i)), with its leading term $0.4135(\delta/\varepsilon)^2 = (3\sqrt{3}/(4\pi))(\delta/\varepsilon)^2$ dashed; the total variation of an omitted pair (part (ii)); and the negative mass per displaced zero (part (iii)). For $\delta > \varepsilon$, outside the range of part (i), the kernels of a displaced zero have total mass zero and curve (i) is flat at 1. 63
3.6 Left: the counting error $M_\varepsilon(\theta) - N(\theta)$ on $20 < \theta < 60$ at $\varepsilon = 0.5$, 0.1 and 0.02. At $\varepsilon = 0.5$ it is a sawtooth, like $-S$; as ε decreases it vanishes between the ordinates and concentrates in Cauchy steps of width ε at the zeros. Right: $M_\varepsilon(200)$, $79 + \tau_\varepsilon(200)$ and $M_\varepsilon(200) + \tau_\varepsilon(200)$, which equals 79 at every resolution (Proposition 3.4.2). . . 66
3.7 Left: absolute error of Selberg's smoothed prime side of $v_{3/4}$, at resolution 0.25, for three cutoffs, and (dotted) with the sum over the first 1700 zeros added. Right: the maximal absolute and relative errors against the cutoff, with $x^{-\varepsilon}$ for comparison. Both decay faster than $x^{-\varepsilon}$; the absolute error follows the single zero law $(x^{-\varepsilon} - x^{-2\varepsilon})/(\varepsilon^2 \log x)$. 72

4.1 Local Thorin density near γ_5 when the fifth zero is displaced to $\frac{1}{2} \pm 0.1 + i\gamma_5$ with its count kept, as in Proposition 3.3.1(i), for $\varepsilon = 0.3, 0.1, 0.05, 0.02$; the dotted curve is the density with the fifth zero on the line, at $\varepsilon = 0.3$. At $\varepsilon = \delta = 0.1$ the kernel contains $\frac{1}{2}P_0$, a point mass of weight $\frac{1}{2}$ at γ_5, which the plotted curve omits. The negative part shrinks like $\varepsilon/(\pi\delta)$ and the vague limit is positive. . . 79
4.2 Normalised alternating derivatives $(-1)^k \partial_w^k H_u(w)/H_u(w)$ of $H_u(w) = \varphi_\alpha(uv)\varphi_\alpha(u/v)$ for $1 \le k \le 6$, $u \in \{1, 25, 400\}$ and $w \in \{3, 20\}$, computed from the Pólya density, at the centre $\alpha = \frac{1}{2}$ and at the unconditional basepoint $\alpha = 1$. HCM requires positivity at every order, for every $u > 0$ and every $w \ge 2$; the figure shows finitely many cases. 81
4.3 Left: the sliver $\mathcal{E}_{T_0}$ of Theorem 4.3.10 for $T_0 = 2$ (dashed) and the discs D_γ of six hypothetical zeros $a + ib$ with $a > 2$ and $0 < b < \frac{1}{2}$, with the poles $-\gamma^2$ and $-\bar{\gamma}^2$ (dots) at the ends of their vertical diameters; the dot at 0 marks the origin. Right: for the toy reciprocal Xi function with the first 200 zeros of ζ and the pair $60 \pm 0.3i$, the set where $\operatorname{Im}\psi > 0$ (shaded) on a grid of 481×300 points, which lies inside the upper half of D_γ (solid); the cross is the pole $-\bar{\gamma}^2 = -3599.91 + 36i$. Drawn by `pakes_disc.py`. 98

5.1 Left: normalised Hankel minors of the heat trace at two times; positivity persists but falls below any fixed precision as t grows. Right: the first six $(-1)^k W^{(k)}(t)$, divided by their values at $t = 0.005$, from 1700 zeros. 121
5.2 Top: the Williamson measures U_k of the heat trace for $k = 10^4$, 10^5, 10^6, computed from 1700 zeros, as densities in $\theta = \sqrt{z}$. Bottom: the Poisson smoothed Thorin densities ρ_ε with $\varepsilon = 30/(2\sqrt{k})$, so that the widths match at $\theta = 30$. 129
5.3 Relative change $|W_F - W|/W$ of the heat trace, on a $\log_{10}$ scale, when the closest pair of zeros between heights 1000 and 2000 (near 1977.22) is moved to distance $\frac{1}{4}$ from the line, computed in forty digit arithmetic with W replaced by its sum over the first 1700 zeros. The change is at most 4×10^{-11}, near $t = 7 \times 10^{-7}$, and falls below 10^{-1000} by $t = 10^{-3}$. The sum over 1700 zeros is smaller than W, by 1.3 per cent at $t = 7 \times 10^{-7}$ and by a factor of 5.4 at $t = 10^{-8}$, so the curve bounds the relative change from above; with W from the explicit formula (1.2.5) the maximum is 3.66×10^{-11}, at $t = 7.2 \times 10^{-7}$. 139
5.4 Left: the heat trace from 1700 zeros, its leading exponential $e^{-\gamma_1^2 t}$, and the three arithmetic pieces, the archimedean one in absolute value. Right: digits lost to cancellation in the explicit formula; for $t \geq 0.05$ they lie $\log_{10} 2 + t/(4 \log 10)$ above the dashed line $\gamma_1^2 t/\log 10$. 147
5.5 First order cell errors $(\mu_\varepsilon(C_k) - 1)/\varepsilon$ for the 79 cells below height 200, computed from the first 1700 ordinates. Their absolute sum, 5.415, falls short of $\kappa_{\text{cell}} = 5.43665$ because the zeros above γ_{1700} are omitted. 150
5.6 The 1699 unfolded spacings of the first 1700 zeros against the GUE Wigner surmise and the Poisson law. 151

6.1 The function $M(\tau^2) = N(\tau) - \theta(\tau)/\pi = 1 + S(\tau)$ against $\tau = \sqrt{t}$, with the zero ordinates marked; under RH it is the distribution function of the Thorin measure of φ. The dashed curve is the gamma factor contribution $-\theta(\tau)/\pi$ alone. The function M rises on $(0, \tau_0)$; beyond τ_0, and so between consecutive ordinates, it decreases with slope $-\theta'(\tau)/\pi$, and at each ordinate it jumps by one. The mean level is 1. 164
6.2 The Thorin density of the gamma factor in the variable $\tau = \sqrt{t}$, namely $-\theta'(\tau)/\pi$, together with the first three Poisson kernels $c_n/(\pi(c_n^2 + \tau^2))$ of Theorem 6.2.1. The density is positive for $\tau < \tau_0 \approx 6.29$ and negative beyond, where the flat compensators outweigh the kernels. 171

6.3 Left: the Meixner density $p(\tau) \propto |\Gamma(\frac{1}{4} + \frac{i\tau}{2})|^2$. Right: the real and imaginary parts of the Bernstein function $\Phi_\Gamma(i\tau)$ of Proposition 6.2.11, written $\Phi(i\tau)$ in the legend. The real part is $2(\theta'(\tau) - \theta'(0))$ and grows like $\log\tau$; the imaginary part is the Meixner score $-\frac{d}{d\tau}\log p$ and tends to $\pi/2$, the exponential tail rate of p. 180

7.1 The function $w = W - W_0$ on a logarithmic scale, with its two components: the heat trace W over the zeros and the negative smooth part $-W_0$. The vertical line is $x_* \approx 0.00322$, where W_0 changes sign. To the left, w differs from $e^{x/4}$ by less than 2×10^{-13}; to the right, $-W_0 > 0$, and from $x_1 = 0.004$ on it dominates any contribution of off line zeros to W. The faint curve is the asymptote $c/(2\sqrt{\pi x})$. 194

7.2 The density f of the infinitely divisible law X_φ with $\mathbb{E}e^{-sX_\varphi} = \varphi(s)$, computed by Talbot inversion, on logarithmic axes. It starts at $f(0^+) = -\zeta(\frac{1}{2})/4$ and decays like $\frac{c}{2\sqrt{\pi}}x^{-3/2}$. In the legend, X is X_φ. 197

7.3 The Cardon ratio for the random harmonic law H_{κ^*}. Left: $\log r(t)$ on the real axis, asymptotically linear with slope $\frac{7}{2}$ (dashed). Centre: $L'(s)$ for $L(s) = \log(r(\sqrt{s})/r(0))$, positive, decreasing and convex as a Stieltjes function must be. Right: the continuation $G(x) = r(-ix)$ on $[0, \pi/4)$, with a simple zero at $x = 0.6388$. 214

7.4 Left: real parts of the zeros of F_n near $\rho_j/2$, in the variable $u = w + 1$, against $1/n$ for $n = 4$ to 64, with the first order lines $\frac{5}{4} + \operatorname{Re}\kappa(\rho_j)/n$ (dotted). The zeros approach $\operatorname{Re} u = \frac{5}{4}$, the image of the critical line, from the left. Right: $n(\frac{5}{4} - \operatorname{Re} u)$ against n, converging to $-\operatorname{Re}\kappa(\rho_j)$. 221

8.1 The law of the clock variable computed from the Pólya density by Talbot inversion (points) and by the spectral expansion with Pólya residues (lines): density, survival function, and the distance of the hazard from γ_1^2, whose logarithm has slope $-(\gamma_2^2 - \gamma_1^2)$ (dashed). Beyond $t \approx 0.17$ the last curve reflects the double precision of the plotting script, not the hazard. 245

8.2 $|\Xi(z)|$ against the Monte Carlo standard error with 10^6 samples of Williams' variable S_2, at the sampled values of z and at the first zero, joined by straight lines. 247

8.3 Relative error of Kent approximants to the clock transform, without (dashed) and with (solid) the Weyl tail. 249

8.4 The Kent approximants at large w, with the dotted lines at $w = \Theta_N^2$ (labelled T_N^2 in the plot). Moment matching gains ten to twelve digits at $w = 1$ and about two as w approaches Θ_N^2. Beyond Θ_N^2 the Weyl tail levels off at a floor set by S, while the error of the matched tail, whose density is rescaled by λ, grows like $|\lambda - 1|\sqrt{w}\log w$. 251

8.5 Modulus of the clock transform along the parabolas on which every first passage mixture with drift c is bounded by one (dashed). Each curve crosses the bound, so the clock is in none of these classes. The abscissa is the parameter q of the parabola $u = \frac{1}{2}(-q^2 + 2icq)$, for $c = 0$, 10, 30, 50 and 100. 253
8.6 Left: the tangent boundary and its asymptotic slope $-\sqrt{2}\gamma_1$. Right: the distribution function of the clock against that of the simulated crossing time of the boundary. 255
8.7 Left: the Brownian boundary of the clock computed from the Volterra equation, the Roberts Shortland tangent boundary, and the right tail asymptote with its logarithmic correction, over the three zones. Right: the simulated distribution function of the crossing time minus that of the clock, for both boundaries; the grey band is $\pm 1/\sqrt{n}$, twice the largest standard error of a simulated distribution function. 258
8.8 The first hundred Keiper Li coefficients from the Pólya density and Keiper's asymptotic. 261
8.9 The Li test functions $4\sin^2(n \arctan \frac{1}{2\theta})$ for $n = 1, 10, 50, 100, 200$, in the square root chart $\theta = \sqrt{z}$, against the zeros below height 200 (grey vertical lines). Larger n probes higher zeros. 262
8.10 Zeros of the closest pair below height 200 along the de Bruijn Newman flow, refined by Newton's method as in Table 8.7, for $t \geq 0$ and $t < 0$, and the computed collision time (dashed line). 266
8.11 The function $\theta \mapsto H_t(2\theta)e^{\pi\theta/4}$ near the Lehmer pair at height 7005, divided by its maximum absolute value on the window, for four times of the de Bruijn Newman flow, computed with the tilted Pólya route. At $t = 0$ it has two sign changes; at $t = -7 \times 10^{-4}$, slightly above the collision time -7.11×10^{-4}, they nearly coincide; at $t = -10^{-3}$ the pair has left the real axis. 267
8.12 The Thorin density $\rho_{0.1}$ of $L(s, \chi_{12})$ for $0 \leq \theta \leq 40$ on a logarithmic scale, computed from the Dedekind eta kernel, with the zeros dashed. 290
8.13 Left: the prime pieces of the sine squared decomposition at the basepoint $\alpha = 2$, written σ in the plot, for ζ, $L(s, \chi_{12})$ and $L(s, \Delta)$, and the pole piece of ζ. For ζ the prime piece is positive and the pole piece negative; for the two eta based functions there is no pole piece and the prime piece takes both signs. Right: the Thorin densities $\rho_{0.1}$ of $L(s, \chi_{12})$ and $L(s, \Delta)$ on a logarithmic scale, peaking at their zeros (vertical lines). 294
8.14 The Hermite Biehler test with the cut at $y = 1$: shaded where $|A(1-s)| > |A(s)|$ for $s = \sigma + i\theta$, $\frac{1}{2} < \sigma \leq 2.5$, with the curves of equal modulus and the zeros of $\xi(\frac{1}{2} + i\theta, \Delta)$. Drawn by `maxexp_figs.py`. 300

8.15 Left: the Bondesson function N_1 of M_1, nonnegative, with its asymptote $\frac{1}{3}\sqrt{3z/\pi}$; each drop is a zero of the transform at $-b_{6k}$. Right: the poles $-2\pi m$, $m \le 60$, and the zeros of $\psi^*(\theta) = \mathbb{E}e^{-\theta M^*}$: the 25 real zeros above $-2\pi\times 61$, one in each gap with $\tau(m+1)\tau(m+2) > 0$, and the five pairs of nonreal zeros, the only ones with $|\operatorname{Re}\theta| \le 2\pi \times 60.5$ and $|\operatorname{Im}\theta| \le 100$. Drawn by `maxexp_figs.py` from `maxexp_psizeros.json`. 312

8.16 Left: the kernel $K_\Delta(v) = e^{6v}\Delta(ie^v)$ and Pólya's kernel $e^{-2\pi\cosh v}$, each divided by its value at 0. Right: the logarithm of the multiplier $H(v) = \sqrt{F(e^v)F(e^{-v})}$; $-\log H$ expands with the positive coefficients $12\sigma(m)/m$. Drawn by `maxexp_figs.py`. 315

8.17 The real zeros of H_t^Δ below height 52 for $-1.1 \le t \le 0.5$, and the seven collisions of Table 8.19; all zeros move down as t grows. Drawn by `maxexp_figs.py` from `maxexp_flow.json`. 317

8.18 Left: $v_\alpha(\theta) = \operatorname{Re}(\xi'/\xi)(\alpha+i\theta,\chi_4) = \pi\rho_{\alpha-1/2}(\theta)$ for $\alpha = 1, 0.75, 0.55$ on a logarithmic scale, with the ordinates of the zeros of $L(s,\chi_4)$ (dashed). As $\alpha \downarrow \frac{1}{2}$ the density concentrates on the ordinates, while its minimum b_α, at the origin, tends to zero. Right: at $\alpha = \frac{3}{2}$, the drift and gamma piece $b_\alpha + G_\alpha$, the prime piece ϖ_α, of both signs, and their sum v_α. Drawn by `dbeta_figs.py`. 329

8.19 Left: the cumulative Thorin mass $\frac{1}{\pi}\arg\xi(\alpha+i\theta,\chi_4) = \frac{1}{\pi}\int_0^\theta v_\alpha$ for $\alpha = 1$, 0.75 and 0.55, against the zero count $N(\theta)$ of $L(s,\chi_4)$; at $\theta = 30$ the masses are 9.7747, 9.8405 and 9.9621, and $N(30) = 10$. The argument is continued through the quotients of values at steps of 0.005, each of argument below 0.11 in modulus. Right: $v_{0.75}(\theta)$, its minimum $b_{0.75} = 0.0389784$ at the origin, and the lower bound of Corollary 9.1.9 with $\alpha' = 3.15$, which exceeds $b_{0.75}$ for $\theta > 2.434$ (`dbeta_minimum.py`). Drawn by `dbeta_figs.py`. 335

8.20 Left: the weights of the nonnegative solution for $\rho_{0.1}$ (stems) and the zeros (dashed), with the data window $[15, 55]$ shaded. Each zero in the data window carries adjacent weights of total mass one to within half a per cent; the zeros at 14.13, 56.45, 59.35 and 60.83, outside it, carry none. Right: the excess residual against the displacement δ of the fifth zero, the δ^2 law through the point at $\delta = 0.05$, and the off grid baseline $r_0 = 0.106$. 339

8.21 Left: the kernel Φ of (1.2.3), Pólya's asymptote Φ^* and the kernel $e^{-2\pi\cosh 2u}$ of P, each divided by its value at $u = 0$. Right: the zero counting functions of Ξ, Ξ^* and P on $(0, 100]$, against $N_0(T) + \frac{7}{8}$ and $N_0(T) - \frac{1}{4}$, the mean values of the Riemann von Mangoldt formula and of Proposition 8.7.8, with $N_0(T) = \frac{T}{2\pi}\log\frac{T}{2\pi e}$. Drawn by `biane_figs.py` from `biane_polya.json`. 342

8.22 The Wick rotation of the Laplace variable. Left: $1/\mathbb{E}e^{-uJ_1^{(\kappa)}}$ for $\kappa = 1, 0, -1$, positive and increasing for $u > 0$ and a damped sine for $u < \frac{\kappa}{2}$, vanishing at minus the Thorin atoms $\pi^2 n^2/2 - \kappa/2$ (marks). Right: $1/\mathbb{E}e^{-uH_1^{(1)}} = e^{\sqrt{1+2u}-1}$, real part and modulus; beyond the branch point $u = -\frac{1}{2}$ (dotted) the modulus is constant, e^{-1}, and there are no zeros, the signature of a Thorin density. Drawn by `biane_figs.py`. 349

8.23 The clock from $R_0 = 2\pi e^{2.5}$. Left: $\mathbb{E}e^{-u\tau_{R_0}}$, exact and from the bridge corrected Monte Carlo with two standard errors, on a logarithmic scale; the dotted lines mark the first three Thorin atoms $\nu_1^2 = 95.4289$, $\nu_2^2 = 154.965$, $\nu_3^2 = 215.648$ of τ_∞. Right: the histogram of the corrected hitting times against the exact densities of τ_{R_0} and of τ_∞, by Talbot inversion. Drawn by `biane_figs.py` from `biane_clock.json`. 359

8.24 The real zeros in $(0, 90)$ of P_t (left) and of Ξ_t (right) for $-60 \le t \le 0$, on the grid of Table 8.32 (step 0.5 above $t = -20$, step 1 below). The zeros of P_t drift upward and leave the window one at a time; those of Ξ_t collide in pairs. Drawn by `biane_figs.py` from `biane_flow.json`. 365

9.1 Left: the Thorin density $\rho_{1/2}(\theta) = v_1(\theta)/\pi$ of T_1 on $[0, 100]$, with the ordinates of the zeros as thin vertical lines. Right: the cumulative Thorin mass $\int_0^T \rho_{1/2}$ against the zero count $N(T)$. The plot writes ρ_1 for $\rho_{1/2}$ and s for u. 379

9.2 Error of the compensated sine squared sum at the basepoint one . 382

9.3 Left: the signed canonical measure $x\Pi_1(dx)$ of $\log \xi(1+s)$, made of the gamma density minus the pole, $\big(\frac{e^{-3x}}{1-e^{-2x}} - 1\big)dx$, which is negative beyond $x = 0.28120$, and the prime atoms $\Lambda(n)/n$ at $\log n$ for $n \le 60$. Right: the bracket $v_1(\theta) = \pi\rho_{1/2}(\theta)$ on $[0, 60]$ assembled by the sine squared lemma from the drift b_1, the gamma piece $A(\theta)$ and the compensated prime piece $Q(\theta)$; the plot writes ρ_1 for $\rho_{1/2}$ and $P(\theta)$ for $Q(\theta)$. Drawn by `figs3.py`. 385

9.4 Left: absolute error of the compensated formula (9.1.8) at $\theta = \gamma_1$ against the prime cutoff X for $\alpha = 1$, 0.9, 0.75 and 0.6, with the rates $X^{1/2-\alpha}$ (dotted). Right: at the centre, $\alpha = \frac{1}{2}$, the compensated sum at the first ordinate grows like $\log X$ (dashed line), the atom of the RH measure forming. Drawn by `figs4.py`. 392

9.5 Left: the empirical Laplace transform of $\widetilde{T} + H_{\widetilde{T}/\sqrt{2}}$ from 10^6 draws, with three standard errors, against $\varphi_1(u) = \xi(1)/\xi(1+\sqrt{u})$; the plot writes s for u. Right: $\sqrt{x}\,P(T_1 > x)$ from the simulation, written with X_1 in the plot, and the limit $\sum_\rho \rho^{-1}/\sqrt{\pi} = 0.01303$ of (9.1.3). Drawn by `figs.py`. 410

9.6 Left: the density of T_1 by Talbot inversion of φ_1 and by the subordination formula (9.2.1), with the tail $0.00652\,x^{-3/2}$. Right: the densities v_α/π at the basepoints $\alpha = 0.6$, 0.75 and 1, with the ordinates of the zeros; they sharpen onto the zeros as α decreases, and for $\alpha < 1$ they are Thorin densities of T_α only if no zero lies to the right of the line $\operatorname{Re} s = \alpha$. The plots write X_1 for T_1, ρ_α for v_α/π and s for u. Drawn by `figs2.py`. 411

9.7 Left: the exact Brownian boundary of T_1, written X_1 in the plot, from the Volterra equation over the three zones of Section 8.2.2.2 (shaded: small time, bulk, right tail), the tangent approximation, which has no solution beyond $t = 0.091$, the flat level -0.19277, and the value $\mathbb{E}\,b(T_1) = \sum_\rho \rho^{-1}/\sqrt{2} = 0.0163$. Right: survival of 1.5×10^5 simulated Brownian paths across the exact boundary against the exact survival function of T_1 and the tail $0.01303\,t^{-1/2}$; the Kolmogorov distance is 0.0019. 415

10.1 Left: the Thorin densities v_α/π at $\alpha = 0.6$, 0.75, 1, written ρ_α in the plot, and their Cauchy floors b_α/π (dotted). Middle: $\pi y^2 f_{Y_\alpha}(y)/b_\alpha$, tending to one. Right: the Poisson traces K_1 and $K_{1/2}$ against $b_1/(\pi y)$. Drawn by `figs2.py` from `strip.json` and `comp.json`. . . 431

10.2 Left: the weight $\gamma^n e^{-\gamma y}$ of the condition (n, y) in (10.1.3), normalised to maximum one, for $(n, y) = (100, 1)$ and $(300, 10)$, peaking at $\gamma = n/y$; the marks are the ordinates below 200. Right: the sign factor $\cos(n\varphi - \delta y)$ of a hypothetical zero off the line for $(\gamma, \delta, y) = (50, 0.25, 1)$ and $(14.13, 0.5, 0.5)$, positive below the orders 365 and 52 of Table 10.5, at which it is first negative. Drawn by `biane_figs.py` from `biane_traces.json`. 435

10.3 Left: $2\pi y K_{1/2}(y)$ and $2\pi y K^P(y)$ against $\log \frac{1}{2\pi y} - \gamma_E$ for $10^{-4} \le y \le 0.05$, with the main term $2\pi y M(y)$ (dotted). Right: the difference $D(y)$, the limit $\frac{9}{8}$ (dotted) and $\frac{9}{8} - cy \log \frac{1}{y}$ (dashed). Drawn by `biane_figs.py` from `biane_traces.json`. 439

10.4 Left: best L^1 error against the number of translates, with the barrier at the centre. Right: the Fourier transform of the best approximant at the first ordinate, against the target $\hat{g}(\gamma_1)$. The data, which include $N = 30$ and $N = 60$, are in `wiener.json`, written by `wiener.py`, and `figs_wiener.py` draws the figure. 442

10.5 Left: $|\eta(\sigma+i\theta)|$, the Fourier transform of Salem's kernel divided by $\Gamma(\sigma+i\theta)$, at $\sigma=\frac{1}{2}$, 0.6, 0.75, 1. Right: the Riesz function $R(x)/x^{1/4}$ from the Möbius series and from the explicit formula (10.1.7) over 60 zeros. The data are in `salem_riesz.json`, written by `salem_riesz.py`; `figs_wiener.py` draws the figure and evaluates the explicit formula with the first 60 ordinates in `g_1_400.npy` and the trivial terms in `riesz_explicit.json`. 443

10.6 Left: the weights $|\Gamma(1-\rho/2)/(2\zeta'(\rho))|$ of the Riesz function and $|\Gamma(1-\rho/2+ih)/(2\zeta'(\rho))|$ of the twisted Riesz function with $h=50$, for the first 300 zeros. Right: $\operatorname{Re} R_{50}(x)/x^{1/4}$ from the Möbius series against the explicit formula (10.2.1) over 300 zeros. The plot writes T for h. Data from `twisted.py`; drawn by `figs_strategy.py`. 450

10.7 Left: $R(x)/x^{1/4}$ from the Möbius series and from the explicit formula over 60 zeros, as in the right panel of Figure 10.5. Middle: $1-1/\zeta(2k)$ and its Monte Carlo value from the clock (Proposition 10.2.3), against 4^{-k}. Right: base ten logarithm of the largest term in Riesz's power series and in the clock series of Proposition 10.2.4 for $T=0.01$, 0.023 and 0.05. Drawn by `figs_riesz.py` from `salem_riesz.json`, `riesz_explicit.json` and `riesz_thorin.json`. 452

10.8 The normalised error $\Delta(x)/x^{1/4}$ of the squarefree count on $[100, 3000]$ (grey) and the explicit formula (10.3.11) with the first 60 zeros and the trivial terms at $s=-1$, -3 and -5, divided by $x^{1/4}$ (dark): the residue at $s=0$ plus the contributions of the first sixty Thorin atoms γ_k^2. Pearson correlation 0.915. Drawn by `sqfree_atoms.py`. 472

11.1 The densities of the centre clock T and of the shifted clock T_1, written X_1 in the plot, on a logarithmic axis. The density of T is drawn on a grid of spacing 0.0005 on $[0.004, 0.4]$, from the Talbot inversion and the spectral expansion of Section 8.1; that of T_1 on the 40 point geometric grid of the Talbot inversion behind Table 9.11, which is why its curve is piecewise linear. The legend gives, for each, the grid point of largest density. Assembled by `comp.py` and drawn by `figs.py`, both in the folder `sato_bondesson`. 492

11.2 Left: the Lévy density $h = -W'$ of the driving process of the centre clock on $[0.002, 0.2]$, with the heat time t written x in the plot, from the first 1700 zeros (curve) and from the explicit formula at the five points of Table 11.1 (dots). Right: the survival function of T_1, written $P(X_1 > x)$ in the plot, its Lévy tail $\nu_{T_1}((x,\infty))$, written $\nu((x,\infty))$, and their common asymptote $b_1(\pi x)^{-1/2} = 0.01303\,x^{-1/2}$, at $x = 0.02$, 0.05, 0.1, 0.5, 1, 5, 10, 50 and 100. Computed by `weil.py` and `comp.py` and drawn by `figs.py`, all in the folder `sato_bondesson`. 494

11.3 Four processes of the map. Top left: paths of a three dimensional Bessel process run until they reach the level one; the hitting time has mean $\frac{1}{3}$, and two independent copies make up the variable S_2 of Section 9.1.1. Top right: ten units of time of the Riemann Ornstein Uhlenbeck process F, written Y in the plot, whose stationary law is the centre clock, with the mean $\mathbb{E}T$, written $\sum \gamma^{-2}$ in the plot. Bottom left: the exact Brownian boundaries of T_α at $\alpha = \frac{1}{2}$, 0.6, 0.75 and 1, those at 0.6 and 0.75 computed under RH beyond $t = 0.03$ (Section 9.2.6), asymptotically linear at the centre and flat above it. Bottom right: Brownian paths crossing the boundary of T_1, written X_1 in the plot (Section 9.2.4). Drawn by `figs.py` in the folder `basepoint_one/process_map` from the output of `ou.py`, the boundaries of `bnd_alpha.py` at $\alpha = 0.6$ and 0.75, and those of Sections 8.2 and 9.2 at the centre and at the basepoint one. 505

11.4 Processes attached to $\xi(\alpha + \cdot)$, with $a = \alpha - \frac{1}{2}$. By rows from the top: the Mellin side, the centre clock, the basepoint clocks, and two rows of further representations. Drawn in TikZ from Table 11.6. . 506

11.5 Left: the distribution function $W(\theta^{-2})$ of the Thorin measure of $\Upsilon(T)$ in the chart $\theta = \sqrt{u}$, against the zero count $N(\theta)$, which under RH is the distribution function of the Thorin measure of T in the same chart. Right: the densities of T, of $\Upsilon(T)$ and of $\Lambda_{\mathrm{BP}}(T)$ on $[0, 0.1]$; the last vanishes to the left of $c = 0.0153$. Computed by `wigfree_upsilon.py` and drawn by `wigfree_plots.py`. 518

11.6 The Jacobi parameters α_n and $2\beta_n$ of ν, $1 \le n \le 49$, on logarithmic axes, against $\frac{1}{2}L_n$ and $\frac{1}{2}L'_n$ from (11.2.13) with the smooth density and against the cruder rule $\frac{1}{2}\gamma^{-2}_{2n/\pi}$. Computed by `wigfree_jacobi.py` and drawn by `wigfree_plots.py`. 521

11.7 The flow $e^{-(t/2)D^2}p$ for p with the roots -3, -1, $\frac{1}{2}$, 2 and $1 \pm 0.8i$. Left: the real parts of the six roots for $0 \le t \le 0.4$; the complex pair has one real part until it meets the axis at $t^* = 0.3203$ (dashed), where its two roots split. Right: the imaginary part of the pair, against $\sqrt{0.64 - t}$ for the pair on its own, which reaches the axis at $t = 0.64$. Computed by `wigfree_heat.py` and drawn by `wigfree_plots.py`. 529

11.8 Left: the Thorin densities at the basepoint one for $L(s, \chi_{12})$ and for ζ on $[0, 60]$, with the ordinates of the zeros of $L(s, \chi_{12})$ (vertical lines). Right: the cumulative Thorin mass for $L(s, \chi_{12})$ against its zero count $N_{\chi_{12}}(T)$, written $N(T)$ in the plot. Drawn by `figs.py` of `basepoint_one/eta`. 532

11.9 Left: the canonical measure $x\Pi(dx)$ of $\log \xi(1 + s, \chi_{12})$, written with $\Lambda(1+s, \chi_{12})$ in the plot: the positive gamma density $e^{-x}/(1 - e^{-2x})$ and the prime atoms $\Lambda_{\chi_{12}}(n)/n$ at $\log n$, of both signs, the positive ones drawn upward and the negative ones downward; there is no pole. Right: $v_1(\theta)$ on $[0, 40]$, written $\pi\rho_1$ in the plot, as the sum of the drift b_1, written b, the gamma piece $A(\theta)$ and the prime piece $Q(\theta)$, written $P(\theta)$ in the plot. Drawn by `figs.py` of `basepoint_one/eta`. 534

11.10 Left: absolute error at $\theta = \gamma_1 = 3.8046$ for $L(s, \chi_{12})$ against the prime cutoff X, for $\alpha = 1$, 0.9, 0.75 and 0.6, from primes alone (solid) and after the zero correction over 297 zeros (dashed). Right: at the centre and the first ordinate the sum grows like $\log X$, the atom of the GRH measure forming. Drawn by `figs.py` of `basepoint_one/eta`. 535

11.11 Left: the exact Brownian boundary of T_1 for $L(s, \chi_{12})$, written X_1 in the plot, over the three zones of Section 8.2.2.2 (shaded), the tangent approximation, which fails at $t = 1.83$, and the flat level -0.2747. Right: survival of 1.5×10^5 simulated Brownian paths across the exact boundary against the exact survival function; the Kolmogorov distance is 0.0014. Drawn by `figs.py` of `basepoint_one/eta`. 537

11.12 Left: negative mass of the density on $[0, 2000]$ against the basepoint α for fifteen hypothetical zeros at real part β (`more2.py`). Middle and right: the density $v_{0.6}/\pi$ near the heights 300 and 10^5, alone and with a hypothetical pair of zeros at $\beta = 0.7$ or 0.9 at the centre of the window (`high.py`). Drawn by `figs_almost.py`. 549

11.13Left: the zero density exponents of Theorem 11.4.2 and the region where they give a power saving; the exponent of Guth and Maynard is drawn on $[\frac{7}{10}, 1)$, where it holds by their estimate and, above $\frac{4}{5}$, by Huxley's [100]. Middle and right: the density v_α/π at $\alpha = 0.6$ and 0.7 near the hypothetical off line zero at height 292.4 (dotted), with and without the hypothetical zeros, evaluated as in `almost.py` with step 0.005. Drawn by `figs_almost.py`. 549
11.14Positivity margins of the Thorin density at midpoints: the 1% quantile and the median of $\bar{\rho}_\kappa$ against the resolution κ, for the zeros below height 2198 and for CUE eigenvalues. Drawn by `figs_strategy.py` from the output of `sens.py`. 550

12.1 The Thorin path to RH. The sine squared lemma connects the arithmetic to the Thorin density. For $\alpha \geq 1$ the shifted clock is a GGC with no hypothesis, and Chapters 9 to 11 describe it explicitly; for $\frac{1}{2} < \alpha < 1$ each of their constructions is equivalent to a zero free half plane, and unconditionally the shifted clock is an almost GGC; at the centre the GGC property is RH. The programme must carry the GGC property across the strip. 599

List of Tables

1.1 Principal results of Chapters 2 to 5. 6
1.2 Principal results of Chapters 6 to 8. 7
1.3 Principal results of Chapters 9 to 11. All are unconditional, and several use verified computations: the verification of RH below height 1000, through Theorem 5.1.1, or below the height $T_0 = 3 \times 10^{12}$, Platt's verification of GRH for small moduli, or computations in ball arithmetic. 9
1.4 The van Dantzig pair, computed by `vdpair.py`. The last column is the larger relative difference between a member and its Hadamard product, $\prod(1-\theta^2/\gamma^2)$ or $\prod(1+\theta^2/\gamma^2)^{-1}$, over the 1700 zeros below height 2198, with the zeros above entering through $\sum_k a_k^{-1}$ and $\sum_k a_k^{-2}$; it is the size of the neglected term of order θ^6. On $(0, 30]$ the first member changes sign at $\gamma_1 = 14.13$, $\gamma_2 = 21.02$ and $\gamma_3 = 25.01$; the second is positive and decreasing. Both are $1 - \theta^2\mathbb{E}T + O(\theta^4)$. 14

1.5 The heat trace from the zeros and from the explicit formula, after [200, Table 4.1]. At $t = 1$ the cancellation between the arithmetic terms reaches about eighty seven digits. 16
1.6 The bracket $v_\alpha(\theta)$ of the sine squared lemma computed from the canonical measure (drift, archimedean integral, pole term and primes up to 2×10^6) and from the Pólya density; $\theta = 14.1347$ is the first ordinate γ_1, rounded. The Pólya values agree with an independent evaluation of $\mathrm{Re}(\xi'/\xi)$ through Γ and ζ to about 10^{-15}. 20
1.7 The Bondesson hierarchy on the clock, whose Lévy density is $W(t)/t$. 24
1.8 Diagnostics for the Thorin condition. The pole test, the Thorin density and the zero density scan locate a failure in height; the others detect it only globally, and the Bernstein test cannot fail for the clock. 25
1.9 GGC Laplace transforms in the variable u, their Thorin measures in z, and the corresponding laws. 26

2.1 Size of the Pólya kernel on lines inside its strip of analyticity, computed on a grid of step 0.002 over $|x| \le 6$; the first two rows are the lines $\eta = 0.1$ and $\eta = \pi/8$. The growth is close to $\delta^{-5/2}$, as the majorant in the proof of Proposition 2.1.1 predicts. 36
2.2 The tilted Pólya route at $\varepsilon = 0.05$, compared with `mpmath` at 50 digits. Errors are absolute errors of ξ'/ξ, which bound those of its real part, and $<$ marks the entries at the floor set by rounding the tilted value to 45 significant digits. The rows with 2001 nodes use the step $h = 0.004$ and the truncation $L = 4$, those with 8401 nodes $h = 0.001$ and $L = 4.2$. The last column is the number of digits, $0.341\,\theta$, that the untilted route loses to cancellation at the same height. 38
2.3 Working precision needed by the untilted and the tilted Pólya route for D target digits. The untilted bits are $(D + \pi\theta/(4\log 10))\log_2 10$ and the tilted bits $(D + \delta\theta/\log 10 + 5)\log_2 10$, where the 5 digits cover $\log_{10} K_\eta < 5$ for $\delta \ge 0.01$. At $\theta = 7005$ the second condition in (2.1.4) asks for $h \le 8.8 \times 10^{-4}$, so 8400 nodes give 25 digits there only with a truncation near $|x| \le 3.7$, the smallest that suffices. . . 38
2.4 The Riemann Siegel formula with Turing's method, the Odlyzko Schönhage algorithm and the Pólya route: what each computes, where, with what certificate, at what cost per point, and how far it has reached. 42

3.1 The four pieces of the sine squared decomposition (3.1.1), with the prime sum over $n \le 2 \times 10^6$, and the bracket $v_\alpha(\theta)$ computed by the Pólya route; it differs from the sum of the four columns by the prime tail beyond 2×10^6 (1.3×10^{-3} at $\alpha = 1.5$, $\theta = 2$). The pole column is the signed contribution, minus the pole piece. The pole and prime pieces grow like $1/(\alpha - 1)$ and cancel to leave a bracket of order one or smaller; κ_α measures the loss. 54

3.2 Distances between the Pólya computed Thorin measure μ_ε and U_{RH} on $0 < z \le 200^2$. Mass is $M_\varepsilon(200)$. Raw TV is $\|\mu_\varepsilon - U_{\mathrm{RH}}\|_{\mathrm{TV}}$, forced to equal mass plus 79 by Proposition 3.2.1. Matched TV is $\tilde{d}_{\mathrm{match}}(\varepsilon) = \|\mu_\varepsilon - P_\varepsilon * \widetilde{U}\|_{\mathrm{TV}}$, and sup diff is the maximum of $|\rho_\varepsilon - P_\varepsilon * \widetilde{U}|$ over $0 < \theta \le 200$. Cell TV is $d_{\mathrm{cell}}(\varepsilon)$, L^1 in θ is $\int_0^{200} |M_\varepsilon - N|\, d\theta$, and $W_\varepsilon(0.01)$ is the smoothed heat trace at $t = 0.01$, with the value from the zeros in the last row. 58

3.3 Proposition 3.4.2 at $T = 200$, computed by the Pólya route with the argument followed on a grid (computed, not certified). The cumulative Thorin mass plus the short horizontal correction recovers the integer $N(200) = 79$ at every resolution, so the zero density $\nu(\frac{1}{2} + \varepsilon, 200)$ vanishes for all eight values of ε. 68

3.4 Selberg's smoothed prime side of the Thorin density bracket $v_{3/4}(\theta)$, at resolution $\varepsilon = 0.25$, against the Pólya values on 501 equally spaced points of $10 \le \theta \le 60$, with the primes alone and with the sum over the first 1700 zeros added. The relative error is $|\mathrm{error}|/v_{3/4}(\theta)$. 71

4.1 First order n with $(-1)^n h^{(n)}(w_0) < 0$ for the half Cauchy pair function, with $u = e^a$. The entries agree with the closed form (4.2.3), which gives 1645 for the entry beyond 400. For $\beta = 2$ the pair function is proportional to $1/(w + 2\cosh a)$, which is completely monotone at every order. 84

4.2 The dictionary. In the Riemann column the Mellin transform (Section 1.2 and Theorem 4.3.26) and the reciprocal Xi function are unconditional, the log variable and the Thorin measure presume RH, and the exit time, the hitting time of Kent's theorem (Section 8.1.3), presumes RH and simple zeros; Section 9.2 gives hitting times at the other basepoints. The last entry is the necessary condition of Theorem 4.3.28, which assumes RH, simple zeros and a growth condition on $1/|\zeta'(\rho)|$; sufficiency is open (Question 4.3.29). 86

4.3 Bosch's family and the reciprocal Xi function $\varphi(s) = \xi(\frac{1}{2})/\xi(\frac{1}{2} + \sqrt{s})$. For the Bosch column $q = 1/|\beta|$; for the Riemann column $p = 1/|\beta|$ and $\varrho e^{i\delta}$ runs over the zeros of Ξ. 101

4.4 Where the lower bound (4.3.4) becomes positive. On a logarithmic grid from t_e to about $10^{14}t_e$, $m_p(t) > 0$ at every grid point $t > t_1$; the ray reaches t_1 at $\tau_1 = t_1^{2/p}$. The entries are computed, not proved. 104

4.5 HCM threshold for powers of W in logistic models with frequencies removed, computed as $\beta_* = \pi/h_*$ from the width h_* of the largest strip on which $\mathrm{Im}(-g'/g) \geq 0$. The half Cauchy row is Bosch's theorem, not a computation. 111

5.1 Normalised leading Hankel minors of the heat trace, $\det[c_{i+j}]_{i,j<N} / \prod_{i<N} c_{2i}$ for the even block and $\det[c_{i+j+1}]_{i,j<N} / \prod_{i<N} c_{2i+1}$ for the odd block, from the first 1700 zeros in sixty digit arithmetic. . 120

5.2 The lowest ordinates from the Stieltjes moments $m_0, \ldots, m_{2N-1}$, computed from the Taylor coefficients of Ξ at the centre: the number returned by the Jacobi matrix of order N to within 10^{-6}, the errors of the first and of the last of them, the last Cholesky pivot of $\mathcal{H}_N$, and the least number of digits, in steps of 10, of the Taylor coefficients and of the arithmetic from which on the same ordinates are returned. 122

5.3 The Gaussian smoothing of the zeros from the zeros and from the primes, by (5.1.3). $\mathcal{G}(u,t)$ is summed over the zeros below height 2197.26, the rest adding less than 10^{-70000}; the polar term and $\mathcal{G}(-u,t)$ are below 10^{-8000}. P_N is P with the sum cut at $n \leq N$, and the P column is P_{10^9}. The bound on the rest of the prime sum that ignores its phases, from $\psi(x) < 1.03883x$ [226, Theorem 12], is 1.6×10^{-20}, 2.1×10^{-10}, 1.0×10^{-8} and 1.0×10^{-4} at $N = 10^9$ for the four rows; in the first row the difference at $N = 10^9$ is the rounding error of the sums, in extended precision. 133

5.4 Growth of Ξ along the imaginary axis against the saddle point value from the Gumbel tail of the Pólya density. 135

5.5 The first failing order of the derivative signs when the closest pair of consecutive zeros in a window of height is moved to the quadruple $\frac{1}{2} \pm \delta \pm ig$, g the midpoint of the pair. k^* is the least order at which $c_k^F(t) < 0$ for some $t > 0$, found by a scan of every order up to 500, a grid of ratio 1.002 above it and every order of the last grid interval (computed, not certified); $c_{k^*}^F(t^*) < 0$ is certified in ball arithmetic. The last column is $c_{k^*}^F(t^*)$ divided by $(k^*/t^*)^{k^*} e^{-k^*}$, the value at $x = k^*/t^*$ of the weight $x^{k^*} e^{-xt^*}$. 140

5.6 What finite tests establish and what they cannot exclude. 142

5.7 The heat trace from the explicit formula and from the zeros. For $t \geq 0.05$ the digits lost are $\log_{10} 2 + (\gamma_1^2 + \frac{1}{4})t/\log 10 \approx 0.30 + 86.88\,t$. 147

6.1 Thorin measures of φ, G and R and of the building blocks used in this chapter and in Chapter 7, with the convention $\log f(s) = -\int \log(1+s/t)\,U(dt)$ after normalising $f(0)=1$. Here $b = \frac{1}{2}(\gamma_E + \log\pi)$, $\kappa > 0$ and $c = -\theta'(0) = \zeta'(\frac{1}{2})/\zeta(\frac{1}{2})$. The rows for φ and R are Theorem 6.1.10, the row for G is Theorem 6.2.1, the fourth row is Lemma 6.1.7, the fifth and sixth appear in Table 1.9, and the last is Theorem 6.2.12. 158

6.2 The two questions across the critical strip. Middle column: the Lévy representation, in the variable τ for $\sigma > 1$ and in the heat kernel form of Theorem 7.1.12 for $0 < \sigma < 1$, where the Lévy measure is positive when $w_\sigma > 0$. Right column: whether R_σ is a GGC (Theorem 7.1.14). 169

6.3 Meixner Pollaczek coefficients of $\frac{1}{4}s(s-1)\zeta(s)$, equivalently Hermite coefficients of $\Omega(x) = \sum_{n\geq 1} g(nx) = \frac{1}{2}x^{-1/2}\Phi(\log x)$. 183

7.1 Argument principle counts for Conjecture T: rectangles searched and zeros of Q_n found with $\operatorname{Re} u > \frac{5}{4}$. 219

7.2 Zeros of F_n near $\rho_j/2$, in the variable $u = w + 1$. The row "limit" is $\rho_j/2 + 1$; the last row, $n(\frac{5}{4} - \operatorname{Re} u)$ at $n = 64$, is to be compared with $-\operatorname{Re}\kappa(\rho_j) = 4.762$, 8.673, 10.149, 15.165. 221

7.3 Zeros of the symmetrised truncations K_n in $-4 < \operatorname{Re} w < 4.5$: total found, how many lie on the line, and the number of zeros of ξ in the doubled window for comparison. In the window $195 < \operatorname{Im} w < 215$ the count for K_8 exceeds that for ξ by one; see the text. 224

7.4 Bounds for the torus extremes over the band $1.2 \leq x \leq 1.3$ and for $r(1.2)$, $r'(1.2)$, rounded outwards, and the smallest integer height at which the first condition of Theorem 7.2.10 holds with these values; the second condition holds from a lower height. The bounds and the second condition are proved by a branch and bound whose final boxes are enclosed in ball arithmetic (Remark 7.2.12); the programs are in `verification_reciprocal_zeta/torus/`. 228

7.5 As Table 7.4, for the single band variant: bounds for the extremes over the band $1 \leq x \leq \frac{3}{2}$ and for $r(1)$, $r'(1)$, and the smallest integer height at which its condition holds with these values. 228

7.6 Forms equivalent to RH, against what is proved in each case with no unproved hypothesis, and what computation that proof consumes. The first row consumes none. 231

7.7 Smallest order n^* at which an inserted off line quadruple at height γ and distance δ makes $\min_t L_n[W](t)$ negative. 234

7.8 Numerical verification of identities of Chapter 6: left and right sides, at the stated arguments. Where the two sides differ, one of them is a truncated sum or a quadrature. 238

7.9 Constants used in Sections 6.1.4 and 7.1.1. 238

8.1 Prony extraction of the atoms from $P(T > t)$ on $t \in [0.03, 0.108]$. Identical results for relative noise 10^{-16}, 10^{-30} and 10^{-45}: the error is model truncation, not noise. 246
8.2 Monte Carlo estimates of $\Xi(z)$ from Williams' representation with 10^6 samples. The last column is the sample size at which the standard error equals $|\Xi(z)|$. 246
8.3 Relative error of the clock transform: partial product over N zeros, $|G/G_N - 1|$, and with the Weyl tail, $|G_N^{\mathrm{W}}/G - 1|$. The two measures of error agree to first order, and $\gg 1$ marks an error above 10^3. The partial product values agree with [200, Section 18.2.1] where both are given. 248
8.4 Relative error $|G_{\mathrm{a}}/G - 1|$ of the approximant G_{a} of G with the Weyl tail and with the moment matched tail. The cutoffs are $\Theta_N^2 = 3.99 \times 10^4$, 2.95×10^5 and 4.83×10^6 for $N = 79$, 300 and 1700. . . 250
8.5 The Roberts Shortland boundary whose tangent hazard equals the hazard of the clock. 255
8.6 Keiper Li coefficients from the Pólya density (all digits shown are correct) and Keiper's leading asymptotic, which becomes accurate only for large n. 261
8.7 The closest pair below height 200 under the de Bruijn Newman flow, located by the sign changes on a grid of step 0.011 and refined by Newton's method; t is the de Bruijn Newman time. 266
8.8 Zeros $\theta_n(t)$ of $\theta \mapsto H_t(2\theta)$ returned by the Jacobi matrix $J_{30}(t)$ of the moments $m_0(t), \ldots, m_{59}(t)$; at $t = 0$ they are γ_1, γ_2, γ_3, γ_5 and γ_{10}. At each of the four times $J_{30}(t)$ returns $\theta_1(t), \ldots, \theta_{12}(t)$ to within 3.1×10^{-7} of the zeros that Newton's method finds on H_t, and $\theta_{13}(t)$ only to within 5.2×10^{-4}. Backward in time each of these zeros moves up. 268
8.9 The first failing order $n_{\mathrm{fail}}(t) \le 50$ of the Hankel test along the flow on the grid of step 0.01, so that $t_N^* = \sup\{t < 0 : n_{\mathrm{fail}}(t) \le N\}$, and, at one time on each step, the least order at which the pair that starts at (γ_k, γ_{k+1}) breaks the test alone, with the contributions of the other complex pairs removed from the moments (real: the pair has not yet collided; > 50: no order up to 50 fails). The five pairs collide at $t = -1.285$, -1.276, -1.144, -0.881 and -0.385. The other pairs below index 59 that have collided by $t = -1.30$, those with $k = 40$, 45, 53 and 57, break no order up to 50 alone. 270

8.10 The Hankel test further back along the flow. At each time: the lowest conjugate pair, labelled $(k, k+1)$ by the zeros γ_k, γ_{k+1} it starts from, and its position θ; the number of real zeros below it; the number of collisions so far among $\gamma_1, \dots, \gamma_{24}$, the zeros that start below $\theta = 88$; the least orders $n_{\text{fail}}(t)$ and $n'_{\text{fail}}(t)$ up to 30 at which $\mathcal{H}_N(t)$ and $\mathcal{H}'_N(t)$ are not positive definite; and the numbers $\iota_N(t)$ and $\iota'_N(t)$ of their negative eigenvalues at $N = 10$ and 30. At $t = -2$ and -3 the first failure registers the pair that starts at (γ_9, γ_{10}), from $t = -4$ to -12 the one that starts at (γ_4, γ_5), and at $t = -15$ and -20 no single pair. 272

8.11 The 28 collisions among the first 120 gaps at $t > -1.6$, latest first: the collision time t_{coll} from the census of Section 8.4.3 (`dbn_collisions.py`), against predictions computed from H_0: the isolated pair estimate $-D_0^2/8 = -(\gamma_{k+1} - \gamma_k)^2/2$, the critical point and Biane forms τ_c and τ_B, and the frozen predictor of Section 8.4.5; $6Q/m$ is the share $6Q(0)/m(0)$ of the other zeros in Biane's functional. Times in the units of H_t. 276

8.12 Laguerre's quantity at the critical point, $\ell(s)$, and its minimum over the gap, $m(s)$, along the backward flow for the pairs $(13, 14)$ and $(19, 20)$, which collide at $s = 1.2853$ and 1.2755, against the bounds of Theorems 8.4.6 and 8.4.8; a bound is shown only where it is finite, for $s < 1/\ell(0)$ (0.9064 and 0.7719) or $s < 1/m(0)$ (1.0334 and 0.9573). Units of H_t; `wigbiane_flow.py`. 285

8.13 Two sided brackets for the collision time of seven pairs, in the units of H_t: the lower bound $1/m(0)$ of Corollary 8.4.12, whose hypothesis holds on the grid, $\bar{\delta} < 0$, and the upper bound s^* of Corollary 8.4.14, with ρ the largest value of Q and $\bar{\kappa} = 0$ for these pairs. The correction $\bar{\eta}$ of Theorem 8.4.11 comes from the tracked pairs and the model above the census (almost entirely from the model for the last three pairs), and "pairs" is the number of conjugate pairs of the census with positive real part that are off the axis at the collision. `wigbiane_flow.py`. 285

8.14 Bounds on the de Bruijn Newman constant, ordered by the bound: lower bounds from [224, Table 1], upper bounds from [65], [125], [217] and [196]. Odlyzko's bound is the one computed in [174, Section 3], which that paper states in rounded form as $\Lambda > -2.7 \times 10^{-9}$. The years are years of publication. 288

8.15 The zero density identity for $L(s, \chi_{12})$ at height 100, computed from the eta kernel. The last column is $N_{\chi_{12}}(100) - 2\nu(\frac{1}{2} + \varepsilon, 100)$; the row $\varepsilon = \frac{1}{2}$ gives $N_{\chi_{12}}(100)$. 290

8.16 The trapezoid evaluators for ζ, $L(s,\chi_{12})$ and $L(s,\Delta)$ at 60 digits, with the settings given in the text. The decay of $\xi(\frac{1}{2}+it)$ is shown up to a power of t; the digits lost per unit height are the slope of $\log_{10}|\xi(\frac{1}{2}+it)|$ between heights 10.3 and 40.3; the times are per point, in `mpmath`. 292
8.17 The probabilities that a Poisson clock of rate 2π rings fewer than k times before M^*, and the identities (8.5.4): the last column is symmetric about $k=6$. Computed by `maxexp_checks.py`. 298
8.18 Checks of the max representation and of the laws of M_1 and M^* . 313
8.19 The collisions of zeros of H_t^{Δ} below height 52, in the order in which they occur as t decreases: the pair at $t=0$, the isolated pair estimate with D the gap of the pair, and the time t_* and height θ_* of the collision, computed by `dbn_delta.py`. 316
8.20 The Pólya kernel Φ_K of $K=\mathbb{Q}(\sqrt{d})$ for the discriminants named in the first author's note: the characters at 2 and 3, the sign of the kernel, the least value of H_K on $[1,\sqrt{|d|}/\pi]$, which decides it by Theorem 8.5.17(iv), with the point where it is attained, and $\zeta_K(\frac{1}{2})=\zeta(\frac{1}{2})L(\frac{1}{2},\chi_d)$. Computed by `sqfree_quadratic.py`. 321
8.21 Positive Pólya kernels Φ_K among the 6079 fundamental discriminants $-20000\le d<0$, by the splitting of 2 and 3 in $K=\mathbb{Q}(\sqrt{d})$ ($\chi_d(p)=1$ split, 0 ramified, -1 inert): the number of positive kernels over the number of discriminants in each class, 2738 positive in all. The empty corner is Theorem 8.5.18. Computed by `sqfree_quadratic.py`. 322
8.22 The Lévy form of Proposition 8.5.23: the gamma piece and the prime powers up to X, whose sum (total) is compared with $\log(\xi(\alpha+s,\chi_4)/\xi(\alpha,\chi_4))-b_\alpha s$ (exact); the drift term $b_\alpha s$ is shown separately. At $(3,2)$ the sums to 2×10^5 and to 10^7 agree in all digits shown. Computed by `dbeta_thorin.py`. 327
8.23 The sine squared decomposition (8.5.9) of $v_\alpha(\theta)=\mathrm{Re}(\xi'/\xi)(\alpha+i\theta,\chi_4)$, with the prime powers up to X, against the closed form; $\gamma_1=6.0209489$. Computed by `dbeta_thorin.py`. 328
8.24 The height $t_0(\alpha)$ above which the bound of Proposition 8.5.25 certifies $v_\alpha>0$, for basepoints below $\alpha_c(\chi_4)=2.2823$; the last column tends to $-2\gamma_E-\log\frac{8}{\pi}=-2.0891$ (Proposition 8.5.26). Computed by `dbeta_thorin.py`. 331
8.25 The brackets of $\xi_{\mathbb{Q}(i)}$, ξ and $\xi(\cdot,\chi_4)$ at the basepoint $\frac{3}{2}$; the last row is the difference of the first two (Proposition 8.5.27). Computed by `dbeta_thorin.py`. 332
8.26 The clusters of the nonnegative solution for $\rho_{0.1}$ on $[15,55]$: for each zero γ_k in the data window, the centre of mass c_k of its cluster, the error of the centre and the total mass of the cluster. 339

8.27 Nonnegative deconvolution of $\rho_{0.1}$ with the signature g of a displaced fifth zero added to the data. Here r is the residual norm and $r_0 = 0.1058$ its value at $\delta = 0$; the excess is $\sqrt{r^2 - r_0^2}$. 340
8.28 Left: the positive zeros below 60 of Ξ, of Pólya's Ξ^* and of $P(\theta) = K_{i\theta/2}(2\pi)$. Right: the numbers of zeros in $(0, T]$ against $N_0(T) = \frac{T}{2\pi}\log\frac{T}{2\pi e}$. Computed by `biane_polya.py`. 343
8.29 The two sides of the Wick rotation. 350
8.30 Index side products for P, Ξ^* and Ξ 356
8.31 The clock from $R_0 = 2\pi e^{2.5}$: exact moments against 100000 Euler paths with step 4×10^{-6}, with and without the bridge correction, and the Laplace transform, with the z scores (Monte Carlo − exact)/standard error. The standard error of the mean is 4.75×10^{-5}; over $0 < u \le 400$ the largest $|z|$ of the corrected transform is 0.72. Computed by `biane_clock.py`. 359
8.32 The real zeros in $(0, 90)$ of P_t and Ξ_t along the de Bruijn Newman flow: their numbers, the first zero of P_t and the first three of Ξ_t. Computed by `biane_flow.py`. 364
8.33 Pólya's function and Ξ. Apart from the computations of the last row, the left column is a list of theorems; in the right column the entries that concern the zeros are equivalent to RH or open. 368

9.1 The identity (9.1.1) of Biane, Pitman and Yor, checked with 4×10^5 draws of S_2 by `comp.py`. The sample mean of S_2 is 0.6664 against $\mathbb{E}S_2 = \frac{2}{3}$. 377
9.2 The Thorin density $\rho_{1/2} = v_1/\pi$ of T_1 at the basepoint one, computed from ξ'/ξ, from the arithmetic form $v_1(\theta) = \frac{1}{1+\theta^2} - \frac{1}{2}\log\pi + \frac{1}{2}\operatorname{Re}\psi(\frac{1+i\theta}{2}) + \operatorname{Re}\frac{\zeta'}{\zeta}(1+i\theta)$ with ζ'/ζ evaluated exactly, and from the Poisson sum over the first 1700 zeros with $\beta = \frac{1}{2}$. The zero sum misses the zeros above height 2198, whose contribution is a nearly constant 1.6×10^{-4}. 379
9.3 The sine squared lemma at the basepoint one 381
9.4 The Lévy form (9.1.5) against $\log(\xi(1+s)/\xi(1))$: the drift term, the gamma piece, the prime atoms up to X, their compensator, the total and the exact value. 383
9.5 The compensated sine squared formula (9.1.8) below the basepoint one at $\theta = \gamma_1$, with primes up to X, against the directly computed $v_\alpha(\gamma_1) = \operatorname{Re}(\xi'/\xi)(\alpha + i\gamma_1)$; at $\alpha = \frac{1}{2}$ the point $\frac{1}{2} + i\gamma_1$ is a zero. . 392
9.6 The error of the compensated sum (9.1.8) at $\theta = \gamma_1$ with primes up to X, its prediction from the explicit formula (9.1.9) over the first 1700 zeros, and what remains after subtracting the prediction. . . 393

9.7 The ratio $v_\alpha(\theta)/(\alpha-\frac12)$ of Theorem 9.1.8 along rows of basepoints. The last column is at the comparison line α' of Table 9.8 taken from the first author's note: 4.75 for ξ, 3.15 for χ_4 and 2.30 for χ_{12}, which lies below 2.5. Every row decreases, and at $\theta=0$ the rows increase to V_f as $\alpha\downarrow\frac12$ (Table 9.8). Computed by `dbeta_minimum.py`. 396

9.8 Constants of Theorem 9.1.11. For a comparison line $\alpha'>1$, t_1 is the root of $A_f^\infty(\alpha',t)=S_f(\alpha')+(\alpha'-\frac12)V_f$, found by bisection and rounded up to three decimals, at which the strict inequality is certified in ball arithmetic; the proof needs the convexity margin to be positive. The first comparison line is that of the first author's note; the minimiser of t_1 is located on a grid of step 0.005; $\alpha_{\max}$ is the largest α' in steps of 0.05 keeping the margin above 0.02. For ξ the range $\alpha>\alpha_{\max}$ is covered by Proposition 9.1.5. Computed by `dbeta_minimum.py`; the ordinates of $L(s,\chi_4)$ and $L(s,\chi_{12})$ are certified by `dbeta_zeros.py`. 400

9.9 What the primes show about the clocks T_α 405

9.10 Monte Carlo check of Theorem 9.2.2: the empirical Laplace transform of 10^6 draws of $\widetilde{T}+H_{\widetilde{T}/\sqrt{2}}$ against φ_1. 410

9.11 The density of T_1 by two independent methods: Talbot inversion of φ_1, and the subordination formula (9.2.1) averaged over 4×10^5 draws of $\widetilde{T}$. 411

9.12 Basepoints above the centre: the drift $\frac{\xi'}{\xi}(\alpha)$, the tail constant of T_α, and the minimum on $[0,60]$, attained at the origin, of the density $\rho_{\alpha-1/2}=v_\alpha/\pi$, the Thorin density of T_α when every zero lies to the left of the line $\operatorname{Re}s=\alpha$. At $\alpha=1$ the minimum at the origin holds at every height (Proposition 9.1.4), as it does at every $\alpha\ge1$ (Proposition 9.1.5); for $\frac12<\alpha<1$ the minimum at the origin holds on the verified window $|\theta|\le3\times10^{12}-1.21$ (Theorem 9.1.11(ii)), and at every height for every such α it is equivalent to RH (part (vi) of that theorem). 417

9.13 Exact Brownian boundaries above the centre: the flat level, the mean boundary value $\mathbb{E}\,b(T_\alpha)$ at the crossing, obtained by integrating the computed boundary against the law of T_α, against the Wald type identity, and the computed $\sqrt{100}\,P(T_\alpha>100)$ against the tail constant of Theorem 9.2.5. Computed by `bnd_alpha.py` for $\alpha=0.6$ and 0.75 and by `rs1.py` for $\alpha=1$. 419

10.1 Proposition 10.1.1 by simulation, computed by `comp.py`. At the centre the sample variance of $Y_{1/2}$ is 0.04637 against 0.04621. The Cauchy tails $y\,P(|Y_\alpha|>y)$ at $y=10$ are 0.0144 at $\alpha=1$ and 0.0066 at $\alpha=0.75$, against $2b_\alpha/\pi=0.0147$ and 0.0074. 424

10.2 The Thorin form (10.1.2) against the reciprocal Xi function on the real axis, computed by `comp.py`. 427
10.3 The EGGC representation in the strip, computed by `strip.py`: the Thorin integral over $\theta \leq 200$ plus the Weyl tail, and the product of the factors of Theorem 10.1.3(3) over the 1700 computed zeros plus the Weyl tail. 430
10.4 The Poisson traces at the basepoint one and at the centre, computed by `comp.py`. 432
10.5 The first order at which the term of a hypothetical zero $\frac{1}{2} - \delta + i\gamma$ in (10.1.3) is negative, against Proposition 10.1.5. The last row is a zero at the edge of the strip at the height of the first ordinate. Computed by `biane_traces.py`. 434
10.6 The Poisson trace $K_{1/2}$ of the zeros of ζ, Pólya's trace K^P, their distances from the common main term M, which tend to $\frac{7}{8}$ and $-\frac{1}{4}$, and their difference D, which tends to $\frac{9}{8}$, against the prediction of Proposition 10.1.8 under RH. Computed by `biane_traces.py`. . . 438
10.7 Best L^1 error in approximating a Gaussian of width 0.15 by N translates of Φ_α, computed by `wiener.py`. At the centre the error cannot fall below $|\hat{g}(\gamma_1)| = 0.1056$, and $\hat{h}(\gamma_1)$ vanishes exactly; the computed value is 1.2×10^{-8}. 441
10.8 Left: the Riesz coefficients from the clock, Proposition 10.2.3. Right: the Riesz function from the Möbius series, which agrees with the power series to fourteen digits, and from the clock average of Proposition 10.2.4. 451
10.9 Top: the Riesz coefficients from the Thorin form of Proposition 10.2.5, truncated as described in the text, and the factors of (10.2.2): the stable factor $e^{-b_1(2k-1)}$, the GGC factor computed from the truncated Thorin form, and the exact GGC factor $c_k e^{b_1(2k-1)}$. Bottom: $\log(\xi(2k)/\xi(1))$ from the Lévy form (9.1.5) at $s = 2k-1$ with primes to 10^7, against its exact value. 454
10.10 Riesz coefficients as killing probabilities of the basepoint one clock, and the stable factor as a Brownian hitting probability; E is an independent exponential clock of rate $(2k-1)^2$, and the errors are one standard error of 10^6 draws. 458
10.11 The sharp counts: the number $Q(x)$ of squarefree integers up to x, the normalised error $\Delta(x)/x^{1/4}$, the error $\mathcal{E}_1(x)$ of (10.3.9), which confirms the constant C, also multiplied by $x^{3/4}$, and the error $\mathcal{E}_{3/4}(x)$ of (10.3.4) at $\alpha = \frac{3}{4}$ multiplied by $x^{1/4}$. The first two errors are normalised by the sizes $x^{1/4}$ and $x^{-3/4}$ that their poles at the zeros predict under RH, the last by its bound in (10.3.4). Computed by `sqfree_sieve.py` from a sieve to 2×10^8; the sums of reciprocals are checked in extended precision. 466

10.12 The zero part $b_1 - 2b_2 = -0.11503675409388031864$ of the constant C, as the sum (10.3.10) over the first N zeros plus its tail from the smooth density of the zeros. Computed by `sqfree_constants.py` with the zeros from Arb [110] at 40 digits. 468

10.13 The exponentially smoothed count of Theorem 10.3.9: the smoothed error, the contribution $2\,\mathrm{Re}\sum_{k\le 40} A_{\rho_k} x^{\rho_k/2}$ of the first forty atoms, the residues at $s = -1, \dots, -6$, of which the first contributes $(c_1 \log x + c_2)/x$, the term $c_1 \log x/x$ alone, and the residual. The atoms beyond the fortieth change the sum by less than 10^{-40}. Computed by `sqfree_smooth.py` at 45 digits from the sum over d; the direct sum over n agrees within 3×10^{-12} up to $x = 10^5$. 475

10.14 Root mean square and largest absolute value of the squarefree error Δ on the ranges $[X, 2X]$, divided by $X^{1/4}$, over real x; the largest value on $[10^7, 2 \times 10^7]$ is the left limit at 17199381, and over the integers alone it is 0.982. Computed by `sqfree_sieve.py`. 476

10.15 The Gamma(2) smoothed count: $Q_2(x) - 6x/\pi^2$, the sum (10.3.13) truncated at $d \le 2 \times 10^5$, the normalised error, the contribution $2\,\mathrm{Re}\sum_{k\le 100} B_{\rho_k} x^{\rho_k/2}$ of the first hundred atoms, the residues at $s = -1, \dots, -6$, whose leading part is $2c_1/x$, and the residual. The constant 1 is the residue $D(0)$ at $s = 0$. Computed by `sqfree_smooth.py` at 45 digits. 481

10.16 Gamma(ν) smoothing above the threshold of Theorem 10.3.18: the limit $-\zeta(-\nu)$ of q_ν at 0, the least value of $1 - \mathcal{R}_\nu$, which has the sign of f_ν, with the point where it is attained (grid search refined by golden section), and the maximal runs of grid points on which $1 - \mathcal{R}_\nu < 0$. Computed by `sqfree_gammanu.py`. 484

11.1 The heat trace W and the Lévy density $h = -W'$ of the driving process of the centre clock, from the first 1700 zeros and from the explicit formula (1.2.5) and its derivative with the primes below 2×10^4. The explicit columns take the weights $\log p$ in double precision, which accounts for their departure from the zero columns at $t = 0.1$. Computed by `weil.py` in the folder `sato_bondesson`. . 490

11.2 The subexponential principle for T_1: the tail of the law is asymptotic to the tail of its Lévy measure. The survival function is that of Section 9.2.4, the Lévy tail is computed from (11.1.1) with the Thorin density $\rho_{1/2}$ tabulated on $[0, 100]$ by `compute.py` of Section 9.1, the ratio is that of the second column to the third, and the last column is the common asymptote of (11.1.2). Computed by `comp.py` in the folder `sato_bondesson`. 493

11.3 What the books of Sato and Bondesson, and the papers around them, give the reciprocal Xi function: each result used in this section, its source, its consequence for the centre clock T, the shifted clocks T_α and the tilted Pólya variables V_α, and its status. Theorem numbers, here and in the text, refer to the first edition of Sato's book [232]. 501

11.4 The Wiener gamma representation (11.1.4) of T_1 against the inverse of the smoothed zero count $M_{1/2}$: the empirical Laplace transform of 2×10^5 draws, discretised as described in the text, against φ_1. Computed by `comp2.py` in the folder `basepoint_one/process_map`. 503

11.5 The characteristic function $\xi(\alpha)/\xi(\alpha + |u|/\sqrt{2})$ of subordinated Brownian motion, estimated by the mean of $\cos(uB_{T_\alpha})$ over 3×10^5 draws at each basepoint, and its Cauchy type tail; the tail estimates rest on 72, 149 and 284 exceedances of $y = 10$. Computed by `comp2.py` in the folder `basepoint_one/process_map`. 506

11.6 The processes by basepoint, with $a = \alpha - \frac{1}{2}$. (RH) marks statements that hold under RH, and the Kent strings need the zeros to be simple as well. The Mellin row gives the size bias of S_2 that produces V_α (Section 9.1.1); $\widetilde{T}_\alpha$, H and the sticky clock are those of Theorem 9.2.5 and Corollary 9.2.3, with $\widetilde{T} = \widetilde{T}_1$; the boundaries b and their Wald defects are those of Sections 9.2.4 and 9.2.6; $Y_\alpha = \sqrt{2}B_{T_\alpha}$ in law (Proposition 10.1.1); and $k_\alpha(x)\,dx/x$ is the Lévy measure of T_α. The boundary row reports computations at $\alpha = \frac{1}{2}$, 0.6, 0.75 and 1, those at 0.6 and 0.75 made under RH beyond $t = 0.03$, and its entry for $\alpha > 1$ is extrapolated, not computed; the flatness is observed, not proved, and above the centre the Wald defect $b_\alpha/\sqrt{2}$ requires the boundary to be bounded at infinity (Lemma 9.2.4). All other entries hold as stated. 507

11.7 The Laplace exponent of $\Upsilon(T)$, which is minus the free cumulant transform of $\Lambda_{\mathrm{BP}}(T)$ at $-s$: the integral of (11.2.8) from ξ alone, and the sums over the first 300 and 1700 zeros, with no tail, with the tail of order one and with the full smooth tail, as differences from the integral. Computed by `wigfree_upsilon.py`. 516

11.8 The clock T, its Upsilon image and its free counterpart, from the densities of Figure 11.5. The exact means are $m_0 = 0.023104993$; the variances $m_1 = 3.71726 \times 10^{-5}$ for T and $\Lambda_{\mathrm{BP}}(T)$ and $2m_1 = 7.43452 \times 10^{-5}$ for $\Upsilon(T)$; the third central moments $2m_2 = 2.8835 \times 10^{-7}$ for T and $\Lambda_{\mathrm{BP}}(T)$ and $12m_2 = 1.7301 \times 10^{-6}$ for $\Upsilon(T)$. The modes of T and $\Upsilon(T)$ are grid points of spacing 0.0005, that of $\Lambda_{\mathrm{BP}}(T)$ of spacing about 7×10^{-5}. Computed by `wigfree_upsilon.py`. 517

11.9 The Jacobi parameters of ν against the local semicircle model: α_n and $2\beta_n$; half the right end $\frac{1}{2}L_n$ of the local interval from (11.2.13) with the smooth density and $n(L_n) = n$, and the ratios to it; the same with the midpoint rule $n(L'_n) = n - \frac{1}{2}$; the cruder rule $\frac{1}{2}\gamma^{-2}_{2n/\pi}$, with $N_0(\gamma_r) = r$; and $\beta_n^2/(\alpha_n\alpha_{n+1})$, which is $\frac{1}{4}$ in the local model. Computed by `wigfree_jacobi.py`. 521

11.10 The first order N at which the moments $\mu_k = \mathbb{E}Y^k/(a)_k$ fail the Hankel test for a Gamma(a) scale mixture, for $Y = |X|$, X^2 and S_2: the Hamburger test uses $[\mu_{i+j}]_{i,j<N}$ alone, the Stieltjes test also $[\mu_{i+j+1}]_{i,j<N}$. Computed by `wigfree_gamma2.py`. 527

11.11 The Thorin density v_1/π of $\xi(1,\chi_{12})/\xi(1+\sqrt{u},\chi_{12})$, computed from $\xi'/\xi(\cdot,\chi_{12})$ and from the Poisson sum over the 297 zeros of $L(s,\chi_{12})$ below height 340 with $\beta = \frac{1}{2}$; the difference is the contribution of the zeros above height 340. 531

11.12 The Lévy form of Proposition 11.3.2 against $\log(\xi(1+s,\chi_{12})/\xi(1,\chi_{12}))$, with primes up to X. The prime piece stays of order one; there is nothing to cancel. 533

11.13 The sine squared formula below the basepoint one for $L(s,\chi_{12})$, with primes up to X: the prime piece, the error against $v_\alpha(\theta)$, and the error left after subtracting the explicit formula over the 297 computed zeros; $\gamma_1 = 3.8046276$. 535

11.14 The two basepoint one examples side by side: the reciprocal Xi function of Chapter 9 and the reciprocal completed L function of χ_{12} of this section. 538

11.15 The zero density exponents $e(\alpha)$ of Theorem 11.4.2, with $N(\alpha,T) \le T^{e(\alpha)+o(1)}$, the power saving $1-e(\alpha)$, and the exponent $2(1-\alpha)$ of the density hypothesis. Computed by `almost.py`. 542

11.16 A single hypothetical pair of zeros at real parts β and $1-\beta$ inserted into the density $v_{0.6}/\pi$, computed from ξ'/ξ on a window of width 8 at the heights 300 and 10^5: the minimum of the density and the negative mass $\int \max(-v_{0.6}/\pi, 0)$ on the window. Computed by `high.py`. 547

11.17 Negative and positive mass of the density v_α/π on $[0,T]$ with fifteen hypothetical quadruples of zeros at real part 0.75. Computed by `almost.py`. 548

11.18 Quantiles of the normalised Thorin density $\bar{\rho}_\kappa$ at the midpoints between consecutive unfolded ordinates, for the zeros below height 2198 and for CUE eigenvalues. Computed by `sens.py`. 550

11.19 Moments of the characteristic polynomial of Haar unitary matrices: the exact formula, direct simulation with 2×10^4 matrices, and the beta decomposition of Bourgade, Hughes, Nikeghbali and Yor with 2×10^5 samples. Computed by `killing2.py`. 552

11.20 Moments of ζ on the critical line against the asymptotic of Hardy, Littlewood and Ingham for $k = 1$ and the leading Keating and Snaith terms for $k = 2$ and 3. Computed by `ks.py`. 553

11.21 The ladder functions of R_t along the de Bruijn Newman flow, on the window $0 \le -\log x \le 8$: the first zero $z_1(t)$ of H_t, the ratio a_2/a_1 of the first two Thorin atoms, the ratios $W^{[2]}/W$ and $W^{[3]}/W$ at $-\log x = 8$, the sets of $-\log x$ on which $W^{[3]}(x;t) < 0$ and $W^{[4]}(x;t) < 0$, to the grid step 0.001, and the largest n with $W^{[1]}, \dots, W^{[n+1]} \ge 0$ on the window: R_t lies in no class above L_n, and in L_n if the signs persist for $x < e^{-8}$. On the window W, $W^{[1]}$ and $W^{[2]}$ are positive and $W^{[5]}$ takes negative values for every t. Numerical for $t > 0$; the row $t = 0$ is Theorem 11.5.8. Computed by `heat_ladder.py`. 575

11.22 The Taylor coefficients of $g = \Psi_Z \circ H$, the only candidate, up to scaling, for the Laplace exponent of a generating process of the centre clock (Theorem 11.5.13), to twelve digits; a star marks a sign that violates (11.5.6). Every digit shown is certified in ball arithmetic at 3200 bits from the Taylor series of ξ at $\frac{1}{2}$, and the signs are certified for $k \le 60$: the wrong signs are those with $16 \le k \le 21$, $29 \le k \le 35$, $43 \le k \le 49$ and $56 \le k \le 60$. Computed by `branch_coeffs.py`. 577

11.23 The GGC test for the tilted variables $\mathcal{B}_t$ of Section 11.5.5.3: the least order $n \le 12$ at which $\mathcal{K}'_n(\lambda)$ has a negative eigenvalue, and the least eigenvalue of the scaled matrix $\mathcal{K}'_n(2)$ at that order; for $t = 0$, where no order fails, the last column gives $\mathcal{K}'_7(2)$. At $\lambda = 200$ no matrix of order at most 12 fails, for any t. Every negative entry of the last column is certified, and so is the entry for $\mathcal{K}'_9(0)$ at $t = -\frac{1}{100}$, -2.899×10^{-12}. Computed by `pakes_filter.py` and `pakes_filter_cert.py`. 584

11.24 The GGC test for the variables $\mathcal{B}_a$ of Section 11.5.5.4 and their reciprocals: the sum S_a of Lemma 11.5.19, the least eigenvalues of the scaled matrices $\mathcal{K}'_7(0)$ and $\mathcal{K}'_7(2)$ of $\mathcal{B}_a$, and the verdicts. "Passes": no matrix $\mathcal{K}_n(\lambda)$ or $\mathcal{K}'_n(\lambda)$ with $n \le 12$ has a negative eigenvalue at $\lambda = 0$, $\frac{1}{2}$, 2, 10, 20, 50 and 200. "Fails at $\mathcal{K}'_n(\lambda)$": the least failing order, certified at that λ. "Zero": the Laplace transform vanishes on $(-\infty, 0)$ (Proposition 11.5.21). GGC for ξ by (11.5.8), for $1/\mathcal{B}_a$ with χ_{-3} and χ_{-4} by Proposition 11.5.21. All the functions but f_{DH} have an Euler product. Computed by `pakes_filter.py` and `pakes_filter_cert.py`. 588

12.1 The path, step by step. 600

12.2 Work packages and milestones of the computational programme. Package 1 is described in Section 10.2.1, where R is Riesz's function and R_h its twist; packages 3 and 4 are described in Sections 11.4.5 and 11.4.2. 606

13.1 Ferromagnetic laws and their clocks. Here $j_{\nu,k}$ are the positive zeros of the Bessel function J_ν, and τ_ν is the hitting time of 1 by a Bessel process of dimension $2\nu+2$ started at 0. In the fifth row ν_k are the zeros of $\nu \mapsto K_{i\nu}(2\pi)$ and T_P is the clock of Theorem 8.7.9. 620

13.2 Zeros of $F_b(z) = \int_0^1 (1-x^b)\cos(zx)\,dx$ in the box $(0.3, 60) \times (-12, 12)$, counted by the argument principle, and real zeros in $(0.3, 60)$, all in ball arithmetic by `fm_intcert.py`. For $b > 1$, certified sign changes on a grid of step 0.002 give 18 real zeros, so every zero in the box is real; for $b < 1$, the box $(0.3, 60) \times (\frac{1}{20}, 12)$ holds 9 zeros and its mirror image 9 more, so none is real. At each b of the table, $F_b(2\pi)$ is positive for $b < 1$ and negative for $b > 1$. . 626

13.3 Which laws on $[-1, 1]$ are ferromagnetic. Every entry is proved in this section or in Section 13.1. 627

13.4 Even cumulants of the Pólya law, by quadrature of the moments of Φ, from the Taylor coefficients of $\log \xi(\frac{1}{2}+s)$, and from the sums $(-1)^{m+1}\frac{(2m)!}{m}\sum_k \gamma_k^{-2m}$ over the first 1700 ordinates with the smooth count beyond them. The difference 1.3×10^{-10} in κ_2 is the error of the smooth count, which ignores the oscillation of $S(t)$ above height 2197. Computed by `fm_polya.py`. 628

13.5 The moment test (13.2.2) for the Pólya law: the scaled moments $b_j = \mathbb{E}X^{2j}/(2j)!$ and the Turán ratios, against the bound $1 + \frac{1}{j}$ that every ferromagnetic law satisfies. All 24 computed ratios exceed the bound. Computed by `fm_polya.py`. 629

13.6 The zero completed laws f_K. For $K \le 16$, f_K is a ferromagnetic probability density by Theorem 13.4.3(i) ($s_K \le 2\beta_K$). For every $K \ge 1$ in the table, $Q_{K,2\beta_K} > 0$ on the line is proved by the cover in ball arithmetic described in the text, so f_K is a ferromagnetic probability density by Theorem 13.4.3(ii), and $\min Q_{K,2\beta_K}$ is the value at $y = 0$, where it is attained. The threshold $v^*(K)$ of Proposition 13.4.4 lies in a certified interval of width at most 1.2×10^{-7}, within 6×10^{-7} of the value shown, and the ratio $v^*(K)/2\beta_K$ is correctly rounded. The distances are enclosed in ball arithmetic, as in Table 13.7. The row $K = 0$ is exact apart from $2\beta_0 = \operatorname{Var} X$ and the distance: $Q_{0,v} = 1$, so f_0 is normal and $v^*(0) = 0$. Computed by `fm_qcert.py` and `fm_tvcert.py` from the certified zeros and variance, except $v^*(1) = s_1$. 637

13.7 Ferromagnetic approximations of the Pólya law and their total variation distances to it, enclosed in ball arithmetic and rounded. A tilt $w\,p$ stands for the normalised law $w\,p/\mathbb{E}w(X)$, and Pólya's kernels are normalised in the same way. Theorem 13.4.3(ii) applies to f_{150} because $f_{150} > 0$, which `fm_qcert.py` proves. Computed by `fm_tvcert.py`. 639

13.8 $-\log|\mathbb{E}e^{iye^{2X}}|^2$ computed four ways: by a direct Fourier integral of the density $y^{1/4}f_0(y)/(2\xi(\frac{1}{2}))$; in the Thorin form with the atoms $\frac{9}{4}$ at the points πm^2 summed in closed form and the continuous part from the computed $\theta_{1/4}$; in the theta Lévy form $\int 4\sin^2(\frac{yx}{2})k_{1/4}(x)\,dx/x$; and from the atoms alone, which is $-\log|\mathbb{E}e^{iy\frac{\pi}{2}S_{9/4}}|^2$. Computed by `fm_sizebias.py`. 648

A.1 Scripts in `code/` (unless another folder is named), the data files they write, and the tables and figures they produce, in book order. 659

A.2 Continuation of Table A.1. 660

A.3 Constants used in several sections, with the file of the archive in which each can be checked. 661

A.4 Scripts of Chapter 9 and Section 10.1, the data files they write, and the tables and figures they produce. The folders `xi`, `process_map`, `scale_mixture` and `wiener` are in `basepoint_one/`; the scripts of `riesz_program` are in its folder `code/` and their data files in its folder `data/`, and those of Section 9.1.6 are in `code/` and `data/` of the archive. 664

A.5 Scripts of Sections 10.2 and 10.3 and Chapter 11, as in Table A.4; the folders `process_map` and `eta` are in `basepoint_one/`, and the scripts of Sections 10.3, 11.2 and 11.5 are in `code/` and their data files in `data/`. 665

A.6 Scripts of Chapter 13, all in the folder `ferro/`, the data files they write, and the tables and results they produce. 666

B.1 Landmarks in the computation of the zeros of ζ. The zeros counted in each row were located on the critical line. For the computations up to 1969 see [72, Chapters 6 to 8]. 672

D.1 The blocks of the verification script: their lines in the script and in the output, what they check, and the statements of Section 4.3 they support. Output lines 37 to 39 and 109 come from code that is not in the script, and lines 93 to 102 from an earlier version of block 13. 684

Chapter 1

Introduction and foundations

The book rests on one equivalence, between the Riemann Hypothesis and the positivity of the Thorin measure of a function built from the completed zeta function. Section 1.1 states it as the Thorin condition, explains why computation can measure and calibrate that positivity but cannot prove it, and gives the plan of the book, its principal results with their status, and its conventions. Sections 1.2 and 1.3 set up the objects and the language on which the later chapters build. The first introduces the Pólya density, the clock and its heat trace, and the Thorin density at the basepoint $\frac{1}{2}+\varepsilon$; the second treats, from a computational point of view, the generalised gamma convolutions, the class of laws that the Thorin condition asks the clock to enter.

1.1 The Thorin condition and what this book shows

Write $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ for the completed zeta function. It is entire of order one, real on the real axis and on the critical line, symmetric under $s \mapsto 1-s$, and its zeros are the nontrivial zeros ρ of ζ. The Riemann Hypothesis (RH) asserts that every ρ has real part $\frac{1}{2}$.

This book works with a reformulation. Put

$$G(u) = \frac{\xi(\frac{1}{2}+\sqrt{u})}{\xi(\frac{1}{2})}.$$

Because $\xi(\frac{1}{2}+s)$ is even in s, G is an entire function of u with no branch point, of order one half, and Hadamard's theorem gives

$$G(u) = \prod_k \Big(1+\frac{u}{a_k}\Big), \qquad a_k = -(\rho_k - \tfrac{1}{2})^2,$$

one factor for each pair $\{\rho, 1-\rho\}$ (Section 1.2 derives the product as (1.2.1)). A zero on the line has $a_k = \gamma_k^2 > 0$; a zero off the line has a_k complex. The reciprocal $1/G$ is the *clock*. The classical theory of generalised gamma convolutions

(GGC) of Thorin and Bondesson [250, 251, 35] says that a function of the form $\exp\{-\int \log(1+u/z)\,U(dz)\}$ with a positive measure U is the Laplace transform of a positive random variable, and that U, the Thorin measure, is unique. The rigidity theorem of [200], which Section 1.2 restates with its source as Proposition 1.2.1, says that the clock $u \mapsto \xi(\frac{1}{2})/\xi(\frac{1}{2}+\sqrt{u})$ is the Laplace transform of a generalised gamma convolution if and only if RH holds, and that its Thorin measure is then

$$U_{\mathrm{RH}} = \sum_{\gamma>0} \delta_{\gamma^2}, \tag{1.1.1}$$

while the clock variable is $T = \sum_{\gamma>0} e_\gamma/\gamma^2$ with independent standard exponentials e_γ.

We call the requirement that the Thorin measure of the clock be a positive measure the *Thorin condition*. It has several equivalent faces, each of which is a sign condition. The logarithmic derivative $G'/G(u) = \sum_k (u+a_k)^{-1}$ must be a Stieltjes transform. The heat trace $W(t) = \sum_k e^{-a_k t}$ must be completely monotone. The Thorin density $\rho_\varepsilon(\theta) = \pi^{-1}\,\mathrm{Re}(\xi'/\xi)(\frac{1}{2}+\varepsilon+i\theta)$ at every resolution $\varepsilon > 0$ must be nonnegative, which is Lagarias's criterion [133]; for $\varepsilon \geq \frac{1}{2}$, where the basepoint $\frac{1}{2}+\varepsilon$ is at least one, it is positive with no hypothesis (Section 9.1). Weil's explicit formula functional must be positive on positive definite test functions [266]. The clock must be hyperbolically completely monotone (Corollary 4.1.3). All of these say the same thing, and all of them are, at a finite resolution and on a finite window, computable. Section 1.3.3 turns them into tests and compares how well each locates a failure (Table 1.8).

1.1.1 What computation can and cannot say

Computation has a long history in the study of RH. Riemann computed the first zeros by hand [238]. Gram [83], Backlund [9], Hutchinson [99] and Titchmarsh [253] extended the tables [72, Chapters 6 to 8]; Turing gave the method that certifies a count without evaluating off the line [262]; Lehmer found the near coincident pairs [143] that bound the de Bruijn Newman constant [62]; Odlyzko computed zeros near the 10^{12}th zero, at height about 2.7×10^{11}, and tested their statistics against random matrix theory [171], and later computed zeros near the 10^{20}th, at height about 1.5×10^{19} [173], and beyond [175]; Platt and Trudgian verified RH below height 3×10^{12} [196]. Appendix B tells this history, with the landmarks in Table B.1. None of it proves RH, and the reason is structural rather than technological: every computation sees finitely many zeros, and there are functions with the order, type and symmetry of ξ that pass a given finite set of heat trace tests and yet have zeros off the line just above any sufficiently large height R for which RH has been verified up to $R+50$ (Theorem 5.2.1 [198, Theorem D]).

The Thorin condition gives computation a different role. Instead of locating zeros one at a time, it asks for the sign of a measure, and the measure can be

computed at every positive resolution without knowing where a single zero is. The questions then become quantitative. How close, in which metric, is the computed Thorin measure to the discrete RH measure? At what resolution does an off line zero become visible, and with what signature? What does the explicit formula say about the cost of computing the Thorin measure from the primes? Which limiting procedures preserve positivity and which do not? And since positivity is a theorem once the basepoint is moved from $\frac{1}{2}$ to one, how much of it survives as the basepoint moves back across the critical strip? The book answers these questions, the last of them in Chapters 9 to 11.

Three ways of evaluating ξ appear throughout. The first is Pólya's integral, $\Xi(\theta) = \int \Phi(u)\cos(\theta u)\,du$, with the Pólya kernel Φ, a positive, doubly exponentially decaying function built from the Jacobi theta function. The integral is exact and needs no primes and no zeros, and since Φ is analytic in a strip the trapezoid rule applied to it is spectrally accurate; the cost is the cancellation $e^{-\pi\theta/4}$, which the tilted contour of Section 2.1 largely removes. The second is the Riemann Siegel formula and its fast multi point descendant, the Odlyzko Schönhage algorithm [172], the tools of choice at large height. The third is the explicit formula, which evaluates the same objects from the primes and the gamma factor; the sine squared lemma is its Thorin form.

The principle that ties them together is that the Thorin measure at resolution ε is the Poisson smoothing of the zeros, and simultaneously the sine squared image of the primes. The first description makes it computable from any evaluator of ξ off the line. The second explains why the primes alone cannot compute it near the line. In Selberg's smoothed explicit formula the prime cutoff x plays the part of the resolution: if no zero has real part greater than $\frac{1}{2}$, the error at resolution ε is of order $x^{-\varepsilon}/(\varepsilon^2 \log x)$, so that k correct digits need $x \approx 10^{k/\varepsilon}$, with primes up to x^2 (Section 3.5.1). Figure 1.1 shows the first description at work: the Thorin measure at decreasing resolution, sharpening onto the zeros.

1.1.2 Plan of the book and its principal results

Two ideas set out in Sections 1.2 and 1.3 carry the rest of the book. The first, stated above, is that the Thorin measure at resolution ε is the Poisson smoothing of the zeros and so is computable from any evaluator of ξ off the critical line; as $\varepsilon \downarrow 0$ it converges to the atoms on the line. The second is that the hierarchy of infinitely divisible laws becomes a hierarchy of positivity conditions on one function, the heat trace, of which the lower rungs are theorems and the top rung is RH. Every later chapter computes, tests or interprets one of these two ideas.

Chapter 2 evaluates ξ and ξ' in the complex plane by the Pólya route, compares it with the classical evaluators and Turing's method, and follows one small computation from the theta series to a Thorin mass and a zero. Chapters 3 to 5 are the core of the book. Chapter 3 measures the distance from the computed Thorin measure to U_{RH}, computes the signature of an off line zero, reads the zero

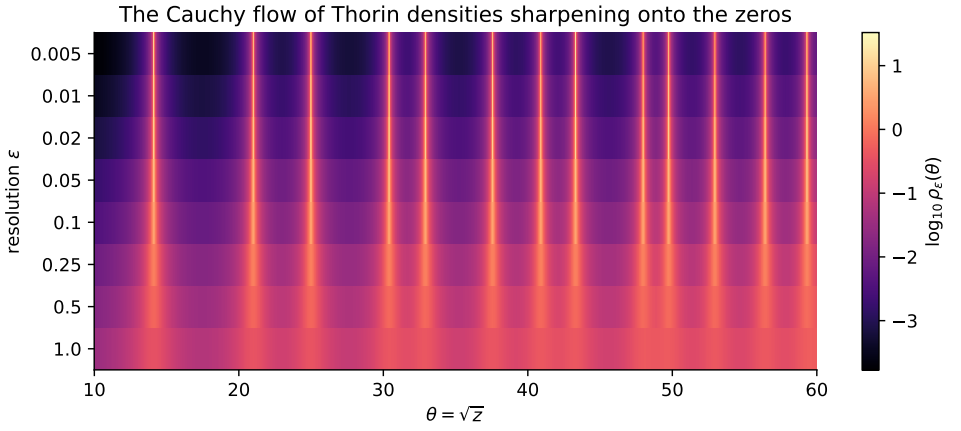


Figure 1.1: The Thorin density $\rho_\varepsilon(\theta)$ on $10 \le \theta \le 60$, computed from the Pólya density at eight resolutions from $\varepsilon = 1$ (bottom) to $\varepsilon = 0.005$ (top), on a logarithmic colour scale. As the resolution sharpens, the density concentrates on bright vertical lines at the ordinates of the zeros, all of which lie on the critical line in this window: the Cauchy flow converging to the RH measure $\sum \delta_{\gamma^2}$.

count and the zero density from the Thorin mass, and gives further readings of the sine squared lemma. Chapter 4 treats weak convergence and hyperbolic complete monotonicity, brings in Bosch's theorem on powers of the half Cauchy law, and proves a theorem of the same shape for the clock, with a threshold set by the angle of the zeros. Chapter 5 climbs the hierarchy of positivity on the heat trace, explains what finite information cannot decide, computes the Thorin functional from the primes and relates the cell distance of Section 3.2 to the spacings of the zeros.

Chapters 6 and 7 study the reciprocal zeta function in the variable u of the clock. The first gives its Thorin type forms and treats the gamma factor, the Meixner law and the sine squared lemma; the second proves that the normalised reciprocal zeta function is infinitely divisible and places it in the Bondesson hierarchy, then truncates the representation, due to Biane, Pitman and Yor, of ξ as the Mellin transform of a GGC with Thorin atoms on a quadratic lattice and studies the conditions on the zeros of the truncations that would imply RH. Chapter 8, with the Pólya route as its evaluator, treats the clock as a random variable, through its law, its Kent approximants and its Brownian boundaries; it then turns to criteria other than the Thorin condition, the Keiper Li coefficients and the de Bruijn Newman flow, carries the evaluator to the L functions of the character χ_{12}, whose theta series is the Dedekind eta function, and of Ramanujan's Δ, and recovers the zeros from the Thorin density by nonnegative deconvolution.

Chapters 9 to 11 move the basepoint. The shifted clock T_α has Laplace trans-

form $\xi(\alpha)/\xi(\alpha+\sqrt{u})$; at $\alpha=\frac{1}{2}$ it is the clock, and from $\alpha=1$ on its Thorin measure is positive with no hypothesis. These chapters carry both ideas from the basepoint one, the resolution $\varepsilon=\frac{1}{2}$, into the critical strip. Chapter 9 works out the basepoint one, from the Mellin side, where 2ξ is a moment function of Bessel hitting times, to the Thorin measure and the law of T_1, and constructs T_α as a hitting time at every basepoint $\alpha>\frac{1}{2}$. Chapter 10 extends the hierarchy to symmetric laws on the line, where the reciprocal of ξ on the real axis is the characteristic function of a normal scale mixture, reads zero free half planes as completeness statements after Wiener, Salem and Riesz, and writes Riesz's criterion as the statement that every T_α with $\alpha>\frac{1}{2}$ is a GGC. Chapter 11 reads the hierarchy through the books of Sato and Bondesson, repeats the calculations at the basepoint one for $L(s,\chi_{12})$, uses zero density estimates to bound how far each T_α can be from a GGC, and closes with the work of Anthony Pakes on size biasing, the Lambert law and ladders of self decomposable laws.

Chapter 12 assembles the results of the preceding chapters into a path towards RH on which every step is marked proved, computed, announced or open, sets out the computations that remain as one programme with its milestones, and draws the lessons of the computations for the Thorin programme. Chapter 13 reads RH as a statement about one law, that the Pólya variable is ferromagnetic, in the sense that its moment generating function vanishes only on the imaginary axis; it develops the class of such laws, checks the necessary conditions on the Pólya law, builds, with no hypothesis, provably ferromagnetic laws within total variation distance 1.3×10^{-4} of it and, by a computer assisted proof, within 2×10^{-6}, and shows that e^{2X}, a size biased form of the variable of Biane, Pitman and Yor, is self decomposable but not a generalised gamma convolution. Appendix A describes the code, the scripts and the data files behind the tables and figures, Appendix B gives a short history of computations on the zeros, Appendix C lists the notation, and Appendix D gives the verification code for the Bosch type theorem.

Tables 1.1, 1.2 and 1.3 collect the principal results of Chapters 2 to 11, each with the section where it is developed. A result is marked proved when that section proves it or cites a proof, and computed when it rests on a computation; the section states how far the computation is certified.

Result	Status	Where
Pólya route: ξ and ξ' from two fixed polynomials in ball arithmetic, about 60 correct digits at height 200 at 420 bits, 0.7 ms per point	computed	Section 2.1
Tilted contour: 25 digits of $\rho_{0.05}$ at height 5000 against `mpmath`, in 28 ms per point; to leading exponential order the loss to cancellation falls from about $0.341\,\theta$ digits to about $\delta\theta/\log 10$	proved and computed	Section 2.1
The Lehmer pair near height 7005 resolved by the tilted Pólya route at 34 ms per point	computed	Section 2.2.3.1

Result	Status	Where
$N(1000) = 649$ from the Thorin mass at resolution $\frac{1}{2}$ and the argument along a horizontal segment	computed	Section 2.2.3.2
Raw total variation to U_{RH} maximal; matched total variation at the floor $3.7 \times 10^{-9}\varepsilon$ of the tail model; cell distance $5.44\,\varepsilon$	computed	Section 3.2
Mass defect $-(\varepsilon/\pi)\Xi'(\Theta)/\Xi(\Theta)$ to first order; at $\Theta = 200$ the observed slope agrees with it to five digits at $\varepsilon = 0.005$	proved and computed	Section 3.2
Signatures of an off line pair in total variation and negative mass, exact for an omitted pair and for the negative mass, to leading order for a displaced pair	proved	Section 3.3
Zero density $\nu(\frac{1}{2} + \varepsilon, 200) = 0$ at eight resolutions, from the argument principle identity of Proposition 3.4.2	computed	Section 3.4
Prime side of the sine squared lemma: when no zero has real part greater than $\frac{1}{2}$, Selberg's smoothed sum computes ρ_ε with relative error at most $(x^{-\varepsilon} + x^{-2\varepsilon})/(\varepsilon \log x)$, so with error of order $x^{-\varepsilon}/(\varepsilon^2 \log x)$ at the peaks, the single zero law; tested at $\varepsilon = \frac{1}{4}$	proved and computed	Section 3.5
The Thorin measures converge vaguely to the measure of the zeros on the line, whether or not RH holds	proved	Section 4.1
The clock is HCM if and only if RH holds, and the shifted clock at any basepoint $\alpha \geq 1$ is HCM unconditionally; the 216 normalised derivatives of a finite HCM test are all positive, certified in ball arithmetic	proved and computed	Section 4.1
Bosch type threshold: $1/G(x^{1/\beta})$ is HCM if and only if $\lvert\beta\rvert \geq 1/(1-\alpha_\xi)$, with α_ξ set by the largest angle of the zeros; unconditionally $1/G(x^p)$ is HCM for $\lvert p\rvert \leq 1 - 1/(\pi T_0)$, where $T_0 = 3 \times 10^{12}$ is the verified height, in particular for $\lvert p\rvert \leq 1 - 1.1 \times 10^{-13}$	proved	Section 4.3
The clock is infinitely divisible and self decomposable with no hypothesis, and $(-1)^k W^{(k)} > 0$ for every $k \leq 1.16 \times 10^{25}$, that is, the Williamson measures, inverse gamma smoothings of the Thorin measure, are positive to that order; the two proofs use the verification of RH below heights 1000 and 3×10^{12} respectively [196]; Laplace's method on the Gumbel tail of the Pólya kernel and a Tauberian argument give the leading term of the Weyl law	proved	Section 5.1

Table 1.1: Principal results of Chapters 2 to 5.

Result	Status	Where
The normalised reciprocal zeta function is infinitely divisible with no hypothesis, and it is not self decomposable, not a GGC and not in the Bondesson class	proved and computed	Section 7.1

Result	Status	Where
Conjecture S, that the symmetrised truncations of the Biane, Pitman and Yor lattice have their zeros in the strip on the line, holds for the sizes $n \leq 24$ and $n = 32, 36$ above explicit heights and numerically below them; for infinitely many n it would imply RH	proved and computed	Section 7.2
Law of the clock by Talbot inversion, with no zero as input; its hazard tends to γ_1^2, reached to eleven digits at $t = 0.25$	computed	Section 8.1
Kent approximants with 1700 zeros and a moment matched Weyl tail: relative error 3.5×10^{-23} at $u = 1$ and 7.0×10^{-13} at $u = 10^4$; for large u the error of the plain Weyl tail levels off at a constant set by S	computed	Section 8.1
The clock is not the first passage time of Brownian motion to an independent random level; the boundary from the Volterra equation, solved in three zones, reproduces its law to Monte Carlo accuracy (Kolmogorov distance 0.0020, against 0.0059 for the tangent boundary)	proved and computed	Section 8.2
Keiper Li coefficients $\lambda_1, \ldots, \lambda_{100}$ from the Pólya density, λ_1 to 25 digits	computed	Section 8.3
Closest pair below height 200 collides at de Bruijn Newman time -0.288; the Lehmer pair near 7005 at -7.11×10^{-4}	computed	Section 8.4
For $L(s, \chi_{12})$ and $L(s, \Delta)$ the sine squared lemma has no pole piece but a signed prime piece	proved	Section 8.5
The eta kernel of $L(s, \chi_{12})$: GRH below height 100 from the zero density identity and the 67 sign changes	computed	Section 8.5
For an imaginary quadratic field the Pólya kernel of ζ_K is positive exactly when an explicit theta sum is positive on a finite interval, which gives $L(\frac{1}{2}, \chi_d) > 0$ for $d = -3, -4, -7, -8$; it takes negative values for every $d \equiv 5 \pmod{24}$	proved	Section 8.5
For the Dirichlet beta function $L(s, \chi_4)$, whose kernel is the cube of the eta function: GRH below height 100; a positive Thorin density at every basepoint $\alpha \geq 1$, which the absolute values of the prime weights certify only for $\alpha > 2.2823$; the density as the difference of those of $\xi_{\mathbb{Q}(i)}$ and ξ, and as the derivative of the argument of the completed function on the line	proved and computed	Section 8.5.7
Nonnegative deconvolution of $\rho_{0.1}$ on $[15, 55]$, with the zeros outside $[5, 65]$ as background, recovers every zero in the window to within 10^{-5}	computed	Section 8.6
For Pólya's Bessel function $K_{i\theta/2}(2\pi)$ the index side clock is the hitting time of an explicit diffusion, its Thorin atoms are the squared zeros and its trace is completely monotone; its de Bruijn Newman constant is finite, by Newman's theorem, although none of its real zeros below height 90 collide down to $t = -60$	proved and computed	Section 8.7

Table 1.2: Principal results of Chapters 6 to 8.

Result	Status	Where
T_1, with Laplace transform $\xi(1)/\xi(1+\sqrt{u})$, is a GGC with positive Thorin density, smallest at the origin, by a computation of the zeros below height 2000 in ball arithmetic and explicit zero counts above it, so that T_1 is a positive stable one half variable plus an independent GGC; its law has infinite mean and a stable one half tail, and the Lévy form of $\xi(1+s)$, which converges by the prime number theorem, gives its Thorin density from the primes	proved and computed	Section 9.1
Below the basepoint one, $v_\alpha(\theta)/(\alpha-\frac{1}{2})$ is nonincreasing in α away from the zeros off the line; the Thorin density is smallest at the origin, and only there, for every $\alpha \in (\frac{1}{2}, 7.35]$ on $\lvert\theta\rvert \le 3\times 10^{12}-1.21$, for $L(s,\chi_4)$ and $L(s,\chi_{12})$ at every height for every $\alpha \ge 1$, and at every height for every $\alpha \in (\frac{1}{2},1)$ exactly when RH holds; for every $\alpha > \frac{1}{2}$ it is positive on $\lvert\theta\rvert \le 3\times 10^{12} - \frac{1}{2}$	proved and computed	Sections 9.1.6, 11.4
Every T_α, $\alpha > \frac{1}{2}$, exists as the tilted centre clock run through the inverse local time of a sticky Brownian motion, and it is a GGC if and only if no zero has real part greater than α	proved	Section 9.2
For real u and $\alpha \ge \frac{1}{2}$, $\xi(\alpha)/\xi(\alpha+\lvert u\rvert)$ is the characteristic function of an infinitely divisible normal scale mixture; for $\frac{1}{2} < \alpha < 1$ it is an extended GGC, and the translates of every tilted Pólya kernel $\Phi_{\alpha'}$ with $\alpha' > \alpha$ are complete in L^1, exactly when no zero has real part greater than α	proved	Section 10.1
The complete monotonicity conditions of the Poisson trace $K_{1/2}$, which together are RH, hold for every order $n < T_0(\pi - y)$ at the point y; $K_{1/2}$ and the trace of Pólya's Bessel function differ by $\frac{9}{8} + O(y\log\frac{1}{y})$ as $y \to 0$	proved and computed	Section 10.1.5
$1/\zeta(2k)$ is, up to an elementary factor, a value of the Laplace transform of the centre clock and of T_1; for $\frac{1}{2} \le \alpha < 1$ the bound $O(x^{\alpha/2+\delta})$ on Riesz's function, for every $\delta > 0$, holds exactly when every $T_{\alpha'}$ with $\alpha' > \alpha$ is a GGC	proved	Section 10.2
The constant of $\sum_{n\le x}\mu(n)^2/n$ is an elementary term plus $\mathbb{E}V_1 - 2\,\mathbb{E}V_2$, two tilted Pólya means; RH holds exactly when the squarefree count smoothed by an exponential or a Gamma(2) law, or the count in mean square, has error $O(x^{1/4+\varepsilon})$; the error is $\Omega_\pm(x^{1/4})$; Gamma(2) smoothing writes it as an average of the Mertens function under a positive law, and two is the largest shape for which this holds	proved	Section 10.3
The centre clock is driven by a subordinator with Lévy density $-W'$ and is unimodal, as is T_α for $\alpha \ge 1$; the tilted Pólya variable is quasi infinitely divisible for every $\alpha > 1$ but not at $\alpha = 1$	proved	Section 11.1
For $L(s,\chi_{12})$ the Thorin measure at the basepoint one is positive, and its Lévy form there has no pole to compensate and prime atoms of both signs	proved	Section 11.3

Result	Status	Where
For $\frac{1}{2} < \alpha < 1$, in the density $\rho_{\alpha-1/2}$ the Cauchy kernels of the zeros to the right of α carry at heights $\theta \le T$ at most $T^{e(\alpha)-1+o(1)}$ times the mass of the kernels of the other zeros, with $e(\alpha) < 1$, by the zero density estimates of Ingham, of Guth and Maynard and of Huxley [104, 88, 100]	proved	Section 11.4
With no hypothesis, the size biased centre clock is the clock plus an independent variable with density $W/\mathbb{E}T$, and RH holds exactly when this variable is a mixture of exponential laws, with weight $\gamma^{-2}/\mathbb{E}T$ at the rate γ^2, about a fifth at the lowest zero; the Lambert law, with Laplace transform $\mathcal{W}(u)/u$, is a GGC whose Thorin measure is the image of the uniform law on $(0,\pi)$ under $b \mapsto b\,e^{-b\cot b}/\sin b$, by a proof of which only one step fails for the clock, and that step is RH	proved and computed	Sections 11.5.2, 11.5.3
The centre clock lies in Urbanik's class L_2 of multiply self decomposable laws and not in L_3; it is not the martingale limit of a supercritical continuous state branching process with finite Malthusian parameter	proved and computed	Section 11.5.4
A property of even kernels that also holds for the damped Pólya kernel $e^{tv^2/4}\Phi(v)$ for some $t < 0$ cannot by itself imply RH, by the theorem of Rodgers and Tao; the GGC property of the variable of Biane, Pitman and Yor fails for the damped kernel at every $t < 0$ tested and, provably, for small $\lvert t\rvert$, so the filter does not exclude it there	proved and computed	Section 11.5.5

Table 1.3: Principal results of Chapters 9 to 11. All are unconditional, and several use verified computations: the verification of RH below height 1000, through Theorem 5.1.1, or below the height $T_0 = 3 \times 10^{12}$, Platt's verification of GRH for small moduli, or computations in ball arithmetic.

1.1.3 Conventions and how to read this book

The clock is a function of $u = s^2$, so that the atoms of its Thorin measure are at γ^2; earlier papers of the programme used $s^2/2$, which rescales Thorin measures by $z \mapsto z/2$. Thorin measures are measures in the variable z, and we write $\theta = \sqrt{z}$ for the chart in which the Thorin measure at the basepoint $\frac{1}{2} + \varepsilon$ has density $\rho_\varepsilon(\theta) = \pi^{-1}\operatorname{Re}(\xi'/\xi)(\frac{1}{2} + \varepsilon + i\theta)$; ε is the resolution. The Pólya kernel Φ is taken in Titchmarsh's normalisation, $\Xi(\theta) = \int_{\mathbb{R}} \Phi(u)\cos(\theta u)\,du$. $N(T)$ counts zeros with $0 < \operatorname{Im}\rho \le T$ and $\nu(\sigma, T)$ those with, in addition, $\operatorname{Re}\rho > \sigma$. Total variation of a signed measure is the total variation norm $\lvert\mu\rvert(E)$ on the stated window. Chapters 6 and 7, and Sections 4.2 and 4.3, write s for u, and the Kent approximants of Section 8.1 write w; each declares its variables where it starts, and Appendix C lists the notation of the book with the place where each symbol is defined. Numerical values are quoted to the accuracy at which they have been checked, and the section in which a value is computed says how.

Readers mainly interested in computation can read Sections 1.2, 2.1 and 2.3

and then go straight to Chapters 3 to 5, where the Thorin condition is analysed numerically. Readers interested in the probabilistic side can start with Sections 1.3, 8.1 and 8.2 and continue with Chapters 9 to 11, where the shifted clocks are hitting times and normal scale mixtures; these chapters build on Sections 1.2 and 1.3 and draw on Section 3.4 for the zero count. Chapters 6 and 7 can be read after Chapter 1 and the statements of Theorems 5.1.1 and 5.1.4 in Section 5.1; they form the most technical part of the book, and they prove unconditional theorems about the reciprocal zeta function. Readers looking for the conclusions can go directly to Sections 12.1 and 12.2.6. Throughout, statements are marked as proved, computed, announced or open, and Appendix A names the script behind each computed table and figure or records that the archive has none.

1.2 The Xi function, the Pólya density and the clock

Section 1.1 states RH as a positivity condition on the clock $1/G$, and computation with that condition rests on four objects. The Hadamard product of ξ in the variable $u = s^2$ turns RH into a statement about the zeros of an entire function of genus zero. The Pólya kernel makes $\xi(\frac{1}{2}+s)/\xi(\frac{1}{2})$ the moment generating function of a probability density, and it underlies the evaluator of Section 2.1. The heat trace of the zeros turns the classes of infinitely divisible laws into sign conditions on a single function, and the explicit formula expresses it through the primes. At the basepoint $\frac{1}{2} + \varepsilon$ the Thorin density is the Poisson smoothing of the zeros at every resolution $\varepsilon > 0$, and it is the density of the Thorin measure of a GGC when the basepoint lies to the right of every zero; the sine squared lemma produces the same density from the primes. The rest of the book computes these objects and measures how close they come to the discrete measure that RH demands. Most of the material is classical or taken from [200].

1.2.1 The completed zeta function and the clock

The completed zeta function ξ of Section 1.1 is entire of order one and maximal type, satisfies $\xi(s) = \xi(1-s)$ and $\overline{\xi(\bar{s})} = \xi(s)$, and is positive on the real axis. On the critical line $\Xi(\theta) = \xi(\frac{1}{2} + i\theta)$ is real and even. The zeros $\rho = \beta + i\gamma$ of ξ lie in the open strip $0 < \operatorname{Re} s < 1$ and are symmetric under $\rho \mapsto 1-\rho$ and $\rho \mapsto \bar{\rho}$. As in Section 1.1, $G(u) = \xi(\frac{1}{2} + \sqrt{u})/\xi(\frac{1}{2})$, and $a_k = -(\rho_k - \frac{1}{2})^2$, one for each pair $\{\rho, 1-\rho\}$.

Hadamard's theorem applied to G, which is entire of order $\frac{1}{2}$, gives, with $u = s^2$,

$$\frac{\xi(\frac{1}{2}+s)}{\xi(\frac{1}{2})} = \prod_k \Big(1 - \frac{s^2}{(\rho_k - \frac{1}{2})^2}\Big) = \prod_k \Big(1 + \frac{s^2}{a_k}\Big), \tag{1.2.1}$$

with $\sum |a_k|^{-1} < \infty$. Logarithmic differentiation gives the partial fraction expan-

sion

$$\frac{\xi'}{\xi}(s) = \sum_{\rho} \frac{1}{s-\rho}, \tag{1.2.2}$$

in which each ρ is grouped with $1-\rho$, as the differentiation of (1.2.1) produces, or with $\bar{\rho}$. Grouped either way the series converges absolutely, since each group is $O(\gamma^{-2})$ for fixed s and $\sum_\rho \gamma^{-2} < \infty$; without grouping it converges only when summed symmetrically in $\operatorname{Im}\rho$. Its real part $\sum_\rho \operatorname{Re}(s-\rho)/|s-\rho|^2$ converges absolutely without grouping.

Passing to $u = s^2$ folds each pair $\pm(\rho_k - \frac{1}{2})$ of zeros of $\xi(\frac{1}{2}+s)$ into the single zero $-a_k$ of G, so the product has genus zero and needs no exponential factors. Since ξ has no real zeros, a_k is real only when ρ_k lies on the critical line, and then $a_k = \gamma_k^2 > 0$. RH is therefore the statement that every zero of G lies on the negative real axis. The reciprocal of a product of genus zero with negative zeros is a product of gamma Laplace transforms, and this is how generalised gamma convolutions enter.

A random variable $H \geq 0$ is a generalised gamma convolution (GGC) if $\mathbb{E}e^{-uH} = \exp\{-du - \int_{(0,\infty)} \log(1+u/z)\, U(dz)\}$ for $u > 0$, with a drift $d \geq 0$ and a positive measure U, the Thorin measure, such that $\int_{(0,1)} |\log z|\, U(dz) + \int_{(1,\infty)} z^{-1} U(dz) < \infty$. The drift and the Thorin measure are unique, and the Laplace transform of a GGC extends to a holomorphic nonvanishing function on $\mathbb{C} \setminus (-\infty, 0]$. Section 1.3 develops the class from the same definition, (1.3.1).

Unconditionally, $1/G$ is the Laplace transform of a probability law on $[0,\infty)$ [200, Theorem A]; Section 5.1 states this as Theorem 5.1.1 and proves it. A random variable T with $\mathbb{E}e^{-uT} = 1/G(u)$ for $u \geq 0$ is called the clock variable. Its mean and variance are $\mathbb{E}T = \sum_k a_k^{-1} = G'(0) = \xi''(\frac{1}{2})/(2\xi(\frac{1}{2})) = 0.0231050$ and $\operatorname{Var} T = \sum_k a_k^{-2} = 3.71726 \times 10^{-5}$; these constants, and $\xi(\frac{1}{2}) = 0.4971207781883141$, recur throughout the book. The Thorin condition asks whether T is a GGC, and the answer is the rigidity theorem stated in Section 1.1, which we restate with its source.

Proposition 1.2.1 (Rigidity, [200, Proposition 3.4]). *The clock $1/G$ is the Laplace transform of a GGC if and only if RH holds, and then its Thorin measure is U_{RH} of (1.1.1) and the clock variable is $T = \sum_{\gamma>0} e_\gamma/\gamma^2$ with independent standard exponentials e_γ.*

The name refers to the absence of any freedom in the Thorin measure. If $1/G$ is the Laplace transform of a GGC with drift d and Thorin measure U, then $d + \int U(dz)/(u+z) = G'/G(u) = \sum_k (u+a_k)^{-1}$ for $u > 0$. The left side is holomorphic on $\mathbb{C}\setminus(-\infty,0]$, so every pole $-a_k$ of the right side lies on the negative real axis. Hence every a_k is positive, which is RH, and uniqueness of the Stieltjes representation forces $d = 0$ and $U = \sum_k \delta_{a_k}$. Conversely, under RH, (1.2.1) writes $1/G$ as a product of gamma Laplace transforms.

1.2.2 The Pólya kernel and the Pólya density

Riemann's continuation of ζ writes ξ as a Mellin transform of a combination of $\omega(x) = \sum_{n\geq 1} e^{-\pi n^2 x}$ and its derivatives, and uses the theta transformation $1 + 2\omega(1/x) = \sqrt{x}\,(1 + 2\omega(x))$, $x > 0$. The substitution $x = e^{2u}$ turns the Mellin transform into a Fourier transform, and in Titchmarsh's normalisation [255, Section 10.1] the result is

$$\begin{aligned} \Phi(u) &= 2\sum_{n\geq 1} \bigl(2\pi^2 n^4 e^{9u/2} - 3\pi n^2 e^{5u/2}\bigr) \exp(-\pi n^2 e^{2u}), \\ \Xi(\theta) &= \int_{\mathbb{R}} \Phi(u) \cos(\theta u)\, du. \end{aligned} \tag{1.2.3}$$

Here u is the variable of the kernel, not the argument of the clock. Pólya studied this representation [213], and Φ is called the Pólya kernel.

The kernel is even and strictly positive, and $\Phi(u) \sim 4\pi^2 e^{9|u|/2} \exp(-\pi e^{2|u|})$ as $|u| \to \infty$. Evenness is the functional equation read through the theta transformation (Exercise 1.2). For $u \geq 0$ every term of (1.2.3) is positive, because $2\pi^2 n^4 e^{9u/2} - 3\pi n^2 e^{5u/2} = \pi n^2 e^{5u/2}(2\pi n^2 e^{2u} - 3)$ and $2\pi n^2 e^{2u} \geq 2\pi > 3$; positivity for $u < 0$ follows by evenness. Evenness also decides how Φ is computed. Term n is $n^{-1/2}$ times the first term translated to the left by $\log n$, so for $u \geq 0$ the first term dominates, while for $u < 0$ the terms are individually of order $n^{-1/2}$ and cancel almost completely. The theta transformation turns their sum into the mirror image of the side $u > 0$, and computations therefore evaluate $\Phi(|u|)$.

Each term of (1.2.3) is entire in u, and since $\operatorname{Re} e^{2u} = e^{2x}\cos 2y$ for $u = x + iy$, the series converges locally uniformly in the strip $|\operatorname{Im} u| < \pi/4$, so Φ is analytic there; the boundary lines $\operatorname{Im} u = \pm\pi/4$ are a natural boundary inherited from the theta function [219, Chapter 11, Section 1.4, and Chapter 5, Section 3.2]. The strip of analyticity makes the trapezoid rule spectrally accurate and permits the tilted contour of Section 2.1.3. Figure 1.2 shows $|\Phi|$ on lines inside the strip.

Since $\Xi(0) = \xi(\frac{1}{2}) = \int_{\mathbb{R}} \Phi$, the Pólya density $p = \Phi/\Xi(0)$ is the law of a symmetric random variable X with all exponential moments, and

$$\frac{\xi(\frac{1}{2} + s)}{\xi(\frac{1}{2})} = \mathbb{E}\, e^{sX} = \mathbb{E}\cosh(sX), \qquad s \in \mathbb{C}. \tag{1.2.4}$$

The zeros of $\xi(\frac{1}{2} + s)$ are therefore the complex s at which the moment generating function of X vanishes, and the ordinates of the zeros on the critical line are the real zeros of its characteristic function $\mathbb{E}\cos(\theta X)$. The clock is $1/\mathbb{E}\cosh(\sqrt{u}\,X)$, and at $u = \theta^2$ it forms with the characteristic function of X a van Dantzig pair, that is, a pair (φ_1, φ_2) of characteristic functions with $\varphi_2(\theta) = 1/\varphi_1(i\theta)$; the name goes back to a problem of van Dantzig, which Lukacs studied [150]. Here $\varphi_1(\theta) = \Xi(\theta)/\Xi(0)$ is the characteristic function of X, and $1/\varphi_1(i\theta) = \xi(\frac{1}{2})/\xi(\frac{1}{2} + \theta) = \mathbb{E}e^{-\theta^2 T}$ is the characteristic function of $Z\sqrt{2T}$, with Z standard normal and independent of T

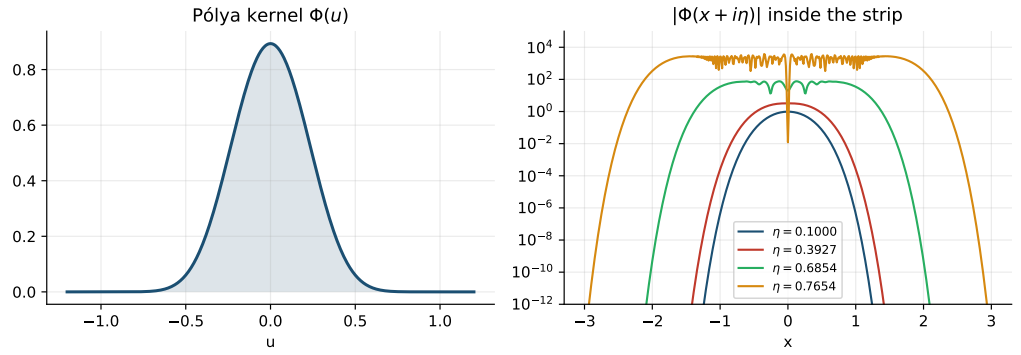


Figure 1.2: Left: the Pólya kernel $\Phi(u)$ on the real line. Right: $|\Phi(x+i\eta)|$, on a logarithmic scale, on the lines $\eta = 0.1000$, 0.3927, 0.6854 and 0.7654 inside the strip of analyticity, that is at distances 0.6854, 0.3927, 0.1 and 0.02 below its edge $\eta = \pi/4$ (the tilts of Section 2.1.3). As the line approaches the natural boundary $\eta = \pi/4$ the kernel grows and spreads (compare Table 2.1), and it still decays doubly exponentially in $|x|$.

[200, Theorem A]. G. Letac asked whether φ_1 is the first member of a van Dantzig pair (private communication), and Theorem 5.1.1 shows that it is, unconditionally. Section 10.1 reads the reciprocal of ξ along the real axis in the same way at every basepoint, and it shows by Wiener's closure theorem that the translates of Φ are not dense in $L^1(\mathbb{R})$, because Ξ has real zeros (Proposition 10.1.9).

Two classical pairs show the shape to expect. The first is $(\cos\theta, 1/\cosh\theta)$, the symmetric Bernoulli law on $\{-1,1\}$ with the hyperbolic secant law, and the second is $(\sin\theta/\theta, \theta/\sinh\theta)$, the uniform law on $[-1,1]$ with the logistic law scaled by $1/\pi$. In both the first member has only real zeros and the second is the product of the reciprocal factors: $1/\cosh\theta = \prod_{k\geq 1}(1+\theta^2/((k-\frac{1}{2})\pi)^2)^{-1}$ is the characteristic function of $\sum_{k\geq 1} L_k/((k-\frac{1}{2})\pi)$, where the L_k are independent standard Laplace variables, with density $\frac{1}{2}e^{-|x|}$ and characteristic function $1/(1+\theta^2)$, and likewise $\theta/\sinh\theta$ is that of $\sum_{k\geq 1} L_k/(k\pi)$. The product form alone does not make a pair, since the first member must be a characteristic function in its own right, and $\prod_k(1-\theta^2/c_k^2)$ over finitely many real c_k is an unbounded polynomial.

The Xi pair has the classical shape exactly when RH holds. If RH holds, $T = \sum_{\gamma>0} e_\gamma/\gamma^2$ (Proposition 1.2.1), and since $Z\sqrt{2e}$ is a standard Laplace variable when e is a standard exponential variable independent of Z, $Z\sqrt{2T}$ has the law of $\sum_{\gamma>0} L_\gamma/\gamma$, the sum that Section 4.2 calls $\log W$. Conversely, if $Z\sqrt{2T}$ has the law of $\sum_k L_k/c_k$ for real c_k with $\sum_k c_k^{-2} < \infty$, then $\xi(\frac{1}{2}+\theta)/\xi(\frac{1}{2}) = \prod_k(1+\theta^2/c_k^2)$ for real θ, hence for complex θ, and every zero of ξ lies on the critical line. Konstantopoulos, Patie and Sarkar state the same equivalence in the language of the Laguerre Pólya class and of Pólya frequency functions [127, Theorem 2.3 and

Corollary 2.4]. Table 1.4 computes the two members and compares them with their Hadamard products over the first 1700 zeros, all of which lie on the line; the zeros above them enter through $\mathbb{E}T = \sum_k a_k^{-1}$ and $\operatorname{Var} T = \sum_k a_k^{-2}$, which are read off from the Taylor coefficients of $\log \xi(\frac12 + s)$ at $s = 0$ with no hypothesis on the zeros. Section 10.1 adds that the second member is self decomposable, that a zero off the line would contribute a factor that is not itself a characteristic function, and that away from the centre there is no such pair.

θ	$\Xi(\theta)/\Xi(0)$	$\xi(\frac12)/\xi(\frac12+\theta)$	products, relative difference
1	0.97714167	0.97717800	1×10^{-18}
2	0.91144824	0.91199057	8×10^{-17}
5	0.55429185	0.56739462	2×10^{-14}
10	0.076375505	0.11524595	1×10^{-12}
15	$-1.4195704 \times 10^{-3}$	1.0310555×10^{-2}	1×10^{-11}
20	$-7.3735457 \times 10^{-5}$	4.8509905×10^{-4}	8×10^{-11}
25	2.7809283×10^{-8}	1.3591692×10^{-5}	3×10^{-10}
30	$-3.0207191 \times 10^{-8}$	2.4728825×10^{-7}	9×10^{-10}

Table 1.4: The van Dantzig pair, computed by `vdpair.py`. The last column is the larger relative difference between a member and its Hadamard product, $\prod(1-\theta^2/\gamma^2)$ or $\prod(1+\theta^2/\gamma^2)^{-1}$, over the 1700 zeros below height 2198, with the zeros above entering through $\sum_k a_k^{-1}$ and $\sum_k a_k^{-2}$; it is the size of the neglected term of order θ^6. On $(0, 30]$ the first member changes sign at $\gamma_1 = 14.13$, $\gamma_2 = 21.02$ and $\gamma_3 = 25.01$; the second is positive and decreasing. Both are $1 - \theta^2\mathbb{E}T + O(\theta^4)$.

The Pólya variable also has a probabilistic construction. Let $S_2 = \sum_{k\ge1} \frac{2}{\pi^2 k^2}(e_k + e_k')$ with independent standard exponentials e_k, e_k'; it has the law of the sum of two independent hitting times of level 1 by a three dimensional Bessel process started at 0. By the theorem of Biane, Pitman and Yor [28], going back to Williams [271], $2\xi(s) = \mathbb{E}[(\pi S_2/2)^{s/2}]$. With $Y = \frac12 \log(\pi S_2/2)$ this reads $\mathbb{E}e^{sY} = 2\xi(s)$, so X is Y tilted by $e^{Y/2}$: $\mathbb{E}f(X) = \mathbb{E}[f(Y)e^{Y/2}]/\mathbb{E}e^{Y/2}$. Section 8.1 uses this to estimate Ξ from samples of S_2 and shows why sampling cannot locate the zeros. Tilting Y by $e^{\alpha Y}$ in place of $e^{Y/2}$ gives, at every real basepoint α and with no hypothesis on the zeros, a variable V_α with $\mathbb{E}e^{sV_\alpha} = \xi(\alpha+s)/\xi(\alpha)$ and $V_{1/2} = X$ (Section 9.1.1).

1.2.3 The heat trace: spectral and arithmetic forms

The heat trace of the zeros is

$$W(t) = \sum_k e^{-a_k t} = \tfrac12 \sum_\rho e^{(\rho-\frac12)^2 t}, \qquad t > 0,$$

and the series converges absolutely for every $t > 0$ wherever the zeros lie, since $\operatorname{Re} a_k = \gamma_k^2 - (\beta_k - \frac12)^2 > \gamma_k^2 - \frac14$ for $\rho_k = \beta_k + i\gamma_k$. Integrating $e^{-(u+a_k)t}$ term

by term, and using Frullani's integral $\log(1+u/a)=\int_0^\infty(1-e^{-ut})e^{-at}t^{-1}dt$ for $\operatorname{Re}a>0$, gives, unconditionally for $u>0$ [200, Theorem 4.3],

$$\frac{G'}{G}(u)=\int_0^\infty e^{-ut}W(t)\,dt,\qquad \log G(u)=\int_0^\infty \bigl(1-e^{-ut}\bigr)\frac{W(t)}{t}\,dt.$$

The second formula shows that $W(t)/t$ is the only possible Lévy density of the clock variable T.

The classes of infinitely divisible laws of Section 1.3.2 become sign conditions on W [200, Theorem 4.6]: T is infinitely divisible if and only if $W\ge 0$, self decomposable if and only if W is nonincreasing, and a GGC if and only if W is completely monotone, which by Proposition 1.2.1 happens if and only if RH holds. The first two conditions hold unconditionally, in the strict forms $W>0$ and $W'<0$ on $(0,\infty)$ (Theorem 5.1.1, [200, Theorem A]), by a proof that uses the verification of RH below height 1000; the third is the Thorin condition, and Table 1.7 records the rungs proved between them.

The explicit formula expresses W through the primes. For even test functions h holomorphic and rapidly decreasing in a strip $|\operatorname{Im}r|\le\frac12+c$ with $c>0$, Weil's explicit formula [266], [108, Theorem 5.12], reads

$$\sum_\rho h(\gamma_\rho)=h(\tfrac{i}{2})+h(-\tfrac{i}{2})+\frac{1}{2\pi}\int_{\mathbb{R}}h(r)\bigl(\operatorname{Re}\psi(\tfrac14+\tfrac{ir}{2})-\log\pi\bigr)dr \\ -2\sum_{n\ge2}\frac{\Lambda(n)}{\sqrt{n}}\,g(\log n),$$

where $\rho=\frac12+i\gamma_\rho$, $g(x)=\frac{1}{2\pi}\int_{\mathbb{R}}h(r)e^{-irx}dr$ and $\psi=\Gamma'/\Gamma$ is the digamma function. The Gaussian $h(r)=e^{-r^2t}$ has $g(x)=e^{-x^2/4t}/(2\sqrt{\pi t})$ and $\sum_\rho h(\gamma_\rho)=2W(t)$, which gives the arithmetic form [200, Theorem 4.4]

$$\begin{aligned}W(t)&=e^{t/4}+\frac{1}{4\pi}\int_{\mathbb{R}}e^{-r^2t}m(r)\,dr-\frac{1}{2\sqrt{\pi t}}\sum_{n\ge2}\frac{\Lambda(n)}{\sqrt{n}}e^{-(\log n)^2/4t},\\ m(r)&=\operatorname{Re}\psi\bigl(\tfrac14+\tfrac{ir}{2}\bigr)-\log\pi.\end{aligned}\tag{1.2.5}$$

The three terms of (1.2.5) are the polar, archimedean and prime contributions. The polar term $e^{t/4}$ comes from $h(\pm\frac{i}{2})$, that is, from the poles of $\pi^{-s/2}\Gamma(s/2)\zeta(s)$ at $s=1$ and $s=0$, which the factor $s(s-1)$ of ξ removes. The archimedean term averages the digamma function of the gamma factor against a Gaussian and is negative for moderate t. The prime term is the series $\sum\Lambda(n)n^{-1/2}$ smoothed by a Gaussian of standard deviation $\sqrt{2t}$ in $\log n$: for small t only the smallest primes contribute, and for large t many primes contribute with nearly equal weight. As $t\to\infty$ the prime term grows like the polar term: $W(t)\to0$ and the archimedean term tends to 0, so (1.2.5) forces the prime term to be $e^{t/4}+o(1)$, and W is what remains after the two cancel.

Table 1.5 reproduces the comparison of [200, Table 4.1], and the two sides agree to fifteen digits. The polar and archimedean terms are of order one and the prime term rises from 8.4×10^{-6} at $t = 0.01$ to 0.30 at $t = 0.2$, while $W(t)$ decays like $e^{-\gamma_1^2 t}$ with $\gamma_1^2 = 199.79$. An estimate of W from (1.2.5) at moderate t must therefore survive a cancellation whose depth grows linearly in t. Section 5.3.2 extends the comparison to smaller and larger t and gives that depth explicitly.

t	$W(t)$ from zeros	polar + archimedean − prime	archimedean	prime
0.01	0.149698011253780	0.149698011253780	−0.8527967	8.3995×10^{-6}
0.02	$1.85412917313195 \times 10^{-2}$	$1.85412917313195 \times 10^{-2}$	−0.9840612	0.0024101
0.05	$4.58783504857357 \times 10^{-5}$	$4.58783504857357 \times 10^{-5}$	−0.9546186	0.0579140
0.10	$2.10479979582235 \times 10^{-9}$	$2.10479979582235 \times 10^{-9}$	−0.8625096	0.1628055
0.20	$4.43018218022364 \times 10^{-18}$	$4.43018218022364 \times 10^{-18}$	−0.7497538	0.3015173

Table 1.5: The heat trace from the zeros and from the explicit formula, after [200, Table 4.1]. At $t = 1$ the cancellation between the arithmetic terms reaches about eighty seven digits.

1.2.4 The Thorin density at a shifted basepoint

Whether or not the Thorin condition holds at the centre, at a basepoint α to the right of every zero the shifted clock $u \mapsto \xi(\alpha)/\xi(\alpha + \sqrt{u})$ is the Laplace transform of a GGC with an explicit Thorin density. The same formula defines a density at every basepoint $\alpha > \frac{1}{2}$, whatever the zeros do, so it can be computed at every positive resolution $\alpha - \frac{1}{2}$, and it is the starting point of the computations in Chapters 3 to 5. For a real basepoint α write $v_\alpha(r) = \mathrm{Re}(\xi'/\xi)(\alpha + ir)$, and let U_α be the signed measure on $(0, \infty)$ given by

$$U_\alpha(dz) = \frac{v_\alpha(\sqrt{z})}{2\pi\sqrt{z}}\, dz. \tag{1.2.6}$$

At $\alpha = \frac{1}{2} + \varepsilon$, where $\varepsilon > 0$ is the resolution, the chart $\theta = \sqrt{z}$ gives U_α the density

$$\rho_\varepsilon(\theta) = \frac{1}{\pi}\, v_{1/2+\varepsilon}(\theta) = \frac{1}{\pi}\, \mathrm{Re}\, \frac{\xi'}{\xi}\Big(\frac{1}{2} + \varepsilon + i\theta\Big), \tag{1.2.7}$$

so that $U_\alpha(dz) = \rho_\varepsilon(\theta)\, d\theta$. The family $\varepsilon \mapsto \rho_\varepsilon$ is the Cauchy flow of [200, Chapter 8].

Theorem 1.2.2 ([200, Theorem 3.7 and Remark 3.10]). *If $\alpha > \mathrm{Re}\,\rho$ for every zero, in particular for every $\alpha \geq 1$, then $u \mapsto \xi(\alpha)/\xi(\alpha + \sqrt{u})$ is the Laplace transform of a GGC T_α with Thorin density* (1.2.6)*, and $v_\alpha > 0$. It is the Laplace transform of a GGC if and only if $\alpha \geq \sup_\rho \mathrm{Re}\,\rho$.*

Since no zero lies on or to the right of the line $\operatorname{Re} s = 1$, the hypothesis of the theorem holds at $\alpha = 1$, and Section 9.1 works out the shifted clock there in full. Until 2026 no smaller basepoint was known to satisfy it; the zero free half plane $\operatorname{Re} s > \frac{7}{8}$ announced in October 2026 would give every $\alpha > \frac{7}{8}$ (Section 12.1.5). At every basepoint $\alpha > \frac{1}{2}$ the shifted clock is still the Laplace transform of a probability law, with no hypothesis on the zeros (Theorem 9.2.5); what the zeros decide is whether that law is a GGC. Let $P_a(x) = a/(\pi(a^2 + x^2))$ denote the Poisson kernel; for $a > 0$ it is the density of the Cauchy law with scale a, and $P_{-a} = -P_a$. Write μ_ε for the signed measure $U_{1/2+\varepsilon}$, and use the same symbol for its image in the chart $\theta = \sqrt{z}$, where it has the density (1.2.7). By Theorem 1.2.2, μ_ε is the Thorin measure of the GGC $T_{1/2+\varepsilon}$ when $\frac{1}{2} + \varepsilon > \operatorname{Re} \rho$ for every zero, in particular for every $\varepsilon \geq \frac{1}{2}$, and we call it the Thorin measure at resolution ε in general. Grouping each zero with its partners under $\rho \mapsto 1 - \bar{\rho}$ and $\rho \mapsto \bar{\rho}$ gives the unconditional structure of its density.

Proposition 1.2.3 (Poisson structure). *For $\varepsilon > 0$ and $\theta \geq 0$ such that $\frac{1}{2} + \varepsilon + i\theta$ is not a zero of ξ,*

$$\rho_\varepsilon(\theta) = \sum_{\gamma > 0,\ \operatorname{Re}\rho = 1/2} \big[P_\varepsilon(\theta - \gamma) + P_\varepsilon(\theta + \gamma)\big] + \sum_{\text{off line } \rho = \frac{1}{2} \pm \delta + i\gamma,\ \gamma > 0} \Big(\big[P_{\varepsilon+\delta} + P_{\varepsilon-\delta}\big](\theta - \gamma) + \big[P_{\varepsilon+\delta} + P_{\varepsilon-\delta}\big](\theta + \gamma)\Big), \quad (1.2.8)$$

where the second sum counts each off line pair $\{\frac{1}{2} + \delta + i\gamma, \frac{1}{2} - \delta + i\gamma\}$, $0 < \delta < \frac{1}{2}$, once. Equivalently, with $\alpha = \frac{1}{2} + \varepsilon$, $\rho_\varepsilon(\theta)$ is the sum of $P_{\alpha-\beta}(\theta - \gamma) + P_{\alpha-\beta}(\theta + \gamma)$ over the zeros $\rho = \beta + i\gamma$ with $\gamma > 0$. In particular, under RH, $\mu_\varepsilon = P_\varepsilon * U^{\pm}_{\mathrm{RH}}$ restricted to $\theta > 0$, where $U^{\pm}_{\mathrm{RH}} = \sum_{\gamma > 0} (\delta_\gamma + \delta_{-\gamma})$.

Proof. Take real parts in (1.2.2) at $s = \alpha + i\theta$ with $\alpha = \frac{1}{2} + \varepsilon$: a zero $\rho = \beta + i\gamma$ contributes $\pi P_{\alpha-\beta}(\theta - \gamma)$ to $v_\alpha(\theta)$, and $\alpha - \beta = \varepsilon \mp \delta$ when $\beta = \frac{1}{2} \pm \delta$. Dividing by π and grouping each zero ρ with $\bar{\rho}$ gives the second form, and collecting the zeros ρ, $\bar{\rho}$, $1 - \rho$ and $1 - \bar{\rho}$ gives (1.2.8). The series converges absolutely, and uniformly for θ in compact sets avoiding the zeros, since each term is $O(\gamma^{-2})$ there and $\sum_\rho \gamma^{-2} < \infty$. □

Each zero on the line thus contributes a Poisson kernel of width ε centred at its ordinate, and a small reflected kernel centred at minus the ordinate. An off line pair contributes kernels of widths $\varepsilon + \delta$ and $\varepsilon - \delta$, and the second is negative once $\varepsilon < \delta$, that is, once the line $\operatorname{Re} s = \frac{1}{2} + \varepsilon$ lies to the left of the zero $\frac{1}{2} + \delta + i\gamma$. If every zero lies to the left of the line, every kernel is positive and so is v_α; a negative value of v_α therefore proves a zero to the right of α.

Remark 1.2.4 (Positivity on one line). The criterion of Theorem 1.2.2 concerns the analytic continuation of the shifted clock, not the sign of v_α on the single line

$\operatorname{Re} s = \alpha$. Once a zero ρ lies to the right of the line, the shifted clock ceases to be a GGC, since $\xi(\alpha+\sqrt{u})$ vanishes at $u = (\rho-\alpha)^2 \in \mathbb{C}\setminus(-\infty,0]$, but the density need not change sign, because the positive kernels of the other zeros can outweigh the negative one. Exercise 1.7 gives an entire function with the order, symmetry and reality of ξ whose density is positive on a line with a zero to its right; for ξ itself it is not known, without assuming RH, whether positivity on one line excludes zeros to its right. Section 5.2.2 gives, to first order in ε, the condition under which an off line zero makes ρ_ε negative at its ordinate, and Section 11.4 bounds the negative part of μ_ε with no hypothesis, through zero density estimates. Positivity on every line to the right of $\operatorname{Re} s = \sigma_0$ does exclude zeros there: if $v_\alpha \geq 0$ for every $\alpha > \sigma_0$, then no zero has $\operatorname{Re}\rho > \sigma_0$, because $\operatorname{Re}(\xi'/\xi)(\rho - b) \sim -m/b$ as $b \downarrow 0$ at a zero ρ of multiplicity m. With $\sigma_0 = \frac{1}{2}$ this is Lagarias's criterion [133]: RH holds if and only if $v_\alpha \geq 0$ for every $\alpha > \frac{1}{2}$.

Integrated over $\theta > 0$, a line zero contributes mass one, which concentrates at $\theta = \gamma$ as $\varepsilon \downarrow 0$. An off line pair contributes mass two when $\varepsilon > \delta$, and when $\varepsilon < \delta$ a signed measure of mass zero whose total variation tends to zero as $\varepsilon \downarrow 0$. These limits can be summed over the zeros: if f is continuous with support in $[a,b] \subset (0,\infty)$, then for $0 < \varepsilon \leq 1$ each term of (1.2.8) with $\gamma > b+1$ integrates f to a number of modulus at most $C_f\gamma^{-2}$, where C_f depends only on f, because the widths c in (1.2.8) satisfy $|c| < \varepsilon + \frac{1}{2} \leq \frac{3}{2}$, $|P_c(x)| \leq |c|/(\pi x^2)$, and $|\theta \pm \gamma| \geq \gamma - b \geq \gamma/(b+1)$ for $\theta \in [a,b]$; as $\sum_\rho \gamma^{-2} < \infty$, dominated convergence applies to the sum over the zeros. Hence $\mu_\varepsilon \to U_{\text{line}} = \sum_{\text{line zeros}} \delta_{\gamma^2}$ vaguely, whatever the truth of RH [198, Proposition 9.5]. Sections 3.2, 3.3, 3.4, 4.1 and 8.6 build on this description.

1.2.5 The sine squared lemma: the Thorin density from the primes

The sine squared lemma produces the same Thorin density from the Lévy side, where the primes appear as atoms. For $x > 0$ and $s \geq 0$,

$$e^{-sx} + sx - 1 = x \int_0^\infty \log\Big(1 + \frac{s^2}{z}\Big) \frac{\sin^2(x\sqrt{z}/2)}{\pi\sqrt{z}}\, dz. \tag{1.2.9}$$

To prove it, substitute $z = \theta^2$, which turns the right side into $\frac{x}{\pi}\int_0^\infty \log(1 + s^2/\theta^2)(1-\cos x\theta)\, d\theta$. Both sides vanish at $s = 0$, and their derivatives in s agree because $\int_0^\infty (1-\cos x\theta)\, 2s(\theta^2+s^2)^{-1} d\theta = \pi(1-e^{-sx})$. Thus a single compensated exponential $e^{-sx} + sx - 1$, the compensated Laplace exponent of one jump of size x, is a positive superposition of the logarithms $\log(1 + s^2/z)$ that define Thorin measures in the variable s^2. The weight $\sin^2(x\sqrt{z}/2)$ vanishes when $\sqrt{z}$ is a multiple of $2\pi/x$ and oscillates with that period, so a jump at $x = \log n$ becomes an oscillation in $\sqrt{z}$ with frequency $\log n$. This is how the prime atoms of the explicit formula turn into the oscillating Thorin density of the reciprocal.

REMARK 1.2.5 (A first passage proof). The original proof of (1.2.9) [198, Lemma 8.1], as recalled in [201], goes through the first passage time τ_x of a standard Brownian motion to the level $x > 0$, which has the density $x(2\pi t^3)^{-1/2}e^{-x^2/(2t)}$ and the Laplace transform $\mathbb{E}e^{-s^2\tau_x/2} = e^{-sx}$. As $x \mapsto \tau_x$ is a positive stable subordinator of index one half with Lévy density $(2\pi t^3)^{-1/2}$, also $\int_0^\infty (1-e^{-s^2t/2})(2\pi t^3)^{-1/2}\,dt = s$, and the three facts combine into

$$e^{-sx} + sx - 1 = \int_0^\infty \bigl(1 - e^{-s^2t/2}\bigr)\bigl(1 - e^{-x^2/(2t)}\bigr)\frac{x\,dt}{\sqrt{2\pi t^3}}, \qquad s, x > 0, \quad (1.2.10)$$

an integral of a positive function. By the Fourier transform of the Gaussian, $1 - e^{-x^2/(2t)} = 2\int_0^\infty (1 - \cos x\theta)\sqrt{t/(2\pi)}\,e^{-t\theta^2/2}\,d\theta$. Substituting this, exchanging the integrals, which Tonelli's theorem allows, and using Frullani's integral $\int_0^\infty \bigl(e^{-t\theta^2/2} - e^{-t(\theta^2+s^2)/2}\bigr)\,dt/t = \log(1 + s^2/\theta^2)$ gives

$$e^{-sx} + sx - 1 = \frac{x}{\pi}\int_0^\infty \log\Bigl(1 + \frac{s^2}{\theta^2}\Bigr)(1 - \cos x\theta)\,d\theta,$$

which is (1.2.9) in the variable $\theta = \sqrt{z}$. In this proof the jump size x enters only through the factor $1 - e^{-x^2/(2t)}$, whose Gaussian Fourier transform produces the weight $1 - \cos x\theta = 2\sin^2(x\theta/2)$, and the Laplace variable only through the factor $1 - e^{-s^2t/2}$, which Frullani's integral turns into the logarithm. The script `dbeta_thorin.py` evaluates both sides of (1.2.10) at $s = 1.3$ and $x = 0.8$, where they agree in all fourteen digits printed, 0.39345468195878.

Integrating (1.2.9) against a measure turns Lévy representations into Thorin representations. A Lévy representation $\log f(\alpha + s)/f(\alpha) = cs + \int (e^{-sx} + sx - 1)L(dx)$ for $s \geq 0$, with c real and L a real measure on $(0, \infty)$ such that $\int \min(x, x^2)\,|L|(dx) < \infty$, becomes a Thorin representation with density $(2\pi\sqrt{z})^{-1}[c + \int 2\sin^2(x\sqrt{z}/2)\,xL(dx)]$, and the bracket equals $\mathrm{Re}(f'/f)(\alpha + i\sqrt{z})$ [198, Theorem 8.2]. For $f = \xi$ at $\alpha > 1$ the input is the signed canonical measure $xL(dx)$, with atoms $\Lambda(n)n^{-\alpha}$ at $\log n$, an archimedean density and a pole density [200, Theorem 8.1].

Table 1.6 compares the two routes: the bracket computed from the canonical measure with the primes truncated at $N = 2 \times 10^6$, and $v_\alpha(\theta)$ computed by the Pólya route of Section 2.1. At $\alpha = 1.5$ and $\alpha = 2$ the difference is the prime tail $\sum_{n>N} \Lambda(n)n^{-\alpha}(1 - \cos(\theta \log n))$, of order $N^{1-\alpha}/(\alpha - 1)$, that is 1.4×10^{-3} and 5×10^{-7}. At $\alpha = 3$ the tail is of order 10^{-13}, the error in the row $\theta = 14.1347$, and the larger discrepancy at $\theta = 50$ comes from the double precision quadrature of the archimedean integral, not from the primes. The Pólya values carry only the spectrally small discretisation error of the trapezoid rule.

Three continuations enter the book and should be kept apart [200, Section 3.3]. Riemann's continuation of ζ by the theta transformation makes ξ entire, and the Pólya representation (1.2.3) is this continuation in Fourier form. The continuation of the Laplace transform $\mathbb{E}e^{-uT}$ from $u \geq 0$ to the meromorphic function $1/G$

α	θ	sine squared, primes $\le 2\times10^6$	Pólya route	error of prime side
3.0	14.1347	0.561447238789	0.561447238789	1.1×10^{-13}
3.0	50	1.120967014740	1.120967015289	5.5×10^{-10}
2.0	14.1347	0.766398694727	0.766399169607	4.7×10^{-7}
2.0	50	1.224475714058	1.224476216908	5.0×10^{-7}
1.5	14.1347	1.065762504228	1.067140080666	1.4×10^{-3}
1.5	50	1.385288214458	1.386706213218	1.4×10^{-3}

Table 1.6: The bracket $v_\alpha(\theta)$ of the sine squared lemma computed from the canonical measure (drift, archimedean integral, pole term and primes up to 2×10^6) and from the Pólya density; $\theta = 14.1347$ is the first ordinate γ_1, rounded. The Pólya values agree with an independent evaluation of $\mathrm{Re}(\xi'/\xi)$ through Γ and ζ to about 10^{-15}.

uses no arithmetic; it gives the Lévy form of the clock and places the atoms of the Thorin measure at the points a_k. The series $-\zeta'/\zeta(s) = \sum \Lambda(n)n^{-s}$ converges absolutely only for $\mathrm{Re}\, s > 1$, and although each of its terms is entire the sum is not; the explicit formula reaches the critical line only after the terms are paired with a test function whose transform decays, as in (1.2.5). The difference between continuing the terms and continuing the sum reappears in Section 3.5 as the cost of computing from the primes.

1.3 Generalised gamma convolutions and the Thorin condition

By Proposition 1.2.1, RH holds if and only if the clock $1/G$ is the Laplace transform of a generalised gamma convolution, and by Theorem 1.2.2 so is the clock shifted to any basepoint to the right of every zero, for instance to any $\alpha \ge 1$. The Thorin condition is therefore a question of membership in this class. A generalised gamma convolution can be described through its Lévy measure, through the Stieltjes transform of its Thorin measure or, when that measure is discrete, through an infinite product, and the Bondesson hierarchy records the larger classes to which the clock provably belongs. Each description yields a test of the Thorin condition. The tests differ widely in conditioning; the Thorin density of Section 1.2 locates a failure in height without a search for zeros. A catalogue of laws with explicit Thorin measures supplies solvable examples, and one of them, the sinh clock, is the benchmark for the distances of Section 3.2.

1.3.1 The definition and its three readings

As in Section 1.2, a random variable $H \geq 0$, or its law, is a generalised gamma convolution (GGC) if

$$\mathbb{E}e^{-uH} = \exp\Big\{ -du - \int_{(0,\infty)} \log\Big(1 + \frac{u}{z}\Big) U(dz) \Big\}, \qquad u > 0, \tag{1.3.1}$$

with drift $d \geq 0$ and a positive Radon measure U on $(0,\infty)$, the Thorin measure, such that $\int_{(0,1)} |\log z|\, U(dz) + \int_{(1,\infty)} z^{-1} U(dz) < \infty$ [250, 251], [35, p. 29]. The drift and the Thorin measure are unique [35, p. 30]. Since $\log(1 + u/z)$ is the Laplace exponent of the exponential law with rate z, a discrete measure $U = \sum_j c_j \delta_{z_j}$ makes H the sum of d and independent gamma variables with shapes c_j and rates z_j, and a general U superposes a continuum of them. The class is the closure of such sums under weak limits (Thorin's closure theorem, Section 4.1.1) and is closed under convolution and positive scaling. The drift is needed for the closure, since Gamma(n, n), the gamma law with shape n and rate n, tends to the point mass at one. If RH holds, the clock variable $T = \sum_{\gamma>0} e_\gamma/\gamma^2$ of Proposition 1.2.1 is such a sum, with a unit Thorin atom at each squared ordinate. The definition has three equivalent forms, and each leads to a computation.

The first reading is through the Lévy measure. Frullani's integral $\log(1 + u/z) = \int_0^\infty (1 - e^{-ut}) e^{-zt} t^{-1} dt$ turns (1.3.1) into

$$-\log \mathbb{E}e^{-uH} = du + \int_0^\infty \big(1 - e^{-ut}\big) \frac{k(t)}{t}\, dt, \qquad k(t) = \int e^{-zt}\, U(dz),$$

so H is infinitely divisible with Lévy density $k(t)/t$, and its canonical function k, t times the Lévy density, is completely monotone. Conversely, by Bernstein's theorem, an infinitely divisible law on $[0,\infty)$ with drift d and Lévy density $k(t)/t$, k completely monotone, is a GGC whose Thorin measure is the measure representing k [109, Section 1.2]. For the clock, the unconditional formula for $\log G$ in Section 1.2 gives $k = W$, the heat trace, so the Thorin condition is the complete monotonicity of W.

The second reading is through the Stieltjes transform of the Thorin measure. Differentiating the logarithm of (1.3.1) gives

$$f_H(u) = -\frac{d}{du} \log \mathbb{E}e^{-uH} = d + \int \frac{U(dz)}{u + z},$$

a nonnegative constant plus the Stieltjes transform of a positive measure, holomorphic on $\mathbb{C}\backslash(-\infty, 0]$ and mapping the upper half plane into the closed lower half plane. At distance $\eta > 0$ below the cut, $\pi^{-1} \operatorname{Im} f_H(-x - i\eta) = \int P_\eta(x - z)\, U(dz)$ is the Poisson smoothing of U, with P_η the Poisson kernel of Section 1.2, and letting $\eta \downarrow 0$ recovers U by Stieltjes inversion: for continuity points $a < b$ of U,

$$U\big((a,b]\big) = \lim_{\eta \downarrow 0} \frac{1}{\pi} \int_a^b \operatorname{Im} f_H(-x - i\eta)\, dx. \tag{1.3.2}$$

For the clock, $G'/G = -\frac{d}{du}\log(1/G)$ takes the place of f_H whether or not RH holds. The chain rule applied to $G(s^2) = \xi(\frac{1}{2}+s)/\xi(\frac{1}{2})$ gives $(\xi'/\xi)(\frac{1}{2}+s) = 2s\,(G'/G)(s^2)$. At $s = \varepsilon + i\theta$ we have $\bar{s}^2 = -x - i\eta$ with $x = \theta^2 - \varepsilon^2$ and $\eta = 2\varepsilon\theta$, and since G is real on the real axis, (1.2.7) becomes

$$\rho_\varepsilon(\theta) = 2\theta \cdot \frac{1}{\pi}\operatorname{Im}\frac{G'}{G}(-x - i\eta) + \frac{2\varepsilon}{\pi}\operatorname{Re}\frac{G'}{G}(s^2).$$

The first term is the integrand of (1.3.2) at distance $\eta = 2\varepsilon\theta$ from the cut, multiplied by the Jacobian $dx/d\theta = 2\theta$; the second carries the factor ε. The Thorin density at resolution ε is therefore a regularised Stieltjes inversion, taken along the parabola $u = (\varepsilon - i\theta)^2$ rather than along a horizontal line, and as $\varepsilon \downarrow 0$ it recovers only the atoms on the critical line (Section 1.2).

Under RH the logarithm gives the counting function directly: for $y > 0$ not of the form γ^2, $U_{\mathrm{RH}}((0,y]) = -\pi^{-1}\operatorname{Im}\log G(-y - i0) = N(\sqrt{y})$, with $\log G$ continued from $(0,\infty)$ through the lower half plane, since the factor $1 + u/\gamma^2$ of G has argument $-\pi$ at $u = -y - i0$ when $\gamma^2 < y$ and 0 otherwise. For the Lambert function $\mathcal{W}$ the same inversion gives the distribution function $-\pi^{-1}\operatorname{Im}\mathcal{W}(-y - i0) = \pi^{-1}\operatorname{Im}\mathcal{W}(-y + i0)$ of its Thorin measure [117], [116, Theorem 3.3] (Section 11.5.3).

The third reading is as a product. If $U = \sum_j c_j \delta_{z_j}$ is discrete, (1.3.1) is the product $e^{-du}\prod_j (1 + u/z_j)^{-c_j}$ of gamma Laplace transforms. Conversely, if F is entire of genus zero with $F(0) = 1$ and all its zeros real and negative, then $1/F$ is the Laplace transform of a GGC with $d = 0$ and a unit atom at $-u_j$ for each zero u_j of F, counted with multiplicity. For the clock $F = G$, the product is Hadamard's (1.2.1), and the Thorin condition says that every zero $-a_k$ of G is real and negative, that is, $a_k > 0$.

1.3.2 The Bondesson hierarchy and the clock

Between the infinitely divisible laws on $[0,\infty)$ (ID) and the GGCs lie two classes defined by conditions on the Lévy density $k(t)/t$: the self decomposable laws (SD), for which k is nonincreasing, and the Bondesson class (BO), for which $k(t)/t$ is completely monotone. The GGCs are the laws for which k itself is completely monotone, and inside them lie the laws with a hyperbolically completely monotone density (HCM, defined below). The inclusions are

$$\mathrm{HCM} \subset \mathrm{GGC} \subset \mathrm{SD} \subset \mathrm{ID}, \qquad \mathrm{GGC} \subset \mathrm{BO} \subset \mathrm{ID},$$

and SD and BO are not nested. The Lévy density $e^{-t}(1 + \frac{1}{2t})$ is completely monotone, but $k(t) = e^{-t}(t + \frac{1}{2})$ increases on $(0, \frac{1}{2})$, so the law is in BO and not in SD [198, Section 2]; the Lévy density $t^{-1}\mathbf{1}_{t<1}$ gives a law in SD and not in BO, since a completely monotone function that vanishes on $(1,\infty)$ vanishes identically.

A positive continuous function φ on $(0,\infty)$ is hyperbolically completely monotone if for every $u > 0$ the function $\varphi(uv)\varphi(u/v)$ is completely monotone in

$w = v + v^{-1}$, and a law is HCM if it has an HCM density. By Bondesson's theorem [35, Theorem 6.1.1], a function φ with $\varphi(0+) = 1$ is the Laplace transform of a GGC if and only if it is HCM, so the Thorin condition becomes a family of sign conditions on real derivatives, which Section 4.1 tests on the clock. An HCM law is a GGC, so the HCM property of a law is stronger than that of its Laplace transform, and strictly so: the GGC of Theorem 4.3.21 has an HCM Laplace transform and a density that is not HCM. HCM laws are also closed under products and quotients of independent variables and under powers X^q with $|q| \geq 1$ [35, Section 5.1].

The extended generalised gamma convolutions (EGGC), introduced by Thorin [252], are the weak limits of convolutions of gamma laws and their reflections [35, Section 7.1]. A symmetric law is an EGGC if and only if it is the law of $Z\sqrt{2H}$ with Z standard normal and H an independent GGC [35, Theorem 7.3.1]; if H has drift d and Thorin measure U, the symmetric law has Gaussian variance $2d$ and Lévy density $|x|^{-1} \int e^{-\sqrt{z}|x|} U(dz)$. Since $\mathbb{E}e^{i\theta Z\sqrt{2T}} = \mathbb{E}e^{-\theta^2 T}$ determines the law of T, the variable $Z\sqrt{2T}$ of the van Dantzig pair of Section 1.2 is an EGGC if and only if T is a GGC, that is, by Proposition 1.2.1, if and only if RH holds. Section 10.1 asks the same question at the other basepoints of the critical strip.

For the clock $k = W$, so each class becomes a sign condition on the heat trace [200, Theorem 4.6], and Table 1.7 records what is known. The proved rows are unconditional theorems whose proofs, given or sketched in Section 5.1, combine explicit estimates with a finite verification of the zeros. All but one involve finitely many derivatives of W, and Section 5.2 explains why no finite number of such conditions can reach the last row; the exception involves every derivative, but only for $t \leq 1.29$ (Section 5.1.3). Section 11.1 collects what the general theory of these classes, after Sato and Bondesson, gives for the clock.

Urbanik's classes of multiply self decomposable laws give conditions on W of another kind: with L_0 the class of self decomposable laws and L_n that of the self decomposable laws whose driving process has its law in L_{n-1}, the clock lies in L_n exactly when $(-t\frac{d}{dt})^j W \geq 0$ for $j \leq n+1$ (Section 11.5.4), and for $n \geq 1$ this does not follow from the GGC property, since the exponential law is a GGC whose canonical function e^{-t} has $(-t\frac{d}{dt})^2 e^{-t} = -t(1-t)e^{-t} < 0$ for $t < 1$. For the clock these conditions hold for $n = 2$ and fail for $n = 3$, with no hypothesis (Theorem 11.5.8), while membership of every L_n would imply RH, since $\bigcap_n L_n$ lies inside the GGCs [23].

1.3.3 Diagnostics for the Thorin condition

RH can be stated as a GGC identity on a real interval. The right side of (1.3.1) extends to a holomorphic nonvanishing function on $\mathbb{C} \setminus (-\infty, 0]$ (Exercise 1.14), so RH holds if and only if there is a GGC H with $\xi(\frac{1}{2})/\xi(\frac{1}{2}+s) = \mathbb{E}e^{-s^2 H}$ for real $0 < s < 1$ alone [198, Proposition 3.3]. Indeed, the identity $1/G(u) = \mathbb{E}e^{-uH}$ then holds for $0 < u < 1$ and continues to $\mathbb{C} \setminus (-\infty, 0]$, where its right side is

class or property	condition on W	status for the clock
ID	$W \geq 0$	proved, with the zeros verified to height 1000 (Theorem 5.1.1)
SD	W nonincreasing	proved, with the zeros verified to height 1000 (Theorem 5.1.1)
strict log convexity	$WW'' > W'^2$ and $W'W''' > W''^2$	proved, with the zeros verified to height 1000 (Section 5.1)
derivative signs to order 1.16×10^{25}	$(-1)^n W^{(n)} > 0$ for $n \leq 1.16 \times 10^{25}$	proved, with the zeros verified to height 3×10^{12} (Theorem 5.1.10)
complete monotonicity near the origin	$(-1)^n W^{(n)}(t) > 0$ for every n and $t \leq 1.29$	proved, with the zeros verified to height 10^{12} (Theorem 5.1.7)
BO	$W(t)/t$ completely monotone	equivalent to RH (Section 8.5.4); open
GGC	W completely monotone	equivalent to RH; open

Table 1.7: The Bondesson hierarchy on the clock, whose Lévy density is $W(t)/t$.

holomorphic, so $1/G$ has no pole there and every zero of G lies on the negative axis; the converse is Proposition 1.2.1. The Thorin formulation thus asks for an identity on a real interval and the positivity of a measure, and for no inequality at complex arguments.

As it stands the identity is of no use to a computation, because analytic continuation from an interval is unstable. A unit Thorin atom at z_0 multiplies the transform by $(1 + u/z_0)^{-1}$, which differs from one by less than $1/z_0$ on $(0, 1)$, so transforms that agree to high accuracy on $(0, 1)$ can have very different Thorin measures at large z, and a numerical GGC representation on a real interval says nothing about the zeros at large height. Riesz's criterion, which states RH as a growth bound on an entire function whose Taylor coefficients involve $1/\zeta(2k)$, meets the same obstruction: these numbers are, up to explicit factors, values of the clock at the real points $(2k - \frac{1}{2})^2$, and they bear on the zeros only through the analytic interpolation of $k \mapsto 1/\zeta(2k)$ into the strip $\frac{1}{4} < \operatorname{Re} k < \frac{1}{2}$ (Section 10.2.6). The information about the zeros enters through the analyticity that the representation forces, and a computation has to supply it in another way: through values of ξ and ξ' at complex arguments, which Chapter 2 computes, or through sign conditions on derivatives of every order. Each characterisation of Sections 1.3.1 and 1.3.2 yields such a test, and Table 1.8 lists them with what a failure proves.

The diagnostics differ in how strongly they weight the higher zeros. Those built from W see a zero at height γ through $e^{-\gamma^2 t}$: in $(-1)^n W^{(n)}(t) = \sum_j a_j^n e^{-a_j t}$ it enters with weight $(\gamma/\gamma_1)^{2n} e^{-(\gamma^2 - \gamma_1^2)t}$ relative to the first zero, and when W is

diagnostic	computes	a failure proves
pole test	poles of G'/G off $(-\infty, 0]$	not GGC; for the clock, a zero off the line
Thorin density at resolution ε	$\rho_\varepsilon = \pi^{-1}\operatorname{Re}(\xi'/\xi)$ at $\frac{1}{2}+\varepsilon+i\theta$	a zero with $\operatorname{Re}\rho > \frac{1}{2}+\varepsilon$ (Proposition 1.2.3, [133])
zero density scan	$M_\varepsilon(T)+\tau_\varepsilon(T)$ against $N(T)$ (Proposition 3.4.2)	$\frac{1}{2}[N(T)-M_\varepsilon(T)-\tau_\varepsilon(T)]$ zeros with $\operatorname{Re}\rho > \frac{1}{2}+\varepsilon$ below height T
heat trace signs	$(-1)^n W^{(n)}(t)$	not GGC; finitely many orders are consistent with off line zeros (Section 5.2)
Hankel blocks	$\det[(-1)^{i+j}W^{(i+j)}(t)]$	not GGC (Weil's criterion at one scale, Section 5.1)
HCM test	$(-1)^n\partial_w^n[\varphi(uv)\varphi(u/v)]$	not GGC (Section 4.1)
Bernstein test	$(-1)^{n-1}L^{(n)}$, $L=\log G$	not ID, which Theorem 5.1.1 excludes for the clock
Stieltjes moments	Hankel matrices $[m_{i+j}]$ and $[m_{i+j+1}]$, $m_n=\sum_k a_k^{-n-1}$	not GGC

Table 1.8: Diagnostics for the Thorin condition. The pole test, the Thorin density and the zero density scan locate a failure in height; the others detect it only globally, and the Bernstein test cannot fail for the clock.

computed from the explicit formula its terms, of order one, must also cancel down to the size $e^{-\gamma_1^2 t}$ of W (Table 1.5). These tests certify easily and discriminate poorly. The Bernstein test on $L(u)=\log G(u)$, computed for $u>\frac{1}{4}$, where the Dirichlet series of $\log\zeta(\frac{1}{2}+\sqrt{u})$ converges, is better conditioned [200, Section 18.2.2]: since $(-1)^{n-1}L^{(n)}(u)=(n-1)!\sum_k(u+a_k)^{-n}$, a zero at height γ enters with relative weight of order $(\gamma_1/\gamma)^{2n}$. It too sees the higher zeros only through subdominant terms, and for the clock it cannot fail: by the formula for $\log G$ in Section 1.2, $(-1)^{n-1}L^{(n)}(u)=\int_0^\infty t^{n-1}e^{-ut}W(t)\,dt$, which is positive because $W>0$ (Theorem 5.1.1). Read at $u=0$ through (1.2.4), which gives $\log\mathbb{E}e^{hX}=\log G(h^2)$, the same formula makes the even cumulants of the Pólya variable X alternate in sign with no hypothesis: $(-1)^{n-1}\kappa_{2n}(X)=\frac{(2n)!}{n!}\int_0^\infty t^{n-1}W(t)\,dt>0$ for every $n\ge 1$. Under RH this is the sign rule that Newman proved for laws of Lee Yang type [162]. The test therefore checks infinite divisibility, which is proved, and not the Thorin condition. Complete monotonicity of L' is weaker than the Thorin condition, which asks that L' be the Stieltjes transform $\int U(dz)/(u+z)$ of a positive measure U: the function $\frac{1}{u+1}-\frac{1}{u+2}$ is completely monotone, but the measure $\delta_1-\delta_2$ that represents it is not positive.

The Thorin density, and the zero density scan built from it, are the diagnostics of Table 1.8 that are local in the ordinate without a search for zeros: by Proposition 1.2.3, a zero at height γ contributes about $1/(\pi\varepsilon)$ to ρ_ε at $\theta=\gamma$ and about

$\varepsilon/(\pi d^2)$ at distance $d \gg \varepsilon$. That locality is why Chapters 3 to 5 are built on it.

1.3.4 Explicit Thorin measures and the sinh benchmark

Table 1.9 lists GGCs whose Thorin measures are explicit, in the variable u of the clock. They are models for the clock, test cases for code, and the building blocks of the Vere-Jones representation of Section 6.1 (Theorem 6.1.8 and Table 6.1).

Laplace transform	Thorin measure	law
$(1+u/z_0)^{-a}$	$a\,\delta_{z_0}$	Gamma(a, z_0)
$\pi\sqrt{u}/\sinh(\pi\sqrt{u})$	$\sum_{k\geq 1}\delta_{k^2}$	$2K^2$, K Kolmogorov
$(\sqrt{u}/\sinh\sqrt{u})^2$	$2\sum_{k\geq 1}\delta_{\pi^2k^2}$	$S_2/2$ (Williams)
$\operatorname{sech}\sqrt{u}$	$\sum_{k\geq 1}\delta_{(k-\frac{1}{2})^2\pi^2}$	exit time of $\sqrt{2}B$ from $(-1,1)$
$(\operatorname{sech}\sqrt{u})^{1/2}$	$\frac{1}{2}\sum_{k\geq 1}\delta_{(k-\frac{1}{2})^2\pi^2}$	Meixner time
$e^{-c\sqrt{u}}$	$c\,(2\pi\sqrt{z})^{-1}dz$	positive stable, index $\frac{1}{2}$
$(1+\sqrt{u}/c)^{-1}$	$c\,\{2\pi\sqrt{z}(z+c^2)\}^{-1}dz$	E^2S/c^2
$e^{x(\alpha-\sqrt{\alpha^2+2u})}$	$x\{\pi\sqrt{2z-\alpha^2}\}^{-1}\mathbf{1}_{z>\alpha^2/2}dz$	inverse Gaussian
$\exp\{1-x\sqrt{u}-e^{-x\sqrt{u}}\}$	$x\sin^2(x\sqrt{z}/2)(\pi\sqrt{z})^{-1}dz$	one compensated jump, by (1.2.9)
$\mathcal{W}(u)/u$	image of db/π on $(0,\pi)$ under $r(b)=b\,e^{-b\cot b}/\sin b$	Lambert law (Theorem 11.5.3)
$K_0(2\pi)/K_{\sqrt{u}}(2\pi)$	$\sum_{n\geq 1}\delta_{\nu_n^2}$, ν_n the zeros of $\nu\mapsto K_{i\nu}(2\pi)$	hitting time of a diffusion (Theorem 8.7.10)
$\xi(\alpha)/\xi(\alpha+\sqrt{u})$, $\alpha\geq 1$	$v_\alpha(\sqrt{z})(2\pi\sqrt{z})^{-1}dz$	shifted clock (Theorem 1.2.2)
$\xi(\frac{1}{2})/\xi(\frac{1}{2}+\sqrt{u})$	$\sum_{\gamma>0}\delta_{\gamma^2}$ if and only if RH holds	the clock (Proposition 1.2.1)

Table 1.9: GGC Laplace transforms in the variable u, their Thorin measures in z, and the corresponding laws.

In the table, K is the Kolmogorov variable, the maximum of $|b|$ for a standard Brownian bridge b on $[0,1]$; S_2 is the sum of two independent hitting times of level 1 by a three dimensional Bessel process started at 0 (Section 1.2); B is a standard Brownian motion; the Meixner time H has $\mathbb{E}e^{-uH}=(\operatorname{sech}\sqrt{u})^{1/2}$, so that $Z\sqrt{2H}$, with Z standard normal and independent of H, has the Meixner law of Section 6.2, whose time in Proposition 6.2.10 is $2H$; and S is the positive stable variable with $\mathbb{E}e^{-uS}=e^{-\sqrt{u}}$, independent of the standard exponential variable E. Rows two to five are the classical hyperbolic laws [28]. At $u=-t^2$ the reciprocals of rows two and four become $\sin(\pi t)/(\pi t)$ and $\cos t$, the characteristic functions of the uniform law on $(-\pi,\pi)$ and of a symmetric sign. Like Ξ, both have infinitely many real zeros, so real zeros do not obstruct the Thorin property; the property requires them. In these models the Thorin atoms lie in $(0,\infty)$ by construction,

while for the clock the positivity of the points a_k is the conclusion to be proved [198, Section 10].

The sinh clock $1/G_0$, with $G_0(u) = \sinh(\pi\sqrt{u})/(\pi\sqrt{u}) = \prod_{k\ge1}(1 + u/k^2)$, is the second row of the table. It has the structure that the clock has under RH, a GGC with unit Thorin atoms, but its ordinates are the integers, equally spaced, and no arithmetic enters. Its Thorin density at resolution ε, the density in $\theta = \sqrt{z}$ of a measure $\mu_\varepsilon^{(0)}$, is

$$\rho_\varepsilon^{(0)}(\theta) = \frac{1}{\pi}\,\mathrm{Re}\Big[\pi\coth\big(\pi(\varepsilon + i\theta)\big) - \frac{1}{\varepsilon + i\theta}\Big] = \sum_{k\ne0} P_\varepsilon(\theta - k),$$

exactly the Poisson smoothing of the atoms and their reflections, so the matched total variation of Section 3.2 vanishes identically. The cumulative mass is explicit. As for ξ (Section 3.2), it is π^{-1} times the change in the argument of $\sinh(\pi s)/(\pi s)$ along $s = \varepsilon + i\theta$, and at a half integer $\Theta = m + \frac12$ the continuous argument of $\sinh(\pi(\varepsilon + i\theta))$ has grown from 0 to $\pi\Theta$ and that of $\varepsilon + i\theta$ to $\frac{\pi}{2} - \arctan(\varepsilon/\Theta)$, so that

$$\mu_\varepsilon^{(0)}\big((0, \Theta^2]\big) = m + \frac{1}{\pi}\arctan\frac{\varepsilon}{\Theta}.$$

The mass defect $\mu_\varepsilon^{(0)}((0,\Theta^2]) - m$ is therefore $\varepsilon/(\pi\Theta) + O(\varepsilon^3)$. Its first order term is the term $-(\varepsilon/\pi)f'(\Theta)/f(\Theta)$ of Proposition 3.2.2 with $f(\theta) = \sin(\pi\theta)/(\pi\theta)$ in place of Ξ, because $f'/f(\Theta) = \pi\cot(\pi\Theta) - 1/\Theta = -1/\Theta$. On the window $0 < \theta \le 50.5$, which holds 50 atoms, the defect is $0.00630\,\varepsilon$ to first order and 3.151×10^{-3} at $\varepsilon = 0.5$.

The cell distance, the sum over the cells C of $|\mu_\varepsilon^{(0)}(C) - 1|$ as in Section 3.2, is explicit as well. We use the cells $(k - \frac12, k + \frac12]$, $k = 1, \dots, 50$, and add the mass of the strip $(0, \frac12]$, which lies in no cell, as unmatched. For the full lattice $\mathbb{Z}$ each cell would have mass exactly one. The sinh clock lacks the atom at 0, so each of its cells is short by the mass that P_ε puts there, and the strip carries $\pi^{-1}\arctan 2\varepsilon$; Section 5.4 computes these masses. Summing,

$$d_{\mathrm{cell}}^{(0)}(\varepsilon) = \frac{1}{\pi}\Big[2\arctan 2\varepsilon - \arctan\frac{\varepsilon}{50.5}\Big] = \frac{4 - 1/50.5}{\pi}\,\varepsilon + O(\varepsilon^3),$$

which is 0.4968, 0.1250, 0.0631, 0.01267 and 0.00633 at $\varepsilon = 0.5$, 0.1, 0.05, 0.01 and 0.005, and $1.267\,\varepsilon$ in the limit. All of it is due to the atom missing at 0, and it stays below $4\varepsilon/\pi$ however many atoms the window holds.

For the 79 zeros below height 200 the constant is 5.44 (Section 3.2), and it has a different origin. The zeros are irregularly spaced, so the mass that leaks out of a cell is no longer replaced by the mass that leaks in from its neighbours, and every cell contributes; Section 5.4 reads the constant as a statistic of the spacings. Figure 1.3 divides both distances by the number of atoms, which gives $1.267\,\varepsilon/50 = 0.0253\,\varepsilon$ for the sinh clock and $5.44\,\varepsilon/79 = 0.069\,\varepsilon$ for the zeros. Their ratio, nearly three, depends on the window, since for the lattice the distance per

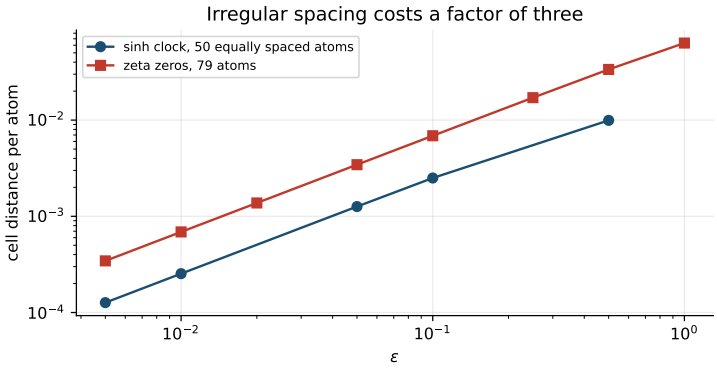


Figure 1.3: Cell distance divided by the number of atoms in the window, for the sinh clock (50 atoms, cells $(k-\frac{1}{2}, k+\frac{1}{2}]$) and for the 79 zeros below height 200 (Voronoi cells of Section 3.2). Both vanish linearly in ε. For the sinh clock the undivided distance stays below $4\varepsilon/\pi$ whatever the number of atoms, so the gap between the curves depends on the window, and the factor of three in the title of the figure holds for these two windows only.

atom falls like $1/N$ as the window grows to N atoms. The benchmark separates the behaviour of the method on perfectly regular atoms, where the only error is the end effect at $\theta = 0$, from the effect of the spacing of the zeros. Williams' Thorin measure $2\sum_k \delta_{\pi^2 k^2}$ of Table 1.9 needs no separate benchmark: it is the same lattice rescaled by π, with atoms of mass two, so its Thorin density at resolution ε is $\frac{2}{\pi}\rho^{(0)}_{\varepsilon/\pi}(\theta/\pi)$ and its distances follow by scaling from the closed forms for the sinh clock.

The diagnostics that locate a failure in height need ξ and ξ' at complex arguments, and Chapter 2 computes them, beginning with the Pólya route of Section 2.1.

1.4 Exercises

On Section 1.2.

1.1. Show from the Hadamard product (1.2.1) that $\sum_k a_k^{-1} = \xi''(\frac{1}{2})/(2\xi(\frac{1}{2}))$, and compute it to ten digits from the Pólya density by differentiating (1.2.4) twice under the integral.

1.2. Prove that Φ is even by applying the theta transformation $1 + 2\omega(1/x) = \sqrt{x}\,(1 + 2\omega(x))$ to $\omega(x) = \sum_{n\geq 1} e^{-\pi n^2 x}$ and differentiating twice.

1.3. Show that if X has the Pólya density then $\mathbb{E}X^2 = 2\sum_k a_k^{-1}$, and deduce that the Pólya law has variance 0.0462.

1.4. Verify the first row of Table 1.5 by summing $e^{-\gamma^2 t}$ over the first forty zeros, and estimate the error of truncation using $N(T) \approx \frac{T}{2\pi}\log\frac{T}{2\pi e}$.

1.5. Show that an off line quadruple $\frac12 \pm \delta \pm i\gamma$ contributes $2e^{-(\gamma^2-\delta^2)t}\cos(2\delta\gamma t)$ to $W(t)$, and find the first time at which that contribution is negative.

1.6. Show that at a basepoint $\alpha > 1$ the Thorin density $v_\alpha(\sqrt{z})/(2\pi\sqrt{z})$ is bounded below by a positive multiple of $z^{-1/2}\log(2+z)$ for large z. Use Proposition 1.2.3 with the Riemann von Mangoldt formula, or, more simply, $v_\alpha(r) = \mathrm{Re}[\frac1s + \frac{1}{s-1} - \frac12\log\pi + \frac12\psi(\frac{s}{2}) + \frac{\zeta'}{\zeta}(s)]$ at $s = \alpha + ir$ with Stirling's formula.

1.7. Let $\delta = 0.49$, $\gamma_0 = 10.5$ and

$$F(s) = \frac{\sinh(\pi(s-\frac12))}{\pi(s-\frac12)}\Big(1 - \frac{(s-\frac12)^2}{(\delta+i\gamma_0)^2}\Big)\Big(1 - \frac{(s-\frac12)^2}{(\delta-i\gamma_0)^2}\Big).$$

Show that F is entire of order one, positive on the real axis and symmetric under $s \mapsto 1-s$, with zeros $\frac12 \pm ik$ $(k \ge 1)$ and $\frac12 \pm \delta \pm i\gamma_0$. Show that $\mathrm{Re}(F'/F)(0.6+ir) > 0$ for every real r, although F vanishes at $0.99+10.5i$. (Put $\varepsilon = 0.1$. The zeros $\frac12 \pm \delta + i\gamma_0$ together contribute $2\varepsilon(t^2-\delta^2+\varepsilon^2)/((t^2+(\delta-\varepsilon)^2)(t^2+(\delta+\varepsilon)^2))$ at $t = r-\gamma_0$, and their conjugates the same expression at $t = r+\gamma_0$; it is at least $-2\varepsilon/(\delta^2-\varepsilon^2)$ and is nonnegative for $t^2 \ge \delta^2-\varepsilon^2$; the zeros on the line contribute $\pi\,\mathrm{Re}\coth(\pi(\varepsilon+ir)) - \varepsilon/(\varepsilon^2+r^2) > 0$, and $\pi\,\mathrm{Re}\coth(\pi(\varepsilon+ir)) \ge \pi\tanh(\pi\varepsilon)$.)

On Section 1.3.

1.8. Show that the Gamma(a,b) law, with shape a and rate b, has Thorin measure $a\delta_b$, and that the sum of independent Gamma(a_j,b_j) variables has Thorin measure $\sum_j a_j\delta_{b_j}$.

1.9. Derive the Thorin measure of sech $\sqrt{u}$ from the product $\cosh x = \prod_{k\ge1}(1+4x^2/((2k-1)^2\pi^2))$.

1.10. Prove that $\int_0^\infty \log(1+u/z)\,(2\pi\sqrt{z})^{-1}dz = \sqrt{u}$ (integrate by parts), and deduce the entry for the positive stable law of index $\frac12$ in Table 1.9.

1.11. Show that $e^{-t}(1+\frac{1}{2t})$ is completely monotone while $k(t) = e^{-t}(t+\frac12)$ is not monotone, and conclude that the infinitely divisible law with Lévy density $e^{-t}(1+\frac{1}{2t})$ and no drift is in BO but not in SD.

1.12. For the sinh clock, derive $\mu_\varepsilon^{(0)}((0,\Theta^2]) = m + \pi^{-1}\arctan(\varepsilon/\Theta)$ at $\Theta = m+\frac12$ from the periodicity of $\sum_{k\in\mathbb{Z}} P_\varepsilon(\theta-k)$, and check the first order value $0.00630\,\varepsilon$ of the mass defect at $\Theta = 50.5$.

1.13. Show that the HCM property of $\varphi(u) = a/(a+u)$ reduces to complete monotonicity of $a^2/(a^2+auw+u^2)$ in w, and deduce that finite products of such functions are HCM.

1.14. Show that the right side of (1.3.1) extends to a holomorphic nonvanishing function on $\mathbb{C} \setminus (-\infty, 0]$. Deduce that if an entire function F with $F(0) = 1$ satisfies $1/F(u) = \mathbb{E}e^{-uH}$ for $0 < u < 1$, with H a GGC, then F has no zeros in $\mathbb{C} \setminus (-\infty, 0]$.

Chapter 2

Evaluating the Xi function

Values of ξ and ξ' at complex arguments are the raw material of every later computation. The computations of Chapters 3 to 5, and many of those of Chapter 8, start from such values at points $\frac{1}{2}+\varepsilon+i\theta$ with $\varepsilon \geq 0$. The reason lies in Chapter 1: the information about the zeros enters the Thorin condition through analyticity, which a computation must supply by evaluating ξ in the complex plane (Section 1.3.3). Section 1.1 named three ways of evaluating ξ. The third, the explicit formula, works from the primes and is a poor evaluator near the critical line: as the basepoint decreases to one its prime sum becomes ill conditioned, below one it diverges (Section 3.1), and Selberg's smoothed form reaches resolution ε only at a cost exponential in $1/\varepsilon$ (Section 3.5).

Section 2.1 develops the first, the Pólya route: the trapezoid rule on the whole line applied to Pólya's integral, organised as two fixed polynomials evaluated in ball arithmetic, and made practical at large height by tilting the contour into the strip where the kernel is analytic. Section 2.2 describes the second, the Riemann Siegel formula and the Odlyzko Schönhage algorithm, together with Turing's method for certifying a count, and compares them with the Pólya route on a Lehmer pair near height 7005 and on the count at height 1000, which the Pólya route reads from the Thorin mass at the basepoint one and the argument along a horizontal segment. The comparison ends in a division of labour, stated at the end of Section 2.2.3.2: the classical evaluators, faster by orders of magnitude, count and locate the zeros at height, and the Pólya route supplies what Chapters 3 to 5 need, Thorin densities off the line and their distances to the measure demanded by RH. Section 2.3 then runs the untilted route once at low height and prints every intermediate number.

2.1 The Pólya route

Pólya's integral (1.2.4) evaluates ξ in the complex plane from the kernel alone, without primes or zeros. Its accuracy rests on one analytic fact, the strip of analyticity of Φ, and its cost on one numerical fact, the cancellation $e^{-\pi\theta/4}$ at

height θ, through which the trapezoid rule loses about $0.341\,\theta$ decimal digits. We build the evaluator and run it below height 200, then tilt the line of integration into the strip, to within a distance δ of its edge, which cuts the loss to about $\delta\theta/\log 10$ digits at the price of more nodes and a larger kernel. In this chapter, δ always denotes this distance, not the displacement of an off line zero.

2.1.1 The trapezoid rule on Pólya's integral

Recall the Pólya kernel (1.2.3) in Titchmarsh's normalisation [255, Section 10.1],

$$\Phi(u) = 2\sum_{n\ge 1}\left(2\pi^2 n^4 e^{9u/2} - 3\pi n^2 e^{5u/2}\right)\exp(-\pi n^2 e^{2u}), \tag{2.1.1}$$

even and positive on the real line, analytic in the strip $|\operatorname{Im} u| < \pi/4$ (Section 1.2) and doubly exponentially decaying in every closed substrip (proof of Proposition 2.1.1). With the Pólya variable X of density $p = \Phi/\Xi(0)$, the moment generating function (1.2.4) and its derivative under the integral sign give, for every complex s,

$$\frac{\xi(\frac12+s)}{\xi(\frac12)} = \mathbb{E}\,e^{sX} = \mathbb{E}\cosh(sX), \qquad \frac{\xi'(\frac12+s)}{\xi(\frac12)} = \mathbb{E}\,Xe^{sX}, \qquad X\sim p, \tag{2.1.2}$$

so the Thorin density (1.2.7) is $\pi^{-1}\operatorname{Re}\{\mathbb{E}Xe^{sX}/\mathbb{E}e^{sX}\}$ at $s=\varepsilon+i\theta$, where the quotient is the logarithmic derivative of the Pólya moment generating function. Throughout, log is the natural logarithm.

Analyticity and decay in the strip make the trapezoid rule with nodes $u_j = jh$ converge geometrically in $1/h$ [256, Theorem 5.1] (Exercise 2.1): for $s=\varepsilon+i\theta$,

$$\Big|\int_{\mathbb{R}}\Phi(u)e^{su}\,du - h\sum_{j\in\mathbb{Z}}\Phi(jh)e^{sjh}\Big| \le \frac{2K_\eta(\varepsilon)\,e^{\eta|\theta|}}{e^{2\pi\eta/h}-1}, \qquad 0<\eta<\pi/4,$$

where $K_\eta(\varepsilon) = \int_{\mathbb{R}}|\Phi(x\pm i\eta)|\,e^{\varepsilon|x|}\,dx$, the same for both signs because Φ is real on the real axis. We write $K_\eta = K_\eta(0)$. Since $|\xi(\frac12+\varepsilon+i\theta)|$ decays like $e^{-\pi\theta/4}$, the relative error is at most of order $\exp\{(\eta+\pi/4)\theta - 2\pi\eta/h\}$. With $h=1/200$ the bound is best for η close to $\pi/4$, and at $\theta=200$ it is about 10^{-289}; halving h changes the computed values by less than 10^{-64}.

The limit in practice is the cancellation in (2.1.2): the integrand is of order one and the integral of order $e^{-\pi\theta/4}$, so at b bits the rounding error is of order $2^{-b}e^{\pi\theta/4}$ relative to $|\xi|$, a loss of $\pi\theta/(4\log 10) = 0.341\,\theta$ decimal digits. Discretisation, rounding and the truncation of the u axis at $|u|\le L$ can be budgeted separately. To reach D correct digits at height θ one needs

$$b \;\ge\; \Big(D + \frac{\pi\theta}{4\log 10}\Big)\log_2 10, \qquad h \;\le\; \frac{2\pi\eta}{(D+0.682\,\theta)\log 10},$$

where the condition on h uses $\eta + \pi/4 \le \pi/2$. The truncation needs $\pi e^{2L} \ge (D + 0.341\,\theta)\log 10 + |\varepsilon|L$, up to lower order terms.

We take $h = 1/200$ and $L = 3$, beyond which Φ is below 10^{-500}, and work at 420 bits (126 decimal digits), comfortably inside the budget up to height 200: at height 200, where $|\xi|$ is about 10^{-63}, the cancellation consumes about half of them and leaves about 60. Truncating the sum at $|j| \le J = L/h$ and folding it with the evenness of Φ gives, with $E = e^{sh}$,

$$\xi(\tfrac12 + s) \approx P(E) + P(E^{-1}), \qquad \xi'(\tfrac12 + s) \approx Q(E) - Q(E^{-1}),$$

$$P(x) = \tfrac{h}{2}\Phi_0 + h\sum_{j=1}^{J} \Phi_j x^j, \quad Q(x) = h^2 \sum_{j=1}^{J} j\Phi_j x^j,$$

with $\Phi_j = \Phi(jh)$ and $J = 600$. The node values, the only place where the theta series is summed, are computed once for a given step, truncation and precision and serve every resolution and every height; one evaluation of ξ and ξ' then costs four polynomial evaluations in ball arithmetic [110], about 0.7 ms at 420 bits. The ball radii certify the rounding error only, and they overestimate it: at height 200 they certify about 38 digits of ξ and, at $\varepsilon = 0.005$, about 36 of $\mathrm{Re}(\xi'/\xi)$, while comparison with `mpmath` at 90 digits shows that the values themselves carry about 63. The discretisation and truncation errors and the tail of the theta series in each Φ_j have explicit bounds (the kernel constant, for instance, is bounded by integrating the majorant (2.1.3)), but the runs of this section do not add them to the radii.

2.1.2 Zeros and Thorin masses below height 200

With the parameters of Section 2.1.1, the zeros and the Thorin masses below height $\Theta = 200$ are computed at the eight resolutions $\varepsilon = 1, 0.5, 0.25, 0.1, 0.05, 0.02, 0.01, 0.005$ in five steps.

1. Compute Φ_j, $0 \le j \le J$, from (2.1.1) at 420 bits.

2. For each resolution ε, evaluate $\rho_\varepsilon(\theta_k)$ and $M_\varepsilon(\theta_k) = \pi^{-1}\arg\xi(\frac12 + \varepsilon + i\theta_k)$, with the branch of the argument that vanishes at $\theta = 0$, on the grid $\theta_k = k\varepsilon/8$ ($k\varepsilon/4$ at $\varepsilon = 0.005$), $0 \le \theta_k \le \Theta$. Since $\partial_\theta \arg\xi(\alpha + i\theta) = v_\alpha(\theta)$ with $\alpha = \frac12 + \varepsilon$, $M_\varepsilon(\theta) = \mu_\varepsilon((0, \theta^2])$ is exactly the cumulative Thorin mass.

3. Locate the line zeros as sign changes of $\Xi(\theta) = \xi(\frac12 + i\theta)$ on a grid of step 0.05, reading each sign from a ball that excludes zero, and refine them by Newton's method using $\Xi'(\theta) = -\operatorname{Im}\xi'(\frac12 + i\theta)$.

4. Count all zeros with $0 < \operatorname{Im}\rho \le \Theta$ by the argument principle, $N(\Theta) = \pi^{-1}\Delta\arg\xi(s)$ along $2 \to 2 + i\Theta \to \frac12 + i\Theta$ [255, Section 9.3].

5. Compare with U_{RH} in the metrics of Section 3.2.

Steps 3 and 4, which use the same two polynomials, together bound the number of off line zeros. Each sign change of Ξ on the grid encloses at least one zero on the line, so $N(\Theta)$ minus the number of sign changes is an upper bound for the number of off line zeros with $0 < \operatorname{Im}\rho \le \Theta$. When the two numbers agree, every zero below height Θ is on the line and simple.

In step 2 the argument is accumulated from the increments $\arg(\xi(s_{k+1})/\xi(s_k))$ along $\operatorname{Re} s = \frac{1}{2} + \varepsilon$. The principal value is the true increment only if $\arg\xi$ changes by less than π between consecutive points, which the samples cannot show; the grids are chosen so that every observed increment is below $\pi/2$ in modulus, a margin checked after the fact rather than a guarantee, and the largest observed was 0.29. At large height ξ underflows double precision ($|\xi|$ is about 10^{-335} at height 1000), so the quotient must be formed in ball arithmetic before the argument is taken; the count at height 1000 in Section 2.2.3.2 fails otherwise.

With $\Theta = 200$ there are 79 sign changes, and at none of the 4001 grid points does the ball for Ξ contain zero. The argument principle gives $N(200) = 79.0000000000004$, with a largest increment of 0.02 radians along the contour, and $\Theta = 200$ lies between $\gamma_{79} = 198.0153$ and $\gamma_{80} = 201.2648$, so it is not an ordinate. The Newton refined roots agree to 5.0×10^{-48} with the ordinates computed independently by `mpmath` [159] at 50 digits. The computation thus recovers, with no input but (2.1.1), the classical fact that the 79 zeros below height 200 lie on the line and are simple [10, 196]. It is not yet a proof: the argument increments are summed in double precision from samples, and the analytic error bounds are not added to the radii (Section 12.2.2 lists what a certified version requires). Figures 2.1 and 2.2 show the computed density in the variables θ and $z = \theta^2$.

2.1.3 Tilting the contour into the strip

Since Φ is analytic in $|\operatorname{Im} u| < \pi/4$, the integrals behind (2.1.2) can be taken along any horizontal line inside the strip.

Proposition 2.1.1. *Let $0 < \delta < \pi/4$ and $\eta = \pi/4 - \delta$. For every complex s,*

$$\xi\Big(\frac{1}{2} + s\Big) = e^{i\eta s} \int_{\mathbb{R}} \Phi(x + i\eta)\, e^{sx}\, dx,$$
$$\xi'\Big(\frac{1}{2} + s\Big) = e^{i\eta s} \int_{\mathbb{R}} (x + i\eta)\, \Phi(x + i\eta)\, e^{sx}\, dx.$$

For real x, $|\Phi(x + i\eta)| \le C_\delta$ and $|(x + i\eta)\Phi(x + i\eta)| \le C_\delta$, with $C_\delta = O(\delta^{-5/2})$ as $\delta \to 0$.

Proof. By Section 1.2, Φ is analytic in $|\operatorname{Im} u| < \pi/4$, and by the identity theorem its evenness $\Phi(u) = \Phi(-u)$ extends from the real line to the strip. For $u = x + iy$

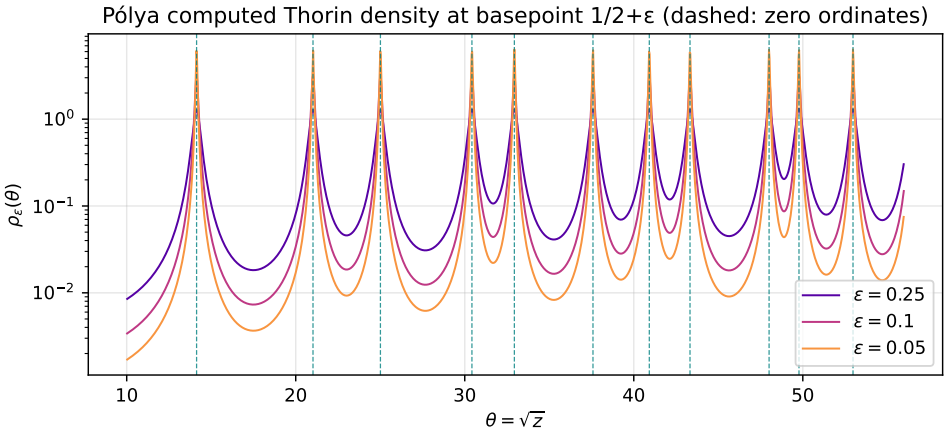


Figure 2.1: The Thorin density $\rho_\varepsilon(\theta)$ at basepoint $\frac{1}{2}+\varepsilon$ computed from the Pólya density, for $\varepsilon = 0.25, 0.1, 0.05$ and $10 < \theta < 56$, on a logarithmic scale; dashed lines mark the ordinates. Each ordinate carries a Cauchy kernel of height $1/(\pi\varepsilon)$ and mass one (Proposition 1.2.3).

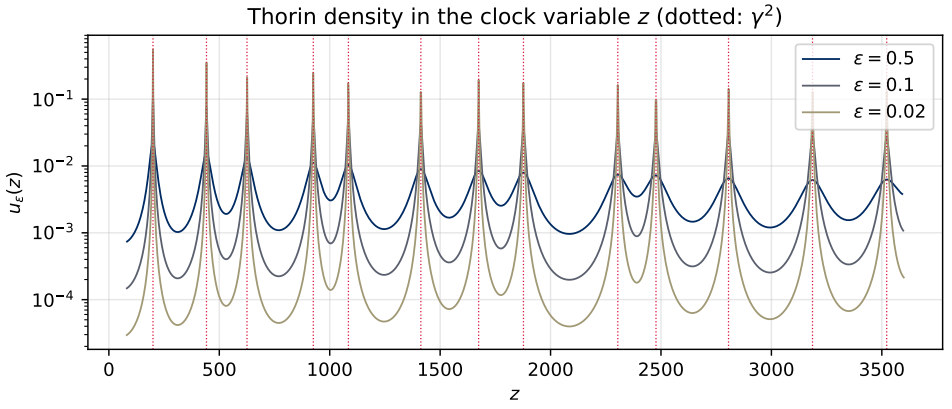


Figure 2.2: The Thorin measure at $\varepsilon = 0.5, 0.1, 0.02$ in the variable $z = \theta^2$, for $81 < z < 3600$, with density $u_\varepsilon(z) = v_{1/2+\varepsilon}(\sqrt{z})/(2\pi\sqrt{z})$ as in (1.2.6). Dotted lines mark the atoms γ^2 of U_{RH}.

with $|y| \le \eta$ and $x \ge 0$ we have $\operatorname{Re} e^{2u} = e^{2x}\cos 2y \ge e^{2x}\sin 2\delta$, so

$$|\Phi(x+iy)| \le 2\sum_{n\ge 1}\left(2\pi^2 n^4 e^{9x/2} + 3\pi n^2 e^{5x/2}\right)\exp(-\pi n^2 e^{2x}\sin 2\delta), \tag{2.1.3}$$

and Φ decays doubly exponentially as $x \to +\infty$, uniformly in the closed strip $|y| \le \eta$; by evenness it does so as $x \to -\infty$ as well. Cauchy's theorem therefore

moves the line of integration in $\xi(\frac12+s)=\int_{\mathbb{R}}\Phi(u)e^{su}\,du$, which is (2.1.2), from $\operatorname{Im}u=0$ to $\operatorname{Im}u=\eta$, and the second identity follows in the same way from $\xi'(\frac12+s)=\int_{\mathbb{R}}u\Phi(u)e^{su}\,du$. Finally, $\sum_{n\ge1}n^ke^{-bn^2}=O(b^{-(k+1)/2})$ uniformly in $b>0$, so the majorant (2.1.3) is $O(e^{-x/2}\delta^{-5/2})$ for $x\ge0$, of order $\delta^{-5/2}$ at $x=0$; with evenness this gives both bounds. □

The geometric content of the proposition is that moving the line of integration upward trades cancellation for size. On the real axis the integrand $\Phi(u)e^{su}$, with $s=\varepsilon+i\theta$, oscillates with frequency θ, and its integral, of order $\theta^{7/4+\varepsilon/2}e^{-\pi\theta/4}|\zeta(\frac12+\varepsilon+i\theta)|$ by Stirling's formula, is exponentially smaller than the integrand. On the line $\operatorname{Im}u=\eta$ the factor $e^{i\theta u}$ has constant modulus $e^{-\theta\eta}$ and leaves the integral as the prefactor $e^{i\eta s}$, which carries almost all of the decay of ξ; the integral that remains is of order $e^{-\delta\theta}$ up to the same factors, against an integrand of size at most $C_\delta e^{\varepsilon x}$. The cancellation thus drops from $e^{-\pi\theta/4}$ to $e^{-\delta\theta}$. The price is a larger kernel, by the factor K_η/K_0 with $K_0=\xi(\frac12)=0.497$, and a finer step, because the natural boundary at $\operatorname{Im}u=\pi/4$ [219, Chapter 11, Section 1.4, and Chapter 5, Section 3.2] is now only δ away. In probabilistic terms the tilt is a rotation of Williams' variable within the half plane of analyticity of its density (Section 8.2.3).

On the tilted line the kernel constant measures the size of the integrand and costs $\log_{10}K_\eta$ digits of precision. By Table 2.1 it grows like $\delta^{-5/2}$, from $\delta=0.1$ to $\delta=0.01$ by a factor 344 against $10^{5/2}=316$, but $\log_{10}K_\eta$ stays under five even at $\delta=0.01$, which is negligible against the saving $(\pi/4-\delta)\theta/\log 10$ at any height where tilting is worthwhile. The maximum of $|\Phi(x+i\eta)|$ grows at about the same rate, and for $\delta\le0.02$ it is attained at the points $x=\pm\frac12\log2$, where e^{2u} is close to $2i$ and the terms of the theta series add in phase. At $x=0$, where e^{2u} is close to i, the terms alternate in sign and nearly cancel (at $\delta=0.02$, $|\Phi(i\eta)|=0.012$ against a maximum of 3938).

$\delta=\pi/4-\eta$	K_η	$\max_x\lvert\Phi(x+i\eta)\rvert$
0.6854	0.554	0.981
0.3927	2.52	3.24
0.2	19.0	15.4
0.1	126	76.5
0.05	774	415
0.02	7836	3938
0.01	43503	22207

Table 2.1: Size of the Pólya kernel on lines inside its strip of analyticity, computed on a grid of step 0.002 over $|x|\le6$; the first two rows are the lines $\eta=0.1$ and $\eta=\pi/8$. The growth is close to $\delta^{-5/2}$, as the majorant in the proof of Proposition 2.1.1 predicts.

The trapezoid rule on the tilted line converges geometrically under two conditions (Exercise 2.2). By Poisson summation its error is $\sum_{k\ne0}I(\theta+2\pi k/h)$, where

$I(t) = \int_{\mathbb{R}} \Phi(x+i\eta) e^{\varepsilon x} e^{itx}\,dx = e^{\eta t - i\eta\varepsilon}\xi(\frac{1}{2}+\varepsilon+it)$. The integrand is analytic in a strip that extends δ above the line and $\pi/2-\delta$ below it, and correspondingly $|I(t)|$ is of order $e^{-\delta t}$ as $t \to +\infty$ and $e^{-(\pi/2-\delta)|t|}$ as $t \to -\infty$, up to the factors of Stirling's formula. Relative to $I(\theta)$ the terms with $k \geq 1$ are of order $e^{-2\pi\delta k/h}$, and those with $k \leq -1$ are small only when the step resolves the oscillation $e^{i\theta x}$ with a margin. For D correct digits this gives

$$\frac{1}{h} \;\geq\; \frac{D\log 10}{2\pi\delta}, \qquad \frac{2\pi}{h} \;\geq\; \theta + \frac{\delta\theta + D\log 10}{\pi/2-\delta}, \tag{2.1.4}$$

up to terms of order $\log\theta + \log(1/h)$. The first condition asks for about $D\log 10/(2\pi\delta)$ nodes per unit length, 458 at $D = 25$ and $\delta = 0.02$, that is $h \leq 2.2\times 10^{-3}$. The second is the binding one at large height: at $\delta = 0.02$ and 25 digits it is the stronger from about $\theta = 2800$ on, and it asks for $h \leq 1.23\times 10^{-3}$ at $\theta = 5000$ and $h \leq 8.8\times 10^{-4}$ at $\theta = 7005$.

Precision and node count thus trade against each other: taking δ proportional to $1/\theta$ holds the loss $\delta\theta/\log 10$ fixed, so the working precision grows only through the $\log_{10} K_\eta$ digits of the kernel constant, about $\frac{5}{2}\log_{10}(1/\delta)$, while both conditions (2.1.4) make the number of nodes, and with it the cost per point, grow linearly in θ up to a logarithmic factor from the truncation.

2.1.4 Precision and cost of the tilted route

The tilt makes height 5000 accessible. With $\delta = 0.02$ and 220 bits the tilted route gives 25 digits of $\rho_{0.05}$ at height 5000 in 28 ms per point, where the untilted route would lose about 1700 digits to cancellation (Table 2.2). The reference is `mpmath`'s ξ'/ξ through Γ and ζ at 50 digits, but the comparison stops near 10^{-45}, because the tilted value is rounded to 45 significant digits before it is compared; this floor, not the reference, limits the entries marked $<$.

On the tilted line the rounding term of Section 2.1.1 becomes $2^{-b}e^{\delta\theta}$ times the kernel constant, and the step obeys (2.1.4). Table 2.3 compares the working precision of the two routes, and Figure 2.3 plots the digits lost against height and the kernel constant against δ. The formula of the table for the tilted bits is conservative: it ignores the factor $\theta^{7/4+\varepsilon/2}$ in $|\xi(\frac{1}{2}+\varepsilon+i\theta)|$, worth about six digits at $\theta = 5000$, and it adds five digits for the kernel constant. The run at $\theta = 5000$ in Table 2.2 used 220 bits, fewer than the 244 of Table 2.3, and returned 25 digits; the Lehmer pair of Section 2.2.3.1 was computed with 256 bits at $\delta = 0.01$, where the formula asks for 201.

Because the tilt reduces the cancellation to $e^{-\delta\theta}$, ordinary double precision suffices at moderate height. With $\delta = 0.15$ and the node values rounded to double, a vectorised evaluation agrees with the ball arithmetic values of $\rho_{0.05}$ to within an absolute error of 2.8×10^{-12} for $\theta \leq 50$, 1.3×10^{-9} for $\theta \leq 100$ and 7.6×10^{-7} for $\theta \leq 150$, at 93 μs per point. This is about eight times faster than

δ	bits	nodes	θ	error of ξ'/ξ	ms per point	untilted digits needed
0.10	200	2001	200	$< 2 \times 10^{-45}$	7	68
0.10	200	2001	500	9.8×10^{-42}	7	171
0.10	200	2001	1000	1.8×10^{-20}	7	341
0.02	220	8401	1000	$< 6 \times 10^{-45}$	28	341
0.02	220	8401	2000	$< 4 \times 10^{-45}$	28	682
0.02	220	8401	5000	2.9×10^{-25}	28	1705

Table 2.2: The tilted Pólya route at $\varepsilon = 0.05$, compared with `mpmath` at 50 digits. Errors are absolute errors of ξ'/ξ, which bound those of its real part, and $<$ marks the entries at the floor set by rounding the tilted value to 45 significant digits. The rows with 2001 nodes use the step $h = 0.004$ and the truncation $L = 4$, those with 8401 nodes $h = 0.001$ and $L = 4.2$. The last column is the number of digits, $0.341\,\theta$, that the untilted route loses to cancellation at the same height.

height θ	target digits	bits (untilted)	bits (tilted, $\delta = 0.02$)	nodes (tilted)
100	30	213	119	8400
200	30	326	122	8400
1000	30	1233	145	8400
5000	25	5748	244	8400
7005	25	8020	302	8400

Table 2.3: Working precision needed by the untilted and the tilted Pólya route for D target digits. The untilted bits are $(D + \pi\theta/(4\log 10))\log_2 10$ and the tilted bits $(D + \delta\theta/\log 10 + 5)\log_2 10$, where the 5 digits cover $\log_{10} K_\eta < 5$ for $\delta \geq 0.01$. At $\theta = 7005$ the second condition in (2.1.4) asks for $h \leq 8.8 \times 10^{-4}$, so 8400 nodes give 25 digits there only with a truncation near $|x| \leq 3.7$, the smallest that suffices.

the untilted evaluator in ball arithmetic and adequate for plotting and scanning. The untilted route fails in double precision above $\theta \approx 40$ (Exercise 2.5).

Batch evaluation would lower the cost further; we have not implemented it. The sums $\sum_j c_j e^{(\varepsilon + i\theta)x_j}$ on a uniform grid $\theta_k = k\Delta$ are a chirp transform, so a whole block of K grid values costs $O((J+K)\log(J+K))$ operations by Bluestein's algorithm [30] instead of $O(JK)$. This is the Pólya counterpart of the Odlyzko Schönhage batching (Section 2.2). Since J grows linearly in θ on the tilted line, blocks with K of order J or more would give the Thorin density on long windows at small ε at a cost per point nearly independent of the grid size.

2.1.5 Ball arithmetic on the tilted line

Horner's rule, which serves the untilted evaluator well, fails in ball arithmetic on the tilted line. The tilted evaluator writes the trapezoid sum as one polynomial

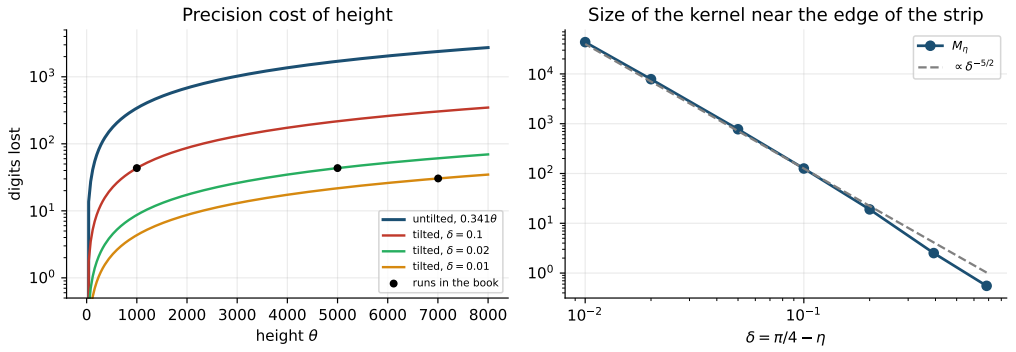


Figure 2.3: Left: decimal digits lost to cancellation as a function of height, untilted and tilted ($\delta = 0.1, 0.02, 0.01$), with the runs $(\theta, \delta) = (1000, 0.1)$, $(5000, 0.02)$ and $(7005, 0.01)$ of this book marked. Right: the kernel constant K_η of Table 2.1 (labelled M_η in the plot) against δ, following $\delta^{-5/2}$ for small δ.

of degree $2J$ in $E = e^{sh}$, with coefficients $h\Phi(x_k + i\eta)$ for x_k running from $-L$ to L, times the prefactor $e^{i\eta s}E^{-J}$. With 2001 nodes, $\delta = 0.1$ and 200 bits, the radius of the Horner recursion grew by a factor of about 1.3 per step, and the final ball at $\theta = 100$ had radius 10^{109} although its midpoint was correct to many digits. Evaluating each exponential e^{sx_k} directly and summing, at the cost of one exponential per node, gives balls of radius about 10^{-55} at the same point.

The cause is the wrapping effect of rectangular complex balls. Arb stores a complex number as a real ball and an imaginary ball [110], so a multiplication by $E = e^{\varepsilon h}e^{i\theta h}$ rotates the midpoint but multiplies the radii by $e^{\varepsilon h}(|\cos\theta h| + |\sin\theta h|)$, up to rounding; at $\theta h = 0.4$ this factor is 1.31. In Horner's rule the coefficient of E^k passes through k such multiplications. On the untilted line the effect is contained. The polynomials are folded at $x = 0$, so the coefficient at $x = jh$ passes through at most 600 multiplications, and the coefficients $h\Phi(jh)$ decay like $\exp(-\pi e^{2jh})$, so those that pass through many multiplications are negligible and the radii stay far below $|\xi|$ up to height 200. On the tilted line the coefficients decay only like $\exp(-\pi e^{2x}\sin 2\delta)$, and the coefficient at x passes through $(x+L)/h$ multiplications. With $h = 0.004$ and $L = 4$, which give 2001 nodes, the coefficients near $x = 2$ are still of order 10^{-12} at $\delta = 0.1$, and they pass through about 1500 multiplications, which enlarge their radii by a factor of about 10^{176}. In complex ball arithmetic, long recursions that rotate the midpoint are to be avoided.

2.2 Classical evaluators, Turing's method and the Pólya route

The Pólya route of Section 2.1 evaluates ξ and ξ' anywhere in the plane from one set of node values, but at height t it loses digits to cancellation and needs a step that shrinks like $1/t$. The classical evaluators on the critical line pay for height differently and, at large height, far less than the Pólya route, tilted or not. The Riemann Siegel formula evaluates ζ from about $\sqrt{t/2\pi}$ bounded terms in double precision, the Odlyzko Schönhage algorithm spreads that cost over many points, and Turing's method turns the sign changes they find into a certified count. We describe these tools as far as the comparison with the Pólya route requires.

2.2.1 The Riemann Siegel formula, Gram points and certified counts

On the critical line $\zeta(\frac12+it) = e^{-i\vartheta(t)}Z(t)$, where $\vartheta(t) = \operatorname{Im}\log\Gamma(\frac14+\frac{it}{2}) - \frac{t}{2}\log\pi$ is the Riemann Siegel theta function and $Z(t)$ is real for real t. Write $m = \lfloor\sqrt{t/2\pi}\rfloor$ and $q = \sqrt{t/2\pi} - m$. For every $K \ge 0$,

$$Z(t) = 2\sum_{n\le m}\frac{\cos(\vartheta(t) - t\log n)}{\sqrt{n}} + (-1)^{m-1}\Big(\frac{t}{2\pi}\Big)^{-1/4}\sum_{j=0}^{K} C_j(q)\Big(\frac{t}{2\pi}\Big)^{-j/2} + R_K(t),$$

where $R_K(t) = O(t^{-(2K+3)/4})$ as $t\to\infty$, $C_0(q) = \cos(2\pi(q^2 - q - \frac{1}{16}))/\cos(2\pi q)$, and the higher C_j are given in [72, Section 7.5] and [79]. This asymptotic expansion was found in Riemann's unpublished notes and published by Siegel [238].

The main sum is the approximate functional equation $\zeta(s) = \sum_{n\le x} n^{-s} + \chi(s)\sum_{n\le y} n^{s-1} + \cdots$, where $\zeta(s) = \chi(s)\zeta(1-s)$ and $2\pi xy = t$ [255, Chapter 4], taken at $x = y = \sqrt{t/2\pi}$ and written for Z. Since $\chi(\frac12 + it) = e^{-2i\vartheta(t)}$, the two sums become complex conjugates after multiplication by $e^{i\vartheta(t)}$, which gives the factor 2; the C_j correct for the abrupt truncation at $n = m$. With about $\sqrt{t/2\pi}$ terms, the cost grows like $\sqrt{t}$. The terms are bounded by 2 and there is no exponential cancellation of the kind that costs the Pólya route $0.341\,t$ digits, so double precision suffices at every height computed in this book; near height 10^{12} and above, the phases $\vartheta(t) - t\log n$ must be reduced modulo 2π in higher precision. Gabcke's thesis gives explicit bounds for the remainders R_K with $K \le 10$ [79], and with them the formula can be used in rigorous computation.

To evaluate Z at many points near height T, Odlyzko and Schönhage [172] compute the main sum on a grid by fast Fourier transforms and rational function evaluation, and interpolate between grid points by band limited interpolation. About $T^{1/2}$ values then cost $T^{1/2+o(1)}$ operations, that is $T^{o(1)}$ per point. Odlyzko compared the spacings of the zeros near zero number 10^{12}, at height about 2.7×10^{11}, with the Gaussian unitary ensemble [171], and with the algorithm he later took such computations to zero number 10^{20}, at height about 1.5×10^{19} [173],

and to zero number 10^{22}, at height about 1.4×10^{21} [175]. The chirp transform batching of the Pólya route proposed in Section 2.1.4 is modelled on the algorithm.

The classical searches for zeros on the line sample Z at the Gram points g_n, defined by $\vartheta(g_n) = n\pi$ [83, pp. 301 and 302]. There $\zeta(\frac{1}{2} + ig_n) = (-1)^n Z(g_n)$ is real, and the first term of the main sum contributes $2\cos\vartheta(g_n) = 2(-1)^n$ to $Z(g_n)$, while the other terms are smaller and of varying sign. So $Z(g_n)$ usually has the sign of $(-1)^n$, and each Gram interval $[g_{n-1}, g_n)$ usually contains exactly one zero. This is Gram's law, and it fails with positive frequency [258, Corollary 5.4]. Call g_n good when $(-1)^n Z(g_n) > 0$, and call an interval $[g_j, g_{j+k})$ between consecutive good Gram points a Gram block. Rosser's rule asserts that such a block contains exactly k zeros [227], [258, Section 7.3]. It holds far longer than Gram's law, since its first exception is the block that begins at $g_{13\,999\,525}$ [43, Table 3], but it too fails with positive frequency [258, Corollary 7.8]. Counting sign changes of Z gives a lower bound for the number of zeros on the line. The difficulty is to show that none is missing, whether off the line or in a close pair that produces no sign change.

Turing's answer [262] uses the counting function $N(T) = \vartheta(T)/\pi + 1 + S(T)$, where, for T not an ordinate, $\pi S(T) = \arg\zeta(\frac{1}{2} + iT)$, the argument being defined by continuous variation along $2 \to 2 + iT \to \frac{1}{2} + iT$. The function S is small on average: Littlewood proved $\int_0^T S(t)\,dt = O(\log T)$ [255, Theorem 9.9(A)], and Turing gave explicit constants for $|\int_{t_1}^{t_2} S(t)\,dt|$, later sharpened by Trudgian [257]. If the sign changes found near height T were fewer than $N(T)$, then S would be displaced by at least 2 over an interval long enough to violate the integral bound. The count is therefore certified by locating the sign changes in a range slightly beyond T and checking them against Turing's bound for $\int S$. The method never evaluates ζ off the line, an advantage in 1953, when every evaluation was expensive. Turing's published argument contained errors, corrected by Lehman [142], and the corrected method underlies every large verification since, including Platt and Trudgian's to height 3×10^{12} [196].

Before Turing, counts were certified by Backlund's method. Since $\operatorname{Re}\zeta(2+it) \geq 2 - \zeta(2) > 0$ for every t, it suffices to check that $\operatorname{Re}\zeta(\sigma + iT) \neq 0$ for $\frac{1}{2} \leq \sigma \leq 2$: then $\arg\zeta$ stays in $(-\frac{\pi}{2}, \frac{\pi}{2})$ along the whole path, so $|S(T)| < \frac{1}{2}$ and $N(T)$ is the integer nearest to $\vartheta(T)/\pi + 1$. Backlund carried this out for $T = 200$ [10] and Hutchinson for $T = 300$ [99, pp. 54 to 57], [257].

2.2.2 The Pólya route beside the classical evaluators

The Pólya route differs from these methods in its object and in its regime. It evaluates ξ and ξ' on and off the line, at $\frac{1}{2} + \varepsilon + i\theta$, from the Pólya kernel, and so delivers the Thorin density ρ_ε, its mass and the zero count from one evaluator. Its count has the logic of Turing's certificate: the sign changes of Ξ bound the number of zeros on the line from below, the argument principle gives $N(T)$, and when the

	Riemann Siegel and Turing	Odlyzko Schönhage	Pólya route
object	sign changes of $Z(t)$	zeros in blocks, statistics	Thorin density ρ_ε, mass, count
where evaluated	critical line	critical line	line $\operatorname{Re} s = \frac{1}{2} + \varepsilon$ and contour
count certificate	Gram points and $\int S$	Turing's method	argument principle count equal to the number of sign changes; computed, not yet certified
cost per point	$\sqrt{t/2\pi}$ terms, double	amortised, $t^{o(1)}$	$0.341\,t$ extra digits and step below π/t (untilted); $\delta t/\log 10$ digits and step below $2\pi/t$ (tilted)
accuracy	asymptotic remainder	asymptotic remainder	spectral; rounding certified by balls
reach	$t \approx 1540$ in 1953	$t \approx 10^{20}$ and beyond [173, 175]	$t \approx 7005$ (Section 2.2.3.1); 25 digits at $t = 5000$ (Table 2.2)

Table 2.4: The Riemann Siegel formula with Turing's method, the Odlyzko Schönhage algorithm and the Pólya route: what each computes, where, with what certificate, at what cost per point, and how far it has reached.

two agree every zero up to height T is on the line and simple (Section 2.1.2).

Both classical certificates have Thorin counterparts, which Section 3.4 develops. Under RH the cumulative Thorin mass M_ε of Section 2.1.2 is the Cauchy average of width ε of N, extended as an odd function (Section 3.4.1), and the L^1 distance $\int |M_\varepsilon - N|\, d\theta$ of Table 3.2 measures how far the average lies from N. Where Turing bounds an average of S to certify a count, the zero density identity, Proposition 3.4.2, reads the count exactly from $M_{1/2}$ and a short horizontal correction, which is Backlund's method with the vertical side moved from $\operatorname{Re} s = 2$ to $\operatorname{Re} s = 1$ (Section 2.2.3.2).

The methods differ most in how their cost grows with height (Table 2.4). The Riemann Siegel formula needs about $\sqrt{t/2\pi}$ bounded terms and no extra precision. The untilted Pólya route computes $\xi(\frac{1}{2} + \varepsilon + it)$, of modulus about $e^{-\pi t/4}$, from integrands of order one, so it needs about $0.341\,t$ extra digits; and by the budget of Section 2.1.1 its step must be below π/t, so that its number of nodes grows linearly with t as well. The cause is aliasing. By Poisson summation the trapezoid sum at $s = \varepsilon + it$ equals $\sum_{k\in\mathbb{Z}} \xi(\frac{1}{2} + s - 2\pi ik/h)$, and the term $k = 1$, at height $t - 2\pi/h$, is exponentially larger than $\xi(\frac{1}{2} + s)$ once $t > \pi/h$. The tilted contour of Section 2.1.3 cuts the loss of digits to about $\delta t/\log 10$, but aliasing still requires $2\pi/h > t$, by the second condition in (2.1.4), so there too the number of nodes grows linearly with t.

The distances of Section 3.2 do not depend on how ξ is evaluated, and this suggests a hybrid at large height. The approximate functional equation holds off the critical line [255, Chapter 4], so Riemann Siegel type expansions of ζ [4,

Theorems 3.1 and 4.2] and of ζ', and the Odlyzko Schönhage algorithm for blocks of points, apply at $\frac{1}{2}+\varepsilon+i\theta$ as well. With them as the evaluator and Odlyzko's zeros [176] as the RH reference, one could compute ρ_ε, the matched total variation and the positivity margin at the heights of Odlyzko's computations, from about 10^{11} to beyond 10^{20}, while the tilted Pólya route supplies values with explicitly bounded errors on low windows and checks the fast evaluator where the two overlap. We have not implemented this hybrid.

2.2.3 Two computations beyond the untilted route

Both computations use the tilted contour of Section 2.1.3, at heights where the untilted rule would need hundreds or thousands of extra digits. The first finds the resolution at which the Thorin density separates two close zeros; the second reads a zero count from the Thorin mass and compares its cost with the classical count.

2.2.3.1 A Lehmer pair near height 7005

Lehmer found two zeros that are unusually close, $\gamma_{6709} = 7005.0629$ and $\gamma_{6710} = 7005.1006$ [143, pp. 297 and 298], at distance 0.0377, the classical example of a Lehmer pair; such pairs give lower bounds for the de Bruijn Newman constant [62] (Section 8.4.6). The untilted Pólya route cannot reach this height: it would need about 2400 digits and a step below 4.5×10^{-4}. The tilted route with $\delta = 0.01$, 256 bits, 13001 nodes, step 6×10^{-4} and truncation $|x| \le 3.9$ computes $\Xi(\theta)e^{\pi\theta/4}$ at 34 milliseconds per point, against 30 milliseconds for `mpmath`'s Riemann Siegel evaluation of Z at thirty digits in the same Python environment. On a grid of step 0.005 it finds sign changes in $(7005.060, 7005.065)$ and $(7005.100, 7005.105)$, the two members of the pair (Figure 2.4).

The Thorin density shows what resolution separates them. A single zero on the line produces a Cauchy peak of height $1/(\pi\varepsilon)$, that is 6.4, 15.9 and 31.8 at $\varepsilon = 0.05$, 0.02 and 0.01. The largest values of ρ_ε on the grid are 11.18, 19.58 and 33.89. At $\varepsilon = 0.05$ the two peaks have merged into one of nearly double height; at $\varepsilon = 0.02$ they are partly separated; at $\varepsilon = 0.01$ each carries its own peak, lifted slightly by its neighbour. The Cauchy kernel predicts this: two kernels with half width at half maximum ε at distance g have a single maximum when $g \le 2\varepsilon/\sqrt{3}$ and two maxima when $g > 2\varepsilon/\sqrt{3}$. With $g = 0.0377$ the pair is resolved as two peaks for $\varepsilon < \sqrt{3}\,g/2 \approx 0.033$, which separates $\varepsilon = 0.05$ from the two smaller resolutions. The criterion concerns two isolated kernels; Exercise 2.11 proves it and asks how much the other zeros move the threshold.

2.2.3.2 The zero count at height 1000

Proposition 3.4.2 of Section 3.4 gives the unconditional count $N(T) = M_{1/2}(T) + \tau_{1/2}(T)$, where $M_{1/2}(T) = \pi^{-1}\arg\xi(1+iT)$ is the cumulative Thorin mass at

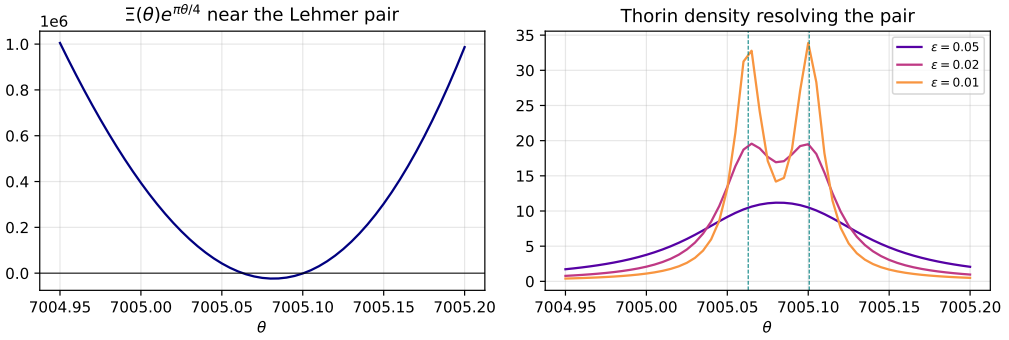


Figure 2.4: The Lehmer pair near 7005 computed by the tilted Pólya route: $\Xi(\theta)e^{\pi\theta/4}$ (left) and the Thorin density at three resolutions (right), drawn through values on a grid of step 0.005, with the two ordinates dashed.

resolution $\frac{1}{2}$, that is at the basepoint one of Section 9.1, and $\tau_{1/2}(T)$ is π^{-1} times the change of $\arg \xi$ along the segment from $1+iT$ to $\frac{1}{2}+iT$. With the tilted route ($\delta = 0.1$, 200 bits, step 0.004 and truncation $|x| \le 4.2$, that is 2101 nodes), 10000 steps of 0.1 along $\operatorname{Re} s = 1$ and 2000 steps along the segment, we find

$$M_{1/2}(1000) = 648.7562315041018, \qquad \tau_{1/2}(1000) = 0.2437684958980,$$

$$M_{1/2}(1000) + \tau_{1/2}(1000) = 648.9999999999999,$$

so $N(1000) = 649$, with a largest argument increment of 0.43 radians. The computation took 77.5 seconds; `mpmath`'s `nzeros`, which combines Riemann Siegel evaluation with Gram blocks and Turing's method, returns 649 in 0.017 seconds. Where Backlund bounds the argument, this computation accumulates it from samples, and ball arithmetic certifies the rounding in each sampled value but not the absence of winding between samples. The count is therefore computed rather than certified. Certifying it would need, besides the analytic error bounds of Section 2.1, an enclosure of ξ between consecutive sample points s_k and s_{k+1}, for instance a check that $\operatorname{Re} \xi(s)/\xi(s_k) > 0$ on a ball containing the segment between them.

For counting and locating zeros at height, the Riemann Siegel family with Turing's method is faster by orders of magnitude, and the Pólya route is not a competitor there. It reads the count as an integer from a smooth quantity, the Thorin mass, with no Gram point bookkeeping; it gives the zero density strip by strip (Section 3.4); and it evaluates objects that the classical methods, used on the critical line, do not compute: Thorin densities and their distances to U_{RH}. The hybrid of Section 2.2.2 would combine the two.

2.3 A worked computation from first principles

Where Section 2.2 took the tilted route to large height, we return to the untilted route and follow one small computation with it at low height, from the theta series to a Thorin mass and a zero. Its intermediate numbers show each ingredient of the method at work: a kernel analytic in a strip, a trapezoid rule that is spectrally accurate for that reason, ball arithmetic that certifies the rounding, the Thorin density as a superposition of Cauchy kernels, the Thorin mass as an argument, and the zero count as an integer read from a smooth quantity. Every value can be reproduced in a few seconds on a laptop with the evaluator listed in Appendix A.

The parameters are those of Section 2.1.1, step $h = 1/200$ and truncation $|u| \le 3$, and the working precision is 240 bits, about 72 decimal digits, the default of the code. As there, s is the displacement from $\frac12$: with $E = e^{sh}$ and $\Phi_j = \Phi(jh)$, the evaluator returns $\xi(\frac12 + s) \approx P(E) + P(E^{-1})$ and $\xi'(\frac12 + s) \approx Q(E) - Q(E^{-1})$, from which $\rho_\varepsilon(\theta) = \pi^{-1}\operatorname{Re}(\xi'/\xi)(\frac12 + \varepsilon + i\theta)$.

Step one: the kernel at a single point. At $u = 0$ the kernel (1.2.3) is

$$\Phi(0) = 2\sum_{n\ge1} \bigl(2\pi^2 n^4 - 3\pi n^2\bigr) e^{-\pi n^2}.$$

The nth term is of order $n^4 e^{-\pi n^2}$, and the terms with $n \ge 8$ add up to less than 10^{-82}, so seven terms already give more than eighty digits:

$$\Phi(0) = 0.893393800934246888173969334109418226440848734189 65\ldots.$$

The ball returned by the arithmetic has radius 3×10^{-71}. The kernel takes its largest value at the origin [60, Theorem A], and $\Phi(0)/\Xi(0) = 1.797$ is the value there of the Pólya density $p = \Phi/\Xi(0)$, close to the value 1.856 at the origin of a normal density with the same variance $\mathbb{E}X^2 = 2\sum_k a_k^{-1} = 0.0462$ (Section 1.2), that is with standard deviation $\sqrt{0.0462} = 0.215$.

Step two: the coefficients. The truncation leaves 601 nodes $u_j = jh$, $0 \le j \le 600$. The coefficients of P are $\frac{h}{2}\Phi_0$ and $h\Phi_j$, $1 \le j \le 600$, and those of Q are $h^2 j\Phi_j$; the first is

$$\tfrac{h}{2}\Phi(0) = 0.00223348450233561722043492333527354556610\ldots.$$

The coefficients decay doubly exponentially. By the asymptotic formula of Section 1.2, $\Phi(2.7) \approx 4\pi^2 e^{12.15}\exp(-\pi e^{5.4}) \approx 6 \times 10^{-296}$, so the coefficients beyond $j \approx 540$ are below 10^{-297}, and at the working precision the polynomial has effective degree well below 600. Keeping the full degree costs nothing and removes any question about truncation.

Step three: $\xi(\frac{1}{2})$ as a trapezoid sum. At $s = 0$, $E = e^{sh} = 1$, and $\xi(\frac{1}{2}) = 2P(1)$ is the trapezoid sum of the kernel:

$$\xi(\tfrac{1}{2}) = 0.49712077818831410991277373968539771980729360955 77\ldots,$$

with ball radius 5×10^{-71}. `mpmath`'s evaluation from the definition, $\frac{1}{2}z(z-1)\pi^{-z/2}\Gamma(z/2)\zeta(z)$ at $z = \frac{1}{2}$ in thirty digit arithmetic, gives

$$0.497120778188314109912773739685,$$

agreeing to every digit. By the bound of Section 2.1.1 the quadrature error is below 10^{-400}, so the seventy digits are set by the working precision, not by the rule; this is spectral accuracy in its simplest form.

Step four: the Thorin density at the first ordinate. At

$$s = 0.1 + 14.134725141734693790i,$$

that is at resolution $\varepsilon = 0.1$ and the first ordinate γ_1, the evaluator returns $\xi(\frac{1}{2}+s)$ and $\xi'(\frac{1}{2}+s)$, and

$$\rho_{0.1}(\gamma_1) = \frac{1}{\pi}\operatorname{Re}\frac{\xi'}{\xi}\Big(\frac{1}{2}+s\Big) = 3.1852530937633722596753 8\ldots.$$

A single Cauchy kernel of width 0.1 contributes exactly $1/(\pi\varepsilon) = 3.18309886183791$ at its centre. By Proposition 1.2.3 the excess, 0.00215423, is the sum of the contributions of every other zero, $\sum_{j\neq 1}[P_{0.1}(\gamma_1-\gamma_j)+P_{0.1}(\gamma_1+\gamma_j)]$, plus the reflected kernel of γ_1 itself. The nearest neighbour, at distance 6.887, contributes $0.1/(\pi \times 6.887^2) = 6.7 \times 10^{-4}$; the reflected kernel contributes $0.1/(\pi \times 28.27^2) = 4 \times 10^{-5}$; the rest comes from the many distant zeros, whose contributions decay like the inverse square of the distance and add up to the remaining 1.4×10^{-3}. Figure 2.5 shows the same decomposition along the whole stretch between the first two zeros.

Between zeros the density is small but strictly positive. At $\theta = 17$, between γ_1 and $\gamma_2 = 21.022$,

$$\rho_{0.1}(17) = 0.007685770789620594 9783\ldots,$$

close to the minimum of $\rho_{0.1}$ between the two zeros (Figure 2.5). Lagarias's criterion [133] asks for this positivity at every θ and every resolution $\varepsilon > 0$.

Step five: the Thorin mass up to height 20. The cumulative Thorin mass of Section 2.1.2 is $M_\varepsilon(\theta) = \mu_\varepsilon((0,\theta^2]) = \pi^{-1}\arg\xi(\frac{1}{2}+\varepsilon+i\theta)$, with the argument continued from the real axis. Stepping along the line $\frac{1}{2}+0.1+i\theta$ from $\theta = 0$ to $\theta = 20$ in 1600 steps of $\varepsilon/8 = 0.0125$, as in that section, and accumulating

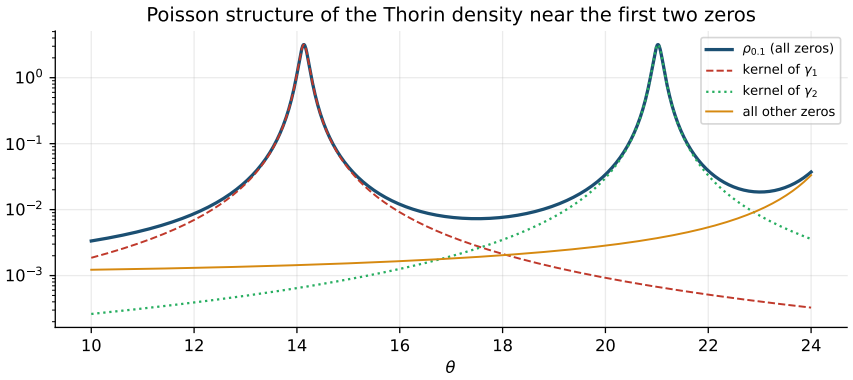


Figure 2.5: The Thorin density $\rho_{0.1}$ near the first two zeros, split into the kernel of γ_1, the kernel of γ_2 and the rest, that is all other zeros and the reflected kernels at $-\gamma_j$, computed from the first 1700 zeros without a tail correction.

the arguments of the quotients $\xi(\frac{1}{2}+s_{k+1})/\xi(\frac{1}{2}+s_k)$, where $s_k = 0.1 + i\theta_k$ and $\theta_k = 0.0125\,k$, gives

$$\mu_{0.1}\big((0,400]\big) = M_{0.1}(20) = 1.0509304\ldots,$$

in 0.77 seconds. There is exactly one zero below height 20. By Proposition 3.2.2 the excess 0.0509 over the count agrees to $O(\varepsilon^2)$ with the first order mass defect $-(\varepsilon/\pi)\Xi'(20)/\Xi(20)$ (Exercise 2.16): it is the mass carried across the edge $\theta = 20$ by the Cauchy tails of the first zero and of the zeros above 20, while what crosses $\theta = 0$ is returned exactly by the reflected kernels. Adding $\tau_{0.1}(20)$, which is π^{-1} times the change of $\arg\xi$ along the segment from $\frac{1}{2}+0.1+20i$ to $\frac{1}{2}+20i$, returns the integer 1, as Proposition 3.4.2 requires.

Step six: locating the first zero. Steps one to five used no zero of ζ except as a place to evaluate. The ordinate γ_1 in step four was only a convenient point, and the density would have revealed it, up to the small shift caused by the other zeros, as the location of its first peak. To locate it from the Pólya route alone, one evaluates $\Xi(\theta) = \xi(\frac{1}{2}+i\theta)$ on a grid, finds the sign change in $(14.10, 14.15)$, and refines by Newton's method with $\Xi'(\theta) = -\operatorname{Im}\xi'(\frac{1}{2}+i\theta)$; six iterations give γ_1 to forty digits. Section 2.1.2 refines all 79 zeros below height 200 in the same way, at 420 bits.

These six steps are the method in its simplest form. Much of the computation in Chapters 3 to 5 repeats them on grids, at other resolutions and, with the tilted contour, at greater heights, and Section 8.5 repeats them with the kernel of $L(s,\chi_{12})$.

2.4 Exercises

On Section 2.1.

2.1. Let f be analytic in a strip containing $|\operatorname{Im} u| \le \eta$, with $\int_{\mathbb{R}} |f(x \pm i\eta)|\,dx \le M$ and $f(u) \to 0$ uniformly as $|\operatorname{Re} u| \to \infty$ in $|\operatorname{Im} u| \le \eta$. Prove that $|\int_{\mathbb{R}} f(u)\,du - h\sum_j f(jh)| \le 2M/(e^{2\pi\eta/h}-1)$ by writing the error as the sum of the integrals, from left to right, of $f(u)(e^{-2\pi i u/h}-1)^{-1}$ along $\operatorname{Im} u = \eta$ and of $f(u)(e^{2\pi i u/h}-1)^{-1}$ along $\operatorname{Im} u = -\eta$ [256, Theorem 5.1]. Deduce the bound of Section 2.1.1.

2.2. Let $g(x) = \Phi(x + i\eta)e^{\varepsilon x}$ with $\eta = \pi/4 - \delta$ and $\varepsilon \ge 0$, and $I(t) = \int_{\mathbb{R}} g(x)e^{itx}\,dx$. Show by Poisson summation that $h\sum_{j\in\mathbb{Z}} g(jh)e^{i\theta jh} - I(\theta) = \sum_{k\neq 0} I(\theta + 2\pi k/h)$. For $0 < a < \delta$, $0 < b < \pi/2 - \delta$ and $0 < \theta < 2\pi/h$, shift the line of integration up by a when $k \ge 1$ and down by b when $k \le -1$ to bound the error by $Ae^{-a\theta}/(e^{2\pi a/h}-1) + Be^{b\theta}/(e^{2\pi b/h}-1)$, where A and B are the suprema of $\int_{\mathbb{R}} |g(x+iy)|\,dx$ over $0 \le y \le a$ and over $-b \le y \le 0$. Derive (2.1.4) from $|I(t)| = e^{\eta t}|\xi(\frac12 + \varepsilon + it)|$ and Stirling's formula.

2.3. Using the formulas in the caption of Table 2.3, estimate the number of bits the untilted route needs at height 2000 for twenty digits, and compare with the tilted route at $\delta = 0.02$.

2.4. Show that on the tilted line $\operatorname{Im} u = \eta$ the factor e^{su} with $s = \varepsilon + i\theta$ has modulus $e^{\varepsilon x - \theta\eta}$. Explain why the prefactor $e^{i\eta s}$ carries all of the exponential decay of ξ except the factor $e^{-\delta\theta}$, and where that factor comes from.

2.5. Implement the two polynomial evaluator in double precision and find the height at which it first returns the wrong sign of Ξ.

2.6. Show that in midpoint radius arithmetic, with a complex number stored as a real and an imaginary ball, multiplication by an exact $w = |w|e^{i\varphi}$ rotates the midpoint and maps real and imaginary radii r to $|w|(|\cos\varphi| + |\sin\varphi|)\,r$, up to rounding. Use this to explain why Horner's rule loses ball accuracy on the tilted line but not on the untilted line.

2.7. Use the tilted route to compute $\Xi(1000)$ and check its sign against $Z(1000)$ from the Riemann Siegel formula, with Z as in Section 2.2 and $\Xi(t) = -\frac12\big(t^2 + \frac14\big)\pi^{-1/4}|\Gamma(\frac14 + \frac12 it)|\,Z(t)$.

On Section 2.2.

2.8. Compute the Gram points $g_0, \dots, g_{20}$ and check Gram's law for them; find the first Gram interval that contains no zero (it lies beyond g_{100}).

2.9. Show that $N(T) = \vartheta(T)/\pi + 1 + S(T)$ follows from the argument principle applied to ξ on a rectangle.

2.10. Explain why a close pair of zeros that falls inside one Gram interval produces no sign change there, and how Turing's method detects that zeros are missing.

2.11. Prove that $P_\varepsilon(x - g/2) + P_\varepsilon(x + g/2)$ has a single maximum if $g \le 2\varepsilon/\sqrt{3}$ and two maxima if $g > 2\varepsilon/\sqrt{3}$, and estimate how the other zeros near height 7005 shift the threshold 0.033 for the Lehmer pair.

2.12. Estimate the cost of counting zeros to height 10^4 by the Thorin mass at resolution $\frac{1}{2}$ with the tilted route, including the growth of the number of nodes with height, and compare with the Riemann Siegel count.

On Section 2.3.

2.13. Compute $\Phi(0)$ with two, three, four and five terms of the series and explain the number of correct digits in each case.

2.14. Repeat step three with $h = 1/5$, $1/10$ and $1/20$ and plot $\log_{10}$ of the error against $1/h$. Show by Poisson summation that the error of the untruncated sum is exactly $2\sum_{k\ge1} \Xi(2\pi k/h)$, and confirm that it falls like $e^{-\pi^2/(2h)}$, the rate set by the strip half width $\pi/4$, up to slowly varying factors.

2.15. Compute $\rho_{0.1}(\gamma_1) - 1/(0.1\pi)$ directly as a sum over the first 1700 zeros with the smooth tail model of Section 3.2, and compare with step four.

2.16. Predict $M_{0.1}(20) - 1$ from the first order formula of Proposition 3.2.2, with $\Xi'(20)/\Xi(20)$ computed by the evaluator of step four, and compare with step five.

Chapter 3

The Thorin measure at finite resolution

Chapter 2 evaluates ξ and ξ' off the critical line from the Pólya kernel alone, and so computes the Thorin density ρ_ε of Section 1.2, the Poisson smoothing of the zeros, at every resolution $\varepsilon > 0$. The Thorin condition is RH (Proposition 1.2.1), and by Lagarias's criterion RH is the positivity of ρ_ε at every resolution (Remark 1.2.4); the question is what finite computations of ρ_ε can establish about it. The answer, which Chapters 4 and 5 complete, has two sides. On every window where the zeros are verified, the computed Thorin measure μ_ε converges to U_{RH}, the discrete measure demanded by RH, at explicit rates as $\varepsilon \downarrow 0$. Above the computed range every finite computation of the kinds in this book remains consistent with a failure of the Thorin condition, and the step that remains, positivity at every resolution and every height, is not a computational one (Section 5.2).

Section 3.1 asks what computation can reach and what the sine squared lemma adds, and sets the limits within which the later computations work. On the window below height 200, where the zeros have been counted, Section 3.2 measures the distance between μ_ε and U_{RH} in several metrics and explains why total variation must be used at matched resolution, and Section 3.3 computes the signatures of an off line zero and calibrates the comparison against them. Section 3.4 turns the argument principle into an identity that reads the zero count and the zero density, strip by strip, from the Thorin mass. Section 3.5 develops further readings of the sine squared lemma, including Selberg's correspondence between resolution and prime cutoff near the critical line and Weil's criterion as a domination on variograms.

3.1 What computation can reach, and what the sine squared lemma adds

Two facts govern what computation can establish about the Thorin condition and which evaluator it should use. A finite computation that finds no zero off the line cannot by itself measure a distance to RH: on a window where the zeros have been counted it reproduces U_{RH}, and at a stated resolution it measures the positivity

margin of ρ_ε up to a stated height, but neither result constrains the zeros above the window (Section 3.1.1). The sine squared lemma identifies the Thorin density from the primes but is a poor way to compute it: its pole and prime pieces cancel as the basepoint approaches one, below one its prime sum diverges, and near the critical line the smoothed prime sum that replaces it needs a number of primes exponential in $1/\varepsilon$ (Section 3.1.2). The Pólya density has none of these defects, and it is the evaluator in the rest of this chapter and in Chapters 4 and 5. The lemma still shapes the computations: it makes the cumulative Thorin mass an argument, it poses positivity certificates as linear programs whose only negative input is the pole, and it gives an exact consistency check at the centre (Section 3.1.3).

3.1.1 What a finite computation can measure

On a window where the zeros have been counted, computation identifies the Thorin functional with U_{RH}. Below height 200 the Pólya route finds 79 sign changes of Ξ and an argument principle count $N(200) = 79.0000000000004$ (Section 2.1.2). So, numerically and in agreement with [10, 196], the 79 zeros below height 200 are simple and on the line, and the Thorin functional $U = \sum_k \delta_{a_k}$ of Section 5.3 coincides with U_{RH} on $(0, 200^2]$. The matched total variation distance of Section 3.2, computed against a model of the zero list, is $3.7 \times 10^{-9}\,\varepsilon$ at every resolution computed, the numerical floor of the comparison.

None of this constrains the zeros above the window. Platt and Trudgian [196] have verified every zero below height 3×10^{12}, yet the perturbation of ξ in Theorem 5.2.1 [200, Chapter 11] keeps the order, type and symmetry of $\xi(\frac{1}{2} + s)$, puts zeros off the line just above any large enough height R for which RH has been verified up to $R + 50$, and still passes the finite tests of the heat trace, from Hankel blocks and derivative signs (to order 10^{17}) to the small time expansion. Tests that count zeros find such a perturbation only when they reach its height. So a computation from the zeros verified up to a given height, or from finitely many tests of the heat trace, cannot by itself measure a scalar distance to RH, although a finite computation could refute RH by certifying a zero off the line. What it can measure is a margin at a stated resolution and height.

The margin is the minimum of ρ_ε, and positivity on one line is weaker than a zero free half plane. If every zero has real part less than $\frac{1}{2} + \varepsilon$, then $\rho_\varepsilon > 0$ and the reciprocal at basepoint $\frac{1}{2} + \varepsilon$ is a GGC transform with Thorin measure μ_ε (Theorem 1.2.2), so a negative value of ρ_ε proves a zero to the right of the line. The converse is not known for ξ, because the positive kernels of the other zeros can outweigh the negative kernel of a zero to the right of the line (Remark 1.2.4); Section 5.2.2 states, to first order in ε, when they cannot. Positivity on every line $\operatorname{Re} s = \frac{1}{2} + \varepsilon'$ with $\varepsilon' > \varepsilon$ does exclude such zeros, and positivity for every $\varepsilon > 0$ is Lagarias's criterion [133], which is RH.

Resolution is the first limitation. On $0 \le \theta \le 200$ the computed minimum is $\min \rho_\varepsilon = 0.0147\,\varepsilon$ at each of the eight resolutions of Section 2.1.2, attained at

$\theta = 0$, where $\rho_\varepsilon(0) = \pi^{-1}\xi'(\frac{1}{2}+\varepsilon)/\xi(\frac{1}{2}+\varepsilon) = 2\varepsilon\pi^{-1}\sum_k a_k^{-1} + O(\varepsilon^3)$, because $(\xi'/\xi)(\frac{1}{2}+\varepsilon) = \sum_k 2\varepsilon/(\varepsilon^2 + a_k)$ and $\sum_k a_k^{-1} = 0.0231050$ (Section 1.2). So $\inf_\theta \rho_\varepsilon(\theta) \le \rho_\varepsilon(0) \to 0$ as $\varepsilon \downarrow 0$: near the critical line positivity is never won by a fixed margin, and nothing in a finite window prevents it from failing above the window. On the verified window the origin is in fact the minimum: for every $\varepsilon \in (0, 6.85]$ the minimum of ρ_ε over $0 \le \theta \le 3\times 10^{12} - 1.21$ is $\rho_\varepsilon(0)$, attained only at the origin, and that this holds at every height for every $\varepsilon \in (0, \frac{1}{2})$ is equivalent to RH (Theorem 9.1.11 and Remark 9.1.12).

Height is the second limitation. The untilted Pólya route loses about $\pi\theta/(4\log 10) = 0.341\,\theta$ decimal digits at height θ, because $|\xi(\frac{1}{2}+\varepsilon+i\theta)|$ decays like $e^{-\pi\theta/4}$, up to a power of θ, while the integrands are of order one; height 10^4 would require about 3400 digits per point. The tilted contour of Section 2.1.3, at distance δ from the edge of the strip of analyticity of Φ, reduces the loss to about $\delta\theta/\log 10$ digits and gives 25 digits of $\rho_{0.05}$ at height 5000 in 28 milliseconds per point (Table 2.2), but the number of nodes, and with it the cost per point, still grows linearly in the height. The Riemann Siegel formula and the Odlyzko Schönhage algorithm evaluate ζ at large height far more cheaply (Section 2.2), so the Pólya route is not a competitor for zero verification; the division of labour between the two is drawn at the end of Section 2.2.3.2. Its value is accuracy at small ε on moderate windows, where the Thorin measure is sharp and the prime side is ill conditioned.

3.1.2 The sine squared decomposition of the Thorin bracket

The sine squared lemma (1.2.9) converts the Lévy description of $\log\xi$, in which the primes are atoms, into the Thorin description of the reciprocal, in which each zero contributes a Poisson kernel (Proposition 1.2.3) and becomes an atom only in the limit $\varepsilon \downarrow 0$ under RH. At a basepoint $\alpha > 1$, applying the kernel $2\sin^2(x\theta/2) = 1 - \cos(x\theta)$ to the canonical measure of [200, Theorem 8.1] writes the bracket $v_\alpha(\theta) = \mathrm{Re}(\xi'/\xi)(\alpha + i\theta)$ as a sum of four explicit pieces:

$$v_\alpha(\theta) = \underbrace{\frac{\xi'}{\xi}(\alpha)}_{\text{drift}} + \underbrace{\int_0^\infty (1-\cos\theta x)\frac{e^{-\alpha x}}{e^{2x}-1}\,dx}_{\text{archimedean } \ge 0} - \underbrace{\frac{\theta^2}{(\alpha-1)\big((\alpha-1)^2+\theta^2\big)}}_{\text{pole } \ge 0} + \underbrace{\sum_{n\ge 2}\frac{\Lambda(n)}{n^\alpha}\big(1-\cos(\theta\log n)\big)}_{\text{primes } \ge 0}. \quad (3.1.1)$$

The pole piece is the closed form of $\int_0^\infty (1-\cos\theta x)e^{-(\alpha-1)x}dx$, and the prime piece equals $-\frac{\zeta'}{\zeta}(\alpha) + \mathrm{Re}\,\frac{\zeta'}{\zeta}(\alpha+i\theta)$.

The drift is the linear coefficient $c = \xi'/\xi(\alpha)$ of the Lévy representation of Section 1.2, not the drift d of a GGC. The archimedean piece is the contribution

α	θ	drift	archimedean	pole	primes	$v_\alpha(\theta)$	$\kappa_\alpha(\theta)$
3.0	2	0.1144	0.0533	−0.2500	0.1988	0.1165	3.85
3.0	50	0.1144	1.2594	−0.4992	0.2464	1.1210	0.67
2.0	2	0.0691	0.0859	−0.8000	0.7154	0.0704	21.5
2.0	50	0.0691	1.3989	−0.9996	0.7561	1.2245	1.43
1.5	2	0.0461	0.1137	−1.8824	1.7683	0.0470	77.6
1.5	50	0.0461	1.4863	−1.9998	1.8527	1.3867	2.78

Table 3.1: The four pieces of the sine squared decomposition (3.1.1), with the prime sum over $n \le 2 \times 10^6$, and the bracket $v_\alpha(\theta)$ computed by the Pólya route; it differs from the sum of the four columns by the prime tail beyond 2×10^6 (1.3×10^{-3} at $\alpha = 1.5$, $\theta = 2$). The pole column is the signed contribution, minus the pole piece. The pole and prime pieces grow like $1/(\alpha - 1)$ and cancel to leave a bracket of order one or smaller; κ_α measures the loss.

of $\Gamma(s/2)$ together with the factor s of $\frac{1}{2}s(s-1)$; it is smooth in θ and grows like $\frac{1}{2}\log\theta$, which accounts for the average density of zeros. The pole piece comes from the pole of ζ at $s = 1$. The prime piece is a sum of oscillations with frequencies $\log n$ and amplitudes $\Lambda(n)n^{-\alpha}$, one for each prime power. The drift is positive, the archimedean and prime pieces are nonnegative, and only the pole piece enters with a minus sign.

Near $\alpha = 1$ the pole piece is almost cancelled by the prime piece. At $\alpha = 1.5$ and $\theta = 2$ the pole contributes -1.8824 and the primes $+1.7683$, while $v_{1.5}(2) = 0.0470$ (Table 3.1 and Figure 3.1); the four columns of the table sum to 0.0457, the difference 1.3×10^{-3} being the prime tail beyond 2×10^6. Both pieces grow like $1/(\alpha-1)$, the pole explicitly and the primes through the pole of $-\zeta'/\zeta$ at one. At each fixed $\theta \neq 0$ their difference stays bounded as $\alpha \downarrow 1$ because ζ has no zero on the line $\operatorname{Re} s = 1$, so the analytic input of the prime number theorem appears in the Thorin description as this cancellation, which [200, Section 8.2] identifies with the collapse onto the zeros. Measured by the condition number

$$\kappa_\alpha(\theta) = \frac{|\text{pole}| + |\text{primes}|}{|v_\alpha(\theta)|},$$

the prime side amplifies relative errors by 3.9 at $\alpha = 3$, 21.5 at $\alpha = 2$ and 77.6 at $\alpha = 1.5$, all at $\theta = 2$. As $\alpha \downarrow 1$ the numerator grows like $2/(\alpha - 1)$ while v_α stays bounded.

The lemma also prices the prime side. By the prime number theorem the truncation tail satisfies, for fixed $\theta \neq 0$,

$$\sum_{n>N} \frac{\Lambda(n)}{n^\alpha}\big(1 - \cos(\theta \log n)\big) \asymp \frac{N^{1-\alpha}}{\alpha - 1},$$

which is 1.4×10^{-3} at $\alpha = 1.5$ with $N = 2 \times 10^6$, the size of the errors in Table 1.6; an absolute error of 10^{-10} at $\alpha = 1.5$ would need a cutoff $N \approx 4 \times 10^{20}$. For

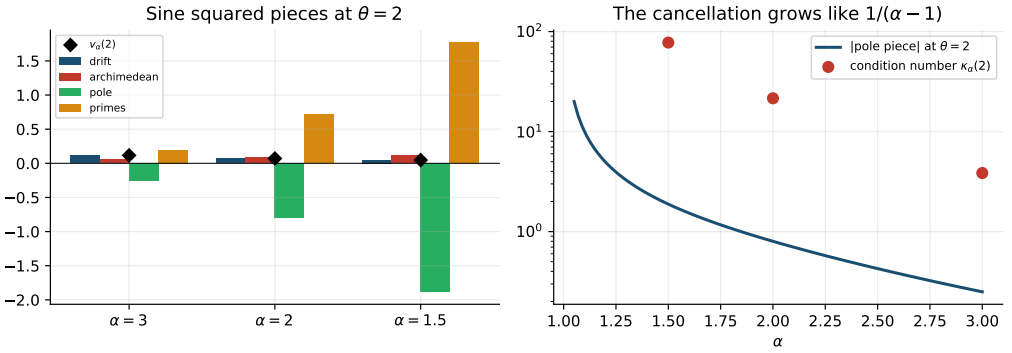


Figure 3.1: Left: the four pieces of the sine squared decomposition at $\theta = 2$ for three basepoints, with the bracket $v_\alpha(2)$. Right: the pole piece and the condition number grow like $1/(\alpha - 1)$ as the basepoint approaches one.

$\alpha \le 1$ the series diverges, and the lemma must be read through the regularised pairing of [200, Chapter 7] and the explicit formula. At the basepoint one the regularisation is explicit: the pole and prime pieces can be compensated against each other, and Section 9.1 converts the resulting Lévy form of $\xi(1+s)$ term by term into the Thorin density. Near the critical line Selberg's smoothed sum over the primes up to x^2 takes the place of the series, and at resolution ε, if no zero lies to the right of the critical line, its error at an ordinate is of order $x^{-\varepsilon}/(\varepsilon^2 \log x)$, so that k correct digits need $x \approx 10^{k/\varepsilon}$ (Section 3.5.1). The Pólya route has none of these costs: its cost per point is polynomial in the height and independent of ε, and at height 200 it reaches $\varepsilon = 0.005$ with about 60 digits (Section 2.1.1).

3.1.3 Consequences for computation

The bracket is the derivative of an argument, $v_\alpha(\theta) = \partial_\theta \arg \xi(\alpha + i\theta)$, because $\partial_\theta \log \xi(\alpha + i\theta) = i(\xi'/\xi)(\alpha + i\theta)$. So the cumulative Thorin mass is an argument,

$$\mu_\varepsilon\big((0, \theta^2]\big) = \frac{1}{\pi} \arg \xi\Big(\frac{1}{2} + \varepsilon + i\theta\Big),$$

continued along the line from the real axis, where $\xi > 0$. It is computed without quadrature, by accumulating the arguments of quotients of consecutive values of ξ, with steps chosen so that each increment can be read from its principal value, a condition checked after the fact rather than guaranteed (Section 2.1.2). Section 3.2 uses it for masses, cell masses, cumulative distances and the mass defect of Proposition 3.2.2, and Section 3.4 for the zero count.

The lemma is linear with a nonnegative kernel, so a nonnegative Lévy input gives a nonnegative Thorin output, and positivity certificates can be posed as linear programs in the sine squared basis [200, Sections 18.2.8 and 18.2.9]. For ξ

the only negative part of the input $xL(dx)$ is the pole density $-e^{-(\alpha-1)x}dx$, so a certificate must show that the positive pieces dominate the pole piece at every θ. For $\alpha > 1$ they do (Theorem 1.2.2), although the proof goes through the zeros and not through the input. For $\frac{1}{2} < \alpha \leq 1$, where the lemma is read through the regularised pairing, domination at every basepoint is Lagarias's criterion, that is RH, and Section 3.5.2 states RH as a domination of the same kind at the centre, Weil's criterion on variograms. Beurling generalised prime systems, in which the sign of the input is controlled by construction, are among the open computations of Section 12.2.3.

The lemma also supplies an exact consistency check at $\alpha = \frac{1}{2}$, where the constant parts of the pole and prime pieces diverge and are continued analytically. The constant part $1/(\alpha-1)$ of the pole piece continues to -2 and enters with a minus sign; the constant part $-\zeta'/\zeta(\alpha)$ of the prime piece continues to $-\zeta'/\zeta(\frac{1}{2})$; and the gamma factor contributes the archimedean constant of [200, Section 6.8], $C_\Gamma = \frac{1}{2}\log\pi - \frac{1}{2}\psi(\frac{1}{4}) - 2 = 0.6860917096$, which is minus the logarithmic derivative of $s\pi^{-s/2}\Gamma(s/2)$ at $s = \frac{1}{2}$. They sum to

$$C_\Gamma + 2 - \frac{\zeta'}{\zeta}\Big(\frac{1}{2}\Big) = 0,$$

which is equivalent to $\xi'(\frac{1}{2}) = 0$, because $\xi'/\xi(s) = 1/s + 1/(s-1) - \frac{1}{2}\log\pi + \frac{1}{2}\psi(s/2) + \zeta'/\zeta(s)$ and $1/s + 1/(s-1)$ vanishes at $s = \frac{1}{2}$. Any implementation of the sine squared route can be tested against this identity, and the Pólya route satisfies its analogue automatically, since

$$\xi'\big(\tfrac{1}{2}\big) = \int_{\mathbb{R}} u\,\Phi(u)\,du = 0$$

by the evenness of Φ.

3.2 Distances to the discrete measure implied by RH

On a window where the zeros are known, the Thorin measure μ_ε computed by the Pólya route can be compared with U_{RH} directly (Section 3.1.1), and how close the two are depends on the metric. Raw total variation is maximal and uninformative; at matched resolution the difference is a numerical floor of at most $3.7 \times 10^{-9}\,\varepsilon$; the cell distance, the mass defect and the error of the heat trace vanish linearly in ε, and the cumulative L^1 distance like $\varepsilon\log(1/\varepsilon)$; and extrapolation in ε gains several orders of magnitude where the dependence on ε is smooth.

Fix $\Theta = 200$ and the window $E = (0, \Theta^2]$ in z, equivalently $(0, \Theta]$ in θ; total variation is invariant under the change of variable $z = \theta^2$. The window contains 79 zeros, all simple and on the line, and Θ is not an ordinate, since $\gamma_{80} = 201.2648$ (Section 2.1.2). For a signed measure ν on E write $\|\nu\|_{\mathrm{TV}} = |\nu|(E)$, and write $M_\varepsilon(\theta) = \mu_\varepsilon((0, \theta^2])$ for the cumulative mass.

3.2.1 Total variation at matched resolution

A smooth approximation to a sum of point masses never converges to it in total variation.

Proposition 3.2.1. *If $\nu \geq 0$ is absolutely continuous and U is purely atomic with N unit atoms in E, then $\|\nu - U\|_{\mathrm{TV}} = \nu(E) + N$.*

Proof. Let A be the finite set of atoms. Then $\nu(A) = 0$, so $|\nu - U|(A) = U(A) = N$ and $|\nu - U|(E \setminus A) = \nu(E \setminus A) = \nu(E)$. □

The Thorin measure is positive on E, where its density has minimum $0.0147\,\varepsilon$ (Section 3.1.1), so $\|\mu_\varepsilon - U_{\mathrm{RH}}\|_{\mathrm{TV}} = \mu_\varepsilon(E) + 79 \approx 158$ for every ε, the largest value possible for two measures of these masses. However narrow a Cauchy kernel is, it gives no mass to the point where the atom sits, so the two measures are mutually singular: total variation does not metrise weak convergence. It becomes informative only after both measures have been brought to the same resolution, or summed over cells.

By Proposition 1.2.3 the object to compare with μ_ε is the Poisson smoothing of the RH measure at the same resolution, and we write $d_{\mathrm{match}}(\varepsilon) = \|\mu_\varepsilon - P_\varepsilon * U^{\pm}_{\mathrm{RH}}\|_{\mathrm{TV}}$ for the matched distance. Under RH the two measures coincide and $d_{\mathrm{match}}(\varepsilon) = 0$, so a computation of the matched distance measures its own error unless some zero is off the line.

A computation replaces $U^{\pm}_{\mathrm{RH}}$ by a model $\widetilde{U}$, which we build from the first 1700 ordinates, computed with `mpmath` [159] (the first 80 at 50 digits, as in Section 2.1.2, and stored to 45), and from the smooth density $\pi^{-1}\vartheta'(t)\,dt$ of the Riemann Siegel theta function beyond the Gram point $T_* = g_{1699} = 2197.8796$ of Section 2.2.1, at which $\vartheta(T_*)/\pi + 1 = 1700$, each with its reflection. We write $\tilde{d}_{\mathrm{match}}(\varepsilon) = \|\mu_\varepsilon - P_\varepsilon * \widetilde{U}\|_{\mathrm{TV}}$ for the matched distance against the model. Since $\gamma_{1700} = 2197.2587 < T_* < \gamma_{1701} = 2198.6875$, we have $N(T_*) = \vartheta(T_*)/\pi + 1$, so the boundary term vanishes when $\int_{T_*}^{\infty} P_\varepsilon(\theta - t)\,d\big(N(t) - \vartheta(t)/\pi - 1\big)$ is integrated by parts, and the error of the tail model is at most of order ε/T_*^2 per unit length. The observed floor is far smaller: $\tilde{d}_{\mathrm{match}}(\varepsilon) = 3.66 \times 10^{-9}\,\varepsilon$ at every resolution from $\varepsilon = 1$ to $\varepsilon = 0.005$ (Table 3.2), that is $1.8 \times 10^{-11}\,\varepsilon$ per unit length, and the pointwise difference stays below 2×10^{-11} (Figure 3.2). The exact proportionality to ε identifies the floor as the residual of the tail model; it is an observed value, not a bound for $d_{\mathrm{match}}(\varepsilon)$: such a bound would need certified bounds on the error of the tail model and on the evaluation of μ_ε. The Pólya values themselves keep about 60 digits even at height 200 (Section 2.1.1), and the densities that enter the comparison are stored in double precision.

ε	mass	raw TV	matched TV	sup diff	cell TV	L^1 in θ	$W_\varepsilon(0.01)$
1	79.36738	158.37	3.66×10^{-9}	1.9×10^{-11}	4.992	59.23	0.29504
0.5	79.19417	158.19	1.83×10^{-9}	9.6×10^{-12}	2.655	41.86	0.22362
0.25	79.09887	158.10	9.14×10^{-10}	4.8×10^{-12}	1.351	28.77	0.18698
0.1	79.03977	158.04	3.66×10^{-10}	1.9×10^{-12}	0.5431	16.01	0.16469
0.05	79.01990	158.02	1.83×10^{-10}	9.6×10^{-13}	0.2718	9.737	0.15721
0.02	79.00796	158.01	7.31×10^{-11}	3.8×10^{-13}	0.1087	4.816	0.15270
0.01	79.00398	158.00	3.66×10^{-11}	2.0×10^{-13}	0.05437	2.756	0.15120
0.005	79.00199	158.00	1.83×10^{-11}	1.5×10^{-13}	0.02718	1.553	0.15045
RH	79	0	0	0	0	0	0.14970

Table 3.2: Distances between the Pólya computed Thorin measure μ_ε and U_{RH} on $0 < z \le 200^2$. Mass is $M_\varepsilon(200)$. Raw TV is $\|\mu_\varepsilon - U_{\mathrm{RH}}\|_{\mathrm{TV}}$, forced to equal mass plus 79 by Proposition 3.2.1. Matched TV is $\tilde{d}_{\mathrm{match}}(\varepsilon) = \|\mu_\varepsilon - P_\varepsilon * \widetilde{U}\|_{\mathrm{TV}}$, and sup diff is the maximum of $|\rho_\varepsilon - P_\varepsilon * \widetilde{U}|$ over $0 < \theta \le 200$. Cell TV is $d_{\mathrm{cell}}(\varepsilon)$, L^1 in θ is $\int_0^{200} |M_\varepsilon - N|\, d\theta$, and $W_\varepsilon(0.01)$ is the smoothed heat trace at $t = 0.01$, with the value from the zeros in the last row.

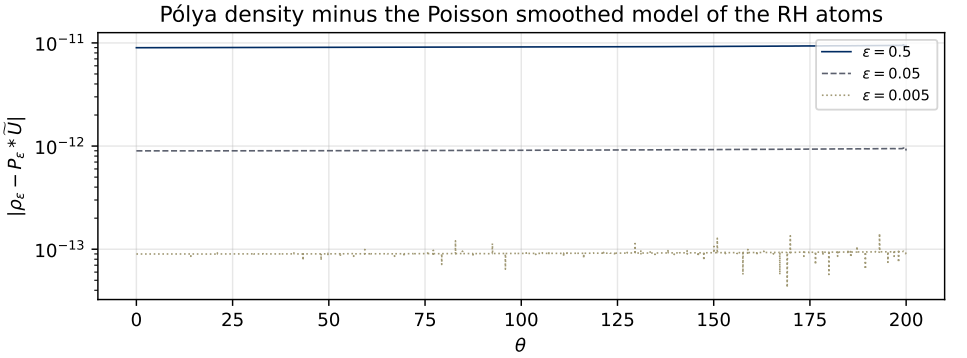


Figure 3.2: Pointwise difference $|\rho_\varepsilon - P_\varepsilon * \widetilde{U}|$ between the Pólya computed density and the Poisson smoothing of the model $\widetilde{U}$ of the RH atoms, at $\varepsilon = 0.5$, 0.05 and 0.005.

3.2.2 Cumulative distances and extrapolation in the resolution

The total mass is the simplest quantity to compare, since $U_{\mathrm{RH}}(E) = N(\Theta) = 79$. By Section 3.1.3 it is an argument, $\mu_\varepsilon(E) = \pi^{-1} \arg \xi(\frac{1}{2} + \varepsilon + i\Theta)$, continued from $\arg \xi(\frac{1}{2} + \varepsilon) = 0$, and its defect has an explicit first order term.

Proposition 3.2.2. *Suppose Θ is not an ordinate and every zero with $0 < \operatorname{Im} \rho \le \Theta$ lies on the line. Then*

$$\mu_\varepsilon\big((0, \Theta^2]\big) = N(\Theta) - \frac{\varepsilon}{\pi} \frac{\Xi'(\Theta)}{\Xi(\Theta)} + O(\varepsilon^2), \qquad \varepsilon \downarrow 0.$$

Proof. Let $\sigma_1 > 1$. By hypothesis, and because ξ is positive on the real axis, ξ has no zero in the closed rectangle with vertices $\frac{1}{2}+\varepsilon$, σ_1, $\sigma_1+i\Theta$ and $\frac{1}{2}+\varepsilon+i\Theta$. So the change of $\arg\xi$ up its left side, which is $\pi\mu_\varepsilon((0,\Theta^2])$, equals the change along the path $\frac{1}{2}+\varepsilon \to \sigma_1 \to \sigma_1+i\Theta \to \frac{1}{2}+\varepsilon+i\Theta$. The classical formula $\pi N(\Theta) = \Delta\arg\xi$ along $\frac{1}{2} \to \sigma_1 \to \sigma_1+i\Theta \to \frac{1}{2}+i\Theta$ [255, Section 9.3] differs from it only by the segment from $\frac{1}{2}+\varepsilon+i\Theta$ to $\frac{1}{2}+i\Theta$, since $\arg\xi$ is constant on the real axis. Put $h(\varepsilon) = \xi(\frac{1}{2}+\varepsilon+i\Theta)$. Since $h(0) = \Xi(\Theta) \neq 0$, a branch of $\arg h$ is real analytic near $\varepsilon = 0$, and

$$\pi\mu_\varepsilon\big((0,\Theta^2]\big) = \pi N(\Theta) + \arg h(\varepsilon) - \arg h(0) = \pi N(\Theta) + \varepsilon\ \mathrm{Im}\,\frac{\xi'}{\xi}\Big(\frac{1}{2}+i\Theta\Big) + O(\varepsilon^2).$$

From $\frac{d}{d\theta}\Xi(\theta) = i\xi'(\frac{1}{2}+i\theta)$ we get $(\xi'/\xi)(\frac{1}{2}+i\Theta) = -i\,\Xi'(\Theta)/\Xi(\Theta)$, and Ξ'/Ξ is real, so the coefficient of ε is $-\Xi'(\Theta)/\Xi(\Theta)$. □

Remark 3.2.3. The remainder in Proposition 3.2.2 is in fact $O(\varepsilon^3)$: $\xi(s) = \xi(1-s)$ and the reality of ξ on the real axis give $h(-\varepsilon) = \overline{h(\varepsilon)}$, so $\arg h(\varepsilon) - \arg h(0)$ is odd in ε and its expansion has no term in ε^2.

The formula says that the mass lost or gained at the edge of the window is governed by the logarithmic derivative of Ξ there. If Θ sits just below a zero, $\Xi'/\Xi(\Theta)$ is large and negative, and the window gains the part of that zero's Cauchy kernel that spills across the edge; if Θ sits just above a zero, the window loses part of its kernel. Choosing Θ midway between zeros makes the correction small, and adding it recovers $N(\Theta)$ up to $O(\varepsilon^3)$; Proposition 3.4.2 makes the recovery exact. At $\Theta = 200$ the Pólya route gives $\Xi'(200)/\Xi(200) = -1.2506879$, so the proposition predicts $\mu_\varepsilon(E) - 79 = -\varepsilon\,\Xi'(200)/(\pi\,\Xi(200)) + O(\varepsilon^3)$, with $-\Xi'(200)/(\pi\,\Xi(200)) = 0.3981063$; the observed $(\mu_\varepsilon(E) - 79)/\varepsilon$ is 0.398105 at $\varepsilon = 0.005$ and 0.398102 at $\varepsilon = 0.01$. Figure 3.3 compares M_ε with the counting function of U_{RH} across the window.

Summing over cells gives a distance that resolves the individual zeros. Let $b_0 = 0$, $b_k = (\gamma_k + \gamma_{k+1})/2$ for $1 \le k \le 78$ and $b_{79} = \Theta$, and let $C_k = (b_{k-1}, b_k]$. These are the Voronoi cells of the ordinates, except that the last ends at Θ rather than midway to γ_{80}. The cell masses $\mu_\varepsilon(C_k) = M_\varepsilon(b_k) - M_\varepsilon(b_{k-1})$ are differences of the argument, and the coarse grained distance

$$d_{\mathrm{cell}}(\varepsilon) = \sum_k |\mu_\varepsilon(C_k) - 1|$$

is the total variation distance between the images of μ_ε and U_{RH} under the map sending each cell to its atom. No b_k is an ordinate, so Proposition 3.2.2 applies at both ends of every cell, with $\Xi'(0) = 0$ at b_0, and with the remainder of Remark 3.2.3 gives $\mu_\varepsilon(C_k) - 1 = -(\varepsilon/\pi)\big[\Xi'/\Xi(b_k) - \Xi'/\Xi(b_{k-1})\big] + O(\varepsilon^3)$. Hence $d_{\mathrm{cell}}(\varepsilon) = \kappa_{\mathrm{cell}}\,\varepsilon + O(\varepsilon^3)$, where $\kappa_{\mathrm{cell}} = \pi^{-1}\sum_k |\Xi'/\Xi(b_k) - \Xi'/\Xi(b_{k-1})|$, the cell constant of this window, is 5.43665, and every cell mass lies in $[0.99888, 1.00126]$ at

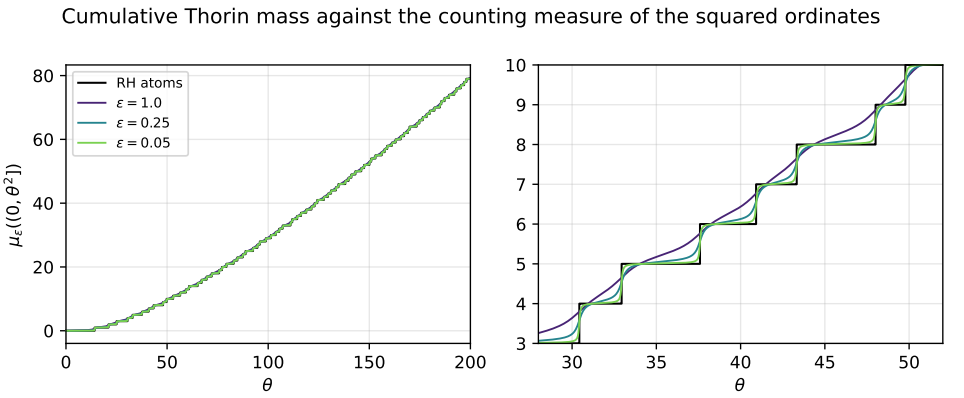


Figure 3.3: Cumulative Thorin mass $\mu_\varepsilon((0,\theta^2]) = \pi^{-1}\arg\xi(\frac{1}{2}+\varepsilon+i\theta)$ at $\varepsilon = 1$, 0.25 and 0.05 against the counting function of U_{RH}, on the whole window (left) and on $28 \le \theta \le 52$ (right).

$\varepsilon = 0.005$. The constant measures the Cauchy leakage across the cell boundaries, and Section 5.4 reads it as a statistic of the irregularity of the spacings.

The L^1 distance $\int_0^\Theta |M_\varepsilon(\theta) - N(\theta)|\,d\theta$ between the cumulative mass and the counting function is a transport distance (the Wasserstein distance when the two masses are equal), and because of the Cauchy tails it behaves like $\varepsilon\log(1/\varepsilon)$: it is 1.55 in θ and 360 in z at $\varepsilon = 0.005$, and Section 3.4.1 derives the rate. The heat trace $W(t) = \int e^{-tz}U_{\mathrm{RH}}(dz)$ is approximated by $W_\varepsilon(t) = \int_0^\infty e^{-t\theta^2}\rho_\varepsilon(\theta)\,d\theta = \int e^{-tz}\mu_\varepsilon(dz)$. At $t = 0.01$ the zeros give $W = 0.1496980112538$ (Table 1.5), while $W_\varepsilon(0.01) = 0.1504500280599$, 0.1512017825455 and 0.1527045050533 at $\varepsilon = 0.005$, 0.01 and 0.02, again with error linear in ε.

The mass and the heat trace are smooth in ε at $\varepsilon = 0$, so Richardson extrapolation applies to them: the mass by Proposition 3.2.2, and the heat trace because by Proposition 1.2.3 it is a sum over the zeros of Poisson integrals of the Gaussian $e^{-t\theta^2}$. The L^1 distance, of order $\varepsilon\log(1/\varepsilon)$, is not smooth there. The combination $2\mu_{0.005}(E) - \mu_{0.01}(E)$ equals $79 + 3.2\times10^{-8}$, against a raw defect of 2.0×10^{-3}. The residual is this small because the remainder in Proposition 3.2.2 is $O(\varepsilon^3)$ (Remark 3.2.3), so the two point combination leaves only the cubic term. For the heat trace the error 7.5×10^{-4} of $W_{0.005}(0.01)$ falls to 2.6×10^{-7} after one Richardson step, the two point combination of $\varepsilon = 0.005$ and 0.01, and to 1.7×10^{-10} after two, the three point combination that adds $\varepsilon = 0.02$. Without extrapolation the same accuracy would need $\varepsilon \approx 8\times10^{-8}$ for the mass, and $\varepsilon \approx 2\times10^{-6}$ and 10^{-9} for the heat trace. Figure 3.4 collects the rates of all the distances.

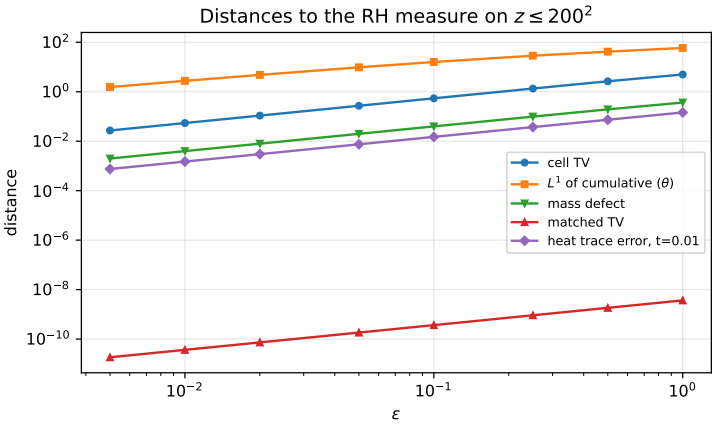


Figure 3.4: The distances of Table 3.2 as functions of ε: the cell distance, the mass defect and the heat trace error at $t = 0.01$ vanish linearly, and the cumulative L^1 distance like $\varepsilon \log(1/\varepsilon)$. The matched total variation $\tilde{d}_{\mathrm{match}}(\varepsilon)$ is the floor $3.66 \times 10^{-9}\,\varepsilon$ of the tail model for the zero list.

3.3 What an off line zero would do

The distances of Section 3.2 compare μ_ε with U_{RH} on a window where the zeros are known to lie on the line, and there the matched distance leaves only a numerical floor. How each distance would change if a zero left the line can be computed in closed form, because by Proposition 1.2.3 an off line zero at distance δ from the line enters the Thorin density, together with its reflection in the line, only through the kernels $P_{\varepsilon+\delta} + P_{\varepsilon-\delta}$ at its ordinate, the second of which is negative once $\varepsilon < \delta$. Proposition 3.3.1 computes the effect for a displacement that keeps the count, for a pair missing from a list of line zeros, and for the negative mass of a displaced zero. Together the three signatures show that an off line zero is most visible at resolutions comparable to its distance from the line, a theme that returns in Section 5.2. Read against the floor of the matched distance, they give detection thresholds below height 200 and a calibration of the method above it.

3.3.1 Three signatures of an off line zero

Off line zeros come in pairs $\frac{1}{2} \pm \delta + i\gamma$ with $0 < \delta < \frac{1}{2}$, together with their conjugates, and near γ a pair contributes $P_{\varepsilon+\delta} + P_{\varepsilon-\delta}$ to ρ_ε where two line zeros would contribute $2P_\varepsilon$. A displacement that keeps the count $N(T)$ therefore moves two line zeros at height γ to such a pair, as a double zero does at a collision in the de Bruijn Newman flow (Section 8.4). Parts (i) and (iii) of Proposition 3.3.1 are normalised per displaced zero, and part (ii) per pair.

PROPOSITION 3.3.1. *Let $\varepsilon > 0$ and $0 < \delta < \frac{1}{2}$; when $\delta = \varepsilon$, P_0 denotes the unit point mass at the origin.*

(i) (Displaced zeros, same count.) If two line zeros at height γ are replaced by the quadruple $\frac{1}{2} \pm \delta \pm i\gamma$, the density near γ changes by $2g$, where $g = \frac{1}{2}(P_{\varepsilon+\delta} + P_{\varepsilon-\delta}) - P_\varepsilon$ is the change per displaced zero, and

$$\|g\|_{\mathrm{TV}} = \frac{3\sqrt{3}}{4\pi}\Big(\frac{\delta}{\varepsilon}\Big)^2 + O\Big(\frac{\delta^4}{\varepsilon^4}\Big), \qquad \delta < \varepsilon.$$

(ii) (Pair omitted from the list.) If a pair at $\frac{1}{2} \pm \delta + i\gamma$ exists but a list of line zeros omits it, the difference is $P_{\varepsilon+\delta} + P_{\varepsilon-\delta}$ and

$$\|P_{\varepsilon+\delta} + P_{\varepsilon-\delta}\|_{\mathrm{TV}} = \begin{cases} 2, & \delta \le \varepsilon, \\ \dfrac{8}{\pi}\arctan\sqrt{\dfrac{\delta+\varepsilon}{\delta-\varepsilon}} - 2, & \delta > \varepsilon. \end{cases}$$

(iii) (Negative mass.) For $\delta > \varepsilon$ the negative part of $\frac{1}{2}(P_{\varepsilon+\delta} + P_{\varepsilon-\delta})$, the density of one displaced zero, has mass

$$\frac{2}{\pi}\arctan\sqrt{\frac{\delta+\varepsilon}{\delta-\varepsilon}} - \frac{1}{2} = \frac{\varepsilon}{\pi\delta} + O\Big(\frac{\varepsilon^3}{\delta^3}\Big).$$

Proof. (i) Taylor expansion in δ gives $g = \frac{\delta^2}{2}\partial_a^2 P_a|_{a=\varepsilon} + O(\delta^4)$ [200, Proposition 19.7], with a remainder of L^1 norm $O(\delta^4/\varepsilon^4)$. Since $\partial_a^2 P_a(x) = (2a/\pi)(a^2 - 3x^2)/(a^2+x^2)^3$, the substitution $x = \varepsilon v$ gives $\partial_a^2 P_a|_{a=\varepsilon} = 2(1-3v^2)/(\pi\varepsilon^3(1+v^2)^3)$ and $dx = \varepsilon\, dv$. The function $(1-3v^2)/(1+v^2)^3$ is the derivative of $v/(1+v^2)^2$, whose total variation is four times its maximum $9/(16\sqrt{3})$, that is $3\sqrt{3}/4$, so the L^1 norm of $\partial_a^2 P_a$ at $a = \varepsilon$ is $(2/(\pi\varepsilon^2)) \cdot 3\sqrt{3}/4 = \frac{3\sqrt{3}}{2\pi\varepsilon^2}$, and multiplying by $\delta^2/2$ gives (i). (ii) For $\delta \le \varepsilon$ both kernels are nonnegative with mass one. For $\delta > \varepsilon$ put $a = \delta + \varepsilon$, $b = \delta - \varepsilon$, so the difference is $P_a - P_b$, which is negative exactly on $|x| < \sqrt{ab}$. Integrating, $\|P_a - P_b\|_{\mathrm{TV}} = \frac{4}{\pi}[\arctan\sqrt{a/b} - \arctan\sqrt{b/a}]$, and $\arctan y + \arctan(1/y) = \pi/2$. (iii) $P_a - P_b$ has total mass zero, so its negative part is half its norm; halve again for the factor $\frac{1}{2}$. □

With $\varepsilon/\delta = \sin\phi$ one has $\arctan\sqrt{(\delta+\varepsilon)/(\delta-\varepsilon)} = \pi/4 + \phi/2$, so for $\delta > \varepsilon$ the norm in (ii) is $(4/\pi)\arcsin(\varepsilon/\delta)$ and the mass in (iii) is $(1/\pi)\arcsin(\varepsilon/\delta)$. In part (i) the list is correct in number but two of its zeros have left the line, and the density differs from the Poisson smoothing of the list by $2g$, which to leading order is $\delta^2\partial_a^2 P_a$ at $a = \varepsilon$, quadratic in the displacement. In part (ii) the list omits the pair; the difference is of order one while the resolution is coarser than the displacement, and fades like $4\varepsilon/(\pi\delta)$ once it is finer. Part (iii) bounds the negative mass that a displaced zero can contribute, which also fades like ε/δ. Figure 3.5 plots the three signatures.

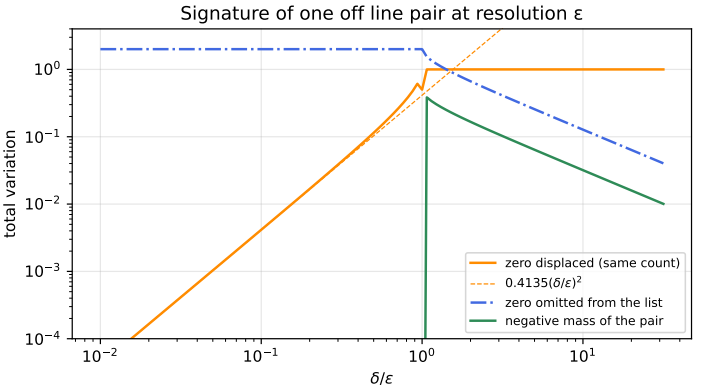


Figure 3.5: The three signatures of Proposition 3.3.1 as functions of δ/ε: $\|g\|_{\mathrm{TV}}$ per displaced zero (part (i)), with its leading term $0.4135(\delta/\varepsilon)^2 = (3\sqrt{3}/(4\pi))(\delta/\varepsilon)^2$ dashed; the total variation of an omitted pair (part (ii)); and the negative mass per displaced zero (part (iii)). For $\delta > \varepsilon$, outside the range of part (i), the kernels of a displaced zero have total mass zero and curve (i) is flat at 1.

3.3.2 Calibration of the matched distance

Proposition 3.3.1 converts the matched distance of Section 3.2 into detection thresholds. By part (ii), a computed matched distance $\tilde{d}_{\mathrm{match}}(\varepsilon)$ of at most $3.7 \times 10^{-9}\,\varepsilon$ excludes any omitted pair with ordinate in $(0, 200)$: such a pair contributes total variation $(4/\pi)\arcsin(\varepsilon/\delta)$, or 2 if $\delta \le \varepsilon$, less only the part of its kernels outside the window. By part (i), a displaced pair that keeps the count adds $2g$ to a difference whose norm is the floor $\tilde{d}_{\mathrm{match}}(\varepsilon)$, so by the triangle inequality the matched distance rises above the floor as soon as $\|g\|_{\mathrm{TV}} > \tilde{d}_{\mathrm{match}}(\varepsilon)$, that is, to leading order, once δ exceeds

$$\delta_{\min}(\varepsilon) = \varepsilon\Big(\frac{4\pi\,\tilde{d}_{\mathrm{match}}(\varepsilon)}{3\sqrt{3}}\Big)^{1/2}. \tag{3.3.1}$$

With the floors of Table 3.2 this threshold is 3.3×10^{-5} at $\varepsilon = 0.5$, 1.1×10^{-6} at $\varepsilon = 0.05$ and 3.3×10^{-8} at $\varepsilon = 0.005$.

The threshold (3.3.1) is the square root law of [200, Proposition 19.7]: the signature grows like the square of the displacement, so the smallest detectable displacement falls like the square root of the achievable distance. The prime side calibration of [200, Section 19.3.6] obeys the same law and reaches $\delta \approx 0.05$ at $\varepsilon = 2$. What changes is the reach: the Pólya route evaluates ρ_ε at small ε to about 60 digits below height 200 (Section 2.1.1), so the floor is set by the tail model for the zero list (Section 3.2.1) rather than by the evaluator, whereas the prime side is limited by the truncation of Table 1.6. A positivity test sees less than the matched distance: only the negative part of the whole density, which by

part (iii) is at most of order ε/δ per displaced zero and which the kernels of the neighbouring zeros can cancel entirely (Section 5.2). Section 11.4 bounds the mass of the negative part of the density below a given height by the number of zeros to the right of the basepoint and, through zero density estimates, shows with no hypothesis that it is a vanishing fraction of the positive part.

These thresholds concern zeros below height 200, where the argument principle count is already decisive, and their use is to calibrate the method above the verified window. There the cost per point of the Pólya route grows with the height (Section 3.1.1), and the count itself must come from the argument of ξ, which Section 3.4 reads from the Thorin mass (Proposition 3.4.2).

3.4 Counting zeros with the Thorin measure

Turing's method certifies the count $N(T)$ without evaluating ζ off the critical line, from the sign changes of Z and an explicit bound on the integral of S, where $\pi S(t) = \arg\zeta(\frac12+it)$; Backlund's method, which it replaced, bounds the argument of ζ along a horizontal segment off the line (Section 2.2.1). The Thorin measure is evaluated off the line, and its cumulative mass M_ε is an argument (Section 3.1.3). Section 3.2 used this for the mass defect of Proposition 3.2.2, assuming that the zeros below the window lie on the line, and Section 3.3 showed how an off line pair enters the density. Without that assumption, the argument principle gives at every resolution an exact identity for $N(T)$ and for the number $\nu(\frac12+\varepsilon, T)$ of zeros to the right of the line of evaluation, so that off line zeros are counted and not only detected. At $\varepsilon = \frac12$ the identity is Backlund's count with the vertical side on $\operatorname{Re} s = 1$, and at $T = 200$ it returns 79 at eight resolutions.

3.4.1 The cumulative Thorin mass as a smoothed counting function

Recall that $M_\varepsilon(\theta) = \mu_\varepsilon((0,\theta^2]) = \pi^{-1}\arg\xi(\frac12+\varepsilon+i\theta)$, with the argument continued along the line $\operatorname{Re} s = \frac12+\varepsilon$ from $\theta = 0$, where $\xi > 0$ (Sections 3.1.3 and 3.2). Since $\xi(s) = \frac12 s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$, the mass splits as

$$M_\varepsilon(\theta) = N_\varepsilon^{\mathrm{sm}}(\theta) + S_\varepsilon(\theta), \qquad S_\varepsilon(\theta) = \frac{1}{\pi}\arg\zeta\Big(\frac12+\varepsilon+i\theta\Big),$$

where, with $s = \frac12+\varepsilon+i\theta$ and $\theta > 0$, the smooth part $N_\varepsilon^{\mathrm{sm}}(\theta) = \pi^{-1}\big[\arg s + \arg(s-1) + \operatorname{Im}\log\Gamma(\frac14+\frac{\varepsilon}{2}+\frac{i\theta}{2}) - \frac{\theta}{2}\log\pi\big]$ comes from $s(s-1)\pi^{-s/2}\Gamma(s/2)$, with the principal arguments of s and $s-1$, which lie in $(0,\pi)$, and the branch of $\log\Gamma$ that is real on the positive axis. Thus S_ε is $\pi^{-1}\arg\zeta$ continued along the vertical line from $S_\varepsilon(0+) = -1$ if $0 < \varepsilon < \frac12$, when $\arg(s-1) \to \pi$ as $\theta \downarrow 0$, and from $S_\varepsilon(0+) = 0$ if $\varepsilon > \frac12$.

As $\varepsilon \downarrow 0$, $N_\varepsilon^{\mathrm{sm}} \to \vartheta/\pi + 1$, where $\vartheta(\theta) = \operatorname{Im}\log\Gamma(\frac14+\frac{i\theta}{2}) - \frac{\theta}{2}\log\pi$ is the Riemann Siegel theta function, and for θ not an ordinate Proposition 3.4.2 below

gives $M_\varepsilon(\theta) \to N(\theta) - 2\nu(\frac{1}{2},\theta)$, where $\nu(\sigma,\theta)$ counts the zeros with real part greater than σ and ordinate in $(0,\theta]$ (Section 1.1). When no zero below height θ lies to the right of the critical line, the limit is therefore the Riemann von Mangoldt formula $N(\theta) = \vartheta(\theta)/\pi + 1 + S(\theta)$ [255, Section 9.3]: M_ε is the counting function moved a distance ε to the right of the line, and S_ε is the analogue of S on the line $\operatorname{Re} s = \frac{1}{2} + \varepsilon$.

Integrating Proposition 1.2.3 in θ writes M_ε as a sum of Cauchy steps. Under RH, for $\theta \geq 0$,

$$M_\varepsilon(\theta) = \sum_{\gamma>0} \big[F_\varepsilon(\theta-\gamma) - F_\varepsilon(-\gamma) + F_\varepsilon(\theta+\gamma) - F_\varepsilon(\gamma)\big],$$
$$F_\varepsilon(x) = \frac{1}{2} + \frac{1}{\pi}\arctan\frac{x}{\varepsilon},$$

with the sum over all positive ordinates: each zero contributes a Cauchy step of width ε in place of a unit step. Since $F_\varepsilon(-\gamma) + F_\varepsilon(\gamma) = 1$, this says that $M_\varepsilon = P_\varepsilon * \tilde{N}$, where $\tilde{N}$ is the odd extension of N; that is, M_ε is the harmonic extension of $\tilde{N}$ to $\operatorname{Re} s > \frac{1}{2}$, evaluated on the line $\operatorname{Re} s = \frac{1}{2} + \varepsilon$. As ϑ is odd, S_ε is then the Poisson average of the odd extension of S up to an explicit smooth function, whereas Turing bounds the box integral $\int_{t_1}^{t_2} S(t)\,dt$ by an explicit inequality. If only the zeros below height $T+B$ are known to lie on the line, the display holds on $[0,T]$ up to an error of order $\varepsilon\delta^2/B^3$ for each off line pair above $T+B$ at distance δ from the line, counted as a double ordinate, because $P_{\varepsilon+\delta}(x) + P_{\varepsilon-\delta}(x) - 2P_\varepsilon(x) = -6\varepsilon\delta^2/(\pi x^4) + O(x^{-6})$. Without such a hypothesis the Poisson average gives way to the exact identity of Proposition 3.4.2.

The tails of the steps set the rate at which M_ε approaches N. For a single step,

$$\int_{-L}^{L} \big|F_\varepsilon(x) - \mathbf{1}_{x\geq 0}\big|\,dx = \frac{2}{\pi}\Big[L\arctan\frac{\varepsilon}{L} + \frac{\varepsilon}{2}\log\Big(1+\frac{L^2}{\varepsilon^2}\Big)\Big]$$
$$= \frac{2\varepsilon}{\pi}\Big(\log\frac{L}{\varepsilon} + 1\Big) + O\Big(\frac{\varepsilon^3}{L^2}\Big),$$

which explains the $\varepsilon\log(1/\varepsilon)$ rate of the L^1 distance in Table 3.2. Taking half of it on each side of each of the 79 zeros below height 200, with L the half gap on that side, gives 1.61 at $\varepsilon = 0.005$, against the computed 1.55; the difference comes from the tails of the other steps. The left panel of Figure 3.6 shows the counting error $M_\varepsilon - N$ at three resolutions.

3.4.2 A Littlewood identity and the zero density

Up to the factor π, the mass is also the derivative in ε of the integral of $\log|\xi|$ along the line of evaluation.

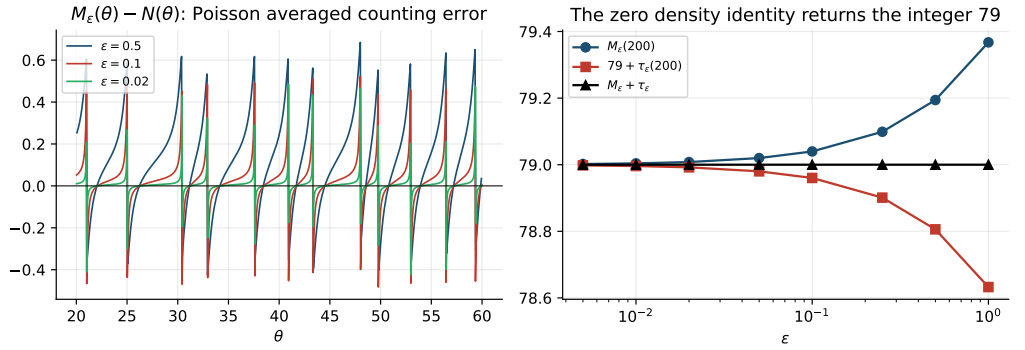


Figure 3.6: Left: the counting error $M_\varepsilon(\theta) - N(\theta)$ on $20 < \theta < 60$ at $\varepsilon = 0.5$, 0.1 and 0.02. At $\varepsilon = 0.5$ it is a sawtooth, like $-S$; as ε decreases it vanishes between the ordinates and concentrates in Cauchy steps of width ε at the zeros. Right: $M_\varepsilon(200)$, $79 + \tau_\varepsilon(200)$ and $M_\varepsilon(200) + \tau_\varepsilon(200)$, which equals 79 at every resolution (Proposition 3.4.2).

PROPOSITION 3.4.1. *Let $T > 0$. For every $\varepsilon > 0$ such that ξ has no zero on the segment from $\frac{1}{2} + \varepsilon$ to $\frac{1}{2} + \varepsilon + iT$,*

$$\pi M_\varepsilon(T) = \frac{\partial}{\partial \varepsilon} \int_0^T \log \left| \xi\Big(\frac{1}{2} + \varepsilon + it\Big) \right| dt,$$

and for every $\varepsilon > 0$

$$\int_0^T \log \left| \xi\Big(\frac{1}{2} + \varepsilon + it\Big) \right| dt = \int_0^T \log \left| \xi\Big(\frac{1}{2} + it\Big) \right| dt + \pi \int_0^\varepsilon M_{\varepsilon'}(T)\, d\varepsilon'.$$

Proof. Since $\partial_\sigma \log|\xi(\sigma+it)| = \mathrm{Re}(\xi'/\xi)(\sigma+it) = v_\sigma(t)$ and $v_\sigma(t) = \partial_t \arg \xi(\sigma+it)$, we have $M_\varepsilon(T) = \pi^{-1} \int_0^T v_{1/2+\varepsilon}(t)\, dt$, and when no zero lies on the segment, differentiation under the integral gives the first identity. The integral $I(\varepsilon)$ on the left of the second is continuous on $[0, \infty)$, because the logarithmic singularities at the zeros are integrable uniformly in ε, and continuously differentiable except at the finitely many ε' at which the segment meets a zero. There I' jumps by 2π for each zero met at an interior point of the segment, since $\int_{-c}^{c} a\, du/(a^2 + u^2) \to \pm\pi$ as $a \to 0\pm$. So I is absolutely continuous, and $I(\varepsilon) - I(0) = \int_0^\varepsilon I'(\varepsilon')\, d\varepsilon'$ is the second identity. □

The jump of 2π in the slope $\pi M_\varepsilon(T)$ when the line crosses a zero is the term 2ν of Proposition 3.4.2 below. Littlewood's lemma [255, Section 9.9] is the same Cauchy Riemann identity with the argument taken along a path through $\mathrm{Re}\, s = 2$; the two arguments differ by $2\pi\nu(\frac{1}{2} + \varepsilon, T)$, and Exercise 3.14 derives the lemma from the two propositions. Let $\tau_\varepsilon(T) = \pi^{-1} \Delta \arg \xi$ along the horizontal segment from $\frac{1}{2} + \varepsilon + iT$ to $\frac{1}{2} + iT$.

PROPOSITION 3.4.2. *Let $\varepsilon > 0$ and $T > 0$, with T not an ordinate and no zero of ξ on the segment from $\frac{1}{2}+\varepsilon$ to $\frac{1}{2}+\varepsilon+iT$. Then*

$$M_\varepsilon(T) + \tau_\varepsilon(T) = N(T) - 2\,\nu\Big(\frac{1}{2}+\varepsilon, T\Big).$$

The segment condition holds at $\varepsilon = \frac{1}{2}$, and $\nu(1,T) = 0$, which gives the unconditional count $N(T) = M_{1/2}(T) + \tau_{1/2}(T)$, so that $\nu(\frac{1}{2}+\varepsilon, T) = \frac{1}{2}\big[N(T) - M_\varepsilon(T) - \tau_\varepsilon(T)\big]$ for every ε satisfying the hypotheses.

Proof. Let $\sigma_1 > 1$. Write Δ_{right} for the change of $\arg\xi$ along the segment from σ_1 to $\sigma_1 + iT$, and $\Delta_{\text{top}}(a \to b)$ for its change along the segment from $a+iT$ to $b+iT$. The argument principle on the rectangle $[\frac{1}{2}+\varepsilon, \sigma_1]\times[0,T]$, on whose bottom edge ξ is positive, gives $\Delta_{\text{right}} + \Delta_{\text{top}}(\sigma_1 \to \frac{1}{2}+\varepsilon) - \pi M_\varepsilon(T) = 2\pi\nu(\frac{1}{2}+\varepsilon, T)$. Subtracting this from the classical formula $\pi N(T) = \Delta_{\text{right}} + \Delta_{\text{top}}(\sigma_1 \to \frac{1}{2})$ [255, Section 9.3] leaves $\Delta_{\text{top}}(\frac{1}{2}+\varepsilon \to \frac{1}{2}) + \pi M_\varepsilon(T) = \pi N(T) - 2\pi\nu(\frac{1}{2}+\varepsilon, T)$, and $\Delta_{\text{top}}(\frac{1}{2}+\varepsilon \to \frac{1}{2}) = \pi\tau_\varepsilon(T)$. At $\varepsilon = \frac{1}{2}$ the segment lies on $\operatorname{Re} s = 1$, where ζ has no zeros, and $\nu(1,T) = 0$ because the Euler product has no zeros in $\operatorname{Re} s > 1$. □

Half the difference of $M_\varepsilon(T) + \tau_\varepsilon(T)$ at two resolutions counts the zeros in the strip between the two lines. The factor two comes from the functional equation: a zero $\frac{1}{2}+\delta+i\gamma$ with $\delta > \varepsilon$ and its partner $\frac{1}{2}-\delta+i\gamma$ add 2 to $N(T)$ and 1 to $\nu(\frac{1}{2}+\varepsilon, T)$, hence nothing to $M_\varepsilon(T) + \tau_\varepsilon(T)$, whereas two zeros on the line at the same ordinate add 2. In the density the pair enters through the kernels $P_{\varepsilon+\delta}$ and $-P_{\delta-\varepsilon}$ of Proposition 1.2.3, whose masses cancel, as in the proof of Proposition 3.3.1(iii) (Exercise 3.12).

For large T the number of zeros to the right of a line is bounded rather than computed. Zero density estimates such as Ingham's [104] bound the number $N(\sigma, T)$ of zeros with real part at least σ and ordinate in $(0,T]$, which is at least $\nu(\sigma, T)$, by $T^{e(\sigma)+o(1)}$, with an exponent $e(\sigma) < 1$ for every $\sigma > \frac{1}{2}$, so that only a vanishing proportion of the zeros can lie to the right of any line $\operatorname{Re} s = \sigma$, and Section 11.4 turns these bounds, with no hypothesis, into a bound on the mass of the Cauchy kernels of the zeros to the right of the basepoint σ, the negative part of the split (11.4.1) (Theorem 11.4.2). Proposition 3.4.2 evaluates ν itself, as an integer, but only up to a finite height.

3.4.3 The count and the zero density below height 200

The right side of the identity is an integer, so a computation of the left side with error below one half determines it, provided no winding of the argument is missed. We follow the argument on a grid, summing the principal values of the arguments of quotients of successive values of ξ, with steps $\varepsilon/8$ ($\varepsilon/4$ at $\varepsilon = 0.005$) up the vertical side from $\theta = 0$ and 4000 steps along the horizontal segment. A certificate would need, besides ball arithmetic at the grid points, a bound on the change of

the argument between them, for instance from a lower bound for $|\xi|$ and an upper bound for $|\xi'|$ on each step. We do not supply that bound, and the conclusions below are computed, not certified.

Table 3.3 gives the identity at $T = 200$, evaluated by the Pólya route. Every row returns 79 to within 6×10^{-13}. The row $\varepsilon = \frac{1}{2}$ gives $N(200) = 79$, and with the 79 sign changes of Ξ on $(0, 200)$ (Section 2.1.2) this places every zero below height 200 on the line. Independently of the sign changes, the rows $\varepsilon < \frac{1}{2}$ give $\nu(\frac{1}{2} + \varepsilon, 200) = 0$ for every $\varepsilon \geq 0.005$, since ν is nonincreasing in σ, so that by the functional equation no zero below height 200 lies farther than 0.005 from the line. The right panel of Figure 3.6 plots the columns of the table against ε.

ε	$M_\varepsilon(200)$	$\tau_\varepsilon(200)$	$M_\varepsilon + \tau_\varepsilon = N - 2\nu$
1	79.36737810665	−0.36737810665	79.00000000000003
0.5	79.19416875933	−0.19416875933	78.99999999999993
0.25	79.09886991193	−0.09886991193	79.00000000000024
0.1	79.03976768776	−0.03976768776	79.00000000000006
0.05	79.01989993142	−0.01989993142	78.99999999999967
0.02	79.00796178162	−0.00796178162	78.99999999999970
0.01	79.00398102016	−0.00398102016	78.99999999999993
0.005	79.00199052625	−0.00199052625	78.99999999999947

Table 3.3: Proposition 3.4.2 at $T = 200$, computed by the Pólya route with the argument followed on a grid (computed, not certified). The cumulative Thorin mass plus the short horizontal correction recovers the integer $N(200) = 79$ at every resolution, so the zero density $\nu(\frac{1}{2} + \varepsilon, 200)$ vanishes for all eight values of ε.

The identity also gives the mass defect of Proposition 3.2.2 exactly. If T is not an ordinate and every zero with ordinate in $(0, T]$ lies on the line, then $\nu(\frac{1}{2} + \varepsilon, T) = 0$ and $M_\varepsilon(T) = N(T) - \tau_\varepsilon(T)$ for every $\varepsilon > 0$. Here $\pi\tau_\varepsilon(T) = \arg \xi(\frac{1}{2}+iT) - \arg \xi(\frac{1}{2}+\varepsilon+iT)$ is the function $\arg h(0) - \arg h(\varepsilon)$ of the proof of that proposition, with $\Theta = T$, real analytic and odd in ε (Remark 3.2.3), so $\tau_\varepsilon(T) = (\varepsilon/\pi)\,\Xi'(T)/\Xi(T) + O(\varepsilon^3)$. At $T = 200$ the first order term is $-0.3981063\,\varepsilon$, minus the mass defect found in Section 3.2.2, and for $\varepsilon \leq 0.1$ the column $\tau_\varepsilon(200)$ of Table 3.3 exceeds it by about $0.043\,\varepsilon^3$.

The count at $\varepsilon = \frac{1}{2}$ is Backlund's, with the vertical side at $\operatorname{Re} s = 1$ and the argument computed rather than bounded (Section 2.2.1). The basepoint $\frac{1}{2} + \varepsilon$ is then one, the last at which positivity of the Thorin density is a theorem, and Section 9.1 studies the Thorin measure there, a sum of Cauchy kernels of widths $1 - \beta$ at the ordinates whose mass up to height T is $N(T) - \tau_{1/2}(T)$. What the Thorin mass adds is the zero density $\nu(\frac{1}{2} + \varepsilon, T)$ at every resolution from the same evaluator and, under RH, a Poisson average of S in place of Turing's box average. It is not a faster count. With the tilted contour of Section 2.1.3 the count at

$T = 1000$ took 77.5 seconds, against 0.017 seconds for Riemann Siegel evaluation with Turing's method (Section 2.2.3.2), and at large height the evaluator would be the hybrid of Section 2.2.2, with the sign changes still placing the zeros on the line.

3.5 Further readings of the sine squared lemma

The count of Section 3.4 uses ξ as a whole, and its arithmetic part is the term $S_\varepsilon = \pi^{-1} \arg \zeta$ of the cumulative mass. Section 3.1 used the sine squared lemma (1.2.9) of Section 1.2 at basepoints $\alpha > 1$, where it is a convergent identity, to isolate the pole as the only negative input, and this section follows the lemma in two directions. On the prime side, integrated over the ordinate it is the term S_ε; near the critical line its prime series diverges, and Selberg's smoothed explicit formula replaces the series and turns the prime cutoff into a resolution, at a cost exponential in $1/\varepsilon$. On the side of positivity, the lemma excludes a Brownian first passage time to a random level and puts Weil's criterion in the form of a domination on variograms.

3.5.1 The prime side near the critical line

For $\alpha > 1$ the integrated lemma is the argument of ζ. In the chart $\theta = \sqrt{z}$ a unit of $xL(dx)$ at x contributes $\pi^{-1}(1-\cos\theta x)$ to the density, and $\pi^{-1}\int_0^\Theta (1-\cos\theta x)\,d\theta = \pi^{-1}(\Theta - \sin(\Theta x)/x)$, so the prime atoms $\Lambda(n)n^{-\alpha}$ at $x = \log n$ return a term linear in Θ plus π^{-1} times

$$-\sum_{n\geq 2} \frac{\Lambda(n)}{n^\alpha} \frac{\sin(\Theta \log n)}{\log n} = \arg \zeta(\alpha + i\Theta),$$

on the branch with $\arg\zeta(\alpha) = 0$, since $\log\zeta(s) = \sum_n \Lambda(n)n^{-s}/\log n$ for $\operatorname{Re} s > 1$. Divided by π, this is $S_{\alpha-1/2}(\Theta)$ in the notation of Section 3.4.1: for $\varepsilon > \frac{1}{2}$ the lemma gives $M_\varepsilon = N_\varepsilon^{\mathrm{sm}} + S_\varepsilon$ with S_ε as a Dirichlet series. The series converges absolutely for $\alpha > 1$ and diverges for $\alpha < 1$, so the function S of Turing's method (Section 2.2.1), at $\alpha = \frac{1}{2}$, is reached only through a smoothing.

Selberg [237] truncates the Dirichlet series of ζ'/ζ smoothly and accounts for the truncation exactly by a sum over the zeros [255, Section 14.20]. With $\Lambda_x(n) = \Lambda(n)$ for $n \leq x$ and $\Lambda_x(n) = \Lambda(n)\log(x^2/n)/\log x$ for $x < n \leq x^2$, and for $x > 1$ and s different from 1, from every zero ρ and from every $-2q$,

$$\begin{aligned}\frac{\zeta'}{\zeta}(s) = &-\sum_{n\leq x^2} \frac{\Lambda_x(n)}{n^s} + \frac{x^{2(1-s)} - x^{1-s}}{(1-s)^2 \log x} \\ &+ \frac{1}{\log x}\sum_{q\geq 1} \frac{x^{-2q-s} - x^{-2(2q+s)}}{(2q+s)^2} + \frac{1}{\log x}\sum_\rho \frac{x^{\rho-s} - x^{2(\rho-s)}}{(s-\rho)^2}.\end{aligned} \tag{3.5.1}$$

At $s = \alpha + i\theta$ with $\alpha = \frac{1}{2} + \varepsilon$, a zero $\rho = \beta + i\gamma$ with $\beta < \alpha$ contributes to the last sum a term of modulus about $x^{\beta-\alpha}/(|s-\rho|^2 \log x)$. For a line zero at its own ordinate the term is $(x^{-\varepsilon} - x^{-2\varepsilon})/(\varepsilon^2 \log x)$, a fraction $x^{-\varepsilon}(1 - x^{-\varepsilon})/(\varepsilon \log x)$ of the contribution $1/\varepsilon$ of that zero to v_α; we call this the single zero law. The same computation bounds the whole sum. If no zero has real part greater than $\frac{1}{2}$, then $|x^{\rho-s}| = x^{-\varepsilon}$ and $|x^{2(\rho-s)}| = x^{-2\varepsilon}$ for every zero, and every zero contributes $\operatorname{Re}(1/(s-\rho)) = \varepsilon/|s-\rho|^2$ to $v_\alpha(\theta)$, so that $\sum_\rho |s-\rho|^{-2} = v_\alpha(\theta)/\varepsilon$ by Proposition 1.2.3. Hence the bracket v_α computed from Selberg's smoothed prime sum has error at most

$$\frac{x^{-\varepsilon} + x^{-2\varepsilon}}{\varepsilon \log x}\, v_\alpha(\theta),$$

which is of order $x^{-\varepsilon}/(\varepsilon^2 \log x)$ at a peak, where v_α is about $1/\varepsilon$, and k correct digits at an ordinate need $x^\varepsilon\, \varepsilon \log x \approx 10^k$, that is

$$x \approx 10^{k/\varepsilon}$$

to leading order, with primes up to x^2. A zero with $\frac{1}{2} < \beta < \alpha$ gives a term that decays like $x^{\beta-\alpha}$, more slowly than $x^{-\varepsilon}$, and one with $\beta > \alpha$ a term that grows like $x^{2(\beta-\alpha)}$, since then $x^{2(\rho-s)}$ dominates. With $B = \sup_\rho \operatorname{Re}\rho$ the error is $O(x^{B-\alpha}/\log x)$ when $B < \alpha$, and when $B > \alpha$ it reaches $x^{2(B-\alpha)+o(1)}$ along a sequence of cutoffs, so that the prime side diverges as $x \to \infty$ if a zero lies to the right of the line $\operatorname{Re} s = \alpha$. Selberg's abscissa $\sigma_{x,t} = \frac{1}{2} + 2\max(\beta - \frac{1}{2}, 2/\log x)$, the maximum over the zeros near height t, takes this into account; when none of these zeros lies to the right of $\frac{1}{2} + 2/\log x$ it equals $\frac{1}{2} + 4/\log x$, where $x^{-\varepsilon} = e^{-4}$ [237]. Section 9.1 meets the same dichotomy without smoothing, by compensating the prime atoms with the pole density: below the basepoint one the compensated lemma converges exactly as far as the zero free half plane reaches.

The terms of the zeros to the right of the line $\operatorname{Re} s = \alpha$ cannot cancel one another. Suppose that $B > \alpha$, write $E_x(s)$ for the last term of (3.5.1), the sum over the zeros divided by $\log x$, and put $y = \log x$, $\lambda_\rho = \rho - s$ and $R(y) = y \operatorname{Re} E_{e^y}(s)$. Bounding each term by its modulus gives $|R(y)| \le \sum_\rho |s-\rho|^{-2}\big(e^{(B-\alpha)y} + e^{2(B-\alpha)y}\big)$, so for $\operatorname{Re} w > 2(B-\alpha)$ the Laplace transform $\int_0^\infty e^{-wy} R(y)\, dy$ may be taken term by term, and it is

$$\frac{1}{2} \sum_\rho \Big[\frac{1}{\lambda_\rho^2} \Big(\frac{1}{w - \lambda_\rho} - \frac{1}{w - 2\lambda_\rho} \Big) + \frac{1}{\bar\lambda_\rho^2} \Big(\frac{1}{w - \bar\lambda_\rho} - \frac{1}{w - 2\bar\lambda_\rho} \Big) \Big].$$

The series converges locally uniformly away from its poles, since $\sum_\rho |\lambda_\rho|^{-2} < \infty$ and $|\lambda_\rho| \to \infty$, and so defines a meromorphic function in the whole plane. A zero ρ_0 with $\operatorname{Re}\rho_0 > (B+\alpha)/2$ gives a pole at $2\lambda_{\rho_0}$, to the right of every pole λ_ρ and $\bar\lambda_\rho$; each fraction with a pole there contributes $-\frac{1}{2}\lambda_{\rho_0}^{-2}$ to the residue, which is therefore not zero. If $R(y) = O(e^{\eta y})$, the Laplace integral converges on $\operatorname{Re} w > \eta$ and continues the series holomorphically there, so $\eta \ge 2(\operatorname{Re}\rho_0 - \alpha)$. Letting $\operatorname{Re}\rho_0$

tend to B, and using the bound on $|R|$, gives $\limsup_{x\to\infty} \log|\operatorname{Re} E_x(s)|/\log x = 2(B-\alpha)$.

The cutoff interpolates between the zeros and the primes. For $x < \sqrt{2}$ the prime sum is empty, and as $x \downarrow 1$ each remaining term tends to its partial fraction, $1/(1-s)$, $1/(s+2q)$ or $1/(s-\rho)$. The limits cannot be taken inside the sums separately: the sum over q grows like $\frac{1}{2}\log(1/\log x)$ and the real part of the sum over the zeros falls like $-\frac{1}{2}\log(1/\log x)$, and only their total, $\zeta'/\zeta(s) - (x^{2(1-s)} - x^{1-s})/((1-s)^2\log x)$, has a limit, $\zeta'/\zeta(s) + 1/(s-1)$. Adding the logarithmic derivative of $s\pi^{-s/2}\Gamma(s/2)$ turns this limit into $\xi'/\xi(s)$, whose real part divided by π is the density $\rho_\varepsilon(\theta)$, the sum of the kernels of Proposition 1.2.3; for large x, when no zero lies to the right of the line $\operatorname{Re} s = \alpha$, the density is carried by the smoothed primes, the Lévy side of the lemma.

Table 3.4 and Figure 3.7 test the formula at $\varepsilon = 0.25$, that is $\alpha = \frac{3}{4}$, on 501 points of $10 \le \theta \le 60$, against the Pólya values of $v_{3/4}$. With the primes alone the maximal relative error is 66%, 20% and 7% for $x = 30$, 300 and 3000, attained where $v_{3/4}$ is small, and the maximal absolute errors 1.14, 0.53 and 0.23, attained at peaks over zeros, follow the single zero law, which predicts 1.15, 0.51 and 0.23; the relative errors stay below the bound $(x^{-\varepsilon} + x^{-2\varepsilon})/(\varepsilon \log x)$, which is 0.72, 0.21 and 0.077. Both errors decrease faster than $x^{-\varepsilon}$. Adding the sum over the first 1700 zeros reduces the error below 1.6×10^{-7}, the residual being the zeros above height 2197. At $\varepsilon = 0.005$ ten digits would need $x \approx 10^{2000}$, about e^{4600}, to leading order (about e^{4000} by the single zero law), while the cost of one Pólya evaluation does not depend on ε (Section 3.1.2). So the sine squared lemma identifies the Thorin density, but near the line the primes compute it only at a cost exponential in $1/\varepsilon$.

	primes only			with 1700 zeros
cutoff x	max |error|	max relative error	$x^{-\varepsilon}$	max |error|
30	1.14	0.66	0.43	1.3×10^{-7}
300	0.53	0.20	0.24	1.5×10^{-7}
3000	0.23	0.072	0.14	7.1×10^{-8}

Table 3.4: Selberg's smoothed prime side of the Thorin density bracket $v_{3/4}(\theta)$, at resolution $\varepsilon = 0.25$, against the Pólya values on 501 equally spaced points of $10 \le \theta \le 60$, with the primes alone and with the sum over the first 1700 zeros added. The relative error is $|\text{error}|/v_{3/4}(\theta)$.

3.5.2 Positivity: Brownian subordination and Weil's criterion

The lemma cannot produce a Brownian first passage time to a random level from a positive input. Brownian motion with variance two per unit time first reaches a level $x > 0$ at a time with Laplace transform $e^{-x\sqrt{u}}$, so its first passage

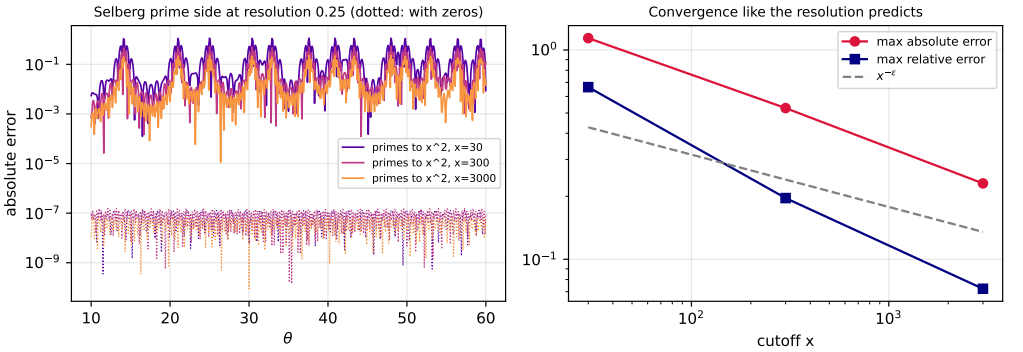


Figure 3.7: Left: absolute error of Selberg's smoothed prime side of $v_{3/4}$, at resolution 0.25, for three cutoffs, and (dotted) with the sum over the first 1700 zeros added. Right: the maximal absolute and relative errors against the cutoff, with $x^{-\varepsilon}$ for comparison. Both decay faster than $x^{-\varepsilon}$; the absolute error follows the single zero law $(x^{-\varepsilon} - x^{-2\varepsilon})/(\varepsilon^2 \log x)$.

time T_Y to an independent level $Y \geq 0$ has $\mathbb{E}e^{-uT_Y} = \mathbb{E}e^{-Y\sqrt{u}}$. The exponent $\log f(\alpha+s)/f(\alpha) = cs + \int (e^{-sx} + sx - 1)L(dx)$ to which the lemma applies (Section 1.2) would have to equal $-\log \mathbb{E}e^{-sY}$, which is concave in s by Hölder's inequality, whereas $L \geq 0$, $L \neq 0$ makes it strictly convex, with second derivative $\int x^2 e^{-sx} L(dx) > 0$. With $c \geq 0$ the lemma gives a GGC with Thorin density $(2\pi\sqrt{z})^{-1}[c + \int 2\sin^2(x\sqrt{z}/2)\, xL(dx)]$, but not a first passage time. Conversely, an infinitely divisible level with $\mathbb{E}e^{-sY} = \exp(-bs - \int (1 - e^{-sx})\Pi(dx))$ enters the lemma with $c = b + \int x\,\Pi(dx)$ and the negative input $L = -\Pi$, and T_Y receives the signed density $(2\pi\sqrt{z})^{-1}[b + \int x\cos(x\sqrt{z})\,\Pi(dx)]$. For a Poisson level ($\Pi = \delta_1$, $b = 0$) this is $\cos(\sqrt{z})/(2\pi\sqrt{z})$, and T_Y is not a GGC: it has an atom e^{-1} at 0, which no GGC other than the point mass at 0 has. A positive input and a random level coexist only for pure drift, with transform $e^{-c\sqrt{u}}$ and Thorin density $c/(2\pi\sqrt{z})$.

For ξ there is no random level at any basepoint. By Stirling's formula and the polynomial growth of ζ in vertical strips, $|\xi(\alpha)/\xi(\alpha+it)|$ is at least $e^{\pi|t|/4}$ divided by a power of $|t|$, while $|\mathbb{E}e^{-itY}| \leq 1$; so $s \mapsto \xi(\alpha)/\xi(\alpha+s)$ is never a Laplace transform in s, although for $\alpha \geq 1$ it is one in $u = s^2$ (Theorem 1.2.2). For $\alpha > 1$ the input of ξ is signed, and the positivity of U_α is created by the zeros through the Poisson kernels of Proposition 1.2.3, not inherited from a positive input (Section 3.1.3); compare [200, Remark 3.8]. At $\alpha = \frac{1}{2}$ the obstruction is the drift zero case of Proposition 8.2.1, whose Brownian motion has variance one, and the curved boundaries of Section 8.2 take the place of the straight boundary with a random intercept: the Roberts Shortland tangent approximation reproduces the law of the clock to 0.0059 in Kolmogorov distance, and the Volterra equation of

Section 8.2.2.2 gives a boundary that reproduces it to Monte Carlo accuracy. The obstruction concerns first passage alone: Section 9.2 shows that at every basepoint $\alpha > \frac{1}{2}$, with no hypothesis, $\xi(\alpha)/\xi(\alpha+\sqrt{u})$ is the Laplace transform of a Brownian first passage time to a random level plus a time proportional to that level.

The same kernel puts Weil's criterion in variogram form. Section 5.3 defines the Thorin functional U on the class $\mathcal{T}$ of functions $\varphi(z) = h(\sqrt{z})$ with h even, holomorphic on a strip $|\operatorname{Im} r| \le \frac{1}{2} + c$ for some $c > 0$, and $O((1+|r|)^{-2-c})$ there. With $g(x) = \frac{1}{2\pi}\int h(r)e^{-irx}\,dr$,

$$\int_0^\infty \varphi(z)\,\frac{2\sin^2(x\sqrt{z}/2)}{2\pi\sqrt{z}}\,dz = g(0) - g(x), \tag{3.5.2}$$

both sides being $\pi^{-1}\int_0^\infty h(r)(1-\cos xr)\,dr$. If $h \ge 0$ on the real line, g is positive definite and $g(0) - g(x) \ge 0$ is a variogram. Weil's criterion [266] states that RH holds if and only if $\langle U, \varphi\rangle \ge 0$ for every such φ, and by the explicit formula U evaluates φ at every zero. Formally, the lemma writes $\langle U, \varphi\rangle$ as the pairing of variograms with the canonical measure of (3.1.1) taken at the centre, the measure ν of Section 5.3, whose archimedean density and prime atoms are nonnegative and whose only negative input is the pole density $-e^{x/2}dx$. So RH says that on every variogram of a positive definite function the pole mass is dominated by the prime and archimedean mass. Both masses are infinite, and the absolutely convergent form of Section 5.3 makes the statement exact, the divergent constants cancelling through $\xi'(\frac{1}{2}) = 0$ (Section 3.1.3).

The linear programming formulation of [200, Sections 18.2.8 and 18.2.9] works on this form, with the variograms as feasible directions. In a space of test functions of finite dimension, $\langle U, \varphi\rangle$ is linear in the coefficients of h and $h \ge 0$ is a family of linear constraints on them, so the regularised domination is a feasibility problem with dual certificates, which Section 12.2.4 proposes to solve at increasing dimension. Weil's criterion concerns U, which sees every zero, whereas the limit $\alpha \downarrow \frac{1}{2}$ of the Thorin measures is U_{line}, which pairs nonnegatively with every such φ whatever the zeros do (Section 5.3). What that limit can give is the question of Section 4.1.

3.6 Exercises

On Section 3.2.

3.1. Prove Proposition 3.2.1 for a positive measure ν on a general measurable space that does not charge the atoms of U.

3.2. Show that the total variation distance between the unit point masses at a and b is 2 for $a \ne b$, and deduce that total variation cannot compare two atomic measures with slightly displaced atoms.

3.3. Compute the L^1 distance between the cumulative functions of a Cauchy kernel of width ε and a unit step on $[-L, L]$, derive the $\varepsilon \log(1/\varepsilon)$ rate, and compare with the formula for a single step in Section 3.4.1.

3.4. Derive the first order formula of Proposition 3.2.2 for the sinh clock of Section 1.3, where it reads $-(\varepsilon/\pi) f'/f$ with $f(\theta) = \sin(\pi\theta)/(\pi\theta)$, and compare it at $\Theta = m + \frac{1}{2}$ with the exact mass $m + \pi^{-1} \arctan(\varepsilon/\Theta)$ of Section 1.3.4; explain why the difference is of order ε^3.

3.5. Explain why the two point Richardson combination removes the linear term in ε and the three point combination the quadratic term, and reproduce the heat trace errors 2.6×10^{-7} and 1.7×10^{-10} from the values of $W_\varepsilon(0.01)$ given in Section 3.2.2.

On Section 3.3.

3.6. Verify the expansion $\frac{1}{2}(P_{\varepsilon+\delta} + P_{\varepsilon-\delta}) - P_\varepsilon = \frac{\delta^2}{2} \partial_a^2 P_a|_{a=\varepsilon} + O(\delta^4)$, show that the remainder has L^1 norm $O(\delta^4/\varepsilon^4)$, and compute the L^1 norm of $\partial_a^2 P_a$ at $a = \varepsilon$.

3.7. Show that for $\delta > \varepsilon$ the kernel $P_{\varepsilon+\delta} - P_{\delta-\varepsilon}$ is negative exactly on $|x| < \sqrt{\delta^2 - \varepsilon^2}$.

3.8. Compute the depth $2\varepsilon/(\pi(\delta^2 - \varepsilon^2))$ of the negative dip of an off line pair at its centre, that is $\varepsilon/(\pi(\delta^2 - \varepsilon^2))$ per displaced zero, compare it with the contribution $\varepsilon/(\pi(\varepsilon^2 + \delta_j^2)) \approx \varepsilon/(\pi\delta_j^2)$ of a line zero at distance δ_j, and derive the local visibility condition of Section 5.2.

3.9. Given the matched floor $\tilde{d}_{\mathrm{match}}(\varepsilon)$, derive the smallest displacement δ of a pair that keeps the count and would be detected, and evaluate it at $\varepsilon = 0.05$ using Table 3.2.

On Section 3.4.

3.10. Show that $I(\varepsilon) = \int_0^T \log|\xi(\frac{1}{2} + \varepsilon + it)|\, dt$ is continuous for $\varepsilon \geq 0$, that its derivative jumps by 2π when the line $\mathrm{Re}\, s = \frac{1}{2} + \varepsilon$ crosses a zero, and deduce the second identity of Proposition 3.4.1.

3.11. Prove Proposition 3.4.2 in detail, paying attention to the orientation of each side of the rectangle.

3.12. Show that the kernels $P_{\varepsilon+\delta}$ and $-P_{\delta-\varepsilon}$ of an off line pair at distance $\delta > \varepsilon$ from the line have total mass zero, and compute to leading order the mass that they and their mirror images at $-\gamma$ put on a window $(0, T]$ containing the ordinate γ. Explain how the factor two in Proposition 3.4.2 accounts for the pair.

3.13. Using the first order formula for $\tau_\varepsilon(T)$ in Section 3.4.3, predict $\tau_{0.05}(200)$ from $\Xi'(200)/\Xi(200) = -1.2506879$ and compare with Table 3.3; estimate the coefficient of the ε^3 term from the rows $\varepsilon = 0.01$ and 0.02.

3.14. Show that Littlewood's lemma follows from Proposition 3.4.1 and Proposition 3.4.2 by integrating in ε.

On Section 3.5.

3.15. Show that $\frac{1}{\pi}\int_0^\Theta (1-\cos\theta x)\,d\theta = \frac{1}{\pi}(\Theta - \sin(\Theta x)/x)$ and, for $\alpha > 1$, derive the Dirichlet series for $\arg\zeta(\alpha + i\Theta)$ from the integrated sine squared lemma.

3.16. Implement Selberg's formula at $s = \frac{3}{4} + i\theta$ with $x = 100$ and check that adding the zero sum over the first hundred zeros reduces the error to the size of the omitted zeros.

3.17. Estimate, from the single zero law, the cutoff x needed for six correct digits of $v_{0.55}(\theta)$ at an ordinate $\theta = \gamma$ from the primes, and compare with the cost of the Pólya route at the same point.

3.18. Show that $g(0) - g(x) \geq 0$ for every real positive definite function g, and that $g(0) - g(x)$ is conditionally negative definite.

Chapter 4

Weak limits, hyperbolic complete monotonicity and a Bosch type theorem

Chapter 3 computes the Thorin measure μ_ε at positive resolutions and measures its distance from U_{RH}, but RH is a statement at $\varepsilon = 0$. Two classical results seem to reach that limit without locating zeros: the class of GGC laws is closed under weak limits, and by Bondesson's theorem (Section 1.3.2) the Laplace transforms of GGC laws are exactly the hyperbolically completely monotone (HCM) functions with limit one at the origin. Section 4.1 settles what each of them can give.

Sections 4.2 and 4.3 then bring in Bosch's theorem on powers of the half Cauchy law. The first explains why his result cannot be transferred as an input, and the second transfers his method instead and proves a threshold theorem for powers of the reciprocal Xi function, in which the angle of the zeros replaces Bosch's drift parameter and RH is the extreme case.

4.1 The limit of vanishing resolution

Neither closure under weak limits nor Bondesson's theorem reaches RH. The Cauchy flow converges vaguely to the measure of the line zeros whether or not RH holds (Section 4.1.1), and HCM of the clock is RH in another form (Section 4.1.2). What survives is a positivity margin at positive resolution and a finite HCM test that the computed clock passes.

4.1.1 Weak limits of Thorin measures

Thorin's closure theorem is the standard way to obtain a GGC as a limit.

Theorem 4.1.1 (Thorin [250, 251]; Bondesson [35, Theorem 3.1.5]; see also [200, Theorem 18.9]). *The class of GGC laws is closed under convergence in law.*

With the rigidity of Proposition 1.2.1, the theorem gives a criterion for RH that mentions no zero.

COROLLARY 4.1.2. *RH holds if and only if there are GGC variables X_N with $\mathbb{E}e^{-uX_N} \to 1/G(u)$ for all u in a set with an accumulation point in $(0,\infty)$ [200, Propositions 18.10 and 18.14].*

Proof. The transforms are analytic and bounded by one on $\operatorname{Re} u > 0$, so by Vitali's theorem they converge locally uniformly there, and the limit is $1/G$ on $(0,\infty)$. Since $1/G(u) \to 1$ as $u \downarrow 0$, the X_N converge in law to a proper limit with transform $1/G$, which is a GGC by the closure theorem, and Proposition 1.2.1 gives RH. Conversely, under RH the clock variable T is a GGC with transform $1/G$; take $X_N = T$. □

The Cauchy flow $\varepsilon \mapsto \rho_\varepsilon$ of Section 1.2 is the candidate at hand. Its measures μ_ε are the Thorin measures of $\xi(\alpha)/\xi(\alpha+\sqrt{u})$ at $\alpha = \frac{1}{2}+\varepsilon$ whenever these are GGC transforms (Theorem 1.2.2), the transforms converge to $1/G$ pointwise as $\varepsilon \downarrow 0$, and the measures converge vaguely. The flow fails for two reasons.

First, the limit is the wrong measure. As shown after Proposition 1.2.3, $\mu_\varepsilon \to U_{\text{line}}$ vaguely whatever the truth of RH [198, Proposition 9.5], because the contribution of an off line pair tends to zero in total variation (Proposition 3.3.1(ii)); the limit omits the off line zeros. It is the Thorin measure of the clock only under RH, since the Thorin functional U of Section 5.3 counts every zero and $U = U_{\text{line}}$ exactly when RH holds (Section 5.3.1 and [200, Proposition 7.6]). Positivity of the vague limit therefore carries no information. In Figure 4.1 the fifth zero is displaced to $\frac{1}{2} \pm \delta + i\gamma_5$ with $\delta = 0.1$, its count kept as in Proposition 3.3.1(i), so that it contributes $\frac{1}{2}(P_{\varepsilon+\delta} + P_{\varepsilon-\delta})(\theta - \gamma_5)$ to ρ_ε. For $\varepsilon < \delta$ the negative part of this kernel has mass $\varepsilon/(\pi\delta) + O(\varepsilon^3/\delta^3)$ (Proposition 3.3.1(iii)) and its value at γ_5 is $-\varepsilon/(\pi(\delta^2 - \varepsilon^2))$. Both vanish as $\varepsilon \downarrow 0$, and an off line pair with the full weight of (1.2.8) doubles both.

Second, the closure theorem needs GGC laws along the sequence. The flow supplies a law at every resolution with no hypothesis, that of the shifted clock T_α of Theorem 9.2.5 at $\alpha = \frac{1}{2}+\varepsilon$, but at small ε it supplies a GGC law only under the hypothesis to be proved: the transform at basepoint $\frac{1}{2}+\varepsilon$ is a GGC transform if and only if no zero lies to the right of the line $\operatorname{Re} s = \frac{1}{2}+\varepsilon$ (Theorem 1.2.2). Positivity of ρ_ε at that one ε is necessary for this but is not known to imply it, since near a cluster of line zeros the negative kernel of an off line pair can be outweighed by the kernels of its neighbours (Exercise 1.7 and Section 5.2.2). Positivity of $\rho_{\varepsilon'}$ for every $\varepsilon' > \varepsilon$ does imply it, as shown in Remark 1.2.4; for $\varepsilon = 0$ this is Lagarias's criterion [133], which is RH. Weak closure adds the endpoint $\varepsilon = 0$ to a family already known to consist of GGCs; it cannot show that the family consists of GGCs [200, Remark 3.10]. Section 11.4 measures how close the flow comes without the hypothesis: at every basepoint above the centre, the negative part of the Thorin measure below a given height is a vanishing fraction of the positive part as the height grows.

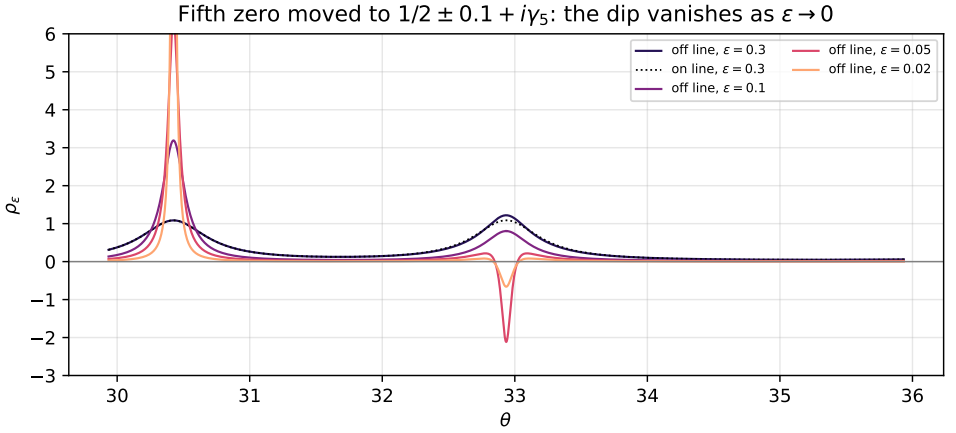


Figure 4.1: Local Thorin density near γ_5 when the fifth zero is displaced to $\frac{1}{2} \pm 0.1 + i\gamma_5$ with its count kept, as in Proposition 3.3.1(i), for $\varepsilon = 0.3, 0.1, 0.05, 0.02$; the dotted curve is the density with the fifth zero on the line, at $\varepsilon = 0.3$. At $\varepsilon = \delta = 0.1$ the kernel contains $\frac{1}{2}P_0$, a point mass of weight $\frac{1}{2}$ at γ_5, which the plotted curve omits. The negative part shrinks like $\varepsilon/(\pi\delta)$ and the vague limit is positive.

The computation adds three quantitative facts. On the window $0 \le \theta \le 200$ the Pólya route computes $\min \rho_\varepsilon = 0.0147\,\varepsilon > 0$ at each of the eight resolutions of Table 3.2, down to $\varepsilon = 0.005$ (Section 3.1.1); with the verification of RH to height 3×10^{12} this is a theorem, since for every $\varepsilon \le 6.85$ the minimum of ρ_ε over $[0, 3\times10^{12} - 1.21]$ is $\rho_\varepsilon(0)$ (Theorem 9.1.11 and Remark 9.1.12), and Section 12.2.2 lists what would make the Pólya computation itself a certificate. The distances of Section 3.2 measure the rate of the vague convergence (Figure 3.4), and the matched total variation of Table 3.2 detects a displaced zero (Proposition 3.3.1(i)).

Weak convergence becomes useful only with approximants whose positivity is proved independently of ξ, such as Beurling generalised prime systems whose zeta functions provably have only real ordinates, finite Krein strings, and the partial products over verified zeros completed by the Weyl tail; the last two are the Kent approximants of Section 8.1.3, without and with the Weyl tail. For such approximants the distances of Section 3.2 are the right measure of the approximation. Verification also removes factors from the question: if the first N zeros are verified to lie on the line, then $1/G_N$, with $G_N(u) = \prod_{k \le N}(1 + u/\gamma_k^2)$, is a GGC transform with no hypothesis, and RH holds if and only if the tail factor G_N/G is a GGC transform as well [200, Corollary 18.11].

4.1.2 The HCM form of the Thorin condition

Recall from Section 1.3.2 that a positive continuous function φ on $(0,\infty)$ is HCM if, for each $u>0$, $\varphi(uv)\varphi(u/v)$ is completely monotone in $w=v+v^{-1}$ on $[2,\infty)$, and that by Bondesson's theorem [35, Theorem 6.1.1] (see also [109]) a function φ on $(0,\infty)$ with $\varphi(0+)=1$ is the Laplace transform of a GGC if and only if it is HCM. With the results of Section 1.2, this gives the following.

Corollary 4.1.3. *(i) For every $\alpha\ge 1$ the function $\varphi_\alpha(u)=\xi(\alpha)/\xi(\alpha+\sqrt{u})$ is HCM, unconditionally. (ii) The reciprocal Xi function $\varphi_{1/2}(u)=\xi(\frac12)/\xi(\frac12+\sqrt{u})$, which is the clock $1/G$, is HCM if and only if RH holds.*

Proof. (i) is Theorem 1.2.2 with Bondesson's theorem, and (ii) is Proposition 1.2.1 with Bondesson's theorem. □

HCM looks like a condition on real functions of a real variable, and it has been proved for many particular densities by real analysis alone [35, 109, 37]. For the clock such a proof would prove RH, and a partial HCM result is a partial result on the Thorin condition of the kind described in Section 5.1.

By the Hadamard product (1.2.1), with $a_k=-(\rho_k-\frac12)^2$ and $w=v+1/v$,

$$\varphi_{1/2}(uv)\,\varphi_{1/2}(u/v)=\prod_k\frac{a_k^2}{a_k^2+a_k\,u\,w+u^2}=\prod_k\frac{a_k/u}{w+c_k(u)},\qquad c_k(u)=\frac{a_k}{u}+\frac{u}{a_k}. \tag{4.1.1}$$

The numbers $c_k(u)$ are local to this subsection, and are not the derivatives $c_k(t)$ of the heat trace in Section 5.1. Under RH every $c_k(u)\ge 2$ is real, each factor is completely monotone in w, and so is the product; this proves the if half of (ii) directly. An off line pair, with values a and $\bar a$ of a_k, contributes $(|a|^2/u^2)/((w+c)(w+\bar c))$ with $c=c(u)$ complex, and $1/((w+c)(w+\bar c))$ is the Laplace transform of the signed function $e^{-t\operatorname{Re}c}\sin(t\operatorname{Im}c)/\operatorname{Im}c$. At a single u the remaining factors may smooth this to a positive function; failure is guaranteed only across all u, by Bondesson's theorem. At $u=|a|$ both values of c equal $2\cos\arg a$, and the pair contributes $(w+2\cos\arg a)^{-2}$, which is completely monotone on $[2,\infty)$. Complete monotonicity in w at finitely many u is therefore necessary for HCM but not sufficient.

We test this necessary condition. From 60 digit values of ξ at real arguments, computed by (2.1.2) without cancellation, numerical differentiation gives the normalised derivatives $d_k=(-1)^k\partial_w^k H_u(w)/H_u(w)$ of $H_u(w)=\varphi_\alpha(uv)\,\varphi_\alpha(u/v)$ for $k\le 6$, $u\in\{1,25,400\}$ and $w\in\{2.05,3,5,10,20,50\}$, at $\alpha=\frac12$ and at the control $\alpha=1$. All 216 values are positive (Figure 4.2 shows $w=3$ and $w=20$), and the signs are certified: from the Taylor coefficients of H_u in w, computed in ball arithmetic [110] from $\xi(s)=\frac12 s(s-1)\pi^{-s/2}\Gamma(\frac{s}{2})\zeta(s)$, and so independently of the Pólya route, every d_k is enclosed in a ball of relative radius below 10^{-70} that

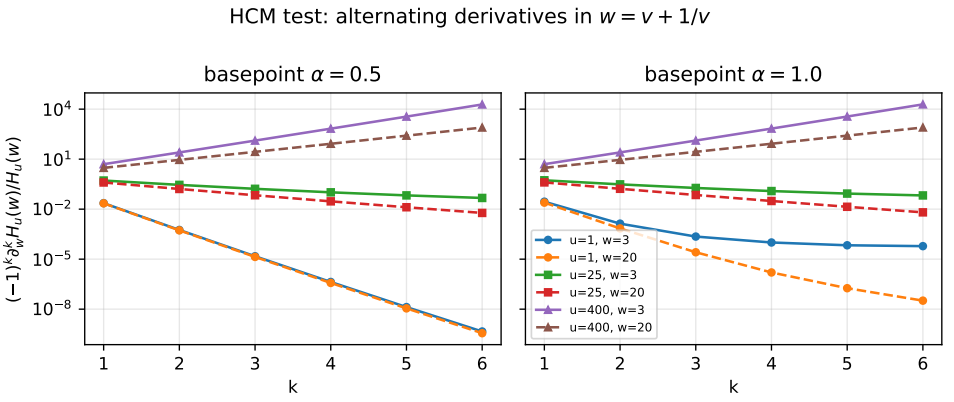


Figure 4.2: Normalised alternating derivatives $(-1)^k \partial_w^k H_u(w)/H_u(w)$ of $H_u(w) = \varphi_\alpha(uv)\varphi_\alpha(u/v)$ for $1 \le k \le 6$, $u \in \{1, 25, 400\}$ and $w \in \{3, 20\}$, computed from the Pólya density, at the centre $\alpha = \frac{1}{2}$ and at the unconditional basepoint $\alpha = 1$. HCM requires positivity at every order, for every $u > 0$ and every $w \ge 2$; the figure shows finitely many cases.

lies in $(0, \infty)$, and these enclosures agree with the values from numerical differentiation to 12 digits. Under RH, (4.1.1) gives $H_u(w+t)/H_u(w) = \mathbb{E}e^{-tY}$ with $Y = \sum_k e_k/(w + c_k(u))$ and independent standard exponentials e_k, so $d_k = \mathbb{E}Y^k$ is of order d_1^k. The smallest value, 2.5×10^{-10} at $\alpha = \frac{1}{2}$, $u = 1$, $w = 50$, $k = 6$, is $2.54\, d_1^6$ with $d_1 = 0.0215$: it reflects the scale of Y, to which many zeros contribute, not closeness to a sign change. The test is consistent with (ii) and cannot decide it (Section 5.2, Table 5.6, and [200, Chapter 11]).

HCM of the law of the clock variable, as opposed to its Laplace transform, is strictly stronger than the GGC property (Section 1.3.2). Under RH the clock variable is $T = \sum_{\gamma>0} e_\gamma/\gamma^2$ (Proposition 1.2.1), and if moreover the zeros are simple and satisfy two further hypotheses, its density is not HCM (Theorem 4.3.23); for the Cauchy analogue of the clock variable this holds unconditionally (Theorem 4.3.21).

4.2 Bosch's half Cauchy theorem and HCM of the reciprocal Xi function

By Corollary 4.1.3(ii) of Section 4.1 the clock $\varphi = 1/G$ is HCM if and only if RH holds; this is the HCM form of the reciprocal Xi programme of [198]. In this section and in Section 4.3 the clock is a function of s, the variable u of Section 1.2: $\varphi(s) = \xi(\frac{1}{2})/\xi(\frac{1}{2} + \sqrt{s})$ is the function $\varphi_{1/2}$ of Corollary 4.1.3. A sharp HCM theorem for a nearby, exactly solvable function might transfer to φ, and the candidate is Bosch's theorem on powers of the half Cauchy law [37].

For a standard Cauchy variable C the Mellin transform $\mathbb{E}|C|^z = \sec(\pi z/2)$ is the reciprocal of $\cos(\pi z/2)$, an entire function whose zeros, the odd integers, are known, just as $\Xi(0)/\Xi(z)$ is the reciprocal of $\Xi(z)/\Xi(0)$. At the half Cauchy point Bosch's theorem says that the density of $|C|$ is not HCM while that of $|C|^2$ is, so its content is sharpness (Section 4.2.1). Set beside the Riemann Xi function (Section 4.2.2), the half Cauchy law is a solved model of Corollary 4.1.3(ii), which Section 4.2.3 restates in the four forms that Bosch's method uses. The model can guide an argument but cannot supply the reality of the zeros (Section 4.2.4): HCM of a product $\prod_n (1 + s/c_n)^{-1}$ records only whether every c_n is positive, and on the Cauchy side the $c_n = (2k-1)^2$ are positive by construction.

4.2.1 HCM and Bosch's theorem at the half Cauchy point

Bosch's theorem concerns densities, so we recall HCM from Section 1.3.2 in the form that applies to them: a continuous function $f : (0, \infty) \to [0, \infty)$ is HCM if for every $u > 0$ the map $v \mapsto f(uv)f(u/v)$, the pair function of f at u, is a completely monotone function of $w = v + v^{-1}$ on $[2, \infty)$, and a positive random variable is HCM if it has an HCM density. In logarithmic coordinates $u = e^a$, $v = e^y$ the condition reads: $y \mapsto f(e^{a+y})f(e^{a-y})$ is completely monotone in $w = 2\cosh y$. Some regularity is needed here: if A is an additive function on $\mathbb{R}$ that is not linear, the pair functions of $x \mapsto e^{A(\log x)}$ are constant, although this function is continuous nowhere and so is not of the form (4.2.1) below. We use the following facts, all due to Bondesson; (F1), (F2), (F5) and (F6) are also stated in Section 1.3.2.

(F1) If X_1, X_2 are independent and HCM then X_1X_2 and X_1/X_2 are HCM [35, Theorem 5.1.1].

(F2) If X is HCM then X^q is HCM for $|q| \geq 1$ [35, pp. 68 and 69]. The mechanism is that for $|p| \leq 1$ the function $v^p + v^{-p}$ is a Bernstein function of $v + v^{-1}$, so if h is an HCM function then so is $x \mapsto h(x^p)$ for $|p| \leq 1$; if X has density f, the density of X^q is $|q|^{-1}x^{1/q-1}f(x^{1/q})$, and the power factor is HCM by (F3).

(F3) Pointwise products of HCM functions are HCM, and powers x^r, $r \in \mathbb{R}$, are HCM [35, p. 68].

(F4) Pointwise limits of HCM functions, when finite and positive, are HCM [35, p. 68 and Theorem 5.1.3].

(F5) Every HCM law is a GGC [35, Theorem 5.1.2].

(F6) A probability law on $[0, \infty)$ is GGC if and only if its Laplace transform is an HCM function [35, Theorem 6.1.1].

(F7) A function f that is not identically zero is HCM if and only if

$$f(x) = Kx^{r-1}h_1(x)h_2(1/x),$$
$$h_j(x) = \exp\Big(-b_j x + \int_{(0,\infty)} \log\frac{1+t}{x+t}\,\Gamma_j(dt)\Big), \tag{4.2.1}$$

with $K > 0$, $r \in \mathbb{R}$, $b_j \geq 0$ and nonnegative measures Γ_j satisfying $\int(1+t)^{-1}\Gamma_j(dt) < \infty$ [35, (5.2.3) and Theorem 5.3.1]. In particular every HCM function that is not identically zero extends to an analytic and zero free function on $\mathbb{C} \setminus (-\infty, 0]$ [35, p. 68].

THEOREM 4.2.1 (Bosch [37, Theorem 1.2]). *Let $T_\alpha = (Z_\alpha/\tilde{Z}_\alpha)^\alpha$ with $Z_\alpha, \tilde{Z}_\alpha$ independent positive α stable (here α is a stability index, not a basepoint, and T_α is not the variable of Theorem 1.2.2). Then T_α has density $\sin(\pi\alpha)/(\pi\alpha(t^2 + 2t\cos\pi\alpha + 1))$, a half Cauchy law with drift, and the density of T_α^β is HCM if and only if $\alpha \leq 1/2$ and $|\beta| \geq 1/(1-\alpha)$.*

At $\alpha = 1/2$ we have $T_{1/2} \stackrel{d}{=} |C|$ with density $2/(\pi(1+t^2))$, and $|C|^2 = Z_{1/2}/\tilde{Z}_{1/2}$ is a ratio of independent Brownian first passage times. The threshold is $|\beta| \geq 2$: $|C|$ is not HCM, while $|C|^2$, the beta prime law with parameters $(1/2, 1/2)$, is. The density of $|C|^\beta$ is

$$f_\beta(x) = \frac{2}{\pi|\beta|}\,\frac{x^{1/\beta - 1}}{1 + x^{2/\beta}}, \qquad x > 0.$$

With $x = e^{a \pm y}$ and $q = 2/|\beta|$ a short computation gives

$$f_\beta(e^{a+y}) f_\beta(e^{a-y}) = \frac{4}{\pi^2\beta^2}\,e^{-2a}\,\frac{1}{2\cosh(qa) + 2\cosh(qy)}. \tag{4.2.2}$$

So at $\alpha = 1/2$ Bosch's theorem is the statement that $1/(c + \cosh(qy))$ is completely monotone in $\cosh y$ for every $c \geq 1$ exactly when $q \leq 1$; here $c = \cosh(qa)$, which ranges over $[1, \infty)$. Sufficiency and the failure at $\beta = 1$ have short proofs.

PROPOSITION 4.2.2. *The half Cauchy density is not HCM.*

Proof. Take $\beta = 1$, so $q = 2$ in (4.2.2). Since $\cosh(2a) + \cosh(2y) = 2\sinh^2 a + w^2/2$ with $w = 2\cosh y$, the pair function is proportional to $h(w) = 1/(c^2 + w^2)$ with $c = 2|\sinh a| > 0$ for $a \neq 0$. For $x > 0$,

$$h(2 + x) = \int_0^\infty e^{-xt}\,e^{-2t}\,\frac{\sin(ct)}{c}\,dt.$$

If h were completely monotone on $(2, \infty)$, Bernstein's theorem would give $h(2 + x) = \int e^{-xt}\mu(dt)$ with $\mu \geq 0$, and uniqueness of the Laplace transform would force $\mu(dt) = e^{-2t}c^{-1}\sin(ct)\,dt$, which is not nonnegative. □

Proposition 4.2.3. *The density of $|C|^\beta$ is HCM for $|\beta| \geq 2$.*

Proof. For $q = 2/|\beta| \leq 1$ the function $\cosh(qy)$ is a Bernstein function of $\cosh y$ by (F2), and $x \mapsto 1/(c+x)$ is completely monotone for $c > 0$, so the composition in (4.2.2) is completely monotone. Alternatively, $Z_{1/2}$ has a density proportional to $x^{-3/2}e^{-b/x}$ for some $b > 0$, an inverse gamma density of the form (F7), so the ratio $|C|^2 = Z_{1/2}/\tilde{Z}_{1/2}$ is HCM by (F1), and (F2) gives all powers $|q| \geq 1$ of it. □

At $\alpha = 1/2$ sufficiency is therefore elementary; the deep part of Bosch's paper is $\alpha < 1/2$. The failure at $\beta = 1$ can be followed order by order. With $u = e^a$, which is not the variable u of the clock, the pair function of Proposition 4.2.2 is proportional to $h(w) = 1/((1-u^2)^2 + u^2w^2) = u^{-2}c^{-1}\operatorname{Im}(w - ic)^{-1}$, where $c = |u - 1/u| = 2|\sinh a|$. Differentiating n times and writing $w - ic = (w^2 + c^2)^{1/2}e^{-i\arctan(c/w)}$ gives, for $w > 0$,

$$(-1)^n h^{(n)}(w) = \frac{n!}{u^2 c}\,\frac{\sin\big((n+1)\arctan(c/w)\big)}{(w^2+c^2)^{(n+1)/2}}. \tag{4.2.3}$$

The first order n with $(-1)^n h^{(n)}(w_0) < 0$ is therefore $\lfloor \pi/\arctan(c/w_0) \rfloor$, about $\pi w_0/c$ for large w_0: complete monotonicity fails at every $u \neq 1$, at an order that grows linearly in w_0 (Table 4.1).

u	$w_0 = 2$	$w_0 = 10$	$w_0 = 100$
1.1	33	164	beyond 400
2	4	21	209
5	2	7	65

Table 4.1: First order n with $(-1)^n h^{(n)}(w_0) < 0$ for the half Cauchy pair function, with $u = e^a$. The entries agree with the closed form (4.2.3), which gives 1645 for the entry beyond 400. For $\beta = 2$ the pair function is proportional to $1/(w + 2\cosh a)$, which is completely monotone at every order.

4.2.2 The half Cauchy law and the Riemann Xi function

On the Cauchy side everything is explicit. The logarithm of $|C|$ has density $g_C(y) = 1/(\pi\cosh y)$ and characteristic function $\operatorname{sech}(\pi t/2)$. The Weierstrass product

$$\operatorname{sech}\Big(\frac{\pi t}{2}\Big) = \prod_{k\geq 1}\Big(1 + \frac{t^2}{(2k-1)^2}\Big)^{-1}$$

says $\log|C| \overset{d}{=} \sum_{k\geq 1} L_k/(2k-1)$ with L_k independent standard Laplace variables. The Mellin transform is $\mathbb{E}|C|^z = \sec(\pi z/2)$ for $|\operatorname{Re} z| < 1$, the reciprocal of

$\cos(\pi z/2)$, a function of the Laguerre Pólya class with real zeros $\pm(2k-1)$; these odd integers are the frequencies of the half Cauchy law. The Cauchy reciprocal Xi function

$$\varphi_C(s) = \mathbb{E}|C|^{i\sqrt{s}} = \operatorname{sech}\Big(\frac{\pi\sqrt{s}}{2}\Big) = \prod_{k\ge1}\Big(1+\frac{s}{(2k-1)^2}\Big)^{-1}$$

is the Laplace transform of the GGC $\sum_k e_k/(2k-1)^2$, with independent standard exponentials e_k and Thorin measure $\sum_k \delta_{(2k-1)^2}$, hence HCM by (F6). The product agrees with $\operatorname{sech}(\pi\sqrt{s}/2)$ to 15 digits at $s = 0.5, 3, 20$; this and the other checks below that name no source used mpmath [159] at 20 to 30 digits. The function φ_C is also the exit time transform $\operatorname{sech}\sqrt{u}$ of Table 1.9 at $u = \pi^2 s/4$: if τ is the exit time of standard Brownian motion from $(-1,1)$ then $\mathbb{E}e^{-\lambda\tau} = 1/\cosh\sqrt{2\lambda}$ and $\varphi_C(s) = \mathbb{E}e^{-(\pi^2/8)s\tau}$. Bourgade, Fujita and Yor [39] derive Euler's formulae for $\zeta(2n)$ from products of independent Cauchy variables, whose Mellin transforms are powers of $\sec(\pi z/2)$. In Section 10.2 their construction becomes the moment identity (10.2.5) for $\log|CC'|$, with C' an independent copy of C, and τ becomes the sech clock (Section 10.2.5).

On the Riemann side the frequencies are the zeros γ of $\Xi(z) = \xi(\frac12 + iz)$. By (1.2.1), whose product converges absolutely because $\sum|\gamma|^{-2} < \infty$, unconditionally

$$\Xi(z) = \Xi(0)\prod_{\gamma}\Big(1-\frac{z^2}{\gamma^2}\Big),$$

the product over zeros γ of Ξ with $\operatorname{Re}\gamma > 0$, counted with multiplicity. No zero is purely imaginary because $\xi(\sigma) > 0$ for real σ. Hence the clock is

$$\varphi(s) = \frac{\Xi(0)}{\Xi(i\sqrt{s})} = \frac{\xi(1/2)}{\xi(1/2+\sqrt{s})} = \prod_{\gamma}\Big(1+\frac{s}{\gamma^2}\Big)^{-1}.$$

Under RH define, in the spirit of Biane, Pitman and Yor [28], $\log W = \sum_\gamma L_\gamma/\gamma$ with independent standard Laplace variables L_γ. Then $\mathbb{E}W^z = \Xi(0)/\Xi(z)$ for $|\operatorname{Re} z| < \gamma_1 \approx 14.1347$, the characteristic function of $\log W$ is $\xi(1/2)/\xi(1/2+t)$, and $\varphi(s) = \mathbb{E}W^{i\sqrt{s}}$. The variable W, not to be confused with the heat trace $W(t)$, exists without RH: $\log W$ has the law of the variable $Z\sqrt{2T}$ of the van Dantzig pair of Section 1.2, where Z is standard normal and independent of the clock variable T, and Theorem 4.3.26 proves its existence a second time, with a lower bound for its density. Only the series for $\log W$ needs RH. Biane, Pitman and Yor represent 2ξ itself, not its reciprocal, as a Mellin transform, that of $(\pi S_2/2)^{1/2}$ with the variable S_2 of Section 1.2, and Section 9.1 describes that Mellin side at every basepoint. Table 4.2 sets out the correspondence between the half Cauchy law and the Riemann Xi function.

On the Riemann side the analogue of g_C is the density of $\log W$. The function $t \mapsto \xi(1/2)/\xi(1/2+t)$ is defined for all real t without any hypothesis (ξ has no

	half Cauchy	Riemann
entire function	$\cos(\pi z/2)$	$\Xi(z)/\Xi(0)$
frequencies	$2k-1$	γ
log variable	$\sum_k L_k/(2k-1)$	$\sum_\gamma L_\gamma/\gamma$
Mellin transform	$\sec(\pi z/2)$	$\Xi(0)/\Xi(z)$
reciprocal Xi	$\operatorname{sech}(\pi\sqrt{s}/2)$	$\xi(1/2)/\xi(1/2+\sqrt{s})$
Thorin measure	$\sum_k \delta_{(2k-1)^2}$	$\sum_\gamma \delta_{\gamma^2}$
exit time	Brownian motion on $(-1,1)$	Krein string with spectrum $\{\gamma^2\}$
density of the power	HCM iff $\lvert\beta\rvert \geq 2$ (Bosch)	HCM only if $\lvert\beta\rvert \geq \gamma_2 - \gamma_1$

Table 4.2: The dictionary. In the Riemann column the Mellin transform (Section 1.2 and Theorem 4.3.26) and the reciprocal Xi function are unconditional, the log variable and the Thorin measure presume RH, and the exit time, the hitting time of Kent's theorem (Section 8.1.3), presumes RH and simple zeros; Section 9.2 gives hitting times at the other basepoints. The last entry is the necessary condition of Theorem 4.3.28, which assumes RH, simple zeros and a growth condition on $1/|\zeta'(\rho)|$; sufficiency is open (Question 4.3.29).

real zeros) and decays like $\exp(-\frac{1}{2}|t|\log|t|)$, so its Fourier inverse

$$g(y) = \frac{1}{\pi}\int_0^\infty \frac{\xi(1/2)}{\xi(1/2+t)}\cos(ty)\,dt$$

is an even function with $\int g = 1$, entire because the integrand decays faster than any exponential; direct quadrature gives $\int g = 0.99996$, which measures the quadrature error. Just as the Pólya kernel is the Fourier inverse of Ξ on the real axis, g is the Fourier inverse of $\Xi(0)/\Xi(i\,\cdot)$, and $g > 0$ unconditionally (Theorem 4.3.26). Under RH, with simple zeros and $\log(1/|\zeta'(\rho_n)|) = O(\gamma_n)$, the residues at the poles $\pm i\gamma_n$ of the integrand give, for $|y| > u_0$ with the constant $u_0 \geq \pi/4$ of Theorem 4.3.28,

$$g(y) = \sum_n \frac{\gamma_n}{2} A_n e^{-\gamma_n|y|}, \qquad A_n = \prod_{m\neq n}\Big(1-\frac{\gamma_n^2}{\gamma_m^2}\Big)^{-1}, \tag{4.2.4}$$

the analogue of $g_C(y) = (2/\pi)\sum_k (-1)^{k-1}e^{-(2k-1)|y|}$, which converges for every $y \neq 0$ because its coefficients are bounded. The Riemann coefficients $A_1 \approx 50.9$, $A_2 \approx -2664.6$, $A_3 \approx 31420$ (Appendix D lists $b_n = \gamma_n A_n/2$) alternate in sign, as the Cauchy coefficients do, but grow roughly like $e^{\pi\gamma_n/4}$ (Theorem 4.3.28), so (4.2.4) diverges for $|y| < \pi/4$. Three terms match the Fourier integral to relative accuracy 3×10^{-3} at $y = 1$ and 1.4×10^{-4} at $y = 1.2$.

4.2.3 The reciprocal Xi function is HCM if and only if RH holds

Bosch's method works with the singularities of a function and with its logarithmic derivative, so we state Corollary 4.1.3(ii) in four forms. Form (iii) is Proposi-

tion 1.2.1, and since $\psi = G'/G$, form (iv) is the Stieltjes form of the Thorin condition in Section 1.1. A Stieltjes function is a function $a/s + b + \int_{(0,\infty)} \sigma(dt)/(s+t)$ with $a, b \geq 0$ and σ a nonnegative measure satisfying $\int (1+t)^{-1}\sigma(dt) < \infty$ [234, Definition 2.1].

THEOREM 4.2.4. *The following are equivalent.*

(i) The Riemann Hypothesis.

(ii) $\varphi(s) = \xi(1/2)/\xi(1/2+\sqrt{s})$ *is HCM on* $(0,\infty)$.

(iii) φ *is the Laplace transform of a GGC, namely* $\sum_\gamma e_\gamma/\gamma^2$.

(iv) $\psi(s) = -\varphi'(s)/\varphi(s) = \dfrac{1}{2\sqrt{s}}\dfrac{\xi'}{\xi}\big(\tfrac{1}{2}+\sqrt{s}\big)$ *is a Stieltjes function.*

Proof. (i) implies (iii): each factor $(1 + s/\gamma^2)^{-1}$ is the Laplace transform of an exponential variable with rate $\gamma^2 > 0$, and the product converges because $\sum \gamma^{-2} < \infty$; the sum $\sum_\gamma e_\gamma/\gamma^2$ converges almost surely. (iii) implies (ii) by (F6). Directly, as in (4.1.1), for $c > 0$ the pair function of $(1 + x/c)^{-1}$ is $(1 + uv/c)^{-1}(1 + u/(vc))^{-1} = (1 + u^2/c^2 + (u/c)w)^{-1}$, completely monotone in w, and (F3), (F4) finish.

(ii) implies (i): by (F7), φ extends analytically to $\mathbb{C} \setminus (-\infty, 0]$. Suppose a zero γ has $\operatorname{Im}\gamma \neq 0$. Since $\operatorname{Re}\gamma > 0$ the point $s_0 = -\gamma^2$ has $\operatorname{Im} s_0 = -2\operatorname{Re}\gamma \operatorname{Im}\gamma \neq 0$, and its principal square root is $\pm i\gamma$, so $\xi(1/2+\sqrt{s_0}) = \Xi(\pm\gamma) = 0$ by evenness. Then φ has a pole at a nonreal point of $\mathbb{C} \setminus (-\infty, 0]$, a contradiction.

(i) is equivalent to (iv): under RH, $\psi(s) = \sum_\gamma 1/(s+\gamma^2)$, a Stieltjes function with measure $\sum_\gamma \delta_{\gamma^2}$. Conversely a Stieltjes function is analytic on $\mathbb{C} \setminus (-\infty, 0]$, and an off line zero produces the nonreal pole s_0 of ψ found above. □

Condition (iv) is a Pick function statement: $\operatorname{Im}\psi(s) \leq 0$ for $\operatorname{Im} s > 0$, so $-\psi$ is a Pick function, that is analytic on the upper half plane with nonnegative imaginary part. It is a companion of Lagarias's criterion, RH if and only if $\operatorname{Re}(\xi'/\xi)(s) > 0$ for $\operatorname{Re} s > 1/2$ [133], and it is the pole test of Table 1.8 in analytic form. To check the formulas we evaluate $\operatorname{Im}\psi$ on 390 points $s = Re^{i\pi j/40}$, $1 \leq R \leq 2000$, $1 \leq j \leq 39$; the maximum is $-2.9 \times 10^{-6} < 0$. The scan reaches only zeros with $\gamma^2 \lesssim 2000$, far below existing verifications. We also compare $\psi(s)$ with $\sum_{n \leq 200} 1/(s+\gamma_n^2)$ plus a Riemann von Mangoldt tail, and they agree to about 5×10^{-6} at $s = 1, 100, 1000$.

The de Bruijn Newman flow embeds φ in a family of the same kind. Let Φ_{dBN} and $H_t(z) = \int_0^\infty e^{tu^2}\Phi_{\mathrm{dBN}}(u)\cos(zu)\,du$ be as in Section 8.4, so that $H_0(z) = \frac{1}{8}\Xi(z/2)$; in Titchmarsh's variable $v = 2u$ of (1.2.3) the Gaussian factor is $e^{tv^2/4}$. Put $\varphi_t(s) = H_t(0)/H_t(2i\sqrt{s})$, the function R_t of Theorem 7.2.15, so that $\varphi_0 = \varphi$. The function H_t is even, real on $\mathbb{R}$ and entire of order one, as in the proof of Theorem 7.2.15, so its Hadamard product in z^2 converges. It is positive on

the imaginary axis, $H_t(iy) = \int_0^\infty e^{tu^2}\Phi_{\mathrm{dBN}}(u)\cosh(yu)\,du > 0$, since $\Phi_{\mathrm{dBN}}(u) = \frac{1}{2}\Phi(2u) > 0$; hence $\varphi_t > 0$ on $(0,\infty)$ and no zero of H_t is purely imaginary. With these facts the argument of Theorem 4.2.4 applies verbatim.

Corollary 4.2.5. *φ_t is HCM if and only if $t \geq \Lambda$, the de Bruijn Newman constant. Hence φ_t is HCM for $t \geq 0.22$ [217], φ_t is not HCM for $t < 0$ [224], and RH holds if and only if φ_t is HCM for every $t > 0$.*

Proof. The zeros of H_t are real if and only if $t \geq \Lambda$ (de Bruijn [65], Newman [163]). If RH holds, H_0 has only real zeros, so $\Lambda \leq 0$ and φ_t is HCM for every $t > 0$. Conversely, $\varphi_t \to \varphi_0$ pointwise as $t \downarrow 0$, so if every φ_t with $t > 0$ is HCM then φ is HCM by (F4), and Theorem 4.2.4 gives RH. □

The bound $\Lambda \leq 0.2$ of Platt and Trudgian [196] (Section 8.4) lowers the unconditional threshold from 0.22 to 0.2. The reciprocal Xi function, if HCM at all, is HCM with no room to spare, since the property fails for every $t < 0$. This is the Riemann counterpart of the boundary $|\beta| = 1/(1-\alpha)$ in Bosch's theorem, which is attained; the difference is that Bosch can compute on the boundary because his kernel is explicit.

4.2.4 What Bosch's theorem can and cannot supply

Bosch's theorem cannot be the input to a proof of Theorem 4.2.4. For products the reason is elementary.

Proposition 4.2.6. *For any positive numbers c_n with $\sum_n c_n^{-1} < \infty$ the function $\prod_n (1 + s/c_n)^{-1}$ is HCM. If one c_n is replaced by a pair $c, \bar{c}$ with c nonreal, the product is still positive on $(0,\infty)$ but is not HCM.*

Proof. The first claim is the argument for (i) implies (ii) in Theorem 4.2.4. For the second, $(1 + s/c)(1 + s/\bar{c}) = |1 + s/c|^2 > 0$ for real s, and the product has a pole at the nonreal point $-c$, contradicting (F7). □

For a product of this type, HCM is equivalent to the reality of its frequencies. Heuristically, every HCM closure operation carries real frequencies to real frequencies and none creates them: on Mellin transforms, products and quotients of independent variables multiply the transforms, and powers rescale the frequencies. Bosch's theorem lives in the class where the frequency set $\{2k-1\}$ is given, so any deduction of Theorem 4.2.4 from it must import the location of the γ from elsewhere.

Nor can the Riemann statement be proved factor by factor. Write $\xi(1/2 + \sqrt{s}) = \frac{1}{2}(s - \frac{1}{4})\,\pi^{-(1/2+\sqrt{s})/2}\,\Gamma(\frac{1}{4} + \frac{\sqrt{s}}{2})\,\zeta(\frac{1}{2} + \sqrt{s})$. The reciprocal factors are not individually HCM. The reciprocal of $\pi^{-(1/2+\sqrt{s})/2}$ is $\pi^{1/4}\pi^{\sqrt{s}/2}$, and the pair function of $\pi^{\sqrt{s}/2}$ is $\exp(\frac{1}{2}\log\pi\,\sqrt{u}\,\sqrt{w+2})$, which increases in w. The factor $1/\zeta(1/2+\sqrt{s})$ changes sign at $s = 1/4$, since $\zeta < 0$ on $(1/2, 1)$, and HCM functions

are positive. The positive combination $1/((s-\frac{1}{4})\zeta(\frac{1}{2}+\sqrt{s}))$ is continuous at $s=1/4$, and by the argument of Theorem 4.2.4 any HCM statement for it would again imply RH. It is the function to which the sech expansion of $1/\zeta(\frac{1}{2}+\sqrt{s})$ in Theorem 6.1.5, built from the Möbius series and the Lai and Vere-Jones identity [135], should be applied.

Bosch's theorem is a density statement, and HCM of a density differs from HCM of a Laplace transform: the density of $|C|$ is not HCM, yet φ_C is. On the Cauchy side the two levels are linked only because sech is self dual, its own Fourier transform up to scaling, so that the density g_C of $\log|C|$ and the transform φ_C are the same function in different coordinates. On the Riemann side there is no self duality: the density level object is g in (4.2.4), the Fourier inverse of $\Xi(0)/\Xi(i\,\cdot)$, and the function level object is φ. A Riemann version of Bosch's theorem would answer the following questions.

Question 4.2.7. The variable W, with $\mathbb{E}W^z=\Xi(0)/\Xi(z)$, exists unconditionally (Section 4.2.2 and Theorem 4.3.26). For which β is the density of W^β HCM? On the Cauchy side the answer is $|\beta|\geq 2$.

Question 4.2.8. Does HCM of the density of W^β for a single β imply RH?

Theorem 4.3.28 answers Question 4.2.7 in part: under RH, simple zeros and a growth condition on $1/|\zeta'(\rho)|$, the density of W^β can be HCM only if $|\beta|\geq\gamma_2-\gamma_1=6.8873$, and Question 4.3.29 asks whether this is also sufficient. At the Cauchy point the same rule gives $3-1=2$, Bosch's threshold.

One might hope that an HCM variable has a logarithm with an EGGC law (Section 1.3.2), which would force the singularities of $\mathbb{E}W^z$, that is the zeros of Ξ, onto the real axis. This implication is false in general. The generalised inverse Gaussian law with density proportional to $x^{-1}e^{-(x+1/x)}$ is HCM by (F7), yet its Mellin transform at $z=it$, $K_{it}(2)/K_0(2)$, has real zeros in t, the first at $t\approx 4.4255$ (by a theorem of Pólya [213], every zero of $t\mapsto K_{it}(2)$ is real). The characteristic function of its logarithm therefore vanishes, and the logarithm is not infinitely divisible. So density level HCM does not by itself give RH, Bosch's density theorem does not transfer to Theorem 4.2.4 without a new idea, and Question 4.2.8 is open.

Three things do transfer. First, the dictionary of Table 4.2 is a solved model in which every quantity (Thorin measure, exit time, Laplace series with alternating coefficients, HCM threshold) is explicit, and it tests any proposed argument: a proof that φ is HCM which does not use the values of the frequencies would also prove HCM for $\cos(\pi z/2)$ with frequencies moved off the axis, which is false by Proposition 4.2.6. Second, Bosch's technique, verifying HCM through a Pick or Stieltjes representation, is condition (iv), so the Riemann problem is already in the form his method needs. Third, the boundary phenomenon of Corollary 4.2.5 gives a target: to show that φ_t is HCM for all $t>0$, which by that corollary is equivalent to $\Lambda\leq 0$, that is, by Rodgers and Tao [224], to $\Lambda=0$.

Bosch's proof places his threshold where zeros of a denominator reach the cut,

and the Riemann problem has a threshold of the same kind: the poles of the reciprocal Xi function, rotated by a power, reach the cut when the power exceeds a value set by the angle of the zeros.

4.3 A Bosch type theorem for the reciprocal Xi function

The observation that closes Section 4.2 becomes a theorem through an elasticity criterion: a positive function h on $(0,\infty)$ is HCM if and only if it continues to a zero free analytic function on the plane cut along $(-\infty,0]$ and its elasticity $-xh'(x)/h(x)$ maps the upper half plane into its closure (Theorem 4.3.2). The criterion reproves Bosch's theorem in a few lines (Theorem 4.3.4). Applied to $\varphi(s)=\xi(\frac12)/\xi(\frac12+\sqrt{s})$ it gives a theorem of the same shape: if δ_* is the largest angle $|\arg\gamma|$ over the zeros γ of Ξ with $\operatorname{Re}\gamma>0$ and $\alpha_\xi=2\delta_*/\pi$, then $\varphi(x^{1/\beta})$ is HCM if and only if $|\beta|\geq 1/(1-\alpha_\xi)$ (Theorem 4.3.6). RH is the statement $\alpha_\xi=0$, and the verification of RH to height $T_0=3\times10^{12}$ [196] makes $\varphi(x^p)$ HCM, unconditionally, for $|p|\leq 1-1/(\pi T_0)$ (Corollary 4.3.8). The same elasticity identifies the Thorin measure of $\varphi(s^p)$: for $p<p_*=1-\alpha_\xi$ it has a density given by ξ'/ξ along a ray (Theorem 4.3.13), and under RH it is p times a sum over the zeros of scaled copies of one law, that of the variable whose pth power is Bosch's variable (Corollary 4.3.14).

The positivity of this Thorin density is where the Riemann problem now sits, and the rest of the section asks what can be said about it without the zeros, and what happens at the level of densities and of Mellin transforms. The primes alone prove the density positive on each ray beyond some point, and that point is asymptotically where the ray leaves the critical strip (Proposition 4.3.19). At the level of densities, the Cauchy analogue of the clock variable has density $(4/\pi)\eta(4it/\pi)^3$, whose natural boundary prevents it from being HCM (Theorem 4.3.21), and under hypotheses on the zeros Pólya's gap theorem gives the same conclusion for the clock variable (Theorem 4.3.23). At the level of Mellin transforms, under RH and two further hypotheses on the zeros, the power W^β of the variable W of Section 4.2 can be HCM only if $|\beta|\geq\gamma_2-\gamma_1$, the first gap between zeros (Theorem 4.3.28); computation suggests that the condition is also sufficient, and at the Cauchy point the same rule gives Bosch's threshold $3-1=2$.

The notation is that of Section 4.2: the clock φ is a function of s, and W is the Mellin variable with $\mathbb{E}W^z=\Xi(0)/\Xi(z)$, not the heat trace $W(t)$ of Section 5.1. In addition X denotes, under RH, the clock variable T of Proposition 1.2.1; p is an exponent, not the Pólya density; ε_h is an elasticity, not a resolution; δ is the angle of a zero; the index α_ξ has nothing to do with the basepoint α of Corollary 4.1.3; and θ and ϑ are angles, not the ordinate and the Riemann Siegel theta function. The numerical checks are documented in Appendix D, which describes the script and its output, both in the archive, block by block, and names the output lines that come from code outside the script; the checks confirm formulas and constants

and prove nothing.

4.3.1 The elasticity criterion

We use HCM as defined in Section 4.2.1, with the facts (F1) to (F7) listed there, above all Bondesson's representation (4.2.1) [35, (5.2.3) and Theorem 5.3.1]. Throughout, $\Omega = \mathbb{C} \setminus (-\infty, 0]$, $\mathbb{C}^+ = \{\operatorname{Im} x > 0\}$, and a Pick function is an analytic map $\mathbb{C}^+ \to \overline{\mathbb{C}^+}$. A law on $[0,\infty)$ is a GGC with drift $b \geq 0$ and Thorin measure U, in the sense of (1.3.1), when its Laplace transform is $h(x) = \exp(-bx - \int \log(1 + x/t)U(dt))$; then $-h'/h = b + \int U(dt)/(x+t)$ is a Stieltjes function. We also need two facts about complete Bernstein functions [234, Theorem 6.2]: a function f on $(0,\infty)$ is complete Bernstein if and only if $f \geq 0$ on $(0,\infty)$ and f continues to a Pick function analytic on Ω, and then $f(x)/x$ is a Stieltjes function.

The criterion below restates (4.2.1) as a sign condition on a logarithmic derivative, a Pick function condition of the kind through which Bosch verifies HCM [37, Section 2.2]. We call it the elasticity criterion.

Definition 4.3.1. For h analytic and zero free on Ω and positive on $(0,\infty)$, let $\log h$ be the branch that is real on $(0,\infty)$ and define the elasticity $\varepsilon_h(x) = -xh'(x)/h(x)$. In logarithmic coordinates $x = e^z$ one has $\varepsilon_h(e^z) = -\frac{d}{dz}\log h(e^z)$.

Theorem 4.3.2. *Let $h > 0$ on $(0,\infty)$. Then h is HCM if and only if h continues to a zero free analytic function on Ω and ε_h is a Pick function, that is* $\operatorname{Im}\varepsilon_h(x) \geq 0$ *for* $\operatorname{Im} x > 0$.

Proof. Necessity. By (4.2.1), h is zero free and analytic on Ω, and its elasticity is

$$\varepsilon_h(x) = 1 - r + b_1 x + \int \frac{x}{x+t}\,\Gamma_1(dt) - \frac{b_2}{x} - \int \frac{\Gamma_2(dt)}{1+tx}.$$

For $\operatorname{Im} x > 0$ each term has nonnegative imaginary part: $\operatorname{Im}\frac{x}{x+t} = \frac{t\operatorname{Im} x}{|x+t|^2}$, $\operatorname{Im}(-\frac{1}{x}) = \frac{\operatorname{Im} x}{|x|^2}$ and $\operatorname{Im}(-\frac{1}{1+tx}) = \frac{t\operatorname{Im} x}{|1+tx|^2}$.

Sufficiency. Fix $u > 0$. The principal branch $y(w) = \operatorname{arccosh}(w/2)$ maps $\mathbb{C} \setminus (-\infty, 2]$ conformally onto the half strip $S = \{\operatorname{Re} y > 0,\ |\operatorname{Im} y| < \pi\}$, and $w = 2\cosh y$. For $y \in S$ both ue^y and ue^{-y} lie in Ω, so

$$F(w) = -\log h\big(ue^{y(w)}\big) - \log h\big(ue^{-y(w)}\big)$$

is analytic on $\mathbb{C} \setminus (-\infty, 2]$ and real on $(2,\infty)$, and $e^{-F(w)}$ is the pair function. Let $\operatorname{Im} w > 0$, so $y = \varrho + i\theta$ with $\varrho > 0$ and $\theta \in (0,\pi)$. Put $r_1 = ue^{\varrho} > r_2 = ue^{-\varrho}$. Since $\log h(\bar{x}) = \overline{\log h(x)}$,

$$\operatorname{Im} F(w) = -\operatorname{Im}\log h(r_1 e^{i\theta}) + \operatorname{Im}\log h(r_2 e^{i\theta}).$$

Along the ray, $\partial_r \operatorname{Im} \log h(re^{i\theta}) = \operatorname{Im}\big(e^{i\theta}h'/h\big) = -r^{-1}\operatorname{Im}\varepsilon_h(re^{i\theta}) \le 0$, so $\operatorname{Im} F(w) \ge 0$. Thus F is a Pick function that continues analytically across $(2,\infty)$, where it is real. By the Nevanlinna representation $F(w) = a + bw + \int(\frac{1}{t-w} - \frac{t}{1+t^2})\sigma(dt)$ with $b \ge 0$ and $\sigma \ge 0$ carried by $(-\infty, 2]$. Hence for $w > 2$,

$$F'(w) = b + \int_{(-\infty,2]} \frac{\sigma(dt)}{(w-t)^2} = b + \int_{(-\infty,2]} \int_0^\infty \lambda e^{-\lambda(w-t)}\, d\lambda\, \sigma(dt),$$

which is completely monotone. A function whose derivative is completely monotone is a Bernstein function up to an additive constant, and e^{-F} is then completely monotone. So every pair function is completely monotone and h is HCM. □

Remark 4.3.3. In words: h is HCM exactly when, along every ray $re^{i\theta}$ with $0 < \theta < \pi$, the argument of h is nonincreasing in r, and h has no zeros or singularities off the cut. The criterion replaces the family of pair functions, one for each $u > 0$, by a single analytic condition. After the change of variable $w = 2\cosh y$, the imaginary part of minus the logarithm of a pair function is the difference of the values of $\arg h$ at two points of one ray, so its sign is the monotonicity of $\arg h$ along rays, and the Nevanlinna representation turns that sign into complete monotonicity. For a GGC Laplace transform $h(x) = \exp(-bx - \int \log(1 + x/t)U(dt))$ the elasticity is $bx + \int \frac{x}{x+t}U(dt)$, visibly Pick, which recovers the HCM property of GGC transforms.

The two conditions play different roles. Analyticity without zeros is global and rigid: a single zero or pole of h anywhere off the negative axis destroys HCM, however small its effect on the real line. The Pick condition is local and quantitative: it can fail along a ray without any singularity. Each example in Sections 4.3.2 and 4.3.3 says which of the two sets the threshold.

4.3.2 Bosch's theorem from the criterion

Bosch's variable $T_\alpha = (Z_\alpha/\tilde{Z}_\alpha)^\alpha$, $0 < \alpha < 1$, of Section 4.2.1 has the density $\sin(\pi\alpha)/(\pi\alpha(t^2 + 2t\cos\pi\alpha + 1))$ of a half Cauchy law with drift. Let $c = \cos\pi\alpha$ and $q = 1/|\beta|$, half the q of (4.2.2). Since $T_\alpha \overset{d}{=} 1/T_\alpha$ we may take $\beta > 0$. The density of T_α^β is proportional to

$$h(x) = \frac{x^{q-1}}{D(x)}, \qquad D(x) = x^{2q} + 2cx^q + 1 = (x^q + e^{i\pi\alpha})(x^q + e^{-i\pi\alpha}).$$

Theorem 4.3.4 (Bosch [37, Theorem 1.2]). *h is HCM if and only if $\alpha \le 1/2$ and $q \le 1 - \alpha$.*

Proof. Zeros. For $x \in \Omega$ the value x^q ranges over the sector $|\arg| < q\pi$. The zeros of D satisfy $x^q = e^{\pm i(1-\alpha)\pi}$, so D is zero free on Ω if and only if $q \le 1 - \alpha$.

Elasticity. For $x = re^{i\theta}$ with $0 < \theta < \pi$ write $x^q = Re^{i\omega}$ with $R = r^q$ and $\omega = q\theta \in (0, q\pi)$. Then $\varepsilon_h = 1 - q + xD'(x)/D(x)$ and $\operatorname{Im}\varepsilon_h = qR\,\partial_R \arg D$. A direct computation of $\operatorname{Im}(\partial_R D \cdot \overline{D})$ gives

$$\operatorname{Im}\varepsilon_h(x) = \frac{2qR\sin\omega\,\big(c(1+R^2) + 2R\cos\omega\big)}{|D(x)|^2}. \tag{4.3.1}$$

If $\alpha \le 1/2$ and $q \le 1-\alpha$ then $c \ge 0$ and $\cos\omega \ge \cos((1-\alpha)\pi) = -c$, so, using $1+R^2 \ge 2R$, the bracket is at least $2R(c+\cos\omega) \ge 0$. Theorem 4.3.2 gives HCM. If $\alpha > 1/2$ then $c < 0$ and the bracket tends to $c < 0$ as $R \to 0$, so ε_h is not Pick. If $q > 1-\alpha$ then h has poles in Ω. In both cases Theorem 4.3.2 rules out HCM. □

In terms of β, the theorem says that the density of T_α^β is HCM if and only if $\alpha \le 1/2$ and $|\beta| \ge 1/(1-\alpha)$. For the half Cauchy variable, $\alpha = 1/2$, the threshold is $|\beta| \ge 2$; there $c = 0$ and (4.3.1) reduces to $\operatorname{Im}\varepsilon_h = 4qR^2\sin\omega\cos\omega/|D|^2$, which is nonnegative for every $\omega \in (0, q\pi)$ exactly when $q \le 1/2$. So the half Cauchy threshold is the condition $\omega \le \pi/2$, equivalently that the zeros $x = e^{\pm i\pi/(2q)}$ of $1 + x^{2q}$ stay on or beyond the cut. For $\beta = 1$ the zeros $\pm i$ of $1 + x^2$ lie in Ω, which gives Proposition 4.2.2 again. Numerically, on a grid of 49×199 points $re^{i\theta}$ with $10^{-6} \le r \le 10^6$, the minimum of $\operatorname{Im}\varepsilon_h$ is positive for $(\alpha, q) = (0.4, 0.6), (0.4, 0.55), (0.5, 0.5), (0.3, 0.7)$, equals -0.0114 for $(0.6, 0.3)$, and is -191 and -1.1×10^{16} near the interior poles for $(0.5, 0.6)$ and $(0.4, 0.75)$, in agreement with the theorem.

The two conditions of the theorem are the two parts of the criterion. The condition $q \le 1-\alpha$ concerns singularities: the poles of the density of T_α^β, where $x^q = e^{\pm i(1-\alpha)\pi}$, rotate towards the positive axis as q grows and reach the cut at $q = 1-\alpha$. The condition $\alpha \le 1/2$ is a Pick condition: when the drift angle is obtuse the argument of the density turns the wrong way near the origin, where nothing is singular. In the Riemann case the rotating poles are the points $-\gamma^2$, and the drift angle becomes the angle that the zeros make with the real axis.

4.3.3 The Riemann analogue

As in Section 4.2.2, γ runs over the zeros of $\Xi(z) = \xi(1/2 + iz)$ with $\operatorname{Re}\gamma > 0$, counted with multiplicity; the nonreal ones come in conjugate pairs $\gamma, \bar\gamma$. By (1.2.1), unconditionally,

$$\varphi(s) = \frac{\xi(1/2)}{\xi(1/2+\sqrt{s})} = \prod_\gamma \Big(1 + \frac{s}{\gamma^2}\Big)^{-1}, \qquad \varepsilon_\varphi(s) = \frac{\sqrt{s}}{2}\,\frac{\xi'}{\xi}\Big(\frac12 + \sqrt{s}\Big) = \sum_\gamma \frac{s}{s+\gamma^2},$$

both absolutely convergent since $\sum |\gamma|^{-2} < \infty$. Write $\gamma = \varrho e^{i\delta}$. Since the zeros of ξ lie in the open critical strip, $|\delta| < \arctan(1/(2\operatorname{Re}\gamma))$, and since every zero has

$\operatorname{Re}\gamma > 14$, also $|\delta| < \pi/4$. Define

$$\delta_* = \max_\gamma |\arg\gamma|, \qquad \alpha_\xi = \frac{2\delta_*}{\pi}, \qquad p_* = 1 - \alpha_\xi.$$

The maximum is attained if some zero is off the line, because for every positive angle only finitely many γ have $|\arg\gamma|$ larger than it. RH is the statement $\delta_* = 0$. Each conjugate pair of zeros contributes to ε_φ a term whose imaginary part has a sign on a sector determined by the angle of the pair.

LEMMA 4.3.5. *Let $\gamma = \varrho e^{i\delta}$ with $0 \le \delta < \pi/4$ and $s = re^{i\theta}$ with $0 < \theta < \pi - 2\delta$. Then*

$$\operatorname{Im}\Big(\frac{s}{s+\gamma^2} + \frac{s}{s+\bar\gamma^2}\Big) \ge 0,$$

with strict inequality. For $\delta = 0$ the single term $\operatorname{Im}\frac{s}{s+\gamma^2} = \frac{\gamma^2 r\sin\theta}{|s+\gamma^2|^2}$ is positive for $0 < \theta < \pi$.

Proof. For $c \in \mathbb{C}$, $\operatorname{Im}\frac{s}{s+c} = \operatorname{Im}\big(-\frac{c}{s+c}\big) = \frac{\operatorname{Im}(\bar c s)}{|s+c|^2}$. With $c = \varrho^2 e^{\pm 2i\delta}$ this is $\varrho^2 r\sin(\theta \mp 2\delta)/|s+c|^2$. The sum is

$$\varrho^2 r\Big(\frac{\sin(\theta-2\delta)}{A_-} + \frac{\sin(\theta+2\delta)}{A_+}\Big), \qquad A_\mp = r^2 + \varrho^4 + 2r\varrho^2\cos(\theta \mp 2\delta).$$

Since $\theta + 2\delta < \pi$ the second sine is positive. If $\theta \ge 2\delta$ both terms are nonnegative. If $\theta < 2\delta$ then $|\theta - 2\delta| < \theta + 2\delta$, so $A_- > A_+$, and $\sin(\theta+2\delta) - \sin(2\delta-\theta) = 2\cos(2\delta)\sin\theta > 0$; the positive term has the larger numerator and the smaller denominator. □

THEOREM 4.3.6. *For $p \in \mathbb{R}$ the function $x \mapsto \varphi(x^p)$ is HCM if and only if $|p| \le p_* = 1 - \alpha_\xi$. Equivalently, $\varphi(x^{1/\beta})$ is HCM if and only if $|\beta| \ge 1/(1-\alpha_\xi)$.*

Proof. Since $h(x)$ is HCM if and only if $h(1/x)$ is, take $p > 0$. Let $h_p(x) = \varphi(x^p)$, positive on $(0,\infty)$ because $\xi > 0$ on $\mathbb{R}$. Its singularities are the poles $x^p = -\gamma^2 = \varrho^2 e^{\pm i(\pi - 2|\delta|)}$, and x^p ranges over $|\arg| < p\pi$ for $x \in \Omega$. Hence h_p is zero free and analytic on Ω if and only if $p\pi \le \pi - 2|\delta|$ for every γ, that is $p \le p_*$. When $p \le p_*$,

$$\varepsilon_{h_p}(x) = p\,\varepsilon_\varphi(x^p) = p\sum_\gamma \frac{x^p}{x^p + \gamma^2},$$

and for $\operatorname{Im} x > 0$ the point $s = x^p$ has $0 < \arg s < p\pi \le \pi - 2\delta_*$. Grouping conjugate pairs, Lemma 4.3.5 makes every term of $\operatorname{Im}\varepsilon_{h_p}$ nonnegative. Theorem 4.3.2 gives HCM. When $p > p_*$ the pole coming from a zero with $|\arg\gamma| = \delta_*$ lies in Ω, and Theorem 4.3.2 rules out HCM. □

COROLLARY 4.3.7. *The following are equivalent: (i) RH; (ii) φ is HCM; (iii) $\varphi(x^p)$ is HCM for every $|p| < 1$; (iv) ε_φ is a Pick function; (v) $\operatorname{Im}\big[z\,(\xi'/\xi)(\frac12 + z)\big] \ge 0$ for all z with $\operatorname{Re} z > 0$ and $\operatorname{Im} z > 0$.*

Proof. By Theorem 4.3.6, each of (i), (ii) and (iii) says that $p_* = 1$; the equivalence of (i) and (ii) is also Corollary 4.1.3(ii), restated as Theorem 4.2.4, so only (iii) to (v) are new. (iv) is the elasticity form of (ii) through Theorem 4.3.2; directly, an off line zero gives a pole of ε_φ at the nonreal point $-\gamma^2$ or its conjugate, and a Pick function has no poles in $\mathbb{C}^+$. (v) is (iv) with $s = z^2$, which maps the open first quadrant onto $\mathbb{C}^+$. □

Criterion (v) is an angular companion to Lagarias's criterion $\operatorname{Re}(\xi'/\xi)(\frac{1}{2}+z) > 0$ for $\operatorname{Re} z > 0$ [133]: it says that the argument of ξ is nondecreasing along every ray $\frac{1}{2} + te^{i\vartheta}$, $0 < \vartheta < \pi/2$, leaving the centre of the critical line into the right half plane.

Corollary 4.3.8 (unconditional). *RH has been verified for all zeros with* $0 < \operatorname{Re}\gamma \le T_0 = 3 \times 10^{12}$ *[196]. Hence* $\delta_* < \arctan(1/(2T_0))$ *and*

$$\varphi(x^p) \text{ is HCM for all } |p| \le 1 - \frac{1}{\pi T_0}, \qquad 1 - \frac{1}{\pi T_0} > 1 - 1.1 \times 10^{-13}.$$

Unconditionally, $\operatorname{Im}\varepsilon_\varphi \ge 0$ *on the sector* $0 < \arg s < \pi - 1/T_0$.

Proof. Any off line zero has $\operatorname{Re}\gamma > T_0$ and $|\operatorname{Im}\gamma| < 1/2$, so $|\arg\gamma| < 1/(2T_0)$ and $2\delta_*/\pi < 1/(\pi T_0)$. The sector statement is Lemma 4.3.5 applied termwise. □

In words, every zero γ contributes a factor $(1 + s/\gamma^2)^{-1}$ to φ, with a pole at $-\gamma^2$. A real zero puts its pole on the cut, which is harmless, and its factor is the Laplace transform of an exponential variable. A nonreal zero puts its pole at angle $2|\arg\gamma|$ from the cut, inside the region where an HCM function must be analytic. In the variable x of $\varphi(x^p)$ that pole sits at argument $(\pi - 2|\delta|)/p$, which reaches the cut when $p = 1 - 2|\delta|/\pi$, and Lemma 4.3.5 shows that once every pole is on or beyond the cut the sign condition holds automatically. RH says that no pole has to be moved at all: the reciprocal Xi function sits at $\alpha = 0$ on Bosch's angular scale. Unconditionally it sits within 1.1×10^{-13} of that end, and an off line zero would have to put its poles in a sector of opening less than 10^{-12} beside the cut. Corollary 4.3.8 brings no new information about the zeros; it translates the verification height into an HCM exponent.

Corollary 4.3.8 is a statement about the elasticity $\varepsilon_\varphi(s) = s\psi(s)$, where $\psi = -\varphi'/\varphi = \sum_\gamma (s+\gamma^2)^{-1}$ is the function of the Stieltjes form (iv) of Theorem 4.2.4. For ψ itself the set where a pair of zeros has the wrong sign is exactly a disc, and the verification confines these discs to a parabola.

Lemma 4.3.9. *Let* $c = c_1 + ic_2$ *with* $c_2 > 0$, *and let* $s = x + iy$ *with* $y > 0$ *and* $s \ne c$. *Then*

$$\operatorname{Im}\Big(\frac{1}{s-c} + \frac{1}{s-\bar c}\Big) = -\frac{2y\,\big(|s-c_1|^2 - c_2^2\big)}{|s-c|^2\,|s-\bar c|^2},$$

so the imaginary part is positive on the open disc $|s - c_1| < c_2$, *whose vertical diameter is the segment from* $\bar c$ *to* c, *zero on its boundary and negative outside it.*

Proof. With $d=(x-c_1)^2$, $\operatorname{Im}\frac{1}{s-c}=-\frac{y-c_2}{d+(y-c_2)^2}$ and $\operatorname{Im}\frac{1}{s-\bar c}=-\frac{y+c_2}{d+(y+c_2)^2}$. Over the common denominator the numerator of the sum is

$$-(y-c_2)\big(d+(y+c_2)^2\big)-(y+c_2)\big(d+(y-c_2)^2\big)=-2y\,(d+y^2-c_2^2),$$

and $d+y^2=|s-c_1|^2$. □

A nonreal zero $\gamma=a+ib$ of Ξ, with $a>0$ and, after conjugation, $b>0$, contributes to ψ the pair of terms $1/(s-c)+1/(s-\bar c)$ with $c=-\bar\gamma^2=b^2-a^2+2iab$. Let

$$D_\gamma=\big\{s:\ |s-(b^2-a^2)|\le 2ab\big\}$$

be the closed disc with diameter from $-\gamma^2$ to $-\bar\gamma^2$. By Lemma 4.3.9 the pair has positive imaginary part only at interior points of D_γ, and negative imaginary part at every point of the upper half plane outside D_γ.

THEOREM 4.3.10 (unconditional). *Let $T_0=3\times10^{12}$, so that every zero γ with $0<\operatorname{Re}\gamma\le T_0$ is real [196], and let*

$$\mathcal{E}_{T_0}=\Big\{x+iy:\ x<-T_0^2+T_0+\tfrac14,\ y^2<\tfrac12-x\Big\}.$$

Split $\psi=S_{\rm on}+R_{\rm off}$, where $S_{\rm on}(s)=\sum_{\gamma\ \rm real}(s+\gamma^2)^{-1}$ runs over the zeros on the critical line and $R_{\rm off}$ is the same sum over the nonreal γ.

(i) $S_{\rm on}$ is a Stieltjes function, with measure $U_{\rm line}=\sum_{\gamma\ \rm real}\delta_{\gamma^2}$.

(ii) $R_{\rm off}$ is meromorphic on $\mathbb{C}$ and nonnegative on $[0,\infty)$. For every nonreal γ the disc D_γ, and with it the poles $-\gamma^2$ and $-\bar\gamma^2$ of $R_{\rm off}$ on its boundary, lies in $\mathcal{E}_{T_0}$, and $\operatorname{Im}R_{\rm off}\le0$ at every point of the upper half plane outside these discs.

(iii) $\psi>0$ on $[0,\infty)$, and $\operatorname{Im}\psi(s)<0$ at every $s=x+iy$ with $y>0$ outside $\mathcal{E}_{T_0}$.

Proof. The proof uses only that the zeros of ξ lie in the open strip $0<\operatorname{Re}s<1$ and the verification [196]. Since $\sum|\gamma|^{-2}<\infty$, the series for ψ, $S_{\rm on}$ and $R_{\rm off}$ converge absolutely and locally uniformly off the points $-\gamma^2$, so $R_{\rm off}$ is meromorphic on $\mathbb{C}$, and (i) holds because $\sum_{\gamma\ \rm real}(1+\gamma^2)^{-1}<\infty$. A real γ contributes $-y/|s+\gamma^2|^2<0$ to $\operatorname{Im}\psi(x+iy)$ for $y>0$, and $(x+\gamma^2)^{-1}>0$ to $\psi(x)$ for $x\ge0$.

A nonreal zero is $\gamma=a+ib$ with $a>T_0$ and, after conjugation, $0<b<\frac12$, because ξ vanishes at $\frac12\pm b+ia$. With $c=b^2-a^2+2iab$ as above, its pair of terms equals $2(x-c_1)/((x-c_1)^2+c_2^2)>0$ at $x\ge0$, since $c_1=b^2-a^2<0$, and its imaginary part is negative at every point of the upper half plane outside D_γ. It remains to show that $D_\gamma\subset\mathcal{E}_{T_0}$. A point $x+iy$ of D_γ has

$$x\le b^2-a^2+2ab=2b^2-(a-b)^2<-T_0^2+T_0+\tfrac14,$$

because the middle term increases in b and, for $a > b$, decreases in a, and equals $-T_0^2 + T_0 + \frac{1}{4}$ at $a = T_0$, $b = \frac{1}{2}$; in particular $x < 0$. With $e = x - (b^2 - a^2)$, so that $a^2 = b^2 - x + e$, it also has

$$y^2 \le 4a^2b^2 - e^2 = 8b^4 - 4b^2x - (e - 2b^2)^2 \le 4b^2(2b^2 - x) < 2b^2 - x < \tfrac{1}{2} - x,$$

because $4b^2 < 1$ and $2b^2 - x > 0$. Hence $D_\gamma \subset \mathcal{E}_{T_0}$, and (ii) follows from the signs of the pairs. At a point of the upper half plane outside $\mathcal{E}_{T_0}$ every pair of terms has negative imaginary part, and so has the term of every real γ; with the signs on $[0, \infty)$, since γ_1 is real, this gives (iii). □

Both bounds that define $\mathcal{E}_{T_0}$ are sharp for zeros with $a > T_0$ and $0 < b < \frac{1}{2}$. The rightmost point $2b^2 - (a - b)^2$ of D_γ tends to $-T_0^2 + T_0 + \frac{1}{4}$ as $b \uparrow \frac{1}{2}$ and $a \downarrow T_0$. For $x < \frac{3}{4} - T_0^2$ the supremum of y^2 over the points $x + iy$ of the discs is $\frac{1}{2} - x$: the bound $8b^4 - 4b^2x$ of the proof is attained, at $e = 2b^2$, by the disc with $a^2 = 3b^2 - x$, which has $a > T_0$ once b is close to $\frac{1}{2}$, and it tends to $\frac{1}{2} - x$ as $b \uparrow \frac{1}{2}$. At the tip $x_0 = -T_0^2 + T_0 + \frac{1}{4}$ the height of $\mathcal{E}_{T_0}$ is exactly $T_0 - \frac{1}{2}$, since $\frac{1}{2} - x_0 = (T_0 - \frac{1}{2})^2$, so the ray from the origin to the corner $x_0 + i(T_0 - \frac{1}{2})$ makes the angle $\theta_0 = \arctan \frac{T_0 - 1/2}{T_0^2 - T_0 - 1/4} = 3.33333 \times 10^{-13}$ with the negative axis, which exceeds $1/T_0$ by the factor $1 + 1.7 \times 10^{-13}$. Further out the angle decreases like $|x|^{-1/2}$, to 10^{-15} at $x = -10^{30}$, and the part of $\mathcal{E}_{T_0}$ in the upper half plane lies in the sector $\pi - \theta_0 < \arg s < \pi$.

Corollary 4.3.8 gives the sign of the elasticity $s\psi(s)$ outside a sector of opening $1/T_0$ beside the cut, through Lemma 4.3.5; Theorem 4.3.10 gives the sign of ψ itself outside a parabola, through the exact region of Lemma 4.3.9, and the parabola is much thinner: it begins at distance $T_0^2 - T_0 - \frac{1}{4}$ from the origin, and its angular width, about $1/T_0$ at the tip, decreases like $|x|^{-1/2}$. The theorem does not make ψ a Stieltjes function off $\mathcal{E}_{T_0}$, since being a Stieltjes function is a global property; it says that if the Stieltjes form fails, it fails through poles of ψ in $\mathcal{E}_{T_0}$, which come from zeros above the verification height. Like the corollary, it brings no new information about the zeros, and nothing in it limits the number of poles in $\mathcal{E}_{T_0}$ (Section 11.4.7).

Remark 4.3.11 (the right end). The right end of $\mathcal{E}_{T_0}$ cannot be moved to $-(T_0 - \frac{1}{2})^2$, the value the argument would give if the rightmost point of D_γ were $-(a-b)^2$: it is $2b^2 - (a - b)^2$, and the extra $2b^2$ reaches into the strip $-(T_0 - \frac{1}{2})^2 \le x < -(T_0 - \frac{1}{2})^2 + \frac{1}{2}$. For $T_0 = 100$, $b = \frac{1}{2} - 10^{-20}$ and $a = T_0 + 10^{-20}$, the point $s = -9899.85 + 0.001i$ lies in D_γ, 0.4 to the right of $-(T_0 - \frac{1}{2})^2 = -9900.25$, and the pair of terms of γ has imaginary part $1.0 \times 10^{-10} > 0$ there; for $T_0 = 3 \times 10^{12}$ the point 0.1 to the left of the rightmost point of D_γ, at height 0.001, gives $3.7 \times 10^{-42} > 0$ in the same way. Whether $\operatorname{Im} \psi$ itself is positive at such points depends on zeros that nobody knows; the termwise argument cannot exclude them.

The script `pakes_disc.py` checks the identities of the two proofs symbolically, Lemma 4.3.9 at 200000 random points in double precision and at 20000 points at 50 digits whose relative distance from the circle is between 10^{-30} and 10^{-3}, and the inclusion $D_\gamma \subset \mathcal{E}_{T_0}$ at 40000 points just inside the boundaries of discs for each of $T_0 = 2$, 100, 10^6 and 3×10^{12}, at 60 digits, concentrated at $b \to \frac{1}{2}$, $a \to T_0$ and at the rightmost points; for $T_0 = 3 \times 10^{12}$, $b = \frac{1}{2} - 10^{-20}$ and $a = T_0 + 10^{-20}$, the rightmost point of D_γ lies 1.2×10^{-7} to the left of the right end of $\mathcal{E}_{T_0}$. Figure 4.3 draws $\mathcal{E}_2$ with six discs, and the set where $\operatorname{Im}\psi > 0$ for a toy reciprocal Xi function whose real zeros are the ordinates of the first 200 zeros of ζ and whose only nonreal pair is $60 \pm 0.3i$, which corresponds to the zeros $\frac{1}{2} \pm 0.3 + 60i$; its disc D_γ has centre -3599.91 and radius 36. On a grid of 7260 points around D_γ in the upper half plane, $\operatorname{Im}\psi > 0$ at 1773 points, all in D_γ, and on the half circle of radius 1.0001×36 the largest value of $\operatorname{Im}\psi$ is -1.9×10^{-4}. The real zeros make $\operatorname{Im}\psi$ negative on a part of D_γ near its boundary.

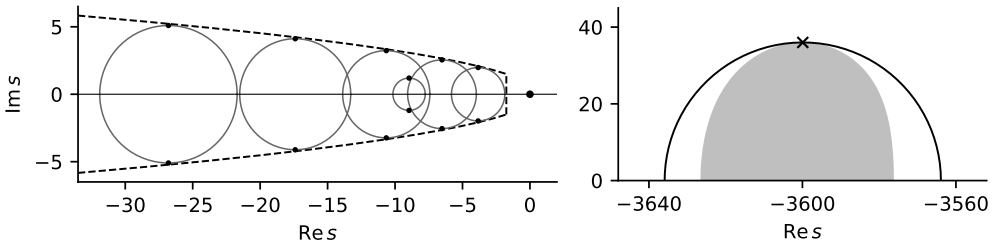


Figure 4.3: Left: the sliver $\mathcal{E}_{T_0}$ of Theorem 4.3.10 for $T_0 = 2$ (dashed) and the discs D_γ of six hypothetical zeros $a + ib$ with $a > 2$ and $0 < b < \frac{1}{2}$, with the poles $-\gamma^2$ and $-\bar{\gamma}^2$ (dots) at the ends of their vertical diameters; the dot at 0 marks the origin. Right: for the toy reciprocal Xi function with the first 200 zeros of ζ and the pair $60 \pm 0.3i$, the set where $\operatorname{Im}\psi > 0$ (shaded) on a grid of 481×300 points, which lies inside the upper half of D_γ (solid); the cross is the pole $-\bar{\gamma}^2 = -3599.91 + 36i$. Drawn by `pakes_disc.py`.

The threshold p_* of Theorem 4.3.6 can be followed along the de Bruijn Newman flow. With H_t and $\varphi_t(s) = H_t(0)/H_t(2i\sqrt{s})$ as in Section 4.2.3, so that $\varphi_0 = \varphi$, let $\delta_*(t)$ be the largest angle $|\arg z|$ over the zeros z of H_t with $\operatorname{Re} z > 0$, and $p_*(t) = 1 - 2\delta_*(t)/\pi$; the angles do not depend on how H_t is normalised. The proof of Theorem 4.3.6 uses only the product over the zeros, positivity on $(0, \infty)$ and the bound $|\arg z| < \pi/4$ required by Lemma 4.3.5, so whenever the zeros of H_t satisfy that bound, $\varphi_t(x^p)$ is HCM if and only if $|p| \le p_*(t)$. By de Bruijn [65] and Newman [163] (Corollary 4.2.5), $p_*(t) = 1$ exactly for $t \ge \Lambda$, so for $t \ge 0.22$ [217] and indeed for $t \ge 0.2$ [196]; by Rodgers and Tao [224], $p_*(t) < 1$ for every

$t < 0$. If RH holds, the heat time t and the Bosch exponent p therefore both place the reciprocal Xi function on the boundary of the HCM region.

QUESTION 4.3.12. Describe $p_*(t) = 1 - \frac{2}{\pi}\delta_*(t)$ for $t < 0$. By Rodgers and Tao $p_*(t) < 1$ for every $t < 0$. Assume RH; how fast does $1 - p_*(t) \to 0$ as $t \uparrow 0$?

Heuristically, the off line zeros of H_t for small $|t|$ come from nearly coincident pairs of real zeros: by the isolated pair estimate of Section 8.4, a pair at distance D_0 collides near $t = -D_0^2/8$ and then has imaginary parts of order $\sqrt{|t|}$, at heights where the gap is of order $\sqrt{|t|}$. Since $1 - p_*(t)$ is governed by $\max \operatorname{Im} z / \operatorname{Re} z$, the answer depends on the lowest height at which such a pair has collided, and so on the small gap statistics of the zeros. Both ingredients of an estimate are missing: de Bruijn's bound on the imaginary parts of the zeros runs only forward in time, from H_0 to H_t with $t > 0$, and no lower bound is known for the heights of the pairs that have collided by time t.

Numerically, $\operatorname{Im} \varepsilon_\varphi(re^{i\theta})/r$ was evaluated at 885 points with $0.5 \le r \le 3000$ and $\theta = j\pi/60$; the minimum was $3.8 \times 10^{-4} > 0$, as Corollary 4.3.8 guarantees in this range. For a toy reciprocal Xi function with zeros $14e^{\pm 0.1i}$ and 21, where $p_* = 1 - 0.2/\pi = 0.9363$, the elasticity of $\varphi(x^p)$ on the grid of Section 4.3.2 has no poles in Ω and positive imaginary part for $p = 0.80, 0.90, 0.93$; for $p = 0.94$ and 0.97 it has poles in Ω, which lie between the grid points or beyond the last ray of the grid, so the scan stays positive; and at $p = 1$ it reaches $\operatorname{Im} \varepsilon = -2.08$ at $x = 178e^{0.965\pi i}$, as Theorem 4.3.6 predicts.

4.3.4 The Thorin measure of $\varphi(s^p)$ and Bosch's variable

By Bondesson's theorem [35, Theorem 6.1.1], recalled in Section 1.3.2, a function with value 1 at 0 is HCM exactly when it is the Laplace transform of a GGC, so Theorem 4.3.6 already makes $\varphi(s^p)$ a GGC transform for $0 < p \le p_*$. The elasticity also identifies its Thorin measure.

THEOREM 4.3.13. *Let $0 < p \le p_*$. Then $\varphi(s^p) = \mathbb{E}e^{-sX_p}$ for a GGC variable X_p with no drift. If $p < p_*$, its Thorin measure has the density*

$$u_p(\tau) = \frac{p}{\pi\tau} \operatorname{Im} \varepsilon_\varphi\big(\tau^p e^{ip\pi}\big) = \frac{p}{2\pi\tau} \operatorname{Im}\Big[z \frac{\xi'}{\xi}\Big(\frac{1}{2} + z\Big)\Big]_{z=\tau^{p/2}e^{ip\pi/2}} \ge 0, \qquad \tau > 0, \tag{4.3.2}$$

which in terms of the zeros is

$$u_p(\tau) = \frac{p}{\pi\tau} \sum_\gamma \operatorname{Im} \frac{\tau^p e^{ip\pi}}{\tau^p e^{ip\pi} + \gamma^2}. \tag{4.3.3}$$

If $p = p_*$, its Thorin measure is $u_{p_*}(\tau)\, d\tau$ plus a unit atom at $|\gamma|^{2/p_*}$ for every zero γ with $\arg \gamma = -\delta_*$, counted with multiplicity, that is, one atom for each conjugate pair of zeros at the extreme angle; when $p_* = 1$ there is an atom at γ^2 for every zero, and $u_1 = 0$.

Proof. Let $h = h_p$ and $\varepsilon_h(x) = p\,\varepsilon_\varphi(x^p)$. By the proof of Theorem 4.3.6, ε_h is analytic on Ω and Pick. On $(0,\infty)$ it is nonnegative: a real zero contributes $x^p/(x^p+\gamma^2) > 0$ and a pair contributes $2\,\mathrm{Re}\,\frac{s}{s+\varrho^2 e^{2i\delta}} = \frac{2s(s+\varrho^2\cos 2\delta)}{|s+\varrho^2 e^{2i\delta}|^2} > 0$. Hence ε_h is complete Bernstein and $\varepsilon_h(x)/x = -(\log h)'(x)$ is a Stieltjes function, $a/x + b + \int U(dt)/(x+t)$. Here $a = \varepsilon_h(0+) = 0$ and $b = \lim_{x\to\infty}\varepsilon_h(x)/x = 0$ because $\varepsilon_\varphi(s) = O(\sqrt{s}\log s)$. Integrating from 0 with $h(0) = 1$ gives $h(x) = \exp(-\int \log(1+x/t)U(dt))$, a GGC transform. For $p < p_*$ the function ε_h is continuous up to the upper edge of the cut, and Stieltjes inversion at $x = \tau e^{i\pi}$ gives $U(d\tau) = u_p(\tau)d\tau$ with $u_p(\tau) = \frac{1}{\pi\tau}\,\mathrm{Im}\,\varepsilon_h(\tau e^{i\pi})$, which is (4.3.2); (4.3.3) follows from the Hadamard form of ε_φ. For $p = p_*$ the same inversion applies away from the points $x_0 = \tau_0 e^{i\pi}$ with $\tau_0 = |\gamma|^{2/p_*}$ and $\arg\gamma = -\delta_*$, at which $x_0^{p_*} = -\gamma^2$. Near such a point the term of γ in ε_h is $p_* x^{p_*}/(x^{p_*}+\gamma^2) = x_0/(x-x_0) + O(1)$, so $\varepsilon_h(x)/x = 1/(x+\tau_0) + O(1)$, and inversion gives a unit atom at τ_0. On the ray itself this term equals $p_*\tau^{p_*}/(\tau^{p_*} - |\gamma|^2)$, which is real, so u_{p_*} does not see the atom. □

Numerically, $-\int_0^\infty \log(1+s/\tau)u_p(\tau)d\tau$, with u_p computed from (4.3.2), reproduces $\log\varphi(s^p)$ to all ten printed digits for $p \in \{0.5, 0.9\}$ and $s \in \{1, 10\}$, for example -0.1823802136 at $p = 0.9$, $s = 10$. The zero sum (4.3.3) converges slowly to (4.3.2): at $p = 0.5$, $\tau = 100$ the sums over 30, 100 and 200 zeros are 2.74, 3.18, 3.35 against the exact 3.67 (units 10^{-4}), and at $p = 0.9$, $\tau = \gamma_1^{2/p}$ they are 3.504, 3.644, 3.696 against 3.797 (units 10^{-3}). On 148 points with $p \in \{0.5, 0.9, 0.99, 0.999\}$ and $10^{-4} \le \tau \le 10^8$ the density (4.3.2) is positive, as Theorem 4.3.13 and Corollary 4.3.8 guarantee.

Under RH the Thorin density has an explicit probabilistic kernel. Write $\mathcal{L}(Y)$ for the law of a random variable Y.

Corollary 4.3.14. *Let $L_p = Z_p/\tilde{Z}_p$ be the ratio of independent positive p stable variables, whose density $\frac{\sin p\pi}{\pi}\frac{t^{p-1}}{t^{2p}+2t^p\cos p\pi+1}$ is Lamperti's [136]; its power L_p^p is Bosch's variable T_α with $\alpha = p$. Under RH the Thorin measure of X_p is*

$$U_p = p\sum_\gamma \mathcal{L}\big(\gamma^{2/p}L_p\big),$$

and $X_p \overset{d}{=} S_p\,X^{1/p}$, where $X = \sum_\gamma e_\gamma/\gamma^2$ is the clock variable of Proposition 1.2.1, with independent standard exponential variables e_γ, and S_p is a positive p stable variable independent of X with $\mathbb{E}e^{-\lambda S_p} = e^{-\lambda^p}$. At $p = 1/2$, $L_{1/2} \overset{d}{=} C^2$ for a standard Cauchy variable C, and $U_{1/2} = \frac{1}{2}\sum_\gamma \mathcal{L}(\gamma^4 C^2)$. As $p \uparrow 1$, $L_p \to 1$ and $U_p \to \sum_\gamma \delta_{\gamma^2}$ vaguely, the Thorin measure of φ itself.

Proof. For real γ the term of (4.3.3) equals $\frac{p}{\pi}\frac{\gamma^2\tau^{p-1}\sin p\pi}{\tau^{2p}+2\gamma^2\tau^p\cos p\pi+\gamma^4}$; substituting $\tau = \gamma^{2/p}t$ turns it into p times Lamperti's density in t. The subordination identity is $\mathbb{E}e^{-sS_pY} = \mathbb{E}e^{-s^pY^p}$ with $Y = X^{1/p}$. □

Bosch's variable is therefore the kernel of the Thorin measure of $\varphi(s^p)$: under RH each zero contributes p times the law of $\gamma^{2/p}L_p$, a scaled pth root of Bosch's variable. For the single factor $(1+s^p)^{-1}$, the Laplace transform of a Mittag Leffler variable, the proof of Corollary 4.3.14 with the single zero $\gamma = 1$ gives the classical fact that the Thorin measure is p times Lamperti's law. The exponent p measures angular smoothing. At $p = 1/2$ each zero contributes a scaled squared Cauchy variable; as p increases to 1 the law of L_p concentrates at 1 and the Thorin measure sharpens to the point masses $\sum_\gamma \delta_{\gamma^2}$. Formula (4.3.2) makes sense whatever the zeros are, and RH is the statement that it stays nonnegative through the whole sharpening (Question 4.3.17 and the argument after it). Table 4.3 sets Bosch's family and the reciprocal Xi function side by side.

	Bosch	Riemann
function	$x^{q-1}/(x^{2q} + 2x^q \cos \pi\alpha + 1)$	$\varphi(x^p)$
singularities of h at	$x^q = e^{\pm i(\pi-\pi\alpha)}$	$x^p = \varrho^2 e^{\pm i(\pi-2\delta)}$
angle parameter	α	$\alpha_\xi = 2\delta_*/\pi$
HCM if and only if	$\alpha \le \frac{1}{2}$ and $q \le 1-\alpha$	$\lvert p\rvert \le 1-\alpha_\xi$
Thorin kernel (under RH)		$p\,\mathcal{L}(\gamma^{2/p}Z_p/\tilde{Z}_p)$ per zero
threshold mechanism	zeros of the denominator reach the cut	poles of φ reach the cut
extremal case	$\alpha = 0$: HCM iff $q \le 1$	$\alpha_\xi = 0$, which is RH

Table 4.3: Bosch's family and the reciprocal Xi function $\varphi(s) = \xi(\frac{1}{2})/\xi(\frac{1}{2} + \sqrt{s})$. For the Bosch column $q = 1/|\beta|$; for the Riemann column $p = 1/|\beta|$ and $\varrho e^{i\delta}$ runs over the zeros of Ξ.

Corollary 4.3.15. *RH holds if and only if X_p exists, that is $\varphi(s^p)$ is a GGC transform, for every $p \in (0,1)$. Unconditionally X_p exists for $0 < p \le 1 - 1/(\pi T_0)$, in particular for $0 < p \le 1 - 1.1 \times 10^{-13}$, with the Thorin density* (4.3.2).

Proof. If RH fails, $p_* < 1$, and for $p \in (p_*, 1)$ the function $\varphi(s^p)$ has a pole in Ω, which no GGC transform can have, since $\exp(-bs - \int \log(1+s/t)U(dt))$ is analytic and zero free on Ω. The rest is Theorem 4.3.13 with Corollary 4.3.8, which gives $1 - 1/(\pi T_0) < p_*$. □

The case $p = 1/2$ is the most concrete. Without any hypothesis,

$$\mathbb{E}e^{-sX_{1/2}} = \frac{\xi(1/2)}{\xi(1/2 + s^{1/4})}$$

defines a GGC variable $X_{1/2}$ with Thorin measure given by (4.3.2). Under RH, $X_{1/2} \stackrel{d}{=} S_{1/2}X^2$ and $U_{1/2} = \frac{1}{2}\sum_\gamma \mathcal{L}(\gamma^4 C^2)$.

Question 4.3.16. Construct $X_{1/2}$ from Brownian motion, Bessel processes or the Biane, Pitman and Yor variables [28] without reference to the zeros. A construction that survived $p \uparrow 1$ would prove RH by Corollary 4.3.15.

4.3.5 Positivity of the Thorin density from the primes

The method reads the threshold off the zeros, as Bosch reads his off the explicit roots $e^{\pm i(1-\alpha)\pi}$: the elasticity has nonnegative imaginary part termwise on the sector $0 < \arg s < \pi - 2\delta_*$, and no information about δ_* is produced. Formula (4.3.2) writes the Thorin density without the zeros, but Theorem 4.3.13 proves its positivity through them. A proof of RH along these lines must show $u_p \geq 0$ for every $p < 1$ directly from (4.3.2), for instance from the Pólya kernel, from the prime sum of Remark 4.3.18 or from a probabilistic model; this is the Thorin measure form of the reciprocal Xi programme of [198]. It cannot be done factor by factor: only the combination $(s-\frac{1}{4})\zeta(\frac{1}{2}+\sqrt{s})$ can enter an HCM or Pick argument (Section 4.2.4).

Question 4.3.17. Prove $u_p \geq 0$ for every $p \in (0,1)$ directly from (4.3.2), without locating the zeros.

The statement for all $p \in (0,1)$ is equivalent to RH. Under RH it holds by Theorem 4.3.13. Conversely, suppose RH fails, let $\gamma = \varrho e^{i\delta_*}$ be a zero of maximal angle, and let $p \in (p_*, 1)$; formula (4.3.2) still defines u_p, except at isolated poles, although it is no longer a Thorin density. At $\tau = \varrho^{2/p}$ the term of $\bar\gamma$ in (4.3.3) equals $-\frac{p}{2\pi\tau}\cot\frac{\pi(p-p_*)}{2}$, which tends to $-\infty$ as $p \downarrow p_*$. The sum of the other terms stays bounded, because the other points $-\gamma'^2$ keep a positive distance from $-\bar\gamma^2$ and the tail of the sum is $O(\sum|\gamma'|^{-2})$. So u_p takes negative values for p slightly above p_*.

Along the ray write $\vartheta = p\pi/2 \in (0,\pi/2)$, $t = \tau^{p/2}$, $z = te^{i\vartheta}$ and $w = \frac{1}{2}+z$, so that $u_p(\tau) = \frac{p}{2\pi\tau}\operatorname{Im}[z\,\frac{\xi'}{\xi}(w)]$ by (4.3.2), and split $\xi'/\xi = \Gamma_\xi + \zeta'/\zeta$, where

$$\Gamma_\xi(w) = \frac{1}{w} + \frac{1}{w-1} - \frac{\log\pi}{2} + \frac{1}{2}\psi\Big(\frac{w}{2}\Big)$$

is the logarithmic derivative of the factor $\frac{1}{2}w(w-1)\pi^{-w/2}\Gamma(w/2)$ of ξ, and $\psi = \Gamma'/\Gamma$ is the digamma function, not the function ψ of Theorem 4.2.4. The ray crosses the critical strip for $t < t_e = 1/(2\cos\vartheta)$ and then enters $\operatorname{Re} w > 1$, where ζ'/ζ is an absolutely convergent Dirichlet series.

Remark 4.3.18 (primes). For $t > t_e$, that is $\tau > (2\cos(p\pi/2))^{-2/p}$, we have $\operatorname{Re} w > 1$ and

$$u_p(\tau) = \frac{p}{2\pi\tau}\operatorname{Im}\Big[z\Big(\frac{1}{w}+\frac{1}{w-1}-\frac{\log\pi}{2}+\frac{1}{2}\psi\Big(\frac{w}{2}\Big)\Big)\Big] - \frac{p}{2\pi\tau}\sum_{n\geq 2}\frac{\Lambda(n)}{\sqrt{n}}\operatorname{Im}\big(z\,n^{-z}\big),$$

with Λ the von Mangoldt function. So for large τ the Thorin density of X_p is a gamma factor term minus an absolutely convergent prime sum. The region of absolute convergence escapes to infinity as $p \uparrow 1$; this is the sense in which the difficulty of RH sits next to the cut.

Far enough beyond the strip, Question 4.3.17 has an unconditional answer.

PROPOSITION 4.3.19. *Let* $\operatorname{Re} w > 1$ *and* $\sigma = \operatorname{Re} w$. *Then*

$$\frac{2\pi\tau}{p}\,u_p(\tau) \;\geq\; m_p(t) := \operatorname{Im}\Big[z\Big(\frac{1}{2w} + \frac{1}{w-1} + \frac{1}{2}\log\frac{w}{2\pi}\Big)\Big] - \frac{t}{6\sigma^2} - t\Big(-\frac{\zeta'}{\zeta}\Big)(\sigma). \tag{4.3.4}$$

Every term of m_p *is explicit, no zero of* ζ *enters, and* $m_p(t) = \frac{t}{2}\sin\vartheta\log t\,(1+o(1))$ *as* $t \to \infty$. *Consequently for each* $p \in (0,1)$ *there is* $\tau_1(p) < \infty$ *with* $u_p(\tau) > 0$ *for all* $\tau > \tau_1(p)$.

Proof. Binet's formula [177, (5.9.13)] gives, for $\operatorname{Re} x > 0$, $\psi(x) = \log x - \frac{1}{2x} - \int_0^\infty b(u)e^{-xu}du$ with $b(u) = \frac{1}{e^u-1} - \frac{1}{u} + \frac{1}{2} \in [0, u/12]$; the bounds follow from the partial fraction expansion $b(u) = \sum_{k\geq1} 2u/(u^2+4\pi^2k^2)$ and $\sum_k k^{-2} = \pi^2/6$. Hence $\psi(x) = \log x - \frac{1}{2x} - R$ with $|R| \leq \frac{1}{12(\operatorname{Re} x)^2}$. With $x = w/2$ this turns $\Gamma_\xi(w)$ into $\frac{1}{2w} + \frac{1}{w-1} + \frac{1}{2}\log\frac{w}{2\pi} - \frac{R}{2}$ with $|R| \leq \frac{1}{3\sigma^2}$, so $\operatorname{Im}[z\Gamma_\xi(w)]$ is at least the first term of m_p minus $t/(6\sigma^2)$. For the prime part, $\operatorname{Im}[z\,\zeta'/\zeta(w)] = -\sum\Lambda(n)\operatorname{Im}(zn^{-w}) \geq -t\sum\Lambda(n)n^{-\sigma}$. The asymptotic follows from $\operatorname{Im}[z\log z] = t(\sin\vartheta\log t + \vartheta\cos\vartheta)$ and $-\zeta'/\zeta(\sigma) \to 0$ as $\sigma \to \infty$. □

Table 4.4 gives the point beyond which $m_p > 0$, computed on a logarithmic grid of 50 points per decade from the strip exit t_e to about $10^{14}t_e$, with $-\zeta'/\zeta(\sigma)$ evaluated directly. The entry t_1 is the last grid point at which $m_p \leq 0$, so the sign change lies within one grid step, a factor $10^{1/50}$, above it. These values are computed, not proved: nothing controls m_p between grid points or beyond the last one, and a proof would need interval arithmetic on the grid cells and a monotonicity argument beyond them.

The prime term of m_p can be made elementary, because $-\zeta'/\zeta(\sigma) < 1/(\sigma-1)$ for every $\sigma > 1$. Delange [66, Théorème] proves the sharper bound $-\zeta'/\zeta(\sigma) < 1/(\sigma-1) - 1/(2\sigma^2)$; the weaker one has a short probabilistic proof. Expanding $(e^v-1)^{-1} = \sum_{n\geq1} e^{-nv}$ gives $\Gamma(\sigma)\zeta(\sigma) = \int_0^\infty v^{\sigma-1}(e^v-1)^{-1}\,dv$, so $(\sigma-1)\zeta(\sigma) = \Gamma(\sigma)\zeta(\sigma)/\Gamma(\sigma-1)$ is the mean of $v/(1-e^{-v})$ under the gamma law of shape $\sigma-1$, and $1/(\sigma-1) + \zeta'/\zeta(\sigma)$, the derivative of its logarithm, is the covariance of $v/(1-e^{-v})$ and $\log v$ under that law divided by the mean; the covariance is positive because both functions increase. Since $-\zeta'/\zeta(\sigma)$ enters m_p with the factor $-t < 0$, replacing it by $1/(\sigma-1)$ lowers m_p, and (4.3.4) still holds. The script also checks the inequality at 80 points with $10^{-5.9} \leq \sigma - 1 \leq 10^2$.

REMARK 4.3.20. The ratio t_1/t_e tends to 1. Near $\vartheta = \pi/2$ write $\sigma = 1 + x$, so $t = t_e(1+2x)$. The dominant terms of m_p/t are $\frac{1}{2}\log\frac{t}{2\pi}$ and $-\frac{1}{x}$, so positivity begins at $x \approx 2/\log(t_e/2\pi)$, that is $t_1(p) = t_e(p)\big(1 + \frac{4+o(1)}{\log(t_e/2\pi)}\big)$. At $p = 0.9999$ the leading term predicts a ratio of about 1.64; the sign change of m_p is at ratio 1.53 (the table lists the grid point 1.51 just below it), and the difference is the $o(1)$ term, which is of order $1/\log t_e$.

p	strip exit $t_e = \frac{1}{2\cos\vartheta}$	t_1	t_1/t_e	$\tau_1 = t_1^{2/p}$
0.5	0.707	4.07	5.75	274
0.9	3.20	11.1	3.47	210
0.99	31.8	69.6	2.19	5.29×10^3
0.999	318	553	1.74	3.10×10^5
0.9999	3183	4818	1.51	2.33×10^7

Table 4.4: Where the lower bound (4.3.4) becomes positive. On a logarithmic grid from t_e to about $10^{14}t_e$, $m_p(t) > 0$ at every grid point $t > t_1$; the ray reaches t_1 at $\tau_1 = t_1^{2/p}$. The entries are computed, not proved.

So Question 4.3.17 reduces, for each p, to the segment $|z| \le t_1(p)$ of the ray, which is asymptotically the part inside the critical strip. That part has length $t_e = 1/(2\cos(p\pi/2)) \approx 1/(\pi(1-p))$, so heights and exponents correspond through $1 - p \approx 1/(\pi T)$, which is the relation in Corollary 4.3.8 between the verification height T_0 and the unconditional range $p \le 1 - 1/(\pi T_0)$. The Bosch exponent is a height in disguise.

4.3.6 The density level and natural boundaries

The results so far concern Laplace transforms. A variable with an HCM density is a GGC and so has an HCM Laplace transform ((F5) and (F6) of Section 4.2.1); at the level of densities the Cauchy side can be solved completely, and it shows that the converse fails. Under RH the clock variable $X = \sum_\gamma e_\gamma/\gamma^2$ is the GGC with transform φ. Its Cauchy counterpart is $X_C = \sum_k e_k/(2k-1)^2$, with transform $\operatorname{sech}(\pi\sqrt{s}/2)$; it is $\frac{\pi^2}{8}$ times the exit time of Brownian motion from $(-1, 1)$ (Section 4.2.2).

Theorem 4.3.21. *The density of X_C is*

$$f_C(t) = \frac{4}{\pi}\sum_{k\ge1}(-1)^{k-1}(2k-1)e^{-(2k-1)^2 t} = \frac{4}{\pi}\,\eta\Big(\frac{4it}{\pi}\Big)^3, \qquad t > 0,$$

with η the Dedekind eta function. The line $\operatorname{Re} t = 0$ is a natural boundary of f_C. Hence X_C is a GGC whose Laplace transform is HCM, but the density of X_C is not HCM.

Proof. The residue of $\operatorname{sech}(\pi\sqrt{s}/2)$ at $s = -(2k-1)^2$ is $\frac{4}{\pi}(-1)^{k-1}(2k-1)$, which gives the series once the Bromwich contour is moved to the left across the circles $|s| = 4k^2$: for $s = 4k^2e^{i\theta}$, $|\cosh(\pi\sqrt{s}/2)|^2 = \sinh^2(\pi k\cos\frac{\theta}{2}) + \cos^2(\pi k\sin\frac{\theta}{2})$ is at least $\frac{1}{4}$ for every θ and at least $\sinh^2(\pi k/2)$ for $|\theta| \le 2\pi/3$, while $|e^{st}| \le e^{-2k^2t}$ for $2\pi/3 \le |\theta| \le \pi$, so for $t > 0$ the integrals over the arcs of these circles to the

left of the Bromwich line tend to 0. Jacobi's identity $\eta(z)^3 = \sum_{n\geq 0}(-1)^n(2n+1)e^{\pi i(2n+1)^2 z/4}$ [90, Theorem 357], at $z = 4it/\pi$, where $e^{\pi i(2k-1)^2 z/4} = e^{-(2k-1)^2 t}$, gives the eta form. As a modular form η has the real axis as natural boundary [219, Chapter 11, Section 1.4, and Chapter 1, Section 4.2], and the real axis in z is the line $\operatorname{Re} t = 0$. An HCM density continues analytically to Ω by (4.2.1), so f_C is not HCM. □

Numerically the series and the eta form agree to fifteen digits at $t = 0.05, 0.3, 1, 3$, and $\int e^{-2t} f_C = \operatorname{sech}(\pi/\sqrt{2}) = 0.214383752348$. The Cauchy side thus separates three levels (Propositions 4.2.2 and 4.2.3): the transform $\operatorname{sech}(\pi\sqrt{s}/2)$ is HCM, the density of X_C is not, because of its natural boundary, and the half Cauchy density is not either, because of its poles at $\pm i$, although its square is.

Question 4.3.22 (density level). Assume RH and that all zeros are simple. Does the density $f_X(t) = \sum_n A_n \gamma_n^2 e^{-\gamma_n^2 t}$ of X, with A_n as in (4.2.4), have $\operatorname{Re} t = 0$ as a natural boundary? If so, X is not HCM.

Pólya's gap theorem for Dirichlet series [212, 214] reduces Question 4.3.22 to two hypotheses on the zeros. The theorem says that if the exponents μ_n of a Dirichlet series $\sum_n c_n e^{-\mu_n t}$ have density $d = \lim n/\mu_n$ and satisfy $\inf_n(\mu_{n+1} - \mu_n) > 0$, then every segment of length greater than $2\pi d$ on the line of convergence carries a singularity. When $\limsup \log n/\mu_n = 0$ the abscissa of convergence equals $\limsup_n \log|c_n|/\mu_n$ [151].

Theorem 4.3.23. *Assume RH and that all zeros are simple, and assume*

(G) $\inf_n(\gamma_{n+1}^2 - \gamma_n^2) > 0$*;*

(D) $\log\bigl(1/|\zeta'(\frac{1}{2} + i\gamma_n)|\bigr) = o(\gamma_n^2)$.

Then for $t > 0$

$$f_X(t) = \sum_n c_n e^{-\gamma_n^2 t}, \qquad c_n = A_n\gamma_n^2 = -\frac{2\gamma_n\,\Xi(0)}{\Xi'(\gamma_n)},$$

the series has abscissa of convergence 0*, and* $\operatorname{Re} t = 0$ *is a natural boundary of* f_X*. In particular the density of* X *is not HCM.*

Proof. Residue expansion. The residue of $\varphi(s) = \Xi(0)/\Xi(i\sqrt{s})$ at $s = -\gamma_n^2$ is c_n as stated. Write f_X as a Bromwich integral and move the contour to the left across the arcs of the circles $|s| = T_k^2$ that lie to the left of the Bromwich line, where, for large k, $T_k \in [k, k+1]$ is chosen at distance at least $c/\log k$ from every ordinate, with a small absolute constant $c > 0$; this is possible because $[k-1, k+2]$ contains $O(\log k)$ ordinates [92]. Then $\zeta'/\zeta(\sigma + iy) = O(\log^2 k)$ uniformly in $-1 \leq \sigma \leq 2$ and $|y - T_k| \leq c/(2\log k)$ [255, Theorem 9.6(A) and Section 14.4], and integration

from $\sigma = 2$ gives $1/|\zeta(\sigma + iy)| \le \exp(O(\log^2 k))$ there. On the part of such an arc within angle $O(1/T_k)$ of the negative axis, $w = \frac{1}{2} + \sqrt{s}$ has $\frac{1}{2} \le \operatorname{Re} w \le 2$ and $|\operatorname{Im} w| = \sqrt{T_k^2 - (\operatorname{Re} w - \frac{1}{2})^2} = T_k + O(1/T_k)$, so for large k it lies in this band or in its reflection in the real axis, and Stirling's formula gives $1/|\Xi(i\sqrt{s})| \le \exp(O(T_k \log T_k))$, while $|e^{st}| \le e^{-tT_k^2/2}$. On the rest of the arc $\operatorname{Re} w \ge 2$, and Stirling's formula gives $|e^{st}\varphi(s)| \le \exp\bigl(tT_k^2 \cos\theta - \frac{T_k}{2} \cos\frac{\theta}{2} \log T_k + O(T_k)\bigr)$ for $s = T_k^2 e^{i\theta}$. For $t > 0$ both bounds tend to zero faster than any power of T_k, so the arcs contribute nothing in the limit and f_X equals the residue series.

Abscissa. The exponents γ_n^2 satisfy $\log n/\gamma_n^2 \to 0$ and have density zero. By Stirling, $|\Xi'(\gamma_n)| = |\xi'(\rho_n)| = \gamma_n^{O(1)} e^{-\pi\gamma_n/4} |\zeta'(\rho_n)|$, so $\log|c_n| = \frac{\pi}{4}\gamma_n + O(\log\gamma_n) + \log(1/|\zeta'(\rho_n)|)$. Since $|\zeta'(\rho_n)| \le \gamma_n^{O(1)}$ always, (D) gives $\log|c_n|/\gamma_n^2 \to 0$, so the abscissa of convergence is 0.

Natural boundary. By (G) and density zero, Pólya's theorem places a singularity in every segment of the imaginary axis. An HCM density continues analytically to Ω, which contains the imaginary axis minus the origin. □

The same argument, with $\mu_k = (2k-1)^2$, gaps $8k$ and $c_k = \frac{4}{\pi}(-1)^{k-1}(2k-1)$, gives a second proof of Theorem 4.3.21 that does not use modularity: by Pólya's gap theorem [212, 214] the line $\operatorname{Re} t = 0$ is a natural boundary of f_C.

Remark 4.3.24 (status of the hypotheses). Hypothesis (D) is far weaker than the conjectured polynomial lower bounds for $|\zeta'(\rho)|$. Hypothesis (G) asks $\gamma_{n+1} - \gamma_n \ge c_0/(\gamma_n + \gamma_{n+1})$ for some $c_0 > 0$, a gap of order $1/\gamma$ against a mean spacing of order $1/\log\gamma$. Under the GUE model the probability that a normalised gap is below x is of order x^3; with $x_n \asymp \log\gamma_n/\gamma_n$ the expected number of violations is $\sum_n (\log\gamma_n/\gamma_n)^3 < \infty$, so by the Borel Cantelli heuristic (G) fails at most finitely often, and then holds with a smaller c_0 if the zeros are simple.

Numerically, for the first 200 zeros the coefficients c_n computed from $\Xi'(\gamma_n)$ alternate in sign without exception; $c_1 = 10163.5$, so $c_1/\gamma_1^2 = 50.87$, in line with $A_1 \approx 50.9$ in (4.2.4); $\log|c_n|/\gamma_n^2$ equals 0.0462, 0.0144, 0.00524, 0.00322, 0.00195 at $n = 1, 10, 50, 100, 200$, tracking $\pi/(4\gamma_n)$; and the smallest gap $\gamma_{n+1}^2 - \gamma_n^2$ is 159, at $n = 4$. The c_n are the residues of the spectral expansion of the clock density in Section 8.1, written R_k there, and Theorem 4.3.23 describes, under its hypotheses, why that expansion cannot be continued past $t = 0$. The residue series with 200 zeros matches a Talbot inversion of φ to all ten printed digits at $t = 0.02$, where both give 80.10419773, and to nine digits at $t = 0.005$, where the series gives $6.537917064 \times 10^{-24}$ (Appendix D). As t decreases, the alternating coefficients, of size about $e^{\pi\gamma_n/4}$, cancel more and more: at $t = 0.003$ the terms reach 2.7×10^{20} while the density is about 3.3×10^{-72}, so about 92 digits cancel, more than the 90 digits carried by that check, whose values at $t = 0.003$ are lost to rounding error. The size of the terms and the density there come from a separate computation at 150 digits, in which the residue series with 250 zeros and a Talbot inversion both

give $3.32007507753 \times 10^{-72}$. This cancellation is the numerical counterpart of the natural boundary.

For small p the density of X_p escapes this obstruction. The transform $\varphi(s^p)$ is analytic on the wider sector $|\arg s| < (\pi - 2\delta_*)/p$ and, by Stirling's formula applied to $\xi(\frac{1}{2} + z)$ for $|\arg z| < \pi/2$, decays along every ray strictly inside it. The inverse Laplace transform then continues the density of X_p to the sector $|\arg t| < (\pi - 2\delta_*)/p - \pi/2$, which contains Ω when $p \leq 2p_*/3$. So the natural boundary of Theorem 4.3.23 disappears for small p.

Question 4.3.25. Determine $\{p : X_p \text{ is HCM}\}$, and the same set for the Cauchy family $X_{C,p}$ with transform $\operatorname{sech}(\pi s^{p/2}/2)$.

Two facts constrain the answer. Under RH, $X_p \stackrel{d}{=} S_p X^{1/p}$, and positive p stable variables are HCM exactly for $p \leq 1/2$, a conjecture of Bondesson [35, Sections 5.7, 5.8 and 7.2] proved by Bosch and Simon [38, p. 32]. Since HCM is preserved under products of independent variables and under powers of modulus at least one ((F1) and (F2) of Section 4.2.1), HCM of X would give HCM of X_p for $p \leq 1/2$; but Theorem 4.3.23 shows that X is not HCM under (G) and (D), so this route is closed unless those hypotheses fail, and one must work with X_p directly. The elasticity criterion reduces Question 4.3.25 to the absence of zeros of the density f of X_p in Ω and the Pick property of $-tf'(t)/f(t)$; by the continuation argument before Question 4.3.25, f is at least analytic on Ω for small p. A numerical scan of this elasticity for $X_{C,1/2}$ would be the first test.

4.3.7 The Mellin variable W and the first gap

Question 4.2.7 asks for which β the density of W^β is HCM; on the Cauchy side the answer is $|\beta| \geq 2$. Let $\chi(t) = \xi(\frac{1}{2})/\xi(\frac{1}{2} + t) = \prod_\gamma (1 + t^2/\gamma^2)^{-1}$ and let $g(y) = \frac{1}{\pi}\int_0^\infty \chi(t)\cos(ty)\,dt$ be the function g of Section 4.2.2, which is entire because $\chi(t) = \exp(-\frac{1}{2}|t|\log|t|(1+o(1)))$. By the van Dantzig pair of Section 1.2, recalled in Section 4.2.2, χ is a characteristic function without RH: the clock variable T exists unconditionally [200, Theorem A], and if Z is a standard normal variable independent of T then $\mathbb{E}e^{itZ\sqrt{2T}} = \mathbb{E}e^{-t^2T} = \chi(t)$. So g is the density of the normal scale mixture $Z\sqrt{2T}$, and $W = \exp(Z\sqrt{2T})$ exists without any hypothesis; Section 10.1 takes up these normal scale mixtures at every basepoint. By the elasticity criterion, HCM of W^β becomes a sign condition on $-g'/g$ in a strip (Proposition 4.3.27), and since g is entire no natural boundary stands in the way.

4.3.7.1 A lower bound for g

Theorem 4.3.26 proves the existence of W a second time, independently of [200, Theorem A], and compares g with the density g_R of the sum over the real zeros.

THEOREM 4.3.26. *Unconditionally, by the verification of RH to height T_0 [196], $g(y) \geq 0.9998\, g_R(y) > 0$ for all real y, where g_R is the density of $\sum_{\gamma\ \mathrm{real}} L_\gamma/\gamma$. Hence χ is a characteristic function and there is a positive random variable W with $\mathbb{E}W^{it} = \chi(t)$ and $\mathbb{E}W^z = \Xi(0)/\Xi(z)$ for $|\operatorname{Re} z| < \gamma_1$.*

Proof. Split $\chi = \chi_R \chi_{\mathrm{off}}$ into the products over real zeros and over conjugate pairs of nonreal zeros. For a pair $\gamma = \varrho e^{i\delta} = a + ib$ with $a, b > 0$, the factor $(1 + t^2/\gamma^2)^{-1}(1 + t^2/\bar\gamma^2)^{-1}$ is the Fourier transform of the even kernel

$$k_\gamma(x) = \frac{\varrho}{2\sin 2\delta}\, e^{-a|x|} \sin(b|x| + \delta),$$

of total mass one, positive for $|x| < (\pi - \delta)/b$ and bounded by $\frac{\varrho}{2\sin 2\delta} e^{-a|x|}$. List the pairs as $\gamma^{(1)}, \gamma^{(2)}, \dots$ in order of $\operatorname{Re}\gamma$, write $k_j = k_{\gamma^{(j)}}$, and let χ_m be the product of the first m pair factors, $\nu_m = k_1 * \cdots * k_m$ and $g_m = g_R * \nu_m$, the function with Fourier transform $\chi_R \chi_m$. We bound g_m uniformly in m and let m grow at the end; if there are finitely many pairs, m stops at their number, and if there are none, $g = g_R$.

Comparison. $g_R = \ell * \tilde g_R$, where $\ell(z) = \frac{\gamma_1}{2} e^{-\gamma_1|z|}$ is the density of L_{γ_1}/γ_1 and $\tilde g_R$ is the density of the sum over the other real zeros, so $e^{-\gamma_1|x|} g_R(y) \leq g_R(y - x) \leq e^{\gamma_1|x|} g_R(y)$. Writing $\nu_m = \nu_m^+ - \nu_m^-$,

$$g_m(y) \geq g_R(y)\Big(\int e^{-\gamma_1|x|}\nu_m(x)dx - 2\int \cosh(\gamma_1 x)\,\nu_m^-(x)dx\Big) =: g_R(y)(A_m - 2B_m).$$

The term A_m. By Parseval, $A_m = \frac{1}{\pi}\int \chi_m(t)\frac{\gamma_1}{\gamma_1^2 + t^2}dt = \mathbb{E}\,\chi_m(\gamma_1 C)$ with C a standard Cauchy variable. For $s \geq 0$ each pair factor is $|1 + s/\gamma^2|^{-2} \geq (1 + s/\varrho^2)^{-2} \geq e^{-2s/\varrho^2}$, so $A_m \geq \mathbb{E}e^{-\kappa C^2} = e^\kappa \operatorname{erfc}(\sqrt\kappa)$ with $\kappa = 2\gamma_1^2\Sigma_{\mathrm{off}}$ and $\Sigma_{\mathrm{off}} = \sum_{\mathrm{off}} |\gamma|^{-2}$. All nonreal zeros have $\operatorname{Re}\gamma > T_0 = 3 \times 10^{12}$ [196], so with $N(T) \leq T\log T$ for $T \geq T_0$, $\Sigma_{\mathrm{off}} \leq 2\int_{T_0}^\infty N(T)T^{-3}dT \leq 2(\log T_0 + 1)/T_0 < 2 \times 10^{-11}$, $\kappa < 8 \times 10^{-9}$ and $A_m > 0.99989$.

The term B_m. For even signed measures $\int \cosh(\gamma_1 x)\, d(|\mu| * |\nu|) = \int \cosh(\gamma_1 x) d|\mu| \cdot \int \cosh(\gamma_1 x) d|\nu|$ exactly, and $(\mu * \nu)^- \leq \mu^+ * \nu^- + \mu^- * \nu^+$. Hence, with $M(\cdot) = \int \cosh(\gamma_1 x)\, d(\cdot)$ and since $M(|k_\gamma|) \geq M(k_\gamma) = |1 - \gamma_1^2/\gamma^2|^{-2} \geq 1$,

$$B_m \leq \sum_{j \leq m} M(k_j^-) \prod_{i \leq m,\ i \neq j} M(|k_i|) \leq SP, \qquad S = \sum_\gamma M(k_\gamma^-), \quad P = \prod_\gamma M(|k_\gamma|),$$

with the sum and the product over all pairs. Here k_γ^- lives on $|x| \geq (\pi - \delta)/b$ and $M(k_\gamma^-) \leq \frac{\varrho}{\sin 2\delta\,(a - \gamma_1)} \exp\big(-(a - \gamma_1)\frac{\pi - \delta}{b}\big)$. As $\varrho/\sin 2\delta = \varrho^3/(2ab)$, this bound is $\frac{\varrho^3}{2a(a-\gamma_1)}\, b^{-1} e^{-K/b}$ with $K = (a - \gamma_1)(\pi - \delta)$. For fixed a it increases with b: ϱ increases, K decreases because $\delta = \arctan(b/a)$ increases, and $b \mapsto b^{-1}e^{-K/b}$ increases on $(0, K)$. So, as $b < \frac12$, it is at most

$$F(a) = \frac{(a^2 + \frac14)^{3/2}}{a(a - \gamma_1)} \exp\Big(-2(a - \gamma_1)\big(\pi - \arctan\tfrac{1}{2a}\big)\Big),$$

and $F(T_0) < \exp(-1.88 \times 10^{13})$, while $-(\log F)' > 6$ on $[T_0, \infty)$. Every pair has $a > T_0$, so Stieltjes integration against $N(T) \le T \log T$ gives

$$S \le F(T_0)\Big(T_0 \log T_0 + \frac{\log T_0 + 1}{6} + \frac{1}{36T_0}\Big) < \exp(-1.8 \times 10^{13}).$$

Moreover $M(|k_\gamma|) = M(k_\gamma) + 2M(k_\gamma^-)$ and $M(k_\gamma) \le \exp(3\gamma_1^2/\varrho^2)$, so $\log P \le 3\gamma_1^2\Sigma_{\text{off}} + 2S < 1.2 \times 10^{-8}$. So $A_m - 2B_m > 0.9998$ for every m.

Limit. For real t each pair factor lies in $(0, 1]$, and χ_R is integrable because it contains the factor $(1 + t^2/\gamma_1^2)^{-1}$. By dominated convergence in the Fourier inversion integral, $g_m \to g$ uniformly on $\mathbb{R}$, so $g \ge 0.9998\, g_R$. The comparison also gives $g_m(y) \le g_R(y) \int e^{\gamma_1|x|}|\nu_m(x)|dx \le 2P\, g_R(y)$, since $e^{\gamma_1|x|} \le 2\cosh(\gamma_1 x)$ and $|\nu_m| \le |k_1| * \cdots * |k_m|$, so $g \le 2P\, g_R$. Hence g is integrable, its Fourier transform is χ, and $\chi(0) = 1$ makes g a probability density. Finally $\int e^{cy} g_R(y)dy = \prod_{\gamma \text{ real}}(1 - c^2/\gamma^2)^{-1} < \infty$ for $|c| < \gamma_1$, so $\int e^{c|y|} g(y)dy < \infty$ there, $\mathbb{E}W^z = \int e^{zy} g(y)dy$ is analytic on $|\operatorname{Re} z| < \gamma_1$, and it equals $\Xi(0)/\Xi(z)$ there because the two agree on the imaginary axis. □

The proof uses only that the zeros with $0 < \operatorname{Re}\gamma \le T_0$ are real. The pair kernel formula and the constants $2(\log T_0 + 1)/T_0$, κ, $e^\kappa \operatorname{erfc}(\sqrt{\kappa})$ and $\log F(T_0)$ of the proof were also checked numerically (Appendix D).

4.3.7.2 Which powers of W are HCM

For real $\beta \ne 0$ the density of W^β is $x^{-1} g(\log x/\beta)/|\beta|$. With $y = \log x$, its elasticity is $1 - \frac{1}{\beta}\frac{g'}{g}(y/\beta)$, and $x \in \mathbb{C}^+$ corresponds to $0 < \operatorname{Im} y < \pi$; for $\beta < 0$ the point $u = y/\beta$ then lies in the lower half of the strip $|\operatorname{Im} u| < \pi/|\beta|$, where the factor $\frac{1}{\beta} < 0$ reverses the sign condition, and $g(\bar{u}) = \overline{g(u)}$, as g is real on $\mathbb{R}$, moves it to the upper half. Theorem 4.3.2 therefore gives the following.

Proposition 4.3.27. *Let $\beta \in \mathbb{R} \setminus \{0\}$. Then W^β is HCM if and only if g has no zeros in the strip $|\operatorname{Im} u| < \pi/|\beta|$ and $\operatorname{Im}\big(-g'(u)/g(u)\big) \ge 0$ for $0 < \operatorname{Im} u < \pi/|\beta|$.*

As $\operatorname{Im} u \downarrow 0$ the condition requires $(\log g)'' \le 0$ on $\mathbb{R}$, that is, log concavity of the density g of $\log W$; numerically $(\log g)'' \le -2.2$ on $[0, 1]$. On the Cauchy side $g_C = 1/(\pi \cosh u)$, $-g_C'/g_C = \tanh u$, which is Pick on $0 < \operatorname{Im} u < \pi/2$ with a pole at $i\pi/2$; Proposition 4.3.27 gives $|\beta| \ge 2$, Bosch's theorem once more.

Theorem 4.3.28. *Assume RH, simple zeros, and $\log(1/|\zeta'(\rho_n)|) = O(\gamma_n)$. Then $u_0 = \frac{\pi}{4} + \limsup_n \gamma_n^{-1} \log(1/|\zeta'(\rho_n)|)$ is finite, and for $\operatorname{Re} u > u_0$*

$$g(u) = -\Xi(0) \sum_n \frac{e^{-\gamma_n u}}{\Xi'(\gamma_n)} = \sum_n b_n e^{-\gamma_n u}, \qquad b_1 > 0 > b_2,$$

and if W^β is HCM then $|\beta| \ge \gamma_2 - \gamma_1 = 6.8873\ldots$.

Proof. The expansion is the residue computation for the Fourier integral $g(u) = \frac{1}{2\pi}\int \chi(t)e^{-itu}\,dt$ closed in the lower half plane. We move the line of integration to $\operatorname{Im} t = -T_k$, where $T_k \in [k, k+1]$ are heights at which $1/\zeta(\sigma \pm iT_k) = T_k^{O(1)}$ uniformly in $-1 \le \sigma \le 2$ [255, Theorem 9.7]; by Stirling's formula and the functional equation the moved integral is $O(e^{(\pi/4-\operatorname{Re} u)T_k + O(\log T_k)})$, which tends to 0 for $\operatorname{Re} u > \pi/4$, so there $g(u)$ is the limit of the sums of the residue terms over $\gamma_n < T_k$. By Stirling, $\log|b_n| = \frac{\pi}{4}\gamma_n + \log(1/|\zeta'(\rho_n)|) + O(\log\gamma_n)$, so the series converges absolutely for $\operatorname{Re} u > u_0$, and $u_0 \ge \pi/4$ because $|\zeta'(\rho_n)| \le \gamma_n^{O(1)}$, so its sum is $g(u)$ there (conjecturally $u_0 = \pi/4$; numerically the series is accurate at $u = 1$); the signs come from $b_n = \frac{\gamma_n}{2}A_n$. Put $\Delta = \gamma_2 - \gamma_1$. As $\operatorname{Re} u \to \infty$ with $\operatorname{Im} u = v$ fixed,

$$\operatorname{Im}\Big(-\frac{g'}{g}\Big)(x+iv) = \Delta\Big|\frac{b_2}{b_1}\Big|e^{-\Delta x}\sin(\Delta v)\,\big(1+O(e^{-cx})\big), \qquad c>0.$$

For $\pi/\Delta < v < 2\pi/\Delta$ this is negative for large x. If $|\beta| < \Delta$ the strip $0 < \operatorname{Im} u < \pi/|\beta|$ contains such v, and Proposition 4.3.27 fails. □

Question 4.3.29 (first gap conjecture). Is W^β HCM exactly when $|\beta| \ge \gamma_2 - \gamma_1$? On the Cauchy side the analogous statement, with frequencies $2k-1$, is Bosch's theorem at $\alpha = 1/2$: $3 - 1 = 2$.

With b_n computed from $\Xi'(\gamma_n)$ ($b_1 = 359.52$, $b_2 = -28007$, $b_3 = 392918$) and g computed by quadrature near the origin (it matches the series to eleven digits at $u = 1$ and $u = 1.2 + 0.3i$, and gives $g(0) = 1.8994210510$), a scan of 4840 points with $0 \le x \le 6$ and $0 < v < 0.99\,v_*$, where $v_* = \pi/\Delta = 0.45614$, found no value of $\operatorname{Im}(-g'/g)$ below -1.3×10^{-15}, which is rounding; in 40 digit arithmetic the minimum over $1.1 \le x \le 9.1$ at $v = 0.999\,v_*$ is $+1.0 \times 10^{-27}$. Just above the threshold the sign flips: the minimum is -6.5×10^{-8} at $v = 1.001\,v_*$, -3.6×10^{-5} at $1.01\,v_*$ and -2.1×10^{-2} at $1.1\,v_*$, so numerically the threshold is sharp to 0.1 per cent. The leading term of the asymptotic formula reproduces $\operatorname{Im}(-g'/g)$ at $x = 8$, $v = v_*/2$ to eight digits.

The Mellin side thus has a Bosch type answer that is conjecturally exact: the HCM exponent of W is the first gap between zeros. Proving sufficiency requires the Pick property of $-g'/g$ in the full strip $0 < \operatorname{Im} u < \pi/\Delta$, which near the origin is a statement about the Pólya type kernel g that does not reduce to the zeros.

4.3.7.3 Solvable models

To see whether the first gap rule is a coincidence of the half Cauchy case, we test it on a family in which everything is explicit. The logistic variable has characteristic function $\pi t/\sinh(\pi t) = \prod_{k\ge1}(1 + t^2/k^2)^{-1}$; it is $\log W$ for $W = U/(1-U)$ with U uniform, and W has density $(1+x)^{-2}$. Removing a finite set R of frequencies multiplies the characteristic function by $\prod_{k\in R}(1 + t^2/k^2)$, which gives

$g = \prod_{k\in R}(1 - D^2/k^2)L$ with $D = d/dy$ and $L(y) = \frac{1}{4}\operatorname{sech}^2(y/2)$; this is still a probability density, because the remaining factors are Laplace characteristic functions. For each model we locate numerically, by bisection over a grid in $0 \le x \le 16$, the largest strip on which $\operatorname{Im}(-g'/g) \ge 0$; by Theorem 4.3.2, as in Proposition 4.3.27, its width h_* gives the HCM threshold $\beta_* = \pi/h_*$. Table 4.5 lists the results.

frequencies	gaps	first gap	largest gap	$\beta_* = \pi/h_*$
$1, 2, 3, \ldots$	$1, 1, 1, \ldots$	1	1	1.000
$1, 3, 5, \ldots$ (half Cauchy)	$2, 2, 2, \ldots$	2	2	2 (Bosch)
$1, 3, 4, 5, \ldots$	$2, 1, 1, \ldots$	2	2	2.000
$1, 3, 5, 6, \ldots$	$2, 2, 1, \ldots$	2	2	2.000
$1, 3, 4, 6, \ldots$	$2, 1, 2, 1, \ldots$	2	2	2.000
$1, 4, 5, 6, \ldots$	$3, 1, 1, \ldots$	3	3	3.000
$1, 4, 5, 7, \ldots$	$3, 1, 2, 1, \ldots$	3	3	3.000
$1, 4, 7, 8, \ldots$	$3, 3, 1, \ldots$	3	3	3.000
$1, 5, 6, 7, \ldots$	$4, 1, 1, \ldots$	4	4	4.000
$1, 2, 4, 5, \ldots$	$1, 2, 1, \ldots$	1	2	1.269
$1, 2, 3, 6, \ldots$	$1, 1, 3, 1, \ldots$	1	3	1.317
$1, 2, 5, 6, \ldots$	$1, 3, 1, \ldots$	1	3	1.483

Table 4.5: HCM threshold for powers of W in logistic models with frequencies removed, computed as $\beta_* = \pi/h_*$ from the width h_* of the largest strip on which $\operatorname{Im}(-g'/g) \ge 0$. The half Cauchy row is Bosch's theorem, not a computation.

Whenever the first gap is at least as large as every later gap, the computed threshold equals the first gap, to the resolution of the computation. When some later gap exceeds the first, the threshold lies strictly between the first gap and the largest gap, and depends on where the large gap sits. The asymptotic argument of Theorem 4.3.28 explains the lower bound in all cases, and in every model of Table 4.5 the bound is attained exactly when the first gap is the largest. For the Riemann zeros the first gap $\gamma_2 - \gamma_1 = 6.8873$ exceeds every later gap among the first 1000 zeros (the next largest is $\gamma_4 - \gamma_3 = 5.414$). The explicit form of the Riemann von Mangoldt formula of Hasanalizade, Shen and Wong, $|N(T) - \frac{T}{2\pi}\log\frac{T}{2\pi e} - \frac{7}{8}| \le 2.5167 + \frac{1}{50T}$ for $e \le T \le X_0 = 30\,610\,046\,000$ and $\le 0.1095\log T + 0.2042\log\log T + 3.0305 + \frac{1}{50T}$ for $T > X_0$ [92, (1.7), (2.7) and Corollary 1.4], shows that every interval $(T, T + 6.887]$ with $T \ge 1000$ contains an ordinate: the lower bound it gives for the number of ordinates there exceeds 0.52 at $T = 1000$, increases with T while $T + 6.887 \le X_0$, and exceeds 11 beyond. So every gap that starts above height 1000 is at most 6.887, less than the first, and every gap that starts below it is a gap between two of the first 650 zeros, since $N(1000) = 649$. The first gap therefore exceeds every later gap, and the Riemann zeros are in the

first gap dominant regime. This refines Question 4.3.29.

Question 4.3.30 (first gap conjecture, refined). Let $Y = \sum_n L_n/\lambda_n$ with independent standard Laplace variables L_n and $0 < \lambda_1 < \lambda_2 < \cdots$, and let $W = e^Y$. If the characteristic function of Y decays faster than every exponential, or if the first gap $\lambda_2 - \lambda_1$ is at least every later gap, is the HCM threshold of W^β exactly $\lambda_2 - \lambda_1$? In particular, is it $\gamma_2 - \gamma_1$ for the Riemann variable?

4.3.8 Where the difficulty lies

Bosch's theorem cannot be the input to a proof that the reciprocal Xi function is HCM, because for a product over the zeros HCM is equivalent to the reality of the zeros (Proposition 4.2.6), and Section 11.1 finds that every GGC criterion in the books of Sato and Bondesson is, at the centre, equivalent to RH. Bosch's method does transfer, and at each of three levels, Laplace transforms, the Mellin variable and densities, it isolates one concrete object on which the question turns.

At the level of transforms the object is the Thorin density u_p of (4.3.2). Its positivity for every $p < 1$ is equivalent to RH; the primes prove it on each ray beyond some point, that point is asymptotically where the ray leaves the critical strip, and what remains is a segment of length about $1/(\pi(1-p))$. The first direction is to prove positivity on that segment from the explicit formula for ξ'/ξ rather than from the zeros; computation alone extends the unconditional range $p \le 1 - 1/(\pi T_0)$ only as fast as the verification height T_0 grows.

At the level of the Mellin variable the object is the sign of $\operatorname{Im}(-g'/g)$ in the strip $0 < \operatorname{Im} u < \pi/(\gamma_2 - \gamma_1)$. The second direction is to prove the first gap conjecture, first in the logistic models of Table 4.5, where every quantity is elementary, and then for W, whose threshold the scan after Question 4.3.29 already locates to 0.1 per cent; the result would be an unconditional HCM statement about W whose threshold is a property of the first two zeros. Question 4.2.8, whether HCM of the density of W^β for a single β implies RH, remains open.

At the level of densities the object is the natural boundary. The third direction is to establish it for the clock variable without the hypotheses (G) and (D) of Theorem 4.3.23, which would show what replaces, on the Riemann side, the modularity that produces the natural boundary on the Cauchy side.

Section 5.1 approaches the Thorin functional $\sum_\gamma \delta_{\gamma^2}$ of the clock, its Thorin measure under RH, from the other end of Bondesson's hierarchy, through its Laplace transform, the heat trace $W(t) = \sum_\gamma e^{-\gamma^2 t}$.

4.4 Exercises

On Section 4.1.

4.1. Show that if X_N are GGCs converging in law to a proper limit, the Thorin measures converge vaguely on $(0, \infty)$, and use the laws Gamma(n, n) of

Section 1.3.1, which converge to the point mass at one, to show that Thorin mass escaping to infinity can reappear in the limit as drift.

4.2. For the sinh clock of Section 1.3.4 with one atom k^2 replaced by the conjugate pair $a, \bar{a}$, $a = -(\delta + ik)^2$ with $0 < \delta < \frac{1}{2}$, as in Proposition 1.2.3, construct the Thorin densities at resolution ε and show that their vague limit is the sinh measure with the atom k^2 removed.

4.3. Verify that $\varphi_{1/2}(uv)\varphi_{1/2}(u/v) = \prod_k a_k^2/(a_k^2 + a_k uw + u^2)$ for the clock, and that at $u = |a|$ an off line pair contributes the completely monotone factor $(w + 2\cos\arg a)^{-2}$.

4.4. Compute $(-1)^k \partial_w^k H_u(w)/H_u(w)$ for $k \le 6$ at $u = 1$ and $w = 3$ from the Pólya density and compare with Figure 4.2.

Chapter 5

The heat trace, the arithmetic and the limits of finite information

Chapter 4 worked at the top of the Bondesson hierarchy, with hyperbolic complete monotonicity, which for a Laplace transform is the GGC property; here we climb the hierarchy from the bottom. Every rung is a sign condition on the heat trace $W(t)$ of Section 1.2, the pairing of the Thorin functional with e^{-tz} (not the Mellin variable W of Section 4.3). The lower rungs are unconditional theorems, proved with a finite verification of the zeros; the top rung is RH. Section 5.1 develops this hierarchy of positivity, and Section 5.2 shows what such finite information cannot decide. Section 5.3 computes the Thorin functional from the primes, and Section 5.4 relates the errors of the Pólya route on a window of finite height to the spacings of the zeros, which it compares with random matrix theory.

Taken together, Chapters 3 to 5 show that on every window where the zeros are verified the computed Thorin measure converges to the RH measure at explicit rates, while above the computed range every finite computation of the kinds in this book remains consistent with a failure of the Thorin condition (Section 5.2). Three of their threads return in Chapters 9 and 11: the count at resolution one half, which is the basepoint one (Section 9.1), the self decomposability of the clock (Section 11.1), and the zero density, whose classical estimates bound the negative Thorin mass below the basepoint one (Section 11.4).

5.1 The heat trace and the hierarchy of positivity

The rungs of the hierarchy proved unconditionally are infinite divisibility and self decomposability of the clock, log convexity of W, positive definite Hankel blocks up to size 3×10^7, the signs of all derivatives, equivalently the positivity of the Williamson measures, up to order 1.16×10^{25}, and near the origin the signs of every derivative: W is completely monotone on $(0, 1.29]$. One scheme proves them all, with a reach in size and order set by the height to which RH has been verified; no verification reaches the top, which is where Section 5.2 begins. The gamma mixing identity behind the Williamson measures also describes the tail of

the Pólya kernel. The results are from [200, Chapters 4 and 5] and [198], except those stated without a citation, which are proved here.

5.1.1 The heat trace and the first rungs

By Section 1.2, unconditionally,

$$\log G(u) = \int_0^\infty \bigl(1 - e^{-ut}\bigr)\frac{W(t)}{t}\,dt, \qquad W(t) = \sum_j e^{-a_j t},$$

where $a_j = -(\rho_j - \frac12)^2$ and the sum runs over one zero of each pair $\{\rho, 1-\rho\}$. So $W(t)/t$ is the only candidate for the Lévy density of the clock, and the classes of Section 1.3 become conditions on W: nonnegativity for infinite divisibility, monotonicity for self decomposability, complete monotonicity for the GGC property.

The function W is analytic in $\operatorname{Re} t > 0$, real on the positive axis, and can be differentiated term by term,

$$c_k(t) = (-1)^k W^{(k)}(t) = \sum_j a_j^k e^{-a_j t}.$$

Under RH every $c_k(t)$ is a sum of positive terms. A quadruple of zeros $\frac12 \pm \delta \pm i\gamma$, $0 < \delta < \frac12$, instead contributes the exponents $a = -(\delta + i\gamma)^2 = \gamma^2 - \delta^2 - 2i\delta\gamma$ and $\bar a$, hence the oscillating term $2|a|^k e^{-\operatorname{Re} a\, t}\cos(2\delta\gamma t - k\phi)$, with $|a| = \gamma^2 + \delta^2$ and $\tan\phi = 2\delta\gamma/(\gamma^2 - \delta^2)$. Its amplitude exceeds the contribution $2\gamma^{2k}e^{-\gamma^2 t}$ of the same two zeros placed at $\frac12 + i\gamma$ by the factor $(1+\delta^2/\gamma^2)^k e^{\delta^2 t} \le \exp\{k\delta^2/\gamma^2 + \delta^2 t\}$ [198, Proposition 11.1].

The proofs use W differently at the two ends of the half line. As $t \downarrow 0$ the archimedean term of (1.2.5) dominates,

$$W(t) = \frac{\log(1/t) - \log(16\pi^2) - \gamma_E}{8\sqrt{\pi t}} + \frac78 + o(1),$$

where γ_E is Euler's constant and $\frac78$ is the constant of the Riemann von Mangoldt formula; at $t = 10^{-3}, 10^{-4}, 10^{-5}$ the error of this two term expansion is 6.5×10^{-4}, 1.5×10^{-4} and 4.0×10^{-5} [198, Corollary 5.5]. As $t \to \infty$ the first zero dominates, $W(t) = e^{-\gamma_1^2 t}(1 + o(1))$ with $\gamma_1^2 = 199.7904548$, and the relative contribution of the next zero is $e^{-(\gamma_2^2 - \gamma_1^2)t} = e^{-242.1t}$. In between, the arithmetic terms cancel to a depth that grows linearly in t (Table 1.5).

Theorem 5.1.1 ([200, Theorem A]). *Unconditionally, that is, using only the verification of RH below height* 1000 *[196], $W(t) > 0$ and $W'(t) < 0$ for every $t > 0$. Consequently the clock is the Laplace transform of an infinitely divisible, self decomposable law with Lévy density $W(t)/t$, and $(\Xi(\theta)/\Xi(0),\ \xi(\frac12)/\xi(\frac12 + \theta))$ is a van Dantzig pair in the sense of Section 1.2.*

Proof. The proof is that of [200, Theorem A], with the constants made explicit. Let $\tau = 1/20000$, a constant of this section only, $\ell = -\frac{1}{2}\log t$, and $s_k(t) = \Gamma(k+\frac{1}{2})t^{-k-\frac{1}{2}}/(4\pi)$, which is $(-1)^k$ times the kth derivative of $s_0(t) = 1/(4\sqrt{\pi t})$. It suffices to show that c_0 and c_1 are positive, and we do so first for $t \le \tau$, from (1.2.5), and then for $t \ge \tau$, from the zeros.

Let A and P be the archimedean and prime terms of (1.2.5). The series $\psi(z) = -\gamma_E + \sum_{n\ge0}\big(\frac{1}{n+1} - \frac{1}{n+z}\big)$ [177, (5.7.6)] shows that m increases on $[0,\infty)$, from $m(0) = \psi(\frac{1}{4}) - \log\pi = -\gamma_E - \frac{\pi}{2} - 3\log 2 - \log\pi = -5.372$. Binet's formula $\psi(z) = \log z - \frac{1}{2z} - 2\int_0^\infty v\,dv/\big((v^2+z^2)(e^{2\pi v}-1)\big)$ [177, (5.9.15)] at $z = \frac{1}{4} + \frac{ir}{2}$, with the integral split at $v = r/4$, gives $|m(r) - \log\frac{r}{2\pi}| \le \frac{9}{8r^2} + 10^{-10} < 0.003$ for $r \ge 20$. Hence $|m| \le 6$ on $[0,20]$ and $\int_0^{20}|m(r) - \log\frac{r}{2\pi}|\,dr \le 120 + 4\pi + 20\log\frac{10}{\pi} - 20 < 136$. Since $\frac{1}{2\pi}\int_0^\infty r^{2k}e^{-r^2t}\log\frac{r}{2\pi}\,dr = s_k(t)(\ell+\alpha_k)$, with $\alpha_0 = -\frac{1}{2}\gamma_E - \log 4\pi = -2.8196$ and $\alpha_1 = \alpha_0 + 1$, it follows that

$$\Big|\frac{(-1)^k A^{(k)}(t)}{s_k(t)} - \ell - \alpha_k\Big| \le \frac{272\cdot 20^{2k}\,t^{k+\frac{1}{2}}}{\Gamma(k+\frac{1}{2})} + 0.003 \qquad (k = 0, 1),$$

and the right side, which increases with t, is below 1.089 for $k = 0$ and 0.047 for $k = 1$ at $t = \tau$. For the prime term put $y_n = (\log n)^2/4t$ and $q = (\log 2)/8t$. Both e^{-y_n} and $|y_n - \frac{1}{2}|e^{-y_n}$ are at most $e^{-y_n/2}$, which is at most n^{-q} because $(\log n)^2 \ge (\log 2)\log n$, and $\Lambda(n) \le \log n < \sqrt{n}$; so $|P^{(k)}(t)| \le (2\sqrt{\pi})^{-1}t^{-k-\frac{1}{2}}\sum_{n\ge2} n^{-q} < t^{-k-\frac{1}{2}}2^{-q}$, that is $|P^{(k)}(t)| < t^{-k-\frac{1}{2}}e^{-\kappa/t}$ for $k = 0, 1$ and $t \le \tau$, where $\kappa = (\log 2)^2/8$ is another constant of this section. Since $\kappa/\tau > 1201$, this is below $10^{-500}s_k(t)$. The polar term adds $e^{t/4}$ to c_0 and subtracts $\frac{1}{4}e^{t/4} < 10^{-5}s_1(t)$ from c_1. So for $t \le \tau$

$$c_0(t) \ge s_0(t)\,(\ell - 3.92), \qquad c_1(t) \ge s_1(t)\,(\ell - 1.87),$$

and both are positive, because $\ell \ge \frac{1}{2}\log 20000 > 4.95$.

For $t \ge \tau$ we use the zeros. Those below height 1000 lie on the line [196] and contribute $\gamma^{2k}e^{-\gamma^2 t} > 0$ to c_k, and by the explicit bound for $N(T)$ of [92] one of them lies in $(20, 50]$ and contributes at least $400^k e^{-2500t}$. A zero above height 1000, wherever it lies, has $\operatorname{Re} a \ge \gamma^2 - \frac{1}{4}$ and $|a| \le 2\gamma^2$. The same bound gives $N(r) \le r^2$ for $r \ge 1000$, and integration by parts against dN bounds the total modulus of the contributions of these zeros to c_0 and c_1 by

$$H_0(t) = e^{-(10^6 - \frac{1}{4})t}\big(10^6 + t^{-1}\big), \qquad H_1(t) = 2e^{-(10^6-\frac{1}{4})t}\big(10^{12} + 2\cdot 10^6 t^{-1} + 2t^{-2}\big).$$

At $t = \tau$ these are below 2×10^{-16} and 4.1×10^{-10}, while $e^{-2500\tau} > 0.88$, and $e^{2500t}H_k(t)$ decreases with t. So $c_k(t) \ge 400^k e^{-2500t} - H_k(t) > 0$ for $t \ge \tau$ and $k = 0, 1$.

Thus $W > 0$ and $W' < 0$ on $(0,\infty)$. Since $0 < W(t) \le \sum_j e^{-t\operatorname{Re} a_j}$, the measure $W(t)\,dt/t$ satisfies $\int_0^\infty (1\wedge t)\,W(t)\,dt/t \le \sum_j 1/\operatorname{Re} a_j < \infty$, so it is a

Lévy measure, and the formula for $\log G$ above makes the clock infinitely divisible with this Lévy measure. A law on the half line whose Lévy density is $W(t)/t$ with W nonincreasing is self decomposable [232, Corollary 15.11]. Finally, $\xi(\frac12)/\xi(\frac12+\theta) = \mathbb{E}e^{-\theta^2 T}$ is the characteristic function of $Z\sqrt{2T}$, with Z standard normal and independent of T, and with the characteristic function $\Xi(\theta)/\Xi(0)$ of the variable X of Section 1.2 it forms a van Dantzig pair. The program `idsd_check.py` (Appendix A) evaluates every numerical inequality of this proof in ball arithmetic. □

The proof needs far fewer zeros than those below height 1000. With the same bounds it goes through with $\tau = 1/8300$ and the zeros below height 400. If the archimedean error is bounded by computation instead, from $\int_0^5 r^{2k}|m(r) - \log\frac{r}{2\pi}|\,dr < 0.836$ and 0.404 for $k = 0, 1$ and $r^2|m(r) - \log\frac{r}{2\pi}| < 0.05$ for $r \geq 5$, it goes through with $\tau = 1/500$ and height 100. If in addition the polar term $e^{t/4}$ is kept in c_0, the prime terms with $n \leq 30$ are taken exactly, and the lower bound for $t \geq \tau$ comes from the first zero, in $(14.1347, 14.1348)$, rather than from a zero in $(20, 50]$, it goes through with $\tau = 1/50$ and height 25, below which there are only two zeros. The program `idsd_height.py` evaluates every numerical inequality of the three versions in ball arithmetic.

The same scheme, which we call the two range method, gives more. With the prime term bounded by $100\,t^{-k-\frac12}e^{-\kappa/t}$ for $k \leq 3$ and a second zero in $(100, 200]$, it shows that W and $-W'$ are strictly log convex [198, Theorem 7.3], that is $c_0c_2 > c_1^2$ and $c_1c_3 > c_2^2$ for every $t > 0$, and that $(-1)^k W^{(k)} > 0$ for $k \leq 3$; the proofs of Theorems 7.1.1 and 6.2.17 use it again.

For the first two rungs the explicit formula can be avoided altogether, because where the size of an off line quadruple cannot be bounded its phase fixes its sign, as the first author's note [208] observes. Suppose that the zeros with ordinate in $(0, H]$ lie on the line, with $H \geq 100$, and let $0 \leq k \leq H/8$. A quadruple $\frac12 \pm \delta \pm i\gamma$ then has $\gamma > H$, and since $0 < \phi < 2\delta/\gamma < 1/\gamma$ its phase satisfies $|2\delta\gamma t - k\phi| < \gamma t + 1$, so its term in $c_k(t)$ is positive unless $\gamma t > \frac{\pi}{2} - 1 > 0.57$. When $\gamma t > 0.57$ the term is at most $2(\gamma^2 + \frac14)^k e^{-(\gamma^2 - \frac14)t}$ in modulus, and since $(\gamma^2 + \frac14)/\gamma_1^2 \leq \gamma^2/99.9$, $\gamma^2 - \frac14 - \gamma_1^2 \geq \gamma^2/2$ and $\gamma^2 t/2 > 0.285\gamma$, it is at most $2(\gamma^2/99.9)^k e^{-0.285\gamma}$ times the term $\gamma_1^{2k} e^{-\gamma_1^2 t}$ of the first zero. A quadruple has two zeros of ordinate γ, so at every $t > 0$

$$c_k(t) \geq \big(1 - S_k(H)\big)\gamma_1^{2k} e^{-\gamma_1^2 t}, \qquad S_k(H) = \sum_{\gamma > H} \Big(\frac{\gamma^2}{99.9}\Big)^k e^{-0.285\gamma},$$

the sum running over the zeros above height H, with multiplicity. The explicit bound for $N(T)$ of [92] gives $N(T+1) - N(T) < 2\log T$ for $T \geq 100$, and summing over unit intervals gives $S_0(100) < 10^{-10.5}$ and $S_1(100) < 10^{-8.5}$. Since Hutchinson placed the zeros below height 300 on the line in 1925 [99], this proves Theorem 5.1.1 without the explicit formula. For larger k the factor $(\gamma^2/99.9)^k$ must be absorbed

by $e^{-0.285\gamma}$ at $\gamma = H$, so the reachable order, about $0.285H/\log(H^2/99.9)$, stays below the height: the verification below 3×10^{12} [196] gives every $k \le 10^{10}$, with $S_k(H) < 10^{-1.4\times 10^{11}}$. The two range method reaches orders proportional to H^2 (Theorems 5.1.4 and 5.1.10), because the explicit formula covers $t \le \tau$ at every order and beyond τ the factor $e^{-\gamma^2\tau}$ beats γ^{2k} up to orders of size τH^2. The program `ladder_bound.py` evaluates these bounds in ball arithmetic.

Positivity of W also limits how the clock factorises. By Mase's theorem [155, Theorems 1 and 2], an infinitely divisible law without Gaussian component whose Lévy measure has a density that is positive on the whole half line, or that has mass on both half lines, has an indecomposable factor, hence a factor that is not infinitely divisible, so it lies outside Linnik's class I_0 [148] of the infinitely divisible laws all of whose factors are infinitely divisible. Since $W > 0$, this applies unconditionally to the clock variable T, to $Z\sqrt{2T}$ with Z standard normal and independent of T, whose characteristic function $\xi(\frac{1}{2})/\xi(\frac{1}{2}+\theta)$ is the second member of the van Dantzig pair (Section 10.1 extends this normal scale mixture to every basepoint, Proposition 10.1.1), and to every nondegenerate GGC [198, Proposition 12.1]. The clock has many factorisations into infinitely divisible laws, one for each splitting of W into two positive functions, but arguments that need every factor to be infinitely divisible, such as those for the normal and Poisson laws, do not apply to it. Section 11.1 draws two further consequences of $W > 0$ and $W' < 0$: by Yamazato's theorem [274] the clock variable is unimodal, and as a self decomposable law it is driven by a subordinator with Lévy density $-W'$.

5.1.2 Hankel blocks and derivative signs

Complete monotonicity concerns every derivative at every time; its Hankel form needs one time. Fix $t > 0$ and form the Hankel matrices $[c_{i+j}(t)]$ and $[c_{i+j+1}(t)]$, $i, j \ge 0$.

Theorem 5.1.2 ([198, Theorem 7.1]). *For any fixed $t > 0$ the following are equivalent: both infinite Hankel matrices are positive semidefinite; Weil's quadratic form is positive semidefinite on the span of $r^m e^{-r^2 t/2}$, $m = 0, 1, 2, \ldots$; RH.*

Under RH the derivatives at one time are the moments of the positive measure $e^{-tz}U(dz)$ on the half line, so both matrices are positive semidefinite. Conversely, if both Hankel matrices are positive semidefinite, Stieltjes' theorem makes the $c_k(t)$ the moments of a positive measure μ on the half line, and since $\sum_k c_k(t)(-s)^k/k! = W(t+s)$ for $|s| < t$, $W(t+s) = \int e^{-sx}\mu(dx)$ there. Both sides are analytic in $\operatorname{Re} s > -t$, the right side because $\int e^{|s|x}\mu(dx) = \sum_k c_k(t)|s|^k/k! = W(t-|s|) < \infty$ for $|s| < t$, so the identity theorem extends the equality to $\operatorname{Re} s > -t$, and W is completely monotone, which is RH. Through the explicit formula, Weil's form on the span of $r^m e^{-r^2 t/2}$ gives the same two matrices, even m the first and odd m the second. The signs of all derivatives at all times are thus concentrated at one time.

The finite blocks are theorems up to a size set by the verification height.

THEOREM 5.1.3 ([198, Theorem 7.4]). *If all zeros below* $T_N = 20000N\sqrt{1+\log N}$ *are on the line, the leading* $N \times N$ *blocks are positive definite at every* $t > 0$. *The verification to height* 3×10^{12} *[196] gives this for every* $N \le 3 \times 10^7$.

Table 5.1 shows normalised leading minors recomputed in sixty digit arithmetic from the first 1700 zeros, whose ordinates are stored in double precision; they agree with [198, Numerical observation 7.2] to every digit printed there.

t	c_0c_2/c_1^2	$N=2$	$N=3$	$N=4$	$N=5$	block
0.01	1.1346548	0.119	2.45×10^{-3}	1.05×10^{-5}	1.04×10^{-8}	even
0.01		0.182	5.87×10^{-3}	3.30×10^{-5}	4.57×10^{-8}	odd
0.05	1.0000081	8.11×10^{-6}	1.77×10^{-14}	5.35×10^{-28}	2.20×10^{-44}	even
0.05		1.79×10^{-5}	1.22×10^{-13}	1.71×10^{-26}	3.81×10^{-42}	odd

Table 5.1: Normalised leading Hankel minors of the heat trace, $\det[c_{i+j}]_{i,j<N}/\prod_{i<N} c_{2i}$ for the even block and $\det[c_{i+j+1}]_{i,j<N}/\prod_{i<N} c_{2i+1}$ for the odd block, from the first 1700 zeros in sixty digit arithmetic.

At $t = 0.01$ the minors are comfortably positive. At $t = 0.05$ the minors of the even block fall like 10^{-5}, 10^{-14}, 10^{-27}, 10^{-44}, and those of the odd block like 10^{-5}, 10^{-13}, 10^{-26}, 10^{-41} (Figure 5.1, left): the first exponents dominate and the others enter only through $e^{-(a_j-a_1)t}$, so the matrix is nearly of rank one. Computed from the zeros, the minors are still sums of positive terms, by the Cauchy Binet formula $\det[c_{i+j}]_{i,j<N} = \sum_{|S|=N}\prod_{l\in S} e^{-a_l t}\prod_{l<l',\, l,l'\in S}(a_l-a_{l'})^2$ over the sets S of N zeros, so what is lost is conditioning, not sign. Computed from the arithmetic side, this loss comes on top of the cancellation of Table 1.5; together the two effects are why the heat trace diagnostics of Table 1.8 certify easily and discriminate poorly. The derivative signs (Figure 5.1, right) reach much further, because a sign needs only that the verified terms dominate, not that a small determinant stay positive.

THEOREM 5.1.4 ([198, Theorem 7.10]). *Unconditionally, that is, using only the verification of RH below height* 10^{12} *[196],* $(-1)^k W^{(k)}(t) > 0$ *for every* $t > 0$ *and every integer* $0 \le k \le 10^{17}$.

The proof again splits at τ. Near the origin, Cauchy's estimate on a circle about t of radius close to t, such as $(1-\frac{1}{k+1})t$, on which $\mathrm{Re}(1/z) \ge 1/(2t)$ and $|z| \ge t/(k+1)$, bounds the kth derivative of the prime term by a multiple of $(k+1)\Gamma(k+\frac{1}{2})t^{-k-\frac{1}{2}}e^{-\kappa/t}$. Against $s_k(t)$ this loses a factor of order k and gains $e^{-\kappa/t} \le e^{-1201}$, and for $k \ge 1$ the archimedean term dominates by a margin of more than $2.9\, s_k(t)$, the case $k = 0$ being Theorem 5.1.1. Away from the origin a zero in $(20, 50]$ dominates all zeros above the verified height $H = 10^{12}$, wherever they lie, provided $tH^2 \ge 2(k+1)$ and a logarithmic condition hold; Section 5.1.7 explains why these allow orders up to about 10^{18}, and how a dominating zero

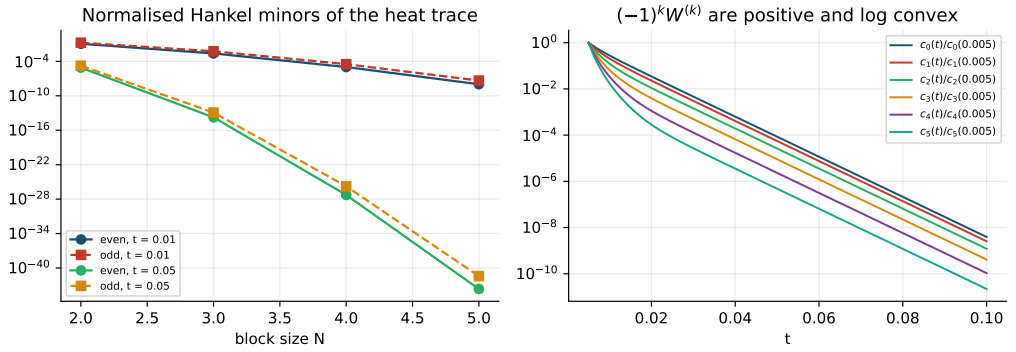


Figure 5.1: Left: normalised Hankel minors of the heat trace at two times; positivity persists but falls below any fixed precision as t grows. Right: the first six $(-1)^k W^{(k)}(t)$, divided by their values at $t = 0.005$, from 1700 zeros.

that follows the peak of the weight removes the logarithmic condition and reaches every order up to 1.16×10^{25} (Theorem 5.1.10).

The Stieltjes moments of Table 1.8 give a Hankel test at the other end of the heat trace, and the test also returns the zeros [208]. They are the moments of W, $m_n = \frac{1}{n!}\int_0^\infty t^n W(t)\,dt = \sum_j a_j^{-n-1}$, and since $\log G(u) = \sum_j \log(1 + u/a_j) = \sum_{n\geq 1}(-1)^{n+1} m_{n-1} u^n/n$ they follow from the Taylor coefficients of $G(u) = \Xi(i\sqrt{u})/\Xi(0) = \sum_k M_k u^k/((2k)!\,M_0)$, where $M_k = \int_{\mathbb{R}} v^{2k}\Phi(v)\,dv$; no zero enters. Under RH the m_n are the moments of the positive measure $\nu = \sum_j a_j^{-1}\delta_{1/a_j}$, so every Hankel matrix $\mathcal{H}_N = [m_{i+j}]_{i,j<N}$ is positive definite. Conversely, if all of them are, Hamburger's theorem makes the m_n the moments of a positive measure, of compact support because $|m_n| \leq \sum_j |a_j|^{-n-1}$ grows at most geometrically, and then $\sum_n m_n w^n = \sum_j 1/(a_j - w)$ has all its poles on the real axis: every a_j is real, and positive because ξ has no zeros in $(0, 1)$, which is RH. The shifted matrices $\mathcal{H}'_N = [m_{i+j+1}]_{i,j<N}$ give the same test. Stieltjes' criterion for a measure on $[0, \infty)$ asks for both families, but here either suffices alone: if every $\mathcal{H}'_N$ is positive definite, Hamburger's theorem applied to (m_{n+1}) and $\sum_n m_n w^n = m_0 + w\sum_n m_{n+1} w^n$ give the same conclusion. Under RH, ν is supported in $[0, \gamma_1^{-2}]$, so its moment problem is a determinate Hausdorff problem. Positivity is tested by the Cholesky factorisation $\mathcal{H}_N = LL^\top$, and the factor gives more: $J_N = L^{-1}\mathcal{H}'_N L^{-\top}$ is the Jacobi matrix of ν, tridiagonal, and its eigenvalues are the nodes of the N point Gauss rule for ν. They approximate the largest atoms $1/a_j$ first, so their inverse square roots return the lowest ordinates (Table 5.2). With $M_0, \ldots, M_{100}$ computed at 1000 digits by the trapezoid rule, every $\mathcal{H}_N$ up to order 50 is positive definite, and J_{50} returns $\gamma_1, \ldots, \gamma_{22}$ to within 10^{-7} and γ_1 to 171 digits.

N	zeros	error of γ_1	last error	$\det \mathcal{H}_N / \det \mathcal{H}_{N-1}$	digits
10	2	4.4×10^{-18}	1.3×10^{-9}	2.0×10^{-61}	60
20	7	1.6×10^{-49}	3.1×10^{-7}	1.2×10^{-138}	140
30	12	8.6×10^{-87}	2.4×10^{-7}	1.0×10^{-221}	230
40	17	6.1×10^{-128}	5.4×10^{-8}	1.3×10^{-308}	310
50	22	5.3×10^{-172}	5.4×10^{-8}	1.4×10^{-398}	400

Table 5.2: The lowest ordinates from the Stieltjes moments $m_0, \dots, m_{2N-1}$, computed from the Taylor coefficients of Ξ at the centre: the number returned by the Jacobi matrix of order N to within 10^{-6}, the errors of the first and of the last of them, the last Cholesky pivot of $\mathcal{H}_N$, and the least number of digits, in steps of 10, of the Taylor coefficients and of the arithmetic from which on the same ordinates are returned.

This is Gauss quadrature, equivalently Padé approximation of G'/G at the origin, and Table 5.2 shows its two limits. The first is precision. The last Cholesky pivot $\det \mathcal{H}_N / \det \mathcal{H}_{N-1}$ falls from 10^{-61} at $N = 10$ to 10^{-398} at $N = 50$, and the computation must resolve it: with the Taylor coefficients rounded to D digits and D digit arithmetic, J_N returns its ordinates only from about that many digits on. The precision thus grows linearly, by about 17 digits for each further zero, and the method suits the lowest zeros, not high ones. The second is reach. An off line quadruple at height γ enters m_n with weight of order γ^{-2n-2}, against γ_1^{-2n-2} for the first zero, so a block of order N sees essentially only the zeros that its Gauss rule resolves; Section 8.4.3 shows this along the de Bruijn Newman flow, where the order 30 test registers the collision of the pair just above the zeros it resolves, and not a collision higher up. The program `jacobi_zeros.py` computes the table.

The Hankel matrices count the zeros off the line as well as test RH. For a real polynomial this is a theorem attributed to Hermite [96]: the Hankel form of the power sums of the roots has as many negative squares as the polynomial has distinct pairs of nonreal roots. Let ι_N and ι'_N be the numbers of negative eigenvalues of $\mathcal{H}_N$ and $\mathcal{H}'_N$.

THEOREM 5.1.5. *Let $K \in \{0, 1, 2, \dots\} \cup \{\infty\}$ be the number of distinct zeros ρ of ξ with $\operatorname{Re}\rho > \frac{1}{2}$ and $\operatorname{Im}\rho > 0$. Then ι_N and ι'_N are nondecreasing in N, $\iota_N \le K$ and $\iota'_N \le K$ for every N, and $\iota_N \to K$ and $\iota'_N \to K$ as $N \to \infty$. In particular RH holds if and only if $\iota_N = 0$ for every N, and if it fails, $\iota_N \ge 1$ from some N on; the same holds for ι'_N.*

Proof. Put $b_k = 1/a_k$, listed with multiplicity, so that $m_n = \sum_k b_k^{n+1}$ and $\sum_k |b_k| < \infty$. For a real vector $x = (x_0, \dots, x_{N-1})$ and $P(w) = \sum_i x_i w^i$,

$$x^\top \mathcal{H}_N x = \sum_k b_k P(b_k)^2, \qquad x^\top \mathcal{H}'_N x = \sum_k b_k^2 P(b_k)^2,$$

the sums converging absolutely. A zero on the line gives $b_k = \gamma_k^{-2} > 0$ and nonnegative terms. A zero $\rho = \frac12 + \delta + i\gamma$ with $\delta > 0$ and $\gamma > 0$, of multiplicity μ, gives $a = -(\delta + i\gamma)^2$ with $\operatorname{Im} a = -2\delta\gamma < 0$, hence $b = 1/a$ with $\operatorname{Im} b > 0$; its mirror image $1 - \bar\rho$ gives $\bar a$ and $\bar b$ with the same multiplicity, and since P has real coefficients the two contribute $2\mu \operatorname{Re}\big(bP(b)^2\big)$ to the first form. With $u = \operatorname{Re} P(b)$ and $v = \operatorname{Im} P(b)$ this is $2\mu\big(\operatorname{Re} b\,(u^2 - v^2) - 2\operatorname{Im} b\,uv\big)$, a quadratic form in (u, v) whose matrix has determinant $-4\mu^2|b|^2 < 0$, so it has exactly one negative square, and composed with the linear map $x \mapsto (u, v)$ it has at most one. The terms of the zeros on the line add up to a positive semidefinite form, and the negative index of a sum of forms is at most the sum of their negative indices, so $\iota_N \le K$. In the second form the weight is b^2, and $2\mu \operatorname{Re}\big(b^2P(b)^2\big)$ is again a form in (u, v) of determinant $-4\mu^2|b|^4 < 0$, so $\iota'_N \le K$. Since $\mathcal{H}_N$ is the leading principal submatrix of $\mathcal{H}_{N+1}$, Cauchy's interlacing theorem shows that ι_N does not decrease, and the same holds for ι'_N.

For the lower bound let $\mathcal{B}$ be the closure of $\{b_k\}$. It is a countable compact set, symmetric under conjugation, whose only possible accumulation point is 0, and a countable compact set does not separate the plane, so $\mathbb{C} \setminus \mathcal{B}$ is connected. Fix a finite $m \le K$ and distinct atoms $\omega_1, \dots, \omega_m$ with $\operatorname{Im} \omega_j > 0$, of multiplicities μ_j; they are isolated points of $\mathcal{B}$. On $\mathbb{C}^m = \mathbb{R}^{2m}$ the form $q(y) = \sum_{j \le m} 2\mu_j \operatorname{Re}(\omega_j y_j^2)$ is a sum of m forms in the pairs $(\operatorname{Re} y_j, \operatorname{Im} y_j)$, each with one negative square, so there are an m dimensional real subspace E and $\lambda_0 > 0$ with $q(y) \le -\lambda_0|y|^2$ on E. Let $y^{(1)}, \dots, y^{(m)}$ be a basis of E and $\lambda_1 > 0$ such that $|\sum_i \alpha_i y^{(i)}|^2 \ge \lambda_1|\alpha|^2$ for $\alpha \in \mathbb{R}^m$. Take open discs D_j about ω_j, and their mirror images D_j^* about $\bar\omega_j$, whose closures are disjoint and meet $\mathcal{B}$ only in their centres, and an open set $O \supset \mathcal{B} \setminus \{\omega_1, \bar\omega_1, \dots, \omega_m, \bar\omega_m\}$, symmetric under conjugation and disjoint from the closed discs, which exists because that set is compact. Let f_i be $y_j^{(i)}$ on D_j, $\overline{y_j^{(i)}}$ on D_j^* and 0 on O. Each f_i is holomorphic on a neighbourhood of $\mathcal{B}$, so by Runge's theorem there are polynomials $\tilde P_i$ with $|\tilde P_i - f_i| < \varepsilon$ on $\mathcal{B}$, and since $\mathcal{B}$ and f_i are symmetric under conjugation, $P_i(w) = \frac12\big(\tilde P_i(w) + \overline{\tilde P_i(\bar w)}\big)$ has real coefficients and satisfies the same bound. For $\alpha \in \mathbb{R}^m$ and $P = \sum_i \alpha_i P_i$ we have $|P(b_k)| \le \varepsilon\sqrt{m}\,|\alpha|$ at every atom other than the ω_j and $\bar\omega_j$, and $P(\omega_j) = \sum_i \alpha_i y_j^{(i)} + \eta_j$ with $|\eta_j| \le \varepsilon\sqrt{m}\,|\alpha|$. The atoms ω_j and $\bar\omega_j$ contribute $\sum_j 2\mu_j \operatorname{Re}\big(\omega_j P(\omega_j)^2\big)$ to the first form, so for $\varepsilon \le 1$ and a constant C that depends on neither ε nor α,

$$\sum_k b_k P(b_k)^2 \le q\Big(\sum_i \alpha_i y^{(i)}\Big) + C\varepsilon|\alpha|^2 \le -(\lambda_0\lambda_1 - C\varepsilon)|\alpha|^2.$$

For $\varepsilon < \lambda_0\lambda_1/C$ the right side is negative for $\alpha \ne 0$, so the coefficient vectors of $P_1, \dots, P_m$ span an m dimensional space on which the form of $\mathcal{H}_N$ is negative definite as soon as N exceeds their degrees, and $\iota_N \ge m$ for those N. For $\mathcal{H}'_N$ the same argument applies with ω_j^2 in place of ω_j in q and the weights b_k^2 in the sum. Since $m \le K$ is arbitrary, $\iota_N \to K$ and $\iota'_N \to K$. □

In determinant form the theorem is classical. For a real entire function of order one with zeros z_j, Grommer [87] proved that all its zeros are real if and only if the determinants $D_n = \det(s_{i+j})_{i,j<n}$ of the power sums $s_k = \sum_j z_j^{-k-2}$ are positive for every $n \geq 1$, and Chebotarev [261] that if the sequence $D_{n-1}D_n$, $n \geq 1$, with $D_0 = 1$, has exactly m negative terms, the function has m distinct pairs of complex conjugate zeros; Baricz and Štampach state both in this form [14, Theorem 10] and note that Grommer's theorem extends one on polynomials attributed to Hermite. For Ξ, whose zeros are the square roots of the a_k, $s_{2l} = 2m_l$ and $s_{2l+1} = 0$, so after the even indices are put first $(s_{i+j})_{i,j<n}$ is block diagonal with the blocks $2\mathcal{H}_{\lceil n/2\rceil}$ and $2\mathcal{H}'_{\lfloor n/2\rfloor}$, whose negative indices add up to $\iota_{\lceil n/2\rceil} + \iota'_{\lfloor n/2\rfloor}$; this tends to $2K$, and each zero ρ counted by K gives two pairs of complex conjugate zeros of Ξ, as Chebotarev's count requires. Counting by eigenvalues rather than by the signs of determinants, Theorem 5.1.5 needs no determinant to be nonzero, allows $K = \infty$ and holds for each family separately. The count is exact in the limit but slow, for the reason of the reach limit above. Along the de Bruijn Newman flow the proof applies at every time to the zeros of H_t, none of which is purely imaginary, and there a pair that starts at (γ_k, γ_{k+1}) is seen soon after its collision only from order about $2k+2$, and from lower orders as it moves away from the axis (Section 8.4.3).

5.1.3 Every order near the origin

Theorem 5.1.4 gives finitely many orders at every time. Near the origin the arithmetic side gives every order. Let $s_k(t) = \Gamma(k + \frac{1}{2})t^{-k-\frac{1}{2}}/(4\pi)$ as in the proof of Theorem 5.1.1, let $E_k(t)$ be the average of m against the weight $r^{2k}e^{-r^2t}$ on $[0,\infty)$, and let $F_k(x) = {}_1F_1(k+\frac{1}{2};\frac{1}{2};-x^2)$. Expanding the cosine in its power series gives $\frac{1}{2\pi}\int_{\mathbb{R}} r^{2k}e^{-r^2t}\cos(ry)\,dr = 2s_k(t)F_k(y/2\sqrt{t})$, whose case $k=0$ is the kernel $e^{-y^2/4t}/(2\sqrt{\pi t})$ of the prime term of (1.2.5). Differentiating (1.2.5) k times term by term, which the Gaussian factors in r and in $\log n$ allow, therefore gives, with no hypothesis on the zeros,

$$\frac{c_k(t)}{s_k(t)} = E_k(t) - 2\sum_{n\geq 2}\frac{\Lambda(n)}{\sqrt{n}}F_k(x_n) + \frac{(-\frac{1}{4})^k e^{t/4}}{s_k(t)}, \qquad x_n = \frac{\log n}{2\sqrt{t}}. \tag{5.1.1}$$

At $t = 0.018$ the two sides agree to twelve digits for $k = 1$, 3 and 10, with c_k summed over the first 1700 zeros and the prime terms over $n \leq 10^4$. In (5.1.1) the archimedean term $E_k(t)$ grows with k, while the next lemma bounds each $F_k(x_n)$ by $B(x_n)e^{-x_n^2/2} = B(x_n)e^{-(\log n)^2/8t}$, uniformly in k; this is why the arithmetic side reaches every order.

Lemma 5.1.6. (a) *For every integer $k \geq 0$ and every $x \geq 0$, $|F_k(x)| \leq B(x)e^{-x^2/2}$, where $B(x) = \max\{2.62,\ (\pi(2ex^2+\frac{1}{2}))^{1/4}\}$.*

(b) *Let $0 < \varepsilon < 1$, and let $k \geq 1$ and $x \geq 0$ satisfy $e^2x^2 \leq 4\varepsilon k$. Then $|F_k(x)| \leq C_\varepsilon(k)e^{-x^2/2}$, where $C_\varepsilon(k) = \sqrt{(k+0.37)/((1-\varepsilon)k+\frac{1}{4})} + 2\sqrt{k+0.37}\,e^{-\varepsilon k}$.*

Proof. Taking $t = 1$ and $y = 2x$ above, $F_k(x) = \Gamma(k+\frac12)^{-1}\int_{\mathbb R} r^{2k}e^{-r^2+2ixr}\,dr$. Put $\mu = x^2/4$. The integrand is entire and decays in every horizontal strip, so the path can be moved to $\operatorname{Im} r = x/2$, where $|r^{2k}| = (\sigma^2+\mu)^k$, $|e^{-r^2}| = e^{-\sigma^2+\mu}$ and $|e^{2ixr}| = e^{-4\mu}$, with $\sigma = \operatorname{Re} r$. Expanding $(\sigma^2+\mu)^k$ by the binomial theorem gives, for every k and x,

$$|F_k(x)| \le e^{-3\mu}\sum_{j=0}^{k}\frac{\mu^j}{j!}P_{k,j}, \qquad P_{k,j} = \frac{\Gamma(k+1)}{\Gamma(k+\frac12)}\,\frac{\Gamma(k-j+\frac12)}{\Gamma(k-j+1)}.$$

Kershaw's inequalities $\sqrt{y+\frac14} \le \Gamma(y+1)/\Gamma(y+\frac12) \le \sqrt{y+0.37}$, proved for $y > 0$ in [124] and true at $y = 0$, where the ratio is $1/\sqrt{\pi}$, give $P_{k,j} \le \sqrt{(k+0.37)/(k-j+\frac14)}$, which is less than $\sqrt2$ for $j \le k/2$ and at most $2\sqrt{k+0.37}$ for every j. For (a), suppose first that $k \ge 2ex^2$. Since $k/2 \ge 4e\mu$, Chernoff's bound $\sum_{j\ge a}\mu^j/j! \le (e\mu/a)^a$ for the Poisson law, valid for $a > \mu$, gives $\sum_{j\ge k/2}\mu^j/j! \le (2e\mu/k)^{k/2} \le 2^{-k}$, and $2\sqrt{k+0.37}\,2^{-k} \le \sqrt{1.37} < 1.2$ for $k \ge 1$. Hence $|F_k(x)| \le e^{-3\mu}(\sqrt2\,e^{\mu}+1.2) \le (\sqrt2+1.2)e^{-x^2/2}$, and $\sqrt2+1.2 < 2.62$; for $k = 0$ the assumption forces $x = 0$, where $F_0 = 1$.

Suppose now that $k < 2ex^2$. Kummer's transformation and the Hermite case of ${}_1F_1$ [177, (13.2.39), (13.6.16)] give $F_k(x) = e^{-x^2}{}_1F_1(-k;\frac12;x^2) = (-1)^k\frac{k!}{(2k)!}e^{-x^2}H_{2k}(x)$, and the bound $e^{-x^2/2}|H_n(x)| \le (2^n n!)^{1/2}$ [177, (18.14.9)] gives

$$|F_k(x)| \le \Big(\frac{4^k(k!)^2}{(2k)!}\Big)^{1/2}e^{-x^2/2} \le \big(\pi(k+\tfrac12)\big)^{1/4}e^{-x^2/2} < \big(\pi(2ex^2+\tfrac12)\big)^{1/4}e^{-x^2/2},$$

since $4^k(k!)^2/(2k)! = \sqrt{\pi}\,\Gamma(k+1)/\Gamma(k+\frac12) \le \sqrt{\pi(k+\frac12)}$ by Kershaw's upper bound.

For (b), split the sum at $j = \varepsilon k$ instead. For $j \le \varepsilon k$, $P_{k,j} \le \sqrt{(k+0.37)/((1-\varepsilon)k+\frac14)}$, and these j contribute at most that times e^{μ}. For $j > \varepsilon k$, $P_{k,j} \le 2\sqrt{k+0.37}$, and since $\varepsilon k \ge e^2\mu > \mu$, Chernoff's bound gives $\sum_{j\ge\varepsilon k}\mu^j/j! \le (e\mu/\varepsilon k)^{\varepsilon k} \le e^{-\varepsilon k}$. Hence $|F_k(x)| \le e^{-3\mu}\big(\sqrt{(k+0.37)/((1-\varepsilon)k+\frac14)}\,e^{\mu} + 2\sqrt{k+0.37}\,e^{-\varepsilon k}\big) \le C_\varepsilon(k)e^{-x^2/2}$. □

THEOREM 5.1.7. (a) *For every integer $k \ge 1$ and every $t \in (0, 0.019]$, $(-1)^kW^{(k)}(t) > 0$. The proof uses no information on the zeros.*

(b) *Unconditionally, that is, using only the verification of RH below height 10^{12} [196], W is completely monotone on $(0, 1.29]$: $(-1)^kW^{(k)}(t) > 0$ for every $t \in (0, 1.29]$ and every integer $k \ge 0$.*

Proof. By (5.1.1) and Lemma 5.1.6(a), for every $k \ge 0$ and $t > 0$,

$$\frac{c_k(t)}{s_k(t)} \ge E_k(t) - S_B(t) - Q_k(t),$$

where

$$S_B(t) = 2\sum_{n\ge 2} \frac{\Lambda(n)}{\sqrt{n}} B(x_n) e^{-(\log n)^2/8t}, \qquad Q_k(t) = \frac{4\pi e^{t/4} t^{k+\frac12}}{4^k\,\Gamma(k+\frac12)}.$$

Above the order $K = 10^{17}$ part (b) of the lemma does better. Take $\varepsilon = 10^{-6}$. The first term of $C_\varepsilon(k)$ decreases in k, because $\frac14 < 0.37(1-\varepsilon)$, and the second decreases for $k > 1/(2\varepsilon)$, so $C_\varepsilon(k) \le C = C_\varepsilon(K+1) < 1 + 6\times 10^{-7}$ for $k > K$, and $e^2x^2 \le 4\varepsilon k$ holds for $x \le 2\sqrt{\varepsilon(K+1)}/e$, which exceeds 2.3×10^5. For $t \ge 0.019$ every prime power $n \le 2^{24}$ has $x_n < 61$. So for $k > K$ and $t \ge 0.019$ the bound holds with S_B replaced by S_C, the sum S_B with $B(x_n)$ replaced by C for $n \le 2^{24}$.

The three terms are monotone. Since m increases on $[0,\infty)$, and the weights $r^{2k}e^{-r^2t}$ have likelihood ratios r^2 from k to $k+1$ and $e^{r^2(t'-t)}$ from t' to $t < t'$, both increasing in r, $E_k(t)$ increases in k and decreases in t. $Q_k(t)$ increases in t, and for $t \le 2$ it decreases in k, the ratio of consecutive terms being $t/(4k+2)$. $B(x)e^{-x^2/2}$ decreases on $(0,\infty)$: the second branch of B increases and is 1.78 at $x = 2^{-1/2}$, so $B = 2.62$ below that point, and above it the logarithmic derivative of the second branch times $e^{-x^2/2}$ is less than $1/(2x) - x < 0$; since x_n decreases in t, S_B and S_C increase in t. So for $k \ge k_0$ and $t \le T \le 2$ each bound is at least its value at (k_0, T), and it suffices to prove $E_1(0.019) - S_B(0.019) - Q_1(0.019) > 0$, which gives (a), and $E_{K+1}(1.29) - S_C(1.29) - Q_{K+1}(1.29) > 0$, which gives the orders $k > K$ on $[0.019, 1.29]$; with (a) for $t \le 0.019$ and Theorem 5.1.4 for the orders $k \le K$, this gives (b).

For the first, lower sums over $[0, 64]$, which the monotonicity of m allows, and the positivity of m beyond 64 give $E_1(0.019) > 0.15924$; the prime powers $n \le 30000$ and the tail bound below give $S_B(0.019) < 0.13628$; and $Q_1(0.019) < 0.00933$. For the second, the weight $r^{2K+2}e^{-r^2t}$ makes $v = r^2t$ a gamma variable of shape $\lambda = K + \frac32$, and Chernoff's bound for the gamma law, $\Pr(v < a) \le (a/\lambda)^\lambda e^{\lambda - a}$ for $a < \lambda$, gives $\Pr(v < a) < e^{-1000}$ for $a = (1 - 2^{-13})^2\lambda$. As m increases, $E_{K+1}(t) \ge m(r_1) - (m(r_1) - m(0))e^{-1000}$ with $r_1 = \sqrt{a/t}$, which gives $E_{K+1}(1.29) > 17.6066$; the prime powers $n \le 2^{24}$ and the tail bound below give $S_C(1.29) < 17.5848$; and $Q_{K+1}(1.29) < e^{-1000}$. In both cases the prime powers above the cutoff n_0, which is 30000 and 2^{24}, add less than $2\big(2.62 + 2.08(2\sqrt{t})^{-1/2}\big)n_0^{1-q}/(q-1)$ with $q = \log n_0/8t$, because $x_n \ge x_{n_0} \ge 1$ and $e^{-(\log n)^2/8t} \le n^{-q}$ for $n > n_0$, $B(x) \le 2.62 + 2.08\sqrt{x}$ for $x \ge 1$, and $\log n$ and $(\log n)^{3/2}$ are at most $\sqrt{n}$ for $n \ge e^{10}$; this is below 10^{-299} and 5×10^{-4}. The program `cm_orders.py` evaluates every numerical inequality of this proof and of the proof of Lemma 5.1.6 in ball arithmetic. □

Part (a) uses the arithmetic side alone. Part (b) uses the verified zeros only through Theorem 5.1.4, so, like that theorem, it holds wherever the zeros above height 10^{12} lie. The two theorems are complementary: Theorem 5.1.4 gives finitely many orders at every time and Theorem 5.1.7 every order up to a finite time,

while RH is every order at every time. Computed in floating point, the bound $E_1 - S_B - Q_1$ of (a) is 0.83 at $t = 0.005$, 0.21 at $t = 0.015$ and 0.014 at $t = 0.019$, and it vanishes at $t = 0.0193$; E_1 itself vanishes at $t = 0.0261$, so no bound on the size of the prime terms carries (a) much further. Part (b) goes further because $E_k(t) = \frac{1}{2}(\psi(k + \frac{1}{2}) - \log t) - \log 2\pi + O(t/k)$ grows like $\frac{1}{2}\log k$, while at such orders part (b) of the lemma bounds the prime terms at almost no cost: the bound $E_{K+1} - S_C$, computed in the same way, is 17.9 at $t = 0.1$, 12.5 at $t = 0.5$ and 5.2 at $t = 1$, and it vanishes at $t = 1.2912$, while $E_{K+1} - S_B$ vanishes at $t = 0.578$.

The ray of Section 5.1.5 shows what the proof uses. On the ray $t = k/u^2$, $e^{-f_0}c_k(t)$ is to leading order the Gaussian smoothing $\mathcal{G}(u, t)$ of the zeros (Proposition 5.1.8), $E_k(t)$ is close to $\log(u/2\pi)$, and for large k the factor $F_k(x_n)$ is close to $e^{-(\log n)^2/8t}\cos(u \log n)$. The proof is therefore the bound of Section 12.2.6 that ignores the phases of the primes, made uniform in the order; with C replaced by one, S_C is twice the sum S of that section, up to the prime powers above 2^{24}. Above the order 10^{17} uniformity costs a factor below $1 + 6 \times 10^{-7}$, so 1.2912, the root of $E_{K+1}(t) = 2S(t)$, is as far as this argument can carry (b) with $K = 10^{17}$: it bounds the prime terms by one envelope for all orders above K, and the archimedean term by its value at the order $K + 1$. A bound that ignores the phases but treats the orders separately is not covered by this limit. At low orders it costs the factor $B \geq 2.62$ of part (a), and part (a) is not sharp: on a grid of $0.5 \leq x \leq 20$ and $k \leq 1200$ the ratio $|F_k(x)|e^{x^2/2}/B(x)$ is at most 0.44, and for $5 \leq x \leq 30$ the maximum of $|F_k(x)|e^{x^2/2}$ over $k \leq 4000$ lies between $0.80x^{1/3}$ and $0.88x^{1/3}$, attained near the turning point $k \approx x^2/4$ of the Hermite function (computed). The ray also says where (b) looks. To leading order, complete monotonicity on $(0, T]$ asks for positive Gaussian smoothings at resolutions $1/(2\sqrt{t})$ of at least $1/(2\sqrt{T})$, which is 0.44 for $T = 1.29$, finer than the 0.46 at which the quadruples of Table 5.5, at distance $\frac{1}{4}$ from the line, first make the smoothing negative, at $t = 1.18$. There is no conflict, because the orders above 10^{17} at $t \leq 1.29$ lie on the rays at heights $u = \sqrt{k/t} > 2.78 \times 10^8$, where the time $t_0(u)$ of Section 12.2.6, below which the phase free bound makes the smoothing positive whatever the zeros, exceeds 1.29. The quadruples lie below height 2000, at heights where t_0 is at most 0.511, and there (b) rests on the verified zeros, through Theorem 5.1.4.

5.1.4 Williamson measures: the Thorin measure smoothed by gamma laws

Complete monotonicity is the limit of k monotonicity, which asks only that the first k derivatives alternate in sign, and by Williamson's theorem [272] a k monotone function that vanishes at infinity is a positive mixture of the truncated powers

$(1-tz/k)_+^{k-1}$. For the heat trace the mixing measure is explicit,

$$W(t)=\int_0^\infty \Big(1-\frac{tz}{k}\Big)_+^{k-1}U_k(dz),\qquad U_k(dz)=\frac{k^k}{(k-1)!}z^{-k-1}c_k\Big(\frac{k}{z}\Big)dz,$$

and we call U_k the Williamson measure of order k. By Theorem 5.1.4 it is positive for every $k\le 10^{17}$ [198, Corollary 7.13], and by Theorem 5.1.10 for every $k\le 1.16\times 10^{25}$. RH holds if and only if $U_k\ge 0$ for infinitely many k, since then W is k monotone for infinitely many k, hence completely monotone; positivity of one U_k is a finite part of the Thorin condition.

The summands of U_k are inverse gamma laws, by an elementary identity: for $k\ge 1$ and $c\ge 0$,

$$e^{-c}=\mathbb{E}\Big(1-\frac{c}{\Gamma_k}\Big)_+^{k-1},\qquad \Gamma_k\sim\mathrm{Gamma}(k,1),\tag{5.1.2}$$

since $\int_c^\infty(\omega-c)^{k-1}e^{-\omega}d\omega/\Gamma(k)=e^{-c}$. With $c=a_jt$ and $z=ka_j/\Gamma_k$, so that $1-a_jt/\Gamma_k=1-tz/k$, the identity shows that U_k is the sum over the zeros of the densities $(ka_j)^kz^{-k-1}e^{-ka_j/z}/(k-1)!$. For real a_j this is the inverse gamma law of ka_j/Γ_k, with mean $ka_j/(k-1)$ when $k\ge 2$; for $k=1$ the mean is infinite. Under RH, therefore, writing $\mathcal{L}(Y)$ for the law of Y,

$$U_k=\sum_\gamma \mathcal{L}\Big(\gamma^2\,\frac{k}{\Gamma_k}\Big),$$

a sum of probability laws: the Thorin measure with each atom γ^2 spread over a relative width of about $1/\sqrt{k}$. As $k\to\infty$ the ratio Γ_k/k concentrates at one and the atoms reappear, which is the passage from Williamson's theorem to Bernstein's; in statistics, Polson, Scott and Windle used k monotone mixtures of this kind to build samplers for the Bayesian bridge [197, Section 2.2, Theorem 2.1] and noted the same limit [197, Section 3]. For an off line quadruple the two summands are complex conjugate densities whose sum is a real signed measure.

The Williamson measures form a second resolution family for the Thorin measure, beside the Poisson family ρ_ε, which is the Cauchy flow of Section 1.2. In the chart $\theta=\sqrt{z}$ the gamma smoothing spreads the atom at γ over a width of about $\gamma/(2\sqrt{k})$, proportional to the height, while the Poisson smoothing spreads every atom over the width ε, and the inverse gamma kernel has tails far lighter than the x^{-2} tails of the Cauchy kernel. In Figure 5.2, where the widths match at $\theta=30$, the Williamson densities fall by many orders of magnitude between neighbouring zeros and the Poisson densities by only one to four, because every zero contributes everywhere through its Cauchy tail.

The contrast decides what each family can do. The Poisson kernel is the real part of the Cauchy kernel $1/(s-\rho)$, so the Poisson smoothing of the zeros is $\pi^{-1}\,\mathrm{Re}(\xi'/\xi)$ off the line and can be computed from ξ alone at every resolution,

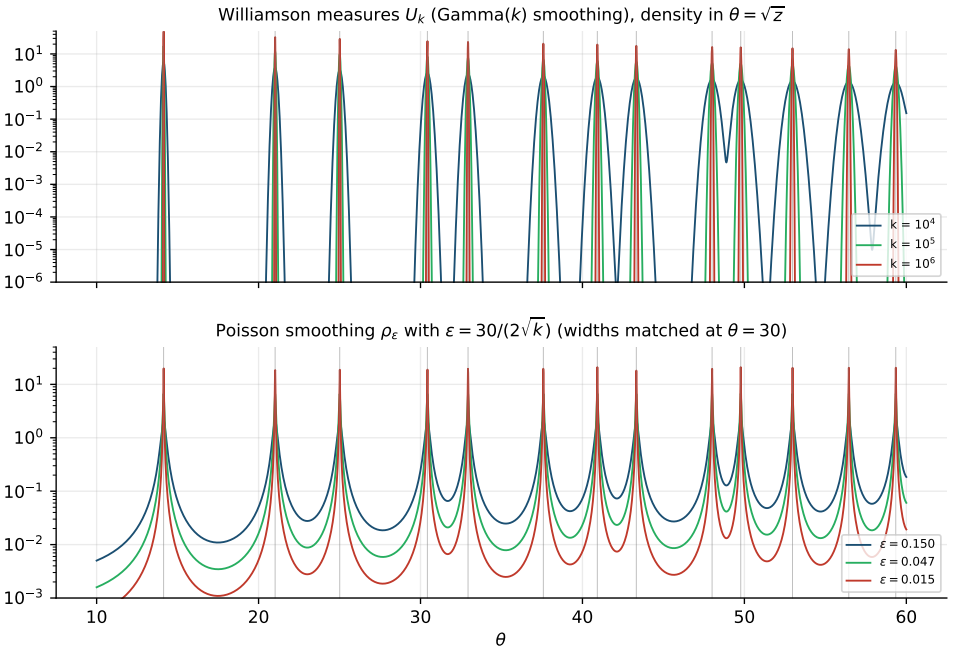


Figure 5.2: Top: the Williamson measures U_k of the heat trace for $k = 10^4$, 10^5, 10^6, computed from 1700 zeros, as densities in $\theta = \sqrt{z}$. Bottom: the Poisson smoothed Thorin densities ρ_ε with $\varepsilon = 30/(2\sqrt{k})$, so that the widths match at $\theta = 30$.

which is why ρ_ε is the central object of Chapter 3 and Section 4.1; the price is the heavy tail, which makes positivity at one resolution a statement about all zeros at once. The gamma smoothing is local, and an off line pair contributes to U_k a signed, oscillating term near its height. U_k is computed here from the zeros; along a ray of the (k, t) plane it is, to leading order, a Gaussian smoothing of the zeros, which the explicit formula gives from the primes, with a number of primes that grows quickly with the resolution (Section 5.1.5). Both families reach RH only in their limits, $k \to \infty$ and $\varepsilon \downarrow 0$.

5.1.5 Along a ray: the Gaussian smoothing of the zeros

To leading order the gamma smoothing of width $\gamma/(2\sqrt{k})$ is a Gaussian smoothing. Fix a height u and move along the ray $t = k/u^2$ of the (k, t) plane, on which the weight $x^k e^{-xt}$ of $c_k(t)$ peaks at $x = u^2$ and the density of U_k is taken at $z = u^2$. Write each zero as $\rho = \frac{1}{2} + \delta + i\gamma$ and put $\tilde{\gamma} = \gamma - i\delta$, so that $a = \tilde{\gamma}^2$; as in every sum of this section, the zero is taken with $\gamma > 0$ from each pair $\{\rho, 1 - \rho\}$. The

Gaussian smoothing of the zeros at resolution $1/(2\sqrt{t})$ is

$$\mathcal{G}(u,t) = \sum e^{-2t(\tilde\gamma-u)^2}.$$

A zero on the line contributes a Gaussian bump at its ordinate. The zeros $\frac12\pm\delta+i\gamma$ of a quadruple $\frac12\pm\delta\pm i\gamma$ have conjugate values of $\tilde\gamma$, so $\mathcal{G}$ is real, and together they contribute $2e^{2t(\delta^2-(\gamma-u)^2)}\cos(4t\delta(\gamma-u))$, a bump amplified by $e^{2t\delta^2}$ and modulated in u with period $\pi/(2t\delta)$. At $u=0$, $\mathcal{G}(0,t)=W(2t)$.

Proposition 5.1.8. *Let $u\ge 2$, $t>0$ and $k=u^2t$, not necessarily an integer, with a^k the principal power in $c_k(t)=\sum a^k e^{-at}$, and let $f_0=k(\log u^2-1)$. Then*

$$e^{-f_0}c_k(t) = \sum_{|\tilde\gamma-u|\le u/2} e^{-2t(\tilde\gamma-u)^2}\big(1+\varepsilon_\rho\big)+R, \qquad |\varepsilon_\rho|\le \exp\Big\{\frac{4t}{3u}|\tilde\gamma-u|^3\Big\}-1,$$

where, with $\ell_\pm = u\pm\frac12\sqrt{u^2-1}$, $H(r)=u^2\log\frac{r^2+1/4}{u^2}-r^2+\frac14+u^2$ and $\eta = 1-u^2/(\ell_+^2+\frac14)>0$,

$$|R| \le N(\ell_-)e^{tH(\ell_-)} + \Big(\ell_+^2+\frac{1}{t\eta}\Big)e^{tH(\ell_+)}.$$

No hypothesis on the zeros is used.

Proof. Put $\tilde\gamma = u+v$ and $w=v/u$. Since $\operatorname{Re}\tilde\gamma>0$ and $|\arg\tilde\gamma|<\frac\pi4$, the principal logarithm gives $e^{-f_0}a^k e^{-at}=e^E$ with

$$E = tu^2\big(2\log(1+w)-2w-w^2\big) = -2tv^2+2tu^2\sum_{m\ge3}\frac{(-1)^{m+1}w^m}{m},$$

and for $|w|\le\frac12$ the series has modulus at most $|w|^3/(3(1-|w|))\le\frac23|w|^3$. This gives $\varepsilon_\rho = e^{E+2tv^2}-1$ and its bound, since $|e^z-1|\le e^{|z|}-1$. For a zero with $|v|>u/2$, $|v|^2=(\gamma-u)^2+\delta^2$ and $\delta^2<\frac14$ give $|\gamma-u|>\frac12\sqrt{u^2-1}$, so that $\gamma<\ell_-$ or $\gamma>\ell_+$, and $|e^E| = \exp t\big(u^2\log(|a|/u^2)-\operatorname{Re}a+u^2\big)\le e^{tH(\gamma)}$, because $|a|=\gamma^2+\delta^2$ and $\operatorname{Re}a=\gamma^2-\delta^2$. The function H increases up to $\sqrt{u^2-\frac14}$, which lies between ℓ_- and ℓ_+, so each of the at most $N(\ell_-)$ zeros below ℓ_- contributes at most $e^{tH(\ell_-)}$. For $r\ge\ell_+$, $\log(1+x)\le x$ gives $H(r)\le H(\ell_+)-\eta(r^2-\ell_+^2)$. The explicit Riemann von Mangoldt formula [92, Corollary 1.2] gives $N(r)\le r^2$ for every r, and partial summation then gives $\sum_{\gamma>\ell_+}e^{-q(\gamma^2-\ell_+^2)}\le \ell_+^2+1/q$ for $q>0$; take $q=t\eta$. □

For $u\ge10$ the ratios $H(\ell_-)/u^2$ and $H(\ell_+)/u^2$ decrease in u, from -0.616 and -0.431 at $u=10$, and $\eta>0.55$, so that $|R|\le(3u^2+2/t)e^{-0.43k}$. What remains is ε_ρ, whose term of order $1/u$ is $\frac{2t}{3u}(\tilde\gamma-u)^3$. Along the ray, therefore, $e^{-f_0}c_k(t)$ is $\mathcal{G}(u,t)$ up to an exponentially small remainder and, in each term, a

relative error at most $e^{4t|\tilde{\gamma}-u|^3/3u}-1$. That error is small, of order $t|\tilde{\gamma}-u|^3/u$, for the terms that carry the sum provided $t|\tilde{\gamma}-u|^3$ is small against u for them; for the zeros within a few multiples of $1/\sqrt{t}$ of u it is of order $1/(\sqrt{t}\,u)$. It is not small in general: at a fixed height, as t grows, the term of the nearest zero dominates, and its relative error grows with t. Where the terms cancel, the errors must be summed in absolute value, since relative errors of the terms do not bound the relative error of the sum. For an integer k the density of U_k in the chart $\theta=\sqrt{z}$ is exactly $\sqrt{2t/\pi}\,e^{-\lambda_k}e^{-f_0}c_k(t)$ at $\theta=u$, where $t=k/u^2$ and $\lambda_k\in(\frac{1}{12k+1},\frac{1}{12k})$ is the remainder in Stirling's formula for $k!$ [221]. The density of U_k at θ is therefore about $\sqrt{2t/\pi}\,\mathcal{G}(\theta,t)$ with $t=k/\theta^2$: the zeros smoothed by normal densities of standard deviation $1/(2\sqrt{t})=\theta/(2\sqrt{k})$, which is the width found above. Section 5.2 uses this to explain where the derivative signs of a function with an off line quadruple first fail.

The Gaussian smoothing detects every off line zero, at a resolution fine enough.

PROPOSITION 5.1.9. *RH holds if and only if $\mathcal{G}(u,t)\ge 0$ for every real u and every $t>0$.*

Proof. Under RH every term is positive. Suppose that $\frac{1}{2}+d+ic$ is a zero with $d>0$, and let $|u-c|<\frac{d}{2}$. The term of a zero has modulus $e^{2t\phi}$ with $\phi=\delta^2-(\gamma-u)^2$. For the zero $\frac{1}{2}+d+ic$, $\phi>\frac{3}{4}d^2$; for a zero with $|\gamma-c|>1$, $\phi<\frac{1}{4}-\frac{9}{16}<0$. Group the finitely many zeros with $|\gamma-c|\le 1$ by their type $(\gamma,|\delta|)$. Two types at the same ordinate have exponents ϕ differing by a nonzero constant, and two at different ordinates have exponents differing by a nonconstant affine function of u, so outside finitely many points one type has the largest ϕ. Fix such a u_*, different from every ordinate, and let (γ_*,d_*) be the maximising type and ϕ_* its exponent. Then $\phi_*>\frac{3}{4}d^2$, so $d_*>0$, and the type consists of $\nu\ge 2$ zeros counted with multiplicity, which contribute $\nu e^{2t\phi_*}\cos(4td_*(\gamma_*-u_*))$. Along $t_j=(2j+1)\pi/(4d_*|\gamma_*-u_*|)$ this is $-\nu e^{2t_j\phi_*}$. The other n zeros with $|\gamma-c|\le 1$ contribute at most $ne^{2t_j(\phi_*-\Delta)}$, with $\Delta>0$ the gap to the next exponent, and the zeros with $|\gamma-c|>1$ at most $32\log(c+2)e^{-5t_j/8}$ once $t_j\ge 1$, since by the explicit Riemann von Mangoldt formula every interval of heights $[y,y+1]$ holds at most $4\log(y+3)$ zeros. Hence

$$e^{-2t_j\phi_*}\mathcal{G}(u_*,t_j)\le -\nu+ne^{-2t_j\Delta}+32\log(c+2)e^{-(2\phi_*+5/8)t_j},$$

which is negative for all large j. □

The proposition restricts Weil's criterion to the modulated Gaussians. For $f(x)=e^{-x^2/4t-iux}$ the transform $\hat{f}(r)=\int f(x)e^{irx}dx=\sqrt{4\pi t}\,e^{-t(r-u)^2}$ gives, summed over all zeros $\rho=\frac{1}{2}+i\gamma_\rho$,

$$\sum_\rho \hat{f}(\gamma_\rho)\overline{\hat{f}(\bar{\gamma}_\rho)}=4\pi t\big(\mathcal{G}(u,t)+\mathcal{G}(-u,t)\big),$$

where $\mathcal{G}(-u,t)$ comes from the zeros $1-\rho$ and is at most $e^{t/2}((u+14)^2+\frac{1}{2t})e^{-2t(u+14)^2}$ in modulus for $u \geq 0$, too little to change the sign in the proof. At $u=0$ the family gives the first rung, $\mathcal{G}(0,t) = W(2t) > 0$ (Theorem 5.1.1). Theorem 5.1.2 uses one time and the whole span of $r^m e^{-r^2 t/2}$; Proposition 5.1.9 uses one modulated Gaussian at a time, but every centre and every resolution. Since the Gaussian is entire, Weil's formula of Section 1.2 applies to $h(r) = e^{-2t(r-u)^2} + e^{-2t(r+u)^2}$, for which $\sum_\rho h(\gamma_\rho) = 2(\mathcal{G}(u,t)+\mathcal{G}(-u,t))$, and gives $\mathcal{G}$ from the primes:

$$\begin{aligned} \mathcal{G}(u,t)+\mathcal{G}(-u,t) &= 2e^{t/2-2tu^2}\cos 2tu + A(u,t) - P(u,t), \\ A(u,t) &= \frac{1}{4\pi}\int_{\mathbb{R}} \big(e^{-2t(r-u)^2}+e^{-2t(r+u)^2}\big) m(r)\,dr, \\ P(u,t) &= \frac{1}{\sqrt{2\pi t}} \sum_{n\geq 2} \frac{\Lambda(n)}{\sqrt{n}} e^{-(\log n)^2/8t}\cos(u\log n), \end{aligned} \tag{5.1.3}$$

with m as in (1.2.5).

Table 5.3 computes both sides, with the zeros of ξ below height 2197.26 and the primes to 10^9. The prime sum converges at every resolution, but more slowly the finer the resolution: since Chebyshev's function $\psi(x) = \sum_{n\leq x}\Lambda(n)$ grows like x, the weight $n^{-1/2}e^{-(\log n)^2/8t}$ acts, as a density in $\log n$, like $e^{t/2}$ times a Gaussian with centre $2t$ and standard deviation $2\sqrt{t}$. At $t=1$ the primes to 10^8 give $\mathcal{G}(100,1)$ to about 10^{-17}. At $t=3$ the value $\mathcal{G}(100,3) = 3.05\times 10^{-4}$ is the difference of an archimedean and a prime term of 0.32 each; the primes to 10^9 give it to 1.1×10^{-6}, and the bound on the rest of the prime sum that ignores its phases is a third of the value. The rest is nearly all main term: with the integral $(2\pi t)^{-1/2}\int_N^\infty x^{-1/2}e^{-(\log x)^2/8t}\cos(u\log x)\,dx$ that the prime number theorem predicts for it, the difference at $N=10^9$ falls to 3×10^{-10}. Where $\mathcal{G}$ is small the two arithmetic terms cancel, the more deeply the finer the resolution, as for W in Table 1.5; these are the points where an off line quadruple would first make $\mathcal{G}$ negative (Section 5.2), and Section 12.2.6 measures how far a bound on P that ignores its phases can reach.

5.1.6 The Gumbel tail of the Pólya kernel and the Weyl law

Every term of the Pólya kernel (1.2.3) contains the factor $\exp(-\pi n^2 e^{2u})$, and for large u only $n=1$ matters:

$$\Phi(u) \approx 2\big(2\pi^2 e^{9u/2} - 3\pi e^{5u/2}\big)\exp\big(-\pi e^{2u}\big), \qquad u\to\infty.$$

Call the right side Φ_1. The term n is $n^{-1/2}\Phi_1(u+\log n)$, so that $\Phi(u) = \sum_{n\geq 1} n^{-1/2}\Phi_1(u+\log n)$ for every real u, and for large u the term $n=2$ is smaller than Φ_1 by a factor of about $16\exp(-3\pi e^{2u})$. The tail is thus a double exponential in u, of Gumbel type.

u	t	$\mathcal{G}(u,t)$	$A(u,t)$	$P(u,t)$	$\lvert\mathcal{G}-A+P_N\rvert$		
					$N=10^4$	$N=10^6$	$N=10^9$
100	1	0.0960845325	0.5519917891	0.4559072567	2.3×10^{-6}	1.2×10^{-10}	9.6×10^{-19}
1977.22	1.75	1.9835406644	0.8672562847	−1.1162843797	1.2×10^{-3}	1.3×10^{-7}	2.2×10^{-13}
1000	2	0.8452691893	0.7150931047	−0.1301760846	5.8×10^{-4}	5.8×10^{-6}	1.9×10^{-11}
100	3	0.0003053958	0.3186935678	0.3183870527	2.0×10^{-3}	5.4×10^{-4}	1.1×10^{-6}

Table 5.3: The Gaussian smoothing of the zeros from the zeros and from the primes, by (5.1.3). $\mathcal{G}(u,t)$ is summed over the zeros below height 2197.26, the rest adding less than 10^{-70000}; the polar term and $\mathcal{G}(-u,t)$ are below 10^{-8000}. P_N is P with the sum cut at $n\le N$, and the P column is P_{10^9}. The bound on the rest of the prime sum that ignores its phases, from $\psi(x)<1.03883x$ [226, Theorem 12], is 1.6×10^{-20}, 2.1×10^{-10}, 1.0×10^{-8} and 1.0×10^{-4} at $N=10^9$ for the four rows; in the first row the difference at $N=10^9$ is the rounding error of the sums, in extended precision.

By (5.1.2) each factor $\exp(-\pi n^2e^{2u})$ is a Gamma(k) mixture of truncated powers $(1-\pi n^2e^{2u}/\omega)_+^{k-1}$, each supported on $u<\frac12\log(\omega/\pi n^2)$, and summed over n these powers form a Riesz mean of order $k-1$ of the counting function of the squares. So the theta function is a Gamma mixture of Riesz means of a lattice count, which tend to it as $k\to\infty$. The mixture is not a positive representation of Φ: the term n of (1.2.3) is $2e^{u/2}(2a^2-3a)e^{-a}$ with $a=\pi n^2e^{2u}$, so every piece is positive for $u>\frac12\log\frac{3}{2\pi}$, where $a>\frac32$ for all n, but the piece $n=1$ is negative further left; and Φ, being bell shaped, is not itself k monotone in u or in e^{2u}. With $k=2$ and $c=e^{-x}$ the identity writes the Gumbel distribution function as a mixture of triangles with a Gamma(2) mixing measure,

$$e^{-e^{-x}}=\int_0^\infty\Big(1-\frac{e^{-x}}{\omega}\Big)_+\omega e^{-\omega}\,d\omega.$$

Gamma(2) is also the shape in Williams' representation of the Pólya kernel (Section 8.2), but the coincidence gives no positive Gamma(2) mixture representation of the Pólya density [200, Section 12.6].

At leading order the tail determines the smooth part of the Thorin measure: it fixes the growth of Ξ along the imaginary axis, and a Tauberian argument turns that growth into the leading term of the Weyl law. By Laplace's method the main contribution to $\Xi(iy)=\int\Phi(u)\cosh(yu)\,du$ comes from the maximum of $yu-\pi e^{2u}$, at $e^{2u}=y/2\pi$, which gives

$$\log\Xi(iy)\approx\frac{y}{2}\log\frac{y}{2\pi e}.$$

Since Ξ is even of order one and $\Xi(0)\ne0$, $\log(\Xi(iy)/\Xi(0))=\sum\log(1+y^2/\gamma^2)$ over the zeros γ of Ξ with $\operatorname{Re}\gamma>0$. Replacing the sum by an integral against

the density $(2\pi)^{-1}\log(T/2\pi)$ of the Weyl law $N(T) \approx \frac{T}{2\pi}\log\frac{T}{2\pi e}$, which Section 8.1.3 uses for the tails of the Kent approximants, reproduces $\frac{y}{2}\log\frac{y}{2\pi e}$, because $\int_0^\infty \log(1+x^{-2})\,dx = \pi$ and $\int_0^\infty \log(1+x^{-2})\log x\,dx = -\pi$. In Table 5.4 the difference between the first two columns grows like $\frac{7}{4}\log y + 0.81$, the next terms of the expansion, which Stirling's formula gives, so the ratio tends to one slowly.

Replacing the sum by an integral assumes the Weyl law, so the computation above goes from the Weyl law to the growth of Ξ. The converse, from the growth of Ξ along the imaginary axis to the leading term of $N(T)$, needs a Tauberian argument, and it holds with no hypothesis on the zeros. Let γ run over the zeros of Ξ with $\operatorname{Re}\gamma > 0$, counted with multiplicity; then $\operatorname{Re}\gamma > 14$ and $|\operatorname{Im}\gamma| < \frac{1}{2}$, $N(T)$ is the number of zeros with $\operatorname{Re}\gamma \le T$, and $\sum_\gamma (\operatorname{Re}\gamma)^{-2} < \infty$ because Ξ has order one. Since $\Xi(iy) > 0$, taking moduli in the Hadamard product (1.2.1) gives $\log(\Xi(iy)/\Xi(0)) = \sum_\gamma \operatorname{Re} f(\gamma)$, where $f(z) = \log(1+y^2/z^2)$ is analytic in $\operatorname{Re} z > 0$ and real on the positive axis. On the segment from $\operatorname{Re}\gamma$ to γ its second derivative $f''(z) = 2(y^2-z^2)/(z^2+y^2)^2 + 2/z^2$ is at most $7/(\operatorname{Re}\gamma)^2$ in modulus for every real y, and the first order term $i\operatorname{Im}\gamma\, f'(\operatorname{Re}\gamma)$ of Taylor's formula is imaginary, so $|\operatorname{Re} f(\gamma) - f(\operatorname{Re}\gamma)| \le \frac{7}{8}(\operatorname{Re}\gamma)^{-2}$ and, uniformly in y,

$$\log\frac{\Xi(iy)}{\Xi(0)} = Q(y^2) + O(1), \qquad Q(x) = \sum_\gamma \log\Big(1 + \frac{x}{(\operatorname{Re}\gamma)^2}\Big).$$

By Stirling's formula the left side, which is $\log(\xi(\frac{1}{2}+y)/\xi(\frac{1}{2}))$, is asymptotic to $\frac{y}{2}\log y$, the leading term of the saddle point value, so $Q(x) \sim \frac{1}{4}\sqrt{x}\log x$ as $x \to \infty$. Its derivative $Q'(x) = \sum_\gamma (x + (\operatorname{Re}\gamma)^2)^{-1}$ decreases, so the monotone density theorem [76, XIII.5] gives $Q'(x) \sim \frac{1}{8}x^{-1/2}\log x$, and Q' is the Laplace transform of the decreasing function $V(v) = \sum_\gamma e^{-v(\operatorname{Re}\gamma)^2}$. The Tauberian theorem for Laplace transforms [76, XIII.5], with the roles of 0 and ∞ interchanged and combined with the monotone density theorem, whose proof applies unchanged at the origin, turns this into $V(v) \sim (8\sqrt{\pi})^{-1}v^{-1/2}\log(1/v)$ as $v \downarrow 0$, and the same theorem applied to $V(v) = \int_0^\infty e^{-vx}\,dN(\sqrt{x})$ gives $N(\sqrt{x}) \sim (4\pi)^{-1}\sqrt{x}\log x$, that is,

$$N(T) \sim \frac{T}{2\pi}\log T, \qquad T \to \infty.$$

The Tauberian argument gives only this leading term. The lower order terms of the Weyl law, and the bound $S(T) = O(\log T)$ [255, Theorem 9.4], come from the argument principle through $N(T) = \vartheta(T)/\pi + 1 + S(T)$ (Section 2.2).

The fluctuation $S(T)$ of the zero count about the Weyl law is absent from this leading behaviour, and the translates of Φ_1 show why. For $y > -\frac{5}{2}$ the substitution $x = e^{2u}$ and Euler's integral give

$$\int_{\mathbb{R}} \Phi_1(u)e^{yu}\,du = \tfrac{1}{2}s(s-1)\pi^{-s/2}\Gamma(\tfrac{s}{2}), \qquad s = \tfrac{1}{2}+y,$$

y	$\log(\Xi(iy)/\Xi(0))$	$\frac{y}{2}\log\frac{y}{2\pi e}$	ratio
50	34.511	26.854	1.285
200	256.128	246.044	1.041
1000	2047.839	2034.939	1.006

Table 5.4: Growth of Ξ along the imaginary axis against the saddle point value from the Gumbel tail of the Pólya density.

and for $y > \frac{1}{2}$, since $\Xi(iy) = \int \Phi(u)e^{yu}\,du$ by evenness, integrating the translates term by term gives $\Xi(iy) = \zeta(s)\cdot\frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(\frac{s}{2})$. So Φ_1, which dominates the tail, is the kernel of the gamma factor and produces, through the argument principle, the smooth part $\vartheta(T)/\pi + 1$ of $N(T)$ (Section 2.2), while its translates by $\log n$, with weights $n^{-1/2}$, produce $\zeta(s) = \sum_n n^{-s}$ and with it $S(T)$, which sets the level at which the error of the Kent approximants with a Weyl tail settles for large arguments of the clock (Section 8.1.4). The absence of $S(T)$ concerns the leading asymptotics only: Φ is real analytic, so its restriction to a half line determines it, and with it every zero.

5.1.7 What a verification height buys

Chapters 6 and 7 reach further rungs by the same method. Adding to the clock variable an independent positive stable variable of index $\frac{1}{2}$, that is, multiplying the clock by $e^{-c\sqrt{u}}$ with $c = \zeta'(\frac{1}{2})/\zeta(\frac{1}{2}) = 2.6860917$, gives the floor $e^{-c\sqrt{u}}\,\xi(\frac{1}{2})/\xi(\frac{1}{2}+\sqrt{u})$ of Section 6.2, whose canonical function, y times its Lévy density, is $k(y) = W(y) + c/(2\sqrt{\pi y})$. By Theorem 5.1.1, $k > W > 0$, so the floor is infinitely divisible unconditionally; Section 6.2 proves this independently, and proves that the floor is a GGC if and only if RH holds and that $(-1)^n k^{(n)}(y) > 0$ for every $n \le 10^{10}$ (Theorems 6.2.15 and 6.2.17). Weil's functional is positive on the cone of test functions completely monotone in r^2 (Theorem 7.2.14); this is the first rung integrated against a positive weight, $\sum_\rho \hat{h}(\gamma_\rho) = 2\int_0^\infty W(y)g(y)\,dy$ with $g \ge 0$, and for it Section 7.2 proves $W > 0$ afresh, from the prime side up to $y = 0.003$ and from the zeros below height 50 beyond.

Every rung is reached by the same mechanism: near the origin the archimedean term dominates the primes, and away from it a known positive term, usually a verified low zero, dominates the unverified high zeros. Only the second limits the order, as the proof of Theorem 5.1.4 shows. The first limits the time instead: it gives every order $k \ge 1$ up to $t = 0.019$, and every order above 10^{17} up to $t = 1.29$, because, relative to the size $s_k(t)$ of the kth derivative of $1/(4\sqrt{\pi t})$, the archimedean term grows like $\frac{1}{2}\log k$ while the bound on the primes does not grow with k (Theorem 5.1.7). In the proof of Theorem 5.1.4, with $\gamma_0 \in (20, 50]$ the height of the dominating zero, a zero at height $\gamma \ge H$ contributes about $(\gamma/\gamma_0)^{2k}e^{-(\gamma^2-\gamma_0^2)t}$ times as much, wherever it lies. The condition $tH^2 \ge 2(k+1)$

makes this ratio decrease in γ, and the logarithmic condition asks, to leading order, that it be less than one at $\gamma = H$ and $t = \tau$, that is $k\log(H^2/\gamma_0^2) < \tau H^2$; larger t only help. The reach is therefore about $\tau H^2/\log(H^2/\gamma_0^2)$, which is 10^{18} for $H = 10^{12}$, so the 10^{17} of the theorem is a safe round number; the height 3×10^{12} of [196] would raise the estimate to about 9×10^{18}.

The logarithm is the price of a fixed dominating zero. The weight $x^k e^{-xt}$ of $c_k(t)$ peaks at $x = k/t$, that is at the height $u = \sqrt{k/t}$, and a zero on the line a few units above u outweighs every zero above H by the Gaussian factor $e^{-t(H-u)^2}$, however many zeros lie there. The zeros are needed only from $t = 1.29$ on, where Theorem 5.1.7(b) hands over, instead of from $\tau = 1/20000$.

THEOREM 5.1.10. *Let $H = 3\times10^{12}$, the height to which RH is verified [196]. Then $(-1)^k W^{(k)}(t) > 0$ for every $t \geq 1.29$ and every integer k with $0 \leq k \leq t(H-23)^2$. Consequently $(-1)^k W^{(k)}(t) > 0$ for every $t > 0$ and every integer k with $0 \leq k \leq 1.29(H-23)^2$, and $1.29(H-23)^2 > 1.16 \times 10^{25}$.*

Proof. Fix $t \geq 1.29$ and k, and put $u = \sqrt{k/t}$, $D = H - u \geq 23$ and $f(v) = v^{2k}e^{-tv^2}$. As in Section 5.1.5, write each zero as $\frac{1}{2} + \delta + i\gamma$ and put $\tilde\gamma = \gamma - i\delta$, so that $c_k(t) = \sum f(\tilde\gamma)$ over one zero of each pair $\{\rho, 1-\rho\}$, taken with $\gamma > 0$. On $(0,\infty)$, $(\log f)'' = -2k/v^2 - 2t$ is at most $-2t$, and at least $-4t$ on $[u,\infty)$, while $(\log f)'(u) = 0$; hence $f(v) \leq f(u)e^{-t(v-u)^2}$ for every $v > 0$, and $f(v) \geq f(u)e^{-2t(v-u)^2}$ for $v \geq u$. The zeros below height H lie on the line and contribute $f(\gamma) > 0$. One of them lies in $[u, u+15]$: for $u \leq 1000$ because $\gamma_1 < 15$ and consecutive ordinates below 1015 differ by less than 7, and for $u \geq 1000$ because the explicit Riemann von Mangoldt formula of [92, (1.7) and Corollary 1.4] gives $N(u+15) - N(u) > 4$. That zero contributes at least $f(u)e^{-450t}$. A zero above H, wherever it lies, has $|\delta| < \frac{1}{2}$ and $|\tilde\gamma| \geq \gamma > u$, so its term has modulus $f(|\tilde\gamma|)e^{2t\delta^2} \leq e^{t/2}f(u)e^{-t(\gamma-u)^2}$. Partial summation with $N(r) \leq r^2$, and $\int_D^\infty e^{-tv^2}dv \leq e^{-tD^2}/(2tD)$, give

$$\sum_{\gamma>H} e^{-t(\gamma-u)^2} \leq e^{-tD^2}\Big(H^2 + \frac{1}{t} + \frac{H}{tD}\Big),$$

and therefore

$$\frac{c_k(t)}{f(u)} \geq e^{-450t} - e^{t/2-tD^2}\Big(H^2 + \frac{1}{t} + \frac{H}{tD}\Big) > 0,$$

since $t(D^2 - 450.5) \geq 1.29 \cdot 78.5 > 101$ while $\log(H^2 + 1 + H/23) < 57.5$. For the second statement, Theorem 5.1.7(b) gives every order for $t \leq 1.29$, and for $t > 1.29$ every $k \leq 1.29(H-23)^2$ satisfies $k \leq t(H-23)^2$. The program `window_orders.py` (Appendix A) computes the zeros below height 1015 in ball arithmetic and evaluates every numerical inequality of the proof. □

The reach is thus $1.29H^2$, up to the 23, against $\tau H^2/\log(H^2/\gamma_0^2)$ for the proof of Theorem 5.1.4. The theorem leaves room for a failure of the signs only at $t > 1.29$ and at orders $k > t(H-23)^2$, whose weights peak within 23 of the verified height or above it, and that is where Proposition 5.1.8 and Table 5.5 place the first failure caused by an off line quadruple: near its own height g, at an order close to t^*g^2. For the floor, $c/(2\sqrt{\pi y})$ is completely monotone, so Theorem 5.1.10 gives $(-1)^n k^{(n)} > 0$ for every $n \le 1.16 \times 10^{25}$; Theorem 6.2.17 proves $n \le 10^{10}$ independently, with a Cauchy circle of radius $y/2$, which loses a factor 2^n and makes its reach grow only linearly in H.

The reachable order thus grows with the verification height, in Theorem 5.1.10 like its square: every finite rung is within reach of a high enough verification, and no verification reaches the top, so the ladder measures how much of RH a given amount of verification buys. Each proof lets the zeros above the verified height lie anywhere, so what it proves is consistent with zeros off the line there, and Section 5.2 gives an explicit example.

5.2 What finite information cannot decide

Every computation in this book is finite: a finite window, finitely many resolutions, finitely many derivatives or matrix blocks. Theorem 5.2.1 replaces two close zeros of ξ on the line by a quadruple off it [200, Chapter 11] and obtains an entire function whose heat trace passes the finite tests of Section 5.1. Below height 2000 the order at which its derivative signs first fail can be computed, and it grows roughly like the square of the height. Section 5.2.2 records what each finite test of the book establishes and what it cannot exclude; the sharpest case is positivity of ρ_ε at one resolution, which can miss zeros far from the line. Finally, a finite computation can refute a sufficient condition for RH: one certified zero of an explicit function closes the Bernstein Pick route through the moments of the Pólya density.

5.2.1 A perturbation that passes the finite tests of the heat trace

Theorem 5.2.1 ([198, Theorem D]). (a) *There are ordinates* $1000 < A < B \le 2000$ *of zeros on the critical line, with* $B - A \le 2$*, such that, with* $g = (A+B)/2$ *and* $w = \frac{1}{4} + ig$*,*

$$F(s) = \frac{\xi(\frac{1}{2}+s)}{\xi(\frac{1}{2})}\,\frac{(1-s^2/w^2)(1-s^2/\bar{w}^2)}{(1+s^2/A^2)(1+s^2/B^2)}$$

is an even entire function with the order, type and reflection symmetry of $\xi(\frac{1}{2}+s)$ *and with four zeros off the critical line, and the following hold. The reciprocal clock* $1/F(\sqrt{u})$ *is the Laplace transform of an infinitely divisible, self decomposable law,*

and its heat trace

$$W_F(t) = W(t) + 2\,\mathrm{Re}\,e^{-at} - e^{-A^2t} - e^{-B^2t}, \qquad a = -w^2 = g^2 - \tfrac{1}{16} - \tfrac{ig}{2},$$

which is t times its Lévy density, is strictly log convex. W_F has the same small time expansion as W through the constant term, and $1/F$ has the same nearest poles as the reciprocal Xi function.

(b) *Let $N \geq 1$ and $R \geq T_N$, where $T_N = 20000N\sqrt{1+\log N}$ as in Section 5.1, and suppose that every zero of ξ with ordinate in $(0, R+50]$ lies on the critical line. Then the pair can instead be chosen with $R < A \leq B \leq R+50$ and $B - A < 2$ (if $A = B$, a multiple zero supplies both), so that F keeps every property listed in (a) and the leading $N \times N$ Hankel blocks of W_F remain positive definite at every $t > 0$. If moreover $R \geq 10^{12}$, the derivative signs of Theorem 5.1.4 are also preserved: $(-1)^k W_F^{(k)}(t) > 0$ for every $t > 0$ and every integer $0 \leq k \leq 10^{17}$. For every $N \leq 3 \times 10^7$ the verification to height 3×10^{12} [196] establishes the hypothesis on the zeros for $R = \max(T_N, 10^{12})$, so that such an F with both properties exists unconditionally.*

The construction moves the zeros at heights A and B, with their conjugates, to the quadruple $\frac{1}{2} \pm \frac{1}{4} \pm ig$. The heat trace changes by the oscillating term $W_F - W$, whose envelope is bounded as in Section 5.1 [198, Proposition 11.1], and at the heights of (b) this change is smaller than the margin by which each of its tests holds. The pair of (b), with the properties of (a) and the Hankel blocks, comes from [198, Theorem 12.5], the derivative signs from [198, Proposition 12.7], which holds for any pair of zeros on the line with $A > 10^{12}$ and $B - A \leq 2$, and the unconditional range from [198, Corollary 12.6], since $T_{3\times10^7} + 50 < 2.6 \times 10^{12}$. Figure 5.3 shows the change for a pair in the range of (a), far below the heights that (b) requires; the pair it moves, near height 1977.22, is the closest pair among the first 1700 zeros, whether or not the spacings are normalised (Section 5.4.2).

Not every finite test is passed. Below any height $T > B$ the argument count of F equals that of ξ, since F loses the zeros at heights A and B and gains two at height g, but these two are off the line, so F has two zeros fewer on the line, counted with multiplicity; when $A < B$ and both zeros are simple, as for the pair of Figure 5.3, it has two sign changes fewer. Turing's comparison of the two counts, the first test of Table 5.6, then detects the perturbation once it passes height B, and the zero density identity at any height $T > g$ returns $\nu(\frac{1}{2}+\varepsilon, T) \geq 1$ for every $\varepsilon < \frac{1}{4}$, whatever the multiplicities. Counting tests are local in height and find the perturbation once their window contains it, whereas a given finite set of heat trace tests, which involve all the zeros at once, misses it when it lies high enough.

The pair of Figure 5.3 lies far below the heights of (b), and at its height the derivative signs of F do fail at a finite order, which can be computed. On the ray $t = k/u^2$ the normalised value $e^{-f_0} c_k^F(t)$ of Proposition 5.1.8, whose proof applies to F because F has at most one zero more than ξ below any height, is a sum of positive terms, one for each zero on the line, the oscillating term of the quadruple

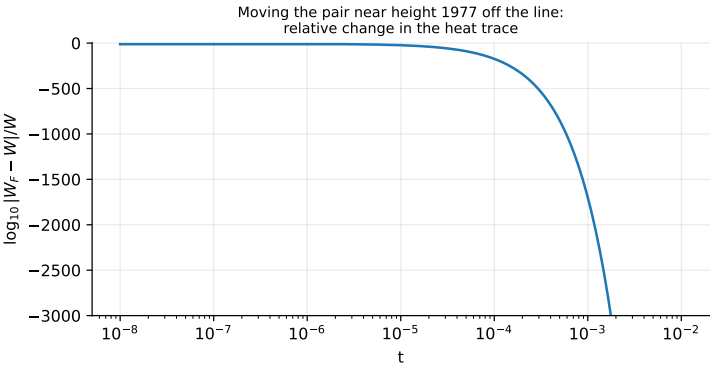


Figure 5.3: Relative change $|W_F - W|/W$ of the heat trace, on a $\log_{10}$ scale, when the closest pair of zeros between heights 1000 and 2000 (near 1977.22) is moved to distance $\frac{1}{4}$ from the line, computed in forty digit arithmetic with W replaced by its sum over the first 1700 zeros. The change is at most 4×10^{-11}, near $t = 7 \times 10^{-7}$, and falls below 10^{-1000} by $t = 10^{-3}$. The sum over 1700 zeros is smaller than W, by 1.3 per cent at $t = 7 \times 10^{-7}$ and by a factor of 5.4 at $t = 10^{-8}$, so the curve bounds the relative change from above; with W from the explicit formula (1.2.5) the maximum is 3.66×10^{-11}, at $t = 7.2 \times 10^{-7}$.

and terms, negligible here, from any zeros off the line above 3×10^{12}, and only the zeros within a few units of u matter. A scan over u and over the orders, at every order up to 500, then on a grid of ratio 1.002 and at every order of the last grid interval, finds that the first order at which $c_k^F(t) < 0$ for some $t > 0$ is $k = 6\,683\,655$, at $t = 1.7113393$. The grid loses nothing, because $c_{k-1}^F(t) = \int_t^\infty c_k^F(s)\,ds$: if $c_k^F \geq 0$ on $(0, \infty)$ then so is c_{k-1}^F, and the orders that fail are all those from the first one on. The negative value is certified in ball arithmetic, with explicit bounds for the zeros below $g - 40$, which lie on the line, and for those above $g + 40$, wherever the ones above 3×10^{12} lie; the positivity at the lower orders is computed, not certified. Table 5.5 repeats the computation for the closest pair of consecutive zeros in five other windows of height, and for the pair near 1977.22 moved to distance 0.1 instead of $\frac{1}{4}$.

The first failing order is $k^* = t^* u_*^2$, where $u_* = \sqrt{k^*/t^*}$ is the height of the failure, between 0.69 and 1.40 from g, so that k^* is within 1.6 per cent of $t^* g^2$, with t^* between 1.18 and 2.51 for $\delta = \frac{1}{4}$: it grows roughly with the square of the height, while t^* shows no trend in the height on this range. Proposition 5.1.8 explains this. Along the ray, $e^{-f_0} c_k^F(t)$ is, up to relative errors of order $t|\tilde{\gamma} - u|^3/u$ in the terms near u, small at these heights and times (Section 5.1.5), the Gaussian smoothing $\mathcal{G}(u,t)$ of the zeros of F, in which the quadruple contributes $2e^{2t(\delta^2 - (g-u)^2)} \cos(4t\delta(g-u))$. This term is negative only where $4t\delta|g-u|$ lies in $(\frac{\pi}{2}, \frac{3\pi}{2})$ modulo 2π, hence at distance at least $\pi/(8t\delta)$ from g, where its modulus is

window	zeros moved	g	δ	k^*	t^*	$c^F_{k^*}(t^*)$, normalised
$(100, 200)$	71, 72	185.24	$\frac{1}{4}$	40 018	1.18413	-6.37×10^{-7}
$(200, 400)$	186, 187	376.08	$\frac{1}{4}$	219 169	1.54072	-6.70×10^{-8}
$(400, 700)$	363, 364	630.64	$\frac{1}{4}$	624 921	1.576635	-1.68×10^{-7}
$(700, 1000)$	453, 454	750.81	$\frac{1}{4}$	792 746	1.410725	-9.26×10^{-8}
$(1000, 1500)$	922, 923	1329.12	$\frac{1}{4}$	4 433 220	2.5121193	-3.37×10^{-8}
$(1500, 2000)$	1496, 1497	1977.22	$\frac{1}{4}$	6 683 655	1.7113393	-7.12×10^{-9}
$(1500, 2000)$	1496, 1497	1977.22	0.1	15 722 219	4.0257254	-6.71×10^{-11}

Table 5.5: The first failing order of the derivative signs when the closest pair of consecutive zeros in a window of height is moved to the quadruple $\frac{1}{2} \pm \delta \pm ig$, g the midpoint of the pair. k^* is the least order at which $c^F_k(t) < 0$ for some $t > 0$, found by a scan of every order up to 500, a grid of ratio 1.002 above it and every order of the last grid interval (computed, not certified); $c^F_{k^*}(t^*) < 0$ is certified in ball arithmetic. The last column is $c^F_{k^*}(t^*)$ divided by $(k^*/t^*)^{k^*} e^{-k^*}$, the value at $x = k^*/t^*$ of the weight $x^{k^*} e^{-xt^*}$.

at most $2e^{2t\delta^2 - \pi^2/(32t\delta^2)}$, while a zero on the line at distance d from u contributes e^{-2td^2}. So $\mathcal{G}(u, t) < 0$ requires every zero on the line to lie at a distance d from u with $2td^2 > \pi^2/(32t\delta^2) - 2t\delta^2 - \log 2$: for $\delta = \frac{1}{4}$ that is $d > 1.43$ at $t = 1$ and $d > 0.62$ at $t = 2$. The first failure time is therefore set by δ and by the gaps between the line zeros near the moved pair, which its removal widens; the mean spacing $2\pi/\log(g/2\pi)$ falls from 1.86 to 1.09 between the first and last rows, but the gaps near the moved pairs vary more, which is why t^* shows no trend. The first t at which $\min_{|u-g| \le 8} \mathcal{G}(u, t) < 0$, computed in floating point, predicts k^* to 0.21 per cent at height 185 and to 0.08 per cent or better at the other heights, and the minimum lies just beyond the distance $\pi/(8t\delta)$ at which the first negative lobe begins, between 0.7 and 1.4 from g. These failure times of $\mathcal{G}$ within 8 of g are certified in ball arithmetic, with the zeros from Arb, for all seven rows of Table 5.5: $\mathcal{G}(u, t) > 0$ at every centre u within 8 of g for every t up to the failure time rounded down to two decimals, and $\mathcal{G} < 0$ at one centre at the failure time rounded up to four decimals. For the pair near 1977.22 that is $\mathcal{G} > 0$ within 8 of g for every $t \le 1.71$ and $\mathcal{G}(g - 0.98587, 1.7112) < 0$. Far from g the bracket says nothing: near or above 3×10^{12}, zeros off the line could make $\mathcal{G}$ negative earlier. One time suffices for the first statement, since $\sqrt{2t/\pi}\, \mathcal{G}(\cdot, t)$ at a smaller t is its value at a larger one convolved with a centred normal density: the sum over the zeros below 3×10^{12} is positive on the whole line at $t = 1.71$, by an interval cover within 8 of g and by domination beyond, so it has a uniform positive lower bound at every smaller t within 8 of g, and the zeros above 3×10^{12}, wherever they lie, can lower $\mathcal{G}$ there by less than $10^{-2 \times 10^{12}}$ (`gsmooth_arb.py`); what they add may be large. Unlike the Hankel condition of Theorem 5.1.2 at one time, positivity of $\mathcal{G}$ at one resolution does not decide whether the zeros are on the line. Moving

the pair near 1977.22 to distance 0.1 raises t^* from 1.71 to 4.03: the closer the quadruple to the line, the finer the resolution needed to see it.

The same mechanism puts the first failure for a quadruple just above the verified height far beyond the orders of (b). There $g^2 \approx 10^{25}$, so the first failing order is about $10^{25}t^*$, and t^* depends on the gaps around the quadruple, which a computation at height 3×10^{12} would have to supply. The mean spacing there is 0.23, and for $\delta = \frac{1}{4}$ a failure at $t \le 1$ would need a stretch of length 2.9, about twelve mean spacings, free of zeros on the line other than the moved pair. A model with the line zeros spread at their mean density gives $t^* \approx 10$ there, but the same model gives 5.9 at height 1977 against the computed 1.71, because it ignores the gaps, so it gives the order of magnitude only. Theorem 5.1.10 certifies the signs of ξ to every order up to $1.29(H-23)^2 > 1.16 \times 10^{25}$, with $H = 3 \times 10^{12}$, so a quadruple of ξ just above H could make a sign fail only at a time $t^* > 1.29$, at an order of about t^*H^2: the orders certified at the time t^* reach that size, and by the model the orders certified at every time fall short of it by a factor of about eight. This is the entry of Table 5.6 on derivative signs.

5.2.2 What finite tests establish, and the local visibility condition

Table 5.6 lists, for each finite test used in this book, where it is made, what it establishes and what it cannot exclude. Each entry of the middle column, made rigorous, is an unconditional theorem on its window; the right column is the reason why no accumulation of such entries proves RH.

The second row rests on a local computation. Suppose that ξ has a pair of zeros $\frac{1}{2} \pm d + i\gamma$ with $d > \varepsilon$, where d is the displacement written δ in Section 5.1 and in Proposition 1.2.3, and that the other zeros near height γ lie on the line at the ordinates $\gamma + \delta_j$. A zero ρ contributes $\mathrm{Re}(s-\rho)^{-1}$ to $\pi\rho_\varepsilon(\theta)$, where $s = \frac{1}{2} + \varepsilon + i\theta$, so at the ordinate, ignoring distant zeros and those at negative heights,

$$\pi\rho_\varepsilon(\gamma) = \varepsilon\Big(\sum_j \frac{1}{\delta_j^2 + \varepsilon^2} - \frac{2}{d^2 - \varepsilon^2}\Big).$$

To first order in ε, the off line zero therefore makes ρ_ε negative at its ordinate if and only if $2/d^2 > \sum_j \delta_j^{-2}$; this is the local visibility condition. For neighbours at the mean spacing $\Delta = 2\pi/\log(\gamma/2\pi)$, that is $\delta_j = j\Delta$ with $j \neq 0$, the sum is $\pi^2/(3\Delta^2)$, and the condition becomes $d < (\sqrt{6}/\pi)\Delta = \sqrt{24}/\log(\gamma/2\pi)$, about 0.78 of the mean spacing. A zero farther from the line leaves ρ_ε positive at its ordinate at every small resolution and makes it negative there only at resolutions comparable to d, while a zero within ε of the line contributes positively and is not seen at all. At a fixed resolution the same scale $1/\log\gamma$ marks, heuristically, how far to the right of $\frac{1}{2} + \varepsilon$ a zero can lie and still make ρ_ε negative (Section 11.4.3).

The condition concerns the ordinate itself. For evenly spaced neighbours the neighbour sum is smallest and the pair term most negative there, so the condition

test	establishes	cannot rule out
sign changes and argument count to height T (Sections 2.1 and 2.2)	all zeros below T on the line	anything above T
$\rho_\varepsilon \geq 0$ on $0 < \theta \leq \Theta$ (Section 3.1)	no zero in the window at distance $d > \varepsilon$ from the line with $2/d^2 > \sum_j \delta_j^{-2}$ (to first order in ε)	zeros within ε of the line; zeros farther from it than about 0.78 of the mean spacing at their height; zeros above the window
zero density scan at T (Section 3.4)	$\nu(\frac{1}{2}+\varepsilon, T) = 0$ for the ε computed	zeros within ε of the line; zeros above T
Hankel blocks of size N (Section 5.1)	Weil positivity on a finite space	off line zeros above T_N
derivative signs to order K (Section 5.1)	finite order Williamson positivity	off line zeros above a height of about $\sqrt{K/1.29}$
HCM derivative signs (Section 4.1)	finitely many CM conditions	failure at other u or orders
matched total variation at resolution ε (Section 3.2)	agreement with the Poisson smoothing of the list	zeros outside the window
Li coefficients λ_n, $n \leq N$ (Section 8.3)	finitely many of Li's inequalities	off line zeros $\frac{1}{2} \pm d + i\gamma$ with Nd/γ^2 small

Table 5.6: What finite tests establish and what they cannot exclude.

also decides whether ρ_ε is negative near γ; in general it does not, because away from the ordinate the neighbour sum can fall faster than the pair term grows. With $d = 0.3$, $\varepsilon = 0.005$ and neighbours at $\gamma + 0.2 + m$ and $\gamma - 4 - m$, $m = 0, 1, 2, \ldots$, the sum $\sum_j \delta_j^{-2} = 26.55$ exceeds $2/d^2 = 22.2$ and $\rho_\varepsilon(\gamma) = 0.0069$, yet $\rho_\varepsilon(\gamma - 0.079) = -0.0060$ (Exercise 5.15). A search for negative values of ρ_ε must therefore cover every point of the window, not only the ordinates, and every resolution: a zero that the condition hides at small resolution still makes ρ_ε negative at its ordinate as ε approaches its distance from the line.

The zero density identity of Proposition 3.4.2 has no blind spot of this kind: one evaluation at resolution ε returns $\nu(\frac{1}{2} + \varepsilon, T)$, the number of zeros with real part greater than $\frac{1}{2} + \varepsilon$ and ordinate in $(0, T]$, however far from the line they lie. The same count controls the negative part of ρ_ε: the negative kernels in its Poisson form, which come from the zeros to the right of $\frac{1}{2} + \varepsilon$, have total mass on $(0, T]$ at most the number of such zeros below height $2T$ plus a tail that tends to zero as $T \to \infty$ when that number is $O(T^c)$ for some $c < 1$ (Lemma 11.4.1). The zero density estimates of Ingham [104] and of Guth and Maynard [88] give such a bound with no hypothesis and make this mass a vanishing fraction of the positive mass (Theorem 11.4.2).

5.2.3 Refuting sufficient routes with one certificate

Not every route to RH through the Pólya density is a positivity question on the clock. Konstantopoulos, Patie and Sarkar [127] introduced a class $\mathcal{D}_P$ of solutions of the van Dantzig problem, the entire functions $\sum_{n\ge0}(-1)^n t^{2n}/(\Psi(1)\cdots\Psi(n))$ in which Ψ is the Laplace exponent of a spectrally negative Lévy process whose largest zero q is at most $\frac12$; then $\Psi(u)=(u-q)\varphi(u)$ with φ a Bernstein function [127, Proposition 5.2]. If $q=0$ and φ is a Bernstein Pick function whose zeros and poles satisfy a separation condition, every zero of the entire function is real [127, Theorem 4.4], so a representation of $\Xi/\Xi(0)$ of this kind would prove RH; this is the Bernstein Pick route. Let X have the Pólya density and put $M(s)=\mathbb{E}|X|^{2s}$. Since $\Xi(t)/\Xi(0)=\sum_{n\ge0}(-1)^n M(n)t^{2n}/(2n)!$, such a representation needs $\Psi(n)=n\varphi_I(n)$ for every $n\ge1$, where $\varphi_I(s)=4(s-\frac12)M(s-1)/M(s)$ is the canonical interpolant of the moment ratios; that is, φ must interpolate $n\varphi_I(n)/(n-q)$ at the integers. The function M is analytic for $\operatorname{Re}s>-\frac12$, and for $-\frac12<\operatorname{Re}s<0$ the classical formula for negative absolute moments gives

$$M(s)=\frac{2\,\Gamma(2s+1)\sin(-\pi s)}{\pi\,\Xi(0)}\int_0^\infty \Xi(\theta)\,\theta^{-2s-1}\,d\theta,$$

whose right side continues M meromorphically to $\operatorname{Re}s<0$.

Theorem 5.2.2 ([198, Theorem E]). *M has exactly one zero in the disc $|s-s_1|<10^{-8}$ with centre $s_1=-8.494182256992559+2.937767068567584i$; the zero is simple, and $M(s-1)$ does not vanish in the disc. Hence φ_I has a nonreal pole, and every Bernstein interpolant of the moment ratios agrees with φ_I on $(1,\infty)$ and cannot be continued through that pole, so none is a complete Bernstein function. For every $0<q<1$ the sequence $n\varphi_I(n)/(n-q)$ has no Bernstein interpolant either.*

Since the Bernstein Pick functions are the complete Bernstein functions, the theorem closes the Bernstein Pick route, and its last statement excludes $\Xi/\Xi(0)$ from $\mathcal{D}_P$ with $0<q\le\frac12$, the case $q>0$ of the question raised in [127, Section 4.1]. Here a finite computation is used in the opposite direction: one interval certificate for a zero of an explicit function refutes a whole sufficient route. The theorem belongs to the arcsine route of [200, Chapter 12], which represents X in law as $2\sqrt{I}$ times the cosine of an independent angle uniform on $[0,\pi]$, with I a positive random variable, so that $\varphi_I(n)=n\mathbb{E}I^{n-1}/\mathbb{E}I^n$ are the moment ratios of I. Its last statement is proved there by a beta multiplier obstruction, and at the single remaining value $q=0$ membership in $\mathcal{D}_P$ reduces to a Hausdorff moment criterion.

Finite information can therefore establish statements on a verified window and refute sufficient conditions by single obstructions, but it cannot establish the Thorin condition: just above any sufficiently large height R for which RH has been verified up to $R+50$, the perturbation theorem places off line zeros that a

given finite set of heat trace tests cannot see, and positivity of ρ_ε at one resolution is compatible with zeros far from the line.

5.3 The Thorin functional from the arithmetic

The Thorin measure has two descriptions: the zeros give its atoms, and the explicit formula gives it from the gamma factor and the primes without reference to any zero. Section 5.1 proves the lower rungs of the positivity ladder with both, the explicit formula near $t = 0$ and the zeros away from it, and Section 5.2 shows that no finite set of such tests decides the Thorin condition. Here the arithmetic description is taken on its own, following [200, Chapter 7]. Defined as a functional on test functions, the Thorin measure needs no hypothesis on the zeros; Weil's explicit formula identifies it with evaluation at the zeros; and on the heat kernel the primes give the heat trace, losing about 87 digits to cancellation per unit of t.

5.3.1 The Thorin functional and the explicit formula

The Thorin measure cannot be introduced as a measure without assuming the conclusion. If W were known to be completely monotone, Bernstein's theorem would supply a unique positive measure U with $W(t) = \int e^{-tz} U(dz)$, but complete monotonicity of W is RH. Nor is $\sum_k \delta_{a_k}$ a measure on $[0,\infty)$ unless every a_k is real and nonnegative, which is RH again; an atom at a nonreal a_k can be paired only with functions defined off the real axis. The remedy is to define U as a linear functional on a class of test functions large enough to determine it and small enough that the arithmetic side converges.

Let $\mathcal{T}$ be the set of functions $\varphi(z) = h(\sqrt{z})$ with h even, holomorphic in $|\operatorname{Im} r| \le \frac12 + c$ for some $c > 0$ and $O((1+|r|)^{-2-c})$ there, and let $g(x) = \frac{1}{2\pi}\int h(r) e^{-irx} dr$. Since h is even, φ does not depend on the choice of square root, and it is holomorphic on the image $\{r^2 : |\operatorname{Im} r| < \frac12 + c\}$ of the strip, the region inside a parabola through $-(\frac12 + c)^2$, which contains $-\frac14$ and every a_k. Define

$$\begin{aligned} \langle U, \varphi\rangle &= \varphi\Big(-\frac14\Big) + \frac{1}{4\pi}\int_{\mathbb{R}} \varphi(r^2)\, m(r)\, dr - \sum_{n\ge 2} \frac{\Lambda(n)}{\sqrt{n}}\, g(\log n), \\ m(r) &= \operatorname{Re}\psi\Big(\frac14 + \frac{ir}{2}\Big) - \log\pi, \end{aligned} \tag{5.3.1}$$

where $\psi = \Gamma'/\Gamma$ and Λ is the von Mangoldt function. No zero appears in this definition. Weil's explicit formula [266], [108, Theorem 5.12] identifies the functional: $\langle U, \varphi\rangle = \sum_k \varphi(a_k)$ for every $\varphi \in \mathcal{T}$, the sum running over the exponents $a_k = -(\rho_k - \frac12)^2$ of Section 1.2, one for each pair $\{\rho, 1-\rho\}$, and in particular $\langle U, e^{-t\cdot}\rangle = W(t)$ [198, Proposition 9.4].

Each term of (5.3.1) has its own source. The term $\varphi(-\frac14) = h(\frac{i}{2})$ comes from the poles of $\pi^{-s/2}\Gamma(s/2)\zeta(s)$ at $s = 1$ and $s = 0$, which the factor $s(s-1)$ of ξ

removes; it is the only evaluation of φ off $[0,\infty)$, and it is not an atom of U but one of three pieces whose sum is the evaluation at the exponents. The integral is the contribution of the gamma factor: with the Riemann Siegel theta function $\vartheta(r) = \operatorname{Im}\log\Gamma(\frac14 + \frac{ir}{2}) - \frac{r}{2}\log\pi$ we have $m = 2\vartheta'$, so the integral is $\pi^{-1}\int_0^\infty h\,d\vartheta$, the integral of h against the smooth part of the zero counting function. The prime sum is the fluctuating part. The half width $\frac12 + c$ of the strip does two things: it makes $g(x) = O(e^{-(1/2+c)|x|})$, so that the prime sum converges absolutely against the weight $n^{-1/2}$, and it contains the ordinates $\pm\frac{i}{2}$ of the poles and the complex ordinate $\gamma_k = -i(\rho_k - \frac12)$ of every zero, since $|\operatorname{Im}\gamma_k| = |\operatorname{Re}\rho_k - \frac12| < \frac12$.

The sine squared lemma (1.2.9) rewrites the functional in terms of variograms. By (3.5.2) its kernel carries each $\varphi \in \mathcal{T}$ to $g(0) - g(x)$, a variogram when $h \geq 0$, and the functional takes the absolutely convergent form [198, Proposition 9.2]

$$\langle U, \varphi\rangle = \int_0^\infty \frac{e^{-x/2}}{e^{2x}-1}\big(g(0)-g(x)\big)\,dx - C_\Gamma g(0) + \int_0^\infty e^{x/2}g(x)\,dx - \sum_{n\geq 2}\frac{\Lambda(n)}{\sqrt{n}}g(\log n),$$

where $C_\Gamma = \frac12\log\pi - \frac12\psi(\frac14) - 2 = 0.6860917096$ is the archimedean constant of Section 3.1.3, equal to $\zeta'(\frac12)/\zeta(\frac12) - 2$ because $\xi'(\frac12) = 0$. To derive this form, split $\varphi(-\frac14) = \int_0^\infty (e^{x/2} + e^{-x/2})g(x)\,dx$; the half with $e^{-x/2}$ combines with the gamma integral into the first two terms.

This form regularises a formal pairing. Formally $\langle U,\varphi\rangle = \int_0^\infty \big(g(0) - g(x)\big)\,\nu(dx)$, where ν is the canonical measure of (3.1.1) at $\alpha = \frac12$,

$$\nu(dx) = \frac{e^{-x/2}\,dx}{e^{2x}-1} - e^{x/2}\,dx + \sum_{n\geq 2}\frac{\Lambda(n)}{\sqrt{n}}\,\delta_{\log n}(dx),$$

but the pole and prime parts of this pairing diverge through their $g(0)$ terms. Part by part, the absolutely convergent form exceeds the formal pairing by multiples of $g(0)$: by $-C_\Gamma$ for the gamma factor, by $\int_0^\infty e^{x/2}dx$ for the pole and by $-\sum_n \Lambda(n)n^{-1/2}$ for the primes. Continued analytically as in Section 3.1.3, the last two are -2 and $\zeta'(\frac12)/\zeta(\frac12)$, and the three constants cancel by the identity $C_\Gamma + 2 - \zeta'(\frac12)/\zeta(\frac12) = 0$ of that section, which is equivalent to $\xi'(\frac12) = 0$. The only negative part of ν is the pole density $-e^{x/2}dx$, and Section 3.5.2 reads Weil's criterion from this sign pattern. At the basepoint one the pole density is $-dx$, and no analytic continuation is needed there: the prime atoms compensated by the pole density have a convergent sine squared integral, and its convergence is the prime number theorem (Proposition 9.1.2). For $L(s,\chi_{12})$, which has no pole, the prime atoms at the basepoint one carry the signs of χ_{12} and converge with no compensation (Section 11.3.2).

The functional is not in general the limit of the Thorin measures as the basepoint approaches the line. As $\alpha \downarrow \frac12$ the measures U_α of (1.2.6) converge vaguely

to $U_{\mathrm{line}} = \sum \delta_{\gamma^2}$, the sum running over the zeros on the line, whatever the truth of RH (Section 1.2 and [198, Proposition 9.5]), whereas unconditionally

$$U = U_{\mathrm{line}} + \sum_{a_k \notin (0,\infty)} \delta_{a_k} \quad \text{on } \mathcal{T},$$

the sum running over the nonreal exponents. These come in conjugate pairs $(\gamma - id)^2$ and $(\gamma + id)^2$, one pair for each off line pair $\frac{1}{2} \pm d + i\gamma$ of Proposition 1.2.3 (written $\frac{1}{2} \pm \delta + i\gamma$ there), so $U = U_{\mathrm{line}}$ exactly when RH holds [198, Proposition 9.6]. The functional evaluates φ at both exponents of an off line pair, while in (1.2.8) the pair contributes kernels $P_{\varepsilon+d} + P_{\varepsilon-d}$ of total mass zero once $\varepsilon < d$, which vanish in the limit. This is why the vague limit in Section 4.1 omits the off line zeros, and why the Thorin mass in Proposition 3.4.2 misses both members of an off line pair, which produces the factor two there.

5.3.2 The heat trace from the primes

For $\varphi(z) = e^{-tz}$, that is $h(r) = e^{-tr^2}$ and $g(x) = e^{-x^2/4t}/(2\sqrt{\pi t})$, the functional is the arithmetic form (1.2.5) of the heat trace, $W = \Pi + A - P$, with the polar term $\Pi = e^{t/4}$, the archimedean term A and the prime term P described in Section 1.2. Table 5.7 extends the comparison of Table 1.5 to smaller and larger t. We evaluate the three terms in sixty digit arithmetic, A by adaptive quadrature and P over $n \le 2 \times 10^5$, and compare $\Pi + A - P$ with the sum over the first 1700 zeros, whose ordinates we take to seventy digits (Appendix A); the last column is the number of decimal digits lost to cancellation, $\log_{10}\{(\Pi + |A| + P)/W\}$. The two columns agree to every digit shown.

The cancellation follows a simple law. For $t \ge 0.005$ the archimedean term is negative (Exercise 5.8), so $\Pi + |A| + P = 2e^{t/4} - W$ and the digits lost are $\log_{10}(2e^{t/4}/W - 1)$. Since $W = e^{-\gamma_1^2 t}\big(1 + O(e^{-(\gamma_2^2 - \gamma_1^2)t})\big)$, this is

$$\log_{10} 2 + \frac{(\gamma_1^2 + \frac{1}{4})\, t}{\log 10} \approx 0.30 + 86.88\, t$$

up to an exponentially small error, below 2×10^{-5} for $t \ge 0.05$; at $t = 0.3$ the law gives $0.30103 + 26.06294 = 26.36397$. The intercept $\log_{10} 2$ records that Π is balanced by $|A| + P$, and the slope exceeds $\gamma_1^2/\log 10 = 86.77$ by $1/(4 \log 10)$, the growth rate of the polar term (Figure 5.4).

The explicit formula holds exactly, and the cancellation costs working precision and nothing more: 26.4 digits at $t = 0.3$, well within sixty digit arithmetic. It is the price of computing an exponentially small quantity as a difference of terms of size $e^{t/4}$, the price the untilted Pólya route pays at height θ, where it loses $0.341\,\theta$ digits (Section 2.1). The zeros give W cheaply at moderate t, the primes only near $t = 0$. Section 11.1 finds the same profile for the Lévy density $-W'$ of the subordinator that drives the clock.

t	polar Π	archimedean A	prime P
0.002	1.000500	0.782881	2.6×10^{-26}
0.005	1.001251	−0.464059	7.2×10^{-11}
0.01	1.002503	−0.852797	8.4×10^{-6}
0.02	1.005013	−0.984061	0.002410
0.05	1.012578	−0.954619	0.057914
0.1	1.025315	−0.862510	0.162806
0.15	1.038212	−0.797634	0.240578
0.2	1.051271	−0.749754	0.301517
0.3	1.077884	−0.681852	0.396032

t	$\Pi + A - P$	W from zeros	digits lost
0.002	1.78338120537913	1.78338120537913	0.0
0.005	0.537191964091041	0.537191964091041	0.4
0.01	0.14969801125378	0.14969801125378	1.1
0.02	0.0185412917313195	0.0185412917313195	2.0
0.05	$4.58783504857357 \times 10^{-5}$	$4.58783504857357 \times 10^{-5}$	4.6
0.1	$2.10479979582235 \times 10^{-9}$	$2.10479979582235 \times 10^{-9}$	9.0
0.15	$9.65642094572633 \times 10^{-14}$	$9.65642094572633 \times 10^{-14}$	13.3
0.2	$4.43018218022364 \times 10^{-18}$	$4.43018218022364 \times 10^{-18}$	17.7
0.3	$9.32464654810622 \times 10^{-27}$	$9.32464654810622 \times 10^{-27}$	26.4

Table 5.7: The heat trace from the explicit formula and from the zeros. For $t \geq 0.05$ the digits lost are $\log_{10} 2 + (\gamma_1^2 + \frac{1}{4})t/\log 10 \approx 0.30 + 86.88\,t$.

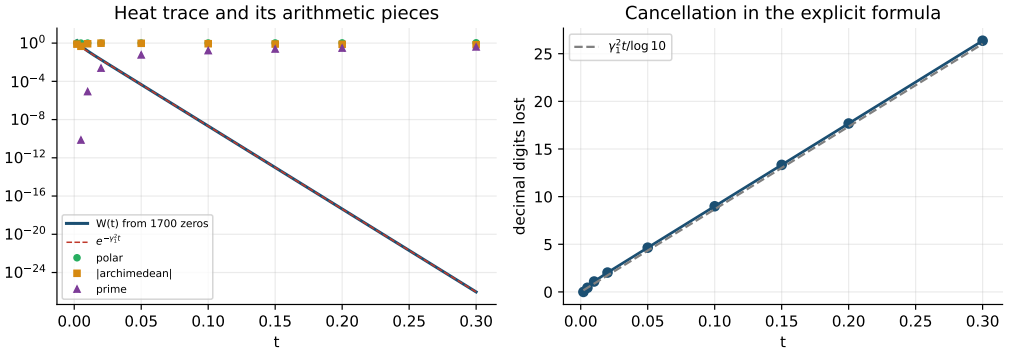


Figure 5.4: Left: the heat trace from 1700 zeros, its leading exponential $e^{-\gamma_1^2 t}$, and the three arithmetic pieces, the archimedean one in absolute value. Right: digits lost to cancellation in the explicit formula; for $t \geq 0.05$ they lie $\log_{10} 2 + t/(4 \log 10)$ above the dashed line $\gamma_1^2 t/\log 10$.

The two descriptions divide the work. The arithmetic side is cheap where g is concentrated near $x = 0$, so that only small n contribute: at small t for the heat kernel, and only at large resolution for the Poisson kernel $h(r) = P_\varepsilon(\theta - r) + P_\varepsilon(\theta + r)$. That kernel is not in $\mathcal{T}$, but for $\varepsilon > \frac{1}{2}$ its transform $g(x) = \pi^{-1} e^{-\varepsilon|x|} \cos \theta x$ makes the prime sum converge absolutely; for smaller ε the prime side is Selberg's smoothed sum, which with cutoff X uses the primes up to X^2 and has error of order $X^{-\varepsilon}/(\varepsilon^2 \log X)$ when no zero lies to the right of the critical line, the single zero law of Section 3.5.1. The spectral side is cheap where the test function is concentrated at the low zeros: at large t, and at small ε on a window. In the proofs of the lower rungs in Section 5.1 the two forms meet at $t = 1/20000$, where the inequalities are checked in interval arithmetic. The Thorin condition requires positivity at every scale, and neither side is cheap at every scale. The Pólya route of Section 2.1 is, but only on a window of finite height, and that restriction is the barrier of Section 5.2.

5.4 Spacings, irregularity and the cell distance

The Pólya route computes the Thorin density at every resolution on a window of finite height, the one computation that Section 5.3 found cheap at every scale, and on such a window its errors are set by the spacings of the zeros. Section 3.2 found that the cell distance between the computed Thorin measure and U_{RH} vanishes linearly, $d_{\mathrm{cell}}(\varepsilon) \approx 5.44\,\varepsilon$ for the 79 zeros below height 200, with a constant κ_{cell} given by the values of Ξ'/Ξ at the cell boundaries. Under RH, $\Xi'/\Xi(b)$ is the sum of $1/(b - x)$ over the ordinates x and their reflections $-x$, which vanishes when these points lie symmetrically about b; so κ_{cell} measures the asymmetry of the ordinates about the ends of the cells. The sinh benchmark of Section 1.3, whose atoms form a regular lattice, has constant 1.27 with its own cells and 0.418 with cells drawn as for the zeros, all of it created at $\theta = 0$ by the missing atom, while for the zeros every cell contributes. The spacings behind these errors follow the random matrix prediction already for the first 1700 zeros, and level repulsion sets both the resolution that separates neighbouring zeros and the scale of the local visibility condition of Section 5.2.

5.4.1 The cell distance to first order

We use the cells of Section 3.2.2: with $\Theta = 200$ and $K = N(\Theta) = 79$, $b_0 = 0$, $b_k = (\gamma_k + \gamma_{k+1})/2$ for $1 \le k < K$, $b_K = \Theta$ and $C_k = (b_{k-1}, b_k]$, the Voronoi cells of the ordinates except that the last ends at Θ rather than at the midpoint 199.64 of γ_{79} and γ_{80}. As there, Proposition 3.2.2, applied at both ends of every cell, gives the first order formula

$$\mu_\varepsilon(C_k) - 1 = -\frac{\varepsilon}{\pi}\Big[\frac{\Xi'}{\Xi}(b_k) - \frac{\Xi'}{\Xi}(b_{k-1})\Big] + O(\varepsilon^3).$$

The remainder is $O(\varepsilon^3)$ because $\arg\xi(\frac12+\varepsilon+ib)-\arg\xi(\frac12+ib)$ is odd in ε (Remark 3.2.3), and Exercise 5.11 gives its leading term. The formula needs only that the zeros below Θ be simple and on the line.

Under RH the formula becomes a sum over the ordinates whose terms have a direct meaning. By Proposition 1.2.3, $\rho_\varepsilon(\theta)=\sum_j[P_\varepsilon(\theta-\gamma_j)+P_\varepsilon(\theta+\gamma_j)]$ over all positive ordinates, and the kernel $P_\varepsilon(\cdot-\gamma)$ has distribution function $F_\varepsilon(y-\gamma)$, where $F_\varepsilon(y)=\frac12+\frac1\pi\arctan(y/\varepsilon)=\mathbf{1}_{y\ge0}-\varepsilon/(\pi y)+O(\varepsilon^3/|y|^3)$ as $\varepsilon/|y|\to0$. No ordinate lies on a cell boundary, so

$$\mu_\varepsilon(C_k)-1=-\frac{\varepsilon}{\pi}\sum_j\Big[\frac{1}{b_k-\gamma_j}-\frac{1}{b_{k-1}-\gamma_j}+\frac{1}{b_k+\gamma_j}-\frac{1}{b_{k-1}+\gamma_j}\Big]+O(\varepsilon^3),$$

the last two terms coming from the kernels $P_\varepsilon(\theta+\gamma_j)$ centred at $-\gamma_j$. Each term is the Cauchy leakage of one kernel through one end of the cell, since a kernel at distance D from a boundary moves mass $\varepsilon/(\pi D)+O(\varepsilon^3/D^3)$ across it. By (1.2.1) with $s=i\theta$, $\Xi'/\Xi(\theta)=\sum_j[(\theta-\gamma_j)^{-1}+(\theta+\gamma_j)^{-1}]$ under RH, so the two expansions agree. Summing absolute values,

$$d_{\mathrm{cell}}(\varepsilon)=\kappa_{\mathrm{cell}}\,\varepsilon+O(\varepsilon^3),$$
$$\kappa_{\mathrm{cell}}=\frac1\pi\sum_k\Big|\sum_j\Big[\frac{1}{b_k-\gamma_j}-\frac{1}{b_{k-1}-\gamma_j}+\frac{1}{b_k+\gamma_j}-\frac{1}{b_{k-1}+\gamma_j}\Big]\Big|.$$

Truncated to the first 1700 ordinates, the zero sum gives 5.415, against $\kappa_{\mathrm{cell}}=5.43665$ from the form in Ξ'/Ξ and the observed slope $d_{\mathrm{cell}}(0.005)/0.005=5.43664$. Since the remainder is $O(\varepsilon^3)$, the observed slope equals κ_{cell} up to $O(\varepsilon^2)$, here 1.3×10^{-5}, and the difference of 0.4 per cent between 5.415 and 5.43665 is the contribution of the zeros above γ_{1700}, which the form in Ξ'/Ξ includes. The truncation also shows in the sum of the cell errors, which telescopes to the coefficient $-\Xi'(200)/(\pi\,\Xi(200))=0.3981063$ of Proposition 3.2.2, while the truncated sum gives 0.3347. In the truncated formula the cell errors range from $-0.225\,\varepsilon$ to $+0.250\,\varepsilon$ (Figure 5.5), and in the form in Ξ'/Ξ from $-0.2243\,\varepsilon$ to $+0.2513\,\varepsilon$, with the two largest positive values at the end cells in both: at the last because Θ is not a midpoint between ordinates, and at the first because below b_1 there is only γ_1 and then the reflections $-\gamma_j$, which do not balance the zeros above b_1.

The sinh benchmark of Section 1.3 shows where the constant of a regular lattice comes from. Its atoms $\theta=k\ge1$ and their reflections form the lattice $\mathbb{Z}$ without the point 0. For the full lattice every cell $(k-\frac12,k+\frac12]$ would have mass exactly one, by periodicity, so the deficit of each cell is the mass that the missing kernel at 0 would have put in it; the atom free interval $(0,\frac12]$, which the benchmark counts as a cell of mass zero, receives the mass $\frac12$ of the full lattice less that of the missing kernel:

$$\mu_\varepsilon\big((k-\tfrac12,k+\tfrac12]\big)=1-\frac1\pi\arctan\frac{\varepsilon}{\varepsilon^2+k^2-\frac14},\qquad\mu_\varepsilon\big((0,\tfrac12]\big)=\frac1\pi\arctan(2\varepsilon).$$

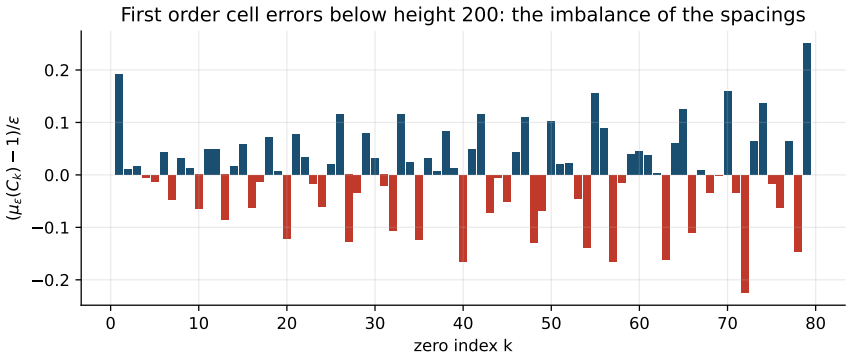


Figure 5.5: First order cell errors $(\mu_\varepsilon(C_k)-1)/\varepsilon$ for the 79 cells below height 200, computed from the first 1700 ordinates. Their absolute sum, 5.415, falls short of $\kappa_{\text{cell}} = 5.43665$ because the zeros above γ_{1700} are omitted.

The interior cells therefore do not balance: each loses $\varepsilon/(\pi(k^2-\frac{1}{4}))+O(\varepsilon^3)$, while $(0,\frac{1}{2}]$ gains $2\varepsilon/\pi+O(\varepsilon^3)$. As $\sum_{k\le N}(k^2-\frac{1}{4})^{-1} = 2-1/(N+\frac{1}{2})$, the cell distance of N atoms is $(4-1/(N+\frac{1}{2}))\,\varepsilon/\pi+O(\varepsilon^3)$, which is $1.26694\,\varepsilon$ for $N=50$, the constant 1.267 of Section 1.3. The constant is less than $4/\pi$ for every N, and with the convention used for the zeros, a first cell $(0,\frac{3}{2}]$, it would be $(\frac{4}{3}-1/(N+\frac{1}{2}))/\pi = 0.418$. For the zeros no cell is balanced, and the errors in Figure 5.5 do not decay along the window.

The constant is a statistic of the irregularity of the spacings, of a specific kind. It vanishes exactly when the sum of $1/(b-x)$ over the ordinates and their reflections vanishes at every cell boundary b, as it does when these points are symmetric about every boundary. The lattice $\gamma_j = (j-\frac{1}{2})h$, whose reflections complete $(\mathbb{Z}+\frac{1}{2})h$, has every cell error zero, whereas the lattice of the benchmark does not; nor is equal spacing needed, since points on the line whose gaps alternate between two lengths are symmetric about every midpoint. For the zeros every ordinate contributes to a cell error, with a weight that decays like $|C_k|/(\gamma_j-b_k)^2$. The two ordinates nearest an interior boundary are symmetric about it, so the asymmetry comes first from the next gaps on either side, and it is small when those gaps are equal.

5.4.2 Spacings of the zeros and the Thorin density

At height γ the mean spacing of the zeros is $\Delta = 2\pi/\log(\gamma/2\pi)$, as in Section 5.2.2. We unfold by $\tilde\gamma = \vartheta(\gamma)/\pi$, with ϑ the Riemann Siegel theta function. The 1699 normalised spacings $\tilde\gamma_{k+1}-\tilde\gamma_k$ of the first 1700 zeros, which lie below height 2198, have mean 0.99998, variance 0.146 and minimum 0.089. Figure 5.6 compares their histogram with the Wigner surmise for the Gaussian unitary ensemble, $p_{\text{W}}(s) = \frac{32}{\pi^2}s^2e^{-4s^2/\pi}$, and with the Poisson law e^{-s} of independent points. The histogram

follows the surmise already at this height, including the quadratic repulsion at small spacings that Montgomery's pair correlation conjecture predicts [158, pp. 183 and 184] and that Odlyzko observed at large height [171]. The variance is below the surmise value $3\pi/8-1 = 0.178$; the spacing distribution approaches the unitary one only slowly with height [158, p. 185], [31, Section 3].

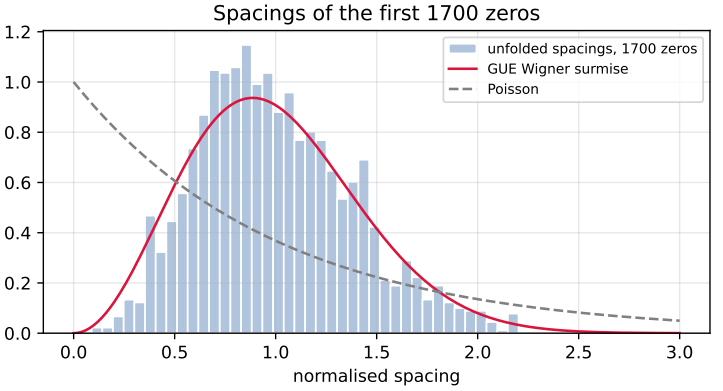


Figure 5.6: The 1699 unfolded spacings of the first 1700 zeros against the GUE Wigner surmise and the Poisson law.

Repulsion affects the Thorin density in two ways. First, it makes very small gaps rare but not absent. The smallest normalised spacing among the first 1700 zeros, 0.089, separates the ordinates 1977.17 and 1977.27, the pair that Section 5.2 moves off the line in Figure 5.3; Lehmer pairs, such as the pair near height 7005 of Section 2.2.3.1, are the rare small gaps at larger height. By the Cauchy kernel criterion of Exercise 2.11, an isolated pair of ordinates at distance g in θ gives two peaks of ρ_ε only for $\varepsilon < \sqrt{3}g/2$. The resolution needed to separate every pair on a window is therefore set by the smallest gap, and repulsion controls it: if the N normalised spacings were independent, with density $p_{\mathrm{W}}(s) \propto s^2$ near zero, the smallest would be of order $N^{-1/3}$, which is 0.084 for $N = 1699$, close to the observed 0.089.

Second, repulsion sets the scale of the local visibility condition of Section 5.2.2: to first order in ε, an off line zero at distance d from the line makes ρ_ε negative at its ordinate if and only if $2/d^2$ exceeds $\sum_j \delta_j^{-2}$, where δ_j are the distances from its ordinate to the other ordinates. The sum is dominated by the nearest neighbours, and repulsion makes it typically of order Δ^{-2}, because $\mathbb{E}s^{-2}$ is finite for the surmise, where it equals $8/\pi$, and infinite for the Poisson law. With neighbours on a lattice of spacing Δ, the density at the ordinate stays positive at small resolution once d exceeds the threshold $(\sqrt{6}/\pi)\Delta \approx 0.78\,\Delta$ of that section. Since $d < \frac{1}{2}$, a zero hidden in this way can exist only where $\Delta < 0.64$, that is above height about 1.1×10^5. Hidden here means hidden at the ordinate: beside

it the density can be negative even when the condition predicts a positive value there (Exercise 5.15).

The behaviour of the cell constant at large height is open (Section 12.2). In unfolded units the cell errors depend mainly on the nearby ordinates, so the sine kernel process of the unitary ensemble would predict their mean and fluctuations per cell; we have not computed that prediction. Computing the cell errors with Odlyzko's zeros near zero number 10^{12} [171], 10^{21} or 10^{22} [175, 176], at heights about 2.7×10^{11}, 1.4×10^{20} and 1.4×10^{21}, would compare the Thorin density with the spacing statistics of the zeros at large height. The computation needs the ordinates, which define the cells, but no sum over zeros: given the ordinates, the hybrid evaluator of Section 2.2.2 has only to evaluate Ξ'/Ξ at the cell boundaries. Section 11.4.5 proposes the same comparison at large height for the positivity margins of the density, its values at the midpoints between consecutive ordinates.

Spacings thus enter the Thorin density of the reciprocal Xi function in three ways: through the cell constant, through the resolution that separates neighbouring zeros and through the scale of local visibility. In all three the zeros are the atoms and the primes play no part; in the reciprocal zeta function the primes enter directly (Section 6.1).

5.5 Exercises

On Section 5.1.

5.1. Show that $c_0c_2 - c_1^2 = \frac{1}{2}\sum_{j,l}(a_j - a_l)^2 e^{-(a_j+a_l)t}$ and deduce that log convexity of W holds under RH.

5.2. Recompute the first row of Table 5.1 with the first 100 zeros and explain why no printed digit changes.

5.3. Show that for a real exponent a the summand of the Williamson measure U_k belonging to a is the inverse gamma density of ka/Γ_k with $\Gamma_k \sim \mathrm{Gamma}(k,1)$, and compute its variance for $k > 2$.

5.4. Let $\widetilde{W}(t) = e^{-t} + 2e^{-10t}\cos t$, the heat trace of the three exponents 1 and $10 \pm i$. Show that $\widetilde{W} > 0$ and $\widetilde{W}' < 0$ on $(0,\infty)$, using $\cos t > 0$ and $\sin t \geq 0$ for $t \leq 1$ and $2\sqrt{101}\,e^{-9} < 1$ for $t \geq 1$, so that the first two rungs of Theorem 5.1.1 hold. With $\theta = \arctan\frac{1}{10}$, show that $\frac{10}{101} < \theta < \frac{1}{10}$, hence $\frac{2\pi}{3} < 24\theta < \pi$, and deduce that $(-1)^{24}\widetilde{W}^{(24)}(0) = 1 + 2\cdot 101^{12}\cos 24\theta < 0$, so that $\widetilde{W}$ is not completely monotone. Then check numerically that the first order k at which $(-1)^k\widetilde{W}^{(k)}(t) < 0$ for some $t > 0$ is 12, near $t = 2.87$, and compare with the first failures of Table 5.5.

5.5. Show that the term of order $1/u$ of ε_ρ in Proposition 5.1.8 is $\frac{2t}{3u}(\tilde\gamma - u)^3$. For $u = 2$, $t = 1$ and $k = 4$, show that no zero lies in the disc $|\tilde\gamma - u| \leq u/2$,

so that $c_4(1) = e^{f_0} R$ is carried entirely by the remainder, and compare the bound on R with the contribution of the first zero.

5.6. Show that $m(0) > -6$ and, comparing the series for ψ with an integral, that $m(60) > 1$. Deduce that for $t \le 10^{-6}$ the archimedean term of (1.2.5) is at least $1/(4\sqrt{\pi t}) - 210/\pi > 55$ and the prime term at most $(\pi t)^{-1/2} e^{-(\log 2)^2/4t} < 0.01$. With the argument of the proof of Theorem 5.1.1 for $t \ge \tau$, taken at height 10^4 and from $t = 10^{-6}$, conclude that $W > 0$ on $(0, \infty)$ follows from the verification of RH below height 10^4 by estimates that can all be checked by hand.

On Section 5.3.

5.7. Verify the arithmetic form of $W(t)$ at $t = 0.01$ with the prime powers up to 10^4, and explain why so few suffice by showing that only $n = 2$ and $n = 3$ affect the first fifteen digits.

5.8. Show that m increases on $[0, \infty)$ and vanishes at a single point r_*, and deduce that $e^{r_*^2 t} A(t)$ decreases in t, so that A changes sign at most once; with Table 5.7, conclude that $A < 0$ for $t \ge 0.005$. Then show that for $t \ge 0.05$ the number of digits lost in the table equals $\log_{10} 2 + (\gamma_1^2 + \frac{1}{4})t / \log 10$ up to an exponentially small error, and identify the source of each term.

5.9. Show that the regularised constants of the gamma, pole and prime parts of the variogram form sum to zero if and only if $\xi'(\frac{1}{2}) = 0$.

5.10. Explain why the test function class $\mathcal{T}$ needs holomorphy in a strip of half width greater than $\frac{1}{2}$.

On Section 5.4.

5.11. Let $\ell = \Xi'/\Xi$. Show that $\xi(\frac{1}{2} + \varepsilon + ib) = \Xi(b - i\varepsilon)$, and deduce from the Taylor series of $\log \Xi$ about b that $\mu_\varepsilon(C_k) - 1 = -\frac{\varepsilon}{\pi}[\ell(b_k) - \ell(b_{k-1})] + \frac{\varepsilon^3}{6\pi}[\ell''(b_k) - \ell''(b_{k-1})] + O(\varepsilon^5)$. Under RH, check that the term in ε^3 agrees with the expansion of F_ε to third order.

5.12. Prove the arctangent formulas for the cell masses of the sinh benchmark, and deduce that its cell distance for N atoms is $(4 - 1/(N + \frac{1}{2}))\,\varepsilon/\pi + O(\varepsilon^3)$, and $(\frac{4}{3} - 1/(N + \frac{1}{2}))\,\varepsilon/\pi + O(\varepsilon^3)$ when the first cell is $(0, \frac{3}{2}]$.

5.13. Show that the Wigner surmise p_{W} has mean 1 and variance $3\pi/8 - 1$, and that $\mathbb{E} s^{-2} = 8/\pi$ under it, while $\mathbb{E} s^{-2} = \infty$ for the Poisson law.

5.14. Show that $\sum_{m \ne 0} (m\Delta)^{-2} = \pi^2/(3\Delta^2)$, deduce the lattice threshold $(\sqrt{6}/\pi)\Delta$ of the visibility condition, and find the height above which the threshold is less than $\frac{1}{2}$.

5.15. In a local model, put an off line pair at distance $d = 0.3$ from the line at the ordinate γ and zeros on the line at $\gamma + 0.2 + m$ and $\gamma - 4 - m$ for $m \geq 0$, and ignore the reflected kernels. At $\varepsilon = 0.005$, show that $2/d^2 < \sum_j \delta_j^{-2}$, so that the visibility condition predicts a positive density at γ, and check that $\rho_\varepsilon(\gamma) > 0$ while $\rho_\varepsilon(\gamma - 0.079) < 0$.

Chapter 6

The reciprocal zeta function

Chapters 1 to 5 studied the Thorin measure of the reciprocal Xi function. The reciprocal zeta function $1/\zeta(\frac{1}{2}+\sqrt{s})$ is, up to the factor $1-4s$ of the pole and a constant, the reciprocal Xi function times the gamma factor, and unlike ξ it has a Dirichlet series and an Euler product, from which we read its Thorin type measures. We study it and its relatives in the variable of the clock, written s, in which a zero $\frac{1}{2}+i\gamma$ sits at $s=-\gamma^2$ on the cut; Section 6.1.1 fixes the coordinates of this chapter and of Chapter 7.

Sections 6.1 and 6.2 compute the Thorin measure of the gamma factor, the Thorin measures of the normalised reciprocal zeta function and of the reciprocal Xi function under RH, and, unconditionally, the formal Thorin measures of these two functions in the Vere-Jones form of Section 6.1.3. The first works from the Dirichlet series and the Euler product, the second from the Weierstrass product of the gamma factor, the Meixner law and the sine squared lemma. Even under RH the Thorin measure of the normalised reciprocal zeta function is not positive, because the Thorin density of the gamma factor changes sign. After the stable tilt $e^{-c\sqrt{s}}$, with $c=\zeta'(\frac{1}{2})/\zeta(\frac{1}{2})$, the reciprocal of the gamma factor is a generalised gamma convolution, and multiplying the normalised reciprocal zeta function by it gives the floor $e^{-c\sqrt{s}}\,\xi(\frac{1}{2})/\xi(\frac{1}{2}+\sqrt{s})$ of the reciprocal Xi function (Section 6.2.4), in which the zeros are the only possible source of negativity.

The infinite divisibility of the floor, as proved here, rests on the positivity of the canonical function of the normalised reciprocal zeta function, which is Theorem 7.1.1 of Chapter 7. It also follows from the infinite divisibility of the clock (Theorem 5.1.1); what the floor adds is a splitting of its canonical function into an arithmetic part and a completely monotone archimedean part. The floor is a generalised gamma convolution exactly when RH holds, and the derivatives of its canonical function alternate in sign up to order 10^{10} by the methods of this chapter and up to order 1.16×10^{25} by the estimates of Section 5.1.

6.1 Thorin type forms of the reciprocal zeta function

Stieltjes inversion gives the Thorin type measure of a single Euler factor, a cosine series of both signs. The Lai and Vere-Jones identity replaces the exponentials of the Dirichlet series by hyperbolic secants, whose Thorin measures are lattices of unit atoms, at the price of Möbius signs in the coefficients. Under RH the Thorin measure of the normalised reciprocal zeta function is dM, where $M-1$ is the argument function $S(\sqrt{t})$. Its heat trace, the canonical function w of Theorem 6.1.11, is explicit in the primes, is an absolutely convergent series of Jacobi theta functions, and gives a criterion for RH (Theorem 6.1.15). Section 6.1.6 follows these objects across the critical strip.

6.1.1 The question and the three functions

Write $\sqrt{s}$ for the principal square root on the cut plane $\mathbb{C}\setminus(-\infty,0]$, so that $\operatorname{Re}\sqrt{s}>0$ there, and evaluate ζ at $z=\frac12+\sqrt{s}$. A zero $\rho=\frac12+i\gamma$ then corresponds to $\sqrt{s}=i\gamma$, that is to $s=-\gamma^2$, a point of the cut, and an atom of a Thorin measure at t produces a singularity at $s=-t$, also on the cut. The Riemann Hypothesis therefore says that every zero sits where a Thorin atom could sit, which is the reason for the variable $\sqrt{s}$ here, as in Section 1.2. The question of this chapter and of Chapter 7 is whether the zeros, the primes and the gamma factor assemble into a Thorin measure, and if so which one. The two chapters have their own coordinates, which hold throughout both unless stated otherwise: s is the variable u of Sections 1.2 and 1.3, t is the variable z of their Thorin measures, x is the time of the heat trace $W(x)$, τ is the ordinate, and z is the argument of ζ and ξ.

The Möbius series and the Euler product converge absolutely to the right of the parabola $x=\frac14-y^2$, on

$$H_{1/2}=\{s:\operatorname{Re}\sqrt{s}>\tfrac12\}=\{x+iy:\ x>\tfrac14-y^2\},$$

and there

$$\frac{1}{\zeta(\frac12+\sqrt{s})}=\sum_{n\ge1}\frac{\mu(n)}{\sqrt{n}}\,e^{-\sqrt{s}\log n}=\prod_p\Big(1-p^{-1/2}e^{-\sqrt{s}\log p}\Big). \tag{6.1.1}$$

Every term is built from $e^{-\lambda\sqrt{s}}$, which is completely monotone in s: it is the Laplace transform of the positive stable law of index $\frac12$, the law of the first passage time of $\sqrt{2}B$ to the level λ, where B is a standard Brownian motion,

$$e^{-\lambda\sqrt{s}}=\int_0^\infty e^{-sx}\,\frac{\lambda}{2\sqrt{\pi}}\,x^{-3/2}e^{-\lambda^2/4x}\,dx. \tag{6.1.2}$$

So (6.1.1) writes $1/\zeta(\frac12+\sqrt{s})$ as a signed series of Laplace transforms of stable laws. The Thorin measure, however, is read from the logarithm. Recall from

(1.3.1) that a generalised gamma convolution (GGC) has Laplace transform

$$f(s) = \exp\Big(-as - \int_{(0,\infty)} \log\big(1 + \tfrac{s}{t}\big)\, U(dt)\Big), \tag{6.1.3}$$

with $a \geq 0$ and $U \geq 0$ the Thorin measure; equivalently $-f'/f(s) = a + \int U(dt)/(s+t)$ is a Stieltjes function [35]. We allow U to be signed, with $\int_{(0,1)} |\log t|\, |U|(dt) + \int_{[1,\infty)} t^{-1} |U|(dt) < \infty$ so that the integral converges, and then call (6.1.3) a Thorin type representation.

Three functions of s run through both chapters. The function $1/\zeta(\frac{1}{2} + \sqrt{s})$ vanishes at $s = \frac{1}{4}$, where ζ has its pole, and equals $1/\zeta(\frac{1}{2}) < 0$ at $s = 0$, so it is not the Laplace transform of a probability law. Both defects are removed by the normalised reciprocal zeta function

$$\varphi(s) = \frac{\zeta(\frac{1}{2})}{(1-4s)\,\zeta(\frac{1}{2} + \sqrt{s})}, \qquad \varphi(0) = 1. \tag{6.1.4}$$

The other two are the gamma factor $G(s) = \Gamma(\frac{1}{4} + \frac{\sqrt{s}}{2})\pi^{-\sqrt{s}/2}/\Gamma(\frac{1}{4})$ and the reciprocal Xi function $R(s) = \xi(\frac{1}{2})/\xi(\frac{1}{2} + \sqrt{s})$, and $\varphi = RG$ (Proposition 6.1.9). In both chapters G is the gamma factor, not the function G of Section 1.2, whose reciprocal is R, except in Section 7.1.3, where G is a Laguerre Pólya function; and $\theta(\tau) = \operatorname{Im} \log \Gamma(\frac{1}{4} + \frac{i\tau}{2}) - \frac{\tau}{2} \log \pi$ is the Riemann Siegel theta function, written ϑ elsewhere in the book (Section 2.2), so that $\theta'(\tau) = \frac{1}{2} \operatorname{Re} \psi(\frac{1}{4} + \frac{i\tau}{2}) - \frac{1}{2} \log \pi$ [72, Section 6.5]. Table 6.1 lists the Thorin measures of the three functions and of the building blocks used below. In it $N(T)$ counts the zeros with $0 < \gamma \leq T$, $S(T) = \pi^{-1} \arg \zeta(\frac{1}{2} + iT)$ is the argument function, $c_n = 2n + \frac{1}{2}$, $c = -\theta'(0)$ is the constant of (6.1.6), and U_λ is the sawtooth measure of Lemma 6.1.7.

6.1.2 Stieltjes inversion and the Euler factors

Three descriptions of a function F on the cut plane recur: its Thorin measure U, the Stieltjes representation $-F'/F(s) = \int U(dt)/(s+t)$, and the boundary values of $-F'/F$ on $(-\infty, 0]$. Lemma 6.1.1 recovers the measure from the boundary values. It computes most of the Thorin measures of this chapter and of Chapter 7, and in each application the work is to evaluate $\operatorname{Im} f(-t+i0)$ for $f = -F'/F$, which amounts to putting $\sqrt{s} = i\sqrt{t}$.

Lemma 6.1.1. *Let f be analytic on $\mathbb{C} \setminus (-\infty, 0]$ with $f(\bar{s}) = \overline{f(s)}$, $f(s) \to 0$ as $|s| \to \infty$ uniformly in $\arg s$, $|f(s)| \leq C|s|^{-\alpha}$ near 0 for some $\alpha < 1$, and such that f extends continuously from the upper half plane to the points of $(-\infty, 0)$ apart from finitely many points $-t_0$, at each of which $f(s) - r/(s+t_0)$ is bounded near $-t_0$ in $\mathbb{C} \setminus (-\infty, 0]$ for some real r; $f(-t+i0)$ denotes the values of this extension and $f(-t-i0) = \overline{f(-t+i0)}$. Then*

$$f(s) = \int_{(0,\infty)} \frac{m(dt)}{s+t}, \qquad m(dt) = -\frac{1}{\pi} \operatorname{Im} f(-t+i0)\, dt,$$

Function	Thorin measure	Sign
φ	$dM(t)$, $M(t) = N(\sqrt{t}) - \theta(\sqrt{t})/\pi = 1 + S(\sqrt{t})$ (RH)	signed
G	$-\dfrac{\theta'(\sqrt{t})}{2\pi\sqrt{t}}dt = \dfrac{1}{2\pi\sqrt{t}}\Big\{b + \sum_{n\ge 0}\Big(\dfrac{c_n}{t+c_n^2} - \dfrac{1}{2n+2}\Big)\Big\}dt$	signed
R	$\sum_{\gamma>0}\delta_{\gamma^2}$ (RH)	positive
$2/(1+e^{-\lambda\sqrt{s}})$	$-U_\lambda = \sum_k \delta_{((2k+1)\pi/\lambda)^2} - \dfrac{\lambda}{4\pi\sqrt{t}}dt$	signed
$(1+\sqrt{s}/\kappa)^{-1}$	$\dfrac{\kappa}{2\pi\sqrt{t}\,(t+\kappa^2)}dt$	positive
$\operatorname{sech}(\lambda\sqrt{s})$	$\sum_k \delta_{((k+\frac12)\pi/\lambda)^2}$	positive
$e^{-c\sqrt{s}}/G$	$\dfrac{\theta'(\sqrt{t}) - \theta'(0)}{2\pi\sqrt{t}}dt$	positive

Table 6.1: Thorin measures of φ, G and R and of the building blocks used in this chapter and in Chapter 7, with the convention $\log f(s) = -\int \log(1+s/t)\,U(dt)$ after normalising $f(0) = 1$. Here $b = \frac12(\gamma_E + \log\pi)$, $\kappa > 0$ and $c = -\theta'(0) = \zeta'(\frac12)/\zeta(\frac12)$. The rows for φ and R are Theorem 6.1.10, the row for G is Theorem 6.2.1, the fourth row is Lemma 6.1.7, the fifth and sixth appear in Table 1.9, and the last is Theorem 6.2.12.

where the integral is the limit of the integrals over $(0,T]$ *as* $T\to\infty$*, absolutely convergent when* $\int_{(1,\infty)} t^{-1}|m|(dt) < \infty$*, and where each of the points* $-t_0$*, which we call a simple pole of* f*, contributes the atom* $r\,\delta_{t_0}$ *to* m*.*

Proof. Fix $s \in \mathbb{C}\backslash(-\infty,0]$ and integrate $f(z)/(z-s)$ over a keyhole contour around $(-\infty,0]$, which by Cauchy's formula gives $2\pi i\, f(s)$; the large circle vanishes by the decay at infinity and the small circle by the bound at the origin. By the continuity of the extension, the integrals along the two sides of the keyhole converge to the integrals of the boundary values as the keyhole closes onto the cut, after subtracting $r/(z+t_0)$ near each pole $-t_0$, where the remainder is bounded. The two sides of the cut combine to $\frac{1}{2\pi i}\{f(-t-i0) - f(-t+i0)\} = -\frac{1}{\pi}\operatorname{Im} f(-t+i0)$. At a simple pole the contour follows small semicircles above and below the cut, which together contribute $r/(s+t_0)$. □

The hypotheses of uniform decay at infinity and of finitely many poles exclude functions with infinitely many poles on the cut, such as $-R'/R$ under RH, which has a pole at every $-\gamma^2$, and $-R'_\sigma/R_\sigma$ for the functions R_σ of Section 6.1.6 when no zero lies to the right of the line $\operatorname{Re} z = \sigma$ and infinitely many lie on it. Such an application needs a direct argument from Hadamard's product, as in the proof of Theorem 6.1.10.

For $f(s) = \lambda e^{-\lambda\sqrt{s}}/(2\sqrt{s}) = -\frac{d}{ds}e^{-\lambda\sqrt{s}}$ one has $\sqrt{-t+i0} = i\sqrt{t}$, and the

lemma gives the density $\frac{\lambda}{2\pi\sqrt{t}}\cos(\lambda\sqrt{t})$, so that

$$e^{-\lambda\sqrt{s}} = 1 - \int_0^\infty \log\bigl(1+\tfrac{s}{t}\bigr)\frac{\lambda\cos(\lambda\sqrt{t})}{2\pi\sqrt{t}}\,dt.$$

This is not the Thorin measure of $e^{-\lambda\sqrt{s}}$, which is the positive density $\lambda/(2\pi\sqrt{t})$ (Table 1.9), but the Thorin type density that an exponential contributes when it sits in an exponent, as it does in the logarithm of every Euler factor of (6.1.1) and in $\log\zeta(\frac12+\sqrt{s})$. It oscillates in sign, and this is the difficulty: every Euler factor has a signed Thorin density, and any positivity must come from the product. The Euler factors show this in closed form.

Proposition 6.1.2. *For $0<q<1$ and $\lambda>0$ the Euler factor $E(s)=1-qe^{-\lambda\sqrt{s}}$ is analytic and zero free on $\mathbb{C}\setminus(-\infty,0]$, and*

$$\log\frac{E(s)}{1-q} = -\int_0^\infty \log\bigl(1+\tfrac{s}{t}\bigr)\,u(t)\,dt,$$

$$u(t) = -\frac{q\lambda}{2\pi\sqrt{t}}\,\frac{\cos(\lambda\sqrt{t})-q}{1-2q\cos(\lambda\sqrt{t})+q^2} = -\frac{\lambda}{2\pi\sqrt{t}}\sum_{k\ge1} q^k\cos(k\lambda\sqrt{t}).$$

Proof. On $\mathbb{C}\setminus(-\infty,0]$ we have $|e^{\lambda\sqrt{s}}|>1>q$, so E has no zeros. Put $f=-E'/E=-\frac{q\lambda}{2\sqrt{s}}(e^{\lambda\sqrt{s}}-q)^{-1}$, which is $O(|s|^{-1/2})$ at 0 and decays at infinity. At $\sqrt{s}=i\tau$, $\tau=\sqrt{t}$, we get $f=\frac{iq\lambda}{2\tau}(e^{i\lambda\tau}-q)^{-1}$ and $\operatorname{Im} f=\frac{q\lambda}{2\tau}\frac{\cos\lambda\tau-q}{1-2q\cos\lambda\tau+q^2}$. Lemma 6.1.1 gives u; integrating $-E'/E=\int u(t)\,dt/(s+t)$ from 0 to s gives the logarithm, and the series is the Poisson kernel expansion. □

With $q=p^{-1/2}$ and $\lambda=\log p$, summing u over p formally gives

$$u_\zeta(t) = -\frac{1}{2\pi\sqrt{t}}\sum_{n\ge2}\frac{\Lambda(n)}{\sqrt{n}}\cos(\sqrt{t}\log n) \quad\text{“}=\text{”}\quad \frac{1}{2\pi\sqrt{t}}\operatorname{Re}\frac{\zeta'}{\zeta}\bigl(\tfrac12+i\sqrt{t}\bigr), \tag{6.1.5}$$

a divergent series. Since $\frac{d}{d\tau}\arg\zeta(\frac12+i\tau)=\operatorname{Re}\frac{\zeta'}{\zeta}(\frac12+i\tau)$, the right side is formally the density of $dS(\sqrt{t})$, but the series converges only in the sense of distributions. Theorem 6.1.10 makes this precise, and Theorem 6.1.11 replaces the divergent cosine series by a convergent Gaussian one.

6.1.3 The Lai and Vere-Jones identity and sawtooth measures

Each exponential in $\log\zeta$ contributes an oscillating Thorin type density (Section 6.1.2). We trade each exponential for a hyperbolic secant, whose Thorin measure is a lattice of unit atoms, at the price of a Möbius inversion. This is done

twice: additively in the Dirichlet series of $1/\zeta$ (Theorem 6.1.5), and logarithmically in $\log\zeta$, where Thorin measures are defined, with $\log(1+e^{-x})$ in place of the secant (Theorem 6.1.8).

Lai and Vere-Jones [135] pose the completeness of the dilates $\operatorname{sech}(\theta x)$, $\theta \geq 0$, as an open problem for the hyperbolic secant law. The Möbius inversion of the alternating exponential series of the secant expands e^{-x} in the dilates $\operatorname{sech}(mx)$. This is Proposition 6.1.3, which we call the Lai and Vere-Jones identity, and the expansions of $1/\zeta$ and $\log\zeta$ built from it form the Vere-Jones representation. Here χ_4 is the nontrivial character modulo 4.

PROPOSITION 6.1.3. *For* $\operatorname{Re} x > 0$,

$$\tfrac{1}{2}\operatorname{sech} x = \sum_{m \text{ odd}} \chi_4(m)\, e^{-mx}, \qquad e^{-x} = \tfrac{1}{2}\sum_{m\geq 1} \mu(m)\chi_4(m)\, \operatorname{sech}(mx).$$

Proof. The first is the geometric series $e^{-x}/(1+e^{-2x})$. Since χ_4 is completely multiplicative, $\sum_m \mu(m)\chi_4(m)\sum_k \chi_4(k)e^{-mkx} = \sum_r e^{-rx}\sum_{mk=r}\mu(m)\chi_4(m)\chi_4(k) = e^{-x}$, the rearrangement being justified by $\sum_{m,k} e^{-mk\operatorname{Re} x} < \infty$ for $\operatorname{Re} x > 0$. □

Applied to each exponential $e^{-\sqrt{s}\log n}$ of the Möbius series, the second identity turns the Dirichlet series of $1/\zeta$ into a series of Brownian exit time transforms.

LEMMA 6.1.4. *For* $\lambda > 0$, $\operatorname{sech}(\lambda\sqrt{s}) = \prod_{k\geq 0}\bigl(1+s/\tau_k\bigr)^{-1}$ *with* $\tau_k = \bigl((k+\frac{1}{2})\pi/\lambda\bigr)^2$, *written* $\tau_k(\lambda)$ *when the dependence on* λ *matters. It is the Laplace transform of* $T_\lambda = \sum_k \tau_k^{-1} e_k$ *with* e_k *independent standard exponentials, the exit time of* $\sqrt{2}B$ *from* $(-\lambda,\lambda)$, *and it is a GGC with atomic Thorin measure* $\sum_k \delta_{\tau_k}$; *see [28] and Table 1.9.*

THEOREM 6.1.5. *For* $s \in H_{1/2}$,

$$\frac{1}{\zeta(\frac{1}{2}+\sqrt{s})} = 1 + \sum_{N\geq 2} A_N \operatorname{sech}(\sqrt{s}\log N) = 1 + \sum_{N\geq 2} A_N\, \mathbb{E}\, e^{-sT_{\log N}},$$

where $A_N = \frac{1}{2}\mu(m)\chi_4(m)\mu(n)\, n^{-1/2}$ *if* $N = n^m$ *with* $n \geq 2$ *squarefree and* m *odd, and* $A_N = 0$ *otherwise. In particular* $A_N = \mu(N)/(2\sqrt{N})$ *for squarefree* N.

Proof. Keep the term $n = 1$ of (6.1.1) and apply Proposition 6.1.3 with $x = \sqrt{s}\log n$ to each $n \geq 2$. The pair (n, m) with n squarefree and $n^m = N$ is unique, since all exponents of N must equal m. Absolute convergence follows from $\sum_n n^{-1/2}\sum_m n^{-m\operatorname{Re}\sqrt{s}} < \infty$ for $\operatorname{Re}\sqrt{s} > \frac{1}{2}$. □

Each $\operatorname{sech}(\sqrt{s}\log N)$ is a solved model, by which we mean a function whose Thorin measure is explicit; it is a GGC with atomic Thorin measure. Its atoms $\tau_k(\log N)$ are units at the squares of the odd multiples of $\pi/(2\log N)$, a lattice of spacing $\pi/\log N$ in the variable $\sqrt{t}$; we call them pseudo zeros. The coefficients

A_N carry the signs of the Möbius function, so positivity is now a question about the coefficients alone. The same Möbius series at the even integers, $1/\zeta(2k) = \sum_n \mu(n) n^{-2k}$, is the starting point of Riesz's criterion (Section 10.1.8), which Section 10.2 reads through the clock of Section 1.2 and the basepoint clocks of Section 9.1. Section 10.1.8 also sets the dilates $\operatorname{sech}(mx)$ of Proposition 6.1.3 among the multiplicative completeness problems of Nyman and Beurling.

Theorem 6.1.5 is additive: it writes $1/\zeta(\frac{1}{2}+\sqrt{s})$ as a signed mixture of explicit GGC transforms, but a mixture of GGCs is not in general a GGC, and the Thorin measure is defined through the logarithm. To reach the logarithm we replace sech by $\log(1+e^{-x})$, whose coefficients have as Dirichlet series Dirichlet's eta function $\eta(1+v) = (1-2^{-v})\zeta(1+v)$, not the Dedekind eta function of Sections 8.5 and 11.3.

Proposition 6.1.6. *Define d_n by $\sum_n d_n n^{-v} = 1/\eta(1+v)$, that is $d_n = \sum_{2^j m=n} \mu(m)/m$. Then $|d_n| \le 1$ and for $\operatorname{Re} x > 0$*

$$e^{-x} = \sum_{m\ge 1} d_m \log\bigl(1+e^{-mx}\bigr).$$

Proof. Expand $\log(1+e^{-y}) = \sum_k (-1)^{k+1} e^{-ky}/k$. The coefficients $(-1)^{k+1}/k$ have Dirichlet series $\eta(1+v)$, so the Dirichlet convolution with d is $\delta_{r,1}$. If $2^\ell \| n$, a squarefree $m = n/2^j$ requires $j \in \{\ell-1, \ell\}$, so $d_n = \mu(n)/n$ for odd n and $d_n = \mu(k)(2^\ell - 2^{\ell-1})/n = \mu(k)/(2k)$ for $n = 2^\ell k$ with $\ell \ge 1$ and k odd. In particular $|d_n| \le 1$ and $\sum_n |d_n|/n < \infty$. □

Lemma 6.1.7. *For $\lambda > 0$ and $s \in \mathbb{C} \setminus (-\infty, 0]$,*

$$\log\bigl(1+e^{-\lambda\sqrt{s}}\bigr) = \log 2 - \int \log\bigl(1+\tfrac{s}{t}\bigr)\, U_\lambda(dt), \qquad U_\lambda(dt) = \frac{\lambda}{4\pi\sqrt{t}}\, dt - \sum_{k\ge 0} \delta_{\sigma_k(\lambda)},$$

with $\sigma_k(\lambda) = \bigl((2k+1)\pi/\lambda\bigr)^2$. Equivalently $U_\lambda(dt) = -d_t \Sigma_\lambda(\sqrt{t})$, where $\Sigma_\lambda(\tau) = \#\{k \ge 0 : (2k+1)\pi/\lambda \le \tau\} - \lambda\tau/2\pi$ is a sawtooth with $|\Sigma_\lambda| \le \frac{1}{2}$.

Proof. Write $1 + e^{-y} = 2e^{-y/2}\cosh(y/2)$ with $y = \lambda\sqrt{s}$. Lemma 6.1.4 gives $\log\cosh(\lambda\sqrt{s}/2) = \sum_k \log(1+s/\sigma_k)$, and $\int_0^\infty \log(1+s/t)\, t^{-1/2} dt = 2\pi\sqrt{s}$ gives $\lambda\sqrt{s}/2 = \int \log(1+s/t) \frac{\lambda}{4\pi\sqrt{t}} dt$. □

The reciprocal of each block has Thorin measure $-U_\lambda$: unit atoms at the pseudo zeros, from $\operatorname{sech}(y/2)$, minus the density $\lambda/(4\pi\sqrt{t})$ that counts them on average, from $e^{y/2}$. This is the shape that Theorem 6.1.10 finds for φ, which contains $1/\zeta$. The function Σ_λ is a toy model of $S(\tau)$, with $1+e^{-\lambda\sqrt{s}}$ playing the role of a zeta function all of whose zeros lie on the line.

THEOREM 6.1.8. *Put* $a_n = \Lambda(n)/(\sqrt{n}\log n)$ *and* $B_N = \sum_{n^m=N} d_m a_n$. *For* $s \in H_{1/2}$,

$$\log\zeta\big(\tfrac12+\sqrt{s}\big) = \sum_{N\ge 2} B_N \log\big(1+N^{-\sqrt{s}}\big)$$

and

$$\frac{d}{ds}\log\zeta\big(\tfrac12+\sqrt{s}\big) = \sum_{N\ge 2} B_N\Big[\sum_{k\ge 0}\frac{1}{s+\sigma_k(\log N)} - \frac{\log N}{4\sqrt{s}}\Big].$$

The first values are $B_p = p^{-1/2}$, $B_4 = \frac14 + \frac{1}{2\sqrt{2}}$, $B_8 = -\frac{1}{6\sqrt{2}}$, $B_9 = \frac16 + \frac{1}{2\sqrt{3}}$.

Proof. For $\operatorname{Re} z > 1$, $\log\zeta(z) = \sum_n a_n n^{-(z-1/2)}$. Apply Proposition 6.1.6 with $x = \sqrt{s}\log n$ and regroup by $N = n^m$; since $|d_m| \le 1$ the double series converges absolutely on $H_{1/2}$. Differentiate with Lemma 6.1.7. □

In the differentiated form each block contributes a Stieltjes transform with poles at $-\sigma_k(\log N)$, and $\log N/(4\sqrt{s})$ is the derivative of the stable exponent $\frac12\sqrt{s}\log N$ that the block carries. Formally the Thorin measure of $1/\zeta(\frac12+\sqrt{s})$ is $-\sum_N B_N U_{\log N} = \sum_N B_N\, d\Sigma_{\log N}(\sqrt{t})$, a superposition of sawtooth measures. The series does not converge as a series of measures, but its Laplace transforms do; Proposition 6.1.13 and Corollary 6.1.14 make this exact.

6.1.4 The normalised reciprocal and its Thorin measure

We pass from $1/\zeta$ to the normalised reciprocal φ of (6.1.4), whose Thorin measure is explicit under RH. The factorisation $\varphi = RG$, with $\xi(z) = \frac12 z(z-1)\pi^{-z/2}\Gamma(z/2)\zeta(z)$ as in Section 1.2, separates the zeros from the archimedean part.

PROPOSITION 6.1.9. *The function* φ *is meromorphic on* $\mathbb{C}\setminus(-\infty,0]$ *and*

$$\varphi(s) = R(s)\,G(s), \qquad R(s) = \frac{\xi(\frac12)}{\xi(\frac12+\sqrt{s})}, \qquad G(s) = \frac{\Gamma(\frac14+\frac{\sqrt{s}}{2})\,\pi^{-\sqrt{s}/2}}{\Gamma(\frac14)}.$$

It is analytic and zero free on $\mathbb{C}\setminus(-\infty,0]$ *if and only if the Riemann Hypothesis holds. On* $(0,\infty)$ *it is analytic and positive, it is continuous at* 0 *with* $\varphi(0)=1$, *and* $-\varphi'/\varphi$ *is analytic in a neighbourhood of* $(0,\infty)$.

Proof. With $z = \frac12+\sqrt{s}$ we have $z(z-1) = s - \frac14$, so $(1-4s)\zeta(z) = -8\,\xi(z)\pi^{z/2}/\Gamma(z/2)$; evaluating at $s=0$ fixes the constant. The gamma factor G is analytic and zero free on $\mathbb{C}\setminus(-\infty,0]$ since $\operatorname{Re}(\frac14+\frac{\sqrt{s}}{2}) > 0$, so φ has no zeros in $\mathbb{C}\setminus(-\infty,0]$, and its poles there are those of R, namely the points $\sqrt{s} = \rho - \frac12$ with $\operatorname{Re}\rho > \frac12$. By the functional equation there are none if and only if RH holds. On the real axis $\zeta(\frac12+x)$ is finite and nonzero for $x>0$, $x\ne\frac12$, and $(1-4s)\zeta(\frac12+\sqrt{s}) \to -4$ as $s\to\frac14$. Since $\zeta(\frac12+x) < 0$ for $0 < x < \frac12$ and

$\zeta(\frac12+x)>0$ for $x>\frac12$, while $1-4s$ changes sign at $s=\frac14$ and $\zeta(\frac12)<0$, the function φ is positive on $(0,\infty)$. □

The reciprocal Xi function R carries the zeros and the gamma factor G the smooth growth. A zero of ζ off the line appears as a pole of φ off the negative axis, and a zero on the line as a singularity on the axis, where a Thorin measure can place an atom.

THEOREM 6.1.10. *Assume RH and let $N(T)$ count the zeros $\frac12+i\gamma$ with $0<\gamma\le T$, with multiplicity, as do the sums over γ below. Then for $s\in\mathbb{C}\setminus(-\infty,0]$*

$$\log\varphi(s)=-\int_{(0,\infty)}\log\bigl(1+\tfrac{s}{t}\bigr)\,dM(t),\qquad M(t)=N(\sqrt t)-\frac{\theta(\sqrt t)}{\pi},$$

that is

$$dM(t)=\sum_{\gamma>0}\delta_{\gamma^2}-\frac{\theta'(\sqrt t)}{2\pi\sqrt t}\,dt,\qquad -\frac{\varphi'}{\varphi}(s)=\int\frac{dM(t)}{s+t}.$$

By the Riemann von Mangoldt formula $M(t)=1+S(\sqrt t)$ for $\sqrt t$ not an ordinate, with $S(0^+)=-1$.

Proof. Under RH, Hadamard's product gives $R(s)=\prod_{\gamma>0}(1+s/\gamma^2)^{-1}$, a GGC with Thorin measure $\sum\delta_{\gamma^2}$, and $-R'/R(s)=\sum_{\gamma>0}1/(s+\gamma^2)$. For G put $g=-G'/G=-\frac{1}{4\sqrt s}\{\psi(\frac14+\frac{\sqrt s}{2})-\log\pi\}$, analytic on $\mathbb{C}\setminus(-\infty,0]$, $O(|s|^{-1/2})$ at 0 and $O(|s|^{-1/2}\log|s|)$ at infinity. At $\sqrt s=i\tau$, $g=\frac{i}{4\tau}\{\psi(\frac14+\frac{i\tau}{2})-\log\pi\}$, so $\operatorname{Im} g=\theta'(\tau)/(2\tau)$, and Lemma 6.1.1 gives the density $-\theta'(\sqrt t)/(2\pi\sqrt t)$. Add the two Stieltjes representations and integrate from 0 to s; Fubini applies because $|dM|$ integrates $\min(1,1/t)(1+|\log t|)$. Finally $N(T)=\theta(T)/\pi+1+S(T)$ [255, Section 9.3]. □

Under RH the Thorin measure of φ therefore has two parts. The first consists of unit atoms at the squared ordinates, the Thorin measure of R, which is a GGC exactly when RH holds (Proposition 1.2.1). The second is the density $-\theta'(\sqrt t)/(2\pi\sqrt t)$ of the gamma factor, whose distribution function is $-\theta(\sqrt t)/\pi$; it is positive for $t<\tau_0^2$ and negative beyond, where $\tau_0\approx6.28984$ is the point at which θ attains its minimum. So $M-1=S(\sqrt t)$ is the fluctuation of the zero count about its smooth part $1+\theta(\sqrt t)/\pi$. Between consecutive ordinates, all of which exceed τ_0, the function M decreases; hence dM is not positive, and φ is not a GGC even under RH. Theorem 7.1.3 proves more, unconditionally.

Near the origin $M(t)\sim c\sqrt t/\pi$, with

$$c=-\theta'(0)=\frac{\log\pi-\psi(\frac14)}{2}=\frac{\zeta'(\frac12)}{\zeta(\frac12)}\approx2.68609,\tag{6.1.6}$$

the last equality from $\xi'(\frac{1}{2}) = 0$. The function M rises to $-\theta(\tau_0)/\pi \approx 1.12394$ at $t = \tau_0^2$, falls to ≈ 0.55025 just before γ_1^2, jumps by one, and thereafter is a staircase with mean 1 whose continuous parts decrease (Figure 6.1; Table 7.9 gives these constants to more digits). The level 1 comes from the pole: the mean value 1 of M matches $\varphi(s) \sim -\zeta(\frac{1}{2})/(4s)$ as $s \to \infty$. At the other end $\varphi(s) = 1 - c\sqrt{s} + O(s)$ as $s \to 0$.

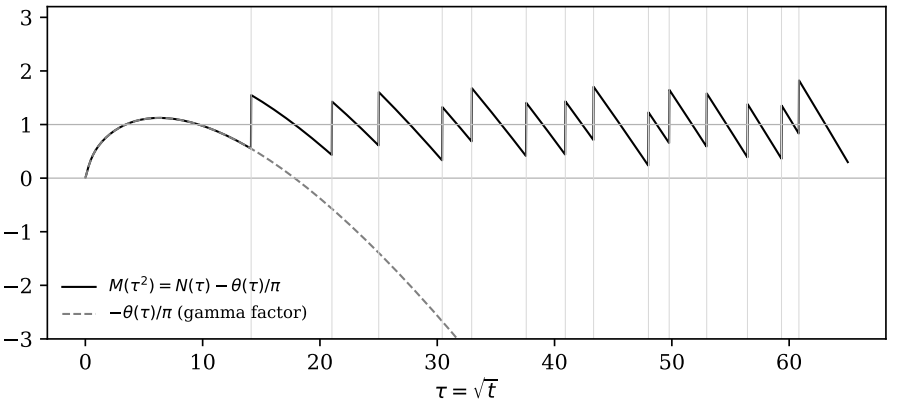


Figure 6.1: The function $M(\tau^2) = N(\tau) - \theta(\tau)/\pi = 1 + S(\tau)$ against $\tau = \sqrt{t}$, with the zero ordinates marked; under RH it is the distribution function of the Thorin measure of φ. The dashed curve is the gamma factor contribution $-\theta(\tau)/\pi$ alone. The function M rises on $(0, \tau_0)$; beyond τ_0, and so between consecutive ordinates, it decreases with slope $-\theta'(\tau)/\pi$, and at each ordinate it jumps by one. The mean level is 1.

6.1.5 Lévy density, theta functions and a criterion for RH

Under RH, integrating by parts in Theorem 6.1.10 and using $s/(s+t) = \int_0^\infty (1 - e^{-sx}) t e^{-tx} dx$ turns the Thorin type representation of φ into a Lévy Khintchine one,

$$\log \varphi(s) = -\int_0^\infty \bigl(1 - e^{-sx}\bigr) \frac{w(x)}{x}\, dx,$$
$$w(x) = \int e^{-tx}\, dM(t), \qquad \frac{w(x)}{x} = \int_0^\infty e^{-tx} M(t)\, dt.$$

The function w, unlike M, can be written down without the zeros, and without RH.

THEOREM 6.1.11. *Define, unconditionally,*

$$w(x) = e^{x/4} - P(x), \qquad P(x) = \frac{1}{\sqrt{4\pi x}} \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}}\, e^{-(\log n)^2/4x}, \qquad x > 0.$$

Then $-\varphi'/\varphi(s) = \int_0^\infty e^{-sx} w(x)\, dx$ *for* $s > \frac{1}{4}$, *and for every* $x > 0$

$$\begin{gathered} w(x) = W(x) - W_0(x), \qquad W(x) = \tfrac{1}{2} \sum_{\rho} e^{x(\rho - \frac{1}{2})^2}, \\ W_0(x) = \frac{1}{\pi} \int_0^\infty e^{-x\tau^2} \theta'(\tau)\, d\tau, \end{gathered} \tag{6.1.7}$$

the sum running over all nontrivial zeros with multiplicity. Here W *is the heat trace of Section 1.2 in the variable* x*; under RH,* $W(x) = \sum_{\gamma > 0} e^{-x\gamma^2}$ *and* $w(x) = \int e^{-tx} dM(t)$. *The function* w *satisfies* $w(0^+) = 1$ *and* $w(x) \sim c/(2\sqrt{\pi x})$ *as* $x \to \infty$.

Proof. By logarithmic differentiation, $-\varphi'/\varphi = \frac{1}{s - 1/4} + \frac{1}{2\sqrt{s}} \frac{\zeta'}{\zeta}(\frac{1}{2} + \sqrt{s})$. The first term is the transform of $e^{x/4}$; for the second use $\zeta'/\zeta = -\sum \Lambda(n) n^{-z}$ and $\frac{1}{2\sqrt{s}} e^{-a\sqrt{s}} = \int_0^\infty e^{-sx} (4\pi x)^{-1/2} e^{-a^2/4x} dx$, with Tonelli justified on $s > \frac{1}{4}$. Identity (6.1.7) is Weil's explicit formula for the even test function $h(r) = e^{-xr^2}$, whose Fourier partner is $g(u) = (4\pi x)^{-1/2} e^{-u^2/4x}$: the pole contributes $h(\pm \frac{i}{2}) = e^{x/4}$, the archimedean place contributes W_0, and the primes contribute P. It is (1.2.5) of Section 1.2 with $t = x$. Under RH, Theorem 6.1.10 gives the same Laplace transform for $\int e^{-tx} dM(t)$ and uniqueness identifies the two. The limit $w(0^+) = 1$ holds because $P(x) = O(x^{-1/2} e^{-(\log 2)^2/4x})$ as $x \to 0$. The behaviour at infinity needs no hypothesis: every term of W has modulus at most $e^{-x(\gamma^2 - 1/4)}$ with $\gamma > 14$, so W decays exponentially, while $W_0(x) \sim -c/(2\sqrt{\pi x})$ by Lemma 6.1.12 below. □

We call w the canonical function of φ: when φ is infinitely divisible, $w(x)/x$ is its Lévy density, and φ is infinitely divisible exactly when $w \geq 0$. Theorem 6.1.11 gives w two faces. On the prime side w is the heat trace $e^{x/4} = h(\pm \frac{i}{2})$ of the pole, the Gaussian evaluated at the imaginary ordinate of $z = 1$, minus the heat trace P of the prime powers, each a Gaussian in $\log n$ of variance $2x$; this series converges absolutely for every $x > 0$, which is the reason for working with the heat kernel e^{-tx}. On the zero side w is the heat trace over the zeros minus that of their smooth count θ/π, which comes from the gamma factor. The proof of Theorem 7.1.1 uses both faces, the prime side for small x and the finitely many verified zeros for large x, and it rests on the following integral formula for W_0.

LEMMA 6.1.12. *For* $x > 0$,

$$W_0(x) = \frac{F(x)}{4\sqrt{\pi x}}, \qquad F(x) = 2\theta'(0) + \int_0^\infty \frac{e^{-u/4}\big(1 - e^{-u^2/16x}\big)}{1 - e^{-u}}\, du.$$

The function F is strictly decreasing, $F(0^+) = +\infty$ and $F(\infty) = 2\theta'(0) = \psi(\frac14) - \log\pi < 0$. Hence W_0 has exactly one zero x_*, with $W_0 > 0$ on $(0, x_*)$ and $W_0 < 0$ on (x_*, ∞).

Proof. Insert $\psi(z) = \int_0^\infty \big(e^{-u}/u - e^{-zu}/(1-e^{-u})\big)du$ with $z = \frac14 + \frac{i\tau}{2}$ into θ', and use $\int_0^\infty e^{-x\tau^2}\cos(\tau u/2)\,d\tau = \frac12\sqrt{\pi/x}\,e^{-u^2/16x}$; the interchange is justified after writing the ψ integrand as $e^{-u}/u - e^{-u/4}/(1-e^{-u})$ plus $e^{-u/4}(1-\cos(\tau u/2))/(1-e^{-u})$, both absolutely integrable against $e^{-x\tau^2}$. The terms free of τ reassemble into $\psi(\frac14) - \log\pi$. The integrand of F is positive and decreasing in x; as $x \to 0$ it increases to $e^{-u/4}/(1-e^{-u})$, which is not integrable at 0. □

The Vere-Jones representation of Theorem 6.1.8 writes w as an absolutely convergent series of theta functions, one for each block. For the sawtooth measure of Lemma 6.1.7 define its heat trace

$$h_\lambda(x) = -\int e^{-tx}\,U_\lambda(dt) = \sum_{k\ge0} e^{-x\sigma_k(\lambda)} - \frac{\lambda}{4\sqrt{\pi x}} = \frac{\lambda}{2\sqrt{\pi x}}\sum_{j\ge1}(-1)^j e^{-j^2\lambda^2/4x},$$

the last form by Poisson summation. The first form is a sum over the pseudo zeros minus their smooth count and the second a sum over the multiples $j\lambda$, so this Jacobi theta identity is the explicit formula of the toy zeta function $1+e^{-\lambda\sqrt{s}}$.

Proposition 6.1.13. *For every $x > 0$, with absolutely convergent sums,*

$$w(x) = e^{x/4} + \sum_{N\ge2} B_N\, h_{\log N}(x).$$

Proof. Substitute the theta form of h_λ and regroup the triple sum over (n, m, j) by $r = mj$. The coefficient of $e^{-r^2(\log n)^2/4x}$ is

$$-(4\pi x)^{-1/2} a_n \log n \sum_{mj=r} m\, d_m (-1)^{j+1},$$

and the inner sum is $\delta_{r,1}$ because its Dirichlet series is $\eta(v)^{-1}\cdot\eta(v)$. Since $a_n \log n = \Lambda(n)/\sqrt{n}$ the sum equals $-P(x)$. The bound $|m\,d_m| \le m$ against Gaussian decay gives absolute convergence. □

Proposition 6.1.13 is the Vere-Jones form of Theorem 6.1.11. Unconditionally, w is the Laplace transform, term by term, of the formal series $\delta_{-1/4} + \sum_N B_N\{\sum_k \delta_{\sigma_k(\log N)} - \frac{\log N}{4\pi\sqrt{t}}dt\}$ of sech atom lattices, with all the arithmetic in the weights B_N; under RH it is also the Laplace transform of dM. The atom at $-\frac14$ comes from the pole of ζ and lies outside $(0,\infty)$, where Thorin measures live, so under RH the sawtooth series must cancel it. Two consequences follow. The first makes the sawtooth expansion of S rigorous after Gaussian averaging.

COROLLARY 6.1.14. *Assume RH. For every $x > 0$,*

$$\int_0^\infty S(\tau)\, 2\tau e^{-x\tau^2}\, d\tau = \frac{e^{x/4}-1}{x} + \sum_{N\geq 2} B_N \int_0^\infty \Sigma_{\log N}(\tau)\, 2\tau e^{-x\tau^2}\, d\tau,$$

and each integral on the right equals $h_{\log N}(x)/x$.

Proof. By Theorem 6.1.11,

$$\int_0^\infty e^{-tx}(1+S(\sqrt{t}))dt = w(x)/x,$$

and $\int_0^\infty e^{-tx}\Sigma_\lambda(\sqrt{t})\, dt = h_\lambda(x)/x$ by parts since $\Sigma_\lambda(0) = 0$. Subtract $\int e^{-tx}dt = 1/x$ and apply Proposition 6.1.13; substitute $t = \tau^2$. □

This is the sawtooth analogue of the classical formal series $S(\tau) = -\pi^{-1}\sum_n a_n \sin(\tau \log n)$, with the pole appearing as the correction $(e^{x/4}-1)/x$. The second consequence is a criterion for RH.

THEOREM 6.1.15. *The following are equivalent.*

(i) The Riemann Hypothesis.

(ii) There is a real signed Borel measure m on $[0,\infty)$ with $\int |m|(dt)/(1+t) < \infty$ and $w(x) = \int e^{-tx} m(dt)$ for all $x > 0$.

In that case $m = dM$.

Proof. If RH holds take $m = dM$; the atoms satisfy $\sum(1+\gamma^2)^{-1} < \infty$ and the density is $O(t^{-1/2}(1+t)^{-1}\log(2+t))$ after division by $1+t$. Conversely, given m, put $J(s) = \int m(dt)/(s+t)$, analytic on $\mathbb{C}\setminus(-\infty,0]$. For $s > \frac{1}{4}$ Fubini gives $\int_0^\infty e^{-sx} w(x)dx = J(s)$, so $-\varphi'/\varphi = J$ on $(\frac{1}{4},\infty)$ and hence on all of $\mathbb{C}\setminus(-\infty,0]$ by analytic continuation. A zero or pole of φ in $\mathbb{C}\setminus(-\infty,0]$ would be a pole of $-\varphi'/\varphi$, so φ has none, and Proposition 6.1.9 gives RH. Uniqueness of the Stieltjes transform gives $m = dM$. □

The criterion is a statement about w alone, which is computable from the primes. In the language of Proposition 6.1.13, RH says that the formal measure $\delta_{-1/4} + \sum_N B_N(-U_{\log N})$ has the Laplace transform of a genuine real measure on $[0,\infty)$. If RH fails, a zero $\rho = \frac{1}{2} + \beta + i\gamma$ with $\beta > 0$ and its conjugate contribute to W the term $e^{x(\beta^2-\gamma^2)}\cos(2\beta\gamma x)$, and the quadruple $\rho, \bar{\rho}, 1-\rho, 1-\bar{\rho}$ contributes twice this. The term is the real part of the Laplace transform of an atom at the complex point $(\gamma - i\beta)^2$, and no real measure on $[0,\infty)$ can reproduce it.

6.1.6 Across the critical strip

Moving the line of evaluation across the critical strip puts the results of this section and of Section 7.1 in one picture. Fix σ and evaluate ζ on the line $\operatorname{Re} z = \sigma$ through $z = \sigma + \sqrt{s}$. A zero $\rho = \beta + i\gamma$ then sits at $s = (\rho - \sigma)^2$: if $\beta > \sigma$ it gives a pole of $1/\zeta(\sigma + \sqrt{s})$ in $\mathbb{C} \setminus (-\infty, 0]$, if $\beta = \sigma$ a point of the cut, and if $\beta < \sigma$ no singularity, only a Poisson kernel of width $\sigma - \beta$ in the Thorin density. On each line one can ask whether a normalised reciprocal of ζ is infinitely divisible, that is, has a Lévy representation with positive Lévy measure, and whether the completed function $R_\sigma(s) = \xi(\sigma)/\xi(\sigma + \sqrt{s})$ has the stronger GGC property.

To the right of $\operatorname{Re} z = 1$ the first question has a classical answer, in the variable τ rather than s. The Dirichlet series converges, $\zeta(\sigma + i\tau)/\zeta(\sigma) = \mathbb{E} e^{-i\tau \log N_\sigma}$ for the zeta distribution $\Pr(N_\sigma = n) = n^{-\sigma}/\zeta(\sigma)$, and the Lévy measure of $\log N_\sigma$ is the prime measure $\sum_n \Lambda(n) n^{-\sigma} (\log n)^{-1} \delta_{\log n}$. The Vere-Jones representation writes this measure as a superposition of the alternating lattices $\sum_j (-1)^{j+1} j^{-1} \delta_{jn \log p}$, the signed Lévy measures of fair coins, one for each prime power p^n, with weight the necklace polynomial $b_n(p^{-\sigma})$ (Theorem 7.1.8). At $\sigma = 0$ the weight $b_n(1)$ is 1 when n is a power of 2 and 0 otherwise, and the identity becomes Euler's binary product, with every block a solved model of unit multiplicity; for $\sigma > 0$ the weights are fractional and some are negative.

Inside the strip $|\zeta(\sigma + i\tau)|$ is unbounded, while a characteristic function is bounded by 1, so the characteristic function picture breaks down; the Vere-Jones representation survives in the variable s. For $0 < \sigma < 1$ put $\varphi_\sigma(s) = \zeta(\sigma)/\{(1 - s/(1-\sigma)^2)\zeta(\sigma + \sqrt{s})\}$, which is φ at $\sigma = \frac{1}{2}$. For $s > (1-\sigma)^2$, $-\varphi_\sigma'/\varphi_\sigma$ is the Laplace transform of a function w_σ, given by the theta series of Proposition 6.1.13 (Section 6.1.5) with $e^{(1-\sigma)^2 x}$ in place of $e^{x/4}$ and the weights $b_n(p^{-\sigma})$ in place of $B_{p^n} = b_n(p^{-1/2})$ (Theorem 7.1.12). When $w_\sigma \geq 0$ this is a Lévy representation, $\log \varphi_\sigma(s) = -\int (1 - e^{-sx}) w_\sigma(x) x^{-1} dx$, with an explicit Lévy density built from the primes. Theorem 7.1.1 proves $w_{1/2} = w > 0$, and numerically $w_\sigma > 0$ on nine lines from $\sigma = 0.05$ to $\sigma = 0.99$, for $10^{-4} \leq x \leq 10^3$ (Section 7.1); whether infinite divisibility holds on every line of the strip is open (Problem 7.8).

The Thorin property is where the zeros enter. The functions φ_σ are never GGCs, for a reason that uses no information about ζ: the factor removing the pole makes w_σ increase at the origin (Proposition 7.1.13). For $\sigma \geq \frac{1}{2}$, by Theorem 7.1.14, the form of Theorem 1.2.2 in the coordinates of Section 6.1.1, the function R_σ is a GGC if and only if $\zeta(z) \neq 0$ for $\operatorname{Re} z > \sigma$, and its Thorin measure is then the zero density smoothed by Poisson kernels of widths $\sigma - \beta$ in the variable $\sqrt{t}$. The smallest such σ is $\Theta = \sup \operatorname{Re} \rho$, and RH is the statement $\Theta = \frac{1}{2}$: the kernels then sharpen all the way to the unit atoms at γ^2 of the Thorin measure of R (Proposition 1.2.3 and the paragraph after Theorem 7.1.14). The Thorin density of the gamma factor is also built from Poisson kernels (Theorem 6.2.1), but their widths $c_n = 2n + \frac{1}{2}$ are fixed and never sharpen, so the gamma factor produces no atoms. Table 6.2 records the picture.

Line	Lévy representation	Thorin representation of R_σ
$\sigma > 1$	$\zeta(\sigma + i\tau)/\zeta(\sigma) = \mathbb{E}e^{-i\tau \log N_\sigma}$; prime Lévy measure as a necklace weighted superposition of fair coin lattices	GGC, unconditionally; Thorin measure a Poisson smoothed zero density
$\sigma = 1$	not treated here	GGC, unconditionally; no atoms, since ζ has no zeros on $\operatorname{Re} z = 1$
$\frac{1}{2} < \sigma < 1$	$w_\sigma > 0$ numerically	GGC if and only if there are no zeros with $\operatorname{Re}\rho > \sigma$
$\sigma = \frac{1}{2}$	$w > 0$ proved	GGC if and only if RH; the Poisson kernels sharpen to atoms at γ^2
$\sigma < \frac{1}{2}$	$w_\sigma > 0$ numerically	never: each zero on the critical line gives a pole of R_σ in $\mathbb{C} \setminus (-\infty, 0]$

Table 6.2: The two questions across the critical strip. Middle column: the Lévy representation, in the variable τ for $\sigma > 1$ and in the heat kernel form of Theorem 7.1.12 for $0 < \sigma < 1$, where the Lévy measure is positive when $w_\sigma > 0$. Right column: whether R_σ is a GGC (Theorem 7.1.14).

Chapters 9 to 11 return to the completed functions R_σ, written there $\varphi_\alpha(u) = \xi(\alpha)/\xi(\alpha + \sqrt{u})$, with the basepoint α in place of σ and the variable u of the book in place of s; they are not the normalised functions φ_σ of this section. Section 9.1 studies the basepoint $\alpha = 1$, the leftmost at which the GGC property is a theorem, Section 10.1 shows that $\xi(\alpha)/\xi(\alpha+|v|)$, as a function of real v, is the characteristic function of a normal scale mixture at every basepoint, and Section 11.4 shows by zero density estimates that for every $\alpha > \frac{1}{2}$ the negative part of the Thorin measure up to height T, if there is one, is a vanishing proportion of the whole as $T \to \infty$.

6.2 The gamma factor, the Meixner law and the sine squared lemma

In Section 6.1 the gamma factor G entered through the factorisation $\varphi = RG$ and its Thorin density $-\theta'(\sqrt{t})/(2\pi\sqrt{t})$, which changes sign at $t = \tau_0^2$. Its Weierstrass product writes this density as a renormalised series of Poisson kernels of widths $c_n = 2n + \frac{1}{2}$, and adding it to Proposition 6.1.13 gives the formal Thorin measure of the reciprocal Xi function R in Vere-Jones form. On the critical line $|G|^2$ is proportional to the density of a Meixner hypergeometric law, whose characteristic function is a GGC transform in the square of its variable and whose orthogonal polynomials satisfy a local Riemann Hypothesis. The sine squared lemma turns the Lévy measure of $\log f$ into the Thorin measure of $f(0)/f(\sqrt{s})$. It shows that

after a stable tilt the reciprocals of $\zeta(\beta+\sqrt{s})$, for $\beta>1$, and of the gamma factor are GGC transforms, and it shows how positivity is lost at the centre. Section 6.2.4 follows the tilt of the gamma factor to the floor $e^{-c\sqrt{s}}R$ of the reciprocal Xi function.

6.2.1 The gamma factor in Vere-Jones form

Recall from Section 6.1.1 the constants $c_n=2n+\frac12$ for $n\ge0$ and $b=\frac12(\gamma_E+\log\pi)\approx0.860973$, where γ_E is Euler's constant. With these constants the Weierstrass product writes G as a product of GGC transforms whose Thorin densities are Poisson kernels, renormalised by stable factors.

Theorem 6.2.1. *For $s\in\mathbb{C}\setminus(-\infty,0]$,*

$$G(s)=e^{-b\sqrt{s}}\prod_{n\ge0}\frac{e^{\sqrt{s}/(2n+2)}}{1+\sqrt{s}/c_n},$$

and the Thorin measure of G is

$$U_G(dt)=\frac{1}{2\pi\sqrt{t}}\Big\{b+\sum_{n\ge0}\Big(\frac{c_n}{t+c_n^2}-\frac{1}{2n+2}\Big)\Big\}dt=-\frac{\theta'(\sqrt{t})}{2\pi\sqrt{t}}\,dt.$$

Each factor $(1+\sqrt{s}/\kappa)^{-1}$ is the Laplace transform of a GGC, with positive Thorin density $q_\kappa(t)=\dfrac{\kappa}{2\pi\sqrt{t}\,(t+\kappa^2)}$, a Poisson kernel in $\sqrt{t}$; each factor $e^{a\sqrt{s}}$ with $a>0$ has the negative Thorin density $-a/(2\pi\sqrt{t})$.

Proof. Weierstrass gives $\log\Gamma(z_0+v)-\log\Gamma(z_0)=-\gamma_E v-\log(1+v/z_0)+\sum_{n\ge1}\{v/n-\log(1+v/(n+z_0))\}$. Take $z_0=\frac14$, $v=\sqrt{s}/2$, note $v/(n+\frac14)=\sqrt{s}/c_n$, and shift the index of v/n. For q_κ: $\frac{d}{ds}\log(1+\sqrt{s}/\kappa)=\frac{1}{2\sqrt{s}(\sqrt{s}+\kappa)}$, and Lemma 6.1.1 with $\sqrt{-t+i0}=i\sqrt{t}$ gives the density $\frac{\kappa}{2\pi\sqrt{t}(t+\kappa^2)}$. For $e^{a\sqrt{s}}$ use $\int_0^\infty\log(1+s/t)t^{-1/2}dt=2\pi\sqrt{s}$. To sum the Thorin densities of the factors, write the bracket with index n in U_G as $(\frac{1}{c_n}-\frac{1}{2n+2})-\frac{t}{c_n(t+c_n^2)}$: both parts are nonnegative, and over n the first sums to a constant and the second to $O(\log(2+t))$, so Tonelli's theorem applies to each part against $\log(1+s/t)\frac{dt}{2\pi\sqrt{t}}$ for $s>0$, and analytic continuation extends the sum to $\mathbb{C}\setminus(-\infty,0]$; the same split justifies the termwise integration against e^{-tx} in Proposition 6.2.2. The identification with θ' is the partial fraction series $\operatorname{Re}\psi(\frac14+\frac{i\tau}{2})=-\gamma_E+\sum_{n\ge0}\{\frac{1}{n+1}-\frac{2c_n}{c_n^2+\tau^2}\}$. □

Formally, the gamma factor is thus a renormalised infinite convolution of GGC laws, and the renormalisation is the only source of negativity in its Thorin density. Write X_φ and X_R for random variables with Laplace transforms φ and R; both laws exist and are infinitely divisible, by Theorem 7.1.1 and by Theorem 5.1.1 [200,

Theorem A]. Write Y_κ for the variable with $\mathbb{E}e^{-sY_\kappa} = \kappa/(\kappa+\sqrt{s})$, which is $Y_\kappa = E_\kappa^2/(4\Gamma_{1/2})$ with E_κ exponential of rate κ and $\Gamma_{1/2}$ an independent gamma variable of shape $\frac12$, and L_a for the positive stable variable of index $\frac12$ with $\mathbb{E}e^{-sL_a} = e^{-a\sqrt{s}}$. Theorem 6.2.1 and Proposition 6.1.9 then say formally

$$X_\varphi \stackrel{d}{=} X_R + L_b + \sum_{n\ge0} \big(Y_{c_n} \ominus L_{1/(2n+2)}\big), \tag{6.2.1}$$

where $A \ominus B$ stands for a formal variable whose Laplace transform is the quotient of those of A and B. Each bracket has Thorin density $\frac{1}{2\pi\sqrt{t}}\{\frac{c_n}{t+c_n^2} - \frac{1}{2n+2}\}$, positive for $t < \frac32 c_n$ and negative beyond. The sum of the Y_{c_n} diverges almost surely because $\sum 1/c_n = \infty$, and the stable deconvolutions are the compensators that make the product of the Laplace transforms converge.

Figure 6.2 shows the Thorin density of G against the first three kernels. In the variable $\tau = \sqrt{t}$ it is $-\theta'(\tau)/\pi$, evaluated from the digamma function, so nothing in it depends on a zero of ζ. It changes sign at the point τ_0 where θ attains its minimum, so the reciprocal $1/G$ has a Thorin density that is negative near the origin, and $1/G$ becomes a GGC only after the stable tilt of Theorem 6.2.12.

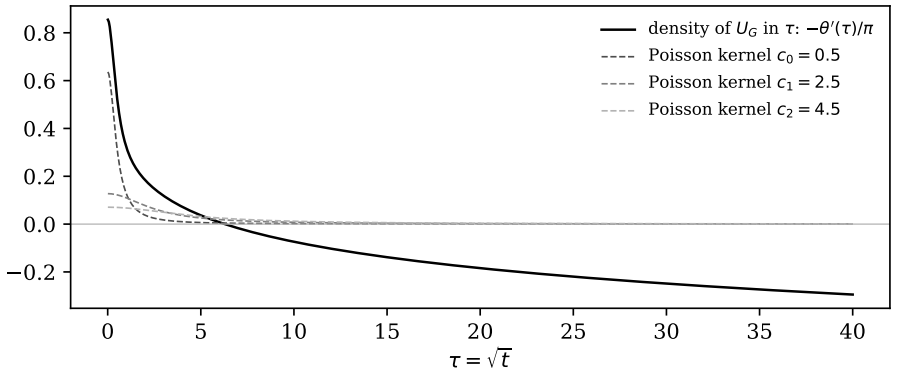


Figure 6.2: The Thorin density of the gamma factor in the variable $\tau = \sqrt{t}$, namely $-\theta'(\tau)/\pi$, together with the first three Poisson kernels $c_n/(\pi(c_n^2+\tau^2))$ of Theorem 6.2.1. The density is positive for $\tau < \tau_0 \approx 6.29$ and negative beyond, where the flat compensators outweigh the kernels.

Proposition 6.2.2. *For $x > 0$,*

$$W_0(x) = -\int e^{-tx}\, U_G(dt) = -\frac{b}{2\sqrt{\pi x}} + \sum_{n\ge0}\Big\{\frac{1}{(4n+4)\sqrt{\pi x}} - \frac12 e^{c_n^2 x}\operatorname{erfc}\big(c_n\sqrt{x}\big)\Big\}.$$

Proof. $\int_0^\infty e^{-tx} q_\kappa(t)\,dt = \frac{\kappa}{\pi}\int_0^\infty \frac{e^{-xv^2}}{v^2+\kappa^2}dv = \frac12 e^{\kappa^2 x}\operatorname{erfc}(\kappa\sqrt{x})$ and $\int_0^\infty e^{-tx}\frac{dt}{2\pi\sqrt{t}} = \frac{1}{2\sqrt{\pi x}}$. □

Both terms of the identity $W = w + W_0$ of (6.1.7) are now explicit, and their sum is the heat trace over the zeros.

THEOREM 6.2.3. *For every $x > 0$, with absolutely convergent sums,*

$$W(x) = e^{x/4} + \sum_{N\ge 2} B_N\, h_{\log N}(x) - \frac{b}{2\sqrt{\pi x}} + \sum_{n\ge 0}\Big\{ \frac{1}{(4n+4)\sqrt{\pi x}} - \frac{1}{2} e^{c_n^2 x}\operatorname{erfc}\big(c_n\sqrt{x}\big)\Big\}.$$

Equivalently, $W(x)$ is the Laplace transform, taken term by term, of the formal series

$$U_R = \delta_{-1/4} + \sum_{N\ge 2} B_N\Big\{\sum_{k\ge 0}\delta_{\sigma_k(\log N)} - \frac{\log N}{4\pi\sqrt{t}}\,dt\Big\} - \frac{1}{2\pi\sqrt{t}}\Big\{ b + \sum_{n\ge 0}\Big(\frac{c_n}{t+c_n^2} - \frac{1}{2n+2}\Big)\Big\}dt,$$

the formal Thorin measure of $R(s) = \xi(\frac12)/\xi(\frac12+\sqrt{s})$. Under RH, R is a GGC, and its Thorin measure $\sum_{\gamma>0}\delta_{\gamma^2} = dN(\sqrt{t})$, where $N(T)$ counts the zeros with $0 < \gamma \le T$, is a positive measure whose Laplace transform is also W.

Proof. $W = w + W_0$ by (6.1.7); substitute Proposition 6.1.13 for w and Proposition 6.2.2 for W_0. Term by term, $\delta_{-1/4}$ has Laplace transform $e^{x/4}$, the bracket with index N has transform $h_{\log N}(x)$ by the definition of h_λ, and the last density is $-U_G$, whose transform is $W_0(x)$ by Proposition 6.2.2. Under RH, Hadamard's product gives $R(s) = \prod_{\gamma>0}(1 + s/\gamma^2)^{-1}$, as in the proof of Theorem 6.1.10, and $W(x) = \sum_{\gamma>0} e^{-x\gamma^2}$ by Theorem 6.1.11. □

The formula is the explicit formula for the heat trace in closed form: the polar term $e^{x/4}$, a sum over prime powers of the block theta functions $h_{\log N}$ weighted by the coefficients B_N of Theorem 6.1.8, which Theorem 7.1.8 of Section 7.1 expresses through necklace polynomials, and a sum of complementary error functions over the poles of the gamma factor. Each term is elementary and computable to high accuracy, and Table 7.8 checks the identity against the zero sum to twenty digits at $x = 0.005$, 0.01 and 0.03.

In U_R the pole gives an atom at $-\frac14$, off the half line $(0,\infty)$ that carries Thorin measures. The primes give lattices of sech atoms at $((2k+1)\pi/\log N)^2$, each compensated by its mean density. The gamma factor gives negative Poisson kernels of widths c_n in the variable $\sqrt{t}$, with poles at $\sqrt{t} = \pm i c_n$, each compensated by a positive flat density. RH is the statement that the three groups combine into unit atoms at the γ^2 and nothing else. They combine only through Laplace transforms: every partial sum of U_R carries the atom at $-\frac14$ and the Thorin measure $\sum_{\gamma>0}\delta_{\gamma^2}$

of R under RH does not, so, like the series after Proposition 6.1.13, U_R does not converge to that measure as a series of measures. By Theorem 6.1.15, R has a Thorin type representation only if RH holds, since adding U_G to its measure would give a measure m as in condition (ii) there.

On the cut the gamma factor has modulus and phase

$$G(-\tau^2 + i0) = \frac{|\Gamma(\frac{1}{4} + \frac{i\tau}{2})|}{\Gamma(\frac{1}{4})}\, e^{i\theta(\tau)}.$$

The phase is the Riemann Siegel theta function, and it is $-\pi$ times the distribution function of the Thorin measure: $U_G((0,\tau^2]) = -\theta(\tau)/\pi$, since $\operatorname{Im}\log(1+s/t) = \pi$ for $t < \tau^2$ at $s = -\tau^2 + i0$. The modulus is the logarithmic potential of the same measure,

$$\log \frac{|\Gamma(\frac{1}{4} + \frac{i\tau}{2})|}{\Gamma(\frac{1}{4})} = -\int \log\Big|1 - \frac{\tau^2}{t}\Big|\, U_G(dt), \tag{6.2.2}$$

the integral converging absolutely.

The square of the modulus is proportional to the density of a Meixner hypergeometric law, with parameter $a = \frac{1}{4}$; Lai and Vere-Jones [135] study this family and its case $a = \frac{1}{2}$, the hyperbolic secant law. Since $\int |\Gamma(a+ix)|^2 e^{iux} dx = 2\pi\Gamma(2a)(2\cosh\frac{u}{2})^{-2a}$,

$$p(\tau) = \frac{|\Gamma(\frac{1}{4} + \frac{i\tau}{2})|^2}{2^{3/2}\pi^{3/2}} \quad \text{is a probability density with} \quad \int p(\tau) e^{iv\tau} d\tau = \operatorname{sech}(v)^{1/2}. \tag{6.2.3}$$

In this section p denotes this Meixner density; the Pólya density $\Phi/\Xi(0)$ of Section 1.2 does not occur. Since $\Xi(\tau) = -\frac{1}{2}(\tau^2 + \frac{1}{4})\pi^{-1/4}|\Gamma(\frac{1}{4} + \frac{i\tau}{2})|Z(\tau)$, with Hardy's function $Z(\tau) = e^{i\theta(\tau)}\zeta(\frac{1}{2} + i\tau)$ of Section 2.2, the Meixner density is the archimedean part of Ξ^2 on the critical line:

$$\Xi(\tau)^2 = \tfrac{1}{4}\big(\tau^2 + \tfrac{1}{4}\big)^2\, 2^{3/2}\pi\, p(\tau)\, Z(\tau)^2.$$

Lai and Vere-Jones call the Meixner hypergeometric distribution the odd man out; the characteristic function $\operatorname{sech}(v)^{1/2}$ of p is a power of the kernel of Proposition 6.1.3. We return to this law in Section 6.2.3.

6.2.2 From Lévy to Thorin: the sine squared lemma

For a function of the form $f(0)/f(\sqrt{s})$ we call f its primal function. The sine squared lemma of [199, Lemma 1] converts the Lévy measure of $\log f$ into the Thorin measure of $f(0)/f(\sqrt{s})$. The Vere-Jones representation produces such Lévy measures, sums of atoms at the points $k\log N$, so the lemma applies to them. We state it in its original scaling and then in the coordinates of Section 6.1.1.

LEMMA 6.2.4 ([199, Lemma 1]). *Suppose that for some measure μ on $(0,\infty)$ and some α with $\int_0^\infty (1\wedge x)e^{-\alpha x}\,\mu(dx)<\infty$,*

$$\log\frac{f(\alpha+s)}{f(\alpha)} - s\frac{f'(\alpha)}{f(\alpha)} = \int_0^\infty \big(e^{-sx}-1+sx\big)e^{-\alpha x}\,\frac{\mu(dx)}{x}, \qquad s\ge 0.$$

Then, for $s\ge 0$,

$$\log\frac{f(\alpha)}{f(\alpha+s)} + s\frac{f'(\alpha)}{f(\alpha)} = -\int_0^\infty \big(1-e^{-\frac{1}{2}s^2t}\big)\frac{\nu_\alpha(t)}{t}\,dt,$$

$$\nu_\alpha(t) = \frac{1}{\sqrt{2\pi}}\int_0^\infty e^{-tz}\Big\{\int_0^\infty 2\sin^2\big(x\sqrt{z/2}\big)e^{-\alpha x}\,\mu(dx)\Big\}\frac{dz}{\sqrt{\pi z}}.$$

The map $\mu\mapsto\nu_\alpha$ is linear and positivity preserving.

In these coordinates, with $s=u^2$ and $u=\sqrt{s}$, the lemma reads as follows. We include the proof by subordination, because its two stages, a Lévy density in s followed by a Thorin density, reappear in Section 6.2.4.

LEMMA 6.2.5 (Lemma 6.2.4 in the scaling $s=u^2$). *Let $\tilde\mu$ be a signed measure on $(0,\infty)$ with $\int(1\wedge x)|\tilde\mu|(dx)<\infty$, and suppose*

$$\log\frac{f(u)}{f(0)} - u\frac{f'(0)}{f(0)} = \int_0^\infty \big(e^{-ux}-1+ux\big)\frac{\tilde\mu(dx)}{x}, \qquad u\ge 0.$$

Then for $s\in\mathbb{C}\setminus(-\infty,0]$

$$\log\frac{f(0)}{f(\sqrt{s})} + \sqrt{s}\,\frac{f'(0)}{f(0)} = -\int_0^\infty \log\Big(1+\frac{s}{t}\Big)U(t)\,dt,$$
$$U(t) = \frac{1}{2\pi\sqrt{t}}\int_0^\infty 2\sin^2\Big(\frac{x\sqrt{t}}{2}\Big)\tilde\mu(dx).$$

If $\tilde\mu\ge 0$ then $U\ge 0$, and $s\mapsto f(0)e^{\sqrt{s}f'(0)/f(0)}/f(\sqrt{s})$ is the Laplace transform of a GGC. Since $\sqrt{s}=\int\log(1+s/t)\,\frac{dt}{2\pi\sqrt{t}}$, the Thorin measure of $f(0)/f(\sqrt{s})$ itself is $U(t)\,dt+\frac{f'(0)}{f(0)}\frac{dt}{2\pi\sqrt{t}}$.

Proof. Subordination by the first passage law (6.1.2) gives, for $x,u>0$,

$$e^{-ux}-1+ux = \int_0^\infty \big(1-e^{-u^2y}\big)\big(1-e^{-x^2/4y}\big)\frac{x}{\sqrt{4\pi y^3}}\,dy,$$

because $\int_0^\infty(1-e^{-u^2y})\frac{x}{\sqrt{4\pi y^3}}dy = ux$. Integrating against $\tilde\mu(dx)/x$ gives a Lévy form in $s=u^2$ with Lévy density $k(y)/y$, where $k(y)=(4\pi y)^{-1/2}\int(1-e^{-x^2/4y})\tilde\mu(dx)$ is the canonical function. Finally $(1-e^{-x^2/4y})/\sqrt{y} = \int_0^\infty e^{-yt}\{1-\cos(x\sqrt{t})\}\frac{dt}{\sqrt{\pi t}}$, so $k(y)=\int e^{-yt}U(t)dt$, and Frullani turns

$\int(1-e^{-sy})k(y)\frac{dy}{y}$ into $\int\log(1+s/t)U(t)dt$. All kernels in these steps are nonnegative, so by Tonelli's theorem the steps hold with $|\tilde\mu|$ in place of $\tilde\mu$, and there every integral is finite, since $0\le(e^{-ux}-1+ux)/x\le\min(\frac{u^2x}{2},u)$ and $\int(1\wedge x)|\tilde\mu|(dx)<\infty$; this justifies each interchange for the signed $\tilde\mu$. The analytic continuation to $\mathbb{C}\setminus(-\infty,0]$ is immediate. □

Lemma 6.2.5 is (1.2.9) of Section 1.2 integrated against $\tilde\mu(dx)/x$; the letters s and u are exchanged between the two sections. We call $\tilde\mu$ the primal Lévy measure of $f(0)/f(\sqrt{s})$. Positivity comes from the kernel. The signed cosine density that Lemma 6.1.1 produces for a single exponential in an exponent (Section 6.1.2) is $-\cos=2\sin^2-1$, and Lemma 6.2.5 separates it into the positive part $2\sin^2$ and a flat part -1, which is absorbed into the linear tilt $u\,f'(0)/f(0)$, a stable law of index $\frac12$.

6.2.2.1 Zeta to the right of the line, the Vere-Jones blocks and the centre

Where the primal Lévy measure is positive the lemma applies directly. The first such case is ζ to the right of $\operatorname{Re}z=1$.

Proposition 6.2.6. *For $\beta>1$ put $\tilde\mu_\beta=\sum_{n\ge2}\Lambda(n)n^{-\beta}\delta_{\log n}$. Then*

$$\frac{\zeta(\beta)}{\zeta(\beta+\sqrt{s})}\,e^{\sqrt{s}\,\zeta'(\beta)/\zeta(\beta)}=\exp\Big(-\int_0^\infty\log\Big(1+\frac{s}{t}\Big)U_\beta(t)\,dt\Big),$$

$$U_\beta(t)=\frac{1}{2\pi\sqrt{t}}\sum_{n\ge2}\frac{\Lambda(n)}{n^\beta}\,2\sin^2\Big(\frac{\sqrt{t}\log n}{2}\Big)\ge0,$$

so the left side is the Laplace transform of a GGC. Adding the tilt, the Thorin measure of $\zeta(\beta)/\zeta(\beta+\sqrt{s})$ is $-\frac{1}{2\pi\sqrt{t}}\sum\Lambda(n)n^{-\beta}\cos(\sqrt{t}\log n)\,dt$, which is the Euler product density (6.1.5) *at β in place of $\frac12$.*

Proof. $\log\zeta(\beta+u)=\sum\Lambda(n)n^{-\beta}e^{-u\log n}/\log n$, so the hypothesis of Lemma 6.2.5 holds with $\tilde\mu_\beta$, which has finite mass for $\beta>1$. □

The tilt is needed: without it the Thorin density

$$-\frac{1}{2\pi\sqrt{t}}\sum\Lambda(n)n^{-\beta}\cos(\sqrt{t}\log n)$$

is negative near $t=0$. For ξ no tilt is needed, since $s\mapsto\xi(\alpha)/\xi(\alpha+\sqrt{s})$ is a GGC transform for every $\alpha\ge1$ (Theorem 1.2.2); at $\alpha=1$, Section 9.1 obtains its Thorin density from a Lévy form of $\log\xi(1+u)$ by the same lemma. As $\beta\downarrow1$ the mass $\sum\Lambda(n)n^{-\beta}$ diverges, and at $\beta=\frac12$, where the pole and the gamma factor also enter, the lemma cannot be applied as it stands. The rest of this chapter and Chapter 7 try to extend the case $\beta>1$ towards the critical line.

The Vere-Jones blocks have signed primal Lévy measures, yet the lemma still gives positive Thorin densities. Apply it to one block $f_\lambda(u) = 1 + e^{-\lambda u}$ of Theorem 6.1.8, shifted by $\alpha > 0$. Since $\log(1 + e^{-\lambda(\alpha+u)}) = \sum_k (-1)^{k+1} q^k e^{-k\lambda u}/k$ with $q = e^{-\lambda\alpha}$, the primal Lévy measure is the alternating lattice $\tilde\mu_{\lambda,\alpha} = \lambda \sum_{k\geq 1} (-1)^{k+1} q^k \delta_{k\lambda}$.

PROPOSITION 6.2.7. *For $\alpha > 0$ the sine squared transform of $\tilde\mu_{\lambda,\alpha}$ is positive:*

$$U_{\lambda,\alpha}(t) = \frac{\lambda}{2\pi\sqrt{t}} \, \frac{q(1-q)\big(1 - \cos\lambda\sqrt{t}\big)}{(1+q)\big(1 + 2q\cos\lambda\sqrt{t} + q^2\big)} \geq 0,$$

so $\frac{1+e^{-\lambda\alpha}}{1+e^{-\lambda(\alpha+\sqrt{s})}} \, e^{-\sqrt{s}\lambda q/(1+q)}$ is the Laplace transform of a GGC. As $\alpha \downarrow 0$, $U_{\lambda,\alpha}(t)\, dt \to \sum_{k\geq 0} \delta_{\sigma_k(\lambda)}$ vaguely, the tilt density tends to $-\frac{\lambda}{4\pi\sqrt{t}}$, and the Thorin measure of $2/(1 + e^{-\lambda\sqrt{s}})$ is recovered as $\sum_k \delta_{\sigma_k(\lambda)} - \frac{\lambda}{4\pi\sqrt{t}} dt = -U_\lambda$, the sawtooth measure of Lemma 6.1.7.

Proof. $\sum_k (-1)^{k+1} q^k = \frac{q}{1+q}$ and $\sum_k (-1)^{k+1} q^k \cos k\omega = \frac{q\cos\omega + q^2}{1 + 2q\cos\omega + q^2}$; their difference is $\frac{q(1-q)(1-\cos\omega)}{(1+q)(1+2q\cos\omega+q^2)} \geq 0$. As $q \uparrow 1$ this is a Poisson kernel in $\omega = \lambda\sqrt{t}$ concentrating at $\omega = (2k+1)\pi$ with mass π per period, which becomes $\sum_k \delta_{\sigma_k(\lambda)}$ after the change of variable. The tilt is $f'(\alpha)/f(\alpha) = -\lambda q/(1+q) \to -\lambda/2$. □

The sine squared image of each block is a Poisson kernel in $\sqrt{t}$ for $\alpha > 0$ and the sech lattice at $\alpha = 0$, so the negative part of the sawtooth measure is entirely the tilt. Summing the blocks with the weights B_N gives $\log\zeta(\frac{1}{2} + \alpha + \sqrt{s})$, and the sum of Thorin densities $\sum_N B_N U_{\log N,\alpha}$ converges absolutely for $\alpha > \frac{1}{2}$, where it equals $U_{1/2+\alpha}$ of Proposition 6.2.6. Positivity is then visible termwise in the prime form. It is not visible blockwise in the Vere-Jones form, because infinitely many weights are negative: $B_{p^3} = \frac{1}{3}(p^{-3/2} - p^{-1/2}) < 0$ for every prime p, and Proposition 7.1.9 of Section 7.1 gives infinitely many more.

At $\beta = \frac{1}{2}$ the prime Lévy measure $\tilde\mu_{1/2} = \sum \Lambda(n) n^{-1/2} \delta_{\log n}$ has infinite mass, and both the image and the tilt in Lemma 6.2.5 diverge: the positive sine squared image has density of order $\frac{1}{2\pi\sqrt{t}} \sum \Lambda(n) n^{-1/2} = \infty$, and the tilt $\zeta'/\zeta(\frac{1}{2} + 0)$ is $-\infty$ along the Dirichlet series. The divergences cancel, and the heat trace makes the cancellation exact. Since $\int_0^\infty e^{-xt}\, 2\sin^2(\frac{\sqrt{t}L}{2}) \frac{dt}{2\pi\sqrt{t}} = \frac{1 - e^{-L^2/4x}}{2\sqrt{\pi x}}$, the heat trace of the sine squared image is formally

$$\frac{1}{2\sqrt{\pi x}} \sum_{n\geq 2} \frac{\Lambda(n)}{\sqrt{n}} \big(1 - e^{-(\log n)^2/4x}\big) = \frac{1}{2\sqrt{\pi x}} \sum_{n\geq 2} \frac{\Lambda(n)}{\sqrt{n}} - P(x).$$

Replace the divergent constant by its analytic continuation $\sum \Lambda(n) n^{-1/2} \to -\zeta'(\frac{1}{2})/\zeta(\frac{1}{2}) = -c$, and add the tilt $\frac{c}{2\sqrt{\pi x}}$ (the heat trace of $\frac{c}{2\pi\sqrt{t}} dt$) and the heat trace $e^{x/4}$ of the pole atom $\delta_{-1/4}$. The result is $e^{x/4} - P(x) = w(x)$, in agreement

with Theorem 6.1.11. So at the centre Lemma 6.2.5 still produces the correct Thorin measure dM, but only after a regularisation that subtracts an infinite positive constant from a positive density, and that is where the sign of dM is lost.

6.2.2.2 The inverse direction: cosine Lévy forms

The Frullani identity behind Lemma 6.2.5 also runs in the opposite direction, from a Thorin measure in s to a symmetric Lévy measure in $u = \sqrt{s}$: for $\beta > 0$ and real u, $\int_0^\infty 2\sin^2(\frac{ux}{2})e^{-\beta x}\frac{dx}{x} = \frac{1}{2}\log(1+u^2/\beta^2)$, hence

$$\int \log\Big(1+\frac{s}{t}\Big)U(dt) = \int_0^\infty 2\sin^2\Big(\frac{ux}{2}\Big)K_U(x)\,\frac{dx}{x}, \qquad K_U(x) = 2\int e^{-\sqrt{t}\,x}\,U(dt). \tag{6.2.4}$$

If $U \geq 0$ then $u \mapsto \exp(-\int \log(1+u^2/t)U(dt))$ is the characteristic function of a symmetric infinitely divisible law with Lévy density $K_U(|x|)/(2|x|)$. Section 10.1 applies this dictionary to $\xi(\alpha)/\xi(\alpha+|u|)$, which is the characteristic function of a normal scale mixture at every basepoint α.

Proposition 6.2.8. *Under* (6.2.4)*:*

(i) the Meixner law (6.2.3) *has Lévy density* $\dfrac{1}{4|x|\sinh(\pi|x|/2)}$ *on* $\mathbb{R}$*, and its characteristic function, as a function of* v^2*, is a GGC transform with Thorin measure* $\frac{1}{2}\sum_{k\geq 0}\delta_{((k+1/2)\pi)^2}$*, since* $\operatorname{sech}(v)^{1/2} = \prod_k (1+v^2/((k+\frac{1}{2})\pi)^2)^{-1/2}$*;*

(ii) the sawtooth measure has $K_{U_\lambda}(x) = \dfrac{\lambda}{\pi x} - \dfrac{1}{\sinh(\pi x/\lambda)}$*;*

(iii) under RH, the Thorin measure $U = \sum_{\gamma>0}\delta_{\gamma^2}$ *of* R *has* $K_U(x) = 2\sum_{\gamma>0}e^{-\gamma x}$*, Cramér's function [57, Chapter I], and* $\xi(\frac{1}{2})/\xi(\frac{1}{2}+u)$ *is the characteristic function of a sum of independent Laplace variables of scales* $1/\gamma$*.*

Proposition 6.2.9. *For every* $x > 0$ *the series* $\sum_N B_N K_{U_{\log N}}(x)$ *diverges.*

Proof. B_N vanishes unless N is a prime power. Grouping the powers of a fixed p, $\sum_k B_{p^k}/k = \sum_{j,m} d_m\, p^{-j/2}/(j^2 m) = \frac{12}{\pi^2}p^{-1/2} + O(p^{-1})$, using $\sum_m d_m/m = 1/\eta(2) = 12/\pi^2$. Since $K_{U_\lambda}(x) = \frac{\pi x}{6\lambda} + O((x/\lambda)^3)$, the group for p equals $\frac{2x}{\pi}\frac{p^{-1/2}}{\log p}(1+o(1))$, and $\sum_p p^{-1/2}/\log p = \infty$. The grouping changes the order of summation, but in the order of increasing N the partial sums are dominated by the prime terms $B_p K_{U_{\log p}}(x) \sim \frac{\pi x}{6}\frac{p^{-1/2}}{\log p}$, so the series diverges in that order as well. □

The inverse direction fails to sum over primes for the reason that Lemma 6.2.5 fails at the centre: without Gaussian damping the weights $p^{-1/2}$ are not summable. The Poisson kernel $e^{-\sqrt{t}\,x}$ is the stable $\frac{1}{2}$ average of the heat kernel, $e^{-\sqrt{t}\,x} =$

$\int_0^\infty e^{-ty}\frac{x}{2\sqrt{\pi}}y^{-3/2}e^{-x^2/4y}dy$, and the average extends over $y \gtrsim (\log N)^2$, where the block heat trace $h_{\log N}(y)$ has lost its Gaussian factor $e^{-(\log N)^2/4y}$ and is close to $-\log N/(4\sqrt{\pi y})$; each prime then contributes of order $p^{-1/2}/\log p$, as in the proof of Proposition 6.2.9. Under the heat kernel alone each block heat trace is a theta function with Gaussian decay in $\log N$, and the series converges (Proposition 6.1.13, Theorem 6.2.3).

6.2.3 The Meixner law and its orthogonal polynomials

The Meixner hypergeometric law of Section 6.2.1, whose density p is proportional to $|G|^2$ on the critical line, is Brownian motion run for an independent GGC time. The score $-p'/p$ of the law and the derivative of the Riemann Siegel theta function are the imaginary and real parts of the boundary values of one Bernstein function, and the positivity of the Lévy measure of that function makes the tilted reciprocal gamma factor a GGC. The orthogonal polynomials of the law are the Meixner Pollaczek polynomials. They give the local Riemann Hypothesis at the archimedean place and two expansions of ζ on the critical line, whose zeros we compare with the ordinates. Finally the law splits the autocorrelation of the Pólya kernel into a Meixner factor and an arithmetic factor.

Let Y have the density p of (6.2.3), so that $\mathbb{E}e^{ivY} = \operatorname{sech}(v)^{1/2}$.

Proposition 6.2.10. *(i) $Y \stackrel{d}{=} B(T)$, where B is a standard Brownian motion and T is independent with $\mathbb{E}e^{-\lambda T} = \operatorname{sech}(\sqrt{2\lambda})^{1/2}$, the variable $C_{1/2}$ of Biane, Pitman and Yor [28], so that Y is the value at time $\frac{1}{2}$ of the process $\widehat{C}_t = B(C_t)$ of Pitman and Yor [194, (1) and (4)]. The time $T = \sum_{k\ge 0}\frac{8\,\Gamma_k}{(2k+1)^2\pi^2}$, with Γ_k independent gamma variables of shape $\frac{1}{2}$ and unit scale, is a GGC with Thorin measure $\frac{1}{2}\sum_k \delta_{(k+1/2)^2\pi^2/2}$.*

(ii) Y is infinitely divisible with Lévy density $\dfrac{1}{4|x|\sinh(\pi|x|/2)}$, one half of the Lévy density $1/(2|x|\sinh(\pi|x|/2))$ of the process $\widehat{C}$ [194, Table 5]; Y has a Meixner law [236, 86]. As $|\tau|\to\infty$, $p(\tau)\sim \pi^{-1/2}|\tau|^{-1/2}e^{-\pi|\tau|/2}$, matching the singularities of $\operatorname{sech}(v)^{1/2}$ at $v = \pm i\pi/2$.

(iii) p is not a Pólya frequency function of infinite order.

Proof. (i) $\mathbb{E}e^{ivB(T)} = \mathbb{E}e^{-v^2T/2} = \operatorname{sech}(v)^{1/2}$, and $\operatorname{sech}(\sqrt{2\lambda})^{1/2} = \prod_k(1+2\lambda/((k+\frac{1}{2})\pi)^2)^{-1/2}$. (ii) is Proposition 6.2.8(i); the tail follows from Stirling, $|\Gamma(\frac{1}{4}+ix)|^2 \sim 2\pi|x|^{-1/2}e^{-\pi|x|}$ with $x = \tau/2$. (iii) By Schoenberg's theorem [235, Theorem 1] the bilateral Laplace transform of a PF_∞ density is the reciprocal of an entire function of Laguerre Pólya class, so the reciprocal of its characteristic function is entire. Here $1/\mathbb{E}e^{ivY} = \cosh(v)^{1/2}$, which has branch points at the simple zeros $v = i(k+\frac{1}{2})\pi$ of cosh and is not entire. □

The time T of (i), which is not the clock variable of Section 1.2, is twice the Meixner time of Table 1.9. The exponent $\frac{1}{2}$ in $\operatorname{sech}(v)^{1/2}$ is the Meixner parameter $2a = \frac{1}{2}$, forced by the factor $\Gamma(z/2)$ of ξ at $z = \frac{1}{2}$. For $2a$ a positive integer the Meixner law is PF_∞; the zeta function sits at a value where it is not, although its characteristic function is still a GGC transform in v^2. This matters because Pólya's approach to RH through the kernel of Ξ is an approach through total positivity, and the archimedean factor alone already fails its strongest form.

Define the measure ν on $(0,\infty)$ by

$$\nu(dy) = \frac{e^{y/2}}{\sinh y}\,dy = 2\sum_{k\ge 0} e^{-c_k y}\,dy,$$

with $c_k = 2k + \frac{1}{2}$ as in Theorem 6.2.1. Its density is completely monotone, the Laplace transform of $2\sum_k \delta_{c_k}$, whose atoms sit at the points c_k of the Weierstrass product of G.

PROPOSITION 6.2.11. *The function* $\Phi_\Gamma(\lambda) = \psi(\frac{1}{4}+\frac{\lambda}{2}) - \psi(\frac{1}{4}) = \int_0^\infty (1-e^{-\lambda y})\,\nu(dy)$ *is a Bernstein function. Its boundary values on the imaginary axis are*

$$\operatorname{Re}\Phi_\Gamma(i\tau) = 2\big(\theta'(\tau) - \theta'(0)\big) = \int_0^\infty (1-\cos\tau y)\,\nu(dy),$$

$$\operatorname{Im}\Phi_\Gamma(i\tau) = -\frac{d}{d\tau}\log p(\tau) = \int_0^\infty \sin(\tau y)\,\nu(dy).$$

Consequently $2\theta(\tau) + \tau\log\pi$ *and* $\log p(\tau)$ *are, up to constants, the imaginary and real parts of* $2\log\Gamma(\frac{1}{4}+\frac{i\tau}{2})$*, the boundary value of a function analytic in the right half plane. So* $2\theta(\tau)+\tau\log\pi$*, which is* 2π *times the smooth part* θ/π *of the zero counting function plus a linear term, is the harmonic conjugate of the Meixner log density. The smooth heat trace is the heat trace of* ν*:*

$$W_0(x) = \frac{\theta'(0)}{2\sqrt{\pi x}} + \frac{1}{4\sqrt{\pi x}}\int_0^\infty \big(1-e^{-y^2/4x}\big)\,\nu(dy),$$

which is Lemma 6.1.12 after the substitution $u = 2y$*; the function* F *of that lemma decreases because* ν *is positive.*

Proof. Substitute $u = 2y$ in $\psi(z) = \int_0^\infty (e^{-u}/u - e^{-zu}/(1-e^{-u}))du$ with $z = \frac{1}{4}+\frac{\lambda}{2}$; note $2e^{-y/2}/(1-e^{-2y}) = e^{y/2}/\sinh y$. Taking $\lambda = i\tau$ gives the boundary values, with $\theta' = \frac{1}{2}\operatorname{Re}\psi(\frac{1}{4}+\frac{i\tau}{2}) - \frac{1}{2}\log\pi$ and $\frac{d}{d\tau}\log|\Gamma(\frac{1}{4}+\frac{i\tau}{2})|^2 = -\operatorname{Im}\psi(\frac{1}{4}+\frac{i\tau}{2})$. For W_0 integrate $\theta'(\tau) = \theta'(0) + \frac{1}{2}\int(1-\cos\tau y)\nu(dy)$ against $\frac{1}{\pi}e^{-x\tau^2}$. □

Figure 6.3 shows the density and $\Phi_\Gamma(i\tau)$, evaluated from Γ and ψ alone, so the increment $\theta'(\tau) - \theta'(0)$, which is π times the increase of the smooth zero density θ'/π, appears with no zero as input.

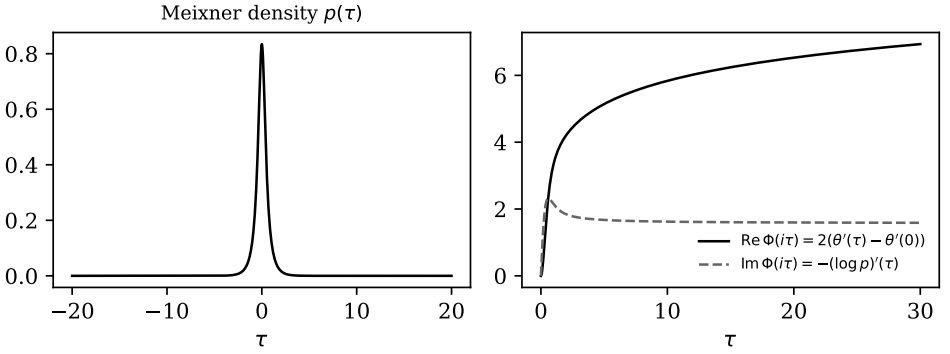


Figure 6.3: Left: the Meixner density $p(\tau) \propto |\Gamma(\frac{1}{4} + \frac{i\tau}{2})|^2$. Right: the real and imaginary parts of the Bernstein function $\Phi_\Gamma(i\tau)$ of Proposition 6.2.11, written $\Phi(i\tau)$ in the legend. The real part is $2(\theta'(\tau) - \theta'(0))$ and grows like $\log \tau$; the imaginary part is the Meixner score $-\frac{d}{d\tau}\log p$ and tends to $\pi/2$, the exponential tail rate of p.

THEOREM 6.2.12. *Let* $c = -\theta'(0) = \frac{1}{2}(\log \pi - \psi(\frac{1}{4})) \approx 2.68609$ *be the constant of* (6.1.6)*. Then*

$$\frac{e^{-c\sqrt{s}}}{G(s)} = \frac{\Gamma(\frac{1}{4})\,\pi^{\sqrt{s}/2}\,e^{-c\sqrt{s}}}{\Gamma(\frac{1}{4} + \frac{\sqrt{s}}{2})} = \exp\Big(-\int_0^\infty \log\Big(1 + \frac{s}{t}\Big) U_\Gamma(t)\,dt\Big),$$

with

$$U_\Gamma(t) = \frac{\theta'(\sqrt{t}) - \theta'(0)}{2\pi\sqrt{t}} = \frac{1}{2\pi\sqrt{t}} \sum_{k\geq 0} \frac{t}{c_k\,(c_k^2 + t)} \geq 0.$$

So $e^{-c\sqrt{s}}/G(s)$ *is the Laplace transform of a GGC, and its Thorin distribution function is* $(\theta(\sqrt{t}) - \theta'(0)\sqrt{t})/\pi$*, a convex increasing function of* $\sqrt{t}$*.*

Proof. With $f(u) = \Gamma(\frac{1}{4} + \frac{u}{2})\pi^{-u/2}$ we have $f'(0)/f(0) = \frac{1}{2}\psi(\frac{1}{4}) - \frac{1}{2}\log\pi = \theta'(0) = -c$, and $\frac{d}{du}[\log f(u) - uf'(0)/f(0)] = \frac{1}{2}\Phi_\Gamma(u) = \frac{1}{2}\int(1 - e^{-uy})\nu(dy)$. Integrating from 0, $\log f(u)/f(0) - uf'(0)/f(0) = \int(e^{-uy} - 1 + uy)\frac{\tilde\mu(dy)}{y}$ with $\tilde\mu = \frac{1}{2}\nu \geq 0$. Lemma 6.2.5 gives $U_\Gamma(t) = \frac{1}{2\pi\sqrt{t}}\int 2\sin^2(\frac{y\sqrt{t}}{2})\frac{1}{2}\nu(dy) = \frac{1}{2\pi\sqrt{t}}\sum_k \int_0^\infty (1 - \cos y\sqrt{t})e^{-c_k y}dy$, which is the series stated, and Proposition 6.2.11 identifies it with θ'. □

Multiplying $1/G$ by $e^{-c\sqrt{s}}$ adds the Thorin density $\frac{c}{2\pi\sqrt{t}}$, and $U_\Gamma = -U_G + \frac{c}{2\pi\sqrt{t}}$, which is the identity $c - b = \sum_k(\frac{1}{c_k} - \frac{1}{2k+2})$, has no negative part. This is the positive counterpart of Theorem 6.2.1, where the gamma factor appeared as a renormalised product with a signed Thorin density: the archimedean factor

becomes positive after one explicit tilt, which leaves the zeros as the only possible source of negativity in the floor of Section 6.2.4.

6.2.3.1 Meixner Pollaczek polynomials and the local Riemann Hypothesis

The Meixner law is also the orthogonality measure of the Meixner Pollaczek polynomials, and through them it controls the zeros of the Mellin transforms of the even Hermite functions. Let $\psi_{2m}(x) = H_{2m}(\sqrt{2\pi}\,x)e^{-\pi x^2}$ be the even Hermite functions, eigenfunctions of the Fourier transform $\hat{f}(y) = \int f(x)e^{-2\pi ixy}dx$ with eigenvalue $(-1)^m$. Define the Mellin transform $\mathcal{M}f(s) = \int_0^\infty x^{s-1}f(x)dx$ and $\Gamma_{\mathbb{R}}(s) = \pi^{-s/2}\Gamma(s/2)$. From here to the end of Section 6.2.3, s is the Mellin variable, the argument of ζ, and not the variable of the clock.

THEOREM 6.2.13. *For each $m \geq 0$, $Q_m(s) = \mathcal{M}\psi_{2m}(s)/\Gamma_{\mathbb{R}}(s)$ is a polynomial of degree m satisfying $Q_m(1-s) = (-1)^m Q_m(s)$, and*

$$Q_m(\tfrac{1}{2} + i\tau) = \kappa_m\, P_m^{(1/4)}\big(\tfrac{\tau}{2}; \tfrac{\pi}{2}\big),$$

where $P_m^{(\lambda)}(x;\phi)$ are the Meixner Pollaczek polynomials and $\kappa_m \neq 0$. These are the orthogonal polynomials of the Meixner law p. Consequently all zeros of Q_m lie on $\operatorname{Re} s = \frac{1}{2}$ and are simple: the Mellin transform of every even Hermite function satisfies the Riemann Hypothesis.

Proof. Expanding $H_{2m}(\sqrt{2\pi}\,x) = \sum_j h_j x^{2j}$ gives

$$\mathcal{M}\psi_{2m}(s) = \frac{1}{2}\Gamma_{\mathbb{R}}(s)\sum_j h_j \pi^{-j}(s/2)_j$$

with Pochhammer symbols, a polynomial of degree m times $\Gamma_{\mathbb{R}}(s)$. Tate's local functional equation [249, Theorem 2.4.1 and Section 2.5], $\mathcal{M}f(s)/\Gamma_{\mathbb{R}}(s) = \mathcal{M}\hat{f}(1-s)/\Gamma_{\mathbb{R}}(1-s)$ for even f, applied to the eigenfunction ψ_{2m} gives $Q_m(1-s) = (-1)^m Q_m(s)$, so $i^{-m}Q_m(\frac{1}{2}+i\tau)$ is a real polynomial $R_m(\tau)$. By Mellin Plancherel,

$$\int_0^\infty \psi_{2m}\psi_{2n}\,dx = \frac{1}{2\pi}\int_{\mathbb{R}} Q_m\overline{Q_n}\,|\Gamma_{\mathbb{R}}(\tfrac{1}{2}+i\tau)|^2 d\tau = \sqrt{2}\,i^{m-n}\int_{\mathbb{R}} R_m(\tau)R_n(\tau)\,p(\tau)\,d\tau,$$

since $|\Gamma_{\mathbb{R}}(\frac{1}{2}+i\tau)|^2 = \pi^{-1/2}|\Gamma(\frac{1}{4}+\frac{i\tau}{2})|^2 = 2^{3/2}\pi\,p(\tau)$. The left side vanishes for $m \neq n$, so (R_m) is the orthogonal polynomial sequence of p, which with $x = \tau/2$ is the Meixner Pollaczek family with $\lambda = \frac{1}{4}$, $\phi = \frac{\pi}{2}$ [126, Section 9.7]. Orthogonal polynomials of a measure with infinite support have real simple zeros. □

The conclusion is the archimedean local Riemann Hypothesis of Bump, Choi, Kurlberg and Vaaler [46], seen through the Meixner law: the squared modulus of the archimedean factor on the line, $|\Gamma_{\mathbb{R}}(\frac{1}{2}+i\tau)|^2 = 2^{3/2}\pi\,p(\tau)$, is the orthogonality measure, and orthogonality forces the zeros of these Mellin transforms onto the

line. We make no priority claim for this route; Meixner Pollaczek expansions of Ξ in the literature are reviewed in Section 6.2.3.2. We checked numerically that the roots of Q_m have real part $\frac{1}{2}$ to 50 digits for $m \le 8$, and that $Q_m(\frac{1}{2} + i\tau)/P_m^{(1/4)}(\tau/2;\pi/2)$ is constant, with $\kappa_2 = -16$, $\kappa_3 = -192i$, $\kappa_5 = 61440i$.

6.2.3.2 Zeta in Meixner coordinates

Expansions of ζ and Ξ in these polynomials have been studied before. Kuznetsov [131, 132] expands Ξ and Dirichlet L functions in Meixner Pollaczek polynomials; Inoue [106] studies the symmetric expansion of the completed zeta function, the multiplicative structure of its coefficients and the zeros of the approximating polynomials; Romik [225] expands Ξ itself in the symmetric Meixner Pollaczek basis $P_n^{(3/4)}(x;\frac{\pi}{2})$, together with the Hermite and continuous Hahn bases, and derives coefficient asymptotics; and Candelpergher [48] gives related polynomial expansions whose polynomials have their roots on the line. The expansions below first divide ξ by the archimedean factor $\Gamma_{\mathbb{R}}$, or by its modulus, which leads to the parameter $\lambda = \frac{1}{4}$ of the Meixner law. They are included for comparison with the Thorin picture and are not claimed as new.

The map $\mathcal{U}f(\tau) = \mathcal{M}f(\frac{1}{2}+i\tau)/\Gamma_{\mathbb{R}}(\frac{1}{2}+i\tau)$ is, by the Mellin Plancherel identity in the proof of Theorem 6.2.13, a unitary map from $L^2(0,\infty)$ onto $L^2(\sqrt{2}\,p(\tau)d\tau)$, carrying the even Hermite functions to the Meixner Pollaczek polynomials. Dilations become multipliers, $\mathcal{U}[f(n\,\cdot)] = n^{-1/2-i\tau}\mathcal{U}f$. The function $g(x) = (2\pi^2x^4 - 3\pi x^2)e^{-\pi x^2} = \frac{1}{2}(xD)(xD+1)e^{-\pi x^2}$, with $D = d/dx$, has $\mathcal{U}g = \frac{1}{4}s(s-1)$, a Meixner Pollaczek combination of degree two, and its dilation sum Ω is, up to the factor $\frac{1}{2}x^{-1/2}$, the Pólya kernel Φ of Section 1.2 in the variable $x = e^u$:

$$\Omega(x) = \sum_{n\ge1} g(nx) = \tfrac{1}{2}x^{-1/2}\Phi(\log x), \qquad \mathcal{U}\Omega = \tfrac{1}{4}s(s-1)\zeta(s)\big|_{s=\frac{1}{2}+i\tau}.$$

So in Meixner coordinates $\frac{1}{4}s(s-1)\zeta(s)$ on the critical line is the image of a function in $L^2(0,\infty)$ and has a convergent Meixner Pollaczek expansion

$$\tfrac{1}{4}s(s-1)\zeta(s) = \sum_{m\ge0} a_{2m}\,Q_m(s), \qquad a_{2m} = \frac{\langle\Omega,\psi_{2m}\rangle}{\langle\psi_{2m},\psi_{2m}\rangle},$$

in $L^2(p)$. Each Q_m satisfies the local Riemann Hypothesis, and ζ is a superposition of them. We ask whether the property survives in the partial sums.

The coefficients decay faster than geometrically (Table 6.3), but the polynomials grow, and the expansion converges slowly as $|\tau|$ grows: at $s = \frac{1}{2} + 5i$ the partial sums with $N = 8, 16, 24$ give -4.4508, -4.4279, -4.4291 for the real part against the exact -4.4302, while near the first zero $\frac{1}{2} + 14.1347i$ they are still far from zero at $N = 24$. The partial sums $\sum_{m\le N} a_{2m}Q_m(s)$ mix polynomials of both parities under $s \mapsto 1-s$, so they have no reason to vanish on the critical

m	a_{2m}	m	a_{2m}	m	a_{2m}
0	4.265×10^{-2}	4	-1.796×10^{-5}	8	8.904×10^{-12}
1	8.707×10^{-2}	5	2.056×10^{-7}	12	-4.088×10^{-19}
2	1.916×10^{-2}	6	5.080×10^{-9}	16	-3.882×10^{-26}
3	6.391×10^{-4}	7	-3.212×10^{-10}	24	5.961×10^{-42}

Table 6.3: Meixner Pollaczek coefficients of $\frac{1}{4}s(s-1)\zeta(s)$, equivalently Hermite coefficients of $\Omega(x) = \sum_{n\geq 1} g(nx) = \frac{1}{2}x^{-1/2}\Phi(\log x)$.

line, and for $N = 12$ and $N = 24$ they do not: all their zeros lie off the line, the nearest to the first zeta zero being $1.92 + 14.13i$ for $N = 12$ and $0.62 + 14.20i$ for $N = 24$. The asymmetric expansion therefore does not test whether the local Riemann Hypothesis survives the superposition; the symmetric expansion does.

The symmetric choice is the real even function $F(\tau) = (\tau^2 + \frac{1}{4})Z(\tau)$, proportional to $\Xi(\tau)/|\Gamma(\frac{1}{4} + \frac{i\tau}{2})|$, expanded in the even Meixner Pollaczek polynomials $P_{2m}^{(1/4)}(\tau/2; \frac{\pi}{2})$, normalised as in [126, Section 9.7]. Its truncations, the orthogonal projections of F in $L^2(p)$ onto the even polynomials of degree at most $2M$, are real even polynomials, and their real zeros can be compared with the ordinates. The first coefficients are -0.642, -1.117, -0.284, -0.027, 0.071; from $m = 3$ on they alternate in sign, and from $m = 4$ on their modulus decreases slowly, to about 0.02 at $m = 10$ and 0.01 at $m = 20$. The slow decay reflects the polynomial growth of F against the exponential decay of p: the expansion converges in $L^2(p)$, but its truncations approximate F poorly at the ordinates. The truncation of degree 40 has real zeros at 14.32, 17.48, 19.98, 23.12, 26.47, 30.09, 34.02, 38.29 in $[0, 40]$, against the ordinates 14.13, 21.02, 25.01, 30.42, 32.94, 37.59. The real zeros of the truncations do not track the ordinates: the zero nearest 14.13 is 14.50, 14.32, 14.16, 13.73, 13.54, 14.84 for the degrees 38, 40, 42, 50, 60, 80, and there are eight to ten real zeros in $[0, 40]$ against six ordinates. Neither expansion suggests a route from the local Riemann Hypothesis to the global one; such a route must use the arithmetic of the full dilation sum, not only the orthogonality of its archimedean pieces.

The factorisation of Ξ^2 in Section 6.2.1 induces a factorisation of the autocorrelation $\Phi * \Phi$ of the Pólya kernel, whose Fourier transform is Ξ^2. It holds exactly for $|\zeta|^2$ and formally for the Dirichlet series, where the arithmetic factor is a positive measure.

PROPOSITION 6.2.14. *The integral* $\mathcal{C}(v) = \int p(\tau)\,|\zeta(\frac{1}{2} + i\tau)|^2 e^{iv\tau}\,d\tau$ *converges absolutely, and*

$$(\Phi * \Phi)(v) = 2^{-3/2}\Big(\frac{d^2}{dv^2} - \frac{1}{4}\Big)^2 \mathcal{C}(v).$$

Formally $\mathcal{C} = \operatorname{sech}(\cdot)^{1/2} * \mathcal{A}$, *with*

$$\mathcal{A} = \sum_{m,n\geq 1} \frac{\delta_{\log(m/n)}}{\sqrt{mn}},$$

where $\operatorname{sech}(v)^{1/2}$ *is the characteristic function of the Meixner law and* $\mathcal{A}$, *whose Fourier transform is formally* $|\zeta(\frac{1}{2}+i\tau)|^2$, *is the autocorrelation of the measure* $\sum_n n^{-1/2}\delta_{\log n}$. *The measure* $\mathcal{A}$ *is positive and positive definite, and the series for* $\mathcal{C}$ *diverges: truncated at* $m,n \leq X$ *it equals* $X\int \operatorname{sech}(v-r)^{1/2}e^{-|r|/2}\,dr + O(X^{1/2})$. *This growth comes from the pole of* ζ, *and the factor* $(\frac{d^2}{dv^2}-\frac{1}{4})^2$ *does not remove it, since* $(\frac{d^2}{dv^2}-\frac{1}{4})e^{-|v|/2} = -\delta_0$.

Proof. Since Φ is even, $\int(\Phi*\Phi)(v)e^{i\tau v}dv = \Xi(\tau)^2 = \frac{1}{4}(\tau^2+\frac{1}{4})^2\pi^{-1/2}|\Gamma(\frac{1}{4}+\frac{i\tau}{2})|^2|\zeta(\frac{1}{2}+i\tau)|^2$, and $\pi^{-1/2}|\Gamma|^2 = 2^{3/2}\pi\,p$. Fourier inversion divides by 2π, the factor $(\tau^2+\frac{1}{4})^2$ becomes $(\frac{d^2}{dv^2}-\frac{1}{4})^2$, and the inverse transform of the even function $p|\zeta|^2$ is $\mathcal{C}/(2\pi)$; the constant is $\frac{1}{4}\,2^{3/2}\pi/(2\pi) = 2^{-3/2}$. The integral $\mathcal{C}$ converges absolutely because $p(\tau) \sim \pi^{-1/2}|\tau|^{-1/2}e^{-\pi|\tau|/2}$ (Proposition 6.2.10) and $\zeta(\frac{1}{2}+i\tau)$ grows at most polynomially. Formally $|\zeta|^2 = |\sum n^{-1/2-i\tau}|^2 = \sum_{m,n}(mn)^{-1/2}e^{i\tau\log(m/n)}$, and $\int p(\tau)e^{i\tau(v+r)}d\tau = \operatorname{sech}(v+r)^{1/2}$, which gives $\mathcal{C} = \operatorname{sech}^{1/2} * \mathcal{A}$. For the truncation put $D_X(\tau) = \sum_{n\leq X} n^{-1/2-i\tau}$; the truncated series is exactly $\int p(\tau)|D_X(\tau)|^2e^{iv\tau}d\tau$. For $|\tau|\leq X$, $D_X(\tau) = \zeta(\frac{1}{2}+i\tau) + X^{1/2-i\tau}/(\frac{1}{2}-i\tau) + O(X^{-1/2})$ [255, Theorem 4.11], and for $|\tau| > X$ the bound $|D_X|^2 \leq 4X$ meets the exponential decay of p. Since $\int e^{-|r|/2}e^{i\tau r}\,dr = (\frac{1}{4}+\tau^2)^{-1}$, the squared pole term contributes $X\int p(\tau)(\frac{1}{4}+\tau^2)^{-1}e^{iv\tau}d\tau = X\int \operatorname{sech}(v-r)^{1/2}e^{-|r|/2}\,dr$, up to an exponentially small error from $|\tau|>X$; the remaining terms contribute $O(X^{1/2})$ uniformly in v, the largest being the cross term of ζ with the pole term. Finally $(\frac{d^2}{dv^2}-\frac{1}{4})^2$ turns $X\int \operatorname{sech}(v-r)^{1/2}e^{-|r|/2}\,dr$ into $-X(\frac{d^2}{dv^2}-\frac{1}{4})\operatorname{sech}(v)^{1/2}$, which is not zero. □

So the Pólya autocorrelation is the Meixner characteristic function smeared over the multiplicative lattice $\{\log(m/n)\}$ with the positive weights $(mn)^{-1/2}$, and then acted on by the pole operator $(\frac{d^2}{dv^2}-\frac{1}{4})^2$. Both pieces are positive, the Meixner factor being a GGC transform in v^2 and $\mathcal{A}$ a positive measure, but convolution with a positive measure does not preserve the GGC property, so the GGC structure of the Meixner factor does not pass to the product. The lattice sum also needs a counterterm, the pole term of the approximate functional equation, which comes from the same pole of ζ that produced the atom at $-\frac{1}{4}$ in Proposition 6.1.13. Problem 7.6 of Section 7.2 asks for the regularised form of $\operatorname{sech}(\cdot)^{1/2} * \mathcal{A}$ and whether it has a Thorin or Lee and Yang structure in v.

6.2.4 An infinitely divisible floor for the reciprocal Xi function

Theorem 6.2.12 makes the archimedean factor positive after one stable tilt. Multiplying $\varphi = RG$ by the GGC transform $e^{-c\sqrt{s}}/G$ gives $e^{-c\sqrt{s}}R$, which we call the floor of the reciprocal Xi function. In the floor the zeros are the only possible source of negativity, and the floor is a GGC exactly when RH holds. Its infinite divisibility, as proved here, rests on the positivity of w, which is proved in Section 7.1.

Theorem 6.2.15. *Unconditionally, $e^{-c\sqrt{s}}\,\xi(\frac{1}{2})/\xi(\frac{1}{2}+\sqrt{s})$ is the Laplace transform of an infinitely divisible law on $(0,\infty)$ with Lévy density*

$$\frac{W(x)+c/(2\sqrt{\pi x})}{x}, \qquad W(x)=\tfrac{1}{2}\sum_{\rho} e^{x(\rho-\frac{1}{2})^2},$$

and $W(x)+c/(2\sqrt{\pi x})>0$ for all $x>0$. The following are equivalent: (i) RH; (ii) this law is a GGC; (iii) $W(x)+c/(2\sqrt{\pi x})$ is completely monotone; (iv) W is completely monotone. The proof uses $w>0$, which is Theorem 7.1.1 of Section 7.1 and rests on the verification of RH to height 50, an explicit bound for $N(T)$ [92] and the enclosures E1 to E3.

Proof. $e^{-c\sqrt{s}}R=\varphi\cdot(e^{-c\sqrt{s}}/G)$ is the product of the transform of an infinitely divisible law (Theorem 7.1.1) and a GGC (Theorem 6.2.12), so it is infinitely divisible, and its Lévy density is the sum $\{w(x)+\int e^{-xt}U_\Gamma(t)dt\}/x$. By Proposition 6.2.2 and Theorem 6.2.12, $\int e^{-xt}U_\Gamma(t)dt = W_0(x)+\frac{c}{2\sqrt{\pi x}}$, and $w+W_0=W$ by (6.1.7). Positivity holds because $w>0$ and $U_\Gamma\geq 0$. For the equivalences: GGC means the Lévy density times x is completely monotone, so (ii)$\Leftrightarrow$(iii). Under RH $W=\sum e^{-x\gamma^2}$ is completely monotone, so (i)$\Rightarrow$(iv)$\Rightarrow$(iii). If (ii) holds, the transform is analytic on $\mathbb{C}\setminus(-\infty,0]$ and agrees with $e^{-c\sqrt{s}}R$ on $(0,\infty)$, so by uniqueness of analytic continuation R has no poles in $\mathbb{C}\setminus(-\infty,0]$; since R has no zeros and G is analytic and zero free on $\mathbb{C}\setminus(-\infty,0]$, $\varphi=RG$ is analytic and zero free there, and Proposition 6.1.9 gives RH. □

Infinite divisibility alone is not new: by Theorem 5.1.1 [200, Theorem A], $W>0$ and $W'<0$ on $(0,\infty)$, so R itself is infinitely divisible and self decomposable, and so is its product with the stable factor $e^{-c\sqrt{s}}$; Section 11.1 draws the consequences of self decomposability. What the floor adds is the splitting of its canonical function $k=W+c/(2\sqrt{\pi x})$ as $k=w+\int e^{-xt}U_\Gamma(t)\,dt$, an arithmetic part plus a completely monotone archimedean part contributed by the Meixner law (Theorem 6.2.12). The tilt, with the constant $c=\zeta'(\frac{1}{2})/\zeta(\frac{1}{2})$, removes the negative part of the Thorin density of $1/G$, which lies at $t<\tau_0^2$; in the Lévy picture it cancels the negative tail of W_0, since $W_0+c/(2\sqrt{\pi x})=\int e^{-xt}U_\Gamma(t)\,dt>0$. In law the floor adds to X_R an independent first passage time L_c of $\sqrt{2}B$ to the level c. At the basepoints $\alpha>\frac{1}{2}$ the clocks are the centre clock tilted and subordinated

by a drift and a stable one half process (Section 9.2), and at the basepoint one the stable factor $e^{-b_1\sqrt{s}}$, with $b_1 = \sum_\rho \rho^{-1}$, plays the role of the tilt (Section 9.1). Everything that could fail therefore lies in w, and RH is the upgrade of k from positivity, the lowest rung of the ladder of Section 5.1, to complete monotonicity, its top.

6.2.4.1 The sine squared lemma at the floor

Theorem 6.2.15 describes the floor through its Lévy density in the variable s. The sine squared lemma works one level earlier, with the primal Lévy measure in the variable $u = \sqrt{s}$. To apply it to the floor we identify the primal function and its Lévy measure, and then follow the two stages of the proof of Lemma 6.2.5: a subordination stage that produces the Lévy density in s, and a Bernstein stage that produces the Thorin measure. As in that proof, from here to the end of the section x is the variable of the primal Lévy measure, and the time of the heat trace is written y.

The floor is $e^{-c\sqrt{s}}R(s) = f(0)/f(\sqrt{s})$ with primal function $f(u) = e^{cu}\xi(\frac{1}{2}+u)$, and $f'(0)/f(0) = c$ because $\xi'(\frac{1}{2}) = 0$. The tilt contributes the linear term cu to $\log f$, which cancels in $\log f(u)/f(0) - uf'(0)/f(0) = \log \xi(\frac{1}{2}+u)/\xi(\frac{1}{2})$, so the primal Lévy measure of the floor is that of ξ at the centre. Collecting the three factors of $\xi(\frac{1}{2}+u) = \frac{1}{2}(u^2-\frac{1}{4})\,\pi^{-1/4} f_\Gamma(u)\,\zeta(\frac{1}{2}+u)$, where $f_\Gamma(u) = \pi^{-u/2}\Gamma(\frac{1}{4}+\frac{u}{2})$, gives formally

$$\tilde{\mu}_\xi(dx) = -2\cosh\big(\tfrac{x}{2}\big)\,dx + \sum_{k\geq 0} e^{-c_k x}\,dx + \sum_{n\geq 2} \frac{\Lambda(n)}{\sqrt{n}}\,\delta_{\log n}(dx). \tag{6.2.5}$$

The first term comes from $u^2 - \frac{1}{4} = (u+\frac{1}{2})(u-\frac{1}{2})$, each linear factor $1 + u/a$ contributing $-e^{-ax}dx$ by Frullani (with $a = -\frac{1}{2}$ by continuation); the second is $\frac{1}{2}\nu$ (Proposition 6.2.11 and the proof of Theorem 6.2.12); the third is the prime measure of Proposition 6.2.6 at $\beta = \frac{1}{2}$. By the explicit formula, as distributions on $x > 0$,

$$\tilde{\mu}_\xi(dx) = -\sum_\rho e^{(\rho-\frac{1}{2})x}\,dx, \qquad \text{which under RH is} \qquad -2\sum_{\gamma>0}\cos(\gamma x)\,dx. \tag{6.2.6}$$

Proposition 6.2.16. *Apply the two stages of Lemma 6.2.5 to $\tilde{\mu}_\xi$, regularising $\int_0^\infty \cos(\gamma x)dx = 0$ and the growing term of* (6.2.5) *by analytic continuation.*

(i) Subordination stage: $(4\pi y)^{-1/2}\int_0^\infty (1 - e^{-x^2/4y})\,\tilde{\mu}_\xi(dx) = W(y)$, the heat trace over the zeros. With the tilt, the canonical function of the floor is $k(y) = W(y) + c/(2\sqrt{\pi y})$, as in Theorem 6.2.15, and its positivity is proved.

(ii) Bernstein stage: $U(t) = \frac{1}{2\pi\sqrt{t}}\int_0^\infty 2\sin^2(\frac{x\sqrt{t}}{2})\,\tilde{\mu}_\xi(dx)$ equals $\sum_{\gamma>0}\delta_{\gamma^2}$ under RH, and the Thorin measure of the floor is $U + \frac{c}{2\pi\sqrt{t}}dt$.

The two stages are linked by $W(y) = \int_0^\infty e^{-yt}U(t)\,dt$. *The floor is a GGC exactly when* $U + \frac{c}{2\pi\sqrt{t}}\,dt \geq 0$, *that is, when* k *is completely monotone; by Theorem 6.2.15 this happens exactly under RH, and then* $U = \sum_{\gamma>0}\delta_{\gamma^2} \geq 0$.

Proof. (i) For $\tilde{\mu} = -2\cos(\gamma x)dx$,

$$\int_0^\infty (1 - e^{-x^2/4y})(-2\cos\gamma x)\,dx = 2\int_0^\infty e^{-x^2/4y}\cos\gamma x\,dx = \sqrt{4\pi y}\,e^{-\gamma^2 y}.$$

Without RH the same computation with $e^{(\rho-\frac{1}{2})x}$ gives $e^{y(\rho-\frac{1}{2})^2}$, so the result is W in general. (ii) $\int_0^\infty (1-\cos x\sqrt{t})(-2\cos\gamma x)\,dx = \pi\,\delta(\sqrt{t}-\gamma)$, and $\frac{1}{2\pi\sqrt{t}}\pi\,\delta(\sqrt{t}-\gamma) = \delta(t-\gamma^2)$. The link is the identity $(1-e^{-x^2/4y})/\sqrt{y} = \int_0^\infty e^{-yt}\{1-\cos x\sqrt{t}\}\frac{dt}{\sqrt{\pi t}}$ used in the proof of Lemma 6.2.5. □

The lemma preserves positivity, so if $\tilde{\mu}_\xi$ were a positive Lévy measure the floor would be a GGC and RH would follow. But $\tilde{\mu}_\xi$ is not positive. Its pole part $-2\cosh(x/2)\,dx$ is negative, and at large x it cancels the mean density $e^{x/2}dx$ of the prime atoms, leaving the oscillating sum over zeros (6.2.6). Its pieces are not admissible either: the pole part grows like $e^{x/2}$ and the prime part has infinite mass, so the hypothesis of Lemma 6.2.5 fails and both stages need regularisation. The positive pieces are those already used: the gamma part, which gives Theorem 6.2.12, and the prime part, which gives Proposition 6.2.6 once it is moved to the right of $\mathrm{Re}\,z = 1$. At the centre the lemma turns the question into one about a signed input whose sine squared image is positive if and only if RH holds.

6.2.4.2 Complete monotonicity to finite order

Positivity at stage (i) of Proposition 6.2.16 is Theorem 6.2.15; positivity at stage (ii) is complete monotonicity of $k(y) = W(y) + c/(2\sqrt{\pi y})$, that is $(-1)^n k^{(n)} \geq 0$ for every n. Between the two lie the same inequalities for finitely many n, and the two range argument of Theorem 7.1.1 proves them for every order up to 10^{10}.

THEOREM 6.2.17. *Let* $k(y) = W(y) + c/(2\sqrt{\pi y})$. *For every* $n \leq 10^{10}$ *and every* $y > 0$,

$$(-1)^n\,k^{(n)}(y) > 0.$$

Consequently, by Williamson's theorem, for each $1 \leq n \leq 10^{10}$ *there is a positive measure* Ω_n *on* $(0,\infty)$ *with* $k(y) = \int (1-ty)_+^{n-1}\,\Omega_n(dt)$. *For* $n \geq 1$ *the proof uses the verification of RH to height* $T_0 = 3\times 10^{12}$ *[196], an explicit bound for* $N(T)$ *[92] and one interval enclosure (E4, Section 7.2.3), and no other information about zeros.*

Proof. The case $n = 0$ is the positivity in Theorem 6.2.15. Fix $1 \leq n \leq 10^{10}$ and put $y_n = 0.05/n$.

Small y. Write $k = w + (W_0 + \frac{c}{2\sqrt{\pi y}})$. By Theorem 6.2.12 the bracket is $\int e^{-ty} U_\Gamma(t)\,dt$ with $U_\Gamma \ge 0$, so $(-1)^n$ times its nth derivative is $L_n(y) = \int t^n e^{-ty} U_\Gamma(t) dt$, decreasing in y. Since $\theta'(\tau) - \theta'(0) \ge \frac{2\tau^2}{\frac14+\tau^2} \ge 1$ for $\tau \ge \frac12$ (the $k=0$ term of the series in Theorem 6.2.12),

$$L_n(y) = \frac{1}{\pi}\int_0^\infty \tau^{2n} e^{-y\tau^2}\big(\theta'(\tau)-\theta'(0)\big)d\tau \ge \frac{\Gamma(n+\frac12)}{2\pi}\,y^{-n-\frac12} - \frac{2^{-2n-1}}{\pi(2n+1)}.$$

On the other side $(-1)^n w^{(n)} = (-1)^n 4^{-n} e^{y/4} - (-1)^n P^{(n)}$. P is analytic in $\operatorname{Re} y > 0$, and on the circle $|z-y| = y/2$ one has $|z| \ge y/2$ and $\operatorname{Re}(1/z) \ge 2/(3y)$, so Cauchy's estimate gives

$$\begin{aligned}|P^{(n)}(y)| &\le n!\,(2/y)^n\,(2\pi y)^{-1/2}\sum_m \Lambda(m) m^{-1/2} e^{-(\log m)^2/6y}\\ &\le C_0\, n!\,(2/y)^n\,(2\pi y)^{-1/2} e^{-(\log 2)^2/6y},\end{aligned}$$

where $C_0 = 0.54993$ bounds $e^{(\log 2)^2/6y}\sum_m \Lambda(m) m^{-1/2} e^{-(\log m)^2/6y}$ for every $y \le 0.05$, each term being increasing in y (enclosure E4). Divide by the leading term of the lower bound for L_n and use Gautschi's inequality $n!/\Gamma(n+\frac12) < \sqrt{n+1}$ [177, (5.6.4)]. For $y \le y_n$ we have $(\log 2)^2/6y \ge 1.6015\,n$, so the ratio is at most

$$r_n = C_0\sqrt{2\pi}\sqrt{n+1}\,e^{(\log 2 - 1.6015)n},$$

which decreases in n, since $r_{n+1}/r_n \le \sqrt{3/2}\,e^{\log 2 - 1.6015} < \frac12$, and satisfies $r_n \le r_1 < 0.786$ (enclosure E4). Hence, for $y \le y_n$,

$$(-1)^n k^{(n)}(y) \ge (1-r_n)\frac{\Gamma(n+\frac12)}{2\pi} y_n^{-n-\frac12} - \frac{2^{-2n-1}}{\pi(2n+1)} - 4^{-n}e^{y/4}.$$

The first term is at least $0.214\,\Gamma(\frac32)\,20^{3/2}/(2\pi) > 2.6$, since $\Gamma(n+\frac12)(20n)^{n+\frac12}$ increases in n, and the two negative terms total less than 0.27. So $(-1)^n k^{(n)} > 0$ on $(0, y_n]$ for every $n \ge 1$.

Large y. Here $(-1)^n k^{(n)} = (-1)^n W^{(n)} + \frac{c\,\Gamma(n+\frac12)}{2\pi} y^{-n-\frac12}$. Zeros with $0 < \gamma \le T_0$ lie on the line and contribute $\gamma^{2n} e^{-y\gamma^2} > 0$. Any other zero has $\gamma > T_0$ and contributes at most $(\gamma+1)^{2n} e^{-y(\gamma^2 - \frac14)}$ in modulus. Let $y \ge y_n$. Since $n/y \le 20n^2 < T_0^2$, the function $(\gamma+1)^{2n} e^{-y\gamma^2}$ decreases for $\gamma \ge T_0$, so by $N(t+1) - N(t) \le 2\log(t+1)$, which follows from [92] as in the proof of Theorem 7.1.1, the zeros with $T_0 + j < \gamma \le T_0 + j + 1$ contribute at most $b_j = 2\log(T_0+j+1)\,(T_0+j+1)^{2n} e^{-y((T_0+j)^2 - \frac14)}$. For $n \le 10^{10}$ we have $b_{j+1}/b_j \le \exp\big(\frac{2n}{T_0} + \frac{1}{T_0 \log T_0} - \frac{2T_0+1}{20n}\big) < 10^{-13}$, so the tail is at most $2b_0 = 4\log(T_0+1)\,(T_0+1)^{2n} e^{-y(T_0^2 - \frac14)}$. Its ratio to the positive term $\frac{c\,\Gamma(n+\frac12)}{2\pi} y^{-n-\frac12}$ is a constant multiple of $y^{n+\frac12} e^{-y(T_0^2 - \frac14)}$, which

decreases for $y \geq y_n$ because $T_0^2 - \frac{1}{4} > 20n(n+\frac{1}{2})$. At $y = y_n$, using $\Gamma(n+\frac{1}{2}) \geq \Gamma(\frac{3}{2})$ and $y_n < 1$, the logarithm of the ratio is at most

$$\log \frac{8\pi \log(T_0+1)}{c\,\Gamma(\frac{3}{2})} + 2n\log(T_0+1) - \frac{T_0^2 - \frac{1}{4}}{20n} < -4.4 \times 10^{13},$$

because the left side increases with n, and at $n = 10^{10}$ its first term is below 6 while $y_n(T_0^2 - \frac{1}{4}) \approx 4.5 \times 10^{13}$ exceeds $2n\log(T_0+1) < 5.8 \times 10^{11}$ by a wide margin. So the tail is smaller than the positive term.

Williamson's theorem [272] characterises n monotone functions on $(0,\infty)$ as mixtures of $(1-ty)_+^{n-1}$ with positive weights; since $k(\infty) = 0$, this gives Ω_n. □

The small y branch works for every n, but only on $(0, 0.05/n]$, the range forced by the Cauchy estimate. The large y branch must cover the rest, and there the bound of the proof lets the positive term $c\,\Gamma(n+\frac{1}{2})y^{-n-\frac{1}{2}}/(2\pi)$ contributed by the tilt dominate the zeros above T_0 whenever $y_n(T_0^2 - \frac{1}{4}) > 2n\log(T_0+1) + 6$. This sufficient condition holds while $n < 8.8 \times 10^{10}$, and we state the safe round number 10^{10}. By Theorem 6.2.15, RH is equivalent to the inequality for every n, and Bernstein's theorem is the limit of Williamson's as $n \to \infty$.

Theorem 6.2.17 is the analogue for the floor of the derivative ladder that Section 5.1 climbs for the clock. It is a consequence of the verification of RH to height T_0 and carries no information about the zeros beyond it: an off line zero above T_0 is invisible to derivatives of order below about yT_0^2, the order at which the peak of $\gamma^{2n}e^{-y\gamma^2}$ passes T_0, and could break complete monotonicity only beyond that order. With the Cauchy estimate of the proof, the reachable order grows roughly linearly with T_0, and sharper estimates reach further. Theorem 5.1.4 [198, Theorem 7.10] proves $(-1)^n W^{(n)} > 0$ for every $n \leq 10^{17}$, and the same range follows for k because $c/(2\sqrt{\pi y})$ is completely monotone. The reach of its proof grows like the square of the verified height up to a logarithm (Section 5.1.7): at $y = 1/20000$, where the proof passes from the primes to the zeros, it reaches orders up to about $yH^2/\log(H^2/\gamma_0^2)$, with $H = 10^{12}$ the verified height that proof uses and $\gamma_0 \in (20, 50]$ the height of the low zero that dominates, a factor $\log(H^2/\gamma_0^2) \approx 48$ below the order yH^2 at which an off line zero above H becomes visible. Theorem 5.1.10 closes that gap with a dominating zero that follows the peak of $\gamma^{2n}e^{-y\gamma^2}$: it proves the signs for every $n \leq y(H-23)^2$ at every $y \geq 1.29$, with $H = 3 \times 10^{12}$, and, with Theorem 5.1.7, for every $n \leq 1.16 \times 10^{25}$ at every $y > 0$, for W and so for k. In every version the reach is set by the verified height and lies below that order: the Thorin measure of the floor is certified positive up to a resolution that grows with the verified height, and RH is the statement that it stays positive at infinite resolution, which no finite number of rungs reaches.

In the floor the archimedean factor supplies the completely monotone part of k but none of the arithmetic: the jumps of N, the negative excursions of S, and the cancellation that must make W completely monotone. The arithmetic inputs

of this section are the verified zeros used in Theorem 6.2.17 and the positivity of w, used in Theorem 6.2.15 and proved in Section 7.1.

Chapter 7

Infinite divisibility and truncations of the reciprocal zeta function

Chapter 6 computed the Thorin measure of the gamma factor and, under RH, those of the normalised reciprocal zeta function φ and of the reciprocal Xi function R; without RH the Vere-Jones form gives only formal series. The proof of Theorem 6.2.15, that the floor $e^{-c\sqrt{s}}R$ is infinitely divisible, used the positivity of the canonical function w of Theorem 6.1.11. Section 7.1 proves this positivity with no hypothesis, so that φ itself is infinitely divisible, and classifies φ in the Bondesson hierarchy: infinitely divisible, but not self decomposable, not in the Bondesson class and not a generalised gamma convolution. None of these failures involves a zero off the line: they come from the factor that removes the pole and from the negative excursions of the argument function S at zeros on the line. The section also reads the weights of the Vere-Jones form of Section 6.1.3 as necklace polynomials, and it ends with the lattice of Biane, Pitman and Yor, the quadratic lattice that carries the Thorin atoms of a generalised gamma convolution whose Mellin transform is ξ.

Section 7.2 studies the truncations of this lattice. One sided truncations fail at zeros where $|S|$ is large, while symmetrised truncations keep their zeros in the strip on the line in every window computed. Conjecture S asserts this for every size; for small sizes it reduces to a finite computation, and for infinitely many sizes it would imply RH. The section ends with an account of what this chapter and Chapter 6 prove, and of what remains open. The coordinates of Chapter 6, declared in Section 6.1.1, hold in this chapter too: in the variable s of the clock a zero $\frac{1}{2} + i\gamma$ sits at $s = -\gamma^2$ on the cut.

7.1 Infinite divisibility, necklaces and solved models

By Theorem 6.1.11, $-\varphi'/\varphi$ is the Laplace transform of the canonical function $w = e^{x/4} - P = W - W_0$. Section 7.1.1 proves that w is positive and studies the law whose Laplace transform is φ. Section 7.1.2 reads the Vere-Jones weights arithmetically, through the prime number theorem and in a pseudo spectral trace

formula, and carries the Lévy and Thorin representations across the critical strip. Section 7.1.3 places the solved models in the theory of Pólya, Cardon, Lee and Yang.

7.1.1 Infinite divisibility and the class of φ

Whether φ is infinitely divisible is a question about the sign of w. A function of the form $\exp(-\int_0^\infty (1-e^{-sx})\,\nu(dx))$, with $\nu \geq 0$ and $\int \min(1,x)\,\nu(dx) < \infty$, is the Laplace transform of an infinitely divisible law on $[0,\infty)$ with Lévy measure ν, and by Theorem 6.1.11 the candidate Lévy density of φ is $w(x)/x$. Recall that $w(0^+) = 1$ and $w(x) \sim c/(2\sqrt{\pi x})$ as $x \to \infty$, where $c = -\theta'(0) \approx 2.68609$ is the constant of (6.1.6). The proof of positivity uses the verification of RH to height 50, classical since Backlund [10] and contained in every later verification [196], the explicit Riemann von Mangoldt bound of Hasanalizade, Shen and Wong [92], and three interval enclosures, $F(0.004) < -0.111$ for the function F of Lemma 6.1.12, $P(0.004) < 2 \times 10^{-13}$ and the tail bound $E(0.004) < 0.0011$ for the sum E defined in the proof, computed in outward rounded interval arithmetic by `verification_reciprocal_zeta/enclosures.py` (enclosures E1 to E3).

Theorem 7.1.1. *$w(x) > 0$ for every $x > 0$. Consequently, unconditionally, φ is the Laplace transform of an infinitely divisible probability law on $(0,\infty)$ with zero drift and Lévy density $w(x)/x$:*

$$\frac{\zeta(\frac12)}{(1-4s)\,\zeta(\frac12+\sqrt{s})} = \exp\Big(-\int_0^\infty \big(1-e^{-sx}\big)\frac{w(x)}{x}\,dx\Big), \qquad \operatorname{Re} s > 0.$$

Proof. Fix $x_1 = 0.004$.

Small x. Each term $x^{-1/2}e^{-(\log n)^2/4x}$ increases in x while $(\log n)^2/4x > \frac12$, which holds for $x \leq x_1$ and $n \geq 2$. Hence $P(x) \leq P(x_1) < 2 \times 10^{-13}$ (enclosure E2: the term $n = 2$ in interval arithmetic, and $n \geq 3$ through $(\log n)^2/4x_1 \geq \frac{\log 3}{4x_1}\log n$), and $w(x) = e^{x/4} - P(x) > 0$ on $(0, x_1]$.

Large x. By Lemma 6.1.12 and $F(x_1) < 0$, for $x \geq x_1$ we have $-W_0(x) \geq |F(x_1)|/(4\sqrt{\pi x})$. Here $F(x_1) < -0.111$ by enclosure E1: an upper Riemann sum of interval suprema of the integrand $e^{-u/4}(1-e^{-u^2/16x_1})/(1-e^{-u})$ of Lemma 6.1.12 on $[10^{-4}, 200]$, with the integrand at most $u(1+u)/16x_1$ on $[0, 10^{-4}]$ and at most $e^{-u/4}/(1-e^{-200})$ beyond 200, and $2\theta'(0) = -\gamma_E - \frac{\pi}{2} - 3\log 2 - \log\pi$. In W the zeros with $0 < \gamma \leq 50$ lie on the line [10] and contribute $e^{-x\gamma^2} > 0$. Any other zero has $\gamma > 50$ and contributes at most $e^{-x(\gamma^2-1/4)}$ in modulus. The explicit Riemann von Mangoldt formula $|N(T) - \frac{T}{2\pi}\log\frac{T}{2\pi e} - \frac78| \leq E(T)$ for $T \geq e$, with $E(T) = 2.5167 + \frac{1}{50T}$ for $T \leq X_0 = 30\,610\,046\,000$ and $E(T) = 0.1095\log T + 0.2042\log\log T + 3.0305 + \frac{1}{50T} < 0.3\log T$ for $T > X_0$ [92, (1.7), (2.7) and Corollary 1.4], gives $N(t+1) - N(t) \leq 2\log(t+1)$ for $t \geq 50$: the main term grows by at most $\frac{1}{2\pi}\log\frac{t+1}{2\pi}$, and while $t+1 \leq X_0$ the two error terms add at most $5.0334 + \frac{1}{25t}$, together less than 5.37 at $t = 50$ against $2\log 51 = 7.86$, with a gap that widens

as t grows, since its derivative exceeds $(2-\frac{1}{2\pi})/(t+1)$; for $t+1>X_0$ the error terms add at most $0.6\log(t+1)$, and $\frac{1}{2\pi}+0.6<2$. So all such zeros together contribute at most

$$E(x)=\sum_{j\ge 0}2\log(51+j)\,e^{-x((50+j)^2-\frac14)}.$$

Enclosure E3 gives $E(x_1)<0.0011$, against $|F(x_1)|/(4\sqrt{\pi x_1})>0.247$. Since $(50+j)^2-\frac14>1/(2x_1)$ for every j, $\sqrt{x}\,E(x)$ decreases on $[x_1,\infty)$, so $E(x)<|F(x_1)|/(4\sqrt{\pi x})\le -W_0(x)$ for all $x\ge x_1$, and $w=W-W_0>0$ on $[x_1,\infty)$.

Representation. By Theorem 6.1.11, $-\varphi'/\varphi(s)=\int_0^\infty e^{-sx}w(x)dx$ for $s>\frac14$. Since $w\ge 0$, Landau's theorem applies [268, Chapter II, Theorem 5b]: if the Laplace transform of a nonnegative function has abscissa of convergence σ_c, then the transform cannot be continued analytically to a neighbourhood of σ_c. Here the transform agrees with $-\varphi'/\varphi$ on $(\frac14,\infty)$, and $-\varphi'/\varphi$ is analytic near every point of $(0,\infty)$ by Proposition 6.1.9. Hence $\sigma_c\le 0$, the integral converges for $\operatorname{Re}s>0$, and equals the continuation of $-\varphi'/\varphi$ there. Integrating over $[a,s]$ for real $0<a<s$ with Tonelli, and letting $a\downarrow 0$ with $\varphi(a)\to 1$ and monotone convergence, gives the formula for real $s>0$; then $\int\min(1,x)\,w(x)x^{-1}dx<\infty$ because the left side is finite, and analytic continuation gives the formula for $\operatorname{Re}s>0$. □

Theorem 7.1.1 is the main unconditional positive result of Chapter 6 and of this chapter, and its proof has a wide margin (Figure 7.1). The two halves of the argument overlap: at $x_1=0.004$ the prime sum $P(x_1)$ is below 2×10^{-13} while $-W_0(x_1)$ already exceeds 0.25, so the positivity of w does not rest on a delicate numerical fact near a single point. The only place where w comes close to a sign change is at infinity, where it decays like $x^{-1/2}$, and there the sign is fixed by $\theta'(0)<0$.

In arithmetic terms, Theorem 7.1.1 is the Gaussian smoothed Chebyshev bound

$$\sum_{n\ge 2}\frac{\Lambda(n)}{\sqrt{n}}\,e^{-(\log n)^2/4x}<\sqrt{4\pi x}\,e^{x/4}\qquad(x>0).$$

The right side is $\int_0^\infty y^{-1/2}e^{-(\log y)^2/4x}\,dy$, the same smoothing applied to dy in place of $d\psi(y)$, where $\psi(y)=\sum_{n\le y}\Lambda(n)$ is Chebyshev's function. The proof uses the zeros only through the ten below height 50, the explicit bound for $N(T)$ and the monotone archimedean term F. To see what the primes alone give, write $\psi(y)=y+\Delta(y)$ and put $I(x)=\frac12 e^{x/4}\operatorname{erfc}(\frac{\sqrt{x}}{2})$.

Proposition 7.1.2. *For $x>0$,*

$$w(x)=I(x)-Q_\Delta(x),\qquad Q_\Delta(x)=\frac{1}{\sqrt{4\pi x}}\int_{1^-}^\infty y^{-1/2}e^{-(\log y)^2/4x}\,d\Delta(y),$$

with $I(x)\sim(\pi x)^{-1/2}$ and $w(x)\sim\frac{c}{2}(\pi x)^{-1/2}$ as $x\to\infty$. Hence $Q_\Delta(x)\sim-(\frac{c}{2}-1)(\pi x)^{-1/2}$ with $\frac{c}{2}-1\approx0.343$. On the other hand, any bound $|\Delta(y)|\le A\sqrt{y}\log^2 y$

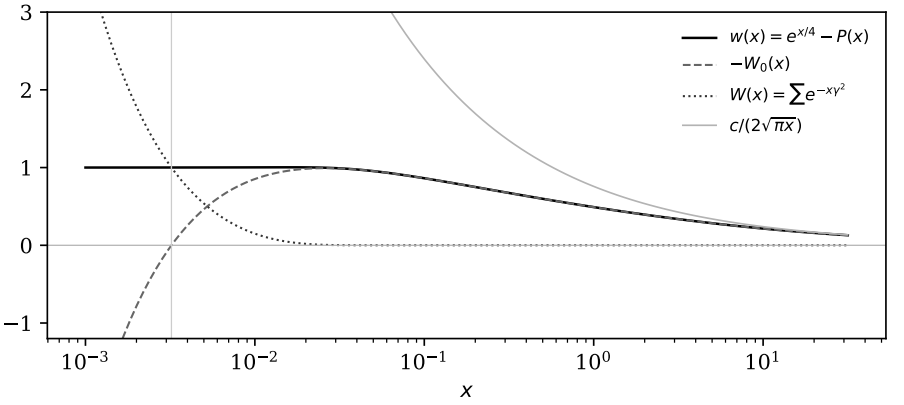


Figure 7.1: The function $w = W - W_0$ on a logarithmic scale, with its two components: the heat trace W over the zeros and the negative smooth part $-W_0$. The vertical line is $x_* \approx 0.00322$, where W_0 changes sign. To the left, w differs from $e^{x/4}$ by less than 2×10^{-13}; to the right, $-W_0 > 0$, and from $x_1 = 0.004$ on it dominates any contribution of off line zeros to W. The faint curve is the asymptote $c/(2\sqrt{\pi x})$.

for $y \geq 2$, including the one that RH implies, yields only $|Q_\Delta(x)| \leq A\,(x/2 + O(\sqrt{x}))$ after integration by parts.

Proof. With $d\Delta = d\psi - dy$, the main term is $(4\pi x)^{-1/2} \int_1^\infty y^{-1/2} e^{-(\log y)^2/4x} dy = (4\pi x)^{-1/2} \int_0^\infty e^{u/2 - u^2/4x} du = \frac{1}{2} e^{x/4}(1 + \operatorname{erf} \frac{\sqrt{x}}{2})$, and $e^{x/4}$ minus this is I. The asymptotics of w are Theorem 6.1.11. For the bound, integrate by parts and use $\frac{d}{dy}\{y^{-1/2} e^{-(\log y)^2/4x}\} = -y^{-3/2} e^{-(\log y)^2/4x}(\frac{1}{2} + \frac{\log y}{2x})$ and $\int_0^\infty u^2 e^{-u^2/4x}(\frac{1}{2} + \frac{u}{2x}) du = \frac{\sqrt{\pi}}{2}(2\sqrt{x})^3 \cdot \frac{1}{4} + 4x$; the boundary term at 1^- and the range $1 \leq y < 2$ contribute $O(x^{-1/2})$. □

In this Gaussian weighting the error term Δ of the prime number theorem has a definite negative bias at large x, $Q_\Delta(x) \approx -0.343(\pi x)^{-1/2}$, which is $I + W_0$ up to an exponentially small term, and $w > 0$ is the statement that $Q_\Delta(x) < I(x)$ for all x. Pointwise bounds on Δ, which oscillates with both signs, cannot give this at large x: the margin that keeps w positive there is of order $x^{-1/2}$, while a pointwise bound on Δ controls Q_Δ only up to order x; the function $D(y) = -A\sqrt{y}\log^2 y$, in place of Δ, gives $Q_D(x) = -A\big(x/2 + 2\sqrt{x/\pi}\big)$ by the computation in the proof, so the bound of Proposition 7.1.2 cannot be improved using the size of Δ alone. Mean square information on $\Delta(y)/\sqrt{y}$ in the logarithmic scale, of the kind Cramér obtained under RH [58], controls Gaussian smoothed quantities much better, and we make no claim that such information is insufficient. For $x \leq 0.004$ the primes

alone suffice (Theorem 7.1.1).

7.1.1.1 The place of φ in the Bondesson hierarchy

Recall from Section 1.3.2 the classes of laws on $[0,\infty)$ defined through the Lévy density $k(x)/x$: the infinitely divisible laws (ID), the self decomposable laws (SD), for which k is nonincreasing, the Bondesson class (BO), for which $k(x)/x$ is completely monotone, and the GGCs, for which k is completely monotone. They are nested as GGC $\subset$ SD $\subset$ ID and GGC $\subset$ BO $\subset$ ID [35, 247].

THEOREM 7.1.3. *$\varphi \in$ ID but $\varphi \notin$ SD, $\varphi \notin$ BO, and φ is not a mixture of exponentials. In particular φ is not a generalised gamma convolution.*

Proof. ID is Theorem 7.1.1.

SD. A law on $(0,\infty)$ with Lévy density $k(x)/x$ is self decomposable if and only if k is nonincreasing (Section 1.3.2). If φ is not infinitely divisible there is nothing to prove, and if it is then $k = w$ by Theorem 6.1.11. Now $w = e^{x/4} - P$, and P together with every one of its derivatives vanishes at 0^+, so $w'(0^+) = \frac{1}{4} > 0$ and w strictly increases at the origin. No numerical value enters, and neither does Theorem 7.1.1: the case split replaces it.

BO. Membership means that the Lévy density $w(x)/x$ equals $\int e^{-tx} m(dt)$ for some $m \geq 0$. Then $-\varphi'/\varphi(s) = \int_0^\infty x e^{-sx} \int e^{-tx} m(dt)\, dx = \int m(dt)/(s+t)^2$ is analytic on $\mathbb{C} \setminus (-\infty, 0]$, so φ has no zeros or poles there and RH follows from Proposition 6.1.9. Under RH, Theorem 6.1.11 gives $w(x) = \int e^{-tx} dM(t)$, with $M(t) = N(\sqrt{t}) - \theta(\sqrt{t})/\pi = 1 + S(\sqrt{t})$ as in Theorem 6.1.10, and since $M(0^+) = 0$ an integration by parts gives $w(x)/x = \int e^{-tx} M(t)\, dt$. Uniqueness of Laplace transforms gives $M\, dt = m \geq 0$, that is $S(\tau) \geq -1$ for almost all τ. But by Selberg's central limit theorem [237], $S(\tau) < -1$ for a positive proportion of $\tau \in [T, 2T]$ once T is large, so $M < 0$ on a set of positive measure. Numerically, for $527.71 \leq \tau \leq 527.90$, just below $\gamma_{289} = 527.9036\ldots$, ball arithmetic gives $N(\tau) = 288$ and $-1.143 < S(\tau) < -1.008$, so $M < 0$ there, and $S(\gamma_{289}^-) = -1.14548\ldots$ (`verification_reciprocal_zeta/certify_k9_k2.py`).

Mixtures of exponentials. The law is a mixture of exponentials exactly when it has a completely monotone density, that is when φ is a Stieltjes function, which requires $\operatorname{Im} \varphi(-t + i0)$ to have one sign. With $\zeta(\frac{1}{2} + i\tau) = Z(\tau) e^{-i\theta(\tau)}$ we get $\operatorname{Im} \varphi(-\tau^2 + i0) = \zeta(\frac{1}{2}) \sin\theta(\tau)/\{(1+4\tau^2) Z(\tau)\}$. On $(0, \gamma_1)$, Z has no zero while θ decreases through $-\pi$ at $\tau_\pi = 3.43622\ldots$, so $\sin\theta$ changes sign.

GGC. Since GGC is contained in SD and in BO, either clause excludes it. The exclusion also follows directly, without computation. If φ were the transform of a GGC then $-\varphi'/\varphi$ would be a Stieltjes function $a + \int U(dt)/(s+t)$ with $U \geq 0$, that is $a + \int_0^\infty e^{-sx} \big(\int e^{-tx} U(dt) \big) dx$. Comparing with $-\varphi'/\varphi(s) = \int_0^\infty e^{-sx} w(x)\, dx$ of Theorem 6.1.11 on $s > \frac{1}{4}$, uniqueness of Laplace transforms forces $a = 0$ and $w(x) = \int e^{-tx} U(dt)$, completely monotone and in particular nonincreasing, against $w'(0^+) = \frac{1}{4} > 0$. □

The theorem is unconditional. Its ID clause rests on Theorem 7.1.1, its BO clause on Selberg's central limit theorem for S, and its SD clause and the conclusion $\varphi \notin \mathrm{GGC}$ on no computation. Since mixtures of exponentials lie in BO [35], the mixture clause also follows from the BO clause, and from $f'(0^+) > 0$ in Theorem 7.1.4; the direct argument, through the computed value of τ_π, is an independent check.

The failures have two causes, and neither involves a zero off the line. Self decomposability and the GGC property fail because of the pole: the factor $1 - 4s$ that removes it puts $e^{x/4}$ into w, so the canonical function increases near the origin. This is a property of the normalisation, and Proposition 7.1.13 shows the same on every line of the strip. The Bondesson property fails because the negative Thorin density of the gamma factor meets the fluctuations of the zeros about their smooth count, measured by S, which has large negative excursions: the first zero with $S(\gamma^-) < -1$ is $\gamma_{127} = 282.4651$, where $S(\gamma_{127}^-) = -1.0063$, and the most negative value among the first 460 zeros is at γ_{289} (the seven such zeros are listed after Theorem 7.2.3, and these values of S are enclosed in ball arithmetic by the program cited in the proof). This failure would occur even if RH were true, and since S is unbounded below, by Selberg and Tsang [237, 260], it recurs infinitely often. Both causes disappear for the completed function R, which is infinitely divisible and self decomposable with no hypothesis, by Theorem 5.1.1 [200, Theorem A]. What Theorem 7.1.1 adds there is the floor of Theorem 6.2.15: the canonical function of $e^{-c\sqrt{s}}R(s)$ splits into the arithmetic part w and a completely monotone archimedean part (Section 6.2).

7.1.1.2 The density of the law

Let X_φ be, as in Section 6.2, a random variable with $\mathbb{E}e^{-sX_\varphi} = \varphi(s)$. Differentiating the representation of Theorem 7.1.1 gives $-\varphi'(s) = \varphi(s)\int_0^\infty e^{-sx}w(x)\,dx$, which says that the measure $x\Pr(X_\varphi \in dx)$ is the convolution of the law of X_φ with $w(x)\,dx$. Since $\varphi(s) \to 0$ as $s \to \infty$, the law has no atom at 0, so it is absolutely continuous, and its density f satisfies Steutel's equation [247, Proposition III.4.16]

$$x\,f(x) = \int_0^x f(x-y)\,w(y)\,dy, \qquad x > 0. \tag{7.1.1}$$

The right side is continuous in x by dominated convergence, since f is integrable and w is bounded and continuous on $[0,\infty)$, so f may be taken continuous on $(0,\infty)$. Near the origin, $w(y) = 1 + O(y)$ gives $f(x) = \bar f(x)(1 + O(x))$ with $\bar f(x) = x^{-1}\int_0^x f$, and $\bar f(x) > 0$ for every $x > 0$, since otherwise $s\varphi(s) \to 0$, whereas $s\varphi(s) \to -\zeta(\tfrac12)/4 > 0$. As $\bar f' = (f - \bar f)/x$, this gives $(\log \bar f)' = O(1)$ near 0, so $\bar f$ and f have a common positive limit $f(0^+)$, and dominated convergence in $s\varphi(s) = \int_0^\infty e^{-y}f(y/s)\,dy$, f being bounded, gives $f(0^+) = \lim_{s\to\infty} s\varphi(s) = -\zeta(\tfrac12)/4 \approx 0.36509$.

At infinity the same equation gives $f(x) \sim w(x)/x \sim \frac{c}{2\sqrt{\pi}} x^{-3/2} \approx 0.75773\, x^{-3/2}$, in three steps. First, $xf(x) \le \sup w$. Second, split the integral in (7.1.1), written as $\int_0^x f(u)w(x-u)\,du$, at $u = x/2$: the part $u \le x/2$ is at most $\sup_{y \ge x/2} w(y) = O(x^{-1/2})$, and the part $u \ge x/2$ is at most $\sup_{[x/2,x]} f \cdot \int_0^{x/2} w = O(x^{-1})\, O(x^{1/2})$, so $f(x) = O(x^{-3/2})$. Third, with this bound the part $u \ge x/2$ is $O(x^{-1}) = o(w(x))$, while the part $u \le x/2$ is $w(x)(1 + o(1))$ by dominated convergence, since $w(x-u)/w(x) \to 1$ for fixed u and stays below 2 for $u \le x/2$ and large x. Hence $xf(x) = w(x)(1 + o(1))$. It follows that $\Pr(X_\varphi > x) \sim \frac{c}{\sqrt{\pi}} x^{-1/2}$ and $\mathbb{E}X_\varphi = \infty$, and since $\varphi(s) = 1 - c\sqrt{s} + O(s)$ gives $\varphi(s/n^2)^n \to e^{-c\sqrt{s}}$, the variable X_φ lies in the domain of attraction of the positive stable law of index $\frac{1}{2}$ that generates the building blocks (6.1.2). Figure 7.2 shows f on logarithmic axes, recovered from φ by numerical Laplace inversion; the value at 0^+ and the $x^{-3/2}$ decay are its two straight portions.

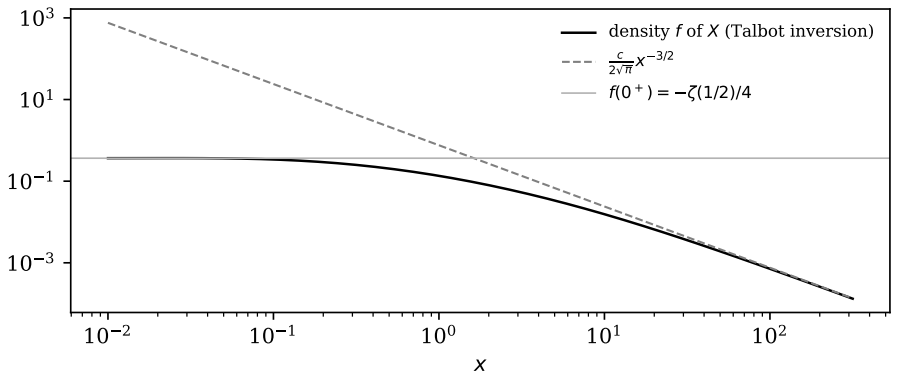


Figure 7.2: The density f of the infinitely divisible law X_φ with $\mathbb{E}e^{-sX_\varphi} = \varphi(s)$, computed by Talbot inversion, on logarithmic axes. It starts at $f(0^+) = -\zeta(\frac{1}{2})/4$ and decays like $\frac{c}{2\sqrt{\pi}} x^{-3/2}$. In the legend, X is X_φ.

THEOREM 7.1.4. *$f(0^+) = -\zeta(\frac{1}{2})/4$ and $f'(0^+) = f(0^+)\, w'(0^+) = -\zeta(\frac{1}{2})/16 \approx 0.09127 > 0$. In particular f is not monotone, so X_φ is not a mixture of uniform laws on intervals $[0, a]$, and f is not log convex and not completely monotone.*

Proof. The value of $f(0^+)$ was found after (7.1.1). Since P and all its derivatives vanish at 0^+, w is continuously differentiable on $[0, \infty)$ with $w(0^+) = 1$ and $w'(0^+) = \frac{1}{4}$. Write (7.1.1) as $xf(x) = \int_0^x f(u)\, w(x-u)\, du$. Since f is continuous on $(0, \infty)$ and bounded near 0, the right side is differentiable, with derivative $f(x)w(0^+) + \int_0^x f(u)w'(x-u)\,du$, so f is differentiable on $(0, \infty)$ and

$$f'(x) = \frac{1}{x} \int_0^x f(u)\, w'(x-u)\, du \longrightarrow f(0^+)\, w'(0^+) \qquad (x \downarrow 0),$$

because f and w' are continuous at 0. A nonincreasing density is required for a mixture of uniforms by Khintchine's theorem, and log convex or completely monotone densities are nonincreasing. □

PROPOSITION 7.1.5. *$w' > 0$ on $(0, 0.004]$ and $w' < 0$ on $[4.52, \infty)$. On $[0.004, 4.6]$ a grid evaluation of w' (computed, not proved) finds exactly one sign change, at $x_m = 0.015851\ldots$, where $w(x_m) = 1.003409\ldots$.*

Proof. On $(0, 0.004]$, $w' = \frac{1}{4}e^{x/4} - P'(x)$ and $0 \le P'(x) \le P'(0.004) < 2 \times 10^{-9}$, since each prime term of P' increases in x there. For $x \ge 0.004$ write $w' = W' - W_0'$. The zeros on the line contribute $-\sum \gamma^2 e^{-x\gamma^2} < 0$ to W'. Every zero below height 3×10^{12} lies on the line [196], so an off line zero contributes at most $(\gamma^2 + \frac{1}{4})e^{-x(\gamma^2 - 1/4)}$ with $\gamma > 3 \times 10^{12}$. With $N(t+1) - N(t) \le 2\log(t+1)$ from the proof of Theorem 7.1.1, these contributions together are less than $e^{-10^{22}}\gamma_1^2 e^{-x\gamma_1^2}$ for $x \ge 0.004$, a negligible fraction of the term of γ_1; hence $W' < 0$. By Proposition 6.2.11, with $\nu(dy) = 2\sum_{k\ge 0} e^{-(2k+\frac{1}{2})y}\,dy$ the measure defined there,

$$-W_0'(x) = \frac{D(x)}{4\sqrt{\pi}\,x^{3/2}}, \qquad D(x) = \theta'(0) + \frac{1}{2}\int_0^\infty g\Big(\frac{y^2}{4x}\Big)\nu(dy),$$
$$g(u) = 1 - (1-2u)e^{-u}.$$

Since $g(0) = 0$, $g'(0) = 3$, g is concave on $[0, \frac{5}{2}]$ and $g \le 1 + 2e^{-3/2} < 3u$ beyond, $g(u) \le 3u$ for all $u \ge 0$. With $\int y^2 \nu(dy) = 4\sum_{k\ge 0}(2k + \frac{1}{2})^{-3} < 32.34$ this gives $D(x) \le \theta'(0) + 12.13/x < 0$ for $x \ge 4.52$, and then both parts of w' are negative. □

The computed statement comes from an evaluation of w' on $[0.004, 4.6]$ from the first 300 zeros and the integral for W_0', on a grid of spacing 10^{-4} up to 0.05 and 5×10^{-3} beyond. On this grid w' has exactly one sign change, at 0.01585, and $|w'| \ge 0.024$ at grid points more than 0.002 from it; the prime and zero evaluations of w' agree to 10^{-11} at $x = 0.01$. Between two grid points h apart at which w' has one sign and $|w'| \ge 0.024$, a further pair of sign changes would need $|w''| \ge 2 \cdot 0.024/h$, that is $|w''| \ge 9.6$ where the spacing is 5×10^{-3} and $|w''| \ge 480$ where it is 10^{-4}. An upper bound on $|w''|$ below these values, which is not derived here, would exclude further sign changes away from x_m. Within 0.002 of x_m the grid gives no such margin, and a proof would also need $w'' < 0$ there, or a bound on w'''. Together these would turn the computation into a proof that w is unimodal.

Numerically f is unimodal as well: it rises from 0.365089 to its maximum 0.366883 at the mode $x = 0.024046$ and then decreases to 0 like $0.75773\,x^{-3/2}$, so its mode lies to the right of the maximum of w. Talbot and de Hoog inversion agree to twelve digits at every point used. Unimodality of f remains open (Section 7.2.5).

Since X_φ is infinitely divisible but not self decomposable, Yamazato's theorem that every self decomposable law is unimodal [274], [232, Theorem 53.1] does not apply, and the rise of f near 0 (Theorem 7.1.4) shows that the question is not trivial. Section 11.1 applies the theorem to the law of X_R, which is self decomposable, and to the shifted clocks T_α of Theorem 1.2.2 with $\alpha \geq 1$ (Corollary 11.1.2).

7.1.1.3 Closure operations and the harmonic mean

GGC is closed under independent sums and weak limits, and, by a theorem of Bondesson [36], under independent products. Any variable assembled from GGC variables by these operations is GGC, so Theorem 7.1.3 gives at once the following.

COROLLARY 7.1.6. *X_φ is not a sum or a product of independent GGC variables, nor a weak limit of such.*

The reciprocal in φ suggests a fourth operation, the harmonic mean. Closure under it would be the statement

(H) if V_1 and V_2 are independent GGC variables, then so is $2V_1V_2/(V_1+V_2)$,

which would have widened Corollary 7.1.6; it is false.

PROPOSITION 7.1.7. *(H) is false. If E_1 and E_2 are independent standard exponentials, then $H = 2E_1E_2/(E_1+E_2)$ has density*

$$f_H(h) = e^{-h}\int_0^\infty \Big(1+\frac{h}{2u}\Big)e^{-u-h^2/4u}\,du, \qquad h>0,$$

and H is not a GGC.

Proof. For $h>0$, $2E_1E_2 > h(E_1+E_2)$ holds if and only if $E_1, E_2 > \frac{h}{2}$ and $(2E_1-h)(2E_2-h) > h^2$: both factors cannot be negative, since $E_1, E_2 < \frac{h}{2}$ forces $|2E_1-h|, |2E_2-h| < h$. Substituting $E_1 = \frac{h}{2}+u$, $E_2 = \frac{h}{2}+v$ gives

$$\Pr(H>h) = e^{-h}\iint_{uv>h^2/4} e^{-u-v}\,du\,dv = e^{-h}\int_0^\infty e^{-u-h^2/4u}\,du,$$

so the survival function is e^{-h} times a Laplace type integral. Differentiating under the integral, which is justified by the bound $\frac{h}{2u}e^{-u-h^2/4u} \leq u^{-1/2}e^{-u}\sup_{y>0} ye^{-y^2}$, locally uniform in h, gives f_H. In terms of the modified Bessel functions of the second kind the two integrals are $h\,\mathrm{K}_1(h)$ and $h\{\mathrm{K}_0(h)+\mathrm{K}_1(h)\}$, by $\int_0^\infty u^{-\nu-1}e^{-u-h^2/4u}\,du = 2(h/2)^{-\nu}\mathrm{K}_\nu(h)$ [177, (10.32.10)]. Splitting the integral at $u=h$ shows that f_H is bounded with $f_H(0^+)=1$.

Suppose H were a GGC with drift a and Thorin measure U of total mass β. Since $f_H > 0$ on $(0,\infty)$ and a GGC satisfies $H \geq a$ almost surely, $a=0$. Since f_H is bounded, $s\,\mathbb{E}e^{-sH} = \int_0^\infty e^{-y}f_H(y/s)\,dy \to f_H(0^+) = 1$, so

$-\log \mathbb{E}e^{-sH} = \log s + o(1)$, while Fatou applied to $\int \log(1 + s/t)\, U(dt)$ gives $\liminf_{s\to\infty}(-\log \mathbb{E}e^{-sH})/\log s \geq \beta$. Hence $\beta \leq 1$.

Now let V be any GGC with zero drift and $\beta \leq 1$, and $\mathcal{L}(s) = \mathbb{E}e^{-sV}$. For $\operatorname{Im} s > 0$ each $1 + s/t$ lies in the upper half plane, so $\operatorname{Im} \log \mathcal{L}(s) = -\int \arg(1 + s/t)\, U(dt) \in [-\pi\beta, 0) \subseteq [-\pi, 0)$, and $\mathcal{L} > 0$ on $(0, \infty)$: $\mathcal{L}$ is a Stieltjes function [234, Corollary 7.4]. In its representation $\mathcal{L}(s) = b_0 + b_1/s + \int (s+r)^{-1} m(dr)$ [234, Definition 2.1] we have $b_0 = \lim_{s\to\infty} \mathcal{L}(s) = 0$ and $b_1 = 0$, since $\mathcal{L}(0^+) = 1$. So $\mathcal{L}(s) = \int_0^\infty e^{-sx}\big(\int e^{-rx} m(dr)\big)dx$ and the density of V is $\int e^{-rx} m(dr)$, completely monotone and in particular nonincreasing.

But f_H is not nonincreasing: $f_H(h) - 1 \sim h \log(1/h)$ as $h \downarrow 0$, and f_H attains 1.11125 at $h = 0.09765$ against $f_H(0^+) = 1$. So H is not a GGC. □

There is also a structural obstruction. The factorisation of φ into R and the gamma factor is a factorisation of Laplace transforms, that is a convolution of laws, while (H) acts on the variables themselves. The gamma factor is not a Laplace transform at all, since it tends to infinity with s, and (6.2.1) shows that it enters only through deconvolution by stable laws. The one place where (H) could have acted is the positive part: R is GGC if and only if RH holds, so an unconditional representation $X_R \stackrel{d}{=} 2V_1V_2/(V_1 + V_2)$ with V_1, V_2 independent GGC variables built from the primes and the gamma factor would have proved RH under (H). Proposition 7.1.7 removes the implication, and we know of no candidate representation. The kernel of the Vere-Jones identity is itself a harmonic mean of functions, $\operatorname{sech}(\lambda\sqrt{s}) = 2/(e^{\lambda\sqrt{s}} + e^{-\lambda\sqrt{s}})$, of the stable transform $e^{-\lambda\sqrt{s}}$ and its reciprocal, which is not a transform; and each zero factor satisfies $1 + u^2/\gamma^2 = 2/\{(1 + iu/\gamma)^{-1} + (1 - iu/\gamma)^{-1}\}$. Neither is a harmonic mean of independent variables, so neither brings (H) into play.

7.1.2 The Vere-Jones weights and the Lévy representation across the strip

The weights B_N of the logarithmic Vere-Jones representation (Theorem 6.1.8) are necklace polynomials evaluated at $p^{-1/2}$, and prime by prime the representation is a deformation of Euler's binary product. In its cyclotomic form the normalisation at the edge of the pole rests on the prime number theorem, and its heat trace form is a trace formula in which the zeros of ζ appear as a signed superposition of arithmetic progressions. The same weights, evaluated at $p^{-\sigma}$, carry the Lévy representation to every vertical line of the critical strip, and on these lines the completed function R_σ has the Thorin property exactly to the right of all zeros. Throughout this subsection σ is the real part of the argument z of ζ.

7.1.2.1 Necklace polynomials and the sign rule

Let $M_k(q) = \frac{1}{k}\sum_{d\mid k}\mu(k/d)\,q^d$ be the necklace polynomial [157], which for integer q counts aperiodic necklaces of length k in q colours.

THEOREM 7.1.8. *$B_N = 0$ unless $N = p^n$ is a prime power, and then*

$$B_{p^n} = b_n\big(p^{-1/2}\big), \qquad b_n(q) = \sum_{i\ge 0,\ 2^i\mid n} M_{n/2^i}(q).$$

Equivalently, for $|y| < 1$ and $0 < q \le 1$,

$$\frac{1}{1-qy} = \prod_{n\ge 1}\big(1+y^n\big)^{b_n(q)}, \tag{7.1.2}$$

so that $\zeta(\frac12+u) = \prod_p\prod_n(1+p^{-nu})^{b_n(p^{-1/2})}$ for $\operatorname{Re} u > \frac12$. At $q = 1$, $b_n(1) = 1$ if n is a power of 2 and 0 otherwise, and (7.1.2) becomes Euler's binary identity $\frac{1}{1-y} = \prod_{j\ge 0}(1+y^{2^j})$.

Proof. From the definition of B_N in Theorem 6.1.8, $B_{p^n} = \sum_{jm=n} d_m\, a_{p^j}$ with $a_{p^j} = q^j/j$. Proposition 6.1.6 gives $y^j = \sum_m d_m \log(1+y^{mj})$; multiplying by q^j/j and summing over j gives $\sum_n b_n(q)\log(1+y^n) = \sum_j q^j y^j/j = -\log(1-qy)$, which is (7.1.2). The cyclotomic identity $\frac{1}{1-qy} = \prod_k(1-y^k)^{-M_k(q)}$ and $\log(1+y^n) = \log(1-y^{2n}) - \log(1-y^n)$ give, comparing coefficients of $\log(1-y^n)$, the recursion $b_n = M_n + b_{n/2}$ for even n and $b_n = M_n$ for odd n, whose solution is the stated sum. At $q = 1$, $M_k(1) = \frac{1}{k}\sum_{d\mid k}\mu(k/d) = \delta_{k,1}$. □

The identity (7.1.2) is a 2 adic variant of the cyclotomic identity of Metropolis and Rota [157]; we have not found it stated in this form, but it may well be known in the combinatorics of necklaces. Its exponents include $b_2 = \frac{q^2+q}{2}$, $b_3 = \frac{q^3-q}{3}$, $b_4 = \frac{q^4}{4} + \frac{q^2}{4} + \frac{q}{2}$, $b_6 = \frac{q^6+q^3-q^2-q}{6}$, which give B_4, B_8 and B_9 of Theorem 6.1.8 and, at $p = 2$, B_{16} and B_{64}, and numerically both sides agree to 25 digits at $q = 2^{-1/2}$, $y = \frac12$. The identity factors a geometric series into products of $1+y^n$, and applied prime by prime it factors the zeta function into solved models. So the Vere-Jones factorisation of each Euler factor is a deformation of Euler's binary product: on the line $\sigma = 0$, where $q = p^{-\sigma} = 1$, every exponent is 0 or 1 and every factor $1 + p^{-2^j u}$ is a solved model with unit multiplicity, while on the critical line the exponents become necklace polynomials evaluated at $p^{-1/2}$.

PROPOSITION 7.1.9. *For every prime p, with $q = p^{-1/2}$: $b_{2^a}(q) > 0$ for all $a \ge 1$, and for odd $n > 1$, $b_n(q) \ne 0$ and the sign of $b_n(q)$ is $(-1)^{\omega(n)}$, where $\omega(n)$ is the number of distinct prime factors of n.*

Proof. For $n = 2^a$, $b_n = q + \sum_{j=1}^{a}\frac{q^{2^j}-q^{2^{j-1}}}{2^j} = \frac{q}{2} + \frac{q^2}{4} + \cdots + \frac{q^{2^{a-1}}}{2^a} + \frac{q^{2^a}}{2^a} > 0$. For odd n, $b_n = M_n$. Put $\ell = \operatorname{rad}(n)$ and $\tilde q = q^{n/\ell} \le q \le 2^{-1/2}$; then $nM_n(q) =$

$\sum_{k|\ell}\mu(\ell/k)\tilde q^k = (-1)^{\omega(n)}\big(\tilde q + \sum_{k|\ell,k>1}\pm\tilde q^k\big)$, and the divisors $k>1$ of the odd squarefree ℓ are distinct odd integers at least 3, so $\sum_{k>1}\tilde q^k < \sum_{m\ \mathrm{odd}\ge 3}\tilde q^m = \tilde q\cdot\frac{\tilde q^2}{1-\tilde q^2}\le\tilde q$. □

For odd n the weights are negative exactly when n has an odd number of distinct prime factors, beginning with $n=3,5,7,9,11,13$; for even n other than powers of 2 the sign depends on how the 2 adic parts combine (for $p=2$, $b_6,b_{10},b_{12},b_{14}<0$ while $b_{30},b_{42}>0$). Among $2\le n\le 60$ there are 42 negative exponents for each of $p=2,3,5,7$. Section 7.1.3 returns to what these fractional and negative multiplicities mean for the zeros.

The weights B_N and d_m also govern the truncation of the sawtooth expansion of S. The sawtooth Σ_λ of Lemma 6.1.7 has the Fourier series $\Sigma_\lambda(\tau)=\sum_{j\ge1}(-1)^j\frac{\sin(j\lambda\tau)}{\pi j}$, so the formal series $\sum_N B_N\Sigma_{\log N}(\tau)$ is a double series in (N,j) whose frequencies are $j\log N$.

Proposition 7.1.10. *Truncate the double series by frequency, keeping the terms with $j\log N\le L$. Then*

$$\sum_{\substack{N,j\\ j\log N\le L}} B_N\,(-1)^j\frac{\sin(j\tau\log N)}{\pi j} = -\frac{1}{\pi}\sum_{n\le e^L}\frac{\Lambda(n)}{\sqrt{n}\log n}\sin(\tau\log n),$$

the classical truncated series for $S(\tau)$. Truncation by $N\le N_0$ instead is not equivalent: it keeps the harmonics $j\log N>\log N_0$ of each retained sawtooth without the terms of larger N that cancel them.

Proof. With $N=n^m$, the frequency is $r\log n$ with $r=mj$, and the constraint $j\log N\le L$ is $r\log n\le L$, which does not separate m from j. The coefficient of $\sin(r\tau\log n)$ is $-\frac{a_n}{\pi}\sum_{mj=r}d_m\frac{(-1)^{j+1}}{j}$, and $\sum_{mj=r}d_m(-1)^{j+1}j^{-1}$ has Dirichlet series $\eta(1+v)^{-1}\eta(1+v)=1$, so it is $\delta_{r,1}$. □

So the sawtooth expansion contains nothing that the classical series lacks once it is truncated consistently, and truncating it by the index N is worse. Numerically, on a uniform grid of step 10^{-3} in $\tau\in[10,50]$ and with the Cesàro weight $1-\log N/\log N_0$, the root mean square error against $S(\tau)$ is 0.228, 0.223, 0.218 for $N_0=10^3,10^4,10^5$, against 0.146, 0.128, 0.115 for the classical series with the same weight, the root mean square of S itself being 0.320; the script `sawtooth_truncation.py` computes these values, and the step 5×10^{-4} changes none of their digits. The bound $|\Sigma_\lambda|\le\frac12$ does not help either, since $\frac12\sum_{N\le N_0}|B_N|$ is 9.0, 18.5, 40.6 at the same N_0 and grows like $\sqrt{N_0}/\log N_0$.

7.1.2.2 The cyclotomic form, the prime number theorem and a trace formula

The cyclotomic identity gives a second factorisation, into the blocks $1-N^{-u}$:

$$\frac{1}{\zeta(\frac{1}{2}+u)} = \prod_p \prod_{k\geq 1} \bigl(1-p^{-ku}\bigr)^{M_k(p^{-1/2})}, \qquad \operatorname{Re} u > \tfrac{1}{2}.$$

Each block $1-e^{-\lambda u} = 2e^{-\lambda u/2}\sinh(\lambda u/2)$ is again a solved model, now with a sinh lattice: with $u=\sqrt{s}$, $\frac{\sinh(\lambda\sqrt{s}/2)}{\lambda\sqrt{s}/2} = \prod_{m\geq 1}\bigl(1+\frac{s}{(2\pi m/\lambda)^2}\bigr)$, a reciprocal GGC with unit atoms at $(2\pi m/\lambda)^2$. The sinh blocks are the local zeta factors on the line $\sigma=0$, and unlike the sech blocks they vanish at $u=0$. Normalising the product at $u\to 0$ therefore requires two identities.

Proposition 7.1.11. *For $0<q<1$, summing over k in increasing order,*

$$\sum_{k\geq 1} M_k(q) = 0, \qquad \sum_{k\geq 1} M_k(q)\log k = \log(1-q).$$

Both follow from the prime number theorem in the forms $\sum_m \mu(m)/m = 0$ and $\sum_m \mu(m)\log m/m = -1$.

Proof. $\sum_{k\leq K} M_k(q) = \sum_d \frac{q^d}{d}\sum_{m\leq K/d}\frac{\mu(m)}{m}$. The inner sums are bounded by 1 in absolute value and tend to 0, so dominated convergence gives 0. Similarly $\sum_{k\leq K} M_k(q)\log k = \sum_d \frac{q^d}{d}\sum_{m\leq K/d}\frac{\mu(m)\log(dm)}{m} \to \sum_d \frac{q^d}{d}(-1) = \log(1-q)$, using $\sum \mu(m)/m = 0$ for the $\log d$ part. □

The two identities are the per prime analogues of these forms of the prime number theorem, and heuristically they are what makes the cyclotomic product of solved models converge at the edge $u\to 0$: $\prod_k (1-y^k)^{-M_k(q)}$ behaves near $y=1$ like $(1-y)^{-\sum M_k}\prod_k k^{-M_k}$, and the finite limit $\frac{1}{1-q}$ requires both. The sech form of Theorem 7.1.8 carries the same information in the value $1/\eta(1) = 1/\log 2$ at $v=0$ of the continuation of $\sum_m d_m m^{-v} = 1/\eta(1+v) = 1/((1-2^{-v})\zeta(1+v))$; the series $\sum_m d_m$ itself diverges, since $d_{2^k} = \frac{1}{2}$ for every $k\geq 1$ (Proposition 6.1.6).

In heat trace form the Vere-Jones representation becomes a trace formula. Proposition 6.1.13 and Poisson summation give, for every $x>0$,

$$\sum_{\gamma>0} e^{-x\gamma^2} - W_0(x) = e^{x/4} + \sum_p \sum_{n\geq 1} b_n(p^{-1/2})\Bigl[\sum_{k\geq 0} e^{-x\left(\frac{(2k+1)\pi}{n\log p}\right)^2} - \frac{n\log p}{4\sqrt{\pi x}}\Bigr], \quad (7.1.3)$$

where the left side is written under RH and is $W(x)-W_0(x)$ in general, and the double series converges absolutely once each bracket is written in its theta form. The left side is the heat trace of the zeros net of its smooth part; the right side is the pole plus the heat traces of the arithmetic progressions $\frac{(2k+1)\pi}{n\log p}$, the pseudo zeros of the blocks, each net of its own mean. So in heat trace the zeros of ζ

are a signed superposition of arithmetic progressions with spacings $2\pi/(n\log p)$, the progression of p^n entering with the necklace weight $b_n(p^{-1/2})$. Heuristically, constructive interference of the progressions places the mass at the ordinates γ, and the negative weights of Proposition 7.1.9 remove the mass from the other points of the progressions.

The trace formula does not give positivity of the Thorin measure of R. The obstruction is explicit: the weights $b_n(p^{-1/2})$ are not nonnegative integers, for odd n their signs follow the parity of $\omega(n)$, and the passage from the binary product at $q=1$, where every multiplicity is 0 or 1, to the critical line at $q=p^{-1/2}$ is a deformation of the exponents in which integrality and positivity are lost together. A proof of RH along these lines would have to show that the interference of the progressions in (7.1.3), with exactly these weights, produces a positive measure; Section 7.2.5 poses the deformation from $q=1$ to $q=p^{-1/2}$ as a problem.

7.1.2.3 The Lévy and Thorin representations across the strip

For $\sigma > 1$ the Vere-Jones representation is a statement about the classical zeta distribution (Section 6.1.6): $\zeta(\sigma+i\tau)/\zeta(\sigma) = \mathbb{E}e^{-i\tau\log N_\sigma}$ where $\Pr(N_\sigma = n) = n^{-\sigma}/\zeta(\sigma)$, and $\log N_\sigma$ is compound Poisson with Lévy measure $\nu_\sigma = \sum_n \frac{\Lambda(n)}{n^\sigma \log n}\delta_{\log n}$. Theorem 7.1.8 decomposes this Lévy measure as

$$\nu_\sigma = \sum_p \sum_{n\geq 1} b_n(p^{-\sigma})\,\nu^{\mathrm{alt}}_{n\log p}, \qquad \nu^{\mathrm{alt}}_\lambda = \sum_{k\geq 1} \frac{(-1)^{k+1}}{k}\,\delta_{k\lambda},$$

where $\nu^{\mathrm{alt}}_\lambda$ is the formal Lévy measure of the fair coin λB, $B\in\{0,1\}$, whose characteristic function $\frac{1}{2}(1+e^{-i\lambda\tau})$ is the sech block. At $\sigma=0$ the weights $b_n(1)$ are 1 at the powers of 2 and 0 otherwise, and the decomposition says, formally (there is no uniform law on the integers), that a uniformly random integer has independent fair binary digits in each prime exponent. For $\sigma>1$ there is also a positive decomposition: by $\frac{1}{1-qy} = \prod_j (1+q^{2^j}y^{2^j})$, the exponent of p in N_σ has independent binary digits with $\Pr(\text{digit } j=1) = \frac{p^{-2^j\sigma}}{1+p^{-2^j\sigma}}$. The Vere-Jones form keeps the coins fair and moves $p^{-\sigma}$ into the multiplicities; the binary digit form keeps unit multiplicities and biases the coins.

Inside the critical strip neither form survives as a characteristic function, since $|\zeta(\sigma+i\tau)|$ is unbounded in τ for $0<\sigma<1$ and the Dirichlet series diverges (Section 6.1.6); the heat kernel supplies the replacement. For $0<\sigma<1$ put

$$\varphi_\sigma(s) = \frac{\zeta(\sigma)}{(1-s/(1-\sigma)^2)\,\zeta(\sigma+\sqrt{s})}, \qquad \varphi_{1/2}=\varphi.$$

THEOREM 7.1.12. *For $0 < \sigma < 1$ and $s > (1-\sigma)^2$, $-\varphi_\sigma'/\varphi_\sigma(s) = \int_0^\infty e^{-sx} w_\sigma(x)\,dx$, where w_σ has the three equivalent forms*

$$\begin{aligned} w_\sigma(x) &= e^{(1-\sigma)^2 x} - \frac{1}{\sqrt{4\pi x}} \sum_{n\ge 2} \frac{\Lambda(n)}{n^\sigma} e^{-(\log n)^2/4x} \\ &= e^{(1-\sigma)^2 x} + \sum_p \sum_{n\ge 1} b_n(p^{-\sigma})\, h_{n\log p}(x) \\ &= \tfrac12 e^{(1-\sigma)^2 x} \operatorname{erfc}\big((1-\sigma)\sqrt{x}\big) + \sum_{\gamma>0} \operatorname{Re}\Big[e^{a_\gamma^2 x} \operatorname{erfc}(a_\gamma\sqrt{x})\Big] \\ &\qquad + \frac{\gamma_E + \log\pi}{4\sqrt{\pi x}} + \sum_{k\ge 0}\Big[\tfrac12 e^{a_k^2 x}\operatorname{erfc}(a_k\sqrt{x}) - \frac{1}{4(k+1)\sqrt{\pi x}}\Big], \end{aligned}$$

with h_λ the Jacobi theta function of Proposition 6.1.13, $a_\gamma = \sigma - \frac12 - i\gamma$ (the last form written under RH; in general γ runs over $-i(\rho - \frac12)$), and $a_k = 2k+2+\sigma$. As $x\to\infty$, $w_\sigma(x) \sim \frac{\zeta'(\sigma)/\zeta(\sigma)}{2\sqrt{\pi x}}$, with $\zeta'(\sigma)/\zeta(\sigma) > 0$ on $(0,1)$. If $w_\sigma \ge 0$, then φ_σ is the Laplace transform of an infinitely divisible law on $(0,\infty)$ with Lévy density $w_\sigma(x)/x$:

$$\log\varphi_\sigma(s) = -\int_0^\infty \big(1 - e^{-sx}\big)\frac{w_\sigma(x)}{x}\,dx, \qquad \operatorname{Re} s > 0.$$

Proof. The first form is the argument of Theorem 6.1.11 with the pole now at $s = (1-\sigma)^2$. For the second, regroup $\sum_n b_n(q)\, n\log p \sum_j (-1)^j e^{-(jn\log p)^2/4x}$ by $r = jn$: with $nb_n = \sum_{ab=n} a\, d_a q^b$ the coefficient of $e^{-(r\log p)^2/4x}$ is $\sum_{b|r} q^b \sum_{ja=r/b} a\, d_a (-1)^j = -q^r$, since the inner sum is $-\delta_{r/b,1}$ (Proposition 6.1.13); this is the prime p part of the first form. For the third, insert the Hadamard product of ζ'/ζ into $-\varphi_\sigma'/\varphi_\sigma = \frac{1}{s-(1-\sigma)^2} + \frac{1}{2\sqrt{s}}\frac{\zeta'}{\zeta}(\sigma+\sqrt{s})$; the pole terms combine to $\frac{1}{2\sqrt{s}(\sqrt{s}+1-\sigma)}$, and each $\frac{1}{\sqrt{s}(\sqrt{s}+a)}$ is the transform of $e^{a^2x}\operatorname{erfc}(a\sqrt{x})$. At $\sigma = \frac12$ this reduces to $w = W - W_0$ with Proposition 6.2.2.

The asymptotic follows from the third form term by term. Uniformly for $|\arg a| \le \frac{3\pi}{4} - \epsilon$ and $|a|\sqrt{x} \ge 1$,

$$\tfrac12 e^{a^2x}\operatorname{erfc}(a\sqrt{x}) = \frac{1}{2a\sqrt{\pi x}} + O\big(|a|^{-3}x^{-3/2}\big);$$

every a in the third form has $|\arg a| < \frac{\pi}{2} + \frac{1}{14}$, and $\sum |a|^{-3}$ converges. So $w_\sigma(x) = C_\sigma/(2\sqrt{\pi x}) + O(x^{-3/2})$ with

$$C_\sigma = \frac{1}{1-\sigma} + \sum_\rho \frac{1}{\sigma-\rho} + \frac{\gamma_E + \log\pi}{2} + \sum_{k\ge 0}\Big(\frac{1}{2k+2+\sigma} - \frac{1}{2k+2}\Big),$$

the sum over ρ taken in conjugate pairs, and $C_\sigma = \zeta'(\sigma)/\zeta(\sigma)$ by the Hadamard product of ξ. Infinite divisibility follows from $w_\sigma \ge 0$ by Landau's theorem as in

Theorem 7.1.1, since φ_σ is analytic near $(0,\infty)$: $\zeta(\sigma+x)\neq 0$ for $x>0$ and the pole at $x=1-\sigma$ is cancelled. □

Numerically w_σ is positive on every line we tried. We evaluated it on 300 points of $x\in[10^{-4},10^3]$, from the prime form for $x\le 0.02$ and the zero form beyond, for $\sigma=0.05,0.1,0.25,0.4,0.5,0.6,0.75,0.9,0.99$; the two forms agree to 5×10^{-6} at $x=0.01$ and 0.05. In every case the minimum is attained at the largest x, and $2\sqrt{\pi x}\,w_\sigma(x)$ at $x=10^3$ equals 1.872, 1.913, 2.087, 2.377, 2.682, 3.154, 4.596, 10.16, 41.1, against $\zeta'(\sigma)/\zeta(\sigma)$, which is 1.873, 1.914, 2.088, 2.379, 2.686, 3.162, 4.628, 10.60, 100.58 at the same nine values of σ. The agreement is close up to $\sigma=0.75$ and breaks down as $\sigma\to1$, where the pole makes the approach slow and the asymptotic has not been reached at $x=10^3$. So $w_\sigma>0$ is checked only for $x\le10^3$ and at nine values of σ. Whether the Lévy density is positive on every vertical line of the strip is open (Problem 7.8).

For the Thorin representation two families have to be distinguished: φ_σ, which carries the gamma factor and the pole, and the completed function $R_\sigma(s)=\xi(\sigma)/\xi(\sigma+\sqrt{s})$. For $\sigma>\frac12$ the second is, with no hypothesis, the Laplace transform of an infinitely divisible law obtained by tilting and subordinating the law of X_R (Theorem 9.2.5 of Section 9.2 and the proof of Proposition 10.1.1 of Section 10.1, where R_σ is written φ_α). The first never has the Thorin property, and the proof uses no computation and nothing about ζ beyond Theorem 7.1.12.

Proposition 7.1.13. *For every $\sigma\in(0,1)$, φ_σ is not the Laplace transform of a GGC, and if it is infinitely divisible it is not self decomposable.*

Proof. The argument is that of Theorem 7.1.3. By Theorem 7.1.12, $-\varphi_\sigma'/\varphi_\sigma(s)=\int_0^\infty e^{-sx}w_\sigma(x)\,dx$ for $s>(1-\sigma)^2$, with

$$w_\sigma=e^{(1-\sigma)^2x}-P_\sigma,\qquad P_\sigma(x)=\frac{1}{\sqrt{4\pi x}}\sum_{n\ge2}\frac{\Lambda(n)}{n^\sigma}e^{-(\log n)^2/4x}.$$

Every derivative of P_σ vanishes at 0^+, so $w_\sigma(x)=1+(1-\sigma)^2x+o(x)$ and w_σ strictly increases near the origin. If φ_σ were a GGC transform, then $-\varphi_\sigma'/\varphi_\sigma=a+\int U(dt)/(s+t)$ with $a\ge0$ and $U\ge0$ satisfying the Thorin integrability conditions, so that $k_U(x)=\int e^{-tx}U(dt)$ is integrable near 0 and, by Tonelli, $\int U(dt)/(s+t)=\int_0^\infty e^{-sx}k_U(x)\,dx$ for $s>(1-\sigma)^2$. Letting $s\to\infty$ gives $a=0$, and uniqueness of the Laplace transform gives $w_\sigma=k_U$, which is completely monotone and so nonincreasing, a contradiction. If φ_σ is infinitely divisible, the same uniqueness identifies its Lévy density as $w_\sigma(x)/x$, and self decomposability would require w_σ to be nonincreasing. □

As on the critical line, the obstruction is the factor $1-s/(1-\sigma)^2$ that removes the pole, and it says nothing about the zeros. Beyond it, for $\frac12<\sigma<1$ with $\sup_\rho\operatorname{Re}\rho<\sigma$, the Thorin density $\frac{1}{2\pi\sqrt{t}}\operatorname{Re}\frac{\zeta'}{\zeta}(\sigma+i\sqrt{t})$ given by

Lemma 6.1.1 also takes negative values on every line we examined (the minimum of $\operatorname{Re}\frac{\zeta'}{\zeta}(\sigma+i\tau)$ over $\tau\in(0,60]$ is -1.049, -0.995, -0.867, -0.769, -0.721 at $\sigma=0.55,0.6,0.75,0.9,0.99$), but the proposition does not need this.

The completed functions R_σ carry the zeros. Theorem 7.1.14 is Theorem 1.2.2 of Section 1.2 in the variable of this chapter.

THEOREM 7.1.14. *Let $\sigma\ge\frac12$. Then $R_\sigma(s)=\xi(\sigma)/\xi(\sigma+\sqrt{s})$ is the Laplace transform of a GGC if and only if ζ has no zero ρ with $\operatorname{Re}\rho>\sigma$. In that case its Thorin measure is the Poisson smoothed zero density*

$$U_\sigma(dt)=\frac{1}{2\pi\sqrt{t}}\sum_\rho\frac{\sigma-\beta_\rho}{(\sigma-\beta_\rho)^2+(\sqrt{t}-\gamma_\rho)^2}\,dt,$$

the sum running over all nontrivial zeros $\rho=\beta_\rho+i\gamma_\rho$, counted with multiplicity, of both signs of γ_ρ, and taken symmetrically; each zero on $\operatorname{Re}\rho=\sigma$ with $\gamma_\rho>0$ contributes a unit atom at γ_ρ^2 instead of its kernel. Consequently R_σ is a GGC for every $\sigma\ge1$, unconditionally, and

$$\Theta:=\sup_\rho\operatorname{Re}\rho=\inf\{\sigma\ge\tfrac12:\ R_\sigma\ \textit{is a GGC}\}.$$

RH is the statement $\Theta=\frac12$, that is, that $R_{1/2}=R$ is a GGC.

Proof. A zero with $\operatorname{Re}\rho>\sigma$ gives $\sqrt{s}=\rho-\sigma$ on the principal branch and hence a pole of R_σ in $\mathbb{C}\setminus(-\infty,0]$, so R_σ is not a GGC. Suppose there is none. Write the zeros with positive ordinate, counted with multiplicity, as $\rho=\beta+i\gamma$, put $a=\sigma-\beta\ge0$, and let $\mathcal{P}_a(x)=a/\{\pi(a^2+x^2)\}$, for $a>0$, be the Cauchy density of scale a (the Poisson kernel P_a of Section 1.2). Hadamard's product [63, Chapter 12], with each ρ paired with $\bar\rho$, gives $\xi(z)=\xi(0)\prod_{\gamma>0}(1-z/\rho)(1-z/\bar\rho)$, absolutely convergent since $\sum_\rho|\rho|^{-2}<\infty$, and hence for $s>0$ and $q=\sqrt{s}$

$$R_\sigma(s)=\prod_{\gamma>0}\frac{a^2+\gamma^2}{(q+a)^2+\gamma^2}.$$

For $a>0$ let $U_{a,\gamma}(dt)=\{\mathcal{P}_a(\sqrt{t}-\gamma)+\mathcal{P}_a(\sqrt{t}+\gamma)\}\,dt/(2\sqrt{t})$, and let $U_{0,\gamma}=\delta_{\gamma^2}$. The substitution $t=\tau^2$, the identity $1/(s+\tau^2)=(\pi/q)\mathcal{P}_q(\tau)$ and $\mathcal{P}_a*\mathcal{P}_q=\mathcal{P}_{a+q}$ give

$$\int_0^\infty\frac{U_{a,\gamma}(dt)}{s+t}=\frac{q+a}{q\{(q+a)^2+\gamma^2\}}=\frac{d}{ds}\log\{(q+a)^2+\gamma^2\},$$

which also holds for $a=0$. Integrating over $(0,s]$ with Tonelli, $\int\log(1+s/t)\,U_{a,\gamma}(dt)=\log\{((q+a)^2+\gamma^2)/(a^2+\gamma^2)\}$, which is nonnegative and at most $(s+2\sigma\sqrt{s})/\gamma^2$. Since $\sum_{\gamma>0}\gamma^{-2}<\infty$, Tonelli's theorem shows that $U_\sigma=\sum_{\gamma>0}U_{a,\gamma}$ is a positive measure with $\int\log(1+s/t)\,U_\sigma(dt)=-\log R_\sigma(s)<\infty$, which gives the Thorin integrability conditions, and that

$R_\sigma(s) = \exp\{-\int \log(1+s/t)\, U_\sigma(dt)\}$ for $s > 0$. So R_σ is the Laplace transform of a GGC with Thorin measure U_σ, whose density is the stated sum, grouped by the pairs $\rho, \bar\rho$, and each zero on $\operatorname{Re}\rho = \sigma$ with $\gamma_\rho > 0$ gives the atom $\delta_{\gamma_\rho^2}$. For $\sigma \geq 1$ the hypothesis holds by the Euler product.

For $\sigma > \Theta$ the density also follows from Lemma 6.1.1: then $-R'_\sigma/R_\sigma(s) = \frac{1}{2\sqrt{s}}\frac{\xi'}{\xi}(\sigma+\sqrt{s})$ is $O(|s|^{-1/2})$ near 0, $O(|s|^{-1/2}\log|s|)$ at infinity uniformly on $\mathbb{C}\setminus(-\infty,0]$ (by the partial fraction formula for ζ'/ζ [63, Chapter 15], since every zero lies at distance at least $\sigma-\Theta$ from the half plane $\operatorname{Re} z \geq \sigma$), and continuous up to the cut away from 0, and at $\sqrt{s} = i\tau$ the lemma gives the density $\frac{1}{2\pi\tau}\operatorname{Re}\frac{\xi'}{\xi}(\sigma + i\tau) = \frac{1}{2\pi\tau}\sum_\rho \operatorname{Re}\frac{1}{\sigma+i\tau-\rho}$, summed symmetrically. □

Moving the line to the right of all zeros replaces the atoms δ_{γ^2} by Poisson kernels of width $\sigma - \beta_\rho$ in $\sqrt{t}$, and the GGC property then holds. The kernels of the zeros with $\gamma_\rho < 0$ are centred at negative values of $\sqrt{t}$ and are positive on $t > 0$; they disappear only in the limit $\sigma \downarrow \beta_\rho$. RH says that the Poisson kernels can be sharpened all the way to atoms at $\sigma = \frac{1}{2}$ without any of them acquiring a negative width. The quasi Riemann Hypothesis $\Theta < 1$, announced in October 2026 with $\Theta \leq \frac{7}{8}$ (Section 12.1.5), is the statement that R_σ is a GGC for some $\sigma < 1$; it is step 7 of the path to RH in Section 12.1 (Table 12.1), and short of it Section 11.4 shows with no hypothesis that for $\frac{1}{2} < \sigma < 1$, in the density $\rho_{\sigma-1/2}$ of (1.2.7), the Cauchy kernels of the zeros to the right of σ, if there are any, carry below height T a vanishing fraction of the mass of the kernels of the other zeros as $T \to \infty$. Numerically $\min_{0<\tau\leq 70} \operatorname{Re}\frac{\xi'}{\xi}(\sigma+i\tau)$ is 4.6×10^{-5}, 0.0046, 0.0116, 0.0231 at $\sigma = 0.501, 0.6, 0.75, 1$, attained as $\tau \to 0$.

7.1.3 Pólya and Cardon theory for the solved models

Pólya's theory of Fourier transforms with only real zeros, and its extension by Cardon, is built from the objects that the Vere-Jones step produces: cosines, hyperbolic cosines and their products. The solved models are special cases of that theory; ζ is not, because its multiplicities are neither integers nor nonnegative; and the theory suggests a construction, the Cardon route, whose difficulty at the boundary of a strip points to the lattice of Biane, Pitman and Yor. Throughout, Φ is the Pólya kernel and $\Xi(\tau) = \int \Phi(t)\cos(\tau t)\, dt$, as in (1.2.3) with the variable written t; Φ is even, analytic in the strip $|\operatorname{Im} t| < \pi/4$, and has a natural boundary on the lines $\operatorname{Im} t = \pm\pi/4$ [219, Chapter 11, Section 1.4, and Chapter 5, Section 3.2]. The Pólya variable X of Section 1.2 has density $\Phi/\Xi(0)$ and moment generating function $\mathbb{E}e^{hX} = \xi(\frac{1}{2}+h)/\xi(\frac{1}{2})$, by (1.2.4). In this subsection G denotes a function of the Laguerre Pólya class, not the gamma factor.

7.1.3.1 Solved models and the Laguerre Pólya class

On the critical line the block $1 + N^{-\sqrt{s}}$ becomes

$$1 + N^{-i\tau} = 2e^{-i\tau\ell/2}\cos\big(\tfrac{\ell\tau}{2}\big), \qquad \ell = \log N,$$

and $\cos(\ell\tau/2) = \mathbb{E}e^{i\tau(\ell/2)\varepsilon}$ with $\varepsilon = \pm 1$ equally likely. A product of blocks with nonnegative integer multiplicities m_N is, up to the phase, the characteristic function of the Rademacher sum $\sum_N \sum_{j \le m_N} \frac{\log N}{2}\varepsilon_{N,j}$, and all its zeros are real, lying on the lattices $(2k+1)\pi/\log N$. Cardon's theorem on exponential sums [49] extends this: if $a_1, \dots, a_n$ and $b_1, \dots, b_n$ are positive and G is a real entire function of genus 0 or 1 with only real zeros, then

$$\sum_{\varepsilon \in \{\pm 1\}^n} G\Big(i\sum_k a_k\varepsilon_k\Big)\exp\Big(iz\sum_k b_k\varepsilon_k\Big),$$

with the same signs in both sums, has only real zeros.

The function ζ is not of this form. Formally $\zeta(\frac{1}{2} + i\tau) = \prod_N (1 + N^{-i\tau})^{B_N}$, but the exponents B_N are not integers, and some are negative: by Theorem 7.1.8, at prime cubes $B_{p^3} = \frac{1}{3}p^{-3/2} - \frac{1}{3}p^{-1/2} < 0$, beginning with the value $B_8 = -\frac{1}{6\sqrt{2}}$ of Theorem 6.1.8, as already noted in Section 6.2.2. Integer exponents would place every zero on the sech lattices of the blocks. Heuristically, the fractional and negative weights are what moves the zeros of ζ off those lattices and onto the ordinates γ, so the reality of the zeros cannot be read off from the Rademacher structure of the individual blocks.

In the kernel variable the solved models act as multipliers. By Pólya's Hilfssatz II [213, Hilfssatz II], if $a > 0$, b is real and G is a real entire function of genus 0 or 1 with only real zeros, at least one of them, then $e^{ib}G(z+ia) + e^{-ib}G(z-ia)$ has only real zeros. With $G = \Xi$ this says that replacing $\Phi(t)$ by $\Phi(t)\cosh(at)$, a solved model in the variable t, cannot create nonreal zeros: if Ξ has only real zeros, so does the transform of the new kernel. For $a \ge \frac{1}{2}$ the conclusion needs no hypothesis: the transform of $\Phi(t)\cosh(at)$ is $\frac{1}{2}(\xi(s+a) + \xi(s-a))$ with $s = \frac{1}{2} + iz$, and Lagarias proved that for $a \ge \frac{1}{2}$ its zeros all lie on the critical line and are simple [134]. Products of such factors converge to Gaussians, $\prod_{k \le n}\cosh(ct/\sqrt{n}) \to e^{c^2t^2/2}$, and this is de Bruijn's route to the Gaussian multipliers $e^{\lambda t^2}$ [65]. In the variable t, multiplying $\Phi(t)$ by $e^{\lambda t^2/4}$ is the de Bruijn Newman flow at time λ (Section 8.4), the flow of Theorem 7.2.15, so the multiplier $e^{\lambda t^2}$ is the flow at time 4λ. The solved models are therefore the elementary steps of the de Bruijn Newman flow, and the bound $\Lambda \le 0.22$ [217], improved to $\Lambda \le 0.2$ by Platt and Trudgian [196], says that the flow at time 0.2, applied to Φ or, in the normalisation of Rodgers and Tao, to the rescaled kernel Φ_{dBN} defined before Theorem 7.2.15, already produces a kernel whose transform has only real zeros.

Proposition 7.1.15. *Let G be a real even entire function of order less than 2 with $G(0) \neq 0$. Then G is in the Laguerre Pólya class if and only if $s \mapsto G(0)/G(i\sqrt{s})$ is the Laplace transform of a GGC whose Thorin measure is $\sum_k m_k \delta_{\alpha_k^2}$ with integers $m_k \geq 1$, where $\pm\alpha_k$ are the zeros of G with multiplicities m_k.*

Proof. By Hadamard, $G(z) = G(0)\prod_k (1 - z^2/\alpha_k^2)^{m_k}$ with no exponential factor, since G is even and of order less than 2. All α_k are real if and only if $G(0)/G(i\sqrt{s}) = \prod_k (1 + s/\alpha_k^2)^{-m_k}$ is a product of gamma transforms, which is the stated GGC. If some α_k^2 is not positive, the product has a zero or a pole off $(-\infty, 0]$, which a GGC transform cannot have. □

The Laguerre Pólya class consists of the locally uniform limits of real polynomials with only real zeros, and a real entire function of order less than 2 belongs to it exactly when its zeros are real. In the language of Proposition 7.1.15, Pólya's multiplier theorem [50, Proposition 1], for even G of order less than 2, reads: if f is even and integrable with $f(t) = O(e^{-|t|^b})$ for some $b > 2$, $\int f(t)e^{izt}dt$ has only real zeros, and $G(0)/G(i\sqrt{s})$ is an integer atom GGC, then $\int G(it)f(t)e^{izt}dt$ has only real zeros. RH is the statement that $R = \xi(\frac{1}{2})/\xi(\frac{1}{2} + \sqrt{s})$ is an integer atom GGC; its atoms are at the γ^2, with mass the multiplicity of the zero. The Thorin measures in the Pólya theory are always integer valued, which is the reason the continuous Thorin densities of the gamma factor and of the sawtooth measures do not produce multipliers directly.

7.1.3.2 Lee and Yang, and Newman

Pólya's Hilfssatz II has a physical reading, to which Kac drew attention in his comments on Pólya's 1926 paper [215, pp. 424 to 426]: it is the one spin case of the Lee and Yang circle theorem for the Ising ferromagnet [141]. Cardon's exponential sums make the correspondence explicit. For $\varepsilon \in \{\pm 1\}^n$ the sum

$$\mathcal{Z}_n(z) = \sum_{\varepsilon} G\Big(i\sum_k a_k\varepsilon_k\Big)\exp\Big(iz\sum_k b_k\varepsilon_k\Big)$$

is a partition function. The spins are weighted by $G(i\sum_k a_k\varepsilon_k)$, a function of a magnetisation, and they feel the purely imaginary external fields izb_k. For the Gaussian member of the Laguerre Pólya class, $G(x) = e^{-\beta x^2/2}$, which has genus 2 and lies outside Cardon's theorem, the weight is $\exp(\beta(\sum_k a_k\varepsilon_k)^2/2)$, the Curie Weiss ferromagnet with couplings $\beta a_j a_k \geq 0$. The zeros of $\mathcal{Z}_n$ in z are then real by the Lee and Yang theorem [141], since off the real axis the fields izb_k have real parts of one sign. A general even G in the class is a limit of products of such Gaussians and of factors $1 + M^2/\alpha^2$ in the magnetisation $M = \sum_k a_k\varepsilon_k$, and Cardon's theorem is the Lee and Yang theorem for this family of ferromagnetic interactions of the magnetisation.

Newman made the connection with the Riemann Hypothesis precise [163]. In a 1974 paper on generalised Ising systems [161] he had characterised the single spin laws with the Lee and Yang property: a symmetric single spin law ρ with $\int e^{bt^2}\rho(dt) < \infty$ for every b has the Lee and Yang property, that ferromagnetic systems of ρ spins have zeros only on the imaginary field axis, exactly when its moment generating function $\int e^{ht}\rho(dt)$ has only purely imaginary zeros, that is when $\int e^{izt}\rho(dt)$ is in the Laguerre Pólya class. In Thorin language this reads as follows.

PROPOSITION 7.1.16. *Let ρ be a symmetric law with Gaussian moments of all orders. Then ρ has the Lee and Yang property if and only if $s \mapsto 1/\int e^{\sqrt{s}t}\rho(dt)$ is the Laplace transform of a GGC with zero drift whose Thorin measure has integer atoms. The atoms are at the squared zeros α_k^2 of $\int e^{izt}\rho(dt)$, with the multiplicities of the zeros.*

Proof. By Newman's theorem [161] the Lee and Yang property holds exactly when $\int e^{izt}\rho(dt)$ is in the Laguerre Pólya class, and then, by Hadamard, $\int e^{ht}\rho(dt) = e^{bh^2}\prod_k(1 + h^2/\alpha_k^2)^{m_k}$ with $b \geq 0$. In fact $b = 0$: for every $\epsilon > 0$, $e^{ht} \leq e^{\epsilon h^2 + t^2/4\epsilon}$ gives $\int e^{ht}\rho(dt) \leq C_\epsilon e^{\epsilon h^2}$ for real h, while the right side is at least e^{bh^2}. With $h = \sqrt{s}$ the reciprocal is $\prod_k(1 + s/\alpha_k^2)^{-m_k}$, a GGC with zero drift and Thorin measure $\sum m_k\delta_{\alpha_k^2}$. Conversely, as in Proposition 7.1.15, a GGC transform with zero drift and integer atoms m_k at α_k^2 is $\prod_k(1 + s/\alpha_k^2)^{-m_k}$, whose reciprocal at $s = -z^2$ is $\prod_k(1 - z^2/\alpha_k^2)^{m_k}$, a function in the Laguerre Pólya class. □

Apply this to the Pólya variable X: since $\mathbb{E}e^{hX} = \xi(\frac{1}{2} + h)/\xi(\frac{1}{2})$, we have $1/\mathbb{E}e^{\sqrt{s}X} = R(s)$, the reciprocal Xi function of Section 6.1.1. Three statements therefore coincide: RH; the law of X is a Lee and Yang single spin law; R is a GGC with no drift whose Thorin measure has an atom at each γ^2, of mass the multiplicity of the zero. The de Bruijn Newman flow at time λ multiplies $\Phi(t)$ by $e^{\lambda t^2/4}$, which in Ising terms adds the ferromagnetic self interaction $e^{\lambda t^2/4}$ to the Pólya variable viewed as a spin, and Λ is the least λ for which the resulting spin is Lee and Yang. Newman's conjecture $\Lambda \geq 0$, proved by Rodgers and Tao [224], says that no antiferromagnetic self interaction can be tolerated.

7.1.3.3 A Cardon route

Cardon's Fourier theorem [50, Theorem 1] gives a constructive sufficient condition. If

$$\Phi(t)\,dt = \Xi(0)\,G(it)\,\Pr\Big(\sum_k a_k\varepsilon_k \in dt\Big), \tag{7.1.4}$$

where the ε_k are independent fair signs, the $a_k > 0$ are nonincreasing, and G is real, even, entire of order less than 2 with only real zeros, then Ξ has only real zeros. In the Ising reading, (7.1.4) says that the Pólya variable is the magnetisation of a

weighted Ising paramagnet with weights a_k, tilted by a ferromagnetic interaction $G(iM)$ of its magnetisation $M = \sum_k a_k\varepsilon_k$. Since $|G(it)| \geq |G(0)|$ for even G in the class, the law of $\sum_k a_k\varepsilon_k$ must carry the doubly exponential decay of Φ, and among Rademacher laws the random harmonic series does.

PROPOSITION 7.1.17. *For $\kappa > -1$ let $H_\kappa = \sum_{k\geq 1} \varepsilon_k/(2(k+\kappa))$, with characteristic function $\hat{f}_\kappa(v) = \prod_k \cos(v/(2(k+\kappa)))$ and density f_κ.*

(i) As $s \to \infty$, $\log \mathbb{E}e^{sH_\kappa} = \frac{s}{2}\log s + A(\kappa)s + o(s)$ with $A(\kappa) = A(0) - \frac{1}{2}(\psi(1+\kappa) + \gamma_E)$ (here ψ is the digamma function), $A(0) = -0.1485757\ldots$, so $\log \Pr(H_\kappa > u) = -\frac{1}{2}e^{2u-2A(\kappa)-1}(1+o(1))$ as $u \to \infty$. Since $\log \Phi(t) = -\pi e^{2t} + \frac{9}{2}t + O(1)$ as $t \to +\infty$, the double exponential terms agree exactly when $A(\kappa) = -\frac{1}{2}(1 + \log 2\pi)$, that is at $\kappa^* = 6.61845\ldots$.

(ii) $\limsup_{|v|\to\infty} |v|^{-1} \log|\hat{f}_\kappa(v)| = -\pi/4$ for every κ. Hence f_κ is analytic in the strip $|\operatorname{Im} t| < \pi/4$, which is exactly the strip of analyticity of Φ.

Proof. (i) Put $\ell_\kappa(s) = \log \mathbb{E}e^{sH_\kappa} = \sum_{k\geq 1} \log\cosh \frac{s}{2(k+\kappa)}$. For $\kappa = 0$ put $\lambda = s/2$ and split the sum at $k = \lambda$, using $\log\cosh y = y - \log 2 + \log(1 + e^{-2y})$ for $k \leq \lambda$. Since $\sum_{k\leq\lambda} 1/k = \log\lambda + \gamma_E + O(1/\lambda)$, and since $k \mapsto \log(1+e^{-2\lambda/k})$ and $k \mapsto \log\cosh(\lambda/k)$ are monotone, so that their sums differ from the corresponding integrals by $O(1)$,

$$\ell_0(s) = \lambda\log\lambda + (\gamma_E - \log 2)\lambda + \lambda\int_1^\infty \frac{\log(1+e^{-2y})}{y^2}\,dy + \lambda\int_0^1 \frac{\log\cosh y}{y^2}\,dy + O(1),$$

which is $\frac{s}{2}\log s + A(0)s + O(1)$ with $A(0) = \frac{1}{2}\big(\gamma_E - 2\log 2 + \int_1^\infty y^{-2}\log(1+e^{-2y})\,dy + \int_0^1 y^{-2}\log\cosh y\,dy\big)$. For general κ the mean value theorem gives $s^{-1}(\ell_\kappa - \ell_0)(s) = \sum_k \tanh(s\xi_k)\big(\frac{1}{2(k+\kappa)} - \frac{1}{2k}\big)$ with ξ_k between $\frac{1}{2(k+\kappa)}$ and $\frac{1}{2k}$, and since $\tanh(s\xi_k) \to 1$, dominated convergence gives the limit $\frac{1}{2}\sum_k(\frac{1}{k+\kappa} - \frac{1}{k}) = -\frac{1}{2}(\psi(1+\kappa)+\gamma_E)$. For the tail put $s_u = e^{2u-2A(\kappa)-1}$. Chebyshev's exponential inequality at $s = s_u$ gives $\log\Pr(H_\kappa > u) \leq \ell_\kappa(s_u) - s_u u = -\frac{1}{2}s_u(1+o(1))$. For the lower bound tilt the law of H_κ by $e^{sx-\ell_\kappa(s)}$. By convexity of ℓ_κ the tilted mean is $\ell_\kappa'(s) = \frac{1}{2}\log s + A(\kappa) + \frac{1}{2} + o(1)$, and the tilted variance is $\ell_\kappa''(s) = \sum_k (2(k+\kappa))^{-2}\operatorname{sech}^2\frac{s}{2(k+\kappa)} = O(1/s)$, by comparison with an integral. Given $\delta > 0$, choose s with $\ell_\kappa'(s) = u + \delta/2$; by Chebyshev's inequality the tilted law gives probability at least $\frac{1}{2}$ to $(u, u+\delta]$ for large u, so $\log\Pr(H_\kappa > u) \geq \ell_\kappa(s) - s(u+\delta) - \log 2 = -\frac{1}{2}s_u e^{\delta}(1+\delta)(1+o(1))$. Letting $\delta \to 0$ gives the tail.

(ii) Since $\hat{f}_\kappa$ is even we take $v \to +\infty$. First, $\int_0^\infty \log|\cos y|\, y^{-2}dy = -\pi/2$: as $\log|\cos|$ is even and π periodic and $\sum_{n\in\mathbb{Z}}(y+n\pi)^{-2} = \csc^2 y$, the integral equals $\int_0^{\pi/2}\log(\cos y)\csc^2 y\,dy$, and an integration by parts against $-\cot y$ turns

this into $-\int_0^{\pi/2} \cot y \tan y \, dy = -\pi/2$. For the upper bound fix $M, Y > 0$ and put $g_M = \max(\log|\cos|, -M)$, which is continuous, of bounded variation on $[0, Y]$ and $O(y^2)$ at 0. The points $y_k = v/(2(k+\kappa))$ have spacing at most $2y_k^2/v$, so $v^{-1}\sum_{y_k \le Y} g_M(y_k) \to \frac{1}{2}\int_0^Y g_M(y) y^{-2} dy$ as a Riemann sum. Since $\log|\cos| \le g_M \le 0$, this gives $\limsup_{v\to\infty} v^{-1}\log|\hat f_\kappa(v)| \le \frac{1}{2}\int_0^Y g_M(y) y^{-2} dy$, and letting $Y \to \infty$ and then $M \to \infty$ gives the bound $-\pi/4$. For the lower bound, the same Riemann sum argument, applied to the bounded continuous function $\Psi(y)/y$ with $\Psi(y) = \int_0^y \log|\cos u| \, du$, gives

$$\begin{aligned}\int_0^V \log|\hat f_\kappa(v)| \, dv &= \sum_k 2(k+\kappa)\Psi\Big(\frac{V}{2(k+\kappa)}\Big) \\ &= \frac{V^2}{2}\int_0^\infty \frac{\Psi(y)}{y^3}\, dy + o(V^2) = -\frac{\pi V^2}{8} + o(V^2),\end{aligned}$$

since an integration by parts gives $\int_0^\infty \Psi(y) y^{-3} dy = \frac{1}{2}\int_0^\infty \log|\cos y| \, y^{-2} dy = -\pi/4$. If $\log|\hat f_\kappa(v)| \le -(\pi/4+\epsilon)v$ for all large v, the left side would be at most $-(\pi/8 + \epsilon/2)V^2 + O(1)$. Finally, $|\hat f_\kappa(v)| \le C_\epsilon e^{-(\pi/4-\epsilon)|v|}$ makes the Fourier integral of $\hat f_\kappa$ converge locally uniformly in $|\operatorname{Im} t| < \pi/4 - \epsilon$, so f_κ is analytic there. □

The coincidence of the strips in (ii) has a common source: both densities are Fourier transforms of functions that decay at the exponential rate $\pi/4$ on the real axis. For f_κ this is (ii), and numerically (`harmonic_decay.py`) the maximum of $\log|\hat f_{\kappa^*}(v)|/v$ over the window $0.95V \le v \le 1.05V$ is -0.717, -0.754, -0.770, -0.773 for $V = 100, 300, 600, 1000$ and -0.783 for $V = 10^4$, approaching $-\pi/4 = -0.785$. For Φ, $|\Xi(\tau)| \le e^{-\pi|\tau|/4 + o(|\tau|)}$ by Stirling's formula, since $\zeta(\frac{1}{2} + i\tau)$ grows at most polynomially, and the strip ends at $|\operatorname{Im} t| = \pi/4$ because Φ is a theta series in e^{2t}, which reaches the imaginary axis there.

With both leading orders matched, the question is the ratio

$$r(t) = \frac{\Phi(t)}{\Xi(0) f_{\kappa^*}(t)},$$

which must equal $G(it)$. We computed f_{κ^*} on the real axis by a saddle point contour integral of $\mathbb{E}e^{sH_{\kappa^*}} = \prod_k \cosh(s/(2(k+\kappa^*)))$, with the tail of the product summed through Hurwitz zeta values, and on the imaginary axis by the Fourier integral $f_{\kappa^*}(ix) = \frac{1}{\pi}\int_0^\infty \hat f_{\kappa^*}(v)\cosh(vx)\, dv$. The two methods agree at the origin, $f_{\kappa^*}(0) = 2.104533$, to seven digits. Figure 7.3 plots what they produce: the ratio on the real axis, the Stieltjes check on L', and the continuation G, whose simple zero at $x = 0.6388$ is the first of the real zeros a Laguerre Pólya G would need.

Proposition 7.1.18. *The ratio r satisfies the following necessary conditions for* (7.1.4) *(computed).*

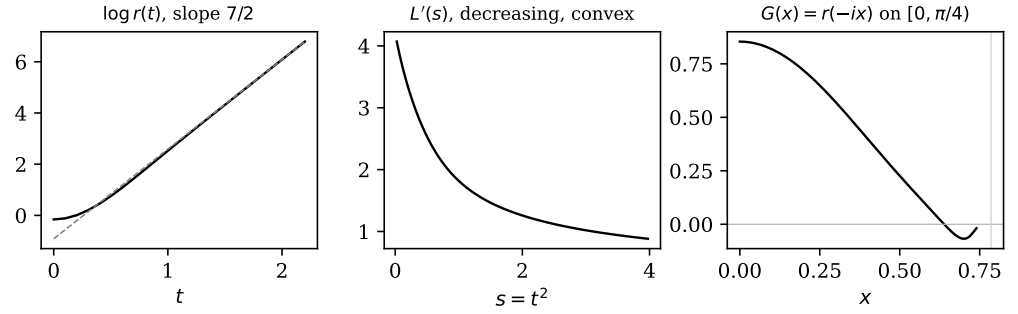


Figure 7.3: The Cardon ratio for the random harmonic law H_{κ^*}. Left: $\log r(t)$ on the real axis, asymptotically linear with slope $\frac{7}{2}$ (dashed). Centre: $L'(s)$ for $L(s) = \log(r(\sqrt{s})/r(0))$, positive, decreasing and convex as a Stieltjes function must be. Right: the continuation $G(x) = r(-ix)$ on $[0, \pi/4)$, with a simple zero at $x = 0.6388$.

(i) Growth. r *is even,* $r(0) = 0.853936$*, and* $\log r(t)$ *becomes linear with slope approaching* $\frac{7}{2}$*: the increments over steps of* 0.1 *fall from* 0.3534 *to* 0.3514 *between* $t = 1.5$ *and* $t = 2.2$*. The slope is explained exactly:* Φ *carries the prefactor* $e^{9t/2}$*, and the saddle point prefactor* $\sqrt{s^*/\pi} \approx e^t$ *of* f_{κ^*} *removes one unit. So* G *would have exponential type* $\frac{7}{2}$*, compatible with order one.*

(ii) Stieltjes signs. With $L(s) = \log r(\sqrt{s}) - \log r(0)$*, a Laguerre Pólya* G *requires* $L'(s) = \sum m_k/(s + \alpha_k^2)$*, a Stieltjes function. On a grid of* 81 *points in* $s \in [0, 4]$ *the difference quotients of* L *give* $L' > 0$*,* $L'' < 0$ *and* $L''' > 0$ *everywhere, and the first order quotients fall from* 4.07 *on the first grid interval to* 0.94 *on the last.*

(iii) The imaginary axis. The continuation $G(x) = r(-ix)$*, computed independently from* $\Phi(ix)$ *and* $f_{\kappa^*}(ix)$*, is real, agrees with* r *at second order (*$\log(G(0.1)/G(0)) = -0.0423$ *against* $\log(r(0.1)/r(0)) = 0.0417$*), decreases from* 0.8539*, has a simple zero at* $\alpha_1 = 0.6388$*, and satisfies the Laguerre inequality* $G'^2 - GG'' \geq 2.4 > 0$ *on* $[0, 0.735]$.

Everything visible inside the strip is consistent with G being a Laguerre Pólya function of exponential type $\frac{7}{2}$: the single zero α_1 accounts for $1/\alpha_1^2 = 2.45$ of the curvature $\sum m_k/\alpha_k^2 \approx 4.2$, and a zero density of about $\frac{7}{2\pi}$ per unit beyond it would account for the linear growth. The decisive question cannot be settled inside the strip.

Proposition 7.1.19. *If* (7.1.4) *holds with* G *entire, then*

$$f_{\kappa^*}(t) = \Phi(t)/(\Xi(0)G(it)),$$

so f_{κ^*} continues meromorphically exactly as far as Φ does, and the two share their singular behaviour on the lines $\operatorname{Im} t = \pm\pi/4$, apart from the zeros of G. Since Φ has a natural boundary on these lines, coming from the Jacobi theta function at the edge of its modular domain, f_{κ^*} must have the same natural boundary, with singular behaviour proportional to that of Φ through a factor that continues to an entire function.

The first sentence of Proposition 7.1.19 is immediate. The second uses the classical fact that the theta function $\omega(x) = \sum_{n\geq 1} e^{-\pi n^2 x}$ has the line $\operatorname{Re} x = 0$ as a natural boundary [219, Chapter 11, Section 1.4, and Chapter 5, Section 3.2], together with $\Phi(t) = 4x^{9/4}\omega''(x) + 6x^{5/4}\omega'(x)$ at $x = e^{2t}$. Nothing is proved here about whether f_{κ^*} has such a boundary; the proposition records what (7.1.4) would require.

So the random harmonic law fits the Pólya kernel inside the strip, and the question is the boundary, where the two functions have different origins: Φ is a modular theta series and f_{κ^*} a Rademacher product whose characteristic function has real zeros at $v = (2j+1)\pi(k+\kappa^*)$, increasingly dense. For (7.1.4) to hold, the Rademacher product would have to reproduce the modular singularities of Φ up to an entire factor. We see no reason for that, and it suggests that the right Ising system for the Pólya variable is not the harmonic paramagnet but one whose magnetisation law carries theta function structure.

7.1.3.4 The Pólya variable as the logarithm of a GGC

Proposition 7.1.19 asks for a Lee and Yang system whose magnetisation law carries the modular boundary of Φ. Jacobi's triple product is the first candidate, since it writes a theta function as the characteristic function of a sum of independent two point variables, but these are not Lee and Yang spins (Proposition 7.1.20); the modular structure turns out to live in a multiplicative magnetisation.

PROPOSITION 7.1.20. *For $0 < q < 1$ the triple product*

$$\sum_{n\in\mathbb{Z}} q^{n^2} e^{inv} = \prod_{m\geq 1} (1-q^{2m})\big(1 + 2q^{2m-1}\cos v + q^{4m-2}\big)$$

exhibits the discrete Gaussian law $\Pr(J=n) \propto q^{n^2}$ as the law of $\sum_m (B_m - B'_m)$ with independent Bernoulli variables of parameter $q^{2m-1}/(1+q^{2m-1})$. Each factor vanishes at $v = \pi \pm i(2m-1)\log(1/q)$, so the characteristic function has no real zeros and the moment generating function has no purely imaginary zeros: the triple product spins are not Lee and Yang spins, and neither is the discrete Gaussian.

Proof. $1 + 2q^{2m-1}\cos v + q^{4m-2} = (1+q^{2m-1}e^{iv})(1+q^{2m-1}e^{-iv})$, and each factor, normalised, is the characteristic function of a Bernoulli variable or its negative. The zeros are those of $1 + q^{2m-1}e^{\pm iv}$. □

So the triple product places the theta function in the field variable, where its zeros lie on horizontal lines rather than on the real axis. For Φ the theta function enters through the modular variable e^{2t}, and the relevant structure is multiplicative. Recall from Section 1.2 the identity of Biane, Pitman and Yor [28], going back to Williams [271], which makes this exact.

THEOREM 7.1.21. *Let $S_2 = \sum_{k\ge1} \frac{2}{\pi^2k^2}(e_k + e_k')$ with independent standard exponentials e_k, e_k', so that $\mathbb{E}e^{-sS_2} = \big(\sqrt{2s}/\sinh\sqrt{2s}\big)^2$ and S_2 is a GGC with Thorin measure $2\sum_k \delta_{\pi^2k^2/2}$. Then*

$$2\xi(z) = \mathbb{E}\Big[\Big(\frac{\pi S_2}{2}\Big)^{z/2}\Big].$$

Consequently, with $Y = \frac12\log(\pi S_2/2)$, $2\Xi(\tau) = \mathbb{E}[e^{Y/2}e^{i\tau Y}]$, and the Pólya variable X is Y tilted by $e^{Y/2}$: $\Phi(t)\,dt = \frac12 e^{t/2}\Pr(Y\in dt)$.

Proof. This is the identity of [28, Section 2]; the density of S_2 is a theta series by the partial fraction expansion of $(x/\sinh x)^2$ and Poisson summation, and its Mellin transform is 2ξ. □

We checked the identity exactly at $z = 2$ and $z = 4$, where $\mathbb{E}S_2 = \frac23$ and $\mathbb{E}S_2^2 = \frac{8}{15}$ give $\frac{\pi}{3} = 2\xi(2)$ and $\frac{2\pi^2}{15} = 2\xi(4)$, and numerically at $z = 0.8 + 3i$, where both sides equal $0.8073665638 + 0.0340962268i$ to ten digits, using $\mathbb{E}V^w = \frac{w}{\Gamma(1-w)}\int_0^\infty s^{-w-1}(1-\mathbb{E}e^{-sV})\,ds$ for positive V and $0 < \operatorname{Re} w < 1$. Table 9.1 in Section 9.1 checks the identity by simulation at real arguments, and Section 9.1.1 there tilts Y to every real basepoint σ: with no hypothesis, $\xi(\sigma+z)/\xi(\sigma)$ is the moment generating function of $\frac12\log(\frac{\pi}{2}S_2)$ under the law of S_2 size biased by $S_2^{\sigma/2}$.

Theorem 7.1.21 shows where the modular boundary comes from. The Pólya variable is a tilted logarithm of a GGC whose Thorin atoms lie on the quadratic lattice $\pi^2k^2/2$, and it is the quadratic lattice, through Jacobi's theta function and Poisson summation, that produces the modular natural boundary at $\operatorname{Im} t = \pm\pi/4$: the density of S_2 is analytic in $\operatorname{Re} x > 0$ with natural boundary on $\operatorname{Re} x = 0$, and $x = \frac{2}{\pi}e^{2t}$. The Thorin structure thus enters twice: R is the reciprocal moment generating function of the Pólya variable in $\sqrt{s}$, and S_2 is a GGC in the usual sense. RH is the statement that every zero of $w \mapsto \mathbb{E}[(\pi S_2/2)^w]$ in the strip $0 < \operatorname{Re} w < \frac12$ lies on the line $\operatorname{Re} w = \frac14$.

This suggests a multiplicative Lee and Yang property. We say that a positive random variable V has *Mellin zeros on a line* if $\mathbb{E}V^w$, continued, vanishes only on one vertical line. The solved models of the Vere-Jones step have exact multiplicative analogues.

PROPOSITION 7.1.22. *Let $V = \sum_{k\le n} c_k e_k$ with independent standard exponentials e_k and distinct $c_k > 0$. Then $\mathbb{E}V^{u-1} = \Gamma(u)\sum_k A_k c_k^u$ with $A_k = c_k^{n-2}/\prod_{j\ne k}(c_k - c_j)$.*

(i) *For $n = 2$ the zeros are $u = 2\pi i m/\log(c_1/c_2)$, $m \neq 0$: all on the line $\operatorname{Re} u = 0$. The two exponential sum is a multiplicative solved model.*

(ii) *For $n = 3$ with $c = (1, \frac{1}{4}, \frac{1}{9})$, a truncation of the quadratic lattice with single atoms, the zeros in $0 < \operatorname{Im} u < 30$ have real parts 0.488, −1.268, 0.300, 0.387, −1.318, 0.457, 0.136, −1.129, 0.500 (computed): they do not lie on a line.*

Proof. The density of V is $\sum_k A_k e^{-x/c_k}$ by partial fractions of $\prod_k (1 + c_k s)^{-1}$, and its Mellin transform is the stated exponential polynomial times $\Gamma(u)$. For $n = 2$ it is proportional to $c_1^u - c_2^u$. For $n = 3$ the zeros are those of $1.5 - 2.4 \cdot 4^{-u} + 0.9 \cdot 9^{-u}$, which has none with $|\operatorname{Re} u| \geq 8$; the argument principle on $[-8, 8] \times [0.01, 30]$ counts nine, none lies in $0 < \operatorname{Im} u < 0.01$, and Newton's method locates them at $0.4881 + 4.4641i$, $-1.2675 + 7.6319i$, $0.2997 + 9.2799i$, $0.3865 + 13.4112i$, $-1.3178 + 15.5448i$, $0.4570 + 18.2547i$, $0.1355 + 22.4710i$, $-1.1294 + 23.3902i$ and $0.4998 + 27.1862i$. □

So the Lee and Yang mechanism survives the passage from additive to multiplicative magnetisations for two components and fails at three, just as a sum of two cosines has real zeros while a generic sum of three need not.

7.2 Truncations, finite conditions and the road to RH

Cutting the series for S_2 in Theorem 7.1.21 after n terms gives variables whose Mellin transforms are explicit, each a gamma factor times an exponential polynomial, with zeros that converge to the points $\rho/2$; a condition on these zeros, holding for infinitely many n, implies RH. We follow this route to the point where it stops. The one sided condition fails because the truncated zeros approach the critical line from a side fixed by an explicit constant $\kappa(\rho)$, and the side changes at zeros where $|S(\gamma)|$ is large, with $S(t) = \pi^{-1} \arg \zeta(\frac{1}{2} + it)$ as in Appendix C; the first such zero is γ_{213}. Symmetrising by the functional equation removes this mechanism. The resulting Conjecture S is proved above explicit heights for $n \leq 6$ and $n = 8$ and checked numerically below them; for the other $n \leq 24$ and for $n = 32, 36$ it follows above explicit heights from torus bounds proved in ball arithmetic, and is checked numerically below them. Section 7.2.2 sets the reductions of Chapter 6 and of this chapter against what is proved unconditionally: Weil's criterion in Thorin coordinates, Weil positivity on the Bernstein cone, the reach of finite order tests, the de Bruijn Newman flow and GGC multipliers. Sections 7.2.3, 7.2.4 and 7.2.5 record the numerical verification of both chapters, say what their results mean for the Thorin approach, and state open problems.

7.2.1 Truncations of the Biane, Pitman and Yor lattice

In this subsection w is the complex variable of a Mellin transform and $u = w + 1$; the canonical function $w(x)$ of Theorem 6.1.11 does not return until Section 7.2.2. The results are of three kinds. The Hurwitz reductions, the first order expansion of the truncated zeros, the formula for its constant and the single term bound for small n are theorems. The argument principle counts are computations in floating point. Corollary 7.2.4 rests in addition on an enclosure of $\operatorname{Re}\kappa(\rho_{213})$ in ball arithmetic, and the results for $n = 7$, $9 \le n \le 24$ and $n = 32, 36$ on bounds over a torus, proved by a branch and bound whose final boxes are enclosed in ball arithmetic.

7.2.1.1 Truncated lattices and Conjecture T

Let e_k, e_k' be independent standard exponential variables, and put $\mathcal{S} = \sum_{k\ge1} k^{-2}(e_k + e_k') = \frac{\pi^2}{2} S_2$, with S_2 the variable of Theorem 7.1.21, and $\mathcal{S}_n = \sum_{k\le n} k^{-2}(e_k + e_k')$, a GGC with finitely many Thorin atoms. Put $F(w) = \mathbb{E}\mathcal{S}^w$ and $F_n(w) = \mathbb{E}\mathcal{S}_n^w$ for $\operatorname{Re} w > -2$. By Theorem 7.1.21,

$$F(w) = 2\pi^w\, \xi(2w),$$

so the zeros of F are the points $w = \rho/2$, and RH says they lie on $\operatorname{Re} w = \frac14$. Each F_n is explicit. Define α_k and β_k, which depend on n, by the partial fraction expansion

$$\prod_{j\le n}\Big(1+\frac{\lambda}{j^2}\Big)^{-2} = \sum_{k\le n} k^{-2}\Big\{\alpha_k\Big(1+\frac{\lambda}{k^2}\Big)^{-1} + \beta_k\Big(1+\frac{\lambda}{k^2}\Big)^{-2}\Big\},$$

so that $\beta_k = k^2 \prod_{j\le n,\, j\ne k}(1-k^2/j^2)^{-2}$ and $\alpha_k = -2\beta_k \sum_{j\le n,\, j\ne k} k^2/(j^2-k^2)$; for $n = 2$ they are $\beta = (\frac{16}{9}, \frac49)$ and $\alpha = (-\frac{32}{27}, \frac{32}{27})$. The left side is the Laplace transform of $\mathcal{S}_n$, whose density is therefore $\sum_k (\alpha_k + \beta_k k^2 x) e^{-k^2 x}$, and taking moments gives, with $u = w + 1$, $F_n(u-1) = \Gamma(u) Q_n(u)$, where $Q_n(u) = \sum_{k\le n} k^{-2u}(\alpha_k + \beta_k u)$. So F_n is, up to Γ, an exponential polynomial with frequencies $2\log k$ and coefficients linear in u.

Theorem 7.2.1. *If for infinitely many n the function F_n has no zeros in the half plane $\operatorname{Re} w > \frac14$, then the Riemann Hypothesis holds.*

Proof. For $\operatorname{Re} w > -2$ the variables $\mathcal{S}_n^w$ are dominated by an integrable variable, since $\mathcal{S}_n \ge e_1 + e_1'$ and $\mathcal{S}_n \le \mathcal{S}$, so $F_n \to F$ locally uniformly on $\operatorname{Re} w > -2$. If F had a zero w_0 with $\operatorname{Re} w_0 > \frac14$, Hurwitz's theorem would give zeros of F_n near w_0 for all large n. Hence every zero $\rho/2$ of F has $\operatorname{Re}\rho \le \frac12$, and the symmetry $\rho \mapsto 1-\rho$ gives $\operatorname{Re}\rho = \frac12$. □

Asking for the hypothesis at every n gives the simplest condition of this kind.

CONJECTURE T. For every $n \geq 1$, $F_n(w) = \mathbb{E}\big[\big(\sum_{k\leq n} k^{-2}(e_k + e'_k)\big)^w\big]$ has no zeros with $\operatorname{Re} w > \frac{1}{4}$.

By Theorem 7.2.1, Conjecture T implies RH. It is a stability statement, of Hermite Biehler or Lee and Yang type [141], for a sequence of explicit exponential polynomials. It is false (Corollary 7.2.4, whose proof uses an enclosure of $\operatorname{Re}\kappa(\rho_{213})$ in ball arithmetic), and we state it because the mechanism of its failure also defeats, for large n, the Hermite Biehler inequality on which the bounds of Theorems 7.2.8 and 7.2.10 for the symmetrised truncations rest.

At small n the evidence supports it: Table 7.1 finds no zero of Q_n with $\operatorname{Re} u > \frac{5}{4}$ in any rectangle searched. The zeros are counted by the argument principle, with 800 to 4000 sample points per side and a working precision of 40 to 340 decimal digits to control the cancellation among the partial fraction coefficients, and located by Newton's method from a grid; the counts and the zeros found agree.

n	region, in $u = w + 1$	zeros with $\operatorname{Re} u > \frac{5}{4}$
$2, 3, 4, 5, 6, 8, 12, 16, 24, 32$	$\frac{5}{4} < \operatorname{Re} u < 6$, $0.3 < \operatorname{Im} u < 40$	none
$8, 16, 32$	$\frac{5}{4} < \operatorname{Re} u < 6$, $40 < \operatorname{Im} u < 100$	none
$48, 64$	$\frac{5}{4} < \operatorname{Re} u < 6$, $0.3 < \operatorname{Im} u < 30$	none
$16, 32, 64$	$6 < \operatorname{Re} u < 40$, $0.3 < \operatorname{Im} u < 40$	none

Table 7.1: Argument principle counts for Conjecture T: rectangles searched and zeros of Q_n found with $\operatorname{Re} u > \frac{5}{4}$.

Rectangles to the left of the line contain the expected zeros: 8, 12, 12, 15, 15, 17, 18, 19, 21, 21 zeros with $-6 < \operatorname{Re} u < \frac{5}{4}$ and $\operatorname{Im} u < 40$ for the ten values of n in the first row. For $n = 2$ the eight zeros found lie on $\operatorname{Re} u = -1$ exactly. On the real axis F_n is positive, and for fixed n and large $|\operatorname{Im} u|$ the zeros move far to the left, because the extreme coefficients β_1 and β_n differ by a factor of order 16^n, up to a power of n. The intermediate heights are therefore where Conjecture T is at risk, and they are the heights that Table 7.1 searches; the failure found in Corollary 7.2.4 lies above them (Remark 7.2.5). An experiment shows that the absence of such zeros at small n is not a consequence of the GGC structure alone: replacing the weights k^{-2} by $k^{-2}(1 + \eta_k)$, with independent η_k uniform on $[-0.15, 0.15]$, one of two draws at $n = 16$ gives a zero with $\operatorname{Re} u > \frac{5}{4}$. The property depends on the exact quadratic lattice.

7.2.1.2 How the truncated zeros reach the line, and why Conjecture T fails

Near a simple zero ρ of ξ, the zeros of F_n converge to $\rho/2$ at rate $1/n$, with an explicit constant.

PROPOSITION 7.2.2. *Let ρ be a simple zero of ξ and put*

$$\kappa(\rho) = \frac{\rho\,\xi(\rho-2)}{2\pi\,\xi'(\rho)} = \frac{\rho\,\xi(3-\rho)}{2\pi\,\xi'(\rho)}.$$

There is a constant C such that for all large n, F_n has exactly one zero $w_n(\rho)$ in the disc $|w - \frac{\rho}{2} - \frac{\kappa(\rho)}{n}| < C/n^2$; thus $w_n(\rho) = \frac{\rho}{2} + \frac{\kappa(\rho)}{n} + O(n^{-2})$.

Proof. Write $\mathcal{S} = \mathcal{S}_n + \mathcal{T}_n$ with $\mathcal{T}_n = \sum_{k>n} k^{-2}(e_k + e_k')$, independent of $\mathcal{S}_n$, so that $\mathbb{E}\mathcal{T}_n = 2\sum_{k>n} k^{-2} = \frac{2}{n} + O(n^{-2})$ and $\mathbb{E}\mathcal{T}_n^2 = O(n^{-2})$. For $a > 0$, $b \ge 0$ and $0 < \operatorname{Re} w < 2$, Taylor's formula with integral remainder and $|(a+tb)^{w-2}| = (a+tb)^{\operatorname{Re} w - 2} \le a^{\operatorname{Re} w - 2}$ give

$$\bigl|(a+b)^w - a^w - wb\,a^{w-1}\bigr| \le \tfrac{1}{2}|w(w-1)|\,b^2 a^{\operatorname{Re} w-2},$$
$$\bigl|(a+b)^{w-1} - a^{w-1}\bigr| \le |w-1|\,b\,a^{\operatorname{Re} w-2}.$$

Since $\mathcal{S}_n \ge e_1 + e_1'$, a $\Gamma(2)$ variable, $\mathbb{E}\mathcal{S}_n^{\operatorname{Re} w-2} \le \Gamma(\operatorname{Re} w)$. Taking expectations with $a = \mathcal{S}_n$ and $b = \mathcal{T}_n$, and using independence,

$$F(w) = F_n(w) + w\,\mathbb{E}\mathcal{T}_n\,F_n(w-1) + O(n^{-2}), \qquad F_n(w-1) = F(w-1) + O(n^{-1}),$$

uniformly on compact subsets of $0 < \operatorname{Re} w < 2$. Hence $F_n(w) = F(w) - \frac{2}{n} wF(w-1) + O(n^{-2})$ uniformly there. Put $w_0 = \rho/2$, so $0 < \operatorname{Re} w_0 < \frac{1}{2}$, and recall $F(w_0) = 0$ and $F'(w_0) = 4\pi^{w_0}\xi'(\rho) \ne 0$. For $|w - w_0| = O(1/n)$, Taylor expansion of F and of $wF(w-1)$ about w_0 gives

$$F_n(w) = F'(w_0)\Bigl(w - w_0 - \frac{\kappa(\rho)}{n}\Bigr) + O(n^{-2}), \qquad \frac{\kappa(\rho)}{n} = \frac{2}{n}\,\frac{w_0 F(w_0-1)}{F'(w_0)},$$

and $F(w_0 - 1) = 2\pi^{w_0-1}\xi(\rho-2)$ turns the second identity into the stated κ. On the circle $|w - w_0 - \kappa/n| = C/n^2$ the linear term has modulus $|F'(w_0)|C/n^2$, which exceeds the $O(n^{-2})$ error once C is larger than its constant divided by $|F'(w_0)|$. Rouché's theorem gives exactly one zero inside for all large n. The second form of κ follows from $\xi(s) = \xi(1-s)$. □

Each simple zero ρ of ξ thus has a companion zero of the truncation at distance of order $1/n$, displaced in the direction $\kappa(\rho)$. If $\operatorname{Re}\kappa(\rho) > 0$ the companion lies to the right of the critical line for all large n. This happens at zeros where $|S(\gamma)|$ is large, the first being γ_{213} (Theorem 7.2.3 and the computation after it), and it is why Conjecture T fails.

For each of the first 100 zeros $\kappa(\rho)$ has negative real part, starting with $\kappa(\rho_1) = -4.762 - 1.944i$, $\kappa(\rho_2) = -8.673 - 0.627i$, $\kappa(\rho_3) = -10.149 - 3.872i$ and $\kappa(\rho_4) = -15.165 + 3.705i$. So these truncated zeros approach the critical line from the left, at distance about $-\operatorname{Re}\kappa(\rho)/n$, which grows with the height: $-\operatorname{Re}\kappa(\rho)/\gamma^2$ lies between 0.011 and 0.024 for the first ten zeros, with 0.024 at γ_1 and 0.013 at γ_{10}.

Figure 7.4 and Table 7.2 show the approach for the first four zeros, computed by locating the zeros of F_n directly at each n rather than from the expansion. The convergence of $n(\frac{5}{4} - \operatorname{Re} u)$ to $-\operatorname{Re}\kappa(\rho_j)$, in the right panel of the figure and the last row of the table, is an independent check on Proposition 7.2.2 at low height.

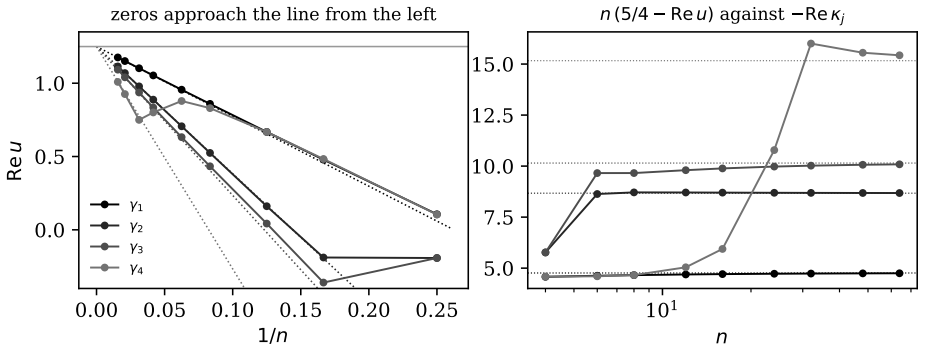


Figure 7.4: Left: real parts of the zeros of F_n near $\rho_j/2$, in the variable $u = w+1$, against $1/n$ for $n = 4$ to 64, with the first order lines $\frac{5}{4} + \operatorname{Re}\kappa(\rho_j)/n$ (dotted). The zeros approach $\operatorname{Re} u = \frac{5}{4}$, the image of the critical line, from the left. Right: $n(\frac{5}{4} - \operatorname{Re} u)$ against n, converging to $-\operatorname{Re}\kappa(\rho_j)$.

	n	near $\rho_1/2$	near $\rho_2/2$	near $\rho_3/2$	near $\rho_4/2$
	8	$0.6678 + 6.8583i$	$0.1610 + 10.6538i$	$0.0426 + 11.9969i$	$0.6676 + 15.7727i$
	16	$0.9558 + 6.9550i$	$0.7062 + 10.5197i$	$0.6321 + 12.2722i$	$0.8787 + 15.7928i$
	32	$1.1020 + 7.0090i$	$0.9785 + 10.5029i$	$0.9368 + 12.3881i$	$0.7498 + 15.4606i$
	64	$1.1758 + 7.0376i$	$1.1144 + 10.5040i$	$1.0924 + 12.4460i$	$1.0089 + 15.2948i$
	limit	$1.25 + 7.0674i$	$1.25 + 10.5110i$	$1.25 + 12.5054i$	$1.25 + 15.2124i$
$n(\frac{5}{4} - \operatorname{Re} u)$	at 64	4.75	8.68	10.09	15.43

Table 7.2: Zeros of F_n near $\rho_j/2$, in the variable $u = w + 1$. The row "limit" is $\rho_j/2+1$; the last row, $n(\frac{5}{4}-\operatorname{Re} u)$ at $n = 64$, is to be compared with $-\operatorname{Re}\kappa(\rho_j) =$ 4.762, 8.673, 10.149, 15.165.

The direction from which a truncated zero reaches the line is not always the same, and the sign of $\operatorname{Re}\kappa(\rho)$ can be read off from Hardy's function.

THEOREM 7.2.3. *Let $\rho = \frac{1}{2} + i\gamma$ be a simple zero on the critical line, let θ be the Riemann Siegel theta function and $Z(t) = e^{i\theta(t)}\zeta(\frac{1}{2} + it)$ Hardy's function, and*

put $a(\gamma) = e^{i\theta(\gamma)}\zeta(\frac{5}{2} + i\gamma)$. *Then*

$$\operatorname{Re}\kappa(\rho) = \frac{(\gamma^2 - \frac{15}{4})\operatorname{Im} a(\gamma) - 4\gamma \operatorname{Re} a(\gamma)}{4\pi^2 Z'(\gamma)},$$
$$\operatorname{Im}\kappa(\rho) = \frac{4\gamma \operatorname{Im} a(\gamma) + (\gamma^2 - \frac{15}{4})\operatorname{Re} a(\gamma)}{4\pi^2 Z'(\gamma)}.$$

Proof. On the line $\xi(\frac{1}{2} + it) = \Xi(t)$ is real, so $\xi'(\rho) = -i\,\Xi'(\gamma)$, and $\Xi(t) = -\frac{1}{2}(t^2 + \frac{1}{4})\pi^{-1/4}|\Gamma(\frac{1}{4} + \frac{it}{2})|Z(t)$ gives $\Xi'(\gamma) = -\frac{1}{2}(\gamma^2 + \frac{1}{4})\pi^{-1/4}|\Gamma(\frac{1}{4} + \frac{i\gamma}{2})|\overline{Z'(\gamma) \text{ at a}}$ zero. Since $\xi(3-\rho) = \xi(\frac{5}{2} - i\gamma)$ and ξ is real on the real axis, $\xi(3-\rho) = \overline{\xi(\frac{5}{2} + i\gamma)}$, and $\Gamma(\frac{5}{4} + \frac{i\gamma}{2}) = (\frac{1}{4} + \frac{i\gamma}{2})\Gamma(\frac{1}{4} + \frac{i\gamma}{2})$ with $\pi^{-i\gamma/2}\Gamma(\frac{1}{4} + \frac{i\gamma}{2}) = |\Gamma(\frac{1}{4} + \frac{i\gamma}{2})|e^{i\theta(\gamma)}$. Substituting, the moduli of Γ cancel, $(\frac{1}{2} + i\gamma)(\overline{\frac{1}{4} - \frac{i\gamma}{2}}) = \frac{1}{2}(\gamma^2 + \frac{1}{4})$ cancels the remaining factor, and $\kappa = -i(\frac{5}{2} - i\gamma)(\frac{3}{2} - i\gamma)\overline{a(\gamma)}/(4\pi^2 Z'(\gamma))$, whose real and imaginary parts are as stated. □

For large γ the term $(\gamma^2 - \frac{15}{4})\operatorname{Im} a(\gamma)$ decides the sign of $\operatorname{Re}\kappa(\rho)$. Since $|\zeta(\frac{5}{2} + i\gamma) - 1| \le \zeta(\frac{5}{2}) - 1 < 0.35$, $a(\gamma)$ is a positive multiple of $e^{i\theta(\gamma)}$ up to a phase below 0.35 in modulus, and up to that phase the sign of $\operatorname{Re}\kappa(\rho)$ is the sign of $\sin\theta(\gamma)/Z'(\gamma)$. Let S_{mid} be the midpoint of the jump of S at the nth zero γ_n. Then $\theta(\gamma_n) = \pi(n - \frac{3}{2} - S_{\mathrm{mid}})$ and the sign of $Z'(\gamma_n)$ is $(-1)^{n+1}$, so the sign of $\sin\theta(\gamma_n)/Z'(\gamma_n)$ is that of $-\cos(\pi S_{\mathrm{mid}})$: negative when $|S_{\mathrm{mid}}| < \frac{1}{2}$, that is when the zero lies within half a spacing of its expected position, and positive when $\frac{1}{2} < |S_{\mathrm{mid}}| < \frac{3}{2}$. The phase of $\zeta(\frac{5}{2} + i\gamma)$ moves the threshold $\frac{1}{2}$ by at most about 0.11 either way. For the first 460 zeros the sign of $\operatorname{Re}\kappa$ agrees with the sign of $\operatorname{Im} a(\gamma)/Z'(\gamma)$ in every case, and $\operatorname{Re}\kappa > 0$ occurs three times, at

$$\gamma_{213} = 415.4552, \quad \gamma_{289} = 527.9036, \quad \gamma_{379} = 650.6687,$$

with $\kappa = 233.96 - 2298.91i$, $604.16 + 3074.08i$, $1164.60 - 4296.91i$.

At these three zeros the jump of S runs from 0.098 to 1.098, from -1.145 to -0.145 and from 0.164 to 1.164; the midpoints of the jumps are 0.598, -0.645 and 0.664. Seven of the first 460 zeros have $S(\gamma^-) < -1$: besides γ_{289} these are γ_{127}, γ_{196}, γ_{233}, γ_{368}, γ_{380} and γ_{401}, with midpoints -0.506, -0.505, -0.537, -0.556, -0.538 and -0.514, and $\operatorname{Re}\kappa$ is negative at every one of them. Three further zeros have midpoints above $\frac{1}{2}$ and $\operatorname{Re}\kappa < 0$: γ_{136}, γ_{256} and γ_{399}, at 0.507, 0.510 and 0.539. The largest $|S_{\mathrm{mid}}|$ with $\operatorname{Re}\kappa < 0$ is 0.556, at γ_{368}, and the smallest with $\operatorname{Re}\kappa > 0$ is 0.598, at γ_{213}, so the threshold on the midpoint lies between about 0.56 and 0.60, inside the range $\frac{1}{2} \pm 0.11$ allowed by the phase of $\zeta(\frac{5}{2} + i\gamma)$.

Corollary 7.2.4. *Conjecture T is false: for all sufficiently large n, F_n has a zero with $\operatorname{Re} w > \frac{1}{4}$ near $\rho_{213}/2$.*

Proof. The first 1.5×10^9 zeros, ρ_{213} among them, lie on the critical line and are simple [264], and ball arithmetic applied to the formula of Theorem 7.2.3

gives $233.95 < \operatorname{Re}\kappa(\rho_{213}) < 233.96$ (Section 7.2.3). By Proposition 7.2.2, for all large n the function F_n has a zero w_n with $|w_n - \frac{\rho_{213}}{2} - \frac{\kappa(\rho_{213})}{n}| < C/n^2$, so $\operatorname{Re} w_n > \frac{1}{4} + \frac{233.95}{n} - \frac{C}{n^2} > \frac{1}{4}$ once $n > C/233.95$. □

In particular the hypothesis of Theorem 7.2.1 fails for every large n.

REMARK 7.2.5. The constants in Proposition 7.2.2 depend on ρ, and the first order regime requires n large compared with $|\kappa(\rho)|$. For the zero ρ_{213} of Corollary 7.2.4, $|\kappa(\rho_{213})| = 2311$, so the phenomenon begins at n of order 10^3 or beyond, larger than every n computed in this section. Table 7.1 accordingly reports no zero with $\operatorname{Re} u > \frac{5}{4}$, and cannot detect this one, since its rectangles reach $\operatorname{Im} u = 100$ while the predicted failure sits at $\operatorname{Im} u = 207.7$.

The failure is intrinsic to the method, not to the particular lattice. Any truncation whose first order error is a multiple of $F(w-1)$ produces the displacement $\kappa(\rho)/n$, and the phase of $\kappa(\rho)$ is governed by $S(\gamma)$. By Selberg's central limit theorem [237] S takes values beyond $\pm\frac{1}{2}$ for a positive proportion of heights, so one expects a positive proportion of zeros to be approached from the right. One sided Hurwitz arguments therefore cannot succeed for truncations of this kind; what is needed is a scheme in which symmetry keeps the zeros on the line.

7.2.1.3 Symmetrised truncations and Conjecture S

The functional equation supplies that symmetry. Put

$$K_n(w) = \pi^{-w}F_n(w) + \pi^{w-\frac{1}{2}}F_n(\tfrac{1}{2} - w), \qquad Z_n(y) = K_n(\tfrac{1}{4} + iy).$$

PROPOSITION 7.2.6. *$K_n(\frac{1}{2} - w) = K_n(w)$, $K_n(\bar{w}) = \overline{K_n(w)}$, so Z_n is real for real y and the zeros of K_n off the line $\operatorname{Re} w = \frac{1}{4}$ come in quadruples. As $n \to \infty$, $K_n \to 4\,\xi(2w)$ locally uniformly on $-\frac{7}{4} < \operatorname{Re} w < \frac{9}{4}$. Every simple zero $\frac{1}{4} + i\frac{\gamma}{2}$ of $\xi(2w)$ on the line is, for all large n, the limit of a zero of K_n lying exactly on the line.*

Proof. The symmetries are immediate from the definition and the reality of F_n on the real axis. The limit follows from $F_n \to F = 2\pi^w\xi(2w)$ and $\xi(1-2w) = \xi(2w)$; since $F_n \to F$ on $\operatorname{Re} w > -2$, the convergence holds on $-2 < \operatorname{Re} w < \frac{5}{2}$, which contains the strip of the statement. Near a simple zero y_0 of the real function $4\xi(\frac{1}{2} + 2iy)$, Z_n converges uniformly and so changes sign near y_0; by Hurwitz, K_n has exactly one zero near $\frac{1}{4} + iy_0$, which must be the real zero of Z_n, since an off line zero would bring its mirror image $\frac{1}{2} - \bar{w}$ with it. □

So for symmetrised truncations the local mechanism that broke Conjecture T is absent: a simple zero on the line cannot leave it under a small symmetric perturbation; it can leave only after colliding with another zero. The Hurwitz argument now gives the following.

THEOREM 7.2.7. *If for infinitely many n all zeros of K_n in the strip $0 < \operatorname{Re} w < \frac{1}{2}$ lie on the line $\operatorname{Re} w = \frac{1}{4}$, then RH holds.*

Proof. A zero $\rho/2$ of $\xi(2w)$ off the line lies in the strip $0 < \operatorname{Re} w < \frac{1}{2}$, since $0 < \operatorname{Re} \rho < 1$, and by Hurwitz it is the limit of zeros of K_n in the strip, which would be off the line for large n. □

CONJECTURE S. For every $n \geq 1$, all zeros of K_n in $0 < \operatorname{Re} w < \frac{1}{2}$ lie on $\operatorname{Re} w = \frac{1}{4}$.

The symmetrised zeros move along the line. The first order argument of Proposition 7.2.2, applied to $K_n = 2\operatorname{Re}[\pi^{-w} F_n(w)]$ on the line, gives

$$w_n^{\mathrm{sym}}(\rho) = \frac{\rho}{2} + \frac{i \operatorname{Im} \kappa(\rho)}{n} + O(n^{-2}) :$$

the symmetrisation removes the real part of the displacement, which is the part that broke Conjecture T, and keeps the imaginary part. Numerically the first two symmetrised zeros are at $\operatorname{Im} w = 7.0328, 7.0072, 7.0231, 7.0413$ and $10.5490, 10.5392, 10.5161, 10.5085$ for $n = 8, 16, 32, 64$, against the first order values 7.0370 and 10.5012 at $n = 64$.

n	window in $\operatorname{Im} w$	zeros in $-4 < \operatorname{Re} w < 4.5$: total, on the line	zeros of ξ
1	(0.5, 40), (40, 70)	23, 21; 27, 27	21; 27
2	(0.5, 40), (40, 70)	23, 21; 27, 27	21; 27
3	(0.5, 40), (40, 70)	23, 21; 27, 27	21; 27
4, 5, 6, 8, 16	(0.5, 40), (40, 70)	21, 21; 27, 27	21; 27
32	(0.5, 40)	21, 21	21
8	(70, 140), (195, 215)	78, 78; 27, 27	78; 26

Table 7.3: Zeros of the symmetrised truncations K_n in $-4 < \operatorname{Re} w < 4.5$: total found, how many lie on the line, and the number of zeros of ξ in the doubled window for comparison. In the window $195 < \operatorname{Im} w < 215$ the count for K_8 exceeds that for ξ by one; see the text.

Table 7.3 counts the zeros of K_n in windows of height. For $n = 1$ the one off line quadruple lies at $\operatorname{Re} w = -2.001, 2.501$, $\operatorname{Im} w = \pm 2.471$, and for $n = 2$ at $\operatorname{Re} w = -2.949, 3.449$, $\operatorname{Im} w = \pm 3.261$, far from the critical strip and near the region where the truncations stop resembling ξ; for $n = 3$ it is similar. These quadruples have $|\operatorname{Re} w - \frac{1}{4}| > 2$, so they lie outside the strip $0 < \operatorname{Re} w < \frac{1}{2}$ of Theorem 7.2.7 and do not bear on Conjecture S. For $n \geq 4$ every zero found lies on the line. In the first three windows the counts on the line equal the numbers of zeros of ξ in the doubled range, as the last column records; the totals for $n = 1, 2, 3$ are larger because of the off line quadruple. The positions converge to $\gamma/2$: for

$n = 16$ the first four zeros, located on a grid of step 0.01, are at 7.005, 10.535, 12.465, 15.295 against 7.067, 10.511, 12.505, 15.212; located more precisely, the first two are at 7.0072 and 10.5392.

The window $195 < \operatorname{Im} w < 215$ matters because it contains $\gamma_{213}/2 = 207.73$, where the unsymmetrised truncations are forced off the line. There the count for K_8 exceeds that for ξ by one: the zeros of ξ with $390 < \gamma < 430$ are γ_{196} to γ_{221}, twenty six of them, the nearest excluded ordinates being $\gamma_{195} = 388.85$ and $\gamma_{222} = 430.33$. The difference is not a contradiction: at this height and $n = 8$ the first order displacement along the line, $\operatorname{Im} \kappa(\rho)/n$, is of order 300 for ρ_{213}, so K_8 is far from its limit and its zeros need not correspond one to one with those of ξ. What matters for Conjecture S is that every zero found in the window lies on the line.

The symmetrised truncations are the lattice analogue of the approximations $\zeta_N(s) = \sum_{m \le N} m^{-s} + \chi(s) \sum_{m \le N} m^{s-1}$ derived from the approximate functional equation, with χ the factor in $\zeta(s) = \chi(s)\zeta(1-s)$. Spira [242, Section 6] and Gonek and Montgomery [81] studied their zeros, and Gonek and Montgomery proved that for $N \le T^{o(1)}$ almost all zeros of ζ_N with ordinates in $[0, T]$ lie on the critical line and are simple. For K_n the off line zeros found disappear from $n = 4$ on, in every window examined. A proof of Conjecture S for infinitely many n would require controlling the zeros of Z_n uniformly in height, including the range of heights comparable with n, where K_n is neither close to $4\xi(2w)$ nor dominated by a single exponential term; this is the same difficulty as for ζ_N, now for an exponential polynomial with gamma lattice coefficients.

7.2.1.4 A single term bound for small n

For small n, Conjecture S reduces to a finite computation, because at large heights the gamma factor makes $\log |A_n|$ increase across the strip. Here $A_n(w) = \pi^{-w} F_n(w)$, so that $K_n(w) = A_n(w) + A_n^*(w)$ with $A_n^*(w) = A_n(\frac{1}{2} - w)$ and $|A_n^*(\sigma + iy)| = |A_n(\frac{1}{2} - \sigma + iy)|$. If $\sigma \mapsto \log |A_n(\sigma + iy)|$ is increasing on $(0, \frac{1}{2})$, then $|A_n| > |A_n^*|$ for $\frac{1}{4} < \sigma < \frac{1}{2}$, and K_n has no zeros there at height y; by symmetry none to the left of the line either. The derivative is $\operatorname{Re} \frac{A_n'}{A_n} = -\log \pi + \operatorname{Re} \psi(w+1) + \operatorname{Re} \frac{Q_n'}{Q_n}(w+1)$, and for large y the digamma term dominates.

THEOREM 7.2.8. *Let $Q_n(u) = \sum_k k^{-2u}(\alpha_k + \beta_k u)$ as above and, for $y > |\alpha_1|/|\beta_1|$, let $\delta_n(y) = \sum_{k \ge 2} k^{-2} \frac{|\beta_k| y + |\alpha_k|}{|\beta_1| y - |\alpha_1|}$ and*

$$\delta_n'(y) = \sum_{k \ge 2} k^{-2} \frac{2 \log k \, (|\beta_k| y + |\alpha_k|) + |\beta_k|}{|\beta_1| y - |\alpha_1|} + \delta_n(y) \, \frac{|\beta_1|}{|\beta_1| y - |\alpha_1|},$$

which bounds a derivative in the proof and is not the derivative of δ_n. If $\delta_n(y) < 1$ and

$$\log \frac{y}{\pi} - \frac{1}{y} - \frac{|\beta_1|}{|\beta_1| y - |\alpha_1|} - \frac{\delta_n'(y)}{1 - \delta_n(y)} > 0,$$

then K_n has no zeros off the line in $0 < \operatorname{Re} w < \frac{1}{2}$ at any height $\geq y$. The condition holds for all $y \geq Y_n$, with $Y_1 = 4.81$, $Y_2 = 5.85$, $Y_3 = 7.11$, $Y_4 = 9.08$, $Y_5 = 12.8$, $Y_6 = 20.8$, $Y_8 = 104.1$ and $Y_{10} = 21232$.

Proof. $|\psi(z) - \log z| \leq 1/|z|$ for $\operatorname{Re} z \geq 1$, so $\operatorname{Re}\psi(w+1) \geq \log y - \frac{1}{y}$ at height y. Write $Q_n(u) = (\alpha_1 + \beta_1 u)(1 + \delta(u))$. On $1 \leq \operatorname{Re} u \leq \frac{3}{2}$, $|u| \geq y$, we have $|k^{-2u}| \leq k^{-2}$, so $|\delta(u)| \leq \delta_n(y)$, and differentiating the quotient $\delta(u)$ term by term gives $|\delta'(u)| \leq \delta_n'(y)$; both bounds decrease in y. Then $\operatorname{Re}\frac{Q_n'}{Q_n} \geq -\frac{|\beta_1|}{|\alpha_1+\beta_1 u|} - \frac{|\delta'|}{1-|\delta|}$, and the condition makes $\operatorname{Re}(A_n'/A_n) > 0$ on the strip at every height $\geq y$. The values of Y_n come from evaluating these bounds with the exact coefficients. □

Corollary 7.2.9. *Conjecture S holds for $n = 1, 2, 3, 4, 5, 6, 8$ above the heights Y_n of Theorem 7.2.8, and is verified numerically below them, by a floating point argument principle count and not by an enclosure.*

Proof. Above Y_n this is Theorem 7.2.8. Below it, the argument principle counts cover the strip $0 < \operatorname{Re} w < \frac{1}{2}$ for $0 < \operatorname{Im} w < 6, 7, 8$ when $n = 1, 2, 3$, and the windows of Table 7.3 cover $0 < \operatorname{Im} w < 70$ for $4 \leq n \leq 6$ and $0 < \operatorname{Im} w < 140$ for $n = 8$, with $0 < \operatorname{Im} w < 0.6$ checked separately and $K_n > 0$ on the real segment; in each case the total count in the strip equals the number of sign changes of Z_n on the line. The counts use 40 or more digits and fine sampling. □

The single term bound of Theorem 7.2.8 deteriorates quickly: the lattice coefficients $|\beta_k/\beta_1|$ grow with n (they are 1.96, 1.31, 0.40 for $k = 2, 3, 4$ at $n = 8$), $Y_{10} = 21232$, and for $n \geq 12$ the bound fails because $\delta_n > 1$. A torus version removes most of this loss.

7.2.1.5 Torus bounds and where they stop

Write $u = x + iy$ and $Q_n(u) = u\,P_n(u) + R_n(u)$ with the Dirichlet polynomials $P_n(u) = \sum_{k \leq n} \beta_k k^{-2u}$ and $R_n(u) = \sum_{k \leq n} \alpha_k k^{-2u}$. At height y the values of P_n depend on y only through the prime phases $(p^{-2iy})_{p \leq n}$, which lie on the torus $\mathbb{T}^{\pi(n)}$. Replacing them by independent phases $\omega = (\omega_p)_{p \leq n} \in \mathbb{T}^{\pi(n)}$ gives the torus function

$$P_n(x, \omega) = \sum_{k \leq n} \beta_k k^{-2x} \exp\Big(-i \sum_{p \leq n} v_p(k)\, \omega_p\Big),$$

where $v_p(k)$ is the exponent of p in k, and in the same way the torus function $P_n'(x, \omega)$ of $P_n'(u) = -\sum_k 2 \log k\, \beta_k k^{-2u}$. Their extremes over the torus bound the values at every height. Put $r(x) = \sum_k |\alpha_k| k^{-2x}$ and $r'(x) = \sum_k 2 \log k\, |\alpha_k| k^{-2x}$.

THEOREM 7.2.10. *Let* $m_n = \min|P_n(x,\omega)|$, $M_n = \max\{-\operatorname{Re}(P_n'/P_n)(x,\omega)\}$ *and* $\ell_n = \max|P_n'/P_n|(x,\omega)$, *over* $\omega \in \mathbb{T}^{\pi(n)}$ *and the band* $1.2 \le x \le 1.3$, *and let*

$$B_n(y) = \max_{0.3\le\sigma\le\frac12,\ \omega} \frac{\log\bigl(|P_n(\frac32-\sigma,\omega)| + r(\frac32-\sigma)/y\bigr) - \log\bigl(|P_n(1+\sigma,\omega)| - r(1+\sigma)/y\bigr)}{2\sigma-\frac12}.$$

If $m_n > r(1.2)/y$,

$$\log\frac{y}{\pi} - \frac{2}{y} - M_n - \frac{\ell_n r(1.2) + r'(1.2) + r(1.2)/y}{y\,(m_n - r(1.2)/y)} > 0 \quad \textit{and} \quad \log\frac{y}{\pi} - \frac{1}{y} > B_n(y),$$

then K_n *has no zeros off the line in* $0 < \operatorname{Re} w < \frac12$ *at height* y, *and the same holds at every larger height.*

Proof. For $\frac14 < \sigma \le 0.3$ both σ and $\frac12 - \sigma$ lie in $[0.2, 0.3]$, and the first condition makes $\operatorname{Re}(A_n'/A_n) > 0$ there: $\operatorname{Re}\psi(w+1) \ge \log y - \frac1y$, $\operatorname{Re}\frac1u > 0$, and $\frac{Q_n'}{Q_n} = \frac1u + \frac{P_n'+R_n'/u - R_n/u^2}{P_n + R_n/u}$ differs from $\frac1u + \frac{P_n'}{P_n}$ by at most the stated correction. So $\log|A_n|$ increases on $[\frac12-\sigma, \sigma]$ and $|A_n| > |A_n^*|$. For $0.3 \le \sigma \le \frac12$ compare directly: $\log|\Gamma(1+\sigma+iy)| - \log|\Gamma(\frac32-\sigma+iy)| \ge (2\sigma-\frac12)(\log y - \frac1y)$ by integrating $\operatorname{Re}\psi$, $|1+\sigma+iy| > |\frac32-\sigma+iy|$, and $|P_n + R_n/u|$ is bounded below on the right by $|P_n| - r/y$ and above on the left by $|P_n| + r/y$; the second condition then gives $|A_n(\sigma+iy)| > |A_n(\frac12-\sigma+iy)|$. Both conditions improve as y increases. □

Table 7.4 lists bounds for the torus extremes, proved by a branch and bound over the band and the torus (Remark 7.2.12), and the smallest integer heights at which the first condition holds with these bounds. The second condition holds there too: the same branch and bound proves it at a height below each threshold, and it improves as y increases. For $n = 7$ and $9 \le n \le 15$ we use a single band variant: if $m_n > r(1)/y$ and the first condition of Theorem 7.2.10 holds with m_n, M_n, ℓ_n the extremes over the band $1 \le x \le \frac32$ and with $r(1)$, $r'(1)$ in place of $r(1.2)$, $r'(1.2)$, then $\operatorname{Re}(A_n'/A_n) > 0$ on $0 < \operatorname{Re} w < \frac12$ at height y, so $\log|A_n|$ increases across the strip and K_n has no zeros off the line there; the condition improves as y increases. The proof is the first part of the proof of Theorem 7.2.10 with the band widened, since r and r' decrease in x and $\operatorname{Re}\psi(w+1) \ge \log y - \frac1y$ on the whole strip. Table 7.5 gives the bounds and thresholds of this variant.

COROLLARY 7.2.11. *Conjecture S holds for* $n = 16, \dots, 24, 32, 36$ *above the thresholds of Table 7.4 and for* $n = 7$ *and* $9 \le n \le 15$ *above those of Table 7.5, and for all these* n *it is verified numerically below the thresholds, by floating point argument principle counts. With Corollary 7.2.9 this covers every* $n \le 24$ *and* $n = 32, 36$.

Proof. Above the thresholds this is Theorem 7.2.10, with the bounds of Table 7.4 and the second condition of Remark 7.2.12, and its single band variant, with the

n	16	17	18	19	20	21	22	23	24	32	36
$m_n \geq$	1.062	1.003	0.947	0.895	0.846	0.800	0.757	0.716	0.677	0.361	0.203
$M_n \leq$	0.862	0.895	0.926	0.956	0.985	1.012	1.039	1.064	1.089	1.259	1.331
$\ell_n \leq$	3.600	3.948	4.310	4.689	5.086	5.504	5.954	6.473	7.075	17.203	33.568
$r(1.2) \leq$	7.522	7.776	8.009	8.223	8.420	8.600	8.834	9.062	9.280	10.675	11.313
$r'(1.2) \leq$	4.648	5.019	5.361	5.675	5.961	6.222	6.650	7.078	7.489	10.181	11.577
threshold	30	34	37	41	46	50	56	62	69	213	558

Table 7.4: Bounds for the torus extremes over the band $1.2 \leq x \leq 1.3$ and for $r(1.2)$, $r'(1.2)$, rounded outwards, and the smallest integer height at which the first condition of Theorem 7.2.10 holds with these values; the second condition holds from a lower height. The bounds and the second condition are proved by a branch and bound whose final boxes are enclosed in ball arithmetic (Remark 7.2.12); the programs are in `verification_reciprocal_zeta/torus/`.

n	7	9	10	11	12	13	14	15
$m_n \geq$	1.403	1.052	0.892	0.744	0.607	0.482	0.367	0.261
$M_n \leq$	0.576	0.715	0.777	0.834	0.887	0.937	0.984	1.028
$\ell_n \leq$	1.796	3.193	4.167	5.428	7.121	9.511	13.140	19.314
$r(1) \leq$	4.848	5.888	6.291	6.629	6.971	7.445	7.886	8.294
$r'(1) \leq$	2.014	3.096	3.520	3.869	4.282	5.032	5.746	6.418
threshold	14	24	31	41	57	84	131	229

Table 7.5: As Table 7.4, for the single band variant: bounds for the extremes over the band $1 \leq x \leq \frac{3}{2}$ and for $r(1)$, $r'(1)$, and the smallest integer height at which its condition holds with these values.

bounds of Table 7.5. The hypothesis $m_n > r/y$ and the first condition, here and in the variant, hold with the true values wherever they hold with these bounds, because both improve as m_n increases and as M_n, ℓ_n and the values of r and r' decrease. Below the thresholds the argument principle count in $0 < \operatorname{Re} w < \frac{1}{2}$ equals the number of sign changes of Z_n on the line, with the following heights and counts:

n	7	9	10	11	12	13	14	15	16	17
height	15	24	32	43	58	86	132	230	500	35
zeros	3	9	14	23	36	64	116	242	649	17
n	18	19	20	21	22	23	24	32	36	
height	250	43	47	51	57	62	110	215	570	
zeros	269	23	26	30	35	40	90	221	763	

Each height is at least the threshold for the same n, with equality for $n = 9$ and $n = 23$, so the counts and the theorem together cover all heights. For $n = 36$ the sampling step on the line is 0.02, cross checked by the argument principle. The

cases $n \le 6$ and $n = 8$ are Corollary 7.2.9. Every count equals the number of zeros of ξ with ordinate below twice the height (for $n = 16$, the 649 zeros with $\gamma < 1000$), except for $n = 9$, where the ninth zero of K_9, at $\operatorname{Im} w = 23.82$, is the image of $\gamma_9/2 = 24.003$ displaced below height 24; such agreement need not hold at heights where n is small compared with $|\kappa|$. The counts are in floating point. □

Remark 7.2.12. The bounds of Tables 7.4 and 7.5 are proved by a branch and bound over the band and the half torus $0 \le \omega_2 \le \pi$, which suffices because $P_n(x,-\omega) = \overline{P_n(x,\omega)}$. On a box with centre (x_c,ω_c) and half widths h in x and h_p in ω_p, the kth term of P_n differs from its value at the centre by at most $\beta_k k^{-2x_c}\{(k^{2h}-1)+k^{2h}\min(2,\sum_p v_p(k)h_p)\}$, and the resulting enclosures of P_n, P_n' and their partial derivatives bound $|P_n|$ from below and $-\operatorname{Re}(P_n'/P_n)$ and $|P_n'/P_n|$ from above on the box, by a mean value form or directly, whichever is sharper; for M_n and ℓ_n the proved bound for m_n serves as a lower bound for $|P_n|$. Since $2\sigma - \frac12 > 0$, the second condition of Theorem 7.2.10 at height y is equivalent to

$$\Big(|P_n(1+\sigma,\omega)| - \frac{r(1+\sigma)}{y}\Big)\exp\Big(\big(2\sigma-\tfrac12\big)\Big(\log\frac{y}{\pi} - \frac{1}{y}\Big)\Big) > |P_n(\tfrac32-\sigma,\omega)| + \frac{r(\frac32-\sigma)}{y}$$

for $0.3 \le \sigma \le \frac12$ and all ω, which also gives the positivity of $|P_n(1+\sigma,\omega)| - r(1+\sigma)/y$ that $B_n(y)$ needs. The same branch and bound, applied to the difference of the two sides, proves this at the heights 29, 33, 36, 40, 45, 49, 55, 61, 68, 211, 555 for $n = 16,\dots,24,32,36$, each below the threshold, and so at every larger height. The searches run in double precision and only choose the boxes: every final box, about 1.1×10^8 in all, is proved again in ball arithmetic with Arb [110], with exact coefficients and on the exact box. The cases $n \le 6$ and $n = 8$ do not depend on the torus (Corollary 7.2.9).

With the band $1.2 \le x \le 1.3$ the method stops between $n = 40$ and $n = 44$. The band minimum m_n falls to 0.20 at $n = 36$, 0.056 at $n = 40$ and 0 at $n = 44$, so at $n = 44$ the hypothesis $m_n > r(1.2)/y$ of Theorem 7.2.10 fails at every height. This reflects the functions and not only the bound: the torus function $P_n(x,\omega)$ then vanishes at a point of the band, and since by Kronecker's theorem the prime phases at height y come arbitrarily close to every point of the torus, A_n has zeros with real part near $[0.2, 0.3]$ at arbitrarily large heights, where $\log|A_n|$ cannot increase across the strip. On the line $x = \frac54$ itself the minimum is still 0.29 at $n = 44$ and 0.19 at $n = 48$, so these torus zeros lie off the line, and a narrower band may postpone the stop. A torus zero with $x > \frac54$ would end every comparison of this kind: it gives zeros of A_n with $\operatorname{Re} w > \frac14$ at arbitrarily large heights, where $|A_n(w)| > |A_n^*(w)|$ fails. On which side of the line the torus zeros at $n = 44$ lie is not determined here.

The monotonicity used in Theorem 7.2.8 implies the Hermite Biehler inequality $|A_n(w)| > |A_n(\frac{1}{2} - \bar{w})|$ for $\frac{1}{4} < \operatorname{Re} w < \frac{1}{2}$, and the inequality holds far beyond the proven range. On a grid of $\sigma \in \{0.26, 0.3, 0.4, 0.49\}$ and $y \le 100$ it holds for $n = 8, 16, 24$ at every height above 3.6, and it fails only below 3.6, where the strip contains no zeros. For $n = 8, 16$ it also holds for $200 \le y \le 212$, around $\gamma_{213}/2$. It cannot hold for all large n: by Corollary 7.2.4, for large n the function A_n has a zero w with $\frac{1}{4} < \operatorname{Re} w < \frac{1}{2}$, where the inequality fails. Quantitatively, near a zero $\rho/2$ the first order margin is governed by the residue of $w\xi(2w-2)/\xi(2w)$, which is a positive multiple of $\kappa(\rho)$, so at zeros with $\operatorname{Re}\kappa(\rho) > 0$ the inequality reverses once n is large compared with $|\kappa(\rho)|$. So Hermite Biehler, like Conjecture T, is broken by the zeros with large $|S(\gamma)|$, while Conjecture S is immune to this mechanism, because a symmetric perturbation cannot move a simple zero off the line; a proof of Conjecture S for infinitely many n must establish real rootedness without a one sided inequality.

7.2.2 What is proved, and how far it reaches

No result of Chapter 6 or of this chapter proves RH. Each unconditional result proves part of a positivity condition that is equivalent to RH. Table 7.6 lists the forms of RH met in the two chapters, the unconditional results that bear on each, and the computation each result uses. Every entry in the middle column is a positivity statement about a transform, or about finitely many derivatives. Every entry on the left is a positivity statement about all derivatives, equivalently about a measure. The question is whether the middle column can be pushed to the left. So far it reaches Weil positivity on the Bernstein cone and the sign conditions $(-1)^n W^{(n)} > 0$ on the heat trace up to a finite order; no finite order reaches the full condition, and neither the de Bruijn Newman flow nor a GGC multiplier closes the gap. Chapters 9 to 11 measure the same gap from the right of the centre: $R_\sigma(s) = \xi(\sigma)/\xi(\sigma + \sqrt{s})$ is a GGC for every $\sigma \ge 1$ with no hypothesis (Theorem 7.1.14; Proposition 9.1.1 computes its Thorin measure at $\sigma = 1$), and for $\frac{1}{2} < \sigma < 1$ zero density estimates show that in the density $\rho_{\sigma-1/2}$ of (1.2.7) the Cauchy kernels of the zeros to the right of σ carry, below height T, a vanishing fraction of the mass of the kernels of the other zeros as $T \to \infty$ (Theorem 11.4.2).

7.2.2.1 Weil's criterion in Thorin coordinates

Proposition 6.2.16 left open which signed primal measures have a positive sine squared image. For $\tilde{\mu}_\xi$ the answer is Weil's criterion.

Proposition 7.2.13. *Let $D(x) = \sum_\rho e^{(\rho-\frac{1}{2})x}$ be Weil's distribution, whose action on a test function h is $\sum_\rho \hat{h}(\gamma_\rho)$ with $\gamma_\rho = -i(\rho - \frac{1}{2})$. Then R is the Laplace transform of a GGC, with Thorin measure the sine squared image of $\tilde{\mu}_\xi$, if and only if D is positive definite, that is, $\sum_\rho \hat{h}(\gamma_\rho) \ge 0$ for every $h = g * \tilde{g}$ with g*

Equivalent form of RH	Proved with no unproved hypothesis	Computation the middle column uses
$R(s) = \xi(\frac{1}{2})/\xi(\frac{1}{2} + \sqrt{s})$ is a GGC, Thorin measure $\sum \delta_{\gamma^2}$ [198, Proposition 3.2]	$e^{-c\sqrt{s}}/G$ is a GGC (Theorem 6.2.12); $e^{\sqrt{s}\zeta'/\zeta(\beta)}\zeta(\beta)/\zeta(\beta + \sqrt{s})$ is a GGC for $\beta > 1$ (Proposition 6.2.6)	none
$W(x) = \frac{1}{2}\sum_\rho e^{x(\rho-\frac{1}{2})^2}$ is completely monotone	$W > 0$ (proof of Theorem 7.2.14) and $(-1)^n W^{(n)} > 0$ for $n \leq 1.16 \times 10^{25}$ (Theorem 5.1.10); $W + c/(2\sqrt{\pi x})$ is positive (Theorem 6.2.15) and n monotone for $n \leq 10^{10}$ (Theorem 6.2.17)	positivity: RH to 50 and enclosures; order 1.16×10^{25}: RH to 3×10^{12} and enclosures; n monotonicity: RH to 3×10^{12}
$e^{-c\sqrt{s}}R$ is a GGC	it is infinitely divisible (Theorem 6.2.15)	RH to 50 and enclosures
w is the Laplace transform of a real measure on $[0, \infty)$ (Theorem 6.1.15)	$w > 0$ (Theorem 7.1.1)	RH to 50 and enclosures
the sine squared image of $\tilde{\mu}_\xi$ is positive, equivalently Weil's distribution is positive definite (Proposition 7.2.13)	its subordination stage is positive; Weil positivity holds on the Bernstein cone (Theorem 7.2.14), against every Gaussian (its proof) and against $r^{2n}e^{-xr^2}$ for $n \leq 1.16 \times 10^{25}$ (Theorem 5.1.10); the canonical function k of the floor is n monotone for every $n \leq 10^{10}$ (Theorem 6.2.17), which is strictly weaker than n monotonicity of W	Bernstein cone: RH to 50 and enclosures; order 1.16×10^{25}: RH to 3×10^{12} and enclosures; n monotonicity: RH to 3×10^{12}

Table 7.6: Forms equivalent to RH, against what is proved in each case with no unproved hypothesis, and what computation that proof consumes. The first row consumes none.

smooth and compactly supported. Both conditions are equivalent to RH.

Proof. By Proposition 1.2.1, R is the Laplace transform of a GGC if and only if RH holds, and its Thorin measure is then $\sum_{\gamma>0}\delta_{\gamma^2}$, which is the sine squared image of $\tilde\mu_\xi$ by Proposition 6.2.16(ii). Weil's theorem [266, 34] states that RH holds if and only if $\sum_\rho \hat h(\gamma_\rho)\ge 0$ for all $h = g*\tilde g$ with g smooth and compactly supported, that is, if and only if D is positive definite. □

A formal computation shows how the two formulations correspond. For a positive measure U on $(0,\infty)$ with $\int U(dt)/(1+t)<\infty$, the primal measure $\tilde\mu_U(dx) = -2\big(\int\cos(\sqrt{t}\,x)\,U(dt)\big)dx$ on $x>0$, understood as a distribution, has sine squared image U: since $\int_0^\infty(1-\cos x\sqrt{z})(-2\cos bx)\,dx = \pi\delta(\sqrt{z}-b)$, $\tilde\mu_U$ maps to $\int\delta(z-t)U(dt)=U$. The map of Lemma 6.2.5 is injective on the class of that lemma: if $\int_0^\infty \sin^2(x\sqrt{z}/2)\,\nu(dx)=0$ for every $z>0$, then $\psi(t)=\int(1-\cos xt)\,\nu(dx)$ vanishes for all real t, and $\psi(t)-\frac12\int_{t-1}^{t+1}\psi(s)\,ds = -\int\cos(xt)\big(1-\frac{\sin x}{x}\big)\nu(dx)$ shows that the finite measure $(1-\frac{\sin x}{x})\nu(dx)$ has vanishing cosine transform, so $\nu=0$. Carried over to distributions, these two facts say that a primal measure has a positive sine squared image if and only if $-\tilde\mu$ is the restriction to $x>0$ of an even positive definite distribution on $\mathbb{R}$, that is, by the Bochner Schwartz theorem, the Fourier transform of a positive measure. For ξ, $-\tilde\mu_\xi$ is the restriction to $x>0$ of D, by (6.2.6); D is even because $\rho\mapsto 1-\rho$ permutes the zeros, and the explicit formula (6.2.5) is its prime side. The computation is formal, for two reasons: Lemma 6.2.5 is proved only for primal measures with $\int(1\wedge x)|\tilde\mu|(dx)<\infty$, and when RH fails D grows exponentially and is not tempered, so the Bochner Schwartz theorem applies only in Weil's own formulation, with compactly supported test functions. At the centre, therefore, the sine squared lemma does not produce a new criterion; it produces Weil's, in Thorin coordinates. Riesz's criterion has a Thorin form of the same kind: Theorem 10.2.7 of Section 10.2 shows that it is equivalent to R_σ being a GGC for every $\sigma>\frac12$, and to the convergence of compensated sine squared sums over the primes.

The unconditional positivity results of Chapter 6, of this chapter and of Section 5.1 are partial positivity statements of Weil type. The gamma part is a positive definite archimedean piece (Theorem 6.2.12). The heat trace results are Weil positivity along the Gaussian family: with h Gaussian, $\sum_\rho\hat h(\gamma_\rho)=2W(x)$, and $W(x)>0$ for every $x>0$ unconditionally (Theorem 5.1.1 [200, Theorem A]; the proof of Theorem 7.2.14 gives a second argument, from the prime side for small x and the zeros below height 50). Weil positivity against $r^{2n}e^{-xr^2}$ is $(-1)^nW^{(n)}\ge 0$, which Theorem 5.1.4 [198, Theorem 7.10] proves for every $n\le 10^{17}$ and Theorem 5.1.10 for every $n\le 1.16\times 10^{25}$. Theorem 6.2.17 proves the corresponding statement for the canonical function $k = W + c/(2\sqrt{\pi x})$ of the floor, for $n\le 10^{10}$, and not for W itself: the two differ at every order $n\ge 1$ by the strictly positive $c\,\Gamma(n+\frac12)x^{-n-\frac12}/(2\pi)$, which is the term the large x branch of that proof uses to dominate the off line tail. Section 6.2 compares the reach of the two theorems

after Theorem 6.2.17. Weil's criterion asks for all $h = g * \tilde{g}$ at once. The known unconditional cases of Weil positivity, due to Yoshida [276, Theorem 1] and to Bombieri [34], concern test functions of small support; the Gaussian family is the opposite regime, of test functions whose Fourier transforms are Gaussian, and it is the regime in which Theorems 5.1.4, 5.1.10 and 6.2.17 apply.

7.2.2.2 The Bernstein cone and the resolution of finite order tests

Proposition 7.2.13 identifies the Thorin positivity of R with Weil positivity for all test functions $\hat{h} = |\hat{g}|^2$. A cone of test functions of another kind, completely monotone in r^2, can be settled unconditionally.

THEOREM 7.2.14. *Let $\hat{h}(r) = \int_0^\infty e^{-yr^2} g(y)\,dy$ with $g \geq 0$, $g \not\equiv 0$ and $\int_0^1 g(y)\, y^{-1/2}\big(1 + \log(1/y)\big)\,dy + \int_1^\infty g(y)\,dy < \infty$, so that $\hat{h}$ is a completely monotone function of r^2. Then*

$$\sum_\rho \hat{h}(\gamma_\rho) = 2\int_0^\infty W(y)\, g(y)\, dy > 0,$$

where $\gamma_\rho = -i(\rho - \frac{1}{2})$ and the sum is over all nontrivial zeros. In particular Weil's functional is positive for every test function of Thorin type $\hat{h}(r) = \prod_k (1+r^2/t_k)^{-1}$, a nonempty product with $t_k > 0$ and $\sum_k 1/t_k < \infty$, for $\hat{h}(r) = \operatorname{sech}(cr)$ with $c > 0$, and for $\hat{h}(r) = (1+r^2/a)^{-m}$ with $a > 0$ and $m > \frac{1}{2}$. The proof uses the verification of RH to height 50 and the interval enclosures of Theorem 7.1.1 together with the enclosures E5 and E6 of Section 7.2.3.

Proof. For each $y > 0$, $\sum_\rho e^{-y\gamma_\rho^2} = \sum_\rho e^{y(\rho-\frac{1}{2})^2} = 2W(y)$, and $W(y) > 0$ for every $y > 0$. For $y \leq 0.003$ this comes from the prime side: $W = w + W_0$ with $w > 0$ (Theorem 7.1.1) and $W_0 \geq 0$, because the function F of Lemma 6.1.12 satisfies $F(0.003) > 0.0329$ (enclosure E6) and is decreasing. For $y \geq 0.003$ it comes from the zeros below height 50, which lie on the line and contribute $e^{-y\gamma^2} > 0$: the first alone contributes more than $e^{-0.003 \cdot 14.14^2} > 0.548$ at $y = 0.003$, the off line tail above height 50 is below 0.0162 there (enclosure E5, with the bound $N(t+1) - N(t) \leq 2\log(t+1)$ of the proof of Theorem 7.1.1), and the ratio of the tail to the first term decreases as y grows. The interchange of sum and integral is justified by Fubini's theorem: $\sum_\rho |e^{y(\rho-\frac{1}{2})^2}| \leq e^{y/4} \sum_\rho e^{-y(\operatorname{Im}\rho)^2}$, which is $O(y^{-1/2}(1 + \log(1/y)))$ for $y \leq 1$ by the Riemann von Mangoldt formula and bounded for $y \geq 1$, and the integrability condition makes this majorant integrable against g. The Thorin type functions are completely monotone in r^2 because $\prod(1 + s/t_k)^{-1}$ and $\operatorname{sech}(c\sqrt{s})$ are GGC transforms. The examples satisfy the integrability condition. A nonempty product $\prod_k (1 + s/t_k)^{-1}$ is the Laplace transform of $\sum_k e_k/t_k$, with e_k independent standard exponential variables, and the density g of this sum is at most t_1, the maximum of the density of e_1/t_1; the function $\operatorname{sech}(c\sqrt{s}) =$

$\prod_{k\ge1}\bigl(1+4c^2s/((2k-1)^2\pi^2)\bigr)^{-1}$ is such a product; and for $(1+r^2/a)^{-m}$ the function $g(y)=a^m y^{m-1}e^{-ay}/\Gamma(m)$ satisfies the condition because $m>\frac12$. $\square$

The Bernstein cone consists of test functions that decrease in $|r|$ and so cannot concentrate near a given height. This is why positivity on it does not imply RH, and it matches the resolution analysis of Table 7.7: an off line zero at height γ and distance δ from the line is detected only by test functions able to localise at γ, such as $r^{2n}e^{-yr^2}$ with n of order γ^2/δ. Weil's full cone $\{|\hat g|^2\}$ contains such functions; the Bernstein cone does not. Theorem 5.1.10 gives $(-1)^nW^{(n)}>0$ for $n\le 1.16\times10^{25}$, which is Weil positivity for $r^{2n}e^{-yr^2}$ and, after integration against g where the integral converges, for the image of the Bernstein cone under multiplication by r^{2n}.

Complete monotonicity of W is Bernstein's condition, and its Post Widder form is

$$L_n[W](t)=\frac{(-1)^n}{n!}\Bigl(\frac{n}{t}\Bigr)^{n+1}W^{(n)}\Bigl(\frac{n}{t}\Bigr)\ge0\qquad\text{for all } n,t.$$

$L_n[W]$ is the Thorin measure $\sum\delta_{\gamma^2}$ seen through a kernel of relative width about $n^{-1/2}$, and an off line quadruple $\rho=\frac12\pm\delta\pm i\gamma$ contributes $2\,\mathrm{Re}\{(n/t)^{n+1}z^ne^{-zn/t}/n!\}$ with $z=\gamma^2-\delta^2-2i\delta\gamma$. Table 7.7 gives the smallest order n^* at which such a quadruple, inserted midway between two consecutive true ordinates among the first 300 zeros, makes $\min_t L_n[W](t)$ negative.

γ	$\delta=0.4$	$\delta=0.2$	$\delta=0.1$	$\delta=0.05$
27.7	724	1448	2896	5792
60.1	11585	23170	46340	92681
120.1	32768	65536	131072	262144

Table 7.7: Smallest order n^* at which an inserted off line quadruple at height γ and distance δ makes $\min_t L_n[W](t)$ negative.

In every row the detection order doubles when δ halves, within the resolution $\sqrt2$ of the scan, and across the rows it grows roughly like γ^2: the ratio $n^*\delta/\gamma^2$ lies between 0.38 and 1.3 in Table 7.7. So an off line zero at height γ first becomes visible at order about γ^2/δ, and a zero above the verified height T_0 of Theorem 6.2.17 at order about T_0^2/δ. Theorem 5.1.10 reaches orders of this size, every order up to $1.29(T_0-23)^2$ for $T_0=3\times10^{12}$ (Section 5.1.7), while the method of Theorem 6.2.17, whose Cauchy estimate loses a factor 2^n, certifies n monotonicity only up to orders proportional to T_0. Finite order tests therefore cannot prove RH: each order sees only a bounded height, and the height it sees tends to infinity only with the order. A proof must supply positivity at all orders at once, and in the Thorin framework that means a structural reason for R to be a GGC.

7.2.2.3 Weak limits and the de Bruijn Newman flow

The class of GGC laws is closed under convolution and weak limits [35, Section 3.1]. So RH is equivalent to exhibiting R as a limit of Laplace transforms of GGC laws, and positivity then passes to the limit at every order simultaneously. The de Bruijn Newman flow gives a family of candidates. Let $\Phi_{\mathrm{dBN}}(u) = \frac{1}{2}\Phi(2u) = \sum_{n\geq 1}(2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u})\exp(-\pi n^2 e^{4u})$, with Φ the Pólya kernel (1.2.3), which Section 7.1.3 uses in the variable t; Φ_{dBN} is the kernel of Section 8.4, in the rescaled variable of Rodgers and Tao [224]. Let $H_t(z) = \int_0^\infty e^{tu^2}\Phi_{\mathrm{dBN}}(u)\cos(zu)\,du$ be the de Bruijn Newman deformation of $H_0(z) = \frac{1}{8}\xi(\frac{1}{2} + \frac{iz}{2})$, with Λ defined in this normalisation, and put

$$R_t(s) = \frac{H_t(0)}{H_t(2i\sqrt{s})}, \qquad R_0 = R.$$

THEOREM 7.2.15. *Let Λ be the de Bruijn Newman constant. For every $t \geq \Lambda$, R_t is the Laplace transform of a GGC with Thorin measure $\sum_k \delta_{(z_k(t)/2)^2}$, where $z_k(t) > 0$ are the zeros of H_t. In particular R_t is a GGC for every $t \geq 0.2$, unconditionally. RH holds if and only if R_t is a GGC for arbitrarily small $t > 0$.*

Proof. For $t \geq \Lambda$ all zeros of H_t are real by definition of Λ, and H_t is even and entire of order one with the Hadamard product $H_t(z) = H_t(0)\prod_k(1 - z^2/z_k^2)$, so $R_t(s) = \prod_k(1 + 4s/z_k^2)^{-1}$, an infinite convolution of exponential laws. The bound $\Lambda \leq 0.2$ is [196], improving $\Lambda \leq 0.22$ [217]. As $t \downarrow 0$, $H_t \to H_0$ locally uniformly, so $R_t \to R$ pointwise on $(0,\infty)$; if each R_t is a GGC transform, so is the limit, and Proposition 6.1.9 gives RH. Conversely RH means $\Lambda \leq 0$, so every R_t with $t \geq 0$ is a GGC. By Rodgers and Tao [224], $\Lambda \geq 0$, so RH is equivalent to $\Lambda = 0$. □

Conversely, for $t < \Lambda$, and so for every $t < 0$ [224], R_t is not the Laplace transform of a GGC (Corollary 4.2.5), and Proposition 11.5.14 turns this into a test for the properties of the Pólya kernel that could imply RH.

Along the flow the functions R_t form a family of clocks indexed by the flow time, and Theorem 7.2.15 places RH at the end of this GGC path: the Thorin measure $\sum \delta_{(z_k(t)/2)^2}$ is positive for $t \geq 0.2$, and RH is the statement that it stays a positive measure as $t \downarrow 0$. The zeros $z_k(t)$ move by the backward heat flow, and the Thorin measure can cease to be positive only by two real zeros colliding and leaving the real line, which is the collision picture of Rodgers and Tao; Section 8.4 computes the flow numerically. What Chapter 6 and this chapter add concerns $t = 0$: any failure there comes from zeros above the verified height, and Table 7.7 shows that it would first be visible at orders of about γ^2/δ. For $t \neq 0$ no prime side is known for the Thorin measure of R_t, and changing the test function does not give one. Replacing the Gaussian e^{-xr^2} of Theorem 6.1.11 by a t dependent test function gives a linear statistic $\sum_\rho h(\gamma_\rho)$ of the zeros of ξ, whereas the heat trace $\sum_k e^{-xz_k(t)^2/4}$ of the Thorin measure of R_t depends on those zeros nonlinearly:

along the equation of motion of the zeros (Section 8.4.1) its derivative in t is a sum over pairs of zeros. No Euler product for H_t is known either.

7.2.2.4 No GGC multiplier can manufacture positivity

One might replace the gamma factor by another function V that makes the Thorin measure positive, and read off RH. Proposition 7.2.16 shows that this is equivalent to RH, whatever V is.

Proposition 7.2.16. *Let V be the Laplace transform of any GGC. Then $V \cdot R$ is the Laplace transform of a GGC if and only if RH holds. For φ only one direction holds for every V: if $V \cdot \varphi$ is a GGC transform then RH holds. The converse depends on V: it fails for $V \equiv 1$ (Theorem 7.1.3) and holds for $V = e^{-c\sqrt{s}}/G$, for which $V\varphi$ is the floor $e^{-c\sqrt{s}}R$ of Theorem 6.2.15.*

Proof. If RH holds, R and V are GGC transforms and so is their product. Conversely, if VR is a GGC transform, then $-(VR)'/(VR)$ and $-V'/V$ are Stieltjes functions, analytic on $\mathbb{C} \setminus (-\infty, 0]$, so $-R'/R$ is analytic there and R has no zeros or poles in $\mathbb{C} \setminus (-\infty, 0]$. By Proposition 6.1.9 this is RH. The same argument applies to $V \cdot \varphi$, since $\varphi = RG$ with G analytic and zero free on $\mathbb{C} \setminus (-\infty, 0]$. For $V = e^{-c\sqrt{s}}/G$, a GGC transform by Theorem 6.2.12, $V\varphi = e^{-c\sqrt{s}}R$, which is a GGC under RH by Theorem 6.2.15. □

Multipliers can remove obstructions that come from normalisation, as the floor does, but not the obstruction that comes from the zeros. Thorin measures add under multiplication, so a multiplier can hide negative Thorin mass. What RH forbids, however, is Thorin mass off the half line, the complex atoms at $-(\rho - \frac{1}{2})^2$ contributed by zeros off the critical line, and no GGC factor can cancel a singularity of R in the cut plane. This is why the floor $e^{-c\sqrt{s}}R$ of Theorem 6.2.15, which uses the best available multiplier $e^{-c\sqrt{s}}/G$, is infinitely divisible but can be shown to be a GGC only by proving RH. Section 11.1 finds the same for the general theory of infinitely divisible laws: each GGC criterion it examines is, for R, equivalent to RH (Section 11.1.5).

7.2.3 Numerical verification

The computations of Chapter 6 and of this chapter have three levels of certainty. The inputs to Theorems 7.1.1, 6.2.15, 6.2.17 and 7.2.14 are enclosed in outward rounded interval arithmetic by `verification_reciprocal_zeta/enclosures.py` (enclosures E1 to E6, recorded in `verification_reciprocal_zeta/enclosures_report.md`; Appendix A lists the scripts). The bounds $233.95 < \operatorname{Re}\kappa(\rho_{213}) < 233.96$ of Corollary 7.2.4 are enclosed in ball arithmetic with Arb [110] by `verification_reciprocal_zeta/certify_k9_k2.py`, which evaluates $\kappa(\rho_{213})$ both from its definition and from the formula of Theorem 7.2.3, and which

also encloses the values of S quoted with Theorem 7.1.3. The torus bounds of Tables 7.4 and 7.5 and the second condition of Theorem 7.2.10 are proved by a branch and bound whose final boxes are enclosed in ball arithmetic with Arb (Remark 7.2.12); the programs are in `verification_reciprocal_zeta/torus`. The identities of Table 7.8 and the constants of Table 7.9 are computed in floating point with `mpmath` at 20 to 30 digits, and the argument principle counts of Section 7.2.1 in floating point with 40 or more digits. The torus minima quoted for $n \geq 40$ after Remark 7.2.12, and the threshold at $n = 40$ in Problem 7.2, are estimated by optimisation and are not certified.

Sums over zeros use the first 200 or 300 zeros: 300 in the rows for Theorems 6.1.11 and 6.2.3, which are checked against $\sum e^{-x\gamma^2}$, and 200 in the rows for Theorem 6.1.10. In the rows for Theorem 6.1.10 the left side is $4/(4s-1) + \frac{1}{2\sqrt{s}}\frac{\zeta'}{\zeta}(\frac{1}{2}+\sqrt{s})$ and the right side $\int M(t)(s+t)^{-2}dt$, the gap being the truncation at γ_{200}; the value $s = 0.1 < \frac{1}{4}$ confirms the representation across the removed pole. The right side of the row for (6.2.2) is a quadrature over $(0,\infty)$ across the logarithmic singularity at $t = \tau^2$, and since (6.2.2) is an identity its gap of 5×10^{-9} is quadrature error. The row for Proposition 6.2.7 uses oscillatory quadrature with error near 2×10^{-5}. The row for Proposition 6.2.6 truncates the sum over prime powers at 3000, and the truncation accounts for the whole of its gap. The omitted terms do not enter with weight one: the identity is $\sum_n \Lambda(n) n^{-\beta}\{(1-n^{-\sqrt{s}})/\log n - \sqrt{s}\}$, so each enters weighted by $\sqrt{s} - (1-n^{-\sqrt{s}})/\log n$, which is 1.607 at $n = 3000$. The missing mass of $\tilde{\mu}_2$ is $-\zeta'(2)/\zeta(2)$ less the partial sum, that is $0.5699610 - 0.5696282 = 3.33 \times 10^{-4}$, and weighted it is 5.35×10^{-4}, against the gap of 5.39×10^{-4} between the two sides of the row.

Table 7.9 collects the constants used in Sections 6.1.4 and 7.1.1. The function F of Lemma 6.1.12 is evaluated by tanh sinh and by Gauss Legendre quadrature, and the two agree to 20 digits. The function w at $x = 10^{-3}, 10^{-2}, 0.1, 1, 10$ is 1.00025, 1.00249, 0.86251, 0.49140, 0.21524; its initial increase rules out self decomposability (Theorem 7.1.3). Talbot inversion of φ gives the density f of the law with Laplace transform φ: $f(0.05) = 0.36222$, $f(1) = 0.13508$, $f(16) = 0.0087321$, and $x^{3/2}f(x) = 0.5589, 0.6382, 0.6908, 0.7220, 0.7392$ at $x = 16, 32, 64, 128, 256$, approaching 0.75773.

7.2.4 What the results say about the Thorin approach

The Thorin approach seeks a function whose Thorin measure is positive if and only if RH holds, together with a reason, independent of the zeros, for that positivity. For $R = \xi(\frac{1}{2})/\xi(\frac{1}{2}+\sqrt{s})$ the first half is Proposition 1.2.1, and the results of Chapter 6 and of this chapter clarify the second half.

The right object is R, not φ. The pole and the gamma factor spoil the positivity of φ for the two reasons given after Theorem 7.1.3, neither of which involves a

Identity	Left side	Right side
Prop. 6.1.3 at $x = 0.3$	0.74081822068171786607	0.74081822068171786607
Prop. 6.1.6 at $x = 0.5$	0.60653065971263342360	0.60653065971263342360
Thm. 6.1.8, $s = 2$, $N \le 10^6$	−0.2311168	−0.2311181
Thm. 6.1.10, $s = 0.1$	2.9071962641	2.9071962669
Thm. 6.1.10, $s = 2$	0.3403105134	0.3403105162
Thm. 6.1.10, $s = 10$	0.0889209011	0.0889209039
Thm. 6.1.11, $x = 0.1$, primes vs zeros	0.8625096	0.8625096
Thm. 6.1.11, $x = 1$, primes vs zeros	0.4914032	0.4914032
Lemma 6.1.12, $W_0(0.1)$, closed form vs direct	−0.86250960	−0.86250960
Prop. 6.1.13, $x = 0.1$, $N \le 10^7$	−0.16280552	−0.16280552
Thm. 6.2.1, $-\theta'(20)$ vs series	−0.57887549742241679994	−0.57887549742241679994
Prop. 6.2.2, $W_0(0.01)$, F vs erfc	−0.85279671685617205031	−0.85279671685617205031
Thm. 6.2.3, $x = 0.005$, zeros vs formula	0.53719196409104089570	0.53719196409104089570
Thm. 6.2.3, $x = 0.01$	0.14969801125377957789	0.14969801125377957789
Thm. 6.2.3, $x = 0.03$	0.00249613734848799360	0.00249613734848799360
(6.2.2), $\tau = 5$	−4.5238630355	−4.5238630405
(6.2.3), $v = 0.7$, $2^{3/2}\pi^{3/2}\operatorname{sech}(v)^{1/2}$	14.057843341066117381	14.057843341066117381
Prop. 6.2.8(ii), $\lambda = 1.3$, $u = 0.9$	0.27030720582643256107	0.27030720582643256107
Prop. 6.2.7, $\lambda = 1.1$, $\alpha = 0.4$, $s = 2.5$	−0.29135440	−0.29133306
Prop. 6.2.6, $\beta = 2$, $s = 3$, $n < 3000$	−0.587879	−0.587340
Thm. 6.2.12, $s = 0.5$	−0.5993903241	−0.5993903238
Thm. 6.2.12, $s = 2$	−1.7275429395	−1.7275429395
Thm. 6.2.12, $s = 7$	−4.1886802718	−4.1886802718
Prop. 6.2.11, $\tau = 3$, $\operatorname{Re}\Phi_\Gamma(i\tau)/2$	2.3139696797	2.3139696797
Prop. 6.2.11, $\tau = 3$, $\operatorname{Im}\Phi_\Gamma(i\tau)$	1.7430641614	1.7430641614

Table 7.8: Numerical verification of identities of Chapter 6: left and right sides, at the stated arguments. Where the two sides differ, one of them is a truncated sum or a quadrature.

Quantity	Value
$c = -\theta'(0) = \zeta'(\frac{1}{2})/\zeta(\frac{1}{2})$	2.6860917096
τ_0, $\theta(\tau_0)$, $\max M = -\theta(\tau_0)/\pi$	6.2898360, −3.5309728, 1.1239436
$M(\gamma_1^{2-})$	0.5502528
τ_π with $\theta(\tau_\pi) = -\pi$	3.4362182
x_* with $W_0(x_*) = 0$	0.0032173906
$F(0.003)$, $F(0.0035)$, $F(0.004)$	0.0366850, −0.0442285, −0.1145624
$P(0.004)$	1.99×10^{-13}
$\min S$ over the first 300 ordinates	$S(\gamma_{289}^{-}) = -1.1454808$
$\max S$ over the first 300 ordinates	1.0975638, at $\gamma = 415.4552$

Table 7.9: Constants used in Sections 6.1.4 and 7.1.1.

zero off the line, and both disappear for the completed function R, whose Thorin measure under RH is $dN(\sqrt{t})$.

The primes enter only through transforms. Every prime side formula converges in heat trace (Proposition 6.1.13, Theorem 6.2.3) but not as a series of measures, and the weaker Poisson regularisation already diverges (Proposition 6.2.9). An argument for positivity that starts from the primes must work with transforms.

Positivity of a transform is not enough. Infinite divisibility, proved in Theorem 7.1.1, concerns the sign of the Lévy density $w(x)/x$; the GGC property concerns all derivatives of w. Finite order information sees only bounded heights (Table 7.7), and multiplication by any GGC cannot help (Proposition 7.2.16). What is needed is a structural source of positivity at all orders. Section 11.4.7 reaches the same conclusion to the right of the centre: at a basepoint $\alpha > \frac{1}{2}$, removing the Cauchy kernels of the zeros to the right of α, which Theorem 11.4.2 shows to carry a vanishing fraction of the mass of the others, is a step from counting zeros to excluding them, and needs input from outside the programme.

The one sided conditions on the truncations fail at the same zeros. Conjecture T and the Hermite Biehler inequality both break down, for large n, at the zeros ρ with $\operatorname{Re}\kappa(\rho) > 0$, which are the zeros where $|S(\gamma)|$ is large (Theorem 7.2.3 and the computation after it); no computation in this chapter reaches values of n at which the failure of Conjecture T would be visible (Remark 7.2.5). The torus bounds, with the band $1.2 \leq x \leq 1.3$, stop much earlier, between $n = 40$ and $n = 44$, and for a different reason: the torus function P_n acquires zeros in the band, so that $\log|A_n|$ can no longer increase across the strip at every large height. The mechanism that survives is symmetry: a symmetric perturbation cannot move a simple zero off the line, only a collision can. Conjecture S isolates this mechanism, and its numerical verification for $n \leq 24$ and $n = 32, 36$ is the strongest evidence in this chapter that it is the right one.

Weil positivity is reached, but only on an archimedean cone. Proposition 7.2.13 shows that Thorin positivity of R is Weil's criterion in other coordinates, so the unconditional fragments proved in the two chapters are unconditional fragments of Weil positivity: the gamma part (Theorem 6.2.12), the Bernstein cone (Theorem 7.2.14), and $W > 0$ itself, which is Weil positivity against every Gaussian. The full family $r^{2n}e^{-xr^2}$ needs $(-1)^n W^{(n)} \geq 0$ for every n, which is proved only up to a finite order (Theorem 5.1.4, and Theorem 6.2.17 for the floor), and no finite order reaches the whole family. The known unconditional cases, of small support [34], sit at the other end of the test function range. The archimedean factor decides which test functions are within reach; the positivity on them comes from the primes at small x and from the verified zeros at large x.

The archimedean factor is positive in the right coordinates. The Meixner law is a GGC in the dual variable (Section 6.2.3), the reciprocal gamma factor is a GGC after a stable tilt (Theorem 6.2.12), and the Meixner Pollaczek polynomials satisfy the local Riemann Hypothesis (Theorem 6.2.13). What the Meixner law cannot

supply is the arithmetic, and it is the arithmetic, through the weights $b_n(p^{-1/2})$ of Theorem 7.1.8 and the values of S, that every route in the two chapters reaches in the end.

Three steps remain open, in increasing order of difficulty. First, write the formal Thorin measure of R_t in Vere-Jones form, extending Theorem 6.2.3 with t dependent weights; no prime side of this kind is known for $t \neq 0$ (Section 7.2.2). Second, prove Conjecture S for infinitely many n, which by Theorem 7.2.7 implies RH (Problems 7.1 and 7.2). Third, find a family of positive admissible primal Lévy measures whose sine squared images converge to the image U of $\tilde{\mu}_\xi$ in Proposition 6.2.16; by Proposition 7.2.13 this is Weil positivity in Thorin coordinates, and a proof along the lines of the two chapters would have to establish it.

Chapter 8 leaves the Thorin measure for the clock itself, the random variable whose Laplace transform is R, and begins in Section 8.1 with its law. Section 9.1 reads the Mellin transform of Theorem 7.1.21 probabilistically, through Bessel hitting times, and Section 10.2 returns to the reciprocal zeta function through the values $1/\zeta(2k)$ that enter Riesz's criterion.

7.2.5 Open problems

The first two problems concern the route to RH through symmetrised truncations; the rest are questions about the structures found along the way.

Problem 7.1. Conjecture S. Prove that the symmetrised truncations K_n have all their zeros in $0 < \operatorname{Re} w < \frac{1}{2}$ on the line $\operatorname{Re} w = \frac{1}{4}$ for infinitely many n; by Theorem 7.2.7 this implies RH. Corollaries 7.2.9 and 7.2.11 give its status for $n \leq 24$ and $n = 32, 36$. Since Conjecture T and the Hermite Biehler inequality for A_n fail for large n, a proof must establish real rootedness without a one sided inequality, for instance by an interlacing argument in n, or by following the zeros of Z_n continuously in a parameter interpolating between n and $n+1$ and excluding collisions of zeros on the line.

Problem 7.2. Beyond the torus method. Corollary 7.2.11 rests on torus bounds proved by a branch and bound up to $n = 36$; at $n = 40$ the threshold, estimated by optimisation, is already near 4900, and with the band $1.2 \leq x \leq 1.3$ the method ends between $n = 40$ and $n = 44$, where the torus minimum of P_n in the band reaches zero. Extend Theorem 7.2.10 with narrower bands around $x = \frac{5}{4}$, and find the first n at which P_n has torus zeros with $x \geq \frac{5}{4}$.

Problem 7.3. From finite order to all orders. Theorem 6.2.17 proves that $k = W + c/(2\sqrt{\pi y})$ is n monotone for $n \leq 10^{10}$, with the bound on n coming only from the height of verified zeros; the small y branch holds for every n, on $(0, 0.05/n]$. Can the two regimes be made to overlap for all n? The large y branch bounds the zeros above T_0 in absolute value, and the proof applies it where $yT_0^2 > 2n \log T_0$. No bound of this kind works where yT_0^2 is below about n: the peak of $\gamma^{2n} e^{-y\gamma^2}$, at

$\gamma^2 = n/y$, then lies above T_0, where the zeros are dense enough that their contributions, taken in absolute value, outweigh the positive term $c\,\Gamma(n+\frac{1}{2})y^{-n-\frac{1}{2}}/(2\pi)$ and the verified zeros together. So this asks for a prime side estimate for $(-1)^n k^{(n)}$ on a range $y \le y_n$ with y_n growing at least linearly in n; a range that decreases more slowly than $1/n$ raises the order reached but reaches only finitely many orders. In the proof of Theorem 6.2.17 the obstruction is the factor 2^n in the Cauchy estimate for $P^{(n)}$ on the circle of radius $y/2$. A circle of radius close to y, as in the proof of Theorem 5.1.4, loses only a factor of order n (Section 5.1.7), but that proof also stops at a finite order, and Proposition 7.1.2 suggests that at large y the obstruction cannot be removed without the explicit formula.

Problem 7.4. Beyond the Bernstein cone. Theorem 7.2.14 proves Weil positivity for test functions completely monotone in r^2, under an integrability condition. Find the largest cone of test functions, closed under the operations that preserve the Thorin structure, on which positivity can be proved unconditionally, and determine how far it can localise in height.

Problem 7.5. Deforming the binary product. By Theorem 7.1.8 each Euler factor at $\operatorname{Re} s = \sigma$ is $\prod_n (1+p^{-nu})^{b_n(p^{-\sigma})}$, with integer multiplicities 0 and 1 at $\sigma = 0$ and necklace multiplicities at $\sigma = \frac{1}{2}$. Follow the pseudo spectral trace formula (7.1.3) along the path $q = p^{-\sigma}$, $0 \le \sigma \le \frac{1}{2}$, and determine where and how the positivity of the interference pattern of the progressions can fail.

Problem 7.6. Regularising the autocorrelation. Proposition 6.2.14 factors $\Phi * \Phi$, formally, into the Meixner characteristic function and the positive lattice measure $\mathcal{A}$. The lattice series diverges: truncated at $m, n \le X$ it grows linearly in X, through the square of the pole term of the partial sums of ζ, and the factor $(\frac{d^2}{dv^2} - \frac{1}{4})^2$ does not remove this growth. Find the regularised form of $\operatorname{sech}^{1/2} * \mathcal{A}$ and decide whether it has a Thorin or Lee and Yang structure in v.

Problem 7.7. Unimodality. Proposition 7.1.5 gives numerical evidence that w is unimodal. Prove it, and prove that f is unimodal, with mode near 0.0240, for instance from Steutel's equation (7.1.1) and the unimodality of w. Yamazato's theorem [274], [232, Theorem 53.1], that every self decomposable law is unimodal, shows with no hypothesis that X_R, the variable with Laplace transform R, is unimodal (Corollary 11.1.2 of Section 11.1), but it does not apply to X_φ, whose density is f, since X_φ is infinitely divisible and not self decomposable (Theorem 7.1.3).

Problem 7.8. Infinite divisibility across the strip. Is $w_\sigma > 0$ for every $\sigma \in (0,1)$ and every $x > 0$? For $\frac{1}{2} < \sigma < 1$ the method of Theorem 7.1.1 may apply: in the third form of Theorem 7.1.12 the terms of the zeros below the verified height are Poisson kernels of positive width, so the question reduces to the sign of the explicit pole and archimedean terms at large x, whose limit is governed by $\zeta'(\sigma)/\zeta(\sigma) > 0$. For the completed functions the answer is known: R_σ is infinitely divisible for every

$\sigma > \frac{1}{2}$ with no hypothesis (Theorem 9.2.5 and the proof of Proposition 10.1.1). Since φ_σ differs from R_σ by the gamma factor and elementary factors in $\sqrt{s}$, this does not settle the question for w_σ.

Problem 7.9. Meixner Pollaczek truncations. The symmetric expansion of $(\tau^2 + \frac{1}{4})Z(\tau)$ captures the first ordinate at degree 40 but converges slowly. Is there a weighting of its coefficients, or a Jensen type construction from them, whose polynomials are real rooted and whose zeros converge to the ordinates?

Problem 7.10. Nyman Beurling. Theorem 6.1.5 writes $1/\zeta(\frac{1}{2}+\sqrt{s})$ in the span of $\{\mathrm{sech}(\sqrt{s}\log N)\}$ with Möbius weights on $H_{1/2}$. Relate the completeness question of Lai and Vere-Jones for the dilates $\{\mathrm{sech}(\lambda x)\}$ to the Nyman Beurling criterion [170, 25], which Section 10.1.8 sets beside it, and decide whether the sech expansion admits an L^2 version on the boundary $\mathrm{Re}\,\sqrt{s} = \frac{1}{2}$.

Problem 7.11. Product representations of X_R. By the closure properties behind Corollary 7.1.6, an unconditional representation of X_R built from independent GGC variables by sums, products and weak limits would make R a GGC and prove RH (Proposition 1.2.1); for X_φ the corollary rules out such a representation. The harmonic mean is not available, because Proposition 7.1.7 shows it is not a closure operation for the class. The deconvolution (6.2.1) of Section 6.2.1 gives X_R, formally, as X_φ minus a stable law and a compensated series of the variables Y_{c_n}. Can the compensation be absorbed into a product?

Chapter 8

Probability, hitting times and other criteria

Chapters 3 to 7 studied the clock $1/G$ through its Thorin measure. The clock is also the Laplace transform of a random variable, whose law the Pólya route of Chapter 2 computes from the same kernel as the Thorin density. The same evaluator computes criteria for RH other than the Thorin condition. Sections 8.1 and 8.2 treat the clock as a random variable, through its law, its Kent approximants and its Brownian boundaries. Sections 8.3 and 8.4 take up two classical equivalents of RH that the Pólya density computes directly, Li's criterion and the de Bruijn Newman flow, and read both in terms of the Thorin measure. Section 8.5 runs the evaluator with kernels built from the Dedekind eta function, for $L(s, \chi_{12})$, for the Dirichlet beta function $L(s, \chi_4)$ and for the L function of Ramanujan's Δ, to see which parts of the Thorin programme change when the L function has no pole, and it reduces the positivity of the Pólya kernel of the Dedekind zeta function of an imaginary quadratic field to a finite interval. Section 8.6 takes up the inverse problem for ζ, the recovery of the zeros from the Thorin density by nonnegative deconvolution, and Section 8.7 closes the chapter with Pólya's Bessel function $K_{i\theta/2}(2\pi)$, for which the clock, its realisation as a hitting time, the complete monotonicity of its trace and the operator behind its zeros are all theorems, and whose de Bruijn Newman flow it compares with that of Ξ.

8.1 The clock as a random variable

The clock variable T of Section 1.2, with $\mathbb{E}e^{-uT} = 1/G(u)$, exists unconditionally and is self decomposable (Theorem 5.1.1). Under RH it is $T = \sum_{\gamma>0} e_\gamma/\gamma^2$ with independent standard exponential variables e_γ (Proposition 1.2.1), and if the zeros are also simple, the theorems of Kent and Krein make it the hitting time of a level by a diffusion whose Dirichlet spectrum is $\{\gamma^2\}$ [122], [200, Section 16.3]. Its law can be computed from the Pólya density with no zero as input, and the first zero appears in it as the rate of decay, γ_1^2 (Section 8.1.1). The other zeros

are much harder to see: in time they enter through exponentially subdominant terms that cost digits of accuracy, and Monte Carlo sampling through Williams' representation of the Pólya variable needs about 10^6 draws to show even the first zero (Section 8.1.2). Computation can instead approximate the clock by explicit generalised gamma convolutions that keep the verified zeros as atoms and replace the others by a smooth tail (Section 8.1.3); at large values of the variable u their error is governed by the argument function $S(t)$ beyond the last zero used (Section 8.1.4).

8.1.1 The law of the clock and its decay rate

The Laplace transform $\mathbb{E}e^{-uT} = \xi(\frac{1}{2})/\xi(\frac{1}{2}+\sqrt{u})$ is available from the Pólya route at every complex u other than its poles, untilted for $|\operatorname{Im}\sqrt{u}| < 120$ and tilted above (Section 2.1.3). Talbot's contour inversion [248], as implemented in `mpmath` [159], then gives the density f of T and its survival function $\bar{F}(t) = P(T > t)$ with no reference to the zeros.

Under RH, with simple zeros, the density also has the spectral expansion [122]

$$f(t) = \sum_k R_k e^{-\gamma_k^2 t}, \qquad P(T>t) = \sum_k \frac{R_k}{\gamma_k^2} e^{-\gamma_k^2 t}, \qquad R_k = \frac{2i\gamma_k\,\xi(\frac{1}{2})}{\xi'(\frac{1}{2}+i\gamma_k)},$$

whose coefficients are the residues of the transform at its poles $u = -\gamma_k^2$. The first step of the proof of Theorem 4.3.23 derives it by a contour shift, with $R_k = -2\gamma_k \Xi(0)/\Xi'(\gamma_k)$ in the role of c_n there, and the Pólya route supplies both the ordinates, as its Newton roots, and the values of ξ' at them. The residues alternate in sign, because Ξ' changes sign at consecutive simple zeros (Exercise 8.1), and they grow like $e^{\pi\gamma_k/4}$ up to powers of γ_k and the factor $1/|\zeta'(\rho_k)|$, because Stirling's formula puts the factor $e^{-\pi\gamma_k/4}$ into $|\Xi'(\gamma_k)| = |\xi'(\rho_k)|$; thus $R_1 = 1.0164\times10^4$ and $R_2 = -1.1775\times10^6$. The series converges for $t > 0$ but cancels heavily at small t, so near the origin Talbot's contour, which uses the transform only away from the poles, is the better tool (Exercise 8.2); under the hypotheses of Theorem 4.3.23 the line $\operatorname{Re} t = 0$ is a natural boundary of the density. Computed with 50 digits from the 79 zeros below 200, the spectral expansion agrees with the Talbot values to 24 digits for $0.010 \le t \le 0.12$. Beyond that range the Talbot values lose relative accuracy as $\bar{F}(t)$ falls, down to six digits at $t = 0.3$, and the spectral expansion is the accurate one (Figure 8.1).

The law is unimodal with no hypothesis (Corollary 11.1.2) and sharply concentrated. Its mean and variance are $\mathbb{E}T = 0.0231050$ and $\operatorname{Var} T = 3.71726\times10^{-5}$ (Section 1.2) and its median is 0.0220; $P(T \le 0.009) = 1.4\times10^{-7}$, the density is 6.5×10^{-24} at $t = 0.005$, and $P(T > 0.1) = 1.07\times10^{-7}$.

The hazard $h = f/\bar{F}$ tends to the principal eigenvalue γ_1^2. Under RH with simple zeros the leading term of the spectral expansion dominates, and $h(t) - \gamma_1^2 =$

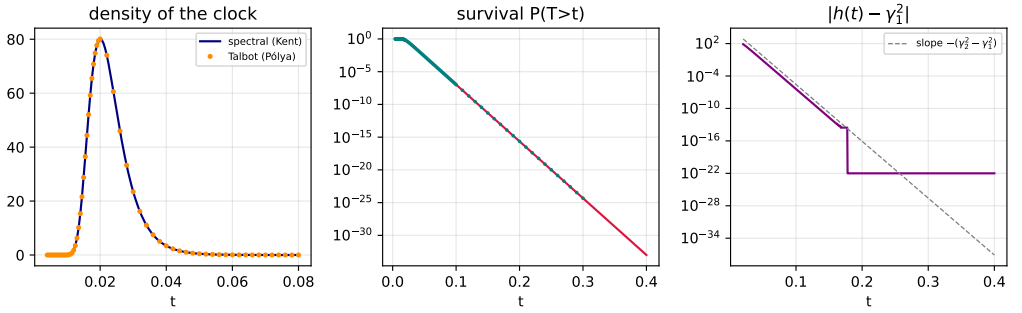


Figure 8.1: The law of the clock variable computed from the Pólya density by Talbot inversion (points) and by the spectral expansion with Pólya residues (lines): density, survival function, and the distance of the hazard from γ_1^2, whose logarithm has slope $-(\gamma_2^2 - \gamma_1^2)$ (dashed). Beyond $t \approx 0.17$ the last curve reflects the double precision of the plotting script, not the hazard.

$O(e^{-(\gamma_2^2-\gamma_1^2)t})$ (Exercise 8.3). This is the quasi stationary behaviour of a hitting time; Roberts [222, Section 2] obtains approximations of Brownian boundary hitting laws from approximations of the conditioned diffusion. The Talbot side shows the limit with no zero as input: $h(0.05) = 199.72$, $h(0.1) = 199.7904544$ and $h(0.25) = 199.79045483$, the last equal to γ_1^2 to the eleven digits shown, which is the accuracy of the inversion that far into the tail. The spectral expansion, which has γ_1 as input, gives $h(0.25) = 199.790454832386859463$, equal to γ_1^2 to all 21 digits. The rate of approach is the spectral gap: a fit of $\log|h(t)-\gamma_1^2|$ on $[0.05, 0.12]$ has slope -242.12, against $-(\gamma_2^2 - \gamma_1^2) = -242.14$.

8.1.2 The cost of the zeros in time and in samples

The time domain sees the higher atoms only through exponentially subdominant terms. We fit sums of p exponentials to $P(T > t)$ by Prony's method, from 40 values at $t = 0.03, 0.032, \ldots, 0.108$ carrying relative noise from 10^{-16} down to 10^{-45} (Table 8.1). With $p = 4$ terms the first atom is correct to all nine digits shown, the second to six, the third to four and the fourth to two per cent, and the results do not improve as the noise falls. The error comes from the neglected terms, whose relative size on a window starting at t_0 is $e^{-(\gamma_{p+1}^2-\gamma_1^2)t_0}$ up to a factor set by the residues. Resolving the kth atom therefore needs data accurate to about $(\gamma_k^2 - \gamma_1^2)t_0/\log 10$ digits, a number that grows without bound in k: a hazard rate method reaches the first atom, and every further atom costs digits.

Sampling meets the same cancellation in another currency. By the identity $2\xi(s) = \mathbb{E}[(\pi S_2/2)^{s/2}]$ of Section 1.2, with $S_2 = \sum_{k\geq 1} \frac{2}{\pi^2 k^2}\Gamma_k$ for independent

terms	γ_1^2	γ_2^2	γ_3^2	γ_4^2
2	199.86679	427.28		
3	199.79031	442.129	618.93	
4	199.790455	441.9255	625.63	907.24
true	199.7904548	441.9261506	625.5429969	925.6730873

Table 8.1: Prony extraction of the atoms from $P(T > t)$ on $t \in [0.03, 0.108]$. Identical results for relative noise 10^{-16}, 10^{-30} and 10^{-45}: the error is model truncation, not noise.

Gamma$(2,1)$ variables Γ_k, the variable $Y = \frac{1}{2}\log(\pi S_2/2)$ satisfies

$$2\,\Xi(z) = \mathbb{E}\big[e^{Y/2}\cos(zY)\big].$$

The Pólya variable is Y tilted by $e^{Y/2}$, and $\Xi(z)$ is the mean of a random variable with finite variance, since $\mathbb{E}e^{Y} = 2\xi(1) = 1$, so a sample mean estimates it at every z. We draw $n = 10^6$ copies of S_2, truncating the series after 2000 terms and replacing the remainder, whose variance is about 3×10^{-12}, by its mean $4/(2000\pi^2)$, and compare the estimates with $\Xi(z)$ in Table 8.2, a check of the identity on the critical line; Table 9.1 checks it at real s (Section 9.1.1).

z	Monte Carlo	standard error	$\Xi(z)$	n with standard error $\lvert\Xi(z)\rvert$
0	0.497061	5.4×10^{-5}	0.497121	1.2×10^{-2}
2	0.453136	7.3×10^{-5}	0.453100	2.6×10^{-2}
5	0.275888	2.4×10^{-4}	0.275550	0.77
8	0.103756	3.4×10^{-4}	0.103303	11
10	0.038401	3.5×10^{-4}	0.037968	86
20	0.000314	3.5×10^{-4}	-3.67×10^{-5}	9.3×10^{7}
30	0.000289	3.5×10^{-4}	-1.50×10^{-8}	5.5×10^{14}

Table 8.2: Monte Carlo estimates of $\Xi(z)$ from Williams' representation with 10^6 samples. The last column is the sample size at which the standard error equals $|\Xi(z)|$.

The standard error settles at 3.5×10^{-4}, the standard deviation of $e^{Y/2}\cos(zY)/2$ over $\sqrt{n}$, while $|\Xi(z)|$ decays like $e^{-\pi z/4}$. The sample size needed to resolve the sign of Ξ therefore grows like $e^{\pi z/2}$: about 10^8 at $z = 20$ and 10^{15} at $z = 30$. Even the first sign change, at $z = 14.13$, needs about 10^6 samples: between the first two zeros $|\Xi|$ is at most 8.0×10^{-4} (at $z = 15.59$), and resolving it at two standard errors takes about 8×10^5 samples. Figure 8.2 shows where $|\Xi|$ falls below the standard error. The Pólya route pays for the same cancellation in digits of working precision and Monte Carlo pays for it in samples, and precision is exponentially cheaper. Sampling is the right tool for the law of the Pólya

variable itself, for its moments and for the Bayesian inverse problems of [200, Section 19.2], and the wrong tool for the zeros. Devroye's exact samplers for theta type laws [68] make sampling efficient, but the sample size needed for the zeros still grows like $e^{\pi z/2}$.

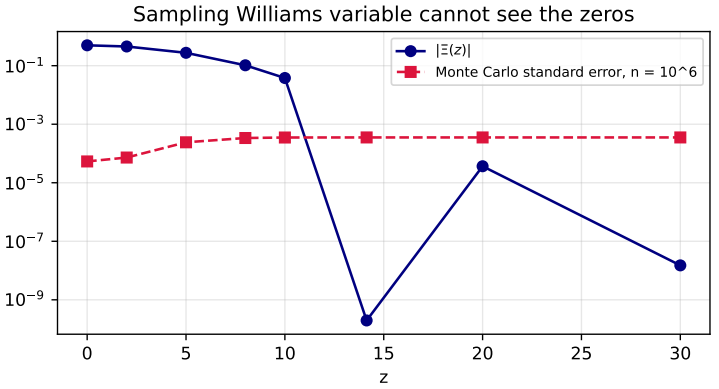


Figure 8.2: $|\Xi(z)|$ against the Monte Carlo standard error with 10^6 samples of Williams' variable S_2, at the sampled values of z and at the first zero, joined by straight lines.

8.1.3 Kent approximants with a Weyl tail

In the rest of this section we write w for the variable u of the clock, as the tables and figures that follow do. Kent's theorem [122] identifies the first hitting time of a level by a diffusion started at a reflecting or entrance end of its interval with the GGC $\prod_n (1+w/\lambda_n)^{-1}$, where the λ_n are the Dirichlet eigenvalues of the interval up to that level; the simplest example is the time a three dimensional Bessel process started at 0 takes to reach the level one, with $\lambda_n = \pi^2 n^2/2$ (Section 9.1.1). Krein's inverse spectral theory [70] builds a string, and so a generalised diffusion, from any increasing sequence $\lambda_n > 0$ with $\sum 1/\lambda_n < \infty$. The GGCs that arise as such hitting times are therefore exactly those without drift whose Thorin measure consists of unit atoms at distinct points. By Proposition 1.2.1 the clock is one of them exactly when RH holds and the zeros are simple, and its diffusion then has Dirichlet spectrum $\{\gamma^2\}$; Section 10.2.4 reads each $1/\zeta(2k)$, up to an explicit factor, as the probability that this diffusion reaches its level before an independent killing. The shifted clocks above the centre are not hitting times of Kent's kind, but with no hypothesis they are the tilted centre clock run through the inverse local time of a sticky Brownian motion (Section 9.2), and Section 11.1.7 places all these processes in one map.

The approximants of the clock that Kent's theorem suggests are the transforms $1/G_N(w)$ of the partial products $G_N(w) = \prod_{k\le N}(1 + w/\gamma_k^2)$, each the transform

of the hitting time of a finite string with N point masses. They are admissible in the closure criterion of Corollary 4.1.2, and they are the approximating sequence, under RH, in the proof of [200, Proposition 18.10]; Section 4.1.1 lists them, with and without the Weyl tail of this section, among the approximants whose positivity is proved independently of ξ. They converge slowly. With $\tau_1(N) = \sum_{k>N} \gamma_k^{-2}$, the relative error $G(w)/G_N(w) - 1$ of the clock transform lies, under RH, between $w\tau_1(N)$ and $e^{w\tau_1(N)} - 1$; it is of order one at $w = 100$ for $N = 20$ and at $w = 1000$ for $N = 300$.

We keep the verified zeros as atoms and replace the others by their average density. By the Riemann von Mangoldt formula $N(t) = \vartheta(t)/\pi + 1 + S(t)$, with the Riemann Siegel theta function ϑ and the argument function $S(t) = \pi^{-1} \arg \zeta(\frac{1}{2} + it)$ of Section 2.2, the zeros above the cutoff Θ_N have density $\vartheta'(\gamma)/\pi \approx \frac{1}{2\pi} \log \frac{\gamma}{2\pi}$, which is positive for $\gamma > 6.3$. So

$$G_N^{\mathrm{W}}(w) = \prod_{k \le N} \Big(1 + \frac{w}{\gamma_k^2}\Big) \exp\Big\{ \int_{\Theta_N}^{\infty} \log\Big(1 + \frac{w}{\gamma^2}\Big) \frac{\vartheta'(\gamma)}{\pi}\, d\gamma \Big\},$$

with W for Weyl, has reciprocal equal to the Laplace transform of a GGC with N unit atoms and the continuous Thorin density $\vartheta'(\sqrt{z})/(2\pi\sqrt{z})$ on $z > \Theta_N^2$. That Thorin measure is not a sum of unit atoms, so, unlike $1/G_N$, the approximant is not a hitting time of the kind in Kent's theorem; the closure criterion needs only the GGC property. For Θ_N we take the Gram point g_{N-1} of Section 2.2.1, where $\vartheta(\Theta_N)/\pi + 1 = N$. For the values of N used here it lies between γ_N and γ_{N+1}, so $S(\Theta_N) = 0$ and the smooth count agrees with the true count at the cutoff, as in Section 3.2, where Θ_{1700} is the point T_*.

Table 8.3 compares the two approximants with the clock computed from ξ. The Weyl tail improves the approximation by at least two orders of magnitude at every N and w tested. With the first 1700 zeros ($\gamma_{1700} = 2197.26$) its error is 3×10^{-11} at $w = 1$ and 3×10^{-7} at $w = 10^4$, where the partial product is wrong by a factor of more than a hundred. Figure 8.3 plots the same data.

N	$w = 1$	$w = 100$	$w = 1000$	$w = 10^4$
1	1.8×10^{-2} / 1.3×10^{-4}	4.8 / 1.0×10^{-2}	$\gg 1$ / 3.3×10^{-2}	$\gg 1$ / 4.3×10^{-2}
20	7.2×10^{-3} / 4.3×10^{-7}	1.04 / 4.2×10^{-5}	$\gg 1$ / 3.7×10^{-4}	$\gg 1$ / 1.5×10^{-3}
79	3.6×10^{-3} / 1.5×10^{-7}	0.43 / 1.5×10^{-5}	34 / 1.5×10^{-4}	$\gg 1$ / 1.2×10^{-3}
300	1.6×10^{-3} / 7.4×10^{-9}	0.17 / 7.4×10^{-7}	3.95 / 7.4×10^{-6}	$\gg 1$ / 7.2×10^{-5}
1700	5.0×10^{-4} / 2.9×10^{-11}	0.051 / 2.9×10^{-9}	0.64 / 2.9×10^{-8}	142 / 2.9×10^{-7}

Table 8.3: Relative error of the clock transform: partial product over N zeros, $|G/G_N - 1|$, and with the Weyl tail, $|G_N^{\mathrm{W}}/G - 1|$. The two measures of error agree to first order, and $\gg 1$ marks an error above 10^3. The partial product values agree with [200, Section 18.2.1] where both are given.

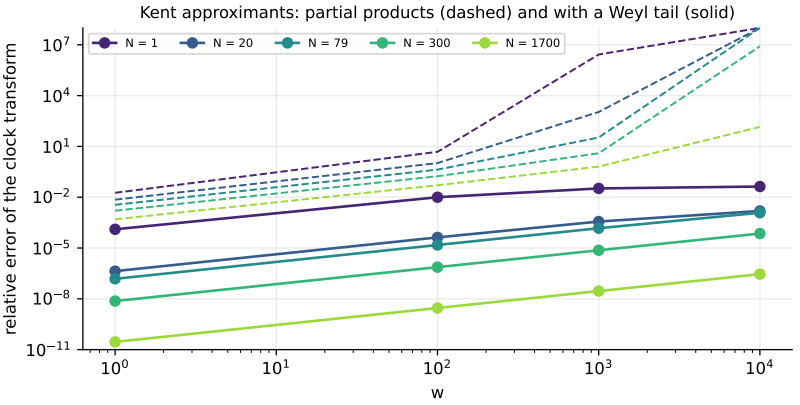


Figure 8.3: Relative error of Kent approximants to the clock transform, without (dashed) and with (solid) the Weyl tail.

At the four points $w = 1, 100, 1000, 10^4$ the table exhibits an explicit positive Thorin measure whose Thorin transform $\int \log(1 + w/z)\, U(dz)$ matches $\log G(w)$ to within 3×10^{-7}. This is a finite instance of the weak Thorin condition of [200, Section 18.2.3], which asks for such a measure, to within any $\eta > 0$, at every finite set of points, and which is equivalent to RH by the closure criterion of Corollary 4.1.2. A finite instance proves nothing: agreement for $w \le 10^4$ cannot detect off line zeros above the verified height, which change $\log G$ there by far less than the error. The error of the Weyl tail grows linearly in w, and to first order it is a weighted average of S beyond the cutoff,

$$\log G(w) - \log G_N^{\mathrm{W}}(w) \approx w\big(\tau_1(N) - \tau_1^{\mathrm{W}}(N)\big) = 2w \int_{\Theta_N}^{\infty} \frac{S(\gamma)}{\gamma^3}\, d\gamma,$$

where $\tau_1^{\mathrm{W}}(N) = \int_{\Theta_N}^{\infty} \gamma^{-2} \vartheta'(\gamma) \pi^{-1} d\gamma$ is the Weyl estimate of $\tau_1(N)$ and the identity is an integration by parts that uses $S(\Theta_N) = 0$. No tail built on the smooth count can follow S without the zeros. The Taylor coefficients of ξ at the centre carry more information, and Section 8.1.4 uses them.

8.1.4 Moment matching and the error at large w

The error of the Weyl tail comes from the first order term of $\log(1 + w/\gamma^2) = w/\gamma^2 - w^2/(2\gamma^4) + \cdots$. For w small compared with Θ_N^2 the error of $\log G_N^{\mathrm{W}}$ is approximately w times $\tau_1(N) - \tau_1^{\mathrm{W}}(N)$, and the second order term is $-w^2/2$ times the corresponding difference for $\tau_2(N) = \sum_{k>N} \gamma_k^{-4}$. Both true tail sums are available without any zero beyond γ_N, because the power sums $p_m = \sum_k a_k^{-m}$, with $a_k = -(\rho_k - \tfrac{1}{2})^2$ as in Section 1.2 and equal to $\sum_{\gamma>0} \gamma^{-2m}$ under RH, are

Taylor coefficients of ξ at the centre: if $\xi(\frac12+s)/\xi(\frac12) = 1 + c_2 s^2 + c_4 s^4 + \cdots$, then

$$p_1 = \sum_k a_k^{-1} = c_2 = 0.023104993115418970 7889,$$

$$p_2 = \sum_k a_k^{-2} = c_2^2 - 2c_4 = 0.000037172599285269 6862,$$

and $\tau_m(N) = p_m - \sum_{k\le N} \gamma_k^{-2m}$. Here $p_1 = \mathbb{E}T$ and $p_2 = \operatorname{Var} T$ (Section 1.2). We therefore replace the Weyl density $\vartheta'(\gamma)/\pi$ on $[\Theta_N, \infty)$ by $\lambda\,\vartheta'(\gamma)/\pi$ on $[\Theta'_N, \infty)$, and choose the two parameters λ and Θ'_N so that the tail reproduces $\tau_1(N)$ and $\tau_2(N)$ exactly. The tail density remains positive, so the approximant is still a GGC. The adjustments are small: for $N = 79$ the cutoff moves from 199.649 to 199.675 and $\lambda = 1.000059$, and for $N = 1700$ it moves from 2197.87959 to 2197.87923 and $\lambda = 0.99999992$.

N	tail	$w=1$	$w=10^2$	$w=10^4$	$w=10^5$	$w=10^6$	$w=10^7$
79	Weyl	1.5×10^{-7}	1.5×10^{-5}	1.2×10^{-3}	4.3×10^{-3}	5.8×10^{-3}	6.0×10^{-3}
	matched	2.6×10^{-17}	2.6×10^{-11}	1.8×10^{-5}	3.1×10^{-3}	6.1×10^{-2}	0.47
300	Weyl	7.4×10^{-9}	7.4×10^{-7}	7.2×10^{-5}	5.5×10^{-4}	1.7×10^{-3}	2.1×10^{-3}
	matched	2.3×10^{-20}	2.3×10^{-14}	2.2×10^{-8}	1.5×10^{-5}	1.8×10^{-3}	2.8×10^{-2}
1700	Weyl	2.9×10^{-11}	2.9×10^{-9}	2.9×10^{-7}	2.8×10^{-6}	2.4×10^{-5}	9.3×10^{-5}
	matched	3.5×10^{-23}	1.1×10^{-17}	7.0×10^{-13}	3.6×10^{-10}	2.5×10^{-7}	5.0×10^{-5}

Table 8.4: Relative error $|G_{\mathrm{a}}/G - 1|$ of the approximant G_{a} of G with the Weyl tail and with the moment matched tail. The cutoffs are $\Theta'^2_N = 3.99\times10^4$, 2.95×10^5 and 4.83×10^6 for $N = 79$, 300 and 1700.

Table 8.4 and Figure 8.4 show two regimes. For w below Θ'^2_N moment matching gains ten to twelve digits at $w = 1$ and about two as w approaches Θ'^2_N: with 1700 zeros the error at $w = 10^4$ falls from 2.9×10^{-7} to 7×10^{-13}, and at $w = 10^5$ from 2.8×10^{-6} to 3.6×10^{-10}. Well beyond Θ'^2_N the gain disappears, and at $w = 10^7$ with $N = 79$ the matched tail is worse than the plain Weyl tail. The cause is the rescaling of the whole tail. For $w \gg \Theta'^2_N$ the matched tail differs from the Weyl tail mainly by $(\lambda-1)\int_{\Theta'_N}^{\infty} \log(1+w/\gamma^2)\,\vartheta'(\gamma)\pi^{-1}d\gamma$, which grows like $(\lambda-1)\frac{\sqrt{w}}{2}\log\frac{\sqrt{w}}{2\pi}$, while the error of the Weyl tail tends to a constant.

That constant is a weighted average of S beyond the cutoff. Under RH, since $S(\Theta_N) = 0$ and no ordinate lies in $(\gamma_N, \Theta_N]$, the true tail sum minus its Weyl estimate is a Stieltjes integral against S, and an integration by parts gives

$$\log G_N^{\mathrm{W}}(w) - \log G(w) = -\int_{\Theta_N}^{\infty} \log\Big(1+\frac{w}{\gamma^2}\Big)\,dS(\gamma) = -\int_{\Theta_N}^{\infty} \frac{2w\,S(\gamma)}{\gamma(\gamma^2+w)}\,d\gamma$$
$$\longrightarrow\ -2\int_{\Theta_N}^{\infty} \frac{S(\gamma)}{\gamma}\,d\gamma \qquad (w\to\infty).$$

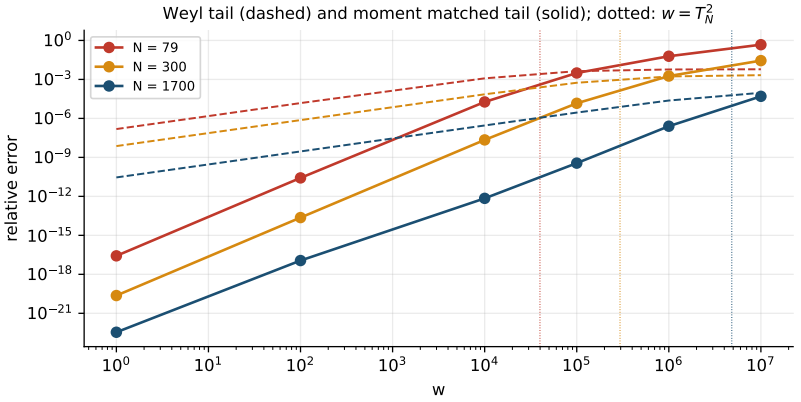


Figure 8.4: The Kent approximants at large w, with the dotted lines at $w = \Theta_N^2$ (labelled T_N^2 in the plot). Moment matching gains ten to twelve digits at $w = 1$ and about two as w approaches Θ_N^2. Beyond Θ_N^2 the Weyl tail levels off at a floor set by S, while the error of the matched tail, whose density is rescaled by λ, grows like $|\lambda - 1|\sqrt{w}\log w$.

The limit is $-\ell_N$, with $\ell_N = 2\int_{\Theta_N}^{\infty} S(\gamma)\gamma^{-1}\,d\gamma$ as in Exercise 8.5. It is finite because $\int_0^t S(u)\,du = O(\log t)$ (Littlewood; Section 2.2.1), and in general it is not zero. The measured errors of the Weyl tail level off at about 6×10^{-3} for $N = 79$ and 2×10^{-3} for $N = 300$; for $N = 1700$ the error is still rising at $w = 10^7$, where it is 9.3×10^{-5}, so its level is not yet reached.

The constant is not an obstruction of principle. It depends only on $\gamma_1, \ldots, \gamma_N$ and on ξ along the real axis, and Stirling's formula gives ℓ_N in closed form (Exercise 8.5). It is also a floor only for the Weyl density. Adding to ϑ'/π a continuous ω with compact support in (Θ_N, ∞) and $\int \omega = 0$ changes the limit by $-2\int \omega(\gamma)\log\gamma\,d\gamma$, and two wide bumps of opposite sign placed far apart give this any value while the density stays positive, so a suitable positive tail removes the constant.

Of the two tails computed here, the matched tail is the better one for w up to about Θ_N^2 and the plain Weyl tail beyond. With them the clock is within a small computed distance of an explicit GGC at every scale computed, and at large w that distance is a weighted integral of S beyond the cutoff, the quantity whose integrals Turing's method bounds (Section 3.4).

8.2 Brownian boundaries and Williams' structure

Under RH with simple zeros, Kent's theorem makes the clock variable T the hitting time of a level by a diffusion, but it describes that diffusion only through its spectrum, the set $\{\gamma^2\}$ that is to be found (Section 8.1.3). We look instead for a

representation by Brownian motion itself. Section 3.5 met Brownian first passage times to a random level through the sine squared lemma, so we ask whether T is a Brownian first passage time. Unconditionally, it is not one to an independent random level for any constant drift (Section 8.2.1). Across a curved boundary it is one: such a boundary exists by a theorem of Ekström and Janson, and we compute it (Section 8.2.2). The Roberts Shortland tangent approximation gives a first boundary, and the Volterra equation of the first passage density a better one, computed in three zones; it descends with the slope $-\sqrt{2}\gamma_1$ set by the first zero and reproduces the law of the clock to Monte Carlo accuracy. Williams' representation of the Pólya kernel through Bessel hitting times explains the tilted contour of Section 2.1.3, but it transfers no positivity to the clock (Section 8.2.3). Section 9.2 returns to these questions at the basepoints above the centre, where the boundary becomes flat.

8.2.1 The clock is not a first passage time to a random level

The first passage time of Brownian motion with drift $c \geq 0$ to a level $a > 0$ has the inverse Gaussian law, with transform $\exp\{a(c - \sqrt{c^2 + 2u})\}$. Mixing over an independent random level with law F gives the transforms $\int \exp\{a(c - \sqrt{c^2 + 2u})\}F(da)$, a class of first passage laws. Such a law inherits a bound from the Brownian motion: on a suitable parabola its transform has modulus at most one, while the reciprocal of ξ grows there.

Proposition 8.2.1. *For no drift $c \geq 0$ and no probability law F on $(0, \infty)$ is the clock variable T equal in law to the first passage time of Brownian motion with drift c to an independent level $A \sim F$.*

Proof. On the parabola $u = \frac{1}{2}(-q^2 + 2icq)$, $q \in \mathbb{R}$, we have $c^2 + 2u = (c + iq)^2$, so $\operatorname{Re}\sqrt{c^2 + 2u} = c$ and every mixture transform has modulus at most one there. Suppose that T had such a law, so that the clock transform $1/G(u) = \mathbb{E}e^{-uT}$ and the mixture transform agree for $u \geq 0$. The mixture transform is analytic and bounded by one on the region $D = \{u : \operatorname{Re}\sqrt{c^2 + 2u} > c\}$, and continuous up to its boundary, which is the parabola. The region D is the image of the half plane $\operatorname{Re} v > c$ under $v \mapsto (v^2 - c^2)/2$, so it is connected, and it contains the right half plane. By the identity theorem $1/G$ coincides with the mixture transform on D, so it has no pole in D and is bounded by one there, and by continuity $|1/G| \leq 1$ on the parabola away from the zeros of $\xi(\frac{1}{2} + \sqrt{u})$. For the clock, however, $\sqrt{u} = (c + iq)/\sqrt{2} + O(1/q)$ as $q \to \infty$ (for $c = 0$ the parabola is the negative axis, approached from above, and $\sqrt{u} = i|q|/\sqrt{2}$), so $|\xi(\frac{1}{2} + \sqrt{u})|$ decays like $\exp(-\pi|q|/(4\sqrt{2}))$ times a power, by Stirling's formula and the polynomial growth of ζ in vertical strips. Hence $|\xi(\frac{1}{2})/\xi(\frac{1}{2} + \sqrt{u})|$ grows exponentially along the parabola, and the transforms cannot agree. □

The proof is unconditional and uses only the existence of T. It covers the

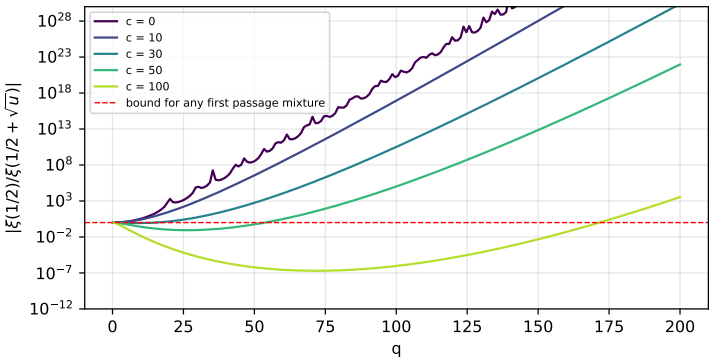


Figure 8.5: Modulus of the clock transform along the parabolas on which every first passage mixture with drift c is bounded by one (dashed). Each curve crosses the bound, so the clock is in none of these classes. The abscissa is the parameter q of the parabola $u = \frac{1}{2}(-q^2 + 2icq)$, for $c = 0$, 10, 30, 50 and 100.

nonincreasing lines $a - ct$, $c \geq 0$; an increasing line, with $c < 0$, gives a passage time that is infinite with probability $1 - \mathbb{E}e^{2cA} > 0$, which the clock is not. For $c = 0$ it is the centre case of the statement in Section 3.5 that no basepoint admits a random level, and Proposition 9.2.1 carries the argument, for every drift $c \geq 0$, to the basepoint one. Near the vertex $u = 0$ the comparison gives an explicit threshold. Expanding $G(u) = 1 + m_1 u + m_2 u^2 + \cdots$ with $m_1 = \mathbb{E}T$ and $m_1^2 - 2m_2 = \operatorname{Var} T$ gives $|G(u)|^2 = 1 - q^2(m_1 - c^2 \operatorname{Var} T) + O(q^3)$ on the parabola, so the clock transform already exceeds one at small q whenever

$$c < c_* = \sqrt{\mathbb{E}T / \operatorname{Var} T} = \sqrt{0.0231050 / 3.71726 \times 10^{-5}} = 24.93.$$

The Pólya route confirms both regimes (Figure 8.5): the modulus exceeds one immediately for $c \leq 24.93$, and first at $q = 15.3$, 53.8 and 172 for $c = 30$, 50 and 100.

8.2.2 A curved Brownian boundary for the clock

Proposition 8.2.1 excludes straight boundaries with a random intercept, and curved boundaries remain. Finding a boundary whose first passage time has a prescribed law is the inverse first passage problem [51, Theorem 1]. By the theorems of Ekström and Janson [73, Theorems 2.3, 8.2 and 9.2], stated in full in Section 9.2.4, the problem has a solution for the clock, which we call the exact boundary. Since the density of the clock is continuous and positive on $(0, \infty)$ (Exercise 8.10), and so bounded below by a positive constant on every compact subinterval of $(0, \infty)$, the same theorems make the exact boundary continuous on $(0, \infty)$ and the unique continuous solution b with $\liminf_{t \downarrow 0} b(t) \geq 0$ of the Volterra integral equation (8.2.1) of the first passage density. We approximate

it in two ways: by the tangent approximation of Roberts and Shortland, which turns the problem into an ordinary differential equation, and by discretising the Volterra equation. The discretised solution, which we call the Volterra boundary, reproduces the law of the clock to Monte Carlo accuracy.

8.2.2.1 The Roberts Shortland tangent boundary

Roberts and Shortland [223] approximate the hazard of a Brownian first passage across a smooth boundary b by the hazard of the tangent line at the current time, for which the Bachelier Lévy formula is exact:

$$h_{\tan}(t) = \frac{\alpha(2\pi t^3)^{-1/2}\exp\{-(\alpha+\beta t)^2/2t\}}{\mathcal{N}\big((\alpha+\beta t)/\sqrt{t}\big) - e^{-2\alpha\beta}\mathcal{N}\big((\beta t-\alpha)/\sqrt{t}\big)},$$
$$\beta = b'(t),\ \alpha = b(t) - tb'(t),$$

where $\mathcal{N}$ is the standard normal distribution function; we write $\overline{\mathcal{N}} = 1-\mathcal{N}$ for its tail. We use the formula in reverse. Setting $h_{\tan}(t) = h(t)$, the computed hazard of the clock, defines an ordinary differential equation for b, which we integrate by the fourth order Runge Kutta method with step 10^{-4} from $t_0 = 0.011$. The starting value $b(t_0) = 0.4023$ is the constant level with the crossing mass of the clock, $2\overline{\mathcal{N}}(b(t_0)/\sqrt{t_0}) = P(T \le t_0) = 1.25\times 10^{-4}$.

Table 8.5 and Figure 8.6 show the result. The boundary falls steeply at first, with slope -87.5 at t_0, and then becomes asymptotically linear with slope tending to $-\sqrt{2}\,\gamma_1 = -19.99$, since an inverse Gaussian hazard tends to $\beta^2/2$; the finite time correction $\beta^2/2 + 3/(2t) = \gamma_1^2$ (Exercise 8.8) predicts $|\beta| = 19.77$ at $t = 0.35$ against the computed 19.81. So the first zero sets the asymptotic slope of the boundary. Simulated against this boundary, the crossing time reproduces the law of the clock to within 0.6 per cent in Kolmogorov distance (Section 8.2.2.2). At the basepoint one, where the survival function decays only like $t^{-1/2}$ instead of exponentially, the tangent approximation, used there in the form that equates the first passage density of the tangent line with the density of the law, agrees with the exact boundary in the bulk but has no solution in the right tail (Section 9.2.4).

8.2.2.2 The boundary from the Volterra equation

The Volterra equation of the first passage density determines the exact boundary (Section 8.2.2 and [73, Theorems 9.1 and 9.2]), and solving its discretisation for the clock needs three zones: small times, where almost no mass has crossed; the bulk, where most of it crosses; and the right tail, where the equation cancels and which is the hardest. The same three zones organise the exact boundaries of the shifted clocks above the centre in Sections 9.2.4 and 9.2.6, and that of the L function of Section 8.5 in Section 11.3.3.

t	$b(t)$	slope $b'(t)$	intercept $b - tb'$
0.011	0.40230	-87.547	1.36532
0.015	0.22920	-35.576	0.76283
0.02	0.07102	-28.478	0.64058
0.03	-0.17850	-22.549	0.49798
0.05	-0.59539	-19.991	0.40417
0.1	-1.57905	-19.601	0.38106
0.2	-3.54492	-19.716	0.39828
0.35	-6.51060	-19.813	0.42396

Table 8.5: The Roberts Shortland boundary whose tangent hazard equals the hazard of the clock.

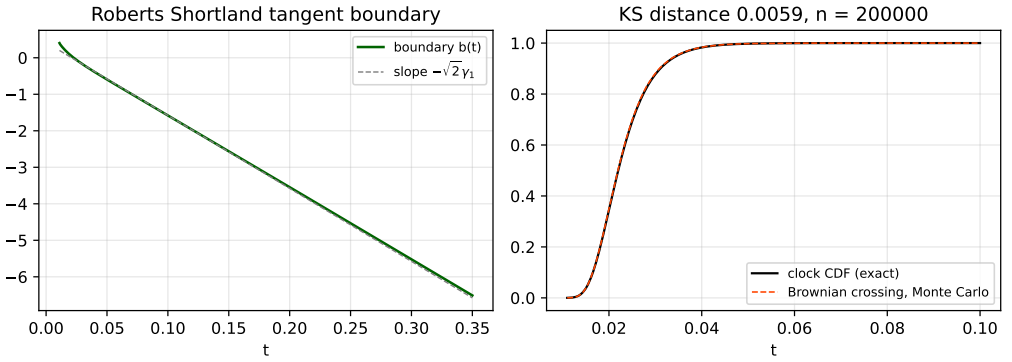


Figure 8.6: Left: the tangent boundary and its asymptotic slope $-\sqrt{2}\gamma_1$. Right: the distribution function of the clock against that of the simulated crossing time of the boundary.

For a continuous boundary b with $b(0+) > 0$, the first passage density f of Brownian motion started at 0 satisfies the Volterra equation of the first kind [192]

$$\begin{aligned} P\big(B_t \geq b(t)\big) &= \int_0^t f(s)\, P\big(B_t \geq b(t) \mid B_s = b(s)\big)\, ds, \\ \text{that is} \quad \overline{\mathcal{N}}\Big(\frac{b(t)}{\sqrt{t}}\Big) &= \int_0^t f(s)\, \overline{\mathcal{N}}\Big(\frac{b(t) - b(s)}{\sqrt{t-s}}\Big) ds, \end{aligned} \tag{8.2.1}$$

with $\overline{\mathcal{N}}$ the standard normal tail. The equation says that a path found above the boundary at time t must have crossed it for the first time at some earlier s, and then moved from $b(s)$ to above $b(t)$. Usually one knows b and solves for f. The inverse problem reverses the roles: we know f, the density of the clock computed in Section 8.1, and solve for b one time step at a time, since once b is known at the earlier grid points the discretised equation at time t is a single scalar equation for $b(t)$. We discretise with 2900 steps on $[0.009, 0.3]$, midpoint quadrature in s,

and a scalar root search at each step.

Before $t_0 = 0.009$ the clock has crossed with probability only $P(T \le t_0) = 1.42 \times 10^{-7}$. Its density is super flat: it vanishes to all orders at $t = 0$, like $\exp(-(1+o(1))\log^2(1/t)/(16t))$, as the growth $\log G(u) \sim \frac{1}{4}\sqrt{u}\log u$ at infinity dictates; de Bruijn's exponential Tauberian theorem [29, Theorem 4.12.9] gives this rate for $P(T \le t)$, and the density has the same rate because the clock is unimodal (Corollary 11.1.2). So the exact boundary is unbounded as $t \downarrow 0$: a path above $b(t)$ at time t has crossed by then, so $\overline{\mathcal{N}}(b(t)/\sqrt{t}) \le P(T \le t)$, which forces $b(t) \ge (1+o(1))\log(1/t)/(2\sqrt{2})$. In this first zone we replace the exact boundary by a constant level, fixed by the reflection principle, $2\overline{\mathcal{N}}(b/\sqrt{t_0}) = P(T \le t_0)$, which gives $b(t_0) = 0.4992$ and places the small mass crossed before t_0 at that level. The Volterra boundary then falls from 0.4992 to 0.3908 at $t = 0.011$. The tangent boundary starts there at 0.4023, the constant level with the same crossing mass by time 0.011; a boundary that decreases on $[0, 0.011]$ with that crossing mass must end below this level, so the gap is no defect of the tangent approximation.

In the bulk, between $t = 0.011$ and $t = 0.05$, where the clock crosses most of its probability, the Volterra and tangent boundaries agree closely: 0.0721 against 0.0710 at $t = 0.02$, and -0.5894 against -0.5954 at $t = 0.05$. Here there is no cancellation: both sides of (8.2.1) are of order one.

In the right tail the equation cancels. Solved in the form (8.2.1), the computation fails at $t = 0.0937$, where the clock has survival probability 3.8×10^{-7}: both sides of the equation are within that amount of $P(T \le t) \approx 1$, and $b(t)$ is determined by the difference of two numbers that agree in their first six digits. In general the absolute error of the quadrature must stay below $P(T > t)$, which is 5×10^{-12} at $t = 0.15$ and 5×10^{-25} at $t = 0.3$, where no computation in double precision can meet it. The cure is to subtract the equation from $P(T \le t) = \int_0^t f$, which gives the complementary form

$$\mathcal{N}\Big(\frac{b(t)}{\sqrt{t}}\Big) - P(T > t) = \int_0^t f(s)\,\mathcal{N}\Big(\frac{b(t) - b(s)}{\sqrt{t-s}}\Big)\,ds,$$

in which every term is small and computed without cancellation, provided the survival function itself is known to relative accuracy. The spectral expansion of Section 8.1 supplies it to full working precision, even where $P(T > t)$ is of order 10^{-25}. With the switch made where $P(T > t)$ falls below one half, the solution runs smoothly to $t = 0.3$.

In this zone the boundary is nearly linear with slope $\beta = -\sqrt{2}\gamma_1 = -19.99$, because a linear boundary with slope β has hazard tending to $\beta^2/2$, which must equal γ_1^2. A linear boundary does not suffice. Its crossing time has survival function of order $t^{-3/2}e^{-\beta^2 t/2}$, while that of the clock is asymptotic to a multiple of $e^{-\gamma_1^2 t}$. The power $t^{-3/2}$ is removed by a logarithmic correction: for $b(t) = \beta t + a + \kappa \log t$ the Gaussian factor $e^{-b(t)^2/2t}$ acquires $t^{-\kappa\beta}$, and $\kappa\beta = -\frac{3}{2}$ cancels

the power. So the right tail boundary is

$$b(t) \approx \beta t + a + \frac{3}{2|\beta|}\log t, \qquad \beta = -\sqrt{2}\,\gamma_1, \qquad \frac{3}{2|\beta|} = 0.0750.$$

Fitting a on $[0.15, 0.3]$ gives $a = 0.578$ with largest residual 0.0095; letting the logarithmic coefficient float gives 0.053 instead of 0.075, the difference coming from higher order terms that the asymptote omits. The slope of the Volterra boundary over $[0.2, 0.3]$ is -19.77. It lies between the slope -19.99 without the logarithm and the slope $-19.99 + 0.075/0.25 = -19.69$ of the asymptote at $t = 0.25$, and it is close to -19.78, the value for the fitted coefficient 0.053.

We simulate $n = 2 \times 10^5$ Brownian paths against both boundaries with step 2×10^{-5} and a Brownian bridge crossing correction; sampling noise alone gives a Kolmogorov distance of about $0.87/\sqrt{n} \approx 0.002$. For the Volterra boundary the Kolmogorov distance to the law of the clock is 0.0020, at the level of the Monte Carlo noise, and the mean crossing time is 0.023085 against $\mathbb{E}T = 0.0231050$. For the tangent boundary the distance is 0.0059 and the mean 0.023021. Figure 8.7 shows the three boundaries and the two errors. To Monte Carlo accuracy, then, the clock is the first passage time of Brownian motion across the computed boundary. The right tail itself cannot be checked by plain simulation, since only about one path in ten million survives past $t = 0.1$; a check there would need importance sampling or splitting, which Section 12.2.3 lists among the open computations.

Wald's identities constrain the exact boundary as a whole. Since $\mathbb{E}T$ is finite, the first passage time H across it satisfies $\mathbb{E}B_H = 0$ and $\mathbb{E}B_H^2 = \mathbb{E}H$, and since $B_H = b(H)$ and H has the law of the clock, $\int_0^\infty b(t) f(t)\,dt = 0$ and $\int_0^\infty b(t)^2 f(t)\,dt = \mathbb{E}T$: the positive values of the boundary at small times and its negative values later balance exactly under the law of the clock. Above the centre the shifted clocks of Section 9.2 have infinite mean, their exact boundaries flatten instead of descending, and the first identity fails by an amount that the zeros fix: at the basepoint one the mean value of the boundary at the crossing is $\frac{1}{\sqrt{2}}\sum_\rho \rho^{-1}$, provided the boundary is bounded at infinity, which is observed in the computation up to $t = 100$ but not proved (Sections 9.2.4 and 9.2.6).

8.2.3 Williams' Gamma(2) structure: the tilt, a benchmark and its limits

Williams [271] and Biane, Pitman and Yor [28] showed that the variable $S_2 = \sum_{n\geq 1} \frac{2}{\pi^2 n^2}\Gamma_n$ of Section 8.1.2 is a sum of two independent hitting times of level one by the three dimensional Bessel process started at 0, and that $2\xi(s) = \mathbb{E}[(\pi S_2/2)^{s/2}]$, so that the Pólya variable is $Y = \frac{1}{2}\log(\pi S_2/2)$ tilted by $e^{Y/2} = (\pi S_2/2)^{1/4}$ (Section 1.2; [200, Section 16.2.1]). The Thorin measure of S_2 has atoms of mass two at the points $\pi^2 n^2/2$, the Gamma(2) shape of a sum of two copies. The structure explains the tilt, supplies an exactly solvable benchmark, and marks what positivity it cannot transfer.

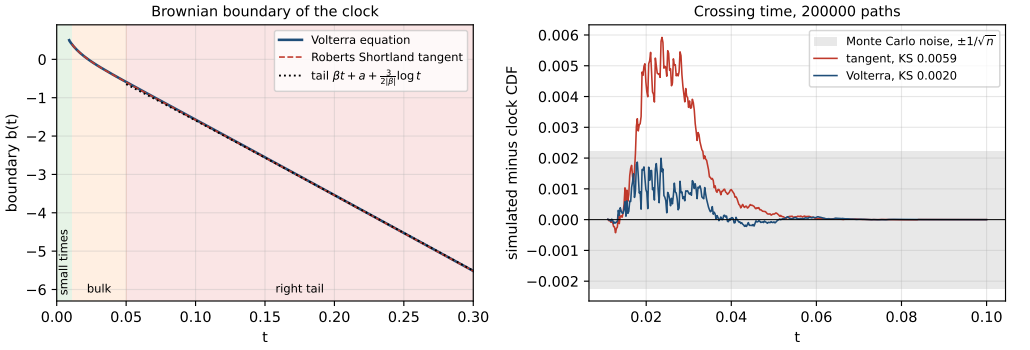


Figure 8.7: Left: the Brownian boundary of the clock computed from the Volterra equation, the Roberts Shortland tangent boundary, and the right tail asymptote with its logarithmic correction, over the three zones. Right: the simulated distribution function of the crossing time minus that of the clock, for both boundaries; the grey band is $\pm 1/\sqrt{n}$, twice the largest standard error of a simulated distribution function.

First, the tilt. Write y for the variable of the Pólya kernel, so that the value y of Y corresponds to the value e^{2y} of $\pi S_2/2$. Moving the line of integration from the real axis to the line $\operatorname{Im} y = \eta$, as in Proposition 2.1.1, rotates e^{2y} by the angle 2η into the complex plane. The density of S_2, a theta series in its argument, continues analytically to the right half plane, and the boundary of that half plane, the natural boundary of the theta function [219, Chapter 11, Section 1.4, and Chapter 5, Section 3.2], is reached exactly at $\eta = \pi/4$. So the admissible tilts of Proposition 2.1.1 are the rotations of Williams' variable within the half plane of analyticity of its density, and the saving $e^{-\eta\theta}$ is the factor gained by that rotation.

Second, a benchmark. Since $\pi^2 S_2/2 = \sum_n \Gamma_n/n^2$, the Thorin measure of $\pi^2 S_2/2$ is $2\sum_n \delta_{n^2}$, and in the square root variable its Cauchy flow $P_\varepsilon * 2\sum_n \delta_n$ is explicit [200, Section 16.4]. On this measure the distances of Section 3.2 separate the effects of the method from those of the arithmetic, and they need no separate computation: $\pi^2 S_2/2$ is the sum of two independent copies of the sinh clock of Section 1.3.4, whose Thorin measure has unit atoms at the squares of the integers, so its distances follow from the closed forms of that section (Section 12.2.3).

Third, the limits. Williams' atoms are the Dirichlet eigenvalues $\pi^2 n^2/2$ of an interval, each counted twice, once for each of the two hitting times, and they live on the Laplace side of the kernel. The zeros of ξ live on the Mellin side of the same variable, and they reappear as unit atoms only on the Laplace side of the reciprocal. The Gamma(2) structure therefore gives positivity and an exact sampler [68] on the kernel side, but it transfers no positivity to the Thorin measure of the clock. In particular it does not give a positive Gamma(2) scale mixture representation of the Pólya density, a route that [200, Section 12.6] excludes.

Section 9.1.1 follows the Mellin side to every basepoint, where size biasing S_2 gives a probability law with no hypothesis and the zeros enter only as zeros of its transform, and Section 11.1.7 draws the Mellin side and the clock side, with the Brownian boundaries of this section, as one map of processes.

8.3 Keiper Li coefficients from the Pólya density

The Pólya density that computes the law of the clock also computes the coefficients of Li's criterion [145], which states that RH holds if and only if

$$\lambda_n = \sum_{\rho} \left[1 - \left(1 - \frac{1}{\rho}\right)^n\right] \geq 0 \qquad \text{for every } n \geq 1,$$

the sum over the zeros taken in conjugate pairs. Keiper [120] introduced the coefficients λ_n and computed them before Li proved the criterion, and Bombieri and Lagarias [33] derived an arithmetic expression for them from the explicit formula and related the criterion to Weil's. The coefficient λ_n is determined by the Taylor coefficients of $\log \xi$ at $s = 1$ up to order n, a point at distance 14.14 from the nearest zero, where the Pólya route loses almost nothing to cancellation. We compute $\lambda_1, \ldots, \lambda_{100}$ in this way and check them against a sum over the zeros. We then show that λ_n is the Thorin functional evaluated at an explicit rational test function, so that Li's criterion, like Weil's, is positivity of that functional on a cone.

8.3.1 Taylor coefficients of $\log \xi$ at $s = 1$

Li [145] expresses the coefficients as derivatives at $s = 1$,

$$\lambda_n = \frac{1}{(n-1)!} \frac{d^n}{ds^n} \Big[s^{n-1} \log \xi(s) \Big]_{s=1}.$$

With $\log \xi(1 + w) = \sum_{k \geq 0} a_k w^k$ and $(1 + w)^{n-1} = \sum_j \binom{n-1}{j} w^j$, comparing the coefficients of w^n gives

$$\lambda_n = n \sum_{j=0}^{n-1} \binom{n-1}{j} a_{n-j}, \tag{8.3.1}$$

so the computation reduces to the Taylor coefficients of $\log \xi$ at $s = 1$. The zeros nearest to $s = 1$ are $\frac{1}{2} \pm 14.1347i$, so $\log \xi(1 + w)$ is analytic in the disc $|w| < R$, where $R = |\rho_1 - 1| = 14.1436$.

The sum (8.3.1) is exact, but its terms can be larger than λ_n. The Hadamard product $\xi(1+w) = \xi(1) \prod_\rho (1 - w/(\rho - 1))$ gives $m a_m = -\sum_\rho (\rho - 1)^{-m}$ for $m \geq 2$, and for large m the two zeros nearest to $s = 1$ dominate this sum, so that $|m a_m|$ is at most about $2R^{-m}$. Since $n\binom{n-1}{m-1} = m\binom{n}{m}$, the terms of (8.3.1) are $\binom{n}{m} m a_m$ with $m = n - j$, and for large n their absolute sum is at most about $2(1 + 1/R)^n$.

Cancellation therefore costs at most about $\log_{10}\big(2(1+1/R)^n/\lambda_n\big)$ digits. This grows like $n\log_{10}(1+1/R)\approx 0.03n$ and is about one digit at $n=100$, where $2(1+1/R)^{100}\approx 1.9\times 10^3$ and $\lambda_{100}=118.6$; the working precision needs to grow only linearly in n, with a small slope.

We compute the a_k by the trapezoid rule for the Cauchy integral on the circle $|w|=6$,

$$a_k \approx \frac{1}{Nr^k}\sum_{j=0}^{N-1}\log\xi\big(1+re^{2\pi ij/N}\big)e^{-2\pi ijk/N}, \qquad r=6,\ N=512,$$

which converges geometrically. The rule returns $a_k + a_{k+N}r^N + a_{k+2N}r^{2N}+\cdots$, so aliasing changes a_k by a relative amount of about $(r/R)^N\approx 10^{-191}$. Rounding errors in the sum are amplified in a_k by about $(R/r)^k$, which is 10^{37} at $k=100$ and is absorbed by the working precision.

Each value $\xi(1+w)=\xi(\frac12+(\frac12+w))$ comes from the Pólya route at 600 bits; the imaginary part of $\frac12+w$ never exceeds 6, so the Pólya sum loses almost nothing to cancellation, and the computation takes a few seconds. The principal branch of the logarithm is the analytic one, because $|\arg\xi(1+w)|<1$ on the circle: in the Hadamard product (1.2.1) with $s=\frac12+w$, so that $|s|\le 6.5$, the factor $1+s^2/\gamma^2$ of the zeros $\frac12\pm i\gamma$ has argument at most $\arcsin(6.5^2/\gamma^2)\le 1.01\cdot 6.5^2/\gamma^2$, and since $\sum_{\gamma>0}\gamma^{-2}=0.0231$ these arguments add up to less than 0.99.

The first coefficient has the closed form $\lambda_1=1+\frac{\gamma_E}{2}-\frac12\log 4\pi$, where γ_E is Euler's constant. The Pólya computation gives

$$\lambda_1 = 0.023095708966121033814310 25,$$

in agreement with the closed form to all twenty five digits shown. The same number $\lambda_1=\sum_\rho \rho^{-1}=(\xi'/\xi)(1)$ reappears at the basepoint one (Section 9.1.2): the shifted clock T_1 there, with Laplace transform $\xi(1)/\xi(1+\sqrt{u})=1-\lambda_1\sqrt{u}+O(u)$ near $u=0$, has a tail of order $x^{-1/2}$, and λ_1/π is its Thorin density at the origin in the chart $\theta=\sqrt{z}$. Table 8.6 lists further values with Keiper's asymptotic $\lambda_n\approx\frac{n}{2}(\log n-1-\log 2\pi+\gamma_E)$ [120], and Figure 8.8 shows all hundred.

8.3.2 Li's criterion as positivity of the Thorin functional

Li's criterion is a statement about the Thorin functional U of Section 5.3, which has a unit atom at $-(\rho-\frac12)^2$ for each pair $\{\rho,1-\rho\}$ of zeros, on the line or off it. For a zero ρ put $q=1-1/\rho$. Then $1-1/(1-\rho)=1/q$, so the pair contributes $2-q^n-q^{-n}$ to λ_n; and if $a=-(\rho-\frac12)^2$, with the square root chosen so that $\rho-\frac12=i\sqrt{a}$, then $q=(2\sqrt{a}+i)/(2\sqrt{a}-i)$. This proves the following.

Proposition 8.3.1. *Without any hypothesis on the zeros,*

$$\lambda_n=\sum_{\{\rho,1-\rho\}}\varphi_n\Big(-\big(\rho-\tfrac12\big)^2\Big)=\langle U,\varphi_n\rangle,$$

n	λ_n (Pólya)	Keiper asymptotic
1	0.023095708966121033814	−1.13
2	0.092345735228046670386	−1.57
3	0.20763892055432480379	−1.74
5	0.57554271446117745243	−1.63
10	2.2793393631931577437	0.21
20	8.7692768720932151296	7.35
50	43.531096488374019299	41.28
100	118.60377537679132993	117.23

Table 8.6: Keiper Li coefficients from the Pólya density (all digits shown are correct) and Keiper's leading asymptotic, which becomes accurate only for large n.

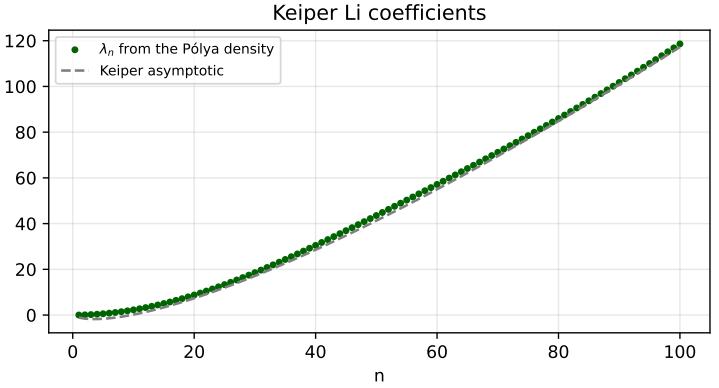


Figure 8.8: The first hundred Keiper Li coefficients from the Pólya density and Keiper's asymptotic.

where

$$\varphi_n(z) = 2 - \left(\frac{2\sqrt{z}+i}{2\sqrt{z}-i}\right)^n - \left(\frac{2\sqrt{z}-i}{2\sqrt{z}+i}\right)^n$$

is a rational function of z*, and* $\varphi_n(z) = 4\sin^2\left(n \arctan \frac{1}{2\sqrt{z}}\right) \geq 0$ *for* $z > 0$.

The two fractions are exchanged when $\sqrt{z}$ changes sign, so φ_n is a rational function of z, with a pole of order n at $z = -\frac{1}{4}$; for example $\varphi_1(z) = (z+\frac{1}{4})^{-1}$ and $\varphi_2(z) = 4z(z+\frac{1}{4})^{-2}$. At infinity $\varphi_n(z) \sim n^2/z$, so the sum converges absolutely.

On the positive axis the test functions are nonnegative: for $z > 0$ the two fractions are conjugate numbers of modulus one, and $\varphi_n(z) = 4\sin^2(n \arctan \frac{1}{2\sqrt{z}}) \geq 0$. On the critical line, for $\rho = \frac{1}{2} + i\gamma$ with $\gamma > 0$, the number $1 - 1/\rho = (\rho - 1)/\rho$ has modulus one and argument $\phi_\gamma = 2 \arctan \frac{1}{2\gamma}$, and the conjugate zero gives $-\phi_\gamma$.

Adding the contributions of ρ and $\bar\rho$,

$$\lambda_n = \sum_{\gamma>0} 2\big(1-\cos(n\phi_\gamma)\big) = \sum_{\gamma>0} 4\sin^2\Big(n\arctan\frac{1}{2\gamma}\Big) \qquad \text{under RH.}$$

In the variable z of the Thorin measure this is the pairing

$$\lambda_n = \Big\langle U_{\mathrm{RH}},\ 4\sin^2\big(n\arctan\tfrac{1}{2\sqrt{z}}\big)\Big\rangle$$

of U_{RH} with the nonnegative function φ_n. Li's criterion thus says that RH holds if and only if $\langle U,\varphi_n\rangle \ge 0$ for every n, that is, the Thorin functional is nonnegative on the cone spanned by the φ_n. Under RH this positivity is automatic, since $U = U_{\mathrm{RH}}$ is a positive measure; the content of the criterion is the converse.

The test function φ_n is about n^2/z for $\sqrt{z} \gg n$ and oscillates between 0 and 4 for $\sqrt{z} \lesssim n/\pi$, so λ_n weighs the zeros below height of order n at full size and the higher zeros by about n^2/γ^2 (Figure 8.9). Every computed λ_n is positive, as it must be: every zero below height 3×10^{12} lies on the line [196] and contributes a nonnegative term, while a zero at height $\gamma \gg n$ contributes only about n^2/γ^2 in absolute value, so the zeros above that height cannot change the sign of λ_n for $n \le 100$.

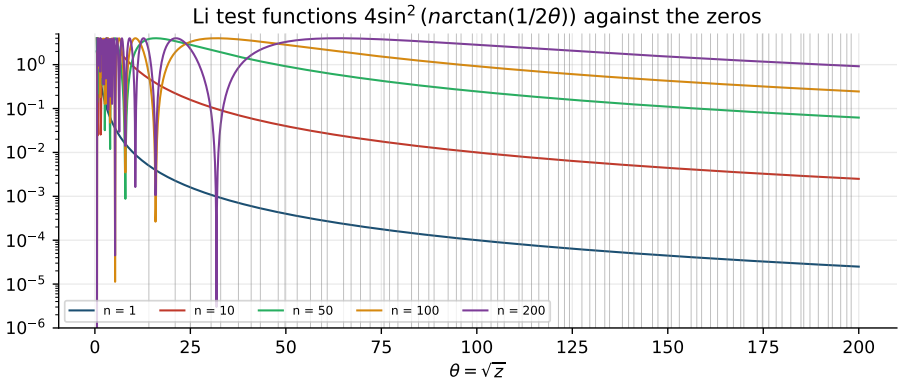


Figure 8.9: The Li test functions $4\sin^2(n\arctan\frac{1}{2\theta})$ for $n = 1, 10, 50, 100, 200$, in the square root chart $\theta = \sqrt{z}$, against the zeros below height 200 (grey vertical lines). Larger n probes higher zeros.

An off line zero makes $|1-1/\rho| \neq 1$. For $\rho = \frac12 - d + i\gamma$ with $d > 0$ one has $|1-1/\rho|^2 = 1 + 2d/((\frac12 - d)^2 + \gamma^2)$, so for large n the term $(1-1/\rho)^n$ grows like e^{nd/γ^2}, and this is how a failure of RH would eventually make λ_n negative (Exercise 8.14). Since the exponent is nd/γ^2, the index at which this happens grows with the height of the zero, so finitely many coefficients, like the finite tests of Section 5.2 (Table 5.6), cannot exclude an off line zero at great height.

A sum over the zeros reproduces the Pólya values. Summing $4\sin^2(n \arctan \frac{1}{2\gamma})$ over the first 1700 zeros, computed with `mpmath` [159] as in Section 3.2, and adding the tail above γ_{1700} with the smooth zero density $\log(\gamma/2\pi)/2\pi$ gives the Pólya values with relative residuals near 5×10^{-6} for $n \le 20$ and 10^{-5} at $n = 100$. The absolute residual grows like n^2 and comes from the cutoff: at γ_{1700} the zero count exceeds its smooth part by about one half, so the smooth tail counts about half a term, $\frac{1}{2}\varphi_n(\gamma_{1700}^2) \approx n^2/(2\gamma_{1700}^2)$, too many. The two computations are independent: one uses the Pólya kernel only, the other the zeros only.

As a diagnostic the coefficients are poorly localised. With the weights of Figure 8.9, the low coefficients are dominated by the low zeros and the high zeros enter only at large n, so λ_n behaves like the heat trace and Bernstein diagnostics of Table 1.8 and unlike the Thorin density, which is local in the ordinate. Their advantage is that they are Taylor data at a single point, 14.14 from the nearest zero, which the Pólya route computes to any precision.

Weil's criterion has the same form as Li's, with the functions of the class $\mathcal{T}$ of Section 5.3 that are nonnegative on $(0, \infty)$ in place of the φ_n (Section 3.5.2), but the two cones are different. The pole of φ_n lies at $-\frac{1}{4}$, the point at which (5.3.1) evaluates its test functions, inside the region where $\mathcal{T}$ requires holomorphy, and φ_n decays only like $1/z$. So φ_n is not in $\mathcal{T}$, $\langle U, \varphi_n \rangle$ is defined through the zeros, and the prime side of λ_n is the regularised arithmetic formula of Bombieri and Lagarias [33]. The sine in φ_n is also unrelated to the sine squared lemma, whose kernel is $\sin^2(x\sqrt{z}/2)$.

8.4 The de Bruijn Newman flow from the Pólya density

Li's criterion reads RH from Taylor data at one point, and it tests one fixed Thorin functional against a sequence of test functions (Section 8.3). The de Bruijn Newman flow reads RH from a deformation of the Pólya kernel, and so deforms the functional instead. Multiplying the kernel by a Gaussian factor turns Ξ into a family of entire functions H_t that evolve by the backward heat equation. RH says that H_0 has only real zeros, and the de Bruijn Newman constant Λ is defined by the property that H_t has only real zeros exactly when $t \ge \Lambda$. In Thorin terms the flow is a path of clocks that are generalised gamma convolutions for $t \ge \Lambda$ (Theorem 7.2.15), and RH says that the path stays in this class down to $t = 0$. The Pólya route computes H_t at the cost of Ξ, since the Gaussian factor only multiplies the node values. We follow two pairs of zeros backward in time to their collisions, the closest pair below height 200 and Lehmer's pair near height 7005, and compare the collision times with the isolated pair estimate and with the lower bound of Csordas, Smith and Varga. The Stieltjes moment test of Section 5.1 runs along the flow at the cost of its moments, and a Hankel test of finite order turns out to be a window on the low zeros that registers their collisions one at a time. Biane's description of the free convolution with a semicircle law singles out

Laguerre's quantity $-(\log H_0)''$, which predicts the collision times from H_0 alone and, with corrections computed along the flow, brackets each of them.

8.4.1 The flow, the motion of its zeros and its Thorin form

In the normalisation of Rodgers and Tao [224], which Theorem 7.2.15 also uses,

$$H_t(z) = \int_0^\infty e^{tu^2}\Phi_{\mathrm{dBN}}(u)\cos(zu)\,du,$$
$$\Phi_{\mathrm{dBN}}(u) = \sum_{n\geq 1}\left(2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u}\right)\exp(-\pi n^2 e^{4u}),$$

so that $H_0(z) = \frac{1}{8}\Xi(z/2)$. If $v = 2u$ denotes the variable of (1.2.3), then $\Phi_{\mathrm{dBN}}(u) = \frac{1}{2}\Phi(v)$ and

$$8H_t(2\theta) = \int_{\mathbb{R}} \Phi(v)\, e^{tv^2/4}\cos(\theta v)\,dv,$$

so the flow multiplies the kernel of (1.2.3) by $e^{tv^2/4}$, and a zero z of H_t corresponds to the ordinate $\theta = z/2$.

De Bruijn showed that H_t has only real zeros for $t \geq \frac{1}{2}$ and that the property persists forward in time [65]. Newman showed that there is a constant $\Lambda > -\infty$, at most $\frac{1}{2}$ by de Bruijn's theorem, such that H_t has only real zeros exactly for $t \geq \Lambda$, and conjectured $\Lambda \geq 0$ [163], which Rodgers and Tao proved [224]. Ki, Kim and Lee sharpened de Bruijn's bound to $\Lambda < \frac{1}{2}$ [125], the Polymath project proved $\Lambda \leq 0.22$ [217], and Platt and Trudgian improved this to $\Lambda \leq 0.2$ [196]. De Bruijn's bound comes from a theorem about strips: if the zeros of H_0 lie in $|\operatorname{Im} z| \leq \Delta$, then H_t has only real zeros for $t \geq \Delta^2/2$ [65, Theorem 13], and the critical strip gives $\Delta = 1$. A zero free half plane $\operatorname{Re} s > \alpha$ gives $\Delta = 2\alpha - 1$, hence $\Lambda \leq 2(\alpha - \frac{1}{2})^2$; for the half plane $\operatorname{Re} s > \frac{7}{8}$ announced in October 2026 (Section 12.1.5) this is $\Lambda \leq \frac{9}{32}$, weaker than 0.2, which a zero free half plane improves only if $\alpha < \frac{1}{2} + \sqrt{0.1} \approx 0.816$. RH is the inequality $\Lambda \leq 0$, so by the theorem of Rodgers and Tao it is equivalent to $\Lambda = 0$: if RH holds, the zeros of H_t are all real from $t = 0$ on and at no earlier time.

While the zeros are simple they move according to

$$\frac{dz_k}{dt} = 2\sum_{j\neq k}\frac{1}{z_k - z_j},$$

the sum running over all other zeros, positive and negative, with z_j and $-z_j$ taken together. Differentiating $H_t(z_k(t)) = 0$ in t and using the backward heat equation $\partial_t H_t = -\partial_z^2 H_t$ gives $dz_k/dt = \partial_z^2 H_t/\partial_z H_t$ at z_k, and the Hadamard product of the even function H_t turns this into the display (Exercise 8.15). The equation is the gradient flow of unit charges on the line with logarithmic interaction, so forward in time the zeros repel each other, spread and become more regular. Backward in

time neighbouring zeros approach each other, and a close pair collides and then leaves the real axis as a conjugate pair. A pair at distance D that is isolated from the other zeros obeys $dD/dt \approx 4/D$, so $D^2 \approx D_0^2 + 8t$ with $D_0 = D(0)$, and the pair collides at $t \approx -D_0^2/8$. RH is the statement that no two zeros collide at a positive time.

Let $R_t(u) = H_t(0)/H_t(2i\sqrt{u})$, where u is the variable of the clock (written s in Chapters 6 and 7), not the variable of integration in H_t, so that R_0 is the clock $1/G$. By Theorem 7.2.15, for every $t \geq \Lambda$, and so for every $t \geq 0.2$, R_t is the Laplace transform of a GGC with Thorin measure $\sum_k \delta_{(z_k(t)/2)^2}$, where $z_k(t)$ runs over the positive zeros of H_t: along the path of clocks the Thorin atoms move with the zeros. The Thorin density at resolution ε follows the flow in the same way. Put $\xi_t(s) = 8H_t\big(-2i(s-\tfrac12)\big)$, so that $\xi_t(\tfrac12 + i\theta) = 8H_t(2\theta)$ and $\xi_0 = \xi$, and

$$\rho_{\varepsilon,t}(\theta) = \pi^{-1}\,\mathrm{Re}(\xi_t'/\xi_t)\big(\tfrac12 + \varepsilon + i\theta\big).$$

Each zero z of H_t gives the zero $\frac12 + iz/2$ of ξ_t, and for every t the function $\rho_{\varepsilon,t}$ is the sum of the Poisson kernels $P_{\frac12+\varepsilon-\beta}(\theta-\gamma)$ over the zeros $\beta+i\gamma$ of ξ_t. For $t \geq \Lambda$ every β equals $\frac12$ and $\rho_{\varepsilon,t}$ is the Thorin density of R_t at resolution ε; the Pólya route computes it by the same multiplication of the node values. As t decreases through a collision, $\rho_{\varepsilon,t}$ shows the off line signature of Section 3.3: two Poisson peaks merge, and after the collision the pair contributes the kernels $P_{\varepsilon+d(t)} + P_{\varepsilon-d(t)}$ at its ordinate, where $d(t)$, half the imaginary part of the pair in z, grows from zero; the second kernel is negative once $d(t) > \varepsilon$. The flow thus produces off line configurations of known geometry, on which the diagnostics of Chapters 3 to 5 can be calibrated; Section 12.2.3 lists this calibration among the open computations.

8.4.2 Two collisions computed from the Pólya kernel

The Pólya route evaluates H_t at any t by multiplying the node values $\Phi(v_j)$ by $e^{tv_j^2/4}$ and summing. The factor is entire, so the error analysis of Section 2.1 carries over, and for $|t| \leq 1$ the factor is at most $e^{9/4}$ on the truncation interval $|v| \leq 3$. Among the 79 zeros below height 200 the closest pair is $\gamma_{71} = 184.8745$, $\gamma_{72} = 185.5988$, at distance 0.7243 in θ, that is $D_0 = 1.4486$ in z. We follow the pair by the sign changes of $H_t(2\theta)$ at 121 equally spaced points of $[\gamma_{71} - 0.3, \gamma_{72} + 0.3]$, computed at 420 bits, and refine each zero by Newton's method; Table 8.7 and Figure 8.10 show the result.

Bisection on t for the disappearance of the two sign changes gives the collision time

$$t_* = -0.2880,$$

with a grid error of at most 6.1×10^{-5}. Near a collision the term $4/D$ dominates the motion of the pair, so $D^2 \approx 8(t - t_*)$ and the gap in θ is about $\sqrt{2(t-t_*)}$; on a grid of step h the sign changes therefore vanish at a time at most $h^2/2$ above t_*.

t	zeros in $\theta = z/2$	sign changes
0.2	184.676, 185.590	2
0.1	184.771, 185.598	2
0.0	184.874, 185.599	2
−0.05	184.931, 185.595	2
−0.1	184.992, 185.587	2
−0.15	185.059, 185.573	2
−0.2	185.136, 185.550	2
−0.3	none	0

Table 8.7: The closest pair below height 200 under the de Bruijn Newman flow, located by the sign changes on a grid of step 0.011 and refined by Newton's method; t is the de Bruijn Newman time.

The isolated pair estimate is $-D_0^2/8 = -0.262$: the neighbouring zeros, which the estimate ignores, delay the collision. Since nonreal zeros persist backward in time, H_t has nonreal zeros for every t below the collision time, so $\Lambda \geq t_* \approx -0.288$; this numerical lower bound is far weaker than the published bounds, as expected for a pair of ordinary spacing.

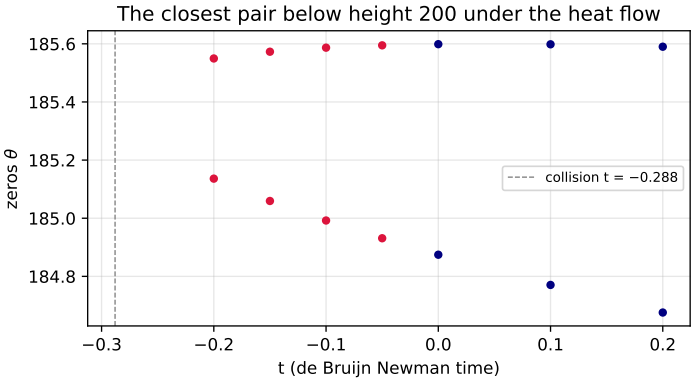


Figure 8.10: Zeros of the closest pair below height 200 along the de Bruijn Newman flow, refined by Newton's method as in Table 8.7, for $t \geq 0$ and $t < 0$, and the computed collision time (dashed line).

Lehmer's pair near height 7005, $\gamma_{6709} = 7005.0629$ and $\gamma_{6710} = 7005.1006$ (Section 2.2.3.1), is much closer: $D_0 = 2 \times 0.0377 = 0.0754$, and its isolated pair estimate $-D_0^2/8 = -7.1 \times 10^{-4}$ is about 400 times closer to zero than the collision time of the pair below height 200. The tilted Pólya route evaluates H_t at that height at the same cost as Ξ, with the Gaussian factor $e^{tv^2/4}$ included in the exponential at each complex node $v = x + i\eta$. With the parameters of Section 2.2.3.1 ($\delta = 0.01$, 256 bits, 13001 nodes) we track the two sign changes of

the pair on a grid of step 0.0025 in θ over $[7005.03, 7005.13]$. Both sign changes are present at $t = 0$ and absent at $t = -0.004$, and twelve bisection steps on this interval give -7.10×10^{-4}, which by the argument above lies above the collision time by at most $0.0025^2/2 \approx 3 \times 10^{-6}$. Newton's method for the double zero, $H_t(2\theta) = \partial_\theta H_t(2\theta) = 0$ in the unknowns θ and t, started from this value, gives

$$t_* = -7.11 \times 10^{-4}$$

at $\theta = 7005.0817$, and Figure 8.11 shows the collision. Here the isolated pair estimate is accurate: for a pair this close the neighbours hardly matter, whereas for the ordinary pair below height 200 they moved the collision from -0.262 to -0.288. The computation takes about half a minute, and the numerical lower bound $\Lambda \geq t_*$ that it gives is much sharper than the one from the pair below height 200.

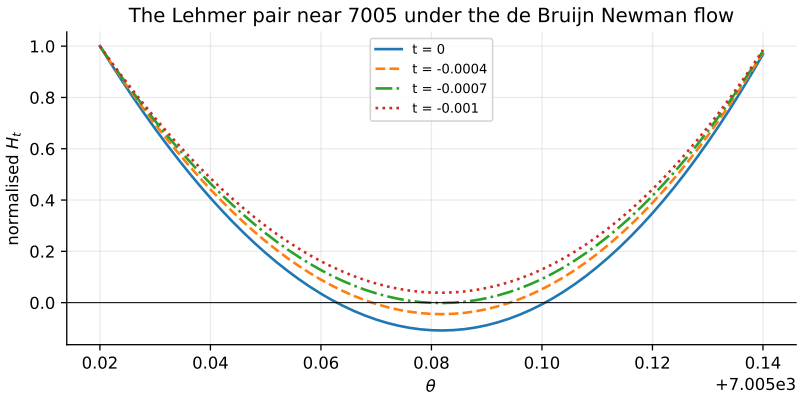


Figure 8.11: The function $\theta \mapsto H_t(2\theta)e^{\pi\theta/4}$ near the Lehmer pair at height 7005, divided by its maximum absolute value on the window, for four times of the de Bruijn Newman flow, computed with the tilted Pólya route. At $t = 0$ it has two sign changes; at $t = -7 \times 10^{-4}$, slightly above the collision time -7.11×10^{-4}, they nearly coincide; at $t = -10^{-3}$ the pair has left the real axis.

Pólya's function $K_{i\theta/2}(2\pi)$ behaves differently under the same flow. Down to $t = -60$ none of its real zeros below height 90 collide, and as t decreases from 0 to -0.5 they rise almost as a rigid block, at speeds within 2% of the mean speed $\pi/8$ that the two functions share, while the zeros of Ξ rise at speeds between 0.004 and 0.735; yet by Newman's theorem the de Bruijn Newman constant of Pólya's function is finite (Section 8.7.6).

8.4.3 The Hankel test along the flow

The Stieltjes moment test of Section 5.1 (Table 5.2) needs only the Taylor coefficients of Ξ at the centre, and along the flow these are moments of the reweighted

kernel [208]. Putting $\theta = i\sqrt{u}$ in the formula for $8H_t(2\theta)$ of Section 8.4.1 gives

$$G_t(u) = \frac{H_t(2i\sqrt{u})}{H_t(0)} = \sum_{k\ge 0} \frac{M_k(t)\,u^k}{(2k)!\,M_0(t)}, \qquad M_k(t) = \int_{\mathbb{R}} v^{2k} e^{tv^2/4} \Phi(v)\,dv,$$

and G_0 is the function G of Section 5.1. Write $\pm\theta_j(t)$ for the zeros of $\theta \mapsto H_t(2\theta)$, so that $\theta_j(0) = \gamma_j$ while the zeros are on the line. Since H_t is even and entire of order one, G_t has order $\frac{1}{2}$ and $G_t(u) = \prod_j (1 + u/a_j(t))$ with $a_j(t) = \theta_j(t)^2$, so the Newton identities turn the $M_k(t)$ into the moments $m_n(t) = \sum_j a_j(t)^{-n-1}$, exactly as at $t = 0$. If every $\theta_j(t)$ is real, the $m_n(t)$ are the moments of the positive measure $\nu_t = \sum_j a_j(t)^{-1}\delta_{1/a_j(t)}$, and every Hankel matrix $\mathcal{H}_N(t) = [m_{i+j}(t)]_{i,j<N}$ is positive definite. The converse is proved as at $t = 0$: if every $\mathcal{H}_N(t)$ is positive definite, every $a_j(t)$ is real, and it is positive because H_t has no zeros on the imaginary axis: $H_t(iy) = \int_0^\infty e^{tu^2}\Phi_{\mathrm{dBN}}(u)\cosh(yu)\,du > 0$, the kernel (1.2.3) being strictly positive. So at every time t all the matrices $\mathcal{H}_N(t)$ are positive definite exactly when H_t has only real zeros, that is, when $t \ge \Lambda$. Each $M_k(t)$ is one sum over the node values of Φ, reweighted by $e^{tv^2/4}$, and the Cholesky factor gives the Jacobi matrix $J_N(t)$, whose eigenvalues return the lowest zeros of H_t as J_N returns the γ_j at $t = 0$ (Table 8.8).

t	$\theta_1(t)$	$\theta_2(t)$	$\theta_3(t)$	$\theta_5(t)$	$\theta_{10}(t)$
0.2	14.07681	20.94669	24.95221	32.88801	49.72988
0	14.13473	21.02204	25.01086	32.93506	49.77383
−0.4	14.25083	21.17365	25.12774	33.02595	49.85213
−1.0	14.42563	21.40341	25.30189	33.15277	49.93350

Table 8.8: Zeros $\theta_n(t)$ of $\theta \mapsto H_t(2\theta)$ returned by the Jacobi matrix $J_{30}(t)$ of the moments $m_0(t), \dots, m_{59}(t)$; at $t = 0$ they are γ_1, γ_2, γ_3, γ_5 and γ_{10}. At each of the four times $J_{30}(t)$ returns $\theta_1(t), \dots, \theta_{12}(t)$ to within 3.1×10^{-7} of the zeros that Newton's method finds on H_t, and $\theta_{13}(t)$ only to within 5.2×10^{-4}. Backward in time each of these zeros moves up.

Going back from $t = 0$, the matrix $\mathcal{H}_N(t)$ stays positive definite until a time t_N^* at which it ceases to be. Since $\mathcal{H}_N(t)$ is the leading block of $\mathcal{H}_{N+1}(t)$, $t_N^* \le t_{N+1}^*$. Under RH every t_N^* is negative, because $\mathcal{H}_N(0)$ is positive definite, and $t_N^* \to 0$: for each $t < 0$ the function H_t has nonreal zeros by the theorem of Rodgers and Tao, so some $\mathcal{H}_N(t)$ is not positive definite, and then $t_{N'}^* \ge t$ for every $N' \ge N$. With the moments from the trapezoid rule of step $1/600$ on $[0, 3.6]$ at $100 + 16N$ digits (Table 5.2 gives the precision that order N needs at $t = 0$), the grid $t = 0, -0.05, \dots, -4$ and bisection to 10^{-7} give no failure of order 10 down to $t = -4$, and

$$t_{20}^* = -1.8991865, \qquad t_{30}^* = -1.2853067,$$
$$t_{40}^* = -1.2760788, \qquad t_{50}^* = -1.0325048;$$

for $N = 30$ a second discretisation (step 1/550, 650 digits) gives a bracket that overlaps the first in $[-1.28530669, -1.28530664]$.

Each failure registers a collision. We follow the real zeros of H_t that start at $2\gamma_j$, $j \le 64$, back to $t = -2.05$ by the equation of motion of Section 8.4.1, with Newton's method on the trapezoid sum for H_t at each step (110 digits, step 1/100), and locate each collision as a double zero, $H_t(z) = \partial_z H_t(z) = 0$, by Newton's method in the two unknowns z and t; the backward heat equation, which holds exactly for the trapezoid sum, gives the derivatives in t. Thirteen of the pairs below index 59 collide in this time range, among them those that start at (γ_k, γ_{k+1}) for $k = 9$, 13, 19, 24, 27 and 34, at

$$t = -1.87791,\ -1.28527,\ -1.27554,\ -1.14387,\ -0.88108,\ -0.38456$$

and the heights $\theta = 49.664$, 60.491, 77.027, 88.466, 95.540 and 111.638; the other seven start at $k = 40$, 43, 45, 48, 51, 53 and 57. To see which collision a failure registers, we continue each double zero into the complex plane, starting from $z_c \pm i\sqrt{2(t_c - t)}$ near the collision, and remove the contribution $2\,\mathrm{Re}\,\theta^{-2n-2}$ of the conjugate pair $\theta = z/2$, $\bar\theta$ from $m_n(t)$. At $t_N^* - \delta$, with $\delta = 0.01$ (for $N = 40$, $\delta = 0.0046$, half the time to the collision of $(\gamma_{13}, \gamma_{14})$), the only one of the thirteen pairs whose removal makes $\mathcal{H}_N(t)$ positive definite again is the one that starts at (γ_9, γ_{10}) for $N = 20$, at $(\gamma_{13}, \gamma_{14})$ for $N = 30$, at $(\gamma_{19}, \gamma_{20})$ for $N = 40$ and at $(\gamma_{27}, \gamma_{28})$ for $N = 50$. The same holds at $t_N^* - 10^{-5}$, except that there removing $(\gamma_{34}, \gamma_{35})$ also restores the order 50 matrix. The failures thus come 0.021, 3.7×10^{-5}, 5.3×10^{-4} and 0.151 after the collisions they register.

A Hankel test of finite order is thus a window on the low part of the spectrum. The order 30 matrix is positive definite at $t = -1.276, -1.278, \ldots, -1.284$ and -1.2852, after the collision of $(\gamma_{19}, \gamma_{20})$ at -1.27554 and before that of $(\gamma_{13}, \gamma_{14})$ at -1.28527: it passes the pair that is too high and too light for its order and registers the lower one. A conjugate pair at height θ enters $m_n(t)$ with weight of order θ^{-2n-2}, against $\theta_1(t)^{-2n-2}$ for the lowest zero, as at $t = 0$, and increasing the order widens the window.

The whole curve $N \mapsto t_N^*$ up to $N = 50$ follows from one elimination per time. Let $n_{\mathrm{fail}}(t)$ be the index of the first nonpositive pivot of Gaussian elimination on $\mathcal{H}_{50}(t)$, that is, the least $N \le 50$ for which $\mathcal{H}_N(t)$ is not positive definite, so that $t_N^* = \sup\{t < 0 : n_{\mathrm{fail}}(t) \le N\}$. On the grid of step 0.01 it is a staircase [208]: no order up to 50 fails on $[-1.03, 0]$, and then n_{fail} is 50 on $[-1.14, -1.04]$, 46 on $[-1.18, -1.15]$, 45 on $[-1.27, -1.19]$, and 39, 28 and 27 at $t = -1.28$, -1.29 and -1.30 (Table 8.9). To see what each step registers, the table gives, at one time on each step, the least order at which each complex pair breaks the test alone, that is, with the contributions of all the other complex pairs removed from the moments. A pair alone breaks the test at a definite order, a higher pair at a higher order, and n_{fail} is close to the least of these orders. The order at which a pair is seen falls as the pair moves away from the axis: $(\gamma_{27}, \gamma_{28})$, which collides at -0.881,

breaks order 50 at $t = -1.09$, order 45 at -1.16 and -1.23, order 41 at -1.28 and order 39 at -1.29, and $(\gamma_{34}, \gamma_{35})$, which collides earlier, at -0.385, but lies higher, breaks the test only from $t = -1.28$ on, at order 50 and then 47. Removing the pair with the least order moves n_{fail} up to about the next one: at $t = -1.30$ from 27 to 35, when $(\gamma_{13}, \gamma_{14})$ is removed, at -1.28 from 39 to 41, when $(\gamma_{19}, \gamma_{20})$ is removed, and at -1.09 from 50 to no failure, when $(\gamma_{27}, \gamma_{28})$ is removed. With all thirteen pairs removed, no order up to 50 fails at any of these times.

			order broken by (γ_k, γ_{k+1}) alone				
t	$n_{\text{fail}}(t)$	at $t =$	$k = 13$	19	24	27	34
$[-1.03, 0]$	none						
$[-1.14, -1.04]$	50	-1.09	real	real	real	50	> 50
$[-1.18, -1.15]$	46	-1.16	real	real	44	45	> 50
$[-1.27, -1.19]$	45	-1.23	real	real	44	45	> 50
-1.28	39	-1.28	real	38	40	41	50
-1.29	28	-1.29	28	35	37	39	47
-1.30	27	-1.30	27	35	37	39	47

Table 8.9: The first failing order $n_{\text{fail}}(t) \le 50$ of the Hankel test along the flow on the grid of step 0.01, so that $t_N^* = \sup\{t < 0 : n_{\text{fail}}(t) \le N\}$, and, at one time on each step, the least order at which the pair that starts at (γ_k, γ_{k+1}) breaks the test alone, with the contributions of the other complex pairs removed from the moments (real: the pair has not yet collided; > 50: no order up to 50 fails). The five pairs collide at $t = -1.285$, -1.276, -1.144, -0.881 and -0.385. The other pairs below index 59 that have collided by $t = -1.30$, those with $k = 40$, 45, 53 and 57, break no order up to 50 alone.

The steps of t_N^* are thus collisions, registered with a delay that depends on the order. A rule suggested in [208] is that the collision of (γ_k, γ_{k+1}) is registered from order about $2k + 2$ on, so that $t_N^* \approx \max\{t_{\text{coll}}(k) : 2k + 2 \le N\}$, where $t_{\text{coll}}(k)$ is the collision time of the pair. It holds for $N = 30$ and 40 ($k = 13$ and 19), roughly for $N = 20$ ($k = 9$, delay 0.021), and fails at $N = 50$: it gives the collision of $(\gamma_{24}, \gamma_{25})$ at -1.144, while the order 50 matrix registers $(\gamma_{27}, \gamma_{28})$, for which $2k+2 = 56$, at -1.0325, before $(\gamma_{24}, \gamma_{25})$ has collided. Table 8.9 shows why: the order at which a pair is seen falls as the pair leaves the axis, so a pair beyond the window of the rule enters it after a delay, and t_N^* is the latest of the collision times of the pairs inside the window and of the times at which pairs above it have moved far enough from the axis.

The curve rises to 0 through the collision times of ever closer pairs. Following every real zero that starts below index 215 in the same way (200 digits, step 1/200), 33 of the first 120 gaps close on $[-2.05, 0]$, 28 of them after $t = -1.6$, and 50 of the first 200 gaps close after $t = -1.6$. The latest collisions among the first 200 gaps are those of $(\gamma_{186}, \gamma_{187})$, at distance 0.498, at $t = -0.1310$, of $(\gamma_{159}, \gamma_{160})$

at -0.1695 and of $(\gamma_{135}, \gamma_{136})$ at -0.2015, and below height 230 those of $(\gamma_{91}, \gamma_{92})$ at -0.2821 and of the closest pair below height 200, $(\gamma_{71}, \gamma_{72})$, at -0.2880, the value of Section 8.4.2. A collision close to $t = 0$ needs a pair as close as Lehmer's, which collides at -7.11×10^{-4} (Section 8.4.2); it is made of the 6709th and 6710th zeros, far beyond the reach of a Hankel test at the centre.

Further back in time the low pairs leave the axis one after another, and Theorem 5.1.5 turns the test into a count. Its proof applies at every time to the zeros of H_t, none of which is purely imaginary: the numbers $\iota_N(t)$ and $\iota'_N(t)$ of negative eigenvalues of $\mathcal{H}_N(t)$ and of $\mathcal{H}'_N(t) = [m_{i+j+1}(t)]_{i,j<N}$ do not decrease with N, never exceed the number of distinct conjugate pairs $\theta, \bar\theta$ of zeros with $\operatorname{Re}\theta > 0$, and tend to it, so that each negative eigenvalue certifies a pair. The signs of the pivots of one Gaussian elimination give $\iota_1(t), \dots, \iota_{30}(t)$, and $n'_{\text{fail}}(t)$ is to $\mathcal{H}'_N(t)$ what $n_{\text{fail}}(t)$ is to $\mathcal{H}_N(t)$. We follow the real zeros below $\theta = 88$, those that start at $\gamma_1, \dots, \gamma_{24}$, back to $t = -20$, from the signs of the trapezoid sum in ball arithmetic on a grid of step 0.02 in θ and 0.05 in t, and locate each collision as a double zero as above. Below $\theta = 88$ the collisions found above are those of (γ_9, γ_{10}), $(\gamma_{13}, \gamma_{14})$ and $(\gamma_{19}, \gamma_{20})$, and after them come those of $(\gamma_{22}, \gamma_{23})$, $(\gamma_{15}, \gamma_{16})$, (γ_4, γ_5), (γ_7, γ_8), $(\gamma_{17}, \gamma_{18})$ and (γ_2, γ_3), at

$$t = -2.32414,\ -2.75904,\ -3.95479,\ -4.03382,\ -7.31401,\ -12.49451$$

and the heights $\theta = 84.875$, 67.406, 33.099, 43.403, 72.746 and 27.321. Each pair is followed into the complex plane as above, and Table 8.10 gives the lowest pair, the first failing orders and the counts at the orders 10 and 30. The jth negative eigenvalue of $\mathcal{H}_N(t)$ appears at $N = 19$ and 25 at $t = -2$, at $N = 9$, 14, 17, 23, 27 and 29 at $t = -5$, and at $N = 5$, 8, 10, 15, 17, 21, 25 and 29 at $t = -20$. Removing the contribution of one pair from the moments shows what the first failure registers: at $t = -2$ and -3 the pair that starts at (γ_9, γ_{10}), whose removal moves n_{fail} from 19 to 21 and from 18 to 21, and from $t = -4$ to -12 the pair that starts at (γ_4, γ_5), whose removal raises n_{fail} by two or three, from 11 to 14 at $t = -4$; at these times the removal of any other pair that starts below $\theta = 88$ leaves n_{fail} unchanged. Both pairs are registered soon after their collisions, at the orders 19 and 11, close to $2k + 2 = 20$ and 10. At $t = -2$ the three pairs that start at (γ_9, γ_{10}), $(\gamma_{13}, \gamma_{14})$ and $(\gamma_{19}, \gamma_{20})$ account for every failure up to order 30: with them removed none remains, and alone they break the test at the orders 19, 21 and 30. From $t = -3$ on, pairs that start above $\theta = 88$ enter as well, since with all the pairs that start below $\theta = 88$ removed the test still fails, from order 24 at $t = -3$, 21 at -4, 15 at -8 and 11 at -15 and -20. At $t = -15$ and -20 the lowest pair is no longer the first to be seen: no single removal raises n_{fail}, and the pair that starts at (γ_4, γ_5) alone breaks the test as early as the lowest pair alone or earlier, at order 4 against 5 at $t = -20$. The two discretisations of the moments, step 1/300 at 600 digits and step 1/450 at 900 digits, give the same counts and the same removal tests, and at the orders 10 and 30 the eigenvalues of the diagonally scaled matrices give the same counts as the pivots.

t	pair	θ	real	collided	n_{fail}	n'_{fail}	$(\iota_{10}, \iota'_{10})$	$(\iota_{30}, \iota'_{30})$
-2	$(9,10)$	$49.715 \pm 0.249i$	8	3	19	19	$(0,0)$	$(2,2)$
-3	$(9,10)$	$50.139 \pm 0.792i$	8	5	18	17	$(0,0)$	$(3,3)$
-4	$(4,5)$	$33.115 \pm 0.151i$	3	6	11	10	$(0,1)$	$(5,5)$
-5	$(4,5)$	$33.478 \pm 0.745i$	3	7	9	9	$(1,1)$	$(6,6)$
-6	$(4,5)$	$33.841 \pm 1.071i$	3	7	9	8	$(1,1)$	$(6,6)$
-8	$(4,5)$	$34.572 \pm 1.592i$	3	8	8	8	$(1,1)$	$(6,6)$
-10	$(4,5)$	$35.306 \pm 2.052i$	3	8	8	8	$(1,1)$	$(6,7)$
-12	$(4,5)$	$36.043 \pm 2.492i$	3	8	8	8	$(1,1)$	$(7,7)$
-15	$(2,3)$	$28.202 \pm 1.166i$	1	9	6	5	$(2,3)$	$(8,8)$
-20	$(2,3)$	$29.967 \pm 2.178i$	1	9	5	5	$(3,3)$	$(8,8)$

Table 8.10: The Hankel test further back along the flow. At each time: the lowest conjugate pair, labelled $(k, k+1)$ by the zeros γ_k, γ_{k+1} it starts from, and its position θ; the number of real zeros below it; the number of collisions so far among $\gamma_1, \dots, \gamma_{24}$, the zeros that start below $\theta = 88$; the least orders $n_{\mathrm{fail}}(t)$ and $n'_{\mathrm{fail}}(t)$ up to 30 at which $\mathcal{H}_N(t)$ and $\mathcal{H}'_N(t)$ are not positive definite; and the numbers $\iota_N(t)$ and $\iota'_N(t)$ of their negative eigenvalues at $N = 10$ and 30. At $t = -2$ and -3 the first failure registers the pair that starts at (γ_9, γ_{10}), from $t = -4$ to -12 the one that starts at (γ_4, γ_5), and at $t = -15$ and -20 no single pair.

The program `jacobi_dbn.py` computes the moments, t^*_N, Table 8.8, the collisions below index 59, the tests with single pairs removed and Table 8.9, `dbn_collisions.py` the census of the first 200 gaps, and `dbn_hankel_long.py` the collisions below $\theta = 88$ down to $t = -20$, the counts, the removal tests and Table 8.10.

8.4.4 Biane's theorem and Laguerre's quantity

The collision times of Section 8.4.3 can be predicted from H_0 alone, before the flow is followed, through a quantity that free probability singles out. This subsection recalls Biane's description of the free convolution of a probability measure with a semicircle law, in which a gap of the support survives for a time equal to the reciprocal of the minimum of $\int \mu(dx)/(u-x)^2$ over the gap, and identifies this integral, for the zeros of Ξ, with Laguerre's quantity $-(\log \Xi)''$. Section 8.4.5 proves that Laguerre's quantity bounds the first collision time of the backward heat flow from below for functions with only real zeros, gives an upper bound, and computes both along the flow of H_t. The material comes from Section 8 of the first author's note [208], except Corollaries 8.4.12 and 8.4.14 and the halved constant of Remark 8.4.10, which are not in the note; the numbers are recomputed here from H_t itself.

Let μ be a probability measure on $\mathbb{R}$, $G_\mu(z) = \int \mu(dx)/(z-x)$ its Cauchy transform, and σ_t the semicircle law of variance $t > 0$, with density $(2\pi t)^{-1}(4t -$

$x^2)^{1/2}$ on $[-2\sqrt{t}, 2\sqrt{t}]$. The free convolution $\mu \boxplus \sigma_t$ is the law of $A+S$ for free random variables A and S with laws μ and σ_t; it is also the limit of the eigenvalue distribution of $A_N + S_N$, where A_N is a Hermitian $N \times N$ matrix whose eigenvalue distribution tends to μ and S_N an independent matrix from the Gaussian unitary ensemble whose eigenvalue distribution tends to σ_t. Put

$$F_\mu(u) = \int \frac{\mu(dx)}{(u-x)^2} \in (0,\infty], \qquad v_t(u) = \inf\Big\{v \ge 0 : \int \frac{\mu(dx)}{(u-x)^2+v^2} \le \frac{1}{t}\Big\},$$
$$\psi_t(u) = u + t\int \frac{(u-x)\,\mu(dx)}{(u-x)^2 + v_t(u)^2}.$$

THEOREM 8.4.1 (Biane [26]). *Let μ be a probability measure on $\mathbb{R}$ and $t > 0$.*

(a) ψ_t is a homeomorphism of $\mathbb{R}$, and $\mu \boxplus \sigma_t$ has a continuous density p_t with $p_t(\psi_t(u)) = v_t(u)/(\pi t)$ for every real u. For $\operatorname{Im} z > 0$, $G_{\mu\boxplus\sigma_t}(z) = G_\mu(\omega_t(z))$, where $\omega_t(z) = z - tG_{\mu\boxplus\sigma_t}(z)$, and ω_t is the inverse of the map $w \mapsto w + tG_\mu(w)$ on the region $\{u+iv : v > v_t(u)\}$, which that map sends onto the upper half plane.

(b) Let $U_t = \{u : F_\mu(u) > 1/t\}$. The interior of the support of $\mu \boxplus \sigma_t$ is $\psi_t(U_t)$, the support is its closure, the density is real analytic on $\psi_t(U_t)$, and the number of components of U_t does not increase with t.

(c) Let (a,b) be a bounded component of $\mathbb{R}\setminus\operatorname{supp}\mu$ with $\mu(\{a\}) > 0$ and $\mu(\{b\}) > 0$. Then F_μ is strictly convex on (a,b) and tends to $+\infty$ at both ends; let u_0 be its minimiser and

$$t_0 = \frac{1}{F_\mu(u_0)} = \Big(\min_{a<u<b} \int \frac{\mu(dx)}{(u-x)^2}\Big)^{-1}. \tag{8.4.1}$$

For $t < t_0$ the set $\{u \in (a,b) : F_\mu(u) \le 1/t\}$ is an interval $[a(t), b(t)]$ with $a(t) < b(t)$, and ψ_t maps $(a(t), b(t))$ onto a component of the complement of the support of $\mu \boxplus \sigma_t$. As t increases to t_0, $a(t)$ increases and $b(t)$ decreases to u_0, and at t_0 the gap closes.

Proof. Part (a) is Lemma 4, Proposition 2 and Corollaries 2 and 3 of [26], where σ_t has R transform tz as here; part (b) is Proposition 3, its proof, and Corollary 4 there. For (c), μ does not charge (a,b), so $F''_\mu(u) = 6\int (u-x)^{-4}\mu(dx) > 0$ there, and the atoms make F_μ tend to $+\infty$ at a and b. The sublevel sets of a strictly convex function are intervals; $\{F_\mu \le 1/t\} \cap (a,b)$ contains u_0 in its interior when $1/t > F_\mu(u_0)$, shrinks as t grows, and reduces to u_0 at $t = t_0$. Its interior is a component of $\mathbb{R} \setminus \overline{U_t}$, because the points of (a,b) just outside it lie in U_t, and ψ_t, a homeomorphism, maps it onto a component of the complement of $\overline{\psi_t(U_t)}$, which is the support by (b). This is the discussion following Corollary 5 of [26]. □

So a gap of the support between two atoms survives exactly until the time (8.4.1), Biane's threshold. For a real entire function f write

$$\mathcal{L}_f(x) = -\frac{d^2}{dx^2}\log|f(x)| = \frac{f'(x)^2 - f(x)f''(x)}{f(x)^2},$$

Laguerre's quantity. If f has genus at most one and zeros z_j, then $\mathcal{L}_f(x) = \sum_j (x - z_j)^{-2}$, a real sum since the nonreal zeros come in conjugate pairs; if all zeros are real, every term is positive, which is Laguerre's inequality $f'^2 - ff'' > 0$ off the zeros. For the normalised counting measure $\mu_p = n^{-1}\sum_j \delta_{z_j}$ of the zeros of a polynomial p of degree n with only real simple zeros, $F_{\mu_p} = \mathcal{L}_p/n$, and by Theorem 8.4.1(c) the gap (z_k, z_{k+1}) survives in $\mu_p \boxplus \sigma_t$ until $t_0 = n/\min_{(z_k,z_{k+1})}\mathcal{L}_p$.

For Ξ, whose zeros are $\pm\gamma_j$, and for $H_0(z) = \frac{1}{8}\Xi(z/2)$,

$$\mathcal{L}_{H_0}(z) = \tfrac{1}{4}\mathcal{L}_\Xi(z/2), \qquad \mathcal{L}_\Xi(x) = \sum_j \Big[\frac{1}{(x-\gamma_j)^2} + \frac{1}{(x+\gamma_j)^2}\Big] = \sum_j \frac{2(x^2+\gamma_j^2)}{(x^2-\gamma_j^2)^2}.$$

Both are computed from Ξ and its first two derivatives, without the zeros, and that is how the predictions below are made. Since $\Xi(x) = \Xi(0)G(-x^2)$, with $G(u) = \xi(\frac{1}{2} + \sqrt{u})/\xi(\frac{1}{2})$ as in Section 1.1,

$$\mathcal{L}_\Xi(x) = 2\frac{G'}{G}(-x^2) - 4x^2\Big(\frac{G'}{G}\Big)'(-x^2) = 2\frac{d}{dx}\Big(x\,\frac{G'}{G}(-x^2)\Big),$$
$$\frac{G'}{G}(u) = \sum_j \frac{1}{u+\gamma_j^2},$$

and G'/G is the Stieltjes transform of the Thorin measure $\sum_j \delta_{\gamma_j^2}$ of the clock $1/G$ (under RH; the identities hold in general with γ_j^2 replaced by $a_j = -(\rho_j - \frac{1}{2})^2$). So Biane's gap functional for the zeros of Ξ is the derivative of $2x\,(G'/G)(-x^2)$, the Stieltjes transform of the Thorin measure read on the negative axis.

The functional also governs the backward heat flow, in which zeros collide. The following elementary lemma is one step of it.

Lemma 8.4.2. *Let p be a real polynomial of degree $n \geq 2$ with only real simple zeros and critical points $c_1 < \cdots < c_{n-1}$, and let s be real with $s\,\mathcal{L}_p(c_i) < 1$ for every i. Then $p + sp''$ has only real simple zeros, one in each of the intervals $(-\infty, c_1)$, (c_1, c_2), $\ldots$, (c_{n-1}, ∞).*

Proof. At a critical point $p'(c_i) = 0$, so $p''(c_i)/p(c_i) = (\log|p|)''(c_i) = -\mathcal{L}_p(c_i)$, and $(p + sp'')(c_i) = p(c_i)\big(1 - s\mathcal{L}_p(c_i)\big)$ has the sign of $p(c_i)$. There is exactly one critical point between consecutive zeros of p, so the signs of $p(c_1), \ldots, p(c_{n-1})$ alternate. The polynomial $p + sp''$ has degree n and the sign of p at $\pm\infty$, which is opposite to the sign of $p(c_1)$ at $-\infty$ and to that of $p(c_{n-1})$ at $+\infty$, since p has one zero to the left of c_1 and one to the right of c_{n-1}. So $p + sp''$ changes sign in each of the n intervals and has n real zeros there. $\square$

Since $e^{sD^2} = \lim_N (1 + \frac{s}{N} D^2)^N$, the lemma suggests, without proving it, that the gap (z_k, z_{k+1}) of H_t stays open while $|t| < 1/\mathcal{L}_{H_0}(c_k)$, where c_k is the critical point of H_0 in the gap: the condition would have to hold at every one of the N steps, for polynomials that change. With $z_k = 2\gamma_k$ and u_k the minimiser of $\mathcal{L}_{H_0}$ over the gap, this gives two predictions of the collision time from H_0 alone, the critical point form and Biane's form,

$$\tau_c = -\frac{1}{\mathcal{L}_{H_0}(c_k)} = -\frac{4}{\mathcal{L}_{\Xi}(c_k/2)},$$
$$\tau_B = -\frac{1}{m(0)}, \qquad m(0) = \min_{(z_k, z_{k+1})} \mathcal{L}_{H_0} = \mathcal{L}_{H_0}(u_k).$$

For an isolated pair, $\mathcal{L} = 8/D_0^2$ at the midpoint of the gap $D_0 = z_{k+1} - z_k$, and both equal the isolated pair estimate $-D_0^2/8$ of Section 8.4.1, which is the exact collision time of $z^2 - a^2$ under e^{sD^2}, since $e^{sD^2}(z^2 - a^2) = z^2 - a^2 + 2s$. Section 8.4.5 turns both predictions into theorems for functions with only real zeros.

Remark 8.4.3. Biane's theorem and the collisions concern opposite directions of the heat flow. For polynomials of degree n the finite free convolution of Marcus, Spielman and Srivastava [154] is $p \boxplus_n q = \hat{q}(D)p$ when $q = \hat{q}(D)x^n$, so $e^{-aD^2} p = p \boxplus_n e^{-aD^2} x^n$. The polynomial $e^{-aD^2} x^n = (2a)^{n/2} H_n\big(x/\sqrt{2a}\big)$, where $H_n = e^{-D^2/2} x^n$ is the Hermite polynomial that is the expected characteristic polynomial of a symmetric Gaussian matrix [154], has zeros with mean square $2a(n-1)$, and for large n their distribution is close to the semicircle law σ_{2an}; in the limit finite free convolution becomes free convolution [154]. So $\mu \mapsto \mu \boxplus \sigma_t$ is the limit of the forward flow e^{-aD^2} with $a \approx t/(2n)$, that is of H_t with t increasing: the zeros repel, and a gap between two groups of zeros fills in as the groups spread, which is what Theorem 8.4.1(c) describes. Zeros collide only in the other direction, under e^{sD^2} with $s = -t > 0$. What carries over is the functional. Biane's threshold for μ_p is $n/\min \mathcal{L}_p$, and Theorem 8.4.8 below shows that under e^{sD^2} a gap of a function with only real zeros cannot close before $s = 1/\min \mathcal{L}$ while all the zeros stay real. For the pair $\pm a$ the two times are a^2 and $a^2/2$, and the factor 2 is the n of the normalisation of μ_p.

Table 8.11 compares the two predictions with the collision times of the census of Section 8.4.3 for the 28 collisions among the first 120 gaps at $t > -1.6$. Every collision of the census, all 58 among the first 200 gaps on $[-2.05, 0]$, comes later in the backward flow than both predictions and the isolated pair estimate: $t_{\mathrm{coll}}/\tau_B \geq 1.083$, $t_{\mathrm{coll}}/\tau_c \geq 1.104$ and $t_{\mathrm{coll}}/(-D_0^2/8) \geq 1.056$, all three for $(186, 187)$, which collides at $t = -0.1310$. Over the 28 pairs of the table, t_{coll}/τ_B lies in $[1.114, 1.719]$, t_{coll}/τ_c in $[1.181, 1.951]$ and $t_{\mathrm{coll}}/(-D_0^2/8)$ in $[1.077, 1.518]$, and $|\tau_B|$ falls short of $|t_{\mathrm{coll}}|$ by 25.5% on average (25.2% over the 50 collisions among the first 200 gaps at $t > -1.6$, 28.3% over all 58). The ratio t_{coll}/τ_B is close to 1 when the two zeros dominate their gap, and it is very nearly an increasing function of the share

$6Q(0)/m(0)$ of the other zeros in Biane's functional (Section 8.4.5): ordered by this share, from 0.200 for $(34, 35)$ to 0.703 for $(48, 49)$, the ratios in the table rise from 1.114 to 1.719 with only small inversions, while the gap alone orders them much less well ($(114, 115)$, with $\gamma_{k+1} - \gamma_k = 0.9307$, has ratio 1.346, and $(13, 14)$, with 1.4847, has 1.244). Real zeros outside the gap pull the two zeros apart while they fall together. Over all 58 collisions t_{coll}/τ_B lies in $[1.083, 2.329]$, and the largest ratios, 2.042 to 2.329, belong to the five collisions with $6Q(0)/m(0) > 0.9$, all at $t < -1.89$.

pair	$\gamma_{k+1} - \gamma_k$	t_{coll}	$-D_0^2/8$	τ_c	τ_B	t_{coll}/τ_B	$6Q/m$	frozen
$(91, 92)$	0.7158	−0.2821	−0.2562	−0.2369	−0.2455	1.149	0.250	−0.2826
$(71, 72)$	0.7243	−0.2880	−0.2623	−0.2305	−0.2517	1.144	0.242	−0.2883
$(97, 98)$	0.7370	−0.3040	−0.2716	−0.2454	−0.2587	1.175	0.286	−0.3047
$(63, 64)$	0.8175	−0.3686	−0.3341	−0.3122	−0.3199	1.152	0.256	−0.3694
$(34, 35)$	0.8451	−0.3846	−0.3571	−0.3115	−0.3452	1.114	0.200	−0.3852
$(53, 54)$	0.8717	−0.4231	−0.3800	−0.3361	−0.3624	1.167	0.277	−0.4244
$(114, 115)$	0.9307	−0.5380	−0.4331	−0.3933	−0.3997	1.346	0.462	−0.5364
$(93, 94)$	0.9763	−0.5533	−0.4766	−0.4179	−0.4471	1.237	0.371	−0.5589
$(106, 107)$	0.9650	−0.5691	−0.4656	−0.3620	−0.4306	1.321	0.451	−0.5724
$(75, 76)$	1.0531	−0.6572	−0.5545	−0.4537	−0.5166	1.272	0.410	−0.6648
$(111, 112)$	1.0745	−0.7102	−0.5772	−0.4715	−0.5310	1.338	0.481	−0.7240
$(117, 118)$	1.0571	−0.7653	−0.5588	−0.4318	−0.5029	1.522	0.599	−0.7676
$(78, 79)$	1.1388	−0.8235	−0.6485	−0.5695	−0.5930	1.389	0.513	−0.8318
$(27, 28)$	1.2193	−0.8811	−0.7433	−0.6313	−0.6937	1.270	0.400	−0.8861
$(45, 46)$	1.2588	−0.9600	−0.7923	−0.6731	−0.7325	1.311	0.452	−0.9751
$(82, 83)$	1.2050	−0.9734	−0.7260	−0.6170	−0.6533	1.490	0.601	−0.9985
$(80, 81)$	1.2288	−1.0044	−0.7550	−0.5459	−0.6808	1.475	0.589	−1.0283
$(87, 88)$	1.1991	−1.0343	−0.7190	−0.5805	−0.6390	1.619	0.667	−1.0540
$(100, 101)$	1.2456	−1.0435	−0.7757	−0.5739	−0.6963	1.499	0.614	−1.0797
$(104, 105)$	1.2476	−1.0485	−0.7783	−0.6591	−0.6983	1.502	0.617	−1.0858
$(57, 58)$	1.2524	−1.1004	−0.7842	−0.6484	−0.7031	1.565	0.621	−1.0984
$(24, 25)$	1.3838	−1.1439	−0.9575	−0.7980	−0.8898	1.286	0.424	−1.1578
$(40, 41)$	1.3100	−1.1519	−0.8580	−0.7345	−0.7771	1.482	0.566	−1.1458
$(65, 66)$	1.3427	−1.2447	−0.9014	−0.6381	−0.8066	1.543	0.631	−1.2737
$(19, 20)$	1.4401	−1.2755	−1.0370	−0.7719	−0.9573	1.332	0.461	−1.2837
$(13, 14)$	1.4847	−1.2853	−1.1022	−0.9064	−1.0334	1.244	0.374	−1.2947
$(51, 52)$	1.4218	−1.3190	−1.0107	−0.7548	−0.9126	1.445	0.582	−1.3684
$(48, 49)$	1.3875	−1.4611	−0.9626	−0.7573	−0.8497	1.719	0.703	−1.4679

Table 8.11: The 28 collisions among the first 120 gaps at $t > -1.6$, latest first: the collision time t_{coll} from the census of Section 8.4.3 (`dbn_collisions.py`), against predictions computed from H_0: the isolated pair estimate $-D_0^2/8 = -(\gamma_{k+1} - \gamma_k)^2/2$, the critical point and Biane forms τ_c and τ_B, and the frozen predictor of Section 8.4.5; $6Q/m$ is the share $6Q(0)/m(0)$ of the other zeros in Biane's functional. Times in the units of H_t.

8.4.5 Collision brackets from Laguerre's quantity

Write $s = -t$ for the backward time, so that $f_s = H_{-s}$ solves $\partial_s f_s = f_s''$; for a real polynomial p, $f_s = e^{sD^2}p = \sum_{k\geq 0} s^k p^{(2k)}/k!$ solves the same equation. In this subsection f_s, $s \geq 0$, is a family of real entire functions of genus at most one, smooth in (z, s), with $\partial_s f_s = f_s''$, and $(z_k(s), z_{k+1}(s))$ is a gap between two adjacent simple real zeros, which move continuously until they meet, if they do. On the gap put $q = f_s'/f_s$ and $\mathcal{L} = \mathcal{L}_{f_s} = -q'$. For a point x of the gap write $x_j = 1/(x - z_j)$, where z_j runs over the zeros of f_s, and $S_k = \sum_j x_j^k$ for $k \geq 2$, a real and absolutely convergent sum; then

$$\mathcal{L}(x) = S_2, \qquad \mathcal{L}'(x) = -2S_3, \qquad \mathcal{L}''(x) = 6S_4.$$

When all zeros are real, q falls from $+\infty$ to $-\infty$ across the gap and vanishes at a single critical point $c(s)$; we write $\ell(s) = \mathcal{L}(c(s))$. In all cases $\mathcal{L} \to +\infty$ at both ends of the gap, and we write $m(s)$ for its minimum over the gap, attained at a minimiser $u(s)$, where $\mathcal{L}'(u) = 0$, that is $S_3 = 0$.

Lemma 8.4.4. *On the gap $\partial_s \mathcal{L} = \mathcal{L}'' + 2q\mathcal{L}' - 2\mathcal{L}^2$. If all zeros of f_s are real and simple, then $c(s)$ is differentiable with $\dot{c} = -\mathcal{L}'(c)/\mathcal{L}(c)$, and*

$$\dot{\ell} = 6S_4 - 2S_2^2 - \frac{4S_3^2}{S_2} \qquad \text{at } x = c(s).$$

Proof. With $w = \log f_s$ the heat equation gives $\partial_s w = f_s''/f_s = w'' + w'^2$, hence $\partial_s q = q'' + 2qq'$ and, with $\mathcal{L} = -q'$, $\partial_s \mathcal{L} = -q''' - 2q'^2 - 2qq'' = \mathcal{L}'' - 2\mathcal{L}^2 + 2q\mathcal{L}'$. Since $\partial_z q = -\mathcal{L} < 0$ at c, the implicit function theorem applied to $q(c(s), s) = 0$ gives $\dot{c} = -\partial_s q/\partial_z q = (q'' + 2qq')/\mathcal{L} = -\mathcal{L}'/\mathcal{L}$ at c, where $q = 0$. Hence $\dot{\ell} = \mathcal{L}'\dot{c} + \partial_s \mathcal{L} = \mathcal{L}'' - 2\mathcal{L}^2 - \mathcal{L}'^2/\mathcal{L}$ at c, and $\mathcal{L} = S_2$, $\mathcal{L}' = -2S_3$, $\mathcal{L}'' = 6S_4$. □

Lemma 8.4.5. *For every real square summable sequence (x_j) with power sums $S_k = \sum_j x_j^k$,*

$$E = 3S_2^3 - 6S_2S_4 + 4S_3^2 \geq 0,$$

and for a nonzero sequence equality holds exactly when the nonzero terms are one pair $\{a, -a\}$.

Proof. By homogeneity take $S_2 = 1$. Let $S_k^+ = \sum_{x_j>0} x_j^k$ and $S_k^- = \sum_{x_j<0} |x_j|^k$, and $\alpha = (S_2^+)^{1/2}$, $\beta = (S_2^-)^{1/2}$, so that $\alpha^2 + \beta^2 = 1$, $S_3 = S_3^+ - S_3^-$ and $S_4 = S_4^+ + S_4^-$. Every positive term is at most α, so $S_4^+ \leq \alpha S_3^+$ and $S_3^+ \leq \alpha S_2^+ = \alpha^3$; likewise $S_4^- \leq \beta S_3^-$ and $S_3^- \leq \beta^3$. Therefore $E \geq g(S_3^+, S_3^-)$ with

$$g(p, n) = 3 - 6(\alpha p + \beta n) + 4(p - n)^2 \qquad \text{on the box } [0, \alpha^3] \times [0, \beta^3].$$

The function g is convex, its Hessian being $\begin{pmatrix} 8 & -8 \\ -8 & 8 \end{pmatrix}$, and its gradient $\big(-6\alpha + 8(p-n), -6\beta - 8(p-n)\big)$, whose components add up to $-6(\alpha + \beta) < 0$, never

vanishes; so its minimum over the box lies on the boundary. On the edge $p = 0$, $g = 3 - 6\beta n + 4n^2 \geq 3 - \frac{9}{4}\beta^2 \geq \frac{3}{4}$, and the edge $n = 0$ is symmetric. At the corner (α^3, β^3), using $\alpha^4 + \beta^4 = 1 - 2\alpha^2\beta^2$ and $1 - 4\alpha^2\beta^2 = (\alpha^2 - \beta^2)^2$,

$$g = 4(\alpha^3 - \beta^3)^2 - 3(\alpha^2 - \beta^2)^2 = (\alpha - \beta)^2\big[4(\alpha^2 + \alpha\beta + \beta^2)^2 - 3(\alpha + \beta)^2\big],$$

and with $\sigma = \alpha + \beta$, $\alpha\beta = (\sigma^2 - 1)/2$, the bracket is $(\sigma^2+1)^2 - 3\sigma^2 = \sigma^4 - \sigma^2 + 1 > 0$. On the edge $p = \alpha^3$, g is a convex quadratic in n with vertex at $n = \alpha^3 + \frac{3}{4}\beta$. If the vertex lies at or beyond β^3, the minimum over the edge is at the corner. Otherwise $\beta^3 - \frac{3}{4}\beta > \alpha^3 \geq 0$ forces $\beta^2 > \frac{3}{4}$ and $\alpha < \frac{1}{2}$, and the vertex value is

$$\tfrac{3}{4} + \tfrac{9}{4}\alpha^2 - 6\alpha^4 - 6\alpha^3\beta \geq h(\alpha) = \tfrac{3}{4} + \tfrac{9}{4}\alpha^2 - 6\alpha^4 - 6\alpha^3;$$

$h'(\alpha) = \alpha(\frac{9}{2} - 18\alpha - 24\alpha^2)$, so h increases up to $\alpha \approx 0.198$ and decreases after it, and on $[0, \frac{1}{2}]$ it is at least $\min(h(0), h(\frac{1}{2})) = \min(\frac{3}{4}, \frac{3}{16}) > 0$. The edge $n = \beta^3$ is symmetric. Hence $E \geq 0$. Equality forces the corner with $\alpha = \beta$, and $S_4^+ = \alpha S_3^+$ with $S_3^+ = \alpha^3$, so every positive term equals α and their squares add up to α^2: there is exactly one positive term, α, and likewise exactly one negative term, $-\alpha$. Conversely $E = 0$ for $\{a, -a\}$. □

Theorem 8.4.6 (critical point bound). *If all zeros of f_σ are real and simple for $0 \leq \sigma < s$, then $\dot{\ell} \leq \ell^2$ on $[0, s)$ and*

$$\ell(\sigma) \leq \frac{\ell(0)}{1 - \sigma\ell(0)} \qquad \big(0 \leq \sigma < \min(s, 1/\ell(0))\big).$$

In particular, if p is a real polynomial with only real simple zeros and the first collision of any two zeros of $e^{sD^2}p$ happens at $s = s_1$, between the two zeros bounding the gap (z_k, z_{k+1}) at $s = 0$, then $s_1 \geq 1/\mathcal{L}_p(c_k)$. The bound is attained by $p(z) = z^2 - a^2$.

Proof. By Lemma 8.4.4, $\dot{\ell} \leq \ell^2$ reads $6S_4 - 2S_2^2 - 4S_3^2/S_2 \leq S_2^2$, which is $E \geq 0$ divided by $S_2 > 0$, Lemma 8.4.5. Then $\frac{d}{d\sigma}(1/\ell) = -\dot{\ell}/\ell^2 \geq -1$, so $1/\ell(\sigma) \geq 1/\ell(0) - \sigma$, which is the bound. For the polynomial, all zeros are real and simple on $[0, s_1)$, and $\ell \geq (c - z_k)^{-2} + (z_{k+1} - c)^{-2} \geq 8/g^2$, where g is the length of the gap. If $s_1 < 1/\ell(0)$, then ℓ stays bounded on $[0, s_1)$, so g stays bounded below and the two zeros cannot meet at s_1. For $z^2 - a^2$, $e^{sD^2}p = z^2 - (a^2 - 2s)$, $c = 0$ and $\ell(s) = 2/(a^2 - 2s) = \ell(0)/(1 - s\ell(0))$, with the collision at $s = a^2/2 = 1/\ell(0)$. □

The theorem applies to every real rooted polynomial, in particular to every truncation $\prod_{j \leq J}(1 - z^2/z_j^2)$ of H_0, and says that the first pair of zeros to collide under the backward flow does so no earlier than its own $1/\ell(0)$. For H_t itself the hypothesis fails at every $t < 0$: by the theorem of Rodgers and Tao [224], H_t has nonreal zeros for every $t < 0$ (Section 8.4.1). The minimum of $\mathcal{L}$ over the gap gives the sharper Biane form; at the minimiser $S_3 = 0$, while $q(u)$ need not vanish.

LEMMA 8.4.7. *For every real square summable sequence with $S_3 = 0$, $2S_4 \le S_2^2$, and for a nonzero sequence equality holds exactly when the nonzero terms are one pair $\{a, -a\}$.*

Proof. In the notation of the proof of Lemma 8.4.5, without the normalisation, $S_3 = 0$ means $S_3^+ = S_3^- = \mu$, and for a nonzero sequence both α and β are then positive. Now $\mu \le \min(\alpha^3, \beta^3)$ and $S_4 \le \alpha S_3^+ + \beta S_3^- = (\alpha + \beta)\mu$. Let $\lambda = \min(\alpha, \beta)$ and $\max(\alpha, \beta) = r\lambda$ with $r \ge 1$. Then $S_4 \le (1 + r)\lambda^4$ and $S_2^2 = (1 + r^2)^2\lambda^4$, and $2(1 + r) \le (1 + r^2)^2$ for $r \ge 1$, with equality only at $r = 1$, since both sides equal 4 there and the right side has derivative $4r(1 + r^2) \ge 8 > 2$. Equality in $2S_4 \le S_2^2$ forces $r = 1$, $\mu = \alpha^3$ and $S_4^+ = \alpha S_3^+$, hence one positive term α and one negative term $-\alpha$. □

THEOREM 8.4.8 (Biane form). *If all zeros of f_σ are real and simple for $0 \le \sigma < s$, then the upper right Dini derivative of m satisfies $D^+m \le m^2$ on $[0, s)$, and*

$$m(\sigma) \le \frac{m(0)}{1 - \sigma m(0)} \qquad \big(0 \le \sigma < \min(s, 1/m(0))\big).$$

In particular, if p is a real polynomial with only real simple zeros and the first collision of any two zeros of $e^{sD^2}p$ happens at $s = s_1$, between the two zeros bounding the gap (z_k, z_{k+1}) at $s = 0$, then $s_1 \ge 1/\min_{(z_k, z_{k+1})} \mathcal{L}_p$. This bound on s_1 contains that of Theorem 8.4.6, since $m(0) \le \ell(0)$, and it is attained by $z^2 - a^2$.

Proof. Since $\mathcal{L} \to +\infty$ at both ends of the gap, the minimisers are interior and m is continuous. If u is a minimiser at time σ, then $m(\sigma + h) \le \mathcal{L}(u, \sigma + h)$ for small $h > 0$, so by Lemma 8.4.4, with $\mathcal{L}'(u) = 0$,

$$D^+m(\sigma) \le \partial_\sigma \mathcal{L}(u, \sigma) = \mathcal{L}''(u) - 2\mathcal{L}(u)^2 = 6S_4 - 2S_2^2 \le S_2^2 = m(\sigma)^2,$$

the last step by Lemma 8.4.7, as $S_3 = 0$ at u. The lower right Dini derivative of the continuous function $1/m(\sigma) + \sigma$ is therefore nonnegative, so the function does not decrease, and $1/m(\sigma) \ge 1/m(0) - \sigma$. A collision forces $m \ge 8/g^2 \to \infty$, as in the proof of Theorem 8.4.6. For $z^2 - a^2$ the minimiser is the critical point 0, so $m = \ell$. □

Both inequalities and both theorems were also tested numerically. For the inequalities, `wigbiane_ineq.py` minimised E/S_2^3 over real sequences of length 2 to 20 by the BFGS method from 2000 random starts for each length, and again, by sequential quadratic programming, from 1000 starts under the constraint $S_1 = \sum_j x_j = 0$, which holds at the critical point c of a polynomial; it maximised $2S_4/S_2^2$ under the constraint $S_3 = 0$ in the same way from 1000 starts for each length. Re-evaluated at 50 digits, every least value of E/S_2^3 was nonnegative, the smallest being 1.5×10^{-31}, and every largest value of $2S_4/S_2^2$ was 1 to 15 digits; in both searches the extremal sequences were pairs $\{a, -a\}$ up to rounding, the

other terms of the minimisers of E/S_2^3 being below 7×10^{-9} for $S_2 = 1$. The program `wigbiane_poly.py` draws 2400 polynomials of degree 3 to 12 with only real zeros, 600 in each of four families (zeros uniform on an interval; Gaussian; uniform with one added zero at 2 to 30% of the mean spacing from another, a Lehmer type pair; and a perturbed lattice), follows the zeros of $e^{sD^2}p$ until at most one real zero is left, and locates each collision as a double zero by Newton's method in (z, s) at 40 digits. For the 2398 polynomials it completed (two failed in the location of a collision), the first collision satisfied $s_1\ell(0) \geq 1.0000136$ and $s_1 m(0) \geq 1.0000055$, both for a Lehmer type pair, with medians 1.061 and 1.039. The bounds of Theorems 8.4.6 and 8.4.8 on $\ell(s)$ and $m(s)$ held at every step before the first collision on every gap. Near a collision the double precision zeros are not accurate enough to test them at the level of 10^{-5}, so wherever a ratio $\ell(s)(1 - s\ell(0))/\ell(0)$ or $m(s)(1 - sm(0))/m(0)$ came within 10^{-4} of 1 (at the 20 closest steps of each polynomial), the zeros were refined at 40 digits; the largest ratios were then $1 - 1.2 \times 10^{-7}$ and $1 - 8 \times 10^{-8}$. At $s = 0$ the derivatives of ℓ and m, by central differences at 40 digits, agree with Lemma 8.4.4 to 18 digits.

The same polynomials say something beyond the theorems. After the first collision conjugate pairs are present, and Theorem 8.4.8 no longer applies; yet each of the 4574 later collisions of two zeros adjacent at $s = 0$ satisfied $s_c m(0) \geq 1.0000326$, although the quantity δ of Corollary 8.4.12 below was positive at some step before 806 of them.

For H_t the hypothesis of Theorem 8.4.8 fails, because pairs at other heights have already left the axis. The argument survives with a correction. At the minimiser u, replace every nonreal pair $a \pm ib$ by the double real point a, its shadow.

LEMMA 8.4.9. *(i) For every real square summable sequence,* $S_4 \leq \frac{1}{2}S_2^2 + S_2^{1/2}|S_3|$.

(ii) If $w = 1/(d - ib)$ *with* $d \neq 0$ *and* b *real, then* $\left|2 \operatorname{Re} w^k - 2d^{-k}\right| \leq k(k+1)\, b^2/|d|^{k+2}$ *for every* $k \geq 1$.

Proof. (i) Replacing the sequence by its negative if necessary, let $S_3 = \varepsilon \geq 0$, so $S_3^+ = S_3^- + \varepsilon$ in the notation of the proof of Lemma 8.4.5. Then $S_4 \leq \alpha S_3^+ + \beta S_3^- = (\alpha + \beta)S_3^- + \alpha\varepsilon$, where $S_3^- \leq \beta^3$ and $S_3^- = S_3^+ - \varepsilon \leq \alpha^3$. As in the proof of Lemma 8.4.7, $(\alpha + \beta)\min(\alpha, \beta)^3 \leq \frac{1}{2}(\alpha^2 + \beta^2)^2 = \frac{1}{2}S_2^2$, and $\alpha\varepsilon \leq S_2^{1/2}|S_3|$. (ii) With $\epsilon = b/d$ and $\theta = \arctan\epsilon$, $w^k = d^{-k}(1 - i\epsilon)^{-k}$ and $\operatorname{Re}(1 - i\epsilon)^{-k} = A\cos k\theta$ with $A = (1 + \epsilon^2)^{-k/2}$. Now $1 - A\cos k\theta = (1 - A) + A(1 - \cos k\theta)$, where $0 \leq 1 - A \leq \frac{k}{2}\epsilon^2$ and $0 \leq 1 - \cos k\theta \leq \frac{1}{2}k^2\theta^2 \leq \frac{1}{2}k^2\epsilon^2$, so $|\operatorname{Re}(1 - i\epsilon)^{-k} - 1| \leq \frac{1}{2}(k + k^2)\epsilon^2$; multiply by $2|d|^{-k}$. □

REMARK 8.4.10. The coefficient of the second term in (i) can be halved, and $\frac{1}{2}$ is then best possible, since a single nonzero term gives $S_4 - \frac{1}{2}S_2^2 = \frac{1}{2}S_2^{1/2}|S_3|$.

Normalise $S_2 = 1$ and $S_3 = \varepsilon \geq 0$ and put $\mu = S_3^-$, so that $S_4 \leq \alpha(\mu+\varepsilon) + \beta\mu$ with $0 \leq \mu \leq \min(\beta^3, \alpha^3 - \varepsilon)$; the bound increases with μ. For $\varepsilon \leq \alpha^3 - \beta^3$ it is at most $(\alpha+\beta)\beta^3 + \alpha\varepsilon$, which is linear in ε, and for $\varepsilon \geq \max(0, \alpha^3 - \beta^3)$ it is at most $\alpha^4 + \alpha^3\beta - \beta\varepsilon$, which decreases. So the bound minus $\frac{1}{2}(1+\varepsilon)$ is largest at $\varepsilon = 0$ or, when $\alpha > \beta$, at $\varepsilon = \alpha^3 - \beta^3$. At $\varepsilon = 0$ the bound is $(\alpha+\beta)\min(\alpha,\beta)^3 \leq \frac{1}{2}$, as in the proof of Lemma 8.4.7. At $\varepsilon = \alpha^3 - \beta^3$ it is $\alpha^4 + \beta^4 = 1 - 2\alpha^2\beta^2$, and $\alpha^4 + \beta^4 - \frac{1}{2} \leq \frac{1}{2}(\alpha^3 - \beta^3)$ reads $(\alpha^2 - \beta^2)^2 \leq \alpha^3 - \beta^3$, that is $(\alpha-\beta)(\alpha+\beta)^2 \leq 1 + \alpha\beta$; with $\nu = 2\alpha\beta \in [0,1]$ this is $(1-\nu)^{1/2}(1+\nu) \leq 1 + \frac{\nu}{2}$, which after squaring becomes $-\nu^2 - \nu^3 \leq \frac{1}{4}\nu^2$. Hence

$$S_4 \leq \tfrac{1}{2}S_2^2 + \tfrac{1}{2}S_2^{1/2}|S_3|,$$

and the coefficient 72 in Theorem 8.4.11 below may be replaced by 36; we keep the form of [208], which the computations below use. A search of `wigbiane_ineq.py` (the Nelder and Mead method from 200 random starts for each length 2 to 20) found no value of $(S_4 - \frac{1}{2}S_2^2)/(S_2^{1/2}|S_3|)$ above $\frac{1}{2}$. The bound (ii) is sharp as $b/d \to 0$, since $\mathrm{Re}(1 - i\epsilon)^{-k} = 1 - \frac{1}{2}k(k+1)\epsilon^2 + O(\epsilon^4)$: at 40 digits, on 4001 values of ϵ from 10^{-6} to 10^4 and for $k \leq 8$, the ratio of the left side of (ii) to the right stays below 1 and comes within 10^{-11} of it at $\epsilon = 10^{-6}$.

THEOREM 8.4.11 (local bound). *Let u be a minimiser of $\mathcal{L}$ over the gap at time s, with $m = \mathcal{L}(u) > 0$, and let $B_p = \sum b^2|u-a|^{-p}$, summed over the conjugate pairs $a \pm ib$, $b > 0$, of nonreal zeros of f_s, each pair once, be finite for $p = 4, 5, 6$. Then*

$$\mathcal{L}''(u) - 3m^2 \leq \eta\, m^2, \qquad \eta = \frac{36\, mB_4 + 108\, B_4^2 + 72\,(m + 6B_4)^{1/2} B_5 + 120\, B_6}{m^2},$$

and $D^+ m \leq (1+\eta)m^2$. Hence, if the two zeros bounding the gap meet at s_c, and $m > 0$ and $\eta \leq \bar{\eta}$ on $[0, s_c)$, then $s_c \geq 1/\big((1+\bar{\eta})m(0)\big)$.

Proof. Let S_k' be the power sums of the real sequence in which every nonreal pair $a \pm ib$ is replaced by its shadow, two terms $1/(u-a)$, and let $D_k = S_k - S_k'$. By Lemma 8.4.9(ii) with $d = u - a$, $|D_2| \leq 6B_4$, $|D_3| \leq 12B_5$ and $|D_4| \leq 20B_6$. Since $S_3 = 0$ at u, $S_3' = -D_3$, and Lemma 8.4.9(i) for the shadow sequence gives $S_4' \leq \frac{1}{2}S_2'^2 + S_2'^{1/2}|D_3|$. With $S_4 = S_4' + D_4$, $0 \leq S_2' = m - D_2 \leq m + 6B_4$ and $S_2'^2 - m^2 = -2mD_2 + D_2^2$,

$$\begin{aligned} 6S_4 - 3m^2 &\leq 3(S_2'^2 - m^2) + 6S_2'^{1/2}|D_3| + 6|D_4| \\ &\leq 6m|D_2| + 3D_2^2 + 6S_2'^{1/2}|D_3| + 6|D_4| \leq \eta m^2, \end{aligned}$$

and $\mathcal{L}''(u) = 6S_4$. As in the proof of Theorem 8.4.8, $D^+ m \leq \mathcal{L}''(u) - 2m^2 \leq (1+\eta)m^2$, so the function $1/m(\sigma) + (1+\bar{\eta})\sigma$ does not decrease on $[0, s_c)$. At the collision $m \to \infty$: the share of the two meeting zeros tends to $+\infty$, while that of the other zeros, which stay away from the point where the two meet, remains

bounded. No nonreal zero tends to that point: if z_c is a zero of multiplicity k of f_{s_c}, then $f_{s_c-\sigma} = e^{-\sigma D^2} f_{s_c}$, so $\sigma^{-k/2} f_{s_c-\sigma}(z_c + \sqrt{\sigma}\, w)$ tends to a nonzero multiple of $H_k(w/\sqrt{2})$ as $\sigma \downarrow 0$, uniformly on compact sets, and by Hurwitz's theorem the k zeros of $f_{s_c-\sigma}$ near z_c lie close to $\sqrt{2\sigma}$ times the zeros of H_k, which are real and simple; so they are real, and real zeros only add to m. □

The proof contains a form of the correction that needs no knowledge of the nonreal zeros.

Corollary 8.4.12. *In the setting of Theorem 8.4.11 put $\delta = \mathcal{L}''(u)/m^2 - 3$, computed from f_s and its first four derivatives at u. Then $D^+ m \le (1+\delta)m^2$ and $\delta \le \eta$, and $\delta \le 0$ when all zeros of f_s are real. Hence, if the two zeros bounding the gap meet at s_c, and $m > 0$ and $\delta \le \bar\delta$ on $[0, s_c)$ with $\bar\delta > -1$, then $s_c \ge 1/\big((1+\bar\delta)m(0)\big)$; if $\delta \le 0$ throughout, the collision cannot come before Biane's time $1/m(0)$, exactly as for functions with only real zeros.*

Proof. The first display of the proof of Theorem 8.4.8 gives $D^+ m \le \mathcal{L}''(u) - 2m^2 = (1+\delta)m^2$, Theorem 8.4.11 gives $\delta \le \eta$, and Lemma 8.4.7 gives $\delta \le 0$ for real zeros. The rest is the comparison argument of the proof of Theorem 8.4.11. □

The same calculation bounds the collision time from above when the two zeros dominate their gap. Write $m = P + Q$, where $P = (u - z_k)^{-2} + (u - z_{k+1})^{-2}$ is the share of the two zeros bounding the gap and Q the share of all the others, and let $W = \sum |u - z|^{-4}$ over the nonreal zeros z of f_s.

Theorem 8.4.13 (a deadline). *While the two zeros bounding the gap are real, the right derivative $m'(s^+)$ exists and is the least value of $\mathcal{L}''(u) - 2m^2$ over the minimisers u, and at every minimiser*

$$\mathcal{L}''(u) - 2m^2 \ge m^2 - 6mQ + 3Q^2 - 6W,$$

with Q and W taken at u. Suppose that $Q \le \rho$ and $W \le \bar W$ at every minimiser for as long as the two zeros bounding the gap are real, and that $m(0) > r_1$, where $r_{1,2} = 3\rho \pm (9\rho^2 + 6\bar W)^{1/2}$. Then the gap closes, one of the two zeros bounding it meeting another zero, at a time

$$s_c \le s^* = \frac{1}{r_1 - r_2} \log \frac{m(0) - r_2}{m(0) - r_1}.$$

If all zeros are real, $\bar W = 0$ and, for $\rho > 0$, $s^* = (6\rho)^{-1} \log\big(m(0)/(m(0) - 6\rho)\big) = 1/m(0) + 3\rho/m(0)^2 + 12\rho^2/m(0)^3 + \cdots$.

Proof. Near a given time the minimisers lie in a fixed compact subinterval of the gap, and by Danskin's theorem the right derivative of the minimum exists and equals the least value of $\partial_s \mathcal{L}$ over the minimisers, which is $\mathcal{L}''(u) - 2m^2 = 6S_4 - 2m^2$ by Lemma 8.4.4. In S_4 the two zeros bounding the gap contribute $x_k^4 + x_{k+1}^4 \ge$

$\frac{1}{2}(x_k^2 + x_{k+1}^2)^2 = \frac{1}{2}P^2$, the other real zeros contribute nonnegative terms, and each nonreal zero z contributes $\mathrm{Re}(u-z)^{-4} \geq -|u-z|^{-4}$. So $\mathcal{L}''(u) - 2m^2 \geq 3P^2 - 6W - 2m^2$, and $P = m - Q$ gives the first inequality. While $m > 0$, $Q \leq \rho$ and $W \leq \bar{W}$, it gives $m'(s^+) \geq m^2 - 6\rho m - 6\bar{W} = (m - r_1)(m - r_2)$, so m stays above the solution y of $y' = (y-r_1)(y-r_2)$, $y(0) = m(0) > r_1 \geq 0$, which increases and tends to $+\infty$ as $s \to s^*$. If the gap were still open on $[0, s^*]$, its two zeros real, simple and apart, then m, being at most $\mathcal{L}$ at the midpoint of the gap, would be bounded on $[0, s^*]$, a contradiction. The first collision of either zero need not be with the other: one of them may first meet the zero beyond it, and the theorem then bounds the time of that collision. For a pair that is known to collide, as in the census, s^* bounds the collision time of the pair. The expansion is that of $x^{-1}\log\frac{1}{1-x} = 1 + \frac{x}{2} + \frac{x^2}{3} + \cdots$ at $x = 6\rho/m(0)$. □

Corollary 8.4.14. *Put $\kappa = \mathcal{L}''(u) - 3P^2$, computed from f_s at u and the two zeros bounding the gap, so that*

$$\kappa = 3\big(x_k^2 - x_{k+1}^2\big)^2 + 6\sum_{j \neq k,k+1} x_j^4 \geq -6W,$$

with $\kappa \geq 0$ when all zeros are real. Theorem 8.4.13 holds with $6\bar{W}$ replaced by $\bar{\kappa} = \max\big(0, \sup(-\kappa)\big)$, the supremum over the minimisers at the times before the collision, and this form is never weaker.

Proof. The identity is $6(x_k^4 + x_{k+1}^4) - 3(x_k^2 + x_{k+1}^2)^2 = 3(x_k^2 - x_{k+1}^2)^2$, and the inequality is that of the proof of Theorem 8.4.13. That proof gives $m'(s^+) = \kappa + 3P^2 - 2m^2 \geq m^2 - 6mQ + 3Q^2 - \bar{\kappa}$, and $\bar{\kappa} \leq 6\bar{W}$. □

The two inequalities behind Theorems 8.4.11 and 8.4.13 were tested on random configurations as well. For 29620 configurations of 3 to 11 real zeros and 1 to 7 conjugate pairs, at the critical point u of $\mathcal{L}$ on a random gap that Newton's method reaches from the least value of $\mathcal{L}$ on a grid, with $\mathcal{L}(u) > 0$, the quantity $6S_4 - 3S_2^2$ never exceeded $0.016\,\eta S_2^2$, and $6S_4 \geq 3P^2 - 6W$ held up to rounding (`wigbiane_ineq.py`). On these samples the correction η is thus far from sharp. On the polynomials of `wigbiane_poly.py`, with ρ the largest value of Q at the minimisers before the collision, the deadline s^* of Theorem 8.4.13 is defined for 2308 of the 2398 first collisions, with $\bar{W} = 0$, and for 3322 of the 4574 later ones (3345 in the form of Corollary 8.4.14), and no collision came after it: the largest ratio s_c/s^* is $1 - 1.5 \times 10^{-11}$ for the first collisions and $1 - 3.1 \times 10^{-10}$ for the later ones.

Corollaries 8.4.12 and 8.4.14 bracket every collision time by quantities along the flow that need only H_t near the gap, while Theorems 8.4.11 and 8.4.13 need the nonreal zeros. The program `wigbiane_flow.py` (Appendix A) computes both kinds for each of the 58 collisions of the census of Section 8.4.3. It evaluates H_t and its first four z derivatives by the trapezoid rule with step $1/360$ on $[0, 1.35]$ in ball

arithmetic at 800 bits (the aliasing terms are below $10^{-100}|H_t(z)|$ for $|\operatorname{Re} z| \le 830$, and the integrand beyond 1.35 is below 10^{-290}), refines the double zero $H_t = \partial_z H_t = 0$ by Newton's method in (z,t), and on a grid of about 140 times s in $[0, |t_{\mathrm{coll}}|)$, uniform up to $0.95|t_{\mathrm{coll}}|$ and geometric towards $|t_{\mathrm{coll}}|$ (the last point $2 \times 10^{-7}|t_{\mathrm{coll}}|$ before it), with the multiples of 0.1 added, it follows the two zeros by Newton's method and computes the minimiser u, the critical point c, and from the logarithmic derivatives of H_t at u and c the quantities ℓ, m, Q, δ and κ, the last three in ball arithmetic, since near the collision m and P are large and nearly equal. For η and W it also follows, by Newton's method in the complex plane, the conjugate pairs born in the census collisions before the time considered (60 pairs, two of them from collisions among the zeros 203 to 206), with their mirror images $-a \pm ib$, and adds a model of the zeros above the census, $z = 2\gamma_j$ for $207 \le j \le 2000$, in which every such zero is half of a pair $a \pm ib$ with $b^2 = 4|t|$. Newton's method reproduces the census collision times to 5×10^{-12}, and at $s = 0$ central differences in s agree with Lemma 8.4.4 to 14 digits. The results are these.

First, $\delta < 0$ at every point of the grid on all 58 trajectories; the largest value is -3.1×10^{-8}, for $(186, 187)$. So the hypothesis of Corollary 8.4.12 with $\bar{\delta} = 0$ holds on the grid in spite of the nonreal zeros (49 conjugate pairs of the census alone are off the axis when $(13, 14)$ collides), and its conclusion, the lower bound $1/m(0)$, holds for all 58 collisions of the census, as the direct comparison of Section 8.4.4 shows ($t_{\mathrm{coll}}/\tau_B \ge 1.083$; Table 8.11). Likewise $\ell(s) \le \ell(0)/(1 - s\ell(0))$ for $s < 1/\ell(0)$ and $m(s) \le m(0)/(1 - sm(0))$ for $s < 1/m(0)$ at every point of the grid; Table 8.12 shows two of the trajectories.

Second, $\kappa \ge 0$ at every point of the grid except on the trajectories of $(108, 109)$ and $(153, 154)$, where $\bar{\kappa} = 0.029$ and 0.071. The upper bound s^* of Corollary 8.4.14, with ρ the largest value of Q on the grid, is defined, $m(0) > r_1$, for 55 of the 58 collisions; it is not for $(153, 154)$, $(178, 179)$ and $(181, 182)$, for which $6\rho/m(0)$ is 0.986, 1.068 and 1.103. Every collision time of the census lies in its bracket $[1/m(0), s^*]$, with $s^* = \infty$ for those three, and Table 8.13 shows the seven pairs of the note [208]. The upper bound is tight when the pair dominates its gap: $s^*/|t_{\mathrm{coll}}|$ ranges from 1.0010, for $(186, 187)$ with $6\rho/m(0) = 0.151$, to 2.021, for $(48, 49)$ with $6\rho/m(0) = 0.965$.

Third, the correction η of Theorem 8.4.11, computed from the tracked pairs and the model above the census, is at most 3.25×10^{-5} for the seven pairs of Table 8.13, but it reaches 0.53 for $(153, 154)$ and exceeds 10^{-2} for four more collisions, all at $t < -1.89$. With $\bar{\eta}$ in place of $\bar{\delta}$ the lower bound still holds for all 58, the closest case being again $(186, 187)$, whose collision time exceeds the bound by the factor 1.083. For $(34, 35)$, $(71, 72)$, $(91, 92)$ and five other late collisions more than half of B_4 at the worst point comes from the model, so $\bar{\eta}$ depends on it, while δ and κ depend on no model. The model is not pessimistic in every respect: the tracked pairs leave the axis faster than an isolated pair would, with b^2 up to 3.39 times $2(t_c - t)$.

$s=-t$	0	0.2	0.4	0.6	0.8	1.0	1.2
$(13,14)$: $\ell(s)$	1.1033	1.2462	1.4539	1.7833	2.3846	3.8293	12.051
$\ell(0)/(1-s\ell(0))$	1.1033	1.4156	1.9748	3.2638	9.3992		
$m(s)$	0.9676	1.1099	1.3170	1.6457	2.2461	3.6898	11.911
$m(0)/(1-sm(0))$	0.9676	1.1998	1.5787	2.3071	4.2837	29.903	
$(19,20)$: $\ell(s)$	1.2955	1.4520	1.6768	2.0293	2.6688	4.2153	13.848
$\ell(0)/(1-s\ell(0))$	1.2955	1.7485	2.6889	5.8171			
$m(s)$	1.0446	1.1904	1.4029	1.7413	2.3644	3.8914	13.500
$m(0)/(1-sm(0))$	1.0446	1.3205	1.7944	2.7990	6.3583		

Table 8.12: Laguerre's quantity at the critical point, $\ell(s)$, and its minimum over the gap, $m(s)$, along the backward flow for the pairs $(13,14)$ and $(19,20)$, which collide at $s=1.2853$ and 1.2755, against the bounds of Theorems 8.4.6 and 8.4.8; a bound is shown only where it is finite, for $s<1/\ell(0)$ (0.9064 and 0.7719) or $s<1/m(0)$ (1.0334 and 0.9573). Units of H_t; `wigbiane_flow.py`.

pair	$m(0)$	$Q(0)$	ρ	$10^8\bar\delta$	$\bar\eta$	pairs	$1/m(0)$	s^*	$\lvert t_{\rm coll}\rvert$
$(13,14)$	0.9676	0.0604	0.0610	-9.0	2.8×10^{-7}	49	1.0334	1.2982	1.2853
$(19,20)$	1.0446	0.0803	0.0879	-12.8	2.4×10^{-6}	48	0.9573	1.3325	1.2755
$(24,25)$	1.1238	0.0794	0.0794	-10.2	3.2×10^{-5}	44	0.8898	1.1578	1.1439
$(27,28)$	1.4415	0.0962	0.1012	-10.2	4.0×10^{-6}	36	0.6937	0.9006	0.8811
$(34,35)$	2.8970	0.0968	0.0969	-4.3	1.7×10^{-9}	11	0.3452	0.3853	0.3846
$(71,72)$	3.9727	0.1605	0.1678	-5.5	2.0×10^{-9}	4	0.2517	0.2903	0.2880
$(91,92)$	4.0736	0.1701	0.1757	-5.7	3.1×10^{-9}	3	0.2455	0.2841	0.2821

Table 8.13: Two sided brackets for the collision time of seven pairs, in the units of H_t: the lower bound $1/m(0)$ of Corollary 8.4.12, whose hypothesis holds on the grid, $\bar\delta<0$, and the upper bound s^* of Corollary 8.4.14, with ρ the largest value of Q and $\bar\kappa=0$ for these pairs. The correction $\bar\eta$ of Theorem 8.4.11 comes from the tracked pairs and the model above the census (almost entirely from the model for the last three pairs), and "pairs" is the number of conjugate pairs of the census with positive real part that are off the axis at the collision. `wigbiane_flow.py`.

Freezing Q at its value at $s=0$, that is, taking $\rho=Q(0)$ and $\bar\kappa=0$ in Corollary 8.4.14, gives a predictor of the collision time from H_0 alone,

$$t_{\rm coll}\approx-\frac{1}{6Q(0)}\log\frac{m(0)}{m(0)-6Q(0)},$$

the last column of Table 8.11. It is not a bound, since Q grows along the flow ($\rho>Q(0)$ for 41 of the 58 collisions) and the predictor falls on both sides of $t_{\rm coll}$, but it is accurate when the pair dominates its gap. Over the 28 collisions of Table 8.11 its relative error is at most 0.32% for the six with $6Q(0)/m(0)<0.3$, 1.01% for the eight with $6Q(0)/m(0)<0.4$, 1.93% for the sixteen with $6Q(0)/m(0)<0.5$, and 3.75% in all, for $(51,52)$; over the 52 collisions of the census with $6Q(0)/m(0)<0.8$

it is at most 5.75%, for $(130, 131)$. It fails when the other zeros carry most of $m(0)$: the six collisions with $6Q(0)/m(0) \geq 0.8$, all at $t < -1.87$, have errors from 10.4%, for $(68, 69)$, to 46.6%, for $(188, 189)$. To first order in $Q(0)$ the predictor is $1/m(0) + 3Q(0)/m(0)^2$ in absolute value, and if the two zeros sit symmetrically about u, so that $P(0) = 8/D_0^2$, it is

$$\frac{D_0^2}{8} + \frac{D_0^4 Q(0)}{32} + O\big(Q(0)^2\big).$$

The bound Λ_k of Csordas, Smith and Varga [62] in Section 8.4.6 has the expansion $-\Delta_k^2/8 - \Delta_k^4 g_k/64 - \cdots$ with $\Delta_k = D_0$: the same two terms with g_k in place of $2Q(0)$, and indeed $g_k \approx 2Q(0)$, since g_k adds up, over the other zeros, the inverse squares of their distances to each of the two zeros of the pair. For $(71, 72)$, $g_{71} = 0.33734$ and $2Q(0) = 0.3210$, and the frozen predictor gives -0.2883, against $t_{\mathrm{coll}} = -0.2880$ and $\Lambda_{71} = -0.3048$.

Both brackets rest on hypotheses along the flow, on δ or η for the lower bound and on Q and κ or W for the upper bound, which the computation checks on the grid of times but which cannot yet be guaranteed in advance.

Problem 8.1. Find a bound for how far the nonreal zeros of H_t, $t < 0$, travel, and for how close to a gap between real zeros they come. Such control would make the brackets unconditional statements about H_t, computed from H_0 alone.

Biane's work meets ξ in two more places. With Pitman and Yor [28], following Williams [271], he wrote $2\xi(s)$ as the Mellin transform of a sum of two Bessel hitting times, the identity behind the Pólya variable of Section 1.2. And in [27] he revisits the paper of Pólya [213] that introduced an approximation ξ^* of ξ with only real zeros. Section 8.7 treats this paper in full: the zeros of $K_{iz}(a)$ as a Dirichlet spectrum, the two hitting times whose Mellin transforms give ξ and the Macdonald function, and their Thorin measures as spectral measures of Brownian motion on $SU(2)$ and on $SL_2(\mathbb{C})/SU(2)$.

8.4.6 Lehmer pairs and lower bounds for Λ

Every collision time is a lower bound for Λ, because nonreal zeros persist backward in time, and Lehmer pairs, pairs with unusually small D_0, collide at times close to zero. Csordas, Smith and Varga [62] turned this into a theorem that uses only the zeros of H_0. Write $x_j = 2\gamma_j$ for the positive zeros of H_0, $x_{-j} = -x_j$, $\Delta_k = x_{k+1} - x_k$ and

$$g_k = \sum_{j \neq k, k+1}{}' \Big[\frac{1}{(x_k - x_j)^2} + \frac{1}{(x_{k+1} - x_j)^2} \Big],$$

where the prime restricts the sum to nonzero integers j, of either sign. The pair (x_k, x_{k+1}) is called a Lehmer pair if $\Delta_k^2 g_k < \frac{4}{5}$, and then [62, Theorem 1]

$$\Lambda \geq \Lambda_k = \frac{\left(1 - \frac{5}{4}\Delta_k^2 g_k\right)^{4/5} - 1}{8 g_k} = -\frac{\Delta_k^2}{8} - \frac{\Delta_k^4 g_k}{64} - \cdots .$$

The leading term is the isolated pair collision time $-D_0^2/8$ with $D_0 = \Delta_k$. The binomial series of $(1-x)^{4/5}$ has only negative coefficients after the constant term, so every further term is negative and decreases in g_k, and so does Λ_k.

For the closest pair below height 200, $\Delta_{71} = 1.44863$ and, from the first 1700 zeros, $g_{71} = 0.33734$, so $\Delta^2 g = 0.708 < \frac{4}{5}$: the pair is a Lehmer pair in this sense, and it gives $\Lambda_{71} = -0.3048$. The omitted zeros above height 2197 increase g by less than 10^{-3}, which lowers Λ_{71} by less than 3×10^{-4}, so with the tail the bound is -0.305. The theorem is proved, but this evaluation of it is a floating point computation with zeros computed by `mpmath` [159]. The bound lies below the computed collision time -0.288, as expected, and below the leading term -0.262 by the correction terms, which are not small for a pair of ordinary spacing. The theorem itself does not order the bound and the collision time: its proof, by contradiction, assumes that H_t has only real zeros for $\Lambda_k \leq t \leq 0$, which fails for every $t < 0$ because $\Lambda \geq 0$.

Table 8.14 lists the published bounds. The last four lower bounds before 2020, among them that of Csordas, Odlyzko, Smith and Varga [61], came from this theorem applied to Lehmer pairs of increasingly high quality, found at heights far greater than 7005, and for such close pairs the correction terms are negligible. The bounds approached zero from below until Rodgers and Tao proved $\Lambda \geq 0$ by a different argument, showing that $\Lambda < 0$ would force the zeros into a local equilibrium incompatible with Montgomery's pair correlation estimates [158].

8.5 Beyond zeta: the Dedekind eta kernel, $L(s,\chi_{12})$ and Ramanujan's Δ

Apart from the sums over zeros computed by `mpmath`, the computations of Sections 8.3 and 8.4 use ζ only through the Pólya kernel, and the Pólya route uses little that is special to ζ. Its evaluator (Section 2.1), which carries most of the computations up to the de Bruijn Newman flow, needs a kernel that is even, analytic in the strip $|\operatorname{Im} u| < \pi/4$ and doubly exponentially decaying; the zero density identity of Section 3.4 needs a completed function that is also positive on the real axis and zero free on $\operatorname{Re} s = 1$; the sine squared decomposition (3.1.1) needs an Euler product. The completed L functions of the real character χ_{12} modulo 12 and of Ramanujan's cusp form Δ have all these properties, and their kernels, built from the Dedekind eta function, are positive by the product formula. From the eta kernel alone, the zero density identity and a count of sign changes place every zero of $L(s,\chi_{12})$ below height 100 on the critical line, which is the generalised

bound	authors	year
$\Lambda > -\infty$	Newman	1976
$\Lambda > -50$	Csordas, Norfolk, Varga	1988
$\Lambda > -5$	te Riele	1991
$\Lambda > -0.385$	Norfolk, Ruttan, Varga	1992
$\Lambda > -0.0991$	Csordas, Ruttan, Varga	1991
$\Lambda > -4.379 \times 10^{-6}$	Csordas, Smith, Varga	1994
$\Lambda > -5.895 \times 10^{-9}$	Csordas, Odlyzko, Smith, Varga	1993
$\Lambda \geq -2.63 \times 10^{-9}$	Odlyzko	2000
$\Lambda > -1.15 \times 10^{-11}$	Saouter, Gourdon, Demichel	2011
$\Lambda \geq 0$	Rodgers, Tao	2020
$\Lambda \leq \frac{1}{2}$	de Bruijn	1950
$\Lambda < \frac{1}{2}$	Ki, Kim, Lee	2009
$\Lambda \leq 0.22$	Polymath	2019
$\Lambda \leq 0.2$	Platt, Trudgian	2021

Table 8.14: Bounds on the de Bruijn Newman constant, ordered by the bound: lower bounds from [224, Table 1], upper bounds from [65], [125], [217] and [196]. Odlyzko's bound is the one computed in [174, Section 3], which that paper states in rounded form as $\Lambda > -2.7 \times 10^{-9}$. The years are years of publication.

Riemann hypothesis (GRH) for χ_{12} on that window, and per zero $L(s, \chi_{12})$ is the cheapest of the three functions to compute with. The sine squared decomposition loses its pole piece, but its prime weights take both signs, so the negative input that ζ concentrates in the pole is spread over the primes. The first author's note on the Thorin measures of the Dedekind eta function asks what the eta function says about GRH for χ_{12} [203, Theorem 12]. Section 11.3 returns to $L(s, \chi_{12})$ at the basepoint one, where its Thorin measure is positive with no hypothesis, its Lévy form converges without the compensation that the pole of ζ needs, and the law of its clock is again that of the first passage of Brownian motion across an exact boundary. Section 8.5.4 reads the product formula as the law of a maximum of exponential variables, which represents $L(s, \Delta)$ with no signs, gives explicit laws in the Bondesson class that are not GGCs, and shows that the law of M^* itself is not in that class, and Section 8.5.5 follows the de Bruijn Newman flow of Δ back to its first collisions. Section 8.5.7 carries the framework over to the Dirichlet beta function $L(s, \chi_4)$, whose kernel is the cube of the eta function: GRH for χ_4 holds below height 100, the Thorin density is positive at every basepoint $\alpha \geq 1$, bounding the prime weights by their absolute values certifies this only for $\alpha > 2.2823$, and through the Gaussian field $\mathbb{Q}(i)$ the density is the difference of the Thorin densities of $\xi_{\mathbb{Q}(i)}$ and ξ.

8.5.1 The eta kernel and GRH for $L(s,\chi_{12})$ below height 100

The character χ_{12} is the even primitive character modulo 12: $\chi_{12}(n) = 1$ for $n \equiv \pm 1$, -1 for $n \equiv \pm 5 \pmod{12}$, and 0 otherwise. Euler's pentagonal number theorem gives

$$\eta(ix) = e^{-\pi x/12}\prod_{n\ge 1}\bigl(1-e^{-2\pi n x}\bigr) = \sum_{n\ge 1}\chi_{12}(n)\,e^{-\pi n^2 x/12}, \qquad x>0,$$

so the theta series of χ_{12} is the Dedekind eta function on the imaginary axis. The completed L function

$$\xi(s,\chi_{12}) = \Bigl(\frac{12}{\pi}\Bigr)^{s/2}\Gamma\Bigl(\frac{s}{2}\Bigr)L(s,\chi_{12}) = \int_0^\infty \eta(ix)\,x^{s/2}\,\frac{dx}{x}$$

satisfies $\xi(s,\chi_{12}) = \xi(1-s,\chi_{12})$ because $\eta(i/x) = \sqrt{x}\,\eta(ix)$, the case $z = ix$ of $\eta(-1/z) = \sqrt{z/i}\,\eta(z)$. With $x = e^{2u}$,

$$\xi\Bigl(\frac12 + s,\chi_{12}\Bigr) = \int_{\mathbb{R}} K(u)\,e^{su}\,du, \qquad K(u) = 2e^{u/2}\eta\bigl(ie^{2u}\bigr),$$

and the modular transformation makes K even. The product formula makes $\eta(ix) > 0$, so K is positive and $K/\xi(\frac12,\chi_{12})$ is a probability density, the analogue of the Pólya density: if Y has this density, then $\mathbb{E}e^{sY} = \xi(\frac12+s,\chi_{12})/\xi(\frac12,\chi_{12})$ for every complex s. Since $\eta(ie^{2u})$ is analytic for $|\operatorname{Im} 2u| < \pi/2$, K is analytic in the same strip $|\operatorname{Im} u| < \pi/4$ as Φ, and like Φ it decays doubly exponentially, so the trapezoid analysis of Section 2.1 applies unchanged.

The construction extends to every real primitive character χ modulo m, whose root number is always one. For even χ the kernel is $2e^{u/2}\sum_{n\ge1}\chi(n)e^{-\pi n^2 e^{2u}/m}$; for odd χ, whose completed function is $(m/\pi)^{(s+1)/2}\Gamma(\frac{s+1}{2})L(s,\chi)$, it is $2e^{3u/2}\sum_{n\ge1} n\chi(n)e^{-\pi n^2 e^{2u}/m}$. Both kernels are even and analytic in $|\operatorname{Im} u| < \pi/4$, but their positivity, which for χ_{12} comes from the product formula, has to be checked case by case.

The two polynomial evaluator of Section 2.1, run with the eta kernel ($h = 1/200$, $|u| \le 3.5$, 320 bits), agrees with `mpmath`'s Dirichlet L function [159] to all sixteen digits of the comparison at $s = 0.6$, $0.6+5i$ and $0.6+20i$; for instance $\xi(0.6+5i,\chi_{12}) = -0.1107052435 - 0.0045876258i$. Sign changes of $\xi(\frac12 + i\theta,\chi_{12})$ on a grid of step 0.02, refined by Newton's method, give 67 zeros below height 100, the first of them at

$$3.8046276,\ 6.6922233,\ 8.8905930,\ 11.1883927,\ 12.9661788,\ 15.1814809,\ \ldots.$$

The Thorin density at resolution ε, $\rho_\varepsilon(\theta) = \pi^{-1}\operatorname{Re}(\xi'/\xi)(\frac12+\varepsilon+i\theta,\chi_{12})$ as in (1.2.7), is again a sum of Poisson kernels over the zeros, by the Hadamard product (Proposition 1.2.3). Figure 8.12 shows $\rho_{0.1}$, with a peak of height about $1/(\pi\varepsilon)$ at

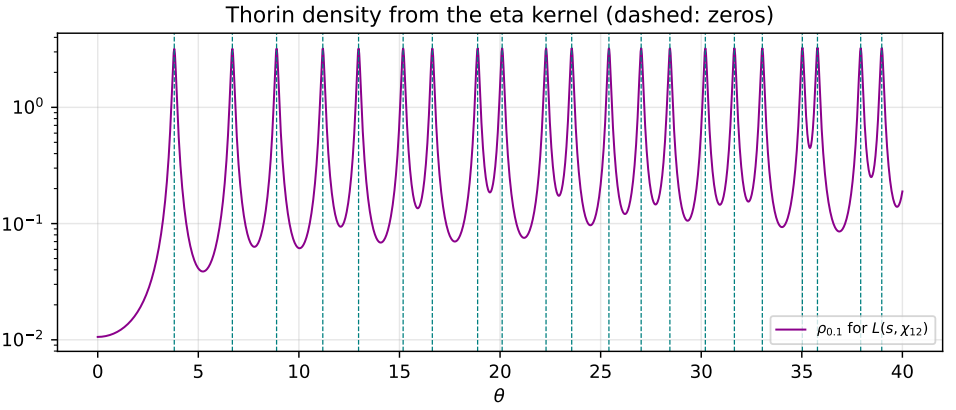


Figure 8.12: The Thorin density $\rho_{0.1}$ of $L(s, \chi_{12})$ for $0 \le \theta \le 40$ on a logarithmic scale, computed from the Dedekind eta kernel, with the zeros dashed.

each zero. The zeros are denser at low height than those of ζ, 67 below height 100 against 29, because the conductor 12 enters the counting function as $\frac{T}{2\pi}\log\frac{12T}{2\pi e}$; the resolution needed to separate neighbours is correspondingly smaller.

The zero density identity, Proposition 3.4.2, holds for $\xi(s, \chi_{12})$ with the same proof, since $\xi(s, \chi_{12})$ is entire, satisfies $\xi(s, \chi_{12}) = \xi(1 - s, \chi_{12})$, is positive on the real axis because K is, and has no zeros with $\operatorname{Re} s \ge 1$. Its hypotheses are also the same: T is not an ordinate, and no zero of $\xi(s, \chi_{12})$ lies on the segment from $\frac{1}{2} + \varepsilon$ to $\frac{1}{2} + \varepsilon + iT$, which is automatic at $\varepsilon = \frac{1}{2}$. In it $N_{\chi_{12}}(T)$ counts the zeros with $0 < \operatorname{Im}\rho \le T$, and $\nu(\sigma, T)$ those among them with $\operatorname{Re}\rho > \sigma$; $M_\varepsilon(T) = \int_0^T \rho_\varepsilon(\theta)\, d\theta$ is the Thorin mass up to height T, and $\tau_\varepsilon(T)$ is the change of argument of $\xi(s, \chi_{12})$ from $\frac{1}{2} + \varepsilon + iT$ to $\frac{1}{2} + iT$, divided by π. Table 8.15 evaluates the identity at $T = 100$.

ε	$M_\varepsilon(100)$	$\tau_\varepsilon(100)$	$M_\varepsilon + \tau_\varepsilon = N - 2\nu$
0.5	67.604974186	−0.604974186	67.00000000000003
0.25	67.476432836	−0.476432836	67.00000000010355
0.1	67.308171973	−0.308171973	67.00000000014663
0.05	67.185437421	−0.185437421	67.00000000009639
0.02	67.080023774	−0.080023774	67.00000000005207

Table 8.15: The zero density identity for $L(s, \chi_{12})$ at height 100, computed from the eta kernel. The last column is $N_{\chi_{12}}(100) - 2\nu(\frac{1}{2} + \varepsilon, 100)$; the row $\varepsilon = \frac{1}{2}$ gives $N_{\chi_{12}}(100)$.

The row $\varepsilon = \frac{1}{2}$ gives $N_{\chi_{12}}(100) = 67$, and with the 67 sign changes every zero below height 100 is simple and lies on the critical line. The other rows give

$\nu(\frac{1}{2}+\varepsilon, 100) = 0$ without using the sign changes. This is GRH for χ_{12} below height 100, read from the eta function alone. It is computed, not certified: the argument is followed on a grid as in Section 3.4, in steps of $\varepsilon/8$ up the line of evaluation and in 2000 steps along the horizontal segment, and its increments are summed in double precision. The deviations from 67, at most 1.5×10^{-10}, are far below the one half that fixes the integer. GRH for small moduli has long been verified to far greater heights by other methods, by Rumely [228] and later by Platt [195].

8.5.2 Ramanujan's Δ and the cost of the three evaluators

The same construction works for Ramanujan's cusp form $\Delta(z) = \eta(z)^{24} = \sum_{n\ge1} \tau(n) q^n$, $q = e^{2\pi i z}$, of weight 12 and level 1. Its L function, normalised so that the critical line is $\operatorname{Re} s = \frac{1}{2}$, is $L(s,\Delta) = \sum_{n\ge1} \tau(n) n^{-s-11/2}$, and the completed function $\xi(s,\Delta) = (2\pi)^{-s-11/2}\Gamma(s+\frac{11}{2})L(s,\Delta)$ satisfies $\xi(s,\Delta) = \xi(1-s,\Delta)$, since the root number of a level one form of weight 12 is $i^{12} = 1$. The Mellin transform of $\Delta(iy)$ gives the Pólya type representation

$$\xi\Big(\frac{1}{2}+s,\Delta\Big) = \int_{\mathbb{R}} K_\Delta(u)\, e^{su}\, du,$$
$$K_\Delta(u) = e^{6u}\,\Delta(ie^u) = e^{6u}e^{-2\pi e^u} \prod_{n\ge1} \big(1-e^{-2\pi n e^u}\big)^{24}.$$

The kernel is even because $\Delta(i/y) = y^{12}\Delta(iy)$, positive by the product formula, and analytic in the strip $|\operatorname{Im} u| < \pi/2$, twice as wide as that of the Pólya kernel.

We computed $\tau(n)$ for $n \le 3000$, the range needed for the prime weights of Section 8.5.3, as the coefficients of η^{24} with integer polynomial arithmetic (checking $\tau(2) = -24$, $\tau(3) = 252$, $\tau(11) = 534612$), and evaluated the kernel by its q expansion for $u \ge 0$ and by evenness for $u < 0$. At 60 digits, with step $1/80$ and truncation $|u| \le 4$, the trapezoid evaluator finds the first zeros on the critical line at

$$9.222379,\ 13.907550,\ 17.442777,\ 19.656513,\ 22.336104,\ \ldots,$$

in agreement with the known values [149, modular form 1.12.a.a], and 20 sign changes below height 50, the number of zeros there.

Table 8.16 compares this evaluator with those for ζ and χ_{12} at the same precision. In the variables of this section their steps are $1/40$ and $1/160$, and their truncations $|u| \le 6$ and $|u| \le 2.5$. The settings are not matched: the truncation is far longer than 60 digits require for ζ, and too short for them for χ_{12}, whose kernel is still about 10^{-16} at $|u| = 2.5$ (Section 8.5.3).

The kernels are evaluated once, before the timing, so the times measure an exponential sum over the nodes and follow the node counts of the table rather than any intrinsic difference: the theta series, the eta series and the q expansion of Δ all converge geometrically, with ratio at most $e^{-2\pi}$, once the modular symmetry is

	ζ	$L(s,\chi_{12})$	$L(s,\Delta)$
kernel	theta series with terms of both signs	η, a positive product	η^{24}, a positive product
decay of ξ on the critical line	$e^{-\pi t/4}$	$e^{-\pi t/4}$	$e^{-\pi t/2}$
pole of L	at $s=1$	none	none
degree, conductor	1, 1	1, 12	2, 1
zeros below height 50	10	28	20
first zero	14.134725	3.804628	9.222379
digits lost per unit height	0.31	0.36	0.59
digits lost per zero near height 40	about 1	about 0.5	about 1
nodes	481	801	641
time per point	9.5 ms	12.8 ms	10.7 ms

Table 8.16: The trapezoid evaluators for ζ, $L(s,\chi_{12})$ and $L(s,\Delta)$ at 60 digits, with the settings given in the text. The decay of $\xi(\frac{1}{2}+it)$ is shown up to a power of t; the digits lost per unit height are the slope of $\log_{10}|\xi(\frac{1}{2}+it)|$ between heights 10.3 and 40.3; the times are per point, in `mpmath`.

used. Per unit height Δ is the most expensive, because its gamma factor $\Gamma(s+\frac{11}{2})$ decays like $e^{-\pi t/2}$ on the critical line, twice as fast as the $e^{-\pi t/4}$ of ζ and χ_{12}: the untilted evaluator loses asymptotically $\pi/(2\log 10) = 0.68$ digits per unit height against $\pi/(4\log 10) = 0.34$. Between heights 10 and 40 the powers of t in the completed functions and the size of $|L|$ at the two ends give the measured rates of the table. Tilting the contour to within δ of the edge of the strip reduces the loss to about $\delta\theta/\log 10$ digits at height θ (Proposition 2.1.1, whose proof applies to both eta kernels), so the wider strip of the Δ kernel offsets its faster decay up to a factor polynomial in $1/\delta$; we have not run the tilted evaluator for Δ. Per zero $L(s,\chi_{12})$ is the cheapest: its conductor packs the zeros at low height, 28 below height 50 against 10 for ζ, and it loses about half a digit per zero where ζ and Δ lose about one.

8.5.3 The sine squared lemma without a pole

The eta based functions differ from ζ in two structural ways. Their kernels are positive for an elementary reason, a product of factors $1-q^n$ with $0<q<1$, where the positivity of the Pólya kernel Φ needs an argument (Section 1.2). And their L functions are entire, so two consequences of the pole of ζ disappear: the negative pole piece of the sine squared decomposition, and the factor that Chapters 6 and 7 need to remove the pole, which by Proposition 7.1.13 alone prevents the normalised reciprocal zeta function from being a GGC. A floor in the sense of Theorem 6.2.15 is still needed for χ_{12}, since its tilt, the factor $e^{-c\sqrt{u}}$ in the variable u of the clock (written $e^{-c\sqrt{s}}$ in Section 6.2), removes the negative part of the Thorin density of the gamma factor, and the gamma factor $(12/\pi)^{s/2}\Gamma(s/2)$ of $\xi(s,\chi_{12})$

is that of ξ times $12^{s/2}$: Theorem 6.2.12 holds for it with $c = 2.6861$ replaced by $c - \frac{1}{2}\log 12 = 1.4436$. The positivity of such a floor would need an analogue of Theorem 7.1.1.

For $f = \chi_{12}$ or Δ write $\xi(s,f) = L_\infty(s,f)L(s,f)$, with L_∞ the gamma factor, and $-(L'/L)(s,f) = \sum_{n\ge 2}\Lambda_f(n)n^{-s}$ for $\operatorname{Re} s > 1$. At a basepoint $\alpha > 1$ the sine squared lemma gives, as (3.1.1) does for ζ,

$$v_\alpha(\theta) = \operatorname{Re}\frac{\xi'}{\xi}(\alpha+i\theta,f) = \underbrace{\frac{\xi'}{\xi}(\alpha,f)}_{\text{drift}} + \underbrace{\operatorname{Re}\frac{L_\infty'}{L_\infty}(\alpha+i\theta,f) - \frac{L_\infty'}{L_\infty}(\alpha,f)}_{\text{archimedean } \ge 0} + \underbrace{\sum_{n\ge 2}\Lambda_f(n)\,n^{-\alpha}\,2\sin^2\Big(\frac{\theta\log n}{2}\Big)}_{\text{primes}},$$

where $v_\alpha(\theta)$ is π times the Thorin density at resolution $\alpha - \frac{1}{2}$, and the archimedean piece is nonnegative because the digamma function $\psi = \Gamma'/\Gamma$ satisfies $\operatorname{Re}\psi(x+iy) \ge \psi(x)$ for $x > 0$. For ζ, whose gamma factor $\pi^{-s/2}\Gamma(1+\frac{s}{2})$ absorbs the factor s of ξ, the remaining factor $s-1$ adds the pole piece $\operatorname{Re}\frac{1}{s-1} - \frac{1}{\alpha-1}$, which is $-\theta^2/\big((\alpha-1)((\alpha-1)^2+\theta^2)\big)$ at $s = \alpha + i\theta$, and the identity becomes (3.1.1).

The prime weights are $\Lambda(n) = \log p$ at $n = p^k$ for ζ, all positive; $\Lambda_{\chi_{12}}(p^k) = \chi_{12}(p)^k \log p$ for χ_{12}, of both signs and zero at $p = 2$ and 3; and $\Lambda_\Delta(p^k) = (\alpha_p^k + \bar\alpha_p^k)\log p$ for Δ, where the Satake parameters α_p and $\bar\alpha_p$ are the roots of $X^2 - \tau(p)p^{-11/2}X + 1$. Deligne's theorem $|\tau(p)| \le 2p^{11/2}$ [67] makes them complex conjugates of modulus one, so the weights of Δ lie in $[-2\log p, 2\log p]$ and take both signs.

We computed the three decompositions at $\alpha = 2$ for $0 \le \theta \le 40$, with prime powers up to 3000, and compared their sum with the left side computed from the kernel. They agree to 5×10^{-4} for ζ and 9×10^{-6} for Δ, the difference being the truncation of the prime sum. For χ_{12} the largest difference, 2.5×10^{-5} at $\theta = 40$, is an error of the kernel evaluator, whose truncation at $|u| \le 2.5$ is too short at that height. Of the 466 prime power weights up to 3000, none is negative for ζ, 222 are negative for χ_{12} (whose 18 weights at powers of 2 and 3 vanish) and 230 for Δ.

On $0 \le \theta \le 40$ the prime piece of ζ ranges over $[0, 0.915]$ and its pole piece over $[-0.999, 0]$ (Figure 8.13, left): the lemma writes the Thorin density as positive arithmetic mass minus one negative term. The prime piece of $L(s,\chi_{12})$ ranges over $[-0.218, 0.053]$, and that of $L(s,\Delta)$ over $[-0.261, 0.222]$. Their left sides are positive, with minima at $\theta = 0$ equal to the drifts, 0.466 and 0.167, so the positivity comes from the drift and the archimedean piece, not from the primes.

For each of the three functions its Riemann hypothesis, RH for ζ and GRH for the other two, is the statement that v_α stays positive for every $\alpha > \frac{1}{2}$; this is Lagarias's criterion [133], whose proof from the Hadamard product applies un-

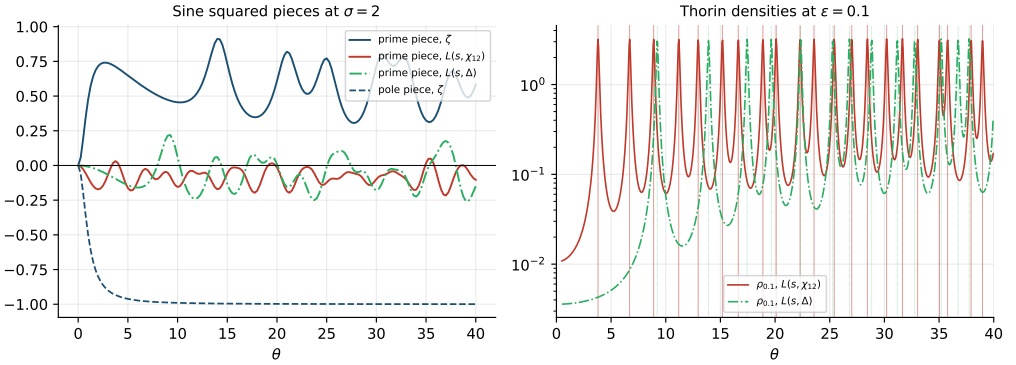


Figure 8.13: Left: the prime pieces of the sine squared decomposition at the basepoint $\alpha = 2$, written σ in the plot, for ζ, $L(s, \chi_{12})$ and $L(s, \Delta)$, and the pole piece of ζ. For ζ the prime piece is positive and the pole piece negative; for the two eta based functions there is no pole piece and the prime piece takes both signs. Right: the Thorin densities $\rho_{0.1}$ of $L(s, \chi_{12})$ and $L(s, \Delta)$ on a logarithmic scale, peaking at their zeros (vertical lines).

changed to $\xi(s, \chi_{12})$ and $\xi(s, \Delta)$. Below $\alpha = 1$ the prime piece no longer converges absolutely. For $L(s, \chi_{12})$, Section 11.3.2 shows that it still converges, as a limit over $n \le X$, when $L(s, \chi_{12})$ has no zeros to the right of some line $\operatorname{Re} s = \alpha'$ with $\alpha' < \alpha$, and that conversely the convergence of the Lévy form at the basepoint α forces the absence of zeros with $\operatorname{Re} s > \alpha$; so the extension of the decomposition down to the centre is GRH for χ_{12}. At resolution 0.1 the Thorin densities of $L(s, \chi_{12})$ and $L(s, \Delta)$ have computed minima 0.011 and 0.004 below height 40 (Figure 8.13, right), positive as GRH predicts.

The distances of Section 3.2, with the zeros of $L(s, \chi_{12})$ or $L(s, \Delta)$ as the reference atoms, Theorem 4.3.6 and the flow of Section 8.4 carry over unchanged, the last two because they use only the Hadamard product, positivity on the real line and the decay of the kernel; we have not computed the first two for these functions, and Section 8.5.5 computes the flow for Δ. Theorem 4.3.6 also needs the bound $|\arg z| < \pi/4$ of Lemma 4.3.5 for the zeros z of $\xi(\frac{1}{2}+iz, f)$ with $\operatorname{Re} z > 0$, and it holds here too: the zeros lie in the open critical strip, so $|\operatorname{Im} z| < \frac{1}{2}$, while the lowest ordinates, 3.80 and 9.22, exceed $\frac{1}{2}$. The arithmetic side, however, offers no termwise positivity, even far to the right of the critical line, so the domination of the pole by positive primes that Sections 3.1 and 3.5 formulate for ζ has no direct analogue. Deligne's bound $|\Lambda_\Delta(n)| \le 2\Lambda(n)$, like $|\Lambda_{\chi_{12}}(n)| \le \Lambda(n)$, bounds the prime piece below only by a multiple of minus the zeta prime piece, which tends to $-\infty$ as $\alpha \downarrow 1$ for every $\theta \neq 0$, and no domination argument follows from it. Far enough to the right the absolute values do suffice: the drift and the archimedean piece outweigh the prime piece bounded by the absolute values of

its weights for $\alpha > 1.8349$ for $L(s,\chi_{12})$ and for $\alpha > 2.2823$ for the Dirichlet beta function $L(s,\chi_4)$, and by Kronecker's theorem no better bound from the sizes of the weights holds at all large heights (Propositions 8.5.25 and 8.5.26). So although $L(s,\chi_{12})$ is the cheapest of the three functions per zero, neither eta based function is theoretically easier than ζ: the sine squared lemma trades the negative pole term for a prime measure of both signs.

8.5.4 Maxima of exponentials

The product formula has a probabilistic reading, which the first author's note [204] develops. The factor $1-e^{-2\pi ny}$ is the distribution function of an exponential variable with rate $2\pi n$, so

$$F(y)=\prod_{n\geq 1}\left(1-e^{-2\pi ny}\right)^{24}=e^{2\pi y}\Delta(iy),\qquad F_1(y)=\prod_{n\geq 1}\left(1-e^{-2\pi ny}\right)=e^{\pi y/12}\eta(iy)$$

are the distribution functions of the maxima M^* and M_1 of independent exponential variables, 24 of each rate $2\pi n$ for M^* and one of each rate for M_1. Section 8.5.4.1 turns the Mellin integral of Δ into an expectation over the law of M^*, a representation of $L(s,\Delta)$ with no signs, and compares it with the incomplete gamma continuations of Riemann and Hecke; Section 8.5.4.2 carries the construction to other products, and Section 8.5.4.3 reads it at real points as a statement about gamma mixtures. The last two parts study the laws themselves. That of M_1 is explicit: it is infinitely divisible, lies in the Bondesson class and is not a GGC (Section 8.5.4.4). That of M^* is not even in the Bondesson class, because its Laplace transform has zeros off the real axis (Section 8.5.4.5).

8.5.4.1 The max representation

Lemma 8.5.1. *Let $E_{n,j}$, $n\geq 1$, $1\leq j\leq 24$, be independent, with $E_{n,j}$ exponential of rate $2\pi n$, and put $M^*=\sup_{n,j}E_{n,j}$ and $M^{(j)}=\sup_n E_{n,j}$. Almost surely these suprema are attained and finite, $\Pr(M^*\leq y)=F(y)$ for $y>0$, and each $M^{(j)}$ has the distribution function F_1. So $M^*=\max_{j\leq 24}M^{(j)}$ is the maximum of 24 independent copies of M_1.*

Proof. For $y>0$, $\sum_{n,j}\Pr(E_{n,j}>y)=24\sum_n e^{-2\pi ny}<\infty$, so by the Borel Cantelli lemma only finitely many $E_{n,j}$ exceed y, almost surely. Taking y rational and below $E_{1,1}$, the supremum is the largest of finitely many values. By independence $\Pr(M^*\leq y)=\lim_N\prod_{n\leq N}(1-e^{-2\pi ny})^{24}=F(y)$, and in the same way each $M^{(j)}$ has the distribution function F_1; the $M^{(j)}$ are independent because they are built from disjoint sets of the $E_{n,j}$. □

The modular relations $\Delta(i/y)=y^{12}\Delta(iy)$ and $\eta(i/y)=\sqrt{y}\,\eta(iy)$ become self dualities of the two laws,

$$F(y)=y^{-12}e^{2\pi y-2\pi/y}F(1/y),\qquad F_1(y)=y^{-1/2}e^{\pi y/12-\pi/(12y)}F_1(1/y),\quad (8.5.1)$$

so the lower tails are thin: $\Pr(M^* \le y) \sim y^{-12} e^{-2\pi/y}$ and $\Pr(M_1 \le y) \sim y^{-1/2} e^{-\pi/(12y)}$ as $y \downarrow 0$. Since $F(1/y) \le 1$, the first relation also gives $F(y) \le y^{-12} e^{2\pi y - 2\pi/y}$ for every $y > 0$, a bound that increases with y, because its logarithmic derivative is $(2\pi y^2 - 12y + 2\pi)/y^2$ and $144 < 16\pi^2$.

The density of M^* is Ramanujan's logarithmic derivative. Expanding $1/(1 - e^{-2\pi n y})$ gives $(\log F)'(y) = 24 \sum_n 2\pi n\, e^{-2\pi n y}/(1 - e^{-2\pi n y}) = 48\pi \sum_{m \ge 1} \sigma(m) e^{-2\pi m y}$, with $\sigma(m)$ the sum of the divisors of m, so that, with Eisenstein's series $E_2(z) = 1 - 24 \sum_{m \ge 1} \sigma(m) e^{2\pi i m z}$,

$$f^*(y) = F(y)\,(\log F)'(y) = 2\pi F(y)\big(1 - E_2(iy)\big). \tag{8.5.2}$$

Taking the logarithmic derivative of (8.5.1) gives the functional equation of the density,

$$f^*(1/y) = y^{12} e^{2\pi/y - 2\pi y} \Big(\big(2\pi y^2 - 12y + 2\pi\big) F(y) - y^2 f^*(y) \Big),$$

which in terms of E_2 is its quasi modularity, $E_2(i/y) = -y^2 E_2(iy) + 6y/\pi$. At $y = 2$ the quotient F'/F of the product, the form (8.5.2) and the functional equation applied at $y = \frac{1}{2}$ give the same value, $f^*(2) = 0.000525840326953580416712$, and the density integrates to one to the 30 digits printed (`maxexp_checks.py`).

Theorem 8.5.2. *For every complex s,*

$$\xi(s, \Delta) = (2\pi)^{-s - \frac{11}{2}}\, \mathbb{E}\, \Gamma\big(s + \tfrac{11}{2}, 2\pi M^*\big), \qquad L(s, \Delta) = \mathbb{E}\, Q\big(s + \tfrac{11}{2}, 2\pi M^*\big), \tag{8.5.3}$$

with $\Gamma(a, x) = \int_x^\infty e^{-v} v^{a-1} dv$ and $Q(a, x) = \Gamma(a, x)/\Gamma(a)$, and both expectations converge absolutely. In particular $L(s, \Delta) = 0$ at the points $s = -\frac{11}{2} - k$, $k = 0, 1, 2, \ldots$, where $Q(-k, x) = 0$.

Proof. Put $a = s + \frac{11}{2}$, so that $\xi(s, \Delta) = \int_0^\infty \Delta(iy) y^{a-1} dy$. Inserting $\Delta(iy) = e^{-2\pi y} \Pr(M^* \le y) = e^{-2\pi y} \mathbb{E}\, \mathbf{1}\{M^* \le y\}$ and integrating in y first gives $\mathbb{E} \int_{M^*}^\infty e^{-2\pi y} y^{a-1} dy = (2\pi)^{-a} \mathbb{E}\, \Gamma(a, 2\pi M^*)$. Fubini's theorem applies: with $\sigma = \operatorname{Re} a$, Tonelli's theorem gives $\mathbb{E} \int_{M^*}^\infty e^{-2\pi y} y^{\sigma - 1} dy = \int_0^\infty \Delta(iy) y^{\sigma - 1} dy$, which is finite for every real σ, since $\Delta(iy)$ decays exponentially as $y \to \infty$ and like $y^{-12} e^{-2\pi/y}$ as $y \downarrow 0$. The same computation gives $\mathbb{E}|\Gamma(a, 2\pi M^*)| \le \mathbb{E}\, \Gamma(\sigma, 2\pi M^*) < \infty$. Dividing by $(2\pi)^{-a} \Gamma(a)$, with $1/\Gamma$ entire, gives the second identity, in which $Q(a, x)$, for fixed $x > 0$, is the entire function $\Gamma(a, x)/\Gamma(a)$ of a. At $a = -k$ it vanishes, since $\Gamma(-k, x)$ is finite and $1/\Gamma(-k) = 0$. □

Corollary 8.5.3 (Gamma mixture and mixed Poisson forms). *Let G_a be a gamma variable of shape $a > 0$ and scale one, and N a Poisson process of unit rate, both independent of M^*. For real $s > -\frac{11}{2}$, with $a = s + \frac{11}{2}$,*

$$L(s, \Delta) = \Pr\big(G_a > 2\pi M^*\big) = \mathbb{E} F\Big(\frac{G_a}{2\pi}\Big) = \mathbb{E} \prod_{n \ge 1} \big(1 - e^{-n G_a}\big)^{24},$$

and for every integer $k \geq 1$

$$L\big(k-\tfrac{11}{2},\Delta\big) = \Pr\big(N(2\pi M^*) \leq k-1\big) = \mathbb{E}\Big[e^{-2\pi M^*}\sum_{j<k}\frac{(2\pi M^*)^j}{j!}\Big].$$

Proof. For real $a>0$, $Q(a,x)=\Pr(G_a>x)$, which gives the first form; conditioning on G_a instead gives $\Pr(M^*<G_a/2\pi)=\mathbb{E}F(G_a/2\pi)$, since the law of M^* has no atoms. For an integer k the kth point of N has the law of G_k, so $\Pr(G_k>x)=\Pr(N(x)\leq k-1)$. As a check, $F(y)=\sum_{n\geq1}\tau(n)e^{-2\pi(n-1)y}$ and $\mathbb{E}e^{-(n-1)G_a}=n^{-a}$ return the Dirichlet series $\sum_n \tau(n)n^{-a}$ term by term when $a>\frac{13}{2}$, where it converges absolutely. □

So $L(-\frac{9}{2},\Delta)=\mathbb{E}e^{-2\pi M^*}=0.03744128126851554174$ is a value of the Laplace transform of M^*, and the central value $L(\frac12,\Delta)=0.79212283864603056936$ is the probability that a Poisson clock of rate 2π rings at most five times before the time M^*. Write $P_k=\Pr(N(2\pi M^*)\leq k-1)=L(k-\frac{11}{2},\Delta)$. The functional equation $\xi(s,\Delta)=\xi(1-s,\Delta)$ at $s=k-\frac{11}{2}$ becomes the family of identities

$$(2\pi)^{-k}(k-1)!\,P_k=(2\pi)^{k-12}(11-k)!\,P_{12-k},\qquad 1\leq k\leq 11, \tag{8.5.4}$$

both sides being $\xi(k-\frac{11}{2},\Delta)$, which are nontrivial statements about the law of a maximum of exponential variables. Table 8.17 lists the probabilities, computed by quadrature against the density (8.5.2); the two sides of (8.5.4) agree to 10^{-43}, and with Hecke's formula below to all 25 digits printed. The gamma mixture form gives $L(0,\Delta)=\mathbb{E}F(G_{11/2}/2\pi)=0.7347229296103482203473$, and the max representation gives $L(\frac{17}{2},\Delta)=0.9985830631627461764919 18$ and $L(\frac{29}{2},\Delta)=0.9999771827993698411748 79$, in agreement with Hecke's formula and with the Dirichlet series.

Remark 8.5.4. The identities (8.5.4) follow from the self duality (8.5.1), since $(2\pi)^{-k}(k-1)!\,P_k=\int_0^\infty e^{-2\pi y}F(y)y^{k-1}dy$ and the substitution $y\mapsto 1/y$ turns this integral into the one with $12-k$ in place of k. They are five relations, since k and $12-k$ give the same one and $k=6$ gives none, and they do not imply the self duality. Let g be smooth and not identically zero, with compact support in $(0,1)$, such that $g(\log y)$ is orthogonal on $(1,e)$ to the five functions $e^{-2\pi y}F(y)(y^{k-1}-y^{11-k})$, $k\leq 5$. For small ε, $\tilde F(y)=F(y)\big(1+\varepsilon g(\log y)\big)$ is again a distribution function, because F' has a positive lower bound and F is below one on $[1,e]$, and it satisfies (8.5.4), because the perturbation changes the difference of the two sides of the kth identity by ε times the integral of $g(\log y)$ against the kth function. It is not self dual: on $(1,e)$ self duality would give $\tilde F(y)=y^{-12}e^{2\pi y-2\pi/y}\tilde F(1/y)=F(y)$, since $g(\log(1/y))=0$. With g even and supported in $(-1,1)$ instead, the same construction gives a self dual distribution function other than F, so self duality does not determine the law of M^* either. What does determine it is self duality together with the form of F as a power

k	$P_k = \Pr\big(N(2\pi M^*) \le k-1\big) = L(k-\frac{11}{2}, \Delta)$	$(2\pi)^{-k}(k-1)!\,P_k = \xi(k-\frac{11}{2}, \Delta)$
1	0.0374412812685	0.00595896498958
2	0.146374542091	0.00370771046495
3	0.315241565881	0.00254175605420
4	0.501617647511	0.00193109920049
5	0.666709188434	0.00163398603484
6	0.792122838646	0.00154487936040
7	0.877354125389	0.00163398603484
8	0.930707030298	0.00193109920049
9	0.962126459694	0.00254175605420
10	0.979809088251	0.00370771046495
11	0.989432913100	0.00595896498958

Table 8.17: The probabilities that a Poisson clock of rate 2π rings fewer than k times before M^*, and the identities (8.5.4): the last column is symmetric about $k = 6$. Computed by `maxexp_checks.py`.

series in $e^{-2\pi y}$. If $F(y) = \sum_{n\ge0} c_n e^{-2\pi n y}$ for $y > 0$, with $c_0 = 1$, and F satisfies (8.5.1), then $f(z) = \sum_{n\ge0} c_n e^{2\pi i(n+1)z}$ is holomorphic on the upper half plane, satisfies $f(z+1) = f(z)$, and satisfies $f(-1/z) = z^{12} f(z)$ on the imaginary axis, hence everywhere by analytic continuation. So f is a cusp form of weight 12 for $\mathrm{SL}_2(\mathbb{Z})$, a multiple of Δ because these cusp forms make up a space of dimension one [108, Chapter 14], and $c_0 = 1$ gives $f = \Delta$. This argument is the simplest case of Hecke's converse theorem [94], which recovers a modular form from the functional equation of its Dirichlet series. The distribution function of a maximum of independent exponential variables whose rates are integer multiples of 2π is a power series of this kind, so M^* is the only such maximum with the self duality (8.5.1).

Cutting the integral at $y = 1$ and folding $(0,1)$ onto $(1,\infty)$ by modularity gives in the same way the symmetric form

$$\xi(s,\Delta) = A(s) + A(1-s), \qquad A(s) = \int_1^\infty \Delta(iy)\, y^{s+\frac{9}{2}} dy, \tag{8.5.5}$$

$$A(s) = (2\pi)^{-s-\frac{11}{2}}\, \mathbb{E}\,\Gamma\big(s+\tfrac{11}{2}, 2\pi\max(M^*,1)\big) = \sum_{n\ge1} \tau(n)\, \frac{\Gamma(s+\frac{11}{2}, 2\pi n)}{(2\pi n)^{s+11/2}},$$

where the last form, Hecke's, comes from expanding $\Delta(iy) = \sum_n \tau(n) e^{-2\pi n y}$ instead. Hecke's form is the analogue for Δ of Riemann's second proof of the functional equation [72, Chapter 1], which cuts the Mellin integral of the theta function $\omega(x) = \sum_{n\ge1} e^{-\pi n^2 x}$ of Section 1.2 at $x = 1$ in the same way.

PROPOSITION 8.5.5 (Riemann). *For every complex $s \neq 0, 1$,*

$$\pi^{-s/2}\Gamma\Big(\frac{s}{2}\Big)\zeta(s) = -\frac{1}{s} - \frac{1}{1-s} + \sum_{n\geq 1}\Big[\frac{\Gamma(\frac{s}{2}, \pi n^2)}{(\pi n^2)^{s/2}} + \frac{\Gamma(\frac{1-s}{2}, \pi n^2)}{(\pi n^2)^{(1-s)/2}}\Big].$$

Proof. For $\operatorname{Re} s > 1$, $\pi^{-s/2}\Gamma(\frac{s}{2})\zeta(s) = \int_0^\infty \omega(x) x^{s/2-1} dx$. On $(0,1)$ substitute $x \mapsto 1/x$ and use $\omega(1/x) = \sqrt{x}\,\omega(x) + \frac{1}{2}\sqrt{x} - \frac{1}{2}$, which is the theta transformation $1 + 2\omega(1/x) = \sqrt{x}\,(1 + 2\omega(x))$:

$$\int_0^1 \omega(x) x^{s/2-1} dx = \int_1^\infty \omega(x) x^{(1-s)/2-1} dx + \frac{1}{2}\int_1^\infty x^{-(s+1)/2} dx - \frac{1}{2}\int_1^\infty x^{-s/2-1} dx,$$

and the last two integrals are $\frac{1}{s-1}$ and $-\frac{1}{s}$. On $(1,\infty)$ integrate ω term by term, with $\int_1^\infty e^{-\pi n^2 x} x^{b-1} dx = (\pi n^2)^{-b}\Gamma(b, \pi n^2)$. The series converges uniformly on compact sets of s, because $\Gamma(b, \pi n^2) \leq C_b e^{-\pi n^2}(\pi n^2)^{\operatorname{Re} b}$, so the identity continues to every $s \neq 0, 1$. □

At $s = 2$ the right side gives $\pi/6$. At $s = \frac{1}{2} + 5i$ it gives a real number, as it must on the critical line, namely $-0.021825742363897361 76$, and both values agree with the left side to 25 digits. The three formulas are incomplete gamma continuations of Mellin integrals, but they randomise different things. In Riemann's and Hecke's the threshold is the fixed point $x = 1$ or $y = 1$, and the rates πn^2 or $2\pi n$ carry weights: one each for ζ, at the price of the pole terms, and the signed coefficients $\tau(n)$ for Δ. In (8.5.3) there is a single rate 2π, coming from the factor $q = e^{-2\pi y}$ of Δ, and the threshold M^* is random with a positive law; the symmetric form (8.5.5) is Riemann's formula with the sum over n replaced by an expectation over $\max(M^*, 1)$, and with no pole terms. The max representation replaces Hecke's sum over the signed coefficients $\tau(n)$ by an expectation over a positive law.

The script `maxlaw.py` evaluates these expectations by quadrature against the density (8.5.2) of M^* for $y \geq 1$, with the self duality for $y < 1$, so that τ enters only through the product. At $s = \frac{1}{2}$, $\frac{1}{2} + 5i$ and 3 the max representation, the symmetric form and Hecke's formula agree to the 25 digits printed, for instance $\xi(\frac{1}{2}, \Delta) = 0.00154487936039502720604$. On the critical line $\xi(\frac{1}{2} + i\theta, \Delta) = 2\operatorname{Re} A(\frac{1}{2} + i\theta)$, and the symmetric form has its first zero at $\theta = 9.2223793999211025222$. It also suggests a Hermite Biehler test, the route that Lagarias studied for ξ [134]: if $|A(s)| > |A(1-s)|$ held throughout $\operatorname{Re} s > \frac{1}{2}$, there could be no zeros there. The test fails. At $\theta = 0$ the half $A(s)$ dominates, with $|A(1-s)|/|A(s)|$ falling from 0.989 at $\operatorname{Re} s = 0.52$ to 0.100 at $\operatorname{Re} s = 4.5$, but on $\operatorname{Re} s = 0.6$ the two halves have equal modulus at $\theta = 10.26$, 14.67, 18.04, 20.36 and 22.99, and the dominant half alternates between these ordinates, which separate the zeros 9.22, 13.91, 17.44, 19.66 and 22.34. The curves of equal modulus are nearly vertical near the critical line and bend towards lower ordinates as $\operatorname{Re} s$ grows (Figure 8.14): on $\operatorname{Re} s = 4.5$ the first crossing is at $\theta = 8.05$, and seven lie below height 25 (`maxlaw.py`, `maxexp_hb.py`). So the cut at $y = 1$ gives no Hermite Biehler function.

The cut can be moved. For every $c > 0$ the same folding gives $\xi(s,\Delta) = A_c(s) + A_{1/c}(1-s)$ with $A_c(s) = \int_c^\infty \Delta(iy) y^{s+9/2} dy$, and c may be chosen separately on each horizontal ray. The script `maxexp_hb.py` computes, for $0 \le \theta \le 25$ in steps of 0.05, the best over 36 cuts $c \in [0.35, 1.2]$ of the least ratio $|A_c(s)|/|A_{1/c}(1-s)|$ over 120 points of the ray $s = \sigma + i\theta$, $\frac{1}{2} < \sigma \le 14.5$. For $\theta \le 12.9$ and $13.95 \le \theta \le 15.2$, and at $\theta = 15.35$ and 15.4, some cut dominates on the whole ray, for instance at $\theta = 5$, 10.5 and 12 with least ratios 566, 4.46 and 2.03, all for the smallest cut tried, $c = 0.35$, where the cut at 1 gives 1.0007, 0.9065 and 0.992. For $12.95 \le \theta \le 13.9$, just below the zero 13.91, at $\theta = 15.25$ and 15.3, and for $15.45 \le \theta \le 25$, no cut in the range dominates, the best least ratio staying between 0.9955 and 1. The freedom is in any case illusory. As $c \to 0$, $A_{1/c}(1-s) \to 0$ and $A_c(s) \to \xi(s,\Delta)$ uniformly on compact sets, so domination for small c only says that $\xi(s,\Delta)$ does not vanish there, and the certificate turns into an evaluation of ξ; and at a zero $\frac{1}{2} + i\gamma$ the two halves have equal modulus for every c, since their sum vanishes. A Hermite Biehler function that could prove GRH for Δ would have to come from the structure of the law of M^*, not from tuning a cut. The representation is exact and has no signs, but like the kernel it restates the problem without locating the zeros.

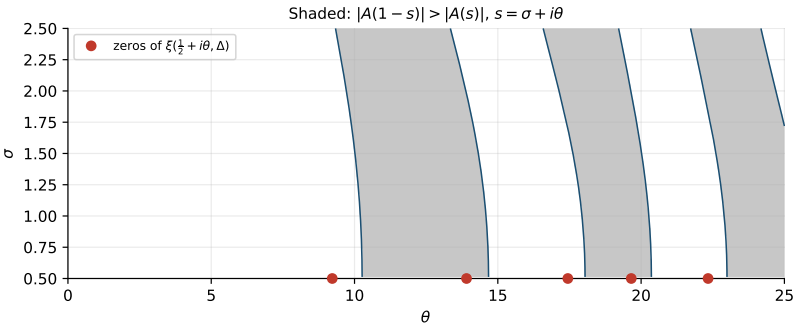


Figure 8.14: The Hermite Biehler test with the cut at $y = 1$: shaded where $|A(1-s)| > |A(s)|$ for $s = \sigma + i\theta$, $\frac{1}{2} < \sigma \le 2.5$, with the curves of equal modulus and the zeros of $\xi(\frac{1}{2} + i\theta, \Delta)$. Drawn by `maxexp_figs.py`.

8.5.4.2 Other theta products

The construction needs a modular form whose expansion on the imaginary axis is a power of q times a product of factors $1 - q^a$, $a > 0$, with nonnegative exponents, so that the product is a distribution function; eta products with nonnegative exponents qualify.

For χ_{12} the Mellin integral of Section 8.5.1, with $\eta(ix) = e^{-\pi x/12} F_1(x)$, gives

in the same way

$$L(s,\chi_{12}) = \mathbb{E}\,Q\Big(\frac{s}{2},\frac{\pi}{12}M_1\Big), \qquad \xi(s,\chi_{12}) = A_{12}(s) + A_{12}(1-s),$$

with $A_{12}(s) = \int_1^\infty \eta(ix)x^{s/2-1}dx = (\frac{\pi}{12})^{-s/2}\mathbb{E}\,\Gamma(\frac{s}{2},\frac{\pi}{12}\max(M_1,1))$; the second exponent comes from $\eta(i/x) = \sqrt{x}\,\eta(ix)$. The factor $q^{1/24}$ of η is what turns Euler's pentagonal exponents $k(3k\mp1)/2$ into the squares $n^2/24$, $n = 6k\mp1$, and so produces the theta series, and the Dirichlet series, of a character. At $s=1$, where $Q(\frac12,x) = \operatorname{erfc}\sqrt{x}$, the class number formula reads $\mathbb{E}\operatorname{erfc}\sqrt{\pi M_1/12} = L(1,\chi_{12}) = \log(2+\sqrt3)/\sqrt3$; at $s=2$, where $Q(1,x) = e^{-x}$, $L(2,\chi_{12}) = \mathbb{E}e^{-\pi M_1/12} = \pi^2/(6\sqrt3)$ is a value of the Laplace transform of M_1, which Section 8.5.4.4 computes in closed form; and at the centre $L(\frac12,\chi_{12}) = \Pr(12G_{1/4} > \pi M_1) = 0.498557002457815436157$. At $s = 0.6+5i$ the max representation, both forms of the symmetric form and `mpmath`'s Dirichlet L function agree to 25 digits, with $\xi(0.6+5i,\chi_{12})$ as in Section 8.5.1, and so do the three values above with their closed forms (`maxexp_checks.py`, `maxlaw.py`).

For the elliptic curve of conductor 11 the newform is $f(z) = \eta(z)^2\eta(11z)^2 = q\prod_{n\ge1}(1-q^n)^2(1-q^{11n})^2 = \sum_{n\ge1}a_nq^n$ [149, modular form 11.2.a.a], so $f(iy) = e^{-2\pi y}F_{11}(y)$, where F_{11} is the distribution function of the maximum M_{11} of independent exponential variables, two of each rate $2\pi n$ and two of each rate $22\pi n$. The Fricke relation $f(i/(11y)) = 11y^2f(iy)$ makes $f(iy)$ decay faster than any power at both ends, and the proof of Theorem 8.5.2 gives, in the arithmetic normalisation $L(s,E) = \sum_n a_nn^{-s}$ with centre $s=1$,

$$L(s,E) = \mathbb{E}\,Q(s,2\pi M_{11}) \quad \text{for every complex } s, \qquad L(1,E) = \mathbb{E}e^{-2\pi M_{11}}.$$

So the central value is a value of the Laplace transform of M_{11}, and the max representation gives $L(1,E) = 0.2538418608559106843378$, equal to all digits to the rapidly convergent series $2\sum_n a_nn^{-1}e^{-2\pi n/\sqrt{11}}$ of root number one.

Jacobi's triple product gives $\vartheta_4(x) = \sum_{n\in\mathbb{Z}}(-1)^ne^{-\pi n^2x} = \prod_{m\ge1}(1-e^{-2\pi mx})(1-e^{-\pi(2m-1)x})^2$, a distribution function with no prefactor: that of the maximum M_4 of independent exponential variables, one of each rate $2\pi m$ and two of each rate $\pi(2m-1)$. The partial sums of the alternating series lie between -1 and 1, so the series can be integrated term by term against $\theta e^{-\theta y}$, and the partial fraction expansion of $z/\sinh z$ gives

$$\mathbb{E}e^{-\theta M_4} = \sum_{n\in\mathbb{Z}}(-1)^n\frac{\theta}{\theta+\pi n^2} = \frac{\sqrt{\pi\theta}}{\sinh\sqrt{\pi\theta}} = \prod_{n\ge1}\Big(1+\frac{\theta}{\pi n^2}\Big)^{-1},$$

so $M_4 \overset{d}{=} \sum_{n\ge1}E_n/(\pi n^2)$ with independent standard exponential variables E_n: a maximum of exponential variables that is also a sum of them, and a GGC with Thorin measure $\sum_n\delta_{\pi n^2}$. Since $\Pr(K\le k) = \sum_n(-1)^ne^{-2n^2k^2} = \vartheta_4(2k^2/\pi)$,

$\pi M_4 = 2K^2$ has the Kolmogorov law of Table 1.9. With no prefactor the Mellin transform gives moments instead of incomplete gamma functions: $\mathbb{E}M_4^{s/2} = \frac{s}{2}\int_0^\infty y^{s/2-1}(1-\vartheta_4(y))\,dy$ for $\operatorname{Re} s > 0$, and integrating $1-\vartheta_4(y) = 2\sum_{n\ge1}(-1)^{n+1}e^{-\pi n^2 y}$ term by term, for $\operatorname{Re} s > 1$ and then by analytic continuation, gives

$$(1-2^{1-s})\,\zeta(s) = \frac{\mathbb{E}\big[(2K^2)^{s/2}\big]}{2\,\Gamma(1+\frac{s}{2})}$$

for every complex s, both sides being entire, since $\vartheta_4(y) = y^{-1/2}\sum_n e^{-\pi(n+\frac12)^2/y}$ makes every moment of M_4 finite. This is a case of the dictionary of Biane, Pitman and Yor [28]. At $s=1$ it gives $\mathbb{E}K = \sqrt{\pi/2}\,\log 2 = 0.868731160636159141830$, which quadrature of $1-\vartheta_4$ reproduces to 2×10^{-23}; at $s=3$ and $\frac12+5i$ the two sides agree to 25 digits, and $\mathbb{E}e^{-3M_4} = \sqrt{3\pi}/\sinh\sqrt{3\pi} = 0.28564527894379183555 1$ by quadrature and in closed form.

For ζ itself the construction stops. The theta function $\vartheta_3(x) = 1+2\omega(x) = \prod_{m\ge1}(1-q^{2m})(1+q^{2m-1})^2$, $q=e^{-\pi x}$, has factors larger than one and is not a distribution function, and the transformation $\vartheta_4(1/x) = \sqrt{x}\,\vartheta_2(x)$ leads to $\vartheta_2(x) = \sum_n e^{-\pi(n+\frac12)^2 x}$, which is not one either. So the max structure reaches ζ only through the factor $1-2^{1-s}$, in moment form. It does not reach it through $\zeta_{\mathbb{Q}(\sqrt3)}(s) = \zeta(s)L(s,\chi_{12})$: the eta law represents the factor $L(s,\chi_{12})$ only, and no product of this kind can represent a Dedekind zeta function, since an eta product with nonnegative exponents decays faster than any power of y at both ends of the imaginary axis, so its Mellin transform is entire, as in the proof of Theorem 8.5.2, while $\zeta_{\mathbb{Q}(\sqrt3)}$ has a pole at $s=1$.

8.5.4.3 Gamma shapes and infinite divisibility at real points

The Pólya kernel of ζ and the kernel K_Δ of Δ differ in their gamma shapes. On the logarithmic scale the function $x^a e^{-x}$ of $x = ce^{ku}$ is proportional to the density of $(\log G_a - \log c)/k$, a log gamma density of shape a, and its Fourier transform $\int_{\mathbb{R}} x^a e^{-x} e^{i\theta u}du = k^{-1}c^{-i\theta/k}\Gamma(a+i\theta/k)$ has no zeros. With $x = 2\pi e^u$, $K_\Delta(u) = e^{6u}\Delta(ie^u) = (2\pi)^{-6}x^6e^{-x}F(e^u)$ is a single gamma kernel of shape 6 times the distribution function of $\log M^*$, and the gamma kernel's own transform is $(2\pi)^{-6-i\theta}\Gamma(6+i\theta)$, so every zero of $\xi(\frac12+i\theta,\Delta)$ is produced by the multiplier $F(e^u)$. The Pólya kernel (1.2.3) is, with $x_n = \pi n^2 e^{2u}$ and the gamma densities $g_k(x) = x^{k-1}e^{-x}/\Gamma(k)$,

$$\Phi(u) = 2e^{u/2}\sum_{n\ge1}\big(2x_n^2 - 3x_n\big)e^{-x_n} = 2e^{u/2}\sum_{n\ge1}\big(4g_3(x_n) - 3g_2(x_n)\big),$$

a signed mixture of Gamma(2) and Gamma(3) densities over the scales πn^2. As a function of $x = e^{2u}$ the sum is, up to the factor $x/2$, the density $2\sum_n(2\pi^2n^4x -$

$3\pi n^2)e^{-\pi n^2 x}$ of $\pi S_2/2$ in the representation $2\xi(s) = \mathbb{E}[(\pi S_2/2)^{s/2}]$ of Williams and of Biane, Pitman and Yor (Section 8.2.3): a positive density that is a signed sum of gamma densities. On the logarithmic scale, with the weight $e^{u/2} = (x_n/\pi n^2)^{1/4}$, the two terms have shapes $\frac{9}{4}$ and $\frac{5}{4}$, while the theta function itself contributes $x_n^{1/4}e^{-x_n}$, of shape $\frac{1}{4}$. The next remark shows where the two extra units come from.

REMARK 8.5.6. Let $D = -x\,d/dx$. Integrating by parts, $\int_0^\infty (Df)(x)x^{s/2-1}dx = \frac{s}{2}\int_0^\infty f(x)x^{s/2-1}dx$ whenever the boundary terms vanish, so the operator $2D(D-\frac{1}{2})$ corresponds to the factor $\frac{1}{2}s(s-1)$ of $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(\frac{s}{2})\zeta(s)$. It annihilates $-\frac{1}{2}$ and $\frac{1}{2}x^{-1/2}$, the two terms by which $\omega(x)$ and $x^{-1/2}\omega(1/x)$ differ, and it maps e^{-x} to $(2x^2-3x)e^{-x}$. Hence $[2D(D-\frac{1}{2})\omega](x)$, which is also $2D(D-\frac{1}{2})$ applied to $x^{-1/2}\omega(1/x)$, decays faster than any power at both ends,

$$\xi(s) = \int_0^\infty \big[2D(D-\tfrac{1}{2})\omega\big](x)\,x^{s/2-1}dx \quad \text{for every } s,$$
$$\Phi(u) = 2e^{u/2}\big[2D(D-\tfrac{1}{2})\omega\big](e^{2u}),$$

the first for $\operatorname{Re} s > 1$ by the integration by parts and then for every s by analytic continuation, and the second by $x = e^{2u}$. So the gamma shapes of the terms $e^{-\pi n^2 x}$ of ω, one in x and $\frac{1}{4}$ on the logarithmic scale, are raised by one and by two, to the Gamma(2) and Gamma(3) of the Pólya kernel, by the operator that removes the poles of ζ. For Δ, whose L function is entire, no such operator is needed and the kernel keeps the single shape 6; the cancellation between the Gamma(2) and Gamma(3) terms of Φ is replaced by the positive multiplier $F(e^u)$.

At real points the Gamma(2) shape enters in a different way, through Kristiansen's theorem [129]: every scale mixture of Gamma(2) laws, the law of G_2V with $V \geq 0$ independent of G_2, is infinitely divisible. Since $G_a \overset{d}{=} G_2B$ for $0 < a < 2$, with B a beta variable of parameters a and $2-a$ independent of G_2, the scale mixtures of Gamma(a) laws with $a \leq 2$ are scale mixtures of Gamma(2) laws.

PROPOSITION 8.5.7. *For real $a > 0$ let $W_a = G_a/(2\pi M^*)$, with G_a independent of M^*. Then W_a is a scale mixture of Gamma(a) laws, the mixing law being that of $1/(2\pi M^*)$, and $L(a-\frac{11}{2},\Delta) = \Pr(W_a > 1)$. For $0 < a \leq 2$, that is $-\frac{11}{2} < s \leq -\frac{7}{2}$, W_a is infinitely divisible. In the same way $L(s,\chi_{12}) = \Pr\big(12G_{s/2}/(\pi M_1) > 1\big)$ for real $s > 0$, where $12G_{s/2}/(\pi M_1)$ is infinitely divisible for $s \leq 4$, and is a mixture of exponential laws for $s \leq 2$, in particular at the centre $s = \frac{1}{2}$.*

Proof. The identities are Corollary 8.5.3 and its analogue for M_1. For $a \leq 2$, $W_a = G_2 \cdot B/(2\pi M^*)$ is a scale mixture of Gamma(2) laws, infinitely divisible by Kristiansen's theorem. For $s \leq 2$, $G_{s/2} \overset{d}{=} G_1B'$ with B' a beta variable of parameters $\frac{s}{2}$ and $1-\frac{s}{2}$ (or $B' = 1$ at $s = 2$), so $12G_{s/2}/(\pi M_1)$ is a scale mixture

of exponential laws, which is infinitely divisible by the theorem of Goldie and Steutel [80, 244]. □

At $a = 2$ the proposition reads $L(-\frac{7}{2}, \Delta) = \Pr(W_2 > 1) = \mathbb{E}\big[(1 + 2\pi M^*)e^{-2\pi M^*}\big] = 0.146374542091266$, the Gamma(2) survival function averaged over $2\pi M^*$ (Table 8.17). The proposition says nothing about zeros: the representation $\Pr(W_a > 1)$ holds for real a only, since on the critical line the shape $6 + i\theta$ is not that of a probability law; for Δ the centre $a = 6$ lies outside Kristiansen's range, while for χ_{12} the centre shape $\frac{1}{4}$ lies inside it. The two Gamma(2) structures are also different in kind: the one in the Pólya kernel is a kernel shape forced by the poles of ζ, which governs the integrand on the critical line, and the one in Proposition 8.5.7 is a mixing law, which governs a value at a real point.

8.5.4.4 The law of M_1

The law of M_1 is explicit. Put $b_n = \pi(n^2 - 1)/12$; the pentagonal number theorem of Section 8.5.1 reads $F_1(y) = \sum_{n\geq 1} \chi_{12}(n) e^{-b_n y}$, so the rates in this expansion are $b_{6k\pm1} = \pi k(3k \pm 1)$, 2π times the generalised pentagonal numbers $k(3k \pm 1)/2$.

PROPOSITION 8.5.8. *For $\theta > 0$ let $w = \sqrt{1 - 12\theta/\pi}$, positive or purely imaginary. Then*

$$\mathbb{E}e^{-\theta M_1} = \frac{\pi(1 - w^2)\sin(\pi w/3)}{2\sqrt{3}\, w \cos(\pi w/2)} = \prod_{k\geq 1} \frac{1 + \theta/b_{6k}}{(1 + \theta/b_{6k-1})(1 + \theta/b_{6k+1})}.$$

Proof. Since $\sum_n \theta/(\theta + b_n) < \infty$, $\theta e^{-\theta y} F_1(y)$ can be integrated term by term, and $\theta + b_n = \pi(n^2 - w^2)/12$ gives $\mathbb{E}e^{-\theta M_1} = \sum_{n\geq 1} \chi_{12}(n)\theta/(\theta + b_n) = (1 - w^2)\sum_{n\geq 1} \chi_{12}(n)/(n^2 - w^2)$. The function $g(w) = \pi \sin(\pi w/3)/(2\sqrt{3}\, w \cos(\pi w/2))$ is even and meromorphic, its only poles are simple and at the odd integers prime to 3, and its residue at $w = n$ is $-\chi_{12}(n)/(2n)$, as for the series, so their difference is entire. On the circles $|w| = 2k$, $|\cos(\pi w/2)|$ is bounded below, so $g = O(1/k)$ there, and the series is $O(k^{-1}\log k)$; by the maximum modulus principle the difference vanishes. For the product, insert $\sin(\pi w/3) = \frac{\pi w}{3}\prod_{m\geq 1}(1 - \frac{w^2}{9m^2})$ and $\cos(\pi w/2) = \prod_{n \text{ odd}}(1 - \frac{w^2}{n^2})$. The factors with $n = 3m$, m odd, cancel, $1 - w^2$ cancels the factor $n = 1$, and what remains is a constant times the factors $1 - w^2/n^2$ with $n = 6k$ above and $n = 6k \pm 1$ below. The value 1 at $\theta = 0$ fixes the constant, and $(1 - w^2/n^2)/(1 - 1/n^2) = 1 + \theta/b_n$. □

At $\theta = \pi/12$, where $w = 0$, the closed form gives $\mathbb{E}e^{-\pi M_1/12} = \pi^2/(6\sqrt{3})$, which is $L(2, \chi_{12})$ by Section 8.5.4.2; the series in the proof reduces to $\sum_n \chi_{12}(n) n^{-2}$ there.

COROLLARY 8.5.9 (A Rényi type identity). *$M_1 \stackrel{d}{=} \sum_{k\ge1} Y_k$ with independent Y_k, where Y_k is exponential of rate b_{6k-1} with probability p_k and of rate b_{6k+1} with probability $1-p_k$, and*

$$p_k = \Big(1-\frac{b_{6k-1}}{b_{6k}}\Big)\frac{b_{6k+1}}{b_{6k+1}-b_{6k-1}} = \frac{(12k-1)(3k+1)}{2(36k^2-1)},$$

so $p_1 = \frac{22}{35}$, $p_2 = \frac{161}{286}$, and $p_k \downarrow \frac12$.

Proof. For $\alpha<\gamma<\beta$ the partial fractions $\frac{1+\theta/\gamma}{(1+\theta/\alpha)(1+\theta/\beta)} = p\frac{\alpha}{\alpha+\theta}+(1-p)\frac{\beta}{\beta+\theta}$ hold with $p=(1-\alpha/\gamma)\beta/(\beta-\alpha)$ and $1-p=\alpha(\beta-\gamma)/(\gamma(\beta-\alpha))$, both in $(0,1)$; take $\alpha=b_{6k-1}=\pi k(3k-1)$, $\gamma=b_{6k}=\pi(36k^2-1)/12$ and $\beta=b_{6k+1}=\pi k(3k+1)$ in each factor of Proposition 8.5.8, and use the continuity theorem for Laplace transforms. □

By Frullani's integral, as in the first reading of Section 1.3.1, the product gives

$$-\log \mathbb{E}e^{-\theta M_1} = \int_0^\infty \big(1-e^{-\theta t}\big)\frac{k_1(t)}{t}\,dt, \quad k_1(t) = \sum_{k\ge1}\big(e^{-b_{6k-1}t}+e^{-b_{6k+1}t}-e^{-b_{6k}t}\big),$$

so M_1 is infinitely divisible, with no drift and Lévy density $k_1(t)/t$; this also follows from Corollary 8.5.9, since mixtures of exponential laws are infinitely divisible [80, 244]. The Lévy density is completely monotone, since $(e^{-b_{6k-1}t}-e^{-b_{6k}t})/t = \int_{b_{6k-1}}^{b_{6k}} e^{-zt}dz$, so M_1 lies in the Bondesson class BO of Section 1.3.2. It is not a GGC: its Thorin measure would have to be $U_1=\sum_k(\delta_{b_{6k-1}}+\delta_{b_{6k+1}}-\delta_{b_{6k}})$, negative at the points b_{6k} where χ_{12} vanishes between $6k-1$ and $6k+1$. Concretely, the transform vanishes at $\theta=-b_{6k}$, where $w=6k$, so $-(\log\mathbb{E}e^{-\theta M_1})'$ has residue -1 there, and if M_1 were a GGC the Stieltjes inversion (1.3.2) of this function would give $U(\{b_{6k}\})=-1$. The poles of the transform are at $\theta=-b_n$, $n\equiv\pm1 \pmod 6$, $n\ge5$, and its zeros at $-b_{6k}$: each zero lies between its own two poles $-b_{6k\pm1}$, but zeros and poles do not alternate, since $-b_7$ and $-b_{11}$, for instance, are consecutive poles. The next lemma, the Frullani identity behind Lemma 6.2.5 and the cosine dictionary (6.2.4) read for a signed Thorin measure, turns this sign structure into a positivity statement.

LEMMA 8.5.10. *Let $X\ge0$ have Laplace exponent $-\log\mathbb{E}e^{-\theta X} = \int_{(0,\infty)}\log(1+\theta/z)\,U(dz)$ for a signed measure U with $\int\log(1+1/z)\,|U|(dz)<\infty$, and put $N(z)=U((0,z])$, with $N(z)\log(1+u^2/z^2)\to0$ as $z\to\infty$ for real u. Then $-\log\mathbb{E}e^{-\theta X} = \int_0^\infty(1-e^{-\theta t})\nu(t)\,dt$ for $\theta\ge0$, with the signed function $\nu(t) = t^{-1}\int e^{-zt}U(dz) = \int_0^\infty e^{-zt}N(z)\,dz$, and for real u*

$$\begin{aligned} -\log\big|\mathbb{E}e^{iuX}\big|^2 &= \int_0^\infty 4\sin^2\Big(\frac{ut}{2}\Big)\nu(t)\,dt \\ &= \int\log\Big(1+\frac{u^2}{z^2}\Big)U(dz) = \int_0^\infty N(z)\,\frac{2u^2}{z(z^2+u^2)}\,dz. \end{aligned} \tag{8.5.6}$$

The law of X is a GGC if and only if $U \geq 0$, and it lies in BO, that is, it is infinitely divisible with a completely monotone Lévy density, if and only if $N \geq 0$.

Proof. Frullani's integral $\log(1+\theta/z) = \int_0^\infty (1-e^{-\theta t})e^{-zt}t^{-1}dt$ gives the Lévy form, the exchange of integrals being justified by $\log(1+\theta/z) \leq C_\theta \log(1+1/z)$, and integrating e^{-zt} against dN by parts gives the second expression of ν. The first equality of (8.5.6) is twice the real part of the Lévy form at $\theta = -iu$, since $1-\cos ut = 2\sin^2(ut/2)$; the second is the Frullani identity $\int_0^\infty 4\sin^2(ut/2)e^{-zt}t^{-1}dt = \log(1+u^2/z^2)$; the third is an integration by parts, with $\frac{d}{dz}\log(1+u^2/z^2) = -2u^2/(z(z^2+u^2))$, whose boundary terms vanish by the assumptions. The signed function ν is determined by the law of X, since $t\nu(t)$ is the inverse Laplace transform of the derivative of $-\log \mathbb{E}e^{-\theta X}$, and so is the signed measure U, by the Stieltjes inversion of $\int U(dz)/(\theta+z)$. If $U \geq 0$ the law is a GGC, and if it is a GGC, its Laplace exponent grows more slowly than θ, so it has no drift, and its Thorin measure is U. If $N \geq 0$, ν is completely monotone and the law is in BO; if the law is in BO, its Lévy density is $\int e^{-zt}\mu(dz)$ with $\mu \geq 0$, it equals ν, and the uniqueness of Laplace transforms gives $\mu(dz) = N(z)\,dz$, so $N \geq 0$, N being right continuous. □

For M_1 the Thorin measure is $U_1 = \sum_{n\geq 2} a(n)\delta_{b_n}$ with weights of sine squared form,

$$a(n) = \sin^2\frac{\pi n}{2} + \frac{4}{3}\sin^2\frac{\pi n}{3} - 1 = \mathbf{1}\{2 \nmid n\} - \mathbf{1}\{3 \mid n\} = \frac{1}{6} - \frac{1}{2}\cos \pi n - \frac{2}{3}\cos\frac{2\pi n}{3},$$

equal to 1 for $n \equiv \pm 1 \pmod 6$, to -1 for $6 \mid n$, and to 0 otherwise; each sine square is a combination of additive characters of $\mathbb{Z}/2$ or $\mathbb{Z}/3$, which makes Poisson summation available.

Proposition 8.5.11. *The canonical function $k_1(t) = t\nu_1(t)$ of M_1 has the theta form*

$$k_1(t) = \tfrac{1}{2}e^{\pi t/12}\Big(1 + \frac{\vartheta_0 - 3\vartheta_{1/2} - 4\vartheta_{1/3}}{\sqrt{3t}}\Big) - 1, \qquad \vartheta_\alpha = \sum_{m\in\mathbb{Z}} e^{-12\pi(m+\alpha)^2/t},$$

so $\nu_1(t) = (2\sqrt{3})^{-1}t^{-3/2} - (2t)^{-1} + O(t^{-1/2})$ as $t \downarrow 0$ and $\nu_1(t) \sim e^{-2\pi t}/t$ as $t \to \infty$. Its Bondesson function $N_1(z) = U_1((0,z])$ is nonnegative, and $N_1(z) \sim \frac{1}{3}\sqrt{3z/\pi}$ as $z \to \infty$.

Proof. Since a is even, $a(0) = -1$ and $a(1) = 1$,

$$k_1(t) = \sum_{n\geq 2} a(n)e^{-b_n t} = \tfrac{1}{2}e^{\pi t/12}\Big(1 + \sum_{n\in\mathbb{Z}} a(n)e^{-\pi n^2 t/12}\Big) - 1.$$

Poisson summation, $\sum_{n\in\mathbb{Z}} \cos(2\pi n\alpha)e^{-\pi n^2 v} = v^{-1/2}\sum_{m\in\mathbb{Z}} e^{-\pi(m+\alpha)^2/v}$ with $v = t/12$, applied to the three terms of the cosine form of $a(n)$, gives the theta form.

As $t \downarrow 0$, $\vartheta_0 = 1 + O(e^{-12\pi/t})$, while $\vartheta_{1/2}$ and $\vartheta_{1/3}$ are $O(e^{-4\pi/(3t)})$, so $k_1(t) = (2\sqrt{3t})^{-1} - \frac{1}{2} + O(\sqrt{t})$; as $t \to \infty$ the first term $e^{-b_5 t} = e^{-2\pi t}$ of the series dominates. Since $b_{6k-1} < b_{6k} < b_{6k+1}$, every negative atom of U_1 is preceded by its own positive atom, so $N_1 \geq 0$. Up to z there are about $X/3$ integers $n \equiv \pm 1 \pmod 6$ and $X/6$ multiples of 6 with $b_n \leq z$, where $X = \sqrt{12z/\pi + 1}$, so $N_1(z) = X/6 + O(1) \sim \frac{1}{3}\sqrt{3z/\pi}$; this agrees with the expansion at 0, since $\nu_1(t) = \int_0^\infty e^{-zt} N_1(z)\, dz$ and $c\sqrt{z}$ gives $c\,\Gamma(\frac{3}{2}) t^{-3/2}$. □

The Lévy density has the index $\frac{1}{2}$ of a Brownian first passage time, matching $-\log \mathbb{E}e^{-\theta M_1} \sim \sqrt{\pi\theta/3}$, which the lower tail $F_1(y) \sim y^{-1/2} e^{-\pi/(12y)}$ gives; this index comes from modularity. By (8.5.6), the modulus of the characteristic function is

$$\left|\mathbb{E}e^{iuM_1}\right|^{-2} = \prod_{k\geq 1} \frac{(1 + u^2/b_{6k-1}^2)(1 + u^2/b_{6k+1}^2)}{1 + u^2/b_{6k}^2} = \exp \int_0^\infty N_1(z)\, \frac{2u^2\, dz}{z(z^2 + u^2)},$$

the exponential of a positive kernel integrated against a nonnegative function. Figure 8.15 (left) shows N_1 against its asymptote.

The sine squared form of $a(n)$ splits U_1 into two positive measures: $\sin^2(\pi n/2)$ selects the odd n, and $1 - \frac{4}{3}\sin^2(\pi n/3)$ the multiples of 3. Each is the Thorin measure of a GGC.

THEOREM 8.5.12. *With independent standard exponential variables E_n and E'_m, let $S = \sum_{n\ odd,\, n\geq 3} E_n/b_n$ and $Z = \sum_{m\geq 1} E'_m/b_{3m}$, with Z independent of M_1. Then S and Z are GGCs,*

$$\mathbb{E}e^{-\theta S} = \frac{\pi}{4}\, \frac{1 - w^2}{\cos(\pi w/2)}, \qquad \mathbb{E}e^{-\theta Z} = \frac{\sqrt{3}\, w/2}{\sin(\pi w/3)},$$

and $M_1 + Z \stackrel{d}{=} S$. Equivalently $U_1 = U_S - U_Z$, with $U_S = \sum_{n\ odd,\, n\geq 3} \delta_{b_n}$ and $U_Z = \sum_{m\geq 1} \delta_{b_{3m}}$, the atoms at b_{3m} with m odd cancelling.

Proof. With $1 + \theta/b_n = (n^2 - w^2)/(n^2 - 1)$, the product $\cos(\pi w/2) = \prod_{n\ \mathrm{odd}}(1 - w^2/n^2)$ gives $\prod_{n\ \mathrm{odd},\, n\geq 3}(1 + \theta/b_n) = \frac{4}{\pi}\cos(\pi w/2)/(1 - w^2)$, the constant $\frac{\pi}{4} = \prod_{n\ \mathrm{odd},\, n\geq 3}(1 - n^{-2})$ being the limit at $w = 1$; and $\sin(\pi w/3)/(\pi w/3) = \prod_m (1 - w^2/9m^2)$ gives $\prod_m (1 + \theta/b_{3m}) = \frac{2}{\sqrt{3}}\sin(\pi w/3)/w$. The quotient of the two transforms is the closed form of Proposition 8.5.8, so $\mathbb{E}e^{-\theta(M_1+Z)} = \mathbb{E}e^{-\theta S}$. □

The rates are shifted eigenvalues of Brownian hitting times [28]. With $\lambda_j = \pi^2(2j-1)^2/8$, the rates of the exit time C of Brownian motion from $(-1,1)$, whose transform is $1/\cosh\sqrt{2\theta} = \prod_j (1 + \theta/\lambda_j)^{-1}$, $b_{2j-1} = \frac{2}{3\pi}(\lambda_j - \lambda_1)$, so $S \stackrel{d}{=} \frac{3\pi}{2}\sum_{j\geq 2} E_j/(\lambda_j - \lambda_1)$: the variable $C = \sum_j E_j/\lambda_j$ with its ground mode $j = 1$ removed and the others shifted by λ_1. The law of C tilted by e^{cC}, $c < \lambda_1$, is

that of $\sum_j E_j/(\lambda_j - c)$, so $\frac{2}{3\pi}S$ is its limit at the critical tilt $c \uparrow \lambda_1$ with the diverging ground mode removed; the tilt itself has no limit, since $\mathbb{E}e^{\lambda_1 C} = \infty$. With $\mu_m = \pi^2 m^2/2$, the rates of the first passage T of the three dimensional Bessel process from 0 to 1, whose transform is $\sqrt{2\theta}/\sinh\sqrt{2\theta}$, $b_{3m} = \frac{3}{2\pi}(\mu_m - \frac{\pi^2}{18})$, so $Z \stackrel{d}{=} \frac{2\pi}{3}T^{(\pi^2/18)}$, where $T^{(c)}$ has the law of T tilted by e^{cT}, a proper law since $\frac{\pi^2}{18} < \mu_1$. The Kolmogorov law is the case with nothing to divide by: comparing transforms, $M_4 = \frac{\pi}{2}T$ exactly. By Kent's theorem with Krein's inverse spectral theory (Section 8.1.3), S and Z, GGCs without drift whose Thorin measures are unit atoms at distinct points, are hitting times of levels by generalised diffusions started at a reflecting or entrance end, so

$$M_1 + \big(\text{a tilted Bessel first passage time}\big) \stackrel{d}{=} \big(\text{the reduced Brownian exit time } S\big),$$

and the transform of M_1 is the quotient of the transforms of two hitting times; the character χ_{12} enters only through the quotient. M_1 itself is not a hitting time from such an end, since those are GGCs. Every first hitting time of a one dimensional diffusion lies in BO [123], as M_1 does, and from an interior point the transform is a quotient of the kind discussed in Section 11.1.4; whether M_1 is the hitting time of some level by some diffusion started at an interior point we do not know.

At $\theta = 2$ and $1 + 3i$, `maxlaw.py` finds the closed form of Proposition 8.5.8, the product and the quadrature of F_1 in agreement to 25 digits, for instance $\mathbb{E}e^{-2M_1} = 0.6942503501538752318025$, and the same for S and Z, with $\mathbb{E}e^{-\theta M_1}\mathbb{E}e^{-\theta Z}/\mathbb{E}e^{-\theta S} - 1$ below 3×10^{-41}. At $u = 5$, $-\log|\mathbb{E}e^{iuM_1}|^2 = 0.40295014525377167405$ from the closed form, from the law of M_1 and from the Thorin sum $\int \log(1 + u^2/z^2)U_1(dz)$, and to 18 digits from the integral of N_1 in (8.5.6); N_1 is nonnegative over the first 60000 atoms, where it reaches 20000 against the asymptote 20000.17; and the theta form of k_1 agrees with the series to 20 digits at $t = 0.01$, 0.05, 0.1, 0.3, 1, 2 and 3. At $\theta = 2$ the derivative $-(\log\mathbb{E}e^{-\theta M_1})' = \int U_1(dz)/(\theta + z)$ is 0.16796840787856222484 in closed form and as the sum over the Thorin atoms, and $\int N_1(z)(2 + z)^{-2}dz$ agrees to 2×10^{-15}, the error of truncating the integral; the Laplace exponent at $\theta = 2$ is 0.36492264843255339528 in closed form and as the Lévy integral $\int_0^\infty (1 - e^{-2t})\nu_1(t)\,dt$. The expansion of ν_1 checks too: $2t(\nu_1(t) - (2\sqrt{3})^{-1}t^{-3/2})$ is -0.9985, -0.99985 at $t = 10^{-4}$, 10^{-6}, and $t\nu_1(t)e^{2\pi t}$ is 0.99685, 0.9999994 at $t = 2$, 5 (`maxlaw.py`, `maxexp_checks.py`).

8.5.4.5 The law of M^*

The law of M^* has no such product, but its transform $\psi^*(\theta) = \mathbb{E}e^{-\theta M^*} = \theta\int_0^\infty e^{-\theta y}F(y)\,dy$ is meromorphic: since $F(y) = \sum_{n\ge1}\tau(n)e^{-2\pi(n-1)y}$, for any $c > 0$

$$\psi^*(\theta) = \theta\int_0^c e^{-\theta y}F(y)\,dy + \theta\sum_{n\ge1}\tau(n)\frac{e^{-(\theta+2\pi(n-1))c}}{\theta + 2\pi(n-1)},$$

with at most simple poles at $\theta = -2\pi m$, $m \geq 1$, of residue $-2\pi m\,\tau(m+1)$. Just to the right of a pole ψ^* has the sign of the residue and just to the left the opposite sign, so it has a real zero between $-2\pi(m+1)$ and $-2\pi m$ whenever $\tau(m+1)\tau(m+2) > 0$. The first such pair is $\tau(6) = -6048$, $\tau(7) = -16744$, and `maxlaw.py`, with $c = 0.3$ at 40 digits, finds the zero at $\theta = -34.4840085134049704852 = -2\pi \times 5.48830041253$. It is the only sign change of ψ^* on $(-15\pi, 0)$, and the argument principle on the rectangle $|\operatorname{Re}\theta| \leq 15\pi$, $|\operatorname{Im}\theta| \leq 60$, which contains the seven poles $-2\pi, \ldots, -14\pi$, gives winding number -6, so it is the only zero there. As for M_1, the zero shows that M^* is not a GGC.

Farther out the zeros leave the real axis. The script `maxexp_psizeros.py` evaluates ψ^* from the cut form with $c = 0.08$, the integral by Gauss Legendre quadrature and the series to $n = 900$, at 230 bits, and checks it against $c = 0.05$ to a relative 10^{-55}. It then certifies the zeros in ball arithmetic, with $c = \frac{3}{100}$, the series to $n = 1500$, its tail bounded by Deligne's bound $|\tau(n)| \leq d(n)n^{11/2} \leq 2n^6$, and the integral bounded by $F(y) \leq y^{-12}e^{2\pi y - 2\pi/y}$, which is increasing.

PROPOSITION 8.5.13. *1. For each of the* 25 *values* $m \leq 60$ *with* $\tau(m+1)\tau(m+2) > 0$, ψ^* *changes sign in an interval of length* 2×10^{-20} *inside the gap* $(-2\pi(m+1), -2\pi m)$*; the zeros begin* $\theta/2\pi = -5.48830041$, -8.66373325, -11.4203774, -13.2533839, -14.7973058, *and the last is* -60.7670708.

2. ψ^* *has exactly one zero in the disc of radius* 10^{-7} *about each of the ten points*

$$
\begin{aligned}
&-155.5940822775511402804 \pm 2.7631188861212609656\,i,\\
&-215.6211277570694988944 \pm 2.9647292148515237348\,i,\\
&-307.4008549057072052743 \pm 3.2283287024583906394\,i,\\
&-322.6135369899940182007 \pm 3.0449254921259967464\,i,\\
&-347.8816009468099010890 \pm 2.7171007701068929577\,i,
\end{aligned}
$$

and it lies within 10^{-19} *of that point.*

Proof. For (1), the values of ψ^* at the two ends of each interval are balls of opposite signs. For (2), let z_0 be the approximation to the zero computed by Newton's method at 230 bits; the point shown is z_0 rounded to nineteen decimals, at distance at most 6.1×10^{-20} from it. With $A \geq |\psi^*(z_0)|$, $B \leq |\psi^{*\prime}(z_0)|$ and $C \geq |\psi^{*\prime\prime}|$ on the square of half side r about z_0, each computed as a ball, $A + 2Cr^2 < Br/2$ holds for $r = 10^{-6}$ and for $r = 10^{-29}$. On the circle $|\theta - z_0| = r$, inside the square, $|\psi^*(\theta) - \psi^{*\prime}(z_0)(\theta - z_0)| \leq A + Cr^2/2 < Br$, so by Rouché's theorem ψ^* has as many zeros in the disc $|\theta - z_0| < r$ as $\psi^{*\prime}(z_0)(\theta - z_0)$, namely one. The disc of radius 10^{-7} about the point shown lies in the disc of radius 10^{-6} about z_0, and the disc of radius 10^{-29} about z_0 lies within 10^{-19} of the point shown. The zeros come in conjugate pairs because ψ^* is real on the real axis. □

The argument principle on the rectangles $|\operatorname{Re}\theta| \le 2\pi(k+\frac12)$, with $|\operatorname{Im}\theta| \le 30$ for $k = 15$, 25, 40 and 60 and $|\operatorname{Im}\theta| \le 100$ for $k = 60$, and on a grid of cells, 2π wide and at most 5 high, covering $-2\pi \times 60.5 \le \operatorname{Re}\theta \le 2\pi \times 5.5$, $\frac12 \le \operatorname{Im}\theta \le 30$, finds exactly these zeros: the winding number on the largest rectangle, which contains the 60 poles $-2\pi, \dots, -120\pi$, is -26, so it contains 34 zeros, the 24 real zeros with $m \le 59$ and the ten nonreal zeros of the proposition. This count, unlike the zeros themselves, is computed and not certified: the change of argument is followed along the contours with steps halved until it is below 0.3, in ball arithmetic whose radii stay below a relative 10^{-46}. Every real zero lies in a gap where the residues force it, every gap with $\tau(m+1)\tau(m+2) < 0$ is free of real zeros, and the five nonreal pairs have their real parts in such gaps, those with $m = 24$, 34, 48, 51 and 55 ($\operatorname{Re}\theta/2\pi = -24.764$, -34.317, -48.924, -51.346, -55.367), at heights between 2.72 and 3.23 (Figure 8.15, right). In a gap where the residues have opposite signs, ψ^* has the same sign at both ends, so its real zeros there come in pairs; the computation finds none, and the five nonreal pairs sit in such gaps, as if a pair of real zeros had collided and left the axis.

Corollary 8.5.14. *M^* is not in the Bondesson class BO; in particular it is not a GGC.*

Proof. If M^* were in BO, its Lévy density would be $\int e^{-zt}\mu(dz)$ with $\mu \ge 0$, so $-(\log\psi^*)'(\theta) = d + \int \mu(dz)(\theta+z)^{-2}$, with $d \ge 0$ the drift, for $\theta > 0$. The right side is holomorphic on $\mathbb{C} \setminus (-\infty, 0]$, since $\int \mu(dz)(1+z)^{-2} < \infty$ and $|\theta + z|$ is bounded below by a multiple of $1 + z$ for θ in a compact subset of $\mathbb{C} \setminus (-\infty, 0]$. Then the meromorphic function $\psi^{*\prime}/\psi^*$ would have no poles in $\mathbb{C} \setminus (-\infty, 0]$, and ψ^* no zeros there, against Proposition 8.5.13(2). □

Nor does M^* have a signed Thorin measure in the sense of Lemma 8.5.10: for a signed measure U on $(0,\infty)$, $-(\log\psi^*)'(\theta) = \int U(dz)(\theta+z)^{-1}$ would again be holomorphic on $\mathbb{C} \setminus (-\infty, 0]$. Unit atoms at the points $2\pi m$ with $\tau(m+1) \ne 0$, less unit atoms at the real zeros, account for the poles and the real zeros of ψ^*, but no measure on the positive axis accounts for the nonreal zeros. Whether M^* is infinitely divisible remains open. If it is, with Lévy measure ν^*, then $-(\log\psi^*)'(\theta) = d + \int t\, e^{-\theta t}\nu^*(dt)$ is completely monotone on $(0,\infty)$: the cumulants $\kappa_j(\theta) = (-1)^j(\log\psi^*)^{(j)}(\theta)$ of M^* tilted by $e^{-\theta M^*}$ are all positive. At $\theta = 0.5$, 2, 8 and 30 the first eight are positive, for instance 0.54374, 0.025649, 0.0045870, 0.0014574 for $j \le 4$ at $\theta = 2$, as the first author's note found. As $\theta \to \infty$ the Laplace method, at the saddle point $y_0 = \sqrt{2\pi/\theta}$ of $e^{-\theta y - 2\pi/y}$ and with the lower tail $F(y) \sim y^{-12}e^{-2\pi/y}$, gives

$$\psi^*(\theta) \sim 2^{-1/2}(2\pi)^{-21/4}\,\theta^{25/4}\,e^{-2\sqrt{2\pi\theta}},$$

so $-\log\psi^*(\theta) = 2\sqrt{2\pi\theta} - \frac{25}{4}\log\theta + \frac{21}{4}\log 2\pi + \frac12\log 2 + o(1)$; the ratio of ψ^* to the right side is 2.33, 1.32 and 1.094 at $\theta = 10^3$, 10^4 and 10^5, consistent

with a correction of order $\theta^{-1/2}$. If M^* is infinitely divisible, it has no drift, and Karamata's Tauberian theorem turns the leading term into $\nu^*((t,\infty)) \sim 2\sqrt{2}\,t^{-1/2}$ as $t \downarrow 0$, the behaviour of the Lévy measure $\sqrt{2}\,t^{-3/2}dt$ of the first passage time of Brownian motion to the level $2\sqrt{\pi}$.

The first author conjectures that every maximum of independent exponential variables is infinitely divisible [204]; M_1, M_4, M^* and M_{11} are of this kind. For finitely many rates $\lambda_1,\dots,\lambda_n$ the transform is rational: with $\lambda_S = \sum_{i\in S}\lambda_i$,

$$\mathbb{E}e^{-\theta M} = \theta\int_0^\infty e^{-\theta y}\prod_{i\le n}(1-e^{-\lambda_i y})\,dy = \sum_{S\neq\emptyset}(-1)^{|S|+1}\frac{\lambda_S}{\theta+\lambda_S},$$

with simple poles at the values $-\lambda_S$ whose grouped coefficients do not vanish, generically 2^n-1 of them, and, since $\mathbb{E}e^{-\theta M} \sim n!\,\lambda_1\cdots\lambda_n\theta^{-n}$, with n fewer zeros. Writing the poles as $-p$ and the zeros as $-z$, Frullani's integral gives $-\log\mathbb{E}e^{-\theta M} = \int_0^\infty(1-e^{-\theta t})k(t)t^{-1}dt$ with $k(t) = \sum_p e^{-pt} - \sum_z e^{-zt}$, as long as every zero has $\operatorname{Re} z > 0$, and by the uniqueness of the Lévy measure the law is infinitely divisible if and only if $k \ge 0$. In particular infinite divisibility forces every zero to have $\operatorname{Re} z$ at least the smallest pole $p_{\min}$, since otherwise k would change sign at large t. For two rates $a<b$ the transform is $ab(a+b+2\theta)/((a+\theta)(b+\theta)(a+b+\theta))$ and the canonical function is $e^{-at}+e^{-bt}+e^{-(a+b)t}-e^{-(a+b)t/2}$, positive by the inequality of arithmetic and geometric means; the signed Thorin measure $\delta_a+\delta_b+\delta_{a+b}-\delta_{(a+b)/2}$ has a nonnegative distribution function, so this maximum is in BO and is not a GGC, and $e^{at}k(t) = 1-x+x^2+x^{2b/(b-a)} > \frac34$ with $x = e^{-(b-a)t/2}$, a bound approached as $b/a \downarrow 1$ (at $b=a$ the zero cancels the pole at $-a$, and $e^{at}k(t) = 1+e^{-at}$). The script `maxexp_checks.py` groups the exponents exactly, finds the zeros at 40 digits and evaluates k at 181 points of $p_{\min}t \in [10^{-3},40]$, for 300 families of random rational rates in $[0.1,5]$ for each $n = 2,3,4,5$ and 40 families with $n=6$. In every family k is positive at every point, $e^{p_{\min}t}k(t) \ge 0.53$, and every zero has $\operatorname{Re} z > p_{\min}$. Nonreal zeros are common: they occur in 140, 222, 266 and 40 of the families with 3, 4, 5 and 6 rates, up to 2, 8, 22 and 52 of them. So nonreal zeros, which exclude BO, do not seem to exclude infinite divisibility. The truncations of M_1, with the rates $2\pi n$, $n \le N$, have only real zeros for $N = 4,\dots,10$, and those of M^*, with each rate $2\pi n$, $n \le N$, taken 24 times, have 24 real zeros for $N=2$ and 72 zeros, 52 of them nonreal, for $N=3$; in each case k is positive on the grid. None of this is a proof.

Question 8.5.15. Is every maximum of independent exponential variables infinitely divisible? In particular, is M^*?

For the clock the relaxation from GGC to BO gains nothing. If its Lévy density $W(t)/t$ were completely monotone, equal to $\int e^{-zt}\mu(dz)$ with $\mu \ge 0$, then the formula $(G'/G)(u) = \int_0^\infty e^{-ut}W(t)\,dt$ of Section 1.2 would give $(G'/G)(u) = \int \mu(dz)(u+z)^{-2}$, holomorphic on $\mathbb{C}\setminus(-\infty,0]$. But G'/G has a pole at every zero $-a_k = (\rho_k-\frac12)^2$ of G, and this point lies in $(-\infty,0]$ only when ρ_k is on the critical

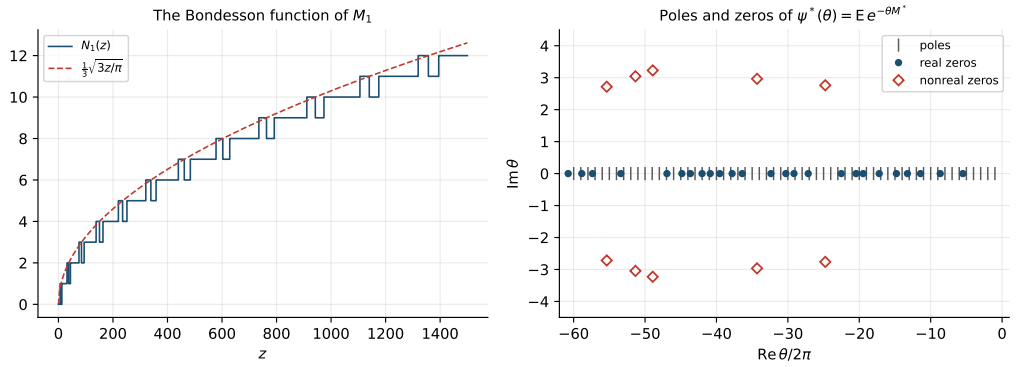


Figure 8.15: Left: the Bondesson function N_1 of M_1, nonnegative, with its asymptote $\frac{1}{3}\sqrt{3z/\pi}$; each drop is a zero of the transform at $-b_{6k}$. Right: the poles $-2\pi m$, $m \le 60$, and the zeros of $\psi^*(\theta) = \mathbb{E}e^{-\theta M^*}$: the 25 real zeros above $-2\pi \times 61$, one in each gap with $\tau(m+1)\tau(m+2) > 0$, and the five pairs of nonreal zeros, the only ones with $|\operatorname{Re}\theta| \le 2\pi \times 60.5$ and $|\operatorname{Im}\theta| \le 100$. Drawn by `maxexp_figs.py` from `maxexp_psizeros.json`.

line. So the clock is in BO if and only if RH holds (Table 1.7). The signed Thorin measures of the max laws come from zeros of their Laplace transforms, and the transform $1/G$ of the clock has none. The two laws of this section show the two ways out of the Thorin class: M_1 leaves it through real zeros, each preceded by its own pole, and stays in BO, while M^* leaves BO too, through zeros off the real axis, the way a zero of ξ off the critical line would take the clock out of both classes.

Table 8.18 collects the checks of this section, in the form of the table of the first author's note, recomputed.

8.5.5 The de Bruijn Newman flow for Δ

The flow of Section 8.4 needs only an even kernel that decays faster than any Gaussian, so it applies to K_Δ:

$$H_t^\Delta(\theta) = \int_0^\infty e^{tu^2} K_\Delta(u)\cos(\theta u)\,du, \qquad 2H_0^\Delta(\theta) = \xi\big(\tfrac{1}{2} + i\theta, \Delta\big).$$

In the variable θ the times of Section 8.4, whose variable is $z = 2\theta$, are divided by four, and the isolated pair estimate for two zeros at distance D is $-D^2/8$. The zeros of H_0^Δ lie in the strip $|\operatorname{Im}\theta| < \frac{1}{2}$, so by de Bruijn's theorem [65] every zero of H_t^Δ is real for $t \ge \frac{1}{8}$; for ζ the same strip gives de Bruijn's bound $\Lambda \le \frac{1}{2}$ in the times of Section 8.4. Dobner's theorem for the extended Selberg class [69] gives the other side: Δ has a de Bruijn Newman constant Λ_Δ, and H_t^Δ has nonreal zeros for every $t < 0$. So $0 \le \Lambda_\Delta \le \frac{1}{8}$, and GRH for Δ is the statement $\Lambda_\Delta = 0$.

quantity	max representation	check
$L(\frac{17}{2},\Delta) = \mathbb{E}\,Q(14, 2\pi M^*)$	0.998583063162746176492	Hecke: same; series to $n = 400$: 18 digits
$L(\frac{29}{2},\Delta) = \mathbb{E}\,Q(20, 2\pi M^*)$	0.999977182799369841175	series: same
$\xi(\frac{1}{2},\Delta)$, symmetric form	0.00154487936039502721	(8.5.3), Hecke: same
(8.5.4), $k = 1$	0.00595896498957823785	$k = 11$: same
(8.5.4), $k = 5$	0.00163398603484069935	$k = 7$: same
$L(\frac{1}{2},\Delta) = \Pr(N(2\pi M^*) \le 5)$	0.792122838646030569	Hecke: same
first zero of $\xi(\frac{1}{2} + i\theta,\Delta)$	9.2223793999211025222	Hecke: same; [149]: 9.222379
first zero of $K_{i\theta}(2\pi)$	9.768770083509977867	moved to 9.2224 by H
$L(1,\chi_{12})$, erfc form	0.760345996300946347531	$\log(2+\sqrt{3})/\sqrt{3}$: same
$L(1,E) = \mathbb{E}e^{-2\pi M_{11}}$	0.253841860855910684338	series of root number one: same
$\mathbb{E}K$ from ϑ_4	0.868731160636159141830	$\sqrt{\pi/2}\,\log 2$: same
$\mathbb{E}e^{-2M_1}$, quadrature	0.694250350153875231803	closed form: same
$\mathbb{E}e^{-3M_4}$, quadrature	0.285645278943791835551	$\sqrt{3\pi}/\sinh\sqrt{3\pi}$: same
zeros of ψ_1	$-b_{6k} = -\frac{36k^2-1}{12}\pi$	closed form vanishes at $k = 1, 2$
$-\log\lvert\mathbb{E}e^{5iM_1}\rvert^2$	0.40295014525377167405	Thorin sum: same; $\int N_1$: 18 digits
$-\psi_1'(2)/\psi_1(2)$	0.16796840787856222484	Thorin sum: same; $\int N_1$: to 2×10^{-15}
$-\log\psi_1(2)$	0.36492264843255339528	Lévy integral: same
first zero of ψ^*	$-2\pi \times 5.48830041253$	certified; the only zero for $\lvert\operatorname{Re}\theta\rvert \le 15\pi$, $\lvert\operatorname{Im}\theta\rvert \le 60$ (computed)
first nonreal zero of ψ^*	$-155.59408228 \pm 2.76311889\,i$	certified

Table 8.18: Checks of Section 8.5.4, by `maxlaw.py`, `maxexp_checks.py` and `maxexp_psizeros.py`, with $\psi_1(\theta) = \mathbb{E}e^{-\theta M_1}$; "same" means agreement to all digits shown. The first six rows are computed from the product for F and the density (8.5.2), independently of the Dirichlet series; the zeros of ψ^* are those of Proposition 8.5.13.

On the critical line the kernel is a probability density up to normalisation, as the eta kernel is in Section 8.5.1: if V has the density $K_\Delta(v)/\xi(\frac12,\Delta)$ on $\mathbb{R}$, then $\mathbb{E}e^{i\theta V} = \xi(\frac12+i\theta,\Delta)/\xi(\frac12,\Delta)$, and GRH for Δ is the statement that this characteristic function has only real zeros. In the language of Section 8.5.4, e^V has the density proportional to $y^5\Delta(iy) = y^5e^{-2\pi y}F(y)$, so V is the logarithm of $G_6/2\pi$ conditioned on $G_6/2\pi \geq M^*$, with G_6 a gamma variable of shape 6 independent of M^*; its law is symmetric by the self duality (8.5.1), with no symmetrisation. The variable V is not infinitely divisible, nor even quasi infinitely divisible, because its characteristic function vanishes at $\theta = 9.2223793999$; this is the argument of Proposition 11.1.3(1) for the Pólya variable. Since $1-\cos x = 2\sin^2(x/2)$ and K_Δ is even,

$$\xi\big(\tfrac12,\Delta\big) - \xi\big(\tfrac12+i\theta,\Delta\big) = 4\int_0^\infty K_\Delta(v)\sin^2\Big(\frac{\theta v}{2}\Big)dv,$$

so the zeros on the critical line are the solutions of $\mathbb{E}[2\sin^2(\theta V/2)] = 1$. At the level of logarithms, the factorisation $\xi(s,\Delta) = (2\pi)^{-s-11/2}\Gamma(s+\frac{11}{2})L(s,\Delta)$ splits $\log|\xi(\frac12+i\theta,\Delta)/\xi(\frac12,\Delta)|^2$ into $\log|\Gamma(6+i\theta)/\Gamma(6)|^2$ and $\log|L(\frac12+i\theta,\Delta)/L(\frac12,\Delta)|^2$. The first has the positive sine squared form given by Binet's formula,

$$\log\Big|\frac{\Gamma(6+i\theta)}{\Gamma(6)}\Big|^2 = -\int_0^\infty 4\sin^2\Big(\frac{\theta u}{2}\Big)\frac{e^{-6u}}{u(1-e^{-u})}\,du,$$

the Lévy Khintchine form of the infinitely divisible variable $\log G_6$, and the second carries every zero. To the right of the critical strip the second has a sine squared form too: for $\sigma > 1$, expanding $\log L(s,f) = \sum_{n\geq2}\Lambda_f(n)n^{-s}/\log n$ gives

$$\log\Big|\frac{L(\sigma+i\theta,f)}{L(\sigma,f)}\Big|^2 = -\sum_{n\geq2}\frac{4\Lambda_f(n)}{n^\sigma\log n}\sin^2\Big(\frac{\theta\log n}{2}\Big),$$

for $f=\chi_{12}$ and Δ, and for ζ with Λ in place of Λ_f. Since $\int_\sigma^\infty n^{-\alpha}d\alpha = n^{-\sigma}/\log n$, the right side is minus twice the integral of the prime piece of the decomposition of Section 8.5.3 over the basepoints $\alpha > \sigma$. For ζ the weights are positive, which is the infinite divisibility of log of the zeta distribution (Section 6.1.6). For Δ they are $8\cos(k\vartheta_p)/(kp^{k\sigma})$ at $n=p^k$, where $\alpha_p = e^{i\vartheta_p}$, $0\leq\vartheta_p\leq\pi$, and they take both signs. The Sato Tate law for Δ, proved by Barnet-Lamb, Geraghty, Harris and Taylor [18], makes the angles ϑ_p equidistributed for the measure $\frac{2}{\pi}\sin^2\vartheta\,d\vartheta$ on $[0,\pi]$, under which $\cos k\vartheta < 0$ with probability $\frac12$ for $k\neq2$ and $\frac12+\frac{1}{\pi}$ for $k=2$. So even where the Euler product converges absolutely the sine squared form of $\log|L(\sigma+i\theta,\Delta)|$ has a signed measure, and on the critical line the series diverges: positivity lives in the kernel and in the gamma factor, not in the arithmetic factor.

The kernel factors as $K_\Delta(u) = e^{-2\pi\cosh u}H(u)$, with $H(u) = \sqrt{F(e^u)F(e^{-u})}$ because $K_\Delta(u)^2 = K_\Delta(u)K_\Delta(-u)$, and $\log H(u) = -12\sum_m\frac{\sigma(m)}{m}(e^{-2\pi me^u} + e^{-2\pi me^{-u}})$. The expansion comes from $-\log F(y) = 24\sum_{m\geq1}\frac{\sigma(m)}{m}e^{-2\pi my}$, so $-\log F$ is completely monotone, the Laplace transform of the positive measure

with mass $24\sigma(m)/m$ at each point $2\pi m$; for η the masses are $\sigma(m)/m$, and every maximum of independent exponential variables has this property, since $-\log(1-e^{-\lambda y}) = \sum_k e^{-k\lambda y}/k$. The transform $K_{i\theta}(2\pi)$ of Pólya's factor has only real zeros [213], the first at $\theta = 9.768770083509977867$ (`maxlaw.py`), and the multiplier H moves it to 9.2223793999. Pólya's theorem on universal factors [65, p. 197], which covers the Gaussian multiplier e^{tu^2} of the flow for $t > 0$, does not apply to H, which is bounded by one and supplies half of the doubly exponential decay of K_Δ. The positivity of the measure of $-\log F$ is automatic and does not by itself decide where the zeros are; whether some multiplier of the same form moves zeros off the line is the following question, and any answer must account for Δ, χ_{12} and the conductor 11 curve, whose kernels have this form. Figure 8.16 shows the two kernels and $\log H$.

QUESTION 8.5.16. For which $a > 0$, $c_m \geq 0$ and $\beta_m > 0$ does the Fourier transform of $e^{-a\cosh u} \exp\big(-\sum_m c_m(e^{-\beta_m e^u} + e^{-\beta_m e^{-u}})\big)$ have only real zeros?

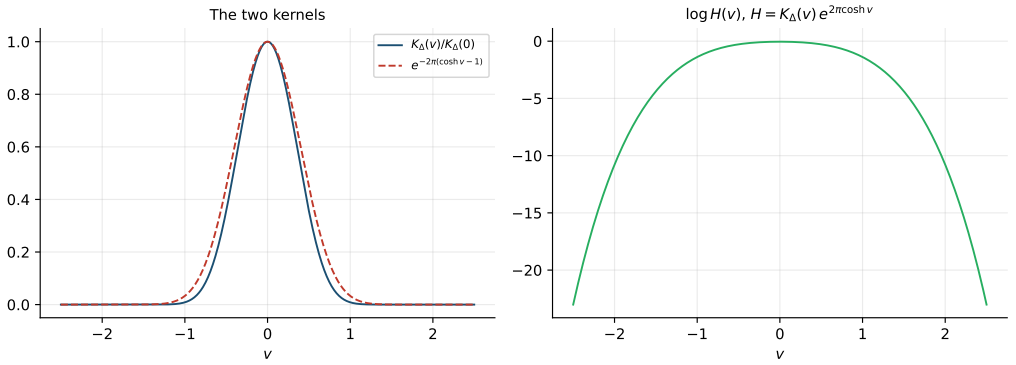


Figure 8.16: Left: the kernel $K_\Delta(v) = e^{6v}\Delta(ie^v)$ and Pólya's kernel $e^{-2\pi\cosh v}$, each divided by its value at 0. Right: the logarithm of the multiplier $H(v) = \sqrt{F(e^v)F(e^{-v})}$; $-\log H$ expands with the positive coefficients $12\sigma(m)/m$. Drawn by `maxexp_figs.py`.

The kernel K_Δ is strictly log concave, as the first author's note [207] shows, and the product formula gives the best constant. For $c > 0$, $\log(1-e^{-ce^u})$ is the logarithm of the distribution function of $\log E$, with E exponential of rate c, and it is strictly concave in u: with $x = ce^u$ its second derivative is $x(e^x - 1 - xe^x)/(e^x-1)^2 < 0$. So $\log F(e^u)$, a sum of such terms, is strictly concave (in the language of Section 8.5.4, $\log M^*$ has a log concave distribution function), and since $\log K_\Delta(u) = 6u - 2\pi e^u + \log F(e^u)$ and K_Δ is even,

$$-\big(\log K_\Delta\big)''(u) > 2\pi e^{|u|}, \qquad u \in \mathbb{R},$$

where the ratio of the two sides tends to 1 as $|u| \to \infty$. The multiplier H is log

concave as well, since $\log H(u) = \frac{1}{2}\big(\log F(e^u) + \log F(e^{-u})\big)$. In the same way $\log K(u) = \log 2 + \frac{u}{2} - \frac{\pi}{12}e^{2u} + \log F_1(e^{2u})$ gives $-(\log K)''(u) > \frac{\pi}{3}e^{2|u|}$ for the kernel of Section 8.5.1. The Pólya kernel Φ is log concave too [53], though it is not a product of this kind. Log concavity alone does not put the zeros of a transform on a line, or in any strip. The function $(1-|u|)_+^2$ is log concave, and its transform $4(\theta - \sin\theta)/\theta^3$ vanishes at the nonzero roots of $\sin\theta = \theta$, which are nonreal, the first at $7.4977 + 2.7687i$, with imaginary parts that grow like $\log|\theta|$. Its convolution with $e^{-\cosh u}$ is even, positive, entire and log concave [233], and decays doubly exponentially, like K_Δ; its transform $8(\theta - \sin\theta)K_{i\theta}(1)/\theta^3$ has the same nonreal zeros.

In the language of Section 8.5.4 the flow is a family of Gaussian tilts: for every real t, $e^{tv^2}K_\Delta(v)$ is integrable, because K_Δ decays doubly exponentially, so after normalisation it is the density of the law of V tilted by e^{tV^2}, and $H_t^\Delta(\theta)/H_t^\Delta(0)$ is its characteristic function. The question $\Lambda_\Delta = 0$ asks whether the smallest tilt whose characteristic function has only real zeros is $t = 0$.

The script `dbn_delta.py` computes H_t^Δ by Gauss Legendre quadrature on $[0, 4]$, beyond which $K_\Delta < 10^{-130}$, with 8 panels of 96 nodes at 40 digits; at $t = 0$ it agrees with 192 nodes per panel and with Hecke's formula to a relative 4×10^{-41} at $\theta = 0$ and 5×10^{-16} at $\theta = 50$. The 21 zeros below height 52, from 9.2224 to 51.1566, stay real for $t = 0.05, 0.1, \ldots, 0.5$ and all move down. Backward in time each pair of neighbours is followed through the critical point of H_t^Δ between them until the critical value changes sign, and each such event is tested by direct sign counts: a collision at (t_*, θ_*) leaves two sign changes on $[\theta_* - 1.2, \theta_* + 1.2]$ at $t_* + 0.005$ and none at $t_* - 0.005$, and the colliding pair is identified by the number of sign changes below $\theta_* - 1.2$, since real zeros keep their order. Table 8.19 lists the seven collisions. The numbers of sign changes on $(0.5, 52]$ at eleven values of t from 0 to -1.5 fall by two at each collision and once by one, between $t = -0.2$ and -0.3, where the highest zero, followed back from 51.1566, passes 52 at $t \approx -0.29$.

pair at $t = 0$	$-D^2/8$	t_*	θ_*
48.3591, 49.2761	−0.1051	−0.12622	49.18
36.7415, 37.7539	−0.1281	−0.15864	37.60
45.0401, 46.1973	−0.1674	−0.24305	46.19
25.2746, 26.8044	−0.2925	−0.40713	27.26
40.2190, 41.7305	−0.2856	−0.48271	42.49
31.1782, 32.7749	−0.3187	−0.48637	33.30
17.4428, 19.6565	−0.6126	−0.98116	21.09

Table 8.19: The collisions of zeros of H_t^Δ below height 52, in the order in which they occur as t decreases: the pair at $t = 0$, the isolated pair estimate with D the gap of the pair, and the time t_* and height θ_* of the collision, computed by `dbn_delta.py`.

The script `maxexp_flow.py` recomputes the table independently, at 70 digits for the kernel and with a different tracking: it evaluates H_t^{Δ} at once on the grid $\theta = 0.5, 0.52, \ldots, 52$, with the cosines kept in fixed point by their three term recurrence, for $t = 0.5, 0.49, \ldots, -1.1$, locates the zeros by inverse interpolation, and refines each fall of the count by two to the time at which the critical value between the closest pair changes sign. It finds the same seven collisions, with the same t_* and θ_* to all digits of the table (for instance $t_* = -0.4071263919$, $\theta_* = 27.257965$), and the highest zero passing 52 between $t = -0.29$ and -0.30. Figure 8.17 shows the trajectories.

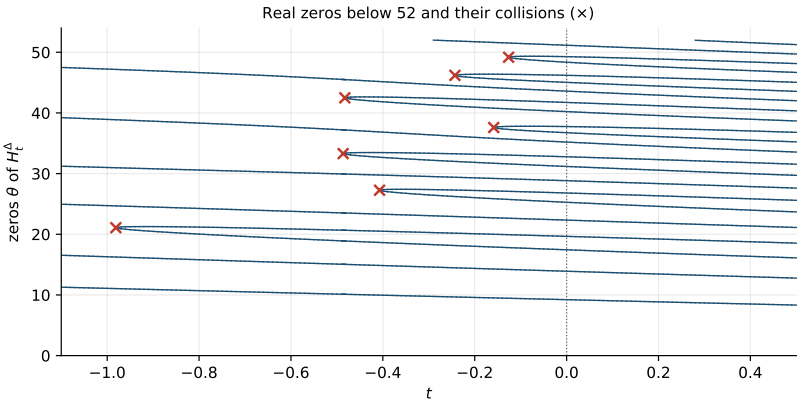


Figure 8.17: The real zeros of H_t^{Δ} below height 52 for $-1.1 \le t \le 0.5$, and the seven collisions of Table 8.19; all zeros move down as t grows. Drawn by `maxexp_figs.py` from `maxexp_flow.json`.

The low zeros of Δ are robust: below height 33, the window of the first author's note, the first pair to leave the line, $(25.27, 26.80)$, does so only at $t = -0.407$. Collisions do not come in order of height, however. The pairs near 37.6 and 49.2 collide at -0.159 and -0.126, and under GRH for Δ Dobner's theorem forces collisions in every interval $(t, 0)$, higher up; so no window of heights decides whether $\Lambda_{\Delta} = 0$ (Dobner's theorem already gives $\Lambda_{\Delta} \ge 0$), and what a computation can show is how fast the collision times approach 0 with height. The neighbours delay every collision beyond the isolated pair estimate, by factors from 1.20 to 1.69. In the variable θ the closest pair of ζ below height 200 collides at -0.072, and Lehmer's pair near height 7005 at -1.8×10^{-4} (Section 8.4.2).

8.5.6 A Pólya kernel for imaginary quadratic fields

Section 8.5.1 left the positivity of the kernels of odd characters to be checked case by case. For the Dedekind zeta function of an imaginary quadratic field the kernel comes from a simple theta series, its positivity reduces to a finite interval, and it

fails for a whole residue class of discriminants, together with the kernel of the odd character. This subsection follows the last part of the first author's note on the squarefree series [206]; the constants of number fields, which these kernels turn into tilted means, are in Section 10.3.3 of Chapter 10.

Let $d<0$ be a fundamental discriminant, $K=\mathbb{Q}(\sqrt{d})$, χ_d the Kronecker symbol of d, a real primitive odd character modulo $|d|$, h the class number of K and w the number of its roots of unity. Then $\zeta_K(s)=\zeta(s)L(s,\chi_d)=\sum_{n\ge1}a_n n^{-s}$, where $a_n=\sum_{m\mid n}\chi_d(m)\ge0$ is the number of ideals of norm n, and we put

$$\Lambda_K(s)=\Big(\frac{\sqrt{|d|}}{2\pi}\Big)^s\Gamma(s)\,\zeta_K(s),\qquad \xi_K(s)=2s(s-1)\Lambda_K(s),$$

the function ξ_K of Section 10.3.3 with $r_1=0$ and $r_2=1$. The duplication formula gives $\xi_K(s)=\frac{2}{\sqrt{|d|}}\,\xi(s)\,\xi(s,\chi_d)$, with $\xi(s,\chi_d)=(|d|/\pi)^{(s+1)/2}\Gamma(\frac{s+1}{2})L(s,\chi_d)$ the completed function of Section 8.5.1, so $\xi_K(s)=\xi_K(1-s)$ because the root number of χ_d is one. Hence $\Lambda_K(s)=\Lambda_K(1-s)$, and Λ_K is meromorphic with simple poles at $s=1$ and $s=0$ only, with residues h/w and $-h/w$, by the class number formula $L(1,\chi_d)=2\pi h/(w\sqrt{|d|})$ and the symmetry. Let

$$\vartheta_K(x)=\sum_{n\ge1}a_n\,e^{-2\pi nx/\sqrt{|d|}},\qquad \Psi(y)=y(y-2)\,e^{-y},\qquad H_K(x)=\big(x^2\vartheta_K'(x)\big)',$$

and $\Phi_K(u)=2e^{u/2}H_K(e^u)$.

Theorem 8.5.17. (i) $\vartheta_K(1/x)=x\,\vartheta_K(x)+\frac{h}{w}\,(x-1)$ *for* $x>0$.

(ii) $H_K(x)=\sum_{n\ge1}a_n\Psi(2\pi nx/\sqrt{|d|})$ *and* $H_K(1/x)=x\,H_K(x)$. *The kernel* Φ_K *is even, analytic in the strip* $|\operatorname{Im}u|<\frac{\pi}{2}$, *and decays doubly exponentially.*

(iii) $\xi_K(\frac12+s)=\int_{\mathbb{R}}\Phi_K(u)\,e^{su}\,du$ *for every* $s\in\mathbb{C}$; *in particular* $\xi_K(\frac12+i\theta)=\int_{\mathbb{R}}\Phi_K(u)\cos(\theta u)\,du$.

(iv) $\Phi_K>0$ *on* $\mathbb{R}$ *if and only if* $H_K>0$ *on* $[1,\sqrt{|d|}/\pi]$. *In particular* $\Phi_K>0$ *when* $|d|<\pi^2$, *that is for* $d=-3,-4,-7,-8$.

(v) *If* $\Phi_K>0$, *then* $\zeta_K(\frac12)<0$ *and* $L(\frac12,\chi_d)>0$. *In particular* $L(\frac12,\chi_d)>0$ *for* $d=-3,-4,-7,-8$.

Proof. (i) For $c>1$, Mellin inversion of $\int_0^\infty e^{-2\pi nx/\sqrt{|d|}}x^{s-1}dx=\Gamma(s)\big(\sqrt{|d|}/2\pi n\big)^s$ gives $\vartheta_K(x)=\frac{1}{2\pi i}\int_{(c)}\Lambda_K(s)x^{-s}ds$. Move the line of integration to $\operatorname{Re}s=1-c$, which the exponential decay of Γ and the polynomial growth of ζ_K on vertical strips allow; the residues at $s=1$ and $s=0$ are $\frac{h}{w}x^{-1}$ and $-\frac{h}{w}$. On the new line substitute $s\mapsto1-s$ and use $\Lambda_K(1-s)=\Lambda_K(s)$: the integral becomes $x^{-1}\vartheta_K(1/x)$. So $\vartheta_K(x)=\frac{h}{w}(x^{-1}-1)+x^{-1}\vartheta_K(1/x)$, which is (i).

(ii) Termwise, $\big(x^2\frac{d}{dx}e^{-\kappa x}\big)'=(\kappa^2x^2-2\kappa x)e^{-\kappa x}=\Psi(\kappa x)$. Differentiating (i) gives $\vartheta_K'(1/x)=-x^2\big(\vartheta_K(x)+x\vartheta_K'(x)+\frac{h}{w}\big)$, so $g(x)=x^2\vartheta_K'(x)$ satisfies $g(1/x)=-\big(\vartheta_K(x)+x\vartheta_K'(x)+\frac{h}{w}\big)$, and differentiating again gives $g'(1/x)=$

$x^2\big(2\vartheta_K'(x)+x\vartheta_K''(x)\big)=x\,g'(x)$, that is $H_K(1/x)=xH_K(x)$. Then $\Phi_K(-u)=2e^{-u/2}H_K(e^{-u})=2e^{u/2}H_K(e^u)=\Phi_K(u)$. Since $a_n\le n$, the series for $H_K(e^u)$ converges locally uniformly where $\operatorname{Re}e^u>0$, that is for $|\operatorname{Im}u|<\frac{\pi}{2}$; and $\Phi_K(u)=O\big(e^{5u/2}\exp(-2\pi e^u/\sqrt{|d|})\big)$ as $u\to\infty$, and likewise as $u\to-\infty$ by evenness.

(iii) By (i), $x\vartheta_K(x)\to\frac{h}{w}$ and $x^2\vartheta_K'(x)\to-\frac{h}{w}$ as $x\to0$, while ϑ_K and ϑ_K' decay exponentially at infinity. For $\operatorname{Re}z>1$ two integrations by parts, whose boundary terms vanish, give $\int_0^\infty H_K(x)x^{z-1}dx=z(z-1)\int_0^\infty\vartheta_K(x)x^{z-1}dx=z(z-1)\Lambda_K(z)$. With $x=e^u$ and $z=\frac12+s$, $\int_{\mathbb R}\Phi_K(u)e^{su}du=2\int_0^\infty H_K(x)x^{z-1}dx=\xi_K(z)$. Both sides are entire in s, the left by (ii), so the identity holds for every s.

(iv) By (ii), $\Phi_K>0$ exactly when $H_K>0$ on $(0,\infty)$, and by the symmetry exactly when $H_K>0$ on $[1,\infty)$. For $x\ge\sqrt{|d|}/\pi$ every argument $2\pi nx/\sqrt{|d|}$ is at least $2n\ge2$, where $\Psi\ge0$, and the term $n=4$ is positive, since $a_4=1+\chi_d(2)+\chi_d(2)^2\ge1$; so $H_K>0$ there. If $|d|<\pi^2$ the interval $[1,\sqrt{|d|}/\pi]$ is empty.

(v) By (iii), $\xi_K(\frac12)=\int_{\mathbb R}\Phi_K>0$, while

$$\xi_K\big(\tfrac12\big)=-\tfrac12\Lambda_K\big(\tfrac12\big)=-\tfrac12\big(\sqrt{|d|}/2\pi\big)^{1/2}\sqrt{\pi}\,\zeta_K\big(\tfrac12\big).$$

So $\zeta_K(\frac12)<0$, and $L(\frac12,\chi_d)=\zeta_K(\frac12)/\zeta(\frac12)>0$ because $\zeta(\frac12)<0$. □

When $\Phi_K>0$, the normalised kernel $\Phi_K/\xi_K(\frac12)$ is the density of a symmetric variable X_K with $\mathbb Ee^{sX_K}=\xi_K(\frac12+s)/\xi_K(\frac12)$, the Pólya variable of K, and its tilts $V_{K,\alpha}$, with $\mathbb Ee^{sV_{K,\alpha}}=\xi_K(\alpha+s)/\xi_K(\alpha)$, have the means $\frac{\xi_K'}{\xi_K}(\alpha)$. The zero part of the constant C_K of Theorem 10.3.7 is then $\mathbb EV_{K,1}-2\,\mathbb EV_{K,2}$, as over $\mathbb Q$ in Theorem 10.3.5; by (iv) this applies to the Gaussian field $d=-4$ of Section 10.3.3. If moreover the Riemann hypothesis holds for ζ_K, that is RH for ζ and for $L(s,\chi_d)$, then $1/\mathbb E\cosh(\sqrt{u}\,X_K)=\xi_K(\frac12)/\xi_K(\frac12+\sqrt{u})$ is the Laplace transform of a GGC, the clock of K (Proposition 10.3.6).

The kernel Φ_K is tied to the kernel of χ_d in Section 8.5.1,

$$K_{\chi_d}(u)=2e^{3u/2}\sum_{n\ge1}n\chi_d(n)\,e^{-\pi n^2e^{2u}/|d|},\qquad \xi\big(\tfrac12+s,\chi_d\big)=\int_{\mathbb R}K_{\chi_d}(u)\,e^{su}\,du.$$

Since the two sided Laplace transform turns convolution into multiplication, the identity $\xi_K=\frac{2}{\sqrt{|d|}}\xi\cdot\xi(\cdot,\chi_d)$ and (iii) give $\Phi_K=\frac{2}{\sqrt{|d|}}\,\Phi*K_{\chi_d}$, with Φ the Pólya kernel (1.2.3). As $\Phi>0$, the kernel Φ_K is positive whenever K_{χ_d} is, and if Φ_K takes a negative value, so does K_{χ_d}. For $d=-4$ Jacobi's identity $\sum_{n\ge0}(-1)^n(2n+1)q^{(2n+1)^2/8}=q^{1/8}\prod_{n\ge1}(1-q^n)^3$ gives $K_{\chi_{-4}}(u)=2e^{3u/2}\eta(ie^{2u})^3>0$, positive by the product formula as the kernel of χ_{12} is.

The script `sqfree_quadratic.py` checks the theorem at 30 digits. For $d=-4$, where $h/w=\frac14$, the trapezoid rule with step $1/200$ gives $\int\Phi_K(u)\cos(\theta u)\,du=0.48753311500024448536$ at $\theta=0$ and 0.25376667901239784985 at $\theta=2.5$, equal

to $\xi_K(\frac12+i\theta)$ in every digit shown; (i) holds at $x=1.5$ and 2, and the evenness of Φ_K at $u=0.3$ and 0.6, within 10^{-31}; and $H_K(1/x)=xH_K(x)$ holds at $x=1.25$ within 3×10^{-31} for $d=-23$ and $d=-163$, where $H_K(1.25)=0.55669258$ and -0.28242077. The values $\zeta_K(\frac12)=-0.70223711$, -0.97506623, -1.6744215 and -1.6070054 for $d=-3,-4,-7,-8$ are negative, as (v) requires.

Since $\int_0^\infty \Psi(y)\,dy=0$, the value $H_K(1)=\sum_n a_n\Psi(2\pi n/\sqrt{|d|})$ is a Riemann sum, with step $2\pi/\sqrt{|d|}$ and weights a_n of mean value $L(1,\chi_d)$, of a function with integral zero. For large $|d|$ it is small compared with the sum of the absolute values of its terms, and its sign, like the sign of H_K elsewhere on $[1,\sqrt{|d|}/\pi]$, depends strongly on the first few weights: Ψ is negative on $(0,2)$ and positive beyond, and the weights $a_2=1+\chi_d(2)$ and $a_3=1+\chi_d(3)$ vanish when 2 and 3 are inert in K. The next theorem makes this precise at the point where the term $n=1$ sits in the negative part of Ψ and the terms $n=2$ and 3, which would compensate it, vanish.

THEOREM 8.5.18. *Let $d<0$ be a fundamental discriminant with $\chi_d(2)=\chi_d(3)=-1$, that is $d\equiv5\pmod{24}$. Then $H_K(x)<0$ at $x=1.42\sqrt{|d|}/2\pi$, so Φ_K takes negative values, and so does the kernel K_{χ_d}. There are infinitely many such d, for instance $d=-p$ for every prime $p\equiv19\pmod{24}$.*

Proof. For a fundamental discriminant $d<0$, $\chi_d(2)=-1$ exactly when $d\equiv5\pmod8$, and $\chi_d(3)=-1$ exactly when $d\equiv2\pmod3$; together they give $d\equiv5\pmod{24}$. The function $n\mapsto a_n$ is multiplicative, with $a_{p^k}=\frac12(1+(-1)^k)$ when $\chi_d(p)=-1$ and $a_{p^k}\le k+1$ for every p. Hence $a_1=1$, $a_2=a_3=0$ and $a_n\le a_n^*$, where $a_n^*=0$ unless 2 and 3 divide n to even powers, and otherwise a_n^* is the number of divisors of n with its factors 2 and 3 removed. At $x=1.42\sqrt{|d|}/2\pi$ the nth argument of Ψ in Theorem 8.5.17(ii) is $1.42n$, which exceeds 2 for $n\ge2$, where $\Psi>0$. So

$$\begin{aligned}H_K(x)&\le\Psi(1.42)+\sum_{n\ge4}a_n^*\,\Psi(1.42n)\\&=-0.19907566+0.13917904=-0.05989662<0,\end{aligned}$$

the sum taken to $n=2000$ and the rest bounded, with $a_n^*\le2\sqrt{n}$, by 10^{-1225} (`sqfree_quadratic.py`). Then $\Phi_K(\log x)<0$, and K_{χ_d} takes negative values by the convolution identity above. If $p\equiv19\pmod{24}$ is prime, then $p\equiv3\pmod4$, so $-p$ is a fundamental discriminant, and $-p\equiv5\pmod{24}$; by Dirichlet's theorem there are infinitely many such primes. □

The bound is negative for every y in $[1.107,1.879]$ on a grid of step 0.001, with least value -0.0598979 at $y=1.418$, so the point $x=y\sqrt{|d|}/2\pi$ can be moved. By the symmetry $H_K(1/x)=xH_K(x)$ it does not matter whether x lies in $[1,\sqrt{|d|}/\pi]$; for $d=-19$ it does not.

Table 8.20 shows the discriminants discussed in the first author's note. The four of class number one with 2 and 3 inert, $d = -19, -43, -67, -163$, take the least value of H_K on $[1, \sqrt{|d|}/\pi]$ at the centre $x = 1$, the two composite ones with $d \equiv 5 \pmod{24}$ inside the interval, and most of the positive kernels at its right end $\sqrt{|d|}/\pi$, where the term $n = 1$ vanishes. All twenty values of $\zeta_K(\frac{1}{2})$ computed are negative, so $L(\frac{1}{2}, \chi_d) > 0$ for all of them whether or not the kernel is positive: Theorem 8.5.17(v) is a sufficient condition only.

d	$\chi_d(2)$	$\chi_d(3)$	$\Phi_K > 0$	least H_K on $[1, \sqrt{\lvert d\rvert}/\pi]$	at $x =$	$\zeta_K(\frac{1}{2})$
-11	-1	1	yes	0.14236	1.0557	-1.4480539
-15	1	0	yes	0.40606	1.2328	-2.6973176
-20	0	1	yes	0.287033	1.4235	-2.4529153
-23	1	1	yes	0.4634	1.5266	-3.5856997
-24	0	0	yes	0.225215	1.2409	-2.2955534
-31	1	-1	yes	0.349008	1.7723	-3.2253855
-39	1	0	yes	0.409694	1.9878	-4.0547932
-47	1	1	yes	0.46368	2.1822	-4.8243519
-71	1	1	yes	0.470667	2.6821	-5.9925928
-191	1	1	yes	0.470667	4.3991	-8.1483722
-19	-1	-1	no	-0.0596605	1	-1.1747378
-43	-1	-1	no	-0.214707	1	-0.76635571
-67	-1	-1	no	-0.251274	1	-0.54460945
-163	-1	-1	no	-0.315818	1	-0.10008066
-427	-1	-1	no	-0.181949	3.1887	-0.20993694
-1003	-1	-1	no	-0.228309	3.9479	-0.64248774

Table 8.20: The Pólya kernel Φ_K of $K = \mathbb{Q}(\sqrt{d})$ for the discriminants named in the first author's note: the characters at 2 and 3, the sign of the kernel, the least value of H_K on $[1, \sqrt{|d|}/\pi]$, which decides it by Theorem 8.5.17(iv), with the point where it is attained, and $\zeta_K(\frac{1}{2}) = \zeta(\frac{1}{2})L(\frac{1}{2}, \chi_d)$. Computed by `sqfree_quadratic.py`.

The script also scans all 6079 fundamental discriminants $-20000 \le d < 0$. It evaluates H_K in double precision on 3000 geometrically spaced points of $[1, \sqrt{|d|}/\pi]$, refines the least value by golden section search, and would recompute at 30 digits any least value within 10^{-8} of zero; there is none. The kernel is positive for 2738 of the discriminants. Table 8.21 sorts them by the splitting of 2 and 3: the proportion of positive kernels falls as either prime goes from split to ramified to inert, and the class $d \equiv 5 \pmod{24}$ of Theorem 8.5.18 has none among its 763 members. By blocks of 2000 in $|d|$ the proportion falls from 337/611 for $|d| \le 2000$ to between 0.41 and 0.45 in each of the five blocks above 10000. Of the 3341 negative kernels, 1546 take the least value of H_K on $[1, \sqrt{|d|}/\pi]$ at the centre $x = 1$, and 1135 the least value of Φ_K on $u \ge 0$ at $u = 0$.

Question 8.5.19. Does the proportion of fundamental discriminants $-X \le d < 0$ with $\Phi_K > 0$ tend to a limit as $X \to \infty$, and is the limit positive? Is Φ_K positive

	$\chi_d(3)=1$	$\chi_d(3)=0$	$\chi_d(3)=-1$
$\chi_d(2)=1$	676/767	334/507	412/757
$\chi_d(2)=0$	489/753	178/514	131/757
$\chi_d(2)=-1$	397/761	121/500	0/763

Table 8.21: Positive Pólya kernels Φ_K among the 6079 fundamental discriminants $-20000 \le d < 0$, by the splitting of 2 and 3 in $K=\mathbb{Q}(\sqrt{d})$ ($\chi_d(p)=1$ split, 0 ramified, -1 inert): the number of positive kernels over the number of discriminants in each class, 2738 positive in all. The empty corner is Theorem 8.5.18. Computed by `sqfree_quadratic.py`.

for infinitely many d, for instance among the d in which 2 and 3 split, where the proportion up to 20000 is 676/767?

Question 8.5.20. For real quadratic fields and for fields of higher degree the gamma factor $\Gamma_{\mathbb{R}}(s)^{r_1}\Gamma_{\mathbb{C}}(s)^{r_2}$ replaces the exponentials in ϑ_K by other kernels, for $\Gamma_{\mathbb{R}}(s)^2$ a Bessel function K_0. Is there a criterion for the positivity of the Pólya kernel of ξ_K as simple as Theorem 8.5.17(iv)?

8.5.7 The Dirichlet beta function and the eta cube kernel

The L function of the odd character χ_4 modulo 4, with $\chi_4(n)=1$ for $n\equiv 1$, -1 for $n\equiv 3 \pmod 4$ and 0 for even n,

$$L(s,\chi_4)=\sum_{n\ge 0}\frac{(-1)^n}{(2n+1)^s}=\prod_p\bigl(1-\chi_4(p)p^{-s}\bigr)^{-1},\qquad \operatorname{Re}s>1,$$

is the Dirichlet beta function, with $L(1,\chi_4)=\frac{\pi}{4}$ and $L(2,\chi_4)$ Catalan's constant. It is the character χ_{-4} of Section 8.5.6, whose kernel is the cube of the Dedekind eta function and positive by the product formula. This subsection carries the Thorin framework over to it, following the first author's note [201]. Against ζ two things change, as for χ_{12}: there is no pole, and the prime weights $\chi_4(n)\Lambda(n)$ take both signs, negative at the primes $3,7,11,19,\dots$. Against χ_{12} one thing changes: the character is odd, so the gamma factor is $\Gamma(\frac{s+1}{2})$, and the archimedean piece of the sine squared decomposition is that of ξ at the basepoint one lower. Section 8.5.7.1 gives the kernel, its reading as a Brownian exit time and the zeros below height 100; Section 8.5.7.2 the Lévy form, the sine squared decomposition, and the exact range of basepoints where bounding the prime weights by their absolute values suffices; Section 8.5.7.3 the factorisation through the Gaussian field $\mathbb{Q}(i)$, which writes the Thorin density as a difference of two Thorin densities; Section 8.5.7.4 the Thorin measure on a vertical line as the argument of the completed function; and Section 8.5.7.5 what transfers to the centre. The comparison across vertical lines of the note, which applies to ξ and $L(s,\chi_{12})$ as well, is in Section 9.1.6.

8.5.7.1 The kernel, the exit time and the zeros

Jacobi's identity $\eta(\tau)^3 = q^{1/8}\prod_{m\ge1}(1-q^m)^3 = \sum_{n\ge0}(-1)^n(2n+1)q^{(2n+1)^2/8}$, with $q = e^{2\pi i\tau}$ (Section 8.5.6), gives at $\tau = ix$ the theta series of χ_4,

$$\eta(ix)^3 = \sum_{n\ge1}\chi_4(n)\,n\,e^{-\pi n^2x/4} = e^{-\pi x/4}\prod_{m\ge1}\bigl(1-e^{-2\pi mx}\bigr)^3, \qquad x>0,$$

and the transformation $\eta(i/x) = \sqrt{x}\,\eta(ix)$ gives $\eta(i/x)^3 = x^{3/2}\eta(ix)^3$. In the normalisation of Section 8.5.1 for odd characters the completed function is

$$\xi(s,\chi_4) = \Bigl(\frac{4}{\pi}\Bigr)^{(s+1)/2}\Gamma\Bigl(\frac{s+1}{2}\Bigr)L(s,\chi_4) = \int_0^\infty \eta(ix)^3\,x^{(s+1)/2}\,\frac{dx}{x},$$

the integral by termwise integration for $\operatorname{Re} s>1$, since $\int_0^\infty e^{-\pi n^2x/4}x^{(s+1)/2-1}dx = \Gamma(\frac{s+1}{2})(4/\pi n^2)^{(s+1)/2}$, and for every s by analytic continuation, since $\eta(ix)^3$ decays like $e^{-\pi x/4}$ as $x\to\infty$ and, by the transformation, like $x^{-3/2}e^{-\pi/(4x)}$ as $x\to0$. The transformation gives $\xi(s,\chi_4) = \xi(1-s,\chi_4)$, the root number of χ_4 being one, and $\xi(s,\chi_4)$ is entire of order one, with no factor $s(s-1)$ to remove. It is $2/\sqrt{\pi}$ times the function Λ of [201], and $\xi(1,\chi_4) = \frac{4}{\pi}L(1,\chi_4) = 1$. Section 8.5.6 notes that the kernel of $\xi(s,\chi_4)$ is positive by the product formula; the next proposition collects its properties.

Proposition 8.5.21. *The kernel $K_4(u) = 2e^{3u/2}\eta(ie^{2u})^3$ satisfies $\xi(\frac12+s,\chi_4) = \int_{\mathbb R}K_4(u)\,e^{su}\,du$ for every complex s. It is even, positive and analytic in the strip $|\operatorname{Im}u|<\pi/4$, it has the symmetric product form*

$$K_4(u) = 2\exp\Bigl(-\frac{\pi}{4}\cosh 2u\Bigr)\prod_{m\ge1}\bigl(1-e^{-2\pi me^{2u}}\bigr)^{3/2}\bigl(1-e^{-2\pi me^{-2u}}\bigr)^{3/2},$$

so that $K_4(u) = 2\exp\bigl(\frac32|u| - \frac{\pi}{4}e^{2|u|}\bigr)\bigl(1+O(e^{-2\pi e^{2|u|}})\bigr)$, and $K_4(0) = 2\eta(i)^3 = \Gamma(\frac14)^3/(4\pi^{9/4}) = 0.906767655167731$.

Proof. With $x = e^{2u}$, $x^{(s+1)/2}dx/x = 2e^{(s+1)u}du$, so the Mellin form with $\frac12+s$ in place of s is the stated integral. The product formula makes every factor of $K_4(u) = 2e^{3u/2}e^{-\frac{\pi}{4}e^{2u}}\prod_m(1-e^{-2\pi me^{2u}})^3$ positive. The transformation gives $K_4(-u) = 2e^{-3u/2}\eta(ie^{-2u})^3 = 2e^{-3u/2}e^{3u}\eta(ie^{2u})^3 = K_4(u)$. Hence $K_4(u) = \sqrt{K_4(u)K_4(-u)}$, and in the product $K_4(u)K_4(-u)$ the factors $e^{\pm3u/2}$ cancel while $e^{-\frac{\pi}{4}(e^{2u}+e^{-2u})} = e^{-\frac{\pi}{2}\cosh 2u}$, which gives the symmetric form. For $u\to+\infty$ the product for $K_4(u)$ itself gives the asymptotic, and evenness gives it for $u\to-\infty$. The function $\eta(ie^{2u})$ is analytic where $\operatorname{Re}e^{2u}>0$, that is for $|\operatorname{Im}u|<\pi/4$. Finally $\eta(i) = \Gamma(\frac14)/(2\pi^{3/4})$. □

Like the kernel K of $L(s,\chi_{12})$, K_4 is analytic in the same strip as the Pólya kernel and decays doubly exponentially, so the evaluator of

Section 2.1 applies unchanged. The script `dbeta_thorin.py` computes $\xi(\frac12+s,\chi_4)$ from K_4 by the trapezoid rule with step $1/64$ on $|u|\le 3$ and agrees with the Dirichlet L function of Arb [110] in all fourteen digits printed at $s=0$, $\frac12$, $0.1+5i$ and $0.1+20i$; for instance $\xi(\frac12,\chi_4)=0.98071361405771$, $\xi(0.6+5i,\chi_4)=0.080871220951695+0.011736751015145i$ and $\xi(0.6+20i,\chi_4)=(-2.3183027026127-0.20613003173368i)\times10^{-6}$. Also $L(\frac12,\chi_4)=0.667691457189609$.

Remark 8.5.22 (The exit time). By Theorem 4.3.21, $\frac4\pi\eta(4it/\pi)^3$ is the density of $X_C=\sum_k e_k/(2k-1)^2$, which is $\frac{\pi^2}{8}$ times the exit time τ of Brownian motion from $(-1,1)$ (Section 4.2.2). With $x=4t/\pi$, $\eta(ix)^3$ is therefore the density on $(0,\infty)$ of $Z=\frac4\pi X_C=\frac\pi2\tau$ [207, Section 3], and

$$\xi(s,\chi_4)=\mathbb{E}\big[Z^{(s-1)/2}\big],\qquad \mathbb{E}e^{-uZ}=\operatorname{sech}\sqrt{\pi u}=\prod_{k\ge1}\Big(1+\frac{4u}{\pi(2k-1)^2}\Big)^{-1}.$$

So Z is a GGC with Thorin measure $\sum_k\delta_{\pi(2k-1)^2/4}$; its total mass $\xi(1,\chi_4)=1$ is the value $L(1,\chi_4)=\frac\pi4$, and its mean $\xi(3,\chi_4)=\frac4\pi\sum_k(2k-1)^{-2}=\frac\pi2$ is the value $L(3,\chi_4)=\pi^3/32$. The transformation $\eta(i/x)^3=x^{3/2}\eta(ix)^3$ says that $1/Z$ has the density $x^{-2}\eta(i/x)^3=x^{-1/2}\eta(ix)^3$, that is $\mathbb{E}g(1/Z)=\mathbb{E}[Z^{-1/2}g(Z)]$ for $g\ge0$, and with $g(z)=z^{(1-s)/2}$ this is the functional equation. The primal variable Y_4, with density $K_4/\xi(\frac12,\chi_4)$ and $\mathbb{E}e^{sY_4}=\xi(\frac12+s,\chi_4)/\xi(\frac12,\chi_4)$, is $\frac12\log Z$ under the size biasing by $Z^{-1/4}$, since $\mathbb{E}[Z^{-1/4}e^{(s/2)\log Z}]/\mathbb{E}[Z^{-1/4}]=\mathbb{E}[Z^{s/2-1/4}]/\mathbb{E}[Z^{-1/4}]$. So the analogue of the Pólya variable is half the logarithm of a Brownian exit time biased by $Z^{-1/4}$, as for ζ it is half the logarithm of the variable $\pi S_2/2$ of Williams and of Biane, Pitman and Yor biased by $(\pi S_2/2)^{1/4}$ (Section 8.2.3). In the notation of Section 11.5.5.4, $Z=1/\mathcal{B}_{\chi_{-4}}$ (Proposition 11.5.21); whether $1/Z$ is a GGC is open, though it passes the test of that section to order 12, and HCM cannot decide it, since the density of Z is not HCM by Theorem 4.3.21. The script checks $\int_0^\infty\eta(ix)^3dx=1$ and $\int_0^\infty e^{-x}\eta(ix)^3dx=\operatorname{sech}\sqrt\pi=0.33029501666335$ to fourteen digits.

The basepoint one has an explicit drift. By the functional equation $b_1=\frac{\xi'}{\xi}(1,\chi_4)=-\frac{\xi'}{\xi}(0,\chi_4)$. With $L(s,\chi_4)=4^{-s}\big(\zeta(s,\frac14)-\zeta(s,\frac34)\big)$, $\zeta(0,a)=\frac12-a$ and Lerch's formula $\partial_s\zeta(0,a)=\log\Gamma(a)-\frac12\log2\pi$ [177, §25.11], one has $L(0,\chi_4)=\frac12$ and $L'(0,\chi_4)=\log\big(\Gamma(\frac14)/2\Gamma(\frac34)\big)$, and with $\psi(\frac12)=-\gamma_E-2\log2$ and $\Gamma(\frac14)\Gamma(\frac34)=\pi\sqrt2$,

$$b_1=3\log2+\frac52\log\pi+\frac{\gamma_E}{2}-4\log\Gamma\Big(\frac14\Big)=0.0777839899618,\tag{8.5.7}$$

and so, since $b_1=\frac12\log\frac4\pi-\frac{\gamma_E}{2}+\frac4\pi L'(1,\chi_4)$, $L'(1,\chi_4)=\frac\pi4\big(\gamma_E+2\log2+3\log\pi-4\log\Gamma(\frac14)\big)=0.192901316797$. At the centre $\xi'(\frac12,\chi_4)=0$, and the variance of the primal variable is $V_4=\operatorname{Var}Y_4=(\log\xi)''(\frac12,\chi_4)=0.156029649890$.

Since $K_4 > 0$, $\xi(\frac12 + s, \chi_4) = \int_{\mathbb{R}} K_4(u) e^{su} du > 0$ for every real s, so $\xi(s,\chi_4)$ has no real zeros; on $(0,\infty)$ this is also seen from the series, which for real $\sigma > 0$ alternates with decreasing terms, so that $L(\sigma,\chi_4) \geq 1 - 3^{-\sigma} > 0$ [201]. In particular $L(\frac12,\chi_4) > 0$, which Theorem 8.5.17(v) also gives. The script `dbeta_zeros.py` computes the zeros of $L(s,\chi_4)$ below height 100 in ball arithmetic with Arb. The function $\xi(\frac12 + i\theta, \chi_4)$ is real; its sign is certified on a grid of step $1/32$, and each sign change is refined by bisection to 10^{-12}. The argument principle along $2 \to 2 + 100i \to \frac12 + 100i$, on pieces halved until the enclosure of $\xi(s,\chi_4)$ over each piece lies in a disc excluding zero (1633 pieces), gives $N(100) = 50$ for the number of zeros with $0 < \gamma \leq 100$, counted with multiplicity, the remaining sides of the rectangle $[-1,2] \times [0,100]$ following by the functional equation and $\xi(\bar s, \chi_4) = \overline{\xi(s,\chi_4)}$. This equals the number of sign changes, so all fifty zeros are simple and on the critical line. Their ordinates begin 6.0209489047, 10.2437703042, 12.9880980123, 16.3426071046, 18.2919931961, 21.4506113440 and end at 98.7553004158, and the largest gap between consecutive ordinates is the first, 4.2228214. The zero counting function satisfies

$$N(T) = \frac{T}{2\pi}\log\frac{2T}{\pi e} + \frac18 + \frac{1}{\pi}\arg L\big(\tfrac12 + iT, \chi_4\big) + \frac{1}{48\pi T} + O(T^{-3}), \qquad (8.5.8)$$

with the argument continued from $L(2,\chi_4) > 0$ along $2 \to 2 + iT \to \frac12 + iT$, for T not an ordinate. Indeed, by the argument principle on the rectangle $[-1,2]\times[0,T]$, the functional equation, the reflection $\xi(\bar s,\chi_4) = \overline{\xi(s,\chi_4)}$ and the positivity of $\xi(s,\chi_4)$ on the real axis, $\pi N(T)$ is the change of $\arg \xi(s,\chi_4)$ along $2 \to 2 + iT \to \frac12 + iT$, and the expansion of the gamma factor by Stirling's series [177, §5.11] in the proof of Proposition 8.5.30 below, at $\alpha = \frac12$, gives (8.5.8); compare [63, Chapter 16], where the main term is derived with an error $O(\log T)$. Its main term $\frac{T}{2\pi}\log\frac{2T}{\pi e} + \frac18$ is 50.3159 at $T = 100$. Under GRH for χ_4, $V_4 = \sum_\gamma \gamma^{-2}$ over all the ordinates of both signs; twice the sum over the fifty computed ones is 0.1395621, and the main term $\frac{1}{\pi}(\log\frac{2T}{\pi} + 1)/T$ of the rest, from $dN(t) \approx \frac{1}{2\pi}\log\frac{2t}{\pi}\,dt$, is 0.0164 at $T = 100$, together 0.1560.

8.5.7.2 The Lévy form and the sine squared decomposition

The Lévy form of Proposition 11.3.2 for χ_{12} has an analogue for χ_4 at every basepoint $\alpha \geq 1$, and for $\alpha > 1$ it converges absolutely on a half plane larger than $\operatorname{Re} s \geq 0$.

Proposition 8.5.23. *For $\alpha \geq 1$,*

$$\log\frac{\xi(\alpha + s,\chi_4)}{\xi(\alpha,\chi_4)} = b_\alpha s + \int_0^\infty \big(e^{-sx} - 1 + sx\big)\,\Pi_\alpha(dx), \qquad b_\alpha = \frac{\xi'}{\xi}(\alpha,\chi_4),$$

with the signed Lévy measure

$$\Pi_\alpha(dx) = \frac{e^{-\alpha x}}{2x\sinh x}\,dx + \sum_{n\geq 2}\frac{\chi_4(n)\Lambda(n)}{n^\alpha \log n}\,\delta_{\log n}(dx),$$

whose canonical measure $x\Pi_\alpha(dx)$ *has the positive density* $e^{-\alpha x}/(2\sinh x)$ *and the atoms* $\chi_4(n)\Lambda(n)n^{-\alpha}$ *at* $\log n$. *For* $\alpha > 1$ *the form holds for* $\operatorname{Re} s > 1-\alpha$, *with absolute convergence. At* $\alpha = 1$ *it holds for* $\operatorname{Re} s > 0$, *the prime sum converging conditionally, as a limit over* $n \le X$.

Proof. The factor $(4/\pi)^{(s+1)/2}$ is linear in log and is absorbed by the drift. Binet's formula [177, §5.9], in the integrated form

$$\log\frac{\Gamma(z+v)}{\Gamma(z)} - v\psi(z) = \int_0^\infty (e^{-vx} - 1 + vx)\frac{e^{-zx}}{x(1-e^{-x})}\,dx$$

for $\operatorname{Re} z > 0$ and $\operatorname{Re}(z+v) > 0$, which one checks by differentiating twice in v, gives at $z = \frac{\alpha+1}{2}$, $v = \frac{s}{2}$ and after $x \mapsto 2x$ the density $\frac{e^{-(\alpha+1)x}}{x(1-e^{-2x})} = \frac{e^{-\alpha x}}{2x\sinh x}$, for $\operatorname{Re} s > -1-\alpha$. For $\operatorname{Re}(\alpha+s) > 1$ the Euler product gives $\log L(\alpha+s,\chi_4) = \sum_n \chi_4(n)\Lambda(n)n^{-\alpha-s}/\log n$, and subtracting the value and the derivative at $s=0$ gives the atoms; for $\alpha > 1$ the sum converges absolutely for $\operatorname{Re} s > 1-\alpha$, because $|e^{-sx}-1+sx| \le e^{-x\operatorname{Re} s}+1+|s|x$ and $\sum_n \Lambda(n)n^{-\alpha}\bigl(n^{-\operatorname{Re} s}+1+|s|\log n\bigr)/\log n < \infty$ there. At $\alpha = 1$ the linear coefficient is $-\sum_n \chi_4(n)\Lambda(n)/n = \frac{L'}{L}(1,\chi_4)$ and the constant term is $\sum_n \chi_4(n)\Lambda(n)/(n\log n) = \log L(1,\chi_4)$; both series converge by the prime number theorem for arithmetic progressions [63, Chapter 20], as in the proof of Proposition 11.3.2. □

The archimedean density is that of ξ shifted by one basepoint: at the basepoint $\alpha+1$ the gamma density $e^{-(\alpha+2)x}/(1-e^{-2x})$ of $x\Pi_{\alpha+1}$ is the gamma density of ξ at the basepoint α (Section 9.1.4), because the gamma factor $\Gamma(\frac{s+1}{2})$ at $\alpha+1$ and the gamma factor $\Gamma(1+\frac{s}{2})$ of ξ, which absorbs the factor s, at α have the same argument $\frac{\alpha+2+s}{2}$. There is no pole term, and the atoms carry the sign of $\chi_4(n)$: of the 466 prime powers up to 3000, 223 have negative weight, 232 positive weight, and the 11 powers of 2 weight zero. As for χ_{12} in Section 11.3.2, the tilted variable $V_\alpha^{(4)}$ with moment generating function $\xi(\alpha+s,\chi_4)/\xi(\alpha,\chi_4)$ is quasi infinitely divisible for $\alpha > 1$, the Euler product writing its characteristic function as that of an infinitely divisible gamma piece times a quotient of two compound Poisson characteristic functions, and it is not infinitely divisible: by the uniqueness of the Lévy measure of a quasi infinitely divisible law [147], the atom $-3^{-\alpha}$ of Π_α at $\log 3$, reflected to $-\log 3$, would have to be positive. At $\alpha = 1$, $V_1^{(4)}$ is not quasi infinitely divisible, since the atoms of each sign have infinite mass, $\sum_{p\equiv 1} 1/p$ and $\sum_{p\equiv 3\,(4)} 1/p$ both diverging, and cancel only conditionally; the argument of Proposition 11.1.3(3) applies. Table 8.22 checks the form.

The sine squared lemma (1.2.9) turns the Lévy form into the Thorin density, the bracket being $\operatorname{Re}(\xi'/\xi)(\alpha+i\theta,\chi_4)$ [198, Theorem 8.2]. With $v_\alpha(\theta) = \operatorname{Re}\frac{\xi'}{\xi}(\alpha+i\theta,\chi_4)$ and $q_c(\theta) = \theta^2/(c(c^2+\theta^2))$ as in (9.1.7),

$$v_\alpha(\theta) = b_\alpha + G_\alpha(\theta) + \varpi_\alpha(\theta), \qquad \alpha \ge 1, \tag{8.5.9}$$

(α,s)	$b_\alpha s$	X	gamma piece	primes	total	exact
$(2,2)$	0.45618502	2×10^5	0.3689751341	−0.1014710984	0.2675040358	0.2675040274
		10^7		−0.1014711067	0.2675040274	
$(3,2)$	0.72952700	2×10^5	0.2703628455	−0.0374815198	0.2328813257	0.2328813257
$(\frac{3}{2},1)$	0.15421515	2×10^5	0.1275974821	−0.0543016088	0.0732958733	0.0732940161
		10^7		−0.0543034673	0.0732940147	

Table 8.22: The Lévy form of Proposition 8.5.23: the gamma piece and the prime powers up to X, whose sum (total) is compared with $\log(\xi(\alpha+s,\chi_4)/\xi(\alpha,\chi_4))-b_\alpha s$ (exact); the drift term $b_\alpha s$ is shown separately. At $(3,2)$ the sums to 2×10^5 and to 10^7 agree in all digits shown. Computed by `dbeta_thorin.py`.

where the gamma piece and the prime piece are

$$\begin{aligned} G_\alpha(\theta) &= \int_0^\infty \frac{1-\cos\theta x}{2\sinh x}\,e^{-\alpha x}dx = \frac12\Big[\operatorname{Re}\psi\Big(\frac{\alpha+1+i\theta}{2}\Big)-\psi\Big(\frac{\alpha+1}{2}\Big)\Big] \\ &= \sum_{k\ge0} q_{\alpha+1+2k}(\theta) \ge 0, \\ \varpi_\alpha(\theta) &= \sum_{n\ge2}\frac{\chi_4(n)\Lambda(n)}{n^\alpha}\big(1-\cos(\theta\log n)\big), \end{aligned}$$

the series for G_α from $1/(2\sinh x)=\sum_{k\ge0}e^{-(2k+1)x}$, and ϖ_α converging conditionally at $\alpha=1$. So $G_\alpha=\sum_{k\ge1}q_{(\alpha-1)+2k}$ is the gamma piece of ξ at the basepoint $\alpha-1$ in (9.1.7), without its pole term $-q_{\alpha-2}$. For $\alpha\ge1$ the Thorin representation is

$$\log\frac{\xi(\alpha+\sqrt{u},\chi_4)}{\xi(\alpha,\chi_4)} = \frac1\pi\int_0^\infty \log\Big(1+\frac{u}{\theta^2}\Big)\,v_\alpha(\theta)\,d\theta, \qquad u\ge0, \tag{8.5.10}$$

by Proposition 8.5.24 below. The bracket identity can also be read off (8.5.10) by Stieltjes inversion: if $\varphi(u)=\xi(\alpha,\chi_4)/\xi(\alpha+\sqrt{u},\chi_4)$, then $-(\log\varphi)'(u)=\int_0^\infty U(dz)/(u+z)$ for the Thorin measure U, and the boundary value at $u=-z+i0$, where $\sqrt{u}=i\sqrt{z}$, returns the density $\operatorname{Re}(\xi'/\xi)(\alpha+i\sqrt{z},\chi_4)/(2\pi\sqrt{z})$ in z, which is $v_\alpha(\theta)/\pi$ in $\theta=\sqrt{z}$.

Table 8.23 checks (8.5.9) against the closed form v_α, including at $\theta=\gamma_1$, where the prime piece is positive. With Gauss Legendre quadrature to $\theta=T$ and the tail $\frac{u}{2\pi}(\log\frac{2T}{\pi}+1)/T$, from $v_\alpha(\theta)\sim\frac12\log\frac{2\theta}{\pi}$, the right side of (8.5.10) at $(\alpha,u)=(2,1)$ is 0.2977961214 for $T=150$ and 0.2977947221 for $T=500$, against $\log(\xi(3,\chi_4)/\xi(2,\chi_4))=0.2977947060$; at $(1,1)$ it is 0.1537912492 and 0.1537878446, against $\log\xi(2,\chi_4)=0.1537879993$. The residual is the oscillating prime tail beyond T.

Figure 8.18 shows the density and the decomposition. On the left, v_α at $\alpha=1$, 0.75 and 0.55. Its minimum is b_α, at the origin (0.0777840, 0.0389784

(α,θ)	$b_\alpha+G_\alpha(\theta)$	X	$\varpi_\alpha(\theta)$	sum	$v_\alpha(\theta)$
$(2,1)$	0.2757938407	2×10^5	-0.0404427501	0.2353510906	0.2353510905
		10^7	-0.0404427502	0.2353510905	
$(2,7)$	0.8545028744	2×10^5	-0.0853835269	0.7691193474	0.7691193461
		10^7	-0.0853835283	0.7691193461	
$(\frac{3}{2},4)$	0.6439624309	2×10^5	-0.3176806015	0.3262818294	0.3262814078
		10^7	-0.3176810099	0.3262814210	
$(\frac{3}{2},\gamma_1)$	0.8321111997	2×10^5	0.3350654907	1.1671766904	1.167178852
		10^7	0.3350675456	1.1671787453	

Table 8.23: The sine squared decomposition (8.5.9) of $v_\alpha(\theta)=\mathrm{Re}(\xi'/\xi)(\alpha+i\theta,\chi_4)$, with the prime powers up to X, against the closed form; $\gamma_1=6.0209489$. Computed by `dbeta_thorin.py`.

and 0.0078013): this is so at all three basepoints on the grid of step 0.005 on $[0,30]$ of the figure, and Theorem 9.1.11 proves it for every $\alpha\in(\frac12,4.70]$ on the verified range and for every $\alpha\ge1$ at every height. Near each ordinate v_α rises to about $1/(\alpha-\frac12)$, the height of the Cauchy kernel of a zero on the line, plus the contribution of the other zeros; its maximum on $[0,30]$ is 2.465, 4.246 and 20.05 at the three basepoints. As $\alpha\downarrow\frac12$ the margin at the origin shrinks linearly, $b_\alpha\approx(\alpha-\frac12)V_4$, and the density concentrates on the ordinates. On the right, at $\alpha=\frac32$, the prime piece ϖ_α is negative on 71% of the grid of step 0.01 on $[0,30]$, with minimum -0.53213 at $\theta=19.89$, where $b_\alpha+G_\alpha=1.41769$ and $v_\alpha=0.88556$, and maximum 0.33562 at $\theta=6.0$, next to γ_1; the minimum of v_α on the grid is $b_{3/2}=0.15421515$, at the origin. For ξ the same figure would show a positive prime piece and the negative pole piece (Figure 8.13): the sine squared decomposition of ξ is not positive term by term either, but its single negative term is the pole, while for χ_4 the negative input is spread over the primes $p\equiv3\pmod4$.

Proposition 8.5.24. *For every $\alpha\ge1$, $u\mapsto\xi(\alpha,\chi_4)/\xi(\alpha+\sqrt{u},\chi_4)$ is the Laplace transform of a GGC with Thorin density $v_\alpha(\theta)/\pi>0$ in the chart $\theta=\sqrt{z}$, and* (8.5.10) *holds. For $\alpha>\frac12$ it is the Laplace transform of a GGC if and only if no zero of $L(s,\chi_4)$ has real part greater than α. GRH for χ_4 holds if and only if $v_\alpha\ge0$ for every $\alpha>\frac12$, and the Thorin measures at resolution $\varepsilon=\alpha-\frac12$ converge vaguely, as $\varepsilon\downarrow0$, to the sum of the point masses δ_{γ^2} over the zeros on the critical line, which is the Thorin measure of $\xi(\frac12,\chi_4)/\xi(\frac12+\sqrt{u},\chi_4)$ under GRH.*

Proof. The proofs of Theorem 1.2.2, of Lagarias's criterion [133] and of the vague limit stated after Remark 1.2.4 use only the Hadamard product of an entire function of order one whose zeros are symmetric under $\rho\mapsto\bar\rho$ and $\rho\mapsto1-\rho$, with $\sum_\rho|\rho|^{-2}<\infty$; they apply to $\xi(s,\chi_4)$ [63, Chapter 12], as Section 8.5.3 and Proposition 11.3.1 apply them to $\xi(s,\chi_{12})$. Positivity at $\alpha\ge1$ needs every zero to have real part less than one, which is the absence of zeros of $L(s,\chi_4)$ on $\mathrm{Re}\,s=1$ [63,

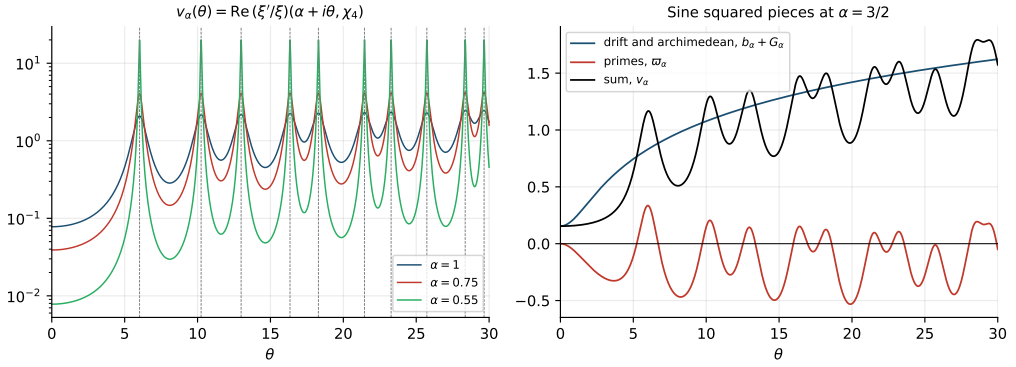


Figure 8.18: Left: $v_\alpha(\theta) = \mathrm{Re}(\xi'/\xi)(\alpha + i\theta, \chi_4) = \pi\rho_{\alpha-1/2}(\theta)$ for $\alpha = 1$, 0.75, 0.55 on a logarithmic scale, with the ordinates of the zeros of $L(s,\chi_4)$ (dashed). As $\alpha \downarrow \frac{1}{2}$ the density concentrates on the ordinates, while its minimum b_α, at the origin, tends to zero. Right: at $\alpha = \frac{3}{2}$, the drift and gamma piece $b_\alpha + G_\alpha$, the prime piece ϖ_α, of both signs, and their sum v_α. Drawn by `dbeta_figs.py`.

Chapter 14] with the Euler product and the functional equation. □

At the basepoint one the clock $T_1^{(4)}$, with Laplace transform $1/\xi(1+\sqrt{u},\chi_4)$, therefore has the expansion $1 - b_1\sqrt{u} + O(u)$ at the origin, and as for ξ in (9.1.3) the Tauberian theorem [76, XIII.5] gives the tail $P(T_1^{(4)} > x) \sim (b_1/\sqrt{\pi})\, x^{-1/2} = 0.0438849\, x^{-1/2}$, with b_1 from (8.5.7).

Positivity at $\alpha \geq 1$ uses the zeros, not the Lévy measure. Bounding the prime weights by their absolute values gives positivity only far to the right, and no such bound does better.

Proposition 8.5.25. *For $\alpha > 1$ and real θ,*

$$v_\alpha(\theta) \geq A_4^\infty(\alpha,\theta) - S_4(\alpha),$$
$$A_4^\infty(\alpha,\theta) = \frac{1}{2}\log\frac{4}{\pi} + \frac{1}{2}\,\mathrm{Re}\,\psi\Big(\frac{\alpha+1+i\theta}{2}\Big),$$
$$S_4(\alpha) = \sum_{n\ \mathrm{odd}} \frac{\Lambda(n)}{n^\alpha} = -\frac{\zeta'}{\zeta}(\alpha) - \frac{\log 2}{2^\alpha - 1},$$

and $A_4^\infty(\alpha,\theta)$ increases in $|\theta|$. The bound is positive at every θ exactly when $\alpha > \alpha_c(\chi_4) = 2.2823106368$, the root of $A_4^\infty(\alpha,0) = S_4(\alpha)$. For χ_{12} the same bound, with $A_{12}^\infty(\alpha,\theta) = \frac{1}{2}\log\frac{12}{\pi} + \frac{1}{2}\,\mathrm{Re}\,\psi(\frac{\alpha+i\theta}{2})$ and $S_{12}(\alpha) = S_4(\alpha) - \log 3/(3^\alpha - 1)$, is positive at every θ exactly when $\alpha > \alpha_c(\chi_{12}) = 1.8348743295$.

Proof. For $\mathrm{Re}\, s > 1$,

$$\frac{\xi'}{\xi}(s,\chi_4) = \frac{1}{2}\log\frac{4}{\pi} + \frac{1}{2}\psi\Big(\frac{s+1}{2}\Big) + \frac{L'}{L}(s,\chi_4),$$

and $|\operatorname{Re}\frac{L'}{L}(\alpha+i\theta,\chi_4)| \le \sum_n |\chi_4(n)|\Lambda(n)n^{-\alpha} = S_4(\alpha)$. The series $\operatorname{Re}\psi(x+iy)-\psi(x) = \sum_{k\ge0} y^2/\big((x+k)((x+k)^2+y^2)\big)$ shows that A_4^∞ increases in $|\theta|$. Since $A_4^\infty(\alpha,0)$ increases in α and $S_4(\alpha)$ decreases, the root is unique, and `dbeta_thorin.py` locates it by bisection with certified signs. For χ_{12} the primes 2 and 3 drop out of the sum. □

In the sine squared form (8.5.9) the same bound reads $b_\alpha + G_\alpha(\theta) - 2\sum_{\chi_4(n)=-1}\Lambda(n)n^{-\alpha}$: the positive atoms are dropped and each negative atom is given its largest value 2 of $1-\cos$. The smaller value of α_c for χ_{12} reflects its larger conductor constant $\frac12\log\frac{12}{\pi}$ and the absence of the primes 2 and 3.

Proposition 8.5.26. *For $\alpha>1$ and every $T>0$,*

$$\inf_{\theta\ge T}\Big(v_\alpha(\theta)-A_4^\infty(\alpha,\theta)\Big) = \inf_{\theta\ge T}\operatorname{Re}\frac{L'}{L}(\alpha+i\theta,\chi_4) = -S_4(\alpha),$$

so no lower bound for $\operatorname{Re}\frac{L'}{L}(\alpha+i\theta,\chi_4)$ better than $-S_4(\alpha)$ holds at all large heights. The bound of Proposition 8.5.25 certifies $v_\alpha(\theta)>0$ exactly for $|\theta|>t_0(\alpha)$, where $A_4^\infty(\alpha,t_0)=S_4(\alpha)$, for $1<\alpha\le\alpha_c$ (so that $t_0(\alpha_c)=0$), and at every real θ for $\alpha>\alpha_c$, and

$$\log t_0(\alpha) = \frac{2}{\alpha-1} - 2\gamma_E - \log\frac{8}{\pi} + O(\alpha-1), \qquad \alpha\downarrow 1,$$

with $-2\gamma_E-\log\frac{8}{\pi} = -2.0891$.

Proof. From the Euler product, $\operatorname{Re}\frac{L'}{L}(\alpha+i\theta,\chi_4) = -\sum_p \log p\,\operatorname{Re}\frac{w_p}{1-w_p}$ with $w_p=\chi_4(p)p^{-\alpha-i\theta}$, and on the circle $|w|=r<1$ the maximum of $\operatorname{Re}\frac{w}{1-w} = \operatorname{Re}\frac{1}{1-w}-1$ is $\frac{r}{1-r}$, at $w=r$; this gives the lower bound $-S_4(\alpha)$. Since the numbers $\log p$ are linearly independent over $\mathbb{Q}$, Kronecker's theorem [90, Chapter 23] gives, for every P and $\eta>0$, a height $\theta\ge T$ at which $|\chi_4(p)p^{-i\theta}-1|<\eta$ for all odd $p\le P$, and the primes beyond P contribute at most $\sum_{p>P}\log p\,p^{-\alpha}/(1-p^{-\alpha})$, which is small for P large. For the asymptotic, $S_4(\alpha)=\frac{1}{\alpha-1}-\gamma_E-\log 2+O(\alpha-1)$ and $A_4^\infty(\alpha,t)=\frac12\log\frac{2t}{\pi}+O(t^{-2})$ as $t\to\infty$, uniformly for $1\le\alpha\le2$, from $\psi(z)=\log z-\frac{1}{2z}+O(|z|^{-2})$; solving $A_4^\infty(\alpha,t_0)=S_4(\alpha)$ gives the formula, since $t_0(\alpha)\to\infty$. □

So for $1<\alpha<\alpha_c$ the absolute values of the weights reduce the positivity of v_α to a computation on $[0,t_0(\alpha)]$, but t_0 grows like $e^{2/(\alpha-1)}$ (Table 8.24). At $\alpha=1$ the prime piece converges only conditionally, by the prime number theorem for progressions, whose analytic input is the absence of zeros of $L(s,\chi_4)$ on $\operatorname{Re}s=1$; the positivity of v_1, which Proposition 8.5.24 draws from that input, cannot come from the sizes of the weights. (Dirichlet's theorem for the modulus 4 itself needs only $L(1,\chi_4)=\frac{\pi}{4}\ne0$.)

α	$S_4(\alpha)$	$t_0(\alpha)$	$\log t_0(\alpha) - \frac{2}{\alpha-1}$
2	0.33891193	2.36723	-1.1383
1.75	0.57950941	4.71677	-1.1155
1.5	1.1261406	14.8728	-1.3005
1.25	2.9637961	589.44	-1.6208
1.1	8.8348984	7.41336×10^7	-1.8786
1.05	18.784554	3.25219×10^{16}	-1.9793

Table 8.24: The height $t_0(\alpha)$ above which the bound of Proposition 8.5.25 certifies $v_\alpha > 0$, for basepoints below $\alpha_c(\chi_4) = 2.2823$; the last column tends to $-2\gamma_E - \log\frac{8}{\pi} = -2.0891$ (Proposition 8.5.26). Computed by `dbeta_thorin.py`.

8.5.7.3 The Gaussian field

The pairing of the primes $p \equiv 1$ and $p \equiv 3 \pmod 4$ is their splitting in $K = \mathbb{Q}(i)$: a prime $p \equiv 1 \pmod 4$ splits into two Gaussian primes of norm p, and a prime $p \equiv 3 \pmod 4$ stays prime, with norm p^2. Accordingly $\zeta_K(s) = \zeta(s)L(s,\chi_4)$, and in the normalisation of Section 10.3.3,

$$\xi_K(s) = 2s(s-1)\pi^{-s}\Gamma(s)\zeta_K(s) = \xi(s)\,\xi(s,\chi_4),$$

by the duplication formula $\Gamma(s) = 2^{s-1}\pi^{-1/2}\Gamma(\frac{s}{2})\Gamma(\frac{s+1}{2})$. Write v_α^K and v_α^ξ for the brackets $\mathrm{Re}(\xi_K'/\xi_K)(\alpha+i\theta)$ and $\mathrm{Re}(\xi'/\xi)(\alpha+i\theta)$, and v_α as before for χ_4.

Proposition 8.5.27. *On every vertical line, $v_\alpha = v_\alpha^K - v_\alpha^\xi$. For $\alpha > 1$ the canonical measure of ξ_K at the basepoint α is*

$$x\Pi_\alpha^K(dx) = \Big(\frac{e^{-(\alpha+1)x}}{1-e^{-x}} - e^{-(\alpha-1)x}\Big)dx + \sum_{n\geq 2}\frac{(1+\chi_4(n))\Lambda(n)}{n^\alpha}\,\delta_{\log n}(dx),$$

and the canonical measure of Proposition 8.5.23 is $x\Pi_\alpha = x\Pi_\alpha^K - x\Pi_\alpha^\xi$, with $x\Pi_\alpha^\xi$ that of Section 9.1.4: the pole densities $-e^{-(\alpha-1)x}$ cancel, the gamma densities differ by $e^{-\alpha x}/(2\sinh x)$, and the atoms by $\chi_4(n)\Lambda(n)n^{-\alpha}$.

Proof. Take logarithmic derivatives in $\xi_K = \xi\cdot\xi(\cdot,\chi_4)$. For the measures write $\xi_K(s) = (s-1)\cdot 2\pi^{-s}\Gamma(s+1)\zeta_K(s)$: Binet's formula for $\Gamma(s+1)$ at α gives the density $e^{-(\alpha+1)x}/(x(1-e^{-x}))$, the factor $s-1$ gives $-e^{-(\alpha-1)x}/x$ as for ξ, and $-\frac{\zeta_K'}{\zeta_K} = -\frac{\zeta'}{\zeta} - \frac{L'}{L}$ has the coefficients $(1+\chi_4(n))\Lambda(n) \geq 0$. Finally $\frac{e^{-(\alpha+1)x}}{1-e^{-x}} - \frac{e^{-(\alpha+2)x}}{1-e^{-2x}} = e^{-\alpha x}\Big(\frac{1}{e^x-1} - \frac{1}{e^{2x}-1}\Big) = \frac{e^{-\alpha x}e^x}{e^{2x}-1} = \frac{e^{-\alpha x}}{2\sinh x}$. □

Corollary 8.5.28. (i) *For every $\alpha \geq 1$ there are independent GGC variables H_α^ξ and $H_\alpha^{(4)}$, with Laplace transforms $\xi(\alpha)/\xi(\alpha+\sqrt{u})$ and $\xi(\alpha,\chi_4)/\xi(\alpha+\sqrt{u},\chi_4)$,*

whose sum $H_\alpha^K = H_\alpha^\xi + H_\alpha^{(4)}$ *is the GGC with Laplace transform* $\xi_K(\alpha)/\xi_K(\alpha+\sqrt{u})$ *and Thorin density* v_α^K/π.

(ii) *GRH for* χ_4 *holds if and only if* $v_\alpha^K \geq v_\alpha^\xi$ *on every line* $\operatorname{Re} s = \alpha > \frac{1}{2}$*: the Thorin density of* ξ_K *dominates that of* ξ *at every basepoint.*

(iii) *Under RH, GRH for* χ_4 *holds if and only if, for every* $\alpha > \frac{1}{2}$, $\xi_K(\alpha)/\xi_K(\alpha+\sqrt{u})$ *is the Laplace transform of* $H_\alpha^\xi + H$ *for some GGC* H *independent of* H_α^ξ.

Proof. (i) By Theorem 1.2.2 and Proposition 8.5.24 both factors are GGC transforms for $\alpha \geq 1$, and transforms of independent sums multiply. (ii) By Proposition 8.5.27, $v_\alpha^K \geq v_\alpha^\xi$ is $v_\alpha \geq 0$, and Proposition 8.5.24 applies. (iii) Under RH, H_α^ξ is a GGC for every $\alpha > \frac{1}{2}$ (Theorem 1.2.2). If GRH holds for χ_4, take $H = H_\alpha^{(4)}$. Conversely, the transform of H is then $\xi(\alpha,\chi_4)/\xi(\alpha+\sqrt{u},\chi_4)$, and if it is a GGC transform for every $\alpha > \frac{1}{2}$, Proposition 8.5.24 gives GRH. □

Neither v_α^K nor v_α^ξ is positive term by term: each has the pole piece $-\theta^2/\big((\alpha-1)((\alpha-1)^2+\theta^2)\big)$ of (3.1.1), and the two cancel in the difference, which is why $\xi(s,\chi_4)$ has none. The factorisation is the right language for the pairing of the primes, but it does not by itself lower α_c: by Proposition 8.5.26 any argument that uses only the sizes of the weights reaches no further. Table 8.25 shows the three brackets at $\alpha = \frac{3}{2}$; `dbeta_thorin.py` checks $v_\alpha^K - v_\alpha^\xi - v_\alpha = 0$ there to 10^{-36}, and the identity $\xi_K = \xi \cdot \xi(\cdot,\chi_4)$ at $s = 0.3 + 4.1i$, where both sides are $0.21807580405390 - 0.037869233276222i$ after division by $\xi(s)$, and at $s = 2.5$.

θ	0	2	5	10	20
$v_{3/2}^K(\theta)$	0.200351	0.225243	0.685189	1.32766	1.49863
$v_{3/2}^\xi(\theta)$	0.0461359	0.0470396	0.0526739	0.102889	0.606114
$v_{3/2}(\theta)$	0.154215	0.178204	0.632515	1.22477	0.892517

Table 8.25: The brackets of $\xi_{\mathbb{Q}(i)}$, ξ and $\xi(\cdot,\chi_4)$ at the basepoint $\frac{3}{2}$; the last row is the difference of the first two (Proposition 8.5.27). Computed by `dbeta_thorin.py`.

The same holds for every real primitive character χ_D, with D a fundamental discriminant and $K = \mathbb{Q}(\sqrt{D})$: ξ_K is a constant multiple of $\xi \cdot \xi(\cdot,\chi_D)$, the constant being $2/\sqrt{|D|}$ for $D < 0$ (Section 8.5.6) and 2 for $D > 0$, since then $\xi_K(s) = s(s-1)D^{s/2}\Gamma_{\mathbb{R}}(s)^2\zeta_K(s)$ and $\xi(s,\chi_D) = D^{s/2}\Gamma_{\mathbb{R}}(s)L(s,\chi_D)$. So $v_\alpha^{(\chi_D)} = v_\alpha^K - v_\alpha^\xi$; the GGC property at every $\alpha \geq 1$, the equivalence of GRH for χ_D with the domination of the Thorin density of ξ by that of ξ_K, and the localisation of Corollary 8.5.31 hold for $\xi(s,\chi_D)$ with the same proofs. For χ_{12}, $K = \mathbb{Q}(\sqrt{3})$, and the numbers are: $\alpha_c(\chi_{12}) = 1.8348743295$ (Proposition 8.5.25); $V_{12} = (\log\xi)''(\frac{1}{2},\chi_{12}) = 0.332923608212$; and at $\alpha = 1$, 0.75 and 0.6 the drift b_α is 0.1650833, 0.0830566 and 0.0332812, the minimum of v_α on a grid of step

0.01 on $[0,60]$, attained at the origin (`dbeta_minimum.py`), as Theorem 9.1.11 proves at every height for $\alpha \geq 1$ and on the verified range for $\frac{1}{2} < \alpha < 1$. The script `dbeta_zeros.py` certifies the first five ordinates of $L(s,\chi_{12})$, 3.8046276331, 6.6922233205, 8.8905929587, 11.1883927451 and 12.9661788080, all simple and on the critical line, in agreement with Section 8.5.1; their largest gap, the first, 2.8876, lies below $2/\sqrt{V_{12}} = 3.4662$, and below the value 3.3189 at $a = \frac{1}{2}$ of the threshold of Remark 9.1.13 for χ_{12}. On the real axis the same script gives $L(\sigma,\chi_{12}) = 0.1246437042$, 0.3344871142, 0.4985570025, 0.6250274399 and 0.7213905658 at $\sigma = 0.1$, 0.3, 0.5, 0.7 and 0.9, positive, as the positivity of the kernel requires.

QUESTION 8.5.29. Is there a construction of $H_\alpha^K = H_\alpha^\xi + H_\alpha^{(4)}$ from the Gaussian primes themselves, two of norm p for each $p \equiv 1$ and one of norm p^2 for each $p \equiv 3 \pmod 4$, that persists below the basepoint one? By Proposition 8.5.26 such a construction must see the phases $p^{-i\theta}$, not only the sizes of the weights; below $\alpha = 1$ its existence is equivalent to GRH for χ_4 under RH, by Corollary 8.5.28(iii). In the language of Section 8.5.7.4, the decomposition is the additivity $\arg \xi_K = \arg \xi + \arg \xi(\cdot,\chi_4)$ on vertical lines, and the domination of Corollary 8.5.28(ii) says that $\arg \xi_K - \arg \xi$ is nondecreasing on every line $\operatorname{Re} s = \alpha > \frac{1}{2}$.

8.5.7.4 The Thorin measure as an argument

The density $v_\alpha(\theta)$ is the derivative in θ of the argument of $\xi(\alpha + i\theta,\chi_4)$, so the Thorin measure on a vertical line is an argument, and at the centre a zero count.

PROPOSITION 8.5.30. *Let $\alpha > \frac{1}{2}$ be such that no zero of $\xi(s,\chi_4)$ lies on the line $\operatorname{Re} s = \alpha$; this holds for every $\alpha \geq 1$. Then for $\theta > 0$*

$$\frac{1}{\pi}\int_0^\theta v_\alpha(t)\,dt = \frac{1}{\pi}\arg \xi(\alpha + i\theta,\chi_4),$$

with the argument continuous along the segment from α to $\alpha + i\theta$ and zero at α; for $\alpha \geq 1$ the left side is the mass of $(0,\theta^2]$ under the Thorin measure of Proposition 8.5.24. As $\theta \to \infty$, with $a = \frac{\alpha+1}{2}$,

$$\frac{1}{\pi}\arg \xi(\alpha+i\theta,\chi_4) = \frac{\theta}{2\pi}\log\frac{2\theta}{\pi e} + \frac{\alpha}{4} + \frac{1}{\pi}\arg L(\alpha+i\theta,\chi_4) - \frac{a(a-1)+\frac{1}{6}}{\pi\theta} + O(\theta^{-3}),$$

with $\arg L$ continuous along the same segment. As $\alpha \downarrow \frac{1}{2}$, for θ not an ordinate, the left side tends to the number of zeros on the critical line with $0 < \gamma < \theta$, which is $N(\theta)$ under GRH; at $\alpha = \frac{1}{2}$ the expansion is the counting formula (8.5.8).

Proof. Since $\frac{d}{dt}\log \xi(\alpha+it,\chi_4) = i\frac{\xi'}{\xi}(\alpha+it,\chi_4)$, the derivative of $\arg \xi(\alpha+it,\chi_4) = \operatorname{Im}\log \xi(\alpha+it,\chi_4)$ is $v_\alpha(t)$, and $\xi(\alpha,\chi_4) > 0$. The expansion is $\frac{\theta}{2}\log\frac{4}{\pi}$ plus Stirling's series $\log\Gamma(z) = (z-\frac{1}{2})\log z - z + \frac{1}{2}\log 2\pi + \frac{1}{12z} + O(|z|^{-3})$ [177, §5.11]

at $z = a + \frac{i\theta}{2}$: with $y = \frac{\theta}{2}$,

$$\operatorname{Im}\log\Gamma(a+iy) = y\log y - y + \frac{\pi}{2}\Big(a-\frac{1}{2}\Big) - \Big(\frac{a(a-1)}{2}+\frac{1}{12}\Big)\frac{1}{y} + O(y^{-3}),$$

from $\arg z = \frac{\pi}{2} - \frac{a}{y} + O(y^{-3})$ and $\log|z| = \log y + \frac{a^2}{2y^2} + O(y^{-4})$; and $\frac{\theta}{2}\log\frac{4}{\pi} + \frac{\theta}{2}\log\frac{\theta}{2} - \frac{\theta}{2} = \frac{\theta}{2}\log\frac{2\theta}{\pi e}$, $\frac{1}{2}(a-\frac{1}{2}) = \frac{\alpha}{4}$. For the limit, write v_α by the Hadamard product, as in Proposition 1.2.3, as the absolutely convergent sum over the zeros $\rho = \beta + i\gamma$ of $(\alpha-\beta)/((\alpha-\beta)^2 + (t-\gamma)^2)$; each term integrates over $[0,\theta]$ to $\arctan\frac{\theta-\gamma}{\alpha-\beta} + \arctan\frac{\gamma}{\alpha-\beta}$, which for a zero on the line tends to π if $0 < \gamma < \theta$ and to 0 if $\gamma > \theta$ or $\gamma < 0$, while the two terms of an off line pair $\frac{1}{2} \pm \delta + i\gamma$ have the limits $\mp\big(\arctan\frac{\theta-\gamma}{\delta} + \arctan\frac{\gamma}{\delta}\big)$, which cancel. The terms are dominated by π and by $C\theta/\gamma^2$ for $|\gamma| > 2\theta + 1$, uniformly for $\frac{1}{2} < \alpha \le 1$, so dominated convergence applies. At $\alpha = \frac{1}{2}$, $a = \frac{3}{4}$ and $a(a-1) + \frac{1}{6} = -\frac{1}{48}$. □

The Thorin measure at resolution $\varepsilon = \alpha - \frac{1}{2}$ is therefore the zero counting function smoothed by the Cauchy kernel of width ε; this is the analogue for χ_4 of the cumulative Thorin mass of ξ as a smoothed counting function (Section 3.4.1). The script `dbeta_thorin.py` continues $\arg\xi(\alpha+i\theta,\chi_4)$ in ball arithmetic, as `dbeta_zeros.py` does, and integrates v_α by quadrature: at $(\alpha,\theta) = (1.5,10)$, $(1.5,25)$ and $(2,30)$ both sides are 1.65731473994, 7.40149777268 and 9.85153586211. At $\alpha = 0.75$ the difference between the left side and the first three terms of the expansion is -3.65×10^{-4}, -1.82×10^{-4} and -9.12×10^{-5} at $\theta = 50$, 100 and 200, that is $-0.0182/\theta$, against $-(a(a-1)+\frac{1}{6})/\pi = -11/(192\pi) = -0.018237$ at $a = \frac{7}{8}$.

Corollary 8.5.31. *GRH for χ_4 holds if and only if, for every $\alpha > \frac{1}{2}$, $\theta \mapsto \arg\xi(\alpha+i\theta,\chi_4)$ is nondecreasing. With no hypothesis, the heights $\theta > 0$ at which it decreases lie in the union of the windows $|\theta-\gamma| < \sqrt{\delta^2-(\alpha-\frac{1}{2})^2}$, each of length less than 1, over the zeros $\frac{1}{2}+\delta+i\gamma$ with $\delta > \alpha - \frac{1}{2}$, and there are none at heights below $2.5\times10^7 - \frac{1}{2}$.*

Proof. The derivative is $v_\alpha(\theta)$, and Proposition 8.5.24 gives the equivalence. By the Hadamard product, v_α is a sum of positive Cauchy kernels, one for each zero on the line and two for each off line pair $\frac{1}{2} \pm \delta + i\gamma$ with $\delta \le \alpha - \frac{1}{2}$, plus, for each pair with $\delta > \alpha - \frac{1}{2}$, the difference $\pi\big(P_{\varepsilon+\delta} - P_{\delta-\varepsilon}\big)(\theta \mp \gamma)$ with $\varepsilon = \alpha - \frac{1}{2}$, which is negative exactly where $|\theta \mp \gamma| < \sqrt{\delta^2-\varepsilon^2}$ (proof of Proposition 3.3.1(ii)); for $\theta > 0$ and $\gamma > 0$ the reflected term at $\theta + \gamma$ is positive, since $\theta + \gamma > \gamma > \delta$. A window has length $2\sqrt{\delta^2-\varepsilon^2} < 2\delta < 1$. Platt's verification of GRH for the modulus 4 to height $10^8/4$ [195] puts every zero off the line above 2.5×10^7, and so every window above $2.5\times10^7 - \frac{1}{2}$. □

A window holds a decrease only if the negative term of its pair outweighs all the positive terms there, and the decrease is bounded: the negative part of the term

$\pi(P_{\varepsilon+\delta}-P_{\delta-\varepsilon})(\theta-\gamma)$ of a pair has integral $4\arctan\sqrt{(\delta+\varepsilon)/(\delta-\varepsilon)}-\pi<\pi$, by Proposition 3.3.1(iii), so over any interval the argument falls by less than π times the number of pairs whose windows meet it, whatever the other zeros, and the fall caused by a pair tends to zero as $(\alpha-\frac12)/\delta\to0$. This answers the quantitative half of a question of the first author's note. No lower bound for the positive terms over a window, such as a mean value estimate for the contribution of the zeros on the line, can exclude this for every pair: at the centre $\theta=\gamma$ of its window the pair contributes $\frac{1}{\varepsilon+\delta}-\frac{1}{\delta-\varepsilon}=-\frac{2\varepsilon}{\delta^2-\varepsilon^2}$, which is unbounded below as $\delta\downarrow\varepsilon$, that is as the zero approaches the line $\operatorname{Re}s=\alpha$ from the right, while by Lemma 11.4.7 a zero far enough to the right of the line cannot produce a decrease against a given positive floor. Outside the strips $|\theta-\gamma|<(1+\sqrt2)\delta$ around the ordinates of the zeros off the line, Theorem 9.1.8 gives the argument a rate of increase that grows like $\log\theta$ (Corollary 9.1.9). Figure 8.19 shows the smoothing, and the minimum at the origin that Section 9.1.6 proves.

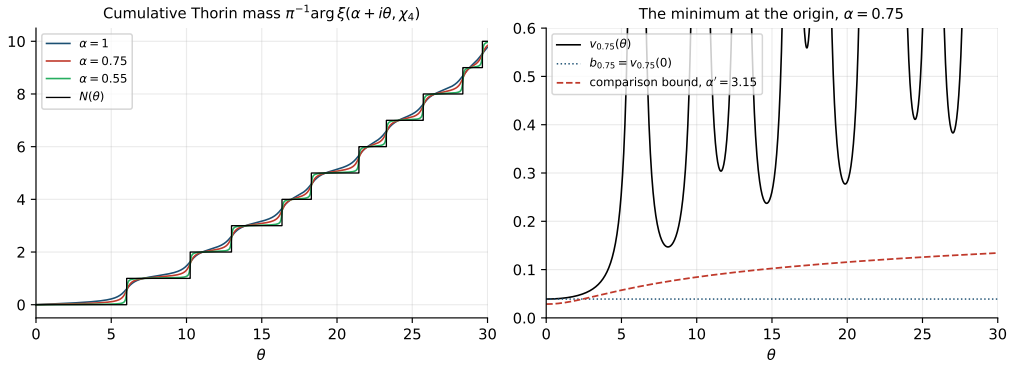


Figure 8.19: Left: the cumulative Thorin mass $\frac1\pi\arg\xi(\alpha+i\theta,\chi_4)=\frac1\pi\int_0^\theta v_\alpha$ for $\alpha=1$, 0.75 and 0.55, against the zero count $N(\theta)$ of $L(s,\chi_4)$; at $\theta=30$ the masses are 9.7747, 9.8405 and 9.9621, and $N(30)=10$. The argument is continued through the quotients of values at steps of 0.005, each of argument below 0.11 in modulus. Right: $v_{0.75}(\theta)$, its minimum $b_{0.75}=0.0389784$ at the origin, and the lower bound of Corollary 9.1.9 with $\alpha'=3.15$, which exceeds $b_{0.75}$ for $\theta>2.434$ (`dbeta_minimum.py`). Drawn by `dbeta_figs.py`.

8.5.7.5 The centre

For ζ the positive structure at the centre is the Thorin measure of the clock, not a structure of the Pólya variable itself (Proposition 11.1.3(1)). The same holds for χ_4. The characteristic function $\xi(\frac12+i\theta,\chi_4)/\xi(\frac12,\chi_4)$ of Y_4 is real and changes sign at $\gamma_1=6.0209489$, where it passes from 0.00607732 at $\theta=5.9$ to -0.00350813 at $\theta=6.1$, and at each of the other ordinates below height 100; Hardy's argument

for ζ [255, Chapter 10], applied to the theta series $\eta(ix)^3$ of χ_4, gives infinitely many such sign changes. So Y_4 is not quasi infinitely divisible, and in particular not infinitely divisible: a quasi infinitely divisible characteristic function never vanishes. Nor is $\xi(\frac{1}{2}+w,\chi_4)/\xi(\frac{1}{2},\chi_4)$ a positive mixture $\int M(d\lambda)/(w+\lambda)^2$ of Gamma(2) transforms, for a more elementary reason: such a mixture decreases on $w>0$, while $\xi(\frac{1}{2}+w,\chi_4)=\int_{\mathbb{R}} K_4(u)\cosh(wu)\,du$ increases there. A one sided object remains, the half kernel $K_4(u)\mathbf{1}\{u>0\}$, and it is not a positive gamma mixture either.

Proposition 8.5.32. *For no $a>0$ is $u^{1-a}K_4(u)$ completely monotone on $u>0$; in particular the half kernel is not a positive mixture of Gamma(a) densities $u^{a-1}e^{-\lambda u}$, for any shape a. The theta function $\eta(ix)^3$ is not completely monotone on $x>0$.*

Proof. If $g(u)=\int_{[0,\infty)} e^{-\lambda u}M(d\lambda)$ with $M\ge 0$ and $M\ne 0$, choose L with $M([0,L])>0$; then $g(u)\ge M([0,L])e^{-Lu}$, so g decays at most exponentially, while $u^{1-a}K_4(u)$ decays like $\exp(-\frac{\pi}{4}e^{2u})$ (Proposition 8.5.21). The Jacobi series $\eta(ix)^3=\sum_{n\ge0}(-1)^n(2n+1)e^{-\pi(2n+1)^2x/4}$ is the Laplace transform of the signed measure with the weights $(-1)^n(2n+1)$ at the points $\pi(2n+1)^2/4$, which by the uniqueness of the Laplace transform is its only representing measure, and it is not positive, so Bernstein's theorem excludes complete monotonicity. □

The decay argument applies to the kernel of every completed L function of degree one, all of which decay like $\exp(-ce^{2|u|})$; for ξ it shows that no positive mixture of Gamma(a) densities represents the Pólya kernel on the half line, for any shape a, as [198, Section 13.6] observes (see also [200, Section 12.6]).

The positive structure is in the reciprocal of the product. With $q=e^{-2\pi x}$,

$$\frac{e^{-\pi x/4}}{\eta(ix)^3}=\prod_{m\ge1}\left(1-q^m\right)^{-3}=\exp\Big(3\sum_{N\ge1}\sigma_{-1}(N)\,e^{-2\pi Nx}\Big),\qquad \sigma_{-1}(N)=\sum_{d\mid N}\frac{1}{d},\tag{8.5.11}$$

from $-\log(1-q^m)=\sum_k q^{mk}/k$; so the logarithm of the left side is completely monotone, and hence so is the left side, the exponential of a completely monotone function being completely monotone. At $x=0.3$ the three forms agree to fifteen digits, 1.77964734678018 (`dbeta_thorin.py`). In the language of Section 8.5.4, $\prod_m(1-e^{-2\pi mx})^3$ is the distribution function of the maximum M_3 of independent exponential variables, three of each rate $2\pi m$, and (8.5.11) is the complete monotonicity of minus the logarithm of such a distribution function noted in Section 8.5.5, with the masses $3\sigma(N)/N$. The representation of Section 8.5.4 also carries over: since $\eta(ix)^3=e^{-\pi x/4}\Pr(M_3\le x)$ and $x^{c-1}e^{-\pi x/4}(\pi/4)^c/\Gamma(c)$ is the density of $\frac{4}{\pi}G_c$, with G_c a gamma variable of shape c independent of M_3, the Mellin form gives $\xi(s,\chi_4)=(\frac{4}{\pi})^c\Gamma(c)\Pr(\frac{4}{\pi}G_c\ge M_3)$ with $c=\frac{s+1}{2}$, that is

$$L(s,\chi_4)=\Pr\left(4G_{(s+1)/2}\ge\pi M_3\right),\qquad s>-1,$$

which at $s = 1$ is $\mathbb{E}e^{-\pi M_3/4} = \frac{\pi}{4}\int_0^\infty \eta(ix)^3 dx = \frac{\pi}{4}$.

The half kernel has the Laplace transform

$$\int_0^\infty e^{-wu} K_4(u)\, du = A_4\big(\tfrac{1}{2} - w\big), \tag{8.5.12}$$

where

$$A_4(s) = \int_1^\infty \eta(ix)^3 x^{(s+1)/2} \frac{dx}{x} = \sum_{n\ge 1} \chi_4(n)\, n\, \frac{\Gamma(\frac{s+1}{2}, \frac{\pi n^2}{4})}{(\pi n^2/4)^{(s+1)/2}}$$

is an incomplete Mellin transform of η^3, and the transformation $\eta(i/x)^3 = x^{3/2}\eta(ix)^3$, applied to the integral over $(0,1)$, gives $\xi(s,\chi_4) = A_4(s) + A_4(1-s)$ for every s. This is the analogue of Riemann's formula (Proposition 8.5.5) and of the symmetric form (8.5.5) for Δ, with no pole terms, since $\eta(ix)^3$ has no constant term. At $w = 0.3$ both sides of (8.5.12) are 0.44608019686738, and $A_4(0.2) + A_4(0.8) = \xi(0.2, \chi_4) = 0.98762005303445$ (`dbeta_thorin.py`).

Question 8.5.33. Can the complete monotonicity of (8.5.11), and of its analogue $\prod_m (1-q^m)^{-1}$ for the kernel of χ_{12}, be turned into a positive representation on the reciprocal side, in the spirit of Thorin's construction for the reciprocal of a GGC transform?

8.6 The inverse problem: recovering the atoms by nonnegative deconvolution

We return to ζ. The distances of Section 3.2 compare its Thorin density with the Poisson smoothing of a known list of zeros; the inverse problem is to recover the zeros from the density alone. Under RH the density is the Poisson smoothing of a positive measure with an atom at each zero, so the recovery is a deconvolution under a positivity constraint, and RH on a window becomes a feasibility question. The monograph [200, Section 19.3] poses it as a Bayesian problem with a positivity prior and a posterior predictive check, and notes that the experiment had not been run because ρ_ε was available only from the primes, at resolutions above one half. The Pólya route of Section 2.1 supplies ρ_ε at small resolution to about 60 digits below height 200 (Section 2.1.1); the experiment uses it in double precision. At $\varepsilon = 0.1$ a nonnegative least squares fit recovers the ten zeros in $[15, 55]$ to within 10^{-5}, with unit masses, and a displaced zero raises the residual roughly like the square of its displacement.

8.6.1 Recovering the zeros by nonnegative least squares

Write $P_\varepsilon(x) = \varepsilon/(\pi(\varepsilon^2 + x^2))$ and $U^{\pm}_{\mathrm{RH}} = \sum_{\gamma>0}(\delta_\gamma + \delta_{-\gamma})$. Under RH, $\rho_\varepsilon = P_\varepsilon * U^{\pm}_{\mathrm{RH}}$ on $\theta > 0$ (Proposition 1.2.3); since P_ε has Fourier transform $e^{-\varepsilon|t|}$, recovering $U^{\pm}_{\mathrm{RH}}$ is severely ill posed. An off line pair $\frac{1}{2} \pm \delta + i\gamma$ with $\delta < \varepsilon$ contributes $P_{\varepsilon+\delta} + P_{\varepsilon-\delta}$ at

γ, where two line zeros would contribute $2P_\varepsilon$, and $P_{\varepsilon-\delta}$ is sharper than the Poisson smoothing at level ε of any positive measure, since $P_{\varepsilon-\delta} = P_\varepsilon * \lambda$ would give λ the exponentially growing Fourier transform $e^{\delta|t|}$. Positivity of the recovered measure is therefore the test.

Fix a data window $[\theta_a, \theta_b]$ with sample points $\theta_1 < \cdots < \theta_n$, and candidate atom locations $x_1 < \cdots < x_m$ on a grid over a candidate window $[x_a, x_b]$ that contains $[\theta_a, \theta_b]$. With $A_{ij} = P_\varepsilon(\theta_i - x_j) + P_\varepsilon(\theta_i + x_j)$ and data $y_i = \rho_\varepsilon(\theta_i) - b_i$, where b_i is the contribution of the zeros outside the candidate window, the problem is

$$\min_{w \geq 0} \ \|Aw - y\|_2.$$

Under a Gaussian likelihood with a flat prior on the positive orthant its solution is the maximum a posteriori estimate, and the minimal residual is the discrepancy statistic of the posterior predictive check: up to the error of the grid, it vanishes when the data are the Poisson smoothing of a positive measure. With data exact to double precision there is no noise level, and the check is a feasibility test. The active set algorithm of Lawson and Hanson [139] solves the problem in finitely many steps; we use its implementation in `scipy`. The generative approach of [200, Section 19.2] would replace the optimisation by a learned map from data to posterior samples, but the optimisation is deterministic, and its output can be compared atom by atom with the known zeros.

We took $\varepsilon = 0.1$, the Pólya computed $\rho_{0.1}$ on $[15, 55]$ at 1601 points (spacing 0.025), and candidates on $[5, 65)$ at spacing 0.02 (3000 columns). The background b_i is the Poisson smoothing of the zeros outside $[5, 65]$ among the first 1700 ordinates, the list of Section 3.2 without its smooth tail, which would add about 3×10^{-5} to each sample; no ordinate inside the candidate window enters the fit. The nonnegative solution concentrates on clusters, runs of adjacent grid points with weights above 10^{-3}, one at each zero in the data window (Table 8.26 and Figure 8.20, left): every ordinate is recovered to within 10^{-5}, with mass one to within half a per cent. The remaining mass, 0.071 of the total 10.094, lies in small clusters at the edges of the data window (0.025), which absorb the tails of the zeros at 14.13 and 56.45 just outside it, and in weights below 10^{-3} (0.046). The residual norm is 0.106, a root mean square of 0.0026 against peaks of height $1/(\pi\varepsilon) = 3.18$. It is set by the grid: the true atoms lie between grid points, and a positive combination of two neighbouring Cauchy kernels reproduces an off grid kernel only to second order in the grid spacing.

8.6.2 A displaced zero and the threshold of detection

To calibrate the check we displaced the fifth zero, $\gamma_5 = 32.935$, in the sense of Proposition 3.3.1(i): we added to the data the change per displaced zero, $g = \frac{1}{2}(P_{\varepsilon+\delta} + P_{\varepsilon-\delta}) - P_\varepsilon$, centred at γ_5, and refitted. Twice g is the change in the density when a double zero at $\frac{1}{2} + i\gamma_5$ leaves the line as the quadruple $\frac{1}{2} \pm \delta \pm i\gamma_5$,

k	γ_k	c_k	$10^6(c_k-\gamma_k)$	mass
2	21.022040	21.022034	−5.5	1.0013
3	25.010858	25.010859	1.6	1.0027
4	30.424876	30.424867	−8.8	1.0020
5	32.935062	32.935070	8.7	1.0020
6	37.586178	37.586171	−7.4	1.0023
7	40.918719	40.918723	3.8	1.0008
8	43.327073	43.327068	−5.5	1.0025
9	48.005151	48.005145	−6.1	1.0024
10	49.773832	49.773840	7.9	1.0028
11	52.970321	52.970330	8.8	1.0042

Table 8.26: The clusters of the nonnegative solution for $\rho_{0.1}$ on $[15, 55]$: for each zero γ_k in the data window, the centre of mass c_k of its cluster, the error of the centre and the total mass of the cluster.

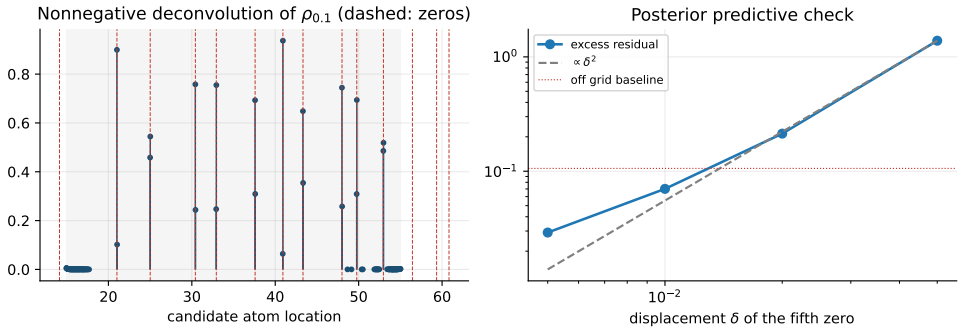


Figure 8.20: Left: the weights of the nonnegative solution for $\rho_{0.1}$ (stems) and the zeros (dashed), with the data window $[15, 55]$ shaded. Each zero in the data window carries adjacent weights of total mass one to within half a per cent; the zeros at 14.13, 56.45, 59.35 and 60.83, outside it, carry none. Right: the excess residual against the displacement δ of the fifth zero, the δ^2 law through the point at $\delta = 0.05$, and the off grid baseline $r_0 = 0.106$.

as at a collision in the de Bruijn Newman flow (Section 8.4). Table 8.27 gives the residuals.

For $\delta \geq 0.02$ the excess residual grows roughly like δ^2, as the signature does (Proposition 3.3.1(i) and [200, Proposition 19.7]): it rises from 0.21 at $\delta = 0.02$ to 1.39 at $\delta = 0.05$, a factor 6.5 for a factor 2.5 in δ, against $2.5^2 = 6.25$. At the smallest displacements it grows more slowly, by a factor 2.4 from $\delta = 0.005$ to 0.01, because there it is dominated by the cross term between the signature and the off grid misfit at γ_5, which has the same shape. The off grid baseline therefore limits detection: the excess reaches it between $\delta = 0.01$ and 0.02 (Figure 8.20,

δ	residual $r = \|Aw - y\|$	excess over baseline	total mass
0	0.1058	0	10.0942
0.005	0.1098	0.029	10.0947
0.01	0.1269	0.070	10.0963
0.02	0.2380	0.213	10.1031
0.05	1.3891	1.385	10.1566

Table 8.27: Nonnegative deconvolution of $\rho_{0.1}$ with the signature g of a displaced fifth zero added to the data. Here r is the residual norm and $r_0 = 0.1058$ its value at $\delta = 0$; the excess is $\sqrt{r^2 - r_0^2}$.

right). This is the square root law that (3.3.1) states for the matched distance (Section 3.3.2): since the misfit grows like the square of the displacement, the smallest detectable displacement falls like the square root of the achievable residual, whatever limits the residual. A finer grid of spacing h lowers the baseline like h^2 and hence the threshold like h, and continuous atom positions would remove the baseline and leave the threshold to the accuracy of the data.

For $\delta \geq \varepsilon$ the check is unreliable. At $\delta = \varepsilon$ the zero $\frac{1}{2} + \delta + i\gamma_5$ lies on the line of evaluation, where its kernel is a point mass that the samples miss, and the residual, 0.099, falls below the baseline. For $\delta > \varepsilon$ the atom at γ_5 leaves the data and is replaced by the signed term $\frac{1}{2}(P_{\varepsilon+\delta} + P_{\varepsilon-\delta})$ of total mass zero, whose negative part, of mass $\varepsilon/(\pi\delta) + O(\varepsilon^3/\delta^3)$ by Proposition 3.3.1(iii), fades as δ grows. The unit of mass that leaves the data is what the zero density identity of Proposition 3.4.2 counts, and zeros farther from the line than ε are left to that identity.

As [200, Section 19.3.7] emphasises, this is a model check at finite height and not a route to a proof. What the Pólya route adds is the small resolution, where the signature $(\delta/\varepsilon)^2$ of Proposition 3.3.1(i) is largest for a given δ. The recovery and the calibration are computed, at one resolution and on one window.

The direction of this section is the hard one: Poisson smoothing, which moves the basepoint to the right, preserves positivity, while deconvolution, which moves it back towards the critical line, does not (Section 11.4.7). Chapter 9 starts instead at the basepoint one, the resolution $\varepsilon = \frac{1}{2}$ (Section 9.1). At the basepoints between one and the centre, Section 11.4 bounds the mass of the negative part of the split (11.4.1), the Cauchy kernels of any zeros to the right of the basepoint (Theorem 11.4.2), and inserts hypothetical off line zeros into the computed density as this section displaced a zero (Section 11.4.4).

8.7 Pólya's Bessel function: hitting times, a clock and the flow

Pólya replaced the kernel Φ of (1.2.3) by its asymptote Φ^* and proved that the transform Ξ^* has only real zeros [213] (Section 13.4.1). The building block of his proof is the Macdonald function

$$P(\theta) = K_{i\theta/2}(2\pi) = \int_{\mathbb{R}} e^{-2\pi\cosh 2u}\, e^{i\theta u}\, du, \tag{8.7.1}$$

which has only real zeros and whose zero counting function has the main term of the Riemann von Mangoldt formula. Biane [27] showed that Ξ and P are Mellin transforms of two Brownian hitting times, and that the Thorin measures of these hitting times are spectral measures of the radial Brownian motions on the unitary group $SU(2)$ and on the hyperbolic space $SL_2(\mathbb{C})/SU(2)$. This section follows the first author's note on Biane's paper [202]. For P every statement of the Thorin programme is a theorem with an explicit witness: the zeros are the Dirichlet spectrum of a Liouville operator, the reciprocal $K_0(2\pi)/K_{\sqrt{u}}(2\pi)$ is a GGC whose Thorin atoms are the squared zeros, its clock is the hitting time of an explicit diffusion, and its Poisson trace is completely monotone because it is the trace of a semigroup. The model also marks where the analogy stops: by a theorem of Newman, Pólya's function has a finite de Bruijn Newman constant, although its zeros show no sign of colliding in the range we compute. Section 8.7.1 sets out the two hitting times and corrects the constants of Biane's dictionary, Section 8.7.2 reads the dictionary as a Wick rotation, Sections 8.7.3 to 8.7.5 treat the zeros of P, the index side clock and a Klein Gordon criterion, Section 8.7.6 follows the zeros of P and of Ξ under the de Bruijn Newman flow, and Section 8.7.7 sets the two functions side by side. The complete monotonicity conditions of the Poisson trace, and its comparison with the trace of P, are in Section 10.1.5. The scripts are `biane_polya.py`, `biane_clock.py`, `biane_flow.py`, `biane_traces.py` and `biane_figs.py`, and every number below that is not quoted from another section comes from one of them.

8.7.1 Two hitting times and Biane's dictionary

As $u \to \pm\infty$ only the term $n = 1$ of (1.2.3) matters, and $\Phi(u) \sim 4\pi^2 e^{9|u|/2}\exp(-\pi e^{2|u|})$. Pólya replaced Φ by the even function with the same asymptotics on both sides,

$$\Phi^*(u) = 8\pi^2 \cosh\tfrac{9u}{2}\, e^{-2\pi\cosh 2u},$$

$$\Xi^*(\theta) = \int_{\mathbb{R}} \Phi^*(u) e^{i\theta u}\, du = 4\pi^2\Big[P\big(\theta - \tfrac{9i}{2}\big) + P\big(\theta + \tfrac{9i}{2}\big)\Big],$$

since the factor $e^{\pm 9u/2}$ shifts the argument of the transform by $\mp\frac{9i}{2}$. This is the form $G(z+ia) + G(z-ia)$ of Pólya's Hilfssatz II (Section 7.1.3) with $G =$

P and $a = \frac{9}{2}$: Pólya proved that P, an entire function of order one, has only real zeros, and the Hilfssatz gives the same for Ξ^*. In the index of the Bessel function the shift is $\frac{9}{4}$, since $P(\theta \mp \frac{9i}{2}) = K_{i\theta/2\pm 9/4}(2\pi)$, which is the form of Ξ^* in Section 13.4.1. Figure 8.21 compares the three kernels. As a pointwise approximation Φ^* is poor near the origin, $\Phi^*(0) = 8\pi^2 e^{-2\pi} = 0.147447$ against $\Phi(0) = 0.893394$, and the kernel $e^{-2\pi\cosh 2u}$ of P is narrower still; what the three share is the doubly exponential tail, which governs the zero counting functions. Table 8.28 lists the zeros below height 60 and the counts up to height 1000, computed by `biane_polya.py` (the zeros of Ξ^* by sign changes of its transform, those of P from the phase of Proposition 8.7.8 below).

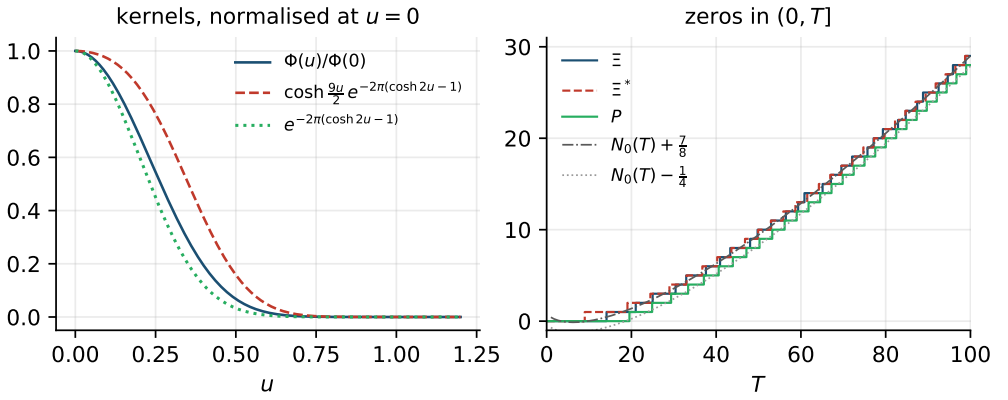


Figure 8.21: Left: the kernel Φ of (1.2.3), Pólya's asymptote Φ^* and the kernel $e^{-2\pi\cosh 2u}$ of P, each divided by its value at $u = 0$. Right: the zero counting functions of Ξ, Ξ^* and P on $(0, 100]$, against $N_0(T) + \frac{7}{8}$ and $N_0(T) - \frac{1}{4}$, the mean values of the Riemann von Mangoldt formula and of Proposition 8.7.8, with $N_0(T) = \frac{T}{2\pi}\log\frac{T}{2\pi e}$. Drawn by `biane_figs.py` from `biane_polya.json`.

The value 2π of the argument is forced. Pólya's asymptotics for the zeros of $y \mapsto K_{iy}(2a)$, which Biane derives by stationary phase [27, Section 3.5], give $\frac{T}{\pi}\log\frac{T}{a} - \frac{T}{\pi} + O(1)$ zeros with $0 < y \le T$; for P the zeros are at $\theta = 2y$ with $a = \pi$, and the count in $(0, T]$ becomes $\frac{T}{2\pi}\log\frac{T}{2\pi} - \frac{T}{2\pi} + O(1) = N_0(T) + O(1)$. The constant $a = \pi$ comes from the factor $e^{-\pi e^{2u}}$ of the term $n = 1$, and the 2 in $\cosh 2u$ turns $\frac{T}{\pi}\log\frac{T}{a}$ into $\frac{T}{2\pi}\log\frac{T}{2\pi}$. Proposition 8.7.8 gives the counting function of P exactly.

Let $H_a^{(c)}$ be the first passage time of $B_t + ct$ to the level $a > 0$, for a standard Brownian motion B and a drift $c \ge 0$, as in Appendix C. Its density is $a(2\pi\tau^3)^{-1/2}e^{-(a-c\tau)^2/2\tau}$, the inverse Gaussian law of Table 1.9, and its Laplace transform is $e^{a(c-\sqrt{c^2+2u})}$.

Proposition 8.7.1. *Let $H = H_a^{(c)}$ with $a = c = \sqrt{2\pi}$.*

n	Ξ	Ξ^*	P
1	14.135	8.993	19.538
2	21.022	19.065	24.897
3	25.011	24.475	29.370
4	30.425	28.975	33.383
5	32.935	33.007	37.099
6	37.586	36.737	40.600
7	40.919	40.249	43.936
8	43.327	43.595	47.140
9	48.005	46.807	50.234
10	49.774	49.908	53.236
11	52.970	52.916	56.158
12	56.446	55.843	59.009
13	59.347	58.700	

T	Ξ	Ξ^*	P	$N_0(T)$
60	13	13	12	11.998
100	29	29	28	28.127
240	102	102	101	100.946
1000	649	649	647	647.741

Table 8.28: Left: the positive zeros below 60 of Ξ, of Pólya's Ξ^* and of $P(\theta) = K_{i\theta/2}(2\pi)$. Right: the numbers of zeros in $(0,T]$ against $N_0(T) = \frac{T}{2\pi}\log\frac{T}{2\pi e}$. Computed by `biane_polya.py`.

(i) *H has the density $\tau^{-3/2}e^{2\pi}e^{-\pi(\tau+1/\tau)}$ on $(0,\infty)$, and $1/H$ has the law of H size biased by H.*

(ii) *$\mathbb{E}H^s = 2e^{2\pi}K_{s-\frac12}(2\pi)$ for every $s\in\mathbb{C}$, so that $\mathbb{E}H^s = \mathbb{E}H^{1-s}$.*

(iii) *$P(\theta) = \frac12 e^{-2\pi}\,\mathbb{E}H^{\frac12+\frac{i\theta}{2}}$ for every $\theta\in\mathbb{C}$.*

(iv) *$U = \frac12\log H$ has the density $2e^{2\pi}e^{-u}e^{-2\pi\cosh 2u}$, and the law of U size biased by e^U has the density $e^{-2\pi\cosh 2u}/K_0(2\pi)$, whose characteristic function is $P(\theta)/P(0)$.*

Proof. (i) With $a=c=\sqrt{2\pi}$, $(a-c\tau)^2/2\tau = \pi(\tau+1/\tau)-2\pi$ and $a/\sqrt{2\pi}=1$. The density of $1/H$ at σ is σ^{-2} times that of H at $1/\sigma$, which is $\sigma^{-1/2}e^{2\pi}e^{-\pi(\sigma+1/\sigma)}$, that is σ times the density of H at σ; in particular $\mathbb{E}H = 1$. (ii) The substitution $\tau = e^w$ and $K_\nu(x) = \int_0^\infty e^{-x\cosh w}\cosh(\nu w)\,dw$ [177, (10.32.9)] give $\int_0^\infty \tau^{\nu-1}e^{-\pi(\tau+1/\tau)}\,d\tau = 2K_\nu(2\pi)$, here with $\nu = s-\frac12$; the integral converges for every complex ν, and $K_\nu = K_{-\nu}$ gives the symmetry. (iii) is (ii) at $s=\frac12+\frac{i\theta}{2}$. (iv) The substitution $\tau = e^{2u}$ turns the density of (i) into $2e^{2\pi}e^{-3u}e^{-2\pi\cosh 2u}e^{2u}$; multiplying by e^u and dividing by $\mathbb{E}H^{1/2} = 2e^{2\pi}K_0(2\pi)$ gives the last density, and its characteristic function is $\int e^{-2\pi\cosh 2u}e^{i\theta u}du/K_0(2\pi) = P(\theta)/P(0)$ by (8.7.1). □

The parallel with Ξ is exact. By the identity (9.1.1) of Biane, Pitman and Yor [28], with $V = \frac{\pi}{2}S_2$, $2\xi(s) = \mathbb{E}V^{s/2}$, so $\Xi(\theta) = \frac12\mathbb{E}V^{\frac14+\frac{i\theta}{2}}$, and the kernel is $\Phi(u) = \frac12 e^{u/2}f_Y(u)$ with $Y = \frac12\log V$: the Pólya law of (1.2.4) is the law of Y size biased by $e^{Y/2}$, as the law of (iv) is that of U size biased by e^U. On the critical

line each function is the Fourier transform, in the logarithm of a hitting time, of the law tilted by a power of that time, and in both cases the evenness of the kernel is a reflection symmetry of the hitting time: $\tau \mapsto 1/\tau$ for H, which is (i), and the functional equation $\mathbb{E}V^{s/2} = \mathbb{E}V^{(1-s)/2}$ for V, which comes from Jacobi's theta transformation. Numerically, `biane_polya.py` integrates the density of (i) against $\tau^{\frac12+3.5i}$ and obtains $e^{-2\pi}\mathbb{E}H^{\frac12+3.5i} = 7.261817522\times10^{-4} = 2P(7)$, with an imaginary part of 2×10^{-50}, and it checks $\mathbb{E}S_2^s = 2(2/\pi)^s\xi(2s)$, which is (9.1.1), by integrating against the density of S_2 at $s = 1$, 0.7 and $\frac12+3i$ (values 0.6666666667, 0.7384660825 and $0.1205355309 - 0.3164244329i$, to ten digits both ways).

Both hitting times are GGCs, with the Thorin measures of Table 1.9. For $H_a^{(c)}$ the Thorin measure is $a\,\{\pi\sqrt{2z-c^2}\}^{-1}\mathbf{1}_{z>c^2/2}\,dz$, and the Lévy density, t^{-1} times its Laplace transform, is $a\,e^{-c^2t/2}(2\pi t^3)^{-1/2}$; at $a = c = 1$ and $t = 0.8$ both expressions give 0.3737294249. For the Bessel hitting time, $\mathbb{E}e^{-uJ_a} = a\sqrt{2u}/\sinh(a\sqrt{2u})$ with $J_a \overset{d}{=} a^2J$, so $J_a + J_a' = a^2S_2$ has Thorin measure $2\sum_{n\ge1}\delta_{\pi^2n^2/2a^2}$ and Lévy density $2t^{-1}\sum_n e^{-\pi^2n^2t/2a^2}$. The Thorin measure can also be read from the continuation of $\Psi(u) = -\log\mathbb{E}e^{-uX}$ across the cut: integrating the Stieltjes inversion (1.3.2),

$$U\big((0,z)\big) = \frac{1}{\pi}\operatorname{Im}\Psi(-z+i0) \qquad (z>0 \text{ not an atom}),$$

because each factor $\log(1+u/w)$ with $w < z$ contributes $i\pi$ at $u = -z+i0$. For $H_a^{(c)}$, $\Psi(-z+i0) = a\big(i\sqrt{2z-c^2}-c\big)$ for $z > \frac{c^2}{2}$, so $U((0,z)) = \frac{a}{\pi}\sqrt{2z-c^2}$, which is 0.7117625434 at $a = c = 1$, $z = 3$, as the integral of the density gives; for $J_a + J_a'$, $\Psi(-\frac{\lambda^2}{2}+i0) = 2\log\frac{\sin\lambda a}{\lambda a}$, which gains $2\pi i$ each time λa passes a multiple of π. Biane's dictionary identifies these Thorin measures with spectral measures. In his normalisation [27, (2.1) to (2.3)] the generator $L_1 = \frac12\frac{d^2}{dx^2}$ on $[0,\pi]$ with Dirichlet conditions has the eigenfunctions $\sin(kx)$, $k = 1,2,\dots$, with eigenvalues $-\frac{k^2}{2}$, and the spectral measure $\sigma_1 = \frac{1}{\pi}\sum_{n\ge1}\delta_{n^2/2}$; the generator $L_2 = \frac12\frac{d^2}{dx^2} + \frac{d}{dx}$ on $[0,\infty)$ with a Dirichlet condition at 0, selfadjoint in $L^2(e^{2x}dx)$, has the generalised eigenfunctions $e^{-x}\sin(kx)$, $k>0$, with eigenvalues $-\frac{k^2+1}{2}$, and the spectral measure $\sigma_2(d\lambda) = d\lambda/(\pi\sqrt{2\lambda-1})$ on $\lambda > \frac12$. In the wave number k the two measures are $\frac{1}{\pi}$ times counting measure and $\frac{1}{\pi}$ times Lebesgue measure.

THEOREM 8.7.2 (Biane's dictionary, with corrected constants). *(i) The Thorin measure of $J_\pi + J_\pi' = \pi^2S_2$ is $2\pi\sigma_1$.*

(ii) The Thorin measure of $H_1^{(1)}$ is σ_2.

(iii) The radial generator $\frac12\frac{d^2}{dx^2} + \cot x\frac{d}{dx}$ of Brownian motion on the unit sphere $S^3 \cong SU(2)$ is the Doob transform of L_1 by its ground state $\sin x$, $\sin^{-1}(L_1+\frac12)\sin$, and the radial generator $\frac12\frac{d^2}{dx^2} + \coth x\frac{d}{dx}$ of Brownian motion on $\mathbb{H}^3 = SL_2(\mathbb{C})/SU(2)$ is the Doob transform of L_2 by the harmonic function $1-e^{-2x}$.

Proof. (i) With $a = \pi$ the atoms $\pi^2 n^2/2a^2$ are at $n^2/2$, with mass $2 = 2\pi \cdot \frac{1}{\pi}$. (ii) With $a = c = 1$ the density $a\{\pi\sqrt{2z - c^2}\}^{-1}$ on $z > \frac{c^2}{2}$ is that of σ_2. (iii) For f smooth, $\frac{1}{2}(f \sin)'' = \sin \cdot(\frac{1}{2}f'' + \cot x\, f' - \frac{1}{2}f)$, so $\sin^{-1}(L_1 + \frac{1}{2})(f \sin) = \frac{1}{2}f'' + \cot x\, f'$; and $L_2 h = 0$ for $h = 1 - e^{-2x}$, while $h^{-1}L_2(hf) = \frac{1}{2}f'' + (1 + h'/h)f'$ with $1 + h'/h = 1 + \frac{2e^{-2x}}{1-e^{-2x}} = \coth x$. The eigenfunctions are checked the same way: $L_2(e^{-x}s) = e^{-x}(\frac{1}{2}s'' - \frac{1}{2}s)$. □

So Riemann's ξ is attached to the compact group and to discrete spectrum, and Pólya's P to the hyperbolic space and to continuous spectrum. For Pólya's function itself the parameters are $a = c = \sqrt{2\pi}$ rather than $a = c = 1$; Brownian scaling gives $H_a^{(c)} \overset{d}{=} \frac{a}{c} H^{(\sqrt{ac})}_{\sqrt{ac}}$, so the level and the drift can always be made equal, which is the reciprocal symmetry of Proposition 8.7.1(i), but their common value $\sqrt{ac}$ is $\sqrt{2\pi}$ for P and 1 in the dictionary. Part (iii) corrects the direction of the Doob transforms in the note [202], which has L_1 and L_2 as the transforms of the radial generators; Biane states them in the direction of (iii) [27, Section 2.4].

Remark 8.7.3 (Constants in Biane's paper). Readers of [27] next to this section should note the following. Biane's $K_\mu(x) = \int_0^\infty t^{\mu-1} e^{-\frac{x}{2}(t+t^{-1})} dt$ is twice the standard Macdonald function, and his kernel (3.1) is the Φ of (1.2.3); with that kernel $\Xi(z) = 2\int_0^\infty \Phi(u)\cos(zu)\, du$ holds for the standard $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(\frac{s}{2})\zeta(s)$, so the factor $\frac{1}{2}$ is missing from his displayed definition of ξ, and his Mellin formula $\mathbb{E}W_a^s = 2(2a^2/\pi)^s \xi(2s)$ also uses the standard ξ (both sides are $\frac{2}{3}$ at $s = 1$, $a = 1$). In his Section 4.3 the Lévy measure of $W_a = J_a + J_a'$ should be $2t^{-1}\sum_n e^{-\pi^2 n^2 t/2a^2} dt$, with the factor t^{-1} and the $\frac{1}{2}$ in the exponent, and that of the drifted first passage time should carry the level x as a factor. His Thorin measure (4.1) should have its atoms at $\pi^2 n^2/2a^2$, not at n^2/a^2; the density (4.2) should be $x\, dc/(\pi\sqrt{2c - a^2})$, which for $x = 1$ is his density divided by $\sqrt{2}$, and the identity offered for it should read $(\pi t)^{-1/2} e^{-a^2 t/2} = \int_{a^2/2}^\infty e^{-ct}\, dc/(\pi\sqrt{c - a^2/2})$, with t in place of t^3. The measure (2.3) lives on $(\frac{1}{2}, \infty)$, not on $[1, \infty)$. In Propositions 2.3 and 2.4 the drift and the harmonic function are oriented for the cone opposite to the one stated: with the cone $\{y_1 \geq \cdots \geq y_n\}$ the drift is $\rho' = (n-1, n-3, \ldots, 1-n)$ and the positive harmonic function is $\prod_{i<j}(1 - e^{-2(y_i - y_j)}) = e^{-\langle \rho', y\rangle} \prod_{i<j} 2\sinh(y_i - y_j)$; the Vandermonde function $\prod_{i>j}(x_i - x_j)$ of his Proposition 2.1 is reversed in the same way, and is positive on the cone as $\prod_{i<j}(x_i - x_j)$. The eigenvalue of Proposition 2.6 is discussed in Remark 8.7.5. With these corrections the dictionary of his Section 4.5 is the exact statement of Theorem 8.7.2.

8.7.2 The Wick rotation and the curvature family

Biane closes his paper by observing that ξ and Pólya's function are related, through $SU(2)$ and $SL_2(\mathbb{C})/SU(2)$, "in this non obvious way" [27, Section 4.5]. The

relation is a Wick rotation, the substitution of iy for a real variable y in a function analytic in y. In quantum mechanics a state evolves by the unitary group e^{-itH}, and Wick's device [267] rotates the time axis, $t = -i\tau$, which turns e^{-itH} into the heat semigroup $e^{-\tau H}$; for $H = -\frac{1}{2}\Delta + V$ the Feynman Kac formula $e^{-\tau H}f(x) = \mathbb{E}_x[\exp(-\int_0^\tau V(B_s)ds)f(B_\tau)]$ makes Brownian motion the Wick rotated form of quantum mechanics. The rotation keeps the spectrum, an eigenvalue E appearing as a frequency e^{-iEt} on one side and as a decay rate $e^{-E\tau}$ on the other, and it exchanges oscillation and positivity: probability laws, Laplace transforms and completely monotone functions live on the Euclidean side, Fourier transforms and their zeros on the real time side. A Thorin measure is spectral data that can be read on either side. The substitution $y \mapsto iy$ exchanges sinh and sin, cosh and cos, and Laplace and Fourier transforms, and through Cartan's duality between a symmetric space G/K of noncompact type and its compact dual U/K it reverses the sign of the curvature. Biane's paper contains the rotation in four places: in space (the sign of the curvature), in the Laplace variable, in time (heat kernels) and in the Mellin index. We take them in turn, after recalling the first part of his paper.

Let $H_n = \{y \in \mathbb{R}^n : \sum_i y_i = 0\}$ and $\mathcal{C} = \{y \in H_n : y_1 \geq \cdots \geq y_n\}$. Biane describes three matrix valued Brownian motions of type A [27, Section 2]. For traceless Hermitian Brownian motion the ordered eigenvalues perform Brownian motion in $\mathcal{C}$ conditioned in Doob's sense never to reach the boundary: the semigroup of Brownian motion killed at $\partial\mathcal{C}$ is the alternating sum $p_t^0(x,y) = \sum_{w\in S_n} \varepsilon(w)p_t(x,wy)$ (the reflection principle), the Vandermonde product $\prod_{i<j}(y_i - y_j)$ is its positive harmonic function vanishing on $\partial\mathcal{C}$, and the Doob transform is Dyson's Brownian motion [71]. For Brownian motion g_t on $SL_n(\mathbb{C})$, the logarithms of the singular values, the radial part of Brownian motion on $SL_n(\mathbb{C})/SU(n)$, perform Brownian motion in $\mathcal{C}$ with generator $\frac{1}{2}\Delta + \langle \nabla \log \tilde{h}, \nabla\rangle$, $\tilde{h}(y) = \prod_{i<j}\sinh(y_i - y_j)$ [112], which is Brownian motion with drift $\rho' = (n-1, n-3, \ldots, 1-n)$, killed at $\partial\mathcal{C}$, transformed by the harmonic function of Remark 8.7.3; here the Martin boundary is large, with a point for each direction in the cone. For Brownian motion on $SU(n)$ the eigenvalues $e^{i\theta_j}$, with $\sum_j \theta_j = 0$, live in the alcove $\theta_1 \geq \cdots \geq \theta_n$, $\theta_1 - \theta_n \leq 2\pi$, a fundamental domain of the affine Weyl group $S_n \ltimes ((2\pi\mathbb{Z})^n \cap H_n)$; the killed semigroup is the alternating sum over that group, and the eigenvalue process is the Doob transform by the Dirichlet ground state $\prod_{j>k}(e^{i\theta_j} - e^{i\theta_k})$, which is a constant multiple of $\prod_{j<k}\sin\frac{\theta_j - \theta_k}{2}$ because $\sum_j \theta_j = 0$.

The three Doob functions are one function at three values of a curvature parameter. For real κ put $s_\kappa(z) = \sin(\sqrt{\kappa}z)/\sqrt{\kappa}$, so that $s_{-1} = \sinh$, $s_0(z) = z$ and $s_1 = \sin$, and

$$h_\kappa(y) = \prod_{i<j} s_\kappa(y_i - y_j), \qquad |\rho|^2 = \sum_{j=1}^n (n+1-2j)^2 = \frac{n(n^2-1)}{3}.$$

PROPOSITION 8.7.4. *For every real κ and every $n \geq 2$, $\Delta h_\kappa = -\kappa|\rho|^2 h_\kappa$ on H_n.*

Proof. For $\kappa = -1$, write $\prod_{i<j} 2\sinh(y_i - y_j) = \prod_{i<j} e^{-(y_i+y_j)}(e^{2y_i} - e^{2y_j})$. The Vandermonde determinant gives $\prod_{i<j}(e^{2y_i} - e^{2y_j}) = \det(e^{2(n-j)y_i})$, and on H_n the factor $\prod_{i<j} e^{-(y_i+y_j)} = e^{-(n-1)\sum_i y_i}$ is one, so the exponents $2(n-j)$ may be replaced by $2(n-j)-(n-1) = n+1-2j$: $\prod_{i<j} 2\sinh(y_i-y_j) = \sum_{w\in S_n} \varepsilon(w) e^{\langle w\rho', y\rangle}$, the Weyl denominator formula. Each $w\rho'$ lies in H_n and has $|w\rho'|^2 = |\rho|^2$, so each exponential is an eigenfunction of the Laplacian of H_n with eigenvalue $|\rho|^2$, and $\Delta h_{-1} = |\rho|^2 h_{-1}$. This identity between entire functions of $y \in H_n \otimes \mathbb{C}$ continues analytically. For $\kappa < 0$, $h_\kappa(y)$ is a constant times $h_{-1}(\sqrt{-\kappa}\, y)$, and the scaling multiplies the Laplacian by $-\kappa$; for $\kappa > 0$, $s_\kappa(z) = \sinh(i\sqrt{\kappa}z)/(i\sqrt{\kappa})$, so $h_\kappa(y)$ is a constant times $h_{-1}(i\sqrt{\kappa}\, y)$, and the substitution multiplies the Laplacian by $(i\sqrt{\kappa})^2 = -\kappa$. At $\kappa = 0$, h_0 is the Vandermonde product, the limit of h_κ as $\kappa \to 0$, since $s_\kappa(z) = z - \kappa z^3/6 + O(\kappa^2)$, and it is harmonic. □

`biane_polya.py` checks the proposition with a symbolic Laplacian at random points of H_n for $n = 2, 3, 4$ and $\kappa = -1, 0, 1$, where $|\rho|^2 = 2$, 8, 20. So Biane's three semigroups are one formula,

$$q_t^{(\kappa)}(x,y) = \frac{h_\kappa(y)}{h_\kappa(x)}\, e^{\kappa|\rho|^2 t/2}\, p_t^0(x,y), \qquad \kappa \in \{-1, 0, 1\},$$

with generator $\frac{1}{2}\Delta + \langle \nabla \log h_\kappa, \nabla\rangle$, where p^0 is the killed kernel of the chamber $\mathcal{C}$ for $\kappa \leq 0$ and of the alcove $\{y_1 \geq \cdots \geq y_n,\ y_1 - y_n \leq \pi\}$ for $\kappa = 1$; for $\kappa = 1$ the coordinates are the half angles $y = \theta/2$, in which h_1 is the positive Dirichlet ground state of the alcove. The rotation $y \mapsto iy$ sends h_κ to a constant times $h_{-\kappa}$, since $s_\kappa(iz) = i s_{-\kappa}(z)$: it exchanges the noncompact and the compact Doob functions, passes through Dyson's Brownian motion at the flat point $\kappa = 0$, and reverses the sign of the ground energy, $\frac{1}{2}\Delta h_\kappa = -\frac{\kappa}{2}|\rho|^2 h_\kappa$, which is $+\frac{1}{2}|\rho|^2$ for $SL_n(\mathbb{C})/SU(n)$ and $-\frac{1}{2}|\rho|^2 = \frac{n-n^3}{6}$ for $SU(n)$ in the half angles.

REMARK 8.7.5 (The eigenvalue of Biane's Proposition 2.6). Biane endows the Lie algebra of $SU(n)$ with the form $\langle A, B\rangle = -\operatorname{Tr}(AB)$, in which the angles θ_j perform standard Brownian motion in H_n, and states that the ground state $\prod_{j<k} \sin\frac{\theta_j - \theta_k}{2}$ has the eigenvalue $\frac{n-n^3}{6}$ for $\frac{1}{2}\Delta$ [27, Proposition 2.6]. By Proposition 8.7.4 in the half angles, the eigenvalue for $\frac{1}{2}\Delta_\theta$ is $\frac{1}{4}\cdot\frac{n-n^3}{6} = \frac{n-n^3}{24}$; more generally, for the form $-c\operatorname{Tr}(AB)$ it is $\frac{n-n^3}{24c}$. Biane's value holds for $c = \frac{1}{4}$, that is when the half angles are standard Brownian motion. His rank one discussion uses a third normalisation, $c = \frac{1}{2}$, in which $SU(2)$ is the unit sphere S^3, the angle $x \in [0, \pi]$ has the generator $\frac{1}{2}\frac{d^2}{dx^2} + \cot x \frac{d}{dx}$ of Theorem 8.7.2(iii), and the ground energy is $\frac{2-8}{12} = -\frac{1}{2}$. `biane_polya.py` confirms $\frac{1}{2}\Delta_\theta h/h = -\frac{1}{4}, -1, -\frac{5}{2}$ for $n = 2, 3, 4$.

In rank one the curvature family is the radial part of Brownian motion on the three dimensional space form of curvature κ, with generator

$\frac{1}{2}\frac{d^2}{dx^2} + \sqrt{\kappa}\cot(\sqrt{\kappa}x)\frac{d}{dx}$: the sphere of radius $\kappa^{-1/2}$ for $\kappa > 0$, the Bessel process of dimension three for $\kappa = 0$, and the hyperbolic space of curvature κ for $\kappa < 0$. Let $J_a^{(\kappa)}$ be its hitting time of the radius a from the centre, with $a < \pi/\sqrt{\kappa}$ when $\kappa > 0$; then $J_a^{(0)} = J_a$, whose Laplace transform is classical [52]. For $\kappa < 0$ the generator is that of the Bessel process of dimension three with drift $\sqrt{-\kappa}$ [230, (2.5)], the norm of a three dimensional Brownian motion with drift of that length, and the case $\kappa < 0$ of the next proposition also follows from the Cameron Martin formula, since for Brownian motion without drift started at the centre the exit place of a ball is uniform and independent of the exit time.

Proposition 8.7.6. *For $u \geq 0$,*

$$\mathbb{E}e^{-uJ_a^{(\kappa)}} = \frac{\sqrt{2u-\kappa}\; s_\kappa(a)}{\sinh\left(a\sqrt{2u-\kappa}\right)},$$

the Laplace transform of a GGC with Thorin measure $\sum_{n\geq 1}\delta_{\pi^2n^2/2a^2-\kappa/2}$. Equivalently, the law of $J_a^{(\kappa)}$ is the exponential tilt $e^{\kappa t/2}\Pr(J_a \in dt)/\mathbb{E}e^{\kappa J_a/2}$ of the law of J_a.

Proof. The generator is the Doob transform of $\frac{1}{2}\frac{d^2}{dx^2}$ by $s = s_\kappa$, since $s'' = -\kappa s$ gives $s^{-1}(\frac{1}{2}\frac{d^2}{dx^2} + \frac{\kappa}{2})(sf) = \frac{1}{2}f'' + \frac{s'}{s}f'$ and $s'/s = \sqrt{\kappa}\cot(\sqrt{\kappa}x)$. Hence $f(x) = \sinh(x\sqrt{2u-\kappa})/s_\kappa(x)$ satisfies $\frac{1}{2}f'' + \frac{s'}{s}f' = s^{-1}(\frac{1}{2}g'' + \frac{\kappa}{2}g) = uf$ with $g = \sinh(x\sqrt{2u-\kappa})$, and f is, up to a constant factor, the positive increasing solution that stays bounded at 0, with $f(0+) = \sqrt{2u-\kappa}$. Hence $\mathbb{E}_x e^{-uJ_a^{(\kappa)}} = f(x)/f(a)$ for $0 < x < a$, and letting $x \downarrow 0$ gives $\mathbb{E}e^{-uJ_a^{(\kappa)}} = f(0+)/f(a)$, which is the formula; for $\kappa > 2u$ the square roots are imaginary and the formula is read through $\sinh(ix)/i = \sin x$. As a function of u the denominator divided by $\sqrt{2u-\kappa}$ is entire of order $\frac{1}{2}$, with zeros where $a\sqrt{2u-\kappa} = i\pi n$, that is at $u = -(\pi^2n^2/2a^2 - \kappa/2)$, all negative under the hypothesis on a; Hadamard's product in genus zero gives the Thorin measure. The tilt is the formula at $u - \frac{\kappa}{2}$ for $\kappa = 0$, divided by its value at $u = 0$, since $\sqrt{-\kappa}/\sinh(a\sqrt{-\kappa}) = 1/s_\kappa(a)$. □

At $u = 0.7$, $a = 0.9$ and $\kappa = 1, -1, \frac{1}{3}$, `biane_polya.py` obtains 0.82508145967889, 0.84051739205883 and 0.83070660638487, each the same from the formula, from the tilt and from the product over the Thorin atoms, and the residual of the differential equation is below 10^{-40}. Three consequences clarify the dictionary. First, Riemann's clock $J_a + J_a'$ is the flat member $\kappa = 0$, and its link with $SU(2)$ in Theorem 8.7.2 is spectral rather than geometric: the atoms $\pi^2n^2/2a^2$ are the Dirichlet spectrum of $\frac{1}{2}\frac{d^2}{dx^2}$ on $[0, a]$, which for $a = \pi$ is the spectrum of the radial generator of $SU(2)$ shifted by its ground energy $\frac{1}{2}$. Second, the hyperbolic member $\kappa = -1$ has the discrete atoms $\pi^2n^2/2a^2 + \frac{1}{2}$; Pólya's continuous measure comes instead from a first passage over a level on the half line, a Dirichlet problem on $[0, \infty)$ with continuous spectrum. Third, the tilt

destroys the reciprocal symmetry of $J_a + J_a'$ that encodes the functional equation. With $W = J_1^{(\kappa)} + J_1'^{(\kappa)}$, the function $(\frac{\pi}{2})^s \mathbb{E}W^s$ equals $2\xi(2s)$ for $\kappa = 0$ and is symmetric under $s \mapsto \frac{1}{2} - s$: it takes the value 0.9963110325 at both $s = 0.1$ and $s = 0.4$. For $\kappa = 1$ the two values are 1.002917197 and 1.023703077. A curvature interpolation between the two clocks must therefore be sought in a class that keeps the symmetry (Question 8.7.21).

Replacing u by a negative value in $\mathbb{E}e^{-uX}$ continues the transform across the origin, and this is the rotation of the Laplace variable used in the inversion formula above. For the curvature family the reciprocal of the transform becomes, for $u < \frac{\kappa}{2}$, the damped sine $\sin(a\sqrt{\kappa - 2u})/(\sqrt{\kappa - 2u}\, s_\kappa(a))$, whose zeros sit at minus the Thorin atoms and slide with κ; for the drifted clock the reciprocal $e^{a(\sqrt{c^2+2u}-c)}$ becomes $e^{-ac}e^{ia\sqrt{2|u|-c^2}}$ beyond the branch point $u = -\frac{c^2}{2}$, which has constant modulus and no zeros. Poles of the transform, which require discrete spectrum, here that of a compact space, give atoms of the Thorin measure; a branch cut gives a density. Figure 8.22 shows both.

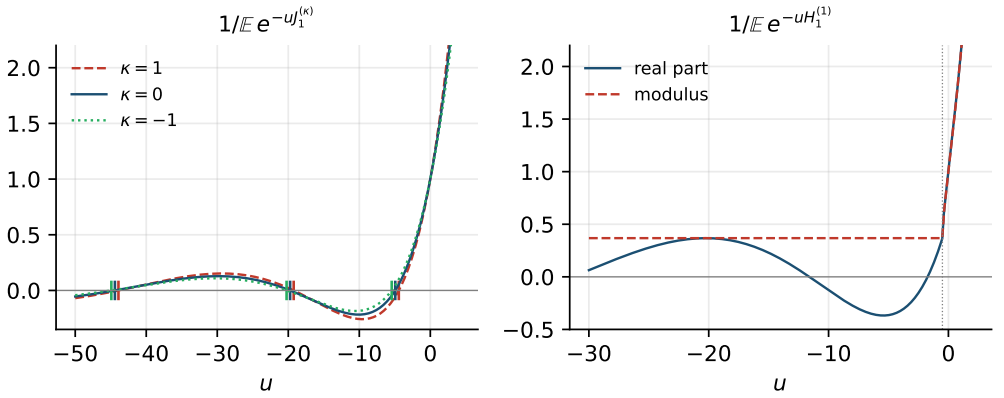


Figure 8.22: The Wick rotation of the Laplace variable. Left: $1/\mathbb{E}e^{-uJ_1^{(\kappa)}}$ for $\kappa = 1, 0, -1$, positive and increasing for $u > 0$ and a damped sine for $u < \frac{\kappa}{2}$, vanishing at minus the Thorin atoms $\pi^2 n^2/2 - \kappa/2$ (marks). Right: $1/\mathbb{E}e^{-uH_1^{(1)}} = e^{\sqrt{1+2u}-1}$, real part and modulus; beyond the branch point $u = -\frac{1}{2}$ (dotted) the modulus is constant, e^{-1}, and there are no zeros, the signature of a Thorin density. Drawn by `biane_figs.py`.

The rotation of time relates the heat kernels of the two dual spaces. Writing $x = -iy$ gives $\frac{d}{dx} = i\frac{d}{dy}$ and $\coth(-iy) = i\cot y$, so $\frac{1}{2}\frac{d^2}{dx^2} + \coth x \frac{d}{dx}$ becomes $-(\frac{1}{2}\frac{d^2}{dy^2} + \cot y \frac{d}{dy})$, and the Doob functions sinh and sin are exchanged. The radial

heat kernels of $\frac{1}{2}\Delta$, relative to the Riemannian volume, are

$$p_t^{\mathbb{H}^3}(r) = \frac{e^{-t/2}}{(2\pi t)^{3/2}} \frac{r}{\sinh r} e^{-r^2/2t},$$
$$p_t^{S^3}(\omega) = \frac{e^{t/2}}{(2\pi t)^{3/2}} \sum_{k\in\mathbb{Z}} \frac{\omega + 2\pi k}{\sin \omega} e^{-(\omega+2\pi k)^2/2t},$$

and the term $k = 0$ of the second is, up to the constant factor $(-1)^{-3/2}$, the first after $r \mapsto i\omega$, $t \mapsto -t$, a genuine Wick rotation of time; the sum over k is the method of images over the affine Weyl group, which makes the kernel periodic. `biane_polya.py` checks that both kernels integrate to one against $4\pi \sinh^2 r\, dr$ and $4\pi \sin^2 \omega\, d\omega$ at $t = 0.3$ and $t = 1$. In rank n the same statement is the formula for $q_t^{(\kappa)}$, whose factor $e^{\kappa|\rho|^2 t/2}$ changes sign with κ as $e^{\mp t/2}$ does here. Compactness replaces the continuous spectrum $[\frac{1}{2}, \infty)$ of $-L_2$ by the lattice $\{\frac{n^2}{2}\}$ of $-L_1$, which is the passage from σ_2 to σ_1 in Theorem 8.7.2.

The fourth rotation acts on the Mellin index. The substitution $s = \frac{1}{2} + i\theta$ turns real Mellin moments into Fourier transforms of a tilted law, and Biane's contour shift $w \mapsto w + \frac{i\pi}{2}$ [27, Section 3.5], with $\cosh(w + \frac{i\pi}{2}) = i \sinh w$, turns the damped integrand $e^{zw - 2a\cosh w}$ into an oscillating one on which stationary phase locates the zeros. For P this is the map $\mu \mapsto i\mu$ on the index of K_μ, and Section 8.7.5 shows that the index is the eigenvalue of a boost, which the rotation turns into an angular momentum. Table 8.29 collects the dictionary. Atoms of a Thorin measure on the Euclidean side become poles after the rotation, and poles require discrete spectrum, which for these radial Brownian motions, with no potential, means a compact space. That is why ξ, whose time side Thorin measure is atomic, is attached to $SU(2)$, while P, whose time side measure has a density, is attached to $\mathbb{H}^3$.

	real time side	Euclidean side
evolution	e^{-itH}, oscillating	$e^{-\tau H}$, positive
geometry	compact, $S^3 = SU(2)$, $\kappa = 1$	noncompact, $\mathbb{H}^3$, $\kappa = -1$
Doob function	$\prod \sin$	$\prod \sinh$
ground energy of $\frac{1}{2}\Delta$	$-\frac{1}{2}\lvert\rho\rvert^2$	$+\frac{1}{2}\lvert\rho\rvert^2$
radial spectrum	discrete, $\{\frac{n^2}{2}\}$	continuous, $[\frac{1}{2}, \infty)$
transform	Fourier, $\mathbb{E}e^{i\lambda X}$	Laplace, $\mathbb{E}e^{-uX}$
Thorin measure	poles and cuts	a positive measure
Bessel function	$K_{i\nu}$, oscillating, with zeros	K_ν, positive
symmetry of Section 8.7.5	boost, rapidity	rotation, angle
zeros of Ξ and P	live here	are Thorin atoms here

Table 8.29: The two sides of the Wick rotation.

8.7.3 The zeros of P as a Dirichlet spectrum

Put $\psi_\mu(x) = K_\mu(e^x)$. Since $\frac{d}{dx} = r\frac{d}{dr}$ for $r = e^x$, the modified Bessel equation $r^2K'' + rK' - (r^2+\mu^2)K = 0$ becomes

$$-\psi_\mu'' + e^{2x}\psi_\mu = -\mu^2\psi_\mu,$$

a Schrödinger equation with the Liouville potential e^{2x}, in which the index enters only through the energy $-\mu^2$. Let $x_0 = \log 2\pi$, and let $A_P = -\frac{d^2}{dx^2} + e^{2x}$ act in $L^2([x_0,\infty))$ with a Dirichlet condition at x_0. Biane recalls this spectral interpretation of the zeros of the Macdonald function as well known, with Titchmarsh's treatise on eigenfunction expansions [254] as its reference, and notes that Pólya does not mention it [27, Section 3.3]. We write $\nu_1 < \nu_2 < \cdots$ for the positive zeros of $\nu \mapsto K_{i\nu}(2\pi)$, so that P vanishes at $\theta = \pm 2\nu_n$.

Proposition 8.7.7. *(i) A_P is selfadjoint, has compact resolvent, and $A_P \geq 4\pi^2$.*

(ii) A number λ is an eigenvalue of A_P if and only if $\lambda = \nu^2$ with $\nu > 0$ and $K_{i\nu}(2\pi) = 0$, and the eigenfunction is then $K_{i\nu}(e^x)$. In particular every zero of P is real, of absolute value at least 4π, and the zeros of P are the numbers $\pm 2\nu_n$ with ν_n^2 running over the spectrum of A_P.

(iii) Every eigenvalue of A_P is simple, and every zero of $\mu \mapsto K_\mu(2\pi)$ and of P is simple. More precisely, for $\nu > 0$ and every $y \in \mathbb{R}$,

$$\psi_{i\nu}'(y)\,\partial_\nu\psi_{i\nu}(y) - \psi_{i\nu}(y)\,\partial_\nu\psi_{i\nu}'(y) = -2\nu\int_y^\infty \psi_{i\nu}(x)^2\,dx < 0. \tag{8.7.2}$$

Proof. (i) The endpoint x_0 is regular, and the potential is bounded below, so the equation is in the limit point case at $+\infty$ and the Dirichlet realisation is selfadjoint; since $e^{2x} \to \infty$, the resolvent is compact. For f in the form domain, $\langle A_P f, f\rangle = \int_{x_0}^\infty (|f'|^2 + e^{2x}|f|^2)\,dx \geq e^{2x_0}\|f\|^2 = 4\pi^2\|f\|^2$. (ii) For every complex μ, $K_\mu(r) = \sqrt{\pi/2r}\,e^{-r}(1 + O(1/r))$ as $r \to \infty$ [177, (10.40.2)], so ψ_μ is a nonzero solution in L^2 near $+\infty$, and in the limit point case it is the only one up to a constant factor. Hence $-\mu^2$ is an eigenvalue exactly when $\psi_\mu(x_0) = K_\mu(2\pi) = 0$. By (i) such an eigenvalue is real and at least $4\pi^2$, so $\mu = \pm i\nu$ with $\nu \geq 2\pi$; since $P(\theta) = K_{i\theta/2}(2\pi)$, the zeros of P are $\theta = \pm 2\nu$. (iii) Two eigenfunctions with the same eigenvalue both vanish at x_0, so they are proportional. For (8.7.2), put $\phi = \partial_\nu\psi_{i\nu}$; differentiating $\psi_{i\nu}'' = (e^{2x} - \nu^2)\psi_{i\nu}$ in ν gives $\phi'' = (e^{2x} - \nu^2)\phi - 2\nu\psi_{i\nu}$, so that $(\psi_{i\nu}'\phi - \psi_{i\nu}\phi')' = 2\nu\psi_{i\nu}^2$. Both functions and their derivatives decay like $\exp(-e^x)$, and integrating from y to ∞ gives the identity. The function $\psi_{i\nu}$ is real valued and not identically zero, so the right side is negative. At $y = x_0$ and $\nu = \nu_n$ the second term on the left vanishes, so $\partial_\nu K_{i\nu}(2\pi) \neq 0$ at ν_n, which is the simplicity of the zero. □

The proposition is Pólya's theorem that P has only real zeros, obtained from selfadjointness, and A_P is an explicit Hilbert Pólya operator for P. Biane also writes the equation in Dirac form [27, Section 3.4]. With $D = \frac{d}{dx}$ and $\mathcal{A} = D + \frac{1}{2} + e^x$, whose formal adjoint is $\mathcal{A}^\dagger = -D + \frac{1}{2} + e^x$, the system $\mathcal{A}f = \gamma g$, $\mathcal{A}^\dagger g = \gamma f$ gives $\mathcal{A}^\dagger\mathcal{A} f = \gamma^2 f$, and a direct computation gives $\mathcal{A}^\dagger\mathcal{A} = A_P + \frac{1}{4}$. With the Dirichlet condition on f the Dirac eigenvalues are therefore $\gamma = \pm(\nu_n^2 + \frac{1}{4})^{1/2}$, not $\pm\nu_n$; the shift $\frac{1}{4}$, which Biane writes as $\gamma^2 - \frac{1}{4}$, is missing from the note [202]. In the variable $u = e^x$ the system reads $(u\frac{d}{du} + \frac{1}{2} + u)f = \gamma g$, $(-u\frac{d}{du} + \frac{1}{2} + u)g = \gamma f$, and since $u\frac{d}{du} + \frac{1}{2}$ is i times the symmetrised dilation operator $\frac{1}{2}(up + pu)$, $p = -i\frac{d}{du}$, it is the Hamiltonian xp of Berry and Keating [24] perturbed by the potential u, as Biane remarks.

The zeros of P obey an exact quantisation rule. Let I_μ be the modified Bessel function of the first kind, and for real ν put

$$\vartheta_P(\nu) = \operatorname{Im}\log\Gamma(1 + i\nu) - \nu\log\pi - \arg {}_0F_1(;1 + i\nu;\pi^2), \tag{8.7.3}$$

with the branches continuous in ν and equal to zero at $\nu = 0$, where ${}_0F_1(;1;\pi^2) = I_0(2\pi) > 0$. Write $N_P(T)$ for the number of zeros of P in $(0, T]$ and

$$x(T) = \frac{1}{\pi}\vartheta_P\big(\tfrac{T}{2}\big), \qquad N_0(T) = \frac{T}{2\pi}\log\frac{T}{2\pi e}.$$

Proposition 8.7.8. *(i) For $\nu > 0$, $I_{i\nu}(2\pi) \neq 0$ and*

$$K_{i\nu}(2\pi) = \frac{\pi\,|I_{i\nu}(2\pi)|\,\sin\vartheta_P(\nu)}{\sinh\pi\nu}.$$

(ii) $\vartheta_P(\nu) > 0$ for $\nu > 0$, $\vartheta_P(\nu_n) = n\pi$ for every n, and the number of zeros of $\nu \mapsto K_{i\nu}(2\pi)$ in $(0, \nu)$ is $\lceil\vartheta_P(\nu)/\pi\rceil - 1$. Equivalently, $N_P(T) = \lfloor x(T)\rfloor$ for every $T > 0$ that is not a zero of P.

(iii) As $\nu \to \infty$,

$$\vartheta_P(\nu) = \nu\log\frac{\nu}{\pi} - \nu + \frac{\pi}{4} + \frac{\pi^2 - \frac{1}{12}}{\nu} + O(\nu^{-3}),$$

so that $x(T) = N_0(T) + \frac{1}{4} + \big(2\pi - \frac{1}{6\pi}\big)\frac{1}{T} + O(T^{-3})$ as $T \to \infty$.

Proof. (i) For real $r > 0$, $I_{-i\nu}(r) = \overline{I_{i\nu}(r)}$, and $K_\mu = \pi(I_{-\mu} - I_\mu)/(2\sin\mu\pi)$ gives $K_{i\nu}(r) = -\pi\operatorname{Im} I_{i\nu}(r)/\sinh\pi\nu$. The series $I_{i\nu}(r) = (r/2)^{i\nu}\,{}_0F_1(;1 + i\nu;r^2/4)/\Gamma(1 + i\nu)$ shows that, for $r > 0$,

$$\vartheta(\nu, r) = \operatorname{Im}\log\Gamma(1 + i\nu) - \nu\log\frac{r}{2} - \arg {}_0F_1\big(;1 + i\nu;\tfrac{r^2}{4}\big)$$

satisfies $I_{i\nu}(r) = |I_{i\nu}(r)|e^{-i\vartheta(\nu,r)}$, and $\vartheta_P(\nu) = \vartheta(\nu, 2\pi)$; this gives the formula. If $I_{i\nu}(r)$ vanished, so would $I_{-i\nu}(r)$, against the Wronskian $I_\mu I'_{-\mu} - I'_\mu I_{-\mu} =$

$-2\sin(\mu\pi)/(\pi r)$ [177, (10.28.1)], which is not zero for $\mu = i\nu$. So ${}_0F_1$ does not vanish on $[0,\infty)\times(0,\infty)$, and ϑ is defined there as a continuous function with $\vartheta(0,r) = 0$. (ii) The same Wronskian, written for $I = I_{i\nu}$ and $\bar I = I_{-i\nu}$, gives $\operatorname{Im}(I'/I) = \sinh(\pi\nu)/(\pi r|I|^2)$, hence $\partial_r\vartheta(\nu,r) = -\sinh(\pi\nu)/(\pi r|I_{i\nu}(r)|^2) < 0$. As $r\to\infty$, $I_{i\nu}(r)\sim e^r/\sqrt{2\pi r}$ and $K_{i\nu}(r)\sim\sqrt{\pi/2r}\,e^{-r}$ uniformly for ν in compact sets, so $\sin\vartheta(\nu,r) = \sinh(\pi\nu)K_{i\nu}(r)/(\pi|I_{i\nu}(r)|)\to 0$ and $\cos\vartheta(\nu,r) > 0$ for large r; by continuity in ν from $\nu = 0$, $\vartheta(\nu,r)\to 0$. So for fixed $\nu > 0$ the function $r\mapsto\vartheta(\nu,r)$ decreases from $\vartheta_P(\nu)$ at $r = 2\pi$ to 0 at $r = \infty$; in particular $\vartheta_P(\nu) > 0$, and $r\mapsto K_{i\nu}(r)$ has exactly $\lceil\vartheta_P(\nu)/\pi\rceil - 1$ zeros in $(2\pi,\infty)$, one at each crossing of a multiple of π. On the other hand, by (8.7.2) at a zero y of $\psi_{i\nu}$, $\psi'_{i\nu}(y)\,\partial_\nu\psi_{i\nu}(y) < 0$, so every zero of $x\mapsto K_{i\nu}(e^x)$ moves to the right as ν increases; no zero comes in from $+\infty$, where $K_{i\nu}(r) > 0$ uniformly for ν in compact sets, and zeros cannot appear in pairs, since they are simple. So the number of zeros in (x_0,∞) is zero for small ν and increases by one exactly when ν passes some ν_n, where a zero enters through x_0: it equals the number of $\nu_n < \nu$. Comparing the two counts gives the statement, and $\vartheta_P(\nu_n)\in\pi\mathbb{Z}$ with the count passing from $n-1$ to n gives $\vartheta_P(\nu_n) = n\pi$. (iii) Stirling's series gives $\operatorname{Im}\log\Gamma(1+i\nu) = \nu\log\nu - \nu + \frac{\pi}{4} - \frac{1}{12\nu} + O(\nu^{-3})$, and with $b = 1+i\nu$, $\log{}_0F_1(;b;z) = z/b - z^2/(2b^2(b+1)) + O(b^{-3})$, so $\arg{}_0F_1(;1+i\nu;\pi^2) = \operatorname{Im}\frac{\pi^2}{1+i\nu} + O(\nu^{-3}) = -\frac{\pi^2}{\nu} + O(\nu^{-3})$. With $\nu = T/2$, $\frac{1}{\pi}(\nu\log\frac{\nu}{\pi} - \nu) = N_0(T)$. □

So the zeros of P are the solutions of $\vartheta_P(\nu_n) = n\pi$, and their counting function is the floor of a smooth function, with no fluctuating remainder. Writing $s_P(T) = \{x(T)\} - \frac12$ for the sawtooth,

$$N_P(T) = N_0(T) - \frac14 - s_P(T) + \Big(2\pi - \frac{1}{6\pi}\Big)\frac{1}{T} + O(T^{-3}),$$

against the Riemann von Mangoldt formula $N(T) = \vartheta(T)/\pi + 1 + S(T)$ of Section 2.2, which with $\vartheta(T) = \frac{T}{2}\log\frac{T}{2\pi} - \frac{T}{2} - \frac{\pi}{8} + \frac{1}{48T} + O(T^{-3})$ reads

$$N(T) = N_0(T) + \frac78 + S(T) + \frac{1}{48\pi T} + O(T^{-3}).$$

The two counting functions share the main term N_0, but their mean constant terms differ by $\frac78 + \frac14 = \frac98$, and the fluctuation of N_P is the bounded sawtooth of a quantisation rule where that of N is $S(T)$. These two facts govern the comparison of the two Poisson traces in Section 10.1.5. Table 8.28 shows the effect: $N(1000) - N_P(1000) = 2$, with $N_0(1000) = 647.741$. The note [202] states that the semiclassical count of A_P is exactly the main term of the Riemann von Mangoldt formula. The count is

$$N_{\mathrm{WKB}}(\nu) = \frac{1}{\pi}\int_{x_0}^{\log\nu}\sqrt{\nu^2 - e^{2x}}\,dx = \frac{1}{\pi}\Big(\nu\operatorname{arccosh}\frac{\nu}{2\pi} - \sqrt{\nu^2 - 4\pi^2}\Big),$$

and its expansion gives $N_{\mathrm{WKB}}(T/2) = N_0(T) + \frac{2\pi}{T} + O(T^{-3})$: it agrees with the main term up to $2\pi/T$, not exactly. Its phase $\varphi(\nu) = \pi N_{\mathrm{WKB}}(\nu) + \frac{\pi}{4}$ is the phase of Debye's expansion of $K_{i\nu}(2\pi)$ for large ν, the term $\frac{\pi}{4}$ being the Maslov correction at the turning point, and by (iii) $\vartheta_P(\nu) - \varphi(\nu) = -\frac{1}{12\nu} + O(\nu^{-3})$.

`biane_polya.py` computes ϑ_P from (8.7.3) with mpmath at 40 digits [159] and solves $\vartheta_P(\nu) = n\pi$ by the secant method for every $\nu_n \le 1000$, 1516 zeros. It checks (i) against Arb's $K_{i\nu}(2\pi)$ [110] at $\nu = 0.5$, 3, 9.5, 25, 101.3 and 700, with relative differences below 4×10^{-37}, encloses the first 40 zeros to 10^{-25} by sign changes of Arb balls, and finds 137 sign changes of $K_{i\nu}(2\pi)$ on a grid of step 0.02 in $(0, 150]$, the number of ν_n below 150. The first zeros are

$$\nu_1 = 9.768770083509978, \qquad \nu_2 = 12.44848789277578,$$
$$\nu_3 = 14.68495977595033, \qquad \nu_4 = 16.69157893829242,$$

and ν_1 agrees with the first zero of $K_{i\theta}(2\pi)$ in Table 8.18, 9.768770083509977867. The phase is increasing on $[8, 1000]$, where its derivative, sampled with step 0.7, is at least 0.76, and the products $\nu(\vartheta_P(\nu) - \nu\log\frac{\nu}{\pi} + \nu - \frac{\pi}{4})$ are 9.88388, 9.79016 and 9.78643 at $\nu = 20$, 100 and 500, against the limit $\pi^2 - \frac{1}{12} = 9.78627$ of (iii); the products $\nu(\vartheta_P - \varphi)$ are -0.1117, -0.0843 and -0.0834, against $-\frac{1}{12}$.

8.7.4 The index side: a GGC with the squared zeros as Thorin atoms, and its clock

Biane's Thorin measures in Theorem 8.7.2 live on the time side: they describe the hitting time itself, in the energy variable of its Laplace transform. The Thorin programme of this book works on the index side, with the reciprocal $1/\mathbb{E}\cosh(\sqrt{u}\,X)$ of the moment generating function of the Pólya law, (1.2.4), whose Thorin atoms would be the squared ordinates. The same hitting time carries both measures, through the Mellin transforms of Proposition 8.7.1 and (9.1.1). For Pólya's function the index side statement is a theorem.

Theorem 8.7.9. *For $u \ge 0$,*

$$\frac{K_0(2\pi)}{K_{\sqrt{u}}(2\pi)} = \frac{\mathbb{E}H^{1/2}}{\mathbb{E}H^{1/2+\sqrt{u}}} = \prod_{n\ge1}\Big(1 + \frac{u}{\nu_n^2}\Big)^{-1},$$

with $H = H^{(\sqrt{2\pi})}_{\sqrt{2\pi}}$ as in Proposition 8.7.1. This is the Laplace transform of the GGC $T_P = \sum_{n\ge1} e_n/\nu_n^2$, with independent standard exponentials e_n, no drift and Thorin measure $\sum_n \delta_{\nu_n^2}$.

Proof. $K_\mu(2\pi) = \int_0^\infty e^{-2\pi\cosh t}\cosh(\mu t)\,dt$ is entire and even in μ, so $w \mapsto K_{\sqrt{w}}(2\pi)$ is entire. Since $|\cosh(\mu t)| \le \cosh(t\,\mathrm{Re}\,\mu)$, $|K_\mu(x)| \le K_{\mathrm{Re}\,\mu}(x)$, and for $\nu > 0$, $K_\nu(x) = \frac12\int_0^\infty s^{\nu-1}e^{-\frac{x}{2}(s+1/s)}ds \le \frac12\Gamma(\nu)(2/x)^\nu$. Hence

$\log|K_{\sqrt{w}}(2\pi)| = O(|w|^{1/2}\log|w|)$, the function has order at most $\frac{1}{2}$ in w, and Hadamard's theorem gives a product of genus zero over its zeros. By Proposition 8.7.7 these are the points $w = -\nu_n^2$, each simple. So $K_{\sqrt{u}}(2\pi) = K_0(2\pi)\prod_n(1 + u/\nu_n^2)$, and Proposition 8.7.1(ii) with $s = \frac{1}{2} + \sqrt{u}$ gives the middle expression. Each factor $(1 + u/\nu_n^2)^{-1}$ is the Laplace transform of e_n/ν_n^2, and $\sum_n \nu_n^{-2} < \infty$. □

The theorem is an instance of two results of the book. In the language of Section 13.1.3, the law U of Proposition 8.7.1(iv), with density $e^{-2\pi\cosh 2u}/K_0(2\pi)$, has the moment generating function $L_U(s) = K_{s/2}(2\pi)/K_0(2\pi)$, which by Proposition 8.7.7 vanishes only at $s = \pm 2i\nu_n$; so U is ferromagnetic, with $c_k = 2\nu_k$ and $b = 0$, and Theorem 13.1.4 gives the clock $T_U = \sum_k e_k/(4\nu_k^2) = T_P/4$, the same variable in the normalisation of the ordinate θ. And Proposition 7.1.15, applied to the even function P of order one, which is in the Laguerre Pólya class by Proposition 8.7.7, gives the GGC $P(0)/P(i\sqrt{s}) = K_0(2\pi)/K_{\sqrt{s}/2}(2\pi)$ with atoms at the squared zeros $4\nu_n^2$. For Ξ the corresponding statement is the rigidity theorem, Proposition 1.2.1: $\xi(\frac{1}{2})/\xi(\frac{1}{2} + \sqrt{u})$ is a GGC, with Thorin measure $\sum_{\gamma>0}\delta_{\gamma^2}$, if and only if RH holds. For P the equivalence has its hypothesis supplied by selfadjointness, and it comes with an explicit random variable, constructed below. The mean and variance of T_P follow from $K_\mu(x) = k_0(x) + \frac{\mu^2}{2}k_2(x) + \frac{\mu^4}{24}k_4(x) + \cdots$, with $k_{2j}(x) = \int_0^\infty t^{2j}e^{-x\cosh t}dt$:

$$\mathbb{E}T_P = \sum_n \frac{1}{\nu_n^2} = \frac{k_2(2\pi)}{2k_0(2\pi)} = 0.0741239453481,$$

$$\operatorname{Var} T_P = \sum_n \frac{1}{\nu_n^4} = 2.25103144196 \times 10^{-4},$$

where `biane_polya.py` obtains 0.0741239452572 for the sum over the 1516 zeros $\nu_n \le 1000$ with the tail described next, and the same variance both ways. For comparison, the clock of Ξ has $\mathbb{E}T = 0.0231050$ (Section 1.2).

Table 8.30 compares the closed forms with the products over the zeros, for P, for Pólya's Ξ^* and for Ξ. The tail beyond the last zero V is $\int_V^\infty \log(1 + u/\nu^2)\,dN(\nu) - \frac{1}{2}\log(1 + u/V^2)$, with dN the smooth density of zeros, $\pi^{-1}\vartheta'_P$ from Proposition 8.7.8(iii) for P and $\frac{1}{2\pi}\log\frac{T}{2\pi}$ for Ξ^* and Ξ; the subtracted half term accounts for the last zero sitting at the lower end of the integral. The note [202] truncated at the zeros below a fixed height and continued with the plain density from the last zero; its agreement to three or four digits, which the last column reproduces, is limited by that tail, not by the identities, which hold to the accuracy of the full computation. For Ξ the agreement is a check, not a test, since RH is verified far beyond this range.

Pólya's Ξ^* belongs to the same class. The law with density proportional to Φ^* is ferromagnetic (Section 13.4.1), so by Theorem 13.1.4, or by Proposition 7.1.15, $\Xi^*(0)/\Xi^*(i\sqrt{u}) = \prod_n(1 + u/\gamma_n^{*2})^{-1}$ is the Laplace transform of a GGC whose

function	u	closed form	product	difference	truncated
P	0.5	0.0370339026997	$\nu_n \le 1000$	-4.6×10^{-11}	1.1×10^{-5}
	2	0.147801988307		-1.8×10^{-10}	4.5×10^{-5}
	5	0.367871890035		-4.6×10^{-10}	1.1×10^{-4}
	100	6.62276702182		-9.1×10^{-9}	2.3×10^{-3}
Ξ^*	10	0.3044209914	$\gamma_n^* \le 1000$	1.6×10^{-8}	8.8×10^{-5}
	60	1.661131318		9.4×10^{-8}	5.3×10^{-4}
	200	4.771881099		3.1×10^{-7}	1.8×10^{-3}
Ξ	10	0.2292377633	1700 zeros	1.6×10^{-7}	3.4×10^{-5}
	60	1.328022599		9.8×10^{-7}	2.0×10^{-4}
	200	4.11056511		3.3×10^{-6}	6.8×10^{-4}

Table 8.30: Index side products. The closed forms are $\log(K_{\sqrt{u}}(2\pi)/K_0(2\pi))$, $\log(\Xi^*(i\sqrt{u})/\Xi^*(0))$ and $\log(\xi(\frac12+\sqrt{u})/\xi(\frac12))$; the next column says which zeros enter the product $\sum\log(1+u/z^2)$, whose difference from the closed form, after the tail of the text is added, is the fifth column. The last column is the difference for the truncation of the note: the 137 zeros $\nu_n<150$, the 102 zeros of Ξ^* below 240 and 300 zeros of Ξ, each with the plain density from the last zero. Computed by `biane_polya.py`.

Thorin measure $\sum_n \delta_{\gamma_n^{*2}}$ sits at the squares of the positive zeros γ_n^* of Ξ^*; the measure is positive. In the language of Pólya's Hilfssatz II this is the closure of the class under the cosine tilt: the kernel of $G(\theta+ia)+G(\theta-ia)$ is $2\cosh(au)$ times that of G, and Pólya's passage from P to Ξ^* is this closure with $a=\frac92$ in the variable θ (Section 7.1.3); the note gives the shift as $\frac94$, which is its value in the index of the Bessel function. As a check of the closed form, `biane_polya.py` computes $\int\Phi^*(u)\cosh(\sqrt{60}\,u)\,du/\int\Phi^*$ from the kernel and $\Xi^*(i\sqrt{60})/\Xi^*(0)$ from the Bessel functions, and both are 5.265264162.

The random variable T_P can be realised pathwise. The index of the Bessel equation enters as a centrifugal term $\mu^2/2r^2$, so a Feynman Kac weight $\exp(-\frac{u}{2}\int dt/R_t^2)$ along a radial path, the clock of the Hartman Watson law and of Brownian winding, shifts the index from 0 to $\sqrt{u}$.

THEOREM 8.7.10. *Let R be the diffusion on $(0,\infty)$ with generator*

$$\mathcal{G}=\frac12\frac{d^2}{dr^2}+\Big(\frac{1}{2r}-\frac{K_1(r)}{K_0(r)}\Big)\frac{d}{dr},$$

the radial part of planar Brownian motion killed at rate $\frac12$ and Doob transformed by K_0. For $r>b>0$ let T_b be the hitting time of b and $\tau_r=\frac12\int_0^{T_b}dt/R_t^2$ under $\Pr_r$. Then $T_b<\infty$ almost surely and, for $u\ge0$,

$$\mathbb{E}e^{-u\tau_r}=\frac{K_{\sqrt{u}}(r)}{K_0(r)}\cdot\frac{K_0(b)}{K_{\sqrt{u}}(b)}.$$

As $r \to \infty$, τ_r converges in law to τ_∞ with $\mathbb{E}e^{-u\tau_\infty} = K_0(b)/K_{\sqrt{u}}(b)$. For $b = 2\pi$, $\tau_\infty \overset{d}{=} T_P$.

Proof. Let $\mathcal{A} = \frac{1}{2}(\frac{d^2}{dr^2} + \frac{1}{r}\frac{d}{dr} - 1)$, the generator of the radial part of planar Brownian motion killed at rate $\frac{1}{2}$. The Bessel equation gives $\mathcal{A}K_\mu = \frac{\mu^2}{2r^2}K_\mu$, in particular $\mathcal{A}K_0 = 0$, and since $K_0' = -K_1$, $\mathcal{G}f = K_0^{-1}\mathcal{A}(K_0 f)$. Hence $v = K_{\sqrt{u}}/K_0$ satisfies $\mathcal{G}v = \frac{u}{2r^2}v$. For real order $K_{\sqrt{u}} \geq K_0$, so $v \geq 1$, and $v(r) \to 1$ as $r \to \infty$ because $K_\mu(r) = \sqrt{\pi/2r}\, e^{-r}(1 + O(1/r))$ uniformly for bounded μ; so v is bounded on $[b, \infty)$. The scale density of $\mathcal{G}$ is $s'(r) = 1/(rK_0(r)^2)$, which is not integrable at ∞, so from $r > b$ the diffusion reaches b before any level $M > r$ with probability $(s(M) - s(r))/(s(M) - s(b)) \to 1$, and $T_b < \infty$ almost surely; the drift is $-1 + \frac{1}{8r^2} + O(r^{-3})$ at infinity. By Itô's formula $v(R_{t\wedge T_b}) \exp(-\frac{u}{2}\int_0^{t\wedge T_b} ds/R_s^2)$ is a bounded martingale, and optional stopping gives $v(b)\mathbb{E}_r e^{-u\tau_r} = v(r)$. The limit follows from $v(r) \to 1$ and the continuity theorem for Laplace transforms, and the last statement from Theorem 8.7.9. □

In the logarithmic coordinate $X = \log R$, run on the clock $\tau = \frac{1}{2}\int dt/R_t^2$, Itô's formula gives

$$dX_\tau = -2e^{X_\tau}\frac{K_1(e^{X_\tau})}{K_0(e^{X_\tau})}\,d\tau + \sqrt{2}\,dW_\tau,$$

a diffusion with generator $\frac{d^2}{dx^2} + 2\frac{\psi'}{\psi}\frac{d}{dx} = -\psi^{-1}A_P\,\psi$, where $\psi(x) = K_0(e^x)$ is the ground solution of $-\psi'' + e^{2x}\psi = 0$. Its drift is about $-2e^x$, so $+\infty$ is an entrance boundary, and τ_∞ is its hitting time of $x_0 = \log 2\pi$ from $+\infty$. This is the Kent form of Section 8.1.3: by Kent's theorem [122] the hitting time from an entrance boundary is $\sum_n e_n/\lambda_n$ over the Dirichlet eigenvalues λ_n of the generator, here those of A_P, which are the ν_n^2. For Pólya's function Theorem 8.7.10 therefore solves the inverse problem that Sections 9.2.1 and 8.1.3 pose for ξ: it exhibits the diffusion, here with the Liouville potential, whose hitting time has its Thorin atoms at the squared zeros, without Krein's inverse spectral theory. For Ξ one wants a half line operator with Dirichlet spectrum $\{\gamma^2/4\}$, the counterpart of $\{\nu_n^2\} = \{\theta_n^2/4\}$ over the zeros $\theta_n = 2\nu_n$ of P; the potential e^{2x} on $[\log 2\pi, \infty)$ has the right Weyl law (Proposition 8.7.8), so it is the natural reference to perturb, and the perturbation must account for the difference between Φ and Pólya's asymptote seen in Figure 8.21 (Question 8.7.18).

Corollary 8.7.11. *For real λ, with τ_∞ the clock of Theorem 8.7.10 for $b = 2\pi$ and a standard Brownian motion B independent of it,*

$$\frac{K_0(2\pi)}{K_\lambda(2\pi)} = \mathbb{E}e^{i\lambda B_{2\tau_\infty}}.$$

Proof. $\mathbb{E}e^{i\lambda B_{2\tau}} = \mathbb{E}e^{-\lambda^2\tau}$ for an independent random time τ; apply Theorem 8.7.10 with $u = \lambda^2$. □

The reciprocal of Pólya's function along the real axis of the index is thus the characteristic function of Brownian motion run at the index side clock, and the clock is a hitting time of killed planar Brownian motion. For Ξ the corresponding identity $\xi(\frac{1}{2})/\xi(\frac{1}{2}+\lambda) = \mathbb{E}e^{i\lambda B_{2T}}$ holds with no hypothesis, with the clock T of Theorem 5.1.1, and the variable B_{2T} is the variable $Y_{1/2}$ of Section 10.1.4; what RH adds is that T is the GGC $\sum_\gamma e_\gamma/\gamma^2$ (Proposition 1.2.1). The note [202] states the identity for Ξ under RH and says that for Riemann's function the random time exists if and only if RH holds; the time exists unconditionally, and only its GGC form is equivalent to RH.

Remark 8.7.12 (The Hartman Watson law). Theorem 8.7.10 is the K counterpart of a formula of Yor [275] for the Hartman Watson law [91]: for planar Brownian motion started at distance r from the origin and conditioned on $|B_t| = \rho$, $\mathbb{E}[\exp(-\frac{u}{2}\int_0^t ds/|B_s|^2)] = I_{\sqrt{u}}(r\rho/t)/I_0(r\rho/t)$. There the endpoint is fixed and the function is I; here the path is killed at rate $\frac{1}{2}$, transformed by K_0 and stopped at a level, and the function is K. Pitman and Yor develop the general theory of such Bessel laws [193].

`biane_clock.py` simulates the clock. With $R_0 = 2\pi e^{2.5} = 76.544867$, so that X starts at $\log 2\pi + 2.5$ as in the note, the exact law of τ_{R_0} has, from the expansion of $\log v$ in u, mean 0.06763390577 and standard deviation 0.01499747823, against 0.07412394535 and 0.01500343775 for τ_∞. The script runs 100000 paths of the Euler scheme for X with step $\Delta = 4\times10^{-6}$, the step of the note, and records two hitting times on the same paths: the plain one, the first step that ends below $\log 2\pi$, and one with the Brownian bridge correction, under which a step from x to x', both above the level $\ell = \log 2\pi$, crosses it with probability $\exp(-(x-\ell)(x'-\ell)/\Delta)$, the variance of the step being 2Δ. The plain scheme overshoots the barrier and is late by about $0.5826\sqrt{2\Delta}/|\beta|$, with $\beta = -13.5317$ the drift at the barrier and $0.5826 = -\zeta(\frac{1}{2})/\sqrt{2\pi}$ the constant of Broadie, Glasserman and Kou [45], that is by 1.218×10^{-4}. Table 8.31 shows that the plain mean is 2.9 standard errors late, by 1.39×10^{-4}, and that the corrected one agrees with the exact mean. The exact Laplace transform is computed with Arb at 704 bits, since Arb's $K_\mu(x)$ loses about $2x/\log 2$ bits to cancellation, 221 bits at $x = R_0$, and the exact densities by the fixed Talbot contour [248, 1] with 32 nodes; doubling the nodes changes the density at 0.07, which is 24.85, by 2×10^{-13}, and the computed densities integrate to 0.9999974 and 0.9999954 on $[0.015, 0.2]$. Figure 8.23 shows both comparisons. The note's own run, 40000 plain paths with mean 0.067695 ± 0.000075, is consistent with the exact mean and with the predicted lateness.

The cosine dictionary (6.2.4) turns each Thorin measure of this section into a Lévy measure of the reciprocal transform, in the variable conjugate to the square root of the energy: a Thorin atom at z becomes the exponential rate $\sqrt{z}$ when the energy is written λ^2, and $\sqrt{2z}$ when it is written $\lambda^2/2$, as for Brownian hitting times. Biane's measures give two classical Lévy measures, and Pólya's index side

	mean	standard deviation	z of the mean	quantiles 0.01, 0.5, 0.99
exact	0.0676339	0.0149975		
plain Euler	0.0677726	0.0150584	2.91	0.04190, 0.06559, 0.11391
bridge corrected	0.0676538	0.0150221	0.42	0.04185, 0.06548, 0.11364
u	25	100	200	400
exact $\mathbb{E}e^{-u\tau_{R_0}}$	0.196345	2.54238×10^{-3}	1.65093×10^{-5}	2.91684×10^{-9}
z, bridge corrected	-0.25	0.36	0.72	0.12
z, plain Euler	-2.81	-1.84	-0.76	-0.48

Table 8.31: The clock from $R_0 = 2\pi e^{2.5}$: exact moments against 100000 Euler paths with step 4×10^{-6}, with and without the bridge correction, and the Laplace transform, with the z scores (Monte Carlo − exact)/standard error. The standard error of the mean is 4.75×10^{-5}; over $0 < u \le 400$ the largest $|z|$ of the corrected transform is 0.72. Computed by `biane_clock.py`.

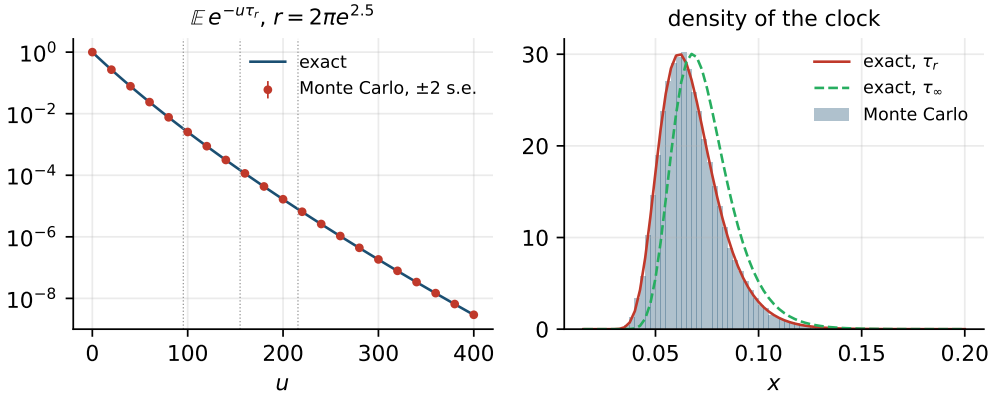


Figure 8.23: The clock from $R_0 = 2\pi e^{2.5}$. Left: $\mathbb{E}e^{-u\tau_{R_0}}$, exact and from the bridge corrected Monte Carlo with two standard errors, on a logarithmic scale; the dotted lines mark the first three Thorin atoms $\nu_1^2 = 95.4289$, $\nu_2^2 = 154.965$, $\nu_3^2 = 215.648$ of τ_∞. Right: the histogram of the corrected hitting times against the exact densities of τ_{R_0} and of τ_∞, by Talbot inversion. Drawn by `biane_figs.py` from `biane_clock.json`.

measure gives a wave trace.

PROPOSITION 8.7.13. *For real λ, $a>0$, $c>0$ and $\ell>0$,*

$$2\log\frac{\sinh\lambda a}{\lambda a}=\int_0^\infty(1-\cos\lambda v)\,\frac{4\,dv}{v\,(e^{\pi v/a}-1)},$$

$$\ell\big(\sqrt{c^2+\lambda^2}-c\big)=\int_0^\infty(1-\cos\lambda v)\,\frac{2\ell c}{\pi}K_1(cv)\,\frac{dv}{v},$$

$$\log\frac{K_\lambda(2\pi)}{K_0(2\pi)}=\int_0^\infty 4\sin^2\Big(\frac{\lambda v}{2}\Big)\,\Theta_P(v)\,\frac{dv}{v},\qquad \Theta_P(v)=\sum_{n\ge1}e^{-\nu_n v}=\operatorname{tr}e^{-v\sqrt{A_P}}.$$

The left sides are $-\log\mathbb{E}e^{-\lambda^2W/2}$ for $W=J_a+J_a'$ and $W=H_\ell^{(c)}$, and $-\log\mathbb{E}e^{-\lambda^2T_P}$.

Proof. By Frullani's integral, $\log(1+\lambda^2/\beta^2)=\int_0^\infty(1-\cos\lambda v)\,2e^{-\beta v}dv/v$. For $W=J_a+J_a'$ the Thorin measure $2\sum_n\delta_{\pi^2n^2/2a^2}$ of Section 8.7.1 gives, at $u=\lambda^2/2$, the rates $\beta=\pi n/a$ with weight 2, and $\sum_n e^{-\pi nv/a}=(e^{\pi v/a}-1)^{-1}$. For $H_\ell^{(c)}$ the density $\ell\{\pi\sqrt{2z-c^2}\}^{-1}$ on $z>\frac{c^2}{2}$ becomes, under $\beta=\sqrt{2z}$, the measure $\ell\beta\,d\beta/(\pi\sqrt{\beta^2-c^2})$ on (c,∞), and $\int_c^\infty e^{-\beta v}\beta(\beta^2-c^2)^{-1/2}d\beta=cK_1(cv)$. For T_P take Theorem 8.7.9 at $u=\lambda^2$, with rates ν_n, and use $1-\cos=2\sin^2(\cdot/2)$. □

`biane_polya.py` checks the first two identities at $\lambda=1.7$, $a=1$, $c=1$, $\ell=0.8$, with the values 0.8845634041934098 and 0.7778466338652816 both ways. The Bessel clock has the Planck measure, a geometric superposition of the frequencies $\pi n/a$, and the drifted clock has the Macdonald measure, absolutely continuous because its spectrum is. The dictionary acts on Thorin measures, not on Biane's Lévy measures in the time variable, which are heat traces. For Pólya's function the third identity is the exact analogue of the Lévy Khintchine form of $\xi(\frac12)/\xi(\frac12+u)$ in Section 10.1.4, with the Poisson trace $K_{1/2}$ replaced by the wave trace Θ_P of the selfadjoint operator A_P, which is completely monotone with no hypothesis. In the variable θ, where the zeros of P are $2\nu_n$, the trace is $K^P(y)=\Theta_P(2y)=\sum_n e^{-2\nu_n y}$, and $\log(K_{\lambda/2}(2\pi)/K_0(2\pi))=\int_0^\infty 4\sin^2(\lambda y/2)K^P(y)\,dy/y$ has the form of the formula for ξ with K^P in place of $K_{1/2}$; Section 10.1.5 compares the two traces. At $\lambda=1.7$ the left side of the third identity is 0.2132910704; the sum over all 1516 zeros $\nu_n\le1000$ with the tail of Table 8.30 differs from it by 3×10^{-10}, and the 101 zeros $\nu_n<120$ with the plain density, as in the note, give 0.2133932327.

8.7.5 Rindler modes and a Klein Gordon criterion

The clock of Theorem 8.7.10 lives in the Euclidean plane. Undoing the Wick rotation of the angle gives a Lorentzian picture in which Pólya's function is a standard object of quantum field theory, and which isolates the structural reason

for the reality of its zeros. Put

$$F(x,w)=\exp(-e^x\cosh w),\qquad \text{so that}\qquad K_\mu(e^x)=\frac{1}{2}\int_{\mathbb{R}}F(x,w)\,e^{\mu w}\,dw.$$

Since $\partial_x^2F=(e^{2x}\cosh^2 w-e^x\cosh w)F$ and $\partial_w^2F=(e^{2x}\sinh^2 w-e^x\cosh w)F$,

$$\Big(\frac{\partial^2}{\partial x^2}-\frac{\partial^2}{\partial w^2}-e^{2x}\Big)F=0.$$

In the coordinates $X_1=e^x\cosh w$, $X_0=e^x\sinh w$ of the Rindler wedge $X_1>|X_0|$, in which $\rho=e^x$ is the proper distance from the edge of the wedge and w the rapidity, $\partial_{X_1}^2-\partial_{X_0}^2=e^{-2x}(\partial_x^2-\partial_w^2)$, so the equation is $(\partial_{X_1}^2-\partial_{X_0}^2-1)F=0$ with $F=e^{-X_1}$: the Klein Gordon equation of unit mass in $1+1$ dimensional Minkowski space, solved by a plane wave. The index μ is the eigenvalue of the boost generator ∂_w, and $K_{i\omega}(\rho)$ is the Rindler mode of boost frequency ω [78]. The zeros of $P(\theta)=K_{i\theta/2}(2\pi)$ are therefore, with $\theta=2\omega$, the boost frequencies of the normal modes of a unit mass scalar field in the Rindler wedge with a Dirichlet mirror on the hyperbola $\rho=2\pi$, the configuration of 't Hooft's brick wall model of a black hole horizon [97]. Biane's Dirac system of Section 8.7.3 is the corresponding Dirac field, close to the Rindler space models of Sierra for the Hamiltonian xp [239].

The Wick rotation $w=i\phi$ turns the wedge into the Euclidean plane in polar coordinates (ρ,ϕ), the boost into a rotation, the Klein Gordon operator into $\Delta-1$, and the plane wave e^{-X_1} into $e^{-\rho\cos\phi}$, which is harmonic for $\frac{1}{2}(\Delta-1)$, the generator of planar Brownian motion killed at rate $\frac{1}{2}$ of Theorem 8.7.10. The boost frequency becomes the angular momentum μ, and the radial operator $\frac{1}{2}(\partial_\rho^2+\rho^{-1}\partial_\rho-\mu^2/\rho^2-1)$ carries the index as the centrifugal term that the Feynman Kac weight of the clock produces. The energy of Biane's dictionary, the rapidity of the Rindler picture and the index of the clock are three readings of one rotation, the last row of Table 8.29. The structure that makes the zeros of P real can be stated for a general even kernel.

THEOREM 8.7.14 (A Klein Gordon criterion). *Let g be even, positive and continuous with $\int g(w)e^{c|w|}dw<\infty$ for every c, and $G(t)=\int g(w)e^{itw}dw$. Suppose that there are $x_0\in\mathbb{R}$, a real continuous function V on $[x_0,\infty)$ bounded below, and $F\in C^2([x_0,\infty)\times\mathbb{R})$ such that*

(a) $(\partial_x^2-\partial_w^2-V(x))F=0$ and $F(x_0,\cdot)=g$;

(b) F and its partial derivatives of order at most two are $O(e^{-c|w|})$ for every $c>0$, uniformly for x in compact subsets of $[x_0,\infty)$, and $\Psi_\mu(x)=\int F(x,w)e^{\mu w}dw$ lies in $L^2((x_0,\infty))$ for every $\mu\in\mathbb{C}$;

(c) Ψ_μ is not identically zero for any $\mu\in\mathbb{C}$.

Then every zero of G is real, and its square is an eigenvalue of $-\frac{d^2}{dx^2}+V$ on $[x_0,\infty)$ with a Dirichlet condition at x_0.

Proof. By (b), Ψ_μ is twice differentiable under the integral sign, and two integrations by parts in w, whose boundary terms vanish, give $\Psi_\mu'' = \int(\partial_w^2 F + VF)e^{\mu w}dw = (\mu^2+V)\Psi_\mu$. Since V is bounded below, the operator $-\frac{d^2}{dx^2}+V$ is in the limit point case at ∞, and its Dirichlet realisation at x_0 is selfadjoint. If $G(t)=0$, then $\Psi_{it}(x_0)=G(t)=0$ by (a), and by (b) and (c) Ψ_{it} is an eigenfunction with eigenvalue $-(it)^2=t^2$, which is therefore real. So t is real or purely imaginary, and $t=-i\mu$ with real μ is excluded because $G(-i\mu)=\int g(w)e^{\mu w}dw>0$. □

For Pólya's function the hypotheses hold in the rapidity variable $w=2u$: with $g(w)=e^{-2\pi\cosh w}$, $G(t)=2K_{it}(2\pi)=2P(2t)$, $V(x)=e^{2x}$, $x_0=\log 2\pi$, $F=\exp(-e^x\cosh w)$ and $\Psi_\mu(x)=2K_\mu(e^x)$, and the theorem gives Proposition 8.7.7(ii) again. `biane_polya.py` checks the equation for F at $(x,w)=(0.7,0.4)$, with residual -5.7×10^{-42} at 40 digits. The criterion says that the reality of the zeros of an even Fourier transform follows from embedding its kernel as the boundary value of a field that satisfies a Klein Gordon equation with a real, time independent potential, which decays fast enough in the rapidity.

For Ξ the corresponding object would be such an extension of Φ, and it would prove RH (Question 8.7.19). The scaling matters. For P the eigenvalues of A_P are $\nu_n^2=\theta_n^2/4$, and the function whose zeros they are, as a function of the spectral parameter μ at the boundary x_0, is the Jost function $\mu\mapsto\Psi_\mu(x_0)=2K_\mu(2\pi)=2P(-2i\mu)$. A potential asymptotic to e^{2x} at $+\infty$ has the Weyl law of Proposition 8.7.8, which matches the zeros of Ξ only in the same scaling, $g(w)=\Phi(w/2)$ and $G(t)=2\Xi(2t)$, and then the Jost function is $\mu\mapsto 2\Xi(-2i\mu)=2\xi(\frac12+2\mu)$. The note [202] proposes $\mu\mapsto\xi(\frac12+\mu)$, which corresponds to the eigenvalues γ^2, whose density is half that of the Weyl law of e^{2x} (a potential growing like e^{4x} would match them). Inverse spectral theory reconstructs an operator from its eigenvalues and norming constants, a potential in the forms of Gelfand and Levitan or of Marchenko and a string in Krein's form, and under RH and the simplicity of the zeros Krein's theory realises the eigenvalues $\gamma^2/4$ (Section 8.1.3); the content of the question is that the operator be of the form $-\frac{d^2}{dx^2}+V$ and that V admit a Klein Gordon solution whose boundary value is $\Phi(w/2)$, which is a statement about the whole function ξ and not only about its zeros.

8.7.6 The de Bruijn Newman flow of P against that of Ξ

For real t put

$$P_t(\theta) = \int_{\mathbb{R}} e^{tu^2/4} e^{-2\pi \cosh 2u}\, e^{i\theta u}\, du,$$
$$\Xi_t(\theta) = \int_{\mathbb{R}} e^{tu^2/4}\, \Phi(u) \cos(\theta u)\, du = 8H_t(2\theta),$$

so that $P_0 = P$, $\Xi_0 = \Xi$, and t is the time of the de Bruijn Newman flow in the normalisation of Section 8.4.1. Decreasing t is the direction in which real zeros can be lost: RH is the statement that Ξ_t keeps only real zeros down to $t = 0$, and by the theorem of Rodgers and Tao [224] it has nonreal zeros for every $t < 0$. For Pólya's kernel the analogous constant is finite.

Proposition 8.7.15. *There is a constant Λ_P with $-\infty < \Lambda_P \le 0$ such that P_t has only real zeros if and only if $t \ge \Lambda_P$.*

Proof. Let S be the set of t for which P_t has only real zeros. By Proposition 8.7.7, $0 \in S$. If $t \in S$ and $t' > t$, then $t' \in S$ by de Bruijn's theorem on the Gaussian factor $e^{\lambda u^2}$, $\lambda > 0$ [65, Theorem 13], which applies to the kernel $e^{tu^2/4}e^{-2\pi\cosh 2u}$ because it is $O(e^{-|u|^b})$ for every $b > 2$. If $t_k \in S$ decrease to t, then $P_{t_k} \to P_t$ locally uniformly on $\mathbb{C}$ by dominated convergence, with $|e^{t_k u^2/4}e^{i\theta u}| \le e^{t_1 u^2/4}e^{|\operatorname{Im}\theta||u|}$, and $P_t(0) > 0$, so by Hurwitz's theorem every zero of P_t is a limit of real zeros and $t \in S$. Hence $S = [\Lambda_P, \infty)$ with $\Lambda_P \in [-\infty, 0]$. Suppose $\Lambda_P = -\infty$. Then, for the even probability measure ρ with density $e^{-2\pi\cosh 2u}/K_0(2\pi)$, the Fourier transform of $e^{\lambda u^2}\rho(du)$ has only real zeros for every real λ, and by Newman's theorem [163], [165, Theorem 10], ρ is either $\frac{1}{2}(\delta_{u_0} + \delta_{-u_0})$ or has a density $Ku^{2m}\exp(-\alpha u^4 - \beta u^2)\prod_j (1 + u^2/a_j^2)e^{-u^2/a_j^2}$, with $K > 0$, $m \ge 0$ an integer, $a_j > 0$, $\sum_j a_j^{-4} < \infty$, and $\alpha > 0$, or $\alpha = 0$ and $\beta + \sum_j a_j^{-2} > 0$. The inequality $(1 + y)e^{-y} \ge e^{-y^2/2}$ for $y \ge 0$ bounds such a density below by $Ku^{2m}\exp(-[\alpha + \sum_j \frac{1}{2a_j^4}]u^4 - \beta u^2)$, while $e^{-2\pi\cosh 2u}$ decays faster than e^{-Cu^4} for every C. This contradiction gives $\Lambda_P > -\infty$. □

The proof is Newman's proof that $\Lambda > -\infty$ [163], Theorem 9 of [165], which uses only the fact that the kernel decays faster than e^{-Cu^4} for every C; it applies verbatim to Φ^*, and to every even, positive kernel with that decay whose Fourier transform has only real zeros. The note [202] conjectures $\Lambda_P = -\infty$, on the evidence of the computation below; Proposition 8.7.15 shows that the conjecture is false. The computation is still informative, because it shows how differently the two flows behave in the range that can be followed.

`biane_flow.py` computes both transforms by the trapezoid rule with step $1/250$ on $[0, 2]$ at 90 digits. The kernels are analytic in $|\operatorname{Im} u| < \pi/4$ (the kernel of P_t is entire) and below 10^{-68} beyond $u = 2$, and at $t = 0$ the sums reproduce

Arb's $K_{i\theta/2}(2\pi)$ at $\theta = 10$, 30, 60, 90 and mpmath's Ξ at $\theta = 14.134725$, 50, 90 with relative errors between 3.5×10^{-62} and 1.4×10^{-45}, the largest at $\theta = 90$. The real zeros in $(0, 90)$ are located by sign changes on a grid of step 0.05 and refined, for $t = 0, -0.5, \ldots, -20$ and $t = -21, \ldots, -60$. Table 8.32 and Figure 8.24 show the result. The real zeros of Ξ_t in the window fall from 25 at $t = 0$ to 17 at $t = -2$, 11 at $t = -4$, 7 at $t = -8$ and 2 at $t = -40$, and they are lost in pairs, neighbouring zeros colliding and leaving the axis: the first collisions in the window, as t decreases, are those of the census of Section 8.4.3, beginning with $(\gamma_{24}, \gamma_{25})$ at $t = -1.144$ and height 88.47 near the top of the window. The pair $(\gamma_4, \gamma_5) = (30.42, 32.94)$ has gone by $t = -4$, and the pair (γ_2, γ_3) by $t = -15$, after which the first zero is isolated. For P_t nothing of the kind occurs down to $t = -60$: every real zero moves monotonically upward, the first from 19.538 at $t = 0$ to 51.881 at $t = -60$, and the number in the window falls only as zeros leave it through $\theta = 90$: from each time of the grid to the next, every zero increases and the count never increases.

t	P_t: zeros	first zero	Ξ_t: zeros	first zeros
0	24	19.538	25	14.135, 21.022, 25.011
−1	23	19.938	25	14.426, 21.403, 25.302
−2	23	20.342	17	14.719, 21.793, 25.589
−4	23	21.163	11	15.311, 22.600, 26.144
−6	23	22.002	9	15.911, 23.452, 26.667
−8	22	22.857	7	16.518, 24.365, 27.139
−10	22	23.730	7	17.133, 25.378, 27.518
−15	21	25.987	5	18.695, 44.574, 58.128
−20	20	28.353	5	20.290, 46.563, 60.344
−25	19	30.832	5	21.911, 48.426, 62.880
−30	18	33.428	3	23.551, 50.245, 89.062
−40	17	38.989	2	26.866, 53.858
−50	15	45.098	2	30.202, 57.488
−60	13	51.881	2	33.539, 61.145

Table 8.32: The real zeros in $(0, 90)$ of P_t and Ξ_t along the de Bruijn Newman flow: their numbers, the first zero of P_t and the first three of Ξ_t. Computed by `biane_flow.py`.

Sign changes count only real zeros, so the script also counts all zeros in the box $0.5 \le \operatorname{Re}\theta \le 60.3$, $|\operatorname{Im}\theta| \le 3$ by the argument principle, on a contour of step 0.04 halved wherever the argument changes by more than 0.5 between neighbours; the largest change that remains is 0.36. For P_t the box holds 12, 11, 8 and 3 zeros at $t = 0$, -10, -30 and -60, all of them real. For Ξ_t it holds 13, 12 and 12 zeros at $t = 0$, -2 and -4, of which 13, 10 and 8 are real: one conjugate pair has appeared by $t = -2$ and two by $t = -4$. Three of the collisions below height 60, those listed in the note [202], are located by bisection in t for the disappearance of two sign

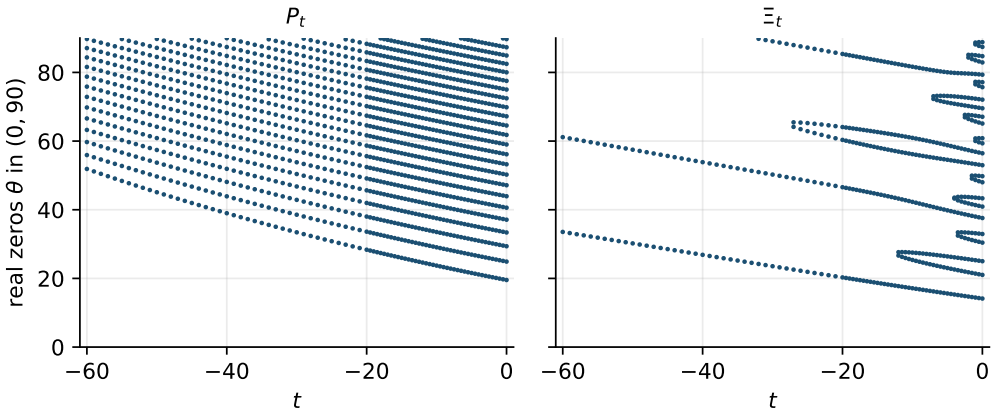


Figure 8.24: The real zeros in $(0, 90)$ of P_t (left) and of Ξ_t (right) for $-60 \le t \le 0$, on the grid of Table 8.32 (step 0.5 above $t = -20$, step 1 below). The zeros of P_t drift upward and leave the window one at a time; those of Ξ_t collide in pairs. Drawn by `biane_figs.py` from `biane_flow.json`.

changes and then by Newton's method for the double zero $\Xi_t = \partial_\theta \Xi_t = 0$ in (θ, t):

$$\begin{aligned}(\gamma_9, \gamma_{10}):&\ t_c = -1.877909073,\ \theta_c = 49.66423465;\\ (\gamma_4, \gamma_5):&\ t_c = -3.954792524,\ \theta_c = 33.09873381;\\ (\gamma_2, \gamma_3):&\ t_c = -12.49451313,\ \theta_c = 27.3212277.\end{aligned}$$

The first agrees with the census of Section 8.4.3 (-1.87791 at height 49.664); the other two lie below its range $[-2.05, 0]$. The note presents the three as the first collisions below height 60, but the computation, on its grid of step 0.5 in t, shows a fourth before $t = -12.5$: the zeros γ_7 and γ_8 are at 43.263 and 43.522 at $t = -4$ and have left the axis at $t = -4.5$, and the isolated pair estimate of Section 8.4.1 puts their collision at about $t = -4.03$, just after that of (γ_4, γ_5), although they are the closer pair at $t = 0$ (2.408 apart, against 2.510). The pair (γ_9, γ_{10}) is the closest pair below height 60, at distance 1.769, and its isolated pair estimate $-\frac{1}{2}(1.769)^2 = -1.565$ of Section 8.4.1 is again an overestimate of the collision time, the neighbours delaying the collision. In the variable θ a simple zero moves by

$$\frac{d\theta_k}{dt} = \frac{\partial_\theta^2 F_t(\theta_k)}{4\,\partial_\theta F_t(\theta_k)} = \frac{1}{2}\sum_{j \ne k} \frac{1}{\theta_k - \theta_j},$$

for $F_t = P_t$ and for $F_t = \Xi_t$, which is the equation of motion of Section 8.4.1 with $z = 2\theta$. At $t = 0$ the formula gives -0.39798088 for the first zero of P and -0.28981529 for the first zero of Ξ, and central differences in t with step 10^{-3} give the same eight digits. As t decreases every two zeros attract each other, and since

the zeros above a given zero are more numerous and denser than those below it, the net pull is upward, which is why both sets of zeros rise; a pair collides when its mutual attraction overcomes the difference of the pulls of all the other zeros on its two members. For P the configuration is regular enough that this does not happen in the window.

The mean motion can be predicted. For $\theta > 4\pi$ the integral $P(\theta)$ is dominated by the conjugate saddle points $u_\pm = \pm\frac{x}{2} + \frac{i\pi}{4}$ of $i\theta u - 2\pi\cosh 2u$, with $x = \operatorname{arccosh}(\theta/4\pi)$, where the exponent is $-\frac{\pi\theta}{4} \pm i(\frac{1}{2}\theta x - 2\pi\sinh x)$. So $P(\theta)$ is $e^{-\pi\theta/4}$ times a slowly varying amplitude times a cosine of phase $\varphi(\theta) = \frac{1}{2}\theta x - 2\pi\sinh x$ plus a constant; since $\varphi'(\theta) = \frac{x}{2}$, the zeros have density $\frac{x}{2\pi} = \frac{1}{2\pi}\log\frac{\theta}{2\pi} + O(\theta^{-2})$. To first order in t the factor $e^{tu^2/4}$ adds to the phase the imaginary part $\frac{\pi t}{16}x$ of $\frac{t}{4}u_+^2$, so the number of zeros below a fixed height changes at the rate $\frac{x}{16}$ per unit of t, and each zero moves with

$$\frac{d\theta}{dt} \approx -\frac{x/16}{x/2\pi} = -\frac{\pi}{8},$$

whatever its height. The equation of motion gives the same value: with the density $\frac{1}{2\pi}\log\frac{|\theta|}{2\pi}$ on both half lines, $\frac{1}{2}\sum_{j\neq k}(\theta_k - \theta_j)^{-1}$ becomes $\frac{1}{4\pi}$ times the principal value integral of $\log|s|/(\theta_k - s)$, which is $-\frac{\pi^2}{2}$ for $\theta_k > 0$. Over $-0.5 \le t \le 0$ the 24 real zeros of P in the window rise at mean speeds that decrease with the height from 0.3990 (the first zero) to 0.3949 (the last), towards $\pi/8 = 0.3927$, while the 25 zeros of Ξ rise at speeds between 0.0040, for γ_{25}, held back by γ_{24}, which is closing on it, and 0.7350, for γ_{19}, which is closing on γ_{20}, with mean 0.3647 (`biane_figs.py`). The zeros of P move almost as a rigid block; those of Ξ fluctuate about the same mean motion, and the fast and slow members of the close pairs are the ones that collide.

The regularity can be measured. In units of the local mean spacing $2\pi/\log(\theta/2\pi)$, taken at the midpoint of each gap, the smallest gap between consecutive real zeros in the window is

t	0	-10	-30	-60
P_t	1.002	1.020	1.059	1.121
Ξ_t	0.534	0.490	three zeros left	two zeros left

(`biane_flow.py` and `biane_figs.py`). The smallest gap of P_t is always that between the two highest zeros of the window, at almost exactly the mean spacing, and it widens as t decreases, while the smallest gaps of Ξ_t, $(59.347, 60.832)$ at $t = 0$ and $(25.378, 27.518)$ at $t = -10$, shrink until the pairs collide. With the density taken at the lower end of each gap, as in the note [202], the values are 0.968, 1.014, 1.053, 1.115 for P_t and 0.531, 0.476 for Ξ_t, and the note's values agree with these to one unit in the third decimal; at $t = 0$ the minimum for P then moves to the first gap $(19.538, 24.897)$, where the density changes fastest, which is an effect of

the convention. The two functions have the same mean density of zeros, $\frac{1}{2\pi}\log\frac{\theta}{2\pi}$ (Proposition 8.7.8); what differs is the fluctuation, the sawtooth of a quantisation rule for P against $S(T)$ for Ξ, whose close pairs (Lehmer pairs, Section 8.4.2) the flow pushes together.

Remark 8.7.16 (One saddle pair against many). For fixed t and large θ the integral $P_t(\theta)$ is governed by a single pair of complex saddle points of $i\theta u - 2\pi\cosh 2u + tu^2/4$, conjugate under $u \mapsto -\bar{u}$ because the kernel is real and even, so that $P_t(\theta) = 2\,\mathrm{Re}(Ae^{i\psi_t(\theta)})(1+o(1))$ with a monotone phase, as in Pólya's asymptotics at $t = 0$; heuristically, all but finitely many zeros of P_t are then real for each fixed t. For Ξ_t the large θ asymptotics is the Riemann Siegel expansion, with a number of interfering terms that grows with θ, and the interference is $S(T)$. The note [202] suggests that a version of the saddle point argument uniform in t, with a finite computation, would prove $\Lambda_P = -\infty$. By Proposition 8.7.15 no such uniform version exists: as t decreases below Λ_P, nonreal zeros appear, outside the box of the argument principle or at times the computation does not reach. The difference between P and Ξ is quantitative, the size of $|\Lambda_P|$ against $\Lambda = 0$ under RH, not a difference between a finite and an infinite constant.

In the language of Lee and Yang [141] (Section 7.1.3 and Proposition 7.1.16), the single spin law with density $e^{-2\pi\cosh 2u}/K_0(2\pi)$ has the Lee and Yang property, since its moment generating function $K_{h/2}(2\pi)/K_0(2\pi)$ vanishes only at imaginary h; it is the ferromagnetic law U of Section 8.7.4. Newman's theorem says that this property survives every Gaussian factor e^{-bu^2}, $b > 0$, only for two point laws and for the densities of the proof of Proposition 8.7.15, which include the φ^4 laws $e^{-au^4-bu^2}$ that Simon and Griffiths realised as limits of Ising ferromagnets [240]. The law with density proportional to $e^{-2\pi\cosh 2u-bu^2}$ has the property exactly when its Fourier transform, a multiple of P_{-4b}, has only real zeros, that is when $b \le -\Lambda_P/4$. So the Simon Griffiths type construction for $e^{-2\pi\cosh 2u-bu^2}$ at every real b that the note proposes cannot exist.

8.7.7 Pólya's function and Ξ side by side

Table 8.33 sets the two functions side by side. It follows the comparison in the note [202], corrected in three rows: the index side clock of Ξ exists with no hypothesis, the de Bruijn Newman constant of P is finite, and the computation of the flow describes a window of heights and times, not the whole flow. The Wick rotation of Section 8.7.2 relates the two entries of each of the second, third and fourth rows. Discreteness moves across the table: P is continuous in the energy and discrete in the angular momentum, while Ξ is discrete in the energy and, under RH, in the angular momentum.

The statements about Ξ in the right column and their relatives, Proposition 1.2.1, Corollary 10.1.4, Proposition 7.1.15 and the GGC form of the clock T discussed after Corollary 8.7.11, are proved by Hadamard factorisation, Bern-

	Pólya's $P(\theta) = K_{i\theta/2}(2\pi)$	Riemann's $\Xi(\theta)$
kernel	$e^{-2\pi\cosh 2u}$, (8.7.1)	$\Phi(u)$, (1.2.3)
hitting time	$H^{(\sqrt{2\pi})}_{\sqrt{2\pi}}$, Brownian motion with drift (Proposition 8.7.1)	$\frac{\pi}{2}S_2$, three dimensional Bessel process, (9.1.1)
symmetric space	$\mathbb{H}^3 = SL_2(\mathbb{C})/SU(2)$	$SU(2) \cong S^3$
Thorin measure of the hitting time (energy)	continuous: σ_2 for $H^{(1)}_1$	discrete: $2\pi\sigma_1$ for $\pi^2 S_2$ (Theorem 8.7.2)
index side Thorin measure (angular momentum)	$\sum_n \delta_{\nu_n^2}$, a theorem (Theorem 8.7.9)	$\sum_\gamma \delta_{\gamma^2}$ if and only if RH (Proposition 1.2.1)
index side clock	τ_∞, a hitting time of an explicit diffusion (Theorem 8.7.10)	a random time T (Theorem 5.1.1); no diffusion is known (Question 8.7.18)
trace of the sine squared form	$\Theta_P = \operatorname{tr} e^{-v\sqrt{A_P}}$, completely monotone (Proposition 8.7.13)	$K_{1/2}$, completely monotone if and only if RH (Corollary 10.1.4)
operator for the zeros	$-\frac{d^2}{dx^2} + e^{2x}$ on $[\log 2\pi, \infty)$ (Proposition 8.7.7)	not known
Klein Gordon extension	$\exp(-e^x \cosh w)$ (Theorem 8.7.14)	not known (Question 8.7.19)
de Bruijn Newman constant	$-\infty < \Lambda_P \le 0$ (Proposition 8.7.15)	$\Lambda \ge 0$ [224]; RH if and only if $\Lambda = 0$
real zeros in $(0, 90)$ for $-60 \le t \le 0$	no collision; mean speeds 0.395 to 0.399 on $[-0.5, 0]$	collisions in pairs from $t = -1.144$; mean speeds 0.004 to 0.735 on $[-0.5, 0]$

Table 8.33: Pólya's function and Ξ. Apart from the computations of the last row, the left column is a list of theorems; in the right column the entries that concern the zeros are equivalent to RH or open.

stein's theorem and integral identities. No property of the primes enters their proofs, and none could, since each is an equivalence between two statements about the zeros of one entire function: they reorganise RH, as Section 12.2 says of the programme as a whole, and do not approach it. The left column shows that the reorganisation is coherent, since for P every entry is a theorem; but the proofs of the entries on the left that concern the zeros all go through the selfadjoint operator A_P, which is exactly what the right column lacks.

The de Bruijn Newman constant explains why a whole family of arguments cannot succeed. The note [202] proves part (a) of the following proposition; part (b) is the strengthening that Newman's theorem gives, and it changes the reading of part (a).

Proposition 8.7.17. *Let $\mathcal{P}$ be a property of even, positive, integrable kernels such*

that

(i) if $\mathcal{P}(g)$ holds, then $\int g(u)e^{i\theta u}du$ has only real zeros, and

(ii) there is $\varepsilon > 0$ such that $\mathcal{P}(g)$ implies $\mathcal{P}(e^{-\varepsilon u^2/4}g)$.

(a) If (ii) holds for $g = \Phi$, then $\mathcal{P}(\Phi)$ is false.

(b) If (ii) holds for all g with the same ε, then $\mathcal{P}(g)$ is false for every kernel g that is $O(e^{-|u|^b})$ for some $b > 2$ and satisfies $g(u)e^{Cu^4} \to 0$ as $|u| \to \infty$ for every C; in particular $\mathcal{P}$ fails for Φ, for Φ^* and for $e^{-2\pi\cosh 2u}$.

Proof. (a) If $\mathcal{P}(\Phi)$ held, then by (ii) and (i) the transform of $e^{-\varepsilon u^2/4}\Phi$, which is $\Xi_{-\varepsilon}$ of Section 8.7.6, would have only real zeros, so $\Lambda \le -\varepsilon < 0$, against the theorem of Rodgers and Tao [224]. (b) If $\mathcal{P}(g)$ held, then by induction $\mathcal{P}(e^{-k\varepsilon u^2/4}g)$ would hold for $k = 0, 1, 2, \dots$, so the transform of $e^{-k\varepsilon u^2/4}g$ would have only real zeros for each k. This kernel is again $O(e^{-|u|^b})$, so by de Bruijn's theorem [65, Theorem 13] the transform of $e^{tu^2/4}g$ would have only real zeros for every $t \ge -k\varepsilon$, and hence for every real t. By Newman's theorem, as in the proof of Proposition 8.7.15, $g/\int g$ would be a density of the form given there, which is bounded below by $Ku^{2m}\exp(-C'u^4 - \beta u^2)$ for some C', against the decay of g. □

Part (a) uses the theorem of Rodgers and Tao and concerns Φ alone. Part (b) uses only de Bruijn's theorem and Newman's theorem of 1976, and it shows that the obstruction has nothing to do with the position of Φ on the boundary of the class of kernels with real zeros: a property that survives a fixed amount of damping for all kernels fails for P as well, although the zeros of P are real for a structural reason, the selfadjointness of A_P. So the reading in the note, that Pólya's kernel tolerates every Gaussian damping, that P has slack and Ξ none, is to be replaced by a quantitative one: P has the finite slack $|\Lambda_P|$, whose size is not known, and Ξ has none if RH holds. Part (b) also limits what the proposition excludes. The note lists among the excluded arguments a ferromagnetic or Lee and Yang construction and a Simon Griffiths representation. The Lee and Yang property of a single spin law is preserved by the ferromagnetic factor e^{bu^2}, $b > 0$, which runs the flow forward, but by (b) it is not preserved by a fixed damping except on Newman's class. A Lee and Yang representation of the law with density proportional to Φ, which is equivalent to RH (Proposition 7.1.16), is therefore not excluded by the proposition, and neither is a Simon Griffiths type representation of that law. What the proposition excludes is a construction with room to spare, one that would apply to $e^{-\varepsilon u^2/4}\Phi$ as well. The criteria of this section are positivity statements equivalent to RH, of the kind whose history Section 12.2.4 recalls with the example of Conrey and Li [54]; their value here is as a dictionary, not as a plan.

Three questions raised by the note are not equivalences and can be studied without assuming what is to be proved. The first two ask for an object, a diffusion or a potential, not for the zeros, and part (b) does not apply to them, because the existence of such an object is not preserved by a fixed Gaussian damping of every kernel: for P it is lost at every $t < \Lambda_P$, where P_t has nonreal zeros. The third is a question about an explicit kernel with no arithmetic in it, and it measures the slack of Pólya's function. A fourth concerns the interpolation between the two clocks of Section 8.7.2.

Question 8.7.18. Is there a diffusion on a half line, with an entrance boundary, whose hitting time of a level from the entrance boundary has the Laplace transform $\xi(\frac{1}{2})/\xi(\frac{1}{2}+2\sqrt{u})$, the transform of the clock T of Theorem 5.1.1 scaled by 4, as τ_∞ has the transform $K_0(2\pi)/K_{\sqrt{u}}(2\pi)$ in Theorem 8.7.10? By Kent's theorem [122] such a hitting time is $\sum_n e_n/\lambda_n$ over the Dirichlet eigenvalues $\lambda_n > 0$ of the generator, so a positive answer implies RH; under RH and the simplicity of the zeros, Krein's inverse spectral theory gives a generalised diffusion with the eigenvalues $\gamma^2/4$ (Section 8.1.3). The question asks for one constructed without RH, for example as a perturbation of the diffusion X of Section 8.7.4, whose potential e^{2x} has the right Weyl law.

Question 8.7.19. Does $g(w) = \Phi(w/2)$ admit a Klein Gordon extension in the sense of Theorem 8.7.14, with a real potential V asymptotic to e^{2x} at $+\infty$? By the theorem a positive answer implies RH. Under RH the question asks whether $\mu \mapsto 2\xi(\frac{1}{2}+2\mu)$ is, up to normalisation, the Jost function of such a potential whose Klein Gordon solution has the boundary value g. A first numerical step is to reconstruct a potential from the first few thousand ordinates and a choice of norming constants by the method of Gelfand and Levitan or of Marchenko, and to test whether it is stable under truncation and close to e^{2x}.

Question 8.7.20. What is the de Bruijn Newman constant Λ_P of Pólya's function, and that of Ξ^*, which is also finite and at most 0 by the proof of Proposition 8.7.15? At what height do the first nonreal zeros of P_t appear as t decreases through Λ_P, and do they appear, as for Ξ_t, through the collision of two real zeros?

Question 8.7.21. Is there a one parameter family of laws joining the drifted first passage time and the sum of two Bessel hitting times, each with a reciprocal symmetry that encodes a functional equation, as $\tau \mapsto 1/\tau$ does for H in Proposition 8.7.1(i) and Jacobi's transformation does for S_2, along which the index side Thorin measure can be followed? The curvature family of Proposition 8.7.6 is an Esscher tilt and breaks the symmetry. Positivity of the index side Thorin measure at the Bessel end is RH.

8.8 Exercises

On Section 8.1.

8.1. Assuming that the zeros are on the critical line and simple, show that the residues of the clock transform at $u = -\gamma_k^2$ are $R_k = 2i\gamma_k \xi(\frac{1}{2})/\xi'(\frac{1}{2} + i\gamma_k)$ and that they alternate in sign. (They are the coefficients c_n of Theorem 4.3.23.)

8.2. Explain why the spectral expansion of the density diverges at $t = 0$ while the Talbot inversion does not, and compare the abscissa of convergence in Theorem 4.3.23.

8.3. Show that the hazard of a finite sum of exponentials with rates $\lambda_1 < \lambda_2 < \cdots$ tends to λ_1 and that $h(t) - \lambda_1$ decays like $e^{-(\lambda_2 - \lambda_1)t}$.

8.4. Sample Williams' variable with 10^5 draws and estimate $\Xi(5)$; compare the standard error with Table 8.2.

8.5. Assume RH, let Θ_N be the cutoff of Section 8.1.3, and assume that $S(\Theta_N) = 0$, as for the values of N used there. Show that $\log G(w) - \log G_N^{\mathrm{W}}(w)$ tends to $\ell_N = 2\int_{\Theta_N}^{\infty} S(\gamma)\gamma^{-1}\,d\gamma$ as $w \to \infty$. Then, using Stirling's formula for $\log \xi(\frac{1}{2} + x)$ as $x \to +\infty$ and $\int_0^{\infty} \log(1 + x^2/t^2) \log \frac{t}{2\pi}\,dt = \pi x \log \frac{x}{2\pi} - \pi x$, show that

$$\ell_N = 2\sum_{k \le N} \log \gamma_k - \log \xi(\tfrac{1}{2}) + \tfrac{1}{4}\log \tfrac{\pi}{2} - \frac{1}{\pi}\int_0^{\Theta_N} \log t \, \log \frac{t}{2\pi}\,dt + \frac{2}{\pi}\int_{\Theta_N}^{\infty} \Big(\vartheta'(t) - \frac{1}{2}\log \frac{t}{2\pi}\Big)\log t\,dt,$$

the terms in $\log x$ cancelling because $\vartheta(\Theta_N) = (N-1)\pi$. So ℓ_N is determined by $\gamma_1, \ldots, \gamma_N$ and $\xi(\frac{1}{2})$.

On Section 8.2.

8.6. Show that on the parabola $u = \frac{1}{2}(-q^2 + 2icq)$ one has $\operatorname{Re}\sqrt{c^2 + 2u} = c$, and hence that every mixture of inverse Gaussian laws with drift c has transform bounded by one there.

8.7. Derive the small q expansion $|G(u)|^2 = 1 - q^2(\mathbb{E}T - c^2 \operatorname{Var} T) + O(q^3)$ on the parabola.

8.8. Show that the hazard of the first passage time of Brownian motion across the line $\alpha + \beta t$, with $\alpha > 0$ and $\beta < 0$, which has the inverse Gaussian law with drift $|\beta|$, tends to $\beta^2/2$ with correction $3/(2t)$.

8.9. Implement the tangent boundary ODE with a different starting time and check that the asymptotic slope is unchanged.

8.10. Show that the density f of the clock is continuous and positive on $(0,\infty)$. (The Laplace transform of T equals $1/G$ on $\operatorname{Re} u \geq 0$, by analytic continuation from $(0,\infty)$ and continuity, so the characteristic function is $1/G(-iy)$; Stirling's formula and the bound $|\zeta(s)| \geq 1/\zeta(\operatorname{Re} s)$ for $\operatorname{Re} s > 1$, applied on the rays $\frac{1}{2} + re^{\pm i\pi/4}$, show that it decays faster than any power of $|y|$. Next, $\log G(u) = o(u)$ as $u \to \infty$, so $P(T < t) > 0$ for every $t > 0$; $1/G$ has poles, so the law of T has unbounded support; and the law is unimodal by Corollary 11.1.2, so f itself, which is the derivative of the distribution function at every point, is nondecreasing up to a mode and nonincreasing after it.)

On Section 8.3.

8.11. Derive formula (8.3.1) from the derivative representation of λ_n.

8.12. Show that $\lambda_1 = \sum_\rho \rho^{-1} = 1 + \frac{\gamma_E}{2} - \frac{1}{2}\log 4\pi$, and that under RH $\lambda_1 = \sum_{\gamma>0} (\gamma^2 + \frac{1}{4})^{-1}$.

8.13. Show that under RH $\lambda_n = \sum_{\gamma>0} 4\sin^2(n \arctan \frac{1}{2\gamma})$, and that the terms are of order n^2/γ^2 for $\gamma \gg n$.

8.14. For a hypothetical zero $\rho = \frac{1}{2} + d + i\gamma$ with $d > 0$, show that $|1 - 1/(1-\bar{\rho})| > 1$, and estimate the index n at which its contribution would make λ_n negative.

On Section 8.4.

8.15. Show that H_t satisfies $\partial_t H_t = -\partial_z^2 H_t$ and derive the equation of motion of a simple zero.

8.16. For an isolated pair at distance D_0, solve $dD/dt = 4/D$ and find the collision time.

8.17. Modify the Pólya evaluator to compute H_t and reproduce Table 8.7.

8.18. Explain why the collision time of a pair of zeros gives numerical evidence for, but not a proof of, a lower bound on Λ.

On Section 8.5.7.

8.19. Show that $K_4(u) = 2e^{3u/2}\eta(ie^{2u})^3$ is even, using $\eta(i/x) = \sqrt{x}\,\eta(ix)$, and deduce the functional equation $\xi(s,\chi_4) = \xi(1-s,\chi_4)$ from $\xi(\frac{1}{2}+s,\chi_4) = \int_{\mathbb{R}} K_4(u) e^{su}\,du$.

8.20. Show that for real $\sigma > 0$ the series $\sum_{n\geq 0}(-1)^n(2n+1)^{-\sigma}$ lies between $1-3^{-\sigma}$ and 1, and deduce that $L(s,\chi_4)$ has no zeros on $(0,1)$ without using the kernel.

8.21. For $0 < \varepsilon < \delta$ show that $\pi(P_{\varepsilon+\delta} + P_{\varepsilon-\delta})$ integrates over the interval $|x| < \sqrt{\delta^2-\varepsilon^2}$, where it is negative, to $\pi - 4\arctan\sqrt{(\delta+\varepsilon)/(\delta-\varepsilon)}$, which lies in $(-\pi, 0)$ and tends to 0 as $\varepsilon/\delta \to 0$; this is the bound on the fall of the argument after Corollary 8.5.31.

8.22. Show that $g(c,u) = 2c(3u^2-c^2)/(c^2+u^2)^3$, the second derivative of $c/(c^2+u^2)$ in u, has $\partial_c g(c,u) = 6(u^4 - 6c^2u^2 + c^4)/(c^2+u^2)^4$, which is positive exactly when $|c| < (\sqrt{2}-1)|u|$ or $|c| > (\sqrt{2}+1)|u|$, and deduce that $g(a+\delta,u) + g(a-\delta,u) > 0$ for $a, \delta > 0$ with $a + \delta < (\sqrt{2}-1)|u|$, the step for the zeros off the line in the proof of Theorem 9.1.11(ii).

On Section 8.7.

8.23. Show that $1/H$ has the law of H size biased by H, for $H = H^{\frac{(\sqrt{2\pi})}{\sqrt{2\pi}}}$ with the density of Proposition 8.7.1(i), and deduce $\mathbb{E}H^s = \mathbb{E}H^{1-s}$ without computing the Mellin transform.

8.24. Show that $F(x,w) = \exp(-e^x \cosh w)$ satisfies $(\partial_x^2 - \partial_w^2 - e^{2x})F = 0$, and that in the coordinates $X_1 = e^x \cosh w$, $X_0 = e^x \sinh w$ one has $\partial_x^2 - \partial_w^2 = e^{2x}(\partial_{X_1}^2 - \partial_{X_0}^2)$, so that $F = e^{-X_1}$ solves the Klein Gordon equation of unit mass.

8.25. Show that $(1+y)e^{-y} \geq e^{-y^2/2}$ for $y \geq 0$, and deduce that each density of Newman's theorem in the proof of Proposition 8.7.15 is bounded below by $Ku^{2m}\exp(-[\alpha + \sum_j \frac{1}{2a_j^4}]u^4 - \beta u^2)$.

8.26. Show that the principal value integral of $\log|s|/(1-s)$ over $\mathbb{R}$, taken with symmetric limits at $\pm\infty$, is $-\frac{\pi^2}{2}$, and deduce the mean speed $-\frac{\pi}{8}$ of the zeros under the flow in Section 8.7.6 from the density $\frac{1}{2\pi}\log\frac{|\theta|}{2\pi}$.

8.27. For $0 < y < \pi$ show, as claimed in Subsection 10.1.5, that $\gamma^n e^{-\gamma y}$, for $\gamma \leq n/\pi$, is at most $\exp(n(\log\frac{y}{\pi} + 1 - \frac{y}{\pi}))$ times its maximum over $\gamma > 0$, and evaluate the exponent for $n = 10^6$ at $y = 1$ and at $y = 3$.

Chapter 9

The basepoint one

Section 8.6 recovered the zeros from the Thorin density at the resolution $\varepsilon = 0.1$, close to the critical line, where the positivity of the density at every height is open. We now work at the resolution $\varepsilon = \frac{1}{2}$ of the Cauchy flow $\varepsilon \mapsto \rho_\varepsilon$ of Section 1.2, the basepoint one. The completed zeta function has two probabilistic readings. On the Mellin side, Williams [271] and Biane, Pitman and Yor [28] showed that $2\xi(s)$ is the Mellin transform of a sum of two Bessel hitting times, which gives the Pólya variable and its exponential tilts at every basepoint with no hypothesis. On the Laplace side, the shifted clocks T_α, with $\mathbb{E}e^{-uT_\alpha} = \xi(\alpha)/\xi(\alpha + \sqrt{u})$, carry the Thorin programme. At the centre, $\alpha = \frac{1}{2}$, the shifted clock is the clock of [198], which is infinitely divisible and self decomposable with no hypothesis (Theorem 5.1.1) and a generalised gamma convolution exactly when RH holds (Proposition 1.2.1). At a basepoint α to the right of every zero, T_α is a GGC whose Thorin density is the Poisson smoothing of the zeros by Cauchy kernels of widths $\alpha - \beta$ (Theorem 1.2.2 and Proposition 1.2.3), and from $\alpha = 1$ on this density is positive with no hypothesis, because no zero lies on the line $\operatorname{Re} s = 1$ or to its right. The basepoint one is therefore the leftmost basepoint at which positivity is a theorem, and the natural place to separate what is unconditional from what is equivalent to RH.

At the basepoint one everything can be computed and most things can be proved, and we work the example out completely, as a test bed for the tools of the book. We ask three questions of T_1: what its Thorin measure and its law are, whether its Thorin density comes from the sine squared lemma, and whether it is a hitting time to which Kent's theorem applies. Section 9.1 answers the first two: it computes the Thorin measure and the law of T_1, and obtains the Thorin density from a Lévy form of $\log \xi(1+s)$ by the sine squared lemma. It also proves, with no hypothesis and with a computation of the zeros below height 2000 in ball arithmetic and explicit zero counts above it, that the density is smallest at the origin, so that the drift of the Lévy form becomes a positive stable one half summand of T_1, independent of a GGC remainder (Proposition 9.1.4). Below the basepoint one it proves the same on the verified window for every $\alpha \in (\frac{1}{2}, 7.35]$,

and shows that the minimum at the origin at every height for every $\alpha \in (\frac{1}{2}, 1)$ is equivalent to RH (Section 9.1.6). Section 9.2 answers the third: every T_α with $\alpha > \frac{1}{2}$ is a hitting time, the tilted centre clock run through the inverse local time of a sticky Brownian motion, with an exact Brownian boundary that, as computed, becomes flat in the right tail.

For the Thorin measure and the Lévy form alike, what is unconditional at the basepoint one extends to every basepoint to the right of the line $\operatorname{Re} s = \alpha$, for a given $\alpha < 1$, exactly when no zero lies to the right of that line; required for every $\alpha > \frac{1}{2}$, this is RH. Chapters 10 and 11 carry these objects across the critical strip and read them in other settings, and Section 11.3 repeats the calculations of this chapter for the function $L(s, \chi_{12})$ of Section 8.5, whose theta series is the Dedekind eta function.

9.1 The Thorin measure and the Lévy form at the basepoint one

On the Mellin side the zeros enter only as zeros of the transforms $\xi(\alpha+s)/\xi(\alpha)$, and only the theory of infinite divisibility singles out the basepoint one (Section 9.1.1). On the Laplace side the Thorin density of T_1 is a sum of Cauchy kernels of widths $1-\beta$ centred at the ordinates, its mass is the zero count of Section 3.4, and the law of T_1 has infinite mean and a stable one half tail (Section 9.1.2). The sine squared decomposition (3.1.1) of Section 3.1 diverges at $\alpha = 1$, but its prime and pole pieces can be compensated against each other, and $\xi(1+s)$ then has an explicit Lévy form, whose convergence is the prime number theorem and whose extension below the line $\operatorname{Re} s = 1$ is exactly a zero free half plane (Sections 9.1.3 to 9.1.5). Below the basepoint one the minimum at the origin persists on the verified window, by a comparison of the densities across vertical lines, and at every height it is equivalent to RH (Section 9.1.6). Averaged over heights against a Gaussian, the compensated sum becomes computable with an explicit error from published bounds for Chebyshev's error $\psi(y) - y$, which certifies the heat trace at the basepoint one from the primes and shows why no such certificate passes below it (Section 9.1.7). The scripts of the chapter are in the folders `basepoint_one/xi` and `basepoint_one/process_map` of the archive, except those of Section 9.1.6, which are in `code/`, and we name them at each table and figure.

9.1.1 The Mellin side: Bessel hitting times and size biasing

Let J be the first time a three dimensional Bessel process started at 0 reaches the level one. Its Laplace transform is

$$\mathbb{E}e^{-uJ} = \frac{\sqrt{2u}}{\sinh\sqrt{2u}} = \prod_{n\geq 1}\Big(1 + \frac{2u}{\pi^2 n^2}\Big)^{-1}, \qquad \mathbb{E}J = \frac{1}{3},$$

so $J \stackrel{d}{=} \sum_n \frac{2}{\pi^2 n^2} e_n$ with independent standard exponentials e_n. This is a GGC with unit Thorin atoms at the points $\pi^2 n^2/2$, which are the Dirichlet eigenvalues of the Bessel process on $[0,1]$, so J is the simplest instance of Kent's theorem (Section 8.1.3): up to a change of scale the Bessel process is the string diffusion with that spectrum [122]. With J' an independent copy of J, the sum $S_2 = J + J' = \sum_n \frac{2}{\pi^2 n^2}\Gamma_n$, with independent Gamma$(2,1)$ variables Γ_n, is Williams' Gamma(2) structure [271] of Section 8.2.3, and Biane, Pitman and Yor [28] proved the identity recalled in Section 1.2 and Theorem 7.1.21,

$$2\xi(s) = \mathbb{E}\Big[\Big(\frac{\pi}{2}S_2\Big)^{s/2}\Big], \qquad s \in \mathbb{C}. \tag{9.1.1}$$

Table 9.1, computed by `comp.py` of `basepoint_one/process_map`, checks (9.1.1) at real s with 4×10^5 draws of S_2, the series for S_2 being truncated after 400 terms and the remainder replaced by its mean; the four values agree with $2\xi(s)$ within 0.6 standard errors. Table 8.2 checks the same identity on the critical line.

s	$\mathbb{E}(\pi S_2/2)^{s/2}$, Monte Carlo	standard error	$2\xi(s)$
0.5	0.99415	0.00017	0.99424
1	0.99981	0.00034	1
2	1.04677	0.00074	1.04720
3	1.14713	0.00127	1.14788

Table 9.1: The identity (9.1.1) of Biane, Pitman and Yor, checked with 4×10^5 draws of S_2 by `comp.py`. The sample mean of S_2 is 0.6664 against $\mathbb{E}S_2 = \frac{2}{3}$.

Put $Y = \frac{1}{2}\log(\frac{\pi}{2}S_2)$, as in Section 1.2. By (9.1.1), $2\xi(\alpha+w) = \mathbb{E}[e^{\alpha Y}e^{wY}]$, so for every real α

$$\frac{\xi(\alpha+w)}{\xi(\alpha)} = \mathbb{E}e^{wV_\alpha}, \qquad P(V_\alpha \in dy) = \frac{e^{\alpha y}P(Y\in dy)}{\mathbb{E}e^{\alpha Y}}.$$

Since the Pólya variable X is Y tilted by $e^{Y/2}$, the variable V_α has the density $e^{(\alpha-\frac12)y}p(y)\,\xi(\frac12)/\xi(\alpha)$, the Esscher tilt of the Pólya density p of (1.2.4); so $V_{1/2} = X$, and we write $V = V_1$ for the tilt at the basepoint one. Equivalently, V_α is $\frac12\log(\frac{\pi}{2}S_2)$ under the law of S_2 size biased by $S_2^{\alpha/2}$: by $S_2^{1/4}$ at the centre, by $S_2^{1/2}$ at the basepoint one, where the weight e^Y needs no normalisation because $\mathbb{E}e^Y = 2\xi(1) = 1$, and by $S_2^{(1-\varepsilon)/2}$ at $1-\varepsilon$. Every basepoint thus gives a probability law with no hypothesis on this side, and nothing distinguishes $\alpha = 1$ except the weight. The law of $-V_\alpha$, with characteristic function $\xi(\alpha - it)/\xi(\alpha)$, is Nakamura's complete Riemann zeta distribution [160]. He wrote its density through Patterson's representation of ξ, and proved that RH holds if and only if, for every $\frac12 < \alpha < 1$, this characteristic function has a Lévy Khintchine form with

a signed measure, with no integrability condition on the measure. The theory of infinite divisibility does distinguish it: V_α is quasi infinitely divisible, though not infinitely divisible, for every $\alpha > 1$, and it is not quasi infinitely divisible at $\alpha = 1$ or at the centre (Proposition 11.1.3; for $\alpha = 1$ see also Section 9.1.4). On this side the zeros enter only as zeros, of the characteristic function $\Xi(t)/\Xi(0)$ of $V_{1/2}$ and of $\xi(\alpha + s)$; the positivity statements equivalent to RH are on the other side, in the reciprocal $\xi(\alpha)/\xi(\alpha + \sqrt{u})$, to which the rest of the chapter is devoted.

9.1.2 The Thorin measure and the law of T_1

For a real basepoint $\alpha \geq \frac{1}{2}$ we write $\varphi_\alpha(u) = \xi(\alpha)/\xi(\alpha+\sqrt{u})$, so that $\varphi_{1/2} = 1/G$ is the clock, and T_α for a random variable with $\mathbb{E}e^{-uT_\alpha} = \varphi_\alpha(u)$ for $u \geq 0$, whenever one exists. Theorem 1.2.2 provides it, as a GGC, for $\alpha \geq 1$, and Theorem 9.2.5 in Section 9.2 provides it for every $\alpha > \frac{1}{2}$. As in Section 1.2, $v_\alpha(\theta) = \mathrm{Re}(\xi'/\xi)(\alpha+i\theta)$, and at the basepoint one, the resolution $\varepsilon = \frac{1}{2}$, the Thorin density in the chart $\theta = \sqrt{z}$ is $\rho_{1/2} = v_1/\pi$ by (1.2.7).

Proposition 9.1.1. *The function $\varphi_1(u) = \xi(1)/\xi(1+\sqrt{u})$ is the Laplace transform of a GGC T_1 with no drift, $\log \varphi_1(u) = -\int_0^\infty \log(1 + u/z)\, U_1(dz)$, and in the variable $\theta = \sqrt{z}$ its Thorin measure has density*

$$\rho_{1/2}(\theta) = \frac{1}{\pi} v_1(\theta) = \frac{1}{\pi} \mathrm{Re}\, \frac{\xi'}{\xi}(1 + i\theta) = \frac{1}{\pi} \sum_\rho \frac{1-\beta}{(1-\beta)^2 + (\theta-\gamma)^2}, \tag{9.1.2}$$

the sum running over all nontrivial zeros $\rho = \beta + i\gamma$ in conjugate pairs. Every term is positive, so U_1 is a positive measure with no hypothesis on the zeros.

Proof. This is Theorem 1.2.2 at $\alpha = 1$, and we record the three facts that it uses there. First, $-\frac{d}{du} \log \varphi_1(u) = \frac{1}{2\sqrt{u}} \frac{\xi'}{\xi}(1 + \sqrt{u})$ is analytic on $\mathbb{C} \setminus (-\infty, 0]$, because the zeros of $\xi(1 + w)$ lie at $w = \rho - 1$ with $\mathrm{Re}\, w = \beta - 1 < 0$ and so are never reached by the principal branch $w = \sqrt{u}$; Stieltjes inversion (1.3.2) of its jump across the cut, where $\sqrt{u} = \pm i\theta$, gives the density (1.2.6) in z, which is (9.1.2) in θ because $dz = 2\theta\, d\theta$. Second, by (1.2.2) with conjugate pairs grouped, $\mathrm{Re}\, 1/(1 + i\theta - \rho) = (1-\beta)/((1-\beta)^2 + (\theta-\gamma)^2) = \pi P_{1-\beta}(\theta - \gamma)$, as in the proof of Proposition 1.2.3, and every term is positive because $\beta < 1$ for all zeros, which is the nonvanishing of ζ on the line $\mathrm{Re}\, s = 1$. Third, there is no drift because $\log \xi(1 + \sqrt{u})$ grows like $\frac{1}{2}\sqrt{u} \log \sqrt{u}$, more slowly than u. □

Under RH every width equals $\frac{1}{2}$, and (1.2.8) at $\varepsilon = \frac{1}{2}$ reads

$$\rho_{1/2}(\theta) = \sum_{\gamma > 0} \Big(P_{1/2}(\theta - \gamma) + P_{1/2}(\theta + \gamma) \Big),$$

the RH measure blurred by the Cauchy kernel of width one half. The value at the origin is explicit, $\rho_{1/2}(0) = \frac{1}{\pi} \sum_\rho \frac{1}{\rho} = \frac{1}{\pi}\big(1 + \frac{\gamma_E}{2} - \frac{1}{2} \log 4\pi\big) = \frac{0.0230957089661}{\pi} =$

0.0073515925, where γ_E is Euler's constant, and it is the minimum of $\rho_{1/2}$, attained only at the origin (Proposition 9.1.4). The cumulative mass $M_{1/2}(T) = \int_0^T \rho_{1/2}(\theta)\,d\theta$ is π^{-1} times the argument of ξ along the line $\operatorname{Re} s = 1$, and since no zero lies on or to the right of that line, Proposition 3.4.2 gives the exact unconditional count $N(T) = M_{1/2}(T) + \tau_{1/2}(T)$, in which $\tau_{1/2}(T)$, the change of argument of ξ from $1 + iT$ to $\frac{1}{2} + iT$ divided by π, is $O(\log T)$, like the argument $\arg\zeta(\frac{1}{2} + iT)$ of Section 2.2.1 [255]. Numerically $\int_0^{100} \rho_{1/2} = 29.111$ against $N(100) = 29$, so that $\tau_{1/2}(100) = -0.111$. Table 9.2 lists values computed from ξ'/ξ, from the arithmetic form at the start of the proof of Proposition 9.1.2, which `compute.py` evaluates on a grid, and from the first 1700 zeros, and Figure 9.1, drawn by `figs.py`, shows the density and its mass.

θ	$\rho_{1/2}(\theta)$ from ξ'/ξ	from the arithmetic form	from 1700 zeros, $\beta = \frac{1}{2}$
0	0.0073515925	0.0073515925	0.0071935
5	0.0084053053	0.0084053053	0.0082472
14.134725	0.6473699273	0.6473699273	0.6472119
20	0.1418590769	0.1418590769	0.1417010
30	0.4070956035	0.4070956035	0.4069375
100	0.2308349819	0.2308349819	0.2306766

Table 9.2: The Thorin density $\rho_{1/2} = v_1/\pi$ of T_1 at the basepoint one, computed from ξ'/ξ, from the arithmetic form $v_1(\theta) = \frac{1}{1+\theta^2} - \frac{1}{2}\log\pi + \frac{1}{2}\operatorname{Re}\psi(\frac{1+i\theta}{2}) + \operatorname{Re}\frac{\zeta'}{\zeta}(1+i\theta)$ with ζ'/ζ evaluated exactly, and from the Poisson sum over the first 1700 zeros with $\beta = \frac{1}{2}$. The zero sum misses the zeros above height 2198, whose contribution is a nearly constant 1.6×10^{-4}.

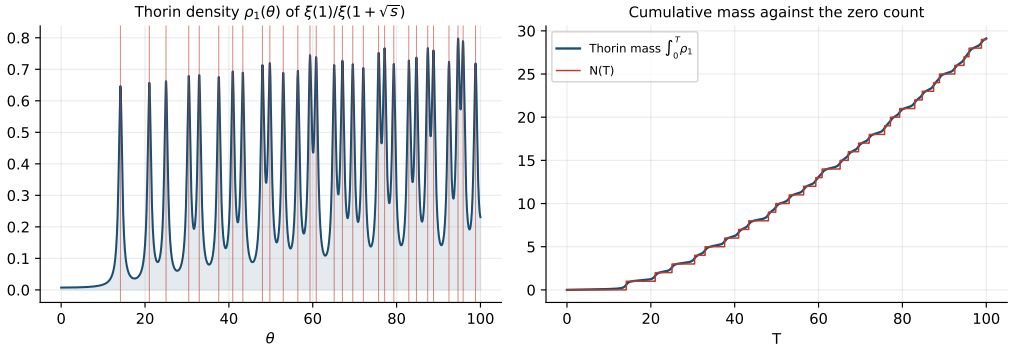


Figure 9.1: Left: the Thorin density $\rho_{1/2}(\theta) = v_1(\theta)/\pi$ of T_1 on $[0, 100]$, with the ordinates of the zeros as thin vertical lines. Right: the cumulative Thorin mass $\int_0^T \rho_{1/2}$ against the zero count $N(T)$. The plot writes ρ_1 for $\rho_{1/2}$ and s for u.

Because $\xi'(1) \neq 0$, the Laplace transform of T_1 is not differentiable at the origin:

$$\varphi_1(u) = 1 - \frac{\xi'}{\xi}(1)\sqrt{u} + O(u) = 1 - \Big(\sum_\rho \frac{1}{\rho}\Big)\sqrt{u} + O(u), \qquad u \downarrow 0.$$

By the Tauberian theorem for Laplace transforms [76, XIII.5], T_1 has infinite mean and

$$P(T_1 > x) \sim \frac{\sum_\rho \rho^{-1}}{\Gamma(\frac{1}{2})}\, x^{-1/2} = 0.01303\, x^{-1/2}, \qquad x \to \infty. \tag{9.1.3}$$

Equivalently, near the origin the Thorin density in z behaves like $\rho_{1/2}(0)/(2\sqrt{z})$, which makes $\int U_1(dz)/z$ diverge. At the centre, by contrast, $\xi'(\frac{1}{2}) = 0$, the clock has the finite mean $\mathbb{E}T = \sum_k a_k^{-1} = 0.0231050$, which is $\sum_{\gamma>0} \gamma^{-2}$ under RH, and the relevant constant is a sum over squared ordinates; at the basepoint one the constant is $\sum_\rho \rho^{-1} = 0.0230957$, and it multiplies $\sqrt{u}$.

9.1.3 The sine squared lemma at the basepoint one

At a basepoint $\alpha > 1$ the sine squared lemma (1.2.9), applied to the canonical measure of $\log \xi$, writes v_α as the sum of a drift, an archimedean piece, a pole piece and a prime piece, (3.1.1). The pole piece is the image under the lemma of the measure $e^{-(\alpha-1)x}dx$ and it enters with a minus sign; the prime piece is positive. As $\alpha \downarrow 1$ both diverge, the pole contribution to $-\infty$ and the prime piece to $+\infty$, because $\sum \Lambda(n)/n$ diverges (Section 3.1.2). So the lemma cannot be applied to the two pieces separately at the basepoint one. It can be applied to their difference. Throughout Section 9.1, $\psi(y) = \sum_{n \le y} \Lambda(n)$ is Chebyshev's function, whose argument is always the real variable $y \geq 1$ or the prime cutoff X, while at every other argument, as in $\psi(\frac{3}{2})$ or $\psi(\frac{1+i\theta}{2})$, $\psi = \Gamma'/\Gamma$ is the digamma function; from here on X always denotes the cutoff, not the Pólya variable of Section 9.1.1, and we write ν_1 for the compensated prime measure of Proposition 9.1.2, which is unrelated to the count $\nu(\sigma, T)$ of Section 3.4.

Proposition 9.1.2. *Let $\nu_1 = \sum_n \Lambda(n) n^{-1} \delta_{\log n} - \mathbf{1}_{(0,\infty)}(x)\, dx$, the prime measure compensated by Lebesgue measure. For every $\theta \neq 0$,*

$$v_1(\theta) = \frac{1}{1+\theta^2} - \frac{1}{2}\log\pi + \frac{1}{2}\operatorname{Re}\psi\Big(\frac{1+i\theta}{2}\Big) + \gamma_E + \lim_{X\to\infty} \int_0^{\log X} 2\sin^2\Big(\frac{\theta x}{2}\Big)\nu_1(dx), \tag{9.1.4}$$

that is, the limit of $\sum_{n\le X} \Lambda(n) n^{-1} 2\sin^2(\frac{1}{2}\theta \log n) - \big(\log X - \sin(\theta \log X)/\theta\big)$. The limit exists by the prime number theorem, and at $\theta = 0$ the formula reduces to $v_1(0) = \sum_\rho \rho^{-1}$.

Proof. On the line $\operatorname{Re} s = 1$ the pole term $1/(s-1) = 1/(i\theta)$ is purely imaginary, so $v_1(\theta) = \operatorname{Re}\frac{1}{1+i\theta} - \frac{1}{2}\log\pi + \frac{1}{2}\operatorname{Re}\psi(\frac{1+i\theta}{2}) + \operatorname{Re}\frac{\zeta'}{\zeta}(1+i\theta)$. For $\operatorname{Re} s \geq 1$ with

$s \neq 1$, partial summation gives

$$-\sum_{n\le X}\frac{\Lambda(n)}{n^s}+\int_1^X y^{-s}dy=-\frac{\psi(X)-X}{X^s}-s\int_1^X(\psi(y)-y)\,y^{-s-1}dy-1,$$

and the prime number theorem with the de la Vallée Poussin error term, $\psi(y)-y=O(y\exp(-c\sqrt{\log y}))$, makes the boundary term vanish and the integral converge absolutely as $X\to\infty$, uniformly for $\operatorname{Re}s\ge 1$. For $\operatorname{Re}s>1$ the limit is $\zeta'(s)/\zeta(s)+1/(s-1)$, and by continuity the same holds on $\operatorname{Re}s=1$, $s\neq 1$. Taking real parts at $s=1+i\theta$, where $\operatorname{Re}\int_1^X y^{-1-i\theta}dy=\sin(\theta\log X)/\theta$, gives $\operatorname{Re}\frac{\zeta'}{\zeta}(1+i\theta)=\lim_X\big[-\sum_{n\le X}\Lambda(n)n^{-1}\cos(\theta\log n)+\sin(\theta\log X)/\theta\big]$. Writing $-\cos=2\sin^2-1$ and using $\sum_{n\le X}\Lambda(n)/n=\log X-\gamma_E+o(1)$, again a consequence of the prime number theorem, gives (9.1.4), since $\int_0^{\log X}2\sin^2(\theta x/2)dx=\log X-\sin(\theta\log X)/\theta$. At $\theta=0$ the bracket vanishes and $1-\frac12\log\pi+\frac12\psi(\frac12)+\gamma_E=1+\frac{\gamma_E}{2}-\frac12\log 4\pi$. □

Table 9.3, computed by `compute.py`, and Figure 9.2, drawn by `figs.py`, check (9.1.4) with primes up to 10^7. The prime piece and the Lebesgue piece each grow like $\log X$, reaching about 16 at $X=10^7$, and their difference converges to the exact value with an error that decreases roughly like $\log^2 X/\sqrt{X}$, the rate the prime number theorem allows under RH.

θ	X	prime piece	Lebesgue piece	formula (9.1.4)	error
5	10^3	6.4288	6.9040	0.022767	-3.6×10^{-3}
5	10^5	10.8715	11.3429	0.026558	1.5×10^{-4}
5	10^7	15.8238	16.2955	0.026345	-6.1×10^{-5}
14.1347	10^3	7.9028	6.9252	1.964685	-6.9×10^{-2}
14.1347	10^5	12.5949	11.5547	2.027421	-6.4×10^{-3}
14.1347	10^7	17.0932	16.0475	2.032931	-8.4×10^{-4}
30	10^3	6.8808	6.9115	1.329190	5.0×10^{-2}
30	10^5	11.4364	11.5191	1.277186	-1.7×10^{-3}
30	10^7	16.0455	16.1267	1.278684	-2.4×10^{-4}

Table 9.3: The sine squared lemma at the basepoint one: the prime piece $\sum_{n\le X}\Lambda(n)n^{-1}2\sin^2(\frac12\theta\log n)$, the Lebesgue piece $\log X-\sin(\theta\log X)/\theta$, the formula (9.1.4) truncated at X, and its error against the exact values $v_1(\theta)=\pi\rho_{1/2}(\theta)=0.026406$, 2.033773 and 1.278929 at $\theta=5$, 14.1347 and 30.

So the Thorin density comes from the sine squared lemma, but its positivity does not. The lemma, applied to the compensated measure ν_1, primes minus the uniform measure that represents the pole of ζ, gives the arithmetic form of the density, and the existence of the sine squared integral of ν_1 is the prime number theorem. Since ν_1 is signed, its sine squared image is not termwise positive. Positivity comes from the zero side, Proposition 9.1.1, through $\beta<1$, and written

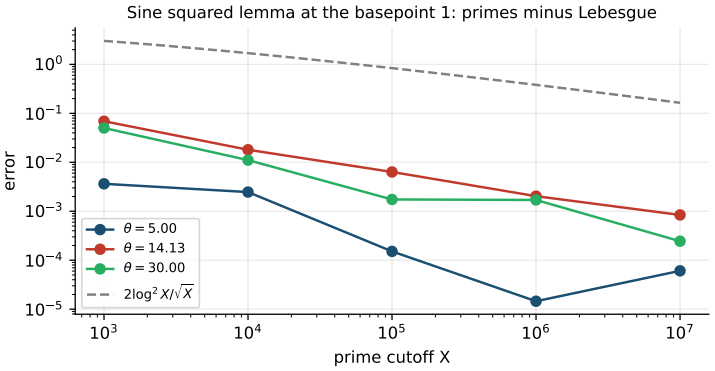


Figure 9.2: Absolute error of the compensated sine squared sum (9.1.4) against the prime cutoff X at $\theta = 5$, 14.13 and 30, with the reference rate $2\log^2 X/\sqrt{X}$ (dashed).

arithmetically it is the inequality

$$-\operatorname{Re}\frac{\zeta'}{\zeta}(1+it) < \frac{1}{1+t^2} - \frac{1}{2}\log\pi + \frac{1}{2}\operatorname{Re}\psi\Big(\frac{1+it}{2}\Big),$$

whose right side grows like $\frac{1}{2}\log(t/2\pi)$: the inequality of Hadamard and de la Vallée Poussin that underlies the proof of the prime number theorem [255]. At the basepoint one the sine squared lemma and the prime number theorem are two readings of the same fact.

9.1.4 The Lévy form of $\xi(1+s)$

Section 9.1.3 started from the arithmetic formula for v_1. The sine squared lemma is designed to act one level earlier, on a Lévy form of $\log\xi$ itself, which it converts into a Thorin form as described after (1.2.9), and at the basepoint one that Lévy form exists and can be written down.

Proposition 9.1.3 (Lévy form). *For* $\operatorname{Re} s > 0$,

$$\log\frac{\xi(1+s)}{\xi(1)} = b_1 s + \int_0^\infty \big(e^{-sx} - 1 + sx\big)\,\Pi_1(dx), \tag{9.1.5}$$

with drift $b_1 = -\frac{1}{2}\log\pi + \frac{1}{2}\psi(\frac{3}{2}) + \gamma_E = \sum_\rho \rho^{-1} = 0.0230957089661$ *and the signed Lévy measure*

$$\Pi_1(dx) = \Big(\frac{e^{-3x}}{1-e^{-2x}} - 1\Big)\frac{dx}{x} + \sum_{n\ge 2}\frac{\Lambda(n)}{n\log n}\,\delta_{\log n}(dx),$$

the integral against the last two pieces being taken as $\lim_{X\to\infty}\int_0^{\log X}$.

Proof. Write $\xi(z) = (z-1)\pi^{-z/2}\Gamma(1+\frac{z}{2})\zeta(z)$ and put $z = 1+s$, so that $\xi(1+s) = \pi^{-(1+s)/2}\Gamma(\frac{3}{2}+\frac{s}{2})\,s\zeta(1+s)$ and $\xi(1) = \frac{1}{2}$. Three classical formulas give the three pieces. The factor $\pi^{-s/2}$ contributes $-\frac{s}{2}\log\pi$. Binet's formula [177, §5.9], $\log\Gamma(a+w) - \log\Gamma(a) = w\psi(a) + \int_0^\infty (e^{-wt} - 1 + wt)\frac{e^{-at}}{t(1-e^{-t})}dt$, with $a = \frac{3}{2}$, $w = \frac{s}{2}$ and $t = 2x$, contributes $\frac{s}{2}\psi(\frac{3}{2})$ and the density $e^{-3x}/(x(1-e^{-2x}))$. For the last factor, the Euler product gives $\log\zeta(1+s) = \sum_n \frac{\Lambda(n)}{n\log n}e^{-s\log n}$ for $\operatorname{Re} s > 0$, and Frullani's integral gives $\log s = \int_0^\infty (e^{-x} - e^{-sx})\frac{dx}{x}$. Their sum is $\log(s\zeta(1+s))$, which is analytic at $s = 0$ with value 0 and derivative γ_E, and subtracting the linear term puts it in the compensated form $\gamma_E s + \int (e^{-sx} - 1 + sx)\,x^{-1}\nu_1(dx)$, where $x^{-1}\nu_1(dx) = \sum \frac{\Lambda(n)}{n\log n}\delta_{\log n} - \frac{dx}{x}$. The compensated integral converges as $X \to \infty$ because the prime atoms and the measure dx/x cancel at first order, by the prime number theorem in the form $\sum_{n\le X}\Lambda(n)/n = \log X - \gamma_E + o(1)$, and at zeroth order, by the logarithm of Mertens' product theorem [90, Theorem 429], $\sum_{p^k\le X} 1/(kp^k) = \log\log X + \gamma_E + o(1)$, which does not need the prime number theorem (the logarithm of the product also contains the terms with $p \le X < p^k$, whose sum is $O(X^{-1/2})$). The drift is $\frac{d}{ds}\log\xi(1+s)|_{s=0} = \frac{\xi'}{\xi}(1)$. □

Table 9.4, computed by `levy.py`, checks (9.1.5) at $s = \frac{1}{2}$, 1 and 2 with primes up to 10^7. The prime piece and its compensator each grow like $s\log X$, reaching 28 at $s = 2$, and their difference approaches the exact value at the rate the prime number theorem allows.

s	X	$b_1 s$	gamma piece	primes	compensator	total	exact
0.5	10^5	0.01155	0.027227	3.40718	3.42845	0.017508	0.017311
0.5	10^7	0.01155	0.027227	5.37344	5.39500	0.017217	0.017311
1	10^5	0.02310	0.102283	8.41323	8.49224	0.046368	0.046118
1	10^7	0.02310	0.102283	12.68129	12.76094	0.045729	0.046118
2	10^5	0.04619	0.368975	19.03582	19.31202	0.138973	0.137917
2	10^7	0.04619	0.368975	27.90836	28.18589	0.137639	0.137917

Table 9.4: The Lévy form (9.1.5) against $\log(\xi(1+s)/\xi(1))$: the drift term, the gamma piece, the prime atoms up to X, their compensator, the total and the exact value.

The variable $V = V_1$ of Section 9.1.1, the Esscher tilt by $e^{y/2}$ of the Pólya density, has $\mathbb{E}e^{sV} = \xi(1+s)/\xi(1)$, so (9.1.5) has the form of a Lévy Khintchine formula for the moment generating function of V, with Π_1, reflected to the negative half line by $x \mapsto -x$, in the place of the Lévy measure. That measure is signed: the absolutely continuous part $\big(\frac{e^{-3x}}{1-e^{-2x}} - 1\big)\frac{1}{x}$ is positive for $x < 0.28120$, where the gamma factor dominates, and negative beyond, where the pole term $-dx/x$ dominates, and the only positive mass beyond is the prime atoms. So the uniqueness theorem for Lévy triplets does not apply to Π_1 directly. Nor is Π_1

the signed Lévy measure of a quasi infinitely divisible law in the sense of Lindner, Pan and Sato [147], a law whose characteristic function is the quotient of two infinitely divisible ones: that would require the positive and negative parts of Π_1 to be Lévy measures, of finite mass away from the origin, while on $(1,\infty)$ the positive part is the prime atoms and the negative part exceeds $0.94\,dx/x$, both of infinite mass, and they cancel only conditionally, which is why (9.1.5) holds only as a compensated limit. Proposition 11.1.3(3) proves that no other signed measure represents the characteristic function of V, so V is not quasi infinitely divisible, and hence, since an infinitely divisible law is quasi infinitely divisible, not infinitely divisible. For every $\alpha>1$, by contrast, V_α is quasi infinitely divisible (Proposition 11.1.3(2)), so the basepoint one is exactly where the property is lost. The left panel of Figure 9.3 shows the canonical measure $x\Pi_1(dx)$.

In the variable $\theta=\sqrt{z}$ the sine squared lemma (1.2.9) reads

$$e^{-sx}+sx-1=\frac{2x}{\pi}\int_0^\infty \log\Big(1+\frac{s^2}{\theta^2}\Big)\sin^2\Big(\frac{x\theta}{2}\Big)d\theta,\qquad \operatorname{Re}s>0,\ x>0,$$

and the drift converts through $\int_0^\infty \log(1+s^2/\theta^2)\,d\theta=\pi s$. Applying both to (9.1.5) and substituting $s=\sqrt{u}$ gives $\log\varphi_1(u)=-\int_0^\infty \log(1+u/\theta^2)\rho_{1/2}(\theta)\,d\theta$ with

$$v_1(\theta)=b_1+\int_0^\infty 2\sin^2\Big(\frac{x\theta}{2}\Big)\,x\Pi_1(dx)=b_1+A(\theta)+Q(\theta),\tag{9.1.6}$$

where the drift b_1 is the stable part, $A(\theta)=\frac12\big[\operatorname{Re}\psi(\frac{3+i\theta}{2})-\psi(\frac32)\big]$ is the gamma piece and $Q(\theta)=\lim_{X\to\infty}\int_0^{\log X}2\sin^2(\frac{x\theta}{2})\,\nu_1(dx)$ is the compensated prime piece. This is Proposition 9.1.2 again, now obtained by converting the Lévy form rather than by manipulating $\operatorname{Re}\xi'/\xi$; the two constant terms $-\frac12\log\pi+\gamma_E$ and $\frac12\psi(\frac32)$ combine into b_1, and $\frac12\operatorname{Re}\psi(\frac{3+i\theta}{2})=\frac12\operatorname{Re}\psi(\frac{1+i\theta}{2})+\frac{1}{1+\theta^2}$ accounts for the term $1/(1+\theta^2)$. With `levy.py` we checked that the sine squared integral of the gamma density equals the digamma expression $A(\theta)$ to ten digits at $\theta=5$, 14.1347 and 30 (0.4749252962, 0.9640604263, 1.3367973267).

The conversion has three readings, each visible in the right panel of Figure 9.3. First, the drift $b_1=\sum_\rho\rho^{-1}$ becomes the constant Thorin density b_1/π, which is the Thorin density of the positive stable one half factor $e^{-b_1\sqrt{u}}$. So $\varphi_1(u)=e^{-b_1\sqrt{u}}\exp\{-\int_0^\infty\log(1+u/\theta^2)(A+Q)(\theta)\,d\theta/\pi\}$, and the second factor is the transform of a GGC with Thorin density $(A+Q)/\pi$, because $v_1=b_1+A+Q$ attains its minimum b_1 only at the origin (Proposition 9.1.4). The stable factor is the source of the tail $b_1(\pi x)^{-1/2}$ of (9.1.3) and of the value $\rho_{1/2}(0)=b_1/\pi$, and it plays the role that the stable tilt plays for the floor of Section 6.2.4. At $u=(2k-1)^2$ the factorisation splits each coefficient $1/\zeta(2k)$ of Riesz's function (10.1.5), up to an explicit factor, into a stable value and a GGC value (Section 10.2.3). Second, the gamma factor contributes $A(\theta)\ge0$, smooth and growing like $\frac12\log(\theta/2)$: the average density of zeros. Third, the primes and the pole contribute $Q(\theta)$, which

oscillates with both signs, between -1.47 and 1.04 on $[0, 100]$ (between -1.29 and 1.04 on the window $[0, 60]$ of the figure), and carries the peaks at the ordinates. Neither part converges alone; their compensated difference is the prime number theorem.

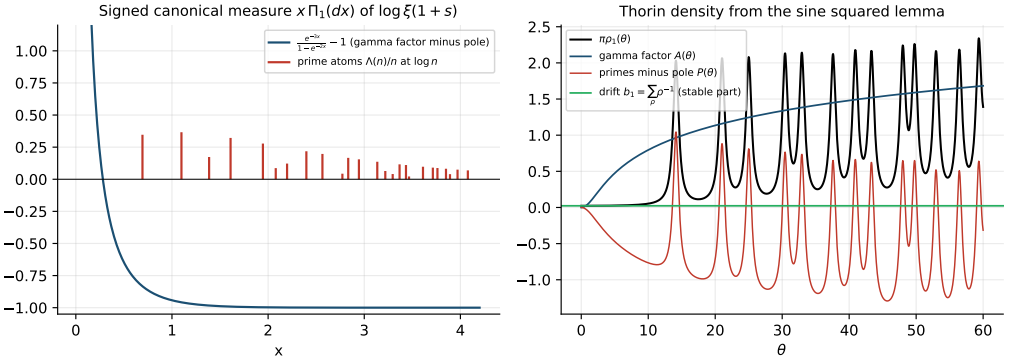


Figure 9.3: Left: the signed canonical measure $x\Pi_1(dx)$ of $\log \xi(1+s)$, made of the gamma density minus the pole, $\big(\frac{e^{-3x}}{1-e^{-2x}} - 1\big)dx$, which is negative beyond $x = 0.28120$, and the prime atoms $\Lambda(n)/n$ at $\log n$ for $n \le 60$. Right: the bracket $v_1(\theta) = \pi\rho_{1/2}(\theta)$ on $[0, 60]$ assembled by the sine squared lemma from the drift b_1, the gamma piece $A(\theta)$ and the compensated prime piece $Q(\theta)$; the plot writes ρ_1 for $\rho_{1/2}$ and $P(\theta)$ for $Q(\theta)$. Drawn by `figs3.py`.

The first reading needs $A + Q \ge 0$, that is, $v_1 \ge b_1$, at every height, and this holds with no hypothesis on the zeros. Near the origin v_1 is convex, at moderate heights a zero on the critical line is always close, and higher up every window of fixed length contains enough zeros for their Cauchy kernels to add up to more than b_1.

PROPOSITION 9.1.4. *With no hypothesis on the zeros, $v_1(\theta) > v_1(0) = b_1$ for every real $\theta \neq 0$; the proof uses a computation of the zeros below height* 2000 *in ball arithmetic and, above that height, explicit bounds for the number of zeros up to a given height [92, 10]. Consequently $T_1 \overset{d}{=} S + T_1^\circ$, where S and T_1° are independent, S is positive stable of index one half with $\mathbb{E}e^{-uS} = e^{-b_1\sqrt{u}}$, and T_1° is a GGC with no drift whose Thorin density in the chart $\theta = \sqrt{z}$ is $(v_1 - b_1)/\pi = (A + Q)/\pi$.*

Proof. By (9.1.2), $v_1(\theta) = \pi \sum_\rho P_{1-\beta}(\theta - \gamma)$ is a sum of positive terms, with $\pi P_a(x) = a/(a^2 + x^2)$, and v_1 is even, because ξ is real on the real axis. We treat three ranges of $\theta \ge 0$.

On $[0, 13.5]$ we use convexity. Every zero has $|\gamma| \ge \gamma_1 = 14.134725$: the certified zeros of the next step, computed in order from the first, show that none has $0 < |\gamma| < \gamma_1$, and none is real, because for $0 < \sigma < 1$ the alternating series

$(1-2^{1-\sigma})\zeta(\sigma)=\sum_{n\ge1}(-1)^{n-1}n^{-\sigma}$ is positive while $1-2^{1-\sigma}<0$, so that $\zeta(\sigma)<0$. Since $\pi P_a''(x)=2a(3x^2-a^2)/(a^2+x^2)^3$, the kernel P_a is strictly convex where $|x|>a/\sqrt{3}$, and for $|\theta|\le 13.5$ every zero has $|\theta-\gamma|\ge\gamma_1-13.5>0.63>1/\sqrt{3}>(1-\beta)/\sqrt{3}$. So v_1 is strictly convex on $[-13.5,13.5]$, and, being even, it has there its unique minimum at $\theta=0$ and increases on $[0,13.5]$. By (1.2.2), differentiated twice,

$$v_1''(0)=-\Big(\frac{\xi'}{\xi}\Big)''(1)=\sum_\rho\frac{2(1-\beta)\big(3\gamma^2-(1-\beta)^2\big)}{\big((1-\beta)^2+\gamma^2\big)^3}>0.$$

On $[13.5,2000]$ we use the nearest zero. The script `v1min.py`, with output `v1min.json` (in the folders `riesz_program/code` and `riesz_program/data` of the archive), computes the first 1800 zeros in ball arithmetic with Arb [110], which isolates them in order by Gram's law and Rosser's rule (Section 2.2.1), each applied only in the range where it is known to hold [43, Section 3], so that none is missed. They reach height 2304.3, all are simple and on the critical line, and consecutive ordinates differ by at most $\gamma_2-\gamma_1<6.8874$. Every $\theta\in[13.5,2000]$ therefore lies within 3.4437 of an ordinate, half the largest gap, and a zero on the critical line at distance x contributes $\pi P_{1/2}(x)=\frac12/(\frac14+x^2)$, which exceeds b_1 for $x<4.6259$. All other terms being positive,

$$v_1(\theta)\ge\frac{1/2}{1/4+3.4437^2}>0.041,\qquad 13.5\le\theta\le2000.$$

For $\theta\ge2000$ we count zeros. Write $N(T)=\vartheta(T)/\pi+1+S(T)$ (Section 2.2.1). Hasanalizade, Shen and Wong [92, (1.7) and Corollary 1.4] give $|S(T)|\le0.2167$ for $0\le T\le X_0=30\,610\,046\,000$, from Platt's database of the zeros, and $|S(T)|\le0.1095\log T+0.2042\log\log T+3.0305$ for every $T\ge e$, and the smooth part $\vartheta(T)/\pi+1$ of $N(T)$ differs from $\frac{T}{2\pi}\log\frac{T}{2\pi e}+\frac78$ by at most $\frac{1}{50T}$ for $T\ge\frac57$ [92, (2.7)]. So

$$\begin{aligned}\Big|N(T)-\frac{T}{2\pi}\log\frac{T}{2\pi e}-\frac78\Big|&\le E(T)\\&=\begin{cases}2.5167+\frac{1}{50T}, & e\le T\le X_0,\\ 0.1095\log T+0.2042\log\log T+3.0305+\frac{1}{50T}, & T>X_0,\end{cases}\end{aligned}$$

and $E(T)<0.3\log T$ for $T>X_0$, since this holds at X_0 and $0.3-0.1095-0.2042/\log X_0>0$. The bound $0.112\log T+0.278\log\log T+2.510+0.2/T$ of Trudgian [259], which is often quoted, has an incomplete proof [19, 92]. The error of Stirling's formula in the smooth part is $(\frac{1}{48T}+O(T^{-3}))/\pi$ [72, Section 6.5], and since the remainder of Stirling's series for $\log\Gamma(z)$ after the term $1/(12z)$ is at most $1/(90|z|^3)$ for $\operatorname{Re}z>0$ [177, §5.11(ii)], it is less than $0.01/T$ for $T\ge14$. The counts below use $\frac{1}{50T}$, as `v1min.py` does when it evaluates them; the sharper $0.01/T$, which the script also checks, would change them by less than 2×10^{-5}. Put $H=15$. The window $(\theta-H,\theta+H]$ holds $N(\theta+H)-N(\theta-H)$ zeros, counted

with multiplicity, and since the main term has derivative $\frac{1}{2\pi}\log\frac{T}{2\pi}$, their number is at least

$$n(\theta) = \frac{H}{\pi}\log\frac{\theta-H}{2\pi} - E(\theta-H) - E(\theta+H).$$

While $\theta+H \le X_0$, n increases, because E decreases there, and $n(2000) > 22.44$; for $\theta+H > X_0$, both values of E are below $0.3\log(\theta+H)$, and $n(\theta) > \frac{H}{\pi}\log\frac{\theta-H}{2\pi} - 0.6\log(\theta+H) > 90.$ So for every $\theta \ge 2000$ the window holds at least 23 zeros. Since $\xi(1-\bar{s}) = \overline{\xi(s)}$, the map $\rho \mapsto 1-\bar{\rho}$ sends each zero to a zero of the same multiplicity; it keeps the ordinate and exchanges β and $1-\beta$, so it permutes the zeros of the window, and the sum of $1-\beta$ over them is half their number. Each of them has $0 < 1-\beta < 1$ and $|\theta-\gamma| \le H$, so its term $(1-\beta)/((1-\beta)^2+(\theta-\gamma)^2)$ in $v_1(\theta)$ exceeds $(1-\beta)/(1+H^2)$, and all other terms are positive. Hence

$$v_1(\theta) > \frac{23}{2(1+H^2)} = \frac{23}{452} > 0.05, \qquad \theta \ge 2000,$$

wherever the zeros lie. Backlund's classical constants 0.137, 0.443 and 4.35, valid for $T \ge 200$ [10], give in the same way the increasing bound $E_B(T) = 0.137\log T + 0.443\log\log T + 4.35 + 0.2/T$ in place of E and at least $\frac{H}{\pi}\log\frac{\theta-H}{2\pi} - 2E_B(\theta+H)$ zeros in the window, a number that increases with θ and exceeds 14.89 at $\theta = 2000$; the window then holds at least 15 zeros, and the bound becomes $15/452 > 0.033$.

It remains to split φ_1. Since $\frac{1}{\pi}\int_0^\infty \log(1+u/\theta^2)\,d\theta = \sqrt{u}$, Proposition 9.1.1 gives

$$\varphi_1(u) = e^{-b_1\sqrt{u}}\exp\Big\{-\frac{1}{\pi}\int_0^\infty \log\Big(1+\frac{u}{\theta^2}\Big)\big(v_1(\theta)-b_1\big)\,d\theta\Big\}, \qquad u \ge 0,$$

where the integral is finite because $0 \le v_1 - b_1 \le v_1$ and $\int_0^\infty \log(1+u/\theta^2)\,v_1(\theta)\,d\theta = -\pi\log\varphi_1(u)$ is finite by Theorem 1.2.2 at $\alpha = 1$. The first factor is the Laplace transform of the positive stable law of index one half with density $\frac{b_1}{2\sqrt{\pi}}t^{-3/2}e^{-b_1^2/(4t)}$. The image U_1° of $\pi^{-1}(v_1-b_1)\,d\theta$ under $\theta \mapsto \theta^2 = z$ is a positive measure with $\int \log(1+u/z)\,U_1^\circ(dz) < \infty$, which is equivalent to the integrability condition in (1.3.1), so the second factor is the Laplace transform of a GGC T_1° with no drift. A product of Laplace transforms is the transform of the sum of independent variables, so $T_1 \stackrel{d}{=} S + T_1^\circ$, and $v_1 - b_1 = A + Q$ by (9.1.6). □

The factorisation separates the tail of T_1 from the rest of its law. The stable summand carries the infinite mean and the whole tail: $P(S > t) \sim b_1/\sqrt{\pi t}$ is the tail (9.1.3) of T_1, while T_1° has finite mean and infinite variance. The Laplace exponent of T_1° is $\log(\xi(1+\sqrt{u})/\xi(1)) - b_1\sqrt{u} = \frac{1}{2}(\log\xi)''(1)\,u + O(u^{3/2})$ as $u \downarrow 0$, so that

$$\mathbb{E}T_1^\circ = \frac{1}{2}(\log\xi)''(1) = -\frac{1}{2}\sum_\rho \rho^{-2} = 0.0230772,$$

and its Thorin density $(v_1 - b_1)/\pi$ vanishes at the origin to exactly second order, since v_1 is even and $v_1''(0) > 0$; for instance $v_1(1) - b_1 = 1.12 \times 10^{-4}$ and $v_1(5) - b_1 = 3.31 \times 10^{-3}$. Near $z = 0$ the Thorin measure U_1° therefore has a density of exact order $\sqrt{z}$, and the variance $\int z^{-2}\, U_1^\circ(dz)$ is infinite. On $[13.5, 2000]$ the smallest value of v_1 on the grid of step 0.02 of `v1min.py` is 0.11326, at $\theta = 17.5$ in the valley between the first two ordinates, about five times b_1. An independent check, `v1_grid_mp.py` in the folder `repro`, evaluates the arithmetic form of v_1 with `mpmath` [159] to 25 digits on the grid of step 0.005 in $(0, 2100]$. It finds $v_1 > b_1$ at all 420000 points, the smallest excess being 2.8×10^{-9}, at $\theta = 0.005$, which is $0.005^2(v_1(1) - b_1)$ to two digits, as the vanishing to second order predicts; the smallest value on $[13.5, 2000]$ is 0.1132554, at $\theta = 17.495$. The first 1850 zeros, computed with `mpmath` by `zeros_mp.py`, agree with those of Arb to double precision.

The same construction applies at every basepoint above one. For every $\alpha \geq 1$ it gives the Lévy form of $\log(\xi(\alpha + s)/\xi(\alpha))$ with drift $b_\alpha = \frac{\xi'}{\xi}(\alpha)$ and

$$x\Pi_\alpha(dx) = \Big(\frac{e^{-(\alpha+2)x}}{1 - e^{-2x}} - e^{-(\alpha-1)x}\Big)dx + \sum_n \frac{\Lambda(n)}{n^\alpha}\, \delta_{\log n}(dx),$$

$$v_\alpha(\theta) = b_\alpha + \int_0^\infty 2\sin^2\Big(\frac{x\theta}{2}\Big) x\Pi_\alpha(dx),$$

which for $\alpha > 1$ is (3.1.1). For $\alpha > 1$ every piece converges on its own, and the pole contribution is the finite negative term $-\theta^2/((\alpha - 1)((\alpha - 1)^2 + \theta^2))$. Expanding $1/(e^{2x} - 1) = \sum_{k \geq 1} e^{-2kx}$ in the archimedean piece of (3.1.1), and writing $q_c(\theta) = \int_0^\infty (1 - \cos\theta x) e^{-cx}\, dx = \theta^2/(c(c^2 + \theta^2))$, gives

$$v_\alpha(\theta) - b_\alpha = \sum_{k \geq 1} q_{\alpha+2k}(\theta) - q_{\alpha-1}(\theta) + \sum_{n \geq 2} \frac{\Lambda(n)}{n^\alpha}\big(1 - \cos(\theta \log n)\big), \qquad \alpha > 1. \tag{9.1.7}$$

Only the pole term is negative. Near the origin $q_c(\theta) \sim \theta^2/c^3$, and the pole term outweighs the gamma terms when $\alpha < 7.789$, where $\frac{1}{8}\zeta(3, \frac{\alpha}{2} + 1) = (\alpha - 1)^{-3}$; below that basepoint the arithmetic form needs the primes to keep $v_\alpha - b_\alpha$ positive near the origin. On the zero side positivity there is again convexity, and the factorisation of Proposition 9.1.4 holds at every basepoint above one.

Proposition 9.1.5. *With no hypothesis on the zeros, $v_\alpha(\theta) > v_\alpha(0) = b_\alpha$ for every $\alpha \geq 1$ and every real $\theta \neq 0$; the proof uses the zeros below height* 2000 *of Proposition 9.1.4, the explicit bounds of [92] for the number of zeros above that height, and a computation in ball arithmetic. Consequently, for every $\alpha \geq 1$, $T_\alpha \stackrel{d}{=} S_\alpha + T_\alpha^\circ$, where S_α and T_α° are independent, S_α is positive stable of index one half with $\mathbb{E}e^{-uS_\alpha} = e^{-b_\alpha\sqrt{u}}$, which is the law of the first passage time of Brownian motion to the level $b_\alpha/\sqrt{2}$, and T_α° is a GGC with no drift, with Thorin density $(v_\alpha - b_\alpha)/\pi$ in the chart $\theta = \sqrt{z}$ and mean $\mathbb{E}T_\alpha^\circ = \frac{1}{2}(\log\xi)''(\alpha)$.*

Proof. At $\alpha = 1$ this is Proposition 9.1.4, so let $\alpha > 1$. Since v_α is even, it suffices to take $\theta > 0$. A zero $\rho = \beta + i\gamma$ contributes $a/(a^2 + (\theta - \gamma)^2)$ to $v_\alpha(\theta)$, with $a = \alpha - \beta \in (\alpha - 1, \alpha)$, and we put $\theta_0(\alpha) = \gamma_1 - \alpha/\sqrt{3}$. Four steps cover all $\alpha > 1$.

Near the origin, for $\alpha \le 9$, we use convexity. If $0 < \theta \le \theta_0(\alpha)$, every zero has $|\theta - \gamma| \ge \gamma_1 - \theta \ge \alpha/\sqrt{3} > a/\sqrt{3}$, so, as in the proof of Proposition 9.1.4, v_α is strictly convex on $[-\theta_0(\alpha), \theta_0(\alpha)]$, and $v_\alpha(\theta) > v_\alpha(0)$ there. Since $\theta_0(9) = 8.94 > 0$, it remains to treat $\theta > \theta_0(\alpha)$ for $\alpha \le 9$, and every $\theta > 0$ for $\alpha > 9$.

For $1 < \alpha \le 3$ we use the zeros. Here $\theta_0(\alpha) \ge \theta_0(3) = 12.40$ and $\gamma_1 - 12.40 < 3.4437$, so by the computation in the proof of Proposition 9.1.4 every $\theta \in [\theta_0(\alpha), 2000]$ lies within 3.4437 of the ordinate of a zero on the critical line, whose term exceeds $r(\alpha) = (\alpha - \frac12)/((\alpha - \frac12)^2 + 3.4437^2)$. For $\theta \ge 2000$ the window $(\theta - 15, \theta + 15]$ holds at least 20 zeros, the sum of a over them is $\alpha - \frac12$ times their number, and each term exceeds $a/(\alpha^2 + 225)$, so that $v_\alpha(\theta) > w(\alpha) = 20(\alpha - \frac12)/(\alpha^2 + 225)$. The functions r and w increase on $[1, 3]$, and so does b_α, whose derivative $\sum_\rho (\gamma^2 - a^2)/(a^2 + \gamma^2)^2$ is positive for $\alpha < \gamma_1$. It therefore suffices that $r(\alpha_k) > b_{\alpha_{k+1}}$ and $w(\alpha_k) > b_{\alpha_{k+1}}$ at the points $\alpha_k = 1 + k/256$, and in ball arithmetic the smallest margins are 0.0180 and 0.0209.

For $3 \le \alpha \le 7$ we use the primes. Adding and subtracting $q_\alpha(\theta)$ in (9.1.7), dropping the prime terms and using $q_c(\theta) < 1/c$ gives

$$\begin{aligned} v_\alpha(\theta) - b_\alpha > L(\alpha, \theta) &= \sum_{k \ge 0} q_{\alpha+2k}(\theta) - \frac{1}{\alpha} - \frac{1}{\alpha - 1} \\ &= \frac12 \Big[\operatorname{Re} \psi\Big(\frac{\alpha + i\theta}{2}\Big) - \psi\Big(\frac{\alpha}{2}\Big) \Big] - \frac{1}{\alpha} - \frac{1}{\alpha - 1}, \end{aligned}$$

which increases in θ. Since $|\psi'(z)| \le \psi'(\operatorname{Re} z)$ and $\psi'(x) \le 1/x + 1/x^2$ for $x > 0$, $|\partial L/\partial \alpha| \le M(\alpha) = 1/\alpha + 3/\alpha^2 + 1/(\alpha - 1)^2$, which decreases in α. It therefore suffices that $L(\alpha_k, \theta_0(\alpha_{k+1})) > M(\alpha_k)/256$ at the points $\alpha_k = 3 + k/256$ of $[3, 7]$; the smallest margin is 0.0228, at $\alpha = 7$, and the same check succeeds up to $\alpha = 7.707$.

For $\alpha \ge 7$ the gamma terms suffice. Since $q_c(\theta)$ decreases in c, $\sum_{k \ge 1} q_{\alpha+2k}(\theta) \ge \frac12 \int_{\alpha+2}^\infty q_c(\theta)\, dc = \frac14 \log(1 + \theta^2/(\alpha + 2)^2)$, so with $A = \alpha + 2$, $B = \alpha - 1$ and $s = \theta^2$,

$$v_\alpha(\theta) - b_\alpha \ge \Phi_\alpha(s) = \frac14 \log\Big(1 + \frac{s}{A^2}\Big) - \frac{s}{B(B^2 + s)}.$$

Here $\Phi_\alpha(0) = 0$, and $\Phi'_\alpha(s) = 1/(4(A^2 + s)) - B/(B^2 + s)^2$ is nonnegative exactly when $(B^2 + s)^2 \ge 4B(A^2 + s)$, the two sides differing by a function that increases in s when $B \ge 2$. For $\alpha \ge 9$ they are in this order already at $s = 0$, because $B^3 - 4A^2 = (\alpha - 1)^3 - 4(\alpha + 2)^2$ is 28 at $\alpha = 9$ and increases beyond, so $\Phi_\alpha(s) > 0$ for every $s > 0$, and no zero enters. For $7 \le \alpha \le 9$ and $\theta > \theta_0(\alpha) \ge \theta_0(9)$, we have $s > s_9 = \theta_0(9)^2 = 79.90$, $6 \le B \le 8$ and $A \le 11$, so $(B^2 + s_9)^2 \ge (36 + s_9)^2 > 32(121 + s_9) \ge 4B(A^2 + s_9)$; hence Φ_α increases on $[s_9, \infty)$, where $\Phi_\alpha(s_9) \ge \frac14 \log(1 + s_9/121) - s_9/(6(36 + s_9)) > 0.0118$.

The script `valpha_min.py` in the folder `repro` carries out the checks of the last three steps in ball arithmetic with Arb [110], computing b_α from $\zeta(\alpha)$ and $\zeta'(\alpha)$. The factorisation follows from the inequality as in the proof of Proposition 9.1.4, Theorem 1.2.2 giving $\int_0^\infty \log(1+u/\theta^2)\,v_\alpha(\theta)\,d\theta<\infty$, and the mean is the coefficient of u in $\log(\xi(\alpha+\sqrt{u})/\xi(\alpha))-b_\alpha\sqrt{u}=\frac12(\log\xi)''(\alpha)\,u+O(u^{3/2})$. $\square$

As at the basepoint one, $v_\alpha-b_\alpha$ starts as $c(\alpha)\theta^2$, with $c(\alpha)=-\frac12(\xi'/\xi)''(\alpha)>0$, so at every basepoint above one the Thorin density of T_α° vanishes to second order at the origin, T_α° has finite mean and infinite variance, and S_α carries the whole tail $P(T_\alpha>t)\sim b_\alpha/\sqrt{\pi t}$. The two summands move very differently with the basepoint. From $\alpha=1$ to $\alpha=9$ the drift b_α grows from 0.0230957 to 0.3567685, while the mean of the remainder falls only slowly, $\mathbb{E}T_\alpha^\circ$ being 0.0230772, 0.0230425, 0.0229942, 0.0228577, 0.0224351, 0.0211082 and 0.0176077 at $\alpha=1$, 1.25, 1.5, 2, 3, 5 and 9; at the last six basepoints $c(\alpha)$ is 1.66, 2.20, 3.25, 5.15, 7.85 and 8.85×10^{-4} (computed by `valpha_min.py`).

For $\alpha<1$ the Euler product no longer converges near $\operatorname{Re}s=\alpha$, and the prime atoms $\Lambda(n)n^{-\alpha}$ can be compensated only if the primes are well enough distributed. So the route from the Lévy form through the sine squared lemma computes the Thorin measure unconditionally on $\alpha\ge1$, the region where it is positive unconditionally; Section 9.1.5 shows exactly how far below the basepoint one it extends.

9.1.5 Below the basepoint one

The construction of Section 9.1.4 can be written down for any basepoint $\alpha=1-\varepsilon$ with $0<\varepsilon<\frac12$, and the question is whether it converges. Formally nothing changes except the pole term, which now grows: with $b_\alpha=\frac{\xi'}{\xi}(\alpha)$,

$$x\Pi_\alpha(dx)=\frac{e^{-(\alpha+2)x}}{1-e^{-2x}}\,dx\;+\;\Big(\sum_n\frac{\Lambda(n)}{n^\alpha}\,\delta_{\log n}(dx)-e^{\varepsilon x}\,dx\Big),$$

and the converted Thorin density would be π^{-1} times

$$b_\alpha+A_\alpha(\theta)+\lim_{X\to\infty}\Big[\sum_{n\le X}\frac{\Lambda(n)}{n^\alpha}\,2\sin^2\Big(\frac{\theta\log n}{2}\Big)-\int_0^{\log X}2\sin^2\Big(\frac{\theta x}{2}\Big)e^{\varepsilon x}dx\Big],\quad(9.1.8)$$

with $A_\alpha(\theta)=\frac12\big[\operatorname{Re}\psi(\frac{\alpha+2+i\theta}{2})-\psi(\frac{\alpha+2}{2})\big]$. The prime piece and the pole piece now each grow like $X^\varepsilon/\varepsilon$, and whether their difference converges is a question about how well $\psi(y)$ is approximated by y.

Proposition 9.1.6. *Let $\Theta=\sup\{\operatorname{Re}\rho\}$ over the nontrivial zeros and let $\frac12<\alpha<1$.*

1. *If $\Theta < \alpha$, the Lévy form* $\log \frac{\xi(\alpha+s)}{\xi(\alpha)} = b_\alpha s + \int_0^\infty (e^{-sx} - 1 + sx)\Pi_\alpha(dx)$ *holds for* $\operatorname{Re} s > 0$*, the compensated limit converging, and the sine squared lemma converts it into* (9.1.8)*, which then equals $v_\alpha(\theta)$ and is positive. The error of the compensated sum with prime cutoff X is $O(X^{\Theta-\alpha+\delta})$ for every $\delta > 0$.*

2. *Conversely, if the Lévy form holds for* $\operatorname{Re} s > 0$*, then $\xi(\alpha + s)$ has no zeros with* $\operatorname{Re} s > 0$*, so $\Theta \le \alpha$.*

Proof. If $\Theta < \alpha$, the explicit formula gives $\psi(y) - y = O(y^{\Theta+\delta})$ for every $\delta > 0$ [255]. Then the Stieltjes integral $\int_1^\infty y^{-w}\, d(\psi(y) - y)$, which equals $-\frac{\zeta'}{\zeta}(w) - \frac{1}{w-1}$ for $\operatorname{Re} w > 1$, converges after one integration by parts for $\operatorname{Re} w > \Theta$ and provides the analytic continuation there, with tail beyond X of order $X^{\Theta-\operatorname{Re} w+\delta}$. Taking $w = \alpha + s$ with $\operatorname{Re} s \ge 0$ and integrating twice in s from $s = 0$ gives the compensated Lévy form, the gamma and polynomial factors being handled exactly as in Proposition 9.1.3. For fixed θ the sine squared lemma may be applied under the limit, because the compensated measure integrated against $2\sin^2(\theta x/2)$ is a combination of the same convergent Stieltjes integrals at $w = \alpha$ and $w = \alpha \pm i\theta$. Positivity follows from the Poisson form of Proposition 1.2.3, as in Proposition 9.1.1, with widths $\alpha - \beta > 0$. Conversely, suppose the Lévy form holds for $\operatorname{Re} s > 0$. Write ν_α for the bracket in $x\Pi_\alpha$, so that the compensated part of the right side is the limit of $\int_0^{\log X} (e^{-sx} - 1 + sx)\, x^{-1}\nu_\alpha(dx)$; its part over $[0, 1]$ is an entire function of s. For real $s > 1 - \alpha$ the integral $\int_1^\infty e^{-sx}x^{-1}\nu_\alpha(dx)$ converges absolutely, so the convergence of the compensated limit at two such values of s gives that of $\int_1^{\log X} \nu_\alpha(dx)$ and of $\int_1^{\log X} x^{-1}\nu_\alpha(dx)$. The function $y \mapsto \int_1^y \nu_\alpha(dx)$ is therefore bounded on $[1, \infty)$, and one integration by parts against it shows that $\int_1^{\log X} e^{-sx}x^{-1}\nu_\alpha(dx)$ converges uniformly on compact subsets of $\operatorname{Re} s > 0$. So the compensated limit converges locally uniformly, the right side of the Lévy form is analytic in $\operatorname{Re} s > 0$, hence so is $\log \xi(\alpha+s)$, and $\xi(\alpha+s)$ cannot vanish there. □

So the extension to $\xi(1 - \varepsilon + s)$ exists exactly when ζ has no zeros in the half plane $\operatorname{Re} s > 1 - \varepsilon$, up to the boundary case $\Theta = 1 - \varepsilon$. At the basepoint one the convergence of the compensated sum was the prime number theorem; at the basepoint $1 - \varepsilon$ it is the estimate $\psi(y) - y = O(y^{1-\varepsilon+\delta})$, which is the zero free half plane $\operatorname{Re} s > 1 - \varepsilon$; and as $\varepsilon \uparrow \frac{1}{2}$ it is $\psi(y) - y = O(y^{1/2+\delta})$, which is RH. The Lévy form route therefore reaches exactly as far as the zero free region does. It is an arithmetic restatement of step 7 of the path in Table 12.1, not a way around it.

Table 9.5, computed by `below.py`, evaluates (9.1.8) with primes up to 10^7 at $\theta = \gamma_1$ and compares it with $v_\alpha(\gamma_1)$ computed directly. On the range of the computation the primes behave as RH predicts, $\psi(y) - y = O(y^{1/2}\log^2 y)$, so the errors should decay like $X^{1/2-\alpha}$, and they do (Figure 9.4, left): the decay is fast at $\alpha = 0.9$, slow at $\alpha = 0.6$, where the prime and pole pieces are about 1500 and their difference is still wrong by about 2, and absent at the centre. At $\alpha = \frac{1}{2}$ and $\theta = \gamma_1$ the compensated sum grows like $\log X$ (Figure 9.4, right), which is the

atom δ_{γ_1} of the RH measure forming as the cutoff increases; away from the zeros it oscillates without converging. The rate of convergence is thus itself a diagnostic: the exponent in $X^{\Theta-\alpha}$ measures the largest real part of a zero, although only up to the range of X that can be computed, so it can never certify Θ.

α	X	prime piece	pole piece	formula (9.1.8)	$v_\alpha(\gamma_1)$
1	10^7	17.093	16.047	2.03293	2.03377
0.9	10^5	23.250	21.754	2.50194	2.52704
0.9	10^7	41.281	39.765	2.52202	2.52704
0.75	10^5	70.611	67.856	3.79129	4.01691
0.75	10^7	219.869	216.972	3.93394	4.01691
0.6	10^5	257.280	251.510	6.84001	10.00677
0.6	10^7	1537.254	1530.446	7.87829	10.00677
0.5	10^5	653.402	642.998	11.497	$+\infty$ (a zero)
0.5	10^7	6114.332	6099.976	15.449	$+\infty$ (a zero)

Table 9.5: The compensated sine squared formula (9.1.8) below the basepoint one at $\theta = \gamma_1$, with primes up to X, against the directly computed $v_\alpha(\gamma_1) = \mathrm{Re}(\xi'/\xi)(\alpha + i\gamma_1)$; at $\alpha = \frac{1}{2}$ the point $\frac{1}{2} + i\gamma_1$ is a zero.

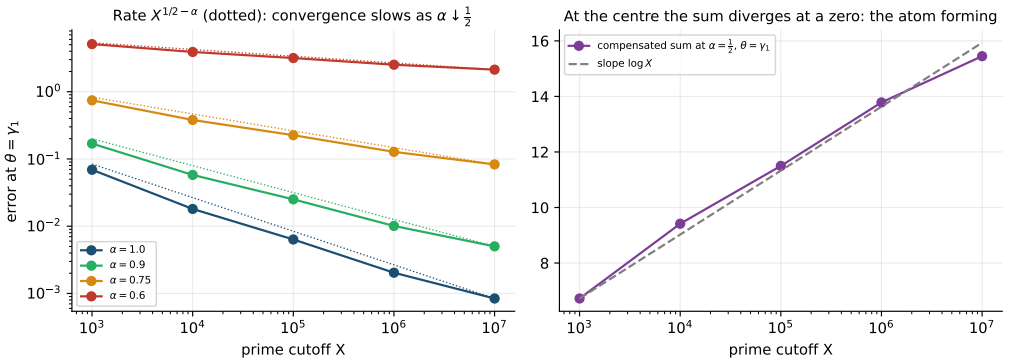


Figure 9.4: Left: absolute error of the compensated formula (9.1.8) at $\theta = \gamma_1$ against the prime cutoff X for $\alpha = 1$, 0.9, 0.75 and 0.6, with the rates $X^{1/2-\alpha}$ (dotted). Right: at the centre, $\alpha = \frac{1}{2}$, the compensated sum at the first ordinate grows like $\log X$ (dashed line), the atom of the RH measure forming. Drawn by `figs4.py`.

The slow convergence can be explained exactly, term by term over the zeros. Writing the tail of the compensated sum beyond X as a Stieltjes integral against $d(\psi(y) - y)$ and inserting the explicit formula $\psi(y) = y - \sum_\rho y^\rho/\rho - \log 2\pi -$

$\frac{1}{2}\log(1-y^{-2})$ [63, Chapter 17] gives, for $\Theta < \alpha$ and X not a prime power,

$$\begin{aligned}&(\text{formula (9.1.8) at cutoff } X) - v_\alpha(\theta)\\ &\quad= \sum_\rho \Big[-\frac{X^{\rho-\alpha}}{\rho-\alpha} + \frac{X^{\rho-\alpha+i\theta}}{2(\rho-\alpha+i\theta)} + \frac{X^{\rho-\alpha-i\theta}}{2(\rho-\alpha-i\theta)}\Big] + O(X^{-1-\alpha}), \quad (9.1.9)\end{aligned}$$

the sum taken over conjugate pairs. Every term has size $X^{\beta-\alpha}$, which makes the rate $X^{\Theta-\alpha}$ visible term by term, and at $\theta=\gamma$ the third term has denominator $\beta-\alpha$, so under RH it approaches $X^{1/2-\alpha}/(1-2\alpha)$ and at the centre it becomes the $\log X$ growth of Figure 9.4. Table 9.6, computed by `errformula.py`, compares the observed error with (9.1.9) truncated to the 1700 zeros below height 2198, and shows the error left after subtracting it. The explicit formula accounts for the error to four or five decimal places for $\alpha \geq 0.75$ and to three at $\alpha = 0.6$, and the hybrid of primes up to X plus the first 1700 zeros reduces the error of the primes alone by factors between about 700 and 11000: at $\alpha = 0.6$ the error falls from 2.13 to 7.7×10^{-4}.

α	X	observed error	predicted by (9.1.9)	error after correction
1	10^5	-6.352×10^{-3}	-6.349×10^{-3}	-2.9×10^{-6}
1	10^7	-8.419×10^{-4}	-8.431×10^{-4}	1.2×10^{-6}
0.9	10^7	-5.014×10^{-3}	-5.020×10^{-3}	6.1×10^{-6}
0.75	10^7	-8.297×10^{-2}	-8.304×10^{-2}	6.8×10^{-5}
0.6	10^5	-3.1668	-3.1665	-2.9×10^{-4}
0.6	10^7	-2.1285	-2.1292	7.7×10^{-4}

Table 9.6: The error of the compensated sum (9.1.8) at $\theta=\gamma_1$ with primes up to X, its prediction from the explicit formula (9.1.9) over the first 1700 zeros, and what remains after subtracting the prediction.

This also sharpens the logic of Proposition 9.1.6. A zero with $\beta > \alpha$ would contribute a term of size $X^{\beta-\alpha}$ that grows with the cutoff, so the compensated sum at the basepoint α would diverge, and the size and phase of that growth would locate the zero. The hybrid formula is a zero detector with the same structure as Turing's method (Section 2.2.1): primes up to X and zeros up to height T must balance to within the tail of both.

The role of the pole also changes below the basepoint one. If instead of compensating one continues the pole contribution analytically, its value at $\alpha = 1-\varepsilon$ is $\operatorname{Re}\frac{1}{-\varepsilon+i\theta} + \frac{1}{\varepsilon} = \frac{\theta^2}{\varepsilon(\varepsilon^2+\theta^2)}$, which is positive, since the pole now lies to the right of the basepoint. The single negative term of the decomposition (3.1.1) above the basepoint one has become positive, and the difficulty has moved entirely into the prime sum, which no longer converges on its own. Below the line $\operatorname{Re} s = 1$, therefore, the negativity that RH must control is carried by the primes, not by the pole.

9.1.6 The minimum at the origin below the basepoint one

Propositions 9.1.4 and 9.1.5 put the minimum of v_α at the origin for every $\alpha \geq 1$, with no hypothesis on the zeros. Below the basepoint one the margin at the origin vanishes linearly, $b_\alpha \approx (\alpha - \frac{1}{2})(\log \xi)''(\frac{1}{2})$, and Table 9.12 found the minimum at the origin on a window by computation. This subsection proves it on the range of heights where the zeros are known to lie on the critical line, up to $\frac{1+\sqrt{2}}{2}$ below its top, for every basepoint $\alpha \in (\frac{1}{2}, 7.35]$, and shows that RH is equivalent to it at every height for every $\alpha \in (\frac{1}{2}, 1)$. The tool is a comparison across vertical lines, from the first author's note on the Dirichlet beta function [201]. It uses only the Hadamard product, the functional equation and an elementary bound on one line to the right of $\operatorname{Re} s = 1$, so it applies equally to the completed L functions of Section 8.5. We state it for $f = \xi$, for $f = \xi(\cdot, \chi_4)$ of Section 8.5.7 and for $f = \xi(\cdot, \chi_{12})$ of Section 8.5.1, and write $v_\alpha(\theta) = \operatorname{Re}(f'/f)(\alpha + i\theta)$, $b_\alpha = v_\alpha(0) = (f'/f)(\alpha)$ and $V_f = (\log f)''(\frac{1}{2})$. For the two L functions the result is the full analogue of Proposition 9.1.5.

Each f is entire of order one, real on the real axis, with $f(s) = f(1-s)$ and no real zeros. Its zeros therefore come in groups: a zero $\frac{1}{2} + i\gamma$ on the critical line, or a pair $\frac{1}{2} \pm \delta + i\gamma$ with $0 < \delta < \frac{1}{2}$. At resolution $\varepsilon = \alpha - \frac{1}{2}$ and offset $u = \theta - \gamma$, a zero on the line contributes $\varepsilon/(\varepsilon^2 + u^2)$ to $v_\alpha(\theta)$, and a pair contributes

$$k_\delta(\varepsilon, u) = \frac{\varepsilon - \delta}{(\varepsilon - \delta)^2 + u^2} + \frac{\varepsilon + \delta}{(\varepsilon + \delta)^2 + u^2} = \frac{2\varepsilon(w + c)}{(w + c)^2 + \kappa^2}, \tag{9.1.10}$$

where $w = \varepsilon^2$, $c = u^2 - \delta^2$ and $\kappa = 2\delta|u|$, since the numerator of the sum is $2\varepsilon(\varepsilon^2 + u^2 - \delta^2)$ and its denominator is $(\varepsilon^2 + \delta^2 + u^2)^2 - 4\varepsilon^2\delta^2 = (w+c)^2 + \kappa^2$. The pair contribution is negative exactly when $u^2 < \delta^2 - \varepsilon^2$, which is Proposition 3.3.1(ii). The sum over the groups converges absolutely, since $\sum_\rho |\rho|^{-2} < \infty$. So $v_\alpha(\theta)/(\alpha - \frac{1}{2})$ is the sum over the groups of the ratios $1/(w + u^2)$ for the zeros on the line and

$$r_\delta(w, u) = \frac{k_\delta(\varepsilon, u)}{\varepsilon} = 2 \operatorname{Re} \frac{1}{w + c + i\kappa}$$

for the pairs, as functions of $w = (\alpha - \frac{1}{2})^2$. The ratio of a zero on the line decreases in w; that of a pair need not.

Lemma 9.1.7 (Comparison). *Let $0 < \delta < \frac{1}{2}$ and let u be real. As a function of $w > 0$, the ratio $r_\delta(w, u)$*

(a) is nonincreasing on $(0, \infty)$ if $|u| \geq (1 + \sqrt{2})\delta$;

(b) is nonincreasing on $[2\delta^2, \infty)$ for every u, in particular on $w \geq \frac{1}{2}$, that is for $\alpha \geq \frac{1}{2} + \frac{1}{\sqrt{2}}$, whatever δ;

(c) if $0 < |u| < (1 + \sqrt{2})\delta$, increases exactly on the interval $\max\big(0, 2\delta^2 - (|u| + \delta)^2\big) < w < 2\delta^2 - (|u| - \delta)^2$, which begins at 0 if and only if $|u| \geq (\sqrt{2} - 1)\delta$; at $u = 0$ it equals $2/(w - \delta^2)$, which has a pole at $w = \delta^2$.

In particular the constant $1+\sqrt{2}$ *in (a) cannot be lowered.*

Proof. For $u \neq 0$, $\partial_w r_\delta = -2\,\mathrm{Re}(w+c+i\kappa)^{-2} = -2\big((w+c)^2-\kappa^2\big)/\big((w+c)^2+\kappa^2\big)^2$, which is positive exactly when $-\kappa < w+c < \kappa$, that is when $2\delta^2 - (|u|+\delta)^2 < w < 2\delta^2 - (|u|-\delta)^2$, since $-\kappa - c = 2\delta^2 - (|u|+\delta)^2$ and $\kappa - c = 2\delta^2 - (|u|-\delta)^2$. The right end is at most $2\delta^2$, which gives (b) for $u \neq 0$; it is positive exactly when $\big||u|-\delta\big| < \sqrt{2}\delta$, that is when $|u| < (1+\sqrt{2})\delta$, which gives (a) and the first part of (c); and the left end is at most 0 exactly when $|u| + \delta \geq \sqrt{2}\delta$. At $u = 0$, $c = -\delta^2$ and $\kappa = 0$, and $2/(w-\delta^2)$ decreases on $w > \delta^2$, which completes (b). Finally $2\delta^2 < \frac{1}{2}$ because $\delta < \frac{1}{2}$. □

So for $(\sqrt{2}-1)\delta \leq |u| < (1+\sqrt{2})\delta$ the ratio of a pair increases from $w = 0$ up to $w = 2\delta^2 - (|u|-\delta)^2$, while for $0 < |u| < (\sqrt{2}-1)\delta$ it first decreases and then increases up to that point; in both cases it decreases beyond it. Summing the lemma over the groups gives the comparison.

THEOREM 9.1.8 (Comparison across vertical lines). *Let* $\frac{1}{2} < \alpha < \alpha'$ *and let* θ *be real. Suppose that every zero* $\frac{1}{2}+\delta+i\gamma$ *of* f *with* $\delta > (\alpha - \frac{1}{2})/\sqrt{2}$ *satisfies* $|\theta - \gamma| \geq (1+\sqrt{2})\delta$. *Then*

$$\frac{v_\alpha(\theta)}{\alpha - \frac{1}{2}} \;\geq\; \frac{v_{\alpha'}(\theta)}{\alpha' - \frac{1}{2}}.$$

The hypothesis holds at every θ *under RH for* ξ, *or GRH for* χ_4 *or* χ_{12}; *at every* θ *when* $\alpha \geq \frac{1}{2} + \frac{1}{\sqrt{2}}$, *with no hypothesis; and at every* θ *with* $|\theta| \leq H_0 - \frac{1+\sqrt{2}}{2}$, *where* H_0 *is a height below which every zero of* f *is known to lie on the critical line:* $H_0 = 3 \times 10^{12}$ *for* ξ *[196], and* $H_0 = 10^8/q$ *for* $\xi(\cdot, \chi_q)$, *that is* 2.5×10^7 *for* χ_4 *and* $10^8/12 > 8.33 \times 10^6$ *for* χ_{12} *[195].*

Proof. A zero on the line has the ratio $1/(w+u^2)$, which decreases in w. A pair with $\delta \leq (\alpha - \frac{1}{2})/\sqrt{2}$ has $2\delta^2 \leq (\alpha - \frac{1}{2})^2$, so its ratio is nonincreasing on $[(\alpha - \frac{1}{2})^2, (\alpha' - \frac{1}{2})^2]$ by Lemma 9.1.7(b), and a pair with $|\theta - \gamma| \geq (1+\sqrt{2})\delta$ by Lemma 9.1.7(a). Summing over the groups, which converges absolutely, gives the inequality. Under RH there are no pairs. If $\alpha \geq \frac{1}{2} + \frac{1}{\sqrt{2}}$, every pair has $\delta < \frac{1}{2} \leq (\alpha - \frac{1}{2})/\sqrt{2}$. If $|\theta| \leq H_0 - \frac{1+\sqrt{2}}{2}$, every zero off the line has $|\gamma| > H_0$, so that $|\theta - \gamma| > \frac{1+\sqrt{2}}{2} > (1+\sqrt{2})\delta$. □

Table 9.7 shows the ratio along rows of basepoints at three heights for each function. Every row decreases, and at $\theta = 0$ the rows approach V_f from below, as part (i) of Theorem 9.1.11 below states.

On a line to the right of $\mathrm{Re}\, s = 1$ the Euler product bounds v_α from below, and the comparison carries the bound to the left.

f	θ	$\alpha = 0.55$	0.6	0.75	1	1.5	2.5	α'
ξ	0	0.0462098	0.0462092	0.0462053	0.0461914	0.0461359	0.0459171	0.0449540
	20	1.07455	1.06774	1.02263	0.891327	0.606114	0.308661	0.142788
	40	1.55119	1.54053	1.47072	1.27476	0.878717	0.478933	0.223196
$\xi(\cdot,\chi_4)$	0	0.156025	0.156011	0.155914	0.155568	0.154215	0.149240	0.144828
	10	16.4457	14.7010	8.49762	3.52441	1.22477	0.492101	0.363147
	40	10.4644	9.84566	7.06888	3.83983	1.77007	0.837764	0.622760
$\xi(\cdot,\chi_{12})$	0	0.332896	0.332812	0.332226	0.330167	0.322371		0.302579
	10	1.93963	1.93082	1.87180	1.69298	1.26262		0.800089
	40	6.07381	5.93675	5.16314	3.71050	2.10812		1.19861

Table 9.7: The ratio $v_\alpha(\theta)/(\alpha-\frac12)$ of Theorem 9.1.8 along rows of basepoints. The last column is at the comparison line α' of Table 9.8 taken from the first author's note: 4.75 for ξ, 3.15 for χ_4 and 2.30 for χ_{12}, which lies below 2.5. Every row decreases, and at $\theta = 0$ the rows increase to V_f as $\alpha \downarrow \frac12$ (Table 9.8). Computed by `dbeta_minimum.py`.

Corollary 9.1.9 (Explicit lower bound). *For $\sigma > 1$ and real θ, $v_\sigma(\theta) \geq A_f^\infty(\sigma,\theta) - S_f(\sigma)$, where*

$$A_\xi^\infty(\sigma,\theta) = -\tfrac12\log\pi + \tfrac12\operatorname{Re}\psi\big(\tfrac{\sigma+i\theta}{2}\big),$$
$$A_{\chi_4}^\infty(\sigma,\theta) = \tfrac12\log\tfrac{4}{\pi} + \tfrac12\operatorname{Re}\psi\big(\tfrac{\sigma+1+i\theta}{2}\big),$$
$$A_{\chi_{12}}^\infty(\sigma,\theta) = \tfrac12\log\tfrac{12}{\pi} + \tfrac12\operatorname{Re}\psi\big(\tfrac{\sigma+i\theta}{2}\big),$$
$$S_f(\sigma) = \sum_{(n,q)=1}\frac{\Lambda(n)}{n^\sigma},$$

with $q = 1$, 4 and 12 for the three functions, so that $S_\xi = -\zeta'/\zeta$, $S_{\chi_4}(\sigma) = S_\xi(\sigma) - \log 2/(2^\sigma-1)$ and $S_{\chi_{12}}(\sigma) = S_{\chi_4}(\sigma) - \log 3/(3^\sigma-1)$; and $A_f^\infty(\sigma,\theta)$ increases in $|\theta|$. Hence, under the hypothesis of Theorem 9.1.8 with $\alpha' > 1$,

$$\begin{aligned} v_\alpha(\theta) &\geq \frac{\alpha-\frac12}{\alpha'-\frac12}\Big(A_f^\infty(\alpha',\theta) - S_f(\alpha')\Big) \\ &= \frac{\alpha-\frac12}{2\alpha'-1}\log|\theta| + O(1), \qquad |\theta|\to\infty. \end{aligned}$$

Proof. For $\operatorname{Re} s > 1$,

$$\frac{\xi'}{\xi}(s) = \frac1s + \frac{1}{s-1} - \frac12\log\pi + \frac12\psi\Big(\frac{s}{2}\Big) + \frac{\zeta'}{\zeta}(s),$$

where $\operatorname{Re}(\frac1s + \frac{1}{s-1}) \geq 0$ and $\operatorname{Re}\frac{\zeta'}{\zeta}(\sigma+i\theta) = -\sum_n \Lambda(n)n^{-\sigma}\cos(\theta\log n) \geq -S_\xi(\sigma)$; for the two L functions the bound is that of Proposition 8.5.25. The monotonicity in $|\theta|$ follows from the series $\operatorname{Re}\psi(x+iy) - \psi(x) = \sum_{k\geq0} y^2/\big((x+k)((x+k)^2+y^2)\big)$, and $\operatorname{Re}\psi(x+iy) = \log|y| + O(1)$ for fixed $x > 0$. Then apply Theorem 9.1.8 between α and α'. □

REMARK 9.1.10 (A bound of order $\log\theta$ outside a sparse set). For fixed $\alpha > \frac{1}{2}$ the hypothesis of Theorem 9.1.8 fails only at heights within $(1+\sqrt{2})\delta < \frac{1+\sqrt{2}}{2}$ of the ordinate of a zero $\frac{1}{2}+\delta+i\gamma$ with $\delta > (\alpha-\frac{1}{2})/\sqrt{2}$. For $f=\xi$ the number of such zeros with $0<\gamma\le T$ is $N(\sigma_0,T)$ with $\sigma_0 = \frac{1}{2}+(\alpha-\frac{1}{2})/\sqrt{2} > \frac{1}{2}$, which is $O(T^{3(1-\sigma_0)/(2-\sigma_0)}\log^5 T) = o(T)$ by Ingham's estimate [104] (Section 11.4.1). So Corollary 9.1.9 with $\alpha' = \alpha+1$ gives $v_\alpha(\theta)/\pi \ge c\log\theta$, for every $c < (\alpha-\frac{1}{2})/(\pi(2\alpha+1))$ and all large θ, outside a set of heights of density zero: this is the lower bound of the first of the four questions of Section 11.4.7, for the density v_α/π itself. The exceptional set, however, contains the window of every zero to the right of the line $\operatorname{Re} s = \alpha$, since such a zero has $\delta > \alpha-\frac{1}{2} > (\alpha-\frac{1}{2})/\sqrt{2}$. So the bound gives no information where Lemma 11.4.7 would need it, and it does not by itself confine a failure of positivity to the thin strips of that question.

THEOREM 9.1.11 (Minimum at the origin). *Let V_f, γ_1, H_0 and $\alpha_{\max}$ be as in Table 9.8.*

(i) For every $\alpha > \frac{1}{2}$, $0 < b_\alpha \le (\alpha-\frac{1}{2})V_f$; the ratio $b_\alpha/(\alpha-\frac{1}{2})$ decreases in α and tends to V_f as $\alpha\downarrow\frac{1}{2}$.

(ii) For $\frac{1}{2} < \alpha \le \alpha_{\max}$, $v_\alpha(\theta) > b_\alpha$ for $0 < |\theta| \le H_0 - \frac{1+\sqrt{2}}{2}$.

(iii) For $\frac{1}{2}+\frac{1}{\sqrt{2}} \le \alpha \le \alpha_{\max}$, $v_\alpha(\theta) > b_\alpha$ for every $\theta \ne 0$.

(iv) For $f=\xi(\cdot,\chi_4)$ and $f=\xi(\cdot,\chi_{12})$, $v_\alpha(\theta) > b_\alpha$ for every $\alpha \ge \frac{1}{2}+\frac{1}{\sqrt{2}}$ and every $\theta \ne 0$.

(v) For $f=\xi(\cdot,\chi_4)$ and $f=\xi(\cdot,\chi_{12})$, $v_\alpha(\theta) > b_\alpha$ for every $\alpha \ge 1$ and every $\theta \ne 0$.

(vi) RH, respectively GRH for χ_4 or for χ_{12}, holds if and only if $v_\alpha(\theta) > b_\alpha$ for every $\alpha \in (\frac{1}{2},1)$ and every $\theta \ne 0$.

Proof. (i) The function $f(\frac{1}{2}+s) = \int_{\mathbb{R}} K(u)e^{su}\,du$ is the two sided Laplace transform of a positive even kernel K: the Pólya kernel Φ for ξ (Section 1.2), the kernel K_4 of Proposition 8.5.21, and the eta kernel of Section 8.5.1. So $\log f(\frac{1}{2}+s)$ is, up to a constant, the cumulant generating function of a nondegenerate law; it is even and strictly convex in real s, and $b_\alpha = (\log f)'(\alpha) = \int_{1/2}^{\alpha}(\log f)''(r)\,dr > 0$. At $\theta=0$ the offset of a group with ordinate γ has $|u| = |\gamma| \ge \gamma_1 > \frac{1+\sqrt{2}}{2} > (1+\sqrt{2})\delta$, so by Lemma 9.1.7(a) the ratio of every group is nonincreasing in w, strictly for the zeros on the line, and $b_\alpha/(\alpha-\frac{1}{2})$, their sum, decreases in α. As $w\downarrow 0$ the ratios increase to γ^{-2} for a zero on the line and to $2\operatorname{Re}(|\gamma|+i\delta)^{-2}$ for a pair, and by monotone convergence the sum tends to the sum of these limits, which is $-\sum_\rho(\frac{1}{2}-\rho)^{-2} = (\log f)''(\frac{1}{2}) = V_f$.

(ii) Put $\alpha' = \alpha_{\max}$ and let t_1 be its value at $\alpha_{\max}$ in Table 9.8, where $A_f^\infty(\alpha',t_1) - S_f(\alpha') > (\alpha'-\frac{1}{2})V_f$ is certified in ball arithmetic; since A_f^∞ increases

in $|\theta|$, the inequality holds for $|\theta| \ge t_1$. For $t_1 \le |\theta| \le H_0 - \frac{1+\sqrt{2}}{2}$, Corollary 9.1.9 and (i) give $v_\alpha(\theta) > (\alpha - \frac12)V_f \ge b_\alpha$ (for $\alpha = \alpha'$ the comparison is an identity). For $|\theta| \le t_1$ we show that v_α is strictly convex on $[-t_1, t_1]$; as it is even, it then exceeds $v_\alpha(0) = b_\alpha$ at $0 < |\theta| \le t_1$. The term $\varphi(\theta-\gamma)$, $\varphi(x) = a/(a^2+x^2)$ with $a = \alpha - \frac12$, of a zero on the line is strictly convex where $|\theta-\gamma| > a/\sqrt{3}$, since $\varphi''(x) = 2a(3x^2-a^2)/(a^2+x^2)^3$. Below H_0 all zeros lie on the line and have $|\gamma| \ge \gamma_1$, and for $|\theta| \le t_1$, $|\theta-\gamma| \ge \gamma_1 - t_1 > (\alpha'-\frac12)/\sqrt{3} \ge a/\sqrt{3}$ by the convexity margin of the table. Every zero above H_0 has $|\theta-\gamma| > H_0 - t_1 > 10^6$. For a pair there, the second derivative in θ of $k_\delta(a, \theta-\gamma)$ is $g(a+\delta, u) + g(a-\delta, u)$ with $g(c,u) = 2c(3u^2-c^2)/(c^2+u^2)^3$, the second derivative of $c/(c^2+u^2)$ in u; since $\partial_c g(c,u) = 6(u^4 - 6c^2u^2 + c^4)/(c^2+u^2)^4 > 0$ for $|c| < (\sqrt{2}-1)|u|$, and $|a \pm \delta| < \alpha_{\max} < (\sqrt{2}-1)10^6$, the odd function $g(\cdot, u)$ increases on an interval containing $\pm(a \pm \delta)$, so that $g(a+\delta,u) > g(\delta - a, u) = -g(a-\delta,u)$. The sum of the second derivatives converges uniformly on $[-t_1, t_1]$, so $v_\alpha'' > 0$ there.

(iii) For $\alpha \ge \frac12 + \frac{1}{\sqrt{2}}$ the hypothesis of Theorem 9.1.8 holds at every θ, so the bound of (ii) holds for every $|\theta| \ge t_1$, and the convexity argument for $|\theta| \le t_1$ is unchanged.

(iv) It remains to treat $\alpha > \alpha_{\max}$, and there we use the primes. For χ_4, $v_\alpha(\theta) - b_\alpha = G_\alpha(\theta) + \varpi_\alpha(\theta)$ by (8.5.9), with $G_\alpha(\theta) = \frac12[\operatorname{Re}\psi(\frac{\alpha+1+i\theta}{2}) - \psi(\frac{\alpha+1}{2})]$; for χ_{12} the same holds with $G_\alpha(\theta) = \frac12[\operatorname{Re}\psi(\frac{\alpha+i\theta}{2}) - \psi(\frac{\alpha}{2})]$ and the weights $\chi_{12}(n)\Lambda(n)$ (Section 8.5.3). Dropping the positive weights and bounding $1-\cos x$ by 2 and by $x^2/2$ gives $\varpi_\alpha(\theta) \ge -R_\alpha(\theta)$ with $R_\alpha(\theta) \le \min\big(2S^-(\alpha), \frac12\theta^2 M_2(\alpha)\big)$, where

$$S^-(\alpha) = \sum_{\chi(n)=-1} \frac{\Lambda(n)}{n^\alpha} = \frac12\Big(S_f(\alpha) + \frac{L'}{L}(\alpha)\Big),$$

$$M_2(\alpha) = \sum_{\chi(n)=-1} \frac{\Lambda(n)\log^2 n}{n^\alpha} = \frac12\Big(S_f''(\alpha) + \Big(\frac{L'}{L}\Big)''(\alpha)\Big).$$

The function $G_\alpha(\theta)$ is a sum of terms $q_c(\theta) = \theta^2/(c(c^2+\theta^2))$, so it increases in $|\theta|$ and decreases in α, and $G_\alpha(\theta)/\theta^2$ decreases in $|\theta|$; and S^- and M_2 decrease in α. So on a cell $s_0 \le \alpha \le s_1$ it suffices that $G_{s_1}(T) > \max\big(2S^-(s_0), \frac12 T^2 M_2(s_0)\big)$ for one $T > 0$: then $G_\alpha(\theta) \ge G_{s_1}(T) > 2S^-(\alpha)$ for $|\theta| \ge T$, and $G_\alpha(\theta) \ge \theta^2 G_{s_1}(T)/T^2 > \frac12\theta^2 M_2(\alpha)$ for $0 < |\theta| \le T$. The script `dbeta_minimum.py` checks this in ball arithmetic on cells of width $\frac{1}{32}$ covering $[\sigma_2, 20]$, with $T \in \{0.25, 0.5, \dots, 4\}$, where $\sigma_2 = 3.71875$ for χ_4 and 2.8125 for χ_{12}; the cells just below σ_2 fail. For $\alpha \ge 20$ take $T = 2$: since q_c decreases in c, $G_\alpha(2) \ge \frac12\int_{\alpha+1}^\infty q_c(2)\,dc = \frac14\log(1 + 4/(\alpha+1)^2) \ge 1/((\alpha+1)^2+4)$, and since every n with $\chi(n) = -1$ is at least 3, $S^-(\alpha)$ and $M_2(\alpha)$ are at most $3^{-(\alpha-20)}$ times their values at 20. With $K = \max(2S^-(20), 2M_2(20))$ it therefore suffices that $K((\alpha+1)^2+4)3^{-(\alpha-20)} < 1$ for $\alpha \ge 20$; the left side decreases in α, and at $\alpha = 20$ it is 3.4×10^{-7} for χ_4 and 3.9×10^{-11} for χ_{12}. As $\alpha_{\max} = 4.70 > 3.71875$ for χ_4 and $3.10 > 2.8125$ for χ_{12}, (iii) and this argument cover every $\alpha \ge \frac12 + \frac{1}{\sqrt{2}}$.

(v) By (ii) and (iv) it remains to treat $1 \le \alpha < \frac{1}{2} + \frac{1}{\sqrt{2}}$ and $|\theta| > H_0 - \frac{1+\sqrt{2}}{2}$, and by symmetry $\theta > 0$. Let n be the number of zeros with $|\gamma - \theta| \le h$, counted with multiplicity, where $h = 15$ for χ_4 and $h = 10$ for χ_{12}. Every zero has $0 < \beta < 1 \le \alpha$, so every term of $v_\alpha(\theta)$ is positive, and a zero in the window contributes $a/(a^2+(\theta-\gamma)^2) \ge a/(\alpha^2+h^2)$, with $a = \alpha - \beta \in (0, \alpha)$; the window contains both members of each pair $\frac{1}{2} \pm \delta + i\gamma$, whose values of a add up to $2(\alpha - \frac{1}{2})$, so the values of a over the window add up to $n(\alpha - \frac{1}{2})$. Hence $v_\alpha(\theta) > n(\alpha - \frac{1}{2})/(\alpha^2 + h^2)$, which is at least $(\alpha - \frac{1}{2})V_f \ge b_\alpha$ as soon as $n \ge V_f(\alpha^2 + h^2)$. Since $\alpha^2 < (\frac{1}{2} + \frac{1}{\sqrt{2}})^2 = 1.4571\ldots$, it suffices that $n > 35.335$ for χ_4 and $n > 33.778$ for χ_{12}. By the explicit counting formula of Bennett, Martin, O'Bryant and Rechnitzer [19, Theorem 1.1], for a primitive character χ of conductor $q > 1$, $T \ge \frac{5}{7}$ and $\ell = \log(q(T+2)/(2\pi)) > 1.567$, the number $N(T, \chi)$ of zeros with $|\gamma| \le T$ satisfies

$$\Big| N(T,\chi) - \frac{T}{\pi}\log\frac{qT}{2\pi e} + \frac{\chi(-1)}{4} \Big| \le E(T) = 0.22737\,\ell + 2\log(1+\ell) - 0.5.$$

As f has no real zeros and its zeros are symmetric about the real axis, $N(\theta + h, \chi) - N(\theta - h, \chi)$ is at most twice n, whatever the convention for zeros at the ends of the window, and since E increases in T,

$$n \;\ge\; \frac{1}{2\pi}\int_{\theta-h}^{\theta+h} \log\frac{qt}{2\pi}\,dt - E(\theta+h) \;\ge\; \frac{h}{\pi}\log\frac{q(\theta-h)}{2\pi} - E(\theta+h).$$

The right side increases in θ, its derivative being at least $\big(h/\pi - 0.22737 - 2/(1+\ell)\big)/(\theta+h+2)$ with $\ell = \ell(\theta+h) > 16$, which is positive. At $\theta = H_0 - \frac{1+\sqrt{2}}{2}$, where $qH_0 = 10^8$ for both characters, $\log(q(\theta-h)/(2\pi)) = 16.58280\ldots$ and $E(\theta+h) = 9.00428\ldots$, so that $n \ge 79.177 - 9.005 > 70$ for χ_4 and $n \ge 52.785 - 9.005 > 43$ for χ_{12}, well above the 36 and 34 zeros needed. The script `dbeta_window.py` checks these inequalities and the hypotheses of the formula in ball arithmetic.

(vi) Under RH, or GRH, the hypothesis of Theorem 9.1.8 holds at every θ, and the proof of (ii) applies with $H_0 = \infty$ for every $\alpha < 1 < \alpha_{\max}$. Conversely, the condition and (i) give $v_\alpha \ge b_\alpha > 0$ on every line $\operatorname{Re} s = \alpha$ with $\frac{1}{2} < \alpha < 1$, and $v_\alpha > 0$ for $\alpha \ge 1$ (Theorem 1.2.2, Proposition 8.5.24, and Proposition 11.3.1 with Section 8.5.3); so $\operatorname{Re}(f'/f)(s) > 0$ for $\operatorname{Re} s > \frac{1}{2}$, which is RH, or GRH, by Lagarias's criterion [133]. □

For the two L functions, part (v) gives the factorisation of Proposition 9.1.5 at every basepoint $\alpha \ge 1$: for the clock T_α with Laplace transform $f(\alpha)/f(\alpha + \sqrt{u})$, $T_\alpha \overset{d}{=} S_\alpha + T_\alpha^\circ$, where S_α and T_α° are independent, S_α is positive stable of index one half with $\mathbb{E}e^{-uS_\alpha} = e^{-b_\alpha\sqrt{u}}$, and T_α° is a GGC with no drift, with Thorin density $(v_\alpha - b_\alpha)/\pi$ in the chart $\theta = \sqrt{z}$ and mean $\frac{1}{2}(\log f)''(\alpha)$. The proof is that of Proposition 9.1.5: the Thorin representation $\log(f(\alpha + \sqrt{u})/f(\alpha)) = \frac{1}{\pi}\int_0^\infty \log(1 + u/\theta^2)\, v_\alpha(\theta)\, d\theta$ holds at every $\alpha \ge 1$ (Proposition 8.5.24 and its proof, which applies to $\xi(\cdot, \chi_{12})$), and $\frac{1}{\pi}\int_0^\infty \log(1 + u/\theta^2)\, d\theta = \sqrt{u}$.

	ξ	$\xi(\cdot,\chi_4)$	$\xi(\cdot,\chi_{12})$
$V_f=(\log f)''(\frac12)$	0.04620998623	0.1560296499	0.3329236082
first ordinate γ_1	14.1347251	6.0209489	3.8046276
verified height H_0	3×10^{12}	2.5×10^{7}	$10^8/12$
comparison line α' of [201]	4.75	3.15	2.30
t_1 at α'	9.236	2.439	1.685
convexity margin $\gamma_1-t_1-(\alpha'-\frac12)/\sqrt3$	2.4450	2.0520	1.0804
α' minimising t_1, and t_1 there	4.770, 9.2353	3.320, 2.4176	2.395, 1.6717
$\alpha_{\max}$	7.35	4.70	3.10
t_1 at $\alpha_{\max}$	10.124	3.574	2.245
convexity margin at $\alpha_{\max}$	0.0559	0.0221	0.0585
cells of the proof of (iv) pass from $\sigma_2=$		3.71875	2.8125

Table 9.8: Constants of Theorem 9.1.11. For a comparison line $\alpha'>1$, t_1 is the root of $A_f^{\infty}(\alpha',t)=S_f(\alpha')+(\alpha'-\frac12)V_f$, found by bisection and rounded up to three decimals, at which the strict inequality is certified in ball arithmetic; the proof needs the convexity margin to be positive. The first comparison line is that of the first author's note; the minimiser of t_1 is located on a grid of step 0.005; $\alpha_{\max}$ is the largest α' in steps of 0.05 keeping the margin above 0.02. For ξ the range $\alpha>\alpha_{\max}$ is covered by Proposition 9.1.5. Computed by `dbeta_minimum.py`; the ordinates of $L(s,\chi_4)$ and $L(s,\chi_{12})$ are certified by `dbeta_zeros.py`.

Part (vi) sharpens Lagarias's criterion, which asks only for $v_\alpha>0$: RH is equivalent to the Thorin density on every line $\operatorname{Re}s=\alpha$ with $\frac12<\alpha<1$ being smallest at the origin and only there, and by (ii) this holds with no hypothesis for $|\theta|\le 3\times10^{12}-1.21$. In the language of Proposition 9.1.5, b_α/π is the density of the positive stable part of the clock, so (vi) says that RH holds exactly when, at every basepoint $\alpha>\frac12$, the Thorin density of T_α exceeds the density of its own stable part away from the origin, that is, when every T_α is a GGC that splits as $S_\alpha+T_\alpha^{\circ}$ with T_α° a GGC with no drift; under RH every T_α is a GGC (Theorem 1.2.2), and the proof of the factorisation is that of Proposition 9.1.5. Part (i) also shows that the stable part is present at every basepoint: in the chart $z=\theta^2$ the density $v_\alpha(\sqrt z)/(2\pi\sqrt z)$ behaves like $b_\alpha/(2\pi\sqrt z)$ as $z\downarrow0$, with $b_\alpha>0$ for every $\alpha>\frac12$. For ξ, part (iii) gives a second proof of Proposition 9.1.5 for $\frac12+\frac{1}{\sqrt2}\le\alpha\le7.35$ that uses of the zeros only the first ordinate and the verification height, not the zeros below height 2000; for $1\le\alpha<\frac12+\frac{1}{\sqrt2}$ the comparison reaches only the verified height, and there and for $\alpha>7.35$ Proposition 9.1.5 is still needed. For the two L functions no zero beyond the first is used: the zeros enter only through the verification heights of Platt and the counting formula of Bennett, Martin, O'Bryant and Rechnitzer.

Remark 9.1.12 (The margin at the origin). For ξ, parts (i) and (ii) explain the margin of Section 3.1. For every $\varepsilon\in(0,6.85]$ the minimum of $\rho_\varepsilon=v_{1/2+\varepsilon}/\pi$

over $0 \le \theta \le 3 \times 10^{12} - 1.21$ is $\rho_\varepsilon(0) = b_{1/2+\varepsilon}/\pi$, attained only at the origin, and $\rho_\varepsilon(0) \le \varepsilon V_\xi/\pi$, with $V_\xi/\pi = 2\pi^{-1}\sum_k a_k^{-1} = 0.0147091$, the constant of the computed minimum $0.0147\,\varepsilon$ (Section 3.1 and Table 9.12). By (i), $\rho_\varepsilon(0)/\varepsilon$ increases to V_ξ/π as $\varepsilon \downarrow 0$; the margin vanishes linearly, as there, and by (vi) its location at the origin at every height for every $\varepsilon < \frac{1}{2}$ is equivalent to RH.

Remark 9.1.13 (The nearest zero under GRH). Under RH, or GRH, there is a shorter route to the minimum at the origin on a finite range of heights, which the first author's note gives first for χ_4 [201]. Let $0 < a \le \frac{1}{2}$ and assume that every zero of f lies on the critical line. Then $v_{1/2+a}(\theta) = \sum_\gamma a/(a^2 + (\theta-\gamma)^2)$ over all ordinates, and $b_{1/2+a} = \sum_\gamma a/(a^2+\gamma^2) \le aV_f$, since $V_f = \sum_\gamma \gamma^{-2}$. If every gap between consecutive positive ordinates up to γ_N is less than $2\sqrt{1/V_f - a^2}$, then $v_{1/2+a}(\theta) > b_{1/2+a}$ for $0 < |\theta| \le \gamma_N$. Indeed, on $|\theta| < \gamma_1 - a/\sqrt{3}$ the sum is even and strictly convex; within $a/\sqrt{3}$ of an ordinate the nearest term alone is at least $3/(4a) > aV_f$, as $a^2 \le \frac{1}{4} < 3/(4V_f)$ for each of the three functions; and otherwise θ lies between two consecutive ordinates at distance at most half their gap g from one of them, whose term is at least $a/(a^2 + g^2/4)$, which exceeds aV_f exactly when $g < 2\sqrt{1/V_f - a^2}$. For χ_4 the threshold is 4.963 at $a = \frac{1}{2}$ and tends to $2/\sqrt{V_{\chi_4}} = 5.063$ as $a \downarrow 0$, and the largest gap below height 100 is the first, 4.2228 (Section 8.5.7.1), so under GRH the conclusion holds for $|\theta| \le 98.75$. Under GRH the gaps tend to zero, since the argument of $L(\frac{1}{2}+iT, \chi_4)$ is $O(\log T/\log\log T)$ by Littlewood's argument for ζ [255, Theorem 14.13], so the gap condition can fail only on a bounded range of heights; the first author's note asked for that range explicitly. Theorem 9.1.11 needs no gaps: with no hypothesis its part (ii) gives the conclusion for $0 < |\theta| \le H_0 - \frac{1+\sqrt{2}}{2}$, and under RH, or GRH, its part (vi) gives it at every height, which settles the question.

Remark 9.1.14 (Complete monotonicity in the resolution). Under RH, or GRH, the ratio of Theorem 9.1.8 is $v_{1/2+\sqrt{w}}(\theta)/\sqrt{w} = \sum_\gamma 1/(w+(\theta-\gamma)^2)$, the sum over all ordinates, a Stieltjes function of $w > 0$ and in particular completely monotone, which is more than the monotonicity of the theorem. With no hypothesis the ratio is then a Stieltjes function at no height, and it fails to be completely monotone near every zero off the line. The ratio of a pair is $r_\delta(w,u) = 2\int_0^\infty e^{-wx}e^{-cx}\cos(\kappa x)\,dx$ for $w > -c$, so the representing function of the whole ratio is $\sum e^{-(\theta-\gamma')^2 x}$, over the zeros γ' on the line, plus $2e^{-cx}\cos\kappa x$ for each pair. If one pair has $\kappa \ne 0$ and $c = u^2 - \delta^2$ smaller than $(\theta-\gamma')^2$ for every other zero and than the value of c of every other pair, its term dominates the representing function as $x \to \infty$ and changes sign, so by Bernstein's theorem the ratio is not completely monotone at that height; at $u = 0$ the pair contributes the pole $2/(w-\delta^2)$. So, in answer to a question of the first author's note [201], no constant plays for complete monotonicity the role that $1+\sqrt{2}$ plays in Lemma 9.1.7: whether the ratio is completely monotone at a height near a zero off the line depends on the neighbouring zeros, not only on the distance to that zero in units of δ.

9.1.7 The heat trace and the Chebyshev error

The compensated sums of Sections 9.1.3 and 9.1.5 converge only conditionally. When $\Theta < \alpha$ the error of a truncation at X is given by the zeros, through (9.1.9), and at $\alpha = 1$ an explicit bound for Chebyshev's error $\psi(y) - y$ also bounds it, but only by an amount that grows linearly with the height θ. Averaging over the height against a Gaussian changes this. For $\alpha \geq 1$ the heat trace of T_α is the Laplace transform $k_\alpha(t) = \pi^{-1} \int_0^\infty e^{-t\theta^2} v_\alpha(\theta)\, d\theta$ of its Thorin measure in the variable $z = \theta^2$, and its Lévy measure is $k_\alpha(t)\, dt/t$, as Section 11.1.6 recalls. The average turns the kernel $2\sin^2(x\theta/2)$ of the sine squared lemma (1.2.9) into a kernel that is positive, increasing and bounded, and one integration by parts then turns an explicit bound for $\psi(y) - y$ into an error bound that holds for all t at once.

Lemma 9.1.15. *For $x \geq 0$ and $t > 0$,*

$$\frac{1}{\pi} \int_0^\infty e^{-t\theta^2}\, 2\sin^2\Big(\frac{x\theta}{2}\Big)\, d\theta = \frac{g_t(x)}{2\sqrt{\pi t}}, \qquad g_t(x) = 1 - e^{-x^2/4t}.$$

Proof. Write $2\sin^2(x\theta/2) = 1 - \cos x\theta$ and use $\int_0^\infty e^{-t\theta^2} \cos x\theta\, d\theta = \frac{1}{2}\sqrt{\pi/t}\, e^{-x^2/4t}$. □

The kernel g_t increases from 0 to 1 on $[0, \infty)$, it satisfies $g_t(x) \leq \min(1, x^2/4t)$, and for fixed x it decreases in t.

Proposition 9.1.16 (Heat kernel form). *For $\alpha \geq 1$ and $t > 0$,*

$$2\sqrt{\pi t}\, k_\alpha(t) = b_\alpha + \bar{A}_\alpha(t) + \int_{[1,\infty)} g_t(\log y)\, y^{-\alpha}\, d\big(\psi(y) - y\big), \tag{9.1.11}$$

where the gamma piece

$$\bar{A}_\alpha(t) = \int_0^\infty g_t(x)\, \frac{e^{-(\alpha+2)x}}{1 - e^{-2x}}\, dx = \sum_{k \geq 1} \Big(\frac{1}{c_k} - \sqrt{\pi t}\, e^{c_k^2 t}\, \operatorname{erfc}\big(c_k\sqrt{t}\big) \Big), \qquad c_k = \alpha + 2k,$$

is positive and the Stieltjes integral converges absolutely after one integration by parts. For $\alpha > 1$ that integral is $\sum_n \Lambda(n) n^{-\alpha} g_t(\log n) - L_\alpha(t)$, with the Lebesgue piece

$$L_\alpha(t) = \int_0^\infty g_t(x)\, e^{-(\alpha-1)x}\, dx = \frac{1}{\alpha - 1} - \sqrt{\pi t}\, e^{(\alpha-1)^2 t}\, \operatorname{erfc}\big((\alpha - 1)\sqrt{t}\big).$$

Proof. Let $\alpha > 1$ and write $v_\alpha(\theta) = b_\alpha + \int_0^\infty 2\sin^2(x\theta/2)\, x\Pi_\alpha(dx)$, as at the end of Section 9.1.4. Near the origin $|x\Pi_\alpha|$ has density about $1/(2x)$, and away from it the gamma and pole densities and the prime atoms have finite mass, so $\int_0^\infty \min(1, x^2)\, |x\Pi_\alpha|(dx) < \infty$. By Lemma 9.1.15 the integral of $e^{-t\theta^2} 2\sin^2(x\theta/2)$

against $d\theta\,|x\Pi_\alpha|(dx)$ is $\frac{\sqrt{\pi}}{2\sqrt{t}}\int g_t\,d|x\Pi_\alpha| < \infty$, so the order of integration may be exchanged, and the lemma gives $2\sqrt{\pi t}\,k_\alpha(t) = b_\alpha + \int_0^\infty g_t(x)\,x\Pi_\alpha(dx)$. The gamma density gives $\bar{A}_\alpha$, whose series follows from $e^{-(\alpha+2)x}/(1-e^{-2x}) = \sum_{k\geq 1} e^{-c_k x}$ and $\int_0^\infty g_t(x)e^{-cx}\,dx = \frac{1}{c} - \sqrt{\pi t}\,e^{c^2 t}\operatorname{erfc}(c\sqrt{t})$ for $c > 0$, the case $c = \alpha - 1$ being L_α; the prime atoms and the pole density give the Stieltjes integral, written in the variable $y = e^x$.

At $\alpha = 1$ the same computation applies to $v_1 = b_1 + A + Q$ of (9.1.6), in which $Q(\theta) = \lim_{X\to\infty} Q_X(\theta)$ with $Q_X(\theta) = \int_{[1,X]} 2\sin^2(\frac{1}{2}\theta\log y)\,y^{-1}d(\psi(y) - y)$. Integration by parts gives

$$Q_X(\theta) = w(X)\big(\psi(X) - X\big) - \int_1^X \big(\psi(y) - y\big)\,w'(y)\,dy, \qquad w(y) = \frac{2\sin^2(\frac{1}{2}\theta\log y)}{y},$$

and since $|w'(y)| \leq (2 + |\theta|)\,y^{-2}$, while $\int_1^\infty |\psi(y) - y|\,y^{-2}dy < \infty$ by the prime number theorem with the de la Vallée Poussin error term, $|Q_X(\theta)| \leq C(1 + |\theta|)$ for all X. By dominated convergence the limit in X may therefore be taken outside the integral over θ, and for each X Lemma 9.1.15, applied to the finitely many atoms and to Lebesgue measure on $[0, \log X]$, gives $\frac{1}{\pi}\int_0^\infty e^{-t\theta^2}Q_X(\theta)\,d\theta = (2\sqrt{\pi t})^{-1}\int_{[1,X]} g_t(\log y)\,y^{-1}d(\psi(y) - y)$; the gamma piece A gives $\bar{A}_1$ as before. For every $\alpha \geq 1$ the same integration by parts, now with $|\frac{d}{dy}(g_t(\log y)\,y^{-\alpha})| \leq (g_t'(\log y) + \alpha)\,y^{-\alpha-1}$ and g_t' bounded, shows that the Stieltjes integral converges absolutely after one integration by parts, the boundary terms vanishing because $g_t(0) = 0$ and $y^{-\alpha}(\psi(y) - y) \to 0$. □

For $\alpha > 1$ every prime power enters (9.1.11) with the positive weight $\Lambda(n)n^{-\alpha}g_t(\log n)$, and the other terms are explicit. Keeping the primes up to X and dropping the rest therefore gives a lower bound with no error term,

$$2\sqrt{\pi t}\,k_\alpha(t) \geq b_\alpha + \bar{A}_\alpha(t) + \sum_{n\leq X} \frac{\Lambda(n)}{n^\alpha}\,g_t(\log n) - L_\alpha(t), \qquad \alpha > 1, \tag{9.1.12}$$

which uses only $\Lambda(n) \geq 0$, the positivity of the coefficients of $-\zeta'/\zeta$. By Table 9.9, the right side is positive for every t once $\alpha \geq 3.191$ with no prime atoms, the primes then entering only through the drift b_α, and once $\alpha \geq 1.533$, 1.367 and 1.284 with the primes up to 10^3, 10^5 and 10^7; the thresholds, to four decimals, are 3.1900, 1.5320, 1.3667 and 1.2840. The same argument pointwise in θ, applied to the sine squared form of v_α with no prime atoms, needs $\alpha \geq 3.575$ (the threshold is 3.5748). These statements are proved in ball arithmetic by `heat_thresholds.py`, after `heat_primes.py` has located the thresholds in floating point. At α equal to the four decimal value less 5×10^{-5} the right side is negative at one t, or one θ. From the smaller of the four decimal value plus 5×10^{-5} and the bound in the table up to $\alpha = 10$, cells cover every α and every $t > 0$, or $\theta \geq 0$, and the right side is positive on each cell: $\bar{A}_\alpha(t)$, the prime sum and $L_\alpha(t)$ are monotone in α

and in t, so each is bounded at a corner of the cell or, near the minimum over t, by a mean value form in t, and b_α by its mean value form in α. The Gaussian average reaches further, for a weaker conclusion, since $v_\alpha > 0$ implies $k_\alpha > 0$: it spreads the pole term, the only negative one, over all heights. No finite X reaches $\alpha = 1$: for fixed t, $L_\alpha(t)$ increases to $\int_0^\infty g_t(x)\,dx = \infty$ as $\alpha \downarrow 1$, while the other terms of (9.1.12) stay bounded.

At the basepoint one the primes and the pole must be kept together. Splitting the Stieltjes integral at X, and writing $\int_0^{\log X} g_t(x)\,dx = \log X - \sqrt{\pi t}\,\mathrm{erf}\,\big(\log X/(2\sqrt{t})\big)$,

$$2\sqrt{\pi t}\,k_1(t) = b_1 + \bar{A}_1(t) + \sum_{n\le X} \frac{\Lambda(n)}{n}\,g_t(\log n) - \Big(\log X - \sqrt{\pi t}\,\mathrm{erf}\,\frac{\log X}{2\sqrt{t}}\Big) + R_X(t), \tag{9.1.13}$$

with $R_X(t) = \int_{(X,\infty)} g_t(\log y)\,y^{-1}d(\psi(y)-y)$. Suppose that $|\psi(y)-y| \le \omega(\log y)\,y$ for $y \ge X$. Integrating by parts, with $|\frac{d}{dy}(g_t(\log y)/y)| \le (g_t'(\log y)+g_t(\log y))\,y^{-2}$, $g_t \le 1$ and $\int_{\log X}^\infty g_t' = 1 - g_t(\log X)$,

$$\begin{aligned} |R_X(t)| &\le g_t(\log X)\,\frac{|\psi(X)-X|}{X} + \big(1-g_t(\log X)\big)\sup_{x\ge \log X}\omega(x) \\ &\quad + \int_{\log X}^\infty \omega(x)\,dx \le \eta_X \end{aligned} \tag{9.1.14}$$

for every $t > 0$, where $\eta_X = \sup_{x\ge\log X}\omega(x) + \int_{\log X}^\infty \omega(x)\,dx$, the last step because $|\psi(X)-X| \le \omega(\log X)X$. The boundary term and the part of the integral carried by g_t' share one supremum, with weights that add up to one. The pointwise weight $2\sin^2(\frac{1}{2}\theta\log y)$ would give instead a bound growing linearly in θ, through its derivative; it is the monotone Gaussian kernel that makes the bound uniform. Explicit forms of the prime number theorem supply ω: $|\psi(y)-y| \le 0.94\sqrt{y}$ for $11 < y \le 10^{19}$ [47]; $|\psi(y)-y| < 1.93378\times10^{-8}\,y$ for $y \ge 10^{19}$ and $|\psi(y)-y| < 6.95\times10^{-9}\,y$ for $y \ge e^{46}$ [44]; and $|\psi(y)-y| \le 8.86\,y(\log y)^{1.514}e^{-0.8288\sqrt{\log y}}$ for $\log y \ge 3000$ [111]. The constants alone would not do, since ω must be integrable, and the decaying bound of Johnston and Yang closes the integral. With $X = 10^7$ they give $\eta_X < 9.2\times10^{-4}$: 3.0×10^{-4} from the supremum, attained at $\log X$, and 6.2×10^{-4} from the integral, almost all of it from Büthe's range.

Proposition 9.1.17. *For every $t > 0$, $2\sqrt{\pi t}\,k_1(t) > 0.021$. The proof uses the primes up to 10^7, the bounds for $\psi(y)-y$ quoted above and a computation in ball arithmetic.*

Proof. By (9.1.13) and (9.1.14) it suffices that the right side of (9.1.13), with $R_X(t)$ replaced by $-\eta_X$, exceed 0.021 for every t. The three terms of it that depend on t, the gamma piece, the prime sum and the bracket, all decrease in t, because g_t does, so on a cell $[t_0,t_1]$ the value obtained by evaluating the first two

at t_1 and the bracket at t_0 is a lower bound. The script `heat_primes.py` evaluates it in ball arithmetic with Arb [110], on 65137 cells that cover $[10^{-10}, 10^8]$ and are refined until each bound exceeds 0.021. It bounds $\bar{A}_1$ from below by the terms of its series with $k \le 1000$ and $c_k\sqrt{t} \le 25$, all terms being positive, and the prime sum from below by grouping the prime powers above 10^4 into blocks of width 5×10^{-4} in $\log n$, each weighted by g_t at its left end. For $t \le 10^{-10}$ the same monotonicity, with the bracket at most $\log X$, gives the lower bound 2.86, and for $t \ge 10^8$ dropping $\bar{A}_1$ and the prime sum gives 0.0221. □

The lower bound of the proof has its minimum, about 0.0220, near $t = 12.7$, and the true value is larger. From the zero side, Proposition 9.1.4 gives $v_1 \ge b_1$, hence $2\sqrt{\pi t}\, k_1(t) > b_1$ for every t, where $b_1 = 0.0230957$. A direct evaluation of the Thorin form in `heat_primes.py` gives $R_X(t) = 1.4 \times 10^{-4}$ at $t = 1$ and at $t = 12$, essentially the boundary term $g_t(\log X)(X - \psi(X))/X$ with $\psi(10^7) = 9998539.403$, against the bound $\eta_X < 9.2 \times 10^{-4}$. So Proposition 9.1.17 is weaker than what the zeros give. Its interest is its input, the primes up to 10^7 and published bounds for $\psi(y) - y$, with no zero computed here. It does not avoid the zeros altogether, since those bounds are themselves proved from them: from the verification of RH to a finite height, now 3×10^{12} [196], and from explicit zero free regions and zero density estimates.

input from the primes	positivity	basepoints
$\Lambda(n) \ge 0$	$v_\alpha(\theta) > 0$ for every θ	$\alpha \ge 3.575$
$\Lambda(n) \ge 0$	$k_\alpha(t) > 0$ for every t	$\alpha \ge 3.191$
primes up to 10^3	$k_\alpha(t) > 0$ for every t	$\alpha \ge 1.533$
primes up to 10^5	$k_\alpha(t) > 0$ for every t	$\alpha \ge 1.367$
primes up to 10^7	$k_\alpha(t) > 0$ for every t	$\alpha \ge 1.284$
primes up to 10^7 and bounds for $\psi(y) - y$	$2\sqrt{\pi t}\, k_1(t) > 0.021$ for every t	$\alpha = 1$

Table 9.9: What the primes show about the Thorin density and the heat trace of T_α, with no information on the zeros beyond what the published bounds for $\psi(y) - y$ contain. The first five rows are proved in ball arithmetic by `heat_thresholds.py`, after `heat_primes.py` has located the thresholds in floating point; the text gives them to four decimals. Positivity is proved on cells covering every α from the bound shown to 10 and every time, or height, and for $\alpha \ge 10$ it is immediate: $b_\alpha = \mathbb{E}V_\alpha$ increases with α, its derivative being $\operatorname{Var} V_\alpha$, $b_{10} = 0.391$, and the pole term is at most $\frac{1}{\alpha-1} \le \frac{1}{9}$ in both forms. The last row is Proposition 9.1.17, also in ball arithmetic. Below $\alpha = 1$ this route needs an explicit quasi Riemann hypothesis as input.

Below the basepoint one this method stops, and the reason is Proposition 9.1.6 again. For $\frac{1}{2} < \alpha < 1$ the integration by parts bounds the tail by a boundary term plus $\int_{\log X}^{\infty} \omega(x)\, e^{(1-\alpha)x}(g_t'(x) + \alpha g_t(x))\, dx$, and since $g_t(x) \to 1$ as $x \to \infty$ for

every t, this is finite only if $\int^\infty \omega(x)e^{(1-\alpha)x}\,dx < \infty$, that is, only if ω decays exponentially. Every explicit unconditional bound decays more slowly, like $e^{-c\sqrt{x}}$ from the zero free region of de la Vallée Poussin or like $e^{-cx^{3/5}(\log x)^{-1/5}}$ from that of Vinogradov and Korobov, and so do the published bounds above beyond their finite ranges. The half plane $\operatorname{Re} s > \frac{7}{8}$ announced in October 2026 (Section 12.1.5) would give $\psi(y)-y = O(y^{7/8+\delta})$ for every $\delta > 0$ [255], enough in principle for every $\alpha > \frac{7}{8}$; its constant has not been made explicit, and an explicit one would let this method pass a little below the basepoint one. A bound with $\int^\infty \omega(x)e^{(1-\alpha)x}\,dx < \infty$ is itself a quasi Riemann hypothesis: it makes $\int_1^\infty y^{-w}\,d(\psi(y)-y) = -\frac{\zeta'}{\zeta}(w) - \frac{1}{w-1}$ converge for $\operatorname{Re} w \geq \alpha$ after one integration by parts, so it forces $\Theta \leq \alpha$, and it follows from $\Theta < \alpha$. Along this route the step below the basepoint one therefore needs a quasi Riemann hypothesis as input, and reaching every $\alpha > \frac{1}{2}$ needs RH itself. The obstruction belongs to the route. Whether $k_\alpha > 0$ just below $\alpha = 1$ needs any hypothesis is not settled by it, since $k_\alpha > 0$ is weaker than $v_\alpha \geq 0$ and only zeros with real part above α contribute negatively to v_α. The heat trace W_α of T_α is positive for every $\alpha > \frac{1}{2}$ (Corollary 9.2.6), but it is k_α only when no zero has real part above α. The Gaussian average makes each rung of Table 9.9 computable with an explicit error, but it cannot supply the next one: that is information about the zeros, not a better kernel.

The first two questions of the chapter are now answered. The Thorin density of T_1 is the Poisson smoothing of the zeros with widths $1-\beta$, positive with no hypothesis, and its law has infinite mean and a stable one half tail with constant $\sum_\rho \rho^{-1}/\sqrt{\pi}$. The density comes from the sine squared lemma, which converts the Lévy form (9.1.5) term by term: its drift becomes the stable one half factor $e^{-b_1\sqrt{u}}$, the transform of an independent summand of T_1 (Proposition 9.1.4), and the same splitting holds at every basepoint above one (Proposition 9.1.5), and the convergence of its compensated prime sum is the prime number theorem. Its signed measure Π_1, made of the gamma density, the prime atoms and the pole, is not the Lévy measure of an infinitely divisible variable, nor that of a quasi infinitely divisible one (Proposition 11.1.3). Positivity is not termwise but is the inequality of Hadamard and de la Vallée Poussin, the same fact read on the zero side. Averaged over heights against a Gaussian, the same bookkeeping certifies the heat trace at the basepoint one from the primes up to 10^7 and published bounds for $\psi(y)-y$ (Proposition 9.1.17), and no such certificate passes below it without a quasi Riemann hypothesis. Below the basepoint one the construction survives exactly as far as a zero free half plane does, and how negative the density can become there with no hypothesis is the subject of Section 11.4.

9.2 Hitting times at the basepoints

The third question, whether T_1 is a hitting time, is probabilistic. At the centre the answers are in Sections 8.1 and 8.2. Under RH with simple zeros, Kent's theorem

makes the clock the hitting time of a level by a string diffusion (Section 8.1.3); with no hypothesis, Proposition 8.2.1 rules out a Brownian first passage to a random level, and Section 8.2.2.2 computes the curved boundary whose Brownian first passage time has the law of the clock. At the basepoint one Kent's theorem does not apply and the first passage to a random level is again impossible (Section 9.2.1), but T_1 is the centre clock, tilted and subordinated by a drift and a stable one half process (Section 9.2.2). The subordination gives the density of T_1 in closed form, and planar Brownian motion accounts for both the Cauchy kernels and the stable part of the subordinator (Section 9.2.3). The subordination also lets us compute the exact Brownian boundary of T_1, which, as computed, becomes flat in the right tail; if it stays bounded, Wald's identity fails by an amount that the zeros determine (Section 9.2.4). The subordination holds at every basepoint above the centre, and the boundary becomes flat at the two further basepoints at which we computed it (Sections 9.2.5 and 9.2.6).

We keep the notation of Section 9.1: $\varphi_\alpha(u) = \xi(\alpha)/\xi(\alpha + \sqrt{u})$ is the Laplace transform of T_α whenever that variable exists, $b_\alpha = \frac{\xi'}{\xi}(\alpha)$, $v_\alpha(\theta) = \mathrm{Re}(\xi'/\xi)(\alpha + i\theta)$, and $\rho_{1/2} = v_1/\pi$ is the Thorin density of T_1 in the chart $\theta = \sqrt{z}$. In addition B is a standard Brownian motion, $H_a^{(c)}$ is the first passage time of $B_t + ct$ to the level $a \geq 0$, and $H_a = H_a^{(0)}$; this H has nothing to do with the de Bruijn Newman flow H_t of Section 8.4.

9.2.1 Kent's theorem and Brownian first passage to a random level

The first candidate is the hitting time of a level by a diffusion. By Kent's theorem [122] and Krein's inverse spectral theory [70], recalled in Section 8.1.3, the first hitting times of levels by diffusions started at a reflecting or entrance end are exactly the GGCs without drift whose Thorin measures consist of unit atoms at distinct points, the Dirichlet eigenvalues of the interval up to the level; the Bessel hitting time J of Section 9.1.1 is the simplest example. The Thorin measure of φ_1 is absolutely continuous, so T_1 is not such a hitting time. It is the clock at the centre that has the discrete form: under RH its Thorin measure has an atom at each γ^2, of mass the multiplicity of the zero, so that it is such a hitting time exactly when the zeros are also simple. Brownian motion offers a second candidate, a first passage to a random level, and Proposition 8.2.1 excludes it for the clock. The same argument excludes it at the basepoint one.

Proposition 9.2.1. *For no drift $c \geq 0$ and no random level $L \geq 0$ independent of the Brownian motion is T_1 equal in law to the first passage time $H_L^{(c)}$. (For $c < 0$ the first passage time is infinite with positive probability, so equality is impossible.)*

Proof. Conditionally on L, the Laplace transform of $H_L^{(c)}$ is $\exp\{-L(\sqrt{c^2 + 2u} - c)\}$, so $\mathbb{E}e^{-uH_L^{(c)}} = \mathbb{E}\exp\{-L(\sqrt{c^2 + 2u} - c)\}$ extends analytically, and bounded by one, to the region $R_c = \{u : \mathrm{Re}\,\sqrt{c^2 + 2u} > c\}$, which contains the right half

plane and lies inside $\mathbb{C}\setminus(-\infty,0]$. The function φ_1 is analytic there as well, by the proof of Proposition 9.1.1, so if the two agreed on $\operatorname{Re} u>0$ they would agree on R_c, and $|\varphi_1|\le 1$ on R_c and on its boundary, the parabola $u_q=\frac12(-q^2+2icq)$, $q\in\mathbb{R}$, on which $\sqrt{c^2+2u_q}=c+iq$, as in the proof of Proposition 8.2.1. On that parabola $\operatorname{Re}\sqrt{u_q}$ stays bounded while $\operatorname{Im}\sqrt{u_q}\sim q/\sqrt{2}$, so by Stirling's formula $|\xi(1+\sqrt{u_q})|$ decays exponentially and $|\varphi_1(u_q)|\to\infty$. For $c=0$ the parabola is the cut itself and $|\varphi_1(-t^2)|=|\xi(1)/\xi(1+it)|$, which equals 13.0, 1.2×10^4, 2.0×10^{10} and 1.5×10^{23} at $t=10,20,40,80$, as computed by `compute.py`. □

This is the same obstruction as at the centre (Proposition 8.2.1): the reciprocal of ξ grows along vertical lines, and no mixture of Brownian first passage transforms can grow. The obstruction concerns Brownian first passage to a random level alone. A subordinator with a drift as well as a stable one half part escapes it: on the cut, where $\sqrt{u}=it$, its exponent $u+\sqrt{u}=-t^2+it$ has negative real part, so mixtures of its transforms can grow, and Theorem 9.2.2 gives an exact representation of this kind.

9.2.2 Subordination of the centre clock

Let T be the clock variable of Section 1.2, with $\mathbb{E}e^{-uT}=\varphi_{1/2}(u)=1/G(u)$, a positive random variable that exists with no hypothesis (Theorem 5.1.1, whose proof uses the verification of RH below height 1000); it is the clock of [198], and we call it the centre clock.

Theorem 9.2.2. *Let $\widetilde{T}$ have the tilted law $P(\widetilde{T}\in dt)=e^{-t/4}P(T\in dt)/\mathbb{E}e^{-T/4}$, and let $(\sigma_\ell)_{\ell\ge0}$ be an independent subordinator with Laplace exponent $u+\sqrt{u}$. Then*

$$T_1\overset{d}{=}\sigma_{\widetilde{T}}\overset{d}{=}\widetilde{T}+H_{\widetilde{T}/\sqrt{2}},$$

where H_a is the first passage time of a standard Brownian motion to the level a, independent of $\widetilde{T}$.

Proof. For $\operatorname{Re}\sqrt{u}\ge0$ the number $\frac12+\sqrt{u}$ has positive real part and $(\frac12+\sqrt{u})^2=\frac14+u+\sqrt{u}$, so $\xi(1+\sqrt{u})=\xi\big(\frac12+\sqrt{\frac14+u+\sqrt{u}}\big)$ with the principal root. Hence

$$\begin{aligned}\varphi_1(u)&=\frac{\xi(1)}{\xi(1+\sqrt{u})}=\frac{\varphi_{1/2}(\frac14+u+\sqrt{u})}{\varphi_{1/2}(\frac14)}\\&=\frac{\mathbb{E}\big[e^{-T/4}e^{-(u+\sqrt{u})T}\big]}{\mathbb{E}e^{-T/4}}=\mathbb{E}e^{-(u+\sqrt{u})\widetilde{T}}=\mathbb{E}e^{-u\sigma_{\widetilde{T}}}.\end{aligned}$$

Finally $\sigma_\ell\overset{d}{=}\ell+H_{\ell/\sqrt{2}}$, because $\mathbb{E}e^{-uH_a}=e^{-a\sqrt{2u}}$. □

The subordinator σ has a classical description: a subordinator with exponent $\kappa_1u+\kappa_2\sqrt{2u}$ is the inverse local time at zero of a Brownian motion that is sticky

at zero, the drift κ_1 measuring the time spent at zero per unit of local time [107, Section 5.7, p. 187]; here $\kappa_1 = 1$ and $\kappa_2 = 1/\sqrt{2}$. So T_1 is the first time the local time at zero of a sticky Brownian motion exceeds an independent level distributed as $\widetilde{T}$.

COROLLARY 9.2.3. *Assume RH and that the zeros are simple. Then $\widetilde{T}$ is Kent's hitting time for a Krein string whose Dirichlet spectrum is $\{\gamma^2 + \frac{1}{4}\} = \{|\rho|^2\}$, and T_1 is the first hitting time of the level y_* by the process $D(\ell_t)$, where D is that string diffusion started at its left end, y_* is the level whose hitting time by D is $\widetilde{T}$, and ℓ_t is the local time at zero of an independent sticky Brownian motion B. Equivalently, T_1 is the first hitting time of the line $\{y = y_*\}$ by the two dimensional Markov process $(D(\ell_t), B_t)$.*

Proof. Under RH, $\mathbb{E}e^{-u\widetilde{T}} = \varphi_{1/2}(\frac{1}{4} + u)/\varphi_{1/2}(\frac{1}{4}) = \prod_{\gamma>0}(\gamma^2 + \frac{1}{4})/(\gamma^2 + \frac{1}{4} + u)$, the product running over the ordinates $\gamma > 0$ repeated according to multiplicity. Since the zeros are simple, this is a GGC without drift with unit atoms at the distinct points $|\rho|^2$, so Kent's theorem and Krein's inverse theory provide the string. The same law arises by killing the centre clock's diffusion at rate $\frac{1}{4}$ and conditioning it to reach its level before it is killed: the killing multiplies the law of the hitting time by $e^{-t/4}$ and subtracts $\frac{1}{4}$ from the generator, and the conditioning, a Doob transform, normalises the law and keeps the spectrum, so every Dirichlet eigenvalue is shifted by $\frac{1}{4}$. The first time $D(\ell_t)$ reaches y_* is the first t with $\ell_t \geq \widetilde{T}$, which is $\sigma_{\widetilde{T}}$. □

The Thorin measure of Proposition 9.1.1 can be read off the theorem: under RH each factor $(\gamma^2 + \frac{1}{4})/(\gamma^2 + \frac{1}{4} + u + \sqrt{u})$ equals $(\frac{1}{4} + \gamma^2)/((\sqrt{u} + \frac{1}{2})^2 + \gamma^2)$, whose Thorin density in θ is the pair of Cauchy kernels of width $\frac{1}{2}$ at $\pm\gamma$. The subordination by $u + \sqrt{u}$ is the probabilistic form of the Poisson smoothing that carries the Thorin measure from the centre to the basepoint one. The tail (9.1.3) also follows: $P(H_a > x) \sim a\sqrt{2/(\pi x)}$, so $P(T_1 > x) \sim \mathbb{E}[\widetilde{T}]/\sqrt{\pi x}$, and $\mathbb{E}\widetilde{T} = -\frac{d}{du}\log\varphi_{1/2}(\frac{1}{4} + u)\big|_{u=0} = \frac{\xi'}{\xi}(1) = \sum_\rho \rho^{-1}$.

We simulated 10^6 draws of $\widetilde{T} + H_{\widetilde{T}/\sqrt{2}}$ with `mc.py`. Under RH, $\widetilde{T} = \sum_k e_k/(\gamma_k^2 + \frac{1}{4})$ with independent standard exponentials e_k; we used the first 200 zeros exactly and replaced the remaining terms, including the zeros above height 2198 through the exact power sums $\sum_{\gamma>0}\gamma^{-2}$ and $\sum_{\gamma>0}\gamma^{-4}$, by a gamma variable with the same mean and variance. The first passage time was drawn as $H_a = a^2/Z^2$ with Z standard normal. Table 9.10 compares the empirical Laplace transform with $\varphi_1(u) = \xi(1)/\xi(1 + \sqrt{u})$, and Figure 9.5, drawn by `figs.py`, shows the transform and the tail $\sqrt{x}\,P(T_1 > x)$ against (9.1.3). All transform values agree within 1.3 standard errors, the sample mean of $\widetilde{T}$ is 0.023108 against $\sum_\rho \rho^{-1} = 0.0230957$, and $\sqrt{x}\,P(T_1 > x)$ is 0.0131 at $x = 10^4$ and 0.0130 at $x = 10^6$ against 0.01303.

u	Monte Carlo	standard error	$\varphi_1(u) = \xi(1)/\xi(1+\sqrt{u})$
0.5	0.972541	0.000095	0.972532
1	0.954946	0.000112	0.954930
10	0.740171	0.000162	0.740164
100	0.094083	0.000061	0.094159
1000	4.018×10^{-8}	3.8×10^{-10}	4.047×10^{-8}

Table 9.10: Monte Carlo check of Theorem 9.2.2: the empirical Laplace transform of 10^6 draws of $\widetilde{T} + H_{\widetilde{T}/\sqrt{2}}$ against φ_1.

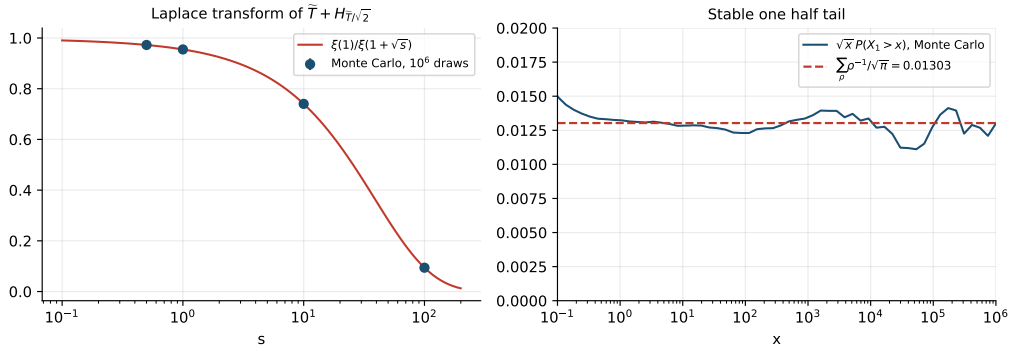


Figure 9.5: Left: the empirical Laplace transform of $\widetilde{T} + H_{\widetilde{T}/\sqrt{2}}$ from 10^6 draws, with three standard errors, against $\varphi_1(u) = \xi(1)/\xi(1+\sqrt{u})$; the plot writes s for u. Right: $\sqrt{x}\,P(T_1 > x)$ from the simulation, written with X_1 in the plot, and the limit $\sum_\rho \rho^{-1}/\sqrt{\pi} = 0.01303$ of (9.1.3). Drawn by `figs.py`.

9.2.3 The density of T_1 and harmonic measure

Conditioning on $\widetilde{T}$ in Theorem 9.2.2 and using the density of H_a gives the density of T_1 as an expectation over the tilted centre clock,

$$f_{T_1}(x) = \mathbb{E}\Big[\frac{\widetilde{T}}{2\sqrt{\pi}}\,(x-\widetilde{T})^{-3/2}\exp\Big(-\frac{\widetilde{T}^2}{4(x-\widetilde{T})}\Big)\,\mathbf{1}\{\widetilde{T} < x\}\Big], \tag{9.2.1}$$

since H_a has density $a(2\pi h^3)^{-1/2}e^{-a^2/2h}$ and $a = \widetilde{T}/\sqrt{2}$. The integrand is smooth, so (9.2.1) can be averaged over draws of $\widetilde{T}$ without the noise of a histogram. Table 9.11, computed by `improve.py`, compares it, with 4×10^5 draws, against numerical inversion of φ_1 by Talbot's method [248], and Figure 9.6, drawn by `figs2.py`, shows the whole density. At the tabulated points the two agree to about 0.1%, except near the mode, at $x = 0.024$, where the Monte Carlo average is within 1%. For large x the density behaves like $\frac{1}{2\sqrt{\pi}}\sum_\rho \rho^{-1}\,x^{-3/2} = 0.00652\,x^{-3/2}$, the derivative of (9.1.3); at $x = 286$ this gives 1.347×10^{-6}, matching both columns.

Near the origin the density is super flat, inherited from the centre clock through $\widetilde{T}$.

x	Talbot inversion	formula (9.2.1)
0.024	5.4512×10^{1}	5.3998×10^{1}
0.1145	2.4453×10^{-1}	2.4471×10^{-1}
0.5476	1.7231×10^{-2}	1.7237×10^{-2}
2.618	1.5601×10^{-3}	1.5607×10^{-3}
12.52	1.4759×10^{-4}	1.4764×10^{-4}
59.83	1.4087×10^{-5}	1.4091×10^{-5}
286.0	1.3470×10^{-6}	1.3474×10^{-6}

Table 9.11: The density of T_1 by two independent methods: Talbot inversion of φ_1, and the subordination formula (9.2.1) averaged over 4×10^5 draws of $\widetilde{T}$.

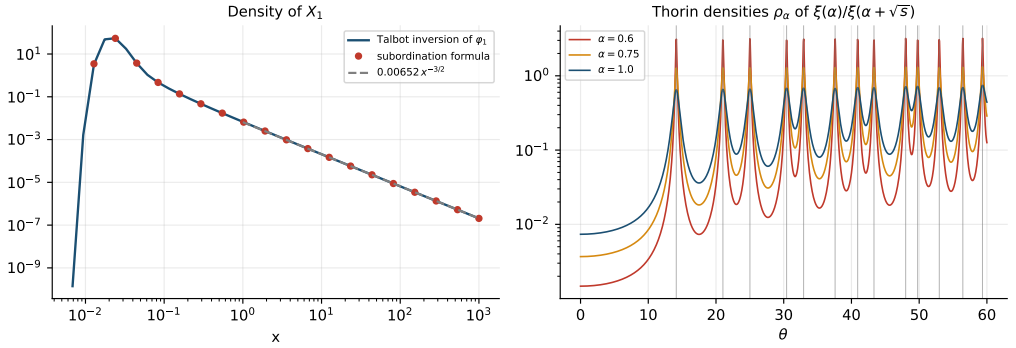


Figure 9.6: Left: the density of T_1 by Talbot inversion of φ_1 and by the subordination formula (9.2.1), with the tail $0.00652\,x^{-3/2}$. Right: the densities v_α/π at the basepoints $\alpha = 0.6$, 0.75 and 1, with the ordinates of the zeros; they sharpen onto the zeros as α decreases, and for $\alpha < 1$ they are Thorin densities of T_α only if no zero lies to the right of the line $\operatorname{Re} s = \alpha$. The plots write X_1 for T_1, ρ_α for v_α/π and s for u. Drawn by `figs2.py`.

The Cauchy kernels of Proposition 9.1.1 and the stable one half subordinator of Theorem 9.2.2 are both laws of planar Brownian motion at the moment it first reaches a line. Let $B = (B^1, B^2)$ be a planar Brownian motion started at a zero $\rho = \beta + i\gamma$, identified with the point (β, γ), and let H be the first time B^1 reaches the line $\operatorname{Re} s = 1$, a distance $1 - \beta$ away, that is, the first passage time $H_{1-\beta}$ of the one dimensional Brownian motion $B^1 - \beta$. Then B^2_H has the Cauchy law with centre γ and width $1 - \beta$, which is the harmonic measure of the half plane, and

H has Laplace transform $e^{-(1-\beta)\sqrt{2u}}$, the stable one half law. So, exactly,

$$\rho_{1/2}(\theta)\,d\theta = \sum_{\rho} P\big(B_H^2 \in d\theta \mid B_0 = \rho\big)\,:$$

the Thorin measure at the basepoint one is the sum over the zeros of the laws of the places where planar Brownian motions started at the zeros first reach the line $\operatorname{Re} s = 1$. The times H have the same stable one half law, up to scale, as the stable part of the subordinator σ in Theorem 9.2.2; the correspondence is of type rather than of scale, since in the theorem the stable part runs per unit of local time $\widetilde{T}$ rather than per unit of distance. Under RH every motion starts at distance $\frac{1}{2}$. This is the probabilistic content of the statement that $\log|\xi|$ is harmonic off the zeros and that $v_1 = \pi\rho_{1/2}$ is its normal derivative on the line $\operatorname{Re} s = 1$.

9.2.4 The exact Brownian boundary and Wald's identity

The inverse first passage question asks for a boundary b such that the first time $H_b = \inf\{t > 0 : B_t \geq b(t)\}$ at which a standard Brownian motion started at zero crosses b has the law of T_1. Proposition 9.2.1 shows that no constant level with a random height works for any drift. A deterministic curved boundary always exists. Anulova proved this for two sided symmetric boundaries [3], and Chen, Cheng, Chadam and Saunders for one sided boundaries when the survival function is continuous and positive [51, Theorem 1]. By the theorems of Ekström and Janson [73, Theorems 2.3, 8.2 and 9.2], every law on $(0,\infty)$ is the law of H_b for a lower semicontinuous boundary b, which is continuous on every interval where the density of the law is bounded below by a positive constant; when that holds on every compact subinterval of $(0,\infty)$, b is the unique continuous solution of the Volterra equation (8.2.1) with $\liminf_{t\downarrow 0} b(t) \geq 0$. The density (9.2.1) of T_1 is continuous and positive on $(0,\infty)$, so the boundary of T_1 is continuous there, and it can be computed from the law of T_1 alone through (8.2.1), with f the density of T_1, used in the complementary form of Section 8.2.2.2, with $P(T_1 > t)$ in place of $P(T > t)$, once the survival probability falls below one half; `rs1.py` solves it. We computed the distribution function of T_1 by Talbot inversion of $\varphi_1(u)/u$ for $t < 0.03$, where T_1 is super flat, and by the subordination representation of Theorem 9.2.2 beyond, in the form

$$P(T_1 > t) = \mathbb{E}\Big[\mathbf{1}\{\widetilde{T} \geq t\} + \mathbf{1}\{\widetilde{T} < t\}\Big(2\mathcal{N}\Big(\frac{\widetilde{T}}{\sqrt{2(t-\widetilde{T})}}\Big) - 1\Big)\Big],$$

with $\mathcal{N}$ the standard normal distribution function, as in Section 8.2.2.2. This form is smooth and gives the survival function to relative accuracy even at $t = 100$, where $\sqrt{t}\,P(T_1 > t) = 0.013037$ against the limit 0.013030. As at the centre, the boundary is computed over the three zones of Section 8.2.2.2: small times, the bulk and the right tail.

In the small time zone the law of T_1 near the origin is that of the tilted centre clock, and the boundary starts from the reflection level 0.505 at $t = 0.009$. In the bulk, from $t = 0.011$ to $t = 0.05$, the boundary falls steeply, to 0.363 at $t = 0.012$ and 0.111 at $t = 0.02$. Here the tangent approximation, taken in its density form, equates the first passage density of the tangent line with the density of T_1, where the boundary of Roberts and Shortland [223] in Section 8.2.2.1 equates hazards. Its solution, the boundary b with $(b - tb')(2\pi t^3)^{-1/2} e^{-b^2/2t} = f(t)$ computed by `rs1c.py`, agrees with the exact boundary to 10^{-4} at $t = 0.012$ and to 1.5×10^{-3} at $t = 0.02$.

The right tail is where the basepoint one differs completely from the centre. At the centre the survival function is exponential and the boundary is asymptotically linear with slope $-\sqrt{2}\gamma_1$ (Section 8.2.2.2). At the basepoint one the survival function is the stable one half tail $0.01303\, t^{-1/2}$, and a Brownian motion survives that long only below a boundary that stops moving. The exact boundary flattens at the level $b_\infty = -0.19277$: it stays within 10^{-3} of this level from $t = 0.056$ on and within 10^{-4} from $t = 1.2$ on, up to the end of the computation at $t = 100$; that $b_\infty = \lim_{t\to\infty} b(t)$ is observed, not proved. The tangent approximation fails here: it overshoots downward, to -0.332 at $t = 0.05$ against the exact -0.188, and has no solution beyond $t = 0.091$. It treats the crossing density at time t as if the path had just started, so a flat boundary at level a would have to produce the density $a\,(2\pi)^{-1/2} t^{-3/2}$, forcing $a = \sqrt{2\pi} \times 0.00652 = 0.0163 > 0$; but by the time the tail is reached the tangent boundary is already negative, and a negative linear boundary can only produce an exponential tail. The heavy tail is carried by the survivors, who have wandered below the boundary, and only the exact equation accounts for them.

The positive number 0.0163 does appear in the exact picture, in a different role. Optional stopping for the martingale $B_{t\wedge H_b}$ gives $\mathbb{E}[b(H_b)\mathbf{1}\{H_b \le t\}] = -\mathbb{E}[B_t \mathbf{1}\{H_b > t\}]$, and once the boundary is flat, a survivor at depth y below it survives a further time r with probability about $y\sqrt{2/(\pi r)}$. The next lemma makes the matching of these two facts exact; it needs the boundary to be continuous and bounded at infinity, but not to converge. For a continuous monotone boundary that is positive at 0 the limit is Theorem 2 of Novikov [168]; the lemma assumes boundedness in place of monotonicity.

LEMMA 9.2.4. *Let b be continuous on $(0,\infty)$ and bounded on $[1,\infty)$, and let $H_b = \inf\{t > 0 : B_t \ge b(t)\}$ be positive almost surely. Then $b(H_b) = B_{H_b}$ is integrable and $\sqrt{t}\, P(H_b > t) \to \sqrt{2/\pi}\, \mathbb{E}\, b(H_b)$ as $t \to \infty$.*

Proof. Since b is bounded above on $[1,\infty)$ and $\limsup_t B_t = \infty$, H_b is finite, and $B_{H_b} = b(H_b)$ by continuity. Choose $u_0 > 0$ with $P(H_b > u_0) > 0$; on that event $b > B$ on $(0, u_0]$, so b, which is also continuous on $[u_0, 1]$ and bounded on $[1,\infty)$, is bounded below on $(0,\infty)$, say by $-K$. Optional stopping gives $A(s) = \mathbb{E}[b(H_b)\mathbf{1}\{H_b \le s\}] = -\mathbb{E}[B_s \mathbf{1}\{H_b > s\}]$. Fix $s \ge 1$, let c_1 and c_2 be the

infimum and the supremum of b on $[s,\infty)$, and let $g(y,\tau) = (2\mathcal{N}(y/\sqrt{\tau}) - 1)^+$, the probability that a Brownian motion started at zero stays below the level y up to time τ. By the Markov property at time s, on $\{H_b > s\}$ the conditional probability of $\{H_b > t\}$ given the path up to time s lies between $g(c_1 - B_s, t-s)$ and $g(c_2 - B_s, t-s)$, and $c_2 - B_s > 0$ there. Since $\sqrt{t}\, g(y, t-s) \to \sqrt{2/\pi}\, y^+$ as $t \to \infty$, and $\sqrt{t}\, g(y, t-s) \le 2y^+$ for $t \ge 2s$, dominated convergence and $y^+ \ge y$ give

$$\liminf_{t\to\infty} \sqrt{t}\, P(H_b > t) \ge \sqrt{\tfrac{2}{\pi}}\big(c_1 P(H_b > s) + A(s)\big),$$
$$\limsup_{t\to\infty} \sqrt{t}\, P(H_b > t) \le \sqrt{\tfrac{2}{\pi}}\big(c_2 P(H_b > s) + A(s)\big).$$

The upper bound at $s = 1$ and the lower bound at every s bound $A(s)$ uniformly, so, since $b(H_b) \ge -K$, monotone convergence makes $b(H_b)$ integrable with $A(s) \to \mathbb{E}\, b(H_b)$. As $s \to \infty$, c_1 and c_2 stay bounded and $P(H_b > s) \to 0$, so both bounds tend to $\sqrt{2/\pi}\, \mathbb{E}\, b(H_b)$. □

The value of $\mathbb{E}\, b(H_b)$ is fixed by the tail constant of the law, not by the level at which the boundary settles. The exact boundary of T_1 is continuous, and as computed it stays within 10^{-3} of b_∞ on $[1, 100]$; its boundedness on $[1,\infty)$, which the lemma needs, is observed, not proved. Granting it, the lemma and the tail (9.1.3) give

$$\mathbb{E}\, b(T_1) = \lim_{t\to\infty} \mathbb{E}\big[(b_\infty - B_t)\mathbf{1}\{H_b > t\}\big] = \sqrt{\frac{\pi}{2}} \times \frac{\sum_\rho \rho^{-1}}{\sqrt{\pi}} = \frac{1}{\sqrt{2}} \sum_\rho \frac{1}{\rho} = 0.0163311,$$

the first equality by optional stopping, since $P(H_b > t) \to 0$. So, if the boundary is bounded on $[1,\infty)$, Wald's identity $\mathbb{E}B_{H_b} = 0$, which holds for every crossing time of finite mean, fails here, and the defect is exactly $\sum_\rho \rho^{-1}/\sqrt{2}$. Numerically, integrating the computed boundary against the law of T_1 gives $\mathbb{E}\, b(T_1) = 0.0163370$, and in the simulation described next $\mathbb{E}[(b(t) - B_t)\mathbf{1}\{H_b > t\}]$, which tends to $\mathbb{E}\, b(T_1)$ for the same reason, is 0.0161, 0.0158 and 0.0161 at $t = 0.25$, 0.5 and 1. The defect is also the level of the stable summand of T_1: since $\mathbb{E}e^{-uH_a} = e^{-a\sqrt{2u}}$, the variable S of Proposition 9.1.4 has the law of the first passage time $H_{b_1/\sqrt{2}}$ (Section 10.2.4).

We simulated 1.5×10^5 Brownian paths against the exact boundary on $[0,1]$, with a Brownian bridge crossing correction, using `rs1d.py`. The Kolmogorov distance between the simulated and the exact survival functions is 0.0019, at the level of the Monte Carlo noise, and the survival probabilities at $t = 0.02$, 0.05, 0.5 and 1 are 0.748, 0.086, 0.0184 and 0.0129 against the exact 0.747, 0.087, 0.0189 and 0.0132. Figure 9.7, drawn by `figs5.py`, shows the boundary in its three zones and the survival check.

So at the basepoint one the shifted clock is, to Monte Carlo accuracy, the first passage time of Brownian motion across a boundary that starts near 0.5, falls

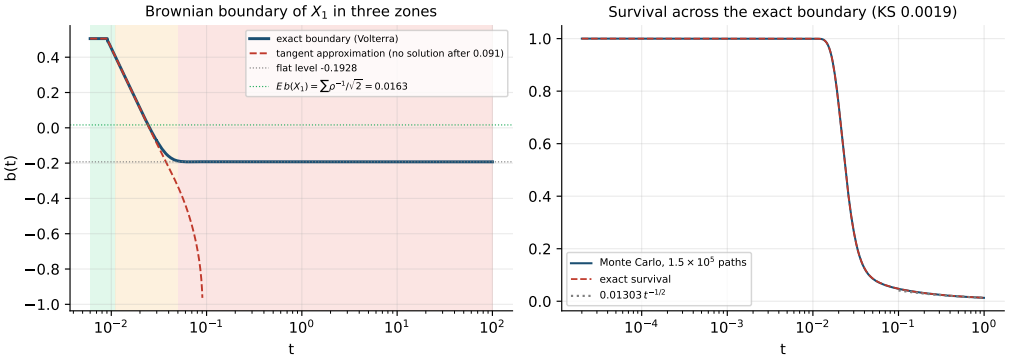


Figure 9.7: Left: the exact Brownian boundary of T_1, written X_1 in the plot, from the Volterra equation over the three zones of Section 8.2.2.2 (shaded: small time, bulk, right tail), the tangent approximation, which has no solution beyond $t = 0.091$, the flat level -0.19277, and the value $\mathbb{E}\,b(T_1) = \sum_\rho \rho^{-1}/\sqrt{2} = 0.0163$. Right: survival of 1.5×10^5 simulated Brownian paths across the exact boundary against the exact survival function of T_1 and the tail $0.01303\,t^{-1/2}$; the Kolmogorov distance is 0.0019.

through the bulk, and becomes flat at -0.19277; the flatness is the Brownian form of the stable one half tail, and, if the boundary stays bounded, its signature is the failure of Wald's identity by exactly $\sum_\rho \rho^{-1}/\sqrt{2}$.

9.2.5 Every basepoint above the centre

Nothing in Theorem 9.2.2 is special to the basepoint one. Since $(\alpha - \frac{1}{2} + \sqrt{u})^2 = (\alpha - \frac{1}{2})^2 + u + 2(\alpha - \frac{1}{2})\sqrt{u}$, the same computation applies at every basepoint to the right of the centre.

Theorem 9.2.5. *For every $\alpha > \frac{1}{2}$, with no hypothesis on the zeros (the existence of the centre clock, Theorem 5.1.1, uses the verification of RH below height 1000), $\varphi_\alpha(u) = \xi(\alpha)/\xi(\alpha + \sqrt{u})$ is the Laplace transform of $T_\alpha \overset{d}{=} \sigma^{(\alpha)}_{\widetilde{T}_\alpha}$, where $\widetilde{T}_\alpha$ is the centre clock tilted by $e^{-(\alpha-\frac{1}{2})^2 T}$ and $\sigma^{(\alpha)}$ is an independent subordinator with exponent $u + 2(\alpha - \frac{1}{2})\sqrt{u}$, that is $T_\alpha \overset{d}{=} \widetilde{T}_\alpha + H_{\sqrt{2}(\alpha-\frac{1}{2})\widetilde{T}_\alpha}$. The law has infinite mean and $P(T_\alpha > x) \sim \frac{\xi'}{\xi}(\alpha)\,(\pi x)^{-1/2}$. The variable T_α is a GGC if and only if no zero has $\beta > \alpha$, in particular for every $\alpha \geq 1$, and when every zero has $\beta < \alpha$ its Thorin density in the chart $\theta = \sqrt{z}$ is $\frac{1}{\pi}\sum_\rho (\alpha - \beta)/((\alpha - \beta)^2 + (\theta - \gamma)^2)$.*

Proof. The subordination is the computation of Theorem 9.2.2 with $\frac{1}{4}$ replaced by $(\alpha - \frac{1}{2})^2$ and $\sqrt{u}$ by $2(\alpha - \frac{1}{2})\sqrt{u}$, and H_a has exponent $a\sqrt{2u}$ with $a = \sqrt{2}(\alpha - \frac{1}{2})\ell$. The tail follows as in Section 9.1.2 from $\varphi_\alpha(u) = 1 - \frac{\xi'}{\xi}(\alpha)\sqrt{u} + O(u)$. The Thorin

density and the GGC criterion are the threshold theorem, Theorem 1.2.2: when every zero has $\beta < \alpha$ each Cauchy term is positive, and a zero with $\beta > \alpha$ puts a pole of φ_α at the point $(\rho-\alpha)^2$ of $\mathbb{C}\setminus(-\infty,0]$ (its principal root is $\rho-\alpha$, since $\operatorname{Re}(\rho-\alpha)>0$), where the Laplace transform of a GGC is holomorphic, whatever the sign of the density near its ordinate (Remark 1.2.4). In the boundary case, when some zero has $\beta=\alpha$ and none has $\beta>\alpha$, every zero has $\beta<\alpha'$ for $\alpha'>\alpha$, so $T_{\alpha'}$ is a GGC; as $\alpha'\downarrow\alpha$, $\varphi_{\alpha'}\to\varphi_\alpha$ pointwise on $[0,\infty)$, so $T_{\alpha'}\to T_\alpha$ in law, and T_α is a GGC by the closure theorem of Section 4.1.1. There the Cauchy kernel of a zero $\alpha+i\gamma$, $\gamma>0$, of multiplicity m becomes an atom: with its conjugate it contributes the factor $(1+u/\gamma^2)^{-m}$ to φ_α, a Thorin atom of mass m at $z=\gamma^2$. □

The subordination also gives the Lévy measure of T_α in closed form, and with it the properties of the centre clock that need no hypothesis.

Corollary 9.2.6. *Let $\alpha>\frac12$ and $a=\alpha-\frac12$. Then T_α is infinitely divisible, with no drift and Lévy measure $W_\alpha(x)\,dx/x$, where*

$$\begin{aligned} W_\alpha(x) &= \frac{ax}{\sqrt{\pi}}\int_0^x (x-s)^{-3/2}e^{-a^2xs/(x-s)}\,W(s)\,ds \\ &= \mathbb{E}\Big[(1+4xV)^{-1/2}\,W\Big(\frac{x}{1+4xV}\Big)\Big], \end{aligned}$$

with $V=a^2/(2Z^2)$ and Z standard normal. Hence $W_\alpha>0$ and $W_\alpha'<0$ on $(0,\infty)$, so that T_α is self decomposable; and on every interval $(0,T]$ on which W is completely monotone, so is W_α.

Proof. By Theorem 9.2.5, T_α is the subordinator $\sigma_s^{(\alpha)}=s+H_{\sqrt{2}as}$ evaluated at the independent time $\widetilde{T}_\alpha$, which is infinitely divisible with no drift and Lévy measure $e^{-a^2s}W(s)\,ds/s$, and $H_{\sqrt{2}as}$ has the density $(as/\sqrt{\pi})y^{-3/2}e^{-a^2s^2/y}$. By Bochner's subordination [232, Theorem 30.1], T_α is infinitely divisible with no drift and Lévy measure $\int_0^\infty P(\sigma_s^{(\alpha)}\in dx)\,e^{-a^2s}W(s)\,ds/s$, which, since $a^2s^2/(x-s)+a^2s=a^2xs/(x-s)$, is $W_\alpha(x)\,dx/x$ with the first form of W_α. The substitution $s=x/(1+4xv)$ turns the integral into an average against the density $\frac{a}{2\sqrt{\pi}}v^{-3/2}e^{-a^2/4v}$ of V, which is the second form. There the derivative in x of $(1+4xv)^{-1/2}W(x/(1+4xv))$ is $-2v(1+4xv)^{-3/2}W+(1+4xv)^{-5/2}W'$, negative because $W>0$ and $W'<0$ (Theorem 5.1.1), and differentiation under the average is legitimate, because $W^{(n)}(s)=O(s^{-n-1/2}\log(1/s))$ as $s\to0$ makes every derivative in x of the integrand $O(\log(1+V))$ on compact subsets of $(0,\infty)$, and $\mathbb{E}\log(1+V)<\infty$. Finally, $x\mapsto x/(1+4xv)$ has a completely monotone derivative and maps $(0,T]$ into itself, and $(1+4xv)^{-1/2}$ is completely monotone, so Faà di Bruno's formula and Leibniz's rule carry complete monotonicity of W on $(0,T]$ to every term of the average. □

With Theorem 5.1.7, W_α is therefore completely monotone on $(0,1.29]$ for every $\alpha>\frac12$. When no zero has real part above α, in particular for every $\alpha\ge1$, T_α is a

GGC and W_α is the Laplace transform k_α of its Thorin measure (Section 11.1.6); when some zero has real part above α, T_α is not a GGC, so W_α is not completely monotone on $(0,\infty)$, although it is on $(0, 1.29]$.

So for every basepoint above the centre the reciprocal Xi function is the Laplace transform of a Brownian hitting time built from the centre clock; what distinguishes the basepoints is not whether a hitting time exists but whether it is a GGC. In particular the variable T_α of Theorem 1.2.2 exists for every $\alpha > \frac{1}{2}$, whether or not it is a GGC, and the signed measure μ_ε that the book computes at the resolution $\varepsilon = \alpha - \frac{1}{2}$ is its Thorin measure when every zero lies to the left of the line $\operatorname{Re} s = \alpha$ (Theorem 1.2.2), while a zero to the right of that line prevents T_α from being a GGC even if μ_ε is positive (Remark 1.2.4), and then v_α/π is not a Thorin density of T_α in any sense (Proposition 11.4.3). Table 9.12, computed by `improve.py`, gives the constants and the minimum of the density v_α/π on $[0, 60]$ for three basepoints, and the right panel of Figure 9.6 shows the densities. The minima, attained at the origin, agree with the margin $0.0147\,\varepsilon$ of Section 3.1 at the resolutions $\varepsilon = \alpha - \frac{1}{2}$. At $\alpha = 1$ the minimum at the origin holds at every height, by Proposition 9.1.4, and so it does at every $\alpha \geq 1$, by Proposition 9.1.5; at $\alpha = 0.6$ and 0.75 it holds on the verified window $|\theta| \leq 3 \times 10^{12} - 1.21$, by Theorem 9.1.11(ii), and that it holds at every height for every $\alpha \in (\frac{1}{2}, 1)$ is equivalent to RH, by part (vi) of that theorem. As $\alpha \downarrow \frac{1}{2}$ the tail constant $\frac{\xi'}{\xi}(\alpha)/\sqrt{\pi}$ tends to zero, because $\xi'(\frac{1}{2}) = 0$, and the law acquires the finite mean $\sum_k a_k^{-1}$ of the centre clock; at the same time the subordinator tends to pure drift and the Cauchy widths to $\frac{1}{2} - \beta$, which is where RH is decided.

α	$\frac{\xi'}{\xi}(\alpha)$	tail constant $\frac{\xi'}{\xi}(\alpha)/\sqrt{\pi}$	$\min_{[0,60]} \rho_{\alpha-1/2}$
0.6	0.00462092	0.002607	0.00147
0.75	0.01155134	0.006517	0.00368
1	0.02309571	0.013030	0.00735

Table 9.12: Basepoints above the centre: the drift $\frac{\xi'}{\xi}(\alpha)$, the tail constant of T_α, and the minimum on $[0, 60]$, attained at the origin, of the density $\rho_{\alpha-1/2} = v_\alpha/\pi$, the Thorin density of T_α when every zero lies to the left of the line $\operatorname{Re} s = \alpha$. At $\alpha = 1$ the minimum at the origin holds at every height (Proposition 9.1.4), as it does at every $\alpha \geq 1$ (Proposition 9.1.5); for $\frac{1}{2} < \alpha < 1$ the minimum at the origin holds on the verified window $|\theta| \leq 3 \times 10^{12} - 1.21$ (Theorem 9.1.11(ii)), and at every height for every such α it is equivalent to RH (part (vi) of that theorem).

9.2.6 Boundaries across the basepoints

Theorem 9.2.5 gives the law of T_α in a form that can be evaluated to relative accuracy far into the tail, so the exact boundary of Section 9.2.4 can be computed

at every basepoint above the centre, and the three zones can be compared across the basepoints. We solved the Volterra equation (8.2.1) [192] at $\alpha = 0.6$ and 0.75 with `bnd_alpha.py` of `basepoint_one/process_map`, which writes `bnd_0.6.json` and `bnd_0.75.json`, forward in time and in the complementary form once the survival probability falls below one half. As at the basepoint one, the distribution function comes from Talbot inversion of $\varphi_\alpha(u)/u$ for $t < 0.03$ and from the subordination identity beyond, here with 3×10^5 draws of $\widetilde{T}_\alpha$, made under RH as a sum of exponential variables over the first 200 zeros and a gamma variable with the mean and variance of the remaining terms. The same folder holds copies of the boundary at the basepoint one, `rs1.json`, and of the Thorin density grid there, `alpha1.json`, both from `basepoint_one/xi`, and of the boundary of the centre clock, `volterra.json`, written by `code/volterra.py` for Section 8.2. The bottom left panel of Figure 11.3 in Section 11.1 draws the four boundaries together.

In the small time zone and the bulk the boundaries at the different basepoints are close: they start near the reflection level 0.50, and at $t = 0.02$ they are at 0.072 for the centre and at 0.080, 0.092 and 0.111 for $\alpha = 0.6$, 0.75 and 1. They differ in the right tail. At the centre the survival function is exponential with rate γ_1^2, and the boundary is asymptotically linear with slope $-\sqrt{2}\gamma_1 = -19.99$ and the logarithmic correction $\frac{3}{2\sqrt{2}\gamma_1} \log t$; the slope of the computed boundary over $[0.2, 0.3]$ is -19.77 (Section 8.2.2.2). Above the centre the tail is stable one half, and the exact boundaries flatten, at -0.33384 for $\alpha = 0.6$, -0.25537 for $\alpha = 0.75$ and -0.19277 for $\alpha = 1$. Each boundary stays within 0.011 of its level at $t = 100$, the end of the computation, from $t = 0.1$ on, and within 10^{-3} from $t = 1$ on. As $\alpha \downarrow \frac{1}{2}$ the flat level moves down, towards the linear boundary of the centre.

The Wald type identity of Section 9.2.4 extends to every basepoint above the centre. The density of T_α is again continuous and positive on $(0, \infty)$, so its boundary is continuous [73, Theorem 8.2], and as computed it is bounded up to $t = 100$. Granting boundedness at infinity, which here too is observed, not proved, Lemma 9.2.4 and the stable tail $P(T_\alpha > t) \sim b_\alpha (\pi t)^{-1/2}$ of Theorem 9.2.5 give $\mathbb{E}\, b(T_\alpha) = b_\alpha/\sqrt{2}$, so Wald's identity $\mathbb{E} B_{H_b} = 0$ fails by exactly $b_\alpha/\sqrt{2}$. Table 9.13 compares the identity with the value obtained by integrating each computed boundary against the law of T_α, and the tail constant with the computed survival function at $t = 100$; all six pairs agree to within 6×10^{-4} in relative terms. The reason given in Section 9.2.4 for the failure of the tangent approximation in the right tail holds at every basepoint above the centre: the approximation matches the local crossing rate as if the path had just started, and it cannot represent the survivors that have wandered below the boundary.

So the shifted clock is a hitting time, though not of the kind Kent's theorem describes directly. Kent's discrete form needs unit atoms at distinct points, and a Brownian first passage to a random level is ruled out by the growth of $1/\xi$ along vertical lines; but the subordination $T_1 = \sigma_{\widetilde{T}}$ of the tilted centre clock by the inverse local time of a sticky Brownian motion holds exactly and with no

α	flat level	$\mathbb{E}\, b(T_\alpha)$	$b_\alpha/\sqrt{2}$	$\sqrt{100}\, P(T_\alpha > 100)$	$b_\alpha/\sqrt{\pi}$
0.6	-0.33384	0.0032687	0.0032675	0.0026083	0.0026071
0.75	-0.25537	0.0081710	0.0081680	0.0065203	0.0065171
1	-0.19277	0.016337	0.016331	0.013037	0.013030

Table 9.13: Exact Brownian boundaries above the centre: the flat level, the mean boundary value $\mathbb{E}\, b(T_\alpha)$ at the crossing, obtained by integrating the computed boundary against the law of T_α, against the Wald type identity, and the computed $\sqrt{100}\, P(T_\alpha > 100)$ against the tail constant of Theorem 9.2.5. Computed by `bnd_alpha.py` for $\alpha = 0.6$ and 0.75 and by `rs1.py` for $\alpha = 1$.

hypothesis, and so does its analogue at every basepoint above the centre. The subordination yields a closed form for the density; the Thorin measure is the sum over the zeros of the laws of the places at which planar Brownian motions started at the zeros first reach the line $\operatorname{Re} s = 1$; and under RH with simple zeros the tilted clock is itself Kent's hitting time, with spectrum $\{|\rho|^2\}$, so that T_1 becomes the hitting time of a level by a string diffusion run on a sticky Brownian local time clock. Its Brownian boundary, as computed, is flat where that of the centre clock is linear, and if it is bounded its Wald defect is fixed by b_α. All of this reads $\varphi_\alpha(u)$ for $u \geq 0$ as a Laplace transform; Section 10.1 evaluates the same functions at u^2, where they become characteristic functions on the whole line.

Chapter 10

The critical strip and Riesz's criterion

Chapter 9 read the reciprocal Xi function at a basepoint α as a Laplace transform on the half line, $\varphi_\alpha(u) = \xi(\alpha)/\xi(\alpha + \sqrt{u})$ for $u \geq 0$. At the basepoint one its Thorin measure is the Poisson smoothing of the zeros and its Lévy form carries the primes, and by Theorem 9.2.5 it is, for every $\alpha > \frac{1}{2}$ and with no hypothesis, the Laplace transform of a positive random variable T_α obtained by subordinating the centre clock. The same function has two further readings, as the characteristic function of a normal scale mixture and through the completeness of the translates of the tilted Pólya kernels. Both run across the whole critical strip, not only to the right of the zeros, and so show how much of what holds with no hypothesis at the basepoint one survives inside the strip. Section 10.1 develops the two readings in turn and ends with Riesz's criterion, which reads the zero free half planes through growth rather than completeness. Section 10.2 writes that criterion in the language of the clocks, where it becomes the statement that every T_α with $\alpha > \frac{1}{2}$ is a GGC, and shows why none of the resulting representations of its coefficients supplies the growth bound that the criterion needs. Section 10.3 turns to the squarefree integers, whose Dirichlet series $\zeta(s)/\zeta(2s)$ carries the same reciprocal $1/\zeta(2s)$: the constant of $\sum_{n \leq x} \mu(n)^2/n$ is a combination of two tilted Pólya means, and smoothing the count by a gamma law gives criteria for RH of the same kind as Riesz's.

For the scale mixtures and the completeness statements, as for the Thorin measure and the Lévy form in Chapter 9, what is unconditional at the basepoint one extends to every basepoint to the right of the line $\operatorname{Re} s = \alpha$, for a given $\alpha < 1$, exactly when no zero lies to the right of that line; required for every $\alpha > \frac{1}{2}$, this is RH. Chapters 9 to 11 prove no new zero free region.

10.1 Normal scale mixtures and completeness across the strip

The first of the two readings evaluates the Laplace transform φ_α at u^2. Since $\varphi_\alpha(u^2) = \xi(\alpha)/\xi(\alpha + |u|)$, the reciprocal of ξ along the real axis is a characteristic function, that of a normal scale mixture, at every basepoint and with no hypothesis

(Section 10.1.1); what the zeros decide is the class of the law. The sine squared lemma turns its Lévy form in $|u|$ into a Thorin form in u^2 (Section 10.1.2), and the law is an extended generalised gamma convolution at and to the right of the basepoint one, at a basepoint inside the strip exactly when the half plane to its right is free of zeros, at the centre exactly when RH holds, and unconditionally up to a remainder of relative size $\log T_0/T_0$, where $T_0 = 3\times 10^{12}$ is the height to which the zeros have been verified (Section 10.1.3). Its Lévy density, multiplied by $|y|$, is the Laplace transform of the Thorin density, a Poisson trace of the zeros that is tied to the heat trace W by stable one half subordination with no hypothesis (Section 10.1.4). Section 10.1.5 writes the complete monotonicity of the Poisson trace at the centre, which is RH, as conditions indexed by an order n and a point y, proves those with $n < T_0(\pi - y)$ with no hypothesis, and compares the trace with that of Pólya's Bessel function, whose complete monotonicity is a theorem.

The second reading is completeness. The Fourier transform of Pólya's kernel tilted to the basepoint α is $\xi(\alpha+i\theta)$, so by Wiener's closure theorem the translates of the tilted kernel are complete in $L^1(\mathbb{R})$ exactly when no zero lies on the line $\operatorname{Re} s = \alpha$. The zero free half planes of Chapter 9 thus become completeness statements, with an explicit obstruction at the centre and an explicit cost of approximation above it (Section 10.1.6). Salem's criterion is the same statement for a Fermi Dirac kernel (Section 10.1.7), and the criteria of Riesz and of Nyman and Beurling are its growth and multiplicative counterparts (Section 10.1.8).

We keep the notation of Chapter 9: T is the centre clock and $T_{1/2} = T$, $\widetilde{T}_\alpha$ is the centre clock tilted by $e^{-(\alpha-\frac12)^2 T}$, H_a is the first passage time of a standard Brownian motion to the level a, $b_\alpha = \frac{\xi'}{\xi}(\alpha)$ and $v_\alpha(\theta) = \operatorname{Re}(\xi'/\xi)(\alpha+i\theta)$. When every zero lies to the left of the line $\operatorname{Re} s = \alpha$, the Thorin density of T_α in the chart $\theta = \sqrt{z}$ is v_α/π, the density $\rho_{\alpha-1/2}$ of (1.2.7) (Theorem 1.2.2); when a zero lies to the right of that line, T_α is not a GGC and v_α/π is not a Thorin density for it (Proposition 11.4.3). Throughout, Z is a standard normal variable independent of everything else. The scripts are in the folders `basepoint_one/scale_mixture` and `basepoint_one/wiener` of the archive, and we name them at each table and figure.

10.1.1 The reciprocal on the real axis is a characteristic function

Evaluating a Laplace transform at the square of a real variable produces the characteristic function of a normal scale mixture: if $X \geq 0$ is independent of Z, then $\mathbb{E}e^{iu\sqrt{2X}Z} = \mathbb{E}e^{-u^2X}$ for real u. The centre clock T of [198], with $\mathbb{E}e^{-uT} = \xi(\frac12)/\xi(\frac12+\sqrt{u})$, is a positive random variable, infinitely divisible and self decomposable with no hypothesis (Theorem 5.1.1, whose proof uses the verification of RH below height 1000). For $\alpha > \frac12$, Theorem 9.2.5 writes $T_\alpha \overset{d}{=} \sigma^{(\alpha)}_{\widetilde{T}_\alpha} \overset{d}{=} \widetilde{T}_\alpha + H_{\sqrt{2}(\alpha-\frac12)\widetilde{T}_\alpha}$, where $\sigma^{(\alpha)}$ is an independent subordinator with Laplace expo-

nent $u + 2(\alpha - \frac{1}{2})\sqrt{u}$, the inverse local time at zero of a sticky Brownian motion. So T_α also exists with no hypothesis, and with $X = T_\alpha$ the evaluation gives the following.

PROPOSITION 10.1.1. *For every $\alpha \geq \frac{1}{2}$ the function $u \mapsto \varphi_\alpha(u^2) = \xi(\alpha)/\xi(\alpha + |u|)$ is the characteristic function of $Y_\alpha = \sqrt{2T_\alpha}\, Z$. The law of Y_α is symmetric, unimodal and infinitely divisible with no hypothesis. At the centre it has variance $2\mathbb{E}T = 2\sum_k a_k^{-1} = 0.046210$. For $\alpha > \frac{1}{2}$ its characteristic function has a corner at the origin, $\xi(\alpha)/\xi(\alpha + |u|) = 1 - b_\alpha|u| + O(u^2)$, so Y_α lies in the domain of attraction of the Cauchy law, with $P(|Y_\alpha| > y) \sim 2b_\alpha/(\pi y)$ as $y \to \infty$.*

Proof. The first statement is the evaluation $\mathbb{E}e^{iu\sqrt{2X}Z} = \mathbb{E}e^{-u^2X}$ with $X = T_\alpha$. Symmetry and unimodality hold for every normal scale mixture, which is a mixture of centred normal laws. The tilt $\widetilde{T}_\alpha$ of the infinitely divisible law of T is infinitely divisible, with Lévy measure $e^{-(\alpha-\frac{1}{2})^2 t} W(t)\, dt/t$, and a subordinator run for an independent infinitely divisible time is infinitely divisible, because a Bernstein function composed with a Bernstein function is again one [234, Corollary 3.8]. So T_α is infinitely divisible, and Y_α, which is a Brownian motion run for the independent time $2T_\alpha$, is infinitely divisible for the same reason. At the centre $\xi'(\frac{1}{2}) = 0$, so $\xi(\frac{1}{2})/\xi(\frac{1}{2} + |u|) = 1 - \mathbb{E}T\, u^2 + O(u^4)$, with $\mathbb{E}T = \sum_k a_k^{-1}$ by (1.2.1). For $\alpha > \frac{1}{2}$ the expansion follows from $\xi(\alpha + |u|) = \xi(\alpha)(1 + b_\alpha|u| + O(u^2))$. The tail comes from that of T_α: by Theorem 9.2.5, $P(T_\alpha > t) \sim b_\alpha(\pi t)^{-1/2}$ as $t \to \infty$, so that, since $P(T_\alpha > t) \leq 1$, also $P(T_\alpha > t) \leq Kt^{-1/2}$ for all $t > 0$ and some constant K. Conditioning on Z, $yP(|Y_\alpha| > y) = \mathbb{E}\big[yP(T_\alpha > y^2/(2Z^2) \mid Z)\big]$, and the integrand is at most $\sqrt{2}K|Z|$ and tends to $\sqrt{2/\pi}\, b_\alpha|Z|$ as $y \to \infty$; by dominated convergence $yP(|Y_\alpha| > y) \to \sqrt{2/\pi}\, b_\alpha \mathbb{E}|Z| = 2b_\alpha/\pi$. □

At the centre $Y_{1/2} = Z\sqrt{2T}$ is the second member of the van Dantzig pair of Section 1.2; it has the same variance as the Pólya variable X, the first member, as the relation $\mathbb{E}e^{-\theta^2 T} = 1/\mathbb{E}e^{\theta X}$ requires. What RH decides is not whether $\xi(\frac{1}{2})/\xi(\frac{1}{2} + |u|)$ is a characteristic function, which it always is, but which class it lies in, and that is the subject of Sections 10.1.2 and 10.1.3. Table 10.1, computed by `comp.py`, checks the proposition by Monte Carlo, drawing T_α from the subordination identity with 4×10^5 draws. The tilted clock $\widetilde{T}_\alpha$ is drawn as a sum of independent exponential variables with rates $\gamma^2 + (\alpha - \frac{1}{2})^2$ over the first 200 zeros, which lie on the critical line, plus a gamma variable with the mean and variance of the remaining factors, and H_a is drawn as a^2/Z'^2 with Z' standard normal.

With no hypothesis the law of $Y_{1/2}$ is also self decomposable. For $0 < c < 1$ the self decomposability of T (Theorem 5.1.1) gives $T \stackrel{d}{=} cT' + R_c$ with $T' \stackrel{d}{=} T$ and $R_c \geq 0$ independent, and then $Y_{1/2} \stackrel{d}{=} \sqrt{c}\, Y' + Z'\sqrt{2R_c}$, where $Y' = Z\sqrt{2T'}$ has the law of $Y_{1/2}$ and Z' is standard normal and independent of (Z, T', R_c).

Under RH each factor $(1 + u^2/\gamma^2)^{-1}$ of $\xi(\frac{1}{2})/\xi(\frac{1}{2} + |u|)$ is a characteristic function, that of L/γ for a standard Laplace variable L, and a zero off the line

α	u	$\mathbb{E}\cos(uY_\alpha)$, Monte Carlo	standard error	$\xi(\alpha)/\xi(\alpha+\lvert u\rvert)$
0.5	1	0.977101	0.000053	0.977178
0.5	5	0.566634	0.000768	0.567395
0.75	1	0.966125	0.000176	0.965980
0.75	5	0.536687	0.000812	0.536769
1	1	0.954736	0.000246	0.954930
1	5	0.507781	0.000851	0.507962

Table 10.1: Proposition 10.1.1 by simulation, computed by `comp.py`. At the centre the sample variance of $Y_{1/2}$ is 0.04637 against 0.04621. The Cauchy tails $y\,P(|Y_\alpha| > y)$ at $y = 10$ are 0.0144 at $\alpha = 1$ and 0.0066 at $\alpha = 0.75$, against $2b_\alpha/\pi = 0.0147$ and 0.0074.

would break this. The quadruple $\frac12 \pm\delta \pm i\gamma$, with $\gamma > 0$ and $0 < \delta < \frac12$, contributes to $\xi(\frac12)/\xi(\frac12+|u|)$, by (1.2.1), the factor $g(u) = |1+u^2/(\gamma-i\delta)^2|^{-2}$, which is positive and integrable with $g(0) = 1$, and

$$\frac{1}{2\pi}\int_{\mathbb{R}} g(u)e^{-iux}\,du = \frac{\gamma^2+\delta^2}{4\gamma\delta}\,e^{-\gamma|x|}\big(\gamma\sin\delta|x| + \delta\cos\delta|x|\big).$$

For $x > 0$ the derivative of the right side is $-\frac{(\gamma^2+\delta^2)^2}{4\gamma\delta}e^{-\gamma x}\sin\delta x$, so its least value, attained at $x = \pi/\delta$, is $-e^{-\pi\gamma/\delta}$ times its value $(\gamma^2+\delta^2)/(4\gamma)$ at $x = 0$. Since g is integrable, it could only be the characteristic function of the law with this density, which is negative there. So g is not a characteristic function, and $\xi(\frac12)/\xi(\frac12+|u|)$ would still be one, as $W > 0$ ensures, but no longer factor by factor. The defect is tiny. Every ordinate is at least γ_1 and $\delta < \frac12$, so the least value exceeds -9.5×10^{-39}, the value of the formula at $\gamma = \gamma_1$ and $\delta = \frac12$; and a zero off the line would lie above T_0, where $e^{-\pi\gamma/\delta} < e^{-2\pi T_0}$. The script `vdpair.py` of Section 1.2 computes these values and checks the transform against quadrature.

Away from the centre neither V_α nor Y_α belongs to a pair. Take $\alpha > \frac12$; for V_α the case $\alpha < \frac12$ follows, since $V_\alpha \overset{d}{=} -V_{1-\alpha}$. The tilted Pólya variable V_α of Section 1.2 has characteristic function $\varphi(\theta) = \xi(\alpha+i\theta)/\xi(\alpha)$, and $1/\varphi(i\theta) = \xi(\alpha)/\xi(\alpha-\theta)$ exceeds 1 for $0 < \theta < 2\alpha-1$, because $|\alpha-\theta-\frac12| < \alpha-\frac12$ there and $\xi(\frac12+s) = \xi(\frac12)\mathbb{E}\cosh(sX)$ increases with $|s|$; at $\alpha = 0.75$ and $\theta = 0.25$ it is 1.00145. So it is not a characteristic function. By its Cauchy tail Y_α has no exponential moments, so it cannot be a first member. As a second member its partner φ would satisfy $\varphi(i\theta) = \xi(\alpha+|\theta|)/\xi(\alpha)$, and a moment generating function finite on the whole line is analytic, while $\xi(\alpha+|\theta|)/\xi(\alpha) = 1 + b_\alpha|\theta| + O(\theta^2)$ has a corner at $\theta = 0$, since $b_\alpha > 0$ (Proposition 10.1.1). At the centre the bound by 1 and the corner both disappear, because $\xi(\frac12-\theta) = \xi(\frac12+\theta)$ and so $b_{1/2} = 0$: the functional equation is what permits the pair, and Theorem 5.1.1 is what supplies it.

10.1.2 From the Lévy form in $|u|$ to the Thorin form in u^2

For $\alpha \geq 1$ the function $\log(\xi(\alpha+s)/\xi(\alpha))$ has, for $\operatorname{Re} s > 0$, the Lévy form (9.1.5) with b_α and Π_α in place of b_1 and Π_1, where $x\Pi_\alpha(dx) = \big(\frac{e^{-(\alpha+2)x}}{1-e^{-2x}} - e^{-(\alpha-1)x}\big)dx + \sum_n \Lambda(n)n^{-\alpha}\delta_{\log n}(dx)$ is the signed measure that Section 9.1.4 gives for every basepoint $\alpha \geq 1$. The integral converges absolutely for $\alpha > 1$, and at $\alpha = 1$ as a compensated limit, by the prime number theorem. Putting $s = |u|$ gives

$$\frac{\xi(\alpha)}{\xi(\alpha+|u|)} = e^{-b_\alpha|u|}\exp\Big\{-\int_0^\infty \big(e^{-|u|x} - 1 + |u|x\big)\Pi_\alpha(dx)\Big\}, \tag{10.1.1}$$

a representation in $|u|$. The first factor is the characteristic function of a Cauchy law of scale b_α. The second is not in Lévy Khintchine form, because $e^{-|u|x}$ is a function of $|u|$ rather than of u. The sine squared lemma (1.2.9) converts it. In the chart $\theta = \sqrt{z}$ and at $s = |u|$, as in the derivation of (9.1.6), it writes $e^{-|u|x}+|u|x-1$ as $\frac{2x}{\pi}\int_0^\infty \log(1+u^2/\theta^2)\sin^2(x\theta/2)\,d\theta$ for $x > 0$, and the drift converts through $|u| = \frac{1}{\pi}\int_0^\infty \log(1+u^2/\theta^2)\,d\theta$. Both identities follow from $\int_0^\infty \log(1+u^2/\theta^2)\cos(x\theta)\,d\theta = \pi(1-e^{-|u|x})/x$.

By Section 1.3.2, a symmetric law is an extended generalised gamma convolution (EGGC) if and only if it is the law of $Z\sqrt{2T'}$ with T' a GGC independent of Z, and at the centre $Y_{1/2} = Z\sqrt{2T}$ is an EGGC exactly when RH holds. The next theorem carries this to every basepoint above the centre; Section 10.1.3 describes the class through its characteristic functions.

Theorem 10.1.2. *Let $\alpha \geq 1$, or more generally let $\alpha > \frac{1}{2}$ be such that $\operatorname{Re}\rho < \alpha$ for every zero ρ. Then*

$$\frac{\xi(\alpha)}{\xi(\alpha+|u|)} = \exp\Big\{-\frac{1}{\pi}\int_0^\infty \log\Big(1+\frac{u^2}{\theta^2}\Big)v_\alpha(\theta)\,d\theta\Big\}, \tag{10.1.2}$$

where

$$v_\alpha(\theta) = b_\alpha + \int_0^\infty 2\sin^2\Big(\frac{x\theta}{2}\Big)x\Pi_\alpha(dx) = \operatorname{Re}\frac{\xi'}{\xi}(\alpha+i\theta) > 0,$$

the first expression being available for $\alpha \geq 1$, and as the compensated limit (9.1.8) *(in which $\varepsilon = 1-\alpha$) when $\sup_\rho \operatorname{Re}\rho < \alpha$, so Y_α is a symmetric EGGC with Thorin density v_α/π in the chart θ. With no hypothesis on the zeros, for $\frac{1}{2} < \alpha < 1$ the law of Y_α is an EGGC if and only if ζ has no zeros with real part greater than α, and at the centre if and only if RH holds. Under RH,*

$$\frac{\xi(\frac{1}{2})}{\xi(\frac{1}{2}+|u|)} = \prod_{\gamma>0}\Big(1+\frac{u^2}{\gamma^2}\Big)^{-1}, \qquad Y_{1/2} \stackrel{d}{=} \sum_k \frac{L_k}{\gamma_k},$$

with independent standard Laplace variables L_k, of density $\frac{1}{2}e^{-|y|}$.

Proof. For $\alpha \geq 1$, insert the two identities of the lemma into (10.1.1) and exchange the order of integration, which is justified for $\alpha > 1$ by absolute convergence and at $\alpha = 1$ by the compensation argument of Proposition 9.1.3. This gives the first expression for v_α, and the second is its identification, (9.1.6) at $\alpha = 1$, which is Proposition 9.1.2 of Section 9.1.3, and (3.1.1) for $\alpha > 1$; for $\frac{1}{2} < \alpha < 1$ with $\sup_\rho \operatorname{Re}\rho < \alpha$ the first expression holds in the compensated form (9.1.8) of Proposition 9.1.6, in which $\varepsilon = 1 - \alpha$. In every case (10.1.2) is the Thorin representation of T_α given by Theorem 1.2.2, $\varphi_\alpha(u) = \exp\{-\int \log(1 + u/z)\, U_\alpha(dz)\}$ with U_α as in (1.2.6), evaluated at u^2 in the chart $z = \theta^2$, and $v_\alpha > 0$ because every Cauchy kernel of Proposition 1.2.3 then has positive width $\alpha - \beta$. Since T_α is a GGC, $Y_\alpha = Z\sqrt{2T_\alpha}$ is an EGGC. Conversely, if $Y_\alpha \stackrel{d}{=} Z\sqrt{2T'}$ with T' a GGC, then $\mathbb{E}e^{-u^2T'} = \mathbb{E}e^{-u^2T_\alpha}$ for all real u, so $T' \stackrel{d}{=} T_\alpha$. Hence Y_α is an EGGC exactly when T_α is a GGC, which by Theorem 1.2.2 happens exactly when $\alpha \geq \sup_\rho \operatorname{Re}\rho$. For $\frac{1}{2} < \alpha < 1$ this is the absence of zeros with real part greater than α, and at the centre, by the symmetry $\rho \mapsto 1 - \rho$, it is RH (Proposition 1.2.1). Under RH the Hadamard product (1.2.1) at $s = |u|$ is the product displayed, and each factor $(1 + u^2/\gamma^2)^{-1}$ is the characteristic function of L/γ, as in Proposition 6.2.8(iii). □

The hypothesis that every zero lies to the left of the line $\operatorname{Re} s = \alpha$ cannot be dropped from (10.1.2). By Proposition 1.2.3, v_α/π is, off the ordinates of the zeros on the line $\operatorname{Re} s = \alpha$ and whatever the zeros do, the sum over the zeros with $\gamma > 0$ of $P_{\alpha-\beta}(\theta - \gamma) + P_{\alpha-\beta}(\theta + \gamma)$, with $P_{-c} = -P_c$, and by the semigroup property of the Cauchy kernels $\int_0^\infty \frac{2s}{s^2+\theta^2}[P_c(\theta - \gamma) + P_c(\theta + \gamma)]\, d\theta = \frac{2(s+|c|)\operatorname{sgn} c}{(s+|c|)^2+\gamma^2}$ for $s > 0$ and real $c \neq 0$. Integrating in s from 0 to $|u|$ shows that a zero $\rho = \beta + i\gamma$ with $\gamma > 0$ contributes to the right side of (10.1.2) the factor $((\alpha-\beta)^2+\gamma^2)/((|u|+\alpha-\beta)^2+\gamma^2)$ when $\beta < \alpha$, which is its factor in $\xi(\alpha)/\xi(\alpha+|u|)$, but the factor $((|u| + \beta - \alpha)^2 + \gamma^2)/((\beta - \alpha)^2 + \gamma^2)$ when $\beta > \alpha$. If no zero lies on the line $\operatorname{Re} s = \alpha$, the integral with the signed density therefore converges and equals $\xi(\alpha)/\xi(\alpha + |u|)$ multiplied by

$$\prod_{\beta>\alpha,\ \gamma>0} \frac{\big((|u| + \beta - \alpha)^2 + \gamma^2\big)\big((|u| - \beta + \alpha)^2 + \gamma^2\big)}{\big((\beta - \alpha)^2 + \gamma^2\big)^2},$$

which exceeds one for $u \neq 0$ as soon as there is such a zero, since each factor equals $1+u^2(u^2+2\gamma^2-2(\beta-\alpha)^2)/((\beta-\alpha)^2+\gamma^2)^2$ and $0 < \beta-\alpha < \frac{1}{2} < \gamma$. The series of Proposition 1.2.3 may be integrated term by term, because the absolute value of the term of a zero integrates to $\log(((|\alpha-\beta|+|u|)^2+\gamma^2)/((\alpha-\beta)^2+\gamma^2)) = O(\gamma^{-2})$. In the variable $s = |u|$, continued to the right half plane, the Poisson integral replaces the pole of $\xi(\alpha)/\xi(\alpha + s)$ at $s = \rho - \alpha$ by a zero at its mirror image $\alpha - \beta + i\gamma$ across the imaginary axis, so it cannot represent φ_α, whose pole at $(\rho - \alpha)^2 \in \mathbb{C} \setminus (-\infty, 0]$ no GGC transform can have. A zero on the line $\operatorname{Re} s = \alpha$

contributes instead an atom of mass one at its ordinate, as every zero does at the centre under RH.

We draw out two features of (10.1.2). The first is the Cauchy factor. The drift b_α becomes, through $|u| = \frac{1}{\pi}\int_0^\infty \log(1+u^2/\theta^2)\,d\theta$, the constant Thorin density b_α/π, which is also the value $v_\alpha(0)/\pi$. At the basepoint one this constant is the minimum of v_1/π, attained only at $\theta = 0$ (Proposition 9.1.4), and `comp.py` confirms it on $[0, 200]$. So Y_1 splits as an independent sum of a Cauchy variable of scale $b_1 = \sum_\rho \rho^{-1} = 0.0230957$ and a symmetric EGGC with Thorin density $(v_1 - b_1)/\pi$ (Figure 10.1, left). This is the image at u^2 of the factorisation $T_1 \stackrel{d}{=} S + T_1^\circ$ of Proposition 9.1.4 into a positive stable variable of index one half and a GGC with Thorin density $(A+Q)/\pi$: since $\mathbb{E}e^{-u^2 S} = e^{-b_1|u|}$, the Cauchy summand is $Z\sqrt{2S}$, and the EGGC summand is $\sqrt{2T_1^\circ}$ times an independent standard normal variable, with characteristic function $e^{b_1|u|}\xi(1)/\xi(1+|u|) = \exp\{-\frac{1}{\pi}\int_0^\infty \log(1 + u^2/\theta^2)(v_1 - b_1)(\theta)\,d\theta\}$. The constant b_1/π is the value at $\theta = 0$ of the harmonic measure density of Section 9.2.3, the sum over the zeros of the Cauchy laws of the points where planar Brownian motions started at the zeros first reach the line $\operatorname{Re} s = 1$, and the Cauchy factor carries the whole tail of Y_1. The second feature is that the passage from $|u|$ to u^2 is exactly the passage from the Lévy form, where the primes appear as atoms at $\log n$, to the Thorin form, where the zeros appear as the centres of the Cauchy kernels in v_α. Table 10.2, computed by `comp.py`, checks (10.1.2) with v_α computed from ξ'/ξ on $[0, 200]$ and the Weyl density $\frac{1}{2\pi}\log\frac{\theta}{2\pi}$ beyond. At $\alpha = 0.75$ the formula presupposes that no zero has real part 0.75 or more; the table is consistent with that to its accuracy, and a zero at a great height γ would change the entries only by a relative amount of order u^2/γ^2.

u	Thorin form, $\alpha = 1$	$\xi(1)/\xi(1+\lvert u\rvert)$	Thorin form, $\alpha = 0.75$	$\xi(0.75)/\xi(0.75+\lvert u\rvert)$
0.5	0.982809	0.982838	0.988506	0.988520
1	0.954904	0.954930	0.965969	0.965980
2	0.871156	0.871171	0.891317	0.891317
5	0.507984	0.507962	0.536812	0.536769
20	0.0003491	0.0003487	0.0004116	0.0004110

Table 10.2: The Thorin form (10.1.2) against the reciprocal Xi function on the real axis, computed by `comp.py`.

10.1.3 Extended generalised gamma convolutions in the critical strip

Thorin introduced generalised gamma convolutions to show that the Pareto law is self decomposable [250] and that the lognormal law is infinitely divisible [251], and extended the idea to laws on the whole line [252]. Bondesson [35, Section 7.1] calls the resulting class the extended generalised gamma convolutions, and Section 1.3.2

recalls it. A symmetric law without Gaussian part is an EGGC exactly when its characteristic function has the form

$$\exp\Big\{-\int_0^\infty \log\Big(1+\frac{u^2}{\theta^2}\Big)m(d\theta)\Big\}$$

with a positive measure m on $(0,\infty)$ that satisfies the integrability conditions of a Thorin measure in the chart $z=\theta^2$; under $z=\theta^2$ it is the Thorin measure of the mixing GGC T' in the representation $Z\sqrt{2T'}$ recalled before Theorem 10.1.2. The building blocks are the symmetric laws with characteristic function $(1+u^2/\theta^2)^{-c}$, $c>0$, which are the laws of the difference of two independent gamma variables of shape c and scale $1/\theta$, and every symmetric EGGC is a weak limit of sums of independent variables of this kind. The symmetric Laplace law, $c=1$, is the block of an atom of mass one, and at the centre under RH it is the only block that occurs. Theorem 10.1.2 places Y_α in this framework: its candidate Thorin measure is $v_\alpha(\theta)\,d\theta/\pi$, and inside the strip the question is whether T_α is a GGC. That is decided by the location of the zeros and not by the sign of v_α on the single line $\operatorname{Re} s=\alpha$: for ξ it is not known whether positivity on one line excludes zeros to its right (Remark 1.2.4).

THEOREM 10.1.3. *Let $\frac12<\alpha<1$ and write $\varepsilon=\alpha-\frac12$.*

1. *With no hypothesis, $\xi(\alpha)/\xi(\alpha+|u|)$ is the characteristic function of the infinitely divisible, symmetric, unimodal law Y_α. If every zero has real part less than α, then* (10.1.2) *holds with the positive density $v_\alpha(\theta)/\pi=\frac1\pi\sum_\rho \frac{\alpha-\beta}{(\alpha-\beta)^2+(\theta-\gamma)^2}$.*

2. *Y_α is an EGGC if and only if no zero has $\beta>\alpha$, that is, if and only if ζ has no zeros in the half plane $\operatorname{Re} s>\alpha$.*

3. *Let $T_0\ge 2$ be a height up to which all zeros are known to lie on the critical line. Then*

$$\frac{\xi(\alpha)}{\xi(\alpha+|u|)}=\prod_{0<\gamma\le T_0}\frac{\varepsilon^2+\gamma^2}{(\varepsilon+|u|)^2+\gamma^2}\times\mathcal{E}_{\alpha,T_0}(u),$$

where the finite product is the characteristic function of an EGGC whose Thorin density $\sum_{0<\gamma\le T_0}[P_\varepsilon(\theta-\gamma)+P_\varepsilon(\theta+\gamma)]$ is positive, and the remainder satisfies $|\log\mathcal{E}_{\alpha,T_0}(u)|\le C(|u|+u^2)\frac{\log T_0}{T_0}$ for an absolute constant C.

Proof. (1) is Proposition 10.1.1 and Theorem 10.1.2, the sum being that of Proposition 1.2.3. (2) is part of Theorem 10.1.2: Y_α is an EGGC exactly when T_α is a GGC, and a zero with $\beta>\alpha$ puts a pole of φ_α at the point $(\rho-\alpha)^2$ of $\mathbb{C}\setminus(-\infty,0]$, where the Laplace transform of a GGC is holomorphic (Theorem 1.2.2); the sign of v_α near the ordinate of that zero plays no role. (3) Group the zeros into those on

the line below T_0 and the rest. Each pair $\frac{1}{2} \pm i\gamma$ contributes the factor $\frac{\varepsilon^2+\gamma^2}{(\varepsilon+|u|)^2+\gamma^2}$, whose Thorin density is the pair of Cauchy kernels of width ε at $\pm\gamma$, by the computation after Theorem 10.1.2, and which is therefore an EGGC characteristic function. Every other zero has $|\gamma| > T_0$, and a pair $\rho, \bar{\rho}$ with $\gamma > T_0$ contributes $-\log(1+x)$ to $\log \mathcal{E}_{\alpha,T_0}(u)$, with $x = (2(\alpha-\beta)|u| + u^2)/((\alpha-\beta)^2+\gamma^2)$. Since $|\alpha-\beta| < 1$, we have $x \geq -(\alpha-\beta)^2/((\alpha-\beta)^2+\gamma^2) > -\frac{1}{2}$ and $|x| \leq 2(|u|+u^2)/\gamma^2$, so the contribution is at most $4(|u|+u^2)/\gamma^2$ in absolute value. Summing over the zeros with $\gamma > T_0$ gives at most $4(|u|+u^2)\sum_{\gamma>T_0}\gamma^{-2}$, and by partial summation this sum is at most $2\int_{T_0}^{\infty} N(t)t^{-3}\,dt$. For $t \geq 2$ we have $N(t) \leq \frac{t}{2\pi}\log t$: no zero has $0 < \gamma < \gamma_1$, and for $t \geq \gamma_1$ the explicit bound of [92], recalled in the proof of Proposition 9.1.4, gives $N(t) \leq \frac{t}{2\pi}\log\frac{t}{2\pi e} + \frac{7}{8} + E(t)$, where $\frac{7}{8} + E(t) < \frac{t}{2\pi}\log(2\pi e)$: up to X_0 the left side is at most 3.40 and the right side is at least 6.38, its value at $t = \gamma_1$, and beyond X_0 the left side is less than $\frac{7}{8} + 0.3\log t$ and the right side exceeds $0.45\,t$. So $\sum_{\gamma>T_0}\gamma^{-2} \leq \frac{1}{\pi}\int_{T_0}^{\infty} t^{-2}\log t\,dt = \frac{1+\log T_0}{\pi T_0} \leq \frac{\log T_0}{T_0}$, the last step because $T_0 \geq 2$, and the bound holds with $C = 4$. □

Part (3) is the unconditional content of the EGGC property in the strip. With the verification height $T_0 = 3\times10^{12}$ of Platt and Trudgian [196], which also enters Corollary 4.3.8, the remainder is of relative size at most 8×10^{-11} for $|u| \leq 1$, so Y_α is, to that accuracy in characteristic function, an EGGC for every $\alpha > \frac{1}{2}$, whether or not RH holds; Theorem 11.4.4 gives the same statement for the Laplace transform of T_α with explicit constants. In the same way, the negative part of $v_\alpha(\theta)/\pi$ that zeros above T_0 could produce is at most about $\frac{\log(T_0/2\pi)}{4\pi^2(T_0-\theta)}$, which is 2.3×10^{-13} for every $\theta \leq 10^9$, since each such zero contributes a kernel of width less than $\frac{1}{2}$ in absolute value, while the verified zeros contribute at least the minimum of their Cauchy smoothing $\sum_{0<\gamma\leq T_0}[P_\varepsilon(\theta-\gamma) + P_\varepsilon(\theta+\gamma)]$; `strip.py` evaluates the bound. Grouping each zero above T_0 with its reflection in the critical line makes the bound unnecessary below $T_0 - \frac{1}{2}$: there every off line pair contributes a nonnegative term, so $v_\alpha > 0$ on $[0, T_0 - \frac{1}{2}]$ for every $\alpha > \frac{1}{2}$ (Corollary 11.4.5). What cannot be removed is the dependence on T_0. A zero above T_0 with $\beta > \alpha$ would stop T_α from being a GGC, and Y_α from being an EGGC, whether or not it made v_α negative near its ordinate (Exercise 1.7), and no finite computation excludes such a zero (Section 3.1.1). What can be said with no hypothesis is that the Cauchy kernels of such zeros carry a vanishing fraction of the mass of v_α/π on $[0, T]$ as $T \to \infty$ (Theorem 11.4.2).

Table 10.3, computed by `strip.py`, checks the product over the 1700 computed zeros, with the Weyl density for the zeros above them, against $\xi(\alpha)/\xi(\alpha+|u|)$ at $\alpha = 0.6$ and 0.75; it agrees to within 4×10^{-7}, better than the Thorin integral over a finite grid. On $[0, 200]$ the Thorin density is positive at both basepoints, and in both cases its minimum is at $\theta = 0$ and equals the Cauchy floor b_α/π: 0.0014709 at $\alpha = 0.6$ and 0.0036769 at $\alpha = 0.75$ (Figure 10.1, left), in agreement with the minima on $[0, 60]$ in Table 9.12 and with the margin $0.0147\,\varepsilon$ of Section 3.1. As

α decreases to $\frac{1}{2}$ the floor tends to zero, because $\xi'(\frac{1}{2}) = 0$, while the densities sharpen onto the ordinates; at the centre the Cauchy component is gone, and what remains, under RH, is the sum of Laplace variables $\sum_k L_k/\gamma_k$.

α	u	Thorin integral (10.1.2)	product over zeros	$\xi(\alpha)/\xi(\alpha+\lvert u\rvert)$
0.6	1	0.9726796	0.9726812	0.9726812
0.6	5	0.5549761	0.5549190	0.5549193
0.6	20	0.0004547	0.0004539	0.0004539
0.75	1	0.9659691	0.9659798	0.9659798
0.75	5	0.5368120	0.5367687	0.5367691
0.75	20	0.0004116	0.0004110	0.0004110

Table 10.3: The EGGC representation in the strip, computed by `strip.py`: the Thorin integral over $\theta \le 200$ plus the Weyl tail, and the product of the factors of Theorem 10.1.3(3) over the 1700 computed zeros plus the Weyl tail.

10.1.4 The density, the Lévy measure and the Poisson trace

Because $\xi(\alpha)/\xi(\alpha+|u|)$ is integrable, Y_α has the density

$$f_{Y_\alpha}(y) = \frac{1}{\pi}\int_0^\infty \cos(uy)\frac{\xi(\alpha)}{\xi(\alpha+u)}du,$$

which `dens2.py` computes by the trapezoid rule on $[0,45]$ with step $h = 10^{-3}$; the characteristic function decays like $\exp(-\frac{u}{2}\log u)$, so the truncation is harmless. At the origin $f_{Y_\alpha}(0) = 1.8994$, 1.8680, 1.8225, 1.7507 for $\alpha = \frac{1}{2}$, 0.6, 0.75, 1. The corner of the characteristic function at the origin gives the tail $f_{Y_\alpha}(y) \sim b_\alpha/(\pi y^2)$ for $\alpha > \frac{1}{2}$, and the computed values of $\pi y^2 f_{Y_\alpha}(y)/b_\alpha$ at the grid points $y = 5.12$, 10.25, 20.50 and 50 are 1.0057, 1.0014, 1.0004 and 1.0003 at each of $\alpha = 0.6$, 0.75 and 1 (Figure 10.1, middle). Of the last excess, about 2×10^{-4} is the relative error $h^2y^2/12$ that the trapezoid rule makes at the corner $u = 0$. At the centre there is no corner, and the density decays like $e^{-\gamma_1|y|}$, the tail of L_1/γ_1 under RH; unconditionally, the characteristic function continues to the meromorphic function $\xi(\frac{1}{2})/\xi(\frac{1}{2}+u)$ of u, whose poles nearest the real axis are $\pm i\gamma_1$.

Frullani's integral gives $\log(1+u^2/\theta^2) = \int_{\mathbb{R}}(1-\cos uy)\,e^{-\theta|y|}\,\frac{dy}{|y|}$, which is the sine squared lemma read in the other direction, since $1-\cos uy = 2\sin^2(uy/2)$, and which is the cosine dictionary (6.2.4) of Section 6.2. Inserting it in (10.1.2) puts the characteristic function of Y_α in Lévy Khintchine form in u:

$$\frac{\xi(\alpha)}{\xi(\alpha+|u|)} = \exp\Big\{-\int_{\mathbb{R}} 2\sin^2\Big(\frac{uy}{2}\Big)\,\frac{K_\alpha(|y|)}{|y|}\,dy\Big\},$$
$$K_\alpha(y) = \frac{1}{\pi}\int_0^\infty e^{-\theta y}v_\alpha(\theta)\,d\theta.$$

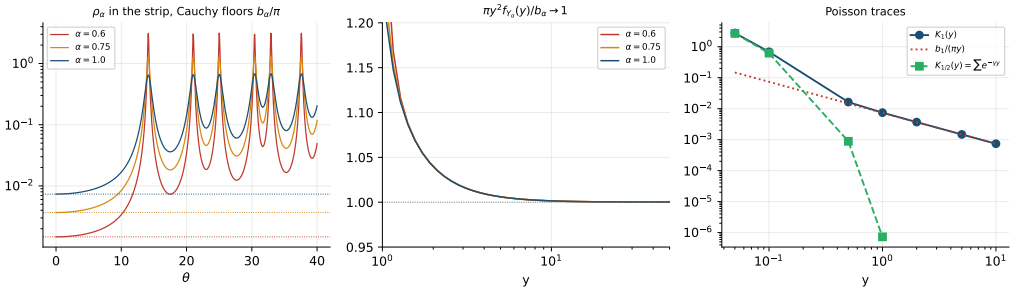


Figure 10.1: Left: the Thorin densities v_α/π at $\alpha = 0.6$, 0.75, 1, written ρ_α in the plot, and their Cauchy floors b_α/π (dotted). Middle: $\pi y^2 f_{Y_\alpha}(y)/b_\alpha$, tending to one. Right: the Poisson traces K_1 and $K_{1/2}$ against $b_1/(\pi y)$. Drawn by `figs2.py` from `strip.json` and `comp.json`.

So the Lévy density of Y_α is $K_\alpha(|y|)/|y|$, as Section 1.3.2 gives for a symmetric EGGC, and K_α is the Laplace transform of the Thorin density; in the normalisation of (6.2.4), $K_\alpha = \frac{1}{2}K_{U_\alpha}$. We call K_α the Poisson trace, because at the centre, where under RH the Thorin measure in the chart θ is $\sum_{\gamma>0}\delta_\gamma$, it is $K_{1/2}(y) = \sum_{\gamma>0} e^{-\gamma y}$, half of Cramér's function [57, Chapter I] of Proposition 6.2.8(iii): the sum over the zeros of the Poisson kernel in place of the heat kernel of the heat trace $W(t) = \sum_k e^{-a_k t}$, which is $\sum_{\gamma>0} e^{-\gamma^2 t}$ under RH. The two traces play parallel roles, and they are linked with no hypothesis. Since $Y_{1/2} = B_{2T}$ for an independent Brownian motion B, and T has Lévy density $W(t)/t$ (Theorem 5.1.1), the Lévy density of $Y_{1/2}$ is $\int_0^\infty (4\pi t)^{-1/2} e^{-y^2/4t}\, W(t)\, t^{-1} dt$, so

$$K_{1/2}(y) = \int_0^\infty \frac{y}{\sqrt{4\pi t^3}}\, e^{-y^2/4t}\, W(t)\, dt = \mathbb{E}\, W(H_{y/\sqrt{2}}), \qquad y > 0.$$

The Poisson trace is the heat trace averaged over the first passage time of Brownian motion to the level $y/\sqrt{2}$, that is, subordinated by the stable one half first passage kernel, whose Laplace transform $e^{-y\sqrt{a}}$ is the stable one half average of the heat kernel recalled after Proposition 6.2.9. This defines $K_{1/2}$ unconditionally, as $K_{1/2}(y) = \sum_k e^{-y\sqrt{a_k}}$ with the principal root, and makes it computable from the explicit formula (1.2.5) for W; under RH the kernel integrates $e^{-\gamma^2 t}$ to $e^{-\gamma y}$. Numerically, `improve.py` evaluates the integral with W from the first 1700 zeros and obtains 0.08566744234217 at $y = 0.2$ and 0.000883746148510 at $y = 0.5$, equal to $\sum e^{-\gamma y}$ over the same zeros to fourteen digits.

Corollary 10.1.4. *RH holds if and only if the Poisson trace $K_{1/2}$ is completely monotone, and if and only if the heat trace W is completely monotone. For $\alpha \geq 1$ the Poisson trace K_α is completely monotone with no hypothesis, and $K_\alpha(y) \sim b_\alpha/(\pi y)$ as $y \to \infty$, so that the Lévy density of Y_α is asymptotic to $b_\alpha/(\pi y^2)$, the Lévy density of the Cauchy component.*

Proof. For $\alpha \geq 1$, K_α is the Laplace transform of the positive density v_α/π, hence completely monotone by Bernstein's theorem, and since v_α is continuous with $v_\alpha(0) = b_\alpha$, $yK_\alpha(y) \to b_\alpha/\pi$ as $y \to \infty$. At the centre, W is completely monotone exactly when T is a GGC (Section 1.2), that is exactly when RH holds (Proposition 1.2.1). Under RH, $K_{1/2}(y) = \sum_{\gamma>0} e^{-\gamma y}$ is completely monotone. Conversely, suppose that $K_{1/2}(y) = \int_{[0,\infty)} e^{-y\theta}\nu(d\theta)$ with a positive measure ν. For $w > 0$ the unconditional formula $K_{1/2}(y) = \sum_k e^{-y\sqrt{a_k}}$ gives

$$\int_0^\infty e^{-wy}\, y\, K_{1/2}(y)\, dy = \sum_k \frac{1}{(w+\sqrt{a_k})^2} = \int_{[0,\infty)} \frac{\nu(d\theta)}{(w+\theta)^2}.$$

The first equality integrates term by term, which is legitimate because $\sqrt{a_k} = \gamma_k - i(\beta_k - \frac{1}{2})$ has real part γ_k when $\rho_k = \beta_k + i\gamma_k$ is taken with $\gamma_k > 0$, so the absolute values of the terms integrate to $\sum_k (w+\gamma_k)^{-2} < \infty$. The second is Tonelli's theorem, and it shows that the right side is finite for $w > 0$. The middle expression is meromorphic in $w \in \mathbb{C}$, with a double pole at each point $-\sqrt{a_k}$, while the right side is holomorphic on $\mathbb{C} \setminus (-\infty, 0]$, since $|w+\theta|$ is bounded below by a positive multiple of $1+\theta$ for w in a compact subset of $\mathbb{C} \setminus (-\infty, 0]$. The two sides agree for $w > 0$, hence on $\mathbb{C} \setminus (-\infty, 0]$ with the poles removed, so no pole $-\sqrt{a_k}$ lies in $\mathbb{C} \setminus (-\infty, 0]$. Since $\operatorname{Re}\sqrt{a_k} > 0$, every $\sqrt{a_k}$ is therefore real and positive, that is, every $a_k = -(\rho_k - \frac{1}{2})^2$ is positive, which is RH. □

Table 10.4, computed by `comp.py`, gives K_1 from v_1 on $[0, 200]$ with the Weyl density beyond, and $K_{1/2}$ as $\sum e^{-\gamma y}$ over the first 1700 zeros (Figure 10.1, right). At large y, $K_1(y)$ approaches $b_1/(\pi y)$, to 0.5% at $y = 10$, while $K_{1/2}(y)$ decays like $e^{-\gamma_1 y}$: at the centre there is no Cauchy component, the tail is Laplace like with rate γ_1, and the variance is finite. At small y both traces are large and close, because there the Poisson kernels see all zeros and the smoothing at the basepoint one matters little.

y	$K_1(y) = \frac{1}{\pi}\int_0^\infty e^{-\theta y} v_1(\theta)\, d\theta$	$b_1/(\pi y)$	$K_{1/2}(y) = \sum e^{-\gamma y}$
0.05	2.7700	0.1470	2.6860
0.1	0.68408	0.07352	0.62104
0.5	0.016327	0.014703	8.84×10^{-4}
1	0.0074306	0.0073516	7.27×10^{-7}
2	0.0036856	0.0036758	5.28×10^{-13}
10	0.00073906	0.00073516	4.1×10^{-62}

Table 10.4: The Poisson traces at the basepoint one and at the centre, computed by `comp.py`.

10.1.5 The complete monotonicity conditions of the Poisson trace

By Corollary 10.1.4, RH is the complete monotonicity of the Poisson trace $K_{1/2}$. This subsection follows the first author's note on Biane's paper [202]. It writes the conditions out zero by zero, proves that a long initial stretch of them holds with no hypothesis, says which zeros each condition sees, gives two further equivalent forms, explains why the explicit formula cannot test the conditions, and compares $K_{1/2}$ with the corresponding trace of Pólya's function $P(\theta) = K_{i\theta/2}(2\pi)$ of Section 8.7, which is completely monotone with no hypothesis. The computations are in `biane_traces.py` and `biane_figs.py`.

Write the zeros as $\rho_k = \beta_k + i\gamma_k$ with $\gamma_k > 0$, so that $\sqrt{a_k} = \gamma_k + i\delta_k$ with $\delta_k = \frac{1}{2} - \beta_k \in (-\frac{1}{2}, \frac{1}{2})$, and put $\varphi_k = \arctan(\delta_k/\gamma_k)$. The series $K_{1/2}(y) = \sum_k e^{-y\sqrt{a_k}}$ converges absolutely, with all its derivatives, for $y > 0$, and the zeros ρ_k and $1-\bar\rho_k$ have conjugate values of $\sqrt{a_k}$. Hence

$$(-1)^n K_{1/2}^{(n)}(y) = \sum_k \mathrm{Re}\left((\sqrt{a_k})^n e^{-y\sqrt{a_k}}\right) = \sum_k |\sqrt{a_k}|^n \, e^{-\gamma_k y} \cos(n\varphi_k - \delta_k y), \tag{10.1.3}$$

with each zero counted once, with its multiplicity; a zero on the line contributes $\gamma_k^n e^{-\gamma_k y} > 0$. The note [202] groups ρ_k with $1-\bar\rho_k$ and writes a factor 2 in front of every term, which counts the zeros on the line twice. So $K_{1/2}$ is completely monotone, and by Corollary 10.1.4 RH holds, exactly when every one of the quantities (10.1.3) is nonnegative, for $n = 0, 1, 2, \ldots$ and $y > 0$. Off the line a term oscillates in n with the small frequency $\varphi_k \approx \delta_k/\gamma_k$, so a zero off the line can spoil its term only at a high order.

Proposition 10.1.5. *The term of a zero with $\delta_k \neq 0$ in* (10.1.3) *is positive whenever*

$$n \arctan\frac{|\delta_k|}{\gamma_k} + |\delta_k|\, y < \frac{\pi}{2}.$$

If $\delta_k > 0$ and $\delta_k y < \frac{\pi}{2}$, the first order n at which the term is negative is the least integer with $n\varphi_k - \delta_k y > \frac{\pi}{2}$, that is the least integer above $(\frac{\pi}{2} + \delta_k y)/\varphi_k$ when this ratio is not an integer. The partner $1-\bar\rho_k$ has the same term.

Proof. $|n\varphi_k - \delta_k y| \le n|\varphi_k| + |\delta_k| y$, and the cosine is positive on $(-\frac{\pi}{2}, \frac{\pi}{2})$. For $\delta_k > 0$ the argument $n\varphi_k - \delta_k y$ increases with n from $-\delta_k y \in (-\frac{\pi}{2}, 0)$, and the cosine is first negative when it passes $\frac{\pi}{2}$. The partner has $-\delta_k$ and $-\varphi_k$, so its argument is the negative of this one. □

`biane_traces.py` finds the first negative order by evaluating the sign factor; Table 10.5 compares it with the formula, and the note's values are the same.

Theorem 10.1.6. *Suppose that every zero with $0 < \gamma_k \le H$ lies on the critical line. Then for $0 < y < \pi$ and every integer n with $0 \le n < H(\pi - y)$, every term of* (10.1.3) *is positive, and $(-1)^n K_{1/2}^{(n)}(y) > 0$. With $H = T_0 = 3 \times 10^{12}$ [196] this*

γ	δ	y	first negative order	$(\frac{\pi}{2}+\delta y)/\varphi$
50	0.25	1	365	364.16
50	0.25	0.3	330	329.16
200	0.1	1	3342	3341.59
14.13	0.5	0.5	52	51.48

Table 10.5: The first order at which the term of a hypothetical zero $\frac{1}{2}-\delta+i\gamma$ in (10.1.3) is negative, against Proposition 10.1.5. The last row is a zero at the edge of the strip at the height of the first ordinate. Computed by `biane_traces.py`.

covers all $n < 9.12\times10^{12}$ *at* $y=0.1$, $n<6.42\times10^{12}$ *at* $y=1$ *and* $n<4.24\times10^{11}$ *at* $y=3$.

Proof. A zero with $\gamma_k \le H$ has $\delta_k = 0$, and its term $\gamma_k^n e^{-\gamma_k y}$ is positive. A zero with $\gamma_k > H$ has $|\varphi_k| \le |\delta_k|/\gamma_k < \frac{1}{2H}$ and $|\delta_k| y < \frac{y}{2}$, so $n|\varphi_k| + |\delta_k| y < \frac{n}{2H} + \frac{y}{2} \le \frac{\pi}{2}$ when $n \le H(\pi - y)$, and its term is positive by Proposition 10.1.5. The series converges absolutely, so its sum is positive. □

The theorem is a statement about the verified height, not about $K_{1/2}$: it improves with every extension of the verification and never reaches the whole family. Every condition that could fail has $y \ge \pi$ or $n \ge H(\pi - y)$, which is above 4×10^{11} for $y \le 3$; there, complete monotonicity is a characterisation of RH, not a test that a computation could run to a failure. The note [202] gives the last bound as 4.3×10^{11}; it is 4.2478×10^{11}.

The weight of the zero γ_k in the condition (n, y) is about $\gamma_k^n e^{-\gamma_k y}$, whose logarithm $n\log\gamma - \gamma y$ is largest at

$$\gamma_* = \frac{n}{y},$$

with a width of order $\sqrt{n}/y$. The condition (n, y) is therefore a smoothed probe of the zeros near the height n/y. Among the first 300 ordinates the dominant one is 21.022 for $(n, y) = (20, 1)$, 98.831 for $(100, 1)$ and for $(50, 0.5)$, with 101.318 a close second (the logarithms of the weights at $(100, 1)$ are 360.51013 and 360.50841), 21.022 for $(100, 5)$ and 30.425 for $(300, 10)$: in each case the ordinate nearest to n/y (`biane_traces.py`). Figure 10.2 shows two weight profiles and two sign factors. The reading explains Theorem 10.1.6 and shows its edge. By Proposition 10.1.5 a zero at height γ can make its term negative only when $n \gtrsim \frac{\pi}{2}\gamma/|\delta| \ge \pi\gamma$, that is when $\gamma \lesssim n/\pi$, below the peak at n/y when $y < \pi$; and since $\gamma^n e^{-\gamma y}$ increases up to n/y, its value at $\gamma \le n/\pi$ is at most $\exp(n(\log\frac{y}{\pi} + 1 - \frac{y}{\pi}))$ times its value at the peak, which is exponentially small in n. Turning this comparison into a proof would need a bound for the number of zeros off the line near each height, and it would push the unconditional range beyond $H(\pi - y)$ without reaching the whole

family: the conditions with n/y beyond every verified height are the content of RH.

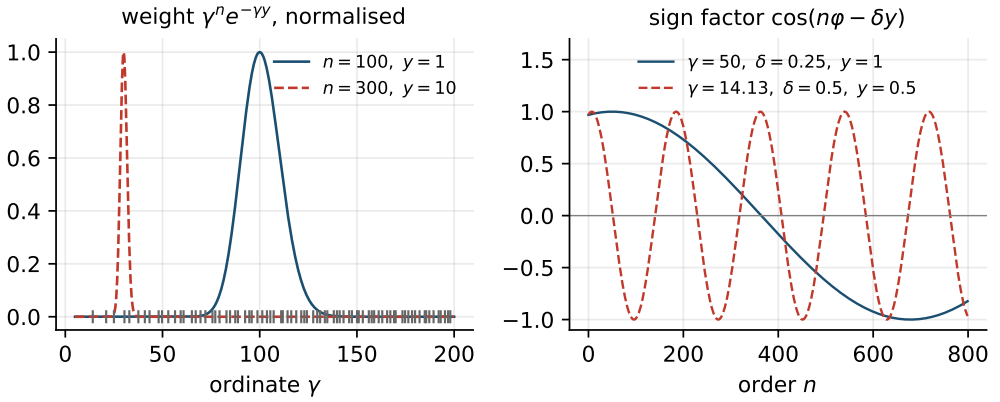


Figure 10.2: Left: the weight $\gamma^n e^{-\gamma y}$ of the condition (n, y) in (10.1.3), normalised to maximum one, for $(n, y) = (100, 1)$ and $(300, 10)$, peaking at $\gamma = n/y$; the marks are the ordinates below 200. Right: the sign factor $\cos(n\varphi - \delta y)$ of a hypothetical zero off the line for $(\gamma, \delta, y) = (50, 0.25, 1)$ and $(14.13, 0.5, 0.5)$, positive below the orders 365 and 52 of Table 10.5, at which it is first negative. Drawn by `biane_figs.py` from `biane_traces.json`.

Proposition 10.1.7. *The following are equivalent:*

(i) RH;

(ii) $K_{1/2}$ is completely monotone on $(0, \infty)$;

(iii) for every $x > 0$ the function $y \mapsto K_{1/2}(x + iy)$ is positive definite on $\mathbb{R}$;

(iv) for some $x > 0$ the function $y \mapsto K_{1/2}(x + iy)$ is bounded on $[0, \infty)$.

Under these conditions $|K_{1/2}(x + iy)| \le K_{1/2}(x)$.

Proof. The series $K_{1/2}(z) = \sum_k e^{-z\sqrt{a_k}}$ converges absolutely and locally uniformly for $\operatorname{Re} z > 0$, since $|e^{-z\sqrt{a_k}}| = e^{-\gamma_k \operatorname{Re} z + \delta_k \operatorname{Im} z} \le e^{-\gamma_k \operatorname{Re} z + |\operatorname{Im} z|/2}$, and defines an analytic function there. (i) implies (ii) by Corollary 10.1.4. If (ii) holds, Bernstein's theorem gives $K_{1/2}(y) = \int_{[0,\infty)} e^{-y\theta} \nu(d\theta)$ with a positive measure ν, the integral converges for $\operatorname{Re} z > 0$ and agrees there with $K_{1/2}(z)$ by analytic continuation, and $K_{1/2}(x + iy) = \int e^{-iy\theta} e^{-x\theta} \nu(d\theta)$ is the Fourier transform of the finite positive measure $e^{-x\theta}\nu(d\theta)$, which gives (iii) and the bound. A positive definite function f satisfies $|f(y)| \le f(0)$, so (iii) implies (iv). Suppose (iv) holds at x. For $\operatorname{Re} w > \frac{1}{2}$

the bound above allows term by term integration, and

$$\int_0^\infty e^{-wy} K_{1/2}(x+iy)\,dy = \sum_k \frac{e^{-x\sqrt{a_k}}}{w+i\sqrt{a_k}}.$$

The right side is a meromorphic function of $w \in \mathbb{C}$, since $\sum_k e^{-x\gamma_k} < \infty$, with a pole at each point $-i\sqrt{a_k} = \delta_k - i\gamma_k$, of residue $e^{-x\sqrt{a_k}}$ times the multiplicity, which is not zero. Since $K_{1/2}(x+iy)$ is bounded, the left side is analytic in $\operatorname{Re} w > 0$, so no pole has $\delta_k > 0$. The zeros ρ_k and $1-\bar\rho_k$ have opposite δ_k, so every δ_k is 0, which is RH. □

Part (iv) replaces the Paley Wiener argument of the note [202], which is not given in detail there, by a pole of a Laplace transform. Written out, (iii) says that for every $x > 0$, all real $y_1, \dots, y_m$ and all complex $c_1, \dots, c_m$,

$$\sum_{j,l} c_j \bar c_l\, K_{1/2}\big(x + i(y_j - y_l)\big) = \sum_k e^{-x\gamma_k} \Big| \sum_j c_j e^{-i\gamma_k y_j} \Big|^2 \ge 0,$$

the equality holding under RH. This is a positivity of the kind in Weil's criterion, with two differences: the test functions are $r \mapsto e^{-rx} |\sum_j c_j e^{-iry_j}|^2$ on $r > 0$, where the damping e^{-rx} does the work of a decay hypothesis, and the sum runs over the ordinates in the upper half plane only.

One would like to turn each condition into a statement about primes with the explicit formula of Section 1.2, the display before (1.2.5). The test function of the condition (n, y) is $h(r) = |r|^n e^{-y|r|}$, with

$$g(x) = \frac{1}{2\pi} \int_{\mathbb{R}} h(r) e^{-irx}\,dr = \frac{n!}{\pi} \operatorname{Re} \frac{1}{(y+ix)^{n+1}},$$

which decays only like a power of $|x|$ and has one sign for large $|x|$. The prime side $-2\sum_{m\ge 2} \Lambda(m) m^{-1/2} g(\log m)$ then diverges, since $\sum_p p^{-1/2} (\log p)^{-j} = \infty$ for every j; the formula holds for g decaying faster than $e^{-|x|/2}$, which goes with h analytic in a strip $|\operatorname{Im} r| < \frac12 + \epsilon$, and this h is not even analytic at $r = 0$. For $n = 0$ the prime side is $-\frac{2}{\pi} \sum_{m\ge 2} \Lambda(m) m^{-1/2} y/(y^2 + \log^2 m)$, which is what the prime terms of (1.2.5) become under the subordination $K_{1/2}(y) = \mathbb{E} W(H_{y/\sqrt{2}})$ of Section 10.1.4, applied term by term, with twice $K_{1/2}$ on the left; its terms have poles at $y = \pm i \log m$, the singularities of Cramér's function [57, Chapter I], and it diverges for the reason of Proposition 6.2.9. Smoothing in the variable of the zeros removes the divergence without changing the problem: the Gaussian weight $e^{-r^2 t}$ in place of $e^{-y|r|}$ gives the heat trace $W(t)$, whose explicit formula (1.2.5) converges absolutely, and whose complete monotonicity is again equivalent to RH (Corollary 10.1.4). The note [202] identifies this smoothing with the de Bruijn Newman flow and concludes that the primes become summable only at times $t > 0$, where the reality of the zeros is known; but the Gaussian weight acts

on the test function, while the flow multiplies the kernel Φ by $e^{tu^2/4}$ and moves the zeros, and the smoothed conditions, those of the heat trace, are still equivalent to RH.

The last comparison is with Pólya's function. By Proposition 8.7.13,

$$\log\frac{K_{\lambda/2}(2\pi)}{K_0(2\pi)} = \int_0^\infty 4\sin^2\Big(\frac{\lambda y}{2}\Big)K^P(y)\,\frac{dy}{y},$$
$$K^P(y) = \sum_n e^{-2\nu_n y} = \operatorname{tr} e^{-2y\sqrt{A_P}},$$

the sum over the zeros $\theta_n = 2\nu_n$ of P, so K^P is the counterpart of $K_{1/2}$ in the formula for $\xi(\frac12)/\xi(\frac12+u)$, and it is completely monotone because A_P is selfadjoint and positive. By Proposition 8.7.8 the two traces have the same main term, and their difference has a limit.

PROPOSITION 10.1.8. *Let* $M(y) = \frac{1}{2\pi y}\big(\log\frac{1}{2\pi y} - \gamma_E\big)$ *and* $c = 2\pi - \frac{1}{6\pi} - \frac{1}{48\pi} = 6.2235022035$. *As* $y \to 0+$*:*

(i) $K_{1/2}(y) = M(y) + \frac78 + O(y\log\frac1y)$, *and under RH* $K_{1/2}(y) = M(y) + \frac78 + \frac{1}{48\pi}y\log\frac1y + o(y\log\frac1y)$;

(ii) $K^P(y) = M(y) - \frac14 + \big(2\pi - \frac{1}{6\pi}\big)y\log\frac1y + O(y)$;

(iii) $D(y) = K_{1/2}(y) - K^P(y) = \frac98 + O(y\log\frac1y)$, *and under RH* $D(y) = \frac98 - c\,y\log\frac1y + o(y\log\frac1y)$.

Proof. (i) The real part of the term $e^{-y\sqrt{a_k}}$ is $e^{-\gamma_k y}\cos(\delta_k y) = e^{-\gamma_k y} + O(y^2e^{-\gamma_k y})$, and the imaginary parts cancel in pairs, so $K_{1/2}(y) = \sum_k e^{-\gamma_k y} + O(y^2\sum_k e^{-\gamma_k y})$, and the sum will turn out to be $O(y^{-1}\log\frac1y)$. By Abel summation $\sum_k e^{-\gamma_k y} = y\int_0^\infty e^{-Ty}N(T)\,dT$, with $N(T) = \vartheta(T)/\pi + 1 + S(T)$ [255, Section 9.3]. The Riemann Siegel function satisfies $\vartheta(T)/\pi = N_0(T) - \frac18 + \frac{1}{48\pi T} + O(T^{-3})$ with $N_0(T) = \frac{T}{2\pi}\log\frac{T}{2\pi e}$, so $N = N_0 + \frac78 + S + E$ with $E(T) = \frac{1}{48\pi T} + O(T^{-3})$ for $T \ge 1$ and E bounded on $(0,1]$. Now $\int_0^\infty Te^{-Ty}\log T\,dT = y^{-2}(1 - \gamma_E - \log y)$ gives $y\int_0^\infty e^{-Ty}N_0(T)\,dT = M(y)$; the constant gives $\frac78$; $y\int_1^\infty e^{-Ty}\frac{dT}{48\pi T} = \frac{1}{48\pi}y\log\frac1y + O(y)$, and the rest of E gives $O(y)$. With $S_1(T) = \int_0^T S(t)\,dt$, an integration by parts gives $y\int_0^\infty e^{-Ty}S(T)\,dT = y^2\int_0^\infty e^{-Ty}S_1(T)\,dT$, which is $O(y\log\frac1y)$ since $S_1(T) = O(\log T)$ [255, Theorem 9.9(A)]. Under RH there are no terms $O(y^2e^{-\gamma_k y})$, and $S_1(T) = O(\log T/(\log\log T)^2)$ [255, Theorem 14.13] makes the last integral $o(y\log\frac1y)$. (ii) By Proposition 8.7.8, the number of zeros $\theta_n \le T$ is $\lfloor x(T)\rfloor$, with $x(T) = N_0(T) + \frac14 + (2\pi - \frac{1}{6\pi})T^{-1} + O(T^{-3})$ for $T \ge 1$. Write $\lfloor x\rfloor = x - \frac12 + (\frac12 - \{x\})$. The terms N_0, $\frac14 - \frac12$ and $(2\pi - \frac{1}{6\pi})T^{-1}$ give $M(y)$, $-\frac14$ and $(2\pi - \frac{1}{6\pi})y\log\frac1y + O(y)$ as in (i). The sawtooth $\frac12 - \{x(T)\}$ has integral over each interval on which x increases by one equal to $O(T^{-1}(\log T)^{-3})$, because such

an interval has length about $2\pi/\log T$ and $x'(T) = \frac{1}{2\pi}\log\frac{T}{2\pi} + O(T^{-2})$ varies over it by $O(T^{-1}(\log T)^{-2})$ relative to its size, so its integral from 0 to T is bounded, and another integration by parts makes its contribution $O(y)$. (iii) is (i) minus (ii). □

`biane_traces.py` tests the proposition. It takes the first 30000 ordinates from Arb, up to 25755.348, cuts at $G = 25754.905$, midway between the last two, and adds the tail $\int_G^\infty e^{-Ty}\vartheta'(T)\,dT/\pi - e^{-Gy}S(G)$, with $S(G) = -0.569$ exact; the neglected term $y\int_G^\infty e^{-Ty}S(T)\,dT$ equals $y^2\int_G^\infty e^{-Ty}\int_G^T S(t)\,dt\,dT$, and Trudgian's bound $|\int_{t_1}^{t_2} S(t)\,dt| \le 2.067 + 0.059\log t_2$ for $t_2 > t_1 > 168\pi$ [257], with Jensen's inequality for the logarithm, bounds it by $ye^{-Gy}(0.059\log(G+1/y) + 2.067)$; the script records twice this bound, 3.5×10^{-14}, 4.1×10^{-5} and 4.3×10^{-5} at $y = 10^{-3}$, 10^{-4} and 10^{-5}, so the entries of Table 10.6 at $y = 10^{-4}$ and 10^{-5} may be off by up to two units in the fifth decimal. The zeros of P come from the phase condition of Proposition 8.7.8, solved by Newton's method for all 30132 zeros to just above the cut, with phase residuals below 4.4×10^{-11}; the 29998 below G are summed and the tail $\int_G^\infty e^{-Ty}\,dx(T) + e^{-Gy}(\{x(G)\} - \frac12)$ is added, with $x(G) = 29998.944$. The factor e^{-Gy} that weights the tails is 2.6×10^{-6} at $y = 5\times10^{-4}$, 0.0058 at 2×10^{-4}, 0.076 at 10^{-4} and 0.77 at 10^{-5}, so the last row of Table 10.6 rests mostly on the smooth tails. The constant $\frac98$ appears clearly, and D approaches it along $\frac98 - cy\log\frac1y$ (Figure 10.3).

y	$K_{1/2}(y)$	$K^P(y)$	$K_{1/2} - M$	$K^P - M$	$D(y)$	$\frac98 - cy\log\frac1y$
0.05	2.686015	2.230883	0.837782	0.382650	0.455132	0.192803
0.02	12.772471	12.033884	0.860279	0.121691	0.738587	0.638070
0.01	35.723862	34.836495	0.867696	−0.019670	0.887366	0.838397
0.005	92.647265	91.663043	0.871374	−0.112848	0.984222	0.960130
0.003	180.932794	179.901902	0.872836	−0.158057	1.030892	1.016540
0.002	303.229389	302.172210	0.873563	−0.183616	1.057179	1.047647
0.001	715.903738	714.816981	0.874286	−0.212471	1.086757	1.082010
5×10^{-4}	1651.569150	1650.465433	0.874645	−0.229071	1.103717	1.101348
2×10^{-4}	4856.772117	4855.656772	0.874859	−0.240486	1.115345	1.114399
10^{-4}	10815.847446	10814.727704	0.874931	−0.244811	1.119742	1.119268
10^{-5}	144797.380092	144796.255761	0.874994	−0.249338	1.124332	1.124283

Table 10.6: The Poisson trace $K_{1/2}$ of the zeros of ζ, Pólya's trace K^P, their distances from the common main term M, which tend to $\frac78$ and $-\frac14$, and their difference D, which tends to $\frac98$, against the prediction of Proposition 10.1.8 under RH. Computed by `biane_traces.py`.

The note [202] states that $K_{1/2}(y) = M(y) + O(\log\frac1y)$, which is true but loses the constant, and it reports $D(y)/\log\frac1y = 0.195,\ 0.194,\ 0.193,\ 0.193$ at $y = 0.003,\ 0.005,\ 0.01,\ 0.02$, from which it concludes that D grows like $0.19\log\frac1y$. Repeating its computation, with the first 300 ordinates and the 312 zeros of P

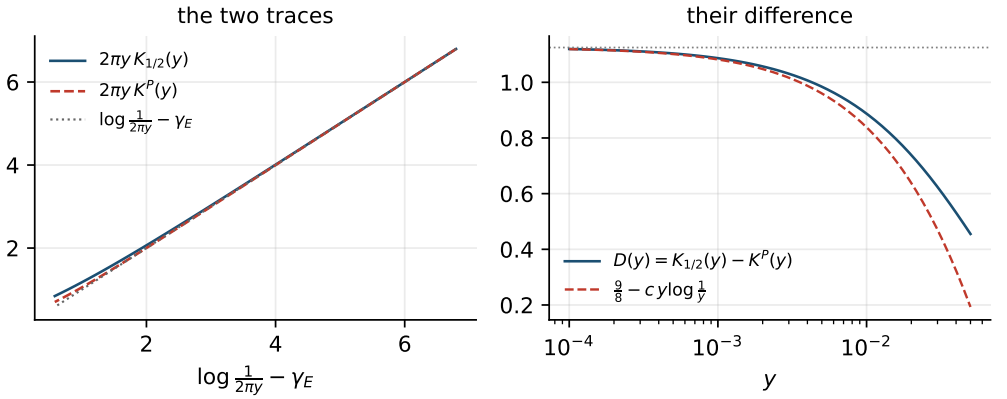


Figure 10.3: Left: $2\pi y K_{1/2}(y)$ and $2\pi y K^P(y)$ against $\log\frac{1}{2\pi y} - \gamma_E$ for $10^{-4} \le y \le 0.05$, with the main term $2\pi y M(y)$ (dotted). Right: the difference $D(y)$, the limit $\frac{9}{8}$ (dotted) and $\frac{9}{8} - cy\log\frac{1}{y}$ (dashed). Drawn by `biane_figs.py` from `biane_traces.json`.

below $\gamma_{300} + 20 = 561.85$, each followed by a plain Weyl tail from the last zero, gives its ratios $K_{1/2}/M = 1.0059$, 1.0102, 1.0250, 1.0722 and $D/\log\frac{1}{y} = 0.195$, 0.192, 0.193, 0.189; the note describes the 312 zeros as lying below $\gamma_{300} = 541.85$, below which there are 298. The accurate values of $D/\log\frac{1}{y}$ in this range, 0.177, 0.186, 0.193, 0.189, are also close to 0.19, but that is a coincidence of the range: by Proposition 10.1.8, D tends to $\frac{9}{8}$, and $D/\log\frac{1}{y}$ is 0.098 at $y = 10^{-5}$.

The comparison shows where the difficulty sits. K^P is completely monotone for a structural reason, as the trace of the semigroup $e^{-2y\sqrt{A_P}}$. By Corollary 10.1.4, RH says that $K_{1/2} = K^P + D$ is completely monotone as well, where D is bounded, tends to $\frac{9}{8}$ while both traces grow like $\frac{1}{2\pi y}\log\frac{1}{y}$, and carries the whole difference between Φ and Pólya's kernel, the arithmetic included. Complete monotonicity constrains the derivatives of every order, and no bound on the size of D can decide it.

10.1.6 Wiener's theorem and the basepoint kernels

The zero free half planes that recur through Chapter 9 and through Sections 10.1.1 to 10.1.4 have a second reading, as completeness statements, through Wiener's closure theorem [269], for which Kac [114] and Korevaar [128] gave short alternative treatments: the translates $f(\cdot - x)$, $x \in \mathbb{R}$, of a function $f \in L^1(\mathbb{R})$ span a dense subspace of $L^1(\mathbb{R})$ if and only if its Fourier transform $\hat f(\theta) = \int f(u)e^{i\theta u}du$ has no real zeros. The L^2 version asks only that $\hat f \neq 0$ almost everywhere; since the zeros of ξ are isolated it never sees them, and it is the L^1 version that is sharp here.

Let Φ be Pólya's kernel (1.2.3), so that $\xi(\frac{1}{2} + s) = \int_{\mathbb{R}} \Phi(u)e^{su}du$ for every

complex s by (1.2.4), and for $\alpha \geq \frac{1}{2}$ put

$$\Phi_\alpha(u) = e^{(\alpha-\frac{1}{2})u}\,\Phi(u), \qquad \hat{\Phi}_\alpha(\theta) = \int_{\mathbb{R}} \Phi_\alpha(u) e^{i\theta u} du = \xi(\alpha + i\theta). \tag{10.1.4}$$

Up to the factor $\xi(\alpha)$, Φ_α is the density of the tilted Pólya variable V_α of Section 9.1.1, with $\mathbb{E}e^{sV_\alpha} = \xi(\alpha+s)/\xi(\alpha)$, and it is integrable for every α because Φ decays doubly exponentially. The real zeros of $\hat{\Phi}_\alpha$ are exactly the ordinates of the zeros of ζ on the line $\operatorname{Re} s = \alpha$, so Wiener's theorem gives the following.

Proposition 10.1.9. *1. The translates of Φ_α are dense in $L^1(\mathbb{R})$ if and only if ζ has no zeros on the line $\operatorname{Re} s = \alpha$.*

2. At the centre they are not dense. For every finite combination $h = \sum_k c_k \Phi(\cdot - x_k)$ and every $g \in L^1(\mathbb{R})$,

$$\|g - h\|_1 \;\geq\; \sup_\gamma |\hat{g}(\gamma)|,$$

the supremum over the ordinates of the zeros on the critical line.

3. At the basepoint one they are dense, with no hypothesis, because $\xi(1+i\theta) \neq 0$.

4. For $\alpha_0 \in [\frac{1}{2}, 1)$, ζ has no zeros in the half plane $\operatorname{Re} s > \alpha_0$ if and only if the translates of Φ_α are dense in $L^1(\mathbb{R})$ for every $\alpha > \alpha_0$. In particular RH holds if and only if they are dense for every $\alpha > \frac{1}{2}$.

5. If the translates of Φ_α are dense, approximation near the zeros is expensive: if $h = \sum_k c_k \Phi_\alpha(\cdot - x_k)$ and $\|g-h\|_1 \leq \delta$, then $\sum_k |c_k| \;\geq\; (|\hat{g}(\gamma)| - \delta)/|\xi(\alpha + i\gamma)|$ for every ordinate γ.

Proof. (1), (3) and (4) are Wiener's theorem applied to $\hat{\Phi}_\alpha(\theta) = \xi(\alpha + i\theta)$, together with the nonvanishing of ζ on the line $\operatorname{Re} s = 1$. For (2) and (5), $\hat{h}(\theta) = \hat{\Phi}_\alpha(\theta) \sum_k c_k e^{i\theta x_k}$, so at the centre $\hat{h}(\gamma) = 0$ at every ordinate of a zero on the line, while $|\hat{g}(\theta) - \hat{h}(\theta)| \leq \|g - h\|_1$ for every θ; for (5), $|\hat{h}(\gamma)| \leq |\xi(\alpha + i\gamma)| \sum_k |c_k|$. □

Wiener's proof of the prime number theorem runs through statement (3), and the proposition shows the whole family interpolating between that theorem at $\alpha = 1$ and RH at $\alpha = \frac{1}{2}$. It is a restatement rather than a new criterion. By statement (4), completeness at every $\alpha' > \alpha$ is the absence of zeros with real part greater than α. That is the condition $\alpha \geq \sup_\rho \operatorname{Re} \rho$ under which T_α is a GGC (Theorem 1.2.2) and Y_α an EGGC (Theorem 10.1.2), and, by Lagarias's argument [133], it is also the positivity of $v_{\alpha'}$ for every $\alpha' > \alpha$ (Remark 1.2.4). A zero exactly on the line $\operatorname{Re} s = \alpha$ is allowed by the GGC property, and contributes an atom to the Thorin measure, but it blocks completeness at α itself; and for ξ it is not known whether positivity of v_α on the single line $\operatorname{Re} s = \alpha$ excludes zeros to its right (Exercise 1.7). What the completeness form adds is an explicit, computable

obstruction at the centre, statement (2), and an explicit cost of approximation above it, statement (5): near the centre $|\xi(\alpha+i\gamma)| \approx (\alpha-\frac12)|\xi'(\frac12+i\gamma)|$ at the ordinate γ of a simple zero, so the coefficients needed to reproduce $\hat g$ at an ordinate grow like $1/(\alpha-\frac12)$. At the first ordinate $|\xi'(\frac12+i\gamma_1)| = 0.00138272$, and $|\xi(\alpha+i\gamma_1)| = 0.000138$, 0.000346, 0.000697 at $\alpha = 0.6$, 0.75, 1, as computed by `wiener.py`.

We tested the dichotomy with `wiener.py`. We approximated the Gaussian $g(u) = (2\pi\kappa^2)^{-1/2}e^{-u^2/2\kappa^2}$ with $\kappa = 0.15$, for which $\hat g(\gamma_1) = e^{-\kappa^2\gamma_1^2/2} = 0.10565$, by combinations of N equally spaced translates of Φ_α on $[-1.5, 1.5]$, minimising the L^1 error on $[-3, 3]$ exactly by linear programming. Table 10.7 and Figure 10.4 show the results. At the centre every approximant has $\hat h(\gamma_1) = 0$ up to rounding, the computed values staying below 1.2×10^{-8} while the coefficient sums grow to 4×10^9, and the best error stalls at 0.1345, above the barrier 0.1056 of statement (2), however many translates are used. Above the centre the error falls below the value at which the centre stalls and keeps decreasing, to below the barrier at $\alpha = 0.75$ and $\alpha = 1$; the barrier binds only at the centre, and at $\alpha = 0.6$ the error is still 0.1139 with 80 translates. The approximants acquire the missing Fourier mass at γ_1: $|\hat h(\gamma_1)|$ reaches 0.017, 0.038 and 0.063 at $\alpha = 0.6$, 0.75 and 1 with 80 translates, against the target 0.106. The remaining error is limited by conditioning rather than by completeness. With 80 translates the coefficient sums reach 3×10^7 to 2×10^8 above the centre, as statement (5) and the rapid decay of $\hat\Phi_\alpha$ require, and double precision linear programming cannot go much further. The experiment therefore shows the qualitative dichotomy clearly and the rate only roughly; the closer α is to $\frac12$, the larger the coefficients and the slower the approach, which is the completeness counterpart of the Cauchy kernels sharpening onto the zeros.

N	$\alpha=\frac12$	$\alpha=0.6$	$\alpha=0.75$	$\alpha=1$
10	0.4705	0.4698	0.4664	0.4540
20	0.1345	0.1221	0.1060	0.0852
40	0.1345	0.1152	0.0907	0.0614
80	0.1365	0.1139	0.0878	0.0570
$\lvert\hat h(\gamma_1)\rvert$ at $N=80$	0	0.0167	0.0377	0.0635

Table 10.7: Best L^1 error in approximating a Gaussian of width 0.15 by N translates of Φ_α, computed by `wiener.py`. At the centre the error cannot fall below $|\hat g(\gamma_1)| = 0.1056$, and $\hat h(\gamma_1)$ vanishes exactly; the computed value is 1.2×10^{-8}.

10.1.7 Salem's criterion and its Pólya analogue

Wiener's theorem has a dual form: for $k \in L^1(\mathbb{R})$, the convolution equation $\int k(x-y)f(y)\,dy = 0$ for all x has a bounded solution $f \neq 0$ if and only if $\hat k$ has a real

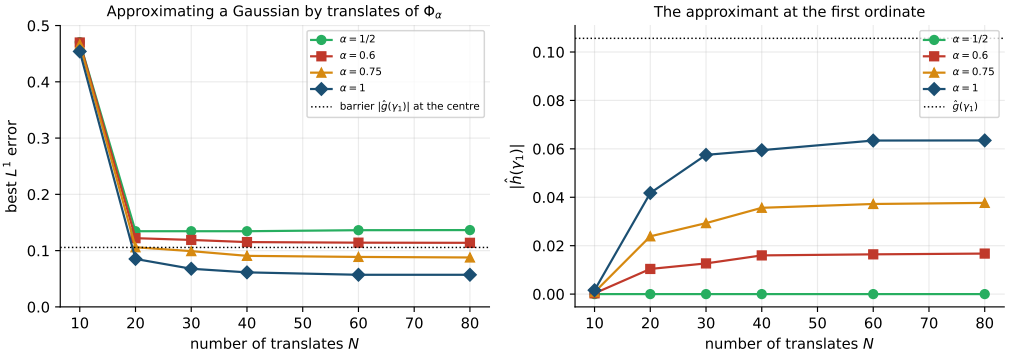


Figure 10.4: Left: best L^1 error against the number of translates, with the barrier at the centre. Right: the Fourier transform of the best approximant at the first ordinate, against the target $\hat{g}(\gamma_1)$. The data, which include $N = 30$ and $N = 60$, are in `wiener.json`, written by `wiener.py`, and `figs_wiener.py` draws the figure.

zero, and then $f(y) = e^{-i\gamma y}$ is a solution for any real zero γ. Salem [229] used it to give one of the classical equivalents of RH: RH holds if and only if, for every $\frac{1}{2} < \sigma < 1$, the equation

$$\int_{\mathbb{R}} \frac{e^{-\sigma y} f(y)}{e^{e^{x-y}} + 1}\, dy = 0 \qquad (x \in \mathbb{R})$$

has no bounded solution other than $f = 0$. Multiplied by $e^{\sigma x}$, it is the convolution equation with the kernel $k_\sigma(u) = e^{\sigma u}/(e^{e^u} + 1)$, the Fermi Dirac function $1/(e^w + 1)$ in the logarithmic variable $w = e^u$, and that substitution gives its Fourier transform,

$$\hat{k}_\sigma(\theta) = \int_{\mathbb{R}} \frac{e^{\sigma u} e^{i\theta u}}{e^{e^u} + 1}\, du = \int_0^\infty \frac{w^{\sigma + i\theta - 1}}{e^w + 1}\, dw = \Gamma(\sigma + i\theta)\, \eta(\sigma + i\theta),$$
$$\eta(s) = (1 - 2^{1-s})\zeta(s),$$

where η is Dirichlet's alternating zeta function, not the Dedekind eta function of Section 11.3. So for $\frac{1}{2} < \sigma < 1$ the real zeros of $\hat{k}_\sigma$ are exactly the ordinates of the zeros of ζ on the line $\operatorname{Re} s = \sigma$, and Salem's criterion is Proposition 10.1.9(4) with the Fermi Dirac kernel in place of the tilted Pólya kernel (10.1.4). Our family gives the Pólya analogue directly.

Corollary 10.1.10. *RH holds if and only if, for every $\frac{1}{2} < \alpha < 1$, the equation $\int_{\mathbb{R}} \Phi_\alpha(x-y) f(y)\, dy = 0$ for all x has no bounded solution $f \neq 0$. At the centre the equation has the explicit bounded solutions $f(y) = \cos(\gamma y)$, one for each ordinate γ of a zero on the critical line, because $\Xi(\gamma) = 0$ there; at the basepoint one it has none.*

Proof. By the dual form of Wiener's theorem with $k = \Phi_\alpha$, a bounded solution $f \neq 0$ exists exactly when $\hat{\Phi}_\alpha(\theta) = \xi(\alpha + i\theta)$ has a real zero, that is, when ζ has a zero on the line $\operatorname{Re} s = \alpha$. That no such zero exists for any $\alpha \in (\frac{1}{2}, 1)$ means that no zero has $\frac{1}{2} < \operatorname{Re}\rho < 1$, which, since no zero lies on the line $\operatorname{Re} s = 1$, is RH by the symmetry $\rho \mapsto 1 - \rho$. At the centre Φ is even, so $\int_{\mathbb{R}} \Phi(x - y)\cos(\gamma y)\, dy = \Xi(\gamma)\cos(\gamma x)$, which vanishes when $\frac{1}{2} + i\gamma$ is a zero; at the basepoint one $\xi(1 + i\theta) \neq 0$ for every real θ. □

The two families differ at the edge of the strip. The factor $1 - 2^{1-s}$ in η vanishes at $s = 1 + 2\pi i k/\log 2$ with $k \neq 0$, so Salem's kernel has real Fourier zeros on the line $\sigma = 1$, at $\theta = 9.0647, 18.1294, 27.1942, \ldots$, and its translates are not complete there, whereas the Pólya family is complete at $\alpha = 1$ because $\xi(1 + i\theta) \neq 0$. Inside the strip the two families have the same zeros and the criteria coincide. The left panel of Figure 10.5, computed by `salem_riesz.py`, shows $|\eta(\sigma + i\theta)|$. At $\sigma = \frac{1}{2}$ it dips to zero at every ordinate; at $\sigma = 0.6$ the dips are shallow, the deepest on $(0, 60]$ being 0.175 near γ_1 and 0.189 near γ_6; at $\sigma = 0.75$ no dip goes below 0.2; and at $\sigma = 1$ the zeros of $1 - 2^{1-s}$ appear. The depth of the dips plays for Salem's kernel the role that $|\xi(\alpha + i\gamma)|$ plays in the cost of approximation of Proposition 10.1.9(5), and it is of order $\sigma - \frac{1}{2}$ near each ordinate.

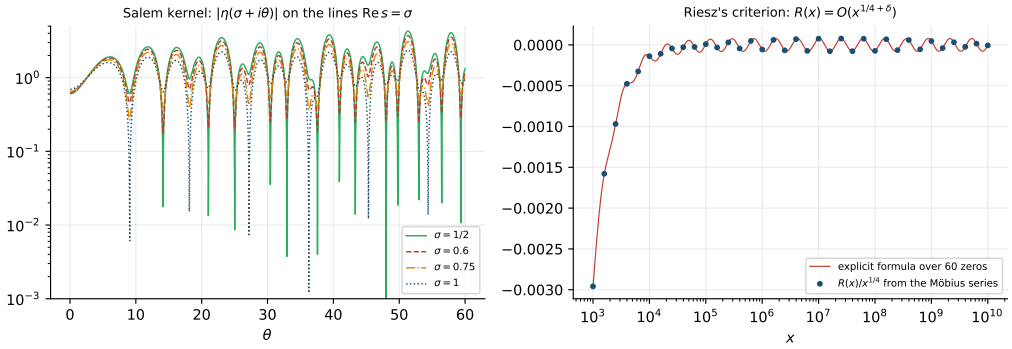


Figure 10.5: Left: $|\eta(\sigma + i\theta)|$, the Fourier transform of Salem's kernel divided by $\Gamma(\sigma + i\theta)$, at $\sigma = \frac{1}{2}$, 0.6, 0.75, 1. Right: the Riesz function $R(x)/x^{1/4}$ from the Möbius series and from the explicit formula (10.1.7) over 60 zeros. The data are in `salem_riesz.json`, written by `salem_riesz.py`; `figs_wiener.py` draws the figure and evaluates the explicit formula with the first 60 ordinates in `g_1_400.npy` and the trivial terms in `riesz_explicit.json`.

10.1.8 Riesz's criterion and multiplicative completeness

Riesz [220] found the first criterion of this type, in terms of growth rather than completeness. With the Möbius function $\mu(n)$, which is unrelated to the Thorin

measure μ_ε of Section 1.2, put

$$R(x) = -\sum_{k\geq 1} \frac{(-x)^k}{(k-1)!\,\zeta(2k)} = x\sum_{n\geq 1} \frac{\mu(n)}{n^2}\, e^{-x/n^2}, \tag{10.1.5}$$

the second form following from $1/\zeta(2k) = \sum_n \mu(n) n^{-2k}$. Riesz proved that RH holds if and only if $R(x) = O(x^{1/4+\delta})$ for every $\delta > 0$. The Mellin transform explains why: for $\frac{1}{2} < \operatorname{Re} s < 1$,

$$\int_0^\infty R(x)\, x^{-s-1} dx = \frac{\Gamma(1-s)}{\zeta(2s)}, \tag{10.1.6}$$

whose poles in the strip $0 < \operatorname{Re} s < \frac{1}{2}$ sit at $s = \rho/2$, and shifting the contour gives, when the zeros are simple, the explicit formula

$$R(x) = \sum_{\rho} \frac{\Gamma(1-\rho/2)}{2\zeta'(\rho)}\, x^{\rho/2} + \sum_{k\geq 1} \frac{\Gamma(1+k)}{2\zeta'(-2k)}\, x^{-k}, \tag{10.1.7}$$

in which the sum over the zeros is the limit of its partial sums over $|\gamma| < T_\nu$ along a suitable sequence of heights $T_\nu \to \infty$. The same argument gives the basepoint version: for $\frac{1}{2} \leq \alpha < 1$, ζ has no zeros in $\operatorname{Re} s > \alpha$ if and only if $R(x) = O(x^{\alpha/2+\delta})$ for every $\delta > 0$. If the bound holds then, since also $|R(x)| \leq \zeta(2)x$, the Mellin integral converges absolutely and is holomorphic on $\frac{\alpha}{2} < \operatorname{Re} s < 1$, so $\Gamma(1-s)/\zeta(2s)$ continues holomorphically there and $\zeta(w) \neq 0$ for $\alpha < \operatorname{Re} w < 2$. Conversely, if ζ has no zeros in $\operatorname{Re} s > \alpha$, the inversion integral $R(x) = \frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty} \Gamma(1-s)\zeta(2s)^{-1} x^s\, ds$, $\frac{1}{2} < c < 1$, moves to the line $\operatorname{Re} s = \frac{\alpha}{2} + \delta$ without crossing a pole and gives $|R(x)| \leq C_\delta x^{\alpha/2+\delta}$, provided $1/\zeta(\sigma + it)$ grows at most like a power of $|t|$ on $\sigma \geq \alpha + 2\delta$, because $\Gamma(1-s)$ decays exponentially in vertical strips. The zero free half plane supplies that bound: the Borel Carathéodory lemma applied to $\log\zeta$ on discs centred at $2 + it$ gives $\log\zeta(\sigma + it) = O_\delta(\log|t|)$ there, as in [255, Section 14.2] with the line $\operatorname{Re} s = \alpha$ in place of the critical line; Proposition 10.2.1 carries out this argument in detail, for a twisted form of R. So Riesz's criterion runs through the same family of half planes as the constructions of Chapter 9, with the growth exponent $\alpha/2$ in place of the basepoint.

In the language of this book, $R(x)/x = \sum_n \mu(n) n^{-2} e^{-x/n^2}$ is the Laplace transform of the signed Möbius measure $\sum_n \mu(n) n^{-2} \delta_{1/n^2}$. It is a heat trace on the arithmetic side, with the Möbius function in place of the zeros, and it is the same Möbius series that underlies the representation of the reciprocal zeta function in Section 6.1 (Theorem 6.1.5). Its sign carries no information; RH is encoded in its rate of growth, exactly as the zero free half planes are encoded in the growth of the error terms of Section 9.1.5. The coefficients $1/\zeta(2k)$ of the power series of R are, up to elementary factors, values of the Laplace transforms of the centre clock T and of T_1 (Section 10.2.2).

We computed $R(x)$ with `salem_riesz.py` from the Möbius series with $n \le N = 10^7$, approximating

$$R(x) \approx x\Big[\sum_{n\le N} \mu(n) n^{-2}\big(e^{-x/n^2} - 1\big) + \frac{6}{\pi^2}\Big]$$

to avoid the slowly converging tail, for $10 \le x \le 10^{12}$. Beyond $x \approx 10^{10}$ the truncation at $N = 10^7$ limits the accuracy: the omitted terms are about $-x^2 \sum_{n>N} \mu(n) n^{-4} \approx 9 \times 10^{-26} x^2$, that is 3×10^{-8} in $R(x)/x^{1/4}$ at $x = 10^{10}$ and 9×10^{-5} at $x = 10^{12}$. For $10^4 \le x \le 10^{10}$ the ratio $R(x)/x^{1/4}$ oscillates with amplitude about 1.1×10^{-4}, and for $10^5 \le x \le 10^{10}$ with amplitude 7.8×10^{-5}, which is the amplitude $|\Gamma(1-\rho_1/2)/\zeta'(\rho_1)|$ of the term of the first zero; the small constant reflects the decay of $\Gamma(1-\rho/2)$ along the critical line, which also makes the term of the second zero about 300 times smaller. The larger swing near $x = 10^4$ comes from the term $\Gamma(2)x^{-1}/(2\zeta'(-2)) = -16.42\,x^{-1}$ of the first trivial zero, which contributes -1.6×10^{-4} to the ratio at $x = 10^4$; the next two trivial zeros enter with the coefficients $\Gamma(1+k)/(2\zeta'(-2k)) = 125.25$ and -508.50 for $k = 2$ and 3. The explicit formula (10.1.7) with the first 60 zeros and these three trivial terms reproduces the computed values to 3.1×10^{-8} on $[10^3, 10^{10}]$, against a signal of size up to 3×10^{-3} (`riesz_explicit.py`, output `riesz_explicit.json`; Figure 10.5, right). The largest difference, at $x = 10^{10}$, is the truncation error just described, and on $[10^3, 10^9]$ the difference is at most 6.2×10^{-10}. The decay of $\Gamma(1-\rho/2)$ hides every zero above the first few from R. Section 10.2.1 twists the Möbius series by $(x/n^2)^{ih}$ so that it responds to the zeros near height $2h$, and computes R to $x = 10^{12}$ and its twists to $x = 10^{10}$.

Wiener's theorem concerns translates. Its multiplicative counterpart concerns dilates, and there completeness is equivalent to RH itself. Nyman and Beurling [170, 25] proved that RH holds if and only if the constant function 1 lies in the $L^2(0,1)$ closure of the span of the functions $r_\lambda(x) = \{\lambda/x\} - \lambda\{1/x\}$, $0 < \lambda \le 1$, where $\{\cdot\}$ is the fractional part, and Báez-Duarte [11] showed that $\lambda = 1/n$ suffices. Under $x = e^{-u}$ this is a closure problem for translates in a weighted L^2 space on the half line, and the Mellin transform of r_λ is $-\zeta(s)(\lambda^s - \lambda)/s$, so the zeros of ζ in the half plane $\operatorname{Re} s > \frac12$ are exactly what obstructs the closure. The same framework governs the dilates $\operatorname{sech}(nx)$ of the Lai and Vere-Jones identity (Proposition 6.1.3), which lies behind the expansion of Theorem 6.1.5: the Mellin transform of sech is $2\Gamma(s)L(s,\chi_4)$, and completeness of systems of dilates is controlled by where the associated Dirichlet series vanishes, in the sense of Beurling's periodic dilation problem and the work of Hedenmalm, Lindqvist and Seip [95]. How the two questions are related is the Nyman Beurling problem of Section 7.2.5. Each of these completeness questions is by itself equivalent to RH or to the generalised Riemann hypothesis. The additive family of Proposition 10.1.9 is the probabilistic one, and it reaches RH only as $\alpha \downarrow \frac12$, in the same way as every other construction of Chapters 9 to 11.

The normal scale mixtures of this section are Brownian motion run at the clock $2T_\alpha$, and their classes are read off those of T_α: infinite divisibility holds with no hypothesis, and the EGGC property holds exactly when T_α is a GGC. The completeness statements are the same zero free half planes read through Fourier transforms, and Riesz's criterion reads them through the growth of $R(x)$. Section 10.2 takes up that criterion.

10.2 Riesz's criterion and the Thorin path

Section 10.1.8 introduced Riesz's criterion [220] and its basepoint form: for $\frac{1}{2} \le \alpha < 1$, the Riesz function (10.1.5), an entire function of x, is $O(x^{\alpha/2+\delta})$ for every $\delta > 0$ exactly when ζ has no zeros in the half plane $\operatorname{Re} s > \alpha$, because its Mellin transform (10.1.6) is $\Gamma(1-s)/\zeta(2s)$; RH is the case $\alpha = \frac{1}{2}$. Hardy and Littlewood [89] gave a companion criterion with $\zeta(2k+1)$ in place of $\zeta(2k)$ and the exponent $-\frac{1}{4}$, and the constructions of this section apply to it in the same way.

We develop the criterion in two directions. The first is computational. Twisting the Riesz function by $(x/n^2)^{ih}$ keeps the criterion equivalent to RH and moves its sensitivity from the first few zeros (Section 10.1.8) to the zeros near height $2h$, and both R and its twists can be computed from the Möbius function alone, independently of any evaluation of ζ (Section 10.2.1). The second reads the coefficients $1/\zeta(2k)$ through the clocks of Chapter 9. With no hypothesis, $1/\zeta(2k)$ is, up to an explicit factor, a value of the Laplace transform of the centre clock at $(2k-\frac{1}{2})^2$, and $R(x)$ is a clock average of an alternating series with Gaussian decay in k, which reduces the cancellation in Riesz's series from about 10^{4344} to about 10^{81} at $x = 10^4$ (Section 10.2.2). At the basepoint one the same computation writes every coefficient as a value of the Laplace transform of the GGC T_1, whose Thorin density the sine squared lemma builds from the primes, and splits it into a positive stable value and a GGC value; Riesz's criterion becomes the statement that the clocks T_α remain GGCs for every $\alpha > \frac{1}{2}$ (Section 10.2.3). The coefficients are also killing probabilities of a sticky Brownian hitting time, and the stable factor is the probability that Brownian motion reaches the level $\sum_\rho \rho^{-1}/\sqrt{2}$ before it is killed; if the Brownian boundary of T_1 is bounded at infinity, as the computation of Section 9.2.4 suggests, this level is its Wald defect (Section 10.2.4). Half Cauchy variables and the exit time of Brownian motion from an interval give a representation of $\zeta(2k)$ dual to that of $1/\zeta(2k)$ (Section 10.2.5). None of this supplies the growth bound. The bound depends on the interpolation of $k \mapsto 1/\zeta(2k)$ into the strip $\frac{1}{4} < \operatorname{Re} k < \frac{1}{2}$, and a GGC representation there is again a zero free half plane (Section 10.2.6).

We keep the notation of Chapter 9 and Section 10.1 for $T = T_{1/2}$, T_α, φ_α, H_a, b_α and v_α; in particular $b_1 = \sum_\rho \rho^{-1} = 0.0230957$, and $\rho_{1/2} = v_1/\pi$ of (9.1.2) is the Thorin density of T_1 in the chart $\theta = \sqrt{z}$. We write $\widetilde{T} = \widetilde{T}_1$ for the centre clock tilted by $e^{-T/4}$, as in Theorem 9.2.2, $\psi(y) = \sum_{n \le y} \Lambda(n)$ for Chebyshev's

function, and

$$\kappa_k = \frac{2k(2k-1)(k-1)!}{\pi^k}, \qquad c_k = \frac{1}{\kappa_k\,\zeta(2k)},$$

so that $\xi(2k) = \frac{1}{2}\kappa_k\zeta(2k)$ and $c_k = \xi(1)/\xi(2k)$. The scripts that we name at each table and figure are in the folder `riesz_program/code` of the archive, and their outputs are in the folder `riesz_program/data`.

10.2.1 Which zeros the Riesz function sees

On the range computed in Section 10.1.8 the Riesz function behaves exactly as RH predicts, and its fluctuations are accounted for by the known zeros, in fact by very few of them. In the explicit formula (10.1.7) a zero ρ enters $R(x)$ through the term $r_\rho x^{\rho/2}$ with the weight $r_\rho = \Gamma(1-\rho/2)/(2\zeta'(\rho))$, and these weights decay exponentially along the critical line. By Stirling's formula $|\Gamma(\frac{3}{4} - \frac{i}{2}\gamma)|$ decays like $e^{-\pi\gamma/4}$, and a fit over the first 120 zeros gives $\log|r_\rho| \approx -0.786\,\gamma + \text{const}$, against $\pi/4 = 0.785$. The weight is 3.9×10^{-5} for the first zero, whose term and that of its conjugate give the amplitude 7.8×10^{-5} found in Section 10.1.8, 2.1×10^{-17} for the tenth, 1.5×10^{-49} for the fiftieth and 2.2×10^{-81} for the hundredth (`sens.py`, output `sens.json`; Figure 10.6, left). The Riesz function is therefore, numerically, a function of the first few zeros only. An off line zero at height 1000 would enter with a weight near 10^{-340}, and no feasible range of x would reveal it. Riesz's criterion is equivalent to RH, but as a computational test it examines only the bottom of the spectrum.

The remedy is to change the kernel. For real h put

$$R_h(x) = x \sum_{n \ge 1} \frac{\mu(n)}{n^2} \Big(\frac{x}{n^2}\Big)^{ih} e^{-x/n^2}, \qquad x > 0,$$

so that $R_0 = R$. The twist changes the Mellin transform only by a shift in the Gamma factor, which has no zeros, and so it keeps everything that makes Riesz's criterion work.

Proposition 10.2.1. *For every real h and $\frac{1}{2} < \operatorname{Re} s < 1$,*

$$\int_0^\infty R_h(x)\, x^{-s-1} dx = \frac{\Gamma(1-s+ih)}{\zeta(2s)}.$$

For $\frac{1}{2} \le \alpha < 1$, ζ has no zeros in the half plane $\operatorname{Re} s > \alpha$ if and only if $R_h(x) = O(x^{\alpha/2+\delta})$ for every $\delta > 0$. In particular RH holds if and only if $R_h(x) = O(x^{1/4+\delta})$ for every $\delta > 0$.

Proof. Term by term, $\int_0^\infty x^{1+ih} n^{-2-2ih} e^{-x/n^2} x^{-s-1} dx = n^{-2s}\Gamma(1-s+ih)$, and summing against $\mu(n)$ gives $\Gamma(1-s+ih)\sum_n \mu(n) n^{-2s}$; the interchange is justified because, with $\sigma = \operatorname{Re} s$, the integrals of the absolute values sum to $\Gamma(1-\sigma)\zeta(2\sigma) <$

∞. Suppose that ζ has no zeros in $\operatorname{Re} s > \alpha$. Then $1/\zeta(2s)$ is analytic in $\operatorname{Re} s > \frac{\alpha}{2}$. For $|t| \geq 10$ the Borel Carathéodory lemma, applied to $\log\zeta$ on the disc $|w-(2+it)| \leq 2-\alpha-\delta$, where $\log|\zeta(w)| \leq A\log|t|$ and $|\log\zeta(2+it)| \leq \log\zeta(2)$, gives $|\log\zeta(w)| \leq C_\delta \log|t|$ on the smaller disc of radius $2-\alpha-2\delta$; this is the argument of [255, Section 14.2] with the line $\operatorname{Re} w = \alpha$ in place of the critical line, as in Section 10.1.8. Hence $|1/\zeta(2s)| \leq |2\operatorname{Im} s|^{C_\delta}$ on $\operatorname{Re} s \geq \frac{\alpha}{2}+\delta$, $|\operatorname{Im} s| \geq 5$, while $|\Gamma(1-s+ih)|$ decays like $e^{-\pi|\operatorname{Im} s - h|/2}$ in vertical strips. So the Mellin inversion integral $R_h(x) = \frac{1}{2\pi i}\int_{(c)} \Gamma(1-s+ih)\zeta(2s)^{-1}x^s\,ds$, with $\frac12 < c < 1$, can be moved to the line $\operatorname{Re} s = \frac{\alpha}{2}+\delta$, which gives $R_h(x) = O(x^{\alpha/2+\delta})$. Conversely, if $R_h(x) = O(x^{\alpha/2+\delta})$ as $x\to\infty$, then, since also $|R_h(x)| \leq \zeta(2)\,x$, the Mellin integral converges absolutely for $\frac{\alpha}{2}+\delta < \operatorname{Re} s < 1$ and continues $\Gamma(1-s+ih)/\zeta(2s)$ analytically there. As Γ has no zeros, $\zeta(2s)$ has no zeros with $\alpha+2\delta < \operatorname{Re} 2s < 2$; since $\delta > 0$ is arbitrary and ζ has no zeros with $\operatorname{Re} s \geq 1$, it has none with $\operatorname{Re} s > \alpha$. At $h = 0$ this is the basepoint version of Riesz's criterion stated in Section 10.1.8. $\square$

Shifting the contour further gives, when the zeros are simple, the twisted form of the explicit formula (10.1.7), with the sum over the zeros again the limit of its partial sums over $|\gamma| < T_\nu$ along a suitable sequence $T_\nu \to \infty$:

$$R_h(x) = \sum_{\rho} \frac{\Gamma(1-\rho/2+ih)}{2\zeta'(\rho)}\,x^{\rho/2} + \sum_{k\geq 1} \frac{\Gamma(1+k+ih)}{2\zeta'(-2k)}\,x^{-k}. \tag{10.2.1}$$

The heights T_ν are those of [255, Theorem 9.7], on which $1/\zeta(\sigma+iT_\nu) = O(T_\nu^A)$ for $-1 \leq \sigma \leq 2$; with the functional equation for $\operatorname{Re} 2s \leq -1$ this makes the integrals over the horizontal segments $\operatorname{Im} 2s = \pm T_\nu$ tend to 0. On the line $\operatorname{Re} s = -q-\frac12$ we write $1/\zeta(2s) = \pi^{1/2-2s}\Gamma(s)/(\Gamma(\frac12-s)\zeta(1-2s))$; since $|\Gamma(1-s+ih)\Gamma(s)| \leq e^{\pi|h|/2}\pi/\cosh(\pi\operatorname{Im} s)$, $|\Gamma(\frac12-s)| \geq q!\,(\pi t/\sinh\pi t)^{1/2}$ with $t = \operatorname{Im} s$, and $|\zeta(1-2s)| \geq 2-\zeta(4)$, the integral over that line is at most $Ce^{\pi|h|/2}\pi^{2q}x^{-q-1/2}/q!$, which tends to 0 as $q\to\infty$.

What the twist changes is the weight. Since $|\Gamma(\frac34 + i(h-\gamma/2))|$ is largest when $\gamma \approx 2h$ and decays like $e^{-\pi|h-\gamma/2|/2}$ away from it, R_h sees the zeros in a band of width about ten around height $2h$. At $h = 50$ the largest weight among the first 300 zeros is 0.124, at $\gamma = 101.3$, where the untwisted weight is 2.9×10^{-35} (Figure 10.6, left). With `twisted.py` (output `twisted.json`) we computed $R_{50}(x)$ from the Möbius series with μ sieved to 10^7, summing the tail through $\sum_n \mu(n)n^{-2-2ih} = 1/\zeta(2+2ih)$, at forty values of x in $[10^4, 10^{10}]$, and compared it with (10.2.1) over the first 300 zeros and three trivial terms. The two agree to 1.2×10^{-8} in $|R_h(x)|/x^{1/4}$, against a signal of size up to 0.27, and the zeros with $|\gamma-100| < 15$ carry 99.9997% of the modulus of the explicit formula, in the median over the forty values of x (Figure 10.6, right).

A family $\{R_h\}$ of twisted Riesz functions is therefore a bank of filters, each an arithmetic quantity computed from the Möbius function alone and each tuned to a

band of heights. An off line zero with real part β near height $2h$ would contribute a term of size $x^{\beta/2}$, growing relative to the background $x^{1/4}$ by the factor $x^{(\beta-1/2)/2}$: a factor 3.2 between $x = 10^{10}$ and $x = 10^{20}$ when $\beta = 0.6$.

Remark 10.2.2. Twisted Riesz functions are not competitive with direct zero finding as a verification method: locating the zeros near height $2h$ by the Riemann Siegel formula is far cheaper than computing $R_h(x)$ for large x. Their value is independence. They are computed from $\mu(n)$ only, with no evaluation of ζ, so agreement between R_h and the explicit formula over computed zeros is a check of the zeros that shares no code or method with their computation.

The cost of $R(x)$ or $R_h(x)$ is dominated by sieving $\mu(n)$ to $N = K\sqrt{x}$. Terms with $n < \sqrt{x}/12$ are below e^{-144} and can be omitted, and the terms with $n > N$ are summed through $\sum_n \mu(n)n^{-2} = 6/\pi^2$, or $1/\zeta(2+2ih)$ for R_h: the script computes $R(x)/x$ as $\sum_{\sqrt{x}/12 \le n \le N} \mu(n)n^{-2}e^{-x/n^2} + \big(6/\pi^2 - \sum_{n\le N} \mu(n)n^{-2}\big)$, and the neglected remainder $\sum_{n>N} \mu(n)n^{-2}(e^{-x/n^2} - 1) \approx -x \sum_{n>N} \mu(n)n^{-4}$ is of order $xN^{-7/2}$ under square root cancellation in the Möbius tail, that is, of order $K^{-7/2}$ in $R(x)/x^{1/4}$. Without that cancellation, $|e^{-u} - 1| \le u$ bounds it in $R(x)/x^{1/4}$ only by about $K^{-3}x^{1/4}/3$, which is 4×10^{-5} at $x = 10^{12}$ and $K = 200$, so the agreement reported next rests on the cancellation. That cancellation is proved in the range used: the Mertens function $\sum_{n\le t} \mu(n)$ is less than $0.571\sqrt{t}$ in absolute value for $33 \le t \le 10^{16}$ [98, Section 6.1], and partial summation then bounds the remainder in $R(x)/x^{1/4}$ by 1.1×10^{-8} for $K = 200$ and $1 \le x \le 10^{12}$. The script `riesz_large.py` implements this with a segmented sieve and extended precision (numpy's long double, which is the 80 bit x87 format on Intel and AMD processors under Linux but only double precision on Apple silicon). With $K = 200$ it reproduces the explicit formula over 60 zeros and three trivial terms to about 2×10^{-9} in $R(x)/x^{1/4}$ at $x = 10^8$, 10^{10}, 10^{11} and 10^{12}, taking 42 seconds at 10^{12} in Python (`check_large.py`, output `check_large.json`). Its twisted option agrees as well: at $x = 10^{10}$ with $h = 50$ it gives $R_{50}(x) = 37.441416 - 26.042495\,i$, within 3.1×10^{-9} of (10.2.1) over 300 zeros in $|R_h(x)|/x^{1/4}$ (`check_large_twisted.py`, output `check_large_twisted.json`). A compiled version would reach $x = 10^{16}$, where $N = 2 \times 10^{10}$, in hours; beyond that the partial sums $\sum_{n\le N} \mu(n)n^{-2}$ should be computed by the combinatorial methods used for the Mertens function rather than by sieving, and the arithmetic moved to quadruple precision. Section 12.2 makes these computations the first work package of a programme of computations for the Thorin path (Table 12.2).

10.2.2 The Riesz coefficients through the centre clock

The Riesz coefficients are values of two Laplace transforms: that of the centre clock T of [198], which is infinitely divisible and self decomposable with no hypothesis (Theorem 5.1.1, whose proof uses the verification of RH below height

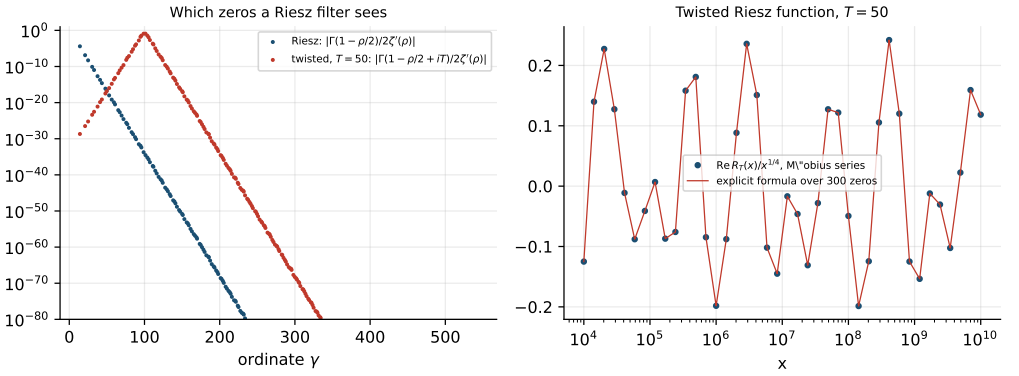


Figure 10.6: Left: the weights $|\Gamma(1-\rho/2)/(2\zeta'(\rho))|$ of the Riesz function and $|\Gamma(1-\rho/2+ih)/(2\zeta'(\rho))|$ of the twisted Riesz function with $h=50$, for the first 300 zeros. Right: $\operatorname{Re} R_{50}(x)/x^{1/4}$ from the Möbius series against the explicit formula (10.2.1) over 300 zeros. The plot writes T for h. Data from `twisted.py`; drawn by `figs_strategy.py`.

1000) and is a GGC, with Thorin measure $\sum_{\gamma>0}\delta_{\gamma^2}$, exactly when RH holds (Proposition 1.2.1), and that of the basepoint one clock T_1, which is a GGC with no hypothesis (Proposition 9.1.1).

Proposition 10.2.3. *For every integer $k\ge 1$, with no hypothesis,*

$$\frac{1}{\zeta(2k)}=\frac{\kappa_k}{2\xi(\frac12)}\,\mathbb{E}\,e^{-(2k-\frac12)^2T}=\kappa_k\,\mathbb{E}\,e^{-(2k-1)^2T_1}.$$

Proof. From the definition of ξ, $\xi(2k)=k(2k-1)\pi^{-k}\Gamma(k)\zeta(2k)=\frac12\kappa_k\zeta(2k)$. Since $2k-\frac12>0$, the Laplace transform of T at $(2k-\frac12)^2$ is $\varphi_{1/2}((2k-\frac12)^2)=\xi(\frac12)/\xi(2k)$; similarly $\mathbb{E}e^{-(2k-1)^2T_1}=\xi(1)/\xi(2k)$, with $\xi(1)=\frac12$. □

Inserting the first form into the power series and exchanging sum and expectation gives a probabilistic formula for the Riesz function.

Proposition 10.2.4. *For every $x>0$, with no hypothesis,*

$$R(x)=-\frac{1}{\xi(\frac12)}\,\mathbb{E}\Big[\sum_{k\ge1}k(2k-1)\Big(-\frac{x}{\pi}\Big)^k e^{-(2k-\frac12)^2T}\Big].$$

Proof. Since $\kappa_k/(2(k-1)!)=k(2k-1)\pi^{-k}$, Proposition 10.2.3 turns the term $-(-x)^k/((k-1)!\,\zeta(2k))$ of Riesz's series into $-\xi(\frac12)^{-1}k(2k-1)(-x/\pi)^k\,\mathbb{E}e^{-(2k-\frac12)^2T}$. The exchange of sum and expectation is justified by Fubini's theorem, because

$$\sum_{k\ge1}k(2k-1)\Big(\frac{x}{\pi}\Big)^k\mathbb{E}\,e^{-(2k-\frac12)^2T}=\xi(\tfrac12)\sum_{k\ge1}\frac{x^k}{(k-1)!\,\zeta(2k)}<\infty.$$

□

The Laplace transform of the clock is thus sampled along the quadratic sequence $(2k-\frac{1}{2})^2$, and the terms of the series averaged in Proposition 10.2.4 decay like e^{-4k^2T} rather than like $1/(k-1)!$. Table 10.8, computed by `riesz_thorin.py` (output `riesz_thorin.json`), checks both propositions with 4×10^5 draws of T, generated under RH as $\sum_{\gamma>0} e_\gamma/\gamma^2$ with independent standard exponentials e_γ: the first 200 zeros are used exactly, and the remaining terms are replaced by a gamma variable with the same mean and variance, computed from the exact power sums $\sum\gamma^{-2}$ and $\sum\gamma^{-4}$. The coefficients agree within about one standard error, and so do the values of R from the Möbius series, the power series and the clock average. The middle panel of Figure 10.7 shows the coefficients through $1-1/\zeta(2k)$, which is $4^{-k}+O(9^{-k})$.

k	$1/\zeta(2k)$	via the clock	standard error
1	0.6079271	0.6079115	0.0000131
2	0.9239384	0.9238153	0.0001046
3	0.9829526	0.9826581	0.0002603
4	0.9959392	0.9954609	0.0004615

x	$R(x)$, Möbius	clock average	standard error
0.5	0.1252199	0.1252287	0.0000074
1	0.0439818	0.0440007	0.0000157
2	−0.2677934	−0.2677918	0.0000026
5	−0.7705597	−0.7706050	0.0000397

Table 10.8: Left: the Riesz coefficients from the clock, Proposition 10.2.3. Right: the Riesz function from the Möbius series, which agrees with the power series to fourteen digits, and from the clock average of Proposition 10.2.4.

The clock average is also far better conditioned than the power series. Riesz's power series is numerically hopeless for large x: its largest term is about $e^x\sqrt{x/2\pi}$, that is $10^{4344.5}$ at $x=10^4$, while $R(10^4)$ is of order 10^{-3}. In the clock average the largest term of the inner series, $\max_k k(2k-1)(x/\pi)^k e^{-(2k-\frac{1}{2})^2T}$, is about 10^{81} at $x=10^4$ for the typical value $T=0.023$, 10^{182} for $T=0.01$ and 10^{39} for $T=0.05$, and its logarithm grows like $(\log x)^2/(16T)$ instead of like x (Figure 10.7, right). The clock average thus reduces the cancellation without removing it, and Section 10.2.6 explains why that is not enough.

10.2.3 The sine squared lemma and a GGC form of Riesz's criterion

Proposition 10.2.3 used the centre clock, a GGC only under RH, and the basepoint one clock, a GGC with no hypothesis, and the same computation works at every basepoint between them. For $\alpha>\frac{1}{2}$ the clock T_α exists with no hypothesis, as

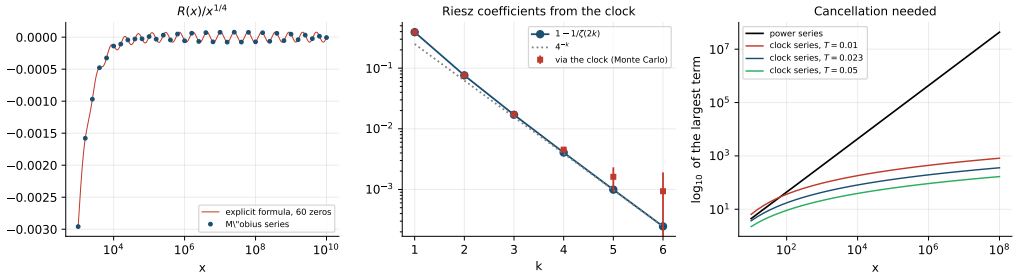


Figure 10.7: Left: $R(x)/x^{1/4}$ from the Möbius series and from the explicit formula over 60 zeros, as in the right panel of Figure 10.5. Middle: $1 - 1/\zeta(2k)$ and its Monte Carlo value from the clock (Proposition 10.2.3), against 4^{-k}. Right: base ten logarithm of the largest term in Riesz's power series and in the clock series of Proposition 10.2.4 for $T = 0.01$, 0.023 and 0.05. Drawn by `figs_riesz.py` from `salem_riesz.json`, `riesz_explicit.json` and `riesz_thorin.json`.

the centre clock tilted and subordinated (Theorem 9.2.5), and it is a GGC exactly when ζ has no zeros with $\operatorname{Re} s > \alpha$ (Theorem 1.2.2), in particular for every $\alpha \geq 1$.

PROPOSITION 10.2.5 (Riesz coefficients as GGC values). *For every $\alpha \in [\frac{1}{2}, 1]$ and every integer $k \geq 1$,*

$$\frac{1}{\zeta(2k)} = \frac{\kappa_k}{2\xi(\alpha)}\, \mathbb{E}\, e^{-(2k-\alpha)^2 T_\alpha}.$$

At $\alpha = 1$, with no hypothesis, T_1 is a GGC with Thorin density $\rho_{1/2} = v_1/\pi$ in the chart $\theta = \sqrt{z}$, and

$$\frac{1}{\zeta(2k)} = \kappa_k \exp\Big\{ -\int_0^\infty \log\Big(1 + \frac{(2k-1)^2}{\theta^2}\Big)\rho_{1/2}(\theta)\, d\theta \Big\}.$$

Proof. Since $2k - \alpha > 0$, the Laplace transform of T_α at $(2k-\alpha)^2$ is $\xi(\alpha)/\xi(2k)$, and $\xi(2k) = \frac{1}{2}\kappa_k \zeta(2k)$. At $\alpha = 1$, $\xi(1) = \frac{1}{2}$, and the Thorin representation $\log \varphi_1(u) = -\int_0^\infty \log(1 + u/\theta^2)\rho_{1/2}(\theta)\, d\theta$ of Proposition 9.1.1, taken at $u = (2k-1)^2$, gives the second formula. □

REMARK 10.2.6. All the Riesz coefficients are samples of one GGC Laplace transform: $c_k = \varphi_1((2k-1)^2)$, where $\varphi_1(u) = \mathbb{E}e^{-uT_1}$. As a Laplace transform φ_1 is completely monotone, and this already makes $\log \varphi_1$ convex in u, by Hölder's inequality; so the slopes $(\log c_{k+1} - \log c_k)/8k$ increase with k. As the transform of a GGC, φ_1 is also hyperbolically completely monotone [35, Theorem 6.1.1]: for each $u > 0$ the function $v \mapsto \varphi_1(uv)\varphi_1(u/v)$ is a completely monotone function of $v + 1/v$. In particular it decreases as v moves away from 1, and differentiating $\log \varphi_1(uv) + \log \varphi_1(u/v)$ in $\log v$ shows that $t \mapsto \log \varphi_1(e^t)$ is concave. At the sample points this gives inequalities in the other direction:

the slopes $(\log c_{k+1} - \log c_k)/\log\frac{2k+1}{2k-1}$ decrease with k, and $c_j c_l \le c_k^2$ whenever $(2j-1)(2l-1) = (2k-1)^2$, for instance $c_1 c_5 \le c_2^2$.

The Thorin density v_1/π is built from the primes by the sine squared lemma (1.2.9). Proposition 9.1.3 writes $\log(\xi(1+s)/\xi(1))$ in the Lévy form (9.1.5), with the drift b_1 and a signed Lévy measure Π_1 made of the gamma density, the pole of ζ and the prime atoms, the last two compensating each other by the prime number theorem, and the lemma converts it term by term into the Thorin form $v_1 = b_1 + A + Q$ of (9.1.6), with the gamma piece A and the compensated prime piece Q. Evaluating at $s = 2k-1$ therefore gives the Riesz coefficients from the primes in two equivalent ways. The Lévy form in $s = 2k-1$ is

$$\frac{1}{\zeta(2k)} = \kappa_k \exp\Big\{ -b_1(2k-1) - \int_0^\infty \big(e^{-(2k-1)x} - 1 + (2k-1)x\big)\,\Pi_1(dx)\Big\},$$

and in the Thorin form in $u = (2k-1)^2$ the drift splits off a positive stable one half factor:

$$\frac{1}{\zeta(2k)} = \kappa_k\, e^{-b_1(2k-1)}\, \exp\Big\{ -\frac{1}{\pi}\int_0^\infty \log\Big(1 + \frac{(2k-1)^2}{\theta^2}\Big)\big(A(\theta) + Q(\theta)\big)\,d\theta\Big\}. \tag{10.2.2}$$

This is the factorisation $T_1 \stackrel{d}{=} S + T_1^\circ$ of Proposition 9.1.4 evaluated at $u = (2k-1)^2$: the middle factor is $\mathbb{E}e^{-(2k-1)^2 S}$ for the positive stable variable S of index one half with $\mathbb{E}e^{-uS} = e^{-b_1\sqrt{u}}$, whose Thorin density is the constant b_1/π because $\int_0^\infty \log(1+u/\theta^2)\,d\theta = \pi\sqrt{u}$, and the last factor is $\mathbb{E}e^{-(2k-1)^2 T_1^\circ}$ for the independent GGC T_1° with Thorin density $(A+Q)/\pi = (v_1 - b_1)/\pi$, which is nonnegative because v_1 attains its minimum b_1 only at the origin. So (10.2.2) splits every coefficient into a stable value and a GGC value with no hypothesis on the zeros.

Inserting Proposition 10.2.5 at $\alpha = 1$ into the power series gives the Riesz function as a GGC mixture,

$$R(x) = -2\,\mathbb{E}\Big[\sum_{k\ge1} k(2k-1)\Big(-\frac{x}{\pi}\Big)^k e^{-(2k-1)^2 T_1}\Big], \tag{10.2.3}$$

where the exchange of sum and expectation is justified as in Proposition 10.2.4, since $\sum_k 2k(2k-1)(x/\pi)^k c_k = \sum_k x^k/((k-1)!\,\zeta(2k)) < \infty$. The sign is that of $R(x) = x\sum_n \mu(n) n^{-2} e^{-x/n^2}$: as $x \downarrow 0$ both sides are $6x/\pi^2 + O(x^2)$. The variable T_1 has two realisations. It is the tilted centre clock run through the inverse local time of a sticky Brownian motion (Theorem 9.2.2 and the paragraph after it), and, like every GGC without drift [109], it is the Wiener gamma integral (11.1.4) against the inverse of the Poisson smoothed zero count $M_{1/2}(\theta) = \pi^{-1}\int_0^\theta v_1$ (Section 11.1.6).

Table 10.9, computed by `ggc_coef.py` (output `ggc_coef.json`), checks both forms. The Thorin integral uses v_1 computed from ξ'/ξ on $[0, 200]$ and the Weyl

density $\frac{1}{2\pi}\log\frac{\theta}{2\pi}$ for $\rho_{1/2}$ beyond, which leaves relative errors between 3×10^{-6} and 3.3×10^{-4} in absolute value for $k\le 6$, the largest at $k=6$. They consist of the error of the Weyl tail, about $3.0\times10^{-6}(2k-1)^2$, and the error of the trapezoid rule on $[0,200]$, between -2.98×10^{-5} at $k=1$ and -3.27×10^{-5} at $k=6$ (`ggc_offset.py`, output `ggc_offset.json`). Of the latter, -2.98×10^{-5} is the same for every k: the rule of step 0.025 starts one step from the logarithmic singularity of the integrand at $\theta=0$, and by Stirling's formula this makes the integral too large by $0.025\,(2-\log 2\pi)\,b_1/\pi = 2.98\times10^{-5}$. At $k=6$ the truncated value exceeds one, which $1/\zeta(12)$ cannot: the error of the Weyl tail, 3.6×10^{-4}, exceeds $1-1/\zeta(12)=2.5\times10^{-4}$. The column GGC factor is the truncated Thorin form divided by κ_k times the stable factor, and it carries the same error; the exact GGC factor is $c_k e^{b_1(2k-1)}$, printed beside it. The Lévy form uses primes to 10^7 and carries the error of the compensated prime sum; its row $k=1$ is the row $s=1$, $X=10^7$ of Table 9.4.

k	$1/\zeta(2k)$	Thorin form	stable factor	GGC factor	exact GGC factor
1	0.6079271	0.6079108	0.9771690	0.9772149	0.9772411
2	0.9239384	0.9239355	0.9330587	0.8144252	0.8144277
3	0.9829526	0.9829961	0.8909397	0.5701667	0.5701415
4	0.9959392	0.9960541	0.8507219	0.3394341	0.3393950
5	0.9990064	0.9992163	0.8123196	0.1742722	0.1742356
6	0.9997540	1.0000823	0.7756508	0.0782553	0.0782296

k	Lévy form, primes to 10^7	$\log\frac{\xi(2k)}{\xi(1)}$
1	0.045729	0.046118
2	0.273882	0.274557
3	0.676388	0.677349

Table 10.9: Top: the Riesz coefficients from the Thorin form of Proposition 10.2.5, truncated as described in the text, and the factors of (10.2.2): the stable factor $e^{-b_1(2k-1)}$, the GGC factor computed from the truncated Thorin form, and the exact GGC factor $c_k e^{b_1(2k-1)}$. Bottom: $\log(\xi(2k)/\xi(1))$ from the Lévy form (9.1.5) at $s=2k-1$ with primes to 10^7, against its exact value.

The identities of Proposition 10.2.5 need no hypothesis; what Riesz's criterion asks is whether the clocks T_α with $\alpha<1$ are themselves GGCs.

THEOREM 10.2.7. *For $\frac12\le\alpha<1$ the following are equivalent:*

1. *$R(x)=O(x^{\alpha/2+\delta})$ for every $\delta>0$;*
2. *ζ has no zeros with $\operatorname{Re}s>\alpha$;*
3. *$T_{\alpha'}$ is a GGC for every $\alpha'>\alpha$;*

4. *for every $\alpha' > \alpha$, $v_{\alpha'}(\theta) = \operatorname{Re} \frac{\xi'}{\xi}(\alpha' + i\theta) \geq 0$ at every real θ for which $\alpha' + i\theta$ is not a zero of ξ;*

5. *for every $\alpha' > \alpha$ and every real θ the compensated sine squared sum*

$$\lim_{L\to\infty} \Big[\sum_{\log n \leq L} \frac{\Lambda(n)}{n^{\alpha'}} \, 2\sin^2\Big(\frac{\theta \log n}{2}\Big) - \int_0^L 2\sin^2\Big(\frac{\theta x}{2}\Big) e^{(1-\alpha')x}\, dx \Big]$$

converges.

In particular RH holds if and only if T_α is a GGC for every $\alpha > \frac{1}{2}$.

Proof. Write $\Theta = \sup \operatorname{Re} \rho$ and $D(y) = \psi(y) - y$. Conditions (1) and (2) are equivalent by the basepoint version of Riesz's criterion (Section 10.1.8, and Proposition 10.2.1 with $h = 0$). Conditions (2) and (3) are equivalent by Theorem 1.2.2: $\varphi_{\alpha'}$ is the Laplace transform of a GGC exactly when $\alpha' \geq \Theta$, and this holds for every $\alpha' > \alpha$ exactly when $\Theta \leq \alpha$, which is (2).

(2) implies (4), because if $\Theta \leq \alpha < \alpha'$ every Cauchy kernel in $v_{\alpha'}(\theta) = \sum_\rho (\alpha' - \beta)/((\alpha' - \beta)^2 + (\theta - \gamma)^2)$ has positive width (Proposition 1.2.3). Conversely, let $\rho_0 = \beta_0 + i\gamma_0$ be a zero of multiplicity m with $\beta_0 > \alpha$. The zeros of ξ are isolated, so only finitely many lie on the segment $\{\sigma + i\gamma_0 : 0 \leq \sigma \leq 1\}$. Let n_0 be the number of those other than ρ_0, counted with multiplicity, and choose $\eta \in (0, \beta_0 - \alpha)$ with $|\beta - \beta_0| \geq 2\eta$ for each of them. For $\alpha' \in (\beta_0 - \eta, \beta_0)$ the point $\alpha' + i\gamma_0$ is not a zero, and Proposition 1.2.3 gives

$$v_{\alpha'}(\gamma_0) = \frac{m}{\alpha' - \beta_0} + \sum_{\rho \neq \rho_0,\ \gamma = \gamma_0} \frac{1}{\alpha' - \beta} + \sum_{\gamma \neq \gamma_0} \frac{\alpha' - \beta}{(\alpha' - \beta)^2 + (\gamma - \gamma_0)^2}.$$

The middle sum is at most n_0/η in absolute value. Since α' and β lie in $(0, 1)$, each term of the last sum is at most $(\gamma - \gamma_0)^{-2}$ in absolute value, and $\sum_{\gamma \neq \gamma_0} (\gamma - \gamma_0)^{-2} < \infty$: finitely many zeros have $0 < |\gamma - \gamma_0| < 1$, and for $|\gamma - \gamma_0| \geq 1$ we have $(\gamma - \gamma_0)^{-2} \leq 5/(4 + (\gamma - \gamma_0)^2) \leq 5(2 - \beta)/((2 - \beta)^2 + (\gamma - \gamma_0)^2)$, a series bounded by $5v_2(\gamma_0) < \infty$. Hence $v_{\alpha'}(\gamma_0) \leq m/(\alpha' - \beta_0) + n_0/\eta + \sum_{\gamma \neq \gamma_0} (\gamma - \gamma_0)^{-2} < 0$ for α' close to β_0, and (4) fails at (α', γ_0). This is why (4) quantifies over every $\alpha' > \alpha$: the kernel of an offending zero dominates the others only as α' approaches its real part.

(2) implies (5). This is part 1 of Proposition 9.1.6 at the basepoint α', since $\Theta \leq \alpha < \alpha'$, and the argument is short. Choose σ with $\Theta < \sigma < \alpha'$, so that $D(y) = O(y^\sigma)$ [255]. With $f(y) = y^{-\alpha'} \, 2\sin^2(\frac{1}{2}\theta \log y)$ the bracket in (5) is

$$\int_{(1, e^L]} f \, dD = f(e^L) D(e^L) - \int_1^{e^L} D(y) f'(y)\, dy,$$

since $f(1) = 0$. The boundary term tends to 0, and the integral converges absolutely because $|f'(y)| \leq (2\alpha' + |\theta|) y^{-\alpha' - 1}$.

(5) implies (2). Suppose (5), and let ρ_0 be a zero of multiplicity m with $\operatorname{Re}\rho_0 > \alpha$. Fix $\alpha' \in (\alpha, \operatorname{Re}\rho_0)$ and $\theta \notin \mathcal{E} = \{\theta : \rho_0 + i\theta \text{ or } \rho_0 - i\theta \text{ is a zero}\}$, a countable set that contains 0, and let $M(t) = \int_{(0,t]}(1-\cos\theta s)\,dD(e^s)$, a function of locally bounded variation. The bracket in (5) is $J(L) = \int_{(0,L]} e^{-\alpha' t}dM(t)$; by hypothesis it converges as $L \to \infty$, so J is bounded. For $\operatorname{Re} w > \alpha'$, integration by parts gives

$$\int_{(0,L]} e^{-wt}\,dM(t) = e^{-(w-\alpha')L}J(L) + (w-\alpha')\int_0^L e^{-(w-\alpha')t}J(t)\,dt,$$

which converges as $L \to \infty$, locally uniformly in w, to an analytic function $F_\theta(w)$ on $\operatorname{Re} w > \alpha'$. For $\operatorname{Re} w > 1$ the integral converges absolutely, and writing $1 - \cos\theta t = 1 - \frac{1}{2}e^{i\theta t} - \frac{1}{2}e^{-i\theta t}$ gives

$$F_\theta(w) = g(w) - \tfrac{1}{2}g(w-i\theta) - \tfrac{1}{2}g(w+i\theta),$$
$$g(w) = \int_{(1,\infty)} y^{-w}\,dD(y) = -\frac{\zeta'}{\zeta}(w) - \frac{1}{w-1}.$$

The right side, call it $\widetilde{F}_\theta(w)$, is meromorphic in the plane with a discrete set $\mathcal{P}$ of poles. On the connected open set $\Omega = \{\operatorname{Re} w > \alpha'\} \setminus \mathcal{P}$ both $\widetilde{F}_\theta$ and F_θ are analytic, and they agree on $\operatorname{Re} w > 1$, so $\widetilde{F}_\theta = F_\theta$ on Ω by the identity theorem. The function g is analytic at 1, and in $\operatorname{Re} w > 0$ its only poles are the zeros of ζ, each simple with residue minus its multiplicity. Since $\theta \notin \mathcal{E}$, $g(w \pm i\theta)$ is analytic at ρ_0, so $\widetilde{F}_\theta$ has a pole at ρ_0 with residue $-m \neq 0$, and $|F_\theta(w)| \to \infty$ as $w \to \rho_0$ in Ω. This contradicts the continuity of F_θ at ρ_0, where $\operatorname{Re}\rho_0 > \alpha'$. Hence no zero has real part greater than α, which is (2). □

The theorem makes precise how far the GGC values reach. At the integers every Riesz coefficient is a GGC value at the basepoint one, with no hypothesis, and the sine squared lemma writes it from the primes. Riesz's criterion, however, concerns the analytic interpolation of $k \mapsto 1/\zeta(2k)$ into $\frac{1}{4} < \operatorname{Re} k < \frac{1}{2}$. The representation through T_α is an expectation only on a cone in the half plane $\operatorname{Re} 2k > \alpha$ (Section 10.2.6), and it extends to the whole half plane through the analytic continuation of φ_α to the cut plane $\mathbb{C} \setminus (-\infty, 0]$, which is analytic exactly when no zero lies to the right of α and which, when every zero has real part less than α, is given there by the Thorin form with the positive density v_α/π (Theorem 1.2.2). Riesz's criterion is therefore the statement that the family of clocks $\{T_\alpha\}_{\alpha>1/2}$ stays inside the GGC class all the way to the centre.

10.2.4 The coefficients as killing probabilities

For a positive random variable X and an independent exponential variable E_λ of rate λ, $\mathbb{E}e^{-\lambda X} = P(X < E_\lambda)$, the probability that X occurs before a killing at

rate λ. Propositions 10.2.3 and 10.2.5 therefore say that every Riesz coefficient is a killing probability, in two ways and with no hypothesis:

$$\frac{1}{\zeta(2k)} = \kappa_k \, P\big(T_1 < E_{(2k-1)^2}\big) = \frac{\kappa_k}{2\xi(\frac{1}{2})} \, P\big(T < E_{(2k-\frac{1}{2})^2}\big). \tag{10.2.4}$$

The clock T_1 is itself a hitting time. By Theorem 9.2.2, $T_1 \stackrel{d}{=} \widetilde{T} + H_{\widetilde{T}/\sqrt{2}}$, and by the paragraph after it T_1 is the first time the local time at zero of a sticky Brownian motion exceeds an independent level distributed as $\widetilde{T}$. So $c_k = 1/(\kappa_k \zeta(2k))$ is the probability that this hitting happens before an independent killing at rate $(2k-1)^2$.

The factorisation (10.2.2) has the same reading. Since $\mathbb{E}e^{-\lambda H_a} = e^{-a\sqrt{2\lambda}}$, the stable factor is

$$e^{-b_1(2k-1)} = P\big(H_{b_1/\sqrt{2}} < E_{(2k-1)^2}\big),$$

the probability that Brownian motion reaches the level $b_1/\sqrt{2} = 0.016331$ before it is killed. In other words the stable variable S of Proposition 9.1.4 has the law of the first passage time $H_{b_1/\sqrt{2}}$, and $T_1 \stackrel{d}{=} H_{b_1/\sqrt{2}} + T_1^\circ$ with T_1° an independent GGC. Each Riesz coefficient is thus κ_k times this hitting probability times a killing probability for the GGC remainder: $c_k = P(H_{b_1/\sqrt{2}} < E_{(2k-1)^2})\, P(T_1^\circ < E'_{(2k-1)^2})$, with independent exponential clocks E and E' of the same rate. If the exact Brownian boundary b of T_1 is bounded at infinity, as Section 9.2.4 observes but does not prove, Lemma 9.2.4 and the tail of T_1 make the level $b_1/\sqrt{2} = \sum_\rho \rho^{-1}/\sqrt{2}$ its Wald defect $\mathbb{E}\, b(T_1)$.

Under RH, if the zeros are also simple, more is true. The centre clock is then a GGC with unit atoms at the distinct points γ^2 (Proposition 1.2.1), hence, by Kent's theorem and Krein's inverse spectral theory (Section 8.1.3), the hitting time of a level by the diffusion of a Krein string with Dirichlet spectrum $\{\gamma^2\}$ (Table 4.2). The second form of (10.2.4) then says that $1/\zeta(2k)$ is proportional to the probability that this Riemann string diffusion reaches its level before it is killed at rate $(2k-\frac{1}{2})^2$. At the basepoint one, Corollary 9.2.3 gives the first form the same reading under the same hypotheses, with T_1 the hitting time of a line by the two dimensional process built from the string diffusion of $\widetilde{T}$ and a sticky Brownian motion.

The Riesz function becomes an alternating sum of killing probabilities,

$$R(x) = -\sum_{k\geq 1} \frac{(-x)^k}{(k-1)!} \, \kappa_k \, P\big(T_1 < E_{(2k-1)^2}\big),$$

and the criterion asks that these alternating sums grow no faster than $x^{1/4+\delta}$. Positivity of each probability says nothing about that cancellation. The killing rates $(2k-\alpha)^2$ that the criterion needs, with k in the strip $\frac{1}{4} < \operatorname{Re} k < \frac{1}{2}$, are

complex, and at a complex rate the Laplace transform is no longer a probability; this is the hitting time form of the obstruction of Section 10.2.6.

We checked these identities with `killing.py` (output `killing.json`), using 10^6 draws of $T_1 = \widetilde{T} + H_{\widetilde{T}/\sqrt{2}}$ generated as in Section 9.2.3: under RH, $\widetilde{T} = \sum_k e_k/(\gamma_k^2+\frac{1}{4})$, with the first 200 zeros used exactly and the remaining terms, those of the zeros up to the 1700th and, through the exact power sums $\sum \gamma^{-2}$ and $\sum \gamma^{-4}$, those above, replaced by a gamma variable with the same mean and variance; and $H_a = a^2/Z^2$ with Z standard normal. Table 10.10 compares the killing probabilities, estimated both by drawing the exponential clock and by averaging $e^{-(2k-1)^2 T_1}$, with c_k, and the Brownian hitting probability with $e^{-b_1(2k-1)}$; all agree within two standard errors. With the averaged killing probabilities for $k \le 30$ from a second run of 10^6 draws (`killing2.py`), the alternating sum gives $R(0.5) = 0.125242$, $R(1) = 0.043990$, $R(2) = -0.267895$ and $R(3) = -0.526639$, against the exact 0.125220, 0.043982, -0.267793 and -0.526356. The relative error grows with x because the alternating sum amplifies the Monte Carlo error of the individual probabilities.

k	$P(T_1 < E)$, drawn	$\mathbb{E}e^{-(2k-1)^2 T_1}$	$c_k = \frac{1}{\kappa_k \zeta(2k)}$	$P(H_{b_1/\sqrt{2}} < E)$	$e^{-b_1(2k-1)}$
1	0.95481 ± 0.00021	0.954928 ± 0.000112	0.954930	0.97699 ± 0.00015	0.97717
2	0.75965 ± 0.00043	0.759835 ± 0.000161	0.759909	0.93352 ± 0.00025	0.93306
3	0.50738 ± 0.00050	0.507865 ± 0.000156	0.507962	0.89098 ± 0.00031	0.89094
4	0.28834 ± 0.00045	0.288646 ± 0.000122	0.288731	0.85059 ± 0.00036	0.85072
5	0.14179 ± 0.00035	0.141473 ± 0.000080	0.141535	0.81195 ± 0.00039	0.81232
6	0.06045 ± 0.00024	0.060638 ± 0.000045	0.060679	0.77555 ± 0.00042	0.77565

Table 10.10: Riesz coefficients as killing probabilities of the basepoint one clock, and the stable factor as a Brownian hitting probability; E is an independent exponential clock of rate $(2k-1)^2$, and the errors are one standard error of 10^6 draws.

10.2.5 Half Cauchy variables, the sech clock and a duality

Bourgade, Fujita and Yor [39] derived Euler's formulae for $\zeta(2n)$ from products of Cauchy variables, and the construction gives a representation of $\zeta(2n)$ dual to that of $1/\zeta(2n)$. Let C and C' be independent standard Cauchy variables, so that $|C|$ and $|C'|$ are half Cauchy. By Section 4.2.2, $\mathbb{E}|C|^s = \sec(\pi s/2)$ for $|\operatorname{Re} s| < 1$, so $\sec^2(\pi s/2) = \mathbb{E}|CC'|^s$ is the moment generating function of $\log|CC'|$ on $|s| < 1$. Differentiating the series $\tan z = \sum_{n\ge1} 2^{2n}(2^{2n}-1)|B_{2n}|z^{2n-1}/(2n)!$ and inserting Euler's formula $\zeta(2n) = (2\pi)^{2n}|B_{2n}|/(2(2n)!)$ gives

$$\sec^2\frac{\pi s}{2} = \sum_{n\ge1} \frac{8(2n-1)(1-2^{-2n})\,\zeta(2n)}{\pi^2}\, s^{2n-2},$$

and comparing coefficients with the moment generating function,

$$\zeta(2n) = \frac{\pi^2}{8(2n-1)(1-2^{-2n})}\,\frac{\mathbb{E}\big[(\log|CC'|)^{2n-2}\big]}{(2n-2)!}, \qquad n \geq 1. \tag{10.2.5}$$

The logarithm of a half Cauchy variable is, in law, a Brownian motion stopped at an exit time. Let τ be the exit time of a standard Brownian motion from $(-1,1)$, so that $\mathbb{E}e^{-\lambda\tau} = 1/\cosh\sqrt{2\lambda}$ (Section 4.2.2), and let B be a Brownian motion independent of τ. Then $\mathbb{E}e^{i\theta B_\tau} = \mathbb{E}e^{-\theta^2\tau/2} = \operatorname{sech}\theta$, and since $\log|C|$ has characteristic function $\operatorname{sech}(\pi t/2)$ (Section 4.2.2),

$$\log|C| \stackrel{d}{=} \frac{\pi}{2}\,B_\tau.$$

We call τ the sech clock. By the product $\cosh\sqrt{2\lambda} = \prod_{k\geq 1}\big(1+8\lambda/((2k-1)^2\pi^2)\big)$ it is a GGC with unit Thorin atoms at $(2k-1)^2\pi^2/8$, the GGC $\sum_k e_k/(2k-1)^2$ of Section 4.2.2 multiplied by $8/\pi^2$, and it is Kent's hitting time for the interval: the hitting time of 1 by reflected Brownian motion started at 0, whose eigenvalues on $[0,1]$, with reflection at 0 and killing at 1, are these atoms. So its spectrum is proportional to the very points $(2k-1)^2$ at which the Riesz coefficients sample the Laplace transform of T_1. For two independent copies, $\log|CC'| \stackrel{d}{=} \frac{\pi}{2}B_{\tau+\tau'}$. Combining (10.2.5) with Proposition 10.2.5 gives a duality between the two clocks,

$$\frac{\mathbb{E}\big[(\log|CC'|)^{2k-2}\big]}{(2k-2)!}\,\mathbb{E}\,e^{-(2k-1)^2T_1} = \frac{8(2k-1)(1-2^{-2k})}{\pi^2\kappa_k}, \tag{10.2.6}$$

in which a moment of Brownian motion stopped at a sum of two independent sech clocks, times a killing probability of the basepoint one clock, is an elementary constant.

We checked the moment formula (10.2.5) exactly through the Taylor coefficients of $\sec^2$, where it reproduces $\zeta(2)$, $\zeta(4)$, $\zeta(6)$ and $\zeta(8)$ to ten digits, and with 10^6 Monte Carlo draws of $\log|CC'|$, whose moments of orders 2, 4 and 6 come out as 4.927 ± 0.009, 97.3 ± 0.5 and 4098 ± 72, against the exact 4.935, 97.41 and 4085.9 (`killing2.py`). The duality (10.2.6) holds with the killing probabilities simulated in the second run of Section 10.2.4: the products are 0.95508, 1.87524, 2.06191 and 1.63889 for $k = 1,\ldots,4$, against 0.95493, 1.87500, 2.06167 and 1.63851.

The duality is structural rather than quantitative: $\zeta(2k)$ as a moment lives at the integers, while Riesz's criterion needs $1/\zeta(2s)$ inside the strip. The moments of ζ on the critical line are a different matter; their random matrix factor, given by the Barnes G function, and their bearing on the zero density exponents are the subject of Section 11.4.6. The half Cauchy law also enters the book through Bosch's theorem that the density of $|C|^\beta$ is hyperbolically completely monotone exactly when $|\beta| \geq 2$ [37] (Section 4.2); so $|C|$ itself is not HCM, while C^2 is. The dictionary of that section, Table 4.2, pairs the odd integers $2k-1$, the frequencies of the half Cauchy law, with the ordinates γ.

10.2.6 Why the representation cannot supply the growth bound

The representations of Sections 10.2.2 to 10.2.5 are exact, and all of them except the Kent string readings of Section 10.2.4 hold with no hypothesis; they connect Riesz's criterion to the clocks, the Kent strings and the Wiener gamma integrals of Chapters 9 to 11, and the clock average of Proposition 10.2.4 is far better conditioned than the power series. They do not supply the bound $R(x) = O(x^{1/4+\delta})$, for four reasons.

1. The integer values carry almost no information. The growth of R is fixed by the Mellin transform $\Gamma(1-s)/\zeta(2s)$, that is, by the analytic interpolation of $k \mapsto 1/\zeta(2k)$ to complex k with $\frac{1}{4} < \operatorname{Re} k < \frac{1}{2}$. At the integers $1/\zeta(2k) = 1 - 4^{-k} + O(9^{-k})$, and these values, which Proposition 10.2.3 represents, determine the interpolation only through its analytic continuation.

2. The representation interpolates only on one side of the line $\operatorname{Re} 2k = \alpha$, and as an expectation only on a cone. For complex k the expectation $\mathbb{E}e^{-(2k-\alpha)^2 T_\alpha}$ is finite exactly when $\operatorname{Re}(2k-\alpha)^2 \geq 0$, because T_α has a tail of order $t^{-1/2}$ (Theorem 9.2.5) and no exponential moments; there it equals $\varphi_\alpha((2k-\alpha)^2) = \xi(\alpha)/\xi(\alpha + \sqrt{(2k-\alpha)^2})$, and the principal square root equals $2k-\alpha$ only when $\operatorname{Re}(2k-\alpha) > 0$. So the representation through T_α returns $\xi(\alpha)/\xi(2k)$ as an expectation on the cone $|\operatorname{Im} 2k| \leq \operatorname{Re} 2k - \alpha$, and on the rest of the half plane $\operatorname{Re} 2k > \alpha$ only through the analytic continuation of φ_α, which is analytic there exactly when no zero lies to the right of α. It is a GGC representation with no hypothesis only for $\alpha \geq 1$, that is, on the half plane $\operatorname{Re} 2k > 1$ of absolute convergence, where the criterion carries no information. To reach $\frac{1}{2} < \operatorname{Re} 2k < 1$ one needs a basepoint $\alpha < 1$, and T_α is a GGC there exactly when ζ has no zeros in $\operatorname{Re} s > \alpha$.

3. The centre clock's Laplace integral converges only up to its first singularity, which under RH is at $-\gamma_1^2$. In the k plane, with $u = (2k - \frac{1}{2})^2$, the region $\operatorname{Re} u > -\gamma_1^2$ is a neighbourhood of the segment $|\operatorname{Im} 2k| < \gamma_1$ of the critical line $\operatorname{Re} 2k = \frac{1}{2}$, not the strip; beyond it one needs analytic continuation, which is again a statement about the zeros.

4. Positivity does not survive. Each coefficient becomes a positive average, but the series is alternating, and the bound requires, at $x = 10^4$ and $T = 0.023$, the cancellation of terms as large as 10^{81} after averaging. No monotonicity or positivity argument controls that cancellation; it is carried by the oscillating terms $x^{\rho/2}$ of the explicit formula.

What would help is what the basepoint version of the criterion (Proposition 10.2.1 with $h = 0$) requires: an analytic, and ideally positive, representation of $1/\zeta(2s)$ on whole vertical lines of the strip $\frac{1}{4} < \operatorname{Re} s < \frac{1}{2}$. A representation of the Thorin kind that is positive there is a GGC at a basepoint below one,

that is, a zero free half plane. The Thorin representation of the Riesz coefficients is therefore an exact reformulation, with better cancellation and a probabilistic reading, that reaches the criterion only through the same step that every other construction of Chapters 9 to 11 requires. At a basepoint $\frac{1}{2} < \alpha < 1$, zero density estimates bound the number of Mellin poles to the right of the line $\operatorname{Re} s = \alpha/2$ up to any given height, as they bound the negative part of v_α/π (Theorem 11.4.2), but a single such pole breaks the bound $R(x) = O(x^{\alpha/2+\delta})$ for small δ.

What the criterion does give is a computational test, the bank of twisted Riesz functions of Section 10.2.1, which check the zeros at any height from the Möbius function alone. Section 11.1, at the start of Chapter 11, turns to the books of Sato and Bondesson on infinitely divisible laws, to see whether their theory can supply the missing step, positivity of the Thorin measure uniformly in the height at a basepoint below one.

10.3 The squarefree series and the clock

Riesz's criterion (Sections 10.1.8 and 10.2) reads RH from a function whose Mellin transform is $\Gamma(1-s)/\zeta(2s)$. The squarefree integers carry the same reciprocal $1/\zeta(2s)$. Their Dirichlet series is

$$D(s) = \sum_{n \geq 1} \frac{\mu(n)^2}{n^s} = \prod_p \left(1 + p^{-s}\right) = \frac{\zeta(s)}{\zeta(2s)}, \qquad \operatorname{Re} s > 1, \tag{10.3.1}$$

whose singularities in the half plane $\operatorname{Re} s > 0$ are the pole at $s = 1$ and the poles at the points $\rho/2$ at which $\zeta(\rho/2) \neq 0$ (all of them under RH), and the number $Q(x)$ of squarefree integers up to x oscillates about $6x/\pi^2$ on the scale $x^{1/4}$ that these poles set when they lie on the line $\operatorname{Re} s = \frac{1}{4}$, that is under RH. This section follows the first author's note [206], which starts from a Math StackExchange thread [156] on the partial sums of $\mu(n)^2 n^{-\alpha}$ over $\mathbb{Q}$ and over number fields. We complete the thread by computing the constant at $\alpha = 1$ in closed form (Section 10.3.1), write D as a gamma factor times a ratio of Pólya moment generating functions and the constant as a difference of two tilted Pólya means, that is of two values of G'/G, over $\mathbb{Q}$ (Section 10.3.2), and over every number field the constant as the same combination of two values of ξ_K'/ξ_K (Section 10.3.3), and then turn to the error $\Delta(x) = Q(x) - 6x/\pi^2$. Its explicit formula is a sum over the zeros, one term for each Thorin atom of the clock (Section 10.3.4). The bound $\Delta(x) = O(x^{1/4+\varepsilon})$ is open even under RH, but it is equivalent to RH once the count is smoothed by an exponential or a Gamma(2) law, and in mean square; the order $x^{1/4}$ is attained with no hypothesis; and the bound for the count smoothed by a Gamma(n) law loses a factor $n^{1/8}$, which still gives $\Delta(x) = O(x^{2/5+\varepsilon})$ under RH (Sections 10.3.5 and 10.3.6). Smoothing by a Gamma(2) law writes the error as an average of the Mertens function under an explicit law, whose density is positive by an inequality of Lazarević, and two is the largest shape for which such

a positive representation exists (Sections 10.3.7 and 10.3.8). The Pólya kernels of the Dedekind zeta functions of imaginary quadratic fields, whose positivity can fail, are in Section 8.5.6 of Chapter 8.

We keep the notation of Section 1.2 for the Pólya variable X, with $\mathbb{E}e^{sX} = \xi(\frac12+s)/\xi(\frac12)$, the clock T, $G(u) = \xi(\frac12+\sqrt{u})/\xi(\frac12)$, the numbers $a_k = -(\rho_k - \frac12)^2$, the heat trace $W(t) = \sum_k e^{-a_k t}$ and the measure U_{RH}, and the notation of Section 9.1.1 for the tilted variables V_α, with $\mathbb{E}e^{sV_\alpha} = \xi(\alpha+s)/\xi(\alpha)$, and $b_\alpha = \frac{\xi'}{\xi}(\alpha) = \mathbb{E}V_\alpha$. Throughout, μ is the Möbius function, $M(t) = \sum_{d\le t}\mu(d)$ is the Mertens function, $\{y\}$ is the fractional part of y, and

$$Q(x) = \sum_{n\le x} \mu(n)^2, \qquad \Delta(x) = Q(x) - \frac{6}{\pi^2}\,x.$$

The scripts named at the tables and the figure are in `code/`, and their data files in `data/`.

10.3.1 The thread: Perron's formula and an elementary method

The question asks for the behaviour as $x\to\infty$ of

$$S_\alpha(x) = \sum_{N(\mathfrak{a})\le x} \frac{\mu(\mathfrak{a})^2}{N(\mathfrak{a})^\alpha},$$

the sum running over the integral ideals $\mathfrak{a}$ of a number field K, with μ the Möbius function of ideals and N the norm, and, in a comment, for its expansion at $\alpha = 1$. Over $\mathbb{Q}$, $S_\alpha(x) = \sum_{n\le x}\mu(n)^2 n^{-\alpha}$, and another comment points to Abel summation from the classical count

$$Q(x) = \frac{6}{\pi^2}\,x + O(\sqrt{x}). \tag{10.3.2}$$

The accepted answer applies Perron's formula over K, and the other answer carries out the elementary method over $\mathbb{Q}$. Neither evaluates the constant at $\alpha = 1$.

Perron's formula says that if $L_f(s) = \sum_n f(n)n^{-s}$ converges absolutely on the line $\operatorname{Re} s = \sigma > 0$, then $\sum_{n\le x} f(n) = \frac{1}{2\pi i}\int_{\sigma-i\infty}^{\sigma+i\infty} L_f(s)x^s\frac{ds}{s}$ for $x>1$ not an integer, the integral taken as the limit of the integrals over $|\operatorname{Im} s|\le T$. Since $\mu(\mathfrak{p}^k)^2$ is 1 for $k\le 1$ and 0 for $k\ge 2$, the Euler factor of $\sum_{\mathfrak{a}}\mu(\mathfrak{a})^2N(\mathfrak{a})^{-s}$ at a prime ideal $\mathfrak{p}$ is $1+N(\mathfrak{p})^{-s} = (1-N(\mathfrak{p})^{-2s})/(1-N(\mathfrak{p})^{-s})$, so the series is $\zeta_K(s)/\zeta_K(2s)$, with ζ_K the Dedekind zeta function of K, and

$$S_\alpha(x) = \frac{1}{2\pi i}\int_{\sigma-i\infty}^{\sigma+i\infty} \frac{\zeta_K(\alpha+s)}{\zeta_K(2\alpha+2s)}\,x^s\,\frac{ds}{s}, \qquad \sigma > \max\{1-\alpha, 0\}. \tag{10.3.3}$$

The accepted answer writes the summand of the Euler factor with an extra factor $1/k$, a slip from the series of the logarithm of an Euler factor, undefined at $k=0$; its next expression, $1+N(\mathfrak{p})^{-\alpha-s}$, is right, and so are its conclusions.

The integrand of (10.3.3) has a simple pole at $s = 1-\alpha$, a pole at $s = 0$, and poles at the points $s = \rho/2 - \alpha$ for the zeros ρ of ζ_K, trivial and nontrivial, at which $\zeta_K(\rho/2) \neq 0$. Moving the line of integration to the left collects the main terms as residues. Let $c_K = \lim_{s\to 1}(s-1)\zeta_K(s) = 2^{r_1}(2\pi)^{r_2}hR/(w\sqrt{|d_K|})$ be the residue of the class number formula, with the usual meanings of r_1, r_2, h, R, w and the discriminant d_K.

The case $\alpha \neq 1$. The residue at $s = 1-\alpha$ is $c_K x^{1-\alpha}/((1-\alpha)\zeta_K(2))$, as the accepted answer states, and the residue at $s = 0$, which it lists among the poles but does not compute, is $\zeta_K(\alpha)/\zeta_K(2\alpha)$, the value of the continued series. Over $\mathbb{Q}$, for $\alpha > \frac12$ and $\alpha \neq 1$, the two residues are the main terms of

$$\sum_{n\le x} \frac{\mu(n)^2}{n^\alpha} = \frac{6}{\pi^2}\,\frac{x^{1-\alpha}}{1-\alpha} + \frac{\zeta(\alpha)}{\zeta(2\alpha)} + O\big(x^{\frac12-\alpha}\big). \tag{10.3.4}$$

For a proof, Abel summation and (10.3.2) give, for $\alpha > \frac12$, $\alpha \neq 1$,

$$\sum_{n\le x} \frac{\mu(n)^2}{n^\alpha} = \frac{6}{\pi^2}\,\frac{x^{1-\alpha}}{1-\alpha} + \Big(\alpha\int_1^\infty \frac{\Delta(t)}{t^{\alpha+1}}\,dt - \frac{6}{\pi^2}\,\frac{\alpha}{1-\alpha}\Big) + \frac{\Delta(x)}{x^\alpha} - \alpha\int_x^\infty \frac{\Delta(t)}{t^{\alpha+1}}\,dt. \tag{10.3.5}$$

The last two terms are $O(x^{\frac12-\alpha})$, and the bracket is $D(\alpha)$, because $D(s) = s\int_1^\infty Q(t)t^{-s-1}dt = \frac{6}{\pi^2}\frac{s}{s-1} + s\int_1^\infty \Delta(t)t^{-s-1}dt$ for $\operatorname{Re} s > 1$ and the right side is analytic in $\operatorname{Re} s > \frac12$ apart from $s = 1$. For $\frac12 < \alpha < 1$ the constant is negative, for example $\zeta(\frac34)/\zeta(\frac32) = -1.31730127858$ (`sqfree_constants.py`): it is the value of the continuation of $\sum_n \mu(n)^2 n^{-\alpha}$ into the half plane where the series diverges, and Corollary 10.3.4 explains its sign. For $\alpha > 1$ the series converges and the same constant is its sum. Write $\mathcal{E}_\alpha(x)$ for the left side of (10.3.4) minus its two main terms, so that by (10.3.5)

$$\mathcal{E}_\alpha(x) = \frac{\Delta(x)}{x^\alpha} - \alpha\int_x^\infty \frac{\Delta(t)}{t^{\alpha+1}}\,dt. \tag{10.3.6}$$

The last column of Table 10.11 shows $\mathcal{E}_{3/4}(x)$ multiplied by $x^{1/4}$, which (10.3.4) bounds: it is at most 0.03 in absolute value, and below 0.006 from $x = 10^5$ on. Under RH the atoms of $\mathcal{E}_{3/4}(x)\,x^{1/4}$ have size $x^{-1/4}$ (Section 10.3.4).

The case $\alpha = 1$. The poles at $1-\alpha$ and at 0 merge into a double pole at $s = 0$ of $\zeta_K(1+s)\zeta_K(2+2s)^{-1}e^{s\log x}/s$. The accepted answer computes the coefficient of $\log x$ and says that the constant term is much more complicated, at least when $K \neq \mathbb{Q}$. It is explicit. Define the Euler Kronecker constant γ_K of K by $\zeta_K(s) = c_K/(s-1) + c_K\gamma_K + O(s-1)$ [101], so that $\gamma_{\mathbb{Q}} = \gamma_E$. With $1/\zeta_K(2+2s) = \zeta_K(2)^{-1}\big(1 - 2s\,\zeta_K'(2)/\zeta_K(2) + O(s^2)\big)$ the residue at $s = 0$ is

$$\frac{c_K}{\zeta_K(2)}\big(\log x + C_K\big), \qquad C_K = \gamma_K - 2\,\frac{\zeta_K'(2)}{\zeta_K(2)}. \tag{10.3.7}$$

Both γ_K and $\zeta_K'(2)/\zeta_K(2)$ are computable, and Theorem 10.3.7 below writes C_K through the zeros of ζ_K. Proposition 10.3.1 turns the residue into an asymptotic formula with a power saving, by the elementary method of the other answer.

The elementary method. The identity $\mu(n)^2 = \sum_{d^2\mid n}\mu(d)$, the coefficientwise form of $D(s) = \zeta(s)\cdot\zeta(2s)^{-1}$, gives

$$Q(x) = \sum_{d\le\sqrt{x}} \mu(d)\Big\lfloor \frac{x}{d^2}\Big\rfloor = x\sum_{d\le\sqrt{x}}\frac{\mu(d)}{d^2} - \sum_{d\le\sqrt{x}}\mu(d)\Big\{\frac{x}{d^2}\Big\}, \tag{10.3.8}$$

and since $\sum_{d>\sqrt{x}} d^{-2} = O(x^{-1/2})$ and the last sum has at most $\sqrt{x}$ terms, each in $[0,1)$, this is (10.3.2). The second answer then sets

$$B(x) = \sum_{n\le x}\Big(\mu(n)^2 - \frac{6}{\pi^2}\Big) = O(\sqrt{x}),$$
$$D_0(s) = \sum_{n\ge1}\frac{\mu(n)^2 - 6/\pi^2}{n^s} = D(s) - \frac{6}{\pi^2}\,\zeta(s).$$

By Abel summation, $D_0(s) = \lim_{x\to\infty}\big(B(x)x^{-s} + s\int_1^x B(t)t^{-s-1}dt\big)$ converges for $\operatorname{Re} s > \frac12$ (the answer has a minus sign in place of the plus), so $\sum_{n\le x}\mu(n)^2/n = \frac{6}{\pi^2}(\log x + \gamma_E) + D_0(1) + o(1)$, and there the answer stops. Since $D_0(s) = \zeta(s)\big(\zeta(2s)^{-1} - \zeta(2)^{-1}\big)$ and ζ has a simple pole with residue 1 at $s=1$,

$$D_0(1) = \lim_{s\to1}\zeta(s)\Big(\frac{1}{\zeta(2s)} - \frac{1}{\zeta(2)}\Big) = -\frac{2\zeta'(2)}{\zeta(2)^2} = 0.69298946940360442669,$$

in agreement with (10.3.7) at $K=\mathbb{Q}$, where $c_{\mathbb{Q}} = 1$ and $\gamma_{\mathbb{Q}} = \gamma_E$.

Proposition 10.3.1. (i) *Let K be a number field of degree $n\ge2$. As $x\to\infty$,*

$$\sum_{N(\mathfrak{a})\le x}\frac{\mu(\mathfrak{a})^2}{N(\mathfrak{a})} = \frac{c_K}{\zeta_K(2)}\big(\log x + C_K\big) + O\big(x^{-1/n}\log x\big).$$

(ii) *Over $\mathbb{Q}$,*

$$\sum_{n\le x}\frac{\mu(n)^2}{n} = \frac{6}{\pi^2}\big(\log x + C\big) + O\big(x^{-1/2}\big),$$
$$C = \gamma_E - 2\,\frac{\zeta'(2)}{\zeta(2)} = 24\log A - \gamma_E - 2\log 2\pi, \tag{10.3.9}$$

where A is the Glaisher Kinkelin constant; $C = 1.71713765109059847340 6241$ and $\frac{6}{\pi^2}C = 1.04389457157119382974$. Moreover $\mathcal{E}_1(x) = \sum_{n\le x}\mu(n)^2/n - \frac{6}{\pi^2}(\log x + C)$ is given by (10.3.6) *with $\alpha = 1$, and*

$$\int_1^\infty \frac{\Delta(t)}{t^2}\,dt = \frac{6}{\pi^2}\,(C-1) = 0.43596741385791166874.$$

Proof. (i) The number of ideals of norm at most y is $c_K y + O(y^{1-1/n})$ [138, Chapter VI], so by Abel summation $L(y) = \sum_{N(\mathfrak{c})\le y} N(\mathfrak{c})^{-1} = c_K(\log y + \gamma_K) + O(y^{-1/n})$ for $y \ge 1$, the constant being the one in the Laurent expansion of ζ_K at $s = 1$. Since $\mu(\mathfrak{a})^2 = \sum_{\mathfrak{b}^2|\mathfrak{a}} \mu(\mathfrak{b})$, as one checks on prime powers,

$$\begin{aligned}\sum_{N(\mathfrak{a})\le x} \frac{\mu(\mathfrak{a})^2}{N(\mathfrak{a})} &= \sum_{N(\mathfrak{b})\le \sqrt{x}} \frac{\mu(\mathfrak{b})}{N(\mathfrak{b})^2}\, L\Big(\frac{x}{N(\mathfrak{b})^2}\Big) \\ &= c_K \sum_{N(\mathfrak{b})\le \sqrt{x}} \frac{\mu(\mathfrak{b})}{N(\mathfrak{b})^2}\big(\log x + \gamma_K - 2\log N(\mathfrak{b})\big) \\ &\quad + O\Big(x^{-1/n} \sum_{N(\mathfrak{b})\le \sqrt{x}} N(\mathfrak{b})^{-2+2/n}\Big).\end{aligned}$$

The last sum is $O(\log x)$, since $-2 + 2/n \le -1$. Completing the sums over $\mathfrak{b}$ costs $O(x^{-1/2}\log x)$, which is $O(x^{-1/n}\log x)$ as $n \ge 2$, since $\sum_{N(\mathfrak{b})>Y} N(\mathfrak{b})^{-2}\log N(\mathfrak{b}) = O(Y^{-1}\log Y)$, and the completed sums are $\sum_{\mathfrak{b}} \mu(\mathfrak{b}) N(\mathfrak{b})^{-2} = 1/\zeta_K(2)$ and $\sum_{\mathfrak{b}} \mu(\mathfrak{b}) \log N(\mathfrak{b})\, N(\mathfrak{b})^{-2} = \zeta_K'(2)/\zeta_K(2)^2$, which give (10.3.7).

(ii) By Abel summation, $\sum_{n\le x} \mu(n)^2/n = Q(x)/x + \int_1^x Q(t)t^{-2}dt$, and inserting $Q(t) = 6t/\pi^2 + \Delta(t)$ gives

$$\sum_{n\le x} \frac{\mu(n)^2}{n} = \frac{6}{\pi^2}\log x + \frac{6}{\pi^2} + \int_1^\infty \frac{\Delta(t)}{t^2}\,dt + \frac{\Delta(x)}{x} - \int_x^\infty \frac{\Delta(t)}{t^2}\,dt,$$

whose last two terms are $O(x^{-1/2})$ by (10.3.2). So the formula holds with some constant, and the constant is identified through D: for $\operatorname{Re} s > 1$, $\int_1^\infty \Delta(t)t^{-s-1}dt = D(s)/s - \frac{6}{\pi^2}\frac{1}{s-1}$, and $D(s) = \frac{6}{\pi^2}\zeta(s) + D_0(s) = \frac{6}{\pi^2}\big(\frac{1}{s-1} + \gamma_E - 2\frac{\zeta'(2)}{\zeta(2)}\big) + O(s-1)$, so that $\int_1^\infty \Delta(t)t^{-2}dt = \frac{6}{\pi^2}(C-1)$ by continuity at $s = 1$. This gives (10.3.9) and the formula for $\mathcal{E}_1$, and the classical evaluation $\zeta'(2)/\zeta(2) = \gamma_E + \log 2\pi - 12\log A$ gives the second form of C. □

The script `sqfree_constants.py` computes C in the two forms of the proposition and in the form of Theorem 10.3.5 below, which agree in all 25 digits printed, and `sqfree_sieve.py` sieves the squarefree integers to 2×10^8, where $Q(2 \times 10^8) = 121585426$. Table 10.11 shows that $\mathcal{E}_1(x)$ is of order $x^{-3/4}$ and $\Delta(x)$ of order $x^{1/4}$, the sizes that the poles at $\rho/2 - 1$ and $\rho/2$ predict under RH (Section 10.3.4). By (10.3.6), $\mathcal{E}_1(x)x^{3/4}$ is $\Delta(x)/x^{1/4}$ minus $\frac{4}{3}$ times an average of $\Delta(t)/t^{1/4}$ over $t > x$ with weight proportional to $t^{-7/4}$, and the two columns differ accordingly.

x	$Q(x)$	$\Delta(x)/x^{1/4}$	$\mathcal{E}_1(x)$	$\mathcal{E}_1(x)\,x^{3/4}$	$\mathcal{E}_{3/4}(x)\,x^{1/4}$
10^3	608	0.0130	-9.130×10^{-4}	-0.162	-0.0291
10^4	6083	0.3729	3.031×10^{-4}	0.303	0.0296
10^5	60794	0.0725	-2.308×10^{-6}	-0.0130	-0.00031
10^6	607926	-0.0348	-1.665×10^{-6}	-0.0526	-0.00177
10^7	6079291	0.3553	1.729×10^{-6}	0.308	0.00561
10^8	60792694	-0.1619	-1.737×10^{-7}	-0.174	-0.00174

Table 10.11: The sharp counts: the number $Q(x)$ of squarefree integers up to x, the normalised error $\Delta(x)/x^{1/4}$, the error $\mathcal{E}_1(x)$ of (10.3.9), which confirms the constant C, also multiplied by $x^{3/4}$, and the error $\mathcal{E}_{3/4}(x)$ of (10.3.4) at $\alpha=\frac34$ multiplied by $x^{1/4}$. The first two errors are normalised by the sizes $x^{1/4}$ and $x^{-3/4}$ that their poles at the zeros predict under RH, the last by its bound in (10.3.4). Computed by `sqfree_sieve.py` from a sieve to 2×10^8; the sums of reciprocals are checked in extended precision.

10.3.2 The Pólya form of the series and the constant

The completed function $\xi(s)=\frac12 s(s-1)\pi^{-s/2}\Gamma(\frac{s}{2})\zeta(s)$ at s and at $2s$, together with the duplication formula, separates the squarefree series into a gamma factor and a ratio of values of ξ.

Lemma 10.3.2. *For all s,*

$$D(s)=\Gamma_{\mathrm{sq}}(s)\,\frac{\xi(s)}{\xi(2s)},\qquad \Gamma_{\mathrm{sq}}(s)=\frac{2s-1}{s-1}\,2^s\pi^{-(s+1)/2}\,\Gamma\Big(\frac{s+1}{2}\Big).$$

Proof. By the definition of ξ, $\zeta(s)/\zeta(2s)=\frac{\xi(s)}{\xi(2s)}\cdot\frac{2(2s-1)}{s-1}\,\pi^{-s/2}\,\frac{\Gamma(s)}{\Gamma(s/2)}$, and the duplication formula $\Gamma(s)=2^{s-1}\pi^{-1/2}\Gamma(\frac{s}{2})\Gamma(\frac{s+1}{2})$ gives the factor Γ_{sq}. □

The factor $(s-1)^{-1}$ carries the pole of D at $s=1$, and the zero of $2s-1$ reflects the pole of $\zeta(2s)$ at $s=\frac12$, so that $D(\frac12)=0$. The poles of $\Gamma(\frac{s+1}{2})$ at $s=-1,-3,-5,\dots$ are the poles of D at the points where $\zeta(2s)$ has a trivial zero and $\zeta(s)$ does not vanish; at $s=-2,-4,\dots$ both vanish and D is regular. The remaining singularities of D, the poles at the points $\rho/2$, all sit in $1/\xi(2s)$. The script `sqfree_constants.py` checks the lemma at $s=3$, $0.3+2.1i$, $\frac34$ and -1.5, with relative errors below 5×10^{-41}.

Proposition 10.3.3. *For every s with $\xi(2s)\neq0$,*

$$\sum_{n\ge1}\frac{\mu(n)^2}{n^s}=\Gamma_{\mathrm{sq}}(s)\,\frac{\mathbb{E}e^{(s-\frac12)X}}{\mathbb{E}e^{(2s-\frac12)X}},$$

the left side read as the continuation $D(s)$ where the series diverges.

Proof. Lemma 10.3.2 and (1.2.4), which gives $\xi(s) = \xi(\frac12)\mathbb{E}e^{(s-\frac12)X}$ and $\xi(2s) = \xi(\frac12)\mathbb{E}e^{(2s-\frac12)X}$. □

The numerator is the moment generating function of X at the point $s-\frac12$ and the denominator at the doubled point $2s-\frac12$. Both are finite and positive at real arguments, so the only obstruction to continuing the right side is the vanishing of $\mathbb{E}e^{(2s-\frac12)X}$ at complex s, which happens exactly at the points $\rho/2$.

COROLLARY 10.3.4. *For real $\alpha \notin \{1, \frac12, -1, -3, -5, \dots\}$ the sign of $\zeta(\alpha)/\zeta(2\alpha)$ is the sign of $\frac{2\alpha-1}{\alpha-1}\Gamma(\frac{\alpha+1}{2})$. It is positive for $\alpha > 1$ and for $-1 < \alpha < \frac12$, negative for $\frac12 < \alpha < 1$, and equal to $(-1)^k$ on the interval $(-2k-1, -2k+1)$ for $k \geq 1$.*

Proof. At real α the ratio of moment generating functions in Proposition 10.3.3 is positive, and so is $2^\alpha \pi^{-(\alpha+1)/2}$. The rational factor is positive except on $(\frac12, 1)$, and $\Gamma(\frac{\alpha+1}{2})$ has the sign $(-1)^k$ when $\frac{\alpha+1}{2} \in (-k, -k+1)$, that is when $\alpha \in (-2k-1, -2k+1)$. □

At the even negative integers both $\zeta(\alpha)$ and $\zeta(2\alpha)$ vanish and the quotient keeps its sign through them. The values 2.4946347, −3.0582242, −2.1462661, 1.0658427, 0.40810034, −0.12665263 and −0.032961215 of $\zeta(\alpha)/\zeta(2\alpha)$ at $\alpha = -0.5, -1.5, \dots, -6.5$ (`sqfree_constants.py`) show the pattern. At $\alpha = \frac34$, inside the interval where the series diverges, $\Gamma_{\mathrm{sq}}(\frac34) = -1.34612167295$, $\mathbb{E}e^{X/4} = 1.00144503253351$ and $\mathbb{E}e^{X} = 1.02335500958202$, the last two by the trapezoid rule on the Pólya kernel (1.2.3); the ratio 0.978590052481 multiplies $\Gamma_{\mathrm{sq}}(\frac34)$ to $-1.31730127858 = \zeta(\frac34)/\zeta(\frac32)$.

The constant C of (10.3.9) is the logarithmic derivative of $(s-1)D(s)$ at $s = 1$, and Proposition 10.3.3 splits that derivative into an elementary part and two tilted Pólya means.

THEOREM 10.3.5. *The constant of* (10.3.9) *is*

$$C = 2 + \log 2 - \tfrac12 \log \pi - \tfrac12 \gamma_E + b_1 - 2b_2 = 2 + \log 2 - \tfrac12 \log \pi - \tfrac12 \gamma_E + \mathbb{E}V_1 - 2\,\mathbb{E}V_2.$$

With no hypothesis on the zeros,

$$\begin{aligned} b_1 - 2b_2 &= \frac{G'}{G}\Big(\frac14\Big) - 6\,\frac{G'}{G}\Big(\frac94\Big) = \sum_k \Big(\frac{1}{\frac14 + a_k} - \frac{6}{\frac94 + a_k}\Big) \\ &= \int_0^\infty \big(e^{-t/4} - 6e^{-9t/4}\big)\, W(t)\, dt, \end{aligned} \tag{10.3.10}$$

the sum running over the pairs $\{\rho, 1-\rho\}$ and converging absolutely. Under RH, $a_k = \gamma_k^2$ and $b_1 - 2b_2 = \int \big((z+\frac14)^{-1} - 6(z+\frac94)^{-1}\big)\, U_{\mathrm{RH}}(dz)$, a combination of two values of the Stieltjes transform of the Thorin measure of the clock.

Proof. By the proof of Proposition 10.3.1, $(s-1)D(s) = \frac{6}{\pi^2}\big(1 + C(s-1) + O((s-1)^2)\big)$, so C is the derivative of $\log((s-1)D(s))$ at $s=1$. By Proposition 10.3.3,

$$\log\big((s-1)D(s)\big) = \log(2s-1) + s\log 2 - \tfrac{s+1}{2}\log\pi + \log\Gamma\big(\tfrac{s+1}{2}\big) + \log \mathbb{E}e^{(s-\frac12)X} - \log \mathbb{E}e^{(2s-\frac12)X}.$$

At $s=1$ the derivatives of the first four terms are 2, $\log 2$, $-\frac12\log\pi$ and $\frac12\psi(1) = -\frac12\gamma_E$, with ψ the digamma function here. The derivative of $\log \mathbb{E}e^{(s-\frac12)X}$ at $s=1$ is $\mathbb{E}Xe^{X/2}/\mathbb{E}e^{X/2}$, the mean of X tilted by $e^{X/2}$, which is $\mathbb{E}V_1 = b_1$ (Section 9.1.1); likewise the last term gives $-2\mathbb{E}V_2 = -2b_2$. For the zero form, differentiating $\log G(w^2) = \log\xi(\frac12+w) - \log\xi(\frac12)$ gives $\frac{\xi'}{\xi}(\frac12+w) = 2w\frac{G'}{G}(w^2)$, so $b_1 = \frac{G'}{G}(\frac14)$ and $b_2 = 3\frac{G'}{G}(\frac94)$. Then $\frac{G'}{G}(u) = \sum_k (u+a_k)^{-1}$ by (1.2.1), and this is $\int_0^\infty e^{-ut}W(t)\,dt$ for $u \ge 0$, by termwise integration, since $\operatorname{Re} a_k \ge \gamma_1^2 - \frac14 > 0$ and $\sum_k |a_k|^{-1} < \infty$. Under RH, $\frac{G'}{G}(u) = \int (u+z)^{-1} U_{\mathrm{RH}}(dz)$ by Proposition 1.2.1. □

The elementary part is $2 + \log 2 - \frac12\log\pi - \frac12\gamma_E = 1.832174405184478792$, and $b_1 = 0.023095708966121033814$, which is $1 + \frac12\gamma_E - \frac12\log 4\pi = \sum_\rho \rho^{-1}$, and $b_2 = 0.069066231530000676225$, by the closed forms of ξ'/ξ and by numerical differentiation, which agree to every digit printed (`sqfree_constants.py`). So $b_1 - 2b_2 = -0.11503675409388031864$, and the elementary part plus the zero part reproduces C to all 25 digits. In the variable $X/2$ of the first author's note the two tilted means are $b_1/2 = 0.0115478544831$ and $b_2/2 = 0.034533115765$. The arithmetic content of C thus sits in two tilted means of the Pólya law, at the basepoints 1 and 2, the images of $s=1$ under $s \mapsto s$ and $s \mapsto 2s$.

Table 10.12 sums (10.3.10) over the lowest zeros. The summand is $4/(1+4\gamma^2) - 24/(9+4\gamma^2) = -5\gamma^{-2} + O(\gamma^{-4})$, so the tail beyond the Nth zero is about $-5\sum_{k>N}\gamma_k^{-2}$; the script integrates the summand against the smooth density $\frac{1}{2\pi}\log\frac{t}{2\pi}$ of the zeros, with the correction for the difference between N and the smooth count $\frac{T}{2\pi}\log\frac{T}{2\pi e} + \frac78$ at the cut. The same tail method applied to $\sum_k \gamma_k^{-2}$ over 10000 zeros reproduces $\mathbb{E}T = \xi''(\frac12)/(2\xi(\frac12)) = 0.02310499311541897$ of Section 1.2 within 5.4×10^{-14}.

N	sum over $\gamma_1, \dots, \gamma_N$	smooth tail	total minus $(b_1 - 2b_2)$
60	−0.0943733259248	−0.020664227	-7.99×10^{-7}
1000	−0.111438864811	−0.0035978885	7.61×10^{-10}
10000	−0.114363258152	−0.00067349594	2.68×10^{-13}

Table 10.12: The zero part $b_1 - 2b_2 = -0.11503675409388031864$ of the constant C, as the sum (10.3.10) over the first N zeros plus its tail from the smooth density of the zeros. Computed by `sqfree_constants.py` with the zeros from Arb [110] at 40 digits.

The variance identity of the first author's note, $\sum_k a_k^{-1} = \frac12 \operatorname{Var} X$, with $a_k =$

γ_k^2 under RH, is the formula $\mathbb{E}T = \sum_k a_k^{-1} = \xi''(\frac12)/(2\xi(\frac12))$ of Section 1.2, and the alternation of the even cumulants of X, which the note derives under RH, holds with no hypothesis (Section 13.1.2).

10.3.3 Number fields

Let K be a number field with r_1 real and r_2 pairs of complex embeddings and discriminant d_K, and put $\Gamma_{\mathbb{R}}(s) = \pi^{-s/2}\Gamma(\frac{s}{2})$, $\Gamma_{\mathbb{C}}(s) = 2(2\pi)^{-s}\Gamma(s)$ and

$$\xi_K(s) = s(s-1)\,|d_K|^{s/2}\,\Gamma_{\mathbb{R}}(s)^{r_1}\Gamma_{\mathbb{C}}(s)^{r_2}\,\zeta_K(s).$$

Then ξ_K is entire of order one, $\xi_K(s) = \xi_K(1-s)$ by Hecke's functional equation [138, Chapter XIII], ξ_K is real on the real axis, and its zeros are the nontrivial zeros of ζ_K, all in the strip $0 < \operatorname{Re} s < 1$; for $K = \mathbb{Q}$, $\xi_{\mathbb{Q}} = 2\xi$. For $K = \mathbb{Q}(i)$, where $\zeta_K(s) = \zeta(s)L(s,\chi_{-4})$, the duplication formula gives $\xi_K(s) = \xi(s)\,\xi(s,\chi_{-4})$ exactly, with $\xi(s,\chi_{-4}) = (4/\pi)^{(s+1)/2}\Gamma(\frac{s+1}{2})L(s,\chi_{-4})$ the completed function of the odd character modulo 4 in the normalisation of Section 8.5.1.

The rigidity theorem of the clock (Proposition 1.2.1) holds over K with the same proof, and its converse half uses nothing about the sign of ξ_K on the real axis, so a real zero is excluded as well.

Proposition 10.3.6. *Suppose $\zeta_K(\frac12) \neq 0$ and let $G_K(u) = \xi_K(\frac12 + \sqrt{u})/\xi_K(\frac12)$. Then $1/G_K$ is the Laplace transform of a GGC if and only if every zero of ζ_K in the critical strip lies on the line $\operatorname{Re} s = \frac12$, the Riemann hypothesis for ζ_K. Its drift is then 0 and its Thorin measure is $\sum_{\gamma>0}\delta_{\gamma^2}$, over the ordinates of the zeros counted with multiplicity. A real zero $\beta \in (\frac12, 1)$ of ζ_K would be a pole of $1/G_K$ at the point $(\beta - \frac12)^2$ of the positive axis.*

Proof. Since $\xi_K(\frac12 + w)$ is even and entire of order one, G_K is entire of order $\frac12$, and Hadamard's theorem gives $G_K(u) = \prod_k (1 + u/a_{K,k})$ with $a_{K,k} = -(\rho_k - \frac12)^2$, one factor for each pair $\{\rho, 1-\rho\}$ of zeros of ξ_K, and $\sum_k |a_{K,k}|^{-1} < \infty$. If every zero lies on the line, every $a_{K,k} = \gamma_k^2$ is positive and $1/G_K$ is a convergent product of gamma Laplace transforms, a GGC with the stated Thorin measure and no drift. Conversely, the Laplace transform of a GGC is holomorphic and zero free on $\mathbb{C} \setminus (-\infty, 0]$, so G_K has no zero there, and its zeros $(\rho - \frac12)^2$ lie in $(-\infty, 0]$; since $\rho \neq \frac12$, this means $\rho - \frac12 \in i\mathbb{R} \setminus \{0\}$, that is $\operatorname{Re}\rho = \frac12$. A real zero β gives the zero $(\beta - \frac12)^2 > 0$ of G_K. □

The hypothesis $\zeta_K(\frac12) \neq 0$ fixes the normalisation. Armitage found number fields whose Dedekind zeta function vanishes at the centre [7]; there the zero of $\xi_K(\frac12 + w)$ at $w = 0$ has even order $2m$, and the proposition applies to $u^{-m}\xi_K(\frac12 + \sqrt{u})$ normalised at $u = 0$. By (10.3.3) the poles of $\zeta_K(s)/\zeta_K(2s)$ in the strip $0 < \operatorname{Re} s < \frac12$ are again the points $\rho/2$ at which $\zeta_K(\rho/2) \neq 0$, under the Riemann hypothesis for ζ_K the images of the Thorin atoms.

THEOREM 10.3.7. *Let*

$$A_K = 2 + \tfrac{1}{2}\log|d_K| + r_1\big(\log 2 - \tfrac{1}{2}\log\pi - \tfrac{1}{2}\gamma_E\big) + r_2\big(2 - \gamma_E - \log 2\pi\big).$$

Then the constant of (10.3.7) *is*

$$C_K = A_K + \frac{\xi_K'}{\xi_K}(1) - 2\,\frac{\xi_K'}{\xi_K}(2) = A_K + \sum_k \Big(\frac{1}{\frac{1}{4}+a_{K,k}} - \frac{6}{\frac{9}{4}+a_{K,k}}\Big),$$

the sum running over the pairs $\{\rho, 1-\rho\}$ *of zeros of* ξ_K*, with multiplicity, a zero of order* $2m$ *at* $s=\frac{1}{2}$ *counting as* m *pairs with* $a_{K,k}=0$*. Under the Riemann hypothesis for* ζ_K*, and* $\zeta_K(\frac{1}{2})\neq 0$*, the zero part is* $\frac{G_K'}{G_K}(\frac{1}{4}) - 6\frac{G_K'}{G_K}(\frac{9}{4})$*, the same combination of values of the Stieltjes transform of the Thorin measure of Proposition 10.3.6 as over* $\mathbb{Q}$*.*

Proof. From the definition of ξ_K,

$$(s-1)\frac{\zeta_K(s)}{\zeta_K(2s)} = \frac{\xi_K(s)}{\xi_K(2s)}\cdot 2(2s-1)\,|d_K|^{s/2}\,\frac{\Gamma_{\mathbb{R}}(2s)^{r_1}\Gamma_{\mathbb{C}}(2s)^{r_2}}{\Gamma_{\mathbb{R}}(s)^{r_1}\Gamma_{\mathbb{C}}(s)^{r_2}},$$

and C_K is the logarithmic derivative of the left side at $s=1$, by the Laurent expansions behind (10.3.7). With $\Gamma_{\mathbb{R}}'/\Gamma_{\mathbb{R}}(s) = -\frac{1}{2}\log\pi + \frac{1}{2}\psi(\frac{s}{2})$, $\Gamma_{\mathbb{C}}'/\Gamma_{\mathbb{C}}(s) = -\log 2\pi + \psi(s)$, $\psi(1)=-\gamma_E$, $\psi(\frac{1}{2})=-\gamma_E-2\log 2$ and $\psi(2)=1-\gamma_E$, the logarithmic derivative of the elementary factor at $s=1$ is A_K, and that of $\xi_K(s)/\xi_K(2s)$ is $\frac{\xi_K'}{\xi_K}(1) - 2\frac{\xi_K'}{\xi_K}(2)$. The Hadamard product of $\xi_K(\frac{1}{2}+w)$, which is even in w, gives $\frac{\xi_K'}{\xi_K}(\frac{1}{2}+w) = \sum_k 2w/(w^2+a_{K,k})$, and the values at $w=\frac{1}{2}$ and $w=\frac{3}{2}$ give the zero form. □

For $K=\mathbb{Q}$, $A_{\mathbb{Q}}$ is the elementary part of Theorem 10.3.5. For the Gaussian field $\mathbb{Q}(i)$, with $r_2=1$ and $d_K=-4$, `sqfree_constants.py` gives $c_K/\zeta_K(2) = 3/(2\pi L(2,\chi_{-4})) = 0.521269392989129$, with $L(2,\chi_{-4})$ Catalan's constant, and

$$\gamma_K = \gamma_E + \frac{L'}{L}(1,\chi_{-4}) = 0.822825249678847033,$$
$$\frac{\zeta_K'}{\zeta_K}(2) = -0.48089570872664776864,$$

and $C_K = 1.784616667132142570284$, where $\frac{L'}{L}(1,\chi_{-4}) = \gamma_E + 2\log 2 + 3\log\pi - 4\log\Gamma(\frac{1}{4}) = 0.24560958477731417239$. In Theorem 10.3.7, $A_K = 4-\gamma_E-\log\pi = 2.2780544492490669653$ and the zero part is the sum of the zero part $b_1-2b_2 = -0.11503675409388$ of ζ and the zero part $\frac{\xi_\chi'}{\xi_\chi}(1) - 2\frac{\xi_\chi'}{\xi_\chi}(2) = -0.378401028023044$ of $\xi_\chi = \xi(\cdot,\chi_{-4})$: the zeros of ζ and of $L(s,\chi_{-4})$ simply add their Thorin atoms, and the two sides of the theorem differ by 4.6×10^{-41}. The 289 zeros of $L(s,\chi_{-4})$ below height 400, the first at 6.020948905, 10.2437703 and 12.98809801, located from the sign changes of the Hardy function of the character in Arb, give -0.365385920572 for the L part, and their smooth tail -0.013015062 brings the sum within 4.6×10^{-8} of the closed form.

10.3.4 The Thorin atoms in the squarefree count

Put $s = \frac14 + \frac12\sqrt{u}$, so that $2s = \frac12 + \sqrt{u}$. Lemma 10.3.2 and the clock give, for $u \geq 0$ and with no hypothesis,

$$D\Big(\frac14 + \frac{\sqrt{u}}{2}\Big) = \Gamma_{\mathrm{sq}}\Big(\frac14 + \frac{\sqrt{u}}{2}\Big)\,\frac{\xi(\frac14 + \frac12\sqrt{u})}{\xi(\frac12)}\,\mathbb{E}e^{-uT},$$

since $1/G$ is the Laplace transform of the clock T (Section 1.2). The poles of D in the strip $0 < \operatorname{Re} s < \frac12$ are therefore among the points where G vanishes, $u = -a_k$, that is $s = \rho_k/2$, and they are all of these points when $\zeta(\rho_k/2) \neq 0$, as under RH; under RH they are the atoms γ_k^2 of U_{RH}, placed at $u = -\gamma_k^2$, that is at $s = \frac14 \pm \frac{i}{2}\gamma_k$. The line $\operatorname{Re} s = \frac14$, the image of the critical line under $s \mapsto s/2$, is the critical line of the squarefree problem, and the strip $0 < \operatorname{Re} s < \frac12$ is the image of the critical strip. These points are the poles $s = \rho/2 - \alpha$ of (10.3.3) at $\alpha = 0$.

At a simple zero ρ, $\zeta(2s) = 2\zeta'(\rho)(s - \rho/2) + O((s-\rho/2)^2)$, so the residue of $D(s)x^s/s$ at $s = \rho/2$ is $R_\rho x^{\rho/2}$, and formally

$$\Delta(x) \approx 1 + \sum_{\rho} R_\rho\, x^{\rho/2} + \sum_{k=1,3,5,\ldots} r_k\, x^{-k}, \qquad R_\rho = \frac{\zeta(\rho/2)}{\rho\,\zeta'(\rho)}, \qquad r_k = \frac{\zeta(-k)}{-2k\,\zeta'(-2k)}, \tag{10.3.11}$$

where $1 = D(0)$ is the residue at $s = 0$ and $r_k x^{-k}$ is the residue at the trivial pole $s = -k$. For the first zero $R_{\rho_1} = 0.028803 - 0.0961835i$ and $|R_{\rho_1}| = 0.100404$, with $\zeta(\rho_1/2) = 1.03142 + 0.452556i$, and for the next four zeros $|R_\rho| = 0.069179$, 0.0338528, 0.0260772 and 0.0464698; $r_1 = -1.3684327776202059$, $r_3 = 0.23541450677954054$ and $r_5 = -0.020962860062791976$ (`sqfree_constants.py`, `sqfree_atoms.py`). Under RH every term of the sum over ρ has size $x^{1/4}$, which is why $\Delta(x) = O(x^{1/4+\varepsilon})$ is the natural conjecture and why $\Delta(x)/x^{1/4}$ stays of order one in Table 10.11; Corollary 10.3.11 shows that the exponent $\frac14$ cannot be lowered. The series over ρ is not expected to converge absolutely, and the bound $O(x^{1/4+\varepsilon})$ is open even under RH: the known conditional exponents are larger than $\frac14$, among them $\frac{7}{22}$ of Baker and Pintz [12], since improved, and the obstacle is the control of $1/\zeta$ near the critical line. In the language of the clock, the oscillating part of $Q(x)$ is the inverse Mellin image of a measure whose atoms are the Thorin atoms of the clock, weighted by R_ρ.

Figure 10.8 compares $\Delta(x)$ on $[100, 3000]$ with the right side of (10.3.11) truncated after the first K zeros, both divided by $x^{1/4}$ and sampled on the grid $x = 100 + j/20$. The Pearson correlation of the two curves is 0.828, 0.866, 0.915, 0.927 and 0.945 for $K = 10$, 20, 60, 100 and 200, and the root mean square of their difference falls from 0.149 to 0.0868, against 0.304 for $\Delta(x)/x^{1/4}$ itself; what remains includes the unit jumps of Q, which no finite sum of atoms reproduces.

For the sum $\sum_{n \leq x} \mu(n)^2/n$ of Proposition 10.3.1 the same atoms appear at $s = \rho/2 - 1$, with residues $\zeta(\rho/2)x^{\rho/2-1}/((\rho-2)\zeta'(\rho))$ of size $x^{-3/4}$ under RH (of

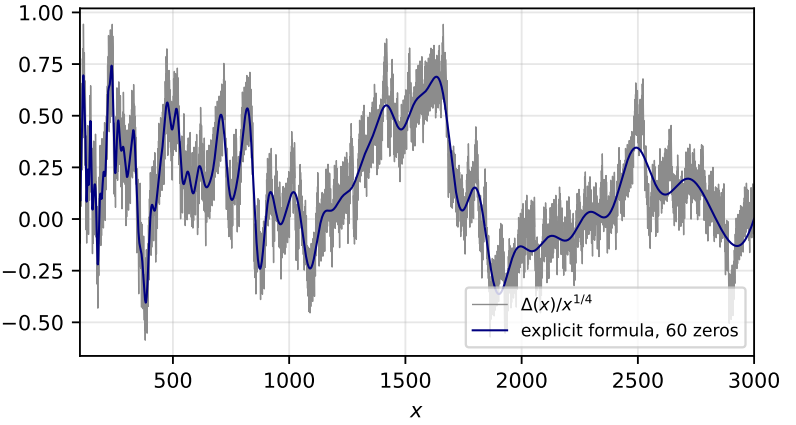


Figure 10.8: The normalised error $\Delta(x)/x^{1/4}$ of the squarefree count on $[100, 3000]$ (grey) and the explicit formula (10.3.11) with the first 60 zeros and the trivial terms at $s = -1$, -3 and -5, divided by $x^{1/4}$ (dark): the residue at $s = 0$ plus the contributions of the first sixty Thorin atoms γ_k^2. Pearson correlation 0.915. Drawn by `sqfree_atoms.py`.

modulus $0.09991\,x^{-3/4}$ for the first zero), inside the error $O(x^{-1/2})$; the column $\mathcal{E}_1(x)x^{3/4}$ of Table 10.11 is of order one. Likewise the error $\mathcal{E}_\alpha$ of (10.3.4), for $\frac{1}{2} < \alpha < 1$, has atoms of size $x^{1/4-\alpha}$. By (10.3.6), a pointwise bound of that order in either case would follow from $\Delta(x) = O(x^{1/4+\varepsilon})$, and it is open for the same reason.

10.3.5 Smoothed counts

Smoothing the count removes the obstacle. For $x > 0$ let

$$Q_1(x) = \sum_{n\ge1} \mu(n)^2 e^{-n/x} = \sum_{d\ge1} \frac{\mu(d)}{e^{d^2/x} - 1}, \qquad Q_2(x) = \sum_{n\ge1} \mu(n)^2 \Big(1 + \frac{2n}{x}\Big) e^{-2n/x},$$

the second form of Q_1 by $\mu(n)^2 = \sum_{d^2\mid n} \mu(d)$. If $G_{1,x}$ is exponential and $G_{2,x}$ has the Gamma(2) law, both with mean x, then $P(G_{1,x} \ge n) = e^{-n/x}$ and $P(G_{2,x} \ge n) = (1 + 2n/x)e^{-2n/x}$, so $Q_1(x) = \mathbb{E}Q(G_{1,x})$ and $Q_2(x) = \mathbb{E}Q(G_{2,x})$: the count is taken at a random point of mean x. Since $\int_0^\infty e^{-n/x} x^{-s-1} dx = \Gamma(s) n^{-s}$ and $\int_0^\infty (1 + 2n/x) e^{-2n/x} x^{-s-1} dx = (1 + s)\Gamma(s)(2n)^{-s}$, the Mellin transforms are

$$\begin{aligned} \int_0^\infty Q_1(x)\, x^{-s-1} dx &= \Gamma(s) D(s), \\ \int_0^\infty Q_2(x)\, x^{-s-1} dx &= (1 + s)\Gamma(s)\, 2^{-s} D(s), \qquad \operatorname{Re} s > 1. \end{aligned} \tag{10.3.12}$$

The kernels $\Gamma(s)$ and $(1+s)\Gamma(s)2^{-s}$ have no zeros in $\operatorname{Re} s > 0$ and decay exponentially on vertical lines. The step from analyticity of D to RH is the same in all the results below.

LEMMA 10.3.8. *If $D(s) = \zeta(s)/\zeta(2s)$ is analytic in the half plane $\operatorname{Re} s > \frac14$ apart from $s = 1$, then RH holds.*

Proof. Let ρ be a zero with $\beta = \operatorname{Re}\rho > \frac12$. Then $\operatorname{Re}(\rho/2) > \frac14$, so the pole of $1/\zeta(2s)$ at $s = \rho/2$ must be cancelled, that is $\zeta(\rho/2) = 0$. Since $\frac14 < \operatorname{Re}(\rho/2) < \frac12$, the functional equation and the reflection $s \mapsto \bar s$ make $\rho' = 1 - \bar\rho/2$ a zero, with $\operatorname{Re}\rho' = 1 - \beta/2 > \frac12$ and $\operatorname{Im}\rho' = \operatorname{Im}\rho/2$. Iterating gives zeros ρ_j whose real parts tend to $\frac23$, the fixed point of $\beta \mapsto 1 - \beta/2$, and whose imaginary parts are $\operatorname{Im}\rho/2^j$. Since $\zeta < 0$ on $(0,1)$, $\operatorname{Im}\rho \neq 0$, so the ρ_j are distinct and accumulate at $\frac23$, where ζ is analytic and not identically zero, a contradiction. (Equally, $|\operatorname{Im}\rho_j|$ eventually falls below $\gamma_1 = 14.13$.) □

THEOREM 10.3.9. (i) *RH holds if and only if $Q_1(x) = \frac{6}{\pi^2}x + O(x^{1/4+\varepsilon})$ for every $\varepsilon > 0$, and if and only if $Q_2(x) = \frac{6}{\pi^2}x + O(x^{1/4+\varepsilon})$ for every $\varepsilon > 0$.*

(ii) *Under RH, if the zeros are simple, then*

$$Q_1(x) = \frac{6}{\pi^2}x + 1 + \sum_\rho A_\rho\, x^{\rho/2} + \frac{c_1 \log x + c_2}{x} + O(x^{-2}),$$

$$Q_2(x) = \frac{6}{\pi^2}x + 1 + \sum_\rho B_\rho\, x^{\rho/2} + \frac{2c_1}{x} + O(x^{-2}),$$

$$A_\rho = \frac{\Gamma(\rho/2)\,\zeta(\rho/2)}{2\zeta'(\rho)}, \qquad B_\rho = \Big(1 + \frac{\rho}{2}\Big)\frac{2^{-\rho/2}\,\Gamma(\rho/2)\,\zeta(\rho/2)}{2\zeta'(\rho)},$$

the sums over ρ taken over $|\operatorname{Im}\rho| < 2T_j$ along a sequence $T_j \to \infty$, with

$$c_1 = -\frac{\zeta(-1)}{2\zeta'(-2)} = -1.36843277762021,$$

$$c_2 = c_1\Big(1 - \gamma_E + \frac{\zeta'(-1)}{\zeta(-1)} - \frac{\zeta''(-2)}{\zeta'(-2)}\Big) = -0.339381226129599.$$

Proof. (i) Suppose RH. By Section 14.2 of [255], $1/\zeta(\sigma + it) = O(|t|^\varepsilon)$ uniformly for $\sigma \geq \frac12 + \varepsilon$, and $\zeta(\sigma + it)$ grows at most polynomially in $|t|$ on vertical strips, while $\Gamma(s)$ decays exponentially. Move the line of integration in $Q_1(x) = \frac{1}{2\pi i}\int_{(2)} \Gamma(s)D(s)x^s\,ds$ to $\operatorname{Re} s = \frac14 + \varepsilon$. The only pole crossed is $s = 1$, with residue $\Gamma(1)/\zeta(2)\cdot x = 6x/\pi^2$; the point $s = \frac12$ is regular because $\zeta(2s)$ has a pole there. On the new line $|\Gamma(s)D(s)|$ is integrable and the integral is $O(x^{1/4+\varepsilon})$. The same argument with the kernel $(1+s)\Gamma(s)2^{-s}$ treats Q_2.

Conversely, suppose $Q_1(x) = \frac{6}{\pi^2}x + O(x^{1/4+\varepsilon})$ for every $\varepsilon > 0$. For $\operatorname{Re} s > 1$, by (10.3.12),

$$\Gamma(s)D(s) - \frac{6}{\pi^2}\,\frac{1}{s-1} = \int_1^\infty \Big(Q_1(x) - \frac{6}{\pi^2}x\Big)x^{-s-1}dx + \int_0^1 Q_1(x)\,x^{-s-1}dx.$$

The first integral converges and is analytic in $\operatorname{Re} s > \frac14$, by the hypothesis, and the second is entire, since $Q_1(x) = O(e^{-1/x})$ as $x \to 0$. As Γ has no zeros, D is analytic in $\operatorname{Re} s > \frac14$ apart from $s = 1$, and Lemma 10.3.8 gives RH. The same argument applies to Q_2.

(ii) By Theorem 9.7 of [255] there are heights $T_j \in [j, j+1]$ at which $1/\zeta(\sigma \pm 2iT_j) = T_j^{O(1)}$ uniformly for $-1 \le \sigma \le 2$, and on the left of the line $\operatorname{Re} s = -\frac12$ the functional equation bounds $1/\zeta(2s)$ polynomially. Move the line of integration to $\operatorname{Re} s = -\frac52$ along the rectangles of height T_j; the exponential decay of Γ makes the horizontal sides vanish as $j \to \infty$, and the integral on the new line is $O(x^{-5/2})$. The poles crossed are $s = 1$; the points $\rho/2$, with the residues $A_\rho x^{\rho/2}$ and $B_\rho x^{\rho/2}$; $s = 0$, where $\Gamma(s)D(s)$ and $(1+s)\Gamma(s)2^{-s}D(s)$ have residue $D(0) = \zeta(0)/\zeta(0) = 1$; $s = -1$; and $s = -2$, a simple pole of both, which gives a multiple of x^{-2}. At $s = -1$, Γ has a simple pole and $1/\zeta(2s)$ another, from the trivial zero -2 of ζ, so $\Gamma(s)D(s) = c_1(s+1)^{-2} + c_2(s+1)^{-1} + O(1)$, with c_1 and c_2 from the expansions $\Gamma(s) = -(s+1)^{-1}(1 + (1-\gamma_E)(s+1) + \cdots)$, $\zeta(s) = \zeta(-1)(1 + \frac{\zeta'(-1)}{\zeta(-1)}(s+1) + \cdots)$ and $\zeta(2s) = 2\zeta'(-2)(s+1)(1 + \frac{\zeta''(-2)}{\zeta'(-2)}(s+1) + \cdots)$, and the residue of $\Gamma(s)D(s)x^s$ is $x^{-1}(c_1 \log x + c_2)$. The factor $1 + s$ makes the pole of $(1+s)\Gamma(s)2^{-s}D(s)$ at $s = -1$ simple, with residue $2c_1$ times x^{-1}. □

At the even negative integers $\Gamma(s)D(s)$ has simple poles, since $\zeta(s)$ vanishes there together with $\zeta(2s)$, and at the odd ones double poles: `sqfree_smooth.py` computes the Laurent coefficients at $s = -1, \dots, -6$ by the trapezoid rule on circles of radius 0.3, finds the coefficients of $(s+2)^{-2}$, $(s+4)^{-2}$ and $(s+6)^{-2}$ below 10^{-47}, and reproduces the closed forms of c_1 and c_2 within 5×10^{-46}; the residue at $s = -2$ is -0.95344364180121, and at $s = -3$ the coefficients of $(s+3)^{-2}$ and $(s+3)^{-1}$ are 0.11770725338977 and 0.230634915663739. For Q_2 the double pole at $s = -1$ becomes a simple one, with residue $2c_1 = -2.73686555524041$, and the next terms are $3.81377456720484\,x^{-2}$ and a double pole at $s = -3$.

The coefficients A_ρ are tiny, $|A_{\rho_1}| = 1.648 \times 10^{-5}$, $|A_{\rho_2}| = 6.838 \times 10^{-8}$, $|A_{\rho_3}| = 1.662 \times 10^{-9}$, and 6.7×10^{-43} at the fortieth zero and 6.29×10^{-82} at the hundredth, because $|\Gamma(\rho/2)|$ decays like $e^{-\pi\gamma/4}$: smoothing by $e^{-n/x}$ damps the atoms exponentially in their height, and up to $x = 10^8$ the atoms beyond the fortieth change the sum by less than 10^{-40}. The Gamma(2) coefficients are larger by the factor $|1 + \rho/2|\,2^{-1/4}$, $|B_{\rho_1}| = 9.944 \times 10^{-5}$. Table 10.13 checks part (ii) for Q_1: the smoothed error is the sum of the atom terms and the trivial terms to the working precision, and at $x = 10^3$ the term $c_2/x = -3.394 \times 10^{-4}$ is needed besides $c_1 \log x/x$. Table 10.15 in Section 10.3.7 does the same for Q_2.

x	$Q_1(x) - \frac{6}{\pi^2}x - 1$	40 atoms	trivial terms	$c_1 \log x/x$	residual
10^3	-9.71103×10^{-3}	8.210×10^{-5}	-9.79313×10^{-3}	-9.4528×10^{-3}	1.5×10^{-26}
10^4	-1.57684×10^{-3}	-2.825×10^{-4}	-1.29432×10^{-3}	-1.26037×10^{-3}	1.8×10^{-33}
$3 \cdot 10^4$	-7.40985×10^{-4}	-2.594×10^{-4}	-4.81551×10^{-4}	-4.70237×10^{-4}	9.0×10^{-37}
10^5	4.22550×10^{-4}	5.835×10^{-4}	-1.60941×10^{-4}	-1.57547×10^{-4}	2.1×10^{-40}
10^6	-8.90390×10^{-4}	-8.711×10^{-4}	-1.92450×10^{-5}	-1.89056×10^{-5}	-1.4×10^{-40}
10^7	7.60273×10^{-4}	7.625×10^{-4}	-2.23959×10^{-6}	-2.20565×10^{-6}	-5.3×10^{-40}
10^8	4.84273×10^{-4}	4.845×10^{-4}	-2.55468×10^{-7}	-2.52075×10^{-7}	-6.9×10^{-39}

Table 10.13: The exponentially smoothed count of Theorem 10.3.9: the smoothed error, the contribution $2\,\mathrm{Re}\sum_{k\le 40} A_{\rho_k} x^{\rho_k/2}$ of the first forty atoms, the residues at $s = -1, \dots, -6$, of which the first contributes $(c_1 \log x + c_2)/x$, the term $c_1 \log x/x$ alone, and the residual. The atoms beyond the fortieth change the sum by less than 10^{-40}. Computed by `sqfree_smooth.py` at 45 digits from the sum over d; the direct sum over n agrees within 3×10^{-12} up to $x = 10^5$.

The theorem isolates what is missing for the sharp count. The Perron kernel x^s/s decays only like $1/|t|$, so the atom series for $Q(x)$ has no exponential damping, and bounding its partial sums would need control of $\sum_{|\gamma|\le T} |\zeta(\rho/2)/(\rho\zeta'(\rho))|$, which is out of reach.

10.3.6 Mean square, and how much smoothing the sharp count needs

Two further results bracket the sharp problem. The first settles it on average; it is presumably folklore, but the proof is short and uses only the analytic continuation behind the atoms. For $\mathrm{Re}\, s > 1$ let

$$F(s) = \int_1^\infty \Delta(x)\, x^{-s-1} dx = \frac{D(s)}{s} - \frac{6}{\pi^2}\,\frac{1}{s-1},$$

the second form because $\int_1^\infty Q(x) x^{-s-1} dx = D(s)/s$.

Theorem 10.3.10. *RH holds if and only if $\int_X^{2X} \Delta(x)^2\, dx = O(X^{3/2+\varepsilon})$ for every $\varepsilon > 0$, that is, if and only if the root mean square of Δ on every range $[X, 2X]$ is $O(X^{1/4+\varepsilon})$.*

Proof. Suppose RH. Then F continues analytically to $\mathrm{Re}\, s > \frac{1}{4}$: the poles of D at the points $\rho/2$ lie on the line $\mathrm{Re}\, s = \frac{1}{4}$, $s = \frac{1}{2}$ is regular, and the pole at $s = 1$ cancels. By Section 14.2 of [255], under RH $\zeta(\sigma + it) = O(|t|^{\max(0, \frac{1}{2}-\sigma)+\varepsilon})$ and $1/\zeta(2\sigma + 2it) = O(|t|^{\varepsilon})$ uniformly for $\sigma \ge \frac{1}{4} + \varepsilon$ and $|t| \ge 1$, so $F(\sigma + it) = O(|t|^{-3/4+2\varepsilon})$ uniformly there. For $w > 2$ let $I(w) = \int_1^\infty \Delta(x)^2 x^{-w-1} dx$, the Laplace transform in $v = \log x$ of the nonnegative function $\Delta(e^v)^2$. Plancherel's theorem for the function $v \mapsto \Delta(e^v) e^{-wv/2}$ on $(0, \infty)$, whose Fourier transform is

$t \mapsto F(\frac{w}{2}+it)$, and the symmetry $F(\bar{s}) = \overline{F(s)}$ give

$$I(w) = \frac{1}{2\pi}\int_{-\infty}^{\infty} F\Big(\frac{w}{2}+it\Big)\,F\Big(\frac{w}{2}-it\Big)\,dt, \qquad w > 2.$$

The right side is analytic in w on $\operatorname{Re} w > \frac12$, since the integrand is analytic there and $O(|t|^{-3/2+4\varepsilon})$ locally uniformly. So I, the Laplace transform of a nonnegative function, continues analytically to a neighbourhood of every point of $(\frac12,\infty)$, and Landau's theorem [268, Chapter II, Theorem 5b] puts its abscissa of convergence at or below $\frac12$. Hence $I(w) < \infty$ for every $w > \frac12$, and $\int_X^{2X} \Delta^2\,dx \le (2X)^{w+1} I(w) = O(X^{3/2+\varepsilon})$ with $w = \frac12+\varepsilon$.

Conversely, by the Cauchy Schwarz inequality the hypothesis gives $\int_X^{2X} |\Delta(x)| x^{-\sigma-1}dx = O(X^{1/4+\varepsilon-\sigma})$, so, summing over $X = 2^j$, the integral defining F converges absolutely for $\operatorname{Re} s > \frac14$. Then $D(s) = s\big(F(s) + \frac{6}{\pi^2}\frac{1}{s-1}\big)$ is analytic in $\operatorname{Re} s > \frac14$ apart from $s = 1$, and Lemma 10.3.8 gives RH. □

Table 10.14 gives the root mean square of Δ on dyadic ranges, computed exactly from the sieve: on $[n, n+1)$ the error decreases linearly from $\Delta_n = Q(n) - 6n/\pi^2$ to $\Delta_n - 6/\pi^2$, so $\int_n^{n+1} \Delta^2 = \Delta_n^2 - \frac{6}{\pi^2}\Delta_n + \frac{12}{\pi^4}$, and the extremes over real x are taken at integers or as left limits there. Over the eighteen ranges $[X, 2X]$ with $X = 10^4 \cdot 2^j$, $j = 0, \dots, 13$, and $X = 10^4, \dots, 10^8$, the root mean square divided by $X^{1/4}$ lies between 0.2337 and 0.3606 and the largest $|\Delta(x)|/X^{1/4}$ between 0.6851 and 1.0361; on $[10^3, 2\times10^8]$ the largest value of $\Delta(x)/x^{1/4}$ is 0.9807, near $x = 47523$, and the least -0.8685, near $x = 80157$.

X	$\big(\frac{1}{X}\int_X^{2X}\Delta^2\big)^{1/2}/X^{1/4}$	$\max_{[X,2X]}\lvert\Delta(x)\rvert/X^{1/4}$	near $x =$
10^4	0.2701	0.8979	18471
10^5	0.2494	0.6851	128599
10^6	0.2779	0.8382	1618611
10^7	0.3541	0.9931	17199381
10^8	0.2812	0.8636	144151023

Table 10.14: Root mean square and largest absolute value of the squarefree error Δ on the ranges $[X, 2X]$, divided by $X^{1/4}$, over real x; the largest value on $[10^7, 2\times10^7]$ is the left limit at 17199381, and over the integers alone it is 0.982. Computed by `sqfree_sieve.py`.

Two consequences follow: a bound off a small set, under RH, and a matching lower bound with no hypothesis.

Corollary 10.3.11. (i) *Under RH, for every $\varepsilon > 0$ the set of $x \in [X, 2X]$ with $|\Delta(x)| > X^{1/4+\varepsilon}$ has measure $O(X^{1-\varepsilon})$.*

(ii) *With no hypothesis,*

$$\limsup_{x\to\infty} \frac{\Delta(x)}{x^{1/4}} \ge |R_{\rho_1}|, \qquad \liminf_{x\to\infty} \frac{\Delta(x)}{x^{1/4}} \le -|R_{\rho_1}|,$$

where $R_{\rho_1} = \zeta(\rho_1/2)/(\rho_1\zeta'(\rho_1))$ *and* $|R_{\rho_1}| = 0.100404$. *The same holds for* $Q_2(x) - \frac{6}{\pi^2}x - 1$ *with* $|B_{\rho_1}| = 9.944 \times 10^{-5}$ *in place of* $|R_{\rho_1}|$.

Proof. (i) is Chebyshev's inequality applied to Theorem 10.3.10. For (ii) suppose that $\Delta(x) \le c\,x^{1/4}$ for $x \ge x_0$, with $0 < c < |R_{\rho_1}|$. The function $L(s) = \int_{x_0}^{\infty}(cx^{1/4} - \Delta(x))x^{-s-1}dx$ is the Laplace transform in $\log x$ of a nonnegative function, and for $\operatorname{Re} s > 1$

$$L(s) = \frac{c\,x_0^{1/4-s}}{s - \frac14} - F(s) + E(s),$$

with E entire. The right side has no singularity on the real half line $(\frac14, \infty)$: a real pole of D there would be a real zero of ζ in $(\frac12, \infty)$, of which there are none, $s = \frac12$ is regular, and the pole at $s = 1$ cancels. By Landau's theorem the integral $L(s)$ converges for $\operatorname{Re} s > \frac14$, so F is analytic there (which already gives RH, by Lemma 10.3.8), and $|L(\sigma + it)| \le L(\sigma)$ for $\sigma > \frac14$. As $\sigma \downarrow \frac14$, $L(\sigma) = c/(\sigma - \frac14) + O(1)$, since F is analytic at $s = \frac14$, where $\zeta(2s) = \zeta(\frac12) \ne 0$. At $s = \sigma + \frac{i}{2}\gamma_1$ the first and last terms of the display are $O(1)$, while $F(s) = R_{\rho_1}/(\sigma - \frac14) + O(1)$, because $D(s)/s$ has a simple pole at $\rho_1/2$ with residue R_{ρ_1} (the zero ρ_1 is simple and $\zeta(\rho_1/2) \ne 0$). Hence $|R_{\rho_1}| \le c$, a contradiction. The lower bound is the same argument applied to $c\,x^{1/4} + \Delta(x)$, and the smoothed case is the same argument with the kernel $(1+s)\Gamma(s)2^{-s}$, whose residue at $\rho_1/2$ is B_{ρ_1}. □

The sieve data of Tables 10.11 and 10.14, with maxima from 0.69 to 1.04, sit well above 0.10. Using many zeros at once, with a Fejér kernel as in Ingham's method for $M(x)$ [105], would raise the constant (Question 10.3.20).

The second result measures the cost of sharpening. Let $G_{n,x}$ have the Gamma(n) law with mean x, so that $G_{n,x}$ concentrates at x in a window of width $x/\sqrt{n}$; $Q_1(x) = \mathbb{E}Q(G_{1,x})$ and $Q_2(x) = \mathbb{E}Q(G_{2,x})$.

Proposition 10.3.12. *Under RH, uniformly in* $n \ge 1$,

$$\mathbb{E}Q(G_{n,x}) = \frac{6}{\pi^2}\,x + O_\varepsilon\big(x^{1/4+\varepsilon}\,n^{1/8+\varepsilon}\big).$$

Proof. With rate $\lambda = n/x$, $P(G_{n,x} \ge m) = \Gamma(n, \lambda m)/\Gamma(n)$, whose Mellin transform in λm is $\Gamma(s+n)/(s\Gamma(n))$, so $\mathbb{E}Q(G_{n,x}) = \frac{1}{2\pi i}\int_{(2)} D(s)\frac{\Gamma(s+n)}{s\Gamma(n)}\big(\frac{x}{n}\big)^s ds$. Move the line to $\operatorname{Re} s = c = \frac14 + \varepsilon$, collecting $6x/\pi^2$ at $s = 1$. On the new line $|\Gamma(n + c + it)|/\Gamma(n) = O\big(n^c \exp(-\kappa \min(t^2/n, |t|))\big)$ for an absolute $\kappa > 0$, since $\Gamma(n+c)/\Gamma(n) = O(n^c)$ and $|\Gamma(n+c+it)/\Gamma(n+c)|^2 = \prod_{k\ge0}\big(1 + t^2/(n+c+k)^2\big)^{-1}$; the factors with $k < n$ give $e^{-\kappa t^2/n}$ when $|t| \le n$, and those with $k \le |t|$ give $e^{-\kappa|t|}$ when $|t| > n$. Under RH, $|D(c + it)| = O(|t|^{1/4+\varepsilon})$ for $|t| \ge 1$ (Section 14.2 of [255] and the functional equation). The remaining integral is therefore

$O\big(x^c \int (1+|t|)^{-3/4+\varepsilon} e^{-\kappa \min(t^2/n,|t|)} dt\big) = O(x^c n^{1/8+\varepsilon})$, the integral being dominated by $|t| \lesssim \sqrt{n}$. □

The exponent is consistent at both ends. At $n = 1$ and $n = 2$ it is Theorem 10.3.9. At $n \asymp x^2$ the window has width $O(1)$ and the bound is $x^{1/2}$, the trivial bound (10.3.2). The loss $n^{1/8}$ is the growth $|t|^{1/4}$ of ζ on the line $\operatorname{Re} s = \frac{1}{4}$, integrated against $dt/|t|$ up to the effective cutoff $|t| \asymp \sqrt{n}$. Balancing the window against the loss gives a bound for the sharp count.

Corollary 10.3.13. *Under RH, $\Delta(x) = O(x^{2/5+\varepsilon})$ for every $\varepsilon > 0$.*

Proof. Since Q is nondecreasing and counts integers, $|Q(y) - Q(x)| \le |y - x| + 1$, so $|\mathbb{E}Q(G_{n,x}) - Q(x)| \le \mathbb{E}|G_{n,x} - x| + 1 \le x/\sqrt{n} + 1$. With Proposition 10.3.12, $\Delta(x) = O(x^{1/4+\varepsilon} n^{1/8+\varepsilon} + x/\sqrt{n})$, and $n = \lceil x^{6/5} \rceil$ balances the two terms at $x^{2/5+O(\varepsilon)}$. □

The exponent $\frac{2}{5}$ is weaker than the known conditional exponents, but it comes in two lines from the smoothed counts, and the argument shows what is missing. If the loss $n^{1/8}$ were replaced by n^{δ}, the same balance would give $\Delta(x) = O(x^{1-3/(8\delta+4)+\varepsilon})$, which tends to $x^{1/4+\varepsilon}$ as $\delta \to 0$; conversely $\Delta(x) = O(x^{1/4+\varepsilon})$ gives $|\mathbb{E}Q(G_{n,x}) - \frac{6}{\pi^2}x| \le \mathbb{E}|\Delta(G_{n,x})| = O(x^{1/4+\varepsilon})$ uniformly in n, by Jensen's inequality. So the sharp conjecture amounts to removing the loss, which needs cancellation in $\int_{|t|\le T} D(\frac{1}{4} + it) x^{it}\, dt/t$ and not only a bound on the size of the integrand.

10.3.7 Gamma(2) smoothing and the Mertens function

The elementary method rests on the terms $\mu(d)(\lfloor x/d^2 \rfloor - x/d^2) = -\mu(d)\{x/d^2\}$ of (10.3.8). Following the idea of Steutel and Thiemann of splitting a random variable into its integer and fractional parts [246], we randomise x by a gamma law. Every term then becomes an explicit expectation, and the whole error becomes an average of the Mertens function. Shape two is also the shape of the gamma variables in Williams' representation of the Pólya variable (Section 8.2.3), a common shape rather than a derivation of one from the other, and Theorem 10.3.18 shows that here it is the largest shape for which the average is taken under a positive law.

Let Y have the Gamma(ν) law with rate λ, let $a > 0$ and $y = \lambda a$. Then $\mathbb{E}\lfloor Y/a \rfloor = \sum_{m\ge1} P(Y \ge ma) = \sum_{m\ge1} \Gamma(\nu, my)/\Gamma(\nu)$ and $\mathbb{E}Y/a = \nu/y$, so the expected fractional part is

$$\mathbb{E}\{Y/a\} = g_\nu(y) = \frac{\nu}{y} - \sum_{m\ge1} \frac{\Gamma(\nu, my)}{\Gamma(\nu)}.$$

For $\nu = 1$, Steutel and Thiemann observed that $\lfloor Y/a \rfloor$ is geometric and $\{Y/a\}$ an independent truncated exponential variable, with $\mathbb{E}\lfloor Y/a \rfloor = 1/(e^y - 1)$ and

$g_1(y) = 1/y - 1/(e^y - 1)$. With Y of mean x, $\mathbb{E}Q(Y)$ is the count $Q_1(x)$ of Section 10.3.5.

LEMMA 10.3.14. *For $\nu = 2$,*

$$\mathbb{E}\lfloor Y/a \rfloor = \frac{1}{e^y - 1} + \frac{y}{4\sinh^2(y/2)}, \qquad g_2(y) = \frac{2}{y} - \frac{1}{e^y - 1} - \frac{y}{4\sinh^2(y/2)}.$$

For $0 < |y| < 2\pi$, with the Bernoulli numbers B_{2k},

$$g_2(y) = \frac{1}{2} + \sum_{k\ge 2} \frac{(2k-2)B_{2k}}{(2k)!}\, y^{2k-1} = \frac{1}{2} - \frac{y^3}{360} + \frac{y^5}{7560} - \cdots,$$

$g_2(y) = 2/y + O(ye^{-y})$ as $y \to \infty$, and for every $\nu > 0$

$$\int_0^\infty g_\nu(y)\, y^{s-1}\, dy = -\frac{\Gamma(s+\nu)\,\zeta(s)}{s\,\Gamma(\nu)}, \qquad 0 < \operatorname{Re} s < 1,$$

which is $-(1+s)\Gamma(s)\zeta(s)$ for $\nu = 2$.

Proof. For $\nu = 2$, $P(Y \ge ma) = (1+my)e^{-my}$, and $\sum_m e^{-my} = 1/(e^y - 1)$, $\sum_m mye^{-my} = y/(4\sinh^2(y/2))$. For the expansion use $1/(e^y - 1) = 1/y - \frac{1}{2} + \sum_{k\ge 1} B_{2k} y^{2k-1}/(2k)!$ and $y/(4\sinh^2(y/2)) = -y\frac{d}{dy}(e^y - 1)^{-1}$. For the Mellin transform write $g_\nu(y) = \mathbb{E}\{Y/y\}$ with Y of law Gamma(ν,1), substitute $u = Y/y$, and use the classical formula $\int_0^\infty \{u\}u^{-s-1}du = -\zeta(s)/s$ for $0 < \operatorname{Re} s < 1$ [255, Section 2.1]: the transform is $\mathbb{E}Y^s \cdot (-\zeta(s)/s)$, with $\mathbb{E}Y^s = \Gamma(s+\nu)/\Gamma(\nu)$. □

The Mellin formula is the gamma analogue of the classical one it is derived from. The terms of the elementary method now read $\mathbb{E}\big[\mu(d)\big(\lfloor G_{2,x}/d^2 \rfloor - G_{2,x}/d^2\big)\big] = -\mu(d)g_2(2d^2/x)$ for $G_{2,x}$ of the Gamma(2) law with mean x, and since $0 \le g_2(y) \le \min(1, 2/y)$ the sum over all d converges absolutely, with no truncation at $d^2 \le x$:

$$Q_2(x) = \sum_{n\ge 1} \mu(n)^2 \Big(1 + \frac{2n}{x}\Big) e^{-2n/x} = \frac{6}{\pi^2}\, x - \sum_{d\ge 1} \mu(d)\, g_2\Big(\frac{2d^2}{x}\Big). \tag{10.3.13}$$

THEOREM 10.3.15. *Let*

$$f_2(y) = -2g_2'(y) = \frac{4}{y^2} - \frac{y\cosh(y/2)}{2\sinh^3(y/2)}, \qquad y > 0.$$

Then $f_2 > 0$, $f_2(y) = y^2/60 + O(y^4)$ as $y \to 0$ and $f_2(y) \sim 4/y^2$ as $y \to \infty$, and f_2 is the density of a probability law $\mathcal{V}_2$ with $P(\mathcal{V}_2 \ge y) = 2g_2(y)$ and

$$\mathbb{E}\mathcal{V}_2^s = -2\,\Gamma(s+2)\,\zeta(s), \qquad -3 < \operatorname{Re} s < 1.$$

For every $x > 0$,

$$Q_2(x) = \frac{6}{\pi^2}\, x - \frac{1}{2}\, \mathbb{E}M\Big(\sqrt{x\mathcal{V}_2/2}\Big).$$

Moreover f_2 is twice the difference between the continuous and the integer mixtures of the Gamma(2) densities $\gamma_{2,\lambda}(y) = \lambda^2 y e^{-\lambda y}$ over their rate:

$$f_2(y) = 2\Big(\int_0^\infty \gamma_{2,\lambda}(y)\, d\lambda - \sum_{m\ge 1} \gamma_{2,m}(y)\Big) = -2\int_0^\infty \Big(\{\lambda\} - \frac{1}{2}\Big)\, \partial_\lambda \gamma_{2,\lambda}(y)\, d\lambda.$$

Proof. With $z = y/2$, $f_2(y) > 0$ is $(\sinh z/z)^3 > \cosh z$, Lazarević's inequality [140], which we prove. Let $\phi(z) = 3\log(\sinh z/z) - \log\cosh z$, so that $\phi(0+) = 0$ and $\phi'(z) = 3\coth z - 3/z - \tanh z$. Multiplying by $z\sinh z\cosh z > 0$ and writing everything at $2z$, $\phi'(z) > 0$ becomes $2z\cosh 2z + 4z - 3\sinh 2z > 0$, and

$$t\cosh t + 2t - 3\sinh t = \sum_{k\ge 2} \frac{(2k-2)\, t^{2k+1}}{(2k+1)!} > 0, \qquad t > 0,$$

since the terms in t and t^3 cancel. So ϕ increases and $\phi > 0$. The expansion at 0 follows from Lemma 10.3.14. The mass is $2(g_2(0+) - g_2(\infty)) = 1$ and the tail is $2g_2(y)$. Integrating by parts in the Mellin formula of Lemma 10.3.14 gives $\mathbb{E}\mathcal{V}_2^s = 2s\int_0^\infty y^{s-1} g_2(y)\, dy = -2s(1+s)\Gamma(s)\zeta(s)$ for $0 < \operatorname{Re} s < 1$, and the behaviour at 0 and at infinity extends the integral and the identity to $-3 < \operatorname{Re} s < 1$. For the representation, $P(\sqrt{x\mathcal{V}_2/2} \ge d) = P(\mathcal{V}_2 \ge 2d^2/x) = 2g_2(2d^2/x)$, so $\frac{1}{2}\mathbb{E}M(\sqrt{x\mathcal{V}_2/2}) = \sum_d \mu(d) g_2(2d^2/x)$, the interchange being justified by absolute convergence, and (10.3.13) gives the formula. Finally $\int_0^\infty \gamma_{2,\lambda}(y)\, d\lambda = 2/y^2$ and $\sum_m \gamma_{2,m}(y) = y\sum_m m^2 e^{-my} = y\cosh(y/2)/(4\sinh^3(y/2))$, and the last form is the Euler Maclaurin formula $\int_0^\infty h - \sum_{m\ge 1} h(m) = -\int_0^\infty (\{\lambda\} - \frac{1}{2}) h'(\lambda)\, d\lambda$ for $h = \gamma_{2,\cdot}(y)$, which vanishes at $\lambda = 0$. □

So the squarefree error, smoothed by a Gamma(2) law, is exactly minus half the Mertens function evaluated at the random point $\sqrt{x\mathcal{V}_2/2}$, and the mixing law $\mathcal{V}_2$ is itself the continuous minus the integer mixture of the Gamma(2) family over its rate. The exponential case is the same with

$$f_1(y) = -2g_1'(y) = \frac{2}{y^2} - \frac{1}{2\sinh^2(y/2)} = 2\Big(\int_0^\infty \lambda e^{-\lambda y} d\lambda - \sum_{m\ge 1} m e^{-my}\Big) > 0,$$

positive because $\sinh z > z$, with $f_1(0+) = \frac{1}{6}$, $\mathbb{E}\mathcal{V}_1^s = -2\Gamma(s+1)\zeta(s)$ for $-1 < \operatorname{Re} s < 1$, and

$$Q_1(x) = \frac{6}{\pi^2}\, x - \frac{1}{2}\, \mathbb{E}M\Big(\sqrt{x\mathcal{V}_1}\Big).$$

Corollary 10.3.16. (i) *With no hypothesis, $Q_\nu(x) = \frac{6}{\pi^2}x + o(x^{1/2})$ for $\nu = 1, 2$.* (ii) *Under RH, $|Q_2(x) - \frac{6}{\pi^2}x| \le \frac{1}{2}\mathbb{E}|M(\sqrt{x\mathcal{V}_2/2})| = O(x^{1/4+\varepsilon})$, and likewise for Q_1; this is a second proof of the forward half of Theorem 10.3.9(i).*

Proof. (i) $|M(t)| \le t$ and $M(t) = o(t)$, and $\mathbb{E}\mathcal{V}_\nu^{1/2} < \infty$, so dominated convergence gives $\mathbb{E}M(\sqrt{x\mathcal{V}_2/2}) = o(x^{1/2})$, and likewise for $\mathcal{V}_1$. (ii) RH gives $M(t) = O(t^{1/2+\varepsilon})$ [255, Section 14.25], and $\mathbb{E}\mathcal{V}_\nu^{1/4+\varepsilon} < \infty$. □

Table 10.15 checks (10.3.13) and Theorem 10.3.9(ii) for Q_2. It computes $Q_2(x) = \sum_d \mu(d)\,\mathbb{E}\lfloor G_{2,x}/d^2 \rfloor = \sum_d \mu(d)\big(1/(e^t-1) + t/(4\sinh^2(t/2))\big)$, $t = 2d^2/x$, whose terms with $t > 140$ are below 10^{-55}, and this agrees with the direct sum over n, in double precision, within 10^{-10} up to $x = 10^6$. The sum (10.3.13) truncated at $d \le 2\times 10^5$, as in the first author's note, loses 2.2×10^{-4} at $x = 10^6$, because $g_2(2d^2/x)$ is close to x/d^2 for large d and the tail $x\sum_{d>2\times10^5}\mu(d)/d^2$ is left out.

x	$Q_2(x) - \frac{6}{\pi^2}x$	$d \le 2\cdot 10^5$	$\frac{Q_2(x)-\frac{6}{\pi^2}x-1}{x^{1/4}}$	100 atoms	trivial terms	residual
10^3	0.997161220898	0.997161002208	-5.048×10^{-4}	-1.0571×10^{-4}	-2.73307×10^{-3}	-1.0×10^{-23}
10^4	1.00094524529	1.00094305839	9.452×10^{-5}	1.21889×10^{-3}	-2.73648×10^{-4}	-1.3×10^{-30}
10^5	0.996620193442	0.99659832449	-1.9006×10^{-4}	-3.35244×10^{-3}	-2.73683×10^{-5}	-1.5×10^{-37}
10^6	1.00610290943	1.00588421991	1.9299×10^{-4}	6.10565×10^{-3}	-2.73686×10^{-6}	-1.8×10^{-41}
10^7	0.992114718377		-1.4022×10^{-4}	-7.88501×10^{-3}	-2.73687×10^{-7}	-3.9×10^{-42}
10^8	1.00440085379		4.4009×10^{-5}	4.40088×10^{-3}	-2.73687×10^{-8}	-2.2×10^{-38}

Table 10.15: The Gamma(2) smoothed count: $Q_2(x) - 6x/\pi^2$, the sum (10.3.13) truncated at $d \le 2\times10^5$, the normalised error, the contribution $2\,\mathrm{Re}\sum_{k\le100} B_{\rho_k} x^{\rho_k/2}$ of the first hundred atoms, the residues at $s = -1,\dots,-6$, whose leading part is $2c_1/x$, and the residual. The constant 1 is the residue $D(0)$ at $s = 0$. Computed by `sqfree_smooth.py` at 45 digits.

10.3.8 Gamma shapes: two is the threshold

For Gamma(n) smoothing with rate λ, $P(Y \ge ma) = e^{-my}\sum_{j<n}(my)^j/j!$, and the Mellin kernel is $\sum_{j<n}\Gamma(s+j)/j! = \Gamma(s+n)/(s\Gamma(n))$; with $\lambda = n/x$ it tends to the Perron kernel x^s/s as $n\to\infty$, which recovers the sharp count, where $\{x/t^2\}$ jumps. For every real shape $\nu > 0$, $f_\nu = -2g_\nu'$ is a signed density of total mass $2g_\nu(0+) = 1$, and the argument of Theorem 10.3.15 gives, for Y of the Gamma(ν) law with rate λ,

$$\mathbb{E}Q(Y) = \frac{6}{\pi^2}\,\mathbb{E}Y - \frac{1}{2}\int_0^\infty M\Big(\sqrt{y/\lambda}\Big) f_\nu(y)\,dy,$$

$$\int_0^\infty y^s f_\nu(y)\,dy = -\frac{2\,\Gamma(s+\nu)\,\zeta(s)}{\Gamma(\nu)}, \qquad -\nu < \mathrm{Re}\,s < 1.$$

The question is for which ν the density f_ν is positive. The poles of the Mellin transform of g_ν in Lemma 10.3.14 at $s = 0$ and $s = -\nu - k$ give the expansion $g_\nu(y) = \frac12 + \zeta(-\nu)y^\nu/\Gamma(\nu+1) + O(y^{\nu+1})$ at 0, and the next lemma makes it exact.

Write $\gamma_a(t) = t^{a-1}e^{-t}/\Gamma(a)$ for the Gamma(a) density, and let

$$\mathcal{R}_\nu(y) = y\sum_{m\ge1}\gamma_{\nu+1}(my) = \frac{y^{\nu+1}}{\Gamma(\nu+1)}\sum_{m\ge1} m^\nu e^{-my},$$
$$q_\nu(y) = \frac{\Gamma(\nu+1)\big(1-\mathcal{R}_\nu(y)\big)}{y^{\nu+1}},$$

so that $\mathcal{R}_\nu(y)$ is the right Riemann sum with step y of the Gamma$(\nu+1)$ density, whose integral is 1.

LEMMA 10.3.17. *For $\nu > 0$ and $y > 0$,*

$$f_\nu(y) = \frac{2\nu\big(1-\mathcal{R}_\nu(y)\big)}{y^2} = \frac{2y^{\nu-1}}{\Gamma(\nu)}\,q_\nu(y),$$

so $f_\nu \ge 0$ exactly when $\mathcal{R}_\nu \le 1$, that is when no Riemann sum of the Gamma$(\nu+1)$ density exceeds its integral. For $0 < y < 2\pi$,

$$q_\nu(y) = -\sum_{k\ge0}\zeta(-\nu-k)\,\frac{(-y)^k}{k!},$$

and $\mathcal{R}_1(y) = (y/2)^2/\sinh^2(y/2)$, $\mathcal{R}_2(y) = (y/2)^3\cosh(y/2)/\sinh^3(y/2)$.

Proof. With Y of the Gamma$(\nu,1)$ law and $c = 1/y$, $g_\nu(y) = \mathbb{E}\{cY\} = c\nu - \sum_{m\ge1}P(Y \ge m/c)$, whose derivative in c is $\nu - \sum_m (m/c^2)p_\nu(m/c)$, with $p_\nu = \gamma_\nu$ the density of Y. Since $tp_\nu(t) = \nu\gamma_{\nu+1}(t)$, the derivative is $\nu(1-\mathcal{R}_\nu(1/c))$, and returning to y gives $g'_\nu(y) = -\nu(1-\mathcal{R}_\nu(y))/y^2$. The series is the expansion $\sum_m m^\nu e^{-my} = \Gamma(\nu+1)y^{-\nu-1} + \sum_{k\ge0}\zeta(-\nu-k)(-y)^k/k!$ of the polylogarithm, valid for $0 < y < 2\pi$, and the closed forms are $\sum_m me^{-my} = 1/(4\sinh^2(y/2))$ and $\sum_m m^2e^{-my} = \cosh(y/2)/(4\sinh^3(y/2))$. □

So $\mathcal{R}_1 \le 1$ is $\sinh z \ge z$ and $\mathcal{R}_2 \le 1$ is Lazarević's inequality. Near the origin f_ν has the sign of $q_\nu(0+) = -\zeta(-\nu)$, and at an even integer $\nu = 2m$, where $\zeta(-2m) = 0$, the sign of $q_{2m}(y) = \zeta(-2m-1)y + O(y^2)$. Since ζ is positive on $(-4,-2)$, $(-8,-6)$, ... and $\zeta(-2m-1) = -B_{2m+2}/(2m+2)$, f_ν is negative near the origin exactly when $\nu \in (4j+2, 4j+4]$ for some $j \ge 0$. At $\nu = 2$ the first term vanishes, $q_2(y) = y/120 + O(y^2)$, and $f_2(y) \sim y^2/60$: the coefficient $-\zeta(-3)/3 = -\frac{1}{360}$ of y^3 in g_2 is why the inequality behind Theorem 10.3.15 has the exponent 3.

THEOREM 10.3.18. $\{\nu > 0 :\ f_\nu \ge 0 \text{ on } (0,\infty)\} = (0,2]$. *Thus two is the largest shape for which the gamma smoothed squarefree error is a positive average of the Mertens function, and the obstruction is the trivial zero of ζ at -2.*

Proof. Let $\nu > 2$ and suppose $f_\nu \geq 0$. Since $f_\nu(y) \sim 2\nu/y^2$ at infinity, $\int_1^\infty y^s f_\nu(y)\,dy$ converges and is analytic for $\operatorname{Re} s < 1$. The function $m(s) = -2\Gamma(s+\nu)\zeta(s)/\Gamma(\nu)$ is analytic in a neighbourhood of $(-\nu, 1)$, so $\int_0^1 y^s f_\nu(y)\,dy = m(s) - \int_1^\infty y^s f_\nu(y)\,dy$, a Laplace transform in $-\log y$ of a nonnegative function, continues analytically along the real interval $(-\nu, 1)$. By Landau's theorem [268, Chapter II, Theorem 5b] it converges on all of $(-\nu, 1)$, and there $m(s) = \int_0^\infty y^s f_\nu(y)\,dy > 0$. But $\Gamma(s+\nu) > 0$ on $(-\nu, 1)$, while $\zeta(s) > 0$ on $(-4, -2)$, which meets $(-\nu, 1)$ because $\nu > 2$, so $m(s) < 0$ there, a contradiction. (Lemma 10.3.17 gives the convergence directly: $f_\nu(y) = O(y^{\nu-1})$ at 0.)

Let $0 < \nu \leq 2$. The cases $\nu = 1$ and $\nu = 2$ are the inequalities above. Otherwise write $\nu = k + \mu$ with $k \in \{0, 1\}$ and $0 < \mu < 1$. Bernstein's formula $t^\mu = \frac{\mu}{\Gamma(1-\mu)} \int_0^\infty (1 - e^{-st}) s^{-\mu-1} ds$, applied to $t = mh$ in $\sum_m (mh)^{k+\mu} e^{-mh}$ and to t in $\int_0^\infty t^{k+\mu} e^{-t} dt$, together with $h \sum_m (mh)^k e^{-mhc} = k!\,\mathcal{R}_k(hc)/c^{k+1}$ and $\int_0^\infty t^k e^{-ct} dt = k!/c^{k+1}$ for $c = 1 + s$, gives

$$\Gamma(\nu+1)\big(\mathcal{R}_\nu(h) - 1\big) = \frac{\mu\, h^{k+1}}{\Gamma(1-\mu)} \int_0^\infty \big(q_k(h(1+s)) - q_k(h)\big)\, s^{-\mu-1}\, ds, \qquad h > 0.$$

So $\mathcal{R}_\nu \leq 1$ follows if q_k is nonincreasing. For $k = 0$, $q_0(y) = 1/y - 1/(e^y - 1) = g_1(y)$, which decreases because $f_1 > 0$. For $k = 1$, $q_1(y) = 1/y^2 - 1/(4\sinh^2(y/2)) = f_1(y)/2$, and with $z = y/2$

$$q_1'(y) = \frac{1}{4}\Big(\frac{\cosh z}{\sinh^3 z} - \frac{1}{z^3}\Big) < 0$$

by Lazarević's inequality. □

The proof is a ladder: positivity for shapes in $(k, k+1)$ follows from the monotonicity of q_k, that is of g_1 for $k = 0$ and of f_1 for $k = 1$. The ladder breaks at the next rung, since $q_2(y) = y/120 + O(y^2)$ increases near 0; that is the same fact, $\zeta(-2) = 0$ and $\zeta(-3) > 0$, seen from the other side. The script `sqfree_gammanu.py` evaluates $1 - \mathcal{R}_\nu$ for twenty shapes at 40 digits, by the series of Lemma 10.3.17 for $y \leq 2.5$ and by the direct sum beyond, which agree at $y = 1$, 2 and 2.5 within 2×10^{-40} for $\nu = 0.5$, 2, 2.5 and 8, on 400 geometrically spaced points in $[10^{-4}, 10^{0.4}]$ and 799 equally spaced points in $(2.5, 60]$. For the ten shapes $\nu = 0.25, 0.5, 0.75, 1, 1.25, 1.5, 1.75, 1.9, 1.99, 2$ no grid point gives a negative value; on the range $[0.02, 15]$ the least values, from 2.68×10^{-3} at $\nu = 0.25$ to 6.90×10^{-10} at $\nu = 2$, are taken at its left end, where $1 - \mathcal{R}_\nu(y)$ is close to $-\zeta(-\nu) y^{\nu+1}/\Gamma(\nu+1)$, or to $y^4/240$ at $\nu = 2$. Table 10.16 shows the shapes above 2: the negative set starts at the origin for the shapes in $(2, 4]$ and in $(6, 8]$, as the sign rule predicts, lies away from it for $\nu = 5$ and 6, and for $\nu = 7$ and 8 consists of two intervals. The least value of $1 - \mathcal{R}_\nu$ moves out roughly like ν.

ν	$-\zeta(-\nu)$	least value of $1-\mathcal{R}_\nu$	at $y=$	grid points with $1-\mathcal{R}_\nu<0$
2.01	-3.012×10^{-4}	-7.310×10^{-10}	0.0273	from 10^{-4} to 0.0362
2.1	-2.7295×10^{-3}	-4.941×10^{-6}	0.265	from 10^{-4} to 0.347
2.5	-8.517×10^{-3}	-1.0763×10^{-3}	1.187	from 10^{-4} to 1.512
3	-8.333×10^{-3}	-7.2304×10^{-3}	2.144	from 10^{-4} to 2.716
3.5	-4.441×10^{-3}	-0.019006	2.9745	from 10^{-4} to 3.723
4	0	-0.035249	3.726	from 10^{-4} to 4.659
5	3.968×10^{-3}	-0.076713	5.084	from 1.716 to 6.386
6	0	-0.12519	6.323	from 3.076 to 7.969
7	-4.167×10^{-3}	-0.17715	7.4915	from 10^{-4} to 1.234, from 4.227 to 9.409
8	0	-0.23064	8.614	from 10^{-4} to 2.269, from 5.307 to 10.85

Table 10.16: Gamma(ν) smoothing above the threshold of Theorem 10.3.18: the limit $-\zeta(-\nu)$ of q_ν at 0, the least value of $1-\mathcal{R}_\nu$, which has the sign of f_ν, with the point where it is attained (grid search refined by golden section), and the maximal runs of grid points on which $1-\mathcal{R}_\nu<0$. Computed by `sqfree_gammanu.py`.

10.3.9 The Euler product of the squarefree law

For $\sigma>1$ let N_σ have the law $P(N_\sigma=n)=\mu(n)^2n^{-\sigma}\zeta(2\sigma)/\zeta(\sigma)$ on the squarefree integers, the squarefree analogue of the zeta distribution of Section 6.1.6. By the Euler product in (10.3.1), $\log N_\sigma=\sum_p B_p\log p$ with independent Bernoulli variables B_p of success probability $p^{-\sigma}/(1+p^{-\sigma})$, and

$$\mathbb{E}e^{-it\log N_\sigma}=\prod_p\frac{1+p^{-\sigma-it}}{1+p^{-\sigma}},\qquad \log\left(1+p^{-s}\right)=\sum_{k\ge1}\frac{(-1)^{k+1}}{k}\,p^{-ks}.$$

So the would be Lévy measure of $\log N_\sigma$ is the signed measure

$$\sum_p\sum_{k\ge1}\frac{(-1)^{k+1}}{k}\,p^{-k\sigma}\,\delta_{k\log p}$$

of finite total variation, a sum over the primes of alternating lattices like the fair coin lattices of Section 6.1.6, here with the geometric weights $p^{-k\sigma}$ of a biased coin: the law is quasi infinitely divisible, and by the uniqueness of the signed Lévy measure it is not infinitely divisible [147]. Its negative masses, at even k, come from the factor $1/\zeta(2s)$, whose weights $-2/k$ at even k outweigh the weights $1/k$ of $\zeta(s)$. This mirrors the picture above: the factor $1/\zeta(2s)$ that spoils infinite divisibility here is the one that, after the change of variable $u=(2s-\frac12)^2$, becomes the clock $1/G(u)$ up to the gamma factor of Lemma 10.3.2, with a positive Thorin measure under RH.

10.3.10 Questions

Question 10.3.19. Prove $\Delta(x)=O(x^{1/4+\varepsilon})$ under RH. Equivalent forms above: remove the loss $n^{1/8}$ in Proposition 10.3.12; or show that the exceptional set of

Corollary 10.3.11(i), with $X^{1/4+\varepsilon}$ replaced by a constant multiple, is empty for large X. By that corollary, the latter would follow from a bound $O(X^{1/4+\varepsilon})$ for the oscillation of Δ on the subintervals of $[X, 2X]$ of length $X^{1-\varepsilon/2}$.

QUESTION 10.3.20. Sharpen the constant $|R_{\rho_1}| = 0.1004$ of Corollary 10.3.11(ii) with a Fejér kernel over many zeros, as Ingham did for $M(x)$ [105], and compare it with the observed $\max|\Delta(x)|/x^{1/4}$, about 1 up to 2×10^8 (Table 10.14). Is $\limsup|\Delta(x)|/x^{1/4} = \infty$, as the linear independence of the ordinates would suggest?

Chapter 11

Infinite divisibility, the eta analogue and almost generalised gamma convolutions

Chapters 9 and 10 attached to the reciprocal Xi function a family of laws indexed by the basepoint: the shifted clocks T_α, obtained by subordinating the tilted centre clock, the normal scale mixtures Y_α and, on the Mellin side, the tilted Pólya variables V_α. From the basepoint one on, the shifted clocks are GGCs and the normal scale mixtures extended GGCs, with no hypothesis, and below it each of these properties is equivalent to a zero free half plane. The five sections of this chapter read these objects in other settings; the question throughout is what can be said below the basepoint one without such a half plane.

Section 11.1 asks what the general theory of infinitely divisible laws, in the books of Sato and of Bondesson, gives these laws, including quasi infinite divisibility of the tilted Pólya variables above the basepoint one and its failure there. It then places the processes that the laws generate, among them a Wiener gamma integral for T_1 and a Riemann Ornstein Uhlenbeck process, on one map. Section 11.2 carries these laws to free probability. The Upsilon transform of Barndorff-Nielsen and Thorbjørnsen turns the centre clock into a GGC with no hypothesis, the Bercovici Pata bijection gives it a free counterpart on the half line, and RH becomes the statement that the clock is itself the Upsilon image of an infinitely divisible law. The same section reads the Jacobi parameters of the Stieltjes moments, Williams' Gamma(2) structure and the heat flow in the language of the semicircle law and of finite free convolution. Section 11.3 repeats the basepoint one calculations for the function $L(s, \chi_{12})$ of Section 8.5, whose completed function has no pole, to see which features of the example of ζ come from its pole. Section 11.4 gives the strongest unconditional statement that the approach delivers: every T_α is a GGC up to a defect carried by the zeros to the right of α, and zero density estimates make the mass of their Cauchy kernels a vanishing fraction of the mass of the others. This falls short of a zero free half plane. Section 11.5 closes the chapter with the work of Anthony Pakes on size biasing, the Lambert law and ladders of self decomposable laws: the Lambert law is a GGC with an explicit Thorin measure, by a proof that transfers to the clock except for one step, which

is RH itself; the size biased clock is the clock plus an independent variable that is a mixture of exponential laws exactly under RH; and the clock lies in Urbanik's class L_2 of multiply self decomposable laws and not in L_3, with no hypothesis. None of the five sections supplies a zero free half plane.

11.1 Infinite divisibility and a map of processes

The general theory of infinitely divisible laws is that of two monographs, the standard references for the classes of laws with which this book works: Sato's book on Lévy processes and infinitely divisible distributions [232] and Bondesson's book on generalised gamma convolutions [35]. From Sato's book we use the representation of a self decomposable law through its background driving Lévy process, Yamazato's theorem that self decomposable laws are unimodal, the moment theorem, the uniqueness of the Lévy Khintchine triplet and, from the later work of Lindner, Pan and Sato [147], the notion of a quasi infinitely divisible law. From Bondesson's book we use the characterisation of the Laplace transforms of GGCs as hyperbolically completely monotone functions, the behaviour of a GGC at the two ends of its Thorin measure and the closure of the class under weak limits, and from the book of Schilling, Song and Vondraček [234] we take the language of Thorin Bernstein functions.

Five consequences come out (Sections 11.1.1 to 11.1.4). The centre clock, which is self decomposable with no hypothesis, is driven by a subordinator whose Lévy density is minus the derivative of the heat trace, and RH is the statement that this density is completely monotone. By Yamazato's theorem the centre clock is unimodal, and so is the shifted clock at every basepoint $\alpha > \frac{1}{2}$. The law at the basepoint one is subexponential, so that its tail is asymptotic to the tail of its Lévy measure, which the Thorin density computes. The tilted Pólya variable V_α is quasi infinitely divisible for every $\alpha > 1$ and ceases to be so at $\alpha = 1$, where the prime number theorem becomes necessary for its Lévy form to converge at all. And the subordination identity of Section 9.2 composes the Laplace exponent of the centre clock, which is known only to be a Bernstein function, with a complete Bernstein function, and the composition is a Thorin Bernstein function for every $\alpha \geq 1$; RH is the statement that no composition is needed. None of these results supplies positivity of the Thorin measure uniformly in the height, and Section 11.1.5 says why. The laws are also the marginals of processes. Self decomposability makes the centre clock the stationary law of an Ornstein Uhlenbeck process driven by the subordinator above, and the Thorin measure at the basepoint one makes T_1 the integral of a gamma process against the inverse of the Poisson smoothed zero count; Section 10.2 quotes the second, and Section 11.1.6 derives both and checks them by simulation. Section 11.1.7 adds Brownian motion run on the clocks and places these processes and those of Chapter 9 on one map, sorted by what each needs: nothing, a zero free half plane, or RH.

We use the notation of Chapter 9. The centre clock T has Laplace transform $\varphi_{1/2}(u) = 1/G(u) = \xi(\frac{1}{2})/\xi(\frac{1}{2} + \sqrt{u})$, and for $\alpha > \frac{1}{2}$ the shifted clock T_α has Laplace transform $\varphi_\alpha(u) = \xi(\alpha)/\xi(\alpha + \sqrt{u})$ and the representation $T_\alpha \stackrel{d}{=} \widetilde{T}_\alpha + H_{\sqrt{2}(\alpha-\frac{1}{2})\widetilde{T}_\alpha}$ of Theorem 9.2.5, where $\widetilde{T}_\alpha$ is the centre clock tilted by $e^{-(\alpha-\frac{1}{2})^2 T}$ and H_a is the first passage time of an independent standard Brownian motion to the level a. For $\alpha \geq 1$, T_α is a GGC whose Thorin density in the chart $\theta = \sqrt{z}$ is v_α/π, with $v_\alpha(\theta) = \mathrm{Re}(\xi'/\xi)(\alpha + i\theta)$ (Theorem 1.2.2 and (1.2.7)); at the basepoint one, where the nonvanishing of ζ on the line $\mathrm{Re}\, s = 1$ leaves no zero on that line to contribute an atom, this is the density $\rho_{1/2} = v_1/\pi$ of (9.1.2). We write $b_\alpha = \frac{\xi'}{\xi}(\alpha)$, so that $b_1 = \sum_\rho \rho^{-1} = 0.0230957089661$, and V_α for the tilted Pólya variable of Section 9.1.1, the Esscher tilt of the Pólya density p of (1.2.4) with $\mathbb{E}e^{sV_\alpha} = \xi(\alpha + s)/\xi(\alpha)$, so that $V_{1/2}$ is the Pólya variable X and V_1 is the variable V of Section 9.1.4. The scripts are in the folders `sato_bondesson` and `basepoint_one/process_map` of the archive, and we name them at each table and figure.

11.1.1 Self decomposability: the driving process and unimodality

The Laplace exponent of the centre clock is $\log G(u) = \int_0^\infty (1 - e^{-ut})\, W(t)\, dt/t$, where $W(t) = \sum_k e^{-a_k t}$, with $a_k = -(\rho_k - \frac{1}{2})^2$, is the heat trace of the zeros, equal to $\sum_{\gamma>0} e^{-\gamma^2 t}$ under RH and to the arithmetic expression (1.2.5) in any case (Section 1.2). Since $W > 0$ and $W' < 0$ on $(0, \infty)$ with no hypothesis (Theorem 5.1.1, whose proof uses the verification of RH below height 1000), T is infinitely divisible with Lévy measure $W(t)\, dt/t$, and it is self decomposable. RH holds exactly when T is a GGC, and then its Thorin measure is $\sum_{\gamma>0} \delta_{\gamma^2}$ (Proposition 1.2.1). Self decomposability has a structural consequence that the book has not used so far. A law on $[0, \infty)$ is self decomposable if and only if it is the law of $\int_0^\infty e^{-t}\, dZ_t$ for a subordinator Z with $\mathbb{E}\log(1+Z_1) < \infty$, its background driving Lévy process [273, 113], [232, Theorem 17.5]. The Lévy measure of a self decomposable law on the half line has the form $k(x)\, dx/x$ with k nonincreasing [232, Corollary 15.11], and the Lévy measure of Z then gives mass $k(x)$ to (x, ∞), so that it has the density $-k'(x)$ when k is differentiable. For the centre clock $k = W$, and self decomposability, which is the unconditional statement that W is nonincreasing, gives the following.

Proposition 11.1.1. *With no hypothesis, $T \stackrel{d}{=} \int_0^\infty e^{-t}\, dZ_t$ for a subordinator Z with Lévy density $h(x) = -W'(x)$, which is $\sum_{\gamma>0} \gamma^2 e^{-\gamma^2 x}$ under RH and is given for every $x > 0$ by the derivative of the explicit formula* (1.2.5) *for W. The condition $\mathbb{E}\log(1 + Z_1) < \infty$ holds because h decays exponentially. Moreover RH holds if and only if h is completely monotone.*

Proof. The representation is the theorem of Wolfe and of Jurek and Vervaat

applied to the self decomposable law of T, and the Lévy density of Z can be checked directly. If $\Psi_Z(u) = -\log \mathbb{E}e^{-uZ_1}$, the Laplace exponent of $\int_0^\infty e^{-t}\,dZ_t$ is $\int_0^\infty \Psi_Z(ue^{-t})\,dt = \int_0^u \Psi_Z(w)\,dw/w$, so that $\Psi_Z(u) = u(\log G)'(u)$. By Section 1.2, $(\log G)'(u) = \int_0^\infty e^{-ut}W(t)\,dt$, and since $W(t) = \int_t^\infty h(x)\,dx$ with $h > 0$, Fubini's theorem gives

$$u(\log G)'(u) = \int_0^\infty h(x) \int_0^x ue^{-ut}\,dt\,dx = \int_0^\infty \bigl(1 - e^{-ux}\bigr)h(x)\,dx.$$

Here $\int_1^\infty h = W(1)$ and $\int_0^1 xh(x)\,dx = \int_0^1 (W(y) - W(1))\,dy$, which is finite because $W(y)\,dy/y$ is a Lévy measure, so h is the Lévy density of a subordinator without drift. It is not integrable at the origin, since $\int_t^1 h = W(t) - W(1)$ and $W(t) \sim \log(1/t)/(8\sqrt{\pi t})$ as $t \downarrow 0$ (Section 5.1), so Z makes infinitely many small jumps. As $x \to \infty$, $h(x) \sim \gamma_1^2 e^{-\gamma_1^2 x}$, which gives $\mathbb{E}\log(1 + Z_1) < \infty$ [232, Theorem 25.3]. For the last statement, h is completely monotone exactly when W is, since $W(x) = \int_x^\infty h$, and complete monotonicity of the heat trace is equivalent to RH (Section 1.2 and Table 1.7). □

So the Thorin question for the clock is a question about a single Lévy process: RH holds exactly when the Lévy density of the driving process of the centre clock is a mixture of exponentials, and the mixing measure is then $\sum_{\gamma>0} \gamma^2 \delta_{\gamma^2}$. Table 11.1 checks the arithmetic computation of h, obtained by differentiating in t the explicit formula (1.2.5), that is Weil's formula with the Gaussian test function e^{-tr^2}, summed over the primes below 2×10^4, against the zero sums over the first 1700 ordinates. The left panel of Figure 11.2 shows h on $[0.002, 0.2]$.

t	$W(t)$, zeros	$W(t)$, explicit formula
0.005	0.537191964091	0.537191964091
0.01	0.149698011254	0.149698011254
0.02	0.0185412917313	0.0185412917313
0.05	$4.587835048574 \times 10^{-5}$	$4.587835048574 \times 10^{-5}$
0.1	$2.104799795822 \times 10^{-9}$	$2.104799798080 \times 10^{-9}$
t	$-W'(t)$, zeros	$-W'(t)$, explicit formula
0.005	165.1022811964	165.1022811964
0.01	33.72924560562	33.72924560562
0.02	3.741068142123	3.741068142123
0.05	$9.166117863113 \times 10^{-3}$	$9.166117863113 \times 10^{-3}$
0.1	$4.205189085540 \times 10^{-7}$	$4.205189085735 \times 10^{-7}$

Table 11.1: The heat trace W and the Lévy density $h = -W'$ of the driving process of the centre clock, from the first 1700 zeros and from the explicit formula (1.2.5) and its derivative with the primes below 2×10^4. The explicit columns take the weights $\log p$ in double precision, which accounts for their departure from the zero columns at $t = 0.1$. Computed by `weil.py` in the folder `sato_bondesson`.

The agreement is to all twelve or thirteen digits shown for $t \le 0.05$, and at $t = 0.1$ to nine digits for W and ten for h. At $t = 0.1$ the terms of the explicit formula cancel to about nine digits (Table 5.7), and `weil.py` takes the weights $\log p$ in double precision; the cancellation magnifies their rounding errors into the differences of 2×10^{-18} in W and 2×10^{-17} in h that the rows at $t = 0.1$ show. With the weights in multiple precision and thirty digit arithmetic (`python3 weil.py mp`) both explicit columns agree with the zero sums at $t = 0.1$ to every digit shown, as (1.2.5) does to fifteen digits in Table 1.5, so the cancellation costs working precision and nothing more. The driving process therefore has the same computational profile as the heat trace: cheap and exact from the primes near the origin, cheap from the zeros away from it.

Yamazato's theorem states that every self decomposable law on the line is unimodal [274], [232, Theorem 53.1]: its distribution function is convex to the left of some point, a mode, and concave to the right of it. Every GGC is self decomposable (Section 1.3.2), and the centre clock and every shifted clock T_α, $\alpha > \frac{1}{2}$, are self decomposable with no hypothesis (Theorem 5.1.1 and Corollary 9.2.6), so we obtain the following.

Corollary 11.1.2. *The centre clock is unimodal with no hypothesis, and so is T_α for every $\alpha > \frac{1}{2}$.*

Unimodality is a new unconditional property of the centre clock. It does not follow from the Thorin picture, which is conditional at the centre, but from self decomposability, which is not. Self decomposability also goes two steps down the chain of driving processes: the driving process Z of Proposition 11.1.1 and the driving process of Z are self decomposable too, so that their marginals are unimodal with no hypothesis, while the next driving process is not, and the clock lies in Urbanik's class L_2 and not in L_3 (Theorem 11.5.8 and Corollary 11.5.10). The computed densities (Figure 11.1) have modes near 0.020 for the centre clock and 0.024 for T_1, and on the computed grids both increase monotonically to the mode and decrease monotonically after it. The grid for the centre clock has spacing 0.0005, while that for T_1 is the geometric grid of the Talbot inversion behind Table 9.11, whose points next to 0.024 are 0.0175 and 0.0328, so the mode of T_1 is located only to that accuracy; the value 54.512 of the density at 0.024 is the largest on that grid. Yamazato's theorem does not reach the reciprocal zeta law of Section 7.1, which is infinitely divisible but not self decomposable (Theorem 7.1.3), and there the unimodality of the density, which the computations following Proposition 7.1.5 support numerically, remains open (Section 7.2.5).

11.1.2 Moments, the tail and the two ends of the Thorin measure

Sato's moment theorem [232, Theorem 25.3] says in particular that an infinitely divisible law has finite mean if and only if its Lévy measure integrates x at infinity. For a GGC with Thorin measure U the Lévy density is $x^{-1} \int e^{-xz}\, U(dz)$

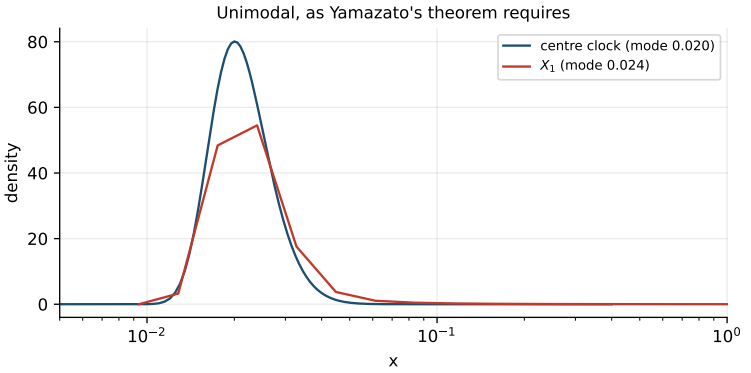


Figure 11.1: The densities of the centre clock T and of the shifted clock T_1, written X_1 in the plot, on a logarithmic axis. The density of T is drawn on a grid of spacing 0.0005 on $[0.004, 0.4]$, from the Talbot inversion and the spectral expansion of Section 8.1; that of T_1 on the 40 point geometric grid of the Talbot inversion behind Table 9.11, which is why its curve is piecewise linear. The legend gives, for each, the grid point of largest density. Assembled by `comp.py` and drawn by `figs.py`, both in the folder `sato_bondesson`.

(Section 1.3.1), so the Lévy measure ν_{T_1} of T_1 and its tail are, in the chart $\theta = \sqrt{z}$,

$$\nu_{T_1}(dx) = \frac{dx}{x}\int_0^\infty e^{-x\theta^2}\rho_{1/2}(\theta)\,d\theta, \qquad \nu_{T_1}((x,\infty)) = \int_0^\infty \rho_{1/2}(\theta)\,E_1(x\theta^2)\,d\theta, \tag{11.1.1}$$

where $E_1(y) = \int_y^\infty e^{-w}\,dw/w$ is the exponential integral. As $x \to \infty$ only θ near zero contributes, and $\int_0^\infty E_1(w^2)\,dw = \sqrt{\pi}$ gives $\nu_{T_1}((x,\infty)) \sim \rho_{1/2}(0)\sqrt{\pi/x} = b_1(\pi x)^{-1/2}$, which is not integrable at infinity, so the Lévy measure does not integrate x at infinity and the mean is infinite, as Section 9.1 found. The tail is regularly varying of index $-\frac{1}{2}$, so the Lévy measure is subexponential. By the theorem of Embrechts, Goldie and Veraverbeke [75], an infinitely divisible law on $[0,\infty)$ is subexponential if and only if its Lévy measure, restricted to $(1,\infty)$ and normalised, is subexponential, and if and only if the tail of the law is asymptotic to the tail of the Lévy measure. Hence

$$P(T_1 > x) \sim \nu_{T_1}((x,\infty)) \sim \frac{b_1}{\sqrt{\pi x}} = 0.01303\,x^{-1/2}, \qquad x \to \infty. \tag{11.1.2}$$

This recovers the tail (9.1.3), which Section 9.1 found by a Tauberian argument, and the same reasoning, with the Thorin density v_α/π and its value b_α/π at the origin, gives $\mathbb{E}T_\alpha = \infty$ and the tail $b_\alpha(\pi x)^{-1/2}$ of T_α for every $\alpha \geq 1$, and for $\frac{1}{2} < \alpha < 1$ whenever no zero lies to the right of α. Theorem 9.2.5 gives the infinite mean and the tail of T_α for every $\alpha > \frac{1}{2}$ by the Tauberian route, and since T_α

is infinitely divisible for every $\alpha > \frac{1}{2}$ (Section 11.1.4), the theorem of Embrechts, Goldie and Veraverbeke, read from the tail of the law to that of the Lévy measure, shows that the Lévy tail of T_α is asymptotic to $b_\alpha(\pi x)^{-1/2}$ as well, wherever the zeros lie.

Table 11.2 and the right panel of Figure 11.2 compare the survival function of T_1, computed in Section 9.2.4 by Talbot inversion near the origin and by the subordination representation beyond, with the Lévy tail computed from (11.1.1) and the Thorin density on $[0, 100]$. In the bulk of the law the two differ widely: at $x = 0.02$, just below the mode, the survival probability is 0.747 while the Lévy tail is 0.101. Further out the survival probability exceeds the Lévy tail by 1.2 per cent at $x = 1$, by 0.15 per cent at $x = 10$ and by 0.05 per cent at $x = 100$, and both approach the asymptote of (11.1.2); the value $\sqrt{100}\,P(T_1 > 100) = 0.013037$ is the one found in Section 9.2.4.

x	$P(T_1 > x)$	$\nu_{T_1}((x,\infty))$	ratio	$b_1(\pi x)^{-1/2}$
0.1	4.7639×10^{-2}	4.1558×10^{-2}	1.1463	4.1206×10^{-2}
0.5	1.8907×10^{-2}	1.8458×10^{-2}	1.0243	1.8428×10^{-2}
1	1.3199×10^{-2}	1.3041×10^{-2}	1.0121	1.3030×10^{-2}
10	4.1272×10^{-3}	4.1209×10^{-3}	1.0015	4.1206×10^{-3}
100	1.3037×10^{-3}	1.3030×10^{-3}	1.0005	1.3030×10^{-3}

Table 11.2: The subexponential principle for T_1: the tail of the law is asymptotic to the tail of its Lévy measure. The survival function is that of Section 9.2.4, the Lévy tail is computed from (11.1.1) with the Thorin density $\rho_{1/2}$ tabulated on $[0, 100]$ by `compute.py` of Section 9.1, the ratio is that of the second column to the third, and the last column is the common asymptote of (11.1.2). Computed by `comp.py` in the folder `sato_bondesson`.

Bondesson's results on the two ends of the Thorin measure explain the shape of T_1: for a GGC the behaviour of the Thorin measure near zero governs the right tail, and its total mass governs the density at the origin [35]. At the basepoint one the Thorin measure U_1 of T_1 (Proposition 9.1.1) has $U_1((0,z]) = \int_0^{\sqrt{z}} \rho_{1/2}(\theta)\,d\theta \sim (b_1/\pi)\sqrt{z}$ as $z \downarrow 0$, since $\rho_{1/2}(0) = b_1/\pi$. The substitution $z = uw$ in $-\log \varphi_1(u) = \int \log(1 + u/z)\, U_1(dz)$, together with $\int_0^\infty \log(1 + 1/w)\, d\sqrt{w} = \pi$, then gives $1 - \varphi_1(u) \sim b_1\sqrt{u}$ as $u \downarrow 0$, the Tauberian form of the tail (11.1.2) and the expansion from which Section 9.1 derived (9.1.3). At the other end the total mass is infinite, since $U_1((0, Y^2]) = M_{1/2}(Y)$ differs from the number $N(Y)$ of zeros with ordinates in $(0, Y]$ by the term $\tau_{1/2}(Y) = O(\log Y)$ of Section 9.1.2, and this is why T_1, like the centre clock (Section 8.2.2.2), is super flat at the origin: its density vanishes faster than any power. Indeed, for every $z_0 > 0$ the Thorin form gives $\varphi_1(u) \le (1 + u/z_0)^{-U_1((0,z_0])}$, and $U_1((0,z_0]) \to \infty$ as $z_0 \to \infty$, so $\varphi_1(u)$ decays faster than any power of u. Markov's inequality $P(T_1 \le x) \le e\,\varphi_1(1/x)$ carries this to the distribution function, and since the density f_{T_1} is nondecreasing up to the mode

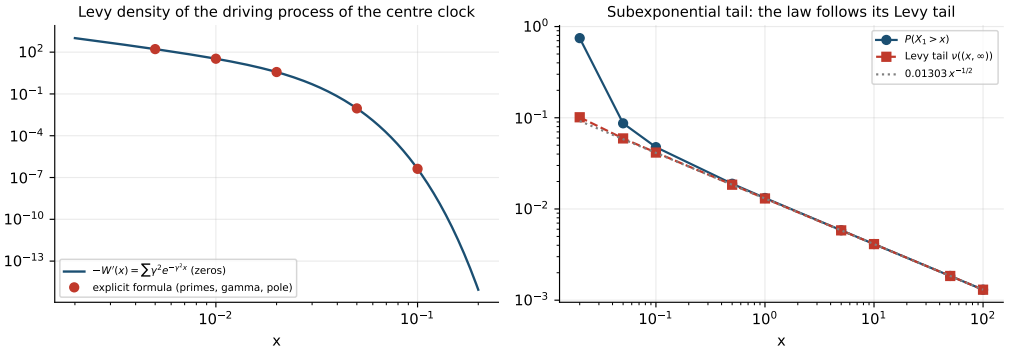


Figure 11.2: Left: the Lévy density $h = -W'$ of the driving process of the centre clock on $[0.002, 0.2]$, with the heat time t written x in the plot, from the first 1700 zeros (curve) and from the explicit formula at the five points of Table 11.1 (dots). Right: the survival function of T_1, written $P(X_1 > x)$ in the plot, its Lévy tail $\nu_{T_1}((x,\infty))$, written $\nu((x,\infty))$, and their common asymptote $b_1(\pi x)^{-1/2} = 0.01303\, x^{-1/2}$, at $x = 0.02,\ 0.05,\ 0.1,\ 0.5,\ 1,\ 5,\ 10,\ 50$ and 100. Computed by `weil.py` and `comp.py` and drawn by `figs.py`, all in the folder `sato_bondesson`.

(Corollary 11.1.2), the bound $x f_{T_1}(x) \le P(T_1 \le 2x)$ for small x carries it to the density.

11.1.3 Uniqueness and quasi infinite divisibility

The Lévy Khintchine triplet of an infinitely divisible law is unique [232, Theorem 8.1], and when the Lévy measure has exponential moments it is also determined by the second derivative of the logarithm of the Laplace transform, through the uniqueness of the inverse Laplace transform. Lindner, Pan and Sato [147] call a law quasi infinitely divisible if its characteristic function is the quotient of two infinitely divisible characteristic functions, equivalently if it has a Lévy Khintchine form with a signed measure $\nu^+ - \nu^-$ whose parts ν^+ and ν^- are both Lévy measures, so in particular of finite mass away from the origin; the Gaussian coefficient is still nonnegative [147, Lemma 2.7], and the triplet is again unique. Their theory places the tilted Pólya variables V_α precisely. The signed measures in question are those of the Lévy forms of Section 9.1.4: for $\alpha \ge 1$,

$$\log \frac{\xi(\alpha+s)}{\xi(\alpha)} = b_\alpha s + \int_0^\infty \left(e^{-sx} - 1 + sx\right) \Pi_\alpha(dx),$$

which at $\alpha = 1$ is (9.1.5), a compensated limit, and which for $\alpha > 1$ converges absolutely when $\operatorname{Re} s > 1 - \alpha$. For $\alpha > 1$, read at $s = it$, it is a Lévy Khintchine form for the characteristic function of V_α, with the measure Π_α carried to the negative half line by the reflection $x \mapsto -x$. We write $x_* = 0.28120$ for the root

of $e^{-3x} = 1 - e^{-2x}$, which is the logarithm of the real root 1.32472 of $y^3 = y + 1$, the plastic number.

PROPOSITION 11.1.3. *1. The Pólya variable $X = V_{1/2}$ is not quasi infinitely divisible, since its characteristic function $\Xi(t)/\Xi(0)$ vanishes at the ordinates.*

2. For every $\alpha > 1$, V_α is quasi infinitely divisible and not infinitely divisible. Its signed Lévy measure is the reflection to the negative half line of

$$\Pi_\alpha(dx) = \Big(\frac{e^{-(\alpha+2)x}}{1-e^{-2x}} - e^{-(\alpha-1)x}\Big)\frac{dx}{x} + \sum_n \frac{\Lambda(n)}{n^\alpha \log n}\,\delta_{\log n}(dx),$$

whose atoms have total mass $\log\zeta(\alpha)$ and whose absolutely continuous part is positive on $(0, x_*)$ and negative beyond. The negative part of Π_α is therefore carried by (x_*, ∞), and its mass on $(1,\infty)$ is $E_1(\alpha-1) - \sum_{k\ge 0} E_1(\alpha+2+2k)$: it is dominated by the pole term $e^{-(\alpha-1)x}dx/x$, of mass $E_1(\alpha-1)$ on $(1,\infty)$, but is not equal to it.

3. At $\alpha = 1$, V_1 is not quasi infinitely divisible. The candidate measure Π_1 of Proposition 9.1.3 has the pole term $-dx/x$ and the prime atoms $\Lambda(n)/(n\log n)$ at $\log n$, each of infinite mass on $(1,\infty)$; since the gamma density $e^{-3x}(1-e^{-2x})^{-1}/x$ is less than $0.06/x$ there, the positive and the negative part of Π_1 both have infinite mass on $(1,\infty)$, and its Lévy form converges only as a compensated limit, which exists by the prime number theorem.

Items (1) and (2) are due to Nakamura [160]: (2), with the same measure, is his Theorem 1.4, and (1) is the remark after it.

Proof. (1) A quasi infinitely divisible characteristic function is the exponential of the finite exponent of its Lévy Khintchine form, so it never vanishes, while $\mathbb{E}e^{itX} = \Xi(t)/\Xi(0)$ by (1.2.4), which vanishes at $t = 14.134725$ and at every other ordinate of a zero on the critical line.

(2) Let $\alpha > 1$. Every piece of the Lévy form then converges absolutely, and each piece is a Lévy measure: the positive density $e^{-(\alpha+2)x}(1-e^{-2x})^{-1}dx/x$, the pole term and the atoms. The factorisation of ξ exhibits the two infinitely divisible laws of the definition. Removing the factor $s(s-1)$ from $\xi(s) = \frac12 s(s-1)\pi^{-s/2}\Gamma(\frac{s}{2})\zeta(s)$ gives, for $\operatorname{Re} s > 1-\alpha$,

$$\frac{\xi(\alpha+s)}{\xi(\alpha)}\cdot\frac{\alpha}{\alpha+s}\cdot\frac{\alpha-1}{\alpha-1+s} = \pi^{-s/2}\,\frac{\Gamma(\frac{\alpha+s}{2})}{\Gamma(\frac{\alpha}{2})}\,\frac{\zeta(\alpha+s)}{\zeta(\alpha)}.$$

Let e_1 and e_2 be standard exponential variables, $\Gamma_{\alpha/2}$ a gamma variable of shape $\frac{\alpha}{2}$ and unit scale, and $\mathcal{Z}_\alpha$ a variable with the zeta law $P(\mathcal{Z}_\alpha = n) = n^{-\alpha}/\zeta(\alpha)$,

all independent of each other and of V_α. Since $\mathbb{E}e^{-se_1/\alpha} = \alpha/(\alpha+s)$, $\mathbb{E}\Gamma_{\alpha/2}^{s/2} = \Gamma(\frac{\alpha+s}{2})/\Gamma(\frac{\alpha}{2})$ and $\mathbb{E}\mathcal{Z}_\alpha^{-s} = \zeta(\alpha+s)/\zeta(\alpha)$, the identity says that

$$V_\alpha - \frac{e_1}{\alpha} - \frac{e_2}{\alpha-1} \stackrel{d}{=} \frac{1}{2}\log\Gamma_{\alpha/2} - \frac{1}{2}\log\pi - \log\mathcal{Z}_\alpha.$$

The right side is infinitely divisible: $\frac{1}{2}\log\Gamma_{\alpha/2}$ has, by Binet's formula [177, §5.9] as in the proof of Proposition 9.1.3, the Lévy measure $e^{-\alpha x}(1-e^{-2x})^{-1}dx/x$ reflected to the negative half line, and by the Euler product $-\log\mathcal{Z}_\alpha$ is compound Poisson with Lévy measure $\sum_n \Lambda(n)(n^\alpha \log n)^{-1}\delta_{-\log n}$, of total mass $\log\zeta(\alpha)$. The variable $-e_1/\alpha - e_2/(\alpha-1)$ is infinitely divisible as well, with Lévy measure $(e^{-\alpha x} + e^{-(\alpha-1)x})\,dx/x$ reflected, so the law of V_α convolved with an infinitely divisible law is infinitely divisible, and V_α is quasi infinitely divisible. Its signed Lévy measure is the difference of the two Lévy measures, which is Π_α reflected, since $e^{-\alpha x}(1-e^{-2x})^{-1} - e^{-\alpha x} = e^{-(\alpha+2)x}(1-e^{-2x})^{-1}$. The absolutely continuous part of Π_α has the density $e^{-(\alpha-1)x}\big(e^{-3x}(1-e^{-2x})^{-1} - 1\big)x^{-1}$, whose sign is that of $e^{-3x} - (1-e^{-2x})$ whatever α is: positive on $(0, x_*)$ and negative beyond, where it is at least $-e^{-(\alpha-1)x}/x$, whose integral over $(1,\infty)$ is $E_1(\alpha-1)$; expanding $(1-e^{-2x})^{-1}$ in powers of e^{-2x} and integrating termwise over $(1,\infty)$ gives the mass of the negative part there. If V_α were infinitely divisible, uniqueness of the triplet [232, Theorem 8.1], applied to the identity in law above, would make its Lévy measure equal to Π_α reflected, which is negative on the reflection of $(x_*, \log 2)$, an interval that carries no atom. So V_α is not infinitely divisible.

(3) Suppose that V_1 is quasi infinitely divisible, so that $\mathbb{E}e^{itV_1} = \exp\Psi_{V_1}(t)$ with

$$\Psi_{V_1}(t) = ict - \frac{A}{2}t^2 + \int_{\mathbb{R}} \big(e^{ity} - 1 - ity\mathbf{1}\{|y|\le 1\}\big)\,\mu(dy),$$

where A and c are real and μ is a signed measure on $\mathbb{R}\setminus\{0\}$ with $\int \min(1,y^2)\,|\mu|(dy) < \infty$. As in (1), the characteristic function never vanishes, so $\xi(1+it) \neq 0$ for real t; and $\xi(1+s) \neq 0$ for $\operatorname{Re} s > 0$ by the Euler product. So $\ell(s) = \log(\xi(1+s)/\xi(1))$, with $\ell(0) = 0$, is a continuous logarithm on the closed half plane $\operatorname{Re} s \ge 0$, holomorphic inside, and $\Psi_{V_1}(t) = \ell(it)$, since both are continuous logarithms of $\xi(1+it)/\xi(1)$ that vanish at $t=0$. For $r>0$ put $\Delta_r F(t) = F(t+r) + F(t-r) - 2F(t)$. Then

$$\Delta_r\Psi_{V_1}(t) = \int_{\mathbb{R}} e^{ity}\,\lambda_r(dy), \qquad \lambda_r(dy) = -Ar^2\delta_0(dy) + (2\cos ry - 2)\,\mu(dy),$$

and λ_r is a finite signed measure because $|2\cos ry - 2| \le \min(4, r^2y^2)$. On the other side, let $\check{\Pi}_1$ be the reflection of Π_1 to the negative half line. For $\sigma > 0$ the variable $V_{1+\sigma}$ has the characteristic function $\exp(\ell(\sigma+it) - \ell(\sigma))$, and by (2) its signed Lévy measure is the reflection of $\Pi_{1+\sigma}(dx) = e^{-\sigma x}\Pi_1(dx)$, that is $e^{\sigma y}\check{\Pi}_1(dy)$. The second difference removes the drift and the compensator, so

$$\Delta_r\ell(\sigma + i\,\cdot\,)(t) = \int_{\mathbb{R}} e^{ity}\,m_\sigma(dy), \qquad m_\sigma(dy) = e^{\sigma y}(2\cos ry - 2)\,\check{\Pi}_1(dy),$$

where m_σ is a finite signed measure, the atoms contributing at most $4\log\zeta(1+\sigma)$ to its total variation. As $\sigma \downarrow 0$, m_σ tends to $m_0 = (2\cos ry - 2)\,\check{\Pi}_1$ in the space of tempered distributions, by dominated convergence, because $|m_0|$ is locally finite and $|m_0|((-R,0)) = O(R)$ as $R \to \infty$: the factor $2\cos ry - 2$ tames $\check{\Pi}_1$ near the origin, and $\Lambda(n) \le \log n$ bounds the atom at $-\log n$ by $1/n$. So the Fourier transforms converge as tempered distributions. On the other hand $\Delta_r \ell(\sigma + it) \to \Delta_r \Psi_{V_1}(t)$ uniformly for t in compact sets, because ℓ is uniformly continuous on compact subsets of the closed half plane. Hence the Fourier transforms of m_0 and λ_r agree on smooth test functions of compact support, which are dense in the Schwartz space, so they agree as tempered distributions, and since the Fourier transform is injective there, $m_0 = \lambda_r$ as measures. The mass at the origin gives $A = 0$. On a compact set $K \subset \mathbb{R} \setminus \{0\}$, $2\cos ry - 2 < 0$ once $r < 2\pi/\max_K |y|$, so $\mu = \check{\Pi}_1$ on K, and hence on $\mathbb{R} \setminus \{0\}$. But the atom of Π_1 at $\log p$ has mass $1/p$, so $|\mu|((-\infty,-1)) \ge \sum_{p \ge 3} 1/p = \infty$, and $|\mu|$ is not a Lévy measure. This contradiction proves (3). □

In the identity in law used for (2) the two infinitely divisible witnesses fail together at $\alpha = 1$: the exponential variable $e_2/(\alpha - 1)$, which carries the pole of ζ, and the zeta law of $\mathcal{Z}_\alpha$, which carries the primes, both cease to exist. So quasi infinite divisibility is lost exactly at the basepoint one, the same place where the prime number theorem becomes necessary and where the Lévy form turns from absolutely to conditionally convergent.

The proof of (3) uses about the primes only the Euler product, and not the prime number theorem: the nonvanishing of ζ on the line $\operatorname{Re} s = 1$, the first step towards that theorem, enters only through the hypothesis, which forces it. The existence of the compensated limit in (3) needs more. Of the two estimates that the proof of Proposition 9.1.3 uses for it, the second is the logarithm of Mertens' product theorem [90, Theorem 429] and gives the cancellation of zeroth order between the prime atoms and dx/x, while the first, which the compensator sx requires, is a form of the prime number theorem, whose analytic proof rests on $\zeta(1+it) \ne 0$ [255]. Below the basepoint one the Lévy form exists only if there is a zero free half plane (Proposition 9.1.6).

Section 9.1.4 reads the conclusion of (3) from the Lévy form itself, and the proof of (3) supplies the uniqueness on which that reading rests, namely that no other signed measure can represent the same characteristic function. Since an infinitely divisible law is quasi infinitely divisible, (3) also shows directly that V_1 is not infinitely divisible, which is how that section concludes. The same argument, with $\xi(1+s,\chi_{12})$ in place of $\xi(1+s)$, shows that the tilted eta variable of Section 11.3.2 is neither quasi infinitely divisible nor infinitely divisible, as that section argues: its Lévy form has the atoms $\chi_{12}(n)\Lambda(n)/(n\log n)$ at $\log n$, of absolute value $1/p$ at $\log p$ for every prime $p \ge 5$, and by Dirichlet's theorem those with $\chi_{12}(n) = 1$ and those with $\chi_{12}(n) = -1$ both have infinite mass.

11.1.4 Bondesson's classes and Thorin Bernstein functions

By Bondesson's characterisation, recalled in Section 1.3.2, a function with limit one at the origin is the Laplace transform of a GGC if and only if it is hyperbolically completely monotone [35, Theorem 6.1.1], and the class of HCM densities is closed under products and quotients of independent variables and under powers X^q with $|q| \geq 1$ [35, Section 5.1]. These are the tools behind the Bosch threshold of Theorem 4.3.6: $\varphi_{1/2}(u^{1/\beta})$ is HCM exactly when $|\beta| \geq 1/(1-\alpha_\xi)$, where $\alpha_\xi = 2\delta_*/\pi$ and δ_* is the largest angle $|\arg\gamma|$ over the zeros γ of Ξ with $\operatorname{Re}\gamma > 0$. RH is the statement $\alpha_\xi = 0$.

The class of GGCs is closed under weak limits (Section 4.1.1), which gives the closure criterion of Corollary 4.1.2 used for the Kent approximants of Section 8.1.3. Bondesson also treats mixtures of exponential laws and their generalised convolutions. This is the natural class for Kent's hitting times started from an interior point: for a diffusion started at x_0 and a level $y_0 > x_0$ the Laplace transform of the first passage time to y_0 is $\psi_u(x_0)/\psi_u(y_0)$, where ψ_u is the increasing positive solution of $\mathcal{A}\psi = u\psi$ for the generator $\mathcal{A}$ of the diffusion, and it is a ratio of two products with interlacing zeros. The law of T_1 is not a mixture of exponentials, since a mixture of exponentials has a completely monotone density while that of T_1 vanishes at the origin and has an interior mode (Figure 11.1). This is consistent with the conclusion of Section 9.2 that T_1 is a subordinated rather than a direct Kent hitting time (Section 9.2.1 and Theorem 9.2.2).

In the language of Schilling, Song and Vondraček [234, Definition 9.10], a law on $[0,\infty)$ is a GGC exactly when its Laplace exponent is a Thorin Bernstein function, equivalently when the derivative of the exponent is a Stieltjes function, the function f_H of Section 1.3.1. The Laplace exponent of the centre clock is $\log G$, a Bernstein function with no hypothesis because T is infinitely divisible; we give it no separate name, since G already denotes the reciprocal of its Laplace transform. The subordination identity of Theorem 9.2.5 reads

$$\begin{aligned} -\log\varphi_\alpha(u) &= \log G\big((\alpha-\tfrac12)^2 + f_\alpha(u)\big) - \log G\big((\alpha-\tfrac12)^2\big), \\ f_\alpha(u) &= u + 2(\alpha-\tfrac12)\sqrt{u}, \end{aligned} \tag{11.1.3}$$

with f_α, the Laplace exponent of the subordinator $\sigma^{(\alpha)}$ of that theorem, a complete Bernstein function. The derivative $\frac{d}{du}[-\log\varphi_\alpha(u)] = \frac{1}{2\sqrt{u}}\frac{\xi'}{\xi}(\alpha+\sqrt{u})$ is a Stieltjes function exactly when no zero lies to the right of α, that is when $\alpha \geq \sup_\rho \operatorname{Re}\rho$, which is the criterion of Theorem 1.2.2: a zero with $\operatorname{Re}\rho > \alpha$ makes $\xi(\alpha+\sqrt{u})$ vanish at the point $u = (\rho-\alpha)^2$ of the cut plane $\mathbb{C}\setminus(-\infty,0]$, where a Stieltjes function is holomorphic. This is a statement about the analytic continuation of the exponent, and not about the sign of the Thorin density v_α/π on the single line $\operatorname{Re} s = \alpha$, which is not known to exclude zeros to the right of that line (Remark 1.2.4).

PROPOSITION 11.1.4. *For every $\alpha \geq 1$ the composition $\log G((\alpha - \frac{1}{2})^2 + f_\alpha) - \log G((\alpha - \frac{1}{2})^2)$ is a Thorin Bernstein function with no hypothesis, while $\log G$ itself is a Thorin Bernstein function if and only if RH holds. For $\frac{1}{2} < \alpha < 1$ the composition is a Thorin Bernstein function if and only if ζ has no zeros with real part greater than α.*

Proof. By (11.1.3) the composition is the Laplace exponent of T_α, so it is a Thorin Bernstein function exactly when T_α is a GGC, which by Theorem 1.2.2 happens exactly when no zero has real part greater than α. For $\alpha \geq 1$ there is none, since ζ has no zeros on or to the right of the line $\operatorname{Re} s = 1$. The exponent $\log G$ is a Thorin Bernstein function exactly when T is a GGC, which is RH by Proposition 1.2.1. □

For every $\alpha > \frac{1}{2}$ the composition is at least a Bernstein function: $u \mapsto \log G((\alpha - \frac{1}{2})^2 + u) - \log G((\alpha - \frac{1}{2})^2)$ is the Laplace exponent of the tilted clock $\widetilde{T}_\alpha$, and a Bernstein function composed with a Bernstein function is again one [234, Corollary 3.8]. So T_α is infinitely divisible for every $\alpha > \frac{1}{2}$ with no hypothesis, and the location of the zeros decides only whether it is a GGC. The proposition says that composing with the complete Bernstein function f_α creates the Thorin property from a function only known to be Bernstein. Analytically this is the Poisson smoothing of the Thorin measure by Cauchy kernels of widths $\alpha - \beta$ (Proposition 1.2.3); probabilistically it is the stable one half subordination of Theorem 9.2.5, by a drift plus a stable subordinator of index one half. RH is the statement that the smoothing was never needed, that is, that $\log G$ is already a Thorin Bernstein function, and the family $\alpha \downarrow \frac{1}{2}$ measures how much smoothing is required at each basepoint.

General composition theorems do not settle this. For a Bernstein function κ and $a \geq 0$ let $\kappa_a(u) = \kappa((\sqrt{u} + a)^2) - \kappa(a^2)$, so that the composition in (11.1.3) is κ_a for $\kappa = \log G$ and $a = \alpha - \frac{1}{2}$. If κ is a Thorin Bernstein function then so is every κ_a, because the shift turns a drift cu into $c(u + 2a\sqrt{u})$ and each factor $(1+u/t)^{-1}$ of the Thorin form into $(a^2+t)/((\sqrt{u}+a)^2+t)$, the Laplace transform of a GGC whose Thorin density in θ is the pair of Cauchy kernels of width a at $\pm\sqrt{t}$, as in Section 9.2.2. Applied to κ_a, with $\kappa_{a+a'} = (\kappa_a)_{a'}$, this shows that the shifts a for which κ_a is a Thorin Bernstein function form a half line, and for $\kappa = \log G$ Proposition 11.1.4 identifies it as $[a_*, \infty)$ with $a_* = \sup_\rho \operatorname{Re}\rho - \frac{1}{2}$; RH is the statement $a_* = 0$. For a fixed shift the implication cannot be reversed. The function $\kappa(u) = 2\log(1 + u) + \log(1 + (120u + 16u^2)/289)$ is a Bernstein function, since κ' is the Laplace transform of $2e^{-x} + 2e^{-15x/4}\cos 2x \geq 0$, and it is not a Thorin Bernstein function, since κ' has poles at $-\frac{15}{4} \pm 2i$, where a Stieltjes function is holomorphic; yet

$$\kappa_{1/2}(u) = \log\left(1 + \frac{u}{4}\right) + 2\log\frac{4((\sqrt{u} + \frac{1}{2})^2 + 1)}{5} + \log\frac{(\sqrt{u} + 1)^2 + 4}{5}$$

is a Thorin Bernstein function, the exponent of an exponential law plus the exponents of GGCs whose Thorin densities are pairs of Cauchy kernels, of width $\frac{1}{2}$ at ± 1 and of width 1 at ± 2; the function $u \mapsto \kappa(u/4a^2)$ gives the same example at any shift $a > 0$.

The Dirichlet mean representation of James, Roynette and Yor [109] writes a GGC whose Thorin measure has finite total mass c as the product $\Gamma_c \mathcal{M}$ of a gamma variable Γ_c of shape c and an independent Dirichlet mean $\mathcal{M}$. It applies to the truncated lattice variables $\mathcal{S}_n$ of Section 7.2.1, whose Thorin measures have total mass $2n$, and to the finite sums of exponential variables of Proposition 7.1.22, but not to T_α or, under RH, to T, whose Thorin measures have infinite mass.

11.1.5 What these results cannot give

Every GGC criterion in these books is, for the reciprocal Xi function at the centre, equivalent to RH: the HCM property of the Laplace transform and the Stieltjes property of the derivative of the exponent (Theorem 4.2.4), the Pick function property (Corollary 4.3.7), and the complete monotonicity of the Lévy density of the driving process (Proposition 11.1.1). The books classify laws and derive consequences once positivity of the Thorin measure is known, and the consequences obtained in Sections 11.1.1 to 11.1.4, the driving process, unimodality, the subexponential tail, quasi infinite divisibility and the Thorin Bernstein compositions, are mostly unconditional. Table 11.3 collects them with their sources.

They are unconditional for one of two reasons. Some use only the infinite divisibility and self decomposability of the centre clock, that is the inequalities $W > 0$ and $W' < 0$, which hold wherever the zeros above height 1000 lie, their proof using only the verification of RH below that height (Theorem 5.1.1). The others use the region $\operatorname{Re} s \geq 1$, through positivity of the Thorin measure at a basepoint $\alpha \geq 1$, which rests on the nonvanishing of ζ on the line $\operatorname{Re} s = 1$ (Theorem 1.2.2), or through the Euler product, as in Proposition 11.1.3. Neither source reaches below the basepoint one: the first cannot detect the higher zeros at all, and the second stops at the line $\operatorname{Re} s = 1$, where the Euler product ceases to converge. So none of the results supplies positivity of the Thorin measure uniformly in the height at a basepoint $\alpha < 1$, in the sense of Theorem 1.2.2 that T_α is a GGC, which is the zero free half plane $\operatorname{Re} s > \alpha$ of step 7 of the path in Table 12.1, and still less at the centre, where it is RH itself. That positivity, uniform in the height, is the one ingredient still missing on the path to RH; the half plane $\operatorname{Re} s > \frac{7}{8}$ announced in October 2026 would supply it for every $\alpha > \frac{7}{8}$, by arithmetic from outside the theory (Section 12.1.5).

What the two books do provide is a vocabulary in which RH can be stated in several equivalent forms, each with its own tools: a subordinator whose Lévy density must be shown to be a mixture of exponentials, a Bernstein function whose derivative must be shown to be a Stieltjes function, and a smoothing by f_α that must be shown to be unnecessary. Section 12.1 collects the steps of the

result	source	consequence here	status
background driving Lévy process	Wolfe; Jurek and Vervaat; Sato, Theorem 17.5	centre clock driven by a subordinator with Lévy density $-W'$; RH if and only if it is completely monotone	proved
Yamazato's theorem	Yamazato; Sato, Theorem 53.1	centre clock and T_α, $\alpha > \frac{1}{2}$, unimodal	proved
Urbanik's classes L_n	Urbanik; Sato 1980; Berg and Forst	centre clock in L_2, not in L_3; its driving process and the driving process of that self decomposable, with unimodal marginals	proved
moment theorem	Sato, Theorem 25.3	$\mathbb{E}T_\alpha = \infty$ for $\alpha \geq 1$	proved
subexponential tails	Embrechts, Goldie, Veraverbeke	$P(T_\alpha > x) \sim b_\alpha(\pi x)^{-1/2}$; ratio 1.0005 at $x = 100$	proved, checked
uniqueness, quasi infinite divisibility	Sato, Theorem 8.1; Lindner, Pan, Sato	V_α quasi infinitely divisible for $\alpha > 1$, not at $\alpha = 1$; Pólya variable not quasi infinitely divisible	proved
HCM characterisation and closure	Bondesson	Bosch threshold for powers	proved
Thorin measure at the two ends	Bondesson	$x^{-1/2}$ tail and super flat origin	proved
Thorin Bernstein composition	Schilling, Song, Vondraček	smoothing by f_α creates the Thorin property for $\alpha \geq 1$	proved
Dirichlet means	James, Roynette, Yor	applies to truncations only	remark

Table 11.3: What the books of Sato and Bondesson, and the papers around them, give the reciprocal Xi function: each result used in this section, its source, its consequence for the centre clock T, the shifted clocks T_α and the tilted Pólya variables V_α, and its status. Theorem numbers, here and in the text, refer to the first edition of Sato's book [232].

Thorin approach, marks each as proved, computed, announced or open, and sets out the path along which positivity would have to be carried from the basepoint one to the centre. The same laws are the marginals of stochastic processes, and Sections 11.1.6 and 11.1.7 collect these processes, with those of Sections 9.2 and 10.1, on one map.

11.1.6 Wiener gamma integrals and the Riemann Ornstein Uhlenbeck process

A GGC is a superposition of gamma laws, and a gamma process realises the superposition pathwise. Let $(\Gamma_t)_{t\geq 0}$ be a standard gamma process, the subordinator with $\mathbb{E}e^{-u\Gamma_t} = (1+u)^{-t}$, so that Γ_t has the gamma law of shape t and unit scale, like the variables $\Gamma_{\alpha/2}$ and Γ_c of Sections 11.1.3 and 11.1.4. For a measurable $f \geq 0$ the Wiener integral $\int_0^\infty f(t)\, d\Gamma_t$ has Laplace exponent $\int_0^\infty \log(1+uf(t))\, dt$. Let U be the Thorin measure of a GGC without drift, $M(\theta) = U((0,\theta^2])$ its cumulative mass in the chart $\theta = \sqrt{z}$, and $M^{-1}(t) = \inf\{\theta : M(\theta) > t\}$ the right continuous inverse. The image of Lebesgue measure on $(0, M(\infty))$ under M^{-1} is dM, so the substitution $t = M(\theta)$ turns $\int_0^\infty \log(1+u/M^{-1}(t)^2)\, dt$ into $\int \log(1+u/z)\, U(dz)$, and the GGC is the Wiener gamma integral

$$\int_0^\infty \frac{d\Gamma_t}{M^{-1}(t)^2}, \tag{11.1.4}$$

which is the Wiener gamma representation of the class [109]. At the centre, under RH, M is the zero counting function N, $M^{-1}(t) = \gamma_k$ for $k-1 \leq t < k$, the ordinates being counted in increasing order and with multiplicity, and (11.1.4) is $\sum_k (\Gamma_k - \Gamma_{k-1})/\gamma_k^2$, the series $\sum_{\gamma>0} e_\gamma/\gamma^2$ of Proposition 1.2.1. At a basepoint α the count is replaced by the Poisson smoothed count $M_{\alpha-1/2}(\theta) = \pi^{-1}\int_0^\theta v_\alpha$ of Section 3.4, which is π^{-1} times the argument of ξ along the line $\operatorname{Re} s = \alpha$. For $\alpha \geq 1$, $v_\alpha > 0$ and T_α is a GGC, so with no hypothesis T_α is (11.1.4) with $M = M_{\alpha-1/2}$; in particular T_1 is a gamma process integrated against the inverse of the Poisson smoothed zero count. For $\frac{1}{2} < \alpha < 1$ the representation holds exactly when T_α is a GGC, that is when no zero has real part greater than α (Theorem 9.2.5), and then M is $M_{\alpha-1/2}$ plus a unit jump at the ordinate of each zero on the line $\operatorname{Re} s = \alpha$ (Theorem 7.1.14). It needs $M_{\alpha-1/2}$ to be nondecreasing, that is $v_\alpha \geq 0$ on the line $\operatorname{Re} s = \alpha$, but for ξ it is not known whether positivity on one line excludes zeros to its right (Remark 1.2.4).

Table 11.4, computed by `comp2.py`, checks the representation of T_1 with 2×10^5 draws. The Thorin mass up to height 100, where the tabulated density $\rho_{1/2}$ of Section 9.1 ends, is cut into cells: 299 cells of a geometric grid on $[10^{-12}, 0.005]$, where $M_{1/2}^{-1}$ is small and the stable one half part of T_1 lives, the cell $[0.005, 0.01]$, and cells of length 0.005 beyond. Each cell contributes an independent gamma variable with shape equal to its length, multiplied by the average of $M_{1/2}^{-1}(t)^{-2}$

over the cell, computed by quadrature, and the mass above height 100 contributes its mean, computed with the Weyl density $\frac{1}{2\pi}\log\frac{\theta}{2\pi}$. At each of the four points the empirical Laplace transform is within one standard error of φ_1.

u	$\mathbb{E}\exp\big(-u\int_0^\infty M_{1/2}^{-1}(t)^{-2}\,d\Gamma_t\big)$, Monte Carlo	standard error	$\varphi_1(u)=\xi(1)/\xi(1+\sqrt{u})$
0.5	0.97240	0.00021	0.97253
1	0.95469	0.00025	0.95493
10	0.73988	0.00036	0.74016
100	0.09410	0.00014	0.09416

Table 11.4: The Wiener gamma representation (11.1.4) of T_1 against the inverse of the smoothed zero count $M_{1/2}$: the empirical Laplace transform of 2×10^5 draws, discretised as described in the text, against φ_1. Computed by `comp2.py` in the folder `basepoint_one/process_map`.

Self decomposability gives a second process. If Z is a subordinator with $\mathbb{E}\log(1+Z_1)<\infty$, the Ornstein Uhlenbeck type process $dF_t=-F_t\,dt+dZ_t$, that is $F_t=e^{-t}F_0+\int_0^t e^{-(t-r)}\,dZ_r$, has a unique stationary law, the law of $\int_0^\infty e^{-t}\,dZ_t$, and every self decomposable law arises in this way from its background driving Lévy process [232, Theorem 17.5 and Corollary 17.9]. By Proposition 11.1.1 the driving process of the centre clock is the subordinator with Lévy density $h=-W'$, which the explicit formula computes from the primes. So there is, with no hypothesis, an Ornstein Uhlenbeck process F whose stationary law is the centre clock, which we call the Riemann Ornstein Uhlenbeck process, and RH is the statement that the jump density of its driver is completely monotone. Under RH the Laplace exponent of Z is $\Psi_Z(u)=u(\log G)'(u)=\sum_{\gamma>0}u/(\gamma^2+u)$, so the driver is a superposition of independent compound Poisson processes, one for each zero counted with multiplicity, each with rate one and exponential jumps of mean γ^{-2}. By the computation in the proof of Proposition 11.1.1, the process driven by the component of γ has the stationary Laplace exponent $\int_0^u dw/(\gamma^2+w)=\log(1+u/\gamma^2)$, that of the exponential law of rate γ^2, and the stationary law of the sum is $\sum_{\gamma>0}e_\gamma/\gamma^2$.

The simulation by `ou.py` is exact for the 1700 computed zeros. Jumps arrive at the times of a Poisson process of rate 1700 on $[0,4000]$, 6.8×10^6 of them, each attached to a zero chosen uniformly and exponential with mean γ^{-2}; the zeros above the 1700th enter as a drift of the driver, which adds to the stationary law the mean $\mathbb{E}T-\sum_{k\le1700}\gamma_k^{-2}$ of their components; between jumps the process decays exponentially, and it is sampled at spacing 0.02 after a burn in of 20. The time average is 0.023094 against $\mathbb{E}T=\sum_k a_k^{-1}=0.023105$ (Section 9.1.2). The empirical Laplace transform of the stationary law is 0.79520 ± 0.00096, 0.32828 ± 0.00180 and 0.016404 ± 0.00030 at $u=10$, 50 and 200, with standard errors from 100 batch means, against $\varphi_{1/2}(u)=\xi(\frac12)/\xi(\frac12+\sqrt{u})=0.79514$, 0.32827 and 0.016399. The stationary process at time $\tau\ge0$ is $e^{-\tau}$ times its value at time 0

plus a variable independent of that value, and the centre clock has finite variance, so the autocorrelation at lag τ is $e^{-\tau}$ with no hypothesis. The estimates at lags 0.5, 1 and 2 are 0.602, 0.352 and 0.119 against 0.607, 0.368 and 0.135; the deficits are within 1.6 standard errors of the estimates, which are about 0.008, 0.010 and 0.013 by batch estimates over 20 blocks of the same run. The top right panel of Figure 11.3 shows ten units of time of the path.

For $\alpha \geq 1$ the shifted clock T_α is a GGC, hence self decomposable, and has its own driver: by (11.1.1), with v_α/π in place of $\rho_{1/2}$, its Lévy measure is $k_\alpha(x)\,dx/x$ with $k_\alpha(x) = \pi^{-1}\int_0^\infty e^{-x\theta^2} v_\alpha(\theta)\,d\theta$, and its driver has the completely monotone Lévy density $-k'_\alpha(x) = \pi^{-1}\int_0^\infty \theta^2 e^{-x\theta^2} v_\alpha(\theta)\,d\theta$. For $\frac{1}{2} < \alpha < 1$ the process exists as well, because T_α is self decomposable with no hypothesis (Corollary 9.2.6); its driver has the Lévy density $-W'_\alpha$, which is completely monotone exactly when T_α is a GGC.

11.1.7 Subordinated Brownian motion and the map of processes

Running an independent standard Brownian motion B on a clock gives a symmetric law with

$$\mathbb{E}e^{iuB_{T_\alpha}} = \mathbb{E}e^{-u^2 T_\alpha/2} = \frac{\xi(\alpha)}{\xi(\alpha + |u|/\sqrt{2})}, \qquad u \in \mathbb{R},$$

so B_{T_α} has the law of $Y_\alpha/\sqrt{2}$, the normal scale mixture of Proposition 10.1.1, and what Section 10.1 proves for Y_α carries over with a factor $\sqrt{2}$. The characteristic function exists at every basepoint with no hypothesis. At the centre B_T has the finite variance $\mathbb{E}T$; above it the characteristic function has a corner at the origin, B_{T_α} lies in the domain of attraction of the Cauchy law, and $P(|B_{T_\alpha}| > y) \sim \sqrt{2}\,b_\alpha/(\pi y)$ as $y \to \infty$. The law is a symmetric EGGC exactly when T_α is a GGC (Theorems 10.1.2 and 10.1.3), and under RH at the centre it is the sum $\sum_k L_k/(\sqrt{2}\gamma_k)$ of scaled Laplace variables. Its Cauchy factor, of scale $b_\alpha/\sqrt{2}$, comes from the constant $b_\alpha/\pi = v_\alpha(0)/\pi$ of the Thorin density wherever the Thorin form (10.1.2) holds, and the planar Brownian reading of Section 9.2.3 holds at every basepoint α to the right of every zero: $v_\alpha(\theta)\,d\theta/\pi = \sum_\rho P(B_H^2 \in d\theta \mid B_0 = \rho)$, where $B = (B^1, B^2)$ is a planar Brownian motion and H is now its first time to reach the line $\operatorname{Re} s = \alpha$. So the constant b_α/π is the density at $\theta = 0$ of this sum of harmonic measures. Table 11.5, computed by `comp2.py`, checks the characteristic function and the tail at three basepoints with 3×10^5 draws of T_α each, from the subordination identity, with the tilted clock drawn as for Table 10.1.

Figure 11.4 and Table 11.6 arrange the processes of Chapter 9 and of this section by basepoint; we write $a = \alpha - \frac{1}{2}$. The Mellin side is the Bessel picture of Williams [271] and of Biane, Pitman and Yor [28]: S_2, the sum of two independent hitting times of the level one by a three dimensional Bessel process, satisfies

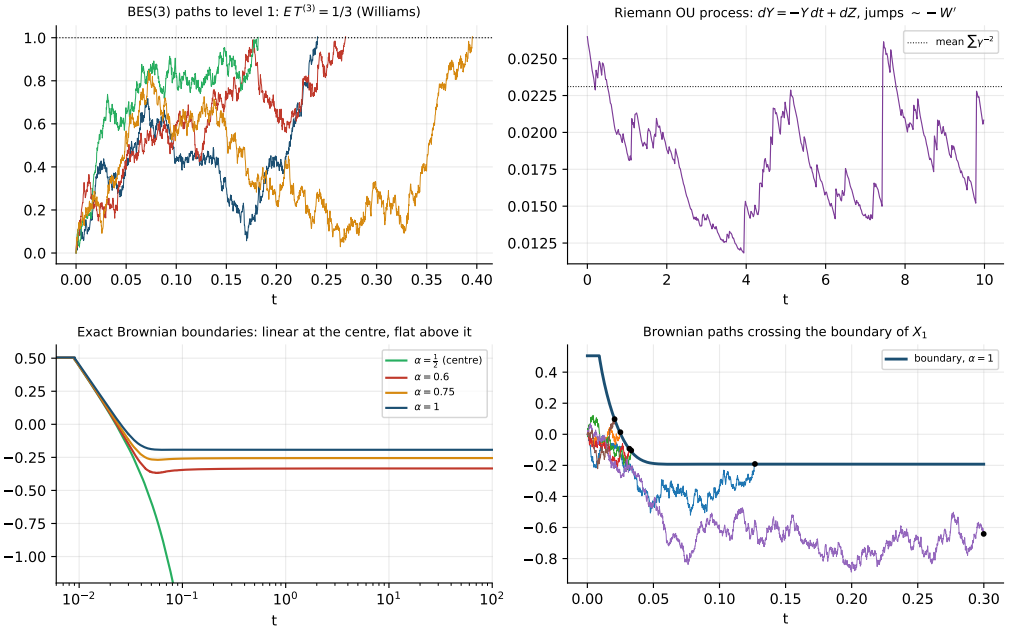


Figure 11.3: Four processes of the map. Top left: paths of a three dimensional Bessel process run until they reach the level one; the hitting time has mean $\frac{1}{3}$, and two independent copies make up the variable S_2 of Section 9.1.1. Top right: ten units of time of the Riemann Ornstein Uhlenbeck process F, written Y in the plot, whose stationary law is the centre clock, with the mean $\mathbb{E}T$, written $\sum \gamma^{-2}$ in the plot. Bottom left: the exact Brownian boundaries of T_α at $\alpha = \frac{1}{2}$, 0.6, 0.75 and 1, those at 0.6 and 0.75 computed under RH beyond $t = 0.03$ (Section 9.2.6), asymptotically linear at the centre and flat above it. Bottom right: Brownian paths crossing the boundary of T_1, written X_1 in the plot (Section 9.2.4). Drawn by `figs.py` in the folder `basepoint_one/process_map` from the output of `ou.py`, the boundaries of `bnd_alpha.py` at $\alpha = 0.6$ and 0.75, and those of Sections 8.2 and 9.2 at the centre and at the basepoint one.

$\mathbb{E}(\pi S_2/2)^{s/2} = 2\xi(s)$ (Section 1.2), and V_α is $\frac{1}{2}\log(\pi S_2/2)$ under the law of S_2 size biased by $S_2^{\alpha/2}$ (Section 9.1.1). The clock side starts from the centre clock, which under RH with simple zeros is Kent's hitting time for a string with spectrum $\{\gamma^2\}$, and passes through the tilt e^{-a^2T} and the sticky Brownian subordination of Theorem 9.2.5 to T_α. From T_α it continues to the Wiener gamma integral of Section 11.1.6, to Brownian motion run on the clock, and to the Brownian boundaries of Section 9.2.6, which are asymptotically linear at the centre and, at the basepoints computed, flat above it, with the Wald defect $\mathbb{E}\,b(T_\alpha) = b_\alpha/\sqrt{2}$ in place of Wald's identity when the boundary is bounded at infinity.

α	$u=1$, Monte Carlo	exact	$u=3$, Monte Carlo	exact	$y\,P(\lvert B_{T_\alpha}\rvert > y)$, $y=10$	$\sqrt{2}\,b_\alpha/\pi$
0.6	0.98546	0.98530	0.89322	0.89286	0.0024	0.0021
0.75	0.98050	0.98049	0.87989	0.87994	0.0050	0.0052
1	0.97273	0.97253	0.85933	0.85888	0.0095	0.0104

Table 11.5: The characteristic function $\xi(\alpha)/\xi(\alpha+|u|/\sqrt{2})$ of subordinated Brownian motion, estimated by the mean of $\cos(uB_{T_\alpha})$ over 3×10^5 draws at each basepoint, and its Cauchy type tail; the tail estimates rest on 72, 149 and 284 exceedances of $y=10$. Computed by `comp2.py` in the folder `basepoint_one/process_map`.

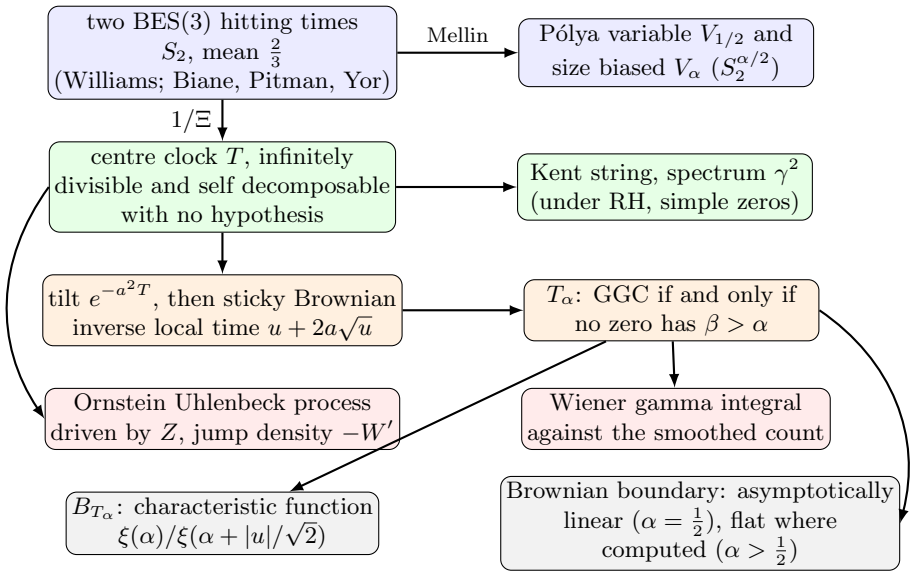


Figure 11.4: Processes attached to $\xi(\alpha+\cdot)$, with $a=\alpha-\frac{1}{2}$. By rows from the top: the Mellin side, the centre clock, the basepoint clocks, and two rows of further representations. Drawn in TikZ from Table 11.6.

Three conclusions come out of the map. First, the Mellin side and the clock side differ in kind. The Bessel picture and the size biased variables V_α built from it are unconditional at every basepoint, and on that side the zeros enter only as zeros, of the characteristic function $\Xi(t)/\Xi(0)$ of the Pólya variable and of $\xi(\alpha+s)$ in the Lévy forms of Section 11.1.3; the positivity statements equivalent to RH, the GGC property of the centre clock and the EGGC property of $Y_{1/2}$, live on the clock side, in the reciprocal, and so does the Kent string, which needs simple zeros as well.

Second, on the clock side the Ornstein Uhlenbeck process at the centre and,

	$\alpha = \frac{1}{2}$	$\frac{1}{2} < \alpha < 1$	$\alpha = 1$	$\alpha > 1$
Mellin side	Pólya variable, $S_2^{1/4}$ bias	$S_2^{\alpha/2}$ bias	$S_2^{1/2}$ bias	$S_2^{\alpha/2}$ bias
Lévy form of $\log \xi(\alpha + s)$	none: the characteristic function vanishes at the ordinates	exists if $\sup \operatorname{Re} \rho < \alpha$, only if $\sup \operatorname{Re} \rho \le \alpha$	compensated limit, by the prime number theorem	absolutely convergent; quasi infinitely divisible
clock law	T, infinitely divisible and self decomposable	$\widetilde{T}_\alpha + H_{\sqrt{2}a\widetilde{T}_\alpha}$	$\widetilde{T} + H_{\widetilde{T}/\sqrt{2}}$	$\widetilde{T}_\alpha + H_{\sqrt{2}a\widetilde{T}_\alpha}$
GGC	if and only if RH	if and only if no zero has $\beta > \alpha$	yes	yes
Thorin measure	$\sum_{\gamma>0} \delta_{\gamma^2}$ (RH)	Cauchy widths $\alpha - \beta$	Cauchy widths $1 - \beta$	Cauchy widths $\alpha - \beta$
Kent string	spectrum γ^2 (RH)	spectrum $\gamma^2 + a^2$ on the sticky clock (RH)	spectrum $\lvert\rho\rvert^2$ on the sticky clock (RH)	spectrum $\gamma^2 + a^2$ on the sticky clock (RH)
tail of the clock	exponential, rate γ_1^2	$b_\alpha(\pi x)^{-1/2}$	$0.01303\, x^{-1/2}$	$b_\alpha(\pi x)^{-1/2}$
Brownian boundary	asymptotically linear, slope $-\sqrt{2}\gamma_1$	flat: -0.3338 ($\alpha = 0.6$), -0.2554 ($\alpha = 0.75$)	flat at -0.19277	flat
Wald defect $\mathbb{E}\, b(T_\alpha)$	0	$b_\alpha/\sqrt{2}$	$\sum_\rho \rho^{-1}/\sqrt{2}$	$b_\alpha/\sqrt{2}$
B_{T_α}	finite variance	Cauchy domain	Cauchy domain, tail $0.0104/y$	Cauchy domain
Ornstein Uhlenbeck driver	jump density $-W'$	jump density $-W'_\alpha$	jump density $-k'_1$	jump density $-k'_\alpha$
symmetric EGGC Y_α	if and only if RH	if and only if no zero has $\beta > \alpha$	yes	yes

Table 11.6: The processes by basepoint, with $a = \alpha - \frac{1}{2}$. (RH) marks statements that hold under RH, and the Kent strings need the zeros to be simple as well. The Mellin row gives the size bias of S_2 that produces V_α (Section 9.1.1); $\widetilde{T}_\alpha$, H and the sticky clock are those of Theorem 9.2.5 and Corollary 9.2.3, with $\widetilde{T} = \widetilde{T}_1$; the boundaries b and their Wald defects are those of Sections 9.2.4 and 9.2.6; $Y_\alpha = \sqrt{2}B_{T_\alpha}$ in law (Proposition 10.1.1); and $k_\alpha(x)\, dx/x$ is the Lévy measure of T_α. The boundary row reports computations at $\alpha = \frac{1}{2}$, 0.6, 0.75 and 1, those at 0.6 and 0.75 made under RH beyond $t = 0.03$, and its entry for $\alpha > 1$ is extrapolated, not computed; the flatness is observed, not proved, and above the centre the Wald defect $b_\alpha/\sqrt{2}$ requires the boundary to be bounded at infinity (Lemma 9.2.4). All other entries hold as stated.

at every basepoint above it, the tilted and subordinated clock, Brownian motion run on the clock and the Brownian boundary exist with no hypothesis. What changes with α is whether T_α is a GGC, that is, whether it has a positive Thorin measure, which decides whether the Wiener gamma representation and the symmetric EGGC property are available: always for $\alpha \geq 1$, for $\frac{1}{2} < \alpha < 1$ exactly when no zero has real part greater than α, whatever the sign of v_α on the line (Remark 1.2.4), and under RH at the centre. The Kent strings need RH at every basepoint, and simple zeros as well, because a Kent hitting time is a GGC whose Thorin measure consists of unit atoms at distinct points (Section 9.2.1). Under RH with simple zeros the tilted clock $\widetilde{T}_\alpha$ is Kent's hitting time for a string with spectrum $\{\gamma^2 + a^2\}$, and T_α is the hitting time of the string diffusion run on the local time of a sticky Brownian motion (Corollary 9.2.3, at $\alpha = 1$). The shifted clock T_α itself is never a Kent hitting time for $\alpha > \frac{1}{2}$: when it is a GGC, each pair of zeros $\rho, \bar{\rho}$ with $\beta < \alpha$, such as those on the critical line, contributes to its Thorin measure the pair of Cauchy kernels of width $\alpha - \beta$ at $\pm\gamma$ (Section 11.1.4), so that measure is not a sum of atoms.

Third, the Bessel process is the template. Its hitting time of the level one is Kent's hitting time for a string with spectrum $\{\pi^2 n^2/2\}$, and under RH with simple zeros the centre clock is Kent's hitting time for a string with spectrum $\{\gamma^2\}$. The processes computed here show what such a string would do, and the basepoints inside the strip show how the picture degrades, continuously and quantitatively, as positivity is pushed towards the centre.

11.2 Free probability and the Upsilon transform

Wigner's semicircle law [270] plays in free probability the part that the Gaussian law plays in classical probability: it is the limit law of the free central limit theorem, for sums of free noncommutative variables [265, 166]. Bercovici and Pata [20] matched the two theories through infinite divisibility. Their bijection sends a classical infinitely divisible law to the freely infinitely divisible law with the same Lévy Khintchine triplet, the Gaussian law to the semicircle law and the Poisson law to the Marchenko Pastur law [152]. Barndorff-Nielsen and Thorbjørnsen [17] showed that the bijection is read off from a classical map, their Upsilon transform, and Barndorff-Nielsen, Maejima and Sato [15] showed that this map sends the self decomposable laws onto the Thorin class. This section, which follows the first author's note [208], applies the two maps to the centre clock T of Section 11.1. With no hypothesis, T has an Upsilon image $\Upsilon(T)$, a GGC whose Thorin measure has the distribution function $u \mapsto W(1/u)$, and a free counterpart $\Lambda_{\mathrm{BP}}(T)$, a freely self decomposable law on $[0, \infty)$; and RH holds exactly when T is itself the Upsilon image of an infinitely divisible law (Theorem 11.2.3). The last three subsections read three computations of the book in the same language: the Jacobi parameters of the Stieltjes moments of Section 5.1 against a local semicircle

model, Williams' Gamma(2) structure against Laguerre polynomials and finite free convolution, and the heat flow of Section 8.4 against finite free convolution with Hermite polynomials. Statements that are heuristic are labelled as such; the others are proved, and every number is computed by a script named in the text.

11.2.1 The Bercovici Pata bijection and the Upsilon transform

As in Section 8.4.4, $G_\mu(z) = \int \mu(dx)/(z-x)$ is the Cauchy transform of a probability measure μ on $\mathbb{R}$ and σ_t the semicircle law of variance t; the subscript distinguishes G_μ from the function G of the clock. The Cauchy transform maps the upper half plane $\mathbb{C}^+$ into the lower half plane $\mathbb{C}^-$ and determines μ, and where μ has a continuous density the density is $-\frac{1}{\pi}\operatorname{Im} G_\mu(x+i0)$. The function $1/G_\mu$ has a right inverse $(1/G_\mu)^{-1}$ on a region $\Gamma_{\alpha,M} = \{z : |\operatorname{Re} z| < \alpha \operatorname{Im} z,\ \operatorname{Im} z > M\}$, and the Voiculescu transform $\varphi_\mu(z) = (1/G_\mu)^{-1}(z) - z$ linearises free additive convolution: $\varphi_{\mu \boxplus \nu} = \varphi_\mu + \varphi_\nu$ on a common region [265, 21]. Here $\mu \boxplus \nu$ is the law of $X+Y$ for free self adjoint variables X and Y with laws μ and ν, and the large size limit of the eigenvalue distribution of the sum of two independent unitarily invariant random matrices whose eigenvalue distributions tend to μ and ν [166]; Section 8.4.4 uses the case $\nu = \sigma_t$. The free cumulant transform $\mathcal{C}_\mu(z) = z\varphi_\mu(1/z)$, defined for $z \in \mathbb{C}^-$ with $1/z \in \Gamma_{\alpha,M}$, is additive under $\boxplus$, and μ is freely infinitely divisible if, for every n, it is the n fold free convolution of a law with itself. A freely infinitely divisible law has a Voiculescu transform that extends analytically to all of $\mathbb{C}^+$ [21], so its free cumulant transform is analytic on all of $\mathbb{C}^-$.

A law μ on $\mathbb{R}$ is infinitely divisible exactly when its cumulant function $C_\mu(u) = \log \int e^{iux} \mu(dx)$ has the Lévy Khintchine form

$$C_\mu(u) = i\eta u - \tfrac{1}{2} A u^2 + \int_{\mathbb{R}} \left(e^{iut} - 1 - iut\, \mathbf{1}_{[-1,1]}(t) \right) \Pi(dt), \qquad u \in \mathbb{R}, \quad (11.2.1)$$

with $\eta \in \mathbb{R}$, $A \geq 0$ and a Lévy measure Π, that is $\Pi(\{0\}) = 0$ and $\int \min(1, t^2)\, \Pi(dt) < \infty$; the triplet (η, A, Π) is unique [232, Theorem 8.1]. A law is freely infinitely divisible exactly when

$$\mathcal{C}_\mu(z) = \eta z + A z^2 + \int_{\mathbb{R}} \Big(\frac{1}{1-tz} - 1 - tz\, \mathbf{1}_{[-1,1]}(t) \Big) \Pi(dt), \qquad z \in \mathbb{C}^-, \quad (11.2.2)$$

with a triplet of the same kind, again unique [21, 20], [5, Proposition 6].

Definition 11.2.1. The Bercovici Pata bijection Λ_{BP} maps the infinitely divisible law with triplet (η, A, Π) in (11.2.1) to the freely infinitely divisible law with the same triplet in (11.2.2) [20], [5, Definition 7].

The map is a homomorphism, $\Lambda_{\mathrm{BP}}(\mu * \nu) = \Lambda_{\mathrm{BP}}(\mu) \boxplus \Lambda_{\mathrm{BP}}(\nu)$, and a homeomorphism for weak convergence [20], [5, Section 3.2]. When Π has moments of all orders, comparing the expansions of e^{iut} and $1/(1-tz)$ in (11.2.1) and (11.2.2)

shows that the free cumulants of $\Lambda_{\mathrm{BP}}(\mu)$ are the classical cumulants of μ: both are $A\mathbf{1}_{\{n=2\}} + \int t^n\,\Pi(dt)$ for $n \ge 2$. The standard examples are these [20], [5, Section 3.2 and Remark 8]. The normal law $N(0,t)$, with triplet $(0,t,0)$, goes to the semicircle law σ_t with density $(2\pi t)^{-1}(4t-x^2)^{1/2}$ on $[-2\sqrt{t}, 2\sqrt{t}]$ and $\mathcal{C}_{\sigma_t}(z) = tz^2$; so the semicircle law of radius R has variance $R^2/4$, and the standard one, $t=1$, has R transform $\mathcal{C}_{\sigma_1}(z)/z = z$ and free triplet $(0,1,0)$. The Poisson law with mean λ, with triplet $(\lambda, 0, \lambda\delta_1)$, goes to the Marchenko Pastur law with rate λ, for which $\mathcal{C}(z) = \lambda z/(1-z)$; for $\lambda = 1$ its density is $(2\pi)^{-1}x^{-1/2}(4-x)^{1/2}$ on $(0,4)$ [152]. The Cauchy law is fixed by Λ_{BP}.

Three classes are matched by Λ_{BP}. Barndorff-Nielsen and Thorbjørnsen [16] proved that it maps the self decomposable laws onto the freely self decomposable laws, those μ that for every $c \in (0,1)$ satisfy $\mu = D_c\mu \boxplus \mu_c$ for some law μ_c, $D_c\mu$ being the law of cX when X has law μ; and Hasebe and Thorbjørnsen [93] proved that freely self decomposable laws are unimodal, the free analogue of Yamazato's theorem (Corollary 11.1.2). Pérez Abreu and Sakuma [191] study the image of the GGCs, the free GGCs. And by [5, Theorem 9] a freely infinitely divisible law μ is the image of an infinitely divisible law on $[0,\infty)$ if and only if $\mu^{\boxplus t}$ is carried by $[0,\infty)$ for every $t > 0$; these are the free regular laws. If μ is infinitely divisible on $[0,\infty)$, with drift $d \ge 0$ and Lévy measure Π on $(0,\infty)$, $\int \min(1,x)\,\Pi(dx) < \infty$, then (11.2.2) reduces to [5, (7)]

$$\mathcal{C}_{\Lambda_{\mathrm{BP}}(\mu)}(z) = dz + \int_{(0,\infty)} \Big(\frac{1}{1-zx} - 1\Big)\,\Pi(dx) = dz + \int_{(0,\infty)} \frac{zx}{1-zx}\,\Pi(dx). \quad (11.2.3)$$

The right side is analytic on $\mathbb{C} \setminus (0,\infty)$, and we use it to extend $\mathcal{C}_{\Lambda_{\mathrm{BP}}(\mu)}$ to the negative half line.

The Upsilon transform [17, 15] maps an infinitely divisible law μ with Lévy process (X_t), $X_1 \sim \mu$, to the law $\Upsilon(\mu)$ of $\int_0^1 \log(1/t)\,dX_t$. It is infinitely divisible, with

$$C_{\Upsilon(\mu)}(u) = \int_0^1 C_\mu\big(u\log(1/t)\big)\,dt = \int_0^\infty C_\mu(ux)\,e^{-x}\,dx. \quad (11.2.4)$$

Barndorff-Nielsen, Maejima and Sato [15] proved that Υ maps the infinitely divisible laws onto the Goldie Steutel Bondesson class (the infinitely divisible laws whose Lévy measure has a completely monotone density in $|x|$ on each half line), and the self decomposable laws onto the Thorin class. On $[0,\infty)$ these are the class BO and the GGCs of Section 1.3.2. Barndorff-Nielsen and Thorbjørnsen [17] found the link with the free side:

$$\mathcal{C}_{\Lambda_{\mathrm{BP}}(\mu)}(i\zeta) = C_{\Upsilon(\mu)}(\zeta), \qquad \zeta < 0. \quad (11.2.5)$$

For $N(0,1)$, $\int_0^\infty (-\frac{1}{2}\zeta^2x^2)e^{-x}dx = -\zeta^2 = \mathcal{C}_{\sigma_1}(i\zeta)$, and for the Poisson law with mean one, $\int_0^\infty (e^{i\zeta x}-1)e^{-x}dx = i\zeta/(1-i\zeta)$, the Marchenko Pastur value. So the free counterpart of a law is fixed by a classical law, its Upsilon image, which

the methods of this book can compute. On the half line the link takes the form $\mathcal{C}_{\Lambda_{\mathrm{BP}}(\mu)}(-s) = -\Psi_{\Upsilon(\mu)}(s)$, $s > 0$, where $\Psi_\mu(s) = -\log \int e^{-sx}\mu(dx)$ is the Laplace exponent (Proposition 11.2.2(a)).

11.2.2 The Upsilon transform on the half line

Let μ be an infinitely divisible law on $[0,\infty)$ with drift $d \geq 0$ and Lévy measure Π on $(0,\infty)$,

$$\Psi_\mu(s) = ds + \int_{(0,\infty)} \bigl(1 - e^{-sx}\bigr)\,\Pi(dx), \qquad \int \min(1,x)\,\Pi(dx) < \infty.$$

If μ is self decomposable, $\Pi(dx) = k(x)\,dx/x$ with k nonincreasing, right continuous and $k(\infty) = 0$, the canonical function of μ; if μ is a GGC with Thorin measure U, then $k(x) = \int e^{-tx}\,U(dt)$ (Section 1.3.1). The cumulants of μ, when finite, are $\kappa_1 = d + \int x\,\Pi(dx)$ and $\kappa_n = \int x^n\,\Pi(dx)$ for $n \geq 2$.

Proposition 11.2.2. *Let μ be as above.*

(a) $\Upsilon(\mu)$ is the infinitely divisible law on $[0,\infty)$ with drift d, Laplace exponent

$$\Psi_{\Upsilon(\mu)}(s) = \int_0^\infty \Psi_\mu(sx)\,e^{-x}\,dx = ds + \int_{(0,\infty)} \frac{sy}{1+sy}\,\Pi(dy) \tag{11.2.6}$$

and completely monotone Lévy density $\ell(z) = \int y^{-1}e^{-z/y}\,\Pi(dy)$. Every infinitely divisible law on $[0,\infty)$ with a completely monotone Lévy density, that is every law in BO, is $\Upsilon(\mu)$ for exactly one μ. Moreover $\mathcal{C}_{\Lambda_{\mathrm{BP}}(\mu)}(-s) = -\Psi_{\Upsilon(\mu)}(s)$ for $s > 0$, and $\kappa_n(\Upsilon(\mu)) = n!\,\kappa_n(\mu)$ whenever $\int x^n\,\Pi(dx) < \infty$.

(b) $\Upsilon(\mu)$ is a GGC if and only if μ is self decomposable. In that case $\Upsilon(\mu)$ has drift d and the Thorin measure $\widetilde{U}$ with $\widetilde{U}((0,u]) = k(1/u)$, the image of the measure $-dk$ under $v \mapsto 1/v$, and Υ is a bijection from the self decomposable laws on $[0,\infty)$ onto the GGCs, with inverse given by $k(v) = \widetilde{U}((0,1/v])$.

(c) If μ is a GGC with Thorin measure U, then $\widetilde{U}$ is the mixture over $U(dt)$ of the laws of t/E, with E standard exponential:

$$\widetilde{U}(du) = u^{-2}(-k')(1/u)\,du, \qquad \Psi_{\Upsilon(\mu)}(s) = ds + \int e^{t/s}E_1(t/s)\,U(dt), \tag{11.2.7}$$

and $\int u^p\,\widetilde{U}(du) = \Gamma(1-p)\int t^p\,U(dt)$ for $p < 1$.

Proof. (a) The first equality in (11.2.6) is (11.2.4) for Laplace transforms: for a subordinator, $\mathbb{E}\exp(-s\int_0^1 f(t)\,dX_t) = \exp(-\int_0^1 \Psi_\mu(sf(t))\,dt)$ for every bounded measurable $f \geq 0$, by approximation with step functions and independence of increments, and the case $f = \log(1/t)$ follows by monotone convergence. Since

$\int_0^\infty (1-e^{-sxy})e^{-x}dx = sy/(1+sy)$, Fubini's theorem gives the second equality, and since

$$\frac{sy}{1+sy} = \int_0^\infty \bigl(1-e^{-sz}\bigr)\, y^{-1}e^{-z/y}\,dz,$$

it gives $\Psi_{\Upsilon(\mu)}(s) = ds + \int_0^\infty (1-e^{-sz})\,\ell(z)\,dz$. The function ℓ is a mixture of exponentials, hence completely monotone, and it is a Lévy density: $\int_0^\infty \min(1,z)\,y^{-1}e^{-z/y}dz = y(1-e^{-1/y})$, which lies between $(1-e^{-1})\min(1,y)$ and $\min(1,y)$, so that $\int \min(1,z)\,\ell(z)\,dz$ is comparable with $\int \min(1,y)\,\Pi(dy) < \infty$. Conversely, let λ be infinitely divisible on $[0,\infty)$ with drift d and completely monotone Lévy density $\ell(z) = \int_{[0,\infty)} e^{-zr}\,\sigma(dr)$ (Bernstein's theorem). Integrability of ℓ on $(1,\infty)$ forces $\sigma(\{0\}) = 0$. Let Π be the image of $r^{-1}\sigma(dr)$ under $r \mapsto 1/r$. Then $\int y^{-1}e^{-z/y}\,\Pi(dy) = \int e^{-zr}\,\sigma(dr) = \ell(z)$, the comparison above shows that Π is a Lévy measure with $\int \min(1,y)\,\Pi(dy) < \infty$, and $\lambda = \Upsilon(\mu)$ for the law μ with drift d and Lévy measure Π. Uniqueness: ℓ determines σ by the uniqueness of Laplace transforms, hence Π, and the drift is read off from Ψ at infinity. By (11.2.3), $\mathcal{C}_{\Lambda_{\mathrm{BP}}(\mu)}(-s) = -ds - \int sx(1+sx)^{-1}\,\Pi(dx) = -\Psi_{\Upsilon(\mu)}(s)$. Finally $\int z^n\ell(z)\,dz = \int \Pi(dy)\int_0^\infty z^n y^{-1}e^{-z/y}dz = n!\int y^n\,\Pi(dy)$, which gives $\kappa_n(\Upsilon(\mu)) = n!\,\kappa_n(\mu)$ for $n \ge 2$; for $n = 1$ the drift, which is the same for both laws, is added on both sides.

(b) Let μ be self decomposable with canonical function k. With $r = 1/y$,

$$z\,\ell(z) = \int_0^\infty \frac{z}{y^2}\,e^{-z/y}\,k(y)\,dy = \int_0^\infty z\,e^{-zr}\,k(1/r)\,dr = \int_{(0,\infty)} e^{-zr}\,d_r\bigl[k(1/r)\bigr],$$

by an integration by parts whose boundary terms vanish: $k(1/r) \to k(\infty) = 0$ as $r \to 0$, and $e^{-zr}k(1/r) \to 0$ as $r \to \infty$ because $yk(y) \le \int_0^y k \to 0$ as $y \to 0$, k being nonincreasing and integrable near 0. The function $r \mapsto k(1/r)$ is nondecreasing, so $\widetilde{U} = d_r[k(1/r)]$ is a positive measure with $\widetilde{U}((0,u]) = k(1/u)$, and $z\ell(z) = \int e^{-zr}\,\widetilde{U}(dr)$ says that $\Upsilon(\mu)$ is a GGC with Thorin measure $\widetilde{U}$ (Section 1.3.1). Conversely, let $\Upsilon(\mu)$ be a GGC with drift d and Thorin measure $\widetilde{U}$, and put $k(v) = \widetilde{U}((0,1/v])$, a nonincreasing function with $k(\infty) = 0$ (its values at its jumps do not matter). By Fubini's theorem $\int_0^1 k(y)\,dy = \int \min(1,1/r)\,\widetilde{U}(dr)$ and $\int_1^\infty k(y)\,dy/y = \int_{(0,1)} \log(1/r)\,\widetilde{U}(dr)$, both finite by the integrability conditions of (1.3.1) and because $\widetilde{U}$ is finite on compact subsets of $(0,\infty)$. So $k(x)\,dx/x$ is a Lévy measure, the self decomposable law μ' with drift d and canonical function k exists, and by the first part $\Upsilon(\mu')$ has drift d and Thorin measure $\widetilde{U}$. Hence $\Upsilon(\mu') = \Upsilon(\mu)$, and $\mu = \mu'$ by the uniqueness in (a). The same construction applied to an arbitrary GGC shows that Υ maps the self decomposable laws onto the GGCs, and (a) gives injectivity.

(c) Now $k(x) = \int e^{-tx}\,U(dt)$, so $\widetilde{U}((0,u]) = k(1/u) = \int e^{-t/u}\,U(dt) = \int \Pr(t/E \le u)\,U(dt)$, and differentiating in u, $\widetilde{U}(du) = u^{-2}\int te^{-t/u}\,U(dt)\,du = u^{-2}(-k')(1/u)\,du$. Since $\Psi_\mu(s) = ds + \int \log(1+s/t)\,U(dt)$ and, by an integration by parts, $\int_0^\infty \log(1+sx/t)\,e^{-x}dx = \int_0^\infty e^{-x}(x+t/s)^{-1}dx = e^{t/s}E_1(t/s)$,

the formula for $\Psi_{\Upsilon(\mu)}$ follows from (11.2.6) by Fubini's theorem. Finally $\int u^p\,\widetilde{U}(du) = \int \mathbb{E}[(t/E)^p]\,U(dt)$ and $\mathbb{E}E^{-p} = \Gamma(1-p)$ for $p<1$. □

So the Thorin measure of $\Upsilon(\mu)$ is the canonical function of μ read backwards, $\widetilde{U}((0,u]) = k(1/u)$, and $\Upsilon(\mu)$ is a GGC exactly when k is nonincreasing; this is the half line reading of $\Upsilon(\mathrm{SD}) = $ Thorin class [15], which the note [208] pointed out through (11.2.7). For the standard exponential law, $U=\delta_1$ and $k(x)=e^{-x}$, so $\Upsilon(\mathrm{Exp}(1))$ is the GGC with Thorin density $u^{-2}e^{-1/u}$, the density of $1/E$, Laplace exponent $e^{1/s}E_1(1/s)$ and Lévy density $\int_0^\infty y^{-2}e^{-z/y-y}dy = 2z^{-1/2}K_1(2\sqrt{z})$.

11.2.3 The Upsilon image and the free counterpart of the clock

We apply Proposition 11.2.2 to the centre clock T, whose Laplace transform is $1/G(u)$ with $G(u) = \xi(\frac12+\sqrt{u})/\xi(\frac12) = \prod_k(1+u/a_k)$ (1.2.1). By Section 11.1.1, T is infinitely divisible with no drift and Lévy measure $W(t)\,dt/t$, where $W(t) = \sum_k e^{-a_k t}$, and it is self decomposable with canonical function W, because $W>0$ and $W'<0$ on $(0,\infty)$ with no hypothesis (Theorem 5.1.1, whose proof uses only zeros below a finite height); $h=-W'$ is the Lévy density of its driving subordinator (Proposition 11.1.1). The Stieltjes moments of Section 5.1 are $m_n = \frac{1}{n!}\int_0^\infty t^n W(t)\,dt = \sum_k a_k^{-n-1}$, and since $-\log G(u) = \sum_{n\ge1}(-1)^n m_{n-1}u^n/n$ near $u=0$, the cumulants of T are $\kappa_n(T) = (n-1)!\,m_{n-1}$. Finally $N(\theta)$ is the number of zeros with ordinate in $(0,\theta]$, and $\gamma_1 = 14.1347\ldots$ is the least ordinate.

Theorem 11.2.3. *(a) With no hypothesis, $\Upsilon(T)$ is a GGC with no drift and Thorin measure $\widetilde{U}(du) = u^{-2}h(1/u)\,du$, so that $\widetilde{U}((0,u]) = W(1/u) > 0$ for every $u>0$. Its Laplace exponent is*

$$\Psi_{\Upsilon(T)}(s) = \int_0^\infty \log G(sx)\,e^{-x}\,dx = \sum_k e^{a_k/s}E_1(a_k/s), \qquad s>0, \quad (11.2.8)$$

with the principal branch of E_1; its Lévy density is the completely monotone function $\ell(z) = z^{-1}\int_0^\infty W(z/w)\,e^{-w}\,dw$; its cumulants are $\kappa_n(\Upsilon(T)) = n!\,(n-1)!\,m_{n-1}$; and $\int u^{-n}\,\widetilde{U}(du) = n!\,m_{n-1}$ for $n\ge1$. Moreover $\mathbb{E}e^{\lambda\Upsilon(T)} = \infty$ for every $\lambda>0$, while $\mathbb{E}e^{\lambda T}<\infty$ exactly when $\lambda<\gamma_1^2$.

(b) With no hypothesis, $\Lambda_{\mathrm{BP}}(T)$ is free regular, freely self decomposable and unimodal. Its free cumulant transform is

$$\mathcal{C}_{\Lambda_{\mathrm{BP}}(T)}(z) = \int_0^\infty \frac{z\,W(x)}{1-zx}\,dx, \qquad z\in\mathbb{C}^-, \quad (11.2.9)$$

where the right side continues it analytically to $\mathbb{C}\backslash(0,\infty)$, with $\mathcal{C}_{\Lambda_{\mathrm{BP}}(T)}(-s) = -\Psi_{\Upsilon(T)}(s)$ for $s>0$; its free cumulants are $(n-1)!\,m_{n-1}$, the cumulants of T. Its Cauchy transform satisfies

$$\Psi_{\Upsilon(T)}\big(-G_{\Lambda_{\mathrm{BP}}(T)}(z)\big) = 1 - z\,G_{\Lambda_{\mathrm{BP}}(T)}(z), \qquad z\in\mathbb{C}^+, \quad (11.2.10)$$

where $\Psi_{\Upsilon(T)}$ is continued analytically to $\mathbb{C} \setminus (-\infty, 0)$ by the right side of (11.2.6)*, and its support lies in $[c, \infty)$, where $c = \max_{s>0}(\Psi_{\Upsilon(T)}(s) - 1)/s > 0$.*

(c) RH holds if and only if T is the Upsilon image of an infinitely divisible law on $[0, \infty)$. In that case $T = \Upsilon(\mu_0)$, where μ_0 is the self decomposable law with no drift and canonical function $N(v^{-1/2})$, that is with Lévy measure $N(x^{-1/2})\, dx/x$; μ_0 is neither a GGC nor in BO. With no hypothesis, T is not the Upsilon image of a law in BO.

Proof. (a) Proposition 11.2.2(b) applies with $k = W$, which is continuously differentiable with $W' = -h$, so $\widetilde{U}((0,u]) = W(1/u)$ and $\widetilde{U}(du) = u^{-2}h(1/u)\, du$; the drift is zero because T has none. The first form of $\Psi_{\Upsilon(T)}$ is (11.2.6) with $\Psi_T = \log G$. For the second, $\operatorname{Re} a_k > 0$ and hence $\operatorname{Re}(1/a_k) > 0$ for every k, so $w = sx/a_k$ has $\operatorname{Re} w \geq 0$, and $\log(1+w) = \int_0^1 w\, d\tau/(1+\tau w)$ with $|1+\tau w| \geq 1$ gives $|\log(1+w)| \leq |w|$. Since $\sum_k |a_k|^{-1} < \infty$, $\sum_k \int_0^\infty |\log(1 + sx/a_k)|\, e^{-x} dx \leq s \sum_k |a_k|^{-1} < \infty$, the series $\log G(sx) = \sum_k \log(1 + sx/a_k)$ can be integrated term by term, and each term gives $e^{a_k/s} E_1(a_k/s)$ by the integration by parts in the proof of Proposition 11.2.2(c), valid for $\operatorname{Re} a_k > 0$ because $e^z E_1(z) = \int_0^\infty e^{-w}(w+z)^{-1} dw$ for $\operatorname{Re} z > 0$. The Lévy density of Proposition 11.2.2(a) is $\ell(z) = \int_0^\infty y^{-2} e^{-z/y} W(y)\, dy$, and the substitution $y = z/w$ gives the stated form. The cumulants follow from $\kappa_n(\Upsilon(T)) = n!\, \kappa_n(T)$, and $\int u^{-n} \widetilde{U}(du) = \int_0^\infty v^n h(v)\, dv = n \int_0^\infty v^{n-1} W(v)\, dv = n!\, m_{n-1}$, by the substitution $v = 1/u$ and an integration by parts whose boundary terms vanish because $W(v) = O(v^{-1/2} \log(1/v))$ at the origin and decays exponentially at infinity. By Sato's moment theorem [232, Theorem 25.17], an infinitely divisible law has $\mathbb{E} e^{\lambda X} < \infty$ exactly when its Lévy measure integrates $e^{\lambda x}$ over $(1, \infty)$. For $\Upsilon(T)$, $z\ell(z) = \int e^{-zr} \widetilde{U}(dr) \geq e^{-z\varepsilon} W(1/\varepsilon)$ for every $\varepsilon > 0$, and with $\varepsilon < \lambda$ the integral $\int_1^\infty e^{\lambda z} \ell(z)\, dz$ diverges. For T, every zero other than the first has $\operatorname{Re} a_k \geq \gamma_2^2 - \frac{1}{4} > \gamma_1^2$, so $W(t) = e^{-\gamma_1^2 t}(1 + o(1))$ as $t \to \infty$, and $\int_1^\infty e^{\lambda t} W(t)\, dt/t$ converges exactly when $\lambda < \gamma_1^2$.

(b) T is infinitely divisible on $[0, \infty)$ with no drift and Lévy measure $\Pi(dx) = W(x)\, dx/x$, $\int \min(1, x)\, \Pi(dx) < \infty$, so $\Lambda_{\mathrm{BP}}(T)$ is free regular by [5, Theorem 9], freely self decomposable by [16], since T is self decomposable, and unimodal by [93]. Formula (11.2.9) is (11.2.3) with this Π, and $\mathcal{C}_{\Lambda_{\mathrm{BP}}(T)}(-s) = -\Psi_{\Upsilon(T)}(s)$ is Proposition 11.2.2(a). The right sides of (11.2.9) and (11.2.6) give $\mathcal{C}_{\Lambda_{\mathrm{BP}}(T)}(w) = -\Psi_{\Upsilon(T)}(-w)$ on all of $\mathbb{C} \setminus (0, \infty)$. Since Π has moments of all orders, the free cumulants are the classical cumulants of T (Section 11.2.1). Write G_Λ for $G_{\Lambda_{\mathrm{BP}}(T)}$. On a region where $(1/G_\Lambda)^{-1}(1/G_\Lambda(z)) = z$, the definition of φ gives $\varphi(1/G_\Lambda(z)) = z - 1/G_\Lambda(z)$, that is $\mathcal{C}_{\Lambda_{\mathrm{BP}}(T)}(G_\Lambda(z)) = zG_\Lambda(z) - 1$; both sides are analytic on $\mathbb{C}^+$, which G_Λ maps into $\mathbb{C}^-$, so the identity holds on $\mathbb{C}^+$, and it is (11.2.10). For the support, put $x(s) = (\Psi_{\Upsilon(T)}(s) - 1)/s$ for $s > 0$. Then $s^2 x'(s) = f(s) = s\Psi'_{\Upsilon(T)}(s) - \Psi_{\Upsilon(T)}(s) + 1$, with $f(0+) = 1$, $f'(s) = s\Psi''_{\Upsilon(T)}(s) < 0$, and $f(s) \to -\infty$

as $s \to \infty$, since $s\Psi' - \Psi = -\int (1 - e^{-sz} - sze^{-sz})\ell(z)\,dz$ and ℓ is not integrable at the origin. So x increases from $-\infty$ to its maximum $c = x(s^*) = \Psi'_{\Upsilon(T)}(s^*)$ on $(0, s^*]$ and decreases to 0 on $[s^*, \infty)$, because $\Psi_{\Upsilon(T)}(s)/s \to 0$; in particular $c > 0$. For $x < 0$, which lies outside the support, $G_\Lambda(x) < 0$ and (11.2.10) extends by continuity, so $S = -G_\Lambda(x) > 0$ solves $x(S) = x$, whose only positive solution is $s(x)$, the inverse of $x(\cdot)$ on $(0, s^*)$. This inverse is analytic on a neighbourhood V of $(-\infty, c)$, a union of discs centred on that half line that do not meet $[c, \infty)$, and $-s$ agrees with G_Λ on $(-\infty, 0)$, hence on the connected set $V \setminus [0, c)$. So G_Λ extends analytically across $(-\infty, c)$ with real values there, and by Stieltjes inversion $\Lambda_{\mathrm{BP}}(T)$ gives no mass to $(-\infty, c)$.

(c) By Proposition 11.2.2(a), the Upsilon images of the infinitely divisible laws on $[0, \infty)$ are the laws of the class BO, and the clock is in BO if and only if RH holds (Section 8.5.4 and Table 1.7). Under RH, $W(z)/z = \sum_{\gamma>0} \int_{\gamma^2}^\infty e^{-zr}\,dr = \int_0^\infty e^{-zr} N(\sqrt{r})\,dr$, so in the proof of Proposition 11.2.2(a) $\sigma(dr) = N(\sqrt{r})\,dr$, and the image of $r^{-1}\sigma(dr)$ under $r \mapsto 1/r$ is $N(x^{-1/2})\,dx/x$. Its canonical function $N(v^{-1/2})$ is nonincreasing, so μ_0 is self decomposable, and by Proposition 11.2.2(b) the Thorin measure of $\Upsilon(\mu_0)$ has the distribution function $u \mapsto N(\sqrt{u})$, that is, it is $\sum_{\gamma>0} \delta_{\gamma^2}$, as Proposition 1.2.1 requires. The canonical function and the Lévy density of μ_0 vanish on (γ_1^{-2}, ∞) without vanishing identically, so neither is completely monotone: μ_0 is not a GGC and not in BO. If $T = \Upsilon(\mu)$ with μ in BO, then T is in BO, RH holds, and $\mu = \mu_0$ by the uniqueness in Proposition 11.2.2(a), which is impossible. □

The Upsilon step thus turns the Thorin question for the clock, whose answer is RH, into a question with a positive answer. The Thorin measure of T is $\sum_{\gamma>0} \delta_{\gamma^2}$ exactly when W is completely monotone, every rung of the hierarchy of Section 5.1; the Thorin measure of $\Upsilon(T)$ is positive as soon as W is nonincreasing, the second rung, because its distribution function is W read backwards. Under RH, Proposition 11.2.2(c) shows how the two are related: $\widetilde{U}$ spreads the atom at γ^2 into the law of γ^2/E, and in the chart $\theta = \sqrt{u}$

$$\widetilde{U}((0, \theta^2]) = W(\theta^{-2}) = \sum_{\gamma>0} e^{-\gamma^2/\theta^2} = \mathbb{E}\, N(\theta\sqrt{E}),$$

with E standard exponential, a smoothing of the zero count. With no hypothesis, the asymptotic $W(t) \sim \log(1/t)/(8\sqrt{\pi t})$ as $t \downarrow 0$ (Section 5.1) and the Riemann von Mangoldt formula give $W(\theta^{-2})/N(\theta) \to \sqrt{\pi}/2 = 0.8862$ as $\theta \to \infty$. Under RH the expectation gives the next term: $\mathbb{E}N_0(\theta\sqrt{E})$, with $N_0(\theta) = (\theta/2\pi)\log(\theta/2\pi e) + \frac{7}{8}$, equals $(\theta/4\sqrt{\pi})\big(\log(\theta/2\pi e) + \frac{1}{2}\psi(\frac{3}{2})\big) + \frac{7}{8}$, where $\mathbb{E}\sqrt{E} = \Gamma(\frac{3}{2}) = \frac{\sqrt{\pi}}{2}$ and $\mathbb{E}[\sqrt{E}\log E] = \Gamma'(\frac{3}{2})$, and $\psi(\frac{3}{2}) = 2 - \gamma_E - 2\log 2 = 0.0365$; so the ratio of $W(\theta^{-2})$ to the smooth count $N_0(\theta)$ exceeds $\sqrt{\pi}/2$ by a relative amount of about $\psi(\frac{3}{2})/(2\log(\theta/2\pi e))$, and the ratio to $N(\theta)$ itself also fluctuates with $N - N_0$. At $\theta = 50$, 100, 200 and 400, $W(\theta^{-2}) = 8.579$, 26.060, 70.798 and

179.827 against $N(\theta) = 10$, 29, 79 and 202; the left panel of Figure 11.5 draws both.

The free counterpart behaves differently at the left end. The clock and its Upsilon image are infinitely divisible with no drift and infinite Lévy measure, so their supports are $[0, \infty)$ [232, Section 24], and their densities are super flat at the origin; the free law has a hard edge at $c > 0$. Since $\Psi''_{\Upsilon(T)}(s^*) \neq 0$, the solution of $x(s) = x$ leaves the real axis as a square root at $x = c$, so the density $\operatorname{Im} s/\pi$ of $\Lambda_{\mathrm{BP}}(T)$ vanishes like $(x - c)^{1/2}$ there, as the semicircle law does at its edges. At the other end, the tail of T is exponential, with rate γ_1^2, while that of $\Upsilon(T)$ is much heavier. As $z \to \infty$ only the first zero matters in $\ell(z) = z^{-1} \int_0^\infty W(z/w) e^{-w} dw$, the saddle point of $e^{-\gamma_1^2 z/w - w}$ lies at $w = \gamma_1 \sqrt{z}$, and

$$\ell(z) \sim 2\gamma_1 z^{-1/2} K_1(2\gamma_1 \sqrt{z}) \sim \sqrt{\pi}\, \gamma_1^{1/2} z^{-3/4} e^{-2\gamma_1 \sqrt{z}};$$

under RH, $\ell(z) = 2z^{-1/2} \sum_{\gamma > 0} \gamma K_1(2\gamma\sqrt{z})$ exactly. The computed density of $\Upsilon(T)$ is 2.552 times ℓ at $x = 0.2$ and 1.828 times ℓ at $x = 0.4$, so the Lévy density takes over the tail only slowly.

Table 11.7 checks (11.2.8), which is also minus the free cumulant transform at $-s$. The integral is computed from ξ alone, by quadrature at 30 digits, and the sum over the zeros from the first 300 or 1700 ordinates, with one of two tails: the tail of order one, $s(\log(\gamma_N/2\pi) + 1)/(2\pi\gamma_N)$, which is $s \int_{\gamma_N}^\infty \theta^{-2} dN_0(\theta)$ and the choice of [208], and the full smooth tail $\int_{\gamma_N}^\infty \phi(\theta^2/s)\, dN_0(\theta) + \phi(\gamma_N^2/s)(N_0(\gamma_N) - N)$, where $\phi(z) = e^z E_1(z)$ and N is the number of zeros used. The note's Table 1, the integral and the sums over 300 zeros without and with the tail of order one at $s = 1$, 10 and 100, is reproduced to every digit shown (0.0230681050, 0.0214685871 and 0.0230714840 at $s = 1$). With 1700 zeros and the full tail the difference is $-6.29 \times 10^{-11} s$ at every s up to 10^3, that is, it comes from the smooth tail of $\sum \gamma^{-2}$ beyond $\gamma_{1700} = 2197.26$, where $\phi(\gamma^2/s) \approx s/\gamma^2$.

		sum over the zeros minus the integral				
s	$\int_0^\infty \log G(sx)\, e^{-x}\, dx$	300 zeros	300, tail 1	300, full tail	1700, tail 1	1700, full tail
1	0.0230681049610841	-0.0016	3.38×10^{-6}	-1.32×10^{-9}	1.2×10^{-7}	-6.29×10^{-11}
10	0.227587470315411	-0.016	3.39×10^{-5}	-1.32×10^{-8}	1.2×10^{-6}	-6.29×10^{-10}
100	2.07560251202393	-0.16	3.54×10^{-4}	-1.32×10^{-7}	1.23×10^{-5}	-6.29×10^{-9}
10^3	15.2403221009594	-1.6	4.96×10^{-3}	-1.31×10^{-6}	1.51×10^{-4}	-6.29×10^{-8}
10^4	87.4852790298693	-15.8	0.187	-1.24×10^{-5}	4.28×10^{-3}	-6.27×10^{-7}

Table 11.7: The Laplace exponent of $\Upsilon(T)$, which is minus the free cumulant transform of $\Lambda_{\mathrm{BP}}(T)$ at $-s$: the integral of (11.2.8) from ξ alone, and the sums over the first 300 and 1700 zeros, with no tail, with the tail of order one and with the full smooth tail, as differences from the integral. Computed by `wigfree_upsilon.py`.

Table 11.8 and the right panel of Figure 11.5 compare the three laws with the Lévy measure $W(t)\, dt/t$. The density of T is the fixed Talbot inversion of $1/G$ at

25 digits on the grid 0.002(0.0005)0.15. That of $\Upsilon(T)$ is the fixed Talbot inversion of $e^{-\Psi_{\Upsilon(T)}}$, with $\Psi_{\Upsilon(T)}$ from the 1700 zeros and the tail $u\tau_1 - u^2\tau_2 + 2u^3\tau_3$, τ_j the smooth tail of $\sum \gamma^{-2j}$ with its boundary term; it runs in double precision with 17 nodes from $x = 0.0125$ on, where 13 nodes change it by at most 9.9×10^{-4} of its maximum, and at 30 digits below, where double precision loses the cancellation (at $x = 0.0125$ the two give 14.23490 and 14.23462, at 0.02 they agree to 1.3×10^{-7} relatively). That of $\Lambda_{\mathrm{BP}}(T)$ is $\operatorname{Im} s/\pi$, with s the solution of $\Psi_{\Upsilon(T)}(s) = 1 + xs$ in the upper half plane given by (11.2.10), found by Newton's method on 500 points of $[c, 0.15]$, spaced quadratically near c, with residuals below 4.6×10^{-15}. Here $c = 0.01529224618$ and $s^* = 387.9402$. The moments come from the trapezoid rule on these grids and reproduce the cumulants of the theorem; on each grid the density rises to its mode and falls after it, as unimodality requires (Corollary 11.1.2 for T; for $\Upsilon(T)$, a GGC and hence self decomposable, Yamazato's theorem [274]; [93] for $\Lambda_{\mathrm{BP}}(T)$). The free counterpart is the most concentrated of the three: it has the same mean and variance as T, a hard edge at c and the highest mode.

	T	$\Upsilon(T)$	$\Lambda_{\mathrm{BP}}(T)$
total mass	1.00000000	1.00000053	1.00000021
mean	0.023104993	0.023104991	0.023104991
variance	3.71726×10^{-5}	7.43455×10^{-5}	3.71726×10^{-5}
third central moment	2.8835×10^{-7}	1.7301×10^{-6}	2.8835×10^{-7}
mode	0.0200	0.0185	0.017711
density at the mode	80.1042	72.6771	91.3878
left end of the support	0	0	0.01529224618

Table 11.8: The clock T, its Upsilon image and its free counterpart, from the densities of Figure 11.5. The exact means are $m_0 = 0.023104993$; the variances $m_1 = 3.71726 \times 10^{-5}$ for T and $\Lambda_{\mathrm{BP}}(T)$ and $2m_1 = 7.43452 \times 10^{-5}$ for $\Upsilon(T)$; the third central moments $2m_2 = 2.8835 \times 10^{-7}$ for T and $\Lambda_{\mathrm{BP}}(T)$ and $12m_2 = 1.7301 \times 10^{-6}$ for $\Upsilon(T)$. The modes of T and $\Upsilon(T)$ are grid points of spacing 0.0005, that of $\Lambda_{\mathrm{BP}}(T)$ of spacing about 7×10^{-5}. Computed by `wigfree_upsilon.py`.

11.2.4 The semicircle as a local model for the Jacobi parameters

Section 5.1 builds from the Stieltjes moments $m_0, \dots, m_{2N-1}$ the Jacobi matrix J_N of the measure $\nu = \sum_j a_j^{-1} \delta_{1/a_j}$, a tridiagonal matrix whose diagonal entries we write $\alpha_1, \dots, \alpha_N$ and whose off diagonal entries, which are positive, we write $\beta_1, \dots, \beta_{N-1}$; they are the recurrence coefficients of the orthogonal polynomials of ν, and they are computed from the Taylor coefficients of Ξ at the centre alone. Free probability supplies a local model for their size.

For constant parameters $\alpha_n = \alpha$ and $\beta_n = \beta$ the Jacobi matrix is $\alpha + \beta(S + S^*)$ on $\ell^2(\{0, 1, 2, \dots\})$, with S the unilateral shift, and the spectral measure of its first basis vector is the semicircle law on $[\alpha - 2\beta, \alpha + 2\beta]$: the moments

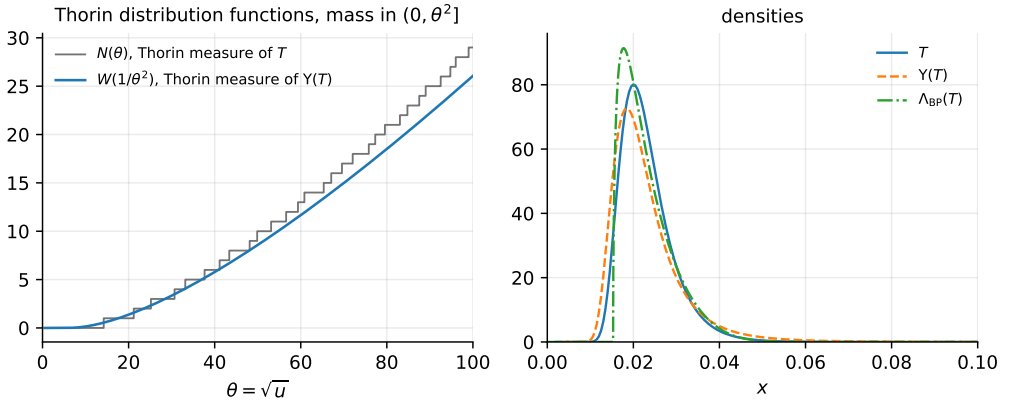


Figure 11.5: Left: the distribution function $W(\theta^{-2})$ of the Thorin measure of $\Upsilon(T)$ in the chart $\theta = \sqrt{u}$, against the zero count $N(\theta)$, which under RH is the distribution function of the Thorin measure of T in the same chart. Right: the densities of T, of $\Upsilon(T)$ and of $\Lambda_{\mathrm{BP}}(T)$ on $[0, 0.1]$; the last vanishes to the left of $c = 0.0153$. Computed by `wigfree_upsilon.py` and drawn by `wigfree_plots.py`.

$\langle e_0, (S + S^*)^k e_0 \rangle$ count the Dyck paths of length k, and these are the Catalan numbers, the moments of σ_1 [166]. The $N \times N$ truncations have the eigenvalues $\alpha + 2\beta\cos(\pi k/(N+1))$, $1 \le k \le N$, whose distribution tends to the arcsine law with density $\big(\pi\sqrt{4\beta^2 - (x-\alpha)^2}\big)^{-1}$ on the same interval, the density of states. If the parameters vary slowly, the eigenvalues are counted by integrating local arcsine laws over the rows. For recurrence coefficients that vary on the scale of the matrix size, $\alpha_{n,N} \to a(n/N)$ and $\beta_{n,N} \to b(n/N)$, this is the theorem of Kuijlaars and Van Assche [130]: the zero distribution of the orthogonal polynomials tends to $\int_0^1 \omega_{[a(t)-2b(t),a(t)+2b(t)]}\,dt$, with $\omega_{[A,B]}$ the arcsine law on $[A, B]$. The parameters of ν decrease to zero with no such scaling, and the count needed below is that of the largest eigenvalues, an edge statement; for ν the local model is a heuristic. Since ν has its atoms in $(0, \gamma_1^{-2}]$ and accumulates only at the origin, the local interval $[\alpha_n - 2\beta_n, \alpha_n + 2\beta_n]$ should reach down to zero, which suggests $\alpha_n \approx 2\beta_n$. Write $L(n) = \alpha_n + 2\beta_n$ for its right end, taken to decrease in n, and $n(L)$ for the inverse function.

The eigenvalues of J_N above a level $x > 0$ approximate the atoms $1/\gamma_j^2$ above x, of which there are $N(x^{-1/2})$. In the local model row n contributes the mass $\frac{2}{\pi}\arccos\sqrt{x/L(n)}$ of the arcsine law on $[0, L(n)]$ above x, and nothing when $x \ge L(n)$, so that

$$N\big(x^{-1/2}\big) = \int_0^\infty \frac{2}{\pi}\arccos\sqrt{x/L(n)}_+\,dn = \frac{1}{\pi}\int_x^\infty n(L)\,\frac{\sqrt{x}\,dL}{L\sqrt{L-x}}, \qquad (11.2.11)$$

the second form by an integration by parts in L, whose boundary terms vanish

because $\arccos 1 = 0$ and $n(L) \to 0$ as $L \to \infty$. With $u = 1/L$ and $w = 1/x$ it becomes Abel's equation

$$\int_0^w \frac{f(u)\,du}{\sqrt{w-u}} = \pi N(\sqrt{w}), \qquad f(u) = u^{-1/2}\,n(1/u). \tag{11.2.12}$$

PROPOSITION 11.2.4. *Let N be nondecreasing and right continuous on $[0,\infty)$, with $N = 0$ on $[0,\theta_0)$ for some $\theta_0 > 0$. Then*

$$n(L) = \int_{(0,L^{-1/2})} \frac{dN(T)}{\sqrt{1-LT^2}} \tag{11.2.13}$$

gives a locally integrable solution f of (11.2.12)*, with $N(\sqrt{w})$ replaced by its left limit at the jumps of N, and it is the only locally integrable solution. If $dN(T) = c\,dT$ on $(0,\infty)$, then $n(L) = \frac{\pi}{2}cL^{-1/2}$, so that $L_n = (\pi c/2n)^2$, which is $\gamma_{2n/\pi}^{-2}$ for the ordinates $\gamma_k = k/c$ of that count.*

Proof. For $0 < a < b$ the substitution $u = 1/L$ gives

$$\begin{aligned}\int_a^b \frac{dL}{L\sqrt{(L-a)(b-L)}} &= \int_{1/b}^{1/a} \frac{du}{\sqrt{(1-au)(bu-1)}} \\ &= \frac{1}{\sqrt{ab}} \int_{1/b}^{1/a} \frac{du}{\sqrt{(1/a-u)(u-1/b)}} = \frac{\pi}{\sqrt{ab}}.\end{aligned}$$

Insert (11.2.13) in the right side of (11.2.11), which is equivalent to (11.2.12), and exchange the integrals, all terms being positive. A point T of the support of dN contributes only when $T < x^{-1/2}$, through $L \in (x, T^{-2})$, and with $b = T^{-2}$ and $\sqrt{1-LT^2} = \sqrt{(b-L)/b}$ its contribution is

$$\frac{\sqrt{x}}{\pi} \int_x^b \frac{\sqrt{b}\,dL}{L\sqrt{(L-x)(b-L)}} = 1.$$

So the right side is N of the open interval $(0, x^{-1/2})$, the left limit of N at $x^{-1/2}$. The function f is locally integrable: n vanishes for $L > \theta_0^{-2}$, and for $0 < a < b$, by Fubini's theorem, $\int_a^b n(L)\,dL \le \int_{(0,a^{-1/2})} dN(T) \int_0^{T^{-2}} dL/\sqrt{1-LT^2} = 2\int_{(0,a^{-1/2})} T^{-2}\,dN(T) < \infty$. If f_1 and f_2 are two locally integrable solutions, $g = f_1 - f_2$ satisfies $\int_0^w g(u)(w-u)^{-1/2}du = 0$ for every w; integrating against $(y-w)^{-1/2}$ over $w \in (0,y)$ and using $\int_u^y dw/\sqrt{(y-w)(w-u)} = \pi$ gives $\pi\int_0^y g = 0$ for every y, so $g = 0$ almost everywhere. Finally $\int_0^{L^{-1/2}} c\,dT/\sqrt{1-LT^2} = cL^{-1/2}\arcsin 1$. □

The local model therefore predicts $\alpha_n \approx 2\beta_n \approx \frac{1}{2}L_n$, with $n(L_n) = n$ [208]. For the zeros of ξ we use the smooth density $dN_0(T) = (2\pi)^{-1}\log(T/2\pi)\,dT$ on

$T > 2\pi$, the derivative of the main term of the Riemann von Mangoldt formula, for which the substitution $T = L^{-1/2}\sin\varphi$ gives

$$n(L) = \frac{1}{2\pi\sqrt{L}} \int_{\varphi_0}^{\pi/2} \log \frac{\sin\varphi}{2\pi\sqrt{L}}\, d\varphi, \qquad \sin\varphi_0 = 2\pi\sqrt{L}.$$

As $L \to 0$, $\int_0^{\pi/2} \log\sin\varphi\, d\varphi = -\frac{\pi}{2}\log 2$ gives $n(L) = (4\sqrt{L})^{-1}\log(1/(4\pi\sqrt{L})) + O(1)$, while the rule of Proposition 11.2.4 for a linear count, applied with the ordinates of the smooth count, $N_0(\gamma_r) = r$ with $N_0(T) = (T/2\pi)\log(T/2\pi e) + \frac{7}{8}$, gives $\frac{1}{2}L_n = \frac{1}{2}\gamma_{2n/\pi}^{-2}$ and corresponds to $n = (4\sqrt{L})^{-1}\log(1/(2\pi e\sqrt{L}))$ to the same order. The logarithmic density thus changes the constant inside the logarithm from $2\pi e$ to 4π, and the two rules differ by a factor that tends to one only logarithmically; this is why the cruder rule $\alpha_n \approx \frac{1}{2}\gamma_{2n/\pi}^{-2}$, which treats the count as linear, is off by a factor near 1.3.

Table 11.9 and Figure 11.6 compare the computed parameters with the two rules. The program `wigfree_jacobi.py` computes $M_0, \dots, M_{100}$ at 1000 digits by the trapezoid rule of step $1/600$ on $[0, 3.6]$ (M_0 agrees with $\xi(\frac{1}{2})$ to 3.6×10^{-1001} relatively), the moments $m_0, \dots, m_{99}$, and the Cholesky factorisation of $\mathcal{H}_{50}$, whose determinant is $10^{-9414.73}$ and whose factor has least diagonal entry 1.18×10^{-199}. The matrix $L^{-1}[m_{i+j+1}]L^{-\top}$ is tridiagonal to 1.3×10^{-836}, and its entries change by at most 2.3×10^{-429} relatively when the factorisation runs at 600 digits. The values L_n solve $n(L) = n$ to within 1.3×10^{-29} at the checked indices $n = 5$, 25 and 49. The first parameters are irregular, $\alpha_1, \dots, \alpha_5 = (1.61, 3.50, 1.80, 1.22, 1.07)\times10^{-3}$ and $\beta_1, \dots, \beta_5 = (1.91, 1.15, 0.697, 0.554, 0.498)\times10^{-3}$, and the model is meant for larger n. Over $17 \le n \le 49$ the ratio $\alpha_n/\frac{1}{2}L_n$ has mean 1.033 and standard deviation 0.060 (range 0.940 to 1.178), and $2\beta_n/\frac{1}{2}L_n$ has mean 1.012 and standard deviation 0.060 (range 0.893 to 1.134); the columns α_n, $2\beta_n$, $\frac{1}{2}L_n$ and these two ratios reproduce Table 6 of [208] to the digits shown there. Over the same range $\alpha_n/(2\beta_n)$ has mean 1.021 and standard deviation 0.047, and $\beta_n^2/(\alpha_n\alpha_{n+1})$, which is $\frac{1}{4}$ in the local model, has mean 0.2502 and standard deviation 0.013 (mean 0.256 and standard deviation 0.059 over $1 \le n \le 49$). The cruder rule falls short: α_n and $2\beta_n$ exceed it by the factors 1.286 and 1.26 on average, with standard deviations 0.081 and 0.078, and its ratio to $\frac{1}{2}L_n$ is 0.781, 0.805 and 0.819 at $n = 17$, 33 and 49.

Two cautions apply. The rule $n(L_n) = n$ and the midpoint rule $n(L_n') = n - \frac{1}{2}$ differ by a shift of n of order one, which the asymptotics cannot see, and they move the mean ratios by two per cent, to 1.011 and 0.991 (standard deviations 0.058, ranges 0.913 to 1.148 and 0.875 to 1.097), within the scatter; the data do not decide between them. And the scatter itself, about six per cent, is not explained by this computation: the note attributes it to the discreteness of the zeros, which the smooth density ignores, and that remains to be tested (Problem 12.3). So the local semicircle model predicts the size of the Jacobi parameters to a few per cent over $17 \le n \le 49$ with no fitted constant; it is a good local model of ν at

its accumulation point, but a heuristic, and the agreement is not a proof of an asymptotic law.

	$10^5\times$			ratio to $\frac12 L_n$		$10^5\times$	ratio to $\frac12 L'_n$		$10^5\times$	
n	α_n	$2\beta_n$	$\frac12 L_n$	α_n	$2\beta_n$	$\frac12 L'_n$	α_n	$2\beta_n$	$\frac12\gamma^{-2}_{2n/\pi}$	$\frac{\beta_n^2}{\alpha_n\alpha_{n+1}}$
5	107.3	99.54	84.84	1.27	1.17	94.11	1.14	1.06	61.8	0.2665
9	56.96	51.12	45.73	1.25	1.12	48.70	1.17	1.05	34.5	0.2548
13	30.20	32.52	30.07	1.00	1.08	31.48	0.959	1.03	23.2	0.2611
17	21.71	21.51	21.83	0.995	0.986	22.64	0.959	0.950	17.0	0.2362
21	18.50	16.55	16.82	1.10	0.983	17.34	1.07	0.954	13.3	0.2479
25	15.90	14.58	13.50	1.18	1.08	13.85	1.15	1.05	10.7	0.2588
29	11.25	11.45	11.16	1.01	1.03	11.41	0.986	1.00	8.93	0.2550
33	9.530	9.336	9.424	1.01	0.991	9.615	0.991	0.971	7.59	0.2647
37	8.704	8.163	8.101	1.07	1.01	8.249	1.06	0.990	6.56	0.2463
41	6.998	7.053	7.062	0.991	0.999	7.179	0.975	0.982	5.74	0.2618
45	6.498	6.150	6.228	1.04	0.987	6.323	1.03	0.973	5.08	0.2481
49	5.957	6.172	5.547	1.07	1.11	5.625	1.06	1.10	4.54	0.2738

Table 11.9: The Jacobi parameters of ν against the local semicircle model: α_n and $2\beta_n$; half the right end $\frac12 L_n$ of the local interval from (11.2.13) with the smooth density and $n(L_n) = n$, and the ratios to it; the same with the midpoint rule $n(L'_n) = n - \frac12$; the cruder rule $\frac12\gamma^{-2}_{2n/\pi}$, with $N_0(\gamma_r) = r$; and $\beta_n^2/(\alpha_n\alpha_{n+1})$, which is $\frac14$ in the local model. Computed by `wigfree_jacobi.py`.

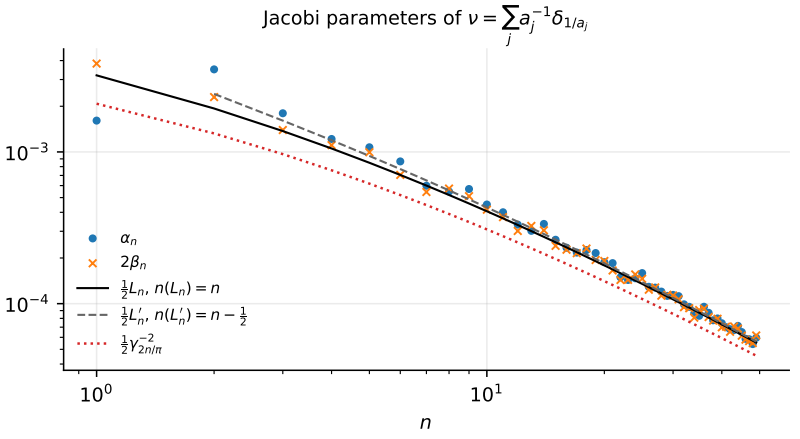


Figure 11.6: The Jacobi parameters α_n and $2\beta_n$ of ν, $1 \le n \le 49$, on logarithmic axes, against $\frac12 L_n$ and $\frac12 L'_n$ from (11.2.13) with the smooth density and against the cruder rule $\frac12\gamma^{-2}_{2n/\pi}$. Computed by `wigfree_jacobi.py` and drawn by `wigfree_plots.py`.

11.2.5 Gamma(2), Laguerre polynomials and finite free convolution

Williams' variable [271] $S_2 = J + J' = \sum_n \frac{2}{\pi^2 n^2}\Gamma_n$, with independent Gamma(2) variables Γ_n, the sum of two independent copies of the hitting time $J = \sum_n \frac{2}{\pi^2 n^2} e_n$ of Section 9.1.1, satisfies $2\xi(s) = \mathbb{E}(\frac{\pi}{2}S_2)^{s/2}$ (9.1.1) [28], and the RH content of this identity lies on its Mellin side (Theorem 7.1.21 and Section 8.2.3). As a check on the normalisation, the cumulants of S_2 are $\kappa_m = 2(m-1)!\,(2/\pi^2)^m\zeta(2m)$, so $\mathbb{E}S_2 = \frac{2}{3}$ and $\mathbb{E}S_2^2 = \frac{8}{15}$, and (9.1.1) at $s = 2$ and 4 gives $2\xi(2) = \frac{\pi}{3}$ and $2\xi(4) = \frac{2\pi^2}{15}$, which are correct. The program `wigfree_gamma2.py` checks (9.1.1) at $s = 2$, 4 and 6 to 1.4×10^{-40}, and at $s = 1$, where $\mathbb{E}(\frac{\pi}{2}S_2)^{1/2} = 2\xi(1) = 1$, through $\mathbb{E}Y^{1/2} = (2\sqrt{\pi})^{-1}\int_0^\infty \lambda^{-3/2}(1-\mathbb{E}e^{-\lambda Y})\,d\lambda$ for $Y \geq 0$. The note asks what the exponent two of the Laplace transform $(\sqrt{2\lambda}/\sinh\sqrt{2\lambda})^2$ of S_2 does, and its answer can be made precise in three ways: through the one dimensional variable J, through a 2×2 random matrix, and through the Laguerre polynomials, the orthogonal polynomials of the gamma laws.

Proposition 11.2.5. *For every complex s,*

$$\mathbb{E}\Big[\Big(\frac{\pi}{2}J\Big)^{s/2}\Big] = \frac{2(1-2^{1-s})\,\xi(s)}{s-1} = s\,\pi^{-s/2}\,\Gamma\Big(\frac{s}{2}\Big)(1-2^{1-s})\,\zeta(s), \qquad (11.2.14)$$

the value at $s = 1$ being $2\xi(1)\log 2 = \log 2$. Equivalently $J \overset{d}{=} V^{-2}S_2$, with V uniform on $[1,2]$ and independent of S_2. The function (11.2.14) *vanishes at the zeros of ξ and at the points $1 + 2\pi ik/\log 2$, $k \neq 0$, of the line $\operatorname{Re} s = 1$, and, unlike the zero set of ξ, its zero set is not symmetric under $s \mapsto 1 - s$.*

Proof. This is the multiplicative relation of Biane, Pitman and Yor between J, their Σ_1, and S_2, their Σ_2 [28, Table 1 and (43) to (45)]; we give the computation. For $\operatorname{Re} p < 0$, $\int_0^\infty \lambda^{-p-1}e^{-\lambda x}d\lambda = \Gamma(-p)x^p$ and $\mathbb{E}e^{-\lambda J} = \sqrt{2\lambda}/\sinh\sqrt{2\lambda}$ give, with $r = \sqrt{2\lambda}$,

$$\mathbb{E}J^p = \frac{2^{p+1}}{\Gamma(-p)}\int_0^\infty \frac{r^{-2p}}{\sinh r}\,dr = \frac{2^{p+2}(1-2^{2p-1})\,\Gamma(1-2p)\,\zeta(1-2p)}{\Gamma(-p)},$$

by $1/\sinh r = 2\sum_{k\geq 0} e^{-(2k+1)r}$, which gives $\int_0^\infty r^{z-1}/\sinh r\,dr = 2(1-2^{-z})\Gamma(z)\zeta(z)$ for $\operatorname{Re} z > 1$. Put $p = s/2$ with $\operatorname{Re} s < 0$. The functional equation $\zeta(1-s) = \pi^{1/2-s}\Gamma(\frac{s}{2})\zeta(s)/\Gamma(\frac{1-s}{2})$, the duplication formula $\Gamma(1-s) = 2^{-s}\pi^{-1/2}\Gamma(\frac{1-s}{2})\Gamma(1-\frac{s}{2})$ and $\Gamma(1-\frac{s}{2})/\Gamma(-\frac{s}{2}) = -\frac{s}{2}$ turn this into $\mathbb{E}J^{s/2} = -2^{1-s/2}s(1-2^{s-1})\pi^{-s}\Gamma(\frac{s}{2})\zeta(s)$, and multiplying by $(\pi/2)^{s/2}$ gives (11.2.14). Both sides are entire, since J has moments of all real orders and the zero of $1-2^{1-s}$ at $s = 1$ cancels the pole of ζ, so the identity holds for every s. Since $\mathbb{E}V^{-s} = \int_1^2 v^{-s}dv = (1-2^{1-s})/(s-1)$, the right side is $\mathbb{E}V^{-s}\,\mathbb{E}(\frac{\pi}{2}S_2)^{s/2}$, and Mellin transforms on a vertical line determine the law of a positive variable. The factor $1-2^{1-s}$ vanishes exactly at $1 + 2\pi ik/\log 2$; on the line $\operatorname{Re} s = 0$, the

image of $\operatorname{Re} s = 1$ under $s \mapsto 1-s$, it equals -1 at the corresponding points $2\pi ik/\log 2$ and ξ has no zeros, so the zero set is not symmetric. □

So the one dimensional variable carries ξ as well, multiplied by the elementary factor $(1-2^{1-s})/(s-1)$, which adds zeros on the line $\operatorname{Re} s = 1$, the first at height $2\pi/\log 2 = 9.0647$, the zeros of the alternating zeta function that Section 10.1.7 meets, and destroys the symmetry. The program checks (11.2.14) at $s = 1$, 2, 4 and 6 (differences 6.6×10^{-23} at $s = 1$ and below 2.3×10^{-41} at the others), together with $\mathbb{E}J = \frac{1}{3}$, $\mathbb{E}J^2 = \frac{7}{45}$ and $\mathbb{E}(\frac{\pi}{2}J)^{1/2} = \log 2$.

PROPOSITION 11.2.6. *Let $z_1, z_2, \dots$ be independent standard complex Gaussian vectors in $\mathbb{C}^2$, with $\mathbb{E}z_n z_n^* = I_2$, let $c_n = 2/(\pi^2 n^2)$ and $W = \sum_n c_n z_n z_n^*$. Then $\operatorname{tr} W \overset{d}{=} S_2$, so that $2\xi(s) = \mathbb{E}(\frac{\pi}{2}\operatorname{tr} W)^{s/2}$, and*

$$\mathbb{E}\det(x - W) = x^2 - 2m_1 x + m_1^2 - m_2, \qquad m_k = \sum_n c_n^k, \quad m_1 = \tfrac{1}{3},\ m_2 = \tfrac{2}{45},$$

with the real roots $m_1 \pm \sqrt{m_2} = 0.122515$ and 0.544152. This is the finite free convolution $\boxplus_2$ of the polynomials $\mathbb{E}\det(x - c_n z_n z_n^*) = x(x - 2c_n)$, and the mixed characteristic polynomial of Marcus, Spielman and Srivastava [153] of the matrices $c_n I_2$.

Proof. $|z_n|^2$ is the sum of two independent standard exponential variables, so $\operatorname{tr} W = \sum_n c_n |z_n|^2 \overset{d}{=} S_2$. For a 2×2 matrix, $\det(x - W) = x^2 - x \operatorname{tr} W + \det W$ and $\det W = \frac{1}{2}((\operatorname{tr} W)^2 - \operatorname{tr} W^2)$. Here $\mathbb{E}\operatorname{tr} W = 2m_1$ and $\mathbb{E}(\operatorname{tr} W)^2 = \operatorname{Var} S_2 + (\mathbb{E}S_2)^2 = 2m_2 + 4m_1^2$, while $\mathbb{E}\operatorname{tr} W^2 = \sum_{n,k} c_n c_k \mathbb{E}|z_n^* z_k|^2 = 6m_2 + 2(m_1^2 - m_2)$, since $\mathbb{E}|z_n|^4 = 6$ and $\mathbb{E}|z_n^* z_k|^2 = \operatorname{tr}(I_2 I_2) = 2$ for $n \neq k$. Hence $\mathbb{E}\det W = m_1^2 - m_2$. The terms $c_n z_n z_n^*$ are independent and unitarily invariant in law, with $\mathbb{E}\det(x - c_n z_n z_n^*) = x^2 - 2c_n x$, so by [154, Theorem 1.2] and bilinearity the expected characteristic polynomial of a finite partial sum is the $\boxplus_2$ convolution of these polynomials; with the coefficient formula of $\boxplus_2$, adding $x(x - 2c)$ to a polynomial $x^2 - Ax + B$ gives $x^2 - (A + 2c)x + B + Ac$, so the constant term is $2\sum_{n<k} c_n c_k = m_1^2 - m_2$ in the limit. The program verifies symbolically that the mixed characteristic polynomial of $c_1 I_2, \dots, c_4 I_2$ is $x^2 - 2m_1 x + m_1^2 - m_2$ with the sums over four terms. □

A Monte Carlo run of `wigfree_gamma2.py`, 2×10^5 draws with the series cut after 300 terms, gives the mean 0.665682 ± 0.0007 of $\operatorname{tr} W$ against 0.665318, the value of $2m_1$ for the first 300 terms, and the mean 0.066221 ± 0.0001 of $\det W$ against 0.066218, that of $m_1^2 - m_2$. In this reading the exponent two of $(\sqrt{2\lambda}/\sinh\sqrt{2\lambda})^2$ is the size of the matrix, the 1×1 model gives J, and the finite free shadow of the Gamma(2) structure is the polynomial of degree two above. The reading is special to the sinh family of [28]. In the cosh family, $\Sigma_h^{\#} = \frac{2}{\pi^2}\sum_n \Gamma_{h,n}/(n - \frac{1}{2})^2$, it is the exponent one whose Mellin transform has a

functional equation, through the Dirichlet L function of the character modulo 4 [28, (39) to (41)], while the Mellin transform of $\Sigma_2^{\#}$ is ξ times a factor of the same kind as in (11.2.14), with zeros on a vertical line [28, Table 1].

Gamma(a) is the orthogonality weight $x^{a-1}e^{-x}$ of the Laguerre polynomials $L_d^{(a-1)}$, and three statements translate scale mixtures of gamma laws into the language of polynomials. In the normalisation of [154] write a polynomial of degree d as $p(x) = \sum_k (-1)^k \binom{d}{k} e_k x^{d-k}$; the multiplicative convolution $\boxtimes_d$ multiplies the normalised coefficients, $e_k(p \boxtimes_d q) = e_k(p) e_k(q)$ [154, Definition 1.4], and by Szegő's theorem it preserves the property of having only nonnegative roots [154, Theorem 1.6].

LEMMA 11.2.7. *Let $a > 0$ and $(a)_k = a(a+1)\cdots(a+k-1)$.*

(i) The reversed Laguerre polynomial $q_a(x) = x^d L_d^{(a-1)}(1/x)$ has d positive simple roots and normalised coefficients $e_k = e_0/(a)_k$, $e_0 = \Gamma(d+a)/(d!\,\Gamma(a))$. Hence $p \boxtimes_d (q_a/e_0)$ has the normalised coefficients $e_k(p)/(a)_k$, and it has only real roots whenever the roots of p are real and of one sign.

(ii) $\{1/(a)_k\}_{k\ge 0}$ is a multiplier sequence: the function

$$\sum_k \frac{x^k}{k!\,(a)_k} = \Gamma(a) x^{(1-a)/2} I_{a-1}(2\sqrt{x})$$

is entire of order $\frac{1}{2}$ with only real negative zeros, at $-j_{a-1,m}^2/4$.

(iii) A density f on $(0,\infty)$ is a scale mixture of Gamma(a) densities if and only if $x^{1-a} f(x)$ is completely monotone; for $a = 2$, if and only if $f(x)/x$ is.

Proof. (i) By [177, (18.5.12)],

$$x^d L_d^{(a-1)}(1/x) = \sum_k (-1)^k \binom{d+a-1}{d-k} \frac{x^{d-k}}{k!},$$

and $\binom{d+a-1}{d-k}/(k!\binom{d}{k}) = \Gamma(d+a)/(d!\,\Gamma(k+a))$, which is $e_0/(a)_k$. The zeros of $L_d^{(a-1)}$ are positive and simple, those of an orthogonal polynomial for the weight $x^{a-1}e^{-x}$ on $(0,\infty)$ [177, Section 18.2(vi)], and the roots of q_a are their reciprocals. The statement on $\boxtimes_d$ is Szegő's theorem for roots of one sign; for nonpositive roots apply it to $(-1)^d p(-x)$, whose normalised coefficients are $(-1)^k e_k(p)$. (ii) The series is that of the modified Bessel function [177, (10.25.2)]. For $x = -y^2/4 < 0$ it equals $\Gamma(a)(y/2)^{1-a} J_{a-1}(y)$, and for $a - 1 > -1$ the zeros of $J_{a-1}(y)/y^{a-1}$ are real [177, Section 10.21(i)]; complex zeros in x would give nonreal zeros in y. A real entire function of order less than one with only real zeros is in the Laguerre Pólya class, and by the theorem of Pólya and Schur [216] a sequence whose exponential generating function is in that class with coefficients of one sign is a multiplier

sequence. (iii) $f(x) = \int_0^\infty \theta^{-a} x^{a-1} e^{-x/\theta}\, \varpi(d\theta)/\Gamma(a)$ says that $x^{1-a} f(x)$ is the Laplace transform of a positive measure in the variable $1/\theta$; conversely Bernstein's theorem writes a completely monotone $x^{1-a} f(x)$ in this form. □

The program checks the coefficients in (i) for $d \le 8$ and $a = \frac{1}{2}, 2, \frac{7}{3}, 6$, finds the roots of $x^8 L_8^{(a-1)}(1/x)$ real with least roots 0.0454871, 0.0406736 and 0.0321239 for $a = \frac{1}{2}, 2, 6$, and checks the Bessel form in (ii) to 3.2×10^{-30} at $x = 3.7$ and its vanishing at $-j_{a-1,1}^2/4$ for $a = \frac{1}{2}, 1, 2, 6.5$ (the points -0.6168502751, -1.445796491, -3.670492661 and -21.88280506).

Let X be the Pólya variable, with density $p = \Phi/\Xi(0)$ for the kernel Φ of (1.2.3) and moment generating function $\mathbb{E}e^{sX} = G(s^2)$ (1.2.4), and write $G(u) = \sum_k b_k u^k$, so that $b_k = \mathbb{E}X^{2k}/(2k)! = M_k/((2k)!\, M_0)$ with $M_k = \int_{\mathbb{R}} v^{2k} \Phi(v)\, dv$, as in Sections 5.1 and 13.3.1.

PROPOSITION 11.2.8. *For $a > 0$ let $G_a(u) = \sum_k b_k u^k/(a)_k$. If RH holds, then for every $a > 0$ the function G_a is in the Laguerre Pólya class with only real negative zeros, and all its Jensen polynomials $\sum_{i=0}^d \binom{d}{i} (n+i)!\, b_{n+i} x^i/(a)_{n+i}$, $d, n \ge 0$, have only real negative zeros. Multiplying the coefficients by $(a)_k$ instead does not preserve real zeros: it turns $(1+u)^2 = 1+2u+u^2$ into $1+2au+a(a+1)u^2$, whose discriminant is $-4a < 0$.*

Proof. Under RH, $G(u) = \prod_j (1+u/\gamma_j^2)$ with $\sum_j \gamma_j^{-2} < \infty$, so G is in the Laguerre Pólya class, and by Pólya's theorem its Jensen polynomials $J_{d,n}(x) = \sum_i \binom{d}{i} (n+i)!\, b_{n+i} x^i$ of Section 13.3.1 have only real zeros. Since $(a)_{n+i} = (a)_n (a+n)_i$, the Jensen polynomial of G_a of degree d and shift n is $1/(a)_n$ times the image of $J_{d,n}$ under the multiplier sequence $\{1/(a+n)_i\}_i$ of Lemma 11.2.7(ii), so it has only real zeros, negative because its coefficients are positive. For $n = 0$, $\binom{d}{i} i!\, d^{-i} \to 1$ gives $\sum_i \binom{d}{i} i!\, b_i x^i/((a)_i d^i) \to G_a(x)$ uniformly on compact sets as $d \to \infty$, so by Hurwitz's theorem G_a has only real negative zeros; since $(a)_k \ge a(k-1)!$, G_a has order at most $\frac{1}{3}$, and it is in the Laguerre Pólya class. The example is a direct computation. □

In terms of Lemma 11.2.7(i), dividing the normalised coefficients by $(a)_k$ is the multiplicative convolution with q_a/e_0, so peeling a Gamma(a) layer, the operation that would pass from the moments of a Gamma(a) scale mixture to those of its mixing law, keeps zeros real, while the inverse operation can destroy them. The note calls the peeling a free Poisson operation; precisely it is $\boxtimes_d$ with the reversed Laguerre polynomial, whose roots are the reciprocals of the Laguerre zeros. The example shows that a step from real zeros of some G_a to real zeros of G would need more than the coefficient map, which does not preserve them; it does not show that no argument could pass in that direction, as the note puts it.

Lemma 11.2.7(iii) makes the Gamma(2) mixture route of [200, Section 12.6], which Section 8.2.3 recalls as excluded, a statement about complete monotonicity,

and the next theorem excludes it for every shape.

THEOREM 11.2.9. *For no $a > 0$ is $|X|$, X^2 or S_2 a scale mixture of Gamma(a) laws.*

Proof. By Lemma 11.2.7(iii) a mixture would make $g(x) = x^{1-a}f(x)$ completely monotone, with f the density. A completely monotone $g \not\equiv 0$ is positive and log convex, so $g(x) \geq g(1)\exp\big((x-1)g'(1)/g(1)\big)$ for $x \geq 1$: it decays at most exponentially. For $v \geq 0$, dropping the negative terms of (1.2.3) and using $e^{2v} \geq 1$ gives

$$\Phi(v) \leq 4\pi^2 e^{9v/2}\exp(-\pi e^{2v})\sum_{n\geq 1} n^4 e^{-\pi(n^2-1)},$$

a constant multiple of $e^{9v/2}\exp(-\pi e^{2v})$. The densities of $|X|$ and X^2 are $2\Phi(v)/\Xi(0)$ and $\Phi(\sqrt{w})/(\Xi(0)\sqrt{w})$, so $v^{1-a}\Phi(v)$ and $w^{1/2-a}\Phi(\sqrt{w})$ would have to be completely monotone, and both decay faster than any exponential. For S_2 the obstruction is at the origin. By Chebyshev's inequality in exponential form, for $\lambda > 0$, $\Pr(S_2 \leq x) \leq e^{\lambda x}\mathbb{E}e^{-\lambda S_2} = e^{\lambda x}\big(\sqrt{2\lambda}/\sinh\sqrt{2\lambda}\big)^2$, and $\lambda = 2/x^2$ gives

$$\Pr(S_2 \leq x) \leq \frac{16\,e^{-2/x}}{x^2(1-e^{-4/x})^2} \leq 17\,x^{-2}e^{-2/x}, \qquad 0 < x \leq 1.$$

A mixture would give, g being nonincreasing and $g(1) > 0$, $f(x) \geq g(1)x^{a-1}$ and $\Pr(S_2 \leq x) \geq g(1)x^a/a$ for $0 < x \leq 1$, which is impossible as $x \downarrow 0$. □

The Hankel test sees the failure at low order. If $Y = \Theta\,\Gamma$ with Γ a Gamma(a) variable independent of $\Theta \geq 0$, then $\mathbb{E}Y^k = (a)_k\mathbb{E}\Theta^k$, so $\mu_k = \mathbb{E}Y^k/(a)_k$ must be a Stieltjes moment sequence, and both $[\mu_{i+j}]_{i,j<N}$ and $[\mu_{i+j+1}]_{i,j<N}$ must be positive semidefinite for every N. Table 11.10 gives the least order N at which each fails to be positive definite, by Cholesky factorisation at 400 digits, with the same orders at 250. The moments of X^2 are M_k/M_0; those of $|X|$ come from a tanh sinh rule, since odd powers of $|v|$ are not smooth at the origin, and agree with the trapezoid values for even powers to 5.4×10^{-212} ($\mathbb{E}|X| = 0.173227179954335$, $\mathbb{E}X^2 = 0.0462099862308379$, $\mathbb{E}|X|^3 = 0.0154158778790498$); those of S_2 follow from its cumulants. The table of [208] is the Hamburger row, the first matrix alone; with the shifted matrix the Stieltjes test fails earlier for $|X|$ at $a = 2$, $\frac{5}{2}$, 6, 12 and 20 and for X^2 at $a = \frac{3}{2}$, 3 and 6.

The theorem also settles which tool applies. Kristiansen proved Steutel's conjecture that scale mixtures of Gamma(2) laws are infinitely divisible [129]; since $x^{\alpha-2}e^{-x}$ is completely monotone for $\alpha \leq 2$, Lemma 11.2.7(iii) makes Gamma(α) itself a scale mixture of Gamma(2) laws, and the theorem covers scale mixtures of Gamma(α) laws for every $\alpha \leq 2$. By Theorem 11.2.9 it cannot be applied to $|X|$, X^2 or S_2. The Gamma(2) structure that is true is Williams' convolution, a GGC and hence infinitely divisible directly.

a	$\frac{1}{2}$	1	$\frac{3}{2}$	2	$\frac{5}{2}$	3	4	6	8	12	20
$\lvert X\rvert$, Stieltjes	2	2	2	2	2	3	3	3	4	4	5
$\lvert X\rvert$, Hamburger	2	2	2	3	3	3	3	4	4	5	6
X^2, Stieltjes	2	3	3	4	4	4	5	5	6	7	9
X^2, Hamburger	2	3	4	4	4	5	5	6	6	7	9
S_2, both	2	2	2	2	2	2	2	3	4	6	12

Table 11.10: The first order N at which the moments $\mu_k = \mathbb{E}Y^k/(a)_k$ fail the Hankel test for a Gamma(a) scale mixture, for $Y = |X|$, X^2 and S_2: the Hamburger test uses $[\mu_{i+j}]_{i,j<N}$ alone, the Stieltjes test also $[\mu_{i+j+1}]_{i,j<N}$. Computed by `wigfree_gamma2.py`.

11.2.6 The heat flow and finite free convolution

Remark 8.4.3 identifies the forward heat flow on polynomials with finite free convolution by a Hermite polynomial and, as the degree grows, with free convolution by a semicircle law. This subsection states the finite facts with proofs, in the normalisation in which the time is the variance of the semicircle law, and follows them on the example of [208]. For a real polynomial p of degree $d \geq 2$ write $p_t = e^{-(t/2)D^2}p$, a finite sum, and

$$q_t(x) = e^{-(t/2)D^2}x^d = t^{d/2}H_d\big(x/\sqrt{t}\big), \qquad t > 0,$$

with $q_0 = x^d$ and $H_d = e^{-D^2/2}x^d$ the Hermite polynomial of Remark 8.4.3, whose zeros are real and simple, those of an orthogonal polynomial for the weight $e^{-x^2/2}$. The time t is twice the time of Section 8.4.1, where $\partial_t H_t = -\partial_z^2 H_t$.

Proposition 11.2.10. *Let p be a real polynomial of degree $d \geq 2$ and $s, t \geq 0$.*

(i) $p_t = p \boxplus_d q_t$.

(ii) $q_s \boxplus_d q_t = q_{s+t}$.

(iii) If p has only real roots, then so has p_t, and the largest root of p_t is at most the largest root of p plus $\sqrt{t}\,h_d$, where h_d is the largest zero of H_d.

(iv) While the roots $z_1(t), \dots, z_d(t)$ of p_t are simple, $\dot{z}_k = \sum_{j\neq k}(z_k - z_j)^{-1}$.

Proof. (i) In operator form, $p \boxplus_d q = \hat{p}(D)\hat{q}(D)x^d$ when $p = \hat{p}(D)x^d$ and $q = \hat{q}(D)x^d$ [154, Definition 1.1 and (2)]. For $q = q_t$ the operator $\hat{q}(D)$ acts on polynomials of degree at most d as $e^{-(t/2)D^2}$, and operators in D commute, so $p \boxplus_d q_t = e^{-(t/2)D^2}\hat{p}(D)x^d = p_t$. (ii) is (i) for $p = q_s$, since $e^{-(t/2)D^2}e^{-(s/2)D^2} = e^{-((s+t)/2)D^2}$; it is the identity $a^d H_d(x/a) \boxplus_d b^d H_d(x/b) = c^d H_d(x/c)$, $c^2 = a^2 + b^2$, of [154, Section 1.2]. (iii) The roots of q_t are $\sqrt{t}$ times those of H_d, and by Walsh's theorem [154, Theorem 1.3] the convolution $p \boxplus_d q$ of two polynomials of degree d with only real roots has only real roots, the largest at most the sum

of the largest roots of p and q. (iv) Differentiate $p_t(z_k(t)) = 0$ in t and use $\partial_t p_t = -\frac{1}{2}p_t''$: $\dot z_k = \frac{1}{2}p_t''(z_k)/p_t'(z_k)$. Writing $p_t = (x - z_k)r$ gives $p_t'(z_k) = r(z_k)$ and $p_t''(z_k) = 2r'(z_k)$, so that $\dot z_k = r'(z_k)/r(z_k) = \sum_{j\neq k}(z_k - z_j)^{-1}$. □

Part (iv) is the motion of the zeros in Section 8.4.1, in the time of this subsection. For a conjugate pair $x \pm iy$, $y > 0$, the partner contributes $-1/(2y)$ to $\dot y$, and each real root r contributes $\operatorname{Im}(x + iy - r)^{-1} = -y/((x - r)^2 + y^2) < 0$: the pair is drawn to the axis by its partner and pulled further by every real root. On its own, $(x - 1)^2 + 0.64$ becomes $(x - 1)^2 + 0.64 - t$, which reaches the axis at $t = 0.64$; in the units of H_t this is $0.8^2/2$, the time of de Bruijn's theorem on strips (Section 8.4.1) for zeros in $|\operatorname{Im} z| \le 0.8$. The example of the note adds four real roots,

$$p(x) = (x + 3)(x + 1)\big(x - \tfrac{1}{2}\big)(x - 2)\big((x - 1)^2 + 0.64\big).$$

The discriminant of p_t in x is a polynomial of degree 15 in t whose only root in $(0, 2)$ is $t^* = 0.3203291143987806$. Before t^* the roots are distinct, so the pair stays off the axis, and since $p_t = e^{-((t-t^*)/2)D^2}p_{t^*}$ for $t > t^*$, part (iii) keeps all roots real afterwards: the pair meets the axis at t^*, at the point 1.25743, in about half the time it needs on its own. The largest imaginary part of the roots is 0.8, 0.583746, 0.384079 and 0.144671 at $t = 0$, 0.1, 0.2 and 0.3; at $t = 0.35$ the pair has split into the real roots 1.11128 and 1.44908; and the real roots repel each other, the outer ones -3 and 2 having moved to -3.45179 and 2.74187 at t^* (Figure 11.7). The program `wigfree_heat.py` verifies (i) and (ii) symbolically for a monic p of degree 5 with generic coefficients, together with $q_t = t^{5/2}H_5(x/\sqrt{t})$, and computes the discriminant exactly.

The Hermite polynomial is the finite free counterpart of the semicircle law: H_d is the expected characteristic polynomial of a symmetric Gaussian matrix [154, Section 1.2], and the roots of $H_d(\sqrt{d}\,x)$, the roots of $q_{1/d}$, are close to the quantiles x_k of σ_1 given by $\sigma_1([-2, x_k]) = (k - \frac{1}{2})/d$. The largest deviation is 0.053008, 0.036497, 0.024531, 0.016224 and 0.010607 for $d = 20$, 40, 80, 160 and 320, and 0.030713, 0.022592, 0.015797, 0.010723 and 0.007140 without the two extreme roots. The largest deviation is at the edge, and its products with $d^{2/3}$, from 0.3906 to 0.4962, grow slowly, as they should at the edge scale $d^{-2/3}$ of the extreme zeros; the largest root for $d = 320$ is 1.951473. In the limit $d \to \infty$ finite free convolution becomes free convolution [6], [154, Section 1.2]: if the root distributions of polynomials p_d with only real roots tend to a law μ in moments, those of $p_d \boxplus_d q_{t/d}$ tend to $\mu \boxplus \sigma_t$. So the polynomial flow run for the time t/d becomes free convolution with σ_t, the identification of Remark 8.4.3.

For H_t the identification is a heuristic. The motion of the zeros is the same, up to the factor two in the time (Section 8.4.1), but H_t has infinitely many zeros, with a density that grows logarithmically, while the limit theorem concerns the distribution of all d roots of a polynomial after scaling; for H_t it would have to be

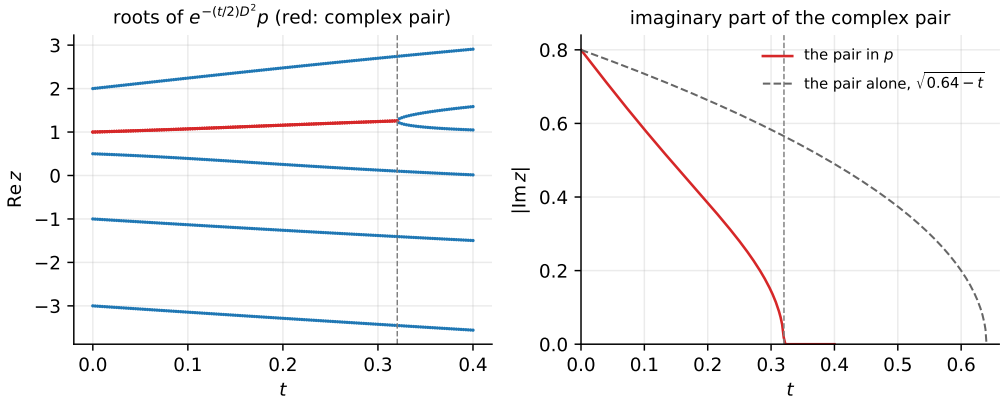


Figure 11.7: The flow $e^{-(t/2)D^2}p$ for p with the roots -3, -1, $\frac{1}{2}$, 2 and $1 \pm 0.8i$. Left: the real parts of the six roots for $0 \le t \le 0.4$; the complex pair has one real part until it meets the axis at $t^* = 0.3203$ (dashed), where its two roots split. Right: the imaginary part of the pair, against $\sqrt{0.64 - t}$ for the pair on its own, which reaches the axis at $t = 0.64$. Computed by `wigfree_heat.py` and drawn by `wigfree_plots.py`.

applied to windows of zeros, with the zeros outside a window acting as an external field, and no theorem of this kind is known. De Bruijn's theorem that real zeros of H_t stay real forward in time [65] is the counterpart of (iii), proved directly. In Thorin terms the forward flow is the path of clocks R_t of Section 8.4.1, in the time of H_t: for $t \ge \Lambda$, and so for $t \ge 0.2$ [196], R_t is a GGC whose Thorin atoms move with the zeros (Theorem 7.2.15), and since $\Lambda \ge 0$ [224], RH is the statement that the Thorin measure is already positive at $t = 0$, that is $\Lambda = 0$.

The Jensen polynomials of Section 13.3.1 meet the Hermite polynomials directly. Griffin, Ono, Rolen and Zagier proved that for each degree d the Jensen polynomials $J_{d,n}$, after an affine change of variable and a normalisation depending on n, tend to the Hermite polynomial of degree d as $n \to \infty$, and this is how they obtain real zeros for all large shifts [84]. In the language of this subsection the Jensen polynomials of large shift are asymptotically finite free Gaussian; Problem 12.1 asks for an exact finite free structure at every shift.

These limits see the zeros only through their distribution. Moving $o(d)$ of the d roots changes no limit law, so a single nonreal pair, or any number of them that is $o(d)$, is invisible to free convolution limits, and these limits cannot decide RH by themselves [208]. What free probability contributes is structure: the transforms and classes of Sections 11.2.1 to 11.2.3, the local model of Section 11.2.4, and the exact finite identities of this subsection and the previous one.

11.3 The Dedekind eta analogue at the basepoint one

At the basepoint one the pole of ζ and the primes act together: the pole term and the prime atoms of the Lévy form both have infinite mass on $(1,\infty)$, the form converges only as a compensated limit, and quasi infinite divisibility is lost (Section 11.1.3). The L function of the real character χ_{12} modulo 12, to which Section 8.5 carried the pipeline of the book, has no pole. Its completed function is entire and is a Mellin transform of the Dedekind eta function, whose product formula makes the kernel positive, and at the basepoint two its sine squared decomposition loses the pole piece while the prime weights take both signs (Section 8.5.3). We repeat for $L(s,\chi_{12})$ the calculations that Chapter 9 made for ζ at the basepoint one: the Thorin measure and the tail of the shifted clock (Section 11.3.1); the Lévy form, its conversion by the sine squared lemma and its extension below the basepoint one, together with the question of Section 11.1 whether the tilted variable is quasi infinitely divisible (Section 11.3.2); and the hitting time with its exact Brownian boundary (Section 11.3.3). Together they redraw for $L(s,\chi_{12})$ the part of the map of Section 11.1.7 at the basepoint one.

Every step goes through with the same structure, and the constants change only through the denser zeros. The comparison of the two examples in Section 11.3.4 therefore separates what in the example of ζ comes from the pole from what comes from the zeros, and it shows that removing the pole does not move the boundary of what is proved. The scripts are in the folder `basepoint_one/eta` of the archive, and we name them at each table and figure.

11.3.1 The completed L function and its Thorin measure

We use the completed function $\xi(s,\chi_{12}) = (12/\pi)^{s/2}\Gamma(s/2)L(s,\chi_{12})$, its kernel K and the variable Y with density $K/\xi(\frac12,\chi_{12})$ of Section 8.5.1; the function satisfies $\xi(s,\chi_{12}) = \xi(1-s,\chi_{12})$ and is entire as it stands, with no factor like the $\frac12 s(s-1)$ of ξ: $L(s,\chi_{12})$ has no pole to remove. We computed its zeros on the critical line by sign changes of $\xi(\frac12+it,\chi_{12})e^{\pi t/4}$ with `zeros12.py`, finding 297 below height 340; Section 8.5.1 lists the first of them. In this section $b_\alpha = (\xi'/\xi)(\alpha,\chi_{12})$ and $v_\alpha(\theta) = \mathrm{Re}(\xi'/\xi)(\alpha+i\theta,\chi_{12})$, and φ_α, T_1, $\widetilde{T}$, Π, A, A_α, Q and $a_k = -(\rho_k-\frac12)^2$ are formed with $\xi(s,\chi_{12})$ and its zeros $\rho=\beta+i\gamma$ in place of ξ. So is the tilted eta variable V_α, the Esscher tilt of Y by $e^{(\alpha-\frac12)Y}$, with $\mathbb{E}e^{sV_\alpha} = \xi(\alpha+s,\chi_{12})/\xi(\alpha,\chi_{12})$. The constants we need, computed by `base.py`, are

$$L(1,\chi_{12}) = \frac{\log(2+\sqrt3)}{\sqrt3} = 0.760346, \qquad L'(1,\chi_{12}) = 0.362495,$$

$$b_1 = \frac{\xi'}{\xi}(1,\chi_{12}) = \frac12\log\frac{12}{\pi} + \frac12\psi\Big(\frac12\Big) + \frac{L'}{L}(1,\chi_{12}) = \sum_\rho \frac1\rho = 0.1650833123,$$

where ψ is the digamma function; b_1 is about seven times the value for ζ. The Taylor coefficients of $\xi(\frac{1}{2}+s,\chi_{12})/\xi(\frac{1}{2},\chi_{12})$ give the power sums $p_1=\sum_k a_k^{-1}=0.1664618$ and $p_2=\sum_k a_k^{-2}=0.0055985$, which are $\sum_{\gamma>0}\gamma^{-2}$ and $\sum_{\gamma>0}\gamma^{-4}$ under GRH; p_1 is the mean of the centre clock.

At the basepoint one the Thorin measure is positive with no hypothesis, exactly as for ζ in Proposition 9.1.1, and it tracks the zero count in the same way.

PROPOSITION 11.3.1. *The function $\varphi_1(u)=\xi(1,\chi_{12})/\xi(1+\sqrt{u},\chi_{12})$ is the Laplace transform of a GGC with no drift, with Thorin density in $\theta=\sqrt{z}$*

$$\frac{1}{\pi}v_1(\theta)=\frac{1}{\pi}\operatorname{Re}\frac{\xi'}{\xi}(1+i\theta,\chi_{12})=\frac{1}{\pi}\sum_{\rho}\frac{1-\beta}{(1-\beta)^2+(\theta-\gamma)^2}>0,$$

the sum over the nontrivial zeros $\rho=\beta+i\gamma$ of $L(s,\chi_{12})$ in conjugate pairs. Positivity is unconditional because $L(1+it,\chi_{12})\neq 0$.

Proof. The proof is that of Proposition 9.1.1, with the Hadamard product of the entire function $\xi(s,\chi_{12})$ in place of that of ξ. □

The minimum on $[0,60]$ is at the origin, $v_1(0)/\pi=b_1/\pi=0.0525477$, and the cumulative mass tracks the zero count: $\int_0^{60}v_1/\pi=35.81$ against $N_{\chi_{12}}(60)=36$, where $N_{\chi_{12}}(T)$ counts the zeros with $0<\gamma\le T$ as in Section 8.5.1. Table 11.11, computed by `base.py`, compares direct values with the Poisson sum over the 297 computed zeros; the nearly constant difference of 1.11×10^{-3} is the contribution of the zeros above height 340. Figure 11.8 compares the density with that of ζ: on this window the zeros of $L(s,\chi_{12})$ are more than twice as dense, and its density is correspondingly higher. As for ζ, $\varphi_1(u)=1-b_1\sqrt{u}+O(u)$, so T_1 has infinite mean and $P(T_1>x)\sim b_1(\pi x)^{-1/2}=0.09314\,x^{-1/2}$.

θ	$v_1(\theta)/\pi$ from $\xi'/\xi(\cdot,\chi_{12})$	from 297 zeros, $\beta=\frac{1}{2}$
0	0.0525477	0.0514359
3.8046276	0.6889076	0.6877958
5	0.1852757	0.1841637
10	0.2694467	0.2683341
30	0.7445558	0.7434362
60	0.7298854	0.7287413

Table 11.11: The Thorin density v_1/π of $\xi(1,\chi_{12})/\xi(1+\sqrt{u},\chi_{12})$, computed from $\xi'/\xi(\cdot,\chi_{12})$ and from the Poisson sum over the 297 zeros of $L(s,\chi_{12})$ below height 340 with $\beta=\frac{1}{2}$; the difference is the contribution of the zeros above height 340.

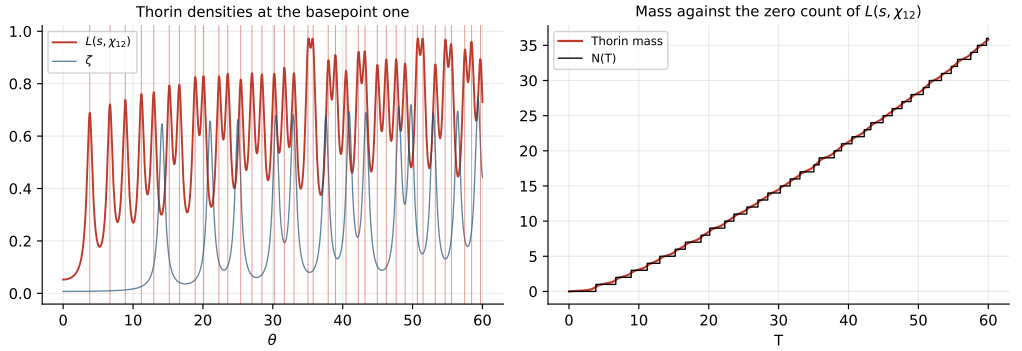


Figure 11.8: Left: the Thorin densities at the basepoint one for $L(s,\chi_{12})$ and for ζ on $[0,60]$, with the ordinates of the zeros of $L(s,\chi_{12})$ (vertical lines). Right: the cumulative Thorin mass for $L(s,\chi_{12})$ against its zero count $N_{\chi_{12}}(T)$, written $N(T)$ in the plot. Drawn by `figs.py` of `basepoint_one/eta`.

11.3.2 The Lévy form without a pole, at and below the basepoint one

The Lévy form of Proposition 9.1.3 has an exact analogue at the basepoint one, with no pole to compensate and with prime atoms of both signs.

Proposition 11.3.2. *For* $\operatorname{Re} s > 0$,

$$\log \frac{\xi(1+s,\chi_{12})}{\xi(1,\chi_{12})} = b_1 s + \int_0^\infty (e^{-sx} - 1 + sx)\,\Pi(dx),$$

where, with $\Lambda_{\chi_{12}}(n) = \chi_{12}(n)\Lambda(n)$,

$$\Pi(dx) = \frac{e^{-x}}{x(1-e^{-2x})}\,dx + \sum_{n\geq 2} \frac{\Lambda_{\chi_{12}}(n)}{n\log n}\,\delta_{\log n}(dx),$$

the prime sum converging conditionally, as a limit over $n \leq X$.

Proof. Binet's formula [177, §5.9] for $\log\Gamma(\frac{1}{2}+\frac{s}{2})$, in the integrated form of the proof of Proposition 9.1.3 with $a = \frac{1}{2}$, gives the gamma density; the factor $(12/\pi)^{s/2}$ and the linear terms combine into b_1; and for $\operatorname{Re} s > 0$ the Euler product gives

$$\log L(1+s,\chi_{12}) = \sum_n \frac{\Lambda_{\chi_{12}}(n)}{n\log n}\,e^{-s\log n}.$$

The linear coefficient is $-\sum_n \Lambda_{\chi_{12}}(n)/n = \frac{L'}{L}(1,\chi_{12})$, which converges by the prime number theorem for arithmetic progressions [63, Chapter 20], in the form $\psi(y,\chi_{12}) = O(y\,e^{-c\sqrt{\log y}})$, where $\psi(y,\chi_{12}) = \sum_{n\leq y}\Lambda_{\chi_{12}}(n)$ is the Chebyshev function of χ_{12}. □

The form differs from that of ζ in two ways. There is no pole, so no compensating measure: the prime atoms converge on their own at the basepoint one. And the measure is signed for a different reason. For ζ the gamma density minus the pole term dx/x is negative beyond $x = 0.281$ and the prime atoms are positive; here the gamma density $e^{-x}/(x(1-e^{-2x}))$ is positive everywhere and the atoms carry the sign of $\chi_{12}(n)$, the first negative atom being $-\frac{1}{5}$ at $\log 5$, with canonical weight $-\frac{\log 5}{5}$ (Figure 11.9, left). Table 11.12, computed by `arith.py`, checks the form.

The tilted eta variable V_1, with moment generating function $\xi(1+s,\chi_{12})/\xi(1,\chi_{12})$, is not quasi infinitely divisible. Its positive and its negative atoms each have infinite total mass, since $\sum_{\chi_{12}(p)=1} 1/p$ and $\sum_{\chi_{12}(p)=-1} 1/p$ both diverge by Dirichlet's theorem, and they cancel only conditionally, so the argument of Proposition 11.1.3(3) applies with $\xi(1+s,\chi_{12})$ in place of $\xi(1+s)$, as the discussion after its proof explains. Since an infinitely divisible law is quasi infinitely divisible, V_1 is not infinitely divisible either. At every basepoint $\alpha > 1$, by contrast, V_α is quasi infinitely divisible: there the atoms of each sign have finite total mass, and the Euler product writes its characteristic function as that of an infinitely divisible gamma piece times a quotient of two compound Poisson characteristic functions.

s	X	$b_1 s$	gamma piece	primes	total	exact
0.5	10^5	0.08254	0.121808	−0.082565	0.121785	0.122329
0.5	10^7	0.08254	0.121808	−0.081931	0.122419	0.122329
1	10^7	0.16508	0.409141	−0.254214	0.320011	0.320099
2	10^7	0.33017	1.270363	−0.689198	0.911332	0.911001

Table 11.12: The Lévy form of Proposition 11.3.2 against $\log(\xi(1+s,\chi_{12})/\xi(1,\chi_{12}))$, with primes up to X. The prime piece stays of order one; there is nothing to cancel.

The sine squared lemma in the form of Section 9.1.4, with $\int_0^\infty \log(1+s^2/\theta^2)\,d\theta = \pi s$, converts the Lévy form, with $s = \sqrt{u}$, into

$$v_1(\theta) = b_1 + A(\theta) + Q(\theta),$$

with

$$A(\theta) = \tfrac{1}{2}\big[\operatorname{Re}\psi(\tfrac{1+i\theta}{2}) - \psi(\tfrac{1}{2})\big], \qquad Q(\theta) = \sum_n \frac{\Lambda_{\chi_{12}}(n)}{n}\, 2\sin^2\Big(\frac{\theta\log n}{2}\Big).$$

The drift becomes the stable one half factor $e^{-b_1\sqrt{u}}$, the gamma factor the smooth positive growth A, and the primes the oscillating Q, now without any divergent partner. With primes to 10^7 the formula gives 2.16385 at $\theta = \gamma_1$ against the exact 2.16427, and 0.84633 at $\theta = 10$ against 0.84649. Figure 11.9 (right) shows the

three pieces: the prime piece ranges over both signs, and the positivity of the sum comes from the drift and the gamma factor, as Section 8.5.3 found at the basepoint two.

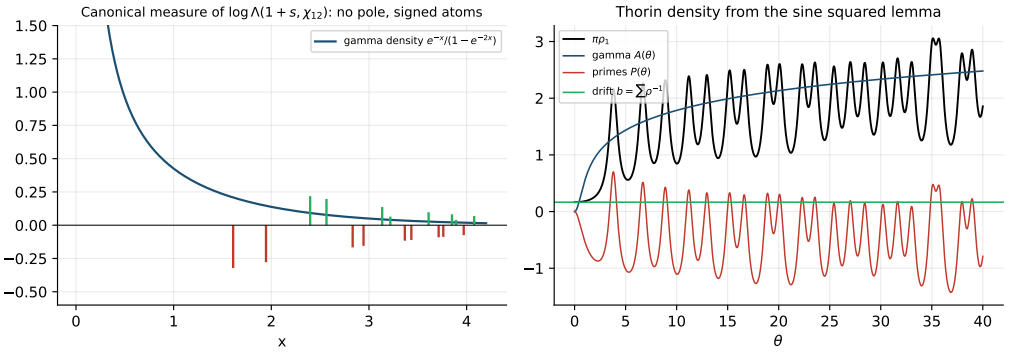


Figure 11.9: Left: the canonical measure $x\Pi(dx)$ of $\log \xi(1+s, \chi_{12})$, written with $\Lambda(1+s, \chi_{12})$ in the plot: the positive gamma density $e^{-x}/(1-e^{-2x})$ and the prime atoms $\Lambda_{\chi_{12}}(n)/n$ at $\log n$, of both signs, the positive ones drawn upward and the negative ones downward; there is no pole. Right: $v_1(\theta)$ on $[0, 40]$, written $\pi\rho_1$ in the plot, as the sum of the drift b_1, written b, the gamma piece $A(\theta)$ and the prime piece $Q(\theta)$, written $P(\theta)$ in the plot. Drawn by `figs.py` of `basepoint_one/eta`.

Below the basepoint one, at $\alpha = 1 - \varepsilon$, the converted Thorin density is π^{-1} times

$$b_\alpha + A_\alpha(\theta) + \lim_{X\to\infty} \sum_{n\le X} \frac{\Lambda_{\chi_{12}}(n)}{n^\alpha}\, 2\sin^2\Big(\frac{\theta \log n}{2}\Big),$$

with $A_\alpha(\theta) = \frac{1}{2}[\operatorname{Re}\psi(\frac{\alpha+i\theta}{2}) - \psi(\frac{\alpha}{2})]$. Exactly as for ζ in Proposition 9.1.6, the limit exists and gives the positive Thorin density v_α/π when $L(s, \chi_{12})$ has no zeros with $\operatorname{Re} s > \alpha'$ for some $\alpha' < \alpha$, and conversely its existence for $\operatorname{Re} s > 0$ in the Lévy form forces the absence of zeros with $\operatorname{Re} s > \alpha$. So the extension below one is, up to the boundary case, a zero free half plane for $L(s, \chi_{12})$, and at the centre it is GRH for χ_{12}. The error at cutoff X is given by the same explicit formula,

$$\sum_\rho \Big[-\frac{X^{\rho-\alpha}}{\rho-\alpha} + \frac{X^{\rho-\alpha+i\theta}}{2(\rho-\alpha+i\theta)} + \frac{X^{\rho-\alpha-i\theta}}{2(\rho-\alpha-i\theta)}\Big] + O(X^{-\alpha}\log X),$$

because $\psi(y, \chi_{12}) = -\sum_\rho y^\rho/\rho + O(\log y)$ has no main term [63, Chapter 19].

The absence of a main term changes the numerics. For ζ the prime piece and the pole piece each grow like $X^{1-\alpha}/(1-\alpha)$ and must cancel; for $L(s, \chi_{12})$ the prime piece stays bounded. At $\alpha = 0.6$, $\theta = \gamma_1$ and $X = 10^7$ the prime piece is 6.07, where for ζ the two pieces are about 1537 each. Table 11.13, computed by `arith.py`, shows the errors, which decay like $X^{1/2-\alpha}$, and what remains after

subtracting the explicit formula over the 297 computed zeros. The correction removes most of the error at $\theta = \gamma_1$, by factors between 17 and 2800; at $\theta = 10$, between zeros, it helps less, because 297 zeros are fewer than the 1700 used for ζ and the missing zeros above height 340 dominate what remains. At the centre and the first ordinate the sum grows like $\log X$, taking the values 6.62, 11.69 and 16.77 at $X = 10^3$, 10^5 and 10^7: the atom δ_{γ_1} forming (Figure 11.10).

α	θ	X	prime piece	error	error after zero correction
1	γ_1	10^5	0.6964	-5.55×10^{-3}	4.2×10^{-6}
1	γ_1	10^7	0.7016	-4.15×10^{-4}	2.5×10^{-5}
0.75	γ_1	10^5	2.0933	-2.13×10^{-1}	1.4×10^{-4}
0.75	γ_1	10^7	2.2471	-5.92×10^{-2}	1.4×10^{-3}
0.6	γ_1	10^5	4.8292	-3.10	1.1×10^{-3}
0.6	γ_1	10^7	6.0670	-1.86	1.6×10^{-2}
0.6	10	10^7	-2.4525	-5.68×10^{-2}	2.6×10^{-2}

Table 11.13: The sine squared formula below the basepoint one for $L(s, \chi_{12})$, with primes up to X: the prime piece, the error against $v_\alpha(\theta)$, and the error left after subtracting the explicit formula over the 297 computed zeros; $\gamma_1 = 3.8046276$.

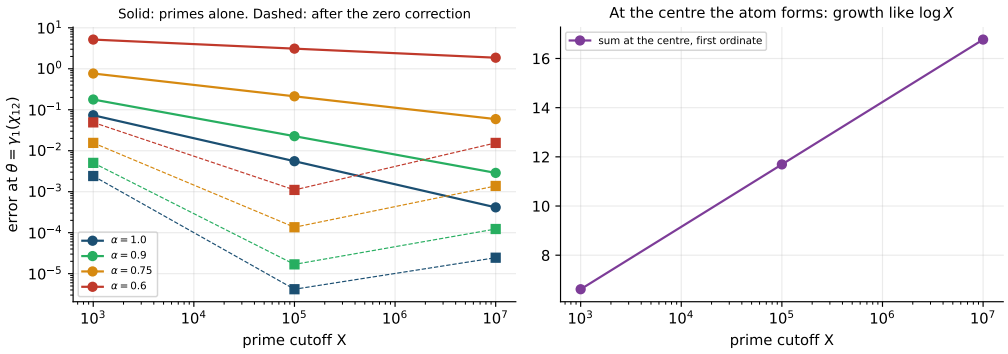


Figure 11.10: Left: absolute error at $\theta = \gamma_1 = 3.8046$ for $L(s, \chi_{12})$ against the prime cutoff X, for $\alpha = 1$, 0.9, 0.75 and 0.6, from primes alone (solid) and after the zero correction over 297 zeros (dashed). Right: at the centre and the first ordinate the sum grows like $\log X$, the atom of the GRH measure forming. Drawn by `figs.py` of `basepoint_one/eta`.

11.3.3 The hitting time and its boundary

The identity $(\frac{1}{2}+\sqrt{u})^2 = \frac{1}{4}+u+\sqrt{u}$ gives, as for ζ,

$$\varphi_1(u) = \frac{\varphi_{1/2}(\frac{1}{4}+u+\sqrt{u})}{\varphi_{1/2}(\frac{1}{4})}, \qquad \varphi_{1/2}(u) = \frac{\xi(\frac{1}{2},\chi_{12})}{\xi(\frac{1}{2}+\sqrt{u},\chi_{12})}.$$

Whenever the centre clock $\varphi_{1/2}$ is the Laplace transform of a probability law, and in particular under GRH for χ_{12}, this reads $T_1 \stackrel{d}{=} \widetilde{T} + H_{\widetilde{T}/\sqrt{2}}$, where $\widetilde{T}$ is the centre clock tilted by $e^{-T/4}$ and H_a is the first passage time of Brownian motion to the level a, as in Theorem 9.2.2. Under GRH, $\widetilde{T} = \sum_k e_k/(\gamma_k^2+\frac{1}{4})$ with independent standard exponentials e_k; if the zeros are also simple, $\widetilde{T}$ is Kent's hitting time for a Krein string with spectrum $\{|\rho|^2\}$ [122], and T_1 is the time for that string diffusion, run on the local time clock of a sticky Brownian motion, to reach its level, as in Corollary 9.2.3. As for ζ, T_1 itself is not a hitting time of the kind in Kent's theorem, because its Thorin measure is absolutely continuous (Proposition 11.3.1).

We simulated 3×10^5 draws of T_1 with `hit.py`, taking $\widetilde{T}$ from this series under GRH: the terms of all 297 computed zeros, and for the zeros above them a gamma variable matched to the tail sums of p_1 and p_2. The sample mean of $\widetilde{T}$ is 0.165108 against $b_1 = 0.165083$, and the empirical Laplace transform agrees with $\varphi_1(u) = \xi(1,\chi_{12})/\xi(1+\sqrt{u},\chi_{12})$ within 1.1 standard errors at $u = 0.1$, 0.5, 1, 5 and 20 (for instance 0.725800 ± 0.000440 against 0.726077 at $u = 1$, and 0.042364 ± 0.000101 against 0.042340 at $u = 20$).

We computed the Brownian boundary $b(t)$ whose crossing time has the law of T_1 from the Volterra equation (8.2.1), with the distribution function from Talbot inversion for $t < 0.12$ and, under GRH as above, from the subordination representation beyond, and the complementary form of the equation once the survival probability falls below one half; `hit.py` solves it and `hit2.py` adds the tangent boundary and the checks. The time scale is about seven times that of ζ, the ratio of the means of the two centre clocks. In the small time zone the boundary starts at the reflection level 1.008 at $t = 0.036$ and falls to 0.611 at $t = 0.08$; through the bulk it falls to 0.400 at $t = 0.12$ and 0.127 at $t = 0.2$, where the survival probability is one half; the tangent approximation agrees to 10^{-3} at $t = 0.08$ and 5×10^{-3} at $t = 0.12$, then drifts, to 0.102 at $t = 0.2$. In the right tail the exact boundary flattens at -0.2747 (-0.26030 at $t = 1$, -0.27461 at $t = 100$, -0.27471 at $t = 1000$), the Brownian form of the stable one half tail. The tangent approximation overshoots, to -0.370 at $t = 0.4$ against the exact -0.166, and has no solution beyond $t = 1.83$, for the reason explained in Section 9.2.4: it cannot represent survivors that have wandered below the boundary.

Wald's identity fails in the same way as for ζ, with the same proviso. The boundary is continuous and, as computed, bounded up to $t = 1000$; if it is bounded on $[1,\infty)$, which here too is observed, not proved, Lemma 9.2.4 and the stable

tail give $\mathbb{E}\, b(T_1) = b_1/\sqrt{2} = \sum_\rho \rho^{-1}/\sqrt{2} = 0.116732$; integrating the computed boundary against the law of T_1 gives 0.116749. We simulated 1.5×10^5 Brownian paths against the exact boundary on $[0, 5]$. Their survival function is within Kolmogorov distance 0.0014 of the exact one (Figure 11.11), and the simulated depth of the survivors below the boundary, $\mathbb{E}[(b(t) - B_t)\mathbf{1}\{H_b > t\}]$, which tends to $\mathbb{E}\, b(T_1)$ by optional stopping, is 0.119, 0.118 and 0.118 at $t = 1$, 2 and 5.

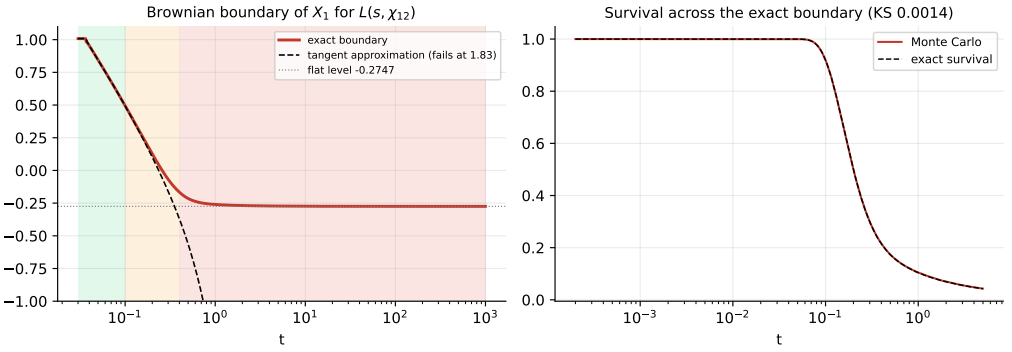


Figure 11.11: Left: the exact Brownian boundary of T_1 for $L(s, \chi_{12})$, written X_1 in the plot, over the three zones of Section 8.2.2.2 (shaded), the tangent approximation, which fails at $t = 1.83$, and the flat level -0.2747. Right: survival of 1.5×10^5 simulated Brownian paths across the exact boundary against the exact survival function; the Kolmogorov distance is 0.0014. Drawn by `figs.py` of `basepoint_one/eta`.

11.3.4 The two examples compared

Table 11.14 sets the two basepoint one examples side by side. Structurally the eta case is the cleaner of the two: its kernel is a positive product, it has no pole, its Lévy form converges at the basepoint one without compensation, and below the basepoint one its prime sums stay bounded, so the computation is far better conditioned. Nothing essential becomes easier, however. The negativity that ζ concentrates in the pole piece is carried here by the signs of $\chi_{12}(p)$; the extension of the Lévy form below the line $\operatorname{Re} s = 1$ is equivalent to a zero free half plane for $L(s, \chi_{12})$; and, under GRH for χ_{12}, the probabilistic picture is the same, with every constant scaled by the denser zeros: a stable one half subordination of the centre clock and, as computed, a boundary that flattens and, if it stays bounded, a Wald defect equal to $\sum_\rho \rho^{-1}/\sqrt{2}$.

The comparison also closes the study of the basepoint one begun in Chapter 9, and it confirms the two conclusions that fix the shape of the path to RH. First, at the basepoint one the constructions for the reciprocal Xi function are, with two exceptions, unconditional and explicit: the Thorin measure is the Poisson

	ζ	$L(s,\chi_{12})$
kernel	theta series	$\eta(ix)$, a positive product
pole	at $s=1$	none
first zero	14.134725	3.804628
drift $b_1=\sum_\rho \rho^{-1}$	0.0230957	0.1650833
tail constant $b_1/\sqrt{\pi}$	0.01303	0.09314
mean of the centre clock $\sum_k a_k^{-1}$	0.0231050	0.1664618
Lévy measure, continuous part	gamma minus pole, negative beyond 0.281	gamma only, positive
Lévy measure, prime atoms	all positive	signs of $\chi_{12}(n)$
compensation at $\alpha=1$	primes against dx/x (prime number theorem)	none needed
prime piece at $\alpha=0.6$, $X=10^7$, $\theta=\gamma_1$	about 1537, cancelled by the pole	6.07
flat level of the exact boundary	-0.19277	-0.27471
$\mathbb{E}\,b(T_1)=b_1/\sqrt{2}$	0.016331	0.116732
tangent approximation fails at	$t=0.091$	$t=1.83$

Table 11.14: The two basepoint one examples side by side: the reciprocal Xi function of Chapter 9 and the reciprocal completed L function of χ_{12} of this section.

smoothing of the zeros, the Lévy form comes from the primes, the gamma factor and the pole, the sine squared lemma converts one into the other with the drift $\sum_\rho \rho^{-1}$ becoming a stable one half factor, and the shifted clock is a subordinated hitting time whose Brownian boundary, as computed, becomes flat. The exceptions are the Krein string representation of Corollary 9.2.3, which assumes RH and simple zeros, and the Wald defect $\sum_\rho \rho^{-1}/\sqrt{2}$: Lemma 9.2.4 gives it for a boundary bounded at infinity, and the boundedness of the computed boundary is observed, not proved. Second, below the basepoint one each positive statement for ζ holds, up to the boundary case, exactly when no zero lies to the right of the line $\operatorname{Re} s=\alpha$: the convergence of the compensated Lévy form, the GGC property of T_α, the extended GGC property of the normal scale mixture, and the completeness in L^1 of the translates of the tilted Pólya kernels on every line to the right of α (Propositions 9.1.6 and 10.1.9, Theorems 9.2.5 and 10.1.3). The eta example shows that the pole changes how the prime sums converge, not where they stop.

The step that none of these constructions supplies is positivity of the Thorin measure at every height at a single basepoint $\alpha<1$, that is, the GGC property of T_α, which by Theorem 1.2.2 is a new zero free half plane, and then all the way to the centre. The variable $T_\alpha=\sigma_{\widetilde{T}_\alpha}^{(\alpha)}$ of Theorem 9.2.5 exists for every $\alpha>\frac{1}{2}$ with no hypothesis, so only its GGC property is conditional; as $\alpha\downarrow\frac{1}{2}$ the subordinator

tends to pure drift, and the positivity must be supplied by the zeros themselves. Reaching any single $\alpha < 1$ is step 7 of the path in Table 12.1, which the results of Sato and Bondesson do not reach either (Section 11.1.5); for the methods of this book the basepoint one is where the path leaves the region of theorems. The half plane $\operatorname{Re} s > \frac{7}{8}$ announced in October 2026 for ζ and every Dirichlet L function, $L(s, \chi_{12})$ included, would move that boundary to $\frac{7}{8}$ by arithmetic from outside the book (Section 12.1.5).

11.4 Almost generalised gamma convolutions

Below the basepoint one, as Section 11.3.4 concluded, every positive object built so far from the shifted clock T_α, whose Laplace transform is $\varphi_\alpha(u) = \xi(\alpha)/\xi(\alpha + \sqrt{u})$, is equivalent, up to the boundary case, to the absence of zeros in the half plane $\operatorname{Re} s > \alpha$. So are Salem's equation, the dual form of completeness (Corollary 10.1.10), and Riesz's growth criterion with exponent $\alpha/2$ (Sections 10.1.8 and 10.2). These equivalences are restatements rather than reductions; their value is that several of them are computable and that they suggest the right intermediate targets. Since no zero free half plane was known before 2026, and none is known to the left of the line $\operatorname{Re} s = \frac{7}{8}$ (Section 12.1.5), the question that remains is quantitative: how negative can the Thorin density be below the basepoint one? Zero density estimates answer it with no hypothesis. The negative part of the density is a sum of Cauchy kernels, one for each zero to the right of α, so its mass is controlled by the number of such zeros, while the positive part grows like the zero count. With the estimates of Ingham, of Guth and Maynard and of Huxley, every T_α with $\alpha > \frac{1}{2}$ is a GGC up to a defect carried by the zeros to the right of α, whose Cauchy kernels carry a vanishing fraction of the mass of the others, with a power saving; we call this property almost GGC.

Section 11.4.1 proves this (Theorem 11.4.2). Section 11.4.2 factorises φ_α exactly into the Laplace transform of a GGC and an explicit defect that carries every zero to the right of α, and bounds the defect with no hypothesis from the verification of RH to the height $T_0 = 3 \times 10^{12}$ [196]. Section 11.4.3 shows by a cancellation lemma where negativity of the density can come from, Section 11.4.4 inserts hypothetical off line zeros into the computed density, and Section 11.4.5 compares the positivity margins of the density with the circular unitary ensemble. Section 11.4.6 turns to the moments of ζ on the critical line. Their random matrix factor is given by the Barnes function, which Nikeghbali and Yor express through GGC variables, and they are exponential moments of the Thorin density integrated across the basepoints; through zero density they bear on the strength of the almost GGC theorem, but not on exact positivity. Section 11.4.7 explains why no count of zeros turns "almost" into "exactly", which is where Chapter 12 takes up the path. With ξ again in place of $\xi(\cdot, \chi_{12})$, we keep the notation of Sections 9.1 and 11.1: $v_\alpha(\theta) = \operatorname{Re}(\xi'/\xi)(\alpha + i\theta)$, and the Thorin density at the basepoint α in the

chart $\theta = \sqrt{z}$ is v_α/π, the density $\rho_{\alpha-1/2}$ of (1.2.7) at the resolution $\alpha - \frac12$; in this section the basepoint, not the resolution, is the index. $P_w(x) = w/(\pi(w^2+x^2))$ is the Poisson kernel of Section 1.2, $P_{-w} = -P_w$, and P_0 is the unit point mass, as in Proposition 3.3.1. A zero is $\rho = \beta + i\gamma$; $N(T)$ counts the zeros with $0 < \gamma \le T$, and $N(\sigma, T)$ those among them with $\beta \ge \sigma$, so that $N(\sigma,T) \ge \nu(\sigma,T)$ in the notation of Proposition 3.4.2. The scripts are in the folder `riesz_program/code` of the archive and their outputs in `riesz_program/data`; we name them at each table and figure.

11.4.1 Mass bookkeeping and the almost GGC theorem

The Poisson form of Proposition 1.2.3 splits according to the side of the line $\operatorname{Re} s = \alpha$ on which each zero lies, $v_\alpha = v_\alpha^+ - v_\alpha^-$, with

$$\begin{aligned} \frac{v_\alpha^+(\theta)}{\pi} &= \sum_{\beta \le \alpha,\ \gamma>0} \big[P_{\alpha-\beta}(\theta-\gamma) + P_{\alpha-\beta}(\theta+\gamma)\big], \\ \frac{v_\alpha^-(\theta)}{\pi} &= \sum_{\beta > \alpha,\ \gamma>0} \big[P_{\beta-\alpha}(\theta-\gamma) + P_{\beta-\alpha}(\theta+\gamma)\big]. \end{aligned} \tag{11.4.1}$$

Both parts are positive, and a zero on the line $\operatorname{Re} s = \alpha$, if there is one, contributes a unit atom at its ordinate to the positive part. Each pair of kernels has mass one on $(0,\infty)$. A zero with $\beta > \alpha$ has the partner $1 - \bar\rho = 1 - \beta + i\gamma$, whose real part $1 - \beta < \frac12 < \alpha$ puts it in the positive part. Write $m_\alpha^\pm(T) = \pi^{-1}\int_0^T v_\alpha^\pm(\theta)\,d\theta$ for the positive and negative masses on $[0,T]$.

Lemma 11.4.1 (Mass bookkeeping). *Let $\alpha > \frac12$, and suppose that $N(\alpha,t) \le t^{e+o(1)}$ as $t \to \infty$ for some $e < 1$. Then*

$$m_\alpha^-(T) \le N(\alpha, 2T) + O\big(T^{e-1+o(1)}\big), \qquad m_\alpha^+(T) = \frac{T}{2\pi}\log\frac{T}{2\pi} + O(T).$$

Proof. A zero with $\beta > \alpha$ and $\gamma \le 2T$ contributes at most its total mass one. For a zero with $\gamma > 2T$ put $w = \beta - \alpha$, so that $0 < w < \frac12$. Since $\int_0^T P_w(\theta+\gamma)\,d\theta = \int_{-T}^0 P_w(\theta-\gamma)\,d\theta$, its pair of kernels has mass on $[0,T]$ at most

$$\int_{-\infty}^{T} P_w(\theta-\gamma)\,d\theta \le \frac{w}{\pi(\gamma - T)} \le \frac{1}{\pi\gamma},$$

and $\sum_{\beta>\alpha,\ \gamma>2T}\gamma^{-1} = \int_{2T}^\infty t^{-1}dN(\alpha,t) \le \int_{2T}^\infty N(\alpha,t)t^{-2}dt \ll T^{e-1+o(1)}$. The positive part contains the kernels of all zeros with $\beta \le \alpha$, among them every zero on the critical line and every partner $1-\bar\rho$. Its mass on $[0,T]$ is therefore $N(T)$ up to the $N(\alpha,T) = O(T^{e+o(1)})$ zeros to the right of α, whose kernels it lacks, and up to the tails of the kernels of the zeros near height T, which change it by $O(\log^2 T)$. Since $N(T) = \frac{T}{2\pi}\log\frac{T}{2\pi} - \frac{T}{2\pi} + O(\log T)$, the second claim follows. □

THEOREM 11.4.2 (Almost GGC). *For every $\frac{1}{2} < \alpha < 1$, with no hypothesis,*

$$\frac{m_\alpha^-(T)}{m_\alpha^+(T)} \le T^{e(\alpha)-1+o(1)}, \qquad e(\alpha) = \begin{cases} \dfrac{3(1-\alpha)}{2-\alpha}, & \frac{1}{2} < \alpha < \frac{7}{10}, \\ \dfrac{15(1-\alpha)}{3+5\alpha}, & \frac{7}{10} \le \alpha < 1, \end{cases}$$

and $e(\alpha) < 1$ throughout, so the negative part v_α^-/π of (11.4.1)*, the Cauchy kernels of the zeros to the right of α, carries on $[0,T]$ a vanishing fraction of the mass of the positive part, with a power saving. For $\alpha \ge 1$ no zero lies to the right of α, and $m_\alpha^- = 0$. Under the density hypothesis $e(\alpha) = 2(1-\alpha)$. RH holds if and only if v_α^- vanishes identically for every $\alpha > \frac{1}{2}$.*

Proof. Combine Lemma 11.4.1 with three zero density estimates: Ingham's $N(\sigma,T) \ll T^{3(1-\sigma)/(2-\sigma)}\log^5 T$ [104]; Guth and Maynard's $N(\sigma,T) \le T^{15(1-\sigma)/(3+5\sigma)+o(1)}$ for $\frac{7}{10} \le \sigma \le \frac{4}{5}$ [88, Theorem 1.2]; and Huxley's $N(\sigma,T) \le T^{3(1-\sigma)/(3\sigma-1)+o(1)}$ for $\sigma \ge \frac{3}{4}$ [100], whose exponent is at most $15(1-\sigma)/(3+5\sigma)$ when $\sigma \ge \frac{4}{5}$. The first two exponents agree at $\sigma = \frac{7}{10}$, and together they give the uniform bound $N(\sigma,T) \le T^{30(1-\sigma)/13+o(1)}$ of Guth and Maynard. Since $3(1-\alpha) < 2-\alpha$ exactly when $\alpha > \frac{1}{2}$, and $15(1-\alpha) < 3+5\alpha$ exactly when $\alpha > \frac{3}{5}$, the exponent is below one for every $\frac{1}{2} < \alpha < 1$. Finally, v_α^- vanishes identically exactly when no zero has $\beta > \alpha$, which is the GGC criterion of Theorem 1.2.2. □

Even the classical estimate of Bohr and Landau [32], $N(\sigma,T) = O(T)$ for fixed $\sigma > \frac{1}{2}$, gives a vanishing fraction $O(1/\log T)$; the zero density estimates improve this to a power of T. Two remarks sharpen the statement. First, the theorem bounds the decomposition mass $m_\alpha^-(T)$, which overcounts the negative part $\int_0^T \max(-v_\alpha/\pi, 0)\,d\theta$ of the density itself, because the positive kernels of nearby zeros cancel much of each negative kernel. The partner of a zero already cancels part of it: with $\varepsilon = \alpha - \frac{1}{2}$ and $\delta = \beta - \frac{1}{2}$, twice the mass in Proposition 3.3.1(iii) gives the negative part of $P_{\alpha+\beta-1} - P_{\beta-\alpha} = P_{\varepsilon+\delta} + P_{\varepsilon-\delta}$ the mass $\frac{4}{\pi}\arctan\sqrt{(\delta+\varepsilon)/(\delta-\varepsilon)} - 1 = \frac{2}{\pi}\arcsin\frac{\varepsilon}{\delta}$, less than the mass one that the decomposition counts, and Section 11.4.4 shows the effect of the other zeros. Second, on the verified range the negative mass is essentially zero. All zeros with $0 < \gamma \le T_0$ lie on the critical line [196], so for $T \le T_0/2$ the only negative contributions come from zeros above T_0, and by the proof of Lemma 11.4.1 their total mass on $[0,T]$ is $O(T_0^{e(\alpha)-1+o(1)})$. Beyond the verified range Theorem 11.4.2 takes over.

Table 11.15, computed by `almost.py`, lists the exponents, and the left panel of Figure 11.13 draws them. The saving $1 - e(\alpha)$ is small near the centre, where Ingham's exponent gives $(2\alpha-1)/(2-\alpha)$, 0.069 at $\alpha = 0.55$, and it grows to 0.8 at $\alpha = 0.9$. The density hypothesis would give $2\alpha - 1$; the difference from the saving of the theorem is largest in the middle of the strip, 0.092 at $\alpha = 0.7$, and vanishes

at $\alpha = 0.9$. Sharper exponents are known near $\alpha = 1$: the density hypothesis itself holds for $\alpha > \frac{25}{32}$ [41, p. 133], and Huxley's exponent [100] is 0.176 at $\alpha = 0.9$; those of the theorem suffice for what follows.

α	Ingham	Guth and Maynard	$e(\alpha)$	saving $1 - e(\alpha)$	density hypothesis
0.55	0.931		0.931	0.069	0.90
0.60	0.857		0.857	0.143	0.80
0.65	0.778		0.778	0.222	0.70
0.70	0.692	0.692	0.692	0.308	0.60
0.75	0.600	0.556	0.556	0.444	0.50
0.80	0.500	0.429	0.429	0.571	0.40
0.90	0.273	0.200	0.200	0.800	0.20

Table 11.15: The zero density exponents $e(\alpha)$ of Theorem 11.4.2, with $N(\alpha, T) \le T^{e(\alpha)+o(1)}$, the power saving $1 - e(\alpha)$, and the exponent $2(1 - \alpha)$ of the density hypothesis. Computed by `almost.py`.

11.4.2 The canonical factorisation and explicit unconditional bounds

The splitting (11.4.1) does more than bookkeeping: it factorises the Laplace transform of the clock. The zeros with $\beta \le \alpha$ give a generalised gamma convolution, and those with $\beta > \alpha$ an explicit factor, which is identically one exactly when there are none.

Proposition 11.4.3 (Canonical factorisation). *For every $\alpha > \frac{1}{2}$ and $u \ge 0$, $\varphi_\alpha(u) = \varphi_\alpha^+(u)\, D_\alpha(u)$, where*

$$\varphi_\alpha^+(u) = \exp\Big\{ -\int_0^\infty \log\Big(1 + \frac{u}{\theta^2}\Big) \frac{v_\alpha^+(\theta)}{\pi}\, d\theta \Big\},$$

$$D_\alpha(u) = \prod_{\beta > \alpha,\ \gamma > 0} \frac{(\beta - \alpha)^2 + \gamma^2}{(\sqrt{u} - \beta + \alpha)^2 + \gamma^2};$$

here φ_α^+ is the Laplace transform of a GGC T_α^+ with Thorin density v_α^+/π in the chart $\theta = \sqrt{z}$, and the product defining the defect D_α runs over the zeros to the right of the line $\operatorname{Re} s = \alpha$ and converges absolutely. The defect is identically one if and only if ζ has no zeros with $\operatorname{Re} s > \alpha$, and then $T_\alpha \overset{d}{=} T_\alpha^+$. Otherwise $D_\alpha(u) \to 0$ as $u \to \infty$, and T_α^+ is not the sum of T_α and an independent nonnegative variable.

Proof. Grouping the Hadamard product of ξ in conjugate pairs, as in the proof of Proposition 1.2.3, gives for $u \ge 0$

$$\varphi_\alpha(u) = \prod_{\gamma > 0} \frac{(\alpha - \beta)^2 + \gamma^2}{(\alpha - \beta + \sqrt{u})^2 + \gamma^2},$$

over the zeros with $\gamma > 0$, counted with multiplicity; each factor is $1 + O((u + \sqrt{u})\gamma^{-2})$ and $\sum \gamma^{-2} < \infty$, so the product converges absolutely. For $\beta < \alpha$, with $w = \alpha - \beta > 0$, the factor is $(w^2 + \gamma^2)/((w + \sqrt{u})^2 + \gamma^2) = \exp\{-\int_0^\infty \log(1 + u/\theta^2)[P_w(\theta - \gamma) + P_w(\theta + \gamma)]\, d\theta\}$, the Laplace transform of a GGC whose Thorin density is the pair of Cauchy kernels of width w at $\pm\gamma$ (the computation after Theorem 10.1.2); for $\beta = \alpha$ it is $\gamma^2/(\gamma^2 + u)$, an exponential law, whose Thorin measure is the atom at γ. The partial products over the zeros with $\beta \le \alpha$ are therefore Laplace transforms of GGCs, they converge for every $u \ge 0$ to a limit continuous at the origin, and by the closure theorem of Section 4.1.1 the limit is the Laplace transform of a GGC, with Thorin density v_α^+/π. The remaining factors form D_α. If a zero lies to the right of α, put $w = \beta - \alpha \in (0, \frac{1}{2})$; its factor tends to zero as $u \to \infty$, and every factor of D_α is at most one for $u \ge 1$, because $(\sqrt{u} - w)^2 \ge w^2$ when $\sqrt{u} \ge 2w$. So $D_\alpha(u) \to 0$. If T_α^+ were $T_\alpha + S$ with $S \ge 0$ independent of T_α, then $\mathbb{E}e^{-uS} = \varphi_\alpha^+(u)/\varphi_\alpha(u) = 1/D_\alpha(u)$ would be unbounded, which is impossible. □

The proposition also says what the signed density v_α/π is. When every zero lies to the left of the line $\operatorname{Re} s = \alpha$ it is the Thorin density of T_α (Theorem 1.2.2); a zero on the line adds to the Thorin measure an atom at its ordinate, the term P_0 of (11.4.1), which the function v_α does not show. When a zero lies to the right of the line, T_α is not a GGC and v_α/π is not a signed Thorin density for it in any sense: the negative kernel $-P_w$ of such a zero integrates to the factor $((w + \sqrt{u})^2 + \gamma^2)/(w^2 + \gamma^2)$, while the zero contributes $(w^2 + \gamma^2)/((\sqrt{u} - w)^2 + \gamma^2)$ to φ_α. So $\varphi_\alpha \ne \varphi_\alpha^+/\varphi_\alpha^-$ for the GGC transform φ_α^- with Thorin density v_α^-/π, and by Proposition 11.4.3 the clock is not T_α^+ minus any independent nonnegative variable. It is a GGC times the defect D_α, which carries all possible failure of RH, and Theorem 11.4.2 says that the zeros entering the defect are sparse in a precise sense: their kernels carry a vanishing, power saving fraction of the mass. The same bookkeeping applies to the other reformulations in Chapters 9 to 11. In the normal scale mixture of Theorem 10.1.3 it bounds the negative part of the candidate EGGC density v_α/π; in Wiener's theorem (Proposition 10.1.9) it bounds the proportion of frequencies at which the Fourier transform of the tilted Pólya kernel can vanish; and in Riesz's criterion (Section 10.1.8) it bounds the number of Mellin poles to the right of $\alpha/2$. Theorem 11.4.4 shows that, with no hypothesis, the defect is extremely close to one at every fixed u.

THEOREM 11.4.4 (Explicit bounds from verification). *Let $T_0 = 3 \times 10^{12}$, below which all zeros lie on the critical line [196], and let $\Sigma_0 = \sum_{\gamma > T_0} \gamma^{-2}$, the sum over all zeros above T_0, counted with multiplicity. By the Riemann von Mangoldt formula with the explicit error term of Hasanalizade, Shen and Wong [92], recalled in the proof of Proposition 9.1.4, Σ_0 differs from $\frac{\log(T_0/2\pi)+1}{2\pi T_0} = 1.4797 \times 10^{-12}$ by less than 2×10^{-24}, so $\Sigma_0 < 1.48 \times 10^{-12}$. Then for every $\alpha > \frac{1}{2}$ and every $u \ge 0$:*

1. $|\log D_\alpha(u)| \le (u+\sqrt{u})\,\Sigma_0$, *so that*

$$e^{-1.5\times10^{-12}(u+\sqrt{u})}\,\varphi_\alpha^+(u) \le \mathbb{E}e^{-uT_\alpha} \le e^{1.5\times10^{-12}(u+\sqrt{u})}\,\varphi_\alpha^+(u);$$

2. *the negative part of the density satisfies* $v_\alpha^-(\theta)/\pi \le 2\Sigma_0/\pi < 9.43\times10^{-13}$ *for* $0 \le \theta \le T_0/2$;

3. *consequently* $v_\alpha(\theta) > 0$ *for all* $0 \le \theta \le T_0/2$ *whenever* $\alpha - \frac12 > 6\times10^{-10}$.

Proof. Every zero to the right of α has $\gamma > T_0$ and $\beta > \frac12$. The zeros above T_0 off the critical line come in pairs $\beta+i\gamma$, $1-\beta+i\gamma$ of equal multiplicity, by $\xi(s)=\xi(1-s)$ and $\xi(\bar s)=\overline{\xi(s)}$, and at most one member of each pair lies to the right of α, so $\sum_{\beta>\alpha,\ \gamma>0}\gamma^{-2} \le \Sigma_0/2$. (1) For a zero to the right of α put $w=\beta-\alpha\in(0,\frac12)$ and $x=(u-2w\sqrt{u})/(w^2+\gamma^2)$, so that its factor in D_α is $1/(1+x)$. Since $u-2w\sqrt{u}\ge -w^2$, we have $x\ge -w^2/\gamma^2 > -\frac12$, and $|x|\le (u+\sqrt{u})/\gamma^2$; for $x\ge-\frac12$, $|\log(1+x)|\le 2|x|$. Hence the logarithm of its factor is at most $2(u+\sqrt{u})/\gamma^2$ in absolute value, and summing gives $|\log D_\alpha(u)| \le (u+\sqrt{u})\Sigma_0$; the inequalities for $\mathbb{E}e^{-uT_\alpha} = \varphi_\alpha^+(u)D_\alpha(u)$ follow. (2) For $\theta\le T_0/2$ and $\gamma>T_0$, $P_w(\theta-\gamma)+P_w(\theta+\gamma)\le \frac{2w}{\pi(\gamma-\theta)^2}\le\frac{4}{\pi\gamma^2}$, and summing over the zeros to the right of α gives at most $\frac{4}{\pi}\cdot\frac{\Sigma_0}{2}$. (3) Write $a=\alpha-\frac12$. If $a\ge\frac12$, then $\alpha\ge1$, no zero lies to the right of α, and $v_\alpha=v_\alpha^+>0$; so let $6\times10^{-10}<a<\frac12$. The zeros below T_0 lie on the line and contribute to v_α^+/π the kernels $P_a(\theta-\gamma)$, and all other kernels of v_α^+ are positive, so $v_\alpha^+(\theta)/\pi \ge \frac{a}{\pi(a^2+\gamma_1^2)}$ at every θ within γ_1 of an ordinate below T_0. Every $\theta\in[0,T_0/2]$ is such a point. On $[0,\gamma_1]$ the ordinate γ_1 serves. On $[\gamma_1,2000]$ one of the first 1800 ordinates serves: they are the certified zeros of the proof of Proposition 9.1.4, they reach height 2304.3, and consecutive ones differ by less than $6.8874<2\gamma_1$. For $2000\le\theta\le T_0/2$, the count of that proof, with $H=14$ in place of 15, shows that the window $(\theta-14,\theta+14]$ holds at least $n(\theta)=\frac{14}{\pi}\log\frac{\theta-14}{2\pi}-E(\theta-14)-E(\theta+14)$ zeros; while $\theta+14\le X_0$, n increases and $n(2000)>20$, and for $\theta+14>X_0$, $n(\theta)>\frac{14}{\pi}\log\frac{\theta-14}{2\pi}-0.6\log(\theta+14)>80$, so the window holds a zero, which lies within $14<\gamma_1$ of θ and, since $\theta+14<T_0$, on the line. The script `explicit_bounds.py`, with output `explicit_bounds.json`, recomputes the 1800 zeros and checks these inequalities in ball arithmetic. Finally, since $a<\frac12$,

$$\frac{a}{\pi(a^2+\gamma_1^2)} > \frac{a}{\pi(\frac14+\gamma_1^2)} > \frac{6\times10^{-10}}{\pi(\frac14+\gamma_1^2)} > 9.5\times10^{-13},$$

which is more than the bound $2\Sigma_0/\pi<9.43\times10^{-13}$ of (2) on v_α^-/π, so $v_\alpha = v_\alpha^+ - v_\alpha^- > 0$ on $[0,T_0/2]$. □

Theorem 11.4.4 restates the verification of RH to height 3×10^{12} in Thorin form: at every basepoint, however close to the centre, the clock is a GGC up to a Laplace

transform factor within $e^{\pm 1.5\times 10^{-12}(u+\sqrt{u})}$, and for $\alpha - \frac{1}{2} > 6 \times 10^{-10}$ the density v_α/π is pointwise positive on the verified window. Part (3) of Theorem 10.1.3 is the same statement for the characteristic function of the normal scale mixture, with a remainder of relative size $\log T_0/T_0$, and the paragraph after its proof bounds the negative part of v_α/π for $\theta \le 10^9$ in the same way as part (2).

Part (3) of Theorem 11.4.4 treats every zero above T_0 as if its kernel could be negative wherever the density is evaluated. As the first author's note on the Dirichlet beta function observes [201], grouping each such zero with its reflection in the critical line does better, because the sum of the two kernels of an off line pair is negative only near its ordinate.

COROLLARY 11.4.5 (Positivity on the verified window). *Let* $T_0 = 3 \times 10^{12}$ *and* $V_\xi = (\log \xi)''(\frac{1}{2}) = 0.04620998623$.

1. *For every* $\alpha > \frac{1}{2}$, $v_\alpha(\theta) > 0$ *for* $|\theta| \le T_0 - \frac{1}{2}$.

2. *For every* $\alpha \in (\frac{1}{2}, 7.35]$, $v_\alpha(\theta) > v_\alpha(0)$ *for* $0 < |\theta| \le T_0 - \frac{1+\sqrt{2}}{2}$, *and* $0 < v_\alpha(0) \le (\alpha - \frac{1}{2})V_\xi$.

Proof. (1) Put $\varepsilon = \alpha - \frac{1}{2}$. By the Hadamard product, as in Proposition 1.2.3, $v_\alpha(\theta)$ is the absolutely convergent sum over the zeros $\rho = \beta + i\gamma$, counted with multiplicity, of $(\alpha-\beta)/((\alpha-\beta)^2+(\theta-\gamma)^2)$. A zero on the critical line contributes $\pi P_\varepsilon(\theta-\gamma) > 0$. The zeros off the line come in pairs $\frac{1}{2} \pm \delta + i\gamma$ with $0 < \delta < \frac{1}{2}$ and equal multiplicities, and a pair contributes $\pi(P_{\varepsilon+\delta} + P_{\varepsilon-\delta})(\theta-\gamma)$, which is nonnegative if $\delta \le \varepsilon$ and, if $\delta > \varepsilon$, negative exactly where $|\theta-\gamma| < \sqrt{\delta^2-\varepsilon^2}$, by the proof of Proposition 3.3.1(ii); and $\sqrt{\delta^2-\varepsilon^2} < \delta < \frac{1}{2}$. Every zero off the line has $|\gamma| > T_0$ [196], so for $|\theta| \le T_0 - \frac{1}{2}$ every pair has $|\theta-\gamma| > \frac{1}{2}$ and contributes a nonnegative term, while the zero $\frac{1}{2} + i\gamma_1$ contributes a positive one.

(2) This is Theorem 9.1.11(i) and (ii) for ξ, with $\alpha_{\max} = 7.35$ and $H_0 = T_0$ (Table 9.8). □

REMARK 11.4.6. Part (1) improves part (3) of Theorem 11.4.4 in both respects: it holds for every $\alpha > \frac{1}{2}$, not only for $\alpha - \frac{1}{2} > 6\times 10^{-10}$, and on $[0, T_0 - \frac{1}{2}]$ rather than on $[0, T_0/2]$. Likewise the bound after the proof of Theorem 10.1.3 on what the zeros above T_0 could subtract from v_α/π is not needed below $T_0 - \frac{1}{2}$, where every pair contributes a nonnegative term. So on the verified window the positivity of the density adds nothing to the verification. The corollary does not touch the defect of part (1) of Theorem 11.4.4: a zero above T_0 to the right of the line $\operatorname{Re} s = \alpha$ stops T_α from being a GGC whatever v_α does on the window (Proposition 11.4.3), and near its own ordinate it can make v_α negative (Section 11.4.3). Part (2), which comes from the comparison across vertical lines of Section 9.1.6, locates the minimum: for every $\alpha \in (\frac{1}{2}, 7.35]$ the density v_α/π is smallest at the origin, and only there, for $0 < |\theta| \le T_0 - \frac{1+\sqrt{2}}{2}$, as the next paragraph finds on $[0, 200]$ at $\alpha = 0.6$. Its proof uses of the zeros only the first ordinate and the verification height.

On the computed range the numbers bear this out. By `more2.py`, the minimum over $[0, 2100]$ of the Poisson sum over the first 1700 ordinates, taken at the midpoints between them and at the origin, is 1.44×10^{-3}, 1.44×10^{-5}, 1.44×10^{-8} and 1.44×10^{-10} at $\alpha-\frac12=10^{-1}$, 10^{-3}, 10^{-6} and 10^{-8}. It is attained at $\theta=0$, it is proportional to $\alpha-\frac12$, it lies above the lower bound $a/(\pi(a^2+\gamma_1^2))$ of the proof, and it is far above 9.4×10^{-13}. The complete density is larger by the contribution of the zeros above height 2198: at $\alpha=0.6$ its minimum on $[0,200]$ is the Cauchy floor $b_\alpha/\pi=0.0014709$, attained at the origin (Section 10.1.3), in agreement with the margin $0.0147\,\varepsilon$ of Section 3.1. The largest gap between consecutive ordinates below height 2198 is 6.89, between γ_1 and γ_2.

The defect is the quantity to track as verification and zero density estimates improve. The proof of part (1) gives, with no hypothesis, $|\log D_\alpha(u)|\le 2(u+\sqrt{u})\sum_{\beta>\alpha,\ \gamma>0}\gamma^{-2}$. With the verification height and the trivial count the sum is at most $\Sigma_0/2$, with $\Sigma_0<1.48\times10^{-12}$. The zero density estimates of Section 11.4.1 [104, 88, 100] improve this to $\sum_{\beta>\alpha}\gamma^{-2}\le\int_{T_0}^\infty t^{-2}dN(\alpha,t)\ll T_0^{e(\alpha)-2+o(1)}$, smaller by a further power of T_0 since $e(\alpha)<1$. What is wanted is a table of explicit defects as functions of α and T_0, with explicit zero density estimates in place of the $o(1)$ in the exponents, updated as verification heights and density exponents improve. Such a table would measure, in one number per basepoint, how close the programme is to its target.

11.4.3 Where negativity can come from: a cancellation lemma

The counting bound treats every zero to the right of α as if its kernel were entirely negative. Pointwise, a negative kernel is cancelled wherever the rest of the density exceeds it, and the kernel of width w has height at most $1/(\pi w)$.

Lemma 11.4.7 (Cancellation). *Let $\rho=\beta+i\gamma$ be a zero with $\beta>\alpha$, and let $m>0$ be a lower bound on an interval I for the rest of the density, $v_\alpha/\pi+P_{\beta-\alpha}(\cdot-\gamma)$. If $\beta-\alpha\ge1/(\pi m)$, the kernel of ρ cannot make v_α negative on I.*

Proof. On I, $v_\alpha/\pi\ge m-\max_x P_{\beta-\alpha}(x)=m-1/(\pi(\beta-\alpha))\ge0$. □

The lemma has a heuristic consequence. At height γ the density v_α/π averages $\frac{1}{2\pi}\log\frac{\gamma}{2\pi}$, the density of the ordinates, and when $\alpha-\frac12$ is large compared with the mean spacing $2\pi/\log\gamma$ it stays close to that average. Taking this average for m in the lemma, a zero to the right of α can then produce negativity only if

$$\beta-\alpha<\frac{2\pi}{\pi\log(\gamma/2\pi)}=\frac{2}{\log(\gamma/2\pi)}.$$

Heuristically, then, positivity of the Thorin density at the basepoint α can be broken at height γ only by zeros in the thin strip $\alpha<\beta<\alpha+2/\log(\gamma/2\pi)$; zeros farther to the right are absorbed by the positive mass of the zeros on the line, and the dangerous strip narrows as the height grows. The local visibility

condition of Section 5.2.2, which concerns small resolutions, has the same scale $1/\log\gamma$. The heuristic proves nothing, because a rigorous lower bound for the background near an arbitrary height would itself control ζ'/ζ in the strip. It does identify where a counterexample would have to live, and it suggests combining zero density estimates for thin strips with lower bounds for $\operatorname{Re}\xi'/\xi$ on average.

We tested the lemma at the heights 300 and 10^5 with `high.py`. The script computes the background $v_{0.6}/\pi$ from ξ'/ξ, through ζ'/ζ and the gamma factor, on a window of width 8 with step 0.01, and adds the kernels of a single hypothetical pair of zeros at $\beta + i\gamma_0$ and $1-\beta+i\gamma_0$, with γ_0 the centre of the window (Figure 11.12, middle and right). At height 300 the background has mean 0.51 and minimum 0.069; at height 10^5 it has mean 1.59, close to the density of ordinates $\frac{1}{2\pi}\log\frac{10^5}{2\pi} = 1.54$, and minimum 0.34. Table 11.16 shows the result. At both heights zeros with $\beta-\alpha \le 0.2$ produce negativity and zeros with $\beta-\alpha \ge 0.3$ are absorbed. At height 10^5 the heuristic threshold $2/\log(10^5/2\pi) = 0.21$ is close to the observed one, while the rigorous guarantee of Lemma 11.4.7 with the window minimum, $1/(\pi m) = 0.93$, is conservative. For a fixed width the negative dip is deeper at the greater height, because the kernel is the same while the local minimum of the background near the hypothetical zero differs.

	height 300		height 10^5	
β	min of $v_{0.6}/\pi$	negative mass	min of $v_{0.6}/\pi$	negative mass
0.62	−13.54	0.521	−13.97	0.567
0.65	−4.16	0.299	−4.59	0.365
0.70	−1.24	0.121	−1.62	0.193
0.75	−0.41	0.041	−0.72	0.097
0.80	−0.10	0.008	−0.32	0.041
0.90	0.072	0	0.023	0
0.95	0.075	0	0.100	0

Table 11.16: A single hypothetical pair of zeros at real parts β and $1-\beta$ inserted into the density $v_{0.6}/\pi$, computed from ξ'/ξ on a window of width 8 at the heights 300 and 10^5: the minimum of the density and the negative mass $\int \max(-v_{0.6}/\pi, 0)$ on the window. Computed by `high.py`.

11.4.4 Hypothetical off line zeros in the computed density

No off line zero is known, so the negative part cannot be observed. To see the mechanism, `almost.py` inserts fifteen hypothetical quadruples of zeros $0.75 \pm i\gamma_j$, $0.25 \pm i\gamma_j$, at the heights $\gamma_j = 20\,j^{5/3}$ below 2000, into the Poisson form of Proposition 1.2.3 together with the first 1700 ordinates, taken on the line; the zeros above height 2198 are omitted. The count of the hypothetical zeros grows like $T^{0.6}$, as a zero density estimate with $e = 0.6$ would allow. Table 11.17 compares the negative mass $\int_0^T \max(-v_\alpha/\pi, 0)\,d\theta$ of the resulting density with its positive mass and with

the counting bound $N_{\mathrm{off}}(2T)$ of Lemma 11.4.1, the number of hypothetical zeros with ordinate at most $2T$. Figure 11.13 shows the density near the hypothetical zero at height 292.4.

α	T	negative mass	positive mass	fraction	counting bound $N_{\mathrm{off}}(2T)$
0.6	250	0.842	108.8	7.7×10^{-3}	6
0.6	1000	1.655	650.5	2.5×10^{-3}	15
0.6	2000	2.278	1519.2	1.5×10^{-3}	15
0.7	250	2.130	110.1	1.9×10^{-2}	6
0.7	1000	4.815	653.6	7.4×10^{-3}	15
0.7	2000	6.938	1523.8	4.6×10^{-3}	15
0.8	any	0		0	0

Table 11.17: Negative and positive mass of the density v_α/π on $[0, T]$ with fifteen hypothetical quadruples of zeros at real part 0.75. Computed by `almost.py`.

Three features are visible. The negative mass is well below the counting bound, because each negative kernel is partly cancelled by its partner and by the positive kernels of the neighbouring zeros on the line: at $\alpha = 0.6$ the negative kernel has width 0.15 and height $1/(0.15\pi) = 2.1$, and most of it is cancelled, while at $\alpha = 0.7$ the width is 0.05 and the kernel, of height 6.4, appears as a sharp negative spike that takes the density to -5.39 at the height 292.40 (Figure 11.13; the minimum is computed by `almost_dip.py`). The fraction decreases with T at the rate the counting predicts. And at $\alpha = 0.8$, to the right of the hypothetical zeros, the density is positive everywhere, as Theorem 1.2.2 requires: a zero with real part β contributes a positive kernel at every basepoint $\alpha > \beta$.

A zero is most visible from basepoints just to its left. The left panel of Figure 11.12, computed by `more2.py`, shows the negative mass on $[0, 2000]$ as a function of α for the same fifteen heights and three real parts $\beta = 0.6$, 0.75 and 0.9. As α increases towards β the negative kernels narrow, cancellation fails, and the negative mass climbs towards the number of hypothetical zeros, reaching 7.55 at $\alpha = 0.85$ for $\beta = 0.9$. At $\alpha = \beta$ the zeros lie on the line of evaluation and the negative mass drops to zero, and for $\alpha > \beta$ they contribute positive kernels. This is the Thorin counterpart of the fact that the translates of the tilted Pólya kernel fail to be complete exactly on the line through the zero (Proposition 10.1.9).

11.4.5 Positivity margins and the circular unitary ensemble

By Lemma 11.4.7, a zero to the right of α at height γ can make the density negative only if its distance to the line $\operatorname{Re} s = \alpha$ is below $1/(\pi m)$, where m is the local minimum of the rest of the density. The distribution of the local minima, the positivity margins, therefore determines how much room RH has at each height. The minima occur between consecutive ordinates. After unfolding the ordinates

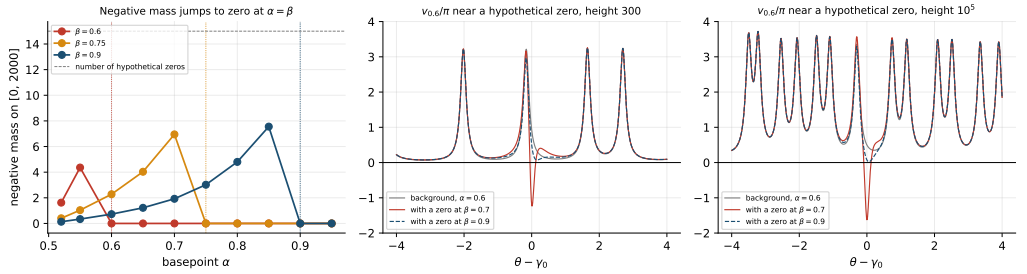


Figure 11.12: Left: negative mass of the density on $[0, 2000]$ against the basepoint α for fifteen hypothetical zeros at real part β (`more2.py`). Middle and right: the density $v_{0.6}/\pi$ near the heights 300 and 10^5, alone and with a hypothetical pair of zeros at $\beta = 0.7$ or 0.9 at the centre of the window (`high.py`). Drawn by `figs_almost.py`.

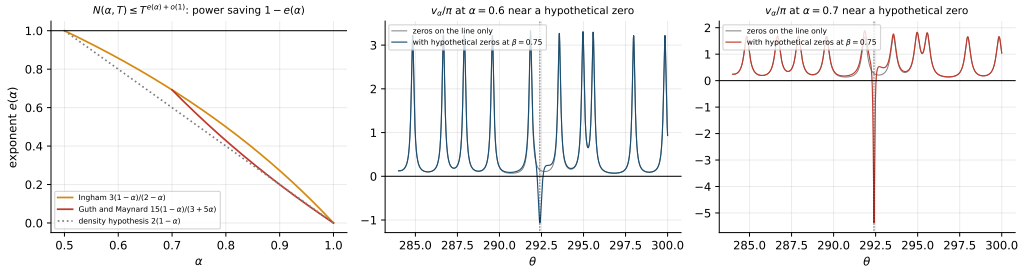


Figure 11.13: Left: the zero density exponents of Theorem 11.4.2 and the region where they give a power saving; the exponent of Guth and Maynard is drawn on $[\frac{7}{10}, 1)$, where it holds by their estimate and, above $\frac{4}{5}$, by Huxley's [100]. Middle and right: the density v_α/π at $\alpha = 0.6$ and 0.7 near the hypothetical off line zero at height 292.4 (dotted), with and without the hypothetical zeros, evaluated as in `almost.py` with step 0.005. Drawn by `figs_almost.py`.

to unit mean spacing, $x_n = \vartheta(\gamma_n)/\pi + 1$ with ϑ the Riemann Siegel theta function, as in Section 5.4.2, the density at the resolution of κ mean spacings is

$$\bar{\rho}_\kappa(x) = \sum_n P_\kappa(x - x_n) = \sum_n \frac{\kappa}{\pi(\kappa^2 + (x - x_n)^2)},$$

whose mean is one. Section 5.4.2 compared the spacings of the zeros with the random matrix prediction; here the object of comparison is the density itself at the points where it is smallest.

The script `sens.py` evaluates $\bar{\rho}_\kappa$ at the 1600 midpoints between consecutive unfolded ordinates below height 2198, leaving out 50 mean spacings at each end, and the same statistic at about 3600 midpoints between the eigenvalues of twelve

Haar distributed unitary matrices of size 400, unfolded and extended periodically. The circular unitary ensemble (CUE) of these matrices models the local statistics of the zeros. Table 11.18 and Figure 11.14 show the result. The medians agree to 1% at every resolution. In the lower tail the zeros have slightly larger margins than the CUE, 0.1177 against 0.1095 at the 1% quantile for $\kappa = 0.1$, a finite height effect in the expected direction, since the low zeros are more rigid than the limit of large matrices.

κ	zeta, 0.1%	CUE, 0.1%	zeta, 1%	CUE, 1%	zeta, median	CUE, median
0.1	0.1075	0.0919	0.1177	0.1095	0.3349	0.3343
0.25	0.2617	0.2178	0.2849	0.2659	0.7116	0.7092
0.5	0.4803	0.4047	0.5150	0.4645	0.9739	0.9796
1	0.7477	0.6552	0.7730	0.7313	1.0264	1.0240

Table 11.18: Quantiles of the normalised Thorin density $\bar{\rho}_\kappa$ at the midpoints between consecutive unfolded ordinates, for the zeros below height 2198 and for CUE eigenvalues. Computed by `sens.py`.

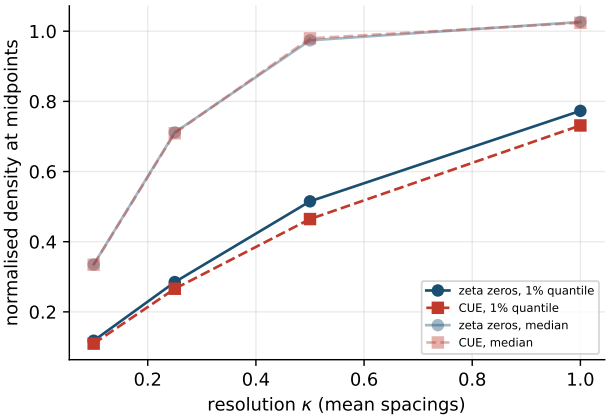


Figure 11.14: Positivity margins of the Thorin density at midpoints: the 1% quantile and the median of $\bar{\rho}_\kappa$ against the resolution κ, for the zeros below height 2198 and for CUE eigenvalues. Drawn by `figs_strategy.py` from the output of `sens.py`.

The next step is to repeat the comparison with Odlyzko's tables of zeros near the 10^{12}th zero [171] and near the 10^{21}st and 10^{22}nd zeros [175, 176], where the finite height corrections are smaller, and to fit the lower tail of the margin distribution. The CUE lower tail is governed by large gaps, whose probability decays like a Gaussian in the gap length, so the smallest margin among the first N midpoints should decrease only like a power of $\sqrt{\log N}$. Converted through the cancellation

lemma, this predicts the width, as a function of the height, of the strip to the right of each basepoint in which an off line zero could hide without being cancelled. That is the quantitative input that question (1) of Section 11.4.7 needs, and a proof along the path would have to reproduce it.

11.4.6 Moments of zeta, the Barnes function and zero density

The strength of Theorem 11.4.2 is set by the zero density exponents, which rest on mean value estimates on the critical line, Ingham's on the fourth moment of ζ. The moments of ζ are also where probabilistic structure of the GGC kind meets the zeros, through random matrix theory. Keating and Snaith [119] modelled $\zeta(\frac{1}{2}+it)$ by the characteristic polynomial $Z_N(\theta) = \det(I - Ae^{-i\theta})$ of a Haar distributed unitary matrix A of size N, a matrix of the circular unitary ensemble of Section 11.4.5, with N in the role of $\log(T/2\pi)$. By analogy with the exact formula

$$\mathbb{E}|Z_N(\theta)|^{2k} = \prod_{j=1}^{N} \frac{\Gamma(j)\Gamma(j+2k)}{\Gamma(j+k)^2} \sim \frac{\mathsf{G}(1+k)^2}{\mathsf{G}(1+2k)}\, N^{k^2}, \qquad N \to \infty,$$

they conjectured that the moments of ζ grow in the same way, with $\log(T/2\pi)$ in place of N and an arithmetic factor a_k in front (Section 11.4.6.1). Here G is the Barnes G function, $\mathsf{G}(1+z) = \Gamma(z)\mathsf{G}(z)$ with $\mathsf{G}(1) = 1$, written in sans serif to keep G for the clock of Section 1.2; Z_N is unrelated to the normal variable Z of Section 10.1. Nikeghbali and Yor [167] gave the random matrix factor a probabilistic meaning in GGC language: the Mellin transform of $|Z_N|$ and the Barnes function are expressed through products and sums of independent gamma, beta and log gamma variables, and the reciprocal of G has a Lévy Khintchine type representation. The source of these representations is the decomposition of Bourgade, Hughes, Nikeghbali and Yor [40],

$$Z_N \stackrel{d}{=} \prod_{j=1}^{N} \left(1 + e^{i\omega_j}\sqrt{B_{1,j-1}}\right),$$

with independent angles ω_j uniform on $[0, 2\pi)$ and independent beta variables $B_{1,j-1}$ with parameters 1 and $j-1$, where $B_{1,0} = 1$.

Table 11.19, computed by `killing2.py`, checks the exact moments against direct simulation of Haar unitary matrices and against the beta decomposition; the $\pm$ are one standard error. The exact values, computed by `rmt_ratios.py`, show how slowly $\mathbb{E}|Z_N|^{2k}/N^{k^2}$ approaches the Barnes constant: for $k = 2$ the ratio is 0.1716, 0.0902, 0.0840 and 0.0834 at $N = 10$, 10^2, 10^3 and 10^4, against $\mathsf{G}(3)^2/\mathsf{G}(5) = \frac{1}{12}$, and for $k = 3$ it is 1.18×10^{-3}, 1.51×10^{-4}, 1.189×10^{-4} and 1.1605×10^{-4}, against $\mathsf{G}(4)^2/\mathsf{G}(7) = 1.1574 \times 10^{-4}$. The Haar estimates of the fourth moment are unreliable at these sample sizes, 1330 ± 103 against 1716 at

$N = 10$, because $|Z_N|^4$ is heavy tailed. The beta decomposition samples the same law more cheaply, so ten times as many samples are drawn, and it agrees with the exact values.

N	k	exact $\mathbb{E}\lvert Z_N\rvert^{2k}$	Haar, 2×10^4	beta decomposition, 2×10^5	$\mathsf{G}(1+k)^2/\mathsf{G}(1+2k)$
10	1	11	10.84 ± 0.25	11.02 ± 0.09	1
10	2	1716	1330 ± 103	1685 ± 77	1/12
20	1	21	20.3 ± 1.0	21.07 ± 0.32	1
20	2	19481	19762 ± 9125	20974 ± 3493	1/12

Table 11.19: Moments of the characteristic polynomial of Haar unitary matrices: the exact formula, direct simulation with 2×10^4 matrices, and the beta decomposition of Bourgade, Hughes, Nikeghbali and Yor with 2×10^5 samples. Computed by `killing2.py`.

11.4.6.1 The Keating and Snaith moments

Write $I_k(T) = \frac{1}{T}\int_0^T |\zeta(\frac12 + it)|^{2k}dt$. The conjecture of Keating and Snaith [119], in the form of Conrey, Farmer, Keating, Rubinstein and Snaith [55], is

$$I_k(T) \sim a_k \,\frac{\mathsf{G}(1+k)^2}{\mathsf{G}(1+2k)}\Big(\log\frac{T}{2\pi}\Big)^{k^2}, \qquad a_k = \prod_p \Big(1-\frac1p\Big)^{k^2}\sum_{m\ge0}\Big(\frac{\Gamma(m+k)}{m!\,\Gamma(k)}\Big)^2 p^{-m},$$

the product of an arithmetic factor from the primes and the random matrix factor. It is a theorem for $k = 1$ and $k = 2$ and open for $k \ge 3$. For $k = 1$ Hardy and Littlewood [89] proved the leading term, and Ingham [103] the mean square with its secondary term, $I_1(T) = \log\frac{T}{2\pi} + 2\gamma_E - 1 + o(1)$; for $k = 2$ Ingham [103] proved the asymptotic with the leading coefficient $\frac{1}{2\pi^2}$. The local factor of a_k is $(1-\frac1p)^{k^2}\,{}_2F_1(k,k;1;\frac1p)$. Since ${}_2F_1(1,1;1;x) = (1-x)^{-1}$ and ${}_2F_1(2,2;1;x) = (1+x)(1-x)^{-3}$, it equals 1 for $k = 1$ and $1-p^{-2}$ for $k = 2$, so that $a_1 = 1$ and $a_2 = \prod_p(1-p^{-2}) = 6/\pi^2$ exactly. The Euler product over the primes below 2×10^5, with each local factor written through ${}_2F_1$, reproduces these: it gives $a_1 = 1$ and $a_2 = 0.6079273$, which agrees with $6/\pi^2 = 0.6079271$ to six digits, the truncation changing it by a relative 3.8×10^{-7}. It gives $a_3 = 0.0493218$; since the local factor for $k = 3$ is $1 - 9p^{-2} + O(p^{-3})$, the omitted primes lower it by a relative $9\sum_{p>2\times10^5} p^{-2} \approx 4\times10^{-6}$, to about 0.0493217. The script `ks.py` hard codes these three values, which `repro/ks_ak.py` recomputes from the product. With $\mathsf{G}(3)^2/\mathsf{G}(5) = \frac{1}{12}$ and $\mathsf{G}(4)^2/\mathsf{G}(7) = \frac{1}{8640}$ the leading coefficients are 1, $0.0506606 = \frac{1}{2\pi^2}$, which is Ingham's constant, and 5.709×10^{-6}, the value $42a_3/9!$ of [55].

Table 11.20, computed by `ks.py`, compares the moments with the leading terms; the moments are computed from $\zeta(\frac12+it)$ on the grid $t = 1, 1.05, 1.1, \ldots$ by

the rectangle rule. For $k = 1$ the full asymptotic is already accurate at $T = 1000$, 5.2113 against 5.2243. For $k = 2$ and $k = 3$ the leading term is a small part of the moment at these heights, 33 against 127 and 13 against 5309 at $T = 1000$, because the lower order terms of the polynomial in $\log T$ predicted in [55] dominate. The $(\log T)^{k^2}$ asymptotics converge very slowly, as the random matrix moments of Table 11.19 converge slowly to the Barnes constant.

T	$I_1(T)$	$\log \frac{T}{2\pi} + 2\gamma_E - 1$	$I_2(T)$	leading, $k = 2$	$I_3(T)$	leading, $k = 3$
125	3.062	3.145	28.0	4.05	396	0.11
250	3.869	3.838	53.1	9.33	1136	0.71
500	4.550	4.531	84.8	18.6	2632	3.36
1000	5.211	5.224	127.2	33.5	5309	12.6

Table 11.20: Moments of ζ on the critical line against the asymptotic of Hardy, Littlewood and Ingham for $k = 1$ and the leading Keating and Snaith terms for $k = 2$ and 3. Computed by `ks.py`.

11.4.6.2 The moments as exponential moments of the Thorin density

The densities v_σ are the derivatives of $\log|\xi|$ across the basepoints: $\frac{\partial}{\partial\sigma}\log|\xi(\sigma + it)| = \operatorname{Re}\frac{\xi'}{\xi}(\sigma + it) = v_\sigma(t)$ wherever $\xi(\sigma + it) \neq 0$. Integrating from the critical line to the basepoint one, for every t such that ξ has no zero on the segment from $\frac{1}{2} + it$ to $1 + it$, which excludes only countably many t,

$$\log|\xi(\tfrac{1}{2} + it)| = \log|\xi(1 + it)| - \int_{1/2}^{1} v_a(t)\,da. \tag{11.4.2}$$

The moments concern ζ, not ξ. With the gamma factor $h(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(\frac{s}{2})$, so that $\xi = h\zeta$, the identity becomes

$$|\zeta(\tfrac{1}{2} + it)|^{2k} = |\zeta(1 + it)|^{2k}\Big|\frac{h(1 + it)}{h(\frac{1}{2} + it)}\Big|^{2k}\exp\Big\{-2k\int_{1/2}^{1} v_a(t)\,da\Big\},$$

and Stirling's formula gives $|h(1 + it)/h(\frac{1}{2} + it)| \sim (t/2\pi)^{1/4}$ as $t \to \infty$. So the Keating and Snaith moments are exponential moments, over t uniform on $[0, T]$, of the density $v_a(t)$ integrated across the basepoints from $\frac{1}{2}$ to 1, with the weight $|\zeta(1 + it)|^{2k}|h(1 + it)/h(\frac{1}{2} + it)|^{2k}$, which involves no zero of ζ. At every such basepoint v_a/π is the Thorin density of T_a exactly when RH holds (Theorem 1.2.2 and Proposition 11.4.3). In the same terms, RH, which is the positivity of v_a for every $a > \frac{1}{2}$ (Lagarias's criterion, Remark 1.2.4), is the statement that $|\xi(\sigma + it)|$ increases with $\sigma \geq \frac{1}{2}$ for every t, a reformulation due to Sondow and Dumitrescu [241].

We checked (11.4.2) with `ks.py` at $t = 10$, 20 and 100, where both sides agree to all fifteen digits computed, $\log|\xi(\frac12+it)|$ being -3.2710155, -10.2139490 and -69.3773014. At $t = 14.134725$, the first ordinate to six decimals, the right side is -22.353, the logarithm of $|\xi(\frac12+it)| \approx 2\times10^{-10}$ at a point within 1.5×10^{-7} of the zero; the script records the left side there as $-\infty$, its value at γ_1 itself, instead of evaluating it. On the grid $\sigma = \frac12, 0.55, \dots, 1$ the modulus $|\xi(\sigma+it)|$ increases at all four heights.

In the language of (11.4.2) the conjecture of Keating and Snaith is a large deviation statement for the integrated Thorin density. Selberg's central limit theorem [237], which Section 7.2 uses for the argument $S(t)$, says that for t uniform on $[T, 2T]$ the variable $\log|\zeta(\frac12+it)|$ is approximately Gaussian with variance $\frac12\log\log T$. Gaussian exponential moments would give $(\log T)^{k^2}$, and the constant $a_k\mathsf{G}(1+k)^2/\mathsf{G}(1+2k)$ is the correction to them. On $[500, 1000]$ the sample variance of $\log|\zeta(\frac12+it)|$ is 1.53, against $\frac12\log\log 750 = 0.95$. The difference is a constant term, not slow convergence: the random matrix model predicts the variance $\frac12\log\log\frac{t}{2\pi} + c + o(1)$ with $c = \frac12\big(\gamma_E + 1 - \sum_p\sum_{m\ge2}(\frac1m - \frac{1}{m^2})p^{-m}\big) = 0.70$ [119], a conjecture, and this gives 1.48 on $[500, 1000]$, close to the sample value.

11.4.6.3 What the moments give

Moments of ζ on the critical line control zero density through mean value theorems, and zero density controls the almost GGC property. Ingham's exponent $3(1-\sigma)/(2-\sigma)$, which gives Theorem 11.4.2 its saving $(2\alpha-1)/(2-\alpha)$ for $\frac12 < \alpha < \frac{7}{10}$, rests on the fourth moment, $k = 2$, proved by Ingham [103]. Bounds for all moments, that is the Lindelöf hypothesis in the form $\int_0^T|\zeta(\frac12+it)|^{2k}dt \ll T^{1+\varepsilon}$ for every k, imply the density hypothesis $N(\sigma,T) \ll T^{2(1-\sigma)+\varepsilon}$, which would give the saving $2\alpha-1$, larger than that of Theorem 11.4.2 for $\frac12 < \alpha < \frac{9}{10}$ (Table 11.15). The Keating and Snaith asymptotics would give this and much more. So the line through the Barnes function is the one place where probabilistic structure of the GGC type bears directly on the strength of the almost GGC theorem.

Two limits should be stated. The GGC structure of Nikeghbali and Yor belongs to the random matrix model, not to ζ: it explains the conjecture without proving it, and only the second and fourth moments of ζ are known asymptotically. And moments, even the Lindelöf hypothesis, bound the number of zeros off the line but do not exclude them. They improve "almost GGC" without reaching "GGC".

11.4.7 From almost to exactly

Theorem 11.4.2 is the strongest statement that the Thorin programme yields with no hypothesis, and in form it is stronger than the positivity at $\alpha \ge 1$: it covers every basepoint above the centre. It cannot be upgraded to exact positivity by improving the exponents, or by the moment bounds of Section 11.4.6. Even the density hypothesis, or the Lindelöf hypothesis which implies it, leaves room for a

sparse set of zeros off the line, and a single off line zero $\beta+i\gamma$ stops T_α from being a GGC for every α in $[\frac{1}{2},\beta)$ (Theorem 1.2.2) and makes v_α negative near γ for every α close enough to β (Section 11.4.3). The step from "almost" to "exactly" is a step from counting zeros to excluding them.

Four facts about the programme say that this step needs input from outside it. The smoothing runs the wrong way: the Cauchy kernels form a semigroup, $P_a * P_b = P_{a+b}$ for $a, b > 0$, so when no zero lies to the right of α the density at a basepoint $\alpha' > \alpha$ is the Poisson smoothing of the density at α by the kernel of width $\alpha' - \alpha$, and positivity propagates to the right; RH asks to propagate it to the left, which is deconvolution, and deconvolution does not preserve positivity. Arithmetic domination requires a zero free region: below the basepoint one the compensated Lévy form, which the sine squared lemma converts into the density, holds if $\psi(x) - x = O(x^{\alpha'})$ for some $\alpha' < \alpha$, and only if $\psi(x) - x = O(x^{\alpha+\delta})$ for every $\delta > 0$, where ψ is Chebyshev's function (Proposition 9.1.6). Either condition is a zero free half plane, none was known for any $\alpha < 1$ before the half plane $\operatorname{Re} s > \frac{7}{8}$ announced in October 2026, which came from outside the programme (Section 12.1.5), and removing the pole does not help: for $L(s, \chi_{12})$ the prime sums need no compensation, and the boundary is the same (Section 11.3.2). There is no margin: along the de Bruijn Newman flow the reciprocal is a GGC for $t \geq 0.2$ [196], which improves $t \geq 0.22$ [217] (Theorem 7.2.15 and Section 8.4), RH is the statement at $t = 0$, and the de Bruijn Newman constant is nonnegative [224], so if RH is true it holds with no slack and a proof must establish a closed condition at the boundary itself. And stronger positivity conditions can be false: Conrey and Li showed that a positivity condition in de Branges' approach to RH fails [54], so any positivity statement stronger than RH should be tested numerically before it is pursued (Section 12.2.4).

In the Stieltjes form (iv) of Theorem 4.2.4 the difference between counting zeros and excluding them is explicit. Split $G'/G = S_{\text{on}} + R_{\text{off}}$, where $S_{\text{on}}(u) = \sum(u+\gamma^2)^{-1}$ runs over the zeros $\frac{1}{2}+i\gamma$ on the line with $\gamma > 0$ and is a Stieltjes function with measure U_{line}, and $R_{\text{off}}(u) = \sum(u-(\rho-\frac{1}{2})^2)^{-1}$ runs over the zeros $\rho = \beta + i\gamma$ off the line with $\gamma > 0$. By Theorem 4.3.10, all the poles of R_{off} lie in a parabolic sliver $\mathcal{E}_{T_0}$ beside the negative axis, which begins at $-T_0^2 + T_0 + \frac{1}{4}$, and in the upper half plane outside it $\operatorname{Im} R_{\text{off}} \leq 0$ and $\operatorname{Im}(G'/G) < 0$. The size of R_{off} away from the sliver needs no count of zeros: if $|u| \leq T_0^2/2$, a disc that misses $\mathcal{E}_{T_0}$, then every term has $|u-(\rho-\frac{1}{2})^2| \geq |\rho-\frac{1}{2}|^2 - |u| > \gamma^2/2$, since $\gamma > T_0$, so $|R_{\text{off}}(u)| \leq 2\Sigma_0 < 3\times 10^{-12}$ by Theorem 11.4.4. Only $\sum \gamma^{-2} < \infty$ and the verified height enter.

The number of terms of R_{off}, on the other hand, is not known to be small. Zero density estimates bound the number of those with $|\beta - \frac{1}{2}| \geq \sigma - \frac{1}{2}$ and $\gamma \leq T$ by $T^{e(\sigma)+o(1)}$ with $e(\sigma) < 1$, for each fixed $\sigma > \frac{1}{2}$ (Theorem 11.4.2), which is $o(T)$; but zeros off the line may approach the line as γ grows, and for all of them together only a proportion is known. At most $(1-\kappa+o(1))N(T)$ of the zeros up to height

T lie off the line, with $\kappa > \frac{5}{12}$ in the last published record [218] and $\kappa > 0.6725$, counting only simple zeros on the line, in two preprints of 2026 [2, 137]; that is not $o(T)$, nor even $o(N(T))$. No count, however strong, touches the Thorin condition, which requires $R_{\mathrm{off}} \equiv 0$, because a single pair puts a pole of G'/G off the cut. Section 11.5.5 gives a test for arguments that claim to exclude such poles: since the de Bruijn Newman flow has nonreal zeros at every $t < 0$ [224], no property that the kernel $e^{tv^2/4}\Phi(v)$ keeps for some $t < 0$ can exclude them, so a proof must use one that fails for every $t < 0$, and the Euler product and the explicit formula are the natural candidates.

The forms that the new input could take are those of Section 12.1: a positive object built from arithmetic, such as a Krein string or a diffusion whose hitting time is the clock, which is the idea of Hilbert and Pólya in probabilistic form, with Williams' Bessel process (Section 8.2) as the model; positive approximants, proved positive independently, converging under the closure theorem; or a direct variogram inequality at the centre. The first new theorem any of them would give is the GGC property of T_α at a single basepoint $\alpha < \frac{7}{8}$, that is $v_\alpha^- \equiv 0$, a zero free half plane beyond the one announced in 2026 (Section 12.1.5); positivity of v_α on the single line $\operatorname{Re} s = \alpha$ is not known to suffice (Remark 1.2.4).

Sections 11.4.2 and 11.4.3 make the remaining problem concrete. The growth of the defect is not part of it. For $u \geq 1$ every factor of D_α is at most one, and a factor with $\gamma \leq \sqrt{u}$ is at least $\gamma^2/(u+\gamma^2) \geq \gamma^2/(2u) \geq 1/u$, since $\gamma \geq \gamma_1 > \sqrt{2}$. Bounding the factors with $\gamma > \sqrt{u}$ as in the proof of Theorem 11.4.4 therefore gives, for $u \geq 1$,

$$\begin{aligned} |\log D_\alpha(u)| &\leq N(\alpha, \sqrt{u}) \log u + 2(u+\sqrt{u}) \sum_{\beta > \alpha,\ \gamma > \sqrt{u}} \gamma^{-2} \\ &\leq u^{e(\alpha)/2 + o(1)}, \qquad u \to \infty, \end{aligned}$$

by the exponents of Theorem 11.4.2. So bounds on the growth of the defect follow from zero density estimates alone, and what remains open is whether it is identically one. Four questions would each be a genuine step.

1. Cancellation on average: show that for fixed $\alpha > \frac{1}{2}$ the background density v_α/π is bounded below by $c \log \gamma$ outside a set of heights of density zero. With Lemma 11.4.7 this would confine any failure of positivity to zeros in the strips $\alpha < \beta < \alpha + O(1/\log \gamma)$. Remark 9.1.10 proves the lower bound for v_α/π itself, but its exceptional set contains a neighbourhood of the ordinate of every zero to the right of α, which is where Lemma 11.4.7 would need it.

2. Thin strip density: bound the number of zeros with $\alpha < \beta < \alpha + 2/\log \gamma$ and $\gamma \leq T$ by $o(T^{1-\delta})$ for some $\delta > 0$, uniformly in α. Combined with (1) this would give a pointwise almost positivity theorem.

3. Explicit defect: bound $\log D_\alpha(u)$ explicitly, with constants depending on α and the verified height, through explicit zero density estimates, as described at the end of Section 11.4.2.

4. Positivity at one basepoint: prove $D_\alpha \equiv 1$, that is $v_\alpha^- \equiv 0$, for a single $\alpha < \frac{7}{8}$, which is a zero free half plane beyond the one announced in 2026 (Section 12.1.5).

What the Thorin framework contributes is a precise target, the GGC property of one explicit clock T_α at one basepoint below $\frac{7}{8}$, together with computational tools that test candidate arguments against it: certified densities on windows, positivity margins calibrated against random matrix theory, and the explicit defect. Chapter 12 takes stock of the whole path, beginning with Section 12.1.

11.5 Pakes, the Lambert law and the Thorin condition

Anthony Pakes has worked on the objects that surround the Thorin condition: size biasing and the classical moment problem, the Laplace transforms of positive infinitely divisible laws, the Lambert function, and ladders of self decomposable laws. This section follows the first author's account of that work and of what it gives for the clock [209], verified, corrected where needed, and written in the notation of the book. In Pakes's language the question of Section 1.3 is whether $\log G$ is a Thorin Bernstein function, that is, a Bernstein function whose Lévy measure has the form $k(t)\,dt/t$ with k completely monotone [234, Chapter 8], or equivalently the Laplace exponent of a GGC. By Theorem 4.2.4 this holds if and only if $(\log G)' = G'/G$ is a Stieltjes function, and if and only if RH holds. For the Lambert function $\mathcal{W}$ the corresponding statement, that $\mathcal{W}'$ is a Stieltjes function, is a theorem, and the Thorin measure of $\mathcal{W}$ is explicit. The section follows this parallel as far as it goes.

Section 11.5.1 reviews the two strands of Pakes's work. Section 11.5.2 writes the Thorin question in his form: with no hypothesis the size biased clock is the clock plus an independent variable whose density is proportional to the heat trace, and RH holds exactly when this variable is a mixture of exponential laws; the moment problem of his first strand becomes the Hankel test of Section 5.1, whose negative eigenvalues count the zeros off the line (Theorem 5.1.5), and Table 8.10 follows the count along the de Bruijn Newman flow. Section 11.5.3 proves the theorem of Kalugin, Jeffrey and Corless that the Lambert law is a GGC with an explicit Thorin measure, by boundary values on the cut, reflection and a growth bound, and isolates the one step of that proof which does not transfer to the clock. The first author's account then localises, with no hypothesis, the points of the upper half plane where $\mathrm{Im}(G'/G)$ could fail to be negative, as it must be for a Stieltjes function, to a thin parabola along the negative axis beyond about $-T_0^2$, where $T_0 = 3 \times 10^{12}$ is the height to which RH is verified; this is Lemma 4.3.9 and

Theorem 4.3.10 in Section 4.3, and what the zeros off the line could contribute there is discussed in Section 11.4.7. Section 11.5.4 places the clock among Urbanik's multiply self decomposable laws, in L_2 and not in L_3 with no hypothesis, and shows that it is not the martingale limit of a supercritical continuous state branching process with finite Malthusian parameter. Section 11.5.5 derives from the theorem of Rodgers and Tao a filter for the properties of the Pólya kernel that could imply RH, and Section 11.5.6 sorts what survives and lists open problems.

We do not prove RH, and several of the equivalences are elementary. What the section adds is a dictionary between Pakes's language and that of the book, a few unconditional facts about the clock, the solved Lambert problem as a template whose single failing step for the clock is RH itself, a list of routes that are closed, and one property, the GGC property of the variable of Biane, Pitman and Yor, that passes the filter of Section 11.5.5 wherever it has been tested, but fails for most of the real Dirichlet L functions tested there, so that it does not detect the Euler product. The programs `pakes_sizebias.py` and `pakes_lambert.py` compute the numbers of Sections 11.5.2 and 11.5.3.

11.5.1 Two strands of Pakes's work

The first strand is the classical moment problem. For a random variable $X \geq 0$ with $0 < \mathbb{E}X < \infty$, the size biased variable $\widehat{X}$, length biased in Pakes's word, has the law $x\,P(X \in dx)/\mathbb{E}X$. Pakes and Khattree [189] characterise laws by in law equations linking $\widehat{X}$ and X, such as $X \stackrel{d}{=} \mathsf{U}\widehat{X}$ with U uniform on $(0,1)$ and independent of $\widehat{X}$, which holds exactly for the exponential laws, and they study the role of the moment problem in such characterisations, which can single out laws that are not determined by their moments. Pakes [180] studies size biasing for the laws with the moments of the log normal law, and in [181] he proves converse forms of the Carleman and Krein criteria and introduces size biasing and the stationary excess operation of order r; Berg [22] showed, correcting a remark in [181], that a law can be determinate in the sense of Stieltjes while its size biased and stationary excess versions of order one are not. Pakes, Hung and Wu [188] give criteria for the determination of a law by its moments, and Pakes [182] describes the structure of the Stieltjes classes of moment equivalent laws; these papers are cited throughout the later literature on the moment problem, for instance by Stoyanov and his coauthors and by Penson and his coauthors.

Size biasing meets infinite divisibility in Steutel's theorem: X is infinitely divisible if and only if $\widehat{X} \stackrel{d}{=} X + A$ for some $A \geq 0$ independent of X, and the law of A is then unique [245], [232, Theorem 51.1], [247]; in this form the theorem is [8, Theorem 11.1], and Pakes takes it for positive laws from Pakes, Sapatinas and Fosam [190]. In Laplace transforms, with $\Psi(u) = -\log \mathbb{E}e^{-uX}$, it reads $\mathbb{E}e^{-u\widehat{X}} = \mathbb{E}e^{-uX}\,\Psi'(u)/\Psi'(0)$, so that A has the Laplace transform $\Psi'(u)/\Psi'(0)$. In [184] Pakes assumes the decomposition and asks when A or $\widehat{X}$ is again infinitely

divisible; he shows that size biasing up to the fourth order preserves infinite divisibility for the Lambert law and for a compound Poisson law, and conjectures that it does so at every order. Section 11.5.2 applies Steutel's theorem to the clock.

For the laws of this book the moment problem is determinate. The moment generating function $\xi(\frac{1}{2}+z)/\xi(\frac{1}{2})$ of the Pólya density is entire (1.2.4). The clock has $\mathbb{E}e^{sT} < \infty$ exactly for $s < \gamma_1^2 = 199.7904548\ldots$: every zero has $|\rho - \frac{1}{2}| \geq |\operatorname{Im}\rho| \geq \gamma_1$, so $1/G$ is analytic in the disc $|u| < \gamma_1^2$ and has a pole at $-\gamma_1^2$, and by Landau's theorem [268, Chapter II, Theorem 5b] the Laplace transform of a positive measure is singular at the real point of its abscissa of convergence, which is therefore $-\gamma_1^2$. Under RH the measure $\nu = \sum_k a_k^{-1}\delta_{1/a_k}$ of the Stieltjes moments m_n is supported in $[0, \gamma_1^{-2}]$, so its moment problem is a determinate Hausdorff problem (Section 5.1), while the Thorin measure U_{RH} itself has unbounded support and infinite mass.

The second strand concerns the Laplace transforms of positive infinitely divisible laws, and the Lambert function $\mathcal{W}$, the principal solution of $\mathcal{W}(u)e^{\mathcal{W}(u)} = u$ (Section 11.5.3). Pakes [183] and, independently, Kalugin and Jeffrey [115] proved that $\mathcal{W}$ is a Bernstein function, from a formula for the derivatives of $\mathcal{W}$ that involves a sequence of polynomials [187, (8)]: Pakes through the positivity of the polynomials, Kalugin and Jeffrey by showing that their coefficients form positive, log concave and unimodal sequences. So $e^{-\mathcal{W}(u)} = \mathcal{W}(u)/u$ is the Laplace transform of an infinitely divisible law, the Lambert law. In the same paper, by Pakes's own account [187, Section 2], a complex variable argument shows that $\mathcal{W}'$ is a Stieltjes function, so that $\mathcal{W}$ is a Thorin Bernstein function and the Lambert law is a GGC. Kalugin, Jeffrey, Corless and Borwein [117] obtained Stieltjes representations independently and showed that $\mathcal{W}$ is a complete Bernstein function; the distribution function of the Thorin measure of $\mathcal{W}$ is $\pi^{-1}\operatorname{Im}\mathcal{W}(-y+i0)$, which vanishes for $y < 1/e$ [117], [116, Theorem 3.3], and the Thorin representation itself is [116, Theorem 6.2], published, with integral representations of other types, in [118]. Pakes gave two further proofs that $\mathcal{W}$ is Thorin Bernstein: from Nuttall's integral for the coefficients $n^n/n!$ of $\mathcal{W}'$, which yields an explicit quantile function of the Thorin measure [185], and from continuous state branching processes, one of whose martingale limits has the cumulant function $\mathcal{W}$ [186], which also gives the self decomposability of the Lambert law.

In [187] Pakes identifies the background driving Lévy process of the Lambert law: it is compound Poisson with unit rate and jump density $\int ze^{-xz}\,U_{\mathcal{W}}(dz)$, where $U_{\mathcal{W}}$ is the Thorin measure of $\mathcal{W}$. With $T_{\mathcal{W}}$ a variable with the Lambert law and J a jump of the driving process, this gives the in law relations $T_{\mathcal{W}} \stackrel{d}{=} \mathsf{U}(T_{\mathcal{W}} + J)$ and $T_{\mathcal{W}} \stackrel{d}{=} \mathsf{U}\mathcal{S}J$, with U uniform on $(0,1)$, $\mathcal{S}$ the stationary excess operator and the factors on the right independent; and if both relations hold for a variable X and some $J \geq 0$ of finite mean $\mu > 0$, then $X \stackrel{d}{=} \mu T_{\mathcal{W}}$, so that the two relations characterise the Lambert law up to a scale factor [187, Theorem 1] (Remark 11.5.5). He embeds the Lambert law in a family of GGCs whose Laplace

transforms solve Lambert's trinomial equation [187, Theorem 6], which gives one more proof that $\mathcal{W}$ is Thorin Bernstein [187, Corollary 2], and he introduces BDLP ladders: from an infinitely divisible $X_0 \geq 0$ with $\mathbb{E}(\log(1+X_0))^n < \infty$ for every n, the law of X_{n+1} is the self decomposable law driven by the Lévy process of X_n [187, Section 5]. In his indexing X_1 is self decomposable; this is a unit shift from Urbanik's classes L_n used in Section 11.5.4, in which L_0 is the class of self decomposable laws [263, 231, 23]. Along a ladder he finds the size biased decomposition $\widehat{X}_{n+1} \stackrel{d}{=} X_{n+1} + A_n$, with A_n of density proportional to the tail of the Lévy measure of X_n [187, (53)], and this is the form in which Section 11.5.2 applies Steutel's theorem to the clock.

11.5.2 The Thorin question in Pakes's form

For the clock the objects of both strands are at hand. The Thorin question asks whether $\log G$ is a Thorin Bernstein function, that is, whether $(\log G)' = G'/G$ is a Stieltjes function (Theorem 4.2.4); then the Thorin measure is $U_{\mathrm{RH}} = \sum_{\gamma>0} \delta_{\gamma^2}$, with no drift (Proposition 1.2.1), and Stieltjes inversion (1.3.2) recovers it from the boundary values of G'/G on the cut, or, in integrated form, from those of $\log G$ (Section 1.3.1). With no hypothesis, $\log G$ is a Bernstein function with Lévy measure $W(t)\,dt/t$, where $W(t) = \sum_k e^{-a_k t}$ is the heat trace (Theorem 5.1.1); under RH, $W(t) = \int e^{-tz}\, U_{\mathrm{RH}}(dz)$, the counterpart for the clock of the formula that gives the canonical function of the Lambert law as the Laplace transform of the density of its Thorin measure (Corollary 11.5.4; [187, (12)]).

The Bernstein property is weaker than the Thorin Bernstein property, but it is not blind to the zeros off the line. A Bernstein function extends holomorphically to $\operatorname{Re} u > 0$, and the zeros $-a$ of G that come from a zero $\rho = \frac{1}{2} + \delta + i\gamma$ off the line have $\operatorname{Re}(-a) = \delta^2 - \gamma^2 < 0$, so holomorphy alone cannot see them. But the Bernstein property of $\log G$ says more, $W(t) \geq 0$ for every $t > 0$, and that would exclude, for instance, a zero off the line below all the other zeros: if $\operatorname{Re} a = \gamma^2 - \delta^2$ were smaller than $\operatorname{Re} a_k$ for every other a_k, which is γ_k^2 for a zero on the line, the four zeros ρ, $1-\rho$, $\bar\rho$, $1-\bar\rho$ would contribute $2e^{-(\gamma^2-\delta^2)t}\cos(2\delta\gamma t)$ to $W(t)$, which dominates as $t \to \infty$ and changes sign. The property holds with no hypothesis (Theorem 5.1.1, whose proof uses the verification of RH below height 1000), so it cannot decide RH; among the Bernstein functions it is the Thorin Bernstein property that is equivalent to RH.

Size biasing gives the Thorin question a form with one variable. The size biased clock $\widehat{T}$ has the law $t\,P(T \in dt)/\mathbb{E}T$, and $\mathbb{E}T = m_0 = \sum_k a_k^{-1} = 0.0231049931\ldots$, the first Stieltjes moment of Section 5.1.

Proposition 11.5.1. *With no hypothesis, $\widehat{T} \stackrel{d}{=} T + A$, with A independent of T and of density $W(t)/m_0$ on $(0,\infty)$. RH holds if and only if A is a mixture of exponential laws, and then $A \stackrel{d}{=} E/V$, with E standard exponential and independent*

of V, and V of law $m_0^{-1}\sum_{\gamma>0}\gamma^{-2}\delta_{\gamma^2}$, the ordinates counted with multiplicity.

Proof. The Laplace transform of $\widehat{T}$ is

$$\frac{\mathbb{E}[Te^{-uT}]}{m_0} = -\frac{1}{m_0}\,\frac{d}{du}\,\frac{1}{G(u)} = \frac{1}{m_0}\,\frac{1}{G(u)}\,\frac{G'}{G}(u).$$

By Section 1.2, $(G'/G)(u) = \int_0^\infty e^{-ut}W(t)\,dt$ for $u>0$ with no hypothesis, and $W>0$ (Theorem 5.1.1); letting $u\downarrow 0$, monotone convergence gives $\int_0^\infty W(t)\,dt = (G'/G)(0) = G'(0) = \sum_k a_k^{-1} = m_0$. So W/m_0 is a probability density with Laplace transform $(G'/G)(u)/m_0$, the Laplace transform of $\widehat{T}$ is the product of those of T and of a variable A with this density, and by the uniqueness of Laplace transforms $\widehat{T}\overset{d}{=}T+A$ with A independent of T. A variable $A>0$ is a mixture of exponential laws, $A\overset{d}{=}E/V'$ with $V'>0$ independent of E, exactly when it has a completely monotone density: the density of E/V' is $\mathbb{E}[V'e^{-V't}]$, and conversely, by Bernstein's theorem, a completely monotone probability density is $\int_{[0,\infty)}e^{-vt}\,\sigma(dv)$ with $\int v^{-1}\sigma(dv)=1$, which forces $\sigma(\{0\})=0$, and this is the density of E/V' with $P(V'\in dv)=v^{-1}\sigma(dv)$. The density W/m_0 is completely monotone exactly when RH holds (Section 1.2 and Table 1.7), and then $W(t) = \sum_{\gamma>0}e^{-\gamma^2 t} = m_0\mathbb{E}[Ve^{-Vt}]$ with V as stated. □

This is Steutel's decomposition of Section 11.5.1 with $\Psi=\log G$, and it is the identity that Pakes derives along BDLP ladders [187, (53)], here for the clock and the subordinator of Proposition 11.1.1 that drives it, whose Lévy density is $h=-W'$. Pakes's variable A_n has the density $x\mapsto\int_x^\infty\omega_n(y)\,dy$, the tail of the Lévy density ω_n of the driving law, normalised by the mean $\int_0^\infty x\,\omega_n(x)\,dx$, and for the clock the tail is $\int_t^\infty h(x)\,dx = W(t)$ and the mean is $\int_0^\infty xh(x)\,dx = \int_0^\infty W(t)\,dt = m_0$. The content of Proposition 11.5.1, RH if and only if W is completely monotone, is that of Proposition 11.1.1; the size biased form puts it in one variable that exists with no hypothesis. Its mean is $\mathbb{E}A = m_1/m_0 = \operatorname{Var}T/\mathbb{E}T = 1.60885567\times10^{-3}$, from the Taylor coefficients of ξ at $\frac12$, with no zero entering. Under RH, A is infinitely divisible, as a mixture of exponential laws [80, 244], and so is $\widehat{T}=T+A$; so under RH both parts of the question that Pakes asks in [184] have a positive answer for the clock. The weights $P(V=\gamma^2)=\gamma^{-2}/m_0$ of the first six ordinates are 0.21663, 0.09794, 0.06919, 0.04676, 0.03990 and 0.03064: the lowest zero carries about a fifth of the law of V, and the first six carry 0.50105, about a half. The first 1700 ordinates carry 0.978508. The rest is $m_0^{-1}\int_{(\gamma_{1700},\infty)}t^{-2}\,dN(t)$; the smooth part $\vartheta(t)/\pi+1$ of the count, with the boundary term $(\vartheta(\gamma_{1700})/\pi+1-1700)/(m_0\gamma_{1700}^2)$ that it leaves at γ_{1700}, accounts for it to within 3×10^{-9}, and the remainder is $2m_0^{-1}\int_{\gamma_{1700}}^\infty S(t)\,t^{-3}\,dt$, where $S(t)$ is the fluctuating part of the count. The program `pakes_sizebias.py` computes these numbers, the moments at 80 digits and the weights from the 70 digit ordinates of `zeros_1700.txt`, checked against mpmath's `zetazero`.

Pakes's first strand, the moment problem, enters through the Taylor coefficients at the origin. For $|u| < \gamma_1^2$ and for $|u| < 1/e$ respectively,

$$(\log G)'(u) = \sum_{n\geq 0} m_n(-u)^n, \qquad \mathcal{W}'(u) = \sum_{n\geq 1} \frac{n^n}{n!}(-u)^{n-1},$$

with the Stieltjes moments $m_n = \sum_k a_k^{-n-1}$ of Section 5.1; the second series is the derivative of the Lagrange series of $\mathcal{W}$ [56, (3.1)]. Under RH, $m_{n-1} = \int z^{-n}\, U_{\mathrm{RH}}(dz)$ for $n \geq 1$, just as $n^n/n! = \int z^{-n}\, U_{\mathcal{W}}(dz)$ (Corollary 11.5.6): both coefficient sequences are then moment sequences of positive measures, the images of $z^{-1}U_{\mathrm{RH}}(dz)$ and of $U_{\mathcal{W}}$ under $z \mapsto 1/z$, and their Hankel matrices are positive definite. For $\mathcal{W}$ this is a consequence of the theorem of Section 11.5.3. For the clock the positive definiteness of every $\mathcal{H}_N = [m_{i+j}]_{i,j<N}$ is equivalent to RH (Section 5.1); the number of negative eigenvalues of $\mathcal{H}_N$ does not decrease with N and tends to the number of distinct zeros ρ with $\operatorname{Re}\rho > \frac{1}{2}$ and $\operatorname{Im}\rho > 0$ (Theorem 5.1.5); and Table 8.10 shows the count at work along the de Bruijn Newman flow, where in the data a pair of zeros that starts at (γ_k, γ_{k+1}) is often registered soon after its collision at an order close to $2k + 2$, although this rule fails at order 50 (Section 8.4.3).

Li's criterion is a positivity condition on the same zeros, read on the Thorin functional through the test functions φ_n of Section 8.3.2. It sees a zero off the line more slowly: the term of $\rho = \frac{1}{2} - d + i\gamma$, with $0 < d < \frac{1}{2}$, grows like e^{nd/γ^2}, so it can dominate only from n of order γ^2/d, which exceeds $2\gamma^2$, while in the data of Section 8.4.3 the Hankel test often registers a pair at an order close to $2k+2$, set by its index k, at a cost in precision that grows linearly with the order (Section 5.1). Both are tests at low height, and in both every zero off the line is eventually seen.

11.5.3 The Lambert law, a Thorin problem solved

Let $\mathcal{W}$ be the principal branch of Lambert's function, the solution of $\mathcal{W}(u)e^{\mathcal{W}(u)} = u$ that is analytic on $\mathbb{C} \setminus (-\infty, -1/e]$, real and increasing on $(-1/e, \infty)$, with $\mathcal{W}(0) = 0$, $\mathcal{W}(-1/e) = -1$ and $|\operatorname{Im}\mathcal{W}| < \pi$. Its Lagrange series $\mathcal{W}(u) = \sum_{n\geq 1}(-n)^{n-1}u^n/n!$ has radius $1/e$, and $\mathcal{W}(u) = \log u - \log\log u + o(1)$ as $u \to \infty$ [56, (3.1), (4.19)]. Since $\mathcal{W}$ is a Bernstein function [183, 115], $e^{-\mathcal{W}(u)} = \mathcal{W}(u)/u$ is the Laplace transform of an infinitely divisible law on $[0, \infty)$, the Lambert law; we write $T_{\mathcal{W}}$ for a variable with this law. It is a GGC with an explicit, absolutely continuous Thorin measure, and the proof of this fact is a template that fails for the clock at exactly one step.

Proposition 11.5.2 ([56, Section 4, (4.4)], [116, (10), (11)]). *For $0 < b < \pi$ put $r(b) = b\, e^{-b\cot b}/\sin b$. Then r increases from $1/e$ to ∞ on $(0, \pi)$, and the boundary values of $\mathcal{W}$ on the cut, from the upper half plane, are $\mathcal{W}(-r(b)+i0) = -b\cot b + ib$.*

Proof. Put $w = \alpha + ib$ with $0 < b < \pi$ in $we^w = -x$, $x > 0$. The imaginary part gives $e^\alpha(\alpha \sin b + b \cos b) = 0$, so $\alpha = -b \cot b$, and then the real part gives $e^\alpha(\alpha \cos b - b \sin b) = -e^{-b \cot b}\, b/\sin b$, so that $x = r(b)$. Next, $(\log r)'(b) = 1/b - 2\cot b + b/\sin^2 b$ and $b \sin^2 b\, (\log r)'(b) = (b - \sin b \cos b)^2 + \sin^4 b > 0$, while $r(0+) = 1/e$ and $r(\pi-) = \infty$; so for each $x > 1/e$ there is exactly one point w of the curve $\{-b \cot b + ib : 0 < b < \pi\}$ with $we^w = -x$, namely the point with $r(b) = x$. This curve bounds the range of the principal branch in the upper half plane [56, Section 4], and $\mathcal{W}$ is continuous on the cut from above, with values on it; so $\mathcal{W}(-r(b) + i0) = -b \cot b + ib$. □

Kalugin, Jeffrey and Corless obtain the next theorem from the Stieltjes representation of $\mathcal{W}'$ [116, Theorem 3.3] and the characterisation of Thorin Bernstein functions in [234, Chapter 8]. The proof below is direct, by boundary values, reflection and a growth bound, and its four steps are compared with the clock at the end of the section.

THEOREM 11.5.3 (Kalugin, Jeffrey and Corless [116, Theorem 6.2], [118]). *Let $U_{\mathcal{W}}$ be the image of the uniform law db/π on $(0, \pi)$ under r, a probability measure on $(1/e, \infty)$. Then for $u \in \mathbb{C} \setminus (-\infty, -1/e]$*

$$\mathcal{W}(u) = \int \log\Big(1 + \frac{u}{z}\Big)\, U_{\mathcal{W}}(dz) = \frac{1}{\pi} \int_0^\pi \log\Big(1 + \frac{u}{r(b)}\Big)\, db.$$

Hence $\mathcal{W}$ is a Thorin Bernstein function, and $T_{\mathcal{W}}$ is a GGC with no drift and Thorin measure $U_{\mathcal{W}}$, of total mass one, whose distribution function is $U_{\mathcal{W}}((1/e, y]) = \pi^{-1} \operatorname{Im} \mathcal{W}(-y + i0)$ for $y > 1/e$.

Proof. Let $\Psi(u)$ be the integral. Since $U_{\mathcal{W}}$ is carried by $(1/e, \infty)$, Ψ is analytic on $\mathbb{C} \setminus (-\infty, -1/e]$, real on $(-1/e, \infty)$, and $\Psi(\bar{u}) = \overline{\Psi(u)}$; so is $\mathcal{W}$. Let $x > 1/e$, and let $b(x) \in (0, \pi)$ be the point with $r(b(x)) = x$. As $\eta \downarrow 0$, $1 + (-x + i\eta)/z$ tends to a negative number from the upper half plane when $z < x$ and to a positive number when $z > x$, and $U_{\mathcal{W}}$ has no atoms, so by dominated convergence $\operatorname{Im} \Psi(-x + i0) = \pi\, U_{\mathcal{W}}((1/e, x)) = b(x)$; by Proposition 11.5.2 also $\operatorname{Im} \mathcal{W}(-x + i0) = b(x)$, which is the formula for the distribution function. Near $1/e$, $r(b) - 1/e \sim b^2/(2e)$, so $U_{\mathcal{W}}$ has a density asymptotic to $(e/2)^{1/2} \pi^{-1} (z - 1/e)^{-1/2}$ there, and a bounded continuous density on $[1/e + c, \infty)$ for every $c > 0$. Hence Ψ extends continuously to the closed upper half plane, and so does $\mathcal{W}$, with $\mathcal{W}(-1/e) = -1$. The difference $D = \mathcal{W} - \Psi$ is therefore continuous on the closed upper half plane and real on the real axis, where on $(-\infty, -1/e)$ the imaginary parts of the two boundary values agree. The Schwarz reflection principle continues D across $(-\infty, -1/e)$, the continuation coincides with D in the lower half plane because $D(\bar{u}) = \overline{D(u)}$, and the singularity at $-1/e$, where D is bounded, is removable. So D is entire, and $D(0) = 0$.

It remains to bound D, and one sided bounds on real parts suffice. First, $\operatorname{Re} \Psi(u) = \int \log|1 + u/z|\, U_{\mathcal{W}}(dz) \le \log(1 + e|u|)$, since $z > 1/e$. Next let $|u| =$

$R \ge e$ and $w = \mathcal{W}(u)$, so that $|w|e^{\operatorname{Re} w} = R$ and $|\operatorname{Im} w| < \pi$. If $|w| > \log R + \pi$, then $|\operatorname{Re} w| > |w| - \pi > \log R$. This is impossible when $\operatorname{Re} w \ge 0$, since then $\operatorname{Re} w = \log R - \log|w| < \log R$; and when $\operatorname{Re} w < 0$ it gives $\log|w| - \log R > |w| - \pi$, that is $|w| - \log|w| < \pi - \log R \le \pi - 1$, which fails for $|w| > 1 + \pi$, since $x - \log x$ increases for $x > 1$ and exceeds $\pi - 1$ at $x = 1 + \pi$. Hence $|w| \le \log R + \pi$ and $\operatorname{Re}\mathcal{W}(u) = \log R - \log|w| \ge \log R - \log(\log R + \pi)$. By continuity both bounds hold on the whole circle, its point on the cut included, so

$$\max_{|u|=R} \operatorname{Re}\big(-D(u)\big) \le \log(1 + eR) - \log R + \log(\log R + \pi), \qquad R \ge e.$$

The Borel Carathéodory lemma for the entire function $-D$, which vanishes at the origin, gives $\max_{|u|=R/2}|D(u)| \le 2\max_{|u|=R}\operatorname{Re}(-D(u))$, which is $O(\log\log R)$ by this bound. By Cauchy's estimates every Taylor coefficient of D of positive order vanishes, so $D \equiv D(0) = 0$. □

The growth step cannot be done by writing the integral as $\log u$ plus a bounded term. The total mass of $U_{\mathcal{W}}$ is one, but $\int \log z\, U_{\mathcal{W}}(dz) = \infty$, since $U_{\mathcal{W}}((y,\infty)) \sim 1/\log y$ as $y \to \infty$, and $\mathcal{W}(u) - \log u = -\log\log u + o(1)$ is unbounded; the one sided bounds avoid this.

Corollary 11.5.4 (first part from [116, Theorem 6.5] and [118]). *The canonical function of $T_{\mathcal{W}}$ is*

$$k_{\mathcal{W}}(x) = \int e^{-xz}\, U_{\mathcal{W}}(dz) = \frac{1}{\pi}\int_0^{\pi} e^{-x\, r(b)}\, db,$$

so that $\mathcal{W}(u) = \int_0^{\infty}(1 - e^{-ux})\, k_{\mathcal{W}}(x)\, dx/x$ for $u \ge 0$. The symmetric law of $Z\sqrt{2T_{\mathcal{W}}}$, with Z standard normal and independent of $T_{\mathcal{W}}$, has characteristic function $e^{-\mathcal{W}(w^2)}$, is an EGGC (Section 1.3.2), and has Lévy density $k^{\mathrm{s}}_{\mathcal{W}}(|x|)/|x|$ with

$$k^{\mathrm{s}}_{\mathcal{W}}(x) = \int e^{-x\sqrt{z}}\, U_{\mathcal{W}}(dz) = \frac{1}{\pi}\int_0^{\pi} e^{-x\sqrt{r(b)}}\, db,$$

that is, $\mathcal{W}(w^2) = \int_0^{\infty} 4\sin^2(wx/2)\, k^{\mathrm{s}}_{\mathcal{W}}(x)\, dx/x$ for real w.

Proof. Insert Frullani's integral $\log(1 + u/z) = \int_0^{\infty}(1 - e^{-ux})e^{-zx}x^{-1}dx$ in Theorem 11.5.3 and exchange the integrals by Tonelli's theorem. For the symmetric law, $\mathbb{E}e^{iwZ\sqrt{2T_{\mathcal{W}}}} = \mathbb{E}e^{-w^2 T_{\mathcal{W}}}$, and the cosine dictionary (6.2.4) with $U = U_{\mathcal{W}}$, for which $K_U = 2k^{\mathrm{s}}_{\mathcal{W}}$, gives the Lévy density $K_U(|x|)/(2|x|)$. □

Remark 11.5.5 (Self decomposability, after Pakes). Since $k_{\mathcal{W}}$ is nonincreasing, $T_{\mathcal{W}}$ is self decomposable, and as in Proposition 11.1.1 its driving subordinator Z has the Lévy density $-k'_{\mathcal{W}}(x) = \int z e^{-xz}\, U_{\mathcal{W}}(dz)$, of total mass $k_{\mathcal{W}}(0+) = 1$ and with mean $\int_0^{\infty} k_{\mathcal{W}}(x)\, dx = \int z^{-1}\, U_{\mathcal{W}}(dz) = 1$ (Corollary 11.5.6). So the driving process is compound Poisson with unit rate and jumps J of mean one; its Laplace

exponent is $u\mathcal{W}'(u) = \mathcal{W}(u)/(1+\mathcal{W}(u))$, and $\mathbb{E}e^{-uJ} = 1/(1+\mathcal{W}(u))$ [187, (5), (6), (18)]. Stopping $T_{\mathcal{W}} = \int_0^\infty e^{-t}\, d\mathsf{Z}_t$ at the first jump of Z, at an exponential time τ, gives the perpetuity $T_{\mathcal{W}} \stackrel{d}{=} \mathsf{U}(T_{\mathcal{W}} + J)$, with $\mathsf{U} = e^{-\tau}$ uniform on $(0,1)$ and the three variables independent. The stationary excess $\mathcal{S}J$ has the density $P(J > x)$, since $\mathbb{E}J = 1$, and the Laplace transform $(1 - \mathbb{E}e^{-uJ})/u = \mathcal{W}'(u)$; since $\mathbb{E}e^{-u\mathsf{U}Y} = u^{-1}\int_0^u \mathbb{E}e^{-sY}ds$ for $Y \ge 0$ independent of U, also $T_{\mathcal{W}} \stackrel{d}{=} \mathsf{U}\,\mathcal{S}J$, and $\mathcal{S}J \stackrel{d}{=} J + T_{\mathcal{W}}$. The converse, part (b) of [187, Theorem 1], has the same short proof, which we repeat in our notation. If $X \stackrel{d}{=} \mathsf{U}(X+J)$ and $X \stackrel{d}{=} \mathsf{U}\,\mathcal{S}J$ for some $J \ge 0$ of mean $\mu \in (0,\infty)$, put $\varphi(u) = \mathbb{E}e^{-uX}$, $j(u) = \mathbb{E}e^{-uJ}$ and $F(u) = u\varphi(u)$. The two relations say that $F' = \varphi j$ and $F' = (1-j)/(\mu u)$; eliminating $j = uF'/F$ gives $F' = F/(u(1+\mu F))$, so $y(u) = \mu F(u/\mu)$ satisfies $y' = y/(u(1+y))$, that is $(\log y + y)' = 1/u$, with $y(u) \sim u$ as $u \downarrow 0$. Hence $ye^y = u$ and $y = \mathcal{W}$ on $(0,\infty)$, which gives $\mathbb{E}e^{-uX} = \mathcal{W}(\mu u)/(\mu u)$, so $X \stackrel{d}{=} \mu T_{\mathcal{W}}$, and $j(u) = 1/(1+\mathcal{W}(\mu u))$, so J/μ has the law of the jumps above.

Corollary 11.5.6 ([116, Corollary 3.4], [117]; Nuttall's integral [169, 42]). *For $n \ge 0$,*

$$\int z^{-n}\, U_{\mathcal{W}}(dz) = \frac{1}{\pi}\int_0^\pi \Big(\frac{\sin b\; e^{b\cot b}}{b}\Big)^n db = \frac{n^n}{n!} \qquad (0^0 = 1),$$

so $(n^n/n!)_{n\ge 0}$ is the moment sequence of the image of $U_{\mathcal{W}}$ under $z \mapsto 1/z$, a probability measure on $(0,e)$. Nuttall conjectured, and Bouwkamp proved, the identity with $\nu^\nu/\Gamma(\nu+1)$ for every real $\nu \ge 0$.

Proof. Differentiating Theorem 11.5.3 gives

$$\mathcal{W}'(u) = \int \frac{U_{\mathcal{W}}(dz)}{u+z} = \sum_{n\ge 1}(-u)^{n-1}\int z^{-n}\, U_{\mathcal{W}}(dz), \qquad |u| < 1/e,$$

the series converging uniformly in z since $z > 1/e$. Compare with the derivative of the Lagrange series $\mathcal{W}(u) = \sum_{n\ge1}(-n)^{n-1}u^n/n!$, whose coefficients are $(-n)^{n-1}/(n-1)! = (-1)^{n-1}n^n/n!$. For $n = 0$ the integral is the total mass, one. □

Conversely, as Kalugin, Jeffrey and Corless observe after their Corollary 3.4 [116], Nuttall's integral, which Bouwkamp proved without the Lambert function, gives back the theorem. It says that the image μ of $U_{\mathcal{W}}$ under $z \mapsto 1/z$ has the moments $n^n/n!$, so $\int t\,\mu(dt)/(1+ut) = \int U_{\mathcal{W}}(dz)/(u+z)$ and $\mathcal{W}'(u)$ have the same Taylor series $\sum_{m\ge0}(-u)^m(m+1)^m/m!$ at the origin; both are analytic on $\mathbb{C}\setminus(-\infty,-1/e]$, so they agree, and integrating from 0 gives Theorem 11.5.3. Pakes's second proof that $\mathcal{W}$ is Thorin Bernstein also goes through Nuttall's integral [185], [187, Section 2].

The program `pakes_lambert.py` checks the section. At 40 digits the boundary values of Proposition 11.5.2 hold to 3×10^{-40} at eight values of b from 0.05 to 3.1 (at 30 digits, at $b=1.3$, to all digits), the identity for $b\sin^2 b\,(\log r)'$ holds to 10^{-39}, and $(r(b)-1/e)/b^2$ is 0.18393972059 at $b=10^{-6}$, against $1/(2e)$. The integral of Theorem 11.5.3 agrees with $\mathcal{W}$ to 2×10^{-40} at seventeen points, ten real ones from -0.3678 to 10^{12} and seven complex ones, some next to the cut, such as $-5+0.001i$ and -10^6+10^3i. At 60 digits Newton's method on $we^w=27/10$, the integral and Arb's Lambert function agree: $\mathcal{W}(2.7)=0.996628734265477466030297598040083309\ldots$. The binary double nearest 2.7 exceeds 2.7 by 1.8×10^{-16}, and $\mathcal{W}$ there is larger by 3.3×10^{-17}, so for digits beyond the sixteenth the input must be passed exactly, as the rational 27/10. The moments of Corollary 11.5.6 agree with $n^n/n!$ for $n\le12$ to 10^{-37}, and with $\nu^\nu/\Gamma(\nu+1)$ at $\nu=0.5$, 1.5, 2.7 and 7.25 to 3×10^{-39}; for $n=0,\ldots,6$ they are 1, 1, 2, 4.5, 10.6667, 26.0417 and 64.8. In double precision the Lévy integrals of Corollary 11.5.4 reproduce $\mathcal{W}(u)$ at $u=1$, 2.7 and 10, and $\mathcal{W}(w^2)$ at $w=1$ and 2, to 6×10^{-16}, and $k_{\mathcal{W}}(x)$ is 0.988 and 0.9988 times $\sqrt{e/2}\,e^{-x/e}/\sqrt{\pi x}$ at $x=100$ and 1000, as the density of $U_{\mathcal{W}}$ at $1/e$ predicts; at $z-1/e=1.8\times10^{-9}$ that density is 0.999999995 times its asymptote. The growth facts behind the proof are visible too: $\mathcal{W}(u)-\log u+\log\log u$ falls from 0.308 at $u=10^2$ to 0.0194 at 10^{128}; $U_{\mathcal{W}}((y,\infty))$ is 0.0290 at $y=10^{16}$, against $1/\log y=0.0271$; and the part of $\int\log z\,U_{\mathcal{W}}(dz)$ from $b<\pi-c$ exceeds $\log(1/c)$ by 0.126, 0.144, 0.145 and 0.145 at $c=10^{-2}$, 10^{-4}, 10^{-8} and 10^{-16}. On the circles $|u|=R$ the lower bound $\operatorname{Re}\mathcal{W}(u)\ge\log R-\log(\log R+\pi)$ of the proof holds with the margins 0.81, 0.76, 0.65, 0.47 and 0.30 at $R=3$, 10, 10^2, 10^4 and 10^8, attained on the cut, and $|\mathcal{W}(u)|\le0.74(\log R+\pi)$ there.

What transfers to the clock. The proof of Theorem 11.5.3 has four steps: the boundary values on the cut, the positivity of the boundary measure, the identification by reflection and a growth bound, and, for the symmetric law, the cosine dictionary (6.2.4). For the clock the first step is the argument count. Continued along the upper side of the negative axis, past the zeros of G on it by small half circles in the upper half plane, $\operatorname{Im}\log G(-x+i0)=\pi N_{\mathrm{line}}(\sqrt{x})$, where $N_{\mathrm{line}}(t)$ counts with multiplicity the zeros on the critical line with ordinate in $(0,t)$, because the four zeros ρ, $1-\rho$, $\bar\rho$, $1-\bar\rho$ of a zero off the line contribute to G the factor $(1+u/a)(1+u/\bar a)=|1+u/a|^2>0$ at real u; so $N_{\mathrm{line}}(\sqrt{x})$, the distribution function of U_{line} (Proposition 1.2.3), is read from the boundary values of $\log G$ with no hypothesis, and it is $N(\sqrt{x})$ under RH. The second step is then automatic, since a counting function is nondecreasing. The fourth step gives the Poisson trace $K_{1/2}(y)=\sum_k e^{-y\sqrt{a_k}}$, which is $\sum_{\gamma>0}e^{-\gamma y}$ under RH, as the Lévy side of $Y_{1/2}=Z\sqrt{2T}$ (Section 10.1.4), while the heat trace W is the canonical function of T itself (Section 1.3.1); the two traces are linked with no hypothesis by subordination. The third step is the obstruction: $\mathcal{W}$ is analytic off $(-\infty,-1/e]$ by construction, whereas $\log G$ is analytic off $(-\infty,0]$ if and only if RH holds

(Proposition 1.2.1 and the argument after it). In the Lambert case the boundary measure is a continuum lying on the cut; for the clock its atoms are the zeros, and whether they all lie on the cut is the whole question.

11.5.4 Multiply self decomposable laws and branching

The centre clock is self decomposable with no hypothesis (Theorem 5.1.1), so it is the integral $\int_0^\infty e^{-t}\,dZ_t$ of its driving subordinator Z (Proposition 11.1.1). The law of Z_1 may be self decomposable in its turn, with a driving process of its own, and so on. Read upwards, from the last driving process to the clock, such a chain is what Pakes calls a BDLP ladder [187, Section 5], after the background driving Lévy process, and how far it goes places a law in one of Urbanik's classes L_n. This subsection shows that for the centre clock the chain has exactly three steps, with no hypothesis on the zeros (Theorem 11.5.8), and that the clock is not the martingale limit of a supercritical continuous state branching process, the construction behind one of Pakes's proofs for the Lambert law (Theorem 11.5.13). Both obstructions come from the bottom of the spectrum: the ladder breaks at a time where the term of the first zero is negative and the other zeros cannot compensate it, and the branching relation breaks at a singularity that a numerical continuation traces to the critical point of Ξ between the first two zeros. Neither says anything about RH. Along the de Bruijn Newman flow, where the zeros move towards the origin and apart, a computation that is numerical for $t > 0$ keeps the index two on its window up to $t = 27.8$ and finds the index one at $t = 27.9$ and at every larger t tested, by the same mechanism (Section 11.5.4.3).

11.5.4.1 Urbanik's classes and the ladder functions

A law μ on $\mathbb{R}$ is self decomposable if for every $c \in (0,1)$ there is a law μ_c, the c component of μ, such that the characteristic functions satisfy $\widehat{\mu}(z) = \widehat{\mu}(cz)\,\widehat{\mu}_c(z)$. Urbanik [263] let L_0 be the class of self decomposable laws, L_n for $n \geq 1$ the class of laws in L_{n-1} all of whose components lie in L_{n-1}, and $L_\infty = \bigcap_n L_n$. Sato [231] proved that L_n is also the class of limit laws of normed sums of independent summands whose laws lie in L_{n-1}, and that L_∞ is the smallest class of laws that contains the stable laws and is closed under convolution and weak limits [231, Theorem 3.5]. In the language of Section 11.1.1, a law lies in L_n, $n \geq 1$, exactly when it is self decomposable and the law at time one of its driving process lies in L_{n-1} [102]. So L_1 consists of the self decomposable laws whose driving process is self decomposable, L_2 of those whose driving process has its law in L_1, and so on. Pakes counts the steps from one: his class L_n^+ consists of the drift free laws on $[0,\infty)$ in L_{n-1}, so that L_1^+ is the self decomposable class [187]. We keep Urbanik's indexing.

For laws on $[0,\infty)$ Berg and Forst put the classes in analytic form. Let f be the Laplace exponent of an infinitely divisible law μ on $[0,\infty)$, so that $\int e^{-ux}\,\mu(dx) =$

$e^{-f(u)}$, and let $(Sf)(u) = uf'(u)$. Then $\mu \in L_n$ if and only if $Sf, S^2f, \dots, S^{n+1}f$ are Bernstein functions [23, Theorem 1.3]. For $n = 0$ this says that μ is self decomposable exactly when Sf is the Laplace exponent of a subordinator, its driving process, as in the proof of Proposition 11.1.1.

For the centre clock $f = \log G$, and S acts on the canonical function W through the ladder functions

$$W^{[0]} = W, \qquad W^{[j+1]}(t) = -t\,\frac{d}{dt}W^{[j]}(t), \qquad \text{so that} \quad W^{[j]} = \Big(-t\frac{d}{dt}\Big)^j W.$$

For one exponential, $(-t\frac{d}{dt})^j e^{-at} = \beta_j(at)\,e^{-at}$, where $\beta_0 = 1$ and $\beta_{j+1}(u) = u\beta_j(u) - u\beta_j'(u)$:

$$\beta_1(u) = u, \quad \beta_2(u) = u^2 - u, \quad \beta_3(u) = u^3 - 3u^2 + u, \quad \beta_4(u) = u^4 - 6u^3 + 7u^2 - u.$$

In general $\beta_j(u) = \sum_{i=1}^{j}(-1)^{j-i}S(j,i)\,u^i = (-1)^j B_j(-u)$, with $S(j,i)$ the Stirling numbers of the second kind and B_j the Touchard polynomials, and since $\beta_j(u)e^{-u} = (-u\frac{d}{du})^j e^{-u}$ and $-u\frac{d}{du}$ multiplies Mellin transforms by s,

$$\int_0^\infty \beta_j(u)\,e^{-u}u^{s-1}\,du = s^j\,\Gamma(s) \qquad (s > 0). \tag{11.5.1}$$

The positive roots of β_3 are $(3 \mp \sqrt{5})/2 = 0.381966$ and 2.618034, and those of β_4 are 0.165757, 1.343380 and 4.490864, so that $\beta_4 < 0$ on $(0, 0.165757) \cup (1.343380, 4.490864)$. Since $W(t) = \sum_k e^{-a_k t}$ with $\operatorname{Re} a_k \geq \gamma_k^2 - \frac{1}{4} > 0$ for every zero (Section 11.1.1),

$$W^{[j]}(t) = \sum_k \beta_j(a_k t)\,e^{-a_k t}, \tag{11.5.2}$$

the series converging absolutely and locally uniformly on $(0,\infty)$; the atoms of an off line zero come in conjugate pairs, so the sum is real.

Lemma 11.5.7. *For every $j \geq 0$ and $u \geq 0$,*

$$\Big(u\frac{d}{du}\Big)^j \log G(u) = \int_0^\infty \big(1 - e^{-ut}\big)\,W^{[j]}(t)\,\frac{dt}{t}. \tag{11.5.3}$$

For $n \geq 0$ the centre clock lies in L_n if and only if $W^{[n+1]} \geq 0$ on $(0,\infty)$, and then $W^{[1]}, \dots, W^{[n]}$ are nonnegative too. In that case the driving processes $Z^{[1]} = Z$, $Z^{[2]}$, …, $Z^{[n+1]}$ exist, each the driving process of the law at time one of the one before; $Z^{[j]}$ has Laplace exponent $(u\frac{d}{du})^j \log G(u)$ and Lévy density $W^{[j]}(t)/t$; and for $1 \leq j \leq n$ the law of $Z_s^{[j]}$ is self decomposable for every $s > 0$.

Proof. By (11.5.2), $W^{[j]}(t) = O(t^j e^{-\gamma_1^2 t})$ as $t \to \infty$, and by the explicit formula (11.5.4) below, $W^{[j]}(t) = O(t^{-1/2}\log(1/t))$ as $t \downarrow 0$. The case $j = 0$ of (11.5.3) is

the Lévy Khintchine form of $\log G$ (Section 11.1.1). If it holds for j, then, since $ue^{-ut} = \frac{d}{dt}(1 - e^{-ut})$,

$$u\frac{d}{du}\int_0^\infty \bigl(1 - e^{-ut}\bigr)W^{[j]}(t)\,\frac{dt}{t} = \int_0^\infty ue^{-ut}\,W^{[j]}(t)\,dt$$
$$= \int_0^\infty \bigl(1 - e^{-ut}\bigr)\,W^{[j+1]}(t)\,\frac{dt}{t},$$

the boundary terms $(1 - e^{-ut})W^{[j]}(t)$ vanishing at both ends by the two bounds. A function of the form (11.5.3) is a Bernstein function exactly when $W^{[j]} \geq 0$: if $W^{[j]} \geq 0$ it is a Lévy Khintchine form, and conversely, if it is a Bernstein function, its derivative $\int_0^\infty e^{-ut}W^{[j]}(t)\,dt$ is completely monotone, hence the Laplace transform of a nonnegative measure, which by the uniqueness of Laplace transforms is $W^{[j]}(t)\,dt$. So by the theorem of Berg and Forst the clock lies in L_n exactly when $W^{[1]}, \ldots, W^{[n+1]}$ are nonnegative. If $W^{[n+1]} = -t\,(W^{[n]})' \geq 0$, then $W^{[n]}$ is nonincreasing, and since it tends to 0 at infinity it is nonnegative; descending in the same way, so are $W^{[n-1]}, \ldots, W^{[1]}$. For the driving processes, the Laplace exponent of $\int_0^\infty e^{-t}\,dY_t$ is $\int_0^u \Psi_Y(w)\,dw/w$ when the subordinator Y has Laplace exponent Ψ_Y (proof of Proposition 11.1.1), so a self decomposable law with Laplace exponent f has a driving process with Laplace exponent $uf'(u)$. Starting from $f = \log G$ and using (11.5.3), $Z^{[j]}$ has Laplace exponent $(u\frac{d}{du})^j \log G$ and Lévy density $W^{[j]}(t)/t$ for as long as the laws at time one are self decomposable, and the law of $Z_s^{[j]}$, with Lévy density $sW^{[j]}(t)/t$, is self decomposable exactly when $W^{[j]}$ is nonincreasing [232, Corollary 15.11], that is, when $W^{[j+1]} \geq 0$. □

For one exponential the second rung already fails: $W^{[2]}(t) = (a^2t^2 - at)e^{-at} < 0$ for $at < 1$. So a gamma law, whose canonical function is a multiple of e^{-at}, is self decomposable but does not lie in L_1, and the question for the clock is whether the sum (11.5.2) over all zeros repairs the signs that each of its terms violates.

11.5.4.2 The ladder index of the centre clock

THEOREM 11.5.8. *Unconditionally, that is, using only the verification of RH below height* 2197.3 *[196], $W^{[3]}(t) > 0$ for every $t > 0$, and $W^{[4]}(e^{-4.15}) < 0$. Consequently W, $W^{[1]}$ and $W^{[2]}$ are positive on $(0, \infty)$; the centre clock lies in L_2 and not in L_3; its driving process $Z = Z^{[1]}$ and the driving process $Z^{[2]}$ of Z are self decomposable at every time, and the driving process $Z^{[3]}$ of $Z^{[2]}$ is self decomposable at no time.*

Proof. By Lemma 11.5.7 it suffices to prove the two inequalities; the lower ladder functions are then positive, not only nonnegative, since $W^{[j]}$ strictly decreases to 0 when $W^{[j+1]} > 0$. As in the proof of Theorem 5.1.1, large t is handled with the zeros and small t with the explicit formula, and the two ranges overlap.

Large t. Let $\tau_1 = (3+\sqrt{5})/(2\gamma_1^2) = 0.0131039$, so that $\gamma_1^2\tau_1 = 2.618034$ is the largest root of β_3; beyond it β_3 is positive and increasing. The zeros with ordinates below $\Gamma = 2197.3$ lie on the line [196]; there are $N(\Gamma) = 1700$ of them, the last at 2197.2587, and `bdlp_ladder.py` encloses each one in ball arithmetic. For $t \ge \tau_1$ each of them contributes $\beta_3(\gamma^2 t)e^{-\gamma^2 t} \ge 0$ to $W^{[3]}(t)$, and the second contributes at least $\beta_3(\gamma_2^2\tau_1)e^{-\gamma_2^2 t} > 99.38\, e^{-\gamma_2^2 t}$, since $\gamma_2^2\tau_1 = 5.7910$. A zero with ordinate $\gamma > \Gamma$, wherever it lies, has $\operatorname{Re} a \ge \gamma^2 - \frac14$ and $|a| \le \gamma^2 + \frac14$, so its term has modulus at most $f(\gamma) = B_3((\gamma^2+\frac14)t)\, e^{-(\gamma^2-\frac14)t}$, where the Touchard polynomial B_j has the moduli of the coefficients of β_j. Since $vB_j'(v) \le jB_j(v)$, the logarithm of $B_j(v)e^{-v}$ decreases at a rate of at least $\frac12$ for $v \ge 2j$, so for $(\Gamma^2+\frac14)t \ge 6$ the function f decreases and $f(\gamma) \le f(\Gamma)e^{-(\gamma^2-\Gamma^2)t/2}$. Let

$$N^+(T) = \frac{T}{2\pi}\log\frac{T}{2\pi e} + 0.1038\log T + 0.2573\log\log T + 9.3675,$$

an upper bound for $N(T)$ for $T \ge e$ [92, Corollary 1.2], whose derivative is at most $\frac{1}{2\pi}\log\frac{T}{2\pi} + \frac{0.4}{T}$ for $T \ge \Gamma$. Partial summation against $N \le N^+$, with $\gamma^2 - \Gamma^2 \ge 2\Gamma(\gamma-\Gamma)$ and $\log\frac{\gamma}{2\pi} \le \log\frac{\Gamma}{2\pi} + \frac{\gamma-\Gamma}{\Gamma}$, bounds the sum of these moduli by

$$\varepsilon(t) = f(\Gamma)\Big(N^+(\Gamma) - 1700 + \frac{L}{\Gamma t} + \frac{1}{2\pi\Gamma^3 t^2}\Big), \qquad L = \frac{1}{2\pi}\log\frac{\Gamma}{2\pi} + \frac{0.4}{\Gamma}.$$

The logarithmic derivative of $e^{\gamma_2^2 t}\varepsilon(t)$ is at most $3/t - (\Gamma^2 - \frac14 - \gamma_2^2) < 0$ for $t \ge \tau_1$, and $e^{\gamma_2^2\tau_1}\varepsilon(\tau_1) < 10^{-27458}$. So $W^{[3]}(t) \ge e^{-\gamma_2^2 t}\big(99.38 - e^{\gamma_2^2 t}\varepsilon(t)\big) > 0$ for $t \ge \tau_1$.

Small t. Applying $(-t\frac{d}{dt})^j$ term by term to (1.2.5), which the Gaussian factors in r and in $\log n$ allow, gives with no hypothesis

$$W^{[j]}(t) = \beta_j\big(-\tfrac{t}{4}\big)e^{t/4} + A_j(t) - P_j(t), \tag{11.5.4}$$

where

$$A_j(t) = \frac{1}{2\pi}\int_0^\infty \beta_j(r^2 t)\, e^{-r^2 t}\, m(r)\, dr,$$

$$P_j(t) = \frac{1}{2\sqrt{\pi t}}\sum_{n\ge 2}\frac{\Lambda(n)}{\sqrt{n}}\, e^{-v_n} q_j(v_n), \qquad v_n = \frac{(\log n)^2}{4t},$$

and $q_0 = 1$, $q_{j+1}(v) = (\frac12 - v)q_j(v) + vq_j'(v)$, because $-t\frac{d}{dt}$ maps $t^{-1/2}$ to $\frac12 t^{-1/2}$ and v_n to v_n; thus

$$q_3(v) = -v^3 + \tfrac92 v^2 - \tfrac{13}{4}v + \tfrac18, \qquad q_4(v) = v^4 - 8v^3 + \tfrac{29}{2}v^2 - 5v + \tfrac{1}{16}.$$

In the notation of the proof of Theorem 5.1.1, $\frac{1}{2\pi}\int_0^\infty e^{-r^2 t}\log\frac{r}{2\pi}\, dr = s_0(t)(\ell+\alpha_0)$, and since $-t\frac{d}{dt}$ multiplies s_0 by $\frac12$ and maps ℓ to $\frac12$, replacing $m(r)$ by $\log\frac{r}{2\pi}$ in A_j gives

$$\bar{A}_j(t) = 2^{-j} s_0(t)\,(\ell + \alpha_0 + j),$$

which also follows from (11.5.1) and its derivative in s at $s=\frac12$. For the difference we split the integral at $r=5$. For $r\le5$ we use $|\beta_j(u)|e^{-u}\le u$, valid for $j=2,3,4$ with equality only in the limit $u\downarrow0$, and the bound $\int_0^5 r^2|m(r)-\log\frac{r}{2\pi}|\,dr<0.404$ stated after the proof of Theorem 5.1.1; for $r\ge5$ we use the bound $r^2|m(r)-\log\frac{r}{2\pi}|<0.05$ stated there, and substitute $u=r^2t$. This gives

$$|A_j(t)-\bar A_j(t)|\le E_j(t)=\frac{1}{2\pi}\Bigl(0.404\,t+0.025\sqrt{t}\,J_j\Bigr),$$

where $J_j=\int_0^\infty|\beta_j(u)|\,e^{-u}u^{-3/2}\,du$ satisfies $J_3<1.3313$ and $J_4<1.8295$. Let $\tau_2=1/75$. For $t\le\tau_2$ the main term $\bar A_3(t)$ decreases in t, because $-t\frac{d}{dt}\bar A_3=\bar A_4$ and $\ell+\alpha_0+4>0$; the modulus $|\beta_3(-\frac{t}{4})|e^{t/4}$ of the polar term and $E_3(t)$ increase; and so does the modulus of each term of $P_3(t)$, because $v_n\ge v_2>9$, the polynomials q_3 and q_4 have no root in $[9,\infty)$, so that $q_3<0<q_4$ there, and $-t\frac{d}{dt}$ maps the nth term of P_3 to the nth term of P_4. At $t=\tau_2$ the four quantities are $\bar A_3(\tau_2)=0.357155$, 0.003378, $E_3(\tau_2)<0.001469$, and at most 0.057882 for the sum of the moduli of the terms of P_3, in which the terms with $n>200$ are bounded through $e^{-(\log n)^2/8t}\le n^{-\sigma}$, $\sigma=(\log200)/8t$, as in the proof of Theorem 5.1.1. Hence $W^{[3]}(t)>0.2944$ for $t\le\tau_2$, and since $\tau_2>\tau_1$, $W^{[3]}>0$ on $(0,\infty)$.

Rung four. Let $t_0=e^{-4.15}$. Then $\gamma_1^2t_0=3.1496$ lies in the interval $(1.343380,4.490864)$ on which $\beta_4<0$. The first five zeros contribute -0.97604, 0.62189, 0.22811, 0.01295 and 0.00216 to $W^{[4]}(t_0)$, and all zeros together, with the tail bound above, give $W^{[4]}(t_0)=-0.11089729616255\pm10^{-23}$ in ball arithmetic. The explicit formula confirms the sign without the zeros: at t_0 the polar term is 0.004066, $\bar A_4(t_0)=0.228563$, $P_4(t_0)=0.344026$ and $E_4(t_0)<0.001928$, so that $W^{[4]}(t_0)<0.004066+0.228563+0.001928-0.344026<-0.109$. □

Remark 11.5.9. (a) The large t range can use the verification of RH below height 3×10^{12} [196] in place of the 1700 zeros, with the crude bound $N(r)\le r^2$ for $r\ge200$ that Backlund's explicit form of the Riemann von Mangoldt formula gives [10]; the zeros above that height then contribute less than $e^{-10^{23}}$ for $t\ge\tau_1$, and [92] is not needed. (b) The constants 0.404 and 0.05 are computed, by `idsd_height.py` and again by `bdlp_ladder.py`, which finds $\int_0^5 r^2|m(r)-\log\frac{r}{2\pi}|\,dr<0.40330$, $r^2|m(r)-\log\frac{r}{2\pi}|<0.0422$ on $[5,10]$ and, from Binet's formula with its leading terms cancelled, <0.0462 beyond. The second has an analytic replacement. Binet's formula [177, (5.9.15)] at $z=\frac14+\frac{ir}{2}$, with the integral split at $v=r/4$ as in the proof of Theorem 5.1.1, gives for $r\ge5$

$$r^2\Bigl|m(r)-\log\frac{r}{2\pi}\Bigr|\le\frac18+\frac12+\frac{4r^2}{3(3r^2-1)}+8r\,T_1\Bigl(\frac{r}{4}\Bigr)\le1.08,$$

where $T_1(a)=\int_a^\infty v\,dv/(e^{2\pi v}-1)$, the four terms coming from $\log|z|$, from $\mathrm{Re}\,\frac{1}{2z}$ and from the two parts of the integral, on which $|v^2+z^2|$ is at least $(3r^2-1)/16$ and at least $r/4$; none of them increases with r, and at $r=5$ they are 0.125,

0.5, 0.45045 and 0.00348. With 1.08 in place of 0.05 the proof goes through unchanged, with $W^{[3]} > 0.28$ on $(0, \tau_2]$ and $W^{[4]}(t_0) < -0.090$. (c) The proof uses no zero above γ_2 except through the signs of their terms, no bound for $S(t)$ and no asymptotic argument. A second certificate in three ranges, in the same program, confirms the theorem: the zeros for $-\log t \le 4.3348$; the explicit formula on 3700 cells of width 0.001 covering $4.30 \le -\log t \le 8$, where the least lower bounds are 0.3943 for $W^{[2]}$ and 0.4096 for $W^{[3]}$; and lower bounds 6.12 for $W^{[2]}$ and 4.02 for $W^{[3]}$ valid for every $t \le e^{-8}$.

The ladder breaks where the first zero dominates. At t_0 its term, $\beta_4(3.1496)e^{-3.1496} = -0.97604$, is only partly cancelled by the others, which leave $W^{[4]}(t_0) = -0.11090$: the second zero, for which $\gamma_2^2 t_0$ already exceeds the largest root 4.490864 of β_4, contributes 0.62189, and the zeros above it are too far out. On the arithmetic side the same failure is the prime 2: its term in $P_4(t_0)$, 0.343448, outweighs the archimedean main term $\bar{A}_4(t_0) = 0.228563$. On $-\log t \in [0, 10]$ a scan in double precision on a grid of step 0.001, with the sign at the ends of every interval certified from the zeros, finds no negative value of W, $W^{[1]}$, $W^{[2]}$ or $W^{[3]}$, and finds $W^{[4]} < 0$ exactly for $-\log t \in (3.946, 4.335)$, $W^{[5]} < 0$ on $(3.443, 4.126)$ and $W^{[6]} < 0$ on $(3.146, 3.773) \cup (4.403, 5.051)$. Every ladder function from the fourth on takes negative values, since $W^{[j+1]} \ge 0$ would force $W^{[j]} \ge 0$ (Lemma 11.5.7). Towards the origin the ladder functions approach fixed fractions of W: by (11.5.4), $W^{[j]}(t) = \bar{A}_j(t) + O(1)$ as $t \downarrow 0$, so that $W^{[2]}/W \to \frac{1}{4}$ and $W^{[3]}/W \to \frac{1}{8}$. The two ratios fall from 5.73 and 5.95 at $-\log t = 4.30$ to 0.614 and 0.404 at 8 and to 0.470 and 0.291 at 10, decreasing throughout $[4.30, 10]$, and the ratios of the main terms are 0.2879 and 0.1535 at $-\log t = 32$. The least values of $W^{[2]}$ and $W^{[3]}$ on $4.30 \le -\log t \le 8$ are 0.396547, at 4.30, and 0.410976, at 4.335.

Corollary 11.5.10. *With no hypothesis, for every $s > 0$ the laws of Z_s and of $Z_s^{[2]}$ are self decomposable, hence unimodal, and the law of Z_s lies in L_1.*

Proof. The first statement is Theorem 11.5.8 with Lemma 11.5.7, and Yamazato's theorem [274], [232, Theorem 53.1] gives unimodality. The Laplace exponent of Z_s is $s\,u(\log G)'(u)$, so by (11.5.3) its images under S and S^2 correspond to $sW^{[2]}$ and $sW^{[3]}$, which are positive, and the theorem of Berg and Forst places Z_s in L_1. □

So Corollary 11.1.2 extends from the clock to its first two driving processes, and Yamazato's theorem gives nothing further along the ladder, since $Z_s^{[3]}$ is not self decomposable.

Remark 11.5.11. The centre clock is not in L_∞, the closure of the stable laws under convolution and weak limits. This follows from Theorem 11.5.8, since $L_\infty \subset L_3$, but it needs no computation. By [23, Theorem 1.6], the Laplace exponents of

the laws of L_∞ on $[0,\infty)$ are the functions $\int_{(0,1]} u^\alpha\,\sigma(d\alpha)$ with σ a finite measure. The atom of σ at $\alpha=1$ is a drift, and for $\alpha<1$, $u^\alpha=\alpha\Gamma(1-\alpha)^{-1}\int_0^\infty(1-e^{-ut})\,t^{-\alpha}\,dt/t$. So a law of L_∞ without drift has the canonical function $k(t)=\int_{(0,1)}\alpha\Gamma(1-\alpha)^{-1}t^{-\alpha}\,\sigma(d\alpha)$, which is at least c/t for $t\ge1$, with $c=\int_{(0,1)}\alpha\Gamma(1-\alpha)^{-1}\,\sigma(d\alpha)>0$ unless the law is degenerate, while $W(t)$ decays like $e^{-\gamma_1^2 t}$.

Remark 11.5.12. The perpetuity $T_{\mathcal W}\stackrel{d}{=}\mathsf U(T_{\mathcal W}+J)$ of Remark 11.5.5 comes from a driving process that is compound Poisson, and the argument there applies to any such process: if the driving process Y of a self decomposable law is compound Poisson with rate c and jumps distributed as J, stopping $X=\int_0^\infty e^{-t}\,dY_t$ at the first jump gives $X\stackrel{d}{=}\mathsf U^{1/c}(X+J)$, with $\mathsf U$ uniform on $(0,1)$ and the three variables independent. The centre clock satisfies no relation of this form, for any $c>0$ and any $J\ge0$. Such a J would have $\mathbb E\log(1+J)<\infty$, since $\log(1+\mathsf U^{1/c}J)\ge\log(1+J)+c^{-1}\log\mathsf U$ and $\mathbb E\log(1+\mathsf U^{1/c}J)\le\mathbb E\log(1+T)\le\mathbb ET=m_0$, and iterating the relation would write T as $\int_0^\infty e^{-t}\,dY_t$ with Y compound Poisson; but the driving process of a self decomposable law is unique in law, its Laplace exponent being $u(\log G)'(u)$, and that of the clock has the infinite Lévy measure $-W'(x)\,dx$ (Proposition 11.1.1). Under RH, $-W'(x)=\sum_{\gamma>0}\gamma^2e^{-\gamma^2x}$ is a mixture of exponentials, as is the jump density $\int ze^{-xz}\,U_{\mathcal W}(dz)$ of the driving process $\mathsf Z$ of the Lambert law (Remark 11.5.5), but its total mass is infinite.

The ladder of this subsection is not the derivative ladder of Theorems 5.1.1, 5.1.4, 5.1.7 and 5.1.10 and Table 1.7, which climbs the signs $c_k=(-1)^kW^{(k)}>0$. Its rungs are alternating combinations of the same c_k, with Stirling numbers as coefficients:

$$W^{[j]}(t)=\sum_{i=1}^{j}(-1)^{j-i}S(j,i)\,t^ic_i(t),$$

$$W^{[3]}=t^3c_3-3t^2c_2+tc_1,\qquad W^{[4]}=t^4c_4-6t^3c_3+7t^2c_2-tc_1.$$

Every rung of the derivative ladder is a consequence of RH, and the whole ladder, complete monotonicity of W, is RH. Of the BDLP ladder RH gives only the first rung, since a GGC is self decomposable; a gamma law is a GGC outside L_1, so the GGC property alone does not give $W^{[2]}\ge0$. In the other direction, L_∞ lies inside the class of GGCs [23, Section 4], so membership of the clock in L_∞ would imply RH, but no finite rung does: for every n the canonical function $(\log\frac1t)_+^n$, which vanishes on $[1,\infty)$ and so is not completely monotone, defines a law in L_n that is not a GGC, its ladder functions being $n!/(n-j)!\,(\log\frac1t)_+^{n-j}$ for $j\le n$ and the measure $n!\,\delta_1(dt)$ for $j=n+1$. So rungs two and three of the BDLP ladder, which Theorem 11.5.8 proves, are unconditional facts about the shape of W, set by the first zeros and their spacing, and the ladder stops at a rung that says nothing about RH.

11.5.4.3 The ladder along the de Bruijn Newman flow

Theorem 7.2.15 places the centre clock in a family of laws. For every $t \geq \Lambda$, and so for every $t \geq 0.2$ with no hypothesis, the function $R_t(u) = H_t(0)/H_t(2i\sqrt{u})$ of Section 8.4.1 is the Laplace transform of a GGC whose Thorin measure has the atoms $a_k(t) = (z_k(t)/2)^2$, where $0 < z_1(t) < z_2(t) < \cdots$ are the zeros of H_t, and $R_0 = 1/G$. Since t now denotes the time of the flow, we write x for the variable of the canonical function $W(x;t) = \sum_k e^{-a_k(t)x}$ of R_t, whose ladder functions are

$$W^{[j]}(x;t) = \Big(- x\frac{\partial}{\partial x}\Big)^j W(x;t) = \sum_k \beta_j\big(a_k(t)x\big)\, e^{-a_k(t)x}.$$

Lemma 11.5.7 holds for these laws with the same proof, the bound for small x now coming from the number of zeros of H_t, so the law with Laplace transform R_t lies in L_n exactly when $W^{[n+1]}(\cdot\,;t) \geq 0$. The flow moves the zeros, and with them the index.

The program `heat_ladder.py` computes H_t by Gauss Legendre quadrature in ball arithmetic, its zeros on $(0, 600]$ by a sign scan of step 0.2 and bisection, and the ladder functions $W^{[j]}(x;t)$, $j \leq 5$, from these zeros on the fixed window $0 \leq -\log x \leq 8$, on a grid of step 0.001, for nine values of t (Table 11.21). Nothing needs computing for $-\log x \leq 0$: the largest roots of $\beta_2, \ldots, \beta_5$ are 1, 2.618034, 4.490864 and 6.510134, and $a_1(t)x \geq a_1(t) \geq 12.47$ in every row. The computation is numerical for $t > 0$. The quadrature is checked against a rule with 1024 nodes, with relative differences below 10^{-46} at $z = 100$, 300 and 590, but its error is not bounded, and the zeros above $z = 600$ are left out; an estimate from the density $\frac{1}{4\pi}\log\frac{z}{4\pi}$ of the zeros of H_0 puts their share of $W^{[3]}$ at $-\log x = 8$ below 2×10^{-9}, and their share of $W^{[5]}$ below 10^{-5}. At $t = 0$ the computed zeros are the numbers $2\gamma_k$ to double precision, and there are $N(300) = 138$ of them, the number that Arb's count of the zeros of ζ gives; H_0 agrees with $\Xi(z/2)/8$, computed from ζ, to 10^{-46}.

The row $t = 0$ repeats from the zeros of H_0 what Theorem 11.5.8 proves for all x, with the interval on which $W^{[4]} < 0$ found by the scan after it. Along the flow the zeros move towards the origin and apart: $z_1(t)$ falls from 28.27 to 7.06 at $t = 50$, the ratio a_2/a_1 of the first two atoms rises from 2.21 to 5.34, and the number of zeros in $(0, 600]$ rises from 138 to 149. The set on which $W^{[4]} < 0$ follows the first atom to smaller values of $-\log x$. It lies where $a_1(t)x$ is in the interval $(1.343380, 4.490864)$ on which $\beta_4 < 0$, less the part on which the other atoms compensate, and at $t = 50$, where it covers most of that interval, the first atom alone almost decides the sign. Up to $t = 20$ the third ladder function stays nonnegative on the window, and R_t keeps the index two of the clock, as far as the window shows. Between $t = 27.8$ and $t = 27.9$ it turns negative. The program finds $W^{[3]} \geq 0$ on the window at $t = 21, \ldots, 27$ and $t = 27.1, \ldots, 27.8$, which gives $W^{[2]} \geq 0$ and $W^{[1]} \geq 0$ there as well, by the descent in the proof of Lemma 11.5.7

t	$z_1(t)$	a_2/a_1	$W^{[2]}/W$	$W^{[3]}/W$	$W^{[3]}<0$	$W^{[4]}<0$	n
0	28.2695	2.2119	0.6144	0.4038	none	$(3.946, 4.335)$	2
0.2	28.1536	2.2142	0.6131	0.4029	none	$(3.935, 4.333)$	2
1	27.6920	2.2239	0.6078	0.3993	none	$(3.893, 4.321)$	2
4	25.9874	2.2678	0.5887	0.3864	none	$(3.737, 4.263)$	2
10	22.7127	2.3915	0.5545	0.3630	none	$(3.426, 4.090)$	2
20	17.7111	2.7188	0.5069	0.3301	none	$(2.884, 3.675)$	2
30	13.3727	3.2530	0.4687	0.3030	$(2.983, 3.343)$	$(2.303, 3.153)$	1
40	9.7966	4.0960	0.4374	0.2807	$(2.236, 3.014)$	$(1.676, 2.593)$	1
50	7.0629	5.3383	0.4115	0.2620	$(1.563, 2.539)$	$(1.022, 2.035)$	1

Table 11.21: The ladder functions of R_t along the de Bruijn Newman flow, on the window $0 \le -\log x \le 8$: the first zero $z_1(t)$ of H_t, the ratio a_2/a_1 of the first two Thorin atoms, the ratios $W^{[2]}/W$ and $W^{[3]}/W$ at $-\log x = 8$, the sets of $-\log x$ on which $W^{[3]}(x;t) < 0$ and $W^{[4]}(x;t) < 0$, to the grid step 0.001, and the largest n with $W^{[1]}, \dots, W^{[n+1]} \ge 0$ on the window: R_t lies in no class above L_n, and in L_n if the signs persist for $x < e^{-8}$. On the window W, $W^{[1]}$ and $W^{[2]}$ are positive and $W^{[5]}$ takes negative values for every t. Numerical for $t > 0$; the row $t = 0$ is Theorem 11.5.8. Computed by `heat_ladder.py`.

applied on $[e^{-8}, \infty)$; at $t = 27.9$ it finds $W^{[3]} < 0$ for $3.230 < -\log x < 3.307$, with least value -0.0015, and at $t = 30$, 40 and 50 on the intervals of the table. So, by this computation, for these t the law with Laplace transform R_t, a GGC with no hypothesis, is not in L_2, and its index on the window drops to one. The mechanism is that of Theorem 11.5.8, one rung lower: the term of the first atom in $W^{[3]}$ is negative where $a_1(t)x$ lies in $(0.381966, 2.618034)$, and once the first two atoms are far enough apart the second no longer compensates it; at the threshold a_2/a_1 is between 3.113 and 3.119, against 2.212 for the clock. At $t = 50$ the ratio $W^{[2]}/W$, which for $t \le 40$ decreases across the whole window, has an interior minimum, 0.276 at $-\log x = 2.682$: the same mechanism has begun to act on the second rung, which a single atom already violates.

Below the window the ratios $W^{[2]}/W$ and $W^{[3]}/W$ tend to $\frac{1}{4}$ and $\frac{1}{8}$, at $t = 0$ by (11.5.4) and for every $t \ge 0.2$ by the count of the zeros of H_t. For fixed $t > 0$ the number $N_t(T)$ of zeros of H_t in $(0, T]$ is $\Psi(T) + \frac{t}{16}\log T + O(1)$, with $\Psi(T) = \frac{T}{4\pi}\log\frac{T}{4\pi} - \frac{T}{4\pi}$: Ki, Kim and Lee proved a formula of this kind for every $t > 0$ [125, Theorem 1.4], as recalled in [224, Section 3], and the Polymath project proved it in this form, with effective constants, for $0 < t \le \frac{1}{2}$ [217, Theorem 1.5]. An error $O(\log T)$ would do. Partial summation against $d\Psi$, with the substitution $u = z^2 x/4$ and (11.5.1) at $s = \frac{1}{2}$, gives

$$W^{[j]}(x;t) = \frac{\log(1/x)}{2^{j+3}\sqrt{\pi x}}\big(1 + o(1)\big) \qquad (x \downarrow 0),$$

which at $t = 0$ is the leading part of $\bar{A}_j$. The approach is slow: at the end of

the window the two ratios are still between 0.41 and 0.62 and between 0.26 and 0.41. So for $t > 0$ the signs below $x = e^{-8}$ are plausible but not proved, and the last column of the table gives the largest index that the window allows. Either way the index is set by the positions and the spacing of the first zeros: it changes along the flow where every law is a GGC with no hypothesis, and it carries no information about RH.

11.5.4.4 No branching process realisation

One of Pakes's proofs that $\mathcal{W}$ is a Thorin Bernstein function obtains the Lambert law as the limit of a branching process [186] (Section 11.5.1). Let Y be a spectrally positive Lévy process with Laplace exponent $\Psi_Y(u) = -\log \mathbb{E}e^{-uY_1}$, with the sign of Proposition 11.1.1, and with finite Malthusian parameter $\Psi_Y'(0) \in (0, \infty)$. Lamperti's time change turns Y into a continuous state branching process $(\mathsf{B}_t)_{t\geq 0}$; started at $\mathsf{B}_0 = z > 0$ it has $\mathbb{E}\mathsf{B}_t = ze^{\Psi_Y'(0)t}$, and the nonnegative martingale $e^{-\Psi_Y'(0)t}\mathsf{B}_t$ converges almost surely to a limit M. By the branching property $\mathbb{E}e^{-u\mathsf{M}} = e^{-z\omega(u)}$ for a function ω that does not depend on z. Conditioning on B_s gives $\omega(u) = v_s\big(\omega(ue^{-\Psi_Y'(0)s})\big)$, where $e^{-zv_s(\theta)} = \mathbb{E}e^{-\theta \mathsf{B}_s}$ and $\partial_s v_s = \Psi_Y(v_s)$, and differentiating at $s = 0$ gives

$$\Psi_Y'(0)\, u\, \omega'(u) = \Psi_Y\big(\omega(u)\big), \tag{11.5.5}$$

which is [187, (68)]. For the Lambert law, Y is the compound Poisson process with unit rate and standard exponential jumps, $\Psi_Y(u) = u/(1+u)$ and $\Psi_Y'(0) = 1$, and $\omega = \mathcal{W}$ solves (11.5.5), since differentiating $\mathcal{W}e^{\mathcal{W}} = u$ gives $u\mathcal{W}' = \mathcal{W}/(1+\mathcal{W})$ [187, Section 7].

For the centre clock, or λT with $\lambda > 0$, to be such a limit we need $z\omega(u) = \log G(\lambda u)$. Then (11.5.5) determines Ψ_Y on the range $[0, \infty)$ of ω:

$$\Psi_Y(w) = \frac{\Psi_Y'(0)}{z}\, g(zw), \qquad g = \Psi_Z \circ H = \frac{H}{H'},$$

where H is the inverse function of $\log G$ and $\Psi_Z(u) = u(\log G)'(u)$ is the Laplace exponent of the driving process of Proposition 11.1.1. The function g is analytic at 0, with $g(w) = \sum_{k\geq 1} g_k w^k$ and $g_1 = 1$. For every spectrally positive Y the function $-\Psi_Y''(u) = \sigma^2 + \int_0^\infty x^2 e^{-ux}\, \Pi(dx)$, with σ^2 the Gaussian coefficient and Π the Lévy measure of Y, is completely monotone (and if Y is a subordinator, so is Ψ_Y'). Since $\Psi_Y^{(k)}(0) = \Psi_Y'(0)\, z^{k-1} k!\, g_k$, this forces

$$(-1)^{k-1} g_k \geq 0 \qquad (k \geq 2). \tag{11.5.6}$$

The coefficients g_k need no hypothesis. The Taylor series of ξ at $\frac{1}{2}$, computed in ball arithmetic, gives that of $\log G(u) = \log \xi(\frac{1}{2} + \sqrt{u}) - \log \xi(\frac{1}{2}) = \sum_{n\geq 1} (-1)^{n+1} m_{n-1} u^n/n$, with the Stieltjes moments $m_n = \sum_k a_k^{-n-1}$ of Section 5.1; reverting the series gives H, and composing gives $g = \Psi_Z \circ H$. The

program `branch_coeffs.py` does this at 3200 bits for $k \le 60$, checks it against the second route $g = H/H'$, which agrees for $k \le 59$ (the last coefficient is cut by the length of the series), and checks $m_1, \dots, m_{11}$ against the sums over the first 1700 zeros. Every sign for $k \le 60$ is certified, and Table 11.22 lists the first thirty coefficients.

k	g_k	k	g_k
1	1	16	$1.49524474945 \times 10^{-15*}$
2	$-3.48161903048 \times 10^{-2}$	17	$-5.70288794250 \times 10^{-16*}$
3	$5.36818625476 \times 10^{-3}$	18	$9.65518218408 \times 10^{-17*}$
4	$-8.70698237260 \times 10^{-4}$	19	$-1.21432018700 \times 10^{-17*}$
5	$1.36973198860 \times 10^{-4}$	20	$1.16783296302 \times 10^{-18*}$
6	$-2.05209454841 \times 10^{-5}$	21	$-6.77018656749 \times 10^{-20*}$
7	$2.91586074297 \times 10^{-6}$	22	$-3.60435568970 \times 10^{-21}$
8	$-3.92530248114 \times 10^{-7}$	23	$1.93592106401 \times 10^{-21}$
9	$4.99813541563 \times 10^{-8}$	24	$-3.85744224396 \times 10^{-22}$
10	$-5.99595770666 \times 10^{-9}$	25	$5.68747574953 \times 10^{-23}$
11	$6.72224097409 \times 10^{-10}$	26	$-6.72011823221 \times 10^{-24}$
12	$-6.93358579492 \times 10^{-11}$	27	$6.03459570343 \times 10^{-25}$
13	$6.37029732062 \times 10^{-12}$	28	$-2.66337804089 \times 10^{-26}$
14	$-4.81096482192 \times 10^{-13}$	29	$-4.09850045758 \times 10^{-27*}$
15	$2.15183811282 \times 10^{-14}$	30	$1.42466491988 \times 10^{-27*}$

Table 11.22: The Taylor coefficients of $g = \Psi_Z \circ H$, the only candidate, up to scaling, for the Laplace exponent of a generating process of the centre clock (Theorem 11.5.13), to twelve digits; a star marks a sign that violates (11.5.6). Every digit shown is certified in ball arithmetic at 3200 bits from the Taylor series of ξ at $\frac{1}{2}$, and the signs are certified for $k \le 60$: the wrong signs are those with $16 \le k \le 21$, $29 \le k \le 35$, $43 \le k \le 49$ and $56 \le k \le 60$. Computed by `branch_coeffs.py`.

THEOREM 11.5.13. *With no hypothesis,* $(-1)^{15} g_{16} < 0$*: in ball arithmetic* $g_{16} = 1.49524474945112 \times 10^{-15} \pm 2 \times 10^{-30}$*. Consequently no positive multiple of the centre clock is the martingale limit of a supercritical continuous state branching process with finite Malthusian parameter, from any initial state.*

Proof. Such a limit would satisfy (11.5.6), which the sign of g_{16} violates. □

Theorem 11.5.13 needs only one certified sign, but the signs fail for a structural reason, and they keep failing. The reason is Pringsheim's theorem, the power series form of Landau's theorem [268, Chapter II, Theorem 5b]: if $(-1)^{k-1} g_k \ge 0$ for all large k and the series $\sum g_k w^k$ has a finite radius of convergence R, then its sum is singular at $w = -R$. But g is analytic on a neighbourhood of the negative axis. With no hypothesis, $(\log G)'(u) = \sum_k 1/(u + a_k) > 0$ for $-\gamma_1^2 < u \le 0$: a zero on the line contributes $1/(u + \gamma^2) > 0$, and the conjugate atoms a, $\bar{a}$ of an off line zero together contribute $2\,\mathrm{Re}(u+a)/|u+a|^2$, with $\mathrm{Re}(u+a) \ge \gamma^2 - \frac{1}{4} - \gamma_1^2 > 0$. At

$u = -1$, -100, -190, -199 and -199.7 the derivative is 0.02314, 0.02957, 0.1236, 1.287 and 11.08. So $\log G$ increases from $-\infty$ to 0 on $(-\gamma_1^2, 0]$, its inverse H is analytic on a neighbourhood of $(-\infty, 0]$, and so is $g = H/H'$. Hence (11.5.6) fails for infinitely many k unless g is entire.

Numerically the radius is finite, $R = 7.1565$, and the singularities that fix it lie off the axis. The function Ξ has a critical point $t^* = 15.5857085898$ between γ_1 and γ_2, certified by bisection on Ξ', and at $u^* = -t^{*2} = -242.914312$ the function $G(u) = \Xi(\sqrt{-u})/\Xi(0)$ has $G'(u^*) = 0$ and $G(u^*) = -0.00161241048$. Continued from 0 around the zero $-\gamma_1^2$ through the upper half plane, $\log G$ reaches u^* with the value

$$\lambda^* = \log|G(u^*)| + i\pi = -6.43002502663 + i\pi,$$

of modulus $|\lambda^*| = 7.156453$ and argument $\arg\lambda^* = 2.687121$, and since $(\log G)'(u^*) = 0$, the inverse H has a square root branch point at λ^*, and by symmetry at $\bar\lambda^*$. Continuing H numerically along the segment from 0 to λ^* in 3000 steps keeps $|G| = e^{\operatorname{Re}\lambda} \geq 0.0016$ on the path, away from the zeros of G, and brings the preimage to within 2.86 of u^*, at a distance proportional to the square root of the distance from λ^* over the last steps, as at a simple critical point; so λ^* and $\bar\lambda^*$ are singularities of the principal branch of g (a numerical continuation, not a certificate). The root test agrees: $|g_k|^{1/k}$ is 0.1269, 0.1274, 0.1296, 0.1288 and 0.1317 at $k = 20$, 30, 40, 50 and 60, approaching $1/|\lambda^*| = 0.13973$ slowly, as the factor $k^{-3/2}$ of a square root singularity predicts, while the consecutive ratios $|g_{k+1}/g_k|$, between 0.120 and 0.162 for $2 \leq k \leq 9$, swing from 0.005 to 3.2 beyond, near each change of sign. The conjugate pair makes $(-1)^{k-1}g_k$ oscillate in k with period $2\pi/(\pi - \arg\lambda^*) = 13.8253$, and the certified wrong signs come in the blocks of Table 11.22, at spacings 13, 14 and 13.

The next critical points of Ξ do not matter. The second, $t^{**} = 22.0979772804$ between γ_2 and γ_3, has $G(-t^{**2}) = 1.56051 \times 10^{-5} > 0$ and $\log G(-t^{**2}) = -11.0679$, and the third, 26.2722473569 between γ_3 and γ_4, has $G = -1.34670 \times 10^{-6}$. On the principal branch g is analytic along the whole negative axis, so the real value -11.0679 is a singular value only of branches reached by winding around the logarithmic points $-\gamma_k^2$ of $\log G$, and it lies outside the disc of convergence in any case. So the branching obstruction, like the end of the ladder, comes from the bottom of the spectrum: the singularity that forces the changes of sign is created by the critical point of Ξ between γ_1 and γ_2.

11.5.5 The criticality filter

Along the de Bruijn Newman flow the function R_t of Theorem 7.2.15 is the Laplace transform of a GGC exactly when $t \geq \Lambda$ (Corollary 4.2.5), and $\Lambda \geq 0$ by the theorem of Rodgers and Tao [224]. So the Pólya kernel sits at the edge of a family of kernels $e^{tv^2/4}\Phi(v)$, $t < 0$, as close to it as one wishes, whose transforms have nonreal zeros and for which the Thorin condition fails. This subsection turns

that fact into a test for the properties of the kernel that might imply RH (Section 11.5.5.1). It applies the test to two representations of the Pólya law as a mixture (Section 11.5.5.2), and to the property that the first author's account singles out, the GGC property of the variable of Biane, Pitman and Yor. That property passes the test at every $t < 0$ tested, and provably for small $|t|$ (Section 11.5.5.3); but it fails for most of the real Dirichlet L functions that we tested, which have an Euler product and, as far as they have been computed, all their zeros on the critical line, so it does not detect the Euler product (Section 11.5.5.4). The programs `pakes_filter.py`, `pakes_filter_cert.py` and `pakes_characters.py` compute the numbers, and the second checks in ball arithmetic [110] every inequality below that is called certified.

11.5.5.1 The filter

For an even, positive, integrable function g on $\mathbb{R}$ put

$$\widehat{g}(\theta) = \int_{\mathbb{R}} g(v)\cos(\theta v)\,dv, \qquad R_g(s) = \frac{\widehat{g}(0)}{\widehat{g}(i\sqrt{s})} = \frac{\int_{\mathbb{R}} g(v)\,dv}{\int_{\mathbb{R}} g(v)\cosh(\sqrt{s}\,v)\,dv}, \quad (11.5.7)$$

for the $s \geq 0$ at which the last integral converges. For the damped Pólya kernel $\Phi_t(v) = e^{tv^2/4}\Phi(v)$, with Φ as in (1.2.3), the transform $\widehat{\Phi_t}$ is the function Ξ_t of Section 8.7.6, and R_{Φ_t} is the function R_t of Theorem 7.2.15, written φ_t in Corollary 4.2.5; at $t = 0$ it is the clock $1/G$.

Proposition 11.5.14. *Let $\mathcal{P}$ be a property of even, positive, integrable functions on $\mathbb{R}$ that Φ_t has for some $t < 0$. Then $\mathcal{P}$ implies neither that $\widehat{g}$ has only real zeros nor that R_g is the Laplace transform of a GGC, for the functions g with $\mathcal{P}$. At every $t \leq 0$ the kernel Φ_t*

(i) is positive and even, and strictly decreasing on $(0, \infty)$;

(ii) is log concave;

(iii) satisfies $\log \Phi_t(v) = -\pi e^{2|v|} + O(v^2)$ as $|v| \to \infty$;

(iv) extends analytically to the strip $|\operatorname{Im} v| < \pi/4$.

So no combination of (i) to (iv) implies that $\widehat{g}$ has only real zeros, or that R_g is the Laplace transform of a GGC.

Proof. Let $t < 0$ be such that Φ_t has $\mathcal{P}$. The zeros of H_t, and so those of $\Xi_t(\theta) = 8H_t(2\theta)$, are all real if and only if $t \geq \Lambda$ (de Bruijn [65], Newman [163]), and $\Lambda \geq 0$ [224], so Ξ_t has a zero $z \notin \mathbb{R}$. It is not on the imaginary axis either, since $\Xi_t(iy) = \int \Phi_t(v)\cosh(yv)\,dv > 0$ for real y. Hence $s_0 = -z^2$ has $\operatorname{Im} s_0 = -2\operatorname{Re} z \operatorname{Im} z \neq 0$. As Ξ_t is even and entire, $s \mapsto \Xi_t(i\sqrt{s})$ is entire, and it vanishes at s_0, so R_t is meromorphic on $\mathbb{C}$ with a pole at $s_0 \notin (-\infty, 0]$. By (F7) of Section 4.2.1 an HCM

function extends analytically to $\mathbb{C} \setminus (-\infty, 0]$, and the extension would agree with R_t off its poles; so R_t is not HCM, and by (F6) it is not the Laplace transform of a GGC. Thus $g = \Phi_t$ has $\mathcal{P}$, while $\widehat{g}$ has nonreal zeros and R_g is not the Laplace transform of a GGC.

For (i), Φ is positive and even, and $\Phi_t'(v) = e^{tv^2/4}\big(\Phi'(v) + \frac{t}{2}v\Phi(v)\big) < 0$ for $v > 0$, since $\Phi' < 0$ on $(0, \infty)$ [60, Theorem A] and $tv\Phi(v) \le 0$. For (ii), $(\log \Phi_t)'' = (\log \Phi)'' + \frac{t}{2} \le (\log \Phi)'' \le 0$, since Φ is log concave [53]. Properties (iii) and (iv) hold for Φ by (1.2.3), and the factor $e^{tv^2/4}$ is entire and adds $tv^2/4$ to the logarithm. □

The proposition puts together the half of Proposition 8.7.17(a) that concerns real zeros and the half of Corollary 4.2.5 that concerns the Thorin condition, and reads them as a test. A property that is to imply RH through the kernel alone must fail for Φ_t at every $t < 0$; we then say that it *passes the filter*. Passing is necessary and far from sufficient: it says only that the theorem of Rodgers and Tao does not rule the property out. For the properties (i), (ii) and (iv) the conclusion needs nothing so deep: the convolution of $(1 - |u|)_+^2$ with $e^{-\cosh u}$ in Section 8.5.5 is even, positive, entire, log concave and doubly exponentially decaying, and its transform has nonreal zeros. What the filter adds is a counterexample that has every property of the list and differs from Φ only by the factor $e^{tv^2/4}$, with t as close to 0 as one wishes.

The filter is the counterpart, for properties of the kernel, of the perturbation test of Section 12.2.6. There the function F of Theorem 5.2.1 keeps the zeros of ξ below a height R, the order, the type and the functional equation, and has four zeros off the line above R; here Φ_t keeps every property of the kernel that a Gaussian factor $e^{tv^2/4}$ with $t < 0$ preserves, and has nonreal zeros by the theorem of Rodgers and Tao, which does not say where. The functions Ξ_t are even and entire of order one, like Ξ. For $t > 0$ the zeros of H_t obey a formula of Riemann and von Mangoldt type, $N_t(T) = \Psi(T) + \frac{t}{16}\log T + O(1)$ (Section 11.5.4.3); for $t < 0$, the case the filter needs, we know of no count in the literature: Rodgers and Tao prove one for $\Lambda < t \le 0$, with the error $O(\log^2 T)$ [224, Section 3], as a step in showing that this range of t is empty. A proof of RH must therefore use, at some step, a property that Φ_t lacks for every $t < 0$. The Euler product and the explicit formula are the natural candidates, as they are for F in Section 12.2.6, but this is a reading and not a theorem: no Euler product for H_t with $t \neq 0$ is known (see the discussion after Theorem 7.2.15), and none is known not to exist. The Lambert proof of Section 11.5.3 is untouched by the filter, since it concerns a single function, whose analyticity off the cut is forced by the equation $\mathcal{W}e^{\mathcal{W}} = u$, and no family of comparison functions sharing its properties is in sight.

11.5.5.2 Mixtures of arcsine laws

For $t \le 0$ let X_t be the variable with density $p_t = \Phi_t/\Xi_t(0)$, so that X_0 is the Pólya variable X. Since p_t is even and decreasing on $(0,\infty)$ by Proposition 11.5.14(i), Khinchine's theorem writes $X_t = UQ_t$ with U uniform on $[-1,1]$ and $Q_t \ge 0$ independent of U, of density $-2rp_t'(r)$; at $t = 0$ this is the mixture of uniform laws of Section 13.3.2. The uniform law is itself a mixture of arcsine laws, and this gives a second representation.

PROPOSITION 11.5.15. *For every $t \le 0$ the function*

$$h_t(\rho) = -2\rho \int_\rho^\infty \frac{p_t'(r)\,dr}{\sqrt{r^2-\rho^2}}, \qquad \rho > 0,$$

is a positive probability density, and

$$p_t(v) = \frac{1}{\pi} \int_{|v|}^\infty \frac{h_t(\rho)\,d\rho}{\sqrt{\rho^2-v^2}}, \qquad v \ne 0.$$

Equivalently, $X_t \overset{d}{=} \varrho_t \cos\Theta$ with Θ uniform on $[0,2\pi)$ and ϱ_t of density h_t independent of Θ: X_t is a mixture of arcsine laws, and it is the first coordinate of the rotation invariant vector $\varrho_t(\cos\Theta,\sin\Theta)$ in $\mathbb{R}^2$, as well as of a rotation invariant vector in $\mathbb{R}^3$.

Proof. By Archimedes' theorem the coordinates of a uniform point of the unit sphere in $\mathbb{R}^3$ are uniform on $[-1,1]$. Writing the point as $(B\cos\Theta, B\sin\Theta, V)$, with V uniform on $[-1,1]$, $B = \sqrt{1-V^2}$ and Θ uniform on $[0,2\pi)$ independent of V, the first coordinate gives $U \overset{d}{=} B\cos\Theta$, where B has density $b/\sqrt{1-b^2}$ on $(0,1)$. Hence $X_t = UQ_t \overset{d}{=} (BQ_t)\cos\Theta$, and Q_t times the uniform point is a rotation invariant vector in $\mathbb{R}^3$ with first coordinate X_t. The density of $\varrho_t = BQ_t$ is

$$\int_\rho^\infty \big(-2rp_t'(r)\big)\,\frac{\rho/r}{\sqrt{1-\rho^2/r^2}}\,\frac{dr}{r} = -2\rho\int_\rho^\infty \frac{p_t'(r)\,dr}{\sqrt{r^2-\rho^2}} = h_t(\rho),$$

positive since $p_t' < 0$ on $(0,\infty)$. Given $\varrho_t = \rho$, the variable $\rho\cos\Theta$ has the arcsine density $1/(\pi\sqrt{\rho^2-v^2})$ on $(-\rho,\rho)$, which gives the formula for p_t. □

The program `pakes_characters.py` checks the proposition at $t = 0$ and $t = -1$ with 20 digits: $\int_0^\infty h_t = 1$ to 15 digits, and p_t rebuilt from h_t agrees with $\Phi_t/\Xi_t(0)$ at $v = 0.1$, 0.4, 0.7 and 1 to a relative 1.4×10^{-20}. At $t = 0$ the proposition is the arcsine route of [200, Chapter 12] recalled in Section 5.2, which writes X as $2\sqrt{I}$ times the cosine of an independent angle; since $\mathbb{E}|\cos\Theta|^s \ne 0$ for $s > -1$, the Mellin transform of $|X|$ determines the law of the radius, so $\varrho_0 \overset{d}{=} 2\sqrt{I}$, and h_0 is the density of the variable I of that route after the change of variable. Both representations, the uniform mixture and the arcsine mixture, hold

at every $t \le 0$, so by Proposition 11.5.14 neither can decide RH, or the Thorin condition, by itself: whatever is to be proved about Q_0 or ϱ_0 must use something that Q_t or ϱ_t lacks for $t < 0$. The scale mixtures of Gamma laws, the other mixture route of [200, Section 12.6], are excluded already at $t = 0$, for every shape (Theorem 11.2.9).

11.5.5.3 The variable of Biane, Pitman and Yor under damping

Let $\mathcal{B} = \frac{\pi}{2} S_2$, where $S_2 = J + J'$ is the sum of two independent hitting times of 1 by a three dimensional Bessel process started at 0, as in (9.1.1) and Theorem 7.1.21. Then

$$\mathbb{E}e^{-\lambda\mathcal{B}} = \Big(\frac{\sqrt{\pi\lambda}}{\sinh\sqrt{\pi\lambda}}\Big)^2 = \prod_{m\ge1}\Big(1+\frac{\lambda}{\pi m^2}\Big)^{-2}, \qquad \mathbb{E}\mathcal{B}^{s/2} = 2\xi(s), \tag{11.5.8}$$

so $\mathcal{B}$ is a GGC with Thorin measure $2\sum_m \delta_{\pi m^2}$, and its density is

$$f(b) = \sum_{m\ge1}\big(4\pi^2 m^4 b - 6\pi m^2\big)e^{-\pi m^2 b} = b^{-5/4}\,\Phi\big(\tfrac{1}{2}\log b\big), \qquad b > 0.$$

More generally, for an even positive kernel g with $\int g(v)e^{|v|/2}\,dv < \infty$ let $\mathcal{B}_g$ have the density proportional to $b^{-5/4}g(\frac{1}{2}\log b)$; the Pólya variable of g, of density $g/\widehat{g}(0)$, is $\frac{1}{2}\log\mathcal{B}_g$ tilted by $\mathcal{B}_g^{1/4}$, and $\mathcal{B}_\Phi = \mathcal{B}$ (Theorem 7.1.21). For the damped kernels write $\mathcal{B}_t = \mathcal{B}_{\Phi_t}$: its density is proportional to $e^{t(\log b)^2/16}f(b)$, finite for every real t because $f(b)$ decays like $e^{-\pi b}$ as $b \to \infty$ and, up to a power of b, like $e^{-\pi/b}$ as $b \downarrow 0$, and $\mathbb{E}\mathcal{B}_t^{s/2} = \Xi_t(i(\frac{1}{2}-s))/\Xi_t(\frac{i}{2})$. Damping the kernel thus tilts $\mathcal{B}$ by a lognormal factor, and "$\mathcal{B}_g$ is a GGC" is a property of the kernel g that Φ has. The filter asks whether Φ_t lacks it for every $t < 0$.

Two tests are available for a positive variable $\mathcal{V}$ with $\ell(\lambda) = \mathbb{E}e^{-\lambda\mathcal{V}}$. Let $\kappa_k(\lambda) = (-1)^k\frac{d^k}{d\lambda^k}\log\ell(\lambda)$, $k \ge 1$, be the cumulants of the law $e^{-\lambda b}\,P(\mathcal{V} \in db)/\ell(\lambda)$. The variable is infinitely divisible if and only if $-\log\ell$ is a Bernstein function, that is, if and only if $\kappa_k(\lambda) \ge 0$ for all $k \ge 1$ and $\lambda > 0$ [234]. If it is a GGC, with $-\log\ell(\lambda) = d\lambda + \int\log(1+\lambda/z)\,U(dz)$, then for $k \ge 0$

$$c_k(\lambda) = \frac{\kappa_{k+1}(\lambda)}{k!} = d\,\delta_{k0} + \int\frac{U(dz)}{(\lambda+z)^{k+1}} = \int x^k\,\sigma_\lambda(dx), \tag{11.5.9}$$

where σ_λ is $d\,\delta_0$ plus the image of $(\lambda+z)^{-1}U(dz)$ under $z \mapsto (\lambda+z)^{-1}$. Hence the Hankel matrices $\mathcal{K}_n(\lambda) = [c_{i+j}(\lambda)]_{i,j=0}^{n-1}$ and $\mathcal{K}_n'(\lambda) = [c_{i+j+1}(\lambda)]_{i,j=0}^{n-1}$ are positive semidefinite for every n and $\lambda \ge 0$, since $\sum_{i,j} y_i y_j c_{i+j+r}(\lambda) = \int x^r\big(\sum_i y_i x^i\big)^2\sigma_\lambda(dx)$ for $r = 0, 1$, and a negative eigenvalue of one of them proves that $\mathcal{V}$ is not a GGC. The smallest case is already useful.

Lemma 11.5.16. *If $\mathcal{V}$ is a GGC with a finite fourth moment, its cumulants $\kappa_k = \kappa_k(0)$ satisfy $2\kappa_2\kappa_4 \ge 3\kappa_3^2$, with equality for the gamma laws.*

Proof. With $c_k = \kappa_{k+1}/k!$ the inequality reads $c_1 c_3 \geq c_2^2$, which is $\det \mathcal{K}_2'(0) \geq 0$. For a gamma law U is a single atom $\alpha\delta_z$ and $c_k = \alpha z^{-k-1}$, so equality holds. □

The program `pakes_filter.py` computes the moments

$$\int_{\mathbb{R}} e^{2kv}\, e^{-\lambda e^{2v}}\, e^{tv^2/4}\, e^{-v/2}\, \Phi(v)\, dv,$$

which are those of $\mathcal{B}_t$ under the law of the cumulants $\kappa_k(\lambda)$ up to normalisation, by the trapezoid rule on $[-5,5]$, where the integrand is analytic in $|\operatorname{Im} v| < \pi/4$ and negligible beyond $|v| = 5$ (below 10^{-30000}). Two settings, 150 digits with step $1/64$ and 200 digits with step $1/96$, agree in every c_k to a relative 7×10^{-81}. The program forms $\kappa_1, \ldots, \kappa_{26}$, at $\lambda = 0$, $\frac{1}{2}$, 2, 10, 20, 50 and 200, and the matrices $\mathcal{K}_n(\lambda)$ and $\mathcal{K}_n'(\lambda)$ of orders $n \leq 12$, whose inertia it reads from the pivots of Gaussian elimination. At $t = 0$ the numbers $c_k(\lambda) = 2\sum_m (\lambda + \pi m^2)^{-k-1}$ are known in closed form, and the computed ones agree with them to a relative 3×10^{-84}. Dividing c_{i+j+r} by c^{i+j+r}, with $c = c_1/c_0$, is a congruence that keeps the inertia, and the eigenvalues quoted below are those of the matrices so scaled. The program `pakes_filter_cert.py` certifies the failures. It encloses the moments in balls with the integration of Arb at 256 bits on $[-5,0]$ and $[0,5]$, adds a bound 2^{-3000} for each tail, computes the cumulants and the scaled matrix M in ball arithmetic, and encloses $y^{\mathsf{T}} M y$ for a rational vector y close to the eigenvector of the least eigenvalue; an enclosure inside $(-\infty, 0)$ proves that M is not positive semidefinite. In every certificate below the cumulants used are certified positive and $y^{\mathsf{T}} M y$ agrees with the computed least eigenvalue to all the digits quoted.

Infinite divisibility shows nothing. In all 301 cases computed, the tilts $\mathcal{B}_t$ for the twenty values of t of the next paragraph, the reciprocals $1/\mathcal{B}_t$ for $t = -1$, 0, $\frac{1}{5}$, and the ten functions other than ξ of Table 11.24 below, on both sides, each at the seven values of λ, none of $\kappa_1, \ldots, \kappa_{26}$ is negative. This is consistent with infinite divisibility of every variable tested, and proves it for none.

The GGC test separates $t = 0$ from every other value. It was run for $t = -20$, -10, -5, -2, -1, $-\frac{1}{2}$, $-\frac{1}{5}$, $-\frac{1}{10}$, $-\frac{1}{20}$, $-\frac{1}{50}$, $-\frac{1}{100}$, 0, $\frac{1}{100}$, $\frac{1}{50}$, $\frac{1}{20}$, $\frac{1}{10}$, $\frac{1}{5}$, $\frac{1}{2}$, 1 and 5. At $t = 0$ no matrix of order at most 12 has a negative eigenvalue, as the theorem requires, though the matrices are nearly singular: the least eigenvalue of $\mathcal{K}_7'(2)$ is 1.2×10^{-13}. For every $t \neq 0$ some $\mathcal{K}_n'(\lambda)$ with $n \leq 12$ has a negative eigenvalue at each $\lambda \leq 50$, at orders 7 to 12 for $t < 0$ and 3 to 10 for $t > 0$. Table 11.23 shows eleven of the twenty values; the other nine give the same picture.

PROPOSITION 11.5.17. *(i) $\mathcal{B}_t$ is not a GGC for $t = -20$, -5, -1, $-\frac{1}{2}$, $-\frac{1}{10}$, $-\frac{1}{100}$, $\frac{1}{100}$, $\frac{1}{5}$, 1 and 5.*

(ii) There is $t_0 > 0$ such that $\mathcal{B}_t$ is not a GGC for $0 < |t| < t_0$.

Proof. (i) For each of these t the matrix $\mathcal{K}_n'(2)$ at the order n of Table 11.23 is certified not to be positive semidefinite.

t	$\lambda=0$	$\frac{1}{2}$	2	10	20	50	eigenvalue at $\lambda=2$
-20	8	9	8	8	7	9	-1.582×10^{-7}
-5	8	8	7	7	8	10	-8.095×10^{-8}
-1	8	8	7	7	8	11	-1.533×10^{-8}
$-\frac{1}{2}$	8	8	7	7	9	11	-4.725×10^{-9}
$-\frac{1}{10}$	8	8	8	8	9	12	-1.800×10^{-9}
$-\frac{1}{100}$	9	9	8	8	10	12	-4.119×10^{-11}
0	none	none	none	none	none	none	1.247×10^{-13}
$\frac{1}{100}$	4	4	5	6	7	10	-2.357×10^{-5}
$\frac{1}{5}$	4	4	4	5	7	9	-1.101×10^{-4}
1	3	3	4	5	6	8	-2.102×10^{-3}
5	3	3	3	5	6	8	-2.141×10^{-3}

Table 11.23: The GGC test for the tilted variables $\mathcal{B}_t$ of Section 11.5.5.3: the least order $n\le12$ at which $\mathcal{K}_n'(\lambda)$ has a negative eigenvalue, and the least eigenvalue of the scaled matrix $\mathcal{K}_n'(2)$ at that order; for $t=0$, where no order fails, the last column gives $\mathcal{K}_7'(2)$. At $\lambda=200$ no matrix of order at most 12 fails, for any t. Every negative entry of the last column is certified, and so is the entry for $\mathcal{K}_9'(0)$ at $t=-\frac{1}{100}$, -2.899×10^{-12}. Computed by `pakes_filter.py` and `pakes_filter_cert.py`.

(ii) Put $\ell_t(\lambda)=\mathbb{E}e^{-\lambda\mathcal{B}_t}$ and split $\int_0^\infty e^{-\lambda b}e^{t(\log b)^2/16}f(b)\,db$, the numerator of ℓ_t, at $b=1$. The part on $(0,1]$ is entire in $(t,\lambda)\in\mathbb{C}^2$, since $|e^{t(\log b)^2/16}|=e^{O((\log b)^2)}$ is negligible against $f(b)=O(b^{-7/2}e^{-\pi/b})$ as $b\downarrow0$, and the denominator, the value at $\lambda=0$, is analytic in t and positive for real t. On $[1,\infty)$ write f as the sum of its terms. Fix $M\ge1$ and x with $\pi M^2<x<\pi(M+1)^2$. The terms with $m>M$ give a function analytic in $t\in\mathbb{C}$ and $\operatorname{Re}\lambda>-\pi(M+1)^2$. For $m\le M$ turn the path $[1,\infty)$ into the ray $b=1+re^{-3\pi i/4}$, $r\ge0$, which avoids $(-\infty,0]$ and along which $(\log b)^2$ grows like $(\log r)^2$. Writing $\lambda+\pi m^2=u+iv$, the integrand decays along every ray $b=1+re^{i\alpha}$ with $-\frac{3\pi}{4}\le\alpha\le0$ when $v>u>0$, so there the rotated integral equals the original one; and it converges, and is analytic in (t,λ), on the half plane $v>u$, which contains $\lambda=-x$. So ℓ_t continues analytically, jointly in (t,λ) near $(0,-x)$, from the upper half plane across the negative axis at $-x$.

Suppose that $\mathcal{B}_t$ is a GGC, with Thorin measure U_t. Then $-\log\ell_t(\lambda)=d_t\lambda+\int\log(1+\lambda/z)\,U_t(dz)$ on $\mathbb{C}\setminus(-\infty,0]$, and as $\varepsilon\downarrow0$ its imaginary part at $\lambda=-y+i\varepsilon$ tends to $\pi\big(U_t((0,y))+\frac{1}{2}U_t(\{y\})\big)$. For t near 0, ℓ_t has no zero near $-x$, since $\ell_0(-x)=\pi x/\sin^2\sqrt{\pi x}>0$ by (11.5.8). So the limit is a real analytic function of y near x, U_t has no atom there, and $U_t((0,y))=-\frac{1}{\pi}\arg\ell_t(-y+i0)$, with the argument continued from 0 on $(0,\infty)$ through the upper half plane, is nondecreasing in y; at $t=0$ it equals $2M$. Write $\ell_t=\ell_0+t\big(\dot\ell-\dot\ell(0)\,\ell_0\big)+O(t^2)$, with $\dot\ell(\lambda)=\frac{1}{16}\int_0^\infty e^{-\lambda b}(\log b)^2f(b)\,db$ continued in the same way; by joint analyticity the remainder is $O(t^2)$ together with its derivative in λ, uniformly near $-x$. Since

ℓ_0 and $\dot\ell(0)$ are real there,

$$U_t((0,y)) = 2M - \frac{t}{\pi}\mathcal{J}(y) + O(t^2), \qquad \mathcal{J}(y) = \frac{\operatorname{Im}\dot\ell(-y+i0)}{\ell_0(-y)},$$

with the same expansion for the derivatives in y. In $\dot\ell(-y+i0)$ the part on $(0,1]$ and the terms with $m > M$ are real, and for $m \le M$ the integral over the rotated ray differs by a real number from $\int_0^\infty e^{-\sigma b}(\log b)^2(4\pi^2m^4b - 6\pi m^2)\,db$ with $\sigma = -s_m + i0$, $s_m = y - \pi m^2$, which is a combination of $\partial_a^2\big[\Gamma(a)\sigma^{-a}\big]$ at $a = 2$ and $a = 1$, with $\log\sigma = \log s_m + i\pi$. This gives

$$\operatorname{Im}\dot\ell(-y+i0) = \frac{\pi}{8}\sum_{m\le M}\Big(\frac{4\pi^2m^4(\log s_m - 1 + \gamma_E)}{s_m^2} + \frac{6\pi m^2(\log s_m + \gamma_E)}{s_m}\Big),$$

where γ_E is Euler's constant. At $y = 1.015\pi$, where $M = 1$, the certificate program gives $\mathcal{J}'(y) = 8.52630079173 \pm 3.4\times10^{-12}$, and at $y = 2.47\pi$ it gives $\mathcal{J}'(y) = -0.0671813667518 \pm 10^{-14}$. So for small $t > 0$ the derivative of $U_t((0,y))$ at $y = 1.015\pi$ is $-\frac{t}{\pi}\mathcal{J}'(1.015\pi) + O(t^2) < 0$, and for small $t < 0$ the derivative at $y = 2.47\pi$ is negative; both contradict monotonicity. □

The program `pakes_characters.py` checks the formula for $\operatorname{Im}\dot\ell$ against the integrals over the rotated rays at $y = 2\pi$, 3.3π, 6π and 12π, to 15 digits, and scans $\mathcal{J}'$ at 399 points in each of the first four gaps $(\pi M^2, \pi(M+1)^2)$: in each gap $\mathcal{J}'$ is positive on a left part and negative on a right part, for y/π in $[1.8925, 3.9925]$, $[5.5125, 8.9875]$, $[11.1, 15.9825]$ and $[18.6775, 24.9775]$, so every one of these gaps excludes both signs of t on its own.

Remark 11.5.18. The variable $\mathcal{B}$ is a GGC, but its density f is not HCM. Indeed f is analytic in $\operatorname{Re} b > 0$ with the natural boundary $\operatorname{Re} b = 0$ (the discussion after Theorem 7.1.21), while an HCM function extends analytically to $\mathbb{C}\setminus(-\infty, 0]$ by (F7). Proposition 11.5.17(i) gives a second proof. For $c \ge 0$ the function $e^{-c(\log b)^2}$ is HCM: with $v = e^y$ and $w = v + v^{-1} = 2\cosh y$, its pair function at u is $e^{-2c(\log u)^2}e^{-2cy^2}$, and y^2 is a Bernstein function of $w - 2$, since

$$\frac{d(y^2)}{dw} = \frac{y}{\sinh y} = \int_0^\infty \frac{d\tau}{w/2 + \cosh\tau}$$

is completely monotone in w; so e^{-2cy^2} is completely monotone in w. If f were HCM, then for $t < 0$ so would be $e^{t(\log b)^2/16}f(b)$ by (F3), and $\mathcal{B}_t$ would be a GGC by (F5), against part (i) at $t = -\frac{1}{100}$.

So the GGC property of $\mathcal{B}_g$ holds for $g = \Phi$, fails for $g = \Phi_t$ at every $t \ne 0$ tested, certifiably at ten of them, and fails for all small $|t|$. It passes the filter wherever it has been tested, and it is the only property in this subsection that does; whether $\mathcal{B}_t$ fails to be a GGC for every $t < 0$ is open (Problem 11.2). It is

not a property of the zeros alone: for $t \geq 0.2$ the zeros of Ξ_t are all real [196] and R_t is the Laplace transform of a GGC (Theorem 7.2.15), yet $\mathcal{B}_t$ is not a GGC at $t = \frac{1}{5}$, 1 and 5, and for $t > 0$ the test fails at lower orders and by wider margins than for $t < 0$. Section 11.5.5.4 shows that it is not a property of Euler products either.

11.5.5.4 Theta kernels, characters and the function of Davenport and Heilbronn

The variable $\mathcal{B}_g$ can be formed for other Dirichlet series with a functional equation, and the question whether its GGC property separates functions with zeros on the critical line from the others can then be asked across functions with and without an Euler product. Let $q \geq 3$, $p \in \{0, 1\}$, and let $a = (a_n)_{n\geq 1}$ be a real sequence of period q with $a_1 = 1$ and $a_{q-n} = (-1)^p a_n$ whose theta series

$$\vartheta_a(x) = \sum_{n\geq 1} n^p a_n e^{-\pi n^2 x/q}, \qquad x > 0,$$

satisfies $\vartheta_a(1/x) = x^{p+1/2}\vartheta_a(x)$. Then

$$\xi(s, a) = \int_0^\infty \vartheta_a(x)\, x^{(s+p)/2}\, \frac{dx}{x} = \Big(\frac{q}{\pi}\Big)^{(s+p)/2} \Gamma\Big(\frac{s+p}{2}\Big) \sum_{n\geq 1} \frac{a_n}{n^s}$$

is entire with $\xi(s, a) = \xi(1 - s, a)$, and $K_a(v) = 2e^{(p+1/2)v}\vartheta_a(e^{2v})$ is an even kernel with $\xi(\frac{1}{2} + s, a) = \int_{\mathbb{R}} K_a(v)e^{sv}\, dv$. The real primitive characters χ_d, d a fundamental discriminant, are of this kind, with $q = |d|$ and $p = 1$ for $d < 0$; for χ_{-4}, written χ_4 in Section 8.5.7, $\vartheta_a(x) = \eta(ix)^3$ and K_a is the kernel K_4 of Proposition 8.5.21. When $\vartheta_a > 0$, the variable $\mathcal{B}_a = \mathcal{B}_{K_a}$ has the density $b^{p/2-1}\vartheta_a(b)/\xi(0, a)$ and $\mathbb{E}\mathcal{B}_a^{s/2} = \xi(s, a)/\xi(0, a)$, and by the functional equation its reciprocal has the density $x^{(p-1)/2}\vartheta_a(x)/\xi(0, a)$. We test both. For χ_{-4} the reciprocal $1/\mathcal{B}_a$ is $\frac{\pi}{2}$ times the exit time of Brownian motion from $(-1, 1)$, the variable written Z in Remark 8.5.22 (not the driving process of Section 11.5.4), and it is a GGC by the product formula for sech. Whether $\mathcal{B}_{\chi_{-4}}$ itself is a GGC is open, and HCM cannot decide it: the density of $1/\mathcal{B}_{\chi_{-4}}$ is not HCM (Theorem 4.3.21), so by (F2) neither is that of $\mathcal{B}_{\chi_{-4}}$.

Lemma 11.5.19. *Let $S_a = \sum_{n\geq 2} n^p |a_n| e^{-\pi(n^2-1)/q}$. If $S_a < 1$, then $\vartheta_a > 0$ on $(0, \infty)$.*

Proof. For $x \geq 1$, $\vartheta_a(x) = e^{-\pi x/q}\big(1 + \sum_{n\geq 2} n^p a_n e^{-\pi(n^2-1)x/q}\big) \geq e^{-\pi x/q}(1 - S_a) > 0$, and for $0 < x < 1$, $\vartheta_a(x) = x^{-p-1/2}\vartheta_a(1/x) > 0$. □

The certificate program encloses S_a in ball arithmetic: it is 0.0019 for χ_{12}, 0.0056 for χ_{-4}, 0.043 for χ_8, 0.045 for χ_{24}, 0.086 for χ_{-3}, 0.092 for the function of Davenport and Heilbronn below, 0.130 for χ_{-8}, 0.158 for χ_5, 0.608 for χ_{-7} and

0.659 for χ_{13}, so these kernels are positive. Positivity can fail: for $d=-19$ and $d=-40$ the program `pakes_filter.py` finds $e^{\pi/q}\vartheta_a(1)=-0.567$ and -0.421. For $d=-19\equiv 5 \pmod{24}$ this is a case of Theorem 8.5.18, and kernels of this kind go back to Pólya's examples recalled in [28, Section 6.1].

The function of Davenport and Heilbronn [64] is $f_{\mathrm{DH}}(s)=\sum_n a_n n^{-s}$ with a of period 5 and $(a_1,\dots,a_5)=(1,\varkappa,-\varkappa,-1,0)$, where

$$\varkappa=\frac{\sqrt{10-2\sqrt5}-2}{\sqrt5-1}=0.2840790438404122960283.$$

With χ the character modulo 5 with $\chi(2)=i$, $f_{\mathrm{DH}}=\frac{1-i\varkappa}{2}L(s,\chi)+\frac{1+i\varkappa}{2}L(s,\bar\chi)$, and $\varkappa$ is chosen so that $(1+i\varkappa)/(1-i\varkappa)$ is the root number

$$\varepsilon(\chi)=0.85065080835203993218+0.52573111211913360603\,i;$$

then $p=1$ and $\vartheta_a(1/x)=x^{3/2}\vartheta_a(x)$, and the program `pakes_characters.py` checks $\xi(s,a)=\xi(1-s,a)$ at $s=0.3+2i$ and $0.8+15.5i$ to 18 digits. The coefficients are not multiplicative, since $a_6=1$ while $a_2a_3=-\varkappa^2$, so f_{DH} has no Euler product, and it has infinitely many zeros in $\operatorname{Re}s>1$ [255, Chapter 10]. Zeros off the critical line inside the strip have been computed by Spira [243] and by Balanzario and Sánchez-Ortiz [13]; the program confirms two of them,

$$0.808517182456637+85.6993484853776\,i,$$
$$0.650830080609737+114.163342730757\,i,$$

where $|f_{\mathrm{DH}}|<10^{-25}$. Modulo 5 there is no other function of the class built from χ and $\bar\chi$: a combination $\alpha L(s,\chi)+\beta L(s,\bar\chi)$ with $a_1=1$ whose completed function is w times its reflection must have $w=\pm1$ and $\beta=w\,\varepsilon(\chi)\alpha$, and the choice $w=-1$ gives the coefficients $(1,-1/\varkappa,1/\varkappa,-1,0)$, whose theta series satisfies $\vartheta(1/x)=-x^{3/2}\vartheta(x)$, so that it vanishes at $x=1$ and takes both signs.

Table 11.24 gives the test for ξ, for nine real characters and for f_{DH}. For the characters and for f_{DH} the moments are computed as for $\mathcal{B}_t$, from the kernels K_a, and at $\lambda=0$ also from the values of L functions alone: $\mathbb{E}\mathcal{B}_a^k=\xi(2k,a)/\xi(0,a)$ and $\mathbb{E}\mathcal{B}_a^{-k}=\xi(2k+1,a)/\xi(0,a)$, with $\sum_n a_n n^{-s}=q^{-s}\sum_{r=1}^q a_r\zeta(s,r/q)$ through the Hurwitz zeta function. The two routes agree, the functional equations of the completed functions hold to a relative 5×10^{-81}, and the failures marked in the table are certified, by the first route for $\lambda>0$ and by the second at $\lambda=0$.

Proposition 11.5.20. *Neither $\mathcal{B}_{\chi_{-8}}$ nor $1/\mathcal{B}_{\chi_{-8}}$ is a GGC.*

Proof. From the values of $L(s,\chi_{-8})$ at $s=0,2,4,6,8$ and at $s=3,5,7,9$, the certificate program gives

$$3\kappa_3^2-2\kappa_2\kappa_4=7.55465911093\pm4\times10^{-12}\qquad\text{and}\qquad 78.8854168657\pm2\times10^{-11}$$

for $\mathcal{B}_{\chi_{-8}}$, where $(\kappa_2,\kappa_3,\kappa_4)=(3.0767,15.997,123.54)$, and for its reciprocal. Both are positive, against Lemma 11.5.16. □

function	S_a	$\mathcal{K}_7'(0)$	$\mathcal{K}_7'(2)$	$\mathcal{B}_a$	$1/\mathcal{B}_a$
ξ		6.2×10^{-16}	1.2×10^{-13}	GGC	fails at $\mathcal{K}_3'(0)$
χ_{-3}	0.086	2.4×10^{-8}	4.9×10^{-8}	passes	GGC
χ_{-4}	0.006	1.2×10^{-8}	4.6×10^{-8}	passes	GGC
χ_5	0.158	7.8×10^{-8}	7.1×10^{-8}	passes	fails at $\mathcal{K}_{10}'(0)$
χ_8	0.043	4.8×10^{-8}	3.8×10^{-8}	passes	fails at $\mathcal{K}_7'(2)$
χ_{12}	0.002	2.1×10^{-8}	4.2×10^{-8}	passes	fails at $\mathcal{K}_9'(10)$
χ_{24}	0.045	7.5×10^{-9}	2.0×10^{-8}	fails at $\mathcal{K}_{11}'(2)$	fails at $\mathcal{K}_6'(0)$
χ_{-7}	0.608	-0.149	-0.221	fails at $\mathcal{K}_3'(0)$	fails, zero
χ_{-8}	0.130	-0.057	-0.477	fails at $\mathcal{K}_2'(0)$	ID, fails, zero
χ_{13}	0.659	-0.0023	-0.256	fails at $\mathcal{K}_3'(0)$	fails at $\mathcal{K}_3'(0)$
f_{DH}	0.092	-1.2×10^{-5}	-1.4×10^{-4}	fails at $\mathcal{K}_6'(0)$	fails, zero

Table 11.24: The GGC test for the variables $\mathcal{B}_a$ of Section 11.5.5.4 and their reciprocals: the sum S_a of Lemma 11.5.19, the least eigenvalues of the scaled matrices $\mathcal{K}_7'(0)$ and $\mathcal{K}_7'(2)$ of $\mathcal{B}_a$, and the verdicts. "Passes": no matrix $\mathcal{K}_n(\lambda)$ or $\mathcal{K}_n'(\lambda)$ with $n\le12$ has a negative eigenvalue at $\lambda=0$, $\frac12$, 2, 10, 20, 50 and 200. "Fails at $\mathcal{K}_n'(\lambda)$": the least failing order, certified at that λ. "Zero": the Laplace transform vanishes on $(-\infty,0)$ (Proposition 11.5.21). GGC for ξ by (11.5.8), for $1/\mathcal{B}_a$ with χ_{-3} and χ_{-4} by Proposition 11.5.21. All the functions but f_{DH} have an Euler product. Computed by `pakes_filter.py` and `pakes_filter_cert.py`.

The same four numbers settle $1/\mathcal{B}_{\chi_{-7}}$ (27.8926758865) and $1/\mathcal{B}_a$ for f_{DH} (0.738104955084), while for $\mathcal{B}$, a GGC, the number is -0.0026, and for $\mathcal{B}_{\chi_{-4}}$ it is -0.70. So $L(s,\chi_{-8})$, which has an Euler product and whose zeros lie on the critical line to height $10^8/8$ at least [195], fails the test already at order two, on both sides. For odd characters the reciprocal side has a closed form, after [28, (96)].

Proposition 11.5.21. *Let $p=1$ and $\vartheta_a>0$, and put $w=\sqrt{\pi q\lambda}$ and $\Sigma_a(w)=\sum_{1\le n<q/2}a_n\sinh\frac{(q-2n)w}{q}$. Then*

$$\int_0^\infty e^{-\lambda x}\vartheta_a(x)\,dx=\frac{\sqrt{q}\,\Sigma_a(w)}{\sinh w},\qquad \mathbb{E}e^{-\lambda/\mathcal{B}_a}=\frac{\sqrt{q}\,\Sigma_a(w)}{\xi(0,a)\sinh w},\tag{11.5.10}$$

and $\xi(0,a)=q^{-1/2}\sum_{1\le n<q/2}(q-2n)a_n$. In particular:

(i) for χ_{-4}, $\mathbb{E}e^{-\lambda/\mathcal{B}_a}=\operatorname{sech}\sqrt{\pi\lambda}$, and $1/\mathcal{B}_a$, which is $\frac\pi2$ times the exit time of Brownian motion from $(-1,1)$, is a GGC;

(ii) for χ_{-3}, $\mathbb{E}e^{-\lambda/\mathcal{B}_a}=3\sinh(w/3)/\sinh w=\prod_{3\nmid m}\bigl(1+3\lambda/\pi m^2\bigr)^{-1}$, so $1/\mathcal{B}_a$ is a GGC with Thorin measure $\sum_{3\nmid m}\delta_{\pi m^2/3}$, and it is $\frac{3\pi}{2}$ times the hitting time of 1 by a three dimensional Bessel process started at $\frac13$;

(iii) for χ_{-8}, $\mathbb{E}e^{-\lambda/\mathcal{B}_a}=\cosh(w/4)/\cosh(w/2)=\prod_{k\ge1}(1+\lambda/4\alpha_k)/(1+\lambda/\alpha_k)$ with $\alpha_k=\frac\pi8(2k-1)^2$, so $1/\mathcal{B}_a$ is infinitely divisible, with Lévy density

$\sum_k (e^{-\alpha_k x} - e^{-4\alpha_k x})/x > 0$, *and it is* π *times the hitting time of* 1 *by a reflected Brownian motion started at* $\frac{1}{2}$*; it is not a GGC, because its Laplace transform vanishes at* $\lambda = -\pi/2$*;*

(iv) if $a = \chi_d$ *with* $d < 0$ *and* $\chi_d(2) = 1$*, that is* $d \equiv 1 \pmod 8$*, the Laplace transform of* $1/\mathcal{B}_a$ *vanishes at a point of* $(-4\pi/q, -\pi/q)$*, so* $1/\mathcal{B}_a$ *is not a GGC; for* χ_{-7} *the point is* $\lambda = -1.033610573$*. The same holds for* f_{DH}*, at* $\lambda = -2.046575671$*.*

Proof. By the functional equation $\vartheta_a(x) = x^{-3/2} \sum_n n a_n e^{-\pi n^2/qx}$, and

$$\int_0^\infty e^{-\lambda x - c/x} x^{-3/2}\, dx = \sqrt{\pi/c}\, e^{-2\sqrt{c\lambda}}$$

with $c = \pi n^2/q$ gives $\int_0^\infty e^{-\lambda x} \vartheta_a(x)\, dx = \sqrt{q} \sum_{n \geq 1} a_n e^{-2nw/q}$ for $\lambda > 0$. Summing over a period and pairing n with $q - n$, where $a_{q-n} = -a_n$, gives the first formula of (11.5.10), which is [28, (96)] in our normalisation; at $\lambda = 0$ it gives $\int_0^\infty \vartheta_a = \xi(1, a) = \xi(0, a)$, and since $1/\mathcal{B}_a$ has the density $\vartheta_a/\xi(0, a)$ the second formula follows. (i) and (ii) follow from the product formulas for cosh and sinh, and [28, (92)] identifies the transforms in (ii) and (iii) with those of the hitting times. In (iii) Frullani's integral $\log(1 + \lambda/c) = \int_0^\infty (1 - e^{-\lambda x}) e^{-cx}\, dx/x$ gives the Lévy density. A GGC transform ℓ cannot vanish on $(-\infty, 0)$ where it extends meromorphically: for $0 < \varepsilon \leq 1$, $|\ell(-x + i\varepsilon)| \geq e^{dx - C}$ with $C = \int \log(1 + (x+1)/z)\, U(dz) < \infty$, because $|1 + (-x + i\varepsilon)/z| \leq 1 + (x+1)/z$. In (iii) the zero is that of $\cosh(w/4)$ at $w = 2\pi i$. In (iv) put $w = iy$; then $\Sigma_a(iy)/\sinh(iy) = \nu(y)/\sin y$ with $\nu(y) = \sum_{1 \leq n < q/2} a_n \sin \frac{(q-2n)y}{q}$, and since $a_{q-n} = -a_n$, $\nu(\pi) = \frac{1}{2} \sum_{n=1}^{q-1} a_n \sin \frac{2\pi n}{q}$ and $\nu(2\pi) = -\frac{1}{2} \sum_{n=1}^{q-1} a_n \sin \frac{4\pi n}{q}$. For an odd real primitive character the Gauss sum is $i\sqrt{q}$, so $\nu(\pi) = \frac{\sqrt{q}}{2}$ and $\nu(2\pi) = -\chi_d(2)\frac{\sqrt{q}}{2}$. If $\chi_d(2) = 1$, ν vanishes at some $y_0 \in (\pi, 2\pi)$, where $\sin y_0 \neq 0$, and the transform vanishes at $\lambda = -y_0^2/\pi q$. For f_{DH}, $\nu(\pi) = \frac{\sqrt{5}}{2}$ and $\nu(2\pi) = -0.317610026505$. □

A survey extends the table. For each of the 26 fundamental discriminants d with $3 \leq |d| \leq 40$, the program `pakes_filter.py` tests the positivity of ϑ_{χ_d} and, where it holds, runs the test at 80 digits to order 10 at $\lambda = 0$, $\frac{1}{2}$, 2, 10 and 20, on both sides. The kernels of $d = -19$ and $d = -40$ are not positive. Of the other 24, $\mathcal{B}_{\chi_d}$ passes only for $d = -4$, -3, 5, 8, 12 and 24 (the last fails at order 11, beyond the survey), and $1/\mathcal{B}_{\chi_d}$ only for $d = -4$ and -3. The 18 failures of $\mathcal{B}_{\chi_d}$ occur at order 2 or 3 for $d = -39$, -35, -31, -24, -23, -20, -15, -11, -8, -7, 13, 21, 29 and 37, and at orders 6 to 8 for $d = 17$, 28, 33 and 40; the failures of $1/\mathcal{B}_{\chi_d}$ for $d = -7$, -15, -23, -31 and -39 are cases of Proposition 11.5.21(iv). The size of the first term does not decide the outcome: the kernels that pass have $S_a \leq 0.159$, but χ_{-8} fails at order 2 with $S_a = 0.130$, and χ_{24} fails with $S_a = 0.045$.

Combinations of L functions do not change the picture. Modulo 5 there is no family, as shown above. Modulo 7 the odd characters are χ_{-7} and a complex pair χ, $\bar\chi$, with $\chi(3) = e^{\pi i/3}$, and for real r

$$f_r = \big(1 - 2r\cos\tfrac{\phi}{2}\big)L(s,\chi_{-7}) + re^{-i\phi/2}L(s,\chi) + re^{i\phi/2}L(s,\bar\chi), \qquad e^{i\phi} = \varepsilon(\chi),$$

has real coefficients, $a_1 = 1$ and $\vartheta(1/x) = x^{3/2}\vartheta(x)$, which the program checks to a relative 10^{-29}. For 17 values of r between -3 and 3 the kernel is positive for $-3 \le r \le 2$ and not for $r = 3$, and for each of the 16 positive kernels both $\mathcal{B}$ and $1/\mathcal{B}$ fail at $\lambda = 0$, at order 2 or 3. None of the characters whose variable passes has a complex character of the same conductor and parity to combine with, so the smallest family through a passing character would need a larger modulus.

The pattern is not that of the Euler product. All nine L functions of Table 11.24 have one, and GRH has been verified for them to height at least $10^8/q$ [195]; yet $\mathcal{B}_{\chi_d}$ fails for χ_{24}, χ_{-7}, χ_{-8} and χ_{13}, and for 18 of the 24 positive kernels of the survey. It fails for f_{DH}, which has zeros off the line, but it fails in the same way for χ_{-7} and χ_{13}, whose zeros are on the line as far as they have been computed. At $\lambda = 0$ the test of order n reads only the moments $\xi(2k,a)/\xi(0,a)$ with $1 \le k \le 2n$, whose numerators are values of Dirichlet series where they converge absolutely, and for $\lambda > 0$ it reads Laplace transforms of the theta series: it sees the shape of the theta transform and not the arithmetic of the coefficients. Where the GGC property is proved, it comes from a product formula for that transform, $(\sqrt{\pi\lambda}/\sinh\sqrt{\pi\lambda})^2$ for ξ and the products of Proposition 11.5.21(i) and (ii) for the reciprocals of χ_{-4} and χ_{-3}. Since the property is a theorem for ξ, an implication from it to zeros on the line, within a class of functions containing ξ, would prove RH; Problem 11.1 asks for such a class. The computations find no function with $\mathcal{B}_a$ a GGC and a zero off the line: f_{DH} and the tilts Φ_t at the eleven tested values of $t < 0$ fail the test. But the Euler product does not imply the property, since χ_{-7}, χ_{-8} and χ_{13} fail with certificates, so a class in which the implication holds is not singled out by the Euler product.

11.5.6 What survives, and open problems

Apart from the Lambert template of Section 11.5.3, the results of this section, and those of the other chapters that it uses, fall into three classes. The first consists of equivalences with RH: the Thorin Bernstein property of $\log G$ (Theorem 4.2.4), the mixture of exponential laws in the size biased clock (Proposition 11.5.1), the vanishing of the Hankel inertia indices (Theorem 5.1.5), and the GGC property of R_t at $t = 0$ (Theorem 7.2.15 and Proposition 1.2.1). They restate the problem and do not solve it. The second consists of facts that hold with no hypothesis: the clock is infinitely divisible and self decomposable (Theorem 5.1.1), it lies in L_2 and not in L_3 (Theorem 11.5.8), and its driving subordinator has self decomposable, hence unimodal, laws (Corollary 11.5.10); the points of the upper half

plane where $\mathrm{Im}(G'/G)$ could fail to be negative lie in a thin parabola beyond about $-T_0^2$ (Theorem 4.3.10); the clock is not the martingale limit of a supercritical continuous state branching process (Theorem 11.5.13); R_t is a GGC for $t \geq 0.2$ [196] and for no $t < 0$ (Corollary 4.2.5); and $\mathcal{B}_t$ is not a GGC for t near 0, $t \neq 0$ (Proposition 11.5.17). Being unconditional does not by itself prevent a statement from implying RH, since a proof of RH would be unconditional too. What prevents it is that none of these facts distinguishes ξ from functions with zeros off the line: the statements about the flow at $t \neq 0$ are removed by the filter of Section 11.5.5, infinite divisibility and self decomposability hold as well for the comparison function F of Section 12.2.6, the ladder index and the branching obstruction are set by the first zeros (Section 11.5.4), and the localisation uses only the zeros below the verification height. The third class consists of the routes that are closed. Branching is closed by Theorem 11.5.13. The ladder is decided with no hypothesis by Theorem 11.5.8, and along the flow its index changes where every law is a GGC (Section 11.5.4.3). The Hankel route is closed for practical purposes, on the heuristic of Table 7.7 that a zero off the line at height γ first becomes visible at an order of about γ^2/δ. Properties of the kernel and its representations as mixtures are removed by the filter (Propositions 11.5.14 and 11.5.15), and the GGC property of the variable of Biane, Pitman and Yor does not detect the Euler product (Section 11.5.5.4).

What remains open is the one step of the Lambert proof that does not transfer: the analyticity of the cumulant function off the cut, which for the clock is RH itself (Proposition 6.1.9; Proposition 1.2.1). Pakes's work, read in the notation of the book, gives a template in which that step is a theorem, a dictionary between his language and that of the book, and a few unconditional facts about the clock. The threshold $\Lambda \leq 0.2$ of the heat flow is Theorem 7.2.15 with the bound of Platt and Trudgian, and is not new here, and the filter follows from the theorem of Rodgers and Tao, not from that bound. Of the four representations tested in Section 11.5.5, the arcsine mixture, the Gamma mixtures, infinite divisibility of $\mathcal{B}_t$ and the GGC property of $\mathcal{B}_t$, the Gamma mixtures fail already at $t = 0$ (Theorem 11.2.9), the arcsine mixture persists for every $t < 0$, infinite divisibility persists at every t tested, as far as the cumulants were computed, and only the last passes the filter, numerically and for small $|t|$ provably. It fails for most of the real Dirichlet L functions tested, which removes the reason to expect it to be sensitive to the Euler product. Other representations of the programme were not tested: the hitting times at the basepoints (Section 9.2), the GGC of the index side (Section 8.7.4) and the structure of Williams (Section 8.2.3).

11.5.6.1 Open problems

Problem 11.1. Find a natural class of functions $\xi(s, a)$ with positive kernels, containing ξ, in which the GGC property of $\mathcal{B}_a$ implies that every zero of $\xi(s, a)$ lies on the critical line, or a function in such a class whose variable $\mathcal{B}_a$ is a GGC

and which has a zero off the line. Since $\mathcal{B}$ is a GGC, the first answer for a class containing ξ proves RH. No candidate for the second is known: the function of Davenport and Heilbronn, the family f_r modulo 7 at the 16 tested values of r with a positive kernel and the tilts $\mathcal{B}_t$ at the eleven tested values of $t < 0$ all fail the test. The question is related to Question 8.7.21 and to the structural reason for R to be a GGC that the discussion of Table 7.7 asks for.

Problem 11.2. Is $\mathcal{B}_t$ a GGC for no $t \neq 0$? For $t < 0$ this would be a property of the kernel that passes the filter at every t. Proposition 11.5.17 proves it for $0 < |t| < t_0$, through the first order change of the Thorin measure in the gaps between its atoms, and at the ten values of part (i).

Problem 11.3. Which of the variables $\mathcal{B}_{\chi_d}$ for $d = -4$, -3, 5, 8 and 12, which pass the test to order 12, are GGCs? (For $d = 24$ the test fails at order 11.) On the reciprocal side, $1/\mathcal{B}_{\chi_d}$ is a GGC for $d = -3$ and -4 (Proposition 11.5.21); for $d = -4$ it is the exit time variable of Remark 8.5.22, whose density is not HCM (Theorem 4.3.21), so the question for $\mathcal{B}_{\chi_{-4}}$ cannot be settled through HCM. For even characters the reciprocal transform is the series $\frac{2}{\xi(0,a)} \sum_n a_n K_0\big(2n\sqrt{\pi\lambda/q}\big)$ of Bessel functions, and no product formula for it is known to us.

Problem 11.4. With no hypothesis, the Li coefficients of Section 8.3.2 are

$$\lambda_n = \int_0^\infty k_n(t) W(t)\, dt = m_0\, \mathbb{E} k_n(A),$$

where W is the heat trace, A is the variable of Proposition 11.5.1, and k_n is the function with $\int_0^\infty e^{-zt} k_n(t)\, dt = \varphi_n(z)$. Since φ_n is a polynomial of degree n in $(z+\frac{1}{4})^{-1}$ with no constant term, $k_n(t) = e^{-t/4} p_n(t)$ with p_n a polynomial of degree $n-1$; for instance $k_1(t) = e^{-t/4}$ and $k_2(t) = (4-t)e^{-t/4}$. Which unconditional properties of W, or of A, force $\mathbb{E} k_n(A) \geq 0$ for every n, which is RH by Li's criterion [145, 33]? Positivity of W (Theorem 5.1.1) gives $n = 1$; complete monotonicity of W, which is RH, gives every n, since $\varphi_n \geq 0$ on $(0, \infty)$.

Problem 11.5. Describe the ladder index of R_t along the de Bruijn Newman flow. The computation of Section 11.5.4.3 finds the index two of the clock up to $t = 27.8$ and one from $t = 27.9$ on, on its window. Is the index nonincreasing in t, and is it one for all large t? The answer carries no information about RH, but it would show how the spacing of the first zeros sets the index.

The first author's account also asked for a quantitative form of the statement that zero density estimates leave only $o(T)$ zeros off the line up to height T. That premise is not known: the estimates bound the number $N(\sigma, T)$ of zeros with real part above $\sigma > \frac{1}{2}$, for instance $N(\sigma, T) \ll T^{1-(\sigma-\frac{1}{2})/4} \log T$ uniformly in σ [237], which is $o(T)$ for each fixed $\sigma > \frac{1}{2}$ but does not reach the zeros within $O(1/\log T)$ of the line, and only a positive proportion of the zeros is known to lie on it. What can be said about the contribution of zeros off the line near the cut

is in Section 11.4.7, and Section 12.2.6 describes what finite information can and cannot reach.

Chapter 12

Conclusions and open problems

Chapters 9 to 11 ended at the edge of the proved region. At the basepoint one the shifted clock is a generalised gamma convolution with no hypothesis, and those chapters describe it explicitly; below the basepoint one each of their constructions is equivalent to a zero free half plane; and Section 11.4 measured how far from a generalised gamma convolution the shifted clocks can be. With that edge located, the results of Chapters 1 to 11 can be placed on a single path, which runs along the basepoint from the region where positivity of the Thorin measure is a theorem to the centre, where it is RH.

Section 12.1 summarises the book and sets out that path, marking each step proved, computed, announced or open. Section 12.2 asks what computation can still contribute to the open steps, in five work packages with their milestones, and ends the main line of the book with the lessons of its computations, the routes now closed, the reach of finite information and what a proof must supply. Chapter 13 follows as a complement on the side of statistical mechanics, where RH is the statement that the Pólya law is ferromagnetic.

12.1 The Thorin approach in summary, and a path to RH

The summary covers what the Thorin formulation offers, what the sine squared lemma achieves and what the computations have shown; Section 12.1.4 then assembles these results into the path towards the Riemann Hypothesis, and Sections 12.1.5 and 12.1.6 take up its open steps and what computation contributes to them. The path is a programme with its gap identified, not a proof.

12.1.1 What the Thorin formulation offers

The formulation turns location into sign. The classical statement of RH is about where the zeros are. The Thorin statement is about the sign of one measure: the clock $u \mapsto \xi(\frac{1}{2})/\xi(\frac{1}{2}+\sqrt{u})$ is a generalised gamma convolution exactly when RH holds, and then its Thorin measure is forced to be $\sum_{\gamma>0}\delta_{\gamma^2}$. Signs can be

certified on windows by interval arithmetic, and positivity is preserved by limits, convolutions and positive kernels, which location statements are not.

The formulation is rigid and needs only a real interval. The Thorin measure of a GGC is unique, so there is no freedom to hide a failure, and a GGC representation of the clock on the real interval $0 < u < 1$ already continues to the cut plane and forces every zero onto the line (Section 1.3.3). The target is therefore an identity in a real variable, with no inequality at complex arguments.

It comes with a resolution parameter. At the basepoint $\frac{1}{2} + \varepsilon$ the Thorin measure exists as a positive measure for every $\varepsilon \geq \frac{1}{2}$, unconditionally, and its density is the Poisson smoothing of the zeros. The family $\varepsilon \mapsto \rho_\varepsilon$ interpolates between an unconditional object and the RH object, it is local in the ordinate, and it is computable from the Pólya density, with no primes and no zeros, to an accuracy that explicit error bounds can certify (Chapter 2).

It gives graded partial results. The Bondesson hierarchy turns into a ladder of positivity conditions on the heat trace: infinite divisibility, self decomposability, log convexity, Hankel blocks up to size 3×10^7, derivative signs to order 1.16×10^{25}. Each rung is an unconditional theorem, and each measures how much of RH a given amount of verification buys (Section 5.1).

It unifies the criteria. Lagarias positivity, Weil positivity, complete monotonicity of the heat trace, hyperbolic complete monotonicity and the Stieltjes property of G'/G are all positivity of the Thorin functional of Section 5.3 on different cones of test functions. Li positivity pairs the same measure with test functions outside the class $\mathcal{T}$ of that section, so that the pairing is defined through the zeros (Section 8.3), and the de Bruijn Newman flow deforms the measure itself rather than the test functions (Section 8.4). The Thorin formulation makes the common structure explicit and shows which form is best conditioned for computation (Section 12.2.6).

It has a probabilistic meaning. The clock is a random variable with a computable law, a quasi stationary rate equal to γ_1^2, a van Dantzig partner and a hitting time interpretation (Sections 8.1 and 8.2), and at every basepoint above the centre the shifted clock is the centre clock tilted and subordinated by a Brownian first passage (Section 9.2). This brings the closure theorems of probability (weak limits and subordination) and the spectral theory of Kent and Krein to bear on the problem.

It counts the zeros from a smooth quantity. The cumulative Thorin mass is an argument, so the zero count is read as an integer from the Thorin measure, by Backlund's method at the resolution $\varepsilon = \frac{1}{2}$, and the zero density $\nu(\frac{1}{2} + \varepsilon, T)$ is read at every resolution from one identity (Section 3.4).

12.1.2 What the sine squared lemma achieves

The lemma is the identity

$$e^{-sx} + sx - 1 = x \int_0^\infty \log\Big(1 + \frac{s^2}{z}\Big) \frac{\sin^2(x\sqrt{z}/2)}{\pi\sqrt{z}}\, dz,$$

extended linearly to measures. It achieves six things.

1. It converts the Lévy description of $\log \xi$, in which the primes are atoms at $\log n$, into the Thorin description of the reciprocal, in which the zeros are atoms at γ^2, by an explicit positive kernel. It is the dictionary between the two sides of the explicit formula in the language of infinitely divisible laws.

2. It identifies the Thorin density: the bracket it produces is $\mathrm{Re}(f'/f)(\alpha+i\sqrt{z})$, which is the derivative in θ of $\arg f(\alpha + i\theta)$. Hence the cumulative Thorin mass is an argument, which is what makes the zero count, the mass defect formula and the zero density identity exact (Sections 3.2 and 3.4).

3. It isolates the difficulty. In the canonical measure of ξ the prime atoms, the archimedean density and the drift are nonnegative, and the pole term $-e^{-(\alpha-1)x}dx$ is the only negative input; for the L function of χ_{12}, which has no pole, the negative input is carried by the signs of $\chi_{12}(p)$ (Sections 8.5.3 and 11.3). RH becomes the statement that, after the kernel acts, the pole mass is dominated by the prime and archimedean mass at every basepoint above the centre. At the centre both masses are infinite, the regularised pairing of Section 5.3 makes the statement exact, and it is Weil's criterion on the cone of variograms (Sections 3.5 and 7.2).

4. It factors through the heat kernel chart. Its first step is Brownian subordination and produces the heat trace W; its second step is a Gaussian cosine transform and produces the Thorin measure. Theorem A of [200], which is Theorem 5.1.1, is positivity after the first step; RH is positivity after the second. The lemma thus locates exactly where the proved results stop.

5. It supplies building blocks. Every compensated exponential jump is a GGC in the square root variable with explicit Thorin density, and so are the Vere-Jones blocks of Chapter 6, which gives exactly solvable models with positive Thorin measures.

6. It prices the prime side. Read through Selberg's formula, the prime side at resolution ε converges like $x^{-\varepsilon}/(\varepsilon^2 \log x)$ in the cutoff x when no zero lies to the right of the critical line, so accuracy 10^{-k} needs a cutoff $x \approx 10^{k/\varepsilon}$, with primes up to x^2 (Section 3.5.1). This is why the lemma identifies the Thorin measure but cannot compute it near the line, and why the Pólya route is needed.

What the lemma does not achieve should be said as clearly. At the centre it reproduces Weil's criterion rather than a new one, and because its input measure is signed, positivity of the output is not inherited from positivity of the input. For $\alpha \geq 1$ positivity comes from the zeros through the Poisson kernels. For $\frac{1}{2} < \alpha < 1$ the compensated form of the lemma converges exactly as far as the zero free half plane reaches (Proposition 9.1.6), and the shifted clock is a GGC exactly when no zero lies to the right of α (Theorem 1.2.2); whether positivity of the density on the single line $\operatorname{Re} s = \alpha$ excludes such zeros is not known (Remark 1.2.4).

12.1.3 What the computations have shown

The book set out to analyse the Thorin condition from a computational point of view, and Chapters 2 to 5 give the first answer. The Thorin measure of the reciprocal Xi function is computable at every positive resolution from the Pólya density alone. The untilted route loses 0.341 decimal digits per unit of height to cancellation and costs about 0.7 ms per point below height 200; the tilted contour cuts the loss and reaches heights of several thousand, with 25 digits at height 5000 and the Lehmer pair near height 7005 resolved (Sections 2.1.3 and 2.2). Measured against the discrete RH measure, the computed measure behaves as the Poisson structure predicts: raw total variation is maximal and uninformative, the difference at matched resolution is a numerical floor, and the cell distance, the mass defect and the error of the heat trace vanish linearly after coarse graining, the mass defect with an exact first order formula (Proposition 3.2.2). The zero density identity reads the count and the strip by strip distribution of the zeros from the Thorin mass, and below height 200 it returns 79 at eight resolutions (Section 3.4). The heat trace ladder, the Hankel blocks, the Williamson measures, the Keiper Li coefficients, the HCM derivatives and the de Bruijn Newman flow are further views of the same measure, each with its own conditioning.

Two structural questions have definite answers. Weak convergence of the Thorin measures holds whatever the zeros do and is therefore uninformative (Section 4.1.1), and hyperbolic complete monotonicity of the reciprocal Xi function is equivalent to RH (Corollary 4.1.3). The reciprocal zeta function of Chapters 6 and 7 shows how much can be proved unconditionally on the Thorin route: infinite divisibility, a precise place in the Bondesson hierarchy, a floor for the reciprocal Xi function whose upgrade to a GGC is RH, and finite order complete monotonicity reaching orders set by the verification height. No property that the damped Pólya kernels $e^{tv^2/4}\Phi(v)$, $t < 0$, share with Φ can imply RH, by the theorem of Rodgers and Tao, and the one property tested in Section 11.5.5 that passes this filter, the GGC property of the variable of Biane, Pitman and Yor, fails for most of the real Dirichlet L functions tested there.

On the probabilistic side the law of the clock is computable, its quasi stationary rate is the square of the first ordinate, its higher atoms are out of reach in the time domain, it is not a Brownian first passage time to a random level, and a curved

Brownian boundary reproduces it to Monte Carlo accuracy (Sections 8.1 and 8.2). Sampling Williams' representation sees the zeros only at exponential cost, because the signal decays like $e^{-\pi z/4}$: even the first zero needs about 10^6 draws. At the basepoint one the shifted clock is, with no hypothesis, a subordinated hitting time (Section 9.2).

The analysis also locates the remaining difficulty. It is a single positivity statement, uniform in the height and in the resolution, and every finite computation in the book is consistent with its failure above the computed range. Computation can calibrate, exclude and falsify; the uniformity has to come from structure. The path of Section 12.1.4 makes this precise.

12.1.4 The path along the basepoint

Figure 12.1 and Table 12.1 set out the path. It runs along the basepoint α from the region where positivity is a theorem to the centre where it is RH, and Chapters 9 to 11 mark where the theorems stop.

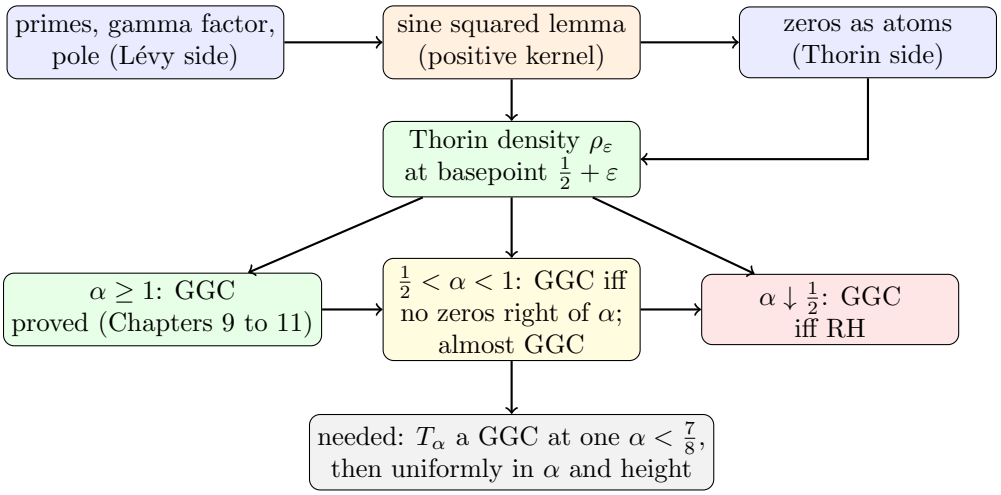


Figure 12.1: The Thorin path to RH. The sine squared lemma connects the arithmetic to the Thorin density. For $\alpha \geq 1$ the shifted clock is a GGC with no hypothesis, and Chapters 9 to 11 describe it explicitly; for $\frac{1}{2} < \alpha < 1$ each of their constructions is equivalent to a zero free half plane, and unconditionally the shifted clock is an almost GGC; at the centre the GGC property is RH. The programme must carry the GGC property across the strip.

Steps 1 to 6 are done. They establish the unconditional structure of the clock, the unconditional GGC property at every basepoint $\alpha \geq 1$, and a computed picture on every window where the zeros are verified. The path therefore runs, as a theorem, from $\alpha = \infty$ down to $\alpha = 1$, and, as a computation, down to $\alpha = \frac{1}{2} + 0.005$ on the windows computed. Step 7 has been announced with $\alpha = \frac{7}{8}$,

step	statement	status
1	The clock is infinitely divisible and self decomposable; the heat trace is positive and decreasing	proved (Theorem 5.1.1, which is Theorem A of [200]), with the verification of RH below height 1000 as input
2	The heat trace is log convex, its leading Hankel blocks of size up to 3×10^7 are positive definite, derivative signs hold to order 1.16×10^{25}	proved, with verification height as input
3	The reciprocal at basepoint $\alpha \geq 1$ is a GGC with Poisson Thorin density, and at $\alpha = 1$ its Lévy form, hitting time, scale mixture and completeness are explicit; for every $\alpha > \frac{1}{2}$ the shifted clock T_α exists and is an almost GGC	proved (Theorems 1.2.2, 9.2.5 and 11.4.2; Chapters 9 to 11), with the verification of RH below height 1000 as input to Theorem 9.2.5
4	The normalised reciprocal zeta function is infinitely divisible; the canonical function of the floor of the reciprocal Xi function is an arithmetic part plus a completely monotone archimedean part, and RH is the complete monotonicity of the sum	proved (Chapters 6 and 7)
5	The Thorin density is computable at every resolution, with an explicit bound for each source of error, and is positive at every sample point of every window computed; the runs do not yet add these bounds to the ball radii (Section 12.2.2)	computed (Chapters 2 to 5)
6	Zero density $\nu(\frac{1}{2}+\varepsilon, 200) = 0$ for every computed ε, down to $\varepsilon = 0.005$	computed (Section 3.4)
7	For some $\alpha < 1$ the shifted clock T_α is a GGC, equivalently, the half plane $\operatorname{Re} s > \alpha$ is zero free (Theorem 1.2.2); then for every $\alpha' > \alpha$ the measure $U_{\alpha'}$ is the Thorin measure of $T_{\alpha'}$ and is positive at all heights	announced for $\alpha = \frac{7}{8}$ in October 2026, with a formal proof in Lean, not yet refereed (Section 12.1.5); open for every $\alpha < \frac{7}{8}$
8	Positivity of U_α for every $\alpha > \frac{1}{2}$ (Lagarias), equivalently the Thorin condition	open; equivalent to RH

Table 12.1: The path, step by step.

by an argument from outside the book (Section 12.1.5); if it stands, the theorem part of the path runs down to $\alpha = \frac{7}{8}$.

Chapters 9 to 11 show what the theorem is at its end point and how far its constructions reach beyond it, and Section 11.3.4 draws the two conclusions that fix the shape of the path. At the basepoint one the constructions attached to the reciprocal Xi function are unconditional and explicit, from the Thorin measure of T_1 and the stable one half tail of its law (Proposition 9.1.1 and (9.1.3)) to the Lévy form, the subordinated hitting time, the normal scale mixture, the completeness of the tilted Pólya kernels (Sections 9.1, 9.2 and 10.1) and the consequences of the theorems of Sato and Bondesson (Table 11.3). There are two exceptions: the Krein string representation of Corollary 9.2.3 assumes RH and simple zeros, and Lemma 9.2.4 gives the Wald defect $\sum_\rho \rho^{-1}/\sqrt{2}$ only for a boundary bounded at infinity, while the boundedness of the computed boundary is observed, not proved. Section 11.3 repeats the construction for $L(s, \chi_{12})$, whose Lévy form needs no compensation because there is no pole. Below the basepoint one each construction holds, up to a boundary case, exactly when no zero lies to the right of α (Theorems 9.2.5 and 10.1.3, Propositions 9.1.6 and 10.1.9), and the theory of Sato and Bondesson reaches nothing there (Section 11.1.5). The theorem part of the path that this book proves therefore stops at $\alpha = 1$.

Riesz's criterion takes the same form. By Theorem 10.2.7, for $\frac{1}{2} \le \alpha < 1$ the growth bound $R(x) = O(x^{\alpha/2+\delta})$ for every $\delta > 0$, where R is Riesz's function, is equivalent to the absence of zeros with $\operatorname{Re} s > \alpha$, to the GGC property of $T_{\alpha'}$ for every $\alpha' > \alpha$, to positivity of the Thorin densities on every line to the right of α, and to the convergence of the compensated sine squared sums there; in particular RH holds if and only if T_α is a GGC for every $\alpha > \frac{1}{2}$. The Riesz coefficients are, up to explicit factors, values of the Laplace transform of T_α at every basepoint $\alpha \in [\frac{1}{2}, 1]$, and at the basepoint one, with no hypothesis, GGC values and killing probabilities of the hitting time T_1 (Proposition 10.2.5 and (10.2.4)); what the criterion asks is whether the clocks themselves remain GGCs below the basepoint one.

What does hold below the basepoint one, with no hypothesis, is an approximate form of positivity. Split the density $\rho_{\alpha-1/2}$ into the Cauchy kernels of the zeros: the kernels of the zeros to the right of α carry a vanishing fraction of its mass on $[0, T]$, with a power saving that comes from the zero density estimates of Ingham, of Guth and Maynard and of Huxley [104, 88, 100]. Every shifted clock T_α with $\alpha > \frac{1}{2}$ is thus an almost GGC (Theorem 11.4.2), and $\rho_{\alpha-1/2}$ is its Thorin density exactly when every zero lies to the left of the line $\operatorname{Re} s = \alpha$. Its Laplace transform factorises exactly into the Laplace transform of a GGC and a defect that carries the zeros to the right of α and is identically one exactly when there are none, and the verification of RH to height 3×10^{12} puts the defect within a factor $e^{\pm 1.5 \times 10^{-12}(u+\sqrt{u})}$ at every basepoint (Proposition 11.4.3 and Theorem 11.4.4). Density arguments cannot upgrade almost to exactly, since even the density hy-

pothesis leaves room for a sparse set of zeros off the line (Section 11.4.7). Every construction of Chapters 9 to 11 therefore reaches the centre only as $\alpha \downarrow \frac{1}{2}$, and only through the same zero free half planes.

12.1.5 The open steps

Step 7 asks for a single basepoint $\alpha < 1$ at which the shifted clock is a GGC, uniformly in the height. By Theorem 1.2.2, that is [200, Theorem 3.7 and Remark 3.10], this is exactly a zero free half plane $\operatorname{Re} s > \alpha$, a quasi Riemann hypothesis. Until 2026 no such half plane was known: the classical zero free regions narrow to the line $\operatorname{Re} s = 1$ as the height grows. In October 2026 OpenAI released a proof, produced with one of its AI models, that $\zeta(s) \neq 0$ for $\operatorname{Re} s > \frac{7}{8}$, and the same for every Dirichlet L function [178], together with a formal proof of both statements in the Lean proof assistant and a separate proof of the weaker half plane $\operatorname{Re} s > \frac{11}{12}$ [179]. The argument proves a power saving for smoothed sums of the Möbius function twisted by Hecke characters of $\mathbb{Q}(\sqrt{-3})$, by comparing them with averages of cubic theta coefficients that large sieve inequalities control, and it uses nothing from the Thorin picture. It has not yet been refereed. We rebuilt the formal proof from the public repository (commit `adc7f12`, Lean 4.34.1, Mathlib `d13f23b`): the Lean kernel accepts both proofs, which use only the axioms `propext`, `Classical.choice` and `Quot.sound`, and the comparator of the Lean Focused Research Organization, run without its sandbox, confirms that the theorems proved are the published challenge statements. If it stands, step 7 holds with $\alpha = \frac{7}{8}$, and the constructions of this book follow. By Theorem 1.2.2 the shifted clock T_α is a GGC for every $\alpha \geq \frac{7}{8}$, with Thorin density (1.2.6) and $v_\alpha > 0$ for $\alpha > \frac{7}{8}$, while at $\alpha = \frac{7}{8}$ each zero $\frac{7}{8} + i\gamma$ with $\gamma > 0$, which the announcement does not exclude, would add an atom at γ^2 (Theorem 7.1.14); for $\alpha > \frac{7}{8}$ the clock T_α is a Brownian first passage time plus an independent GGC with no drift, because the first two steps of the proof of Proposition 9.1.5 go through for $\frac{7}{8} < \alpha < 1$, where every zero has $0 < \alpha - \beta < \alpha$, with smallest margins 0.0137 and 0.0157 in ball arithmetic (`valpha_min.py`); by Proposition 9.1.6 the compensated Lévy form converges for $\frac{7}{8} < \alpha < 1$, with error $O(X^{7/8-\alpha+\delta})$ at prime cutoff X for every $\delta > 0$, and the same holds for $L(s, \chi_{12})$ (Section 11.3.2); by Theorem 10.2.7 Riesz's function satisfies $R(x) = O(x^{7/16+\delta})$ for every $\delta > 0$; and by de Bruijn's theorem $\Lambda \leq \frac{9}{32}$ (Section 8.4), weaker than the bound 0.2, while for the cosine tilts of Section 13.4.2 his theorem for cosine factors gives $\Omega \leq \frac{3}{8}$, against $\frac{1}{2}$ with no hypothesis. The Thorin picture reads off these consequences, but it did not produce the half plane, which came from outside it, as Section 11.4 anticipates. Below $\frac{7}{8}$ nothing changes. Positivity of the Thorin density on the single line $\operatorname{Re} s = \alpha$ is a weaker statement than a zero free half plane, and it is not known to exclude zeros to the right of that line (Remark 1.2.4). So the first theorem the Thorin path needs beyond the announcement, a zero free half plane $\operatorname{Re} s > \alpha$ with $\alpha < \frac{7}{8}$, is an open problem in its own right, and any progress on

it would be a major result independent of RH; the Lévy form, the scale mixture, completeness and Riesz's criterion (Sections 9.1, 10.1 and 10.2) restate it and do not get round it. The Thorin formulation offers two specific tools for it. The sine squared form, compensated as in Proposition 9.1.6, writes the required positivity as domination of the pole piece by the prime and archimedean pieces at basepoint α, a statement about a single explicit function of θ for each α; and the Poisson form shows that a zero to the right of α appears as a Cauchy kernel of negative width, whose signature Section 3.3 calibrates exactly.

Step 8 carries the GGC property from any fixed α to every $\alpha > \frac{1}{2}$. Weak limits cannot do this on their own (Section 4.1), because the limit of the Cauchy flow is positive whether or not RH holds. Three mechanisms could do it, each consistent with everything computed in this book:

1. a positive approximating family whose positivity is proved independently, such as zeta functions of Beurling generalised prime systems that provably have only real ordinates, finite Krein strings, or the partial products over verified zeros completed by the Weyl tail, the last two being the Kent approximants of Section 8.1.3 without and with the Weyl tail, converging to the clock under the closure criterion of Corollary 4.1.2 [200, Propositions 18.10 and 18.14]; once the first N zeros are verified, RH is equivalent to the GGC property of the tail factor left when their factors are removed from the clock [200, Corollary 18.11];

2. an operator whose spectrum is the set of ordinates, where positivity would follow from self adjointness, the Hilbert Pólya idea in Thorin form: if the zeros are also simple, Kent's theorem then makes the clock the hitting time of a level by a diffusion whose Dirichlet spectrum is $\{\gamma^2\}$ (Section 8.1.3);

3. a direct inequality for the variogram pairing that controls the pole by the primes and the gamma factor at every scale, the Weil form of step 8.

12.1.6 What computation contributes

Computation cannot supply the uniformity in the height that steps 7 and 8 require: neither finitely many heat trace tests (Theorem 5.2.1) nor positivity at a fixed resolution (Section 5.2.2) can exclude an off line zero above the verified height. It can do three things that a proof on this path needs, which Section 12.2.1 develops into a programme. It can certify positivity on ever larger windows at every resolution, with the Pólya route at moderate height and a Riemann Siegel evaluator at large height. It tests candidate lemmas for steps 7 and 8 quickly, by computing the relevant margins, as Section 7.2 does for the truncations of the Biane, Pitman and Yor lattice. And it calibrates the size of the positivity margins, which tells a proof how much room it has: the Thorin density at resolution ε has minimum $0.0147\,\varepsilon$ on the window below height 200 (Section 3.1), a margin that

vanishes linearly and must be controlled uniformly. That this margin is the value at the origin is a theorem on the whole verified window, and at every height for every $\varepsilon < \frac{1}{2}$ it is equivalent to RH (Theorem 9.1.11).

The Thorin approach converts RH into the positivity of a single measure that depends on a resolution parameter. At coarse resolution the measure is positive by a theorem, at every resolution it is computable from the Pólya density, and at vanishing resolution its positivity is RH. The sine squared lemma is the bridge from the primes to this measure: it identifies the measure, makes its mass an argument, isolates the pole of ζ as the only negative input, and explains why the primes alone cannot compute it near the line. The path along this bridge is proved from $\alpha = \infty$ to $\alpha = 1$, where Chapters 9 to 11 describe the clock explicitly and with no hypothesis, continues to $\alpha = \frac{7}{8}$ if the half plane announced in 2026 stands (Section 12.1.5), and is computed on windows down to $\alpha = \frac{1}{2} + 0.005$; the remaining step is the GGC property uniformly in the height across the rest of the strip, which begins with a zero free half plane $\operatorname{Re} s > \alpha$ with $\alpha < \frac{7}{8}$ and ends with RH.

Readers who want to pursue the path can start from four places in the book. Section 3.5 gives the variogram form of the domination inequality, on which the linear programming formulation of [200, Sections 18.2.8 and 18.2.9] works, the most concrete target for step 8. Both are posed at the centre, where the variogram form is Weil's criterion and the linear program asks whether the clock $1/G$ is a GGC, so each is equivalent to RH. A direct proof of the variogram form is the third mechanism of Section 12.1.5. Section 4.1 and [200, Section 18.2.3] give the closure criterion in its weakest form, the target for approximating families in step 8. Section 11.4 closes with four questions between almost and exact positivity (Section 11.4.7), the last of which is step 7 below $\frac{7}{8}$, the GGC property at a single basepoint $\alpha < \frac{7}{8}$. Section 12.2 sets out the computations that would calibrate each attempt, from certified windows at large height to the de Bruijn Newman laboratory, and draws the lessons of the book.

12.2 Open problems and lessons for the Thorin programme

Section 12.1 located the gap on the path to RH: positivity uniform in the height, first as the GGC property at a single basepoint below one (step 7 of Table 12.1) and then across the strip. The programme of computations below is aimed at that gap, and the lessons that follow it are drawn for anyone working on the Thorin route to RH, including the first author's own programme [200]. Three of the five work packages of the programme, the twisted Riesz filters, the positivity margins against random matrix theory and the almost GGC defect, are taken up in Sections 10.2 and 11.4; the open problems on the reciprocal zeta function are stated in Section 7.2.5, those on the basepoint one and the critical strip in Section 11.4.7, and those of Pakes's framework in Section 11.5.6.1, and we do not

repeat them.

12.2.1 What computation can and cannot do on the path

On the path of Section 12.1.4, for $\alpha \geq 1$ the Thorin density is positive, the Lévy form of $\log \xi(\alpha + s)$ converges, the normal scale mixtures are extended GGCs and the tilted Pólya kernels are complete in the sense of Wiener, all with no hypothesis (Sections 9.1 to 11.3), and for $\frac{1}{2} < \alpha < 1$ each of these, read as the GGC property of the shifted clock T_α and as convergence, the extended GGC property and completeness at every basepoint to the right of α, is equivalent to a zero free half plane $\operatorname{Re} s > \alpha$.

Computation cannot supply uniformity in height (Section 12.1.6): every finite computation leaves room for zeros above the computed range, and a single off line zero, however high, stops T_α from being a GGC at every basepoint α between $\frac{1}{2}$ and its real part, whether or not it makes the Thorin density negative near its ordinate (Section 5.2.2). What computation can do falls under four aims. It can extend and certify: prove positivity of the Thorin density on explicit windows, at every resolution, with interval arithmetic. It can test intermediate conjectures: the four questions of Section 11.4.7 (a lower bound for the background density of order $\log \gamma$ outside a sparse set, thin strip density bounds, an explicit bound for the defect of the canonical factorisation and positivity at one basepoint) all have numerically testable consequences. It can calibrate: measure how much room positivity has, and compare with random matrix predictions, so that a proof knows which inequalities are sharp. And it can cross check: compute the same quantities from the primes, through the criteria of Riesz and Salem and the compensated Lévy form, independently of evaluations of ζ.

Five work packages serve these aims. The first, twisted Riesz functions as an arithmetic filter bank, is the cross check of Section 10.2.1; the second, certified Thorin densities at height, is the subject of Section 12.2.2; the third and fourth, the positivity margins against the circular unitary ensemble and explicit tables of the almost GGC defect, are described in Sections 11.4.5 and 11.4.2; and the fifth, testing stronger conjectures before pursuing them, is the subject of Section 12.2.4. Table 12.2 lists their milestones and the resources each needs. The computations that remain from the chapters of the book fall under the same aims, and Sections 12.2.2 to 12.2.4 take them together with the packages.

12.2.2 Certified Thorin densities at height

At resolution $\varepsilon = \alpha - \frac{1}{2}$ the Thorin density $\rho_\varepsilon(\theta) = \frac{1}{\pi} \operatorname{Re} \frac{\xi'}{\xi}(\alpha + i\theta)$ is a smooth function of θ. Its positivity on a window of ordinates is what the absence of zeros with $\beta > \alpha$ predicts there; since an off line zero need not make the density negative (Section 5.2.2), a certificate for the window should also count such zeros. Two computations do this. The first evaluates ρ_ε on a grid with rigorous error

package	milestone	resources
1	$R(x)$ and $R_h(x)$ for $x \le 10^{16}$, h on a grid to 500, compared with the explicit formula	compiled sieve, days of CPU
2	certified ρ_ε at $\varepsilon = 10^{-1}, 10^{-2}, 10^{-3}$ on windows at heights 10^6 to 10^{10}	Arb, weeks of CPU
3	margin distributions from Odlyzko's tables near the 10^{12}th [171], 10^{21}st and 10^{22}nd zeros [175, 176]; CUE extreme value fit	hours
4	explicit almost GGC defect tables	hours
5	numerical tests of the four questions of Section 11.4.7 and of HCM below one	ongoing

Table 12.2: Work packages and milestones of the computational programme. Package 1 is described in Section 10.2.1, where R is Riesz's function and R_h its twist; packages 3 and 4 are described in Sections 11.4.5 and 11.4.2.

bounds, using the ball arithmetic Riemann Siegel and Euler Maclaurin evaluators of Arb [110], exposed in python flint, at moderate height and the Pólya route where it is faster, and bounds the variation between grid points by the derivative. The second counts the zeros to the right of the line $\operatorname{Re} s = \alpha$ through the zero density identity $M_\varepsilon(T) + \tau_\varepsilon(T) = N(T) - 2\nu(\frac{1}{2} + \varepsilon, T)$ of Proposition 3.4.2, which returns an integer from a smooth computation at any resolution. It is the count of Section 3.4, which at $\varepsilon = \frac{1}{2}$ is Backlund's method. Section 3.4.3 followed the argument in steps of $\varepsilon/8$ ($\varepsilon/4$ at $\varepsilon = 0.005$) without a bound on its change between grid points, so its counts are computed, not certified; with such a bound the count would certify a whole window, at an estimated cost of a few hundred evaluations per unit height at resolution $\varepsilon \approx 0.1$. The deliverables are certified positivity of ρ_ε for $\varepsilon \in \{10^{-1}, 10^{-2}, 10^{-3}\}$ on windows of length 10^4 at heights 10^6, 10^8 and 10^{10}, together with the minimum density on each window, which feeds the margin statistics of Section 11.4.5. Below the verified height 3×10^{12} the positivity itself needs no new computation: an off line pair makes the density negative only within $\frac{1}{2}$ of its ordinate, so $\rho_\varepsilon > 0$ on $[0, 3 \times 10^{12} - \frac{1}{2}]$ for every $\varepsilon > 0$ (Corollary 11.4.5); there the certificates test the evaluators against the verification and measure the margins.

Below height 200 the certificate is within reach now. The ball arithmetic of Section 2.1 certifies the rounding only. Adding the discretisation bound of Section 2.1.1 and of Proposition 2.1.1, bounds for the truncations of the theta series and of the u axis, a bound on the change of the argument between grid points, and an explicit bound on the remainder of Proposition 3.2.2, which is $O(\varepsilon^3)$ (Remark 3.2.3), turns positivity of ρ_ε on $0 < z \le 200^2$ at each computed ε, the integer identity of Proposition 3.4.2 and the HCM derivative signs of Section 4.1.2 into theorems on that window.

Above that window the evaluator changes. The tilted contour gives 25 digits at height 5000 and resolves the Lehmer pair near height 7005 (Sections 2.1.3 and 2.2.3.1), but its number of nodes grows linearly with the height. Chirp transform batching of the Pólya route, and the hybrid of Section 2.2.2, a Riemann Siegel or Odlyzko Schönhage evaluator at $\frac{1}{2} + \varepsilon + i\theta$ with Odlyzko's zeros as the RH reference, would give the positivity margin, the matched total variation and the zero density scan near the 10^{12}th [171], 10^{21}st and 10^{22}nd zeros [175, 176]. The same zeros would measure the cell constant of Section 5.4 at large height, whose behaviour is open (Section 5.4.2), and allow a comparison of the cell errors with the prediction of the sine kernel process of the unitary ensemble.

12.2.3 Calibration against known configurations

The diagnostics of Chapters 3 to 5 are calibrated when they are run on configurations whose zeros are known exactly, and three such computations remain. Section 8.6 ran the posterior predictive check of [200, Section 19.3.7], with a positivity prior on the measure U in $\rho_\varepsilon = P_\varepsilon * U$, at $\varepsilon = 0.1$ on one window, and calibrated it with a displaced zero; it is the linear inverse problem of [200, Section 19.3], whose data the Pólya route supplies. Three extensions are open: smaller resolutions, where the signature of a displaced zero is largest and the Pólya route supplies ρ_ε to about 60 digits below height 200 (Section 2.1.1), whereas the deconvolution used double precision; continuous atom positions, which would remove the baseline of the grid; and larger heights. The square root law (3.3.1) of Section 3.3.2, which follows from Proposition 3.3.1, sets what the check can detect. The Williams benchmark of Section 8.2.3 needs no separate computation, since its distances follow from the closed forms of Section 1.3.4.

The de Bruijn Newman flow of Section 8.4 manufactures off line pairs in a controlled way. Running these Thorin diagnostics along the flow through the collision of the Lehmer pair near 7005 would calibrate the zero density scan, the matched total variation and the negative mass against configurations that are known exactly, extending the collision time computed in Section 8.4 to the full set of diagnostics.

Generalised prime systems of Beurling whose zeta functions provably have only real ordinates give positive Thorin measures by construction. Measuring how fast their clock transforms approach $1/G$, in the distances of Section 3.2 and along a sequence with an accumulation point as the closure criterion of Corollary 4.1.2 [200, Propositions 18.10 and 18.14] allows, is the computational side of that criterion; the Kent approximants of Section 8.1.3 are the first instance.

Two computations of Chapter 8 remain at their edges. The Brownian boundary of Section 8.2 reproduces the law of the clock in the bulk, to 0.0020 in Kolmogorov distance for the boundary of the Volterra equation and to 0.0059 for the tangent approximation of Roberts and Shortland (Section 8.2.2.2), but the zone of large times t is checked by no simulation, since only about one path in ten million

survives past $t = 0.1$. Importance sampling with an exponentially tilted drift, or multilevel splitting, would check that zone and the logarithmic correction of the asymptote of the boundary directly. The Cauchy integral method of Section 8.3 extends to n in the thousands, at a cost of about $0.03n$ digits lost to cancellation, so that the precision grows linearly in n. Comparing the Pólya values of the Keiper Li coefficients with the sums $\sum_{\gamma>0} 4\sin^2(n \arctan \frac{1}{2\gamma})$ over the zeros to large n would test how the Li functional sees the zeros near height n.

12.2.4 Testing stronger conjectures before pursuing them

Conrey and Li showed that a positivity condition in de Branges' approach to RH is false even though RH may be true [54]. Any intermediate statement proposed on the Thorin path should therefore be tested numerically first. The candidates are a lower bound $\rho_{\alpha-1/2}(\theta) \geq c(\alpha) \log \theta$ outside a sparse set, the first of the four questions of Section 11.4.7; monotonicity or convexity of the margin in α; positivity of the Thorin density for α slightly below one, uniformly on large windows; hyperbolic complete monotonicity of the Laplace transform φ_α of T_α below one; and variogram inequalities at single scales. The first two are partly settled: the lower bound holds for $\rho_{\alpha-1/2}$ itself outside a set of heights of density zero, although not near the zeros where Lemma 11.4.7 would need it (Remark 9.1.10), and the margin divided by $\alpha - \frac{1}{2}$ decreases in α, with the ratio $\rho_{\alpha-1/2}(\theta)/(\alpha - \frac{1}{2})$ nonincreasing in α at every height outside the windows of the zeros off the line (Theorems 9.1.8 and 9.1.11). Each can be evaluated on the computed windows and at the heights of Odlyzko's tables, and any statement that fails there should be discarded. In parallel, the de Bruijn Newman flow, along which the reciprocal of the heat evolved Ξ is a GGC for $t \geq 0.2$ and RH is the statement at $t = 0$ [217, 196, 224], gives a one parameter test bed in which the same diagnostics can be followed as t decreases.

The variogram inequalities can be tested in finite dimension. The reformulation of Section 3.5 poses finite dimensional feasibility problems: is the regularised pole mass dominated by the prime and archimedean mass on the cone of variograms of positive definite functions of a given dimension? Solving them with explicit dual certificates at increasing dimension would measure how the margin behaves.

The cross checks of the fourth aim test the zeros themselves. Riesz's function, computed from the Möbius function alone, sees only the first few zeros, and its twists by $(x/n^2)^{ih}$ move its sensitivity to the zeros near height $2h$ while keeping the criterion equivalent to RH (Proposition 10.2.1); agreement of these twisted functions with the explicit formula over the computed zeros is a check that shares no code or method with the computation of the zeros (Remark 10.2.2). The open problems on the reciprocal zeta function (Section 7.2.5), among them Conjecture S on the symmetrised truncations of the Biane, Pitman and Yor lattice and the Nyman Beurling problem, and the four questions of Section 11.4.7 are the natural targets for all these tests.

12.2.5 Open problems from free probability

The note [208] closes with eight problems, and five of them are settled or taken up in the book. The positivity of the Thorin measure of $\Upsilon(T)$ and the infinite divisibility of T, classical and free, follow from Theorem 5.1.1 through Theorem 11.2.3(a) and (b). The Gamma(2) problem is settled by Propositions 11.2.6 and 11.2.8 and Theorem 11.2.9. The failure curve of the Hankel test along the flow and the collision brackets are the subjects of Sections 8.4.3 and 8.4.5. What remains open is the following.

Problem 12.1. Jensen polynomials as finite free products. The multiplicative convolution $\boxtimes_d$ multiplies normalised coefficients and keeps roots real and of one sign (Lemma 11.2.7(i)), and the normalised coefficients of the reversed Jensen polynomial $x^d J_{d,n}(1/x)$ are, up to the signs $(-1)^i$, the numbers $(n+i)!\,b_{n+i}$, $0 \le i \le d$. A factorisation $(n+i)!\,b_{n+i} = \alpha_i \beta_i$ into the normalised coefficients of two polynomials with only real roots of one sign would therefore give real zeros of $J_{d,n}$. Find such factorisations for every shift n at a fixed degree d, or show that none exists with factors of a given kind. The obvious split, a Gaussian multiplier sequence q^{i^2}, $0 < q \le 1$, times a residual close to the coefficients of $(1+x)^d$, fails, because a root of multiplicity d splits into nonreal roots under a generic perturbation; and by Proposition 11.2.8 the factor $1/(a)_i$ can be multiplied in but not divided out. A factorisation has to put real structure into both factors [208].

Problem 12.2. Laguerre and Meixner Pollaczek expansions. Theorem 11.2.9 excludes scale mixtures of gamma laws for $|X|$ and X^2, and Proposition 11.2.8 shows that removing a gamma layer keeps zeros real while adding one does not. Find a variable in which the Pólya density, or Ξ itself, has an expansion in Laguerre or Meixner Pollaczek polynomials with positive coefficients and truncations with only real zeros [208]. The symmetric Meixner Pollaczek expansions of Section 6.2.3.2 converge slowly, and a real rooted construction from their coefficients is a problem of Section 7.2.5.

Problem 12.3. Jacobi parameters. Under RH, where ν is a positive measure and all its Jacobi parameters exist, prove that $\alpha_n \sim 2\beta_n \sim \frac{1}{2}L_n$, with L_n given by (11.2.13) for the smooth count: a spectral form of the Riemann von Mangoldt formula. Table 11.9 supports it to a few per cent for $n \le 49$, but the local model of Section 11.2.4 is a heuristic, outside the scaling of [130]. Decide between the rules $n(L_n) = n$ and $n(L_n') = n - \frac{1}{2}$, which the data do not separate, and explain the scatter of the ratios, about six per cent, through the individual gaps between the zeros, as [208] proposes.

Problem 12.4. The order at which a collision is seen. Section 8.4.3 computes the failure times t_N^* of the Hankel test along the flow to $N = 50$ and explains their steps by collisions; the rule of [208] that the collision of (γ_k, γ_{k+1}) is registered from order $2k+2$ on fails at $N = 50$ (Table 8.9). Find the rule that gives the

order at which a pair is registered, as a function of the index k of the pair and of its distance from the axis, and follow the staircase of t_N^* to orders of a few hundred, towards the limit 0 that it has under RH.

Problem 12.5. Unconditional collision brackets. Section 8.4.5 brackets the collision times by quantities along the flow (Corollaries 8.4.12 and 8.4.14) and through the nonreal zeros (Theorems 8.4.11 and 8.4.13), and checks the hypotheses on a grid of times for the 58 collisions of the census. Find the a priori control of the nonreal zeros of H_t, $t < 0$, asked for in Problem 8.1, which would make the brackets statements about H_t computed from H_0 alone.

Problem 12.6. Moving the ladder without new zeros. Theorems 5.1.7 and 5.1.10 prove the derivative signs $(-1)^k W^{(k)} > 0$ on ranges that grow with the height to which RH is verified (Section 5.1.7), and complete monotonicity of W on $(0, \infty)$ is RH. Find a mechanism that extends these ranges with no new information on the zeros, for instance a functional equation for the law of T under Υ, as [208] suggests. Theorem 11.2.3(c) restricts the form such an equation can take: T is not the Upsilon image of a law in BO, and under RH it is the image of the law μ_0, which is neither a GGC nor in BO.

12.2.6 Lessons from the computations

The computations of the book distil into lessons for anyone working on the Thorin route to RH. Each is a statement that a computation in the book established or refuted, and each saves effort on a route that cannot succeed or points to one that might.

Equivalent statements are not equally computable. The Thorin condition has at least eight equivalent faces: positivity of the Thorin measure, complete monotonicity of the heat trace, Stieltjes form of G'/G, Hankel positivity at one time, HCM of the clock, Lagarias positivity, Weil positivity and Li positivity. Mathematically they are the same statement. Computationally they differ by many orders of magnitude in conditioning. The heat trace at moderate time loses about $87t$ digits to cancellation (Section 5.3); the normalised minors of the even Hankel block at $t = 0.05$ fall to 10^{-44} by size five (Section 5.1); the Keiper Li coefficients weight a zero at height γ by n^2/γ^2 until n reaches γ (Section 8.3); the Thorin density at resolution ε is local in the ordinate and sees each zero at full size $1/\varepsilon$ (Section 3.2). A computational attack should choose its face by conditioning, and the local face is the Thorin density.

The primes compute the wrong end. The sine squared lemma writes the Thorin density as a sum of positive prime and archimedean pieces and one negative pole piece. It is the right way to understand the structure: the pole is the only term of the wrong sign, and RH is the statement that it is dominated after the kernel acts; Section 8.5.3 shows what changes for an L function without a pole. It is the wrong way to compute near the line, where k digits at resolution ε need primes

up to about $10^{2k/\varepsilon}$ even when no zero lies to the right of the line (Section 3.5.1). Any argument that proposes to establish positivity near the line by bounding prime sums must overcome that exponential, and Section 7.1 shows the same obstruction in another form: the bias that keeps the canonical function positive is of order $x^{-1/2}$, while pointwise bounds on the primes control it only to order x (Proposition 7.1.2).

The heat trace meets the same obstruction in its own terms. By Proposition 5.1.9, RH is the positivity of the Gaussian smoothing $\mathcal{G}(u,t)$ of the zeros at every height u and every $t>0$, and (5.1.3) writes the symmetric smoothing $\mathcal{G}(u,t)+\mathcal{G}(-u,t)$ as a polar term, negligible at large u, plus an archimedean term $A(u,t)$, close to $\log(u/2\pi)/(2\sqrt{2\pi t})$, minus a prime sum $P(u,t)$. Bounding $|\cos(u\log n)|$ by one bounds $P(u,t)$ by $S(t)/\sqrt{2\pi t}$, where $S(t)=\sum_{n\geq 2}\Lambda(n)n^{-1/2}e^{-(\log n)^2/8t}$ is, up to a factor, the prime term of (1.2.5) at time $2t$, and so proves $\mathcal{G}(u,t)+\mathcal{G}(-u,t)>0$ for t below the time $t_0(u)$ at which the bound meets the archimedean and polar terms, which agrees to five digits with the root of $2S(t)=\log(u/2\pi)$ (computed by `phasefree.py`): t_0 is 0.45 at height 10^3, 1.72 at the verified height 3×10^{12} and 3.64 at 10^{50}, to the digits shown (proved in ball arithmetic by `phasefree_arb.py`), and it grows only doubly logarithmically in u, since $S(t)$ is asymptotic to $\sqrt{8\pi t}\,e^{t/2}$. For $\mathcal{G}(u,t)$ itself the bound must also exceed $|\mathcal{G}(-u,t)|$, at most $e^{t/2}((u+14)^2+\frac{1}{2t})e^{-2t(u+14)^2}$ by Section 5.1.5, which is below 10^{-880} for $u\geq 10^3$ and $t\geq 10^{-3}$; on that range the time at which positivity of $\mathcal{G}(u,t)$ is proved differs from $t_0(u)$ only far beyond the digits shown. These resolutions are too coarse to see the quadruples of Table 5.5: at their heights t_0 lies between 0.311 and 0.511, while within 8 of each moved pair they make $\mathcal{G}$ negative only for $t>1.18$ (both proved in ball arithmetic). Made uniform in the order, the same bound proves the signs of all derivatives of order above 10^{17} for $t\leq 1.29$, which with the verified zeros makes W completely monotone there (Theorem 5.1.7). That passes 1.18 only because, along the rays, those orders lie at heights above 2.78×10^8, where t_0 exceeds 1.29; at the heights of the quadruples the theorem uses the verified zeros instead. Nor can any bound on P that holds uniformly in the height do better, since by Dirichlet's theorem on simultaneous approximation the phases $u\log n$ of any finite set of prime powers come arbitrarily close to multiples of 2π at arbitrarily large u, where $P(u,t)$ therefore comes arbitrarily close to $S(t)/\sqrt{2\pi t}$. Nor does positivity at a single resolution give a zero free strip: at fixed t the bound $S(t)/\sqrt{2\pi t}$ on P proves $\mathcal{G}(u,t)+\mathcal{G}(-u,t)>0$ at every height u with $t_0(u)>t$, that is above about $2\pi e^{2S(t)}$, which is 8.1×10^{212} at $t=6$, and $\mathcal{G}(u,6)>0$ itself for every $u\geq 10^{213}$ (proved in ball arithmetic, with the bound on $|\mathcal{G}(-u,6)|$ above), whatever the zeros are, so a strip needs t to grow with the height. A proof along the heat trace must use the phases, uniformly in the height.

Limits do not create positivity. Weak limits of Thorin measures preserve positivity but cannot manufacture it. The Cauchy flow converges vaguely to the measure of the line zeros whether or not RH holds, because an off line pair at

distance d from the line contributes to μ_ε a signed measure whose negative part, for $\varepsilon < d$, has mass only $\frac{2}{\pi}\arcsin\frac{\varepsilon}{d} \approx 2\varepsilon/(\pi d)$, that is $\varepsilon/(\pi d)$ per displaced zero (Proposition 3.3.1(iii) and Section 4.1). Positivity must therefore be proved at every resolution, not in the limit. The closure criterion helps only for approximants whose positivity is proved independently, such as the Beurling systems and the Kent approximants of the first mechanism in Section 12.1.5. Verification removes the factors of the verified zeros from the question and leaves it for the tail factor of the clock [200, Corollary 18.11]. A proposal that reaches the Thorin condition as a limit must say where the positivity of its approximants comes from.

12.2.7 Closed routes and the reach of finite information

Multipliers and reformulations do not help by themselves, and several routes are now closed or reduced to RH itself. Multiplying the clock by any GGC leaves the problem exactly as hard (Proposition 7.2.16). HCM of the reciprocal Xi function is equivalent to the GGC property by Bondesson's theorem, and so to RH (Corollary 4.1.3); it is not a way round it. Weak closure of the Cauchy flow cannot extend the family (Section 4.1.1). The clock is not a Brownian first passage time to a random level for any constant drift, so the simplest hitting time representations are excluded unconditionally (Proposition 8.2.1), although at every basepoint above the centre the shifted clock is a subordinated hitting time (Section 9.2). The Bernstein Pick route through the moments of the Pólya density is closed by a single interval certificate (Section 5.2). And the prime side of the sine squared lemma cannot compute near the line (Section 3.5). Each of these negative results was obtained cheaply, and each removes a route that had looked promising. Sections 10.2 and 11.1 close two more: the exact representations of the Riesz coefficients through the clocks cannot supply Riesz's growth bound (Section 10.2.6), and the theorems of Sato and Bondesson reach nothing below the basepoint one (Section 11.1.5). Section 11.5 closes three more: the clock is not the martingale limit of a supercritical continuous state branching process (Theorem 11.5.13), its index among Urbanik's classes is decided with no hypothesis (Theorem 11.5.8), and no property that the damped kernels $e^{tv^2/4}\Phi(v)$, $t < 0$, share with Φ can imply RH (Proposition 11.5.14).

Finite information has a precise reach. Every finite computation in the book establishes a statement on a window and nothing beyond it. The perturbation theorem makes this exact for the heat trace tests: for every N and every sufficiently large height R for which RH has been verified up to $R + 50$, there are functions with off line zeros just above R that keep the first N Hankel blocks positive and the derivative signs through order 10^{17} (Theorem 5.2.1 and [200, Theorem 11.6 and Proposition 11.8]), and every such result in the book is of that kind. The local visibility condition makes it exact for positivity at a fixed small resolution: to first order in ε, an off line zero at distance d from the line makes ρ_ε negative at its ordinate if and only if $2/d^2 > \sum_j \delta_j^{-2}$, where δ_j are the distances to the

other ordinates, and the condition does not improve as $\varepsilon \downarrow 0$ (Section 5.2.2). For neighbours at the mean spacing near height T it fails once d exceeds about 0.78 of the spacing, which a zero can do only above height about 1.1×10^5 (Section 5.4); such a zero shows itself only at resolutions comparable to d. The zero density identity of Proposition 3.4.2 is the most complete finite test, because it scans every resolution at once and returns integers, but it too is a statement at a finite height. What finite computation buys is a calibration of how much of RH a given amount of work establishes, and that calibration is itself useful: it tells a proof what it must supply.

The perturbation theorem also gives a test that a proposed proof of RH must pass. Replace ξ by the function F of Theorem 5.2.1, with R above every height that the proof uses explicitly. A step that holds for F as well as for ξ cannot be the step that excludes zeros off the line, because F has four. The steps that survive include the Riemann von Mangoldt formula with an explicit error, which F satisfies with the error enlarged by one on $[A, B)$, the only heights where the two counts differ; the verified zeros below R; the small time expansion of the heat trace through the constant term, since $W_F - W = \big(\frac{1}{2}(B-A)^2 + \frac{1}{8}\big)t + O(t^2)$; infinite divisibility, self decomposability and log convexity; the Hankel blocks of size N when $R \geq T_N$ and the derivative signs up to order 10^{17} when $R \geq 10^{12}$; and the order, type and functional equation. What separates ξ from F is arithmetic, the Euler product and the explicit formula with the prime weights $\Lambda(n)$, together with local counts of the zeros on the line in windows that contain the height g of the quadruple (Section 5.2.1). A proof through the derivative signs can do without them only below the order at which the signs of F first fail, which is about $t^* R^2$, with t^* between 1.18 and 4.03 in the cases computed below height 2000 and, by the model of Section 5.2.1, of order one or more at larger heights, far beyond 10^{17}. The function F exists unconditionally when $R + 50 \leq 3 \times 10^{12}$; for larger R the test still applies, because a correct proof of RH makes the hypothesis of Theorem 5.2.1(b) true for every R. The criticality filter of Section 11.5.5 is the companion test for properties of the Pólya kernel: a step that holds for $e^{tv^2/4}\Phi(v)$ with some $t < 0$, whose transform has nonreal zeros by the theorem of Rodgers and Tao, cannot be the step that excludes zeros off the line.

12.2.8 What a proof must supply

What remains is a single positivity statement, in equivalent forms: the Thorin density ρ_ε is nonnegative at every basepoint above the centre; the Gaussian smoothing $\mathcal{G}(u, t)$ of the zeros is nonnegative at every height and every resolution (Proposition 5.1.9); the shifted clock is a GGC at every basepoint above the centre, which by Theorem 10.2.7 is also Riesz's criterion; the pole mass is dominated by the prime and archimedean mass on every variogram of a positive definite function; the zero density $\nu(\frac{1}{2} + \varepsilon, T)$ vanishes for every ε and T. A proof of the Thorin condition must supply this positivity uniformly in the height and in the resolution,

and it needs a structural reason for it. The candidates consistent with everything computed here are the three mechanisms for step 8 in Section 12.1.5: a positive approximating family with independently proved positivity converging under the closure criterion, an operator whose spectrum is the set of ordinates, and a direct inequality for the variogram pairing. The computations of this book cannot supply any of these. What they can do, and have done, is test candidate lemmas quickly, produce counterexamples, exclude routes that fail, and measure the margins that a successful argument will have to respect.

The programme of this section has one organising principle: compute the quantities whose positivity or growth is equivalent to RH, at the places where a proof would have to control them, and compare what is found with what random matrix theory and the explicit formula predict. Riesz's criterion in its classical form is blind above the first few zeros; twisted Riesz functions repair this and give an arithmetic filter bank independent of evaluations of ζ (Section 10.2.1). Thorin densities at small resolution can be certified on windows by ball arithmetic and the zero density identity (Section 12.2.2). Below height 2198 their positivity margins agree with the circular unitary ensemble in the median to 1% and lie slightly above it in the lower tail, which makes the room RH has at each height predictable, and the explicit defect of the almost GGC factorisation measures how far each basepoint is from its target (Sections 11.4.5 and 11.4.2). None of this proves RH; all of it sharpens the target and tests the statements a proof would use. Appendix A lists the scripts behind the computations of the book.

Chapter 13

Ferromagnetic random variables

A symmetric random variable is ferromagnetic if its moment generating function vanishes only on the imaginary axis. The name comes from the Ising model: by the theorems of Lee and Yang and of Newman recalled in Section 7.1.3, for a law with Gaussian moments of all orders this single condition is the Lee and Yang property of the law used as a spin. By (1.2.4) the moment generating function of the Pólya variable X is $\xi(\frac{1}{2}+s)/\xi(\frac{1}{2})$, so RH is the statement that one explicit law, the Pólya law with density $p = \Phi/\Xi(0)$, is ferromagnetic. This chapter treats that statement as a question about laws. Which laws are ferromagnetic, and how can one tell? Which of the necessary conditions does the Pólya law satisfy? How close to it can provably ferromagnetic laws come? And what structure does the positive variable e^{2X}, which lives on the Mellin side of Section 9.1.1, carry without any hypothesis?

Section 13.1 sets up the class: its Hadamard form, the alternating cumulants, the clock of a ferromagnetic law, which is a generalised gamma convolution with one Thorin atom for each pair of zeros and recovers the van Dantzig pair of Section 1.2 for the Pólya law, and closure under weak limits. Section 13.2 decides the question for laws on an interval written as mixtures of uniform laws, where the shape of the density decides it: a concave shoulder makes a law ferromagnetic, and a convex one prevents it unless the density is linear on its support. Section 13.3 checks the necessary conditions on the Pólya law. Section 13.4 builds ferromagnetic approximations to the Pólya law, from Pólya's own approximation to laws that keep the first K zeros exactly. With no hypothesis, the Pólya law lies within total variation distance 1.3×10^{-4} of a law that is ferromagnetic by an explicit construction, and within 2×10^{-6} of one whose positivity, and hence ferromagnetism, rests on a computer assisted proof in ball arithmetic. The section also proves that RH is equivalent to an identity in law between the Pólya variable smoothed by a Gaussian and a random series of Maxwell variables, one for each zero. Section 13.5 turns to e^{2X}, a size biased form of the variable $\frac{\pi}{2}S_2$ of Biane, Pitman and Yor. It is not a generalised gamma convolution, but it factors against $\frac{\pi}{2}S_{9/4}$, it is self decomposable with a Lévy density built from the theta

function, and its powers $e^{\beta X}$ have hyperbolically completely monotone densities for $|\beta| \geq 5.5791422$, by a computer assisted proof, and do not for $|\beta| \leq 5.5791395$. Section 13.6 lists open problems.

The chapter grew out of the first author's note on ferromagnetic random variables, and its results are marked as in the rest of the book. A result with a proof in the text is proved, with no hypothesis unless one is stated. A number in a table is computed, by the script named in the caption, in the folder `ferro` (Section A.4). We use the units of (1.2.3) throughout, in which the zeros of the characteristic function of X are the ordinates γ and $\operatorname{Var} X = 2\mathbb{E}T = 0.0462099862$. The normalisation of Rodgers and Tao in Section 8.4 doubles the variable of the kernel, and statements quoted from the literature have been converted. As in Section 1.2, Z is a standard normal variable, the e_k are standard exponential variables, the L_k are standard Laplace variables, ε and the ε_k are fair random signs, U is uniform on $[-1, 1]$, and all of them are independent of each other and of everything else.

13.1 Ferromagnetic laws and their clocks

This section sets up the class. A ferromagnetic law has a moment generating function of a rigid shape, a Gaussian factor times a product over its zeros, and everything else follows from that shape: the signs of the cumulants, a clock that is a generalised gamma convolution with one Thorin atom for each pair of zeros, a dual law of Pólya frequency type, and closure of the class under convergence in law. For the Pólya law the clock is the clock T of Section 1.2, and the dual pair is the van Dantzig pair found there.

13.1.1 Definition and background

Definition 13.1.1. A real random variable X is *ferromagnetic* if it is symmetric, $\mathbb{E}e^{b_0 X^2} < \infty$ for some $b_0 > 0$, and its moment generating function $L_X(z) = \mathbb{E}e^{zX}$, which is then entire, vanishes only on the imaginary axis. Equivalently, every zero of the characteristic function $\varphi_X(\theta) = \mathbb{E}e^{i\theta X} = L_X(i\theta)$ is real.

These are Newman's laws of type $\mathcal{L}$ [162]. For a symmetric law the condition $\mathbb{E}e^{b_0 X^2} < \infty$ for some $b_0 > 0$ is equivalent to his bound $|\mathbb{E}e^{zX}| \leq Ce^{C'|z|^2}$: in one direction $e^{sx} \leq e^{b_0 x^2 + s^2/4b_0}$, and in the other $\mathbb{E}e^{bX^2} = \mathbb{E}\,\mathbb{E}[e^{\sqrt{2b}ZX} \mid Z] \leq C\,\mathbb{E}e^{2bC'Z^2} < \infty$ for $4bC' < 1$. Bounded laws, Gaussian laws and the Pólya law, whose density decays like $\exp(-\pi e^{2|t|})$, all satisfy it.

The name comes from statistical mechanics. Lee and Yang proved that the partition function of an Ising ferromagnet, with spins ± 1 and a common complex field, vanishes only when the field is purely imaginary [141], and Newman showed that a symmetric single spin law with Gaussian moments of all orders has the same property in every ferromagnetic system exactly when its own moment generating function has only imaginary zeros [161]; Lieb and Sokal gave another proof and

extended the theorem to spins with several components [146]. Section 7.1.3 recalls these results, and Proposition 7.1.16 restates Newman's theorem in the language of Thorin measures. For laws with Gaussian moments of all orders, then, the one variable condition of Definition 13.1.1 carries the whole Lee and Yang property. Definition 13.1.1 admits the Gaussian laws as well, and this is what makes the class closed under weak limits (Theorem 13.1.5). Two smaller classes of single spin laws carry the correlation inequalities of the Ising model. The Griffiths Simon class consists of the weak limits of suitably scaled sums of ferromagnetically coupled Ising spins; it contains the laws with density proportional to $e^{-ax^4-bx^2}$, $a > 0$, and its laws inherit the Lee and Yang property and the inequalities of Griffiths, Kelly and Sherman and of Griffiths, Hurst and Sherman [240, 85]. The class of Ellis, Monroe and Newman consists of the laws with density $e^{-V(x)}$, V even and V' convex on $[0,\infty)$, and their weak limits; the Griffiths, Hurst and Sherman inequality holds for ferromagnets with such single spin laws [74]. Section 13.3.1 shows that the Pólya law belongs to the second class, by a theorem of Newman, and Section 13.6 asks whether it belongs to the first.

13.1.2 The Hadamard form and the cumulants

Theorem 13.1.2 (Newman [162]). *Let X be symmetric with $\mathbb{E}e^{b_0X^2} < \infty$ for some $b_0 > 0$. Then X is ferromagnetic if and only if*

$$L_X(s) = e^{bs^2} \prod_k \Big(1 + \frac{s^2}{c_k^2}\Big), \qquad b \geq 0, \quad c_k > 0, \quad \sum_k c_k^{-2} < \infty, \tag{13.1.1}$$

with each c_k repeated according to the multiplicity of the zero ic_k. If $\mathbb{E}e^{bX^2} < \infty$ for every $b > 0$, then $b = 0$.

Proof. Since $|L_X(z)| \leq \mathbb{E}e^{|\operatorname{Re} z||X|} \leq Ce^{|z|^2/4b_0}$, the function φ_X is entire of order at most 2; it is even and real on $\mathbb{R}$, and $\varphi_X(0) = 1$. Suppose that its zeros are real, at $\pm c_k$ with $c_k > 0$. Hadamard's theorem, with the primary factors of genus 2 paired over $\pm c_k$, gives $\varphi_X(\theta) = e^{-b'\theta^2} \prod_k (1 - \theta^2/c_k^2) e^{\theta^2/c_k^2}$ with b' real and $\sum_k c_k^{-4} < \infty$, so that, for real s,

$$\frac{\log L_X(s)}{s^2} = b' + \sum_k \frac{1}{c_k^2} \, \frac{\log(1 + x_k) - x_k}{x_k}, \qquad x_k = \frac{s^2}{c_k^2}.$$

The function $(\log(1+x)-x)/x$ decreases from 0 to -1 on $(0,\infty)$, so by monotone convergence the sum tends to $-\sum_k c_k^{-2}$ as $s \to \infty$. If this sum diverged, $L_X(s)$ would tend to 0, which is impossible because $L_X(s) = \mathbb{E}\cosh(sX) \geq 1$. Hence $\sum_k c_k^{-2} < \infty$, the factors e^{θ^2/c_k^2} can be collected into the exponential, and (13.1.1) holds with $b = b' - \sum_k c_k^{-2}$. Then $\log L_X(s) = bs^2 + o(s^2)$, since $\log(1+x)/x$ is bounded by 1 and tends to 0, and $L_X \geq 1$ forces $b \geq 0$. The converse is clear, since

the right side of (13.1.1) vanishes only at $\pm ic_k$. If $\mathbb{E}e^{bX^2} < \infty$ for every b, then $L_X(s) \le C_\epsilon e^{\epsilon s^2}$ for every $\epsilon > 0$, so $b = 0$, as in the proof of Proposition 7.1.16. □

The proof shows where the Gaussian factor comes from: it is the part of the variance that no zero accounts for. Taking logarithms in (13.1.1) and expanding for $|s| < c_1$ gives the cumulants.

Corollary 13.1.3 (Newman [162]). *If X is ferromagnetic, with* (13.1.1)*, then* $\operatorname{Var} X = 2b + 2\sum_k c_k^{-2}$*, and for $m \ge 2$*

$$\frac{\kappa_{2m}(X)}{(2m)!} = \frac{(-1)^{m+1}}{m} \sum_k c_k^{-2m}. \tag{13.1.2}$$

So the even cumulants alternate in sign: $\kappa_4 \le 0$, $\kappa_6 \ge 0$, and so on. Moreover $\mathbb{E}e^{sX} \le e^{s^2 \operatorname{Var} X/2}$ for real s, and $\mathbb{E}X^4 \le 3(\operatorname{Var} X)^2$.

Proof. The first two statements are the expansion of $\log L_X(s) = bs^2 + \sum_k \log(1 + s^2/c_k^2)$. The bound on $\mathbb{E}e^{sX}$ follows from $\log(1 + x) \le x$, and the bound on $\mathbb{E}X^4 = \kappa_4 + 3\kappa_2^2$ from $\kappa_4 \le 0$. □

For the uniform law on $[-1, 1]$, with $c_k = \pi k$, (13.1.2) gives $\kappa_4 = -12\zeta(4)/\pi^4 = -\frac{2}{15}$, the familiar value. For the Pólya law the alternation holds with no hypothesis: by the formula for $\log G$ in Section 5.1, $(-1)^{n-1}\kappa_{2n}(X) = \frac{(2n)!}{n!}\int_0^\infty t^{n-1} W(t)\, dt > 0$ for every n (Section 1.3), and under RH this is (13.1.2) with $c_k = \gamma_k$ and $b = 0$. Section 13.3.1 computes the first three even cumulants in three ways.

13.1.3 The clock of a ferromagnetic law

The reciprocal of (13.1.1) is a product of gamma transforms, and this attaches a positive random variable to every ferromagnetic law.

Theorem 13.1.4. *Let X be ferromagnetic, with* (13.1.1)*, and put*

$$T_X = b + \sum_k \frac{e_k}{c_k^2}, \qquad Y_X = Z\sqrt{2T_X},$$

the series converging almost surely and in mean.

(i) $\mathbb{E}e^{-uT_X} = 1/L_X(\sqrt{u})$ *for $u \ge 0$, and T_X is a GGC with drift b and Thorin measure $\sum_k \delta_{c_k^2}$, one unit atom for each pair of zeros $\pm ic_k$.*

(ii) $\mathbb{E}e^{i\theta Y_X} = 1/L_X(\theta)$ *for real θ, so that $(\varphi_X, \varphi_{Y_X})$ is a van Dantzig pair, and $Y_X \stackrel{d}{=} \sqrt{2b}\, Z + \sum_k L_k/c_k$.*

(iii) The moment generating function of Y_X is $1/\varphi_X(t)$ for $|t| < c_1$ and infinite for $|t| \ge c_1$, unless X is Gaussian.

(iv) Unless X is degenerate, Y_X has a Pólya frequency density of infinite order.

(v) $\kappa_m(T_X) = (-1)^{m+1} m!\, \kappa_{2m}(X)/(2m)!$ *for* $m \geq 1$*; in particular* $\mathbb{E}T_X = \frac{1}{2}\operatorname{Var} X$ *and* $\operatorname{Var} T_X = -\kappa_4(X)/12$.

Proof. (i) $\mathbb{E}e^{-ue_k/c_k^2} = (1 + u/c_k^2)^{-1}$; multiply over k and compare with (13.1.1) at $s = \sqrt{u}$. (ii) Conditionally on T_X, Y_X is centred normal with variance $2T_X$, so $\mathbb{E}e^{i\theta Y_X} = \mathbb{E}e^{-\theta^2 T_X} = 1/L_X(\theta)$, which is also the characteristic function of $\sqrt{2b}\, Z + \sum_k L_k/c_k$, since $\mathbb{E}e^{i\theta L} = (1+\theta^2)^{-1}$. (iii) $\mathbb{E}e^{tY_X} = \mathbb{E}e^{t^2 T_X} = e^{bt^2} \prod_k (1 - t^2/c_k^2)^{-1}$ while $t^2 < c_1^2$, and $\mathbb{E}e^{t^2 e_1/c_1^2} = \infty$ for $t^2 \geq c_1^2$. (iv) The Laplace density $\frac{1}{2}e^{-|x|}$ and the normal densities are Pólya frequency functions of infinite order, and the class is closed under convolution and under the limits involved here. Equivalently, the bilateral Laplace transform of Y_X is the reciprocal of $\varphi_X(s) = e^{-bs^2} \prod_k (1 - s^2/c_k^2)$, which has Schoenberg's form [235]. (v) For u near 0, $\log \mathbb{E}e^{-uT_X} = -\log L_X(\sqrt{u}) = -\sum_m \kappa_{2m}(X) u^m/(2m)!$, and comparing coefficients with $\sum_m \kappa_m(T_X)(-u)^m/m!$ gives the formula. □

Part (i) is Proposition 7.1.16 with the Gaussian factor allowed: the drift of the clock is the Gaussian part of X, and the Thorin atoms sit at the squared zeros. For the Pólya law under RH, $b = 0$, $c_k = \gamma_k$ and T_X is the clock T of Proposition 1.2.1, and $(\varphi_X, \varphi_{Y_X})$ is the van Dantzig pair of Section 1.2, with $Y_X = Z\sqrt{2T}$ the variable $Y_{1/2}$ of Proposition 10.1.1. Without RH the clock T still exists (Theorem 5.1.1), and so does the pair; what fails without RH is the product form of the clock. Konstantopoulos, Patie and Sarkar study van Dantzig pairs of this kind through the Laguerre Pólya class and Pólya frequency functions, and connect them with the Lee and Yang property [127]. The correspondence runs one way only: every ferromagnetic law gives a Pólya frequency law, but $\varphi(s) = 1 - s^2$, the reciprocal of the bilateral Laplace transform of the Laplace law, is not a characteristic function.

Part (v) holds for the Pólya law with no hypothesis, with the clock T of Theorem 5.1.1 in place of T_X: the identity $\log \mathbb{E}e^{-uT} = -\log G(u) = -\log \mathbb{E}e^{\sqrt{u}X}$ is all the proof uses. With the cumulants of Table 13.4 it gives $\mathbb{E}T = 0.0231049931$ and $\operatorname{Var} T = -\kappa_4/12 = 3.71725993 \times 10^{-5}$, the constants of Section 1.2.

13.1.4 Examples

Table 13.1 lists the basic ferromagnetic laws with their clocks. The variables are those of Biane, Pitman and Yor [28]: C is the exit time of Brownian motion from $(-1, 1)$, with $\mathbb{E}e^{-uC} = 1/\cosh\sqrt{2u}$, and J is the hitting time of 1 by a Bessel process of dimension 3 started at 0, with $\mathbb{E}e^{-uJ} = \sqrt{2u}/\sinh\sqrt{2u}$, as in Section 9.1.1. The first two rows are the classical van Dantzig pairs of Section 1.2.

The third row rests on Kent's theorem for Bessel processes [121]: for the Bessel process of dimension $2\nu + 2$ started at 0, the hitting time of 1 satisfies $\mathbb{E}e^{-u\tau_\nu} = (\sqrt{2u})^\nu/(2^\nu \Gamma(\nu+1) I_\nu(\sqrt{2u}))$, so that $\mathbb{E}e^{-u\tau_\nu/2} = 1/L_X(\sqrt{u})$ and $T_X = \frac{1}{2}\tau_\nu = \sum_k e_k/j_{\nu,k}^2$. The zeros $j_{\nu,k}$ are real for $\nu > -1$, so every symmetric beta law

law of X	$L_X(s)$	c_k	clock T_X	Y_X
fair sign ε	$\cosh s$	$(k-\frac{1}{2})\pi$	$C/2$	hyperbolic secant
uniform on $[-1,1]$	$\sinh s/s$	$k\pi$	$J/2$	logistic, scale $1/\pi$
$c_\nu(1-x^2)^{\nu-1/2}$, $\nu>-\frac{1}{2}$	$\Gamma(\nu+1)(2/s)^\nu I_\nu(s)$	$j_{\nu,k}$	$\frac{1}{2}\tau_\nu$	Bessel type
$N(0,\sigma^2)$	$e^{\sigma^2 s^2/2}$	none	$\sigma^2/2$	$N(0,\sigma^2)$
$e^{-2\pi\cosh 2x}/K_0(2\pi)$	$K_{s/2}(2\pi)/K_0(2\pi)$	$2\nu_k$	$T_P/4$	$Z\sqrt{T_P/2}$
Pólya, under RH	$\xi(\frac{1}{2}+s)/\xi(\frac{1}{2})$	γ_k	T	$Z\sqrt{2T}$

Table 13.1: Ferromagnetic laws and their clocks. Here $j_{\nu,k}$ are the positive zeros of the Bessel function J_ν, and τ_ν is the hitting time of 1 by a Bessel process of dimension $2\nu+2$ started at 0. In the fifth row ν_k are the zeros of $\nu\mapsto K_{i\nu}(2\pi)$ and T_P is the clock of Theorem 8.7.9.

$c_\nu(1-x^2)^{\nu-1/2}$ with $\nu>-\frac{1}{2}$ is ferromagnetic. The family contains the arcsine law ($\nu=0$), the uniform law ($\nu=\frac{1}{2}$) and the semicircle law ($\nu=1$), and it tends to the law of ε as $\nu\downarrow-\frac{1}{2}$, where $\tau_{-1/2}=C$.

13.1.5 Closure

Products of functions of the form (13.1.1) have the same form, so sums of independent ferromagnetic variables are ferromagnetic. Mixtures are a different matter: Section 13.2 shows that a mixture of ferromagnetic laws need not be ferromagnetic. The class is closed under convergence in law, and this is what the Gaussian factor in Definition 13.1.1 is for: the laws of $n^{-1/2}\sum_{k\le n}\varepsilon_k$ have Gaussian moments of all orders, and their limit $N(0,1)$ does not.

Theorem 13.1.5. *Let X_n be ferromagnetic and suppose that $X_n\to X$ in law. Then X is ferromagnetic,* $\operatorname{Var}X_n\to\operatorname{Var}X$, *and $L_{X_n}\to L_X$ locally uniformly on $\mathbb{C}$.*

Proof. Let $\sigma_n^2=\operatorname{Var}X_n$. By Corollary 13.1.3, $\mathbb{E}X_n^4\le 3\sigma_n^4$, so the Paley Zygmund inequality applied to X_n^2 gives $P(X_n^2>\sigma_n^2/4)\ge(\frac{3}{4})^2\sigma_n^4/\mathbb{E}X_n^4\ge\frac{3}{16}$. A sequence that converges in law is tight, so the σ_n are bounded, say by σ. By Corollary 13.1.3 again, $|L_{X_n}(z)|\le L_{X_n}(\operatorname{Re}z)\le e^{\sigma^2(\operatorname{Re}z)^2/2}$, so the L_{X_n} form a locally bounded, hence normal, family of entire functions, and for each real s the variables e^{sX_n} are uniformly integrable, since their second moments are bounded. Hence $\mathbb{E}e^{sX_n}\to\mathbb{E}e^{sX}$ for real s, which gives $\mathbb{E}e^{sX}\le e^{\sigma^2s^2/2}$ and, averaging over $s=\sqrt{2b}Z$, $\mathbb{E}e^{bX^2}<\infty$ for $b<1/2\sigma^2$. So L_X is entire, every locally uniform limit point of the L_{X_n} agrees with it on $\mathbb{R}$, and by Vitali's theorem $L_{X_n}\to L_X$ locally uniformly. Each L_{X_n} has no zero in the half plane $\operatorname{Re}z>0$, and $L_X(0)=1$, so by Hurwitz's theorem L_X has no zero there either; by symmetry it has none in $\operatorname{Re}z<0$. The fourth moments are bounded, so the X_n^2 are uniformly integrable and the variances converge. □

The theorem is the spin side counterpart of Thorin's closure theorem for GGCs, which Corollary 4.1.2 turned into a criterion for RH. It also gives the Gaussian multipliers of de Bruijn from the hyperbolic cosines, the route that Section 7.1.3 describes.

PROPOSITION 13.1.6. *Let X be ferromagnetic with law ρ, and let $\mathbb{E}e^{b_0 X^2} < \infty$.*

(i) For every $a > 0$ the law $\cosh(ax)\,\rho(dx)/\mathbb{E}\cosh(aX)$ is ferromagnetic.

(ii) For every $0 < \lambda < b_0$ the law $e^{\lambda x^2}\rho(dx)/\mathbb{E}e^{\lambda X^2}$ is ferromagnetic.

Proof. (i) The moment generating function of the tilted law is proportional to $L_X(z+a) + L_X(z-a)$. For w with $\operatorname{Re} w > 0$ and $a > 0$ we have $|w + a| > |w - a|$. Applied to $w = z \pm ic_k$ this gives $|c_k^2 + (z+a)^2| > |c_k^2 + (z-a)^2|$ when $\operatorname{Re} z > 0$, and $|e^{b(z+a)^2}| = e^{4ab\operatorname{Re} z}|e^{b(z-a)^2}|$. So for $\operatorname{Re} z > 0$ every factor of (13.1.1) has larger modulus at $z+a$ than at $z-a$, and $L_X(z+a) + L_X(z-a) \neq 0$ unless L_X is constant, in which case the sum is 2. The sum is even in z, so it has no zeros with $\operatorname{Re} z < 0$ either, and the tilted law satisfies the moment condition because $\cosh(ax) \le e^{a|x|}$. (ii) By (i), applied n times, the law with density proportional to $\cosh(x\sqrt{2\lambda/n})^n$ with respect to ρ is ferromagnetic. Since $\cosh y \le e^{y^2/2}$, these densities are dominated by $e^{\lambda x^2}$, which is ρ integrable, and they converge to $e^{\lambda x^2}$, so the laws converge in total variation to $e^{\lambda x^2}\rho(dx)/\mathbb{E}e^{\lambda X^2}$. Theorem 13.1.5 applies. □

Part (i) is the one spin case of the Lee and Yang theorem, and for the Pólya law, whose transform has genus 1, it is Pólya's Hilfssatz II of Section 7.1.3. For the Pólya law, part (ii) is the forward half of de Bruijn's theorem [65]: multiplying p by $e^{\lambda x^2/4}$ is the de Bruijn Newman flow at time λ (Section 8.4), and once the flow has reached a ferromagnetic law it stays in the class. Both parts preserve the class and neither creates it; the tilts that matter for RH are those which produce ferromagnetic laws from the Pólya law with no hypothesis, and they are the subject of Section 13.4.2.

With Theorem 13.1.5 the criterion for RH is immediate.

COROLLARY 13.1.7. *RH holds if and only if the Pólya law is a limit in law of ferromagnetic laws.*

Proof. If RH holds, the Pólya law is ferromagnetic by (1.2.4), and it is its own limit. Conversely, a limit in law of ferromagnetic laws is ferromagnetic by Theorem 13.1.5, and then every zero of Ξ is real. □

The corollary does not make RH easier, but it gives a distance a meaning. Let δ be the infimum of the total variation distances $\sup_A |P(X \in A) - P(Y \in A)|$ from the Pólya variable X to ferromagnetic variables Y; when Y has a density f the distance is $\mathrm{TV}(f,p) = \frac{1}{2}\int |f-p|$. If RH holds, $\delta = 0$; conversely, ferromagnetic

laws that converge to the Pólya law in total variation converge in law, so $\delta = 0$ implies RH. Section 13.4 bounds δ from above with explicit laws.

13.2 Laws on an interval

The Pólya law is symmetric and unimodal, and every such law is a mixture of uniform laws. This section asks which mixtures are ferromagnetic, for laws on an interval, where Fourier transforms of compactly supported functions are entire of exponential type and the question can be settled. The answer turns on the shape of the density. Mixing towards the centre can destroy the property, as an atom at the origin shows; a concave shoulder guarantees it, and a convex one rules it out unless the density is linear on its support, a scaled triangle; and mixing towards the edges is always safe. The Pólya law falls between the two shoulder cases, which is one way to see why its zeros are hard.

13.2.1 Mixtures of uniform laws

By Khinchine's theorem [76], a random variable X is symmetric and unimodal about 0 if and only if $X \stackrel{d}{=} UR$ with $R \geq 0$ independent of U. Then $L_X(s) = \mathbb{E}[\sinh(sR)/(sR)]$, and if X has a density p that is absolutely continuous on $(0, \infty)$ and tends to 0 at infinity, R has the density $-2rp'(r)$. Williamson's theorem refines the representation [272]: a density on $(0, \infty)$ is k monotone exactly when it is a scale mixture of $k(1-x)_+^{k-1}$, so the symmetric k monotone densities are the scale mixtures of

$$f_k(x) = \tfrac{k}{2}(1 - |x|)_+^{k-1},$$

of which f_1 is the uniform density and f_2 the triangle. The uniform law itself is ferromagnetic, with $L_U(s) = \sinh s/s$, and Viète's product

$$\frac{\sinh s}{s} = \prod_{k \geq 1} \cosh \frac{s}{2^k}, \qquad U \stackrel{d}{=} \sum_{k \geq 1} 2^{-k} \varepsilon_k,$$

shows why: the uniform variable is a sum of independent fair signs, and the property is that of a single spin. So is every sum $\sum_k c_k U_k$ of independent scaled uniform variables. Mixtures behave differently.

Theorem 13.2.1. *For every $0 < q < 1$ the law $(1-q)\,U + q\,\delta_0$ is symmetric and unimodal but not ferromagnetic. Consequently, for every $0 < q < 1$ and every small enough $a > 0$, the mixture of the uniform laws on $[-1, 1]$ and on $[-a, a]$ with weights $1 - q$ and q is not ferromagnetic.*

Proof. The moment generating function $L(s) = (1-q)\sinh s/s + q$ vanishes exactly where $\sinh s/s = -c$, with $c = q/(1-q) > 0$. There is no real zero, since $\sinh s/s \geq 1$ on $\mathbb{R}$, and only finitely many imaginary ones, since $s = iy$ needs

$\sin y/y = -c$ and $|\sin y/y| \le 1/|y|$. But L has infinitely many zeros. It is entire of order 1, so with finitely many zeros it would be $e^{\alpha s}P(s)$ with a polynomial P; evenness forces $\alpha = 0$, and L is not a polynomial, since it grows like e^s/s on the positive axis. So all but finitely many zeros lie off the imaginary axis. The second statement follows from Theorem 13.1.5, since the mixtures converge in law to $(1-q)U + q\delta_0$ as $a \downarrow 0$. □

Both components of the mixture are ferromagnetic, so the class is not closed under mixing, even of unimodal laws with a common mode. With $c = 0.1$ the zeros of L in the upper half plane with imaginary part below 65 are $3.4990638i$ and $5.6792078i$ on the imaginary axis, near $i\pi$ and $2i\pi$, and nine pairs $\pm x_j + iy_j$ off it, from $\pm 0.4325231 + 10.8984730i$ to $\pm 2.4991599 + 61.2197027i$, near $i(2j - \frac{1}{2})\pi$; the argument principle on the rectangle $[-6, 6] \times [0, 65]$, evaluated in ball arithmetic, counts exactly these twenty, each isolated in a box of side 2×10^{-10}, and there is none with $|\operatorname{Re} s| \ge 6$, where $|\sinh s| \ge \sinh|\operatorname{Re} s| > c|s|$ (`fm_intcert.py`). For $c > c_*$, where $c_* = -\min_y \sin y/y = 0.2172336\ldots$, the minimum being at the first positive root $y = 4.4934095\ldots$ of $\tan y = y$, there are no imaginary zeros at all. The same mechanism decides the classical spin one law.

Proposition 13.2.2. *For $0 \le q \le 1$ the law $q\,\delta_0 + \frac{1-q}{2}(\delta_{-1} + \delta_1)$ is ferromagnetic if and only if $q \le \frac{1}{2}$ or $q = 1$.*

Proof. For $q = 1$ the law is δ_0, and $L = 1$ has no zeros. For $q < 1$ the zeros of $L(s) = (1-q)\cosh s + q$ solve $\cosh s = -c$ with $c = q/(1-q)$. If $c \le 1$, write $-c = \cos\theta_0$ with $\theta_0 \in [\frac{\pi}{2}, \pi]$; the solutions $s = \pm i\theta_0 + 2\pi i k$ are imaginary. If $c > 1$ the solutions are $s = \pm \operatorname{arccosh} c + i\pi(2k+1)$, which are not; for $q = 0.6$ the first is $0.9624237 + i\pi$. □

13.2.2 The shoulder theorem

The examples are explained by the slope of the density. Let p be a symmetric density on $[-1, 1]$ that is absolutely continuous and nonincreasing on $[0, 1]$, with $p(1) = 0$, and put $g = -p' \ge 0$ on $(0, 1)$. Integrating $\varphi_X(z) = 2\int_0^1 p(y)\cos(zy)\,dy$ by parts turns the cosine transform of the density into a sine transform of its slope:

$$z\,\varphi_X(z) = 2\int_0^1 g(y)\sin(zy)\,dy. \tag{13.2.1}$$

Pólya proved that if f is positive, integrable and nondecreasing on $(0, 1)$, then $\int_0^1 f(t)\cos(zt)\,dt$ and $\int_0^1 f(t)\sin(zt)\,dt$ have only real zeros [211]. So everything depends on whether the slope g rises or falls.

Theorem 13.2.3. *Let p be a symmetric density on $[-1, 1]$, absolutely continuous and nonincreasing on $[0, 1]$, with $p(1) = 0$, and let $[-c, c]$, $0 < c \le 1$, be its support.*

(i) If p is concave on $[0, 1]$, then X is ferromagnetic.

(ii) If p is convex on $[0,1]$ and not linear on $[0,c]$, then X is not ferromagnetic. Moreover $\varphi_X \ge 0$ on $\mathbb{R}$, and its real zeros, if any, are double and lie in a lattice $(2\pi/s_0)\mathbb{Z}$ with $0 < s_0 < c$.

Proof. (i) If p is concave, then $c = 1$ and g is nondecreasing on $(0,1)$. If $g > 0$ on $(0,1)$, Pólya's theorem in its sine form and (13.2.1) show that φ_X has only real zeros. Otherwise p is constant near 0; then $(p + \epsilon(1-|x|))/(1+\epsilon)$ is concave with a positive slope, so it is ferromagnetic, and Theorem 13.1.5 applies as $\epsilon \downarrow 0$.

(ii) If p is convex, g is nonincreasing on $(0,c)$, so $g(y) = g(c-) + \mu((y,c))$ with a positive measure μ on $(0,c)$, which is not zero because p is not linear. Since $\int_0^s \sin(zy)\,dy = 2\sin^2(zs/2)/z$, Fubini's theorem in (13.2.1) gives, for real $z \ne 0$,

$$\frac{z^2}{4}\,\varphi_X(z) = g(c-)\sin^2\frac{zc}{2} + \int_{(0,c)} \sin^2\frac{zs}{2}\,\mu(ds) \;\ge 0.$$

A real zero z_0 needs $z_0 s \in 2\pi\mathbb{Z}$ for every s in the support of μ, so with any such $s_0 < c$ the real zeros lie in $(2\pi/s_0)\mathbb{Z}$. At a real zero the right side has vanishing first derivative and second derivative $\frac{1}{2}g(c-)c^2 + \frac{1}{2}\int s^2\mu(ds) > 0$, because every cosine $\cos(z_0 s)$ that enters equals 1; so the zero is double. Suppose now that every zero of φ_X were real. Since φ_X is even and entire of exponential type, Hadamard's theorem would give $\varphi_X(z) = \prod_{m\in M}(1 - z^2/z_m^2)^2$ for a set M of positive integers, with $z_m = 2\pi m/s_0$, and hence

$$\varphi_X(iy) \le \prod_{m\ge 1}\Big(1 + \frac{y^2 s_0^2}{4\pi^2 m^2}\Big)^2 = \Big(\frac{\sinh(ys_0/2)}{ys_0/2}\Big)^2 \le e^{s_0|y|}.$$

But $p > 0$ on $[0,c)$, so for $0 < \epsilon < c - s_0$ and $y > 0$, $\varphi_X(iy) = L_X(y) \ge \int_{c-\epsilon}^{c} e^{yx}p(x)\,dx \ge C_\epsilon e^{(c-\epsilon)y}$ with $C_\epsilon > 0$, which contradicts the bound as $y \to \infty$. □

The triangle $p(x) = 1 - |x|$ is the borderline: it is both concave and convex, so (i) applies and (ii) does not, and $\varphi(z) = (\sin(z/2)/(z/2))^2$ has only real zeros, all double, at $2\pi m$. For a convex shoulder that is not linear on its support the proof shows the mechanism: the sine transform of a falling slope is a positive combination of the squares $\sin^2(zs/2)$, which can vanish on the real line only on a lattice, and that is too few zeros for a function that grows like $e^{c|y|}$ on the imaginary axis.

COROLLARY 13.2.4. *(i) The law with density proportional to $1-|x|^b$ on $[-1,1]$, $b > 0$, is ferromagnetic if and only if $b \ge 1$. In Khinchine's form, UR with R distributed as* Beta$(a,1)$ *is ferromagnetic if and only if $a \ge 2$.*

(ii) The symmetric k monotone kernel f_k is ferromagnetic if and only if $k \le 2$.

(iii) The laws with density proportional to $(1-x^2)^m$, $0 < m \le 1$, which include the semicircle law, are ferromagnetic.

(iv) The product U_1U_2 of two independent uniform variables, with density $\frac{1}{2}\log(1/|x|)$, is not ferromagnetic.

Proof. (i) $1-x^b$ is concave on $[0,1]$ for $b \geq 1$ and convex and not linear for $b < 1$; the mixing variable has density $-2rp'(r) \propto r^b$, that of $\mathrm{Beta}(b+1,1)$. This settles $a = b+1 > 1$. For $0 < a \leq 1$ the density of UR is unbounded at 0 and the theorem does not apply, but Proposition 13.2.6 below excludes these laws: with $\mu_j = a/(a+2j)$, its case $j=1$ requires $a \geq \sqrt{10}-2 > 1$. (ii) f_1 is uniform and f_2 the triangle; for $k \geq 3$, $(1-x)^{k-1}$ is convex and not linear. (iii) $\frac{d^2}{dx^2}(1-x^2)^m = -2m(1-x^2)^{m-2}(1-(2m-1)x^2) \leq 0$ on $[0,1]$ when $m \leq 1$. (iv) The density is unbounded at the origin, so Theorem 13.2.3 does not apply, but directly $\varphi(z) = \int_0^1 \sin(zu)/(zu)\,du = \mathrm{Si}(z)/z$, and $\mathrm{Si}(z) > 0$ for $z > 0$. So φ has no real zero, while it has infinitely many zeros, by the argument in the proof of Theorem 13.2.1. □

Item (iii) recovers the symmetric beta laws with $\frac{1}{2} < \nu \leq \frac{3}{2}$ of Table 13.1 without Bessel functions, and item (iv) shows that products of independent ferromagnetic variables need not be ferromagnetic. The argument of item (iv) is a general one: by Theorem 13.1.2, a ferromagnetic law whose characteristic function has no real zero is Gaussian. For the k monotone kernels with $k \geq 3$ it gives a second proof of item (ii), since on the imaginary axis $L_k(iy) = k!\,R_k(y)/y^k$, where $R_3(y) = y - \sin y$, $R_4(y) = \frac{y^2}{2} - 1 + \cos y$ and $R_{k+1}(y) = \int_0^y R_k$, and $R_k(y) > 0$ for $y > 0$ once $k \geq 3$; so $\varphi_k = L_k(i\,\cdot)$ is positive on $\mathbb{R}$. The zeros of $L_3(s) = 6(\sinh s - s)/s^3$ of smallest modulus are $\pm 2.7686783 \pm 7.4976763i$, and those of $L_4(s) = 24(\cosh s - 1 - s^2/2)/s^4$ are $\pm 4.5014572 \pm 8.4247845i$; in each case a count by the argument principle in ball arithmetic shows that this is the only zero of modulus at most 8.01, respectively 9.57, in the closed first quadrant (`fm_intcert.py`).

Table 13.2 shows how sharply the transition in item (i) is seen numerically. At $b = 1$ the transform $\int_0^1 (1-x)\cos(zx)\,dx = (1-\cos z)/z^2$ has the double zeros $2\pi m$. As b crosses 1 each double zero splits, into two real zeros for $b > 1$ and into a conjugate pair for $b < 1$, and the sign of the transform at $2\pi m$ changes with it.

13.2.3 Mixing towards the edges

Mixtures of uniform laws centred at the origin can fail; mixtures of uniform laws on outer shells cannot.

Theorem 13.2.5. *Every symmetric density on $[-1,1]$ that is nondecreasing in $|x|$ is ferromagnetic. Equivalently, every mixture of the uniform laws on the shells $\{a \leq |x| \leq 1\}$, $0 \leq a < 1$, is ferromagnetic.*

Proof. Here $\varphi_X(z) = 2\int_0^1 p(t)\cos(zt)\,dt$ with p nondecreasing on $(0,1)$. If $p > 0$ on $(0,1)$, Pólya's theorem in its cosine form [211] applies; otherwise add ϵ times

$a = b + 1$	1.1	1.3	1.6	1.9	1.99	2.01	2.2	2.5	3	4	6
real zeros	0	0	0	0	0	18	18	18	18	18	18
zeros in the box	18	18	18	18	18	18	18	18	18	18	18

Table 13.2: Zeros of $F_b(z) = \int_0^1 (1 - x^b)\cos(zx)\,dx$ in the box $(0.3, 60) \times (-12, 12)$, counted by the argument principle, and real zeros in $(0.3, 60)$, all in ball arithmetic by `fm_intcert.py`. For $b > 1$, certified sign changes on a grid of step 0.002 give 18 real zeros, so every zero in the box is real; for $b < 1$, the box $(0.3, 60) \times (\frac{1}{20}, 12)$ holds 9 zeros and its mirror image 9 more, so none is real. At each b of the table, $F_b(2\pi)$ is positive for $b < 1$ and negative for $b > 1$.

the uniform density, normalise, and let $\epsilon \downarrow 0$ (Theorem 13.1.5). A density nondecreasing in $|x|$ is a mixture of the shell densities, and conversely. □

Each shell law is itself ferromagnetic, since it is the law of $\frac{1+a}{2}\varepsilon + \frac{1-a}{2}U$, with $L(s) = \cosh(\frac{1+a}{2}s)\sinh(\frac{1-a}{2}s)/(\frac{1-a}{2}s)$. The content of the theorem is that, unlike the central mixtures of Theorem 13.2.1, these mixtures stay in the class. The arcsine law, the symmetric beta law with $\nu = 0$, is an example.

13.2.4 A moment test

Ferromagnetic laws satisfy inequalities between consecutive moments. If X is ferromagnetic, $L_X(\sqrt{x}) = \sum_j b_j x^j$, with $b_j = \mathbb{E}X^{2j}/(2j)!$, equals $e^{bx}\prod_k(1 + x/c_k^2)$, an entire function of x in the Laguerre Pólya class with only negative zeros. Its coefficients satisfy the Turán inequalities $(j!\,b_j)^2 \geq (j-1)!\,b_{j-1}(j+1)!\,b_{j+1}$, the limit as $n, K \to \infty$ of Newton's inequalities for the polynomials $(1 + bx/n)^n \prod_{k\leq K}(1 + x/c_k^2)$, whose zeros are real; that is

$$b_j^2 \geq \Big(1 + \frac{1}{j}\Big) b_{j-1} b_{j+1}, \qquad j \geq 1. \tag{13.2.2}$$

For a mixture $X = UR$ the moments factor as $\mathbb{E}X^{2j} = \mu_j/(2j+1)$ with $\mu_j = \mathbb{E}R^{2j}$, and (13.2.2) becomes a condition on the mixing law.

Proposition 13.2.6. *If UR is ferromagnetic and R is not 0 almost surely, then for every $j \geq 1$*

$$\frac{2j+1}{2j+3} \leq \frac{\mu_j^2}{\mu_{j-1}\mu_{j+1}} \leq 1.$$

Proof. The moments μ_j are finite, since $\mathbb{E}(UR)^{2j} = \mu_j/(2j+1)$ and L_{UR} is entire, and positive, since $P(R > 0) > 0$. Substitute $b_j = \mu_j/(2j+1)!$ in (13.2.2): the factor is $(1 + \frac{1}{j})\frac{(2j+1)!^2}{(2j-1)!(2j+3)!} = \frac{2j+1}{2j+3}$. The upper bound is the Cauchy Schwarz inequality and holds for every R. Cleared of denominators, $(2j+1)\mu_{j-1}\mu_{j+1} \leq$

$(2j+3)\mu_j^2$ and $\mu_j^2 \le \mu_{j-1}\mu_{j+1}$, the inequalities hold also when $R = 0$ almost surely. □

The lower bound says that the even moments of R must be close to log linear, so the mixing law cannot be too spread out. The test is necessary and weak. For R distributed as $\mathrm{Beta}(a,1)$, $\mu_j = a/(a+2j)$; at $j=1$ the condition is $a^2+4a-6 \ge 0$, that is $a \ge \sqrt{10}-2 = 1.1622777$, and for $j \ge 2$ it holds for every $a > 0$. Corollary 13.2.4(i) gives the true threshold $a \ge 2$.

13.2.5 Summary

Table 13.3 collects the answers. The Pólya law is unbounded, so none of them applies to it, but its shape places it between the two shoulder cases: Section 13.3.2 shows that its shoulder is concave near the centre and convex in the tails.

law on $[-1,1]$	ferromagnetic
uniform, and sums $\sum_k c_k U_k$	yes
symmetric beta $c_\nu(1-x^2)^{\nu-1/2}$, $\nu > -\frac{1}{2}$	yes
concave shoulder	yes
nondecreasing in $\lvert x\rvert$ (shell mixtures)	yes
convex shoulder, not linear on the support	no
$1-\lvert x\rvert^b$	if and only if $b \ge 1$
k monotone kernel f_k	if and only if $k \le 2$
uniform with an atom at 0, weight in $(0,1)$	no
product of two uniform variables	no
spin one, $q\delta_0 + \frac{1-q}{2}(\delta_{-1}+\delta_1)$	if and only if $q \le \frac{1}{2}$ or $q = 1$

Table 13.3: Which laws on $[-1,1]$ are ferromagnetic. Every entry is proved in this section or in Section 13.1.

13.3 The Pólya law

The Pólya variable X has the density $p = \Phi/\Xi(0)$, with $\Xi(0) = \xi(\frac{1}{2}) = 0.4971207782$, and $L_X(s) = \xi(\frac{1}{2}+s)/\xi(\frac{1}{2})$ is an entire function of order 1. Its density decays like $\exp(-\pi e^{2|t|})$, so X has Gaussian moments of all orders, and RH is the statement that X is ferromagnetic; by Theorem 13.1.2, with $b=0$, it is the product formula $L_X(s) = \prod_k (1 + s^2/\gamma_k^2)$ over the positive ordinates. The law is symmetric and unimodal, since $\Phi'(t) < 0$ for $t > 0$ [60, Theorem A]. This section checks on X the necessary conditions of Sections 13.1 and 13.2, writes X as a mixture of uniform laws, and proves a limit on the ways in which X could split into independent pieces.

13.3.1 Necessary conditions

If X is ferromagnetic, Corollary 13.1.3 with $c_k = \gamma_k$ gives $\kappa_2 = 2\sum_k \gamma_k^{-2}$, $\kappa_4 = -12\sum_k \gamma_k^{-4}$ and $\kappa_6 = 240\sum_k \gamma_k^{-6}$. Table 13.4 computes the cumulants by quadrature of the moments of Φ and from the Taylor coefficients of $\log \xi(\frac{1}{2}+s)$, and the sums from the first 1700 ordinates, with the smooth Riemann von Mangoldt count $\frac{t}{2\pi}\log\frac{t}{2\pi} - \frac{t}{2\pi} + \frac{7}{8}$ beyond the last of them. The signs alternate, as they must for a ferromagnetic law and as they do for the Pólya law with no hypothesis (Section 13.1.2). The agreement of the columns is not evidence for RH. The Hadamard product $\xi(\frac{1}{2}+s)/\xi(\frac{1}{2}) = \prod_k (1 + s^2/\gamma_k^2)$, with one factor for each pair of zeros $\frac{1}{2} \pm i\gamma_k$, holds with no hypothesis if γ_k is allowed to be complex, and it converges absolutely because $\sum_k |\gamma_k|^{-2} < \infty$; it gives the same formulas for the cumulants. So the table checks the computation and the zero list, nothing more.

	moments of Φ	Taylor coefficients of $\log\xi$	zero sums
κ_2	0.0462099862308	0.0462099862308	0.046209986105
κ_4	$-4.46071191423 \times 10^{-4}$	$-4.46071191423 \times 10^{-4}$	$-4.46071191423 \times 10^{-4}$
κ_6	$3.46017435362 \times 10^{-5}$	$3.46017435362 \times 10^{-5}$	$3.46017435362 \times 10^{-5}$

Table 13.4: Even cumulants of the Pólya law, by quadrature of the moments of Φ, from the Taylor coefficients of $\log\xi(\frac{1}{2}+s)$, and from the sums $(-1)^{m+1}\frac{(2m)!}{m}\sum_k \gamma_k^{-2m}$ over the first 1700 ordinates with the smooth count beyond them. The difference 1.3×10^{-10} in κ_2 is the error of the smooth count, which ignores the oscillation of $S(t)$ above height 2197. Computed by `fm_polya.py`.

The moment test of Section 13.2.4 is sharper. For the Pólya law the scaled moments $b_j = \mathbb{E}X^{2j}/(2j)!$ are the Taylor coefficients of Ξ, up to sign and the factor $\Xi(0)$:

$$\frac{\Xi(t)}{\Xi(0)} = \sum_{j\ge 0} (-1)^j b_j t^{2j}.$$

In this form (13.2.2), $b_j^2 \ge (1+\frac{1}{j})b_{j-1}b_{j+1}$, is the system of Turán inequalities for the moments $\int_0^\infty t^{2j}\Phi(t)\,dt$, which Csordas, Norfolk and Varga proved for every $j \ge 1$, answering a question of Pólya [60]. So the Pólya law passes the moment test with no hypothesis. Table 13.5 shows the margins. They decrease slowly, from 0.150 at $j = 1$ to 0.017 at $j = 24$, while the bound $1 + \frac{1}{j}$ itself approaches 1; the Gaussian law, for which $b_j = (\sigma^2/2)^j/j!$, satisfies every inequality with equality.

The Turán inequalities are the first of a hierarchy. For $d, n \ge 0$ let

$$J_{d,n}(x) = \sum_{i=0}^{d} \binom{d}{i} (n+i)!\, b_{n+i}\, x^i,$$

the Jensen polynomial of degree d and shift n of the sequence $j!\,b_j$. Pólya proved that RH holds if and only if every $J_{d,n}$ has only real zeros, and for $d = 2$ this is

j	b_j	$b_j^2/(b_{j-1}b_{j+1})$	$1+1/j$
1	2.3104993×10^{-2}	2.1496879	2
2	2.4833405×10^{-4}	1.5941155	1.5
3	1.6743526×10^{-6}	1.4057378	1.3333333
4	8.0306974×10^{-9}	1.3100631	1.25
5	2.9401358×10^{-11}	1.2518550	1.2
10	1.1310848×10^{-24}	1.1322525	1.1
15	1.7510030×10^{-39}	1.0907154	1.0666667
20	2.9919818×10^{-55}	1.0693835	1.05
24	2.1176410×10^{-68}	1.0585214	1.0416667

Table 13.5: The moment test (13.2.2) for the Pólya law: the scaled moments $b_j = \mathbb{E}X^{2j}/(2j)!$ and the Turán ratios, against the bound $1+\frac{1}{j}$ that every ferromagnetic law satisfies. All 24 computed ratios exceed the bound. Computed by `fm_polya.py`.

(13.2.2) with $j = n+1$. Griffin, Ono, Rolen and Zagier proved that for each d all but finitely many of the $J_{d,n}$ have only real zeros, and that all of them do when $d \le 8$ [84]. Each degree is a necessary condition for the Pólya law to be ferromagnetic, and none of them is known to fail. The shifts make the hierarchy weaker than it looks. With $F(x) = \sum_{j\ge0} b_j x^j$, so that $F(-t^2) = \varphi_X(t)$, the polynomials $J_{d,n}$ with n fixed are the Jensen polynomials of the derivative $F^{(n)}$, and all of them have only real zeros exactly when $F^{(n)}$ is in the Laguerre Pólya class. For $n \ge 1$ that says nothing about the zeros of φ_X. For the spin one law of Table 13.3 with $q > \frac{1}{2}$, $F(x) = q + (1-q)\cosh\sqrt{x}$ and $F'(x) = (1-q)\sinh\sqrt{x}/(2\sqrt{x})$, whose zeros $-k^2\pi^2$ are real, so every $J_{d,n}$ with $n \ge 1$ has only real zeros, while $\varphi_X(t) = q + (1-q)\cos t$ has no real zero at all. In the other direction, if ξ had only finitely many zeros off the critical line, then F, which has order $\frac{1}{2}$, would have only finitely many nonreal zeros, and $F^{(n)}$ would be in the Laguerre Pólya class for every large n, by the theorem of Craven, Csordas and Smith on real entire functions of order less than two [59]. So real zeros for every $J_{d,N}$ at one shift N and in every degree would follow from finitely many exceptions to RH, and the spin one law shows that for laws in general they do not imply even that.

A necessary condition of a different kind comes from the correlation inequalities of Section 13.1.1. Write $\Phi = e^{-F}$. Newman proved that F' is convex on $[0,\infty)$ [164]; in his normalisation $\Xi(x/2) = 4\int e^{ixt - V(t)}\,dt$, and $V(t) = F(2t) + \log 2$. So the Pólya law belongs to the class of Ellis, Monroe and Newman, and every ferromagnet whose single spins are independent copies of X satisfies the Griffiths, Hurst and Sherman inequality [74]. For one spin that inequality says that the magnetisation $(\log L_X)'(s)$ is concave for $s \ge 0$, which gives $\kappa_4 \le 0$, a weak form of the alternation of Section 13.1.2. Membership in this class gives the correlation inequalities, not the Lee and Yang property, and it does not decide RH; Newman's

paper discusses how the two properties are related. For illustration, $F''(0) = 18.72690493$ and $F''''(0) = 143.0149378$. On the grid $u = j/40$, $1 \le j \le 160$, F''' is positive, with its smallest value 3.57 at the first point, where $F'''(u) \approx F''''(0)u$; it is 185.0 at $u = 1$ and 7.49×10^4 at $u = 4$, close to the value $8\pi e^{2u}$ of the leading term of $F(u) = \pi e^{2u} - \frac{9}{2}u - \log 4\pi^2 + O(e^{-2u})$ (computed by `fm_polya.py`).

13.3.2 A mixture of uniform laws

Since p is symmetric and unimodal, Khinchine's theorem of Section 13.2.1 writes $X \stackrel{d}{=} UR$, with R independent of U and with the density $h(r) = -2rp'(r) = -2r\Phi'(r)/\Xi(0)$ on $(0,\infty)$. Since $\mathbb{E}|X|^m = \mathbb{E}R^m/(m+1)$, the mixing variable has $\mathbb{E}R = 2\mathbb{E}|X| = 0.3464543599$ and $\mathbb{E}R^2 = 3\operatorname{Var} X = 0.1386299587$. Integrating by parts as in (13.2.1), now on the half line,

$$z\,\varphi_X(z) = 2\int_0^\infty \big(-p'(u)\big)\sin(zu)\,du,$$

so RH says that the sine transform of the positive function $-\Phi'$ on $(0,\infty)$ has only real zeros. The shape of this slope is that of neither part of Theorem 13.2.3. Since $p'' = p\,(F'^2 - F'')$, the density is concave near the origin, where $p''(0) = -F''(0)p(0) = -33.65$, and convex for large u, where $F'^2/F'' \to \infty$. So the slope $-p'$ rises from 0 and then falls back to 0. On a bounded support, a slope that rises gives only real zeros, by Pólya's theorem, and one that falls, and is not constant on the support, gives zeros off the real line, by Theorem 13.2.3(ii); the slope of the Pólya law does both, and RH is the statement that it still gives only real zeros. The damped densities, proportional to $e^{tu^2/4}\Phi(u)$ with $t < 0$, are symmetric and unimodal too, hence mixtures of uniform laws, and they are mixtures of arcsine laws as well (Proposition 11.5.15), while their transforms have nonreal zeros, so neither mixture can decide RH by itself (Section 11.5.5).

The mixture does not discretise well. Replacing R by J equally weighted atoms at its quantiles r_j of order $(j - \frac{1}{2})/J$ gives the characteristic function $\varphi_J(z) = J^{-1}\sum_j \sin(r_j z)/(r_j z)$. For $J = 5$ it has 17 zeros in the rectangle $(0.25, 100) \times (-30, 30)$, of which only 7 are real, and for $J = 20$ only 3 of 19 are real, while φ_X has 29 zeros in $(0, 100)$, all real (computed by `fm_polya.py`). As Theorem 13.2.1 suggests, mixing in the R variable destroys the property easily, although not always: the mixture of the uniform laws on $[-1, 1]$ and $[-3, 3]$ with weights $\frac{1}{4}$ and $\frac{3}{4}$ has characteristic function $\frac{1}{4}\frac{\sin z}{z} + \frac{3}{4}\frac{\sin 3z}{3z} = \frac{\sin 2z}{2z}\cos z$, and it is the law of $2U + \varepsilon$, which is ferromagnetic. The approximations of Section 13.4 are built on the Fourier side instead.

13.3.3 No component with finitely many zeros

A sum of independent ferromagnetic variables is ferromagnetic (Section 13.1.5), and one might hope to reach the Pólya law, or to approximate it, as such a sum

of simple pieces. The growth of L_X rules out most pieces.

THEOREM 13.3.1. *Let X be a symmetric random variable with $\mathbb{E}e^{sX} < \infty$ for every real s and $\log \mathbb{E}e^{sX} = o(s^2)$ as $s \to \infty$, and let $X \stackrel{d}{=} D + R$ with D and R independent. Then $\log \mathbb{E}e^{sD} = o(s^2)$ as $|s| \to \infty$. In particular, if D is ferromagnetic and not 0, then $b = 0$ in its Hadamard form* (13.1.1)*, and its characteristic function has infinitely many zeros. The Pólya law satisfies the hypothesis, so it has no Gaussian component, and it is not a finite or convergent infinite sum of independent ferromagnetic variables of which one is not 0 and has finitely many zeros.*

Proof. By independence $\mathbb{E}e^{sX} = \mathbb{E}e^{sD}\,\mathbb{E}e^{sR}$, and both factors are finite because the product is. Jensen's inequality gives $\mathbb{E}e^{sR} \ge e^{s\mathbb{E}R}$, and $\mathbb{E}R = -\mathbb{E}D$ because $\mathbb{E}X = 0$. So $\log \mathbb{E}e^{sD} \le \log \mathbb{E}e^{sX} + s\mathbb{E}D$, and the right side is $o(s^2)$ as $s \to \pm\infty$, by the symmetry of X. If D is ferromagnetic, $\mathbb{E}e^{sD} = e^{bs^2} \prod_k (1 + s^2/c_k^2) \ge e^{bs^2}$, so $b = 0$. If moreover there are finitely many zeros, then $\varphi_D(t) = \prod_{k \le K} (1 - t^2/c_k^2)$ is a polynomial bounded on the real line, hence constant, so $K = 0$ and $D = 0$. For the Pólya law, $\log \mathbb{E}e^{sX} = \log \xi(\frac{1}{2} + s) - \log \xi(\frac{1}{2}) = O(s \log s)$, since ξ has order 1. A summand of an almost surely convergent sum of independent variables is an independent component of the sum. □

The theorem needs no hypothesis, and it says something about every approximation scheme. The laws of Section 13.4 that keep the first K zeros of φ_X exactly carry a Gaussian factor, so none of them is a component of the Pólya law. If RH holds they approach it in law, along every sequence of K for which they are probability densities (Theorem 13.4.3(iii)), but not as partial sums of a decomposition; without RH they do not approach it at all, since a weak limit of ferromagnetic laws is ferromagnetic (Theorem 13.1.5). If RH holds, any decomposition of X into independent ferromagnetic pieces splits the ordinates into infinite sets, each the zero set of a characteristic function $\prod_k (1 - t^2/c_k^2)$ with no Gaussian factor, like those of the uniform law, the fair sign and the symmetric beta laws in Table 13.1.

The simplest such splitting would be a Viète product $L_X(s) = \prod_j \cosh(a_j s)$, which would write $X = \sum_j a_j \varepsilon_j$ as a sum of independent scaled fair signs, a member of the Griffiths Simon class with uncoupled spins, as the uniform law is by Viète's formula. The zeros of $\cosh(a_j s)$ are $\pm i(m - \frac{1}{2})\pi/a_j$, $m \ge 1$, so the ordinates would split into arithmetic progressions, and three consecutive terms of a progression satisfy $\gamma - 2\gamma' + \gamma'' = 0$. The hypothesis that the positive ordinates are linearly independent over the rationals, which Ingham used in his work on the Mertens conjecture [105], forbids such a relation. Under that hypothesis the Pólya law is not a sum of independent scaled fair signs, and if it belongs to the Griffiths Simon class at all, the spins must be coupled. Section 13.6 returns to this question.

13.4 Ferromagnetic approximations of the Pólya law

By Corollary 13.1.7 and the remark after it, RH holds exactly when the total variation distance δ from the Pólya law to the class of ferromagnetic laws is 0. This section bounds δ from above with no hypothesis, by the distances to laws that are proved to be ferromagnetic: Pólya's own approximations, tilts of p by multipliers that shrink the strip of the zeros, and laws that keep the first K zeros exactly and replace the others by a Gaussian factor. The last family leads to an identity in law equivalent to RH. The distances are enclosed in ball arithmetic, with explicit bounds for the discretisation and the tails, so the bounds on δ they give are proved; Table 13.7 at the end of the section collects them.

13.4.1 Pólya's approximations

The Gaussian law $N(0, \operatorname{Var} X)$ is ferromagnetic and lies at total variation distance 1.48×10^{-2} from the Pólya law. Pólya kept the leading part of the first term of Φ and made it even:

$$\Phi^*(t) = 8\pi^2 \cosh \tfrac{9t}{2}\, e^{-2\pi \cosh 2t}, \qquad \int \Phi^*(t) e^{i\theta t}\, dt = 4\pi^2 \Big[K_{\frac{9}{4}+\frac{i\theta}{2}}(2\pi) + K_{\frac{9}{4}-\frac{i\theta}{2}}(2\pi) \Big],$$

by $\int e^{-a\cosh u + \nu u}\, du = 2K_\nu(a)$, and he proved that this transform has only real zeros [213]. So the law with density proportional to Φ^* is ferromagnetic. It is the law of $\frac{\varepsilon}{2} \log G$, where G is generalised inverse Gaussian with density proportional to $g^{5/4} e^{-\pi(g+1/g)}$, since the substitution $g = e^{2t}$ turns $e^{9t/2} e^{-2\pi\cosh 2t}$ into that density; so it can be sampled exactly. Keeping the whole first term gives the positive kernel

$$\Psi(t) = \Big(8\pi^2 \cosh \tfrac{9t}{2} - 12\pi \cosh \tfrac{5t}{2} \Big) e^{-2\pi \cosh 2t},$$

and the same theorem of Pólya covers it.

Proposition 13.4.1. *The law with density proportional to Ψ is ferromagnetic.*

Proof. Write $\Psi(t) = \psi(it) f(t)$ with $f(t) = e^{-2\pi\cosh 2t}$ and $\psi(z) = 8\pi^2 \cos \frac{9z}{2} - 12\pi \cos \frac{5z}{2}$. In the variable $w = z/2$, ψ is the trigonometric polynomial $8\pi^2 \cos 9w - 12\pi \cos 5w$ of degree 9. At $w_j = j\pi/9$, $0 \le j < 18$, the first term is $8\pi^2(-1)^j$ and the second is at most $12\pi < 8\pi^2$ in absolute value, so ψ changes sign at least 18 times in a period, and since a trigonometric polynomial of degree 9 has at most 18 zeros in a period strip, all its zeros are real. So ψ is a real entire function of order 1 with only real zeros. The transform of f is $K_{i\theta/2}(2\pi)$, which has only real zeros, by Pólya's theorem on $\int e^{-\lambda\cosh t} e^{izt} dt$ [65, (1.2)], and f decays faster than $e^{-|t|^b}$ for every b. Pólya's theorem on universal factors [65, p. 197], [50, Proposition 1] then says that the transform of $\psi(it) f(t)$ has only real zeros. The density decays like f, so it has Gaussian moments of all orders. □

The same proof, with $\psi(z) = 8\pi^2 \cos\frac{9z}{2}$, gives Pólya's result for Φ^*. As densities both approximations are poor (Table 13.7): their total variation distances to p are 0.099 and 0.161, their variances 0.0624 and 0.0726 against $\operatorname{Var} X = 0.0462$, and their values at the origin 1.404 and 1.163 against $p(0) = 1.797$, all enclosed in ball arithmetic (`fm_tvcert.py`). Their transforms have 14 zeros each in $(0.25, 62.5)$, all real, and an argument principle count in $(0.25, 62.5) \times (-2, 2)$ finds no others, the same number as the ordinates below 62.5. The positions differ: the first five are 8.99, 19.07, 24.47, 28.97, 33.01 for Φ^* and 7.60, 18.72, 24.28, 28.83, 32.89 for Ψ, against 14.13, 21.02, 25.01, 30.42, 32.94 (computed by `fm_polya.py`). Pólya's point was the count, which these kernels reproduce, and not the positions.

13.4.2 Tilts that shrink the strip

Every zero γ of $\varphi_X = \Xi/\Xi(0)$ satisfies $|\operatorname{Im}\gamma| < \frac{1}{2}$, because $0 < \operatorname{Re}\rho < 1$. Multipliers of the density move the zeros towards the real axis, and de Bruijn measured by how much. For $\lambda_1, \dots, \lambda_N > 0$ and $\mu \geq 0$ put

$$m(t) = e^{\mu t^2} \prod_{k \leq N} \cosh(\lambda_k t), \qquad c(m) = 2\mu + \sum_{k \leq N} \lambda_k^2.$$

Then $m(t) = \mathbb{E}e^{tY} = \mathbb{E}\cosh(tY)$ for $Y = \sqrt{2\mu}\, Z + \sum_k \lambda_k \varepsilon_k$, and $c(m) = \operatorname{Var} Y$.

Proposition 13.4.2. *The zeros of $\int \Phi(t) m(t) e^{izt}\, dt$ lie in the strip $|\operatorname{Im} z| \leq (\max(\frac{1}{4} - c(m), 0))^{1/2}$. The same holds for $N = \infty$ when $\sum_k \lambda_k^2 < \infty$. In particular, with no hypothesis, the law with density proportional to $m\,p$ is ferromagnetic whenever $\operatorname{Var} Y \geq \frac{1}{4}$; for example $\cosh(at)\,p$ for $a \geq \frac{1}{2}$, $\frac{\sinh(ct)}{ct}\,p$ for $c \geq \frac{\sqrt{3}}{2}$, and $e^{\mu t^2} p$ for $\mu \geq \frac{1}{8}$. If no zero of ζ has real part greater than $\frac{7}{8}$, as announced in October 2026 (Section 12.1.5), all of this holds with $\frac{9}{64}$ in place of $\frac{1}{4}$; then $\cosh(at)\,p$ is ferromagnetic for $a \geq \frac{3}{8}$ and $\frac{\sinh(ct)}{ct}\,p$ for $c \geq \frac{3\sqrt{3}}{8}$.*

Proof. For finite N this is de Bruijn's Theorem 13, for the Gaussian factor $e^{\frac{1}{2}(2\mu)t^2}$, followed by his Theorem 12, for the product of cosines [65]; both apply to an even kernel F with $F(t) = O(e^{-|t|^b})$ for some $b > 2$, which Φ and $\Phi e^{\mu t^2}$ are, and both take as input a closed strip containing the zeros, here $|\operatorname{Im} z| \leq \frac{1}{2}$. Under the hypothesis of the last sentence every zero has $\frac{1}{8} \leq \operatorname{Re}\rho \leq \frac{7}{8}$, by $\xi(s) = \xi(1-s)$, so the zeros of φ_X lie in $|\operatorname{Im} z| \leq \frac{3}{8}$, and the same two theorems give the strip with $\frac{9}{64}$ in place of $\frac{1}{4}$. For $N = \infty$ the partial products m_N have $c(m_N) \uparrow c(m)$, and their transforms converge to that of Φm locally uniformly in z, by dominated convergence with $m_N \leq m$. By Hurwitz's theorem the zeros of the limit lie in the intersection of the strips. The tilted densities decay like Φ, so they have Gaussian moments of all orders, and when $c(m) \geq \frac{1}{4}$ their characteristic functions have only real zeros. The three examples have $\operatorname{Var} Y = a^2$, $c^2/3$ (by Viète's product $\frac{\sinh ct}{ct} = \prod_k \cosh\frac{ct}{2^k}$) and 2μ. □

The single cosine is elementary, and the proof shows where the number $\frac{1}{2}$ comes from. The characteristic function of $\cosh(at)p(t)$ is proportional to $\varphi_X(z+ia)+\varphi_X(z-ia)$. Take $z=x+iy$ with $y>0$ and a pair of zeros $\beta\pm i\eta$ of φ_X, with $c=(x-\beta)^2$. De Bruijn's identity [65, Lemma 1]

$$\big(c+(y+a-\eta)^2\big)\big(c+(y+a+\eta)^2\big)-\big(c+(y-a-\eta)^2\big)\big(c+(y-a+\eta)^2\big) \\ =8ay\,(c+y^2+a^2-\eta^2)$$

shows that the pair contributes a factor larger than 1 to $|\varphi_X(z+ia)|^2/|\varphi_X(z-ia)|^2$ as soon as $a>|\eta|$. With $a\ge\frac{1}{2}$ this holds for every zero, by the Hadamard product, so the two terms cannot cancel above the real axis, nor, by symmetry, below it. In the variable $s=\frac{1}{2}+iz$ the transform is $\frac{1}{2}(\xi(s+a)+\xi(s-a))$, and Lagarias proved, in addition, that its zeros are simple [134]. For the de Bruijn Newman flow the multiplier is $e^{\lambda t^2/4}$, with $c(m)=\lambda/2$; the proposition gives de Bruijn's bound $\Lambda\le\frac{1}{2}$, and the bound $\Lambda\le 0.2$ of Platt and Trudgian [196] makes $e^{\lambda t^2/4}p$ ferromagnetic for every $\lambda\ge 0.2$, again with no hypothesis.

The three multipliers of the proposition with $\operatorname{Var}Y=\frac{1}{4}$ give total variation distances 2.731×10^{-3}, 2.739×10^{-3} and 2.751×10^{-3} from p, in the order listed. They agree because $m(t)=1+\frac{t^2}{8}+\frac{\mathbb{E}Y^4}{24}t^4+\dots$ depends on the law of Y only from the fourth order on, and $\mathbb{E}Y^4$ is $\frac{1}{16}$, $\frac{9}{80}$ and $\frac{3}{16}$ for a sign, a uniform variable and a Gaussian of variance $\frac{1}{4}$; the single cosine is the smallest perturbation. The flow at $\lambda=0.2$, with $c(m)=0.1$, gives 1.097×10^{-3}. For comparison, $\cosh(\sqrt{0.1}\,t)\,p$, with the same contraction, gives 1.094×10^{-3}, but whether it is ferromagnetic is not known: the bound of Platt and Trudgian is a statement about the Gaussian flow, which rests on a computation of its zeros, and Proposition 13.4.2 needs $\operatorname{Var}Y\ge\frac{1}{4}$, or $\frac{9}{64}$ if the announced half plane stands, both above 0.1 (the distances are enclosed in ball arithmetic by `fm_tvcert.py`).

The cosine tilts define a constant in the spirit of Λ. Let Ω be the infimum of the $a_0\ge0$ such that $\cosh(at)\,p$ is ferromagnetic for every $a\ge a_0$. Proposition 13.4.2 gives $\Omega\le\frac{1}{2}$, and $\Omega\le\frac{3}{8}$ if the half plane $\operatorname{Re}s>\frac{7}{8}$ announced in October 2026 is free of zeros. If RH holds, Proposition 13.1.6(i) gives $\Omega=0$. Conversely, if $\Omega=0$, the tilted laws converge to p in total variation as $a\downarrow0$, and p is ferromagnetic by Theorem 13.1.5. So RH holds if and only if $\Omega=0$. Unlike the flow, the operators $\varphi\mapsto\varphi(\cdot+ia)+\varphi(\cdot-ia)$ do not form a semigroup, and we do not know whether the set of good values of a is an interval.

13.4.3 Laws that keep the first zeros

Let $0<\gamma_1\le\gamma_2\le\dots$ be the ordinates of the zeros on the critical line, with multiplicity, and put $s_K=\sum_{k\le K}\gamma_k^{-2}$ and $\beta_K=\frac{1}{2}\operatorname{Var}X-s_K$. The idea is to keep the first K factors of the Hadamard product exactly and to replace the others

by the Gaussian factor with the same effect on the variance:

$$\varphi_K(\theta) = e^{-\beta_K \theta^2} \prod_{k \le K} \Big(1 - \frac{\theta^2}{\gamma_k^2}\Big).$$

A symmetric Maxwell variable M, with density $x^2 e^{-x^2/2}/\sqrt{2\pi}$, has $\mathbb{E}e^{i\theta M} = (1 - \theta^2)e^{-\theta^2/2}$ and $\operatorname{Var} M = 3$; it is ferromagnetic, with $L_M(s) = (1 + s^2)e^{s^2/2}$. Let $M_1, M_2, \dots$ be independent copies.

THEOREM 13.4.3. *(i) $\beta_K > 0$ for every K. If $3s_K \le \operatorname{Var} X$, then φ_K is the characteristic function of*

$$X_K = \big(\operatorname{Var} X - 3s_K\big)^{1/2} Z + \sum_{k \le K} \frac{M_k}{\gamma_k},$$

whose law f_K is ferromagnetic, with $\operatorname{Var} X_K = \operatorname{Var} X$ and $\kappa_4(X_K) = -12 \sum_{k \le K} \gamma_k^{-4}$. The condition holds for $K \le 16$ and fails for $K = 17$.

(ii) For every K, if the inverse Fourier transform f_K of φ_K is nonnegative, then f_K is a ferromagnetic probability density.

(iii) RH holds if and only if $\varphi_K \to \varphi_X$ pointwise as $K \to \infty$. Under RH, $f_K \to p$ in law along every sequence of K for which $f_K \ge 0$.

Proof. With no hypothesis, $\frac{1}{2} \operatorname{Var} X = \sum_k \gamma_k^{-2}$, where the sum runs over one zero of each pair $\pm\gamma$ and complex γ are allowed (Section 13.3.1). A zero off the line contributes, with its conjugate, $2 \operatorname{Re} \gamma^{-2} > 0$, since $|\operatorname{Re}\gamma| > 14 > |\operatorname{Im}\gamma|$. So β_K is the sum of γ_k^{-2} over the line zeros with $k > K$ and of these positive terms, and it is positive.

(i) M_k/γ_k has characteristic function $(1 - \theta^2/\gamma_k^2)e^{-\theta^2/2\gamma_k^2}$, so the sum over $k \le K$ accounts for every factor of φ_K except $e^{-(\beta_K - s_K/2)\theta^2}$, a Gaussian factor of variance $2\beta_K - s_K = \operatorname{Var} X - 3s_K$. A sum of independent ferromagnetic variables is ferromagnetic. The cumulants come from $\log \varphi_K$. Since s_K increases with K, the last statement is $3s_{16} < \operatorname{Var} X < 3s_{17}$. In ball arithmetic, with the certified zeros of Arb [110] and $\operatorname{Var} X = (\log \xi)''(\frac{1}{2})$ from the Taylor series of $\log \xi$, $3s_{16} = 0.0456803$, $\operatorname{Var} X = 0.0462100$ and $3s_{17} = 0.0463006$, with margins 5.3×10^{-4} and 9.1×10^{-5} (`fm_cutoff.py`).

(ii) f_K has total mass $\varphi_K(0) = 1$, and its moment generating function $e^{\beta_K s^2} \prod_{k \le K}(1 + s^2/\gamma_k^2)$ vanishes only on the imaginary axis. By Proposition 13.4.4 below, f_K is a Gaussian density times a polynomial, so it has Gaussian moments.

(iii) As $K \to \infty$, $\beta_K \downarrow \beta_\infty$, the sum of the terms $2 \operatorname{Re} \gamma^{-2}$ over the zeros off the line, and $\varphi_K \to \varphi_\infty = e^{-\beta_\infty \theta^2} \prod_k (1 - \theta^2/\gamma_k^2)$, a product over the line zeros, whose zeros are all real. If RH holds, $\beta_\infty = 0$ and $\varphi_\infty = \varphi_X$ by the Hadamard product.

If RH fails, φ_X vanishes at a nonreal point and φ_∞ does not, so $\varphi_\infty \neq \varphi_X$. Along a sequence with $f_K \geq 0$ the φ_K are characteristic functions, and Lévy's continuity theorem gives convergence in law. □

Part (i) needs no hypothesis and no computation beyond the certified inequality $3s_{16} < \operatorname{Var} X$, so f_{16} is a proved ferromagnetic law. Its distance from the Pólya law is $\mathrm{TV}(f_{16}, p) = 1.27\times10^{-4}$, ten times smaller than that of the tilt by the flow at $\lambda = 0.2$. Beyond $K = 16$ the Maxwell construction runs out of Gaussian variance, and positivity must be checked directly. The exact condition comes from a closed form, in which He_n is the Hermite polynomial with $\mathrm{He}_n(y) = \mathbb{E}(y+iZ)^n$, g_v is the $N(0,v)$ density, and e_j is the jth elementary symmetric function of $\gamma_1^{-2}, \dots, \gamma_K^{-2}$.

Proposition 13.4.4. *For $v > 0$ the function $e^{-v\theta^2/2}\prod_{k\leq K}(1-\theta^2/\gamma_k^2)$ is the Fourier transform of*

$$f_{K,v}(x) = g_v(x)\,Q_{K,v}\big(x/\sqrt{v}\big), \qquad Q_{K,v}(y) = \sum_{j=0}^{K} \frac{e_j}{v^j}\,\mathrm{He}_{2j}(y),$$

and $f_K = f_{K,2\beta_K}$. There is a threshold $v^*(K)$ such that $f_{K,v} \geq 0$ if and only if $v \geq v^*(K)$, and $v^*(K) \leq s_K$. Moreover $Q_{K,v}(y) > 0$ for $|y| \geq \sqrt{8K+2}$, so the sign of $f_{K,v}$ is decided on $|x| < \sqrt{(8K+2)v}$.

Proof. Differentiating g_v gives $g_v^{(n)}(x) = (-1)^n v^{-n/2}\mathrm{He}_n(x/\sqrt{v})g_v(x)$, and the transform of $g_v^{(2j)}$ is $(-1)^j\theta^{2j}e^{-v\theta^2/2}$. Expanding $\prod_k(1-\theta^2/\gamma_k^2) = \sum_j(-1)^j e_j\theta^{2j}$ gives the formula. For $v < v'$, $f_{K,v'} = f_{K,v} * g_{v'-v}$, so positivity persists as v increases, and the set of good v is closed. At $v = s_K$ the transform is that of $\sum_{k\leq K} M_k/\gamma_k$, so $v^*(K) \leq s_K$. The zeros of He_n lie in $|y| < \sqrt{4n+2}$, by a Sturm comparison for the Hermite equation, so for $|y| \geq \sqrt{8K+2}$ every He_{2j} with $j \leq K$ is positive, and so are the coefficients $e_j v^{-j}$. □

For $K = 1$, $Q_{1,v}(y) = 1 + (y^2-1)/(v\gamma_1^2)$, so $v^*(1) = \gamma_1^{-2} = s_1$, the Maxwell value. For larger K the threshold is far smaller, and Table 13.6 brackets it for $K \leq 800$. The Hermite sum cancels catastrophically, its terms being many orders of magnitude larger than the sum, so the computation runs in ball arithmetic [110] at thousands of bits (8210 at $K = 800$), from inputs that are certified too: the ordinates $\gamma_1, \dots, \gamma_{800}$ computed by Arb, which isolates the zeros, counts them by Turing's method and refines them, and $\operatorname{Var} X = (\log\xi)''(\frac{1}{2})$ from the Taylor series of $\log\xi$, both at 9600 bits. The recurrence $\mathrm{He}_{n+1} = y\mathrm{He}_n - n\mathrm{He}_{n-1}$ gives $Q_{K,v}$ in the monomial basis. On an interval $[c-r, c+r]$ its Taylor coefficients T_m at c come exactly from one polynomial product, and $Q_{K,v}(c+h) \geq T_0 - \sum_{m\geq1}|T_m|r^m$ for $|h| \leq r$, with no remainder; intervals on which this bound is not positive are bisected. A cover of $[0, \sqrt{8K+2}]$ by such intervals, with Proposition 13.4.4 beyond it and the evenness of $Q_{K,v}$, proves $Q_{K,v} > 0$ on the line. In this way $f_K > 0$

is proved for every $1 \le K \le 200$ and for $K = 300$, 400, 600 and 800; f_0 is the normal density of variance $\operatorname{Var} X$, since $Q_{0,v} = 1$. For all these K the minimum of $Q = Q_{K,2\beta_K}$ is attained only at $y = 0$: near 0 because $Q(y) - Q(0) \ge y^2\bigl(b_2 - \sum_{m\ge 2} |b_{2m}| y^{2m-2}\bigr) > 0$ for $0 < |y| \le 2^{-5}$, the b_n being the coefficients of Q, and elsewhere by covers on which the lower bound exceeds $Q(0)$. Moreover $Q_{K,2\beta_K}(0)$ decreases strictly along these K, to 0.289 at $K = 200$ and 0.184 at $K = 800$. For the column $v^*(K)$, a complete cover at $v = t_+ \cdot 2\beta_K$ gives $v^*(K) \le t_+ \cdot 2\beta_K$, and a certified negative value of $Q_{K,v}$ at $v = t_- \cdot 2\beta_K$ gives $v^*(K) > t_- \cdot 2\beta_K$, for dyadic rationals $t_- < t_+$ at most 2^{-18} apart. Just below $v^*(K)$ these negative values lie near the centre, at $|x| < 0.09$ for every $K \ge 1$ in the table and at $|x| < 0.05$ for $K \ge 10$ (`fm_qcert.py`, and for $K = 1$ the formula above); for $K = 0$ there are none, as $v^*(0) = 0$.

K	$2\beta_K$	s_K	$v^*(K)$	$v^*(K)/2\beta_K$	$\min Q_{K,2\beta_K}$	$\mathrm{TV}(f_K, p)$
0	0.046210	0	0	0	1	1.48×10^{-2}
1	0.036199	0.005005	0.005005	0.138	0.862	3.81×10^{-3}
2	0.031674	0.007268	0.004955	0.156	0.804	2.09×10^{-3}
3	0.028477	0.008867	0.005416	0.190	0.762	1.30×10^{-3}
5	0.024472	0.010869	0.005527	0.226	0.705	6.96×10^{-4}
10	0.019121	0.013544	0.005441	0.285	0.623	2.64×10^{-4}
16	0.015756	0.015227	0.005222	0.331	0.566	1.27×10^{-4}
17	0.015343	0.015434	0.005190	0.338	0.558	1.14×10^{-4}
20	0.014273	0.015969	0.005065	0.355	0.538	8.75×10^{-5}
50	0.009108	0.018551	0.004146	0.455	0.430	1.70×10^{-5}
100	0.006222	0.019994	0.003303	0.531	0.355	4.42×10^{-6}
150	0.004907	0.020652	0.002816	0.574	0.316	1.93×10^{-6}
200	0.004122	0.021044	0.002487	0.603	0.289	
300	0.003199	0.021505	0.002059	0.644	0.255	
400	0.002659	0.021775	0.001784	0.671	0.232	
600	0.002036	0.022087	0.001441	0.708	0.203	
800	0.001677	0.022267	0.001229	0.733	0.184	

Table 13.6: The zero completed laws f_K. For $K \le 16$, f_K is a ferromagnetic probability density by Theorem 13.4.3(i) ($s_K \le 2\beta_K$). For every $K \ge 1$ in the table, $Q_{K,2\beta_K} > 0$ on the line is proved by the cover in ball arithmetic described in the text, so f_K is a ferromagnetic probability density by Theorem 13.4.3(ii), and $\min Q_{K,2\beta_K}$ is the value at $y = 0$, where it is attained. The threshold $v^*(K)$ of Proposition 13.4.4 lies in a certified interval of width at most 1.2×10^{-7}, within 6×10^{-7} of the value shown, and the ratio $v^*(K)/2\beta_K$ is correctly rounded. The distances are enclosed in ball arithmetic, as in Table 13.7. The row $K = 0$ is exact apart from $2\beta_0 = \operatorname{Var} X$ and the distance: $Q_{0,v} = 1$, so f_0 is normal and $v^*(0) = 0$. Computed by `fm_qcert.py` and `fm_tvcert.py` from the certified zeros and variance, except $v^*(1) = s_1$.

The question is the trend of the ratio $v^*(K)/2\beta_K$. While it stays below 1, f_K is a probability density. It increases slowly, to 0.733 at $K = 800$, and its slope in $\log K$ falls from 0.110 between $K = 30$ and 50 to 0.086 between $K = 600$ and 800. A straight line in $\log K$ through the last two points reaches 1 near $K = 1.8 \times 10^4$, while the power law $1 - v^*(K)/2\beta_K \approx 1.65K^{-0.27}$, fitted for $K \geq 100$, never does. The data do not decide between them; a computation near $K = 5000$ would. Either outcome is consistent with RH, since Theorem 13.4.3(iii) asks only for the convergence of φ_K, not for positivity.

13.4.4 The Maxwell series

For $K \leq 16$, Theorem 13.4.3(i) reads as a law the factorisation

$$\varphi_K(\theta) = e^{-(\operatorname{Var} X - 3s_K)\theta^2/2}\, \mathbb{E} e^{i\theta \sum_{k \leq K} M_k/\gamma_k}.$$

It holds for every K, though from $K = 17$ on its Gaussian factor has negative variance, and letting $K \to \infty$ in it, with that factor moved to the other side, gives an identity in law. Let $\mathcal{M} = \sum_k M_k/\gamma_k$, summed over the zeros on the critical line. The series converges almost surely, because its terms are independent and centred with summable variances $3\gamma_k^{-2}$.

Theorem 13.4.5. *Let Z be independent of X. Then RH holds if and only if*

$$X + \left(\tfrac{1}{2} \operatorname{Var} X\right)^{1/2} Z \stackrel{d}{=} \mathcal{M}.$$

Proof. $\mathbb{E} e^{i\theta\mathcal{M}} = \prod_k (1 - \theta^2/\gamma_k^2) e^{-\theta^2/2\gamma_k^2}$, an entire function of θ with only real zeros. Under RH, $\sum_k \gamma_k^{-2} = \frac{1}{2} \operatorname{Var} X$, and the product is $\varphi_X(\theta) e^{-\operatorname{Var} X\, \theta^2/4}$, the characteristic function of the left side. If RH fails, the characteristic function of the left side vanishes at a nonreal point, and that of $\mathcal{M}$ does not. □

In words: smoothing the Pólya law by an independent Gaussian of half its variance gives a sum of independent Maxwell variables, one for each zero, scaled by $1/\gamma$. Each Maxwell variable is ferromagnetic and has one pair of zeros and a Gaussian factor, so by Theorem 13.3.1 none of them is a component of X itself; the smoothing is what makes the decomposition possible. The theorem also explains the range $K \leq 16$ in Theorem 13.4.3(i): each Maxwell piece brings its own Gaussian factor, of variance γ_k^{-2}, and once $3s_K$ exceeds $\operatorname{Var} X$ the Gaussian variance $2\beta_K$ of φ_K no longer covers them.

A simulation checks the identity numerically. With 10^7 draws, the first 1700 terms sampled exactly and the rest replaced by a normal variable of variance $3 \sum_{k>1700} \gamma_k^{-2} = 1.49 \times 10^{-3}$ (computed as $3(\frac{1}{2} \operatorname{Var} X - s_{1700})$, which assumes RH above height 2197), the Kolmogorov distance between the empirical law of $\mathcal{M}$ and $p * N(0, \frac{1}{2} \operatorname{Var} X)$ is 1.72×10^{-4}, that is $0.545/\sqrt{n}$, with p value 0.93. The sample variance is 0.0693239 against $\frac{3}{2} \operatorname{Var} X = 0.0693150$, and the sample fourth

cumulant is $(-4.408 \pm 0.066) \times 10^{-4}$ against $\kappa_4(X) = -4.461 \times 10^{-4}$, within 0.8 standard errors (computed by `fm_maxwell.py`). Like Table 13.4, this checks the computation, not RH: below height 2197 the zeros are known to be on the line.

13.4.5 How close

Table 13.7 collects the approximations.

Proposition 13.4.6. *With no hypothesis, the distance δ of Section 13.1.5 is at most* $\mathrm{TV}(f_{150}, p) < 1.932 \times 10^{-6}$*, since f_{150} is a ferromagnetic law by Theorem 13.4.3(ii) and the positivity proof of Section 13.4.3; without that computer assisted proof, f_{16} gives $\delta < 1.266 \times 10^{-4}$.*

RH is the statement that $\delta = 0$. Every density here is even, and $\mathrm{TV}(q, p) = \int_0^\infty |q - p|$ is enclosed in ball arithmetic. On $[0, 3]$, cut into intervals $[c - r, c + r]$ with $r = 2^{-12}$, the integral of $|d|$, $d = q - p$, differs from the exact integral of $|d(c) + d'(c)h|$ over $|h| \le r$ by at most $|d''(c)|r^3/3 + M_3 r^4/12$, where M_3 bounds $|d'''|$ by interval evaluation (for the laws f_K, by $\frac{1}{\pi}\int_0^\infty \theta^3 |\varphi_K(\theta)|\, d\theta$); beyond 3 each density is bounded by an explicit function of known logarithmic decay rate. The laws f_K are built from the certified zeros and variance of Section 13.4.3, so the distances are those of the true laws (`fm_tvcert.py`).

Approximation	Ferromagnetic by	TV to p
$N(0, \mathrm{Var}\, X)$	Gaussian	1.48×10^{-2}
Pólya's Φ^*	Pólya [213]	9.92×10^{-2}
Pólya's Ψ	Proposition 13.4.1	1.61×10^{-1}
$\cosh(t/2)\, p$	de Bruijn, Lagarias	2.73×10^{-3}
$\frac{\sinh(\sqrt{3}t/2)}{\sqrt{3}t/2}\, p$	de Bruijn	2.74×10^{-3}
$e^{t^2/8}\, p$	de Bruijn	2.75×10^{-3}
$e^{t^2/20}\, p$, the flow at $\lambda = 0.2$	Platt and Trudgian	1.10×10^{-3}
f_{16}	Theorem 13.4.3(i)	1.27×10^{-4}
f_{150}	Theorem 13.4.3(ii)	1.93×10^{-6}
$\cosh(\sqrt{0.1}\, t)\, p$	not known	1.09×10^{-3}
uniform quantile mixtures, $J = 5, 20$	not ferromagnetic (Section 13.3.2)	

Table 13.7: Ferromagnetic approximations of the Pólya law and their total variation distances to it, enclosed in ball arithmetic and rounded. A tilt $w\, p$ stands for the normalised law $w\, p / \mathbb{E} w(X)$, and Pólya's kernels are normalised in the same way. Theorem 13.4.3(ii) applies to f_{150} because $f_{150} > 0$, which `fm_qcert.py` proves. Computed by `fm_tvcert.py`.

13.5 The exponential of the Pólya variable

The positive variable e^{2X} lives on the Mellin side of Section 9.1.1: it is the variable $\frac{\pi}{2}S_2$ of Biane, Pitman and Yor, size biased by its power $\frac{1}{4}$. Size biasing a sum of independent gamma variables by a power $q \in (0,1)$ has a simple effect, which we prove for integer shapes (Theorem 13.5.5): it raises every shape by q, at the cost of an independent generalised gamma convolution that puts mass q on each gap between consecutive rates. For e^{2X} this gives an exact factorization against $\frac{\pi}{2}S_{9/4}$. From it, e^{2X} is self decomposable with a Lévy density built from Riemann's theta function, although it is not a GGC. Along the family $e^{\beta X}$ the picture is then complete except on one interval: $e^{\beta X}$ is not infinitely divisible for $0 < |\beta| < 2$, it is self decomposable and not a GGC at $|\beta| = 2$, and it has a hyperbolically completely monotone density, so it is a GGC, for $|\beta| \geq 5.5791422$, a bound certified in ball arithmetic. None of this involves a hypothesis on the zeros. GGC, HCM and self decomposable laws are as in Section 1.3.2.

13.5.1 Size biased laws and their phase

Write $B_0 = \frac{\pi}{2}S_2 = \sum_{m\geq 1} \Gamma_m/(\pi m^2)$, with independent Gamma$(2,1)$ variables Γ_m as in Section 9.1.1. It is a GGC with Thorin atoms of mass 2 at the points πm^2 and Laplace transform $(\sqrt{\pi\lambda}/\sinh\sqrt{\pi\lambda})^2$, and by (9.1.1) $\mathbb{E}B_0^\sigma = 2\xi(2\sigma)$ for every complex σ. With Riemann's function $\omega(x) = \sum_{n\geq 1} e^{-\pi n^2 x}$ of Section 1.2, its density is

$$f_0(y) = \sum_{m\geq 1} \big(4\pi^2 m^4 y - 6\pi m^2\big) e^{-\pi m^2 y} = 4y\,\omega''(y) + 6\,\omega'(y)\,:$$

two integrations by parts give $\int_0^\infty y^\sigma (4y\omega'' + 6\omega')\,dy = 2\sigma(2\sigma-1)\pi^{-\sigma}\Gamma(\sigma)\zeta(2\sigma) = 2\xi(2\sigma)$. For real q let B_q be B_0 size biased by B_0^q, with density $y^q f_0(y)/(2\xi(2q))$, so that

$$\mathbb{E}B_q^\sigma = \frac{\xi(2\sigma+2q)}{\xi(2q)}, \qquad e^{2X} \stackrel{d}{=} B_{1/4},$$

the second because $\mathbb{E}e^{2\sigma X} = \xi(\frac{1}{2}+2\sigma)/\xi(\frac{1}{2})$. Equivalently, $\Phi(t) = 4x^{9/4}\omega''(x) + 6x^{5/4}\omega'(x)$ at $x = e^{2t}$ turns the density $p(\frac{1}{2}\log y)/(2y)$ of e^{2X} into $y^{1/4} f_0(y)/(2\xi(\frac{1}{2}))$. In the notation of Section 9.1.1, where V_α is the Esscher tilt of the Pólya density at the basepoint α, $B_q = e^{2V_{2q}}$; so the B_q with $0 < q < 1$ correspond to the basepoints $0 < \alpha < 2$, and $e^{2X} = e^{2V_{1/2}}$ to the centre. Since X is symmetric, $e^{-\beta X} \stackrel{d}{=} e^{\beta X}$, and for $\beta > 0$, $e^{\beta X} = B_{1/4}^{\beta/2}$.

For small β the tail of X decides the question of infinite divisibility.

Proposition 13.5.1. *For $0 < |\beta| < 2$ the variable $e^{\beta X}$ is not infinitely divisible.*

Proof. Take $\beta > 0$. The first term of Φ dominates for large t, so $\log P(X > x) = -\pi e^{2x} + O(x)$ and, with $Y = e^{\beta X}$, $\log P(Y > y) = -\pi y^{2/\beta} + O(\log y)$. Since

$2/\beta > 1$, $\mathbb{E}\exp(cY\log Y) < \infty$ for every $c > 0$. By Sato's theorem on the tails of infinitely divisible laws [232, Theorem 26.1], an infinitely divisible law with this property has no Lévy measure, so it is Gaussian or degenerate, and Y is neither. □

The other results of the section come from the continuation of the Laplace transform

$$\Lambda_q(\lambda) = \mathbb{E}e^{-\lambda B_q} = \frac{1}{2\xi(2q)} \int_0^\infty y^q f_0(y)\, e^{-\lambda y}\, dy,$$

which is finite for $\lambda > -\pi$. Let $0 < q < 1$, the range of the results below, fix $M \geq 1$ and let R_M be the tail $\sum_{m>M}$ of the series for f_0. It is bounded near 0, because $f_0(y) = y^{-5/2} f_0(1/y)$ by the evenness of Φ, so that $f_0(y) \to 0$ as $y \downarrow 0$, and it is $O(y\, e^{-\pi(M+1)^2 y})$ at infinity. Integrating the first M terms gives, for $\operatorname{Re}\lambda > -\pi$,

$$2\xi(2q)\,\Lambda_q(\lambda) = \sum_{m \leq M} \Big[\frac{4\pi^2 m^4 \Gamma(q+2)}{(\pi m^2 + \lambda)^{q+2}} - \frac{6\pi m^2 \Gamma(q+1)}{(\pi m^2 + \lambda)^{q+1}} \Big] + \int_0^\infty y^q R_M(y)\, e^{-\lambda y}\, dy, \tag{13.5.1}$$

with principal powers. The last integral is analytic for $\operatorname{Re}\lambda > -\pi(M+1)^2$ and real on the real axis. So (13.5.1) continues Λ_q analytically to the plane cut along $(-\infty, -\pi]$, and Λ_q has boundary values $\Lambda_q(-r+i0)$ that are continuous in $r \neq \pi m^2$, while near each $-\pi m^2$ it behaves like a multiple of $(\pi m^2 + \lambda)^{-q-2}$. Size biasing by y^q has turned the double poles of $(\sqrt{\pi\lambda}/\sinh\sqrt{\pi\lambda})^2$ into branch points.

The boundary values carry the Thorin measure. If G is a GGC with drift b and Thorin measure μ, its Laplace transform $H(\lambda) = \exp\{-b\lambda - \int \log(1 + \lambda/t)\, \mu(dt)\}$ continues analytically to $\mathbb{C} \setminus (-\infty, 0]$, and since $\operatorname{Im}\log(1 + (-r + i\epsilon)/t)$ tends, as $\epsilon \downarrow 0$, to π for $t < r$, to $\frac{\pi}{2}$ for $t = r$ and to 0 for $t > r$, the imaginary part of the exponent of $H(-r + i\epsilon)$ tends to $-\pi\big(\mu((0,r)) + \frac{1}{2}\mu(\{r\})\big)$; at a point r that is not an atom of μ and where $H(-r+i0)$ is finite and not zero, the argument of $H(-r+i0)$ is therefore $-\pi\mu((0,r))$, modulo 2π. We call a continuous determination of $-\frac{1}{\pi}\arg H(-r + i0)$ on an interval a phase of H there. Fourati used the argument of such boundary values to show that certain powers of positive stable variables are not GGC [77]. The next lemma states what the phase determines when the boundary values do not vanish.

Lemma 13.5.2. *Let G be a GGC with drift b, Thorin measure μ and Laplace transform H, and let $I \subset (0, \infty)$ be an open interval such that for each $r \in I$, $H(-r + i\epsilon)$ tends as $\epsilon \downarrow 0$ to a nonzero limit $H(-r + i0)$, continuous in r. Then μ has no atoms in I, and $\mu((0,r)) - \phi(r)$ is constant on I for every phase ϕ of H on I.*

Proof. Let $L(\lambda) = -b\lambda - \int \log(1 + \lambda/t)\, \mu(dt)$, so that $H = e^L$. For $r \in I$ and $0 < \epsilon \leq 1$, $\operatorname{Im}\log(1 + (-r + i\epsilon)/t) = \arg(t - r + i\epsilon)$ lies in $(0, \pi)$, tends to π, $\frac{\pi}{2}$ or 0 according as $t < r$, $t = r$ or $t > r$, and is at most $2\epsilon/t$ for $t \geq 2r$. Since

$\mu((0,2r)) < \infty$ and $\int_{[2r,\infty)} \mu(dt)/t < \infty$, dominated convergence gives $\operatorname{Im} L(-r + i\epsilon) \to -\pi\big(\mu((0,r)) + \frac{1}{2}\mu(\{r\})\big)$. If $\mu(\{r\}) = a > 0$ then, since $|t - r + i\epsilon| \le t + r + 1$,

$$\operatorname{Re} L(-r + i\epsilon) \;\ge\; br - \int_{t \ne r} \log\Big(1 + \frac{r+1}{t}\Big)\,\mu(dt) - a \log\frac{\epsilon}{r} \;\longrightarrow\; \infty,$$

so $|H(-r + i\epsilon)| \to \infty$, which is excluded. Hence μ has no atoms in I, and $e^{i \operatorname{Im} L(-r+i\epsilon)}$, the argument factor of $H(-r + i\epsilon)$, tends to that of $H(-r + i0)$. So $\mu((0,r)) - \phi(r)$ is an even integer for each $r \in I$, and being continuous on I it is constant. □

The same lower bound for $\operatorname{Re} L$ shows that the boundary values of a GGC transform never vanish.

THEOREM 13.5.3. *For $0 < q < 1$ the variable B_q is not a GGC. In particular e^{2X} is not a GGC.*

Proof. Let $r > \pi$, $r \ne \pi m^2$, and take M with $\pi(M+1)^2 > r$ in (13.5.1). At $\lambda = -r + i0$ the terms with $\pi m^2 > r$ and the last integral are real, and $(\pi m^2 - r + i0)^{-p} = (r - \pi m^2)^{-p} e^{-i\pi p}$ for $\pi m^2 < r$, so

$$2\xi(2q)\,\operatorname{Im} \Lambda_q(-r + i0) = -\sin(\pi q) \sum_{\pi m^2 < r} \Big[\frac{4\pi^2 m^4 \Gamma(q+2)}{(r - \pi m^2)^{q+2}} + \frac{6\pi m^2 \Gamma(q+1)}{(r - \pi m^2)^{q+1}}\Big] < 0. \tag{13.5.2}$$

Suppose that B_q is a GGC, and apply Lemma 13.5.2 on $I = (\pi, 4\pi)$. By (13.5.2) a phase ϕ of Λ_q on I takes its values in $(2k, 2k+1)$ for a single integer k. As $r \downarrow \pi$ the first term of (13.5.1) dominates, so that $\Lambda_q(-r + i0) \sim c\,(r - \pi)^{-q-2} e^{-i\pi(q+2)}$ with $c > 0$, and $\phi(r) \to 2k + q$. As $r \uparrow 4\pi$ the term with $m = 2$ dominates and is positive, and $\phi(r) \to 2k$. So ϕ falls by q across I, while $\mu((0,r)) = \phi(r) + \text{const}$ is nondecreasing, a contradiction. □

So size biasing by a power $q \in (0,1)$ destroys the GGC property of B_0. Size biasing a GGC by a negative power gives a GGC: Fourati's Lemma 6.3 [77] states this for every real exponent, and its proof, which multiplies the variable by one plus an independent gamma variable of shape $\beta > 0$ and lets the rate of the gamma variable tend to 0, covers the size bias by $x^{-\beta}$, once slips in the printed normalising constants are corrected. For positive exponents the statement fails. By Theorem 13.5.3 it fails for every exponent in $(0,1)$, and it fails at 1 as well: the size bias of order one of the GGC $e_1 + \frac{1}{2}e_2$ has Laplace transform $\frac{4}{3}(2\lambda + 3)(1 + \lambda)^{-2}(2 + \lambda)^{-2}$, which vanishes at the point $-\frac{3}{2}$ of the cut. With the negative orders, B_q is a GGC for every $q \le 0$ and for no $q \in (0,1)$.

13.5.2 The size bias principle

A gamma variable of shape t size biased by y^q is a gamma variable of shape $t+q$ with the same rate. For a sum of gamma variables with different rates the same happens at every rate, up to an independent GGC that lives between the rates. The proof uses divided differences. For a polynomial Q with real zeros $s_1 \le \cdots \le s_n$, listed with multiplicity, and a function h analytic near $[s_1, s_n]$, put

$$h[Q] = \frac{1}{2\pi i} \oint \frac{h(\zeta)}{Q(\zeta)}\, d\zeta = \sum_{\zeta_0} \mathrm{Res}_{\zeta=\zeta_0} \frac{h(\zeta)}{Q(\zeta)},$$

the contour enclosing the zeros of Q inside the domain of h and the sum running over the distinct zeros ζ_0 of Q, so that $h[Q]$ is the divided difference of h at $s_1, \dots, s_n$ when Q is monic. The residue at a zero of multiplicity j involves only the derivatives of h of orders below j there. By the formula of Hermite and Genocchi, $h[Q]$ is the integral of $h^{(n-1)}(\sum_j w_j s_j)$ over the simplex of weights $w_j \ge 0$, $\sum_j w_j = 1$, with its uniform measure of total mass $1/(n-1)!$; so $h[Q]$ has the sign of $h^{(n-1)}$ when that sign is constant on $[s_1, s_n]$.

Lemma 13.5.4. *Let $Q(\zeta) = \prod_{n=1}^{K} (\zeta - \tau_n)^{t_n}$ with $\tau_1 < \cdots < \tau_K$ and integers $t_n \ge 1$, let $1 \le k < K$ and $\tau_k < x < \tau_{k+1}$, and write $Q = Q_- Q_+$ with $Q_-(\zeta) = \prod_{n \le k} (\zeta - \tau_n)^{t_n}$, of degree N_-, and Q_+ of degree N_+. For $p > 0$ put*

$$D_+(x) = h_+[Q_+], \qquad h_+(\zeta) = \frac{(\zeta - x)^{-p}}{Q_-(\zeta)},$$

$$D_-(x) = h_-[Q_-], \qquad h_-(\zeta) = \frac{(x - \zeta)^{-p}}{Q_+(\zeta)}.$$

Then $(-1)^{N_+ - 1} D_+(x) > 0$ and $(-1)^{N_+} D_-(x) > 0$, and as functions of $x \in (\tau_k, \tau_{k+1})$, $|D_+|$ increases and $|D_-|$ decreases.

Proof. On (x, ∞) the function h_+ is a product of the completely monotone functions $(\zeta - x)^{-p}$ and $(\zeta - \tau_n)^{-1}$, $n \le k$, so $(-1)^j h_+^{(j)} > 0$ there for every j, and the formula of Hermite and Genocchi with $j = N_+ - 1$ gives the first sign. On $(-\infty, x)$ the function $(-1)^{N_+} h_-(\zeta) = (x - \zeta)^{-p} \prod_{n>k} (\tau_n - \zeta)^{-t_n}$ is a product of functions all of whose derivatives are positive, which gives the second. Finally $\partial_x (\zeta - x)^{-p} = p(\zeta - x)^{-p-1}$ and $\partial_x (x - \zeta)^{-p} = -p(x - \zeta)^{-p-1}$, so D_+' and D_-' are p and $-p$ times the same expressions with $p+1$ in place of p: D_+' has the sign of D_+, and D_-' the sign opposite to that of D_-. □

Theorem 13.5.5 (size bias principle). *Let $Y = \sum_{n=1}^{K} \Gamma_n / \tau_n$ with $0 < \tau_1 < \cdots < \tau_K$ and independent gamma variables Γ_n with integer shapes $t_n \ge 1$ and unit scale, let $0 < q < 1$, and let Y_q be Y size biased by Y^q. Then*

$$Y_q + G \overset{d}{=} \sum_{n=1}^{K} \frac{\Gamma_n'}{\tau_n},$$

where the Γ'_n are independent gamma variables with shapes $t_n + q$ and unit scale, and G, independent of Y_q, is a GGC without drift whose Thorin measure has a continuous distribution function θ_G that vanishes on $(0, \tau_1]$, increases by exactly q on each interval $[\tau_k, \tau_{k+1}]$ and is constant on $[\tau_K, \infty)$.

Proof. Let $N = \sum_n t_n$ and $Q(\zeta) = \prod_n (\zeta - \tau_n)^{t_n}$. The density of Y is $c_0(-1)^{N-1}(e^{-\zeta y})[Q]$ with $c_0 = \prod_n \tau_n^{t_n}$, as taking Laplace transforms under the contour integral shows: $(1/(\zeta+\lambda))[Q] = -1/Q(-\lambda)$, by the residue at $\zeta = -\lambda$. Multiplying by y^q before integrating gives, for $\lambda \geq 0$,

$$\Lambda(\lambda) = \mathbb{E}e^{-\lambda Y_q} = c\,(-1)^{N-1} g_\lambda[Q], \qquad g_\lambda(\zeta) = (\zeta+\lambda)^{-1-q},$$

with a constant $c > 0$, and since $g_\lambda^{(N-1)} = (-1)^{N-1}(1+q)\cdots(N-1+q)\,(\zeta+\lambda)^{-N-q}$, the formula of Hermite and Genocchi gives

$$\Lambda(\lambda) = \frac{\mathbb{E}(\bar{s}+\lambda)^{-N-q}}{\mathbb{E}\bar{s}^{\,-N-q}}, \qquad \bar{s} = \sum_{j=1}^{N} w_j s_j,$$

where $s_1, \ldots, s_N$ are the rates listed with multiplicity and (w_j) is uniform on the simplex. This continues Λ analytically to $\mathbb{C} \setminus (-\infty, -\tau_1]$, where the residue form $c(-1)^{N-1}g_\lambda[Q]$ remains valid, and shows that $|\Lambda(\lambda)|$ is of exact order $|\lambda|^{-N-q}$ as $|\lambda| \to \infty$, uniformly up to the cut.

Boundary values. Let $\tau_k < r < \tau_{k+1}$. At points $\zeta < r$ the derivatives of $(\zeta - r + i0)^{-1-q}$ are $e^{-i\pi(1+q)}$ times those of $(r-\zeta)^{-1-q}$. Splitting the residues of g_λ/Q into those at τ_n with $n > k$ and those with $n \leq k$ gives, with $D_\pm$ of Lemma 13.5.4 at $p = 1+q$ and the signs found there,

$$\Lambda(-r+i0) = c\,(-1)^{N-1}\big(D_+(r) + e^{-i\pi(1+q)}D_-(r)\big) = c\,(-1)^{N-}z(r),$$
$$z(r) = |D_+(r)| + e^{-i\pi q}|D_-(r)|.$$

The point $z(r)$ is the sum of two nonzero vectors at the angle $\pi q < \pi$, so its argument lies in $(-\pi q, 0)$. The ratio $|D_-|/|D_+|$ decreases, from ∞ as $r \downarrow \tau_k$, where D_- has the singular term $(r-\tau_k)^{-t_k-q}$, to 0 as $r \uparrow \tau_{k+1}$; so $\arg z(r)$ increases from $-\pi q$ to 0. For $r > \tau_K$ the same computation with $Q_+ = 1$ gives $\Lambda(-r+i0) = c(-1)^{N-1}e^{-i\pi(1+q)}D_-(r)$ with $D_- > 0$.

The factor G. Put $\theta_G = 0$ on $(0, \tau_1]$, $\theta_G(r) = kq + \frac{1}{\pi}\arg z(r)$ for $\tau_k < r < \tau_{k+1}$, and $\theta_G = (K-1)q$ on $[\tau_K, \infty)$. This function is continuous and nondecreasing. Near τ_k, $|D_+/D_-|$ is $(r-\tau_k)^{t_k+q}$ times a function that is continuously differentiable up to τ_k, and similarly near τ_{k+1}; since $t_k + q > 1$, θ_G is continuously differentiable, with derivative 0 at the points τ_n. Let G be the GGC without drift and with Thorin measure $d\theta_G$, with Laplace transform $H_G(\lambda) = \exp\{-\int \log(1+\lambda/t)\,d\theta_G(t)\}$, and let $\Lambda_+(\lambda) = \prod_n (1+\lambda/\tau_n)^{-t_n-q}$, the Laplace transform of the right side. The

function $J = \Lambda H_G/\Lambda_+$ is analytic on $\mathbb{C}\setminus(-\infty,-\tau_1]$ and positive on $(-\tau_1,\infty)$. For $\tau_k < r < \tau_{k+1}$ the arguments of Λ, H_G and $1/\Lambda_+$ at $-r+i0$ are, modulo 2π,

$$\pi N_- + \arg z(r), \qquad -\pi kq - \arg z(r), \qquad \pi(N_- + kq),$$

and for $r > \tau_K$ they are $\pi N - \pi q$, $-\pi(K-1)q$ and $\pi(N+Kq)$. In both cases they add up to a multiple of 2π, so J has real boundary values from above, by symmetry the same boundary values from below, and it continues analytically across the cut between the points $-\tau_n$. Near $-\tau_n$, $\Lambda = O(|\lambda+\tau_n|^{-t_n-q})$ by the residue form, and $1/\Lambda_+$ cancels this, while H_G is bounded because θ_G has a bounded density; so these singularities are removable, and J is entire. As $|\lambda|\to\infty$, $|\Lambda|$, $|1/\Lambda_+|$ and $|H_G|$ are of the exact orders $|\lambda|^{-N-q}$, $|\lambda|^{N+Kq}$ and $|\lambda|^{-(K-1)q}$, so J is bounded. By Liouville's theorem $J \equiv J(0) = 1$, that is $\Lambda H_G = \Lambda_+$, which is the identity in law. □

For two exponential variables everything is explicit. If $K = 2$ and $t_1 = t_2 = 1$, then Λ is proportional to $(\tau_1+\lambda)^{-1-q} - (\tau_2+\lambda)^{-1-q}$, so that

$$H_G(\lambda) = \frac{\tau_2^{1+q} - \tau_1^{1+q}}{(\tau_2+\lambda)^{1+q} - (\tau_1+\lambda)^{1+q}},$$
$$\theta_G(r) = \frac{1}{\pi}\arg\Big((\tau_2 - r)^{1+q} + e^{i\pi q}(r-\tau_1)^{1+q}\Big), \qquad \tau_1 < r < \tau_2.$$

The variable B_0 is a sum of this kind with infinitely many rates πm^2, each of shape 2, and the theorem passes to the limit.

THEOREM 13.5.6. *For $0 < q < 1$,*

$$B_q + G_q \;\stackrel{d}{=}\; \frac{\pi}{2} S_{2+q},$$

where G_q is independent of B_q and is a GGC without drift whose Thorin measure has a continuous distribution function θ_q with $\theta_q = 0$ on $(0,\pi]$ and $\theta_q(\pi m^2) = (m-1)q$ for every $m \ge 1$. In particular $e^{2X} + G_{1/4} \stackrel{d}{=} \frac{\pi}{2} S_{9/4}$.

Proof. Let $B^{(M)} = \sum_{m\le M} \Gamma_m/(\pi m^2)$ and let $B_q^{(M)}$ be its size bias of order q. By Theorem 13.5.5, $B_q^{(M)} + G_M \stackrel{d}{=} \sum_{m\le M} \Gamma'_m/(\pi m^2)$, with shapes $2+q$ on the right. As $M\to\infty$, $\mathbb{E}[(B^{(M)})^q e^{-\lambda B^{(M)}}] \to \mathbb{E}[B_0^q e^{-\lambda B_0}]$ for $\lambda \ge 0$ by dominated convergence, since $B^{(M)} \le B_0$; so $B_q^{(M)} \to B_q$ in law, and the right sides converge in law to $\frac{\pi}{2} S_{2+q}$, whose Laplace transform is $\Lambda_+(\lambda) = (\sqrt{\pi\lambda}/\sinh\sqrt{\pi\lambda})^{2+q}$. Hence the Laplace transforms of the G_M converge to $H = \Lambda_+/\Lambda_q$, which is continuous at 0. So G_M converges in law to a variable G_q with Laplace transform H, G_q is a GGC by the closure theorem of Section 4.1.1, and $B_q + G_q \stackrel{d}{=} \frac{\pi}{2} S_{2+q}$. A GGC with drift b is at least b, while $\frac{\pi}{2} S_{2+q}$ is smaller than any given positive number with positive probability, so G_q has no drift.

It remains to find the Thorin measure μ of G_q. By (13.5.1), near each $-\pi m^2$ both Λ_q and Λ_+ are $(\pi m^2+\lambda)^{-2-q}$ times functions with continuous nonzero boundary values, and between these points $\Lambda_q(-r+i0) \neq 0$ by (13.5.2). So H has continuous nonzero boundary values on all of $(0,\infty)$, and by Lemma 13.5.2 μ has no atoms and $\mu((0,r)) = \phi(r)$, where ϕ is the phase of H that vanishes on $(0,\pi)$, where $H>0$. On $(\pi m^2, \pi(m+1)^2)$ the argument of $\Lambda_+(-r+i0)$ is $-\pi(2+q)m$, and $\Lambda_q(-r+i0)$ lies in the lower half plane by (13.5.2), with argument tending to $-\pi(2+q)$ modulo 2π at the left end and to 0 at the right end, as in the proof of Theorem 13.5.3. So, modulo 2, ϕ takes its values in $(mq-1, mq)$ on this interval and tends to $(m-1)q$ at its left end and to mq at its right end. Since ϕ is continuous and $\phi(\pi)=0$, induction on m gives $\phi(\pi m^2) = (m-1)q$. □

The theorem gives B_q a signed Thorin measure, $(2+q)\sum_m \delta_{\pi m^2} - d\theta_q$: atoms of mass $2+q$ at the points πm^2, as for $\frac{\pi}{2}S_{2+q}$, and a negative continuous part of mass q on each gap, which is what Theorem 13.5.3 detects. Since $\theta_q(r) \le q\,\#\{m : \pi m^2 < r\}$, the phase of B_q lies between those of $\frac{\pi}{2}S_2$ and $\frac{\pi}{2}S_{2+q}$.

13.5.3 Self decomposability and the theta Lévy density

Theorem 13.5.7. *For $0<q<1$ the variable B_q is self decomposable, with no drift and Lévy density $k_q(x)/x$, where*

$$k_q(x) = (2+q)\,\omega(x) - \int_0^\infty e^{-tx}\,d\theta_q(t), \qquad 2\,\omega(x) \le k_q(x) \le (2+q)\,\omega(x).$$

In particular e^{2X} is self decomposable, hence infinitely divisible and unimodal, its Lévy density lies between $2\omega(x)/x$ and $\frac{9}{4}\omega(x)/x$, and it is not a GGC.

Proof. By Theorem 13.5.6, $-\log\Lambda_q = -\log\Lambda_+ + \log H$, and Frullani's integral $\log(1+\lambda/t) = \int_0^\infty (1-e^{-\lambda x})e^{-tx}\,dx/x$, applied to the atoms of $\frac{\pi}{2}S_{2+q}$ and to the Thorin measure of G_q, gives

$$-\log\Lambda_q(\lambda) = \int_0^\infty (1-e^{-\lambda x})\,k_q(x)\,\frac{dx}{x}.$$

On $[\pi m^2, \pi(m+1)^2]$ the measure $d\theta_q$ has mass q and $e^{-tx} \le e^{-\pi m^2 x}$, so $0 \le \int e^{-tx}\,d\theta_q(t) \le q\,\omega(x)$, which gives the bounds. In particular $k_q>0$, so B_q is infinitely divisible with Lévy density $k_q(x)/x$, and it is self decomposable if k_q is nonincreasing (Section 1.3.2), that is, if

$$\int_0^\infty te^{-tx}\,d\theta_q(t) \;\le\; (2+q)\sum_{m\ge1} \pi m^2 e^{-\pi m^2 x}, \qquad x>0.$$

Bound the contribution of $[\pi m^2, \pi(m+1)^2]$ to the left side by $q\,s_m(x)$, with $s_m(x) = \max\{te^{-tx} : \pi m^2 \le t \le \pi(m+1)^2\}$. For $m \ge 2$, $s_m(x) \le \pi(m+1)^2 e^{-\pi m^2 x} \le$

$\frac{9}{4}\pi m^2 e^{-\pi m^2 x}$, and $\frac{9}{4}q < 2+q$, which leaves $(2-\frac{5}{4}q)\pi m^2 e^{-\pi m^2 x}$ of the mth term on the right. It remains to show $q\,s_1(x) \le (2+q)\pi e^{-\pi x} + (2-\frac{5}{4}q)4\pi e^{-4\pi x}$. If $x \ge 1/\pi$, then $s_1(x) = \pi e^{-\pi x}$. If $x \le 1/(4\pi)$, then $s_1(x) = 4\pi e^{-4\pi x}$, and $4q \le 4(2-\frac{5}{4}q) + (2+q)$ because $8q \le 10$. Otherwise $s_1(x) = 1/(ex)$, and $(2+q)\pi x\, e^{1-\pi x} \ge (2+q)\frac{1}{4}e^{3/4} > 1 > q$, because ue^{1-u} increases on $(\frac{1}{4}, 1)$. Self decomposable laws are unimodal [274], and $e^{2X} = B_{1/4}$ is not a GGC by Theorem 13.5.3. □

So e^{2X} is a natural self decomposable law outside the GGC class, with a Lévy density that is explicit up to a correction of at most $\frac{1}{9}$ of its main term: in the order of Lévy densities it lies between $\frac{\pi}{2}S_2$ and $\frac{\pi}{2}S_{9/4}$, whose Lévy densities are $2\omega(x)/x$ and $\frac{9}{4}\omega(x)/x$. Near 0, $\omega(x) \sim 1/(2\sqrt{x})$, and $0 \le q\sqrt{t/\pi} - \theta_q(t) \le 2q$, since $\theta_q = 0$ on $(0, \pi]$ and, on $[\pi m^2, \pi(m+1)^2]$, the first lies between mq and $(m+1)q$ and the second between $(m-1)q$ and mq; so the subtracted fraction $1 - k_q/((2+q)\omega)$ tends to $q/(2+q)$ as $x \to 0$.

The script `fm_sizebias.py` checks the three theorems numerically. It continues Λ_q by (13.5.1) with $M = 5$, at $30 + r/4$ digits, on 1800 points of $r \in [0.05, 90]$, for $q = 0.1$, 0.25, 0.5, 0.75 and 0.9. In every case the phase of Λ_q decreases along the grid on each of the four gaps $(\pi m^2, \pi(m+1)^2)$, $m \le 4$, so that θ_q increases, and at the last grid point below $\pi(m+1)^2$, θ_q is within 10^{-7} of mq, as Theorem 13.5.6 requires. Extending θ_q beyond the scan by $q(\sqrt{t/\pi} - 1)$, the value it takes at the points πm^2, the function k_q is positive and decreasing on 300 points of $[10^{-3}, 3]$, with minimum between 1.69×10^{-4} and 2.34×10^{-4} at $x = 3$, and the subtracted fraction at $x = 10^{-3}$ is 0.046, 0.108, 0.194, 0.264 and 0.300, against the limits 0.048, 0.111, 0.200, 0.273 and 0.310.

The factorization can also be tested away from the cut, through the characteristic function on the real line. For a positive infinitely divisible variable B with no drift and Lévy density $k(x)/x$, $-\log|\mathbb{E}e^{iyB}|^2 = \int_0^\infty 4\sin^2(\frac{yx}{2})\,k(x)\,dx/x$, and Frullani's integral $\int_0^\infty 4\sin^2(\frac{yx}{2})e^{-tx}\,dx/x = \log(1 + y^2/t^2)$ turns this into the Thorin form $\int \log(1+y^2/t^2)$ against the Thorin measure, here the signed one. For B_q the atoms sum in closed form, by $\prod_m (1 + u^4/m^4) = (\cosh\sqrt{2}\pi u - \cos\sqrt{2}\pi u)/(2\pi^2 u^2)$:

$$-\log\left|\mathbb{E}e^{iyB_q}\right|^2 = (2+q)\log\frac{\cosh\sqrt{2\pi y} - \cos\sqrt{2\pi y}}{2\pi y} - \int_0^\infty \log\left(1 + \frac{y^2}{t^2}\right) d\theta_q(t).$$

Table 13.8 compares, for e^{2X}, the direct Fourier integral of the density with the Thorin form and the theta Lévy form, both built from the computed $\theta_{1/4}$. The two forms agree with each other to a relative 2×10^{-7}, and with the direct integral to a relative 1.2×10^{-5}, while the atoms alone, that is $\frac{\pi}{2}S_{9/4}$, are off by 3 to 6 percent. As a last check, $\mathbb{E}G_{1/4} = \mathbb{E}\frac{\pi}{2}S_{9/4} - \mathbb{E}e^{2X} = \frac{3\pi}{8} - \xi(\frac{5}{2})/\xi(\frac{1}{2}) = 0.0815947$, and $\int \theta_{1/4}(t)\,t^{-2}dt = 0.0815351$ from the computed Thorin measure; the extension beyond $t = 90$ differs from $\theta_{1/4}$ by at most q, which can move the integral by at most $q/90 < 0.003$.

y	direct	Thorin form	theta Lévy form	atoms alone
0.5	0.0588838	0.0588840	0.0588840	0.0609722
1	0.227615	0.227616	0.227616	0.235869
2	0.808286	0.808290	0.808290	0.839868
4	2.347416	2.347431	2.347431	2.457076
8	5.261493	5.261552	5.261553	5.575941

Table 13.8: $-\log|\mathbb{E}e^{iye^{2X}}|^2$ computed four ways: by a direct Fourier integral of the density $y^{1/4}f_0(y)/(2\xi(\frac{1}{2}))$; in the Thorin form with the atoms $\frac{9}{4}$ at the points πm^2 summed in closed form and the continuous part from the computed $\theta_{1/4}$; in the theta Lévy form $\int 4\sin^2(\frac{yx}{2})k_{1/4}(x)\,dx/x$; and from the atoms alone, which is $-\log|\mathbb{E}e^{iy\frac{\pi}{2}S_{9/4}}|^2$. Computed by `fm_sizebias.py`.

On the dual side the roles are reversed. Section 10.1.4 puts the characteristic function $\xi(\frac{1}{2})/\xi(\frac{1}{2}+u)$ of $Y_{1/2}$ in the same form, with no hypothesis: $\log(\xi(\frac{1}{2}+u)/\xi(\frac{1}{2})) = 2\int_0^\infty(1-\cos ux)\,K_{1/2}(x)\,dx/x$ for real u. So the logarithm of the moment generating function of X has the form of $-\log|\mathbb{E}e^{iyB_q}|^2$, with the Poisson trace $K_{1/2}$, a sum over the zeros, in place of the theta kernel k_q. The Poisson trace is completely monotone exactly when RH holds (Corollary 10.1.4), while $k_{1/4}$ is positive and decreasing with no hypothesis.

13.5.4 Hyperbolically completely monotone powers

Large powers bring the GGC property back. The criterion is the elasticity criterion of Section 4.3.1, applied as in Proposition 4.3.27 with the density g of $\log W$ replaced by the Pólya density. The kernel Φ is analytic in the strip $|\operatorname{Im} t| < \pi/4$, where $\operatorname{Re} e^{2t} > 0$, and since it is real and even, $F(v) = \Phi(iv)$ is real for real v. We have $F(0) = 0.8933938$; F has a local maximum $F(v_m) = 7.0315837$ at $v_m = 0.5630962646$, and it vanishes at $v = 0.6388301$, which is the zero α_1 of the continuation in Proposition 7.1.18(iii).

Proposition 13.5.8. *The variable $e^{\beta X}$ has an HCM density, and so is a GGC, for $|\beta| \geq 5.5791422$, and it does not have an HCM density for $|\beta| \leq 5.5791395$. The proof of the first statement is computer assisted.*

Proof. Let $\beta > 0$. The density $p(\beta^{-1}\log y)/(\beta y)$ of $e^{\beta X}$ has elasticity $1-\beta^{-1}(\Phi'/\Phi)(\beta^{-1}\log y)$, and the upper half plane in y corresponds to the strip $0 < \operatorname{Im} t < \pi/\beta$. By Theorem 4.3.2, as in Proposition 4.3.27, $e^{\beta X}$ has an HCM density if and only if Φ continues to a zero free analytic function on $|\operatorname{Im} t| < \pi/\beta$ and

$$h(x,v) = \operatorname{Im}\Big(-\frac{\Phi'}{\Phi}\Big)(x+iv) \geq 0 \qquad \text{for } 0 < v < \pi/\beta,$$

and HCM laws are GGC by (F5) of Section 4.2.1. On the imaginary axis $h(0,v) =$

$F'(v)/F(v)$. Ball arithmetic [110] encloses v_m in an interval of width 2×10^{-17} with certified signs $F' > 0$ at its left end and $F > 0 > F'$ at its right end v_b. If $\pi/\beta > v_b$, the strip contains iv_b, where Φ is analytic and $h < 0$, so $e^{\beta X}$ is not HCM. Since $\pi/v_b > 5.5791395$, this proves the second statement.

For the first, let $v_* = 0.563096$, so that $\pi/v_* < 5.5791422$, and let φ_1 be the term $n = 1$ of Φ. Three estimates, in ball arithmetic with rigorous bounds for the tail of the series, show that $h \geq 0$ on $0 < v \leq v_*$. (Z) For $\operatorname{Re} t \geq 0$ and $|\operatorname{Im} t| \leq 0.5634$, $|\Phi/\varphi_1 - 1| \leq 0.6015$, so $\Phi \neq 0$ there, and also for $\operatorname{Re} t \leq 0$, because $\Phi(-\bar{t}) = \overline{\Phi(t)}$. (T) On the segment $v = v_*$, $0 \leq x \leq 2$, adaptive bisection into 2896 intervals encloses h in positive balls. The smallest lower bound, 1.8×10^{-8}, occurs next to $x = 0$, where $h(0, v)$ tends to 0 as $v \uparrow v_m$. Since $h(-x, v) = h(x, v)$, $h > 0$ on the whole segment $|x| \leq 2$. (S) For $x \geq 2$ and $0 < v \leq v_*$, the first term gives $\operatorname{Im}(-\varphi_1'/\varphi_1) \geq 2\pi e^{2x} \sin 2v \geq 8e^4 v$, and Cauchy's estimate on discs of radius 0.1 bounds the imaginary part of the derivative of $\log(\Phi/\varphi_1)$ by $10^{-40}v$; so $h > 0$ there, and by symmetry also for $x \leq -2$. By (Z), h is harmonic on the strip $|\operatorname{Im} t| < 0.5634$. It vanishes on the real axis and, by (T) and (S), is positive on the top and the sides of the rectangle $[-2, 2] \times [0, v_*]$, so $h \geq 0$ inside by the minimum principle, and with (S), $h \geq 0$ on $0 < v \leq v_*$. So $e^{\beta X}$ is HCM for $\beta \geq \pi/v_*$. □

The certificate stops 3×10^{-6} short of $\pi/v_m = 5.5791396$, because h vanishes at iv_m and the strip of height v_m cannot be covered by balls with positive lower bounds; presumably the threshold is exactly π/v_m. The computation is `fm_hcm.py`, which works in the variable $t/2$ of Rodgers and Tao [224], where the heights are halved and the thresholds doubled.

Along the family $e^{\beta X}$, then: no infinite divisibility for $0 < |\beta| < 2$ (Proposition 13.5.1), self decomposability without the GGC property at $|\beta| = 2$ (Theorem 13.5.7), and an HCM density, hence the GGC property, for $|\beta| \geq \pi/v_* = 5.57914219\ldots$, with $v_* = 0.563096$ (Proposition 13.5.8 and its proof). The HCM threshold, the least $|\beta|$ for which $e^{\beta X}$ has an HCM density (these powers form a closed half line, by (F2) and (F4) of Section 4.2.1), lies between 5.5791395 and π/v_*, and for $2 < |\beta|$ below it infinite divisibility and the GGC property are open (Section 13.6). The bracket for the HCM threshold comes from the kernel, not from the zeros, and it involves no hypothesis: the obstruction found is on the imaginary axis, and that it sets the exact threshold, at π/v_m, is a conjecture. It is the Pólya side counterpart of Theorem 4.3.28, where, under RH and two further hypotheses on the zeros, the HCM powers of the Mellin variable W are limited by the first gap $\gamma_2 - \gamma_1$. For the clock, whose Laplace transform $\xi(\frac{1}{2})/\xi(\frac{1}{2} + \sqrt{u})$ is the moment generating function of X inverted at $\sqrt{u}$, the GGC property is RH itself (Proposition 1.2.1).

13.6 Open problems

The problems follow the order of the chapter. The first five ask how far the methods of Sections 13.2 to 13.4 reach towards the Pólya law; the last two concern the variable e^{2X} of Section 13.5, where no hypothesis on the zeros enters.

Problem 13.1. Mixed shoulders. Theorem 13.2.3 decides the symmetric laws on $[-1,1]$ whose density is concave on $[0,1]$ and those whose density is convex there and not linear on its support: by (13.2.1) the question is whether the sine transform of the slope $g = -p'$ has only real zeros, and the theorem answers it when g is monotone. Decide the slopes that first rise and then fall. One class is settled. If g is symmetric about $\frac{1}{2}$, then $p(0) = \int_0^1 g = 1$, because $1 = 2\int_0^1 p = 2\int_0^1 xg(x)\,dx = \int_0^1 g$ by the symmetry of g, so $u \mapsto g(\frac{1}{2}+u)$ is a probability density on $[-\frac{1}{2}, \frac{1}{2}]$; and $X \stackrel{d}{=} \frac{1}{2}U + B$, with B independent of U and distributed with that density, since the density of $\frac{1}{2}U + B$ has slope g on $(0,1)$. As $\sin(z/2)/(z/2)$ has only real zeros, X is then ferromagnetic exactly when B is. For the beta slopes $g(y) \propto y^a(1-y)^b$, $a, b \geq 0$, this decides the diagonal $a = b$, where B is a symmetric beta law and X is ferromagnetic (Section 13.1.4), and Theorem 13.2.3 decides the edges: X is ferromagnetic for $b = 0$, where p is concave, and not for $a = 0 < b$, where p is convex and not linear. The chapter does not decide the rest of the quadrant. On the half line the slope of the Pólya law rises and then falls (Section 13.3.2), and RH is the statement that its sine transform has only real zeros.

Problem 13.2. The Griffiths Simon class. Newman's theorem places the Pólya law in the class of Ellis, Monroe and Newman (Section 13.3.1). Does it belong to the Griffiths Simon class of Section 13.1.1? The laws of that class are ferromagnetic, so a positive answer would imply RH, and under Ingham's hypothesis the spins of any approximating system would have to be coupled (Section 13.3.3).

Problem 13.3. Splitting the zeros. Assume RH, so that $\varphi_X(t) = \prod_\gamma (1 - t^2/\gamma^2)$ over the positive ordinates, with no Gaussian factor by Theorem 13.3.1. Can the ordinates be split into two nonempty sets A and A' such that $\prod_{\gamma\in A}(1 - t^2/\gamma^2)$ and $\prod_{\gamma\in A'}(1 - t^2/\gamma^2)$ are both characteristic functions? This is the same as writing $X = D + R$ with D and R independent, ferromagnetic and not 0, and by Theorem 13.3.1 both sets are then infinite. Under Ingham's hypothesis no three ordinates are in arithmetic progression, so no piece is a scaled fair sign or a scaled uniform variable, whose zeros form arithmetic progressions.

Problem 13.4. Positivity of the zero completed laws. The laws f_K of Section 13.4.3 are proved to be ferromagnetic probability densities for $K \leq 16$ by the Maxwell construction, and for every $K \leq 200$ and for $K = 300$, 400, 600 and 800 by a computer assisted proof in ball arithmetic. Compute the threshold $v^*(K)$ near $K = 5000$, where the linear and the power law fits of the ratio $v^*(K)/2\beta_K$

part; the cancellation there calls for a working precision of tens of thousands of bits. Is $v^*(K) < 2\beta_K$ for every K? If so, every f_K is a ferromagnetic probability density by Theorem 13.4.3(ii), and with no hypothesis their limit in law, with characteristic function $e^{-\beta_\infty \theta^2} \prod_k (1 - \theta^2/\gamma_k^2)$ over the zeros on the critical line, is a ferromagnetic law by Theorem 13.1.5; by Theorem 13.4.3(iii) it is the Pólya law exactly when RH holds.

Problem 13.5. The cosine constant. By Section 13.4.2, $\Omega \le \frac{1}{2}$ with no hypothesis, $\Omega \le \frac{3}{8}$ if the half plane $\operatorname{Re} s > \frac{7}{8}$ announced in October 2026 is free of zeros (Section 12.1.5), and RH holds if and only if $\Omega = 0$. Is the set of $a \ge 0$ for which $\cosh(at)\, p$ is ferromagnetic an interval? Improve these bounds by computation, as computations with the zeros have improved de Bruijn's bound $\Lambda \le \frac{1}{2}$ on the de Bruijn Newman constant, to $\Lambda \le 0.2$ [196]. A first case is $a = \sqrt{0.1}$, below both bounds: the law $\cosh(\sqrt{0.1}\, t)\, p$ is as close to p as the flow at $\lambda = 0.2$, which is ferromagnetic, and deciding it is a computation with the zeros of $\frac{1}{2}(\xi(s+a) + \xi(s-a))$, of the kind that gave the bound 0.2 for the flow.

Problem 13.6. The size bias principle for real shapes. Theorem 13.5.5 is proved for integer shapes. Does it hold for all real shapes $t_n > 0$? For $K = 1$ it is the gamma identity, with $G = 0$. Integer shapes enter the proof twice, through the divided difference formula for the density of Y and through Lemma 13.5.4, and together they show that on each gap (τ_k, τ_{k+1}) the phase of the Laplace transform of Y_q is nonincreasing and falls by exactly q. Is this still true for real shapes? With it, the rest of the proof goes through, provided the estimates of the transform near the points $-\tau_n$ and at infinity can be checked as well.

Problem 13.7. The powers of e^X. For $2 < |\beta|$ below the HCM threshold, which lies between 5.5791395 and $\pi/0.563096 = 5.57914219\ldots$ (Proposition 13.5.8), is $e^{\beta X}$ infinitely divisible, self decomposable or a GGC? Below the threshold it has no HCM density, which leaves the GGC property open, and Theorem 13.5.7 gives self decomposability only at $|\beta| = 2$. Is the HCM threshold exactly $\pi/v_m = 5.5791396$? That requires $h \ge 0$ on the open strip $0 < v < v_m$, where $h(0, v) \to 0$ as $v \uparrow v_m$, so a certificate must replace the balls with positive lower bounds near iv_m by a local argument.

Appendix A

Code and reproducibility

A number in this book can be checked once three things are known: the evaluator that produced it, the precision at which it ran, and the script that turned its output into a table or a figure. The book's archive of scripts and data files is in its repository, `https://github.com/VadimSokolov/computing_rh`, beside the LaTeX source of the book. The scripts of Chapters 1 to 5 and 8 are in the folder `code/` of the book's archive, and the data files they write are in `data/`. The interval enclosures, spot checks and other scripts of Chapters 6 and 7 are in `verification_reciprocal_zeta/`; the verification script of Section 4.3 is in `code_bosch/`, with an identical copy beside the authors' notes in `bosch_hcm_xi/`; the scripts of Chapters 9 to 11 are in `basepoint_one/`, `sato_bondesson/` and `riesz_program/`; and the reruns and reconstructions made for the book are in `repro/`. Section A.1 prints the two Pólya evaluators of Section 2.1 and states what their ball radii certify. Section A.2 gives the software the scripts use and the order in which they run, names the scripts and data files behind the tables and figures of Chapters 1 to 8, and collects the constants that recur in several sections. Section A.3 does the same for Chapters 9 to 11, and Section A.5 reports what a rerun of the whole archive reproduced and which computations still have no script.

A.1 The Pólya evaluators

The first listing is `code/polya_flint.py`, the evaluator of Section 2.1 on the real u axis. Its function `setup` computes the node values $\Phi_j = \Phi(jh)$, $0 \le j \le J$, from the theta series (2.1.1) in ball arithmetic and stores the polynomials P and Q of Section 2.1.1 as `PA` and `PB`; `xi_pair` returns $P(E) + P(E^{-1})$ and $Q(E) - Q(E^{-1})$ with $E = e^{sh}$, the trapezoid approximations to $\xi(\frac{1}{2}+s)$ and $\xi'(\frac{1}{2}+s)$. The defaults are 240 bits, $h = 1/200$ and $|u| \le 3$, so that $J = 600$. Most runs call `setup(420, 200, 3)`, the parameters of Section 2.1.1; the Keiper Li coefficients of Section 8.3 use 600 bits, and the Selberg computation of Section 3.5 uses 300.

code/polya_flint.py

```
"""Polya route to xi(1/2+s) and xi'(1/2+s): trapezoid rule on R for
   xi(1/2+s) = int Phi(u) e^{su} du (Phi even), xi'(1/2+s) = int u Phi(u) e^{su}
       du,
   Phi(u) = 2 sum_n (2 pi^2 n^4 e^{9u/2} - 3 pi n^2 e^{5u/2}) exp(-pi n^2 e^{2u})
       (Titchmarsh 10.1).
   Nodes u_j = j h, |j| <= J. Ball arithmetic (python-flint) for the sums."""
from flint import arb, acb, acb_poly, ctx
def setup(prec_bits=240, h_inv=200, umax=3):
    ctx.prec = prec_bits
    h = arb(1)/h_inv; J = int(umax*h_inv)
    pi = arb.pi()
    Phi=[]
    for j in range(J+1):
        u = j*h; e2=(2*u).exp(); s=arb(0); n=1
        while True:
            t = (2*pi**2*n**4*(9*u/2).exp() - 3*pi*n**2*(5*u/2).exp())*(-pi*n*n*e2)
                .exp()
            s += t
            if n>2 and abs(float(t.mid())) < 2.0**(-prec_bits-20): break
            n += 1
        Phi.append(2*s)
    # P(z) = h*[Phi0/2 + sum_{j>=1} Phi_j z^j]; xi = P(E)+P(1/E)
    cA = [h*Phi[0]/2] + [h*Phi[j] for j in range(1,J+1)]
    cB = [arb(0)] + [h*j*h*Phi[j] for j in range(1,J+1)]
    return dict(h=h, PA=acb_poly(cA), PB=acb_poly(cB))
def xi_pair(S, s):
    s = acb(s) if not isinstance(s, acb) else s
    E = (s*S['h']).exp(); Ei = 1/E
    A = S['PA'](E) + S['PA'](Ei)
    B = S['PB'](E) - S['PB'](Ei) # u Phi odd weight: int u Phi e^{su} over R
    return A, B
```

The second listing is `code/tilted.py`, which implements Proposition 2.1.1. With $\eta = \pi/4 - \delta$ and nodes $u_j = x_j + i\eta$, $x_j = jh$, $|j| \le J$, on the line $\operatorname{Im} u = \eta$, where $J = \lceil x_{\max}/h \rceil$ for the cut $|x| \le x_{\max}$, so that there are $2J+1$ nodes, it returns $e^{i\eta s} h \sum_j \Phi(u_j) e^{s x_j}$ and $e^{i\eta s} h \sum_j u_j \Phi(u_j) e^{s x_j}$, the trapezoid approximations to $\xi(\frac{1}{2}+s)$ and $\xi'(\frac{1}{2}+s)$ on the tilted line; `Phi_c` sums the theta series at complex u, using the evenness of Φ when $\operatorname{Re} u < 0$. Of the two evaluators in the file, `xi_pair` applies Horner's rule and is called only by `ttest.py`, which exhibits the growth of the ball radii described in Section 2.1.5; the production scripts evaluate each exponential directly, through `xi_pair_direct` or an equivalent loop. The Lehmer pair and its de Bruijn Newman flow use $\delta = 0.01$, 256 bits, $h = 6 \times 10^{-4}$ and $|x| \le 3.9$; the count at height 1000 in Section 2.2.3.2 uses $\delta = 0.1$, 200 bits, $h = 0.004$ and $|x| \le 4.2$; the module `clock.py`, which evaluates the Laplace transform of the clock for Sections 8.1 and 8.2, switches to the tilted evaluator with $\delta = 0.1$, 320 bits, $h = 0.003$ and $|x| \le 4.5$ when $|\operatorname{Im} s| \ge 120$; and the runs of Table 2.2 take δ, the precision, h, the cut and the heights from the command line

of `ttest2.py`. The command lines, recorded at the top of the two test scripts, are `python3 ttest2.py 0.10 200 0.004 4 200 500 1000`, with 2001 nodes, for the rows with $\delta = 0.1$, `python3 ttest2.py 0.02 220 0.001 4.2 1000 2000 5000`, with 8401 nodes, for the rows with $\delta = 0.02$, and `python3 ttest.py 0.1 200 0.004 4`, which prints `nan` at the heights 200 to 2000, where the Horner ball of ξ contains zero.

code/tilted.py

```
"""Tilted Polya contour: xi(1/2+s) = int_{Im u = eta} Phi(u) e^{su} du, eta = pi/4
     - delta.
   Phi is analytic in |Im u| < pi/4; on the tilted line |e^{su}| carries e^{-theta
       *eta},
   so the cancellation drops from e^{-pi theta/4} to e^{-delta theta}."""
from flint import arb, acb, acb_poly, ctx
import math
def Phi_c(u, tol_bits):
    # Phi(u) for Re u >= 0 (use evenness otherwise)
    if u.real < 0: u = -u
    pi = arb.pi(); e2 = (2*u).exp(); s = acb(0); n = 1
    e92 = (9*u/2).exp(); e52 = (5*u/2).exp()
    while True:
        t = (2*pi**2*n**4*e92 - 3*pi*n**2*e52)*(-pi*n*n*e2).exp()
        s += t
        if n > 2 and abs(complex(t)) < 2.0**(-tol_bits): break
        n += 1
    return 2*s
def setup(delta, prec_bits, h, xmax):
    ctx.prec = prec_bits
    eta = arb.pi()/4 - arb(delta)
    J = int(math.ceil(xmax/h)); hh = arb(h)
    c0 = []; c1 = []
    for j in range(-J, J+1):
        u = acb(j*hh, eta)
        p = Phi_c(u, prec_bits+20)
        c0.append(hh*p); c1.append(hh*u*p)
    return dict(J=J, h=hh, eta=eta, P=acb_poly(c0), Q=acb_poly(c1))
def xi_pair(S, s):
    s = acb(s)
    E = (s*S['h']).exp(); pref = (s*acb(0, S['eta'])).exp() * E**(-S['J'])
    return pref*S['P'](E), pref*S['Q'](E)
def xi_pair_direct(S, s):
    s = acb(s); J=S['J']; h=S['h']
    cP = S['P'].coeffs(); cQ = S['Q'].coeffs()
    A = acb(0); B = acb(0)
    for k in range(len(cP)):
        e = (s*((k-J)*h)).exp()
        A += cP[k]*e; B += cQ[k]*e
    pref = (s*acb(0, S['eta'])).exp()
    return pref*A, pref*B
```

In both evaluators the radii of the returned balls bound the rounding errors of

the finite sums, and nothing else. Three errors lie outside them: the discretisation error of the trapezoid rule, bounded in Section 2.1.1; the cut of the axis at $|u| \le 3$, where $\Phi < 10^{-500}$, or at $|x| \le x_{\max}$ on the tilted line; and the truncation of the theta series, whose loop stops at the first term with midpoint below 2^{-b-20} at working precision b and does not add the tail to the radius. A certified enclosure of ξ adds bounds for these three errors to the radii. The stopping test compares double precision numbers, so for $b \ge 1055$ its threshold underflows to zero and the test is never met. The untilted route needs more than 1054 bits for thirty digits above height about 840 (Table 2.3), and there the test has to be made in ball arithmetic. The runs in this book use at most 600 bits and are not affected.

A.2 Scripts, data files and reference constants

Ball arithmetic comes from `flint`, the Python interface to FLINT and Arb [110], which also gives the seventy digit ordinates of the first 1700 zeros used by `arith.py`. Reference values of ζ, Γ and ξ, the other ordinates of the zeros, the interval arithmetic of Chapters 6 and 7 and the Talbot inversion of Laplace transforms [248] come from `mpmath` [159]; `numpy` and `scipy` supply arrays, quadrature, interpolation, root finding, differential equations and nonnegative least squares, `sympy` the primes and the Möbius function, and `matplotlib` the figures.

The scripts run from `code/` and read and write their data files in the working directory; the data files named in Tables A.1 and A.2 are in `data/`, except the reports and JSON files of Chapters 6 and 7, which are in `verification_reciprocal_zeta/` and its subfolder `torus/`. The Pólya evaluators use no zeros. Ten scripts, `analyse.py`, `deconv.py`, `deconvplot.py`, `hankel.py`, `kenttail.py`, `kenttail2.py`, `kmono.py`, `perturb.py`, `plots3.py` and `selberg.py`, load the first 1700 ordinates of the zeros from `g_1_400.npy`, `g_401_1000.npy` and `g_1001_1700.npy`. These lists are not in `code/`: `repro/zeros.py` computes them with `mpmath.zetazero` at thirty digits, and identical copies are in `riesz_program/zeros/` and in several of the folders that Section A.3 describes. `spacing.py` reads the lists when they are present and otherwise computes and saves them, and `zeros_hp.py`, which writes the seventy digit ordinates of `zeros_1700.txt` that `arith.py`, `cm_orders.py`, `firstfail.py`, `firstfail_arb.py`, `gaussweil.py`, `phasefree.py` and `vdpair.py` read, uses the lists and `part1.json` only to check its output. Two scripts come first, because others read their output. `part1.py` finds the sign changes of Ξ below height 200 on a grid of step 0.05, refines the zeros by Newton's method and computes $N(200)$ by the argument principle. `part2.py` evaluates $\pi\rho_\varepsilon(\theta) = \mathrm{Re}(\xi'/\xi)(\frac{1}{2}+\varepsilon+i\theta)$ and the continuous argument of $\xi(\frac{1}{2}+\varepsilon+i\theta)$ for $0 \le \theta \le 200$, at each resolution ε given on its command line, and saves them in the files `eps_*.npz` with the largest argument increment, which it also prints. The book uses the eight resolutions 1, 0.5, 0.25, 0.1, 0.05, 0.02, 0.01 and 0.005, and the largest increment over all

of them, 0.29 at $\varepsilon = 1$, is the one quoted in Section 2.1.2. The distances of Section 3.2 (`analyse.py`), the zero density identity of Section 3.4 (`turing_ig.py`) and the deconvolution of Section 8.6 (`deconv.py` and `deconvplot.py`) start from these files, and `analyse.py` writes `plotcurves.npz`, which `plots.py`, `plots3.py` and `plots4.py` read; the `.npz` files are not in the archive. `analyse.py`, `clockgrid.py`, `hazard_spec.py` and `dbn.py` read `part1.json`, and the boundary computations of Section 8.2 start from `hazard_spec.json`. `hcm_arb.py`, which encloses in ball arithmetic the 216 values that `hcm.py` computes, reads `hcm.json` to compare them. Likewise `firstfail_arb.py`, which certifies in ball arithmetic the negative values that `firstfail.py` finds, reads `firstfail.json`, and so do `gsmooth_arb.py` and `phasefree_arb.py`, for the pairs and their heights; these two compute the zeros they need with Arb instead of reading them.

The plotting scripts draw the figures from the saved data. The figures are in colour in the digital edition, the PDF at the DOI on the copyright page, and in grey in the paperback; where a figure tells its curves apart by colour alone, the digital edition is the one to read. `plots.py` takes no arguments; it reads `plotcurves.npz`, `analysis.json`, `hcm.json` and `results_prime.json`, and besides its figures it writes `extra.json` and `sens2.json`. `dbnplot.py` reads `dbn.json`. `plots.py`, `plots2.py`, `perturb.py` and `dbnplot.py` save their figures in `fig/` below the working directory, and `plots3.py`, `plots4.py`, `kmono.py`, `deconvplot.py`, `lplots.py` and `lehmerflowplot.py` in `book/fig/`; of these only `perturb.py` and `dbnplot.py` create their folders. Four scripts of `code/` are reconstructions, written for the book where the archive held a data file or a module but no script; the header of each says so and gives its command line, its inputs and what it reproduces. `hankel.py` and `spacing.py` reproduce `hankel.json` and `spacing.json` byte for byte, and `spacing.py` also draws `spacing.pdf` in the working directory. `sinh_bench.py` reproduces `sinh_bench.json`, whose lattice sums are cut at $|k| \le 400000$, to a relative 1.1×10^{-12}, and `python3 sinh_bench.py closed` writes the closed forms, which Sections 1.3 and 5.4 quote, to `sinh_bench_closed.json`. `polya.py` wraps `setup(420, 200, 3)` and `xi_pair` of `polya_flint.py` for `primeside.py`, which imports it.

In `verification_reciprocal_zeta/`, `enclosures.py` computes the enclosures E1 to E6 of Chapters 6 and 7 in outward rounded interval arithmetic and writes `enclosures_report.md`. `certify_k9_k2.py` encloses in ball arithmetic the value of $\operatorname{Re}\kappa(\rho_{213})$ used in Corollary 7.2.4 (both from its definition and from the formula of Theorem 7.2.3), the sign of $\operatorname{Re}\kappa$ at the first 460 zeros, and the values of S quoted in and after the proof of Theorem 7.1.3, and writes `certify_k9_k2.json`; it runs in a few seconds. `spot_checks.py` recomputes the constants of these chapters that `mpmath` and the first few hundred zeros can settle and prints each beside the value in the text. It writes no file: its report is its output, `python3 spot_checks.py > spot_checks_report.md`, which takes about eight minutes, and it checks the nine rows of Table 7.9 and four of the twenty five rows of

Table 7.8. Two scripts compute numbers quoted in Section 7.1, each in about a minute. `sawtooth_truncation.py` computes the root mean square of S and of the errors of its truncated sawtooth and classical series on uniform grids of steps 10^{-3}, 5×10^{-4} and 2.5×10^{-4} in $[10, 50]$, and the sums $\frac{1}{2}\sum_{N\le N_0}|B_N|$, quoted after Proposition 7.1.10, and writes `sawtooth_truncation.json`. `harmonic_decay.py` computes the maxima of $\log|\hat{f}_{\kappa^*}(v)|/v$ over the windows $[0.95V, 1.05V]$ quoted after Proposition 7.1.17, by root finding in each interval between consecutive zeros of $\hat{f}_{\kappa^*}$, where the quotient has a single maximum, rather than on a grid, and writes `harmonic_decay.json`. The subfolder `torus/` holds the branch and bound of Remark 7.2.12, which proves in ball arithmetic the bounds and thresholds of Tables 7.4 and 7.5 and the second condition of Theorem 7.2.10. It ran as arrays of tasks on a cluster, and its proofs in ball arithmetic took about twenty CPU hours; the folder holds the programs and their aggregated JSON outputs, and `torus/README.md` says what each script does and how to rerun it. Tables A.1 and A.2 list the scripts section by section. Tables that collect statements, or evaluate the formulas given in their captions, need no script; Section A.5 names the other tables and figures of Chapters 1 to 8 that the two tables do not list.

section	scripts (data files)	tables and figures
1.1	`plots4.py`	Figure 1.1
1.2	`primeside.py`, `polya.py` (`results_prime.json`), `arith.py` (`arith.json`), `vdpair.py` (`vdpair.json`), `plots3.py`	Tables 1.4, 1.5, 1.6; Figure 1.2
1.3	`sinh_bench.py` (`sinh_bench.json`), `plots3.py`	Figure 1.3
2.1	`polya_flint.py`, `part1.py` (`part1.json`), `part2.py`, `tilted.py`, `ttest.py`, `ttest2.py`, `plots.py`, `plots3.py`	the zeros and $N(200)$ below height 200; Table 2.2; Figures 2.1, 2.2, 2.3
2.2	`lehmer.py` (`lehmer.json`), `turing1000.py` (`turing1000.json`)	the Lehmer pair near height 7005; the count of Section 2.2.3.2
2.3	`polya_flint.py`, `plots3.py`	Figure 2.5
3.1	`primeside.py` (`results_prime.json`), `plots3.py`	Table 3.1; Figure 3.1
3.2	`part2.py`, `analyse.py` (`analysis.json`), `plots.py` (`extra.json`)	Table 3.2; Figures 3.3, 3.4, 3.2
3.3	`sensitivity.py` (`results_sens.json`), `plots.py` (`sens2.json`)	Figure 3.5
3.4	`turing_ig.py` (`turing_ig.json`), `plots3.py`	Table 3.3; Figure 3.6
3.5	`selberg.py` (`selberg.json`)	Table 3.4
4.1	`hcm.py` (`hcm.json`), `hcm_arb.py` (`hcm_arb.json`), `plots.py`	Figures 4.1, 4.2; the 216 values of Section 4.1.2 and their certified signs
4.3	`code_bosch/verify_bosch_xi_v7.py`, `pakes_disc.py` (`pakes_disc.json`)	Tables 4.4, 4.5; Figure 4.3; the checks of Theorem 4.3.10 and Remark 4.3.11; Appendix D

section	scripts (data files)	tables and figures
5.1	`idsd_check.py` (`idsd_check.json`), `idsd_height.py` (`idsd_height.json`), `ladder_bound.py` (`ladder_bound.json`), `jacobi_zeros.py` (`jacobi_zeros.json`), `cm_orders.py` (`cm_orders.json`), `window_orders.py` (`window_orders.json`), `hankel.py` (`hankel.json`), `saddle.py` (`saddle.json`), `kmono.py` (`kmono.json`), `gaussweil.py` (`gaussweil.json`), `plots3.py`	the proof of Theorem 5.1.1, the remark after it and the one range proof; the check of (5.1.1), Lemma 5.1.6, Theorem 5.1.7 and the numbers after it; Theorem 5.1.10; Tables 5.1, 5.2, 5.3, 5.4; Figures 5.1, 5.2
5.2	`perturb.py` (`perturb.json`), `firstfail.py` (`firstfail.json`), `firstfail_arb.py` (`firstfail_arb.json`), `gsmooth_arb.py` (`gsmooth_arb.json`)	Table 5.5; Figure 5.3; the failure times of $\mathcal{G}$
5.3	`zeros_hp.py` (`zeros_1700.txt`), `arith.py` (`arith.json`), `plots3.py`	Table 5.7; Figure 5.4
5.4	`spacing.py` (`spacing.json`), `plots.py` (`extra.json`), `plots3.py`	the cell constant κ_{cell}; Figures 5.5, 5.6

Table A.1: Scripts in `code/` (unless another folder is named), the data files they write, and the tables and figures they produce, in book order.

section	scripts (data files)	tables and figures
6.1 to 7.2	`verification_reciprocal_zeta/` `enclosures.py`, `spot_checks.py` (`enclosures_report.md`, `spot_checks_report.md`), `sawtooth_truncation.py` (`sawtooth_truncation.json`), `harmonic_decay.py` (`harmonic_decay.json`), `certify_k9_k2.py` (`certify_k9_k2.json`), `torus/` (`certified_bounds.json`, `arb_thresholds.json` and six other JSON files)	the enclosures E1 to E6; Table 7.9 and four rows of Table 7.8; the numbers after Propositions 7.1.10 and 7.1.17; $\operatorname{Re}\kappa(\rho_{213})$ in the proof of Corollary 7.2.4 and the values of S quoted with Theorem 7.1.3; Tables 7.4 and 7.5 and the second condition of Theorem 7.2.10
8.1	`clock.py`, `clockgrid.py` (`clockgrid.json`), `hazard_spec.py` (`hazard_spec.json`), `kenttail.py` (`kenttail.json`), `kenttail2.py` (`kenttail2.json`), `williams_mc.py` (`williams_mc.json`), `plots2.py`	Tables 8.3, 8.4, 8.2; Figure 8.1
8.2	`turing_ig.py`, `rsboundary.py` (`rsboundary.json`), `mc.py` (`mc.json`), `volterra.py` (`volterra.json`), `mc2.py` (`mc2.json`), `hitting.py`, `hitting2.py`, `plots2.py`, `threezones.py`	the check of Proposition 8.2.1; Table 8.5; Figures 8.6 and 8.7; the boundary of Section 8.2.2.2
8.3	`li.py` (`li.json`), `plots3.py`	Table 8.6; Figure 8.9

section	scripts (data files)	tables and figures
8.4	`dbn.py` (`dbn.json`), `lehmer_dbn.py` (`lehmer_dbn.json`), `jacobi_dbn.py` (`jacobi_dbn.json`), `dbn_collisions.py` (`dbn_collisions.json`), `dbn_hankel_long.py` (`dbn_hankel_long.json`), `wigbiane_ineq.py` (`wigbiane_ineq.json`), `wigbiane_poly.py` (`wigbiane_poly.json`), `wigbiane_flow.py` (`wigbiane_flow.json`), `dbnplot.py`, `lehmerflowplot.py`	Tables 8.7, 8.8, 8.9, 8.10, 8.11, 8.12, 8.13; Figures 8.10, 8.11; the Hankel test along the flow and its counts to $t = -20$; the collisions of the first 200 gaps; the tests of the Biane and deadline bounds
8.5	`lchi12.py` (`lchi12.json`), `lfun.py` (`lfun.json`), `lsin2.py` (`lsin2.json`), `lplots.py` (`lrho.json`), `maxlaw.py` (`maxlaw.json`), `dbn_delta.py` (`dbn_delta.json`), `maxexp_checks.py` (`maxexp_checks.json`), `maxexp_psizeros.py` (`maxexp_psizeros.json`), `maxexp_hb.py` (`maxexp_hb.json`), `maxexp_flow.py` (`maxexp_flow.json`), `maxexp_figs.py`, `sqfree_quadratic.py` (`sqfree_quadratic.json`), `dbeta_zeros.py` (`dbeta_zeros.json`), `dbeta_thorin.py` (`dbeta_thorin.json`), `dbeta_minimum.py` (`dbeta_minimum.json`), `dbeta_figs.py` (`dbeta_figs.json`)	Tables 8.15, 8.16, 8.17, 8.18, 8.19, 8.20, 8.21, 8.22, 8.23, 8.24, 8.25; Figures 8.13, 8.14, 8.15, 8.16, 8.17, 8.18, 8.19
8.6	`deconv.py` (`deconv.json`), `deconvplot.py`	Tables 8.26, 8.27; Figure 8.20
8.7	`biane_polya.py` (`biane_polya.json`), `biane_clock.py` (`biane_clock.json`), `biane_flow.py` (`biane_flow.json`), `biane_traces.py` (`biane_traces.json`), `biane_figs.py` (`biane_figs.json`)	Tables 8.28, 8.30, 8.31, 8.32; Figures 8.21, 8.22, 8.23, 8.24; in Section 10.1, Tables 10.5, 10.6 and Figures 10.2, 10.3
9.1 to 11.5	the folders of Section A.3	Tables A.4, A.5
12.2	`phasefree.py` (`phasefree.json`), `phasefree_arb.py` (`phasefree_arb.json`)	the thresholds $t_0(u)$ of Section 12.2.6, and $\mathcal{G}(u, 6) > 0$ for $u \geq 10^{213}$

Table A.2: Continuation of Table A.1.

Table A.3 collects the constants used in several sections and names the file of the archive in which each can be checked; the printed values agree with these files to every digit shown.

A.3 The scripts for the basepoint one and the critical strip

The scripts of Chapters 9 to 11 stay in the folders of the authors' notes, and each reads and writes its data files in its own folder; the exceptions are Sections 9.1.6, 10.3, 11.2 and 11.5, whose scripts are in `code/` and whose data

quantity	value	file
$\xi(\frac{1}{2})$	0.4971207781883141	`williams_mc.json`
$\sum_k a_k^{-1} = \mathbb{E}T$	0.0231050	`kenttail2.json`
$\sum_k a_k^{-2} = \operatorname{Var} T$	3.71726×10^{-5}	`kenttail2.json`
γ_1^2	199.790454832386859463	`hazard_spec.json`
$\gamma_2^2 - \gamma_1^2$	242.1357	`hazard_spec.json`
$\lambda_1 = b_1 = \sum_\rho \rho^{-1}$	0.023095708966121033814	`li.json`
$c = \zeta'(\frac{1}{2})/\zeta(\frac{1}{2})$, the constant of (6.1.6)	2.6860917096	`spot_checks_report.md`
$\Xi'(200)/\Xi(200)$	-1.2506879	`extra.json`
κ_{cell}, the cell constant of the 79 cells below height 200	5.43665	`extra.json`
$N(200)$	79	`part1.json`
$N(1000)$	649	`turing1000.json`
de Bruijn Newman collision time of γ_{71}, γ_{72}, the closest pair below height 200	-0.2880	`dbn.json`
collision time of the Lehmer pair near height 7005	-7.11×10^{-4}	`lehmer_dbn.json`
$N_{\chi_{12}}(100)$, the number of zeros of $L(s, \chi_{12})$ with $0 < \operatorname{Im}\rho \le 100$	67	`lchi12.json`

Table A.3: Constants used in several sections, with the file of the archive in which each can be checked.

files are in `data/`, and Section 10.1.5, computed by the scripts of Section 8.7 in Table A.2. In `basepoint_one/`, the folder `xi` holds the scripts for ξ of Chapter 9, and `eta` those for $L(s, \chi_{12})$ of Section 11.3; `scale_mixture` and `wiener` hold those of the two halves of Section 10.1, and `process_map` those of the map of processes of Section 11.1 and of the boundaries across the basepoints of Section 9.2.6. The folder `sato_bondesson/` holds the other scripts of Section 11.1, and `riesz_program/code/` those of Sections 10.2 and 11.4, whose data files are in `riesz_program/data/`; the script `setup.sh` links these files and the zero lists of `riesz_program/zeros/` into `code/`, from which the scripts run. Tables A.4 and A.5 give the scripts behind each table and figure. Several names recur across the folders, so the tables give the folder of every script: `comp.py` is in `scale_mixture`, `process_map` and `sato_bondesson`, `figs.py` in `xi`, `eta`, `process_map` and `sato_bondesson`, `improve.py` and `figs2.py` in `xi` and `scale_mixture`, and `salem_riesz.py` in `wiener` and `riesz_program/code`, while `mc.py` of `xi` and `arith.py` of `eta` are not the scripts of those names in `code/`. The scripts take no arguments, with three exceptions: `bnd_alpha.py` takes the basepoint, 0.6 or 0.75; `weil.py` takes the weights $\log p$ in double precision, or at thirty digits when called as `python3 weil.py mp`; and `riesz_large.py` takes x, the truncation K and the twist h, as in `python3 riesz_large.py 1e10 200 50`.

In `xi`, `compute.py` writes the Thorin density at the basepoint one, the sine squared check with the primes up to 10^7 and the values of Proposition 9.2.1

to alpha1.json, and each other script that computes writes the data file of its own name, except rs1c.py, which adds the tangent boundary to rs1.json. mc.py also writes surv.json and a sample X_sample.npy, and rs1d.py and errformula.py read rs1.json and below.json. The sample is not in the archive, so figs.py, which draws Figure 9.5 from it, runs only after mc.py. The script heat_primes.py was added for the book: it reads no file, computes the thresholds of Table 9.9 in floating point and certifies Proposition 9.1.17 in ball arithmetic with python-flint, in about a minute and a half on 32 cores. In eta, zeros12.py computes the 297 zeros of $L(s, \chi_{12})$ below height 340 and saves them in g12.npy, which base.py, arith.py and hit.py read; hit2.py adds to hit.json, and figs.py reads alpha1.json of xi as well as the files of its own folder.

In scale_mixture, dens2.py adds the densities of Y_α to strip.json, and figs2.py draws Figure 10.1 from strip.json and comp.json. In wiener, two scripts were added for the book: figs_wiener.py, which draws Figures 10.4 and 10.5, whose drawing script the authors' package did not contain, and riesz_explicit.py, a reconstruction that writes riesz_explicit.json from salem_riesz.json, byte for byte as archived. In process_map, comp.py and comp2.py read a copy of alpha1.json, and figs.py draws Figure 11.3 from ou.json, the files bnd_0.6.json and bnd_0.75.json, and copies of rs1.json from xi and of volterra.json from data/. In sato_bondesson/, comp.py reads alpha1.json, rs1.json and improve.json of basepoint_one/xi and clockgrid.json and hazard_spec.json of data/, and weil.py prints its results and writes no file; figs.py holds as constants the five values of the explicit formula that Figure 11.2 shows as dots.

In riesz_program/code/, killing2.py adds to killing.json, which rmt_ratios.py reads, more2.py reads almost.json, and figs_riesz.py reads riesz_thorin.json, salem_riesz.json and riesz_explicit.json, the last two identical to the files of basepoint_one/wiener. Seven scripts of the folder were added for the book: v1min.py, which certifies Proposition 9.1.4; explicit_bounds.py, which certifies in ball arithmetic the numbers used in the proofs of Theorems 10.1.3(3) and 11.4.4; almost_dip.py, check_large_twisted.py, ggc_offset.py and rmt_ratios.py, which compute numbers of Sections 10.2 and 11.4 that had no archived output; and figs_almost.py, which redraws Figures 11.13 and 11.12 in the notation of the book. Two checks were added in repro/: ks_ak.py recomputes the arithmetic factors that ks.py holds as constants, and zeros_mp.py and v1_grid_mp.py, run as arrays of tasks on a cluster and collected by v1_collect.py in v1_check.json, check Proposition 9.1.4 independently with mpmath. The figures of Chapters 9 to 11 are copies, renamed in fig/, of the PDF files that these scripts write: bp1_*.pdf from xi, scale_mixture, wiener and process_map, bp1eta_*.pdf from eta, sb_*.pdf from sato_bondesson, and ri_*.pdf and ag_*.pdf from riesz_program/code, where figs_strategy.py writes the panels of Figures 10.6

and 11.14. The labels inside the other plots keep the notation of the notes.

section	folder	scripts (data files)	tables and figures
9.1	xi	compute.py (alpha1.json), figs.py	Tables 9.2, 9.3; Figures 9.1, 9.2
	xi	levy.py (levy.json), figs3.py	Table 9.4; Figure 9.3
	xi	below.py (below.json), errformula.py (errformula.json), figs4.py	Tables 9.5, 9.6; Figure 9.4
	xi	heat_primes.py (heat_primes.json)	Table 9.9; Proposition 9.1.17
	xi	heat_thresholds.py (heat_thresholds.json)	the thresholds of Table 9.9, in ball arithmetic
	process_map	comp.py (comp.json)	Table 9.1
	riesz_program	v1min.py (v1min.json)	Proposition 9.1.4 and the values after its proof
	repro	zeros_mp.py, v1_grid_mp.py, v1_collect.py (v1_check.json)	the check of Proposition 9.1.4
	repro	valpha_min.py (valpha_min.json)	Proposition 9.1.5 and the values after its proof
9.1.6	code	dbeta_minimum.py (dbeta_minimum.json), dbeta_window.py (dbeta_window.json)	Tables 9.7, 9.8; the window count in the proof of Theorem 9.1.11(v)
9.2	xi	compute.py (alpha1.json)	Proposition 9.2.1
	xi	mc.py (mc.json, surv.json), figs.py	Table 9.10; Figure 9.5
	xi	improve.py (improve.json), figs2.py	Tables 9.11, 9.12; Figure 9.6
	xi	rs1.py, rs1c.py (rs1.json), rs1d.py (rs1d.json), figs5.py	Figure 9.7
	process_map	bnd_alpha.py (bnd_0.6.json, bnd_0.75.json), with rs1.py of xi	Table 9.13
10.1	scale_mixture	comp.py (comp.json)	Tables 10.1, 10.2, 10.4
	scale_mixture	strip.py, dens2.py (strip.json), improve.py (improve.json), figs2.py	Table 10.3; Figure 10.1; Section 10.1.4
	wiener	wiener.py (wiener.json), figs_wiener.py	Table 10.7; Figure 10.4
	wiener	salem_riesz.py (salem_riesz.json), riesz_explicit.py (riesz_explicit.json), figs_wiener.py	Figure 10.5

	riesz_program	explicit_bounds.py (explicit_bounds.json)	the numbers in the proof of Theorem 10.1.3(3)

Table A.4: Scripts of Chapter 9 and Section 10.1, the data files they write, and the tables and figures they produce. The folders xi, process_map, scale_mixture and wiener are in basepoint_one/; the scripts of riesz_program are in its folder code/ and their data files in its folder data/, and those of Section 9.1.6 are in code/ and data/ of the archive.

section	folder	scripts (data files)	tables and figures
10.2	riesz_program	twisted.py (twisted.json), sens.py (sens.json), figs_strategy.py	Figure 10.6
	riesz_program	riesz_large.py, check_large.py (check_large.json), check_large_twisted.py (check_large_twisted.json)	the Riesz function up to $x = 10^{12}$ in Section 10.2.1
	riesz_program	riesz_thorin.py (riesz_thorin.json), figs_riesz.py	Table 10.8; Figure 10.7
	riesz_program	ggc_coef.py (ggc_coef.json), ggc_offset.py (ggc_offset.json)	Table 10.9
	riesz_program	killing.py, killing2.py (killing.json)	Table 10.10
10.3	code	sqfree_constants.py (sqfree_constants.json)	Table 10.12 and the constants of the section
	code	sqfree_sieve.py (sqfree_sieve.json)	Tables 10.11, 10.14
	code	sqfree_atoms.py (sqfree_atoms.json)	Figure 10.8
	code	sqfree_smooth.py (sqfree_smooth.json)	Tables 10.13, 10.15
	code	sqfree_gammanu.py (sqfree_gammanu.json)	Table 10.16
11.1	sato_bondesson	weil.py	Table 11.1; the dots of Figure 11.2
	sato_bondesson	comp.py (comp.json), figs.py	Table 11.2; Figures 11.1, 11.2
	process_map	comp2.py (comp2.json)	Tables 11.4, 11.5
	process_map	ou.py (ou.json), figs.py	Figure 11.3
11.2	code	wigfree_upsilon.py (wigfree_upsilon.json)	Tables 11.7, 11.8 and the numbers of Section 11.2.3
	code	wigfree_jacobi.py (wigfree_jacobi.json)	Table 11.9
	code	wigfree_gamma2.py (wigfree_gamma2.json)	Table 11.10
	code	wigfree_heat.py (wigfree_heat.json)	the numbers of Section 11.2.6
	code	wigfree_plots.py	Figures 11.5, 11.6, 11.7
11.3	eta	zeros12.py (g12.npy), base.py (base.json)	Table 11.11

section	folder	scripts (data files)	tables and figures
	`eta`	`arith.py` (`arith.json`)	Tables 11.12, 11.13
	`eta`	`hit.py`, `hit2.py` (`hit.json`), `figs.py`	Figures 11.8, 11.9, 11.10, 11.11
11.4	`riesz_program`	`almost.py` (`almost.json`), `almost_dip.py` (`almost_dip.json`)	Tables 11.15, 11.17
	`riesz_program`	`explicit_bounds.py` (`explicit_bounds.json`)	the numbers in Theorem 11.4.4 and its proof
	`riesz_program`	`more2.py` (`more.json`), `high.py` (`high.json`), `figs_almost.py` (`figs_almost.json`)	Table 11.16; Figures 11.12, 11.13
	`riesz_program`	`sens.py` (`sens.json`), `figs_strategy.py`	Table 11.18; Figure 11.14
	`riesz_program`	`killing2.py` (`killing.json`), `rmt_ratios.py` (`rmt_ratios.json`)	Table 11.19 and the exact ratios after it
	`riesz_program`	`ks.py` (`ks.json`)	Table 11.20
	`repro`	`ks_ak.py` (`ks_ak.json`)	the factors a_1, a_2, a_3 of Table 11.20
11.5	`code`	`pakes_sizebias.py` (`pakes_sizebias.json`)	the numbers after Proposition 11.5.1
	`code`	`pakes_lambert.py` (`pakes_lambert.json`)	the numbers of Section 11.5.3
	`code`	`bdlp_ladder.py` (`bdlp_ladder.json`)	Theorem 11.5.8, Remark 11.5.9 and the scan after them
	`code`	`branch_coeffs.py` (`branch_coeffs.json`)	Table 11.22, Theorem 11.5.13 and the numbers after it
	`code`	`heat_ladder.py` (`heat_ladder.json`)	Table 11.21 and the numbers of Section 11.5.4.3
	`code`	`pakes_filter.py` (`pakes_filter.json`)	Tables 11.23, 11.24; the infinite divisibility counts, the survey of discriminants, the family modulo 7 and the values $e^{\pi/q}\vartheta_a(1)$ of Section 11.5.5.4
	`code`	`pakes_filter_cert.py` (`pakes_filter_cert.json`)	the certified entries of Tables 11.23, 11.24; Propositions 11.5.17, 11.5.20; the sums S_a after Lemma 11.5.19
	`code`	`pakes_characters.py` (`pakes_characters.json`)	the checks after Propositions 11.5.15 and 11.5.17; the constants, zeros and functional equation of f_{DH}; Proposition 11.5.21

Table A.5: Scripts of Sections 10.2 and 10.3 and Chapter 11, as in Table A.4; the folders `process_map` and `eta` are in `basepoint_one/`, and the scripts of Sections 10.3, 11.2 and 11.5 are in `code/` and their data files in `data/`.

A.4 The scripts of the ferromagnetic chapter

The scripts of Chapter 13 are in the folder `ferro/`, each with the data file of its own name and the log of its run on the Hopper cluster. They work in the units of (1.2.3), and those that need the zeros read the first 1700 ordinates, to 70 digits, from a copy of `data/zeros_1700.txt` in the working folder. `fm_hermite.py` runs in two modes: with a list of values of K it brackets the thresholds $v^*(K)$ of Table 13.6 and writes `fm_hermite.json`, and with the arguments `all 200` it checks f_K itself for every $K \le 200$ and writes `fm_hermite_all.json`. `fm_maxwell.py` takes the number of draws, 10^7 for the book; `fm_maxwell_1e6.json` is an earlier run with 10^6 draws. `fm_hcm.py` computes in ball arithmetic with python-flint [110], and its output is the certificate of Proposition 13.5.8, in the variable $t/2$. `fm_cutoff.py` also works in ball arithmetic, with the zeros isolated by Arb instead of read from the file, and so do `fm_qcert.py` and `fm_tvcert.py`: in the mode `zeros 800 9600`, `fm_qcert.py` writes $\operatorname{Var} X$ and the first 800 ordinates, computed by Arb at 9600 bits, to `fm_zeros_hp.txt` (not in the archive), which both scripts then read; its modes `table`, `all 200` and `minq` give the brackets of $v^*(K)$, the positivity of f_K for every $K \le 200$ and the minima of $Q_{K,2\beta_K}$. Table A.6 lists the scripts and what they produce.

section	scripts (data files)	tables and results
13.2	`fm_interval.py` (`fm_interval.json`)	Table 13.2; the zeros after Theorem 13.2.1 and after Corollary 13.2.4
	`fm_intcert.py` (`fm_intcert.json`)	the same, in ball arithmetic
13.3	`fm_polya.py` (`fm_polya.json`)	Tables 13.4, 13.5; the values in Sections 13.3.1 and 13.3.2
13.4	`fm_polya.py` (`fm_polya.json`)	the zeros in Section 13.4.1; floating point values of the distances
	`fm_cutoff.py` (`fm_cutoff.json`)	the inequality $3s_{16} < \operatorname{Var} X < 3s_{17}$ in Theorem 13.4.3(i), in ball arithmetic
	`fm_hermite.py` (`fm_hermite.json`, `fm_hermite_all.json`)	the grid search that starts the brackets of $v^*(K)$
	`fm_qcert.py` (`fm_qcert_table.json`, `fm_qcert_all.json`, `fm_qcert_minq.json`)	Table 13.6: $f_K > 0$, $v^*(K)$ and $\min Q_{K,2\beta_K}$, in ball arithmetic
	`fm_tvcert.py` (`fm_tvcert.json`)	the distances, variances and values in Sections 13.4.1 and 13.4.2 and Tables 13.6 and 13.7, in ball arithmetic
	`fm_maxwell.py` (`fm_maxwell.json`)	the simulation after Theorem 13.4.5
13.5	`fm_sizebias.py` (`fm_sizebias.json`)	Table 13.8; the checks of Theorems 13.5.3, 13.5.6 and 13.5.7
	`fm_hcm.py` (`fm_hcm.json`)	Proposition 13.5.8

Table A.6: Scripts of Chapter 13, all in the folder `ferro/`, the data files they write, and the tables and results they produce.

A.5 Reruns and gaps

In September 2026 every script of the archive was run again from a fresh copy on the Hopper cluster of George Mason University, and every data file it wrote was compared with the archived one; the reports are `compare_report.md`, `compare_part6_report.md` and `compare_riesz_report.md` in `repro/`. Apart from the running times that four of the files record, the rerun reproduced every archived data file that a script writes to a relative 10^{-9}, most of them bit for bit, and the report of the enclosures of Chapters 6 and 7 byte for byte. There are two exceptions: in `selberg.json`, 31 of 4020 values differ by up to 2.1×10^{-8} relative, and in `improve.json` of `scale_mixture` the densities agree to 2.9×10^{-8} in absolute terms but not in relative terms in the far tail, where they are below 10^{-7} and no printed number depends on them. The enclosure program was corrected afterwards to bound the remainders of the sums in E3 and E5 and to compare every bound, rounded outward, with the number used in the text; its report replaced the archived one. The scripts changed since then and those added for the book were run on the cluster too, and the data files in the archive are their output or agree with it. On 28 September, after the last of these changes, the driver behind `compare_report.md`, which by then ran every script of `code/`, the Bosch script and the five programs of `verification_reciprocal_zeta/` outside `torus/`, was run once more in an empty folder. All 65 runs ended without error, and every data file that a script writes agreed with the archived one to a relative 10^{-9}, apart from the running times that six of the files record, one host name and the same 31 values of `selberg.json`; the report of the enclosures and the output of `spot_checks.py` were reproduced byte for byte (`repro/compare_report_0928.md`). The scripts `vdpair.py`, `idsd_height.py`, `threezones.py` and `cm_orders.py`, added on 29 September after that run, were run on the cluster on their own, `vdpair.json`, `idsd_height.json`, `fig/threezones.pdf` and `cm_orders.json` are their output, and the driver now includes them.

The rerun also showed what the archive lacked, and each item was followed up on the cluster in a folder of `repro/reconstruct/` with its scripts, its logs and a report. `primeside.py` imports a module `polya` that was not in the archive; with the reconstructed `polya.py`, its `results_prime.json` agrees with the archived file in all 162 values to 7.4×10^{-16} relative, 148 of them bit for bit, and `plots.py` run on it regenerates `extra.json` and `sens2.json` byte for byte. `ttest.py` and `ttest2.py` stopped because they were given no arguments; the driver now passes the command lines of Section A.1, and with them `ttest2.py` reproduces the node counts and the errors of Table 2.2, while its times per point vary with the machine. Two lines of the Bosch script executed a file `gaptest.py` that is not in the archive, which defined the functions of block 12 again and an empty list `out`; they are replaced by `out=[]`, and the script runs as it stands; Appendix D says what it

reproduces. `spot_checks.py` writes no file, so the rerun had nothing to compare; run as in Section A.2, it reproduces the report in the archive byte for byte, and the 112 numbers and statements of Chapters 6 and 7, Section 5.1 and this appendix that the report bears on agree with it at the printed precision. Of the five data files that no script wrote, `hankel.json`, `spacing.json` and `sinh_bench.json` are reproduced by the reconstructions of Section A.2 and `riesz_explicit.json` by `riesz_explicit.py` of `basepoint_one/wiener`; the fifth is `results_main.json`.

Four gaps remain in these items. The printed output of the Bosch script is not the record of a single run: output lines 37 to 39 and 109 come from code that is not in the archive, and lines 93 to 102 from an earlier version of block 13; reconstructions in `repro/reconstruct/bosch_v7` reproduce lines 37 to 39 and 109 and the density at $t = 0.003$ at 150 digits, but the output file is kept as the authors made it. `primeside.py` runs only with the reconstructed `polya.py`, and its Pólya column then differs from the archived one by up to four units in the last place, a difference that none of 196 tested conventions removes and that no printed number shows. `spot_checks.py` still prints, beside the values it computes, three reference values from an earlier version of Chapters 6 and 7, and its docstring gives two minutes for a run that takes about eight; the values it computes agree with the text. Finally, `data/results_main.json` has no script that writes or reads it. A reconstruction in `repro/reconstruct/primeside` reproduces 117 of its 135 numbers, but not its distances `tv` and `sup`, which rest on a tail model that cannot be recovered, nor its timings; the book takes from it only $\gamma_{1700} = 2197.2587$, which the zero lists also give.

Some computations of the book have no script in the archive. In Chapters 1 to 5 and 8 these are Table 2.1, whose values `plots3.py` holds as constants, Tables 4.1 and 8.1, and Figures 2.4, 3.7, 8.12, 8.2, 8.3, 8.4, 8.5 and 8.8, most of whose data files exist although the code that drew them does not; the intermediate values of Section 2.3, which come from calls of `polya_flint.py` at its defaults; the check with the step h halved and the time per evaluation in Section 2.1.1; the double precision run with $\delta = 0.15$ of Section 2.1.4; and the numerical checks in the text of Section 4.2, made with `mpmath` as that section says. The Horner experiment of Section 2.1.5 is reproduced by `horner100.py` in `repro/reconstruct/ttest2`, outside `code/`; `lehmer.py` prints the timings of Section 2.2.3.1 but does not save them; and `kmono.json` holds only minima that underflow to zero, while `kmono.py` draws Figure 5.2 from the values it computes. In Chapters 6 and 7 no script reproduces Tables 6.3, 7.1, 7.2 and 7.7, the zeros of the truncations in Table 7.3, the rows of Table 7.8 that `spot_checks.py` does not check, or the seven figures in `rzfig/`. Other computed statements of Section 7.1 have no script either: the grid evaluation of Proposition 7.1.5, with the mode of f and the agreement of Talbot and de Hoog inversion quoted after it, the values of w_σ in Section 7.1.2.3 and the data of Proposition 7.1.18. In Chapters 9 to 11 no script produces the column of Table 9.2 that sums over the first 1700 zeros, `ou.json` does not store the batch

standard errors of the autocorrelations quoted in Section 11.1, and `comp.py` of `process_map` prints the sample mean 0.6664 of S_2 given with Table 9.1 but does not store it in `comp.json`.

Appendix B turns from the computations of this book to those that came before it.

Appendix B

A short history of computations on the zeros

Table B.1 lists landmarks in the computation of the zeros of ζ, from Riemann's own calculation to the rigorous verification of Platt and Trudgian. Every verification in it from Backlund's on rests on two ingredients, an evaluator of ζ on or near the critical line and a certificate that no zero has been missed; Section 2.2 describes both. The window on which Chapters 3 to 5 measure distances, the 79 zeros below height 200, is the range that Backlund verified by hand in 1914 [9].

The computations of this book do not extend the table. They compute a measure rather than a list of zeros, together with its distance to U_{RH}, and the end of Section 2.2.3.2 sets out how this work divides with the classical evaluators. The table meets the book directly in four places. Where Turing bounds an average of S to certify a count (Section 2.2.1), the zero density identity of Section 3.4 is exact, and its value is an integer at every resolution; at the resolution $\frac{1}{2}$, the basepoint one of Section 9.1, it is Backlund's method with the vertical side moved from $\operatorname{Re} s = 2$ to $\operatorname{Re} s = 1$, and it gives the count $N(T) = M_{1/2}(T) + \tau_{1/2}(T)$ with no hypothesis, which Section 2.2.3.2 computes at height 1000. Lehmer's close pair near height 7005 [143] is computed from the Pólya kernel in Section 2.2.3.1 and followed under the de Bruijn Newman flow in Section 8.4. The Odlyzko Schönhage algorithm, with Odlyzko's zeros as the reference measure, is one of the evaluators that Section 2.2.2 proposes for the Thorin comparison at large height. The verification of Platt and Trudgian supplies the height $T_0 = 3 \times 10^{12}$ behind the unconditional statements of Sections 4.1, 4.3, 6.2, 7.1, 7.2, 10.1 and 11.4. Appendix C lists the notation that several sections share.

year	authors	result
1859	Riemann	the first zeros, in unpublished notes later found by Siegel [238]
1903	Gram	the first 15 zeros; Gram points [83]
1914	Backlund	all 79 zeros below height 200 [9, 10]
1925	Hutchinson	all 138 zeros below height 300 [99]
1936	Titchmarsh	the first 1041 zeros, all below height 1468, by the Riemann Siegel formula, computed with the help of Comrie [253]
1953	Turing	all 1104 zeros below height 1540, on the Manchester computer; Turing's method [262]
1956	Lehmer	the first 25000 zeros; the close pair near height 7005 [143, 144]
1969	Rosser, Yohe, Schoenfeld	the first 3.5×10^6 zeros; Rosser's rule [227]
1986	van de Lune, te Riele, Winter	the first 1.5×10^9 zeros [264]
1987	Odlyzko	10^5 zeros near the 10^{12}th zero, and the statistics of their spacings [171]
1988	Odlyzko, Schönhage	fast evaluation of ζ at many points [172]
2004	Gourdon	the first 10^{13} zeros, not interval certified [82]
2021	Platt, Trudgian	RH verified rigorously below height 3×10^{12} [196]

Table B.1: Landmarks in the computation of the zeros of ζ. The zeros counted in each row were located on the critical line. For the computations up to 1969 see [72, Chapters 6 to 8].

Appendix C

Notation

The tables list the notation used in more than one section and the place where each symbol is defined; the first five cover the book and record the second meanings that some letters carry in particular chapters and sections. The clock is a function of $u = s^2$, and its Thorin measures live in a variable z in which the atoms of U_{RH} sit at the squared ordinates γ^2; the square root chart $\theta = \sqrt{z}$ puts them back at the ordinates. A resolution $\varepsilon > 0$ means the basepoint $\alpha = \frac{1}{2} + \varepsilon$. A zero is written $\rho = \beta + i\gamma$, and sums and products over k run over the pairs $\{\rho, 1-\rho\}$. Sections 4.2 and 4.3 use their own variables, and so do Chapters 6 and 7; the last two tables list them. Other notation used in a single section is defined where it is used.

symbol	meaning	defined in
ξ, Ξ	the completed zeta function $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$; $\Xi(\theta) = \xi(\frac{1}{2} + i\theta)$	Section 1.2
ρ, γ_k	a zero $\rho = \beta + i\gamma$ of ξ, that is a nontrivial zero of ζ; the ordinates $0 < \gamma_1 \le \gamma_2 \le \cdots$ of the zeros with $\gamma > 0$	Section 1.2
a_k	$-(\rho_k - \frac{1}{2})^2$, one for each pair $\{\rho, 1-\rho\}$, equal to γ_k^2 on the line (in Section 8.3, the Taylor coefficients of $\log \xi(1+w)$)	(1.2.1)
Φ, p, X	the Pólya kernel in Titchmarsh's normalisation; the Pólya density $p = \Phi/\Xi(0)$; a random variable with density p (in Section 6.2, p is the Meixner density; in Section 4.3, X is the clock variable; in Sections 9.1 and 11.3, X is the prime cutoff)	(1.2.3), (1.2.4)
S_2, Y, J	$S_2 = \sum_{k \ge 1} \frac{2}{\pi^2 k^2}(e_k + e'_k)$ with independent standard exponential variables, the sum of two independent copies of the first passage time J of the level 1 by a three dimensional Bessel process started at 0; $Y = \frac{1}{2}\log(\pi S_2/2)$, with $\mathbb{E}e^{sY} = 2\xi(s)$; X is Y tilted by $e^{Y/2}$	Section 1.2, Section 9.1.1

symbol	meaning	defined in
u, z, θ	the variable $u = s^2$ of the clock (also the variable of Φ); the variable of Thorin measures; the chart $\theta = \sqrt{z}$, which is also the ordinate (in Chapters 6 and 7 and in Sections 4.2 and 4.3, s and t play the roles of u and z; the Kent approximants of Section 8.1 write w for u)	Section 1.1
$G(u)$	$\xi(\frac{1}{2} + \sqrt{u})/\xi(\frac{1}{2})$; the clock is $1/G$ (in Chapters 6 and 7, G is the gamma factor and the clock is R; in Section 7.1.3, G is a Laguerre Pólya function)	Section 1.1
$\mathcal{G}(u,t)$	the Gaussian smoothing $\sum e^{-2t(\tilde{\gamma}-u)^2}$ of the zeros at resolution $1/(2\sqrt{t})$, with $\tilde{\gamma} = \gamma - i\delta$ for a zero $\frac{1}{2} + \delta + i\gamma$ with $\gamma > 0$	Section 5.1.5
T, e_γ	the clock variable, with $\mathbb{E}e^{-uT} = 1/G(u)$, which exists with no hypothesis and is the centre clock of Chapters 9 to 11; under RH, $T = \sum_{\gamma>0} e_\gamma/\gamma^2$ with independent standard exponential variables e_γ (T is also a height, as in $N(T)$)	Section 1.2, Proposition 1.2.1
U, U_α	the Thorin measure of a GGC H, with $\mathbb{E}e^{-uH} = \exp\{-du - \int \log(1+u/z)\, U(dz)\}$ and drift d; the signed measure $U_\alpha(dz) = v_\alpha(\sqrt{z})\, dz/(2\pi\sqrt{z})$ at a basepoint α; for the clock, the Thorin functional $\sum_k \delta_{a_k}$	Section 1.2, (1.2.6), Section 5.3.1
k	the canonical function of an infinitely divisible law on $[0,\infty)$ with Lévy density $k(t)/t$, that is t times its Lévy density ($k(y)$ in Sections 5.1 and 6.2)	Section 1.3

symbol	meaning	defined in
U_{RH}, U_{line}, $U^{\pm}_{\mathrm{RH}}$	$\sum_{\gamma>0} \delta_{\gamma^2}$; the same sum over the zeros on the line; $\sum_{\gamma>0}(\delta_\gamma + \delta_{-\gamma})$	(1.1.1), Proposition 1.2.3
$\mathcal{W}$, $T_{\mathcal{W}}$, $U_{\mathcal{W}}$, U	the principal branch of Lambert's function, $\mathcal{W}(u)e^{\mathcal{W}(u)} = u$ (not the heat trace W); the Lambert law, with Laplace transform $\mathcal{W}(u)/u = e^{-\mathcal{W}(u)}$; its Thorin measure; a variable uniform on $(0,1)$ (not the Thorin functional U)	Sections 11.5.1, 11.5.3
φ_α, T_α	$\xi(\alpha)/\xi(\alpha + \sqrt{u})$, the clock at a basepoint $\alpha \geq \frac{1}{2}$, with $\varphi_{1/2} = 1/G$; the variable with $\mathbb{E}e^{-uT_\alpha} = \varphi_\alpha(u)$, which exists for every $\alpha > \frac{1}{2}$ and is a GGC if and only if no zero has $\beta > \alpha$, in particular for every $\alpha \geq 1$; $T_{1/2} = T$ (in Section 4.2, T_α is Bosch's ratio of stable variables)	Theorems 1.2.2, 9.2.5

symbol	meaning	defined in
$\widetilde{T}$, $\widetilde{T}_\alpha$, $\sigma^{(\alpha)}$	the centre clock tilted by $e^{-T/4}$, and by $e^{-(\alpha-\frac{1}{2})^2 T}$, so that $\widetilde{T} = \widetilde{T}_1$; the subordinator with Laplace exponent $u + 2(\alpha - \frac{1}{2})\sqrt{u}$, with $T_\alpha \stackrel{d}{=} \sigma^{(\alpha)}_{\widetilde{T}_\alpha}$	Theorems 9.2.2, 9.2.5
H_a, $H_a^{(c)}$	the first passage time of $B_t + ct$ to the level $a \geq 0$, for a standard Brownian motion B, and $H_a = H_a^{(0)}$; unrelated to H_t	Section 9.2
Y_α, Z	the normal scale mixture $Y_\alpha = \sqrt{2T_\alpha}\, Z$, with Z standard normal and independent of T_α, whose characteristic function is $\xi(\alpha)/\xi(\alpha + \lvert u\rvert)$ (in Section 11.1, Z is the subordinator with $T \stackrel{d}{=} \int_0^\infty e^{-t}\, dZ_t$)	Proposition 10.1.1
$W(t)$, $c_k(t)$	the heat trace $\sum_k e^{-a_k t}$, also written $\int e^{-tz} U(dz)$; the signed derivatives $(-1)^k W^{(k)}(t)$ (in Chapters 6 and 7 the time is x, or y where x is a Lévy variable, as in the proof of Lemma 6.2.5 and from Proposition 6.2.16 to the end of Section 6.2; in Sections 4.2 and 4.3, W is also the Mellin variable with $\mathbb{E}W^z = \Xi(0)/\Xi(z)$; the numbers $c_k(u)$ of Section 4.1 are local to it)	Sections 1.2, 5.1
m_n, $\mathcal{H}_N$, $\mathcal{H}'_N$, ι_N, ι'_N	the Stieltjes moments $\sum_k a_k^{-n-1}$, with $m_0 = \mathbb{E}T$; the Hankel matrices $[m_{i+j}]_{i,j<N}$ and $[m_{i+j+1}]_{i,j<N}$; the numbers of their negative eigenvalues, which count the zeros off the line (Section 8.4 uses the same symbols for the moments $m_n(t)$ of H_t)	Section 5.1, Theorem 5.1.5
$W^{[j]}$, β_j, B_j	the ladder functions $(-t\frac{d}{dt})^j W$, with $W^{[0]} = W$ (along the de Bruijn Newman flow $W^{[j]}(x;t)$, with x the time of the heat trace and t that of the flow); the polynomials with $(-t\frac{d}{dt})^j e^{-at} = \beta_j(at)e^{-at}$; the Touchard polynomials, $\beta_j(u) = (-1)^j B_j(-u)$	Section 11.5.4
L_n, L_∞	Urbanik's classes of multiply self decomposable laws, with L_0 the self decomposable laws; unrelated to the Post Widder functionals $L_n[W]$ of Section 7.2.2	Section 11.5.4.1
$Z^{[j]}$	the jth driving process of the centre clock, $Z^{[1]} = Z$, with Laplace exponent $(u\frac{d}{du})^j \log G$	Lemma 11.5.7
g, g_k, H, λ^*	the candidate generating exponent $\Psi_Z \circ H$ and its Taylor coefficients, with H the inverse of $\log G$; the branch point $\log\lvert G(u^*)\rvert + i\pi$ of H	Section 11.5.4.4
Φ_t, $\widehat{g}$, R_g	the damped Pólya kernel $e^{tv^2/4}\Phi(v)$; the cosine transform $\int g(v)\cos(\theta v)\, dv$ of an even kernel; $\widehat{g}(0)/\widehat{g}(i\sqrt{s})$, with $R_{\Phi_t} = R_t$	Section 11.5.5.1

symbol	meaning	defined in
X_t, p_t, Q_t, ϱ_t, h_t	the variable with density $p_t = \Phi_t/\Xi_t(0)$, $X_0 = X$; the radius of Khinchine's mixture, $X_t = UQ_t$; the radius of the arcsine mixture, $X_t = \varrho_t \cos\Theta$, and its density	Section 11.5.5.2
$\mathcal{B}$, f, $\mathcal{B}_g$, $\mathcal{B}_t$	$\frac{\pi}{2}S_2$, with $\mathbb{E}\mathcal{B}^{s/2} = 2\xi(s)$; its density $b^{-5/4}\Phi(\frac{1}{2}\log b)$; the variable with density proportional to $b^{-5/4}g(\frac{1}{2}\log b)$; $\mathcal{B}_{\Phi_t}$	Section 11.5.5.3
ℓ, $\kappa_k(\lambda)$, $c_k(\lambda)$, σ_λ	a Laplace transform $\mathbb{E}e^{-\lambda\mathcal{V}}$; the cumulants $(-1)^k(\log\ell)^{(k)}(\lambda)$; $\kappa_{k+1}(\lambda)/k!$, the moments of σ_λ for a GGC (not the $c_k(t)$ of the heat trace)	(11.5.9)
$\mathcal{K}_n(\lambda)$, $\mathcal{K}'_n(\lambda)$	the Hankel matrices $[c_{i+j}(\lambda)]_{i,j<n}$ and $[c_{i+j+1}(\lambda)]_{i,j<n}$, positive semidefinite for a GGC	Section 11.5.5.3
ℓ_t, $\dot{\ell}$, $\mathcal{J}$	$\mathbb{E}e^{-\lambda\mathcal{B}_t}$; its derivative in t at $t=0$, up to a term in ℓ_0; the first order change of the Thorin measure in a gap, in the proof of Proposition 11.5.17	Section 11.5.5.3
ϑ_a, $\xi(s,a)$, K_a, $\mathcal{B}_a$	the theta series $\sum n^p a_n e^{-\pi n^2 x/q}$ of a periodic sequence; its completed Dirichlet series; the kernel $2e^{(p+1/2)v}\vartheta_a(e^{2v})$; $\mathcal{B}_{K_a}$ (for $a = \chi_{-4}$, $1/\mathcal{B}_a$ is the variable Z of Remark 8.5.22)	Section 11.5.5.4
χ_d, S_a, Σ_a	the real primitive character of discriminant d ($\chi_{-4} = \chi_4$); the positivity sum of Lemma 11.5.19; the numerator of the reflected transform (11.5.10)	Section 11.5.5.4
f_{DH}, $\varkappa$, f_r	the function of Davenport and Heilbronn; its coefficient 0.28407904…; the self dual family modulo 7	Section 11.5.5.4
k_n	the inverse Laplace transform of Li's φ_n, $\lambda_n = \int k_n W$	Problem 11.4

symbol	meaning	defined in
ε	the resolution; the basepoint is $\frac{1}{2}+\varepsilon$	Section 1.1
$v_\alpha(r)$	$\mathrm{Re}(\xi'/\xi)(\alpha+ir)$, which is π times the density of U_α in the chart $\theta = \sqrt{z}$ (Chapters 9 to 11 write $v_\alpha(\theta)$)	Section 1.2
b_α, Π_α	$b_\alpha = \frac{\xi'}{\xi}(\alpha) = v_\alpha(0)$, the drift of the Lévy form $\log\frac{\xi(\alpha+s)}{\xi(\alpha)} = b_\alpha s + \int_0^\infty (e^{-sx} - 1 + sx)\,\Pi_\alpha(dx)$ (for $\alpha \geq 1$, and for $\frac{1}{2} < \alpha < 1$ when $\Theta < \alpha$), with $b_1 = \sum_\rho \rho^{-1} = \lambda_1$; its signed Lévy measure, with atoms $\Lambda(n)/(n^\alpha \log n)$ at $\log n$	Proposition 9.1.3, Section 9.1.5

symbol	meaning	defined in
A, A_α, Q, ν_1	the gamma piece $A(\theta) = \frac{1}{2}[\operatorname{Re}\psi(\frac{3+i\theta}{2}) - \psi(\frac{3}{2})]$ and the compensated prime piece $Q(\theta) = \lim_{X\to\infty} \int_0^{\log X} 2\sin^2(\frac{x\theta}{2})\,\nu_1(dx)$ of $v_1 = b_1 + A + Q$; the gamma piece A_α at α; the prime measure $\nu_1 = \sum_n \Lambda(n)n^{-1}\delta_{\log n} - dx$ on $(0,\infty)$, compensated by Lebesgue measure and unrelated to $\nu(\sigma,T)$ (Section 11.3 uses A, A_α and Q for $L(s,\chi_{12})$)	(9.1.6), Proposition 9.1.2
S, T_1°; S_α, T_α°	the positive stable variable of index one half with $\mathbb{E}e^{-uS} = e^{-b_1\sqrt{u}}$; the GGC with no drift, independent of S, whose Thorin density in the chart $\theta = \sqrt{z}$ is $(A+Q)/\pi$, so that $T_1 \stackrel{d}{=} S + T_1^\circ$ (unrelated to $S(t)$ and S_2); the same at a basepoint $\alpha \geq 1$, with b_α and $(v_\alpha - b_\alpha)/\pi$, so that $T_\alpha \stackrel{d}{=} S_\alpha + T_\alpha^\circ$	Propositions 9.1.4, 9.1.5
V_α, V	the variable with $\mathbb{E}e^{sV_\alpha} = \xi(\alpha+s)/\xi(\alpha)$, whose density $e^{(\alpha-\frac{1}{2})y}p(y)\,\xi(\frac{1}{2})/\xi(\alpha)$ is a tilt of the Pólya density, with no hypothesis; $V_{1/2} = X$ and $V = V_1$	Section 1.2, Section 9.1.1
$\rho_\varepsilon(\theta)$	the Thorin density at basepoint $\frac{1}{2}+\varepsilon$ in the chart $\theta = \sqrt{z}$, $\pi^{-1}v_{1/2+\varepsilon}(\theta)$	(1.2.7)
μ_ε	the signed measure $U_{1/2+\varepsilon}$, with density $\rho_\varepsilon(\theta)\,d\theta$ in the chart θ; the Thorin measure of the GGC $T_{1/2+\varepsilon}$ when $\frac{1}{2}+\varepsilon > \operatorname{Re}\rho$ for every zero, and called the Thorin measure at resolution ε in general ($\mu(n)$ is the Möbius function)	Section 1.2

symbol	meaning	defined in
P_a	the Poisson (Cauchy) kernel $a/(\pi(a^2+x^2))$, with $P_{-a} = -P_a$	Section 1.2
$d_{\mathrm{match}}(\varepsilon)$	$\|\mu_\varepsilon - P_\varepsilon * U_{\mathrm{RH}}^{\pm}\|_{\mathrm{TV}}$, the distance to the RH measure smoothed at the same resolution, which vanishes under RH	Section 3.2
$\tilde{d}_{\mathrm{match}}(\varepsilon)$	$\|\mu_\varepsilon - P_\varepsilon * \widetilde{U}\|_{\mathrm{TV}}$, the matched distance computed against the model $\widetilde{U}$ of the zero list (the first 1700 ordinates and the density $\pi^{-1}\vartheta'$ beyond them, each with its reflection)	Section 3.2
κ_{cell}	the cell constant $\pi^{-1}\sum_k \lvert\Xi'/\Xi(b_k) - \Xi'/\Xi(b_{k-1})\rvert$ of the 79 cells $(b_{k-1}, b_k]$ below height 200, so that $d_{\mathrm{cell}}(\varepsilon) = \kappa_{\mathrm{cell}}\,\varepsilon + O(\varepsilon^3)$	Section 3.2

symbol	meaning	defined in
$M_\varepsilon(\theta)$	the cumulative mass $\mu_\varepsilon((0,\theta^2]) = \pi^{-1}\arg\xi(\frac12+\varepsilon+i\theta)$, with the branch of the argument that vanishes at $\theta=0$ (the Mertens function $M(t)=\sum_{d\le t}\mu(d)$ in Section 10.3)	Section 2.1.2, Section 3.4
$\tau_\varepsilon(T)$	π^{-1} times the change of $\arg\xi$ along the segment from $\frac12+\varepsilon+iT$ to $\frac12+iT$, so that $M_\varepsilon(T)+\tau_\varepsilon(T) = N(T) - 2\nu(\frac12+\varepsilon,T)$ ($\tau(n)$ is Ramanujan's function in Section 8.5, and τ is the ordinate in Chapters 6 and 7)	Proposition 3.4.2
$N(T)$, $\nu(\sigma,T)$	the number of zeros with $0<\operatorname{Im}\rho\le T$; the number of those with $\operatorname{Re}\rho>\sigma$ (Sections 3.4 and 11.4 write $N(\sigma,T)$ for the number with $\operatorname{Re}\rho\ge\sigma$)	Section 1.1
$S(t)$, $S_\varepsilon(t)$	$\pi^{-1}\arg\zeta(\frac12+it)$, with the argument continued along $2\to 2+it\to\frac12+it$; $\pi^{-1}\arg\zeta(\frac12+\varepsilon+it)$, continued along the line $\operatorname{Re}s=\frac12+\varepsilon$ from $S_\varepsilon(0+)=-1$ if $0<\varepsilon<\frac12$ and from 0 if $\varepsilon>\frac12$	Sections 2.2.1, 3.4.1
$\vartheta(t)$	the Riemann Siegel theta function $\operatorname{Im}\log\Gamma(\frac14+\frac{it}{2})-\frac{t}{2}\log\pi$ (written θ in Chapters 6 and 7; in Section 4.3, θ and ϑ are angles)	Section 2.2
$Z(t)$	Hardy's function $e^{i\vartheta(t)}\zeta(\frac12+it)$, real for real t	Section 2.2
Θ	a height, the top of a range of ordinates, with $\Theta=200$ for the window of Chapters 2 to 5; in Chapters 6 and 7 and Sections 9.1 and 10.2, $\sup\operatorname{Re}\rho$	Section 2.1.2, Section 6.1
T_0	the height 3×10^{12} below which RH has been verified [196] (used in Sections 4.1, 4.3, 6.2, 7.1, 7.2, 10.1 and 11.4)	Section 1.1
λ_n	the Keiper Li coefficients $\sum_\rho[1-(1-1/\rho)^n]$	Section 8.3
H_t, Λ, Φ_{dBN}	the de Bruijn Newman flow, normalised by $H_0(z)=\frac18\Xi(z/2)$; the de Bruijn Newman constant; the kernel $\Phi_{\mathrm{dBN}}(u)=\frac12\Phi(2u)$ of the flow ($\Lambda(n)$ is von Mangoldt's function)	Section 8.4
$P(\theta)$, ν_n, A_P, K^P	Pólya's Bessel function $P(\theta)=K_{i\theta/2}(2\pi)$; the zeros of $\nu\mapsto K_{i\nu}(2\pi)$, so that P vanishes at $\theta=\pm2\nu_n$; the operator $-\frac{d^2}{dx^2}+e^{2x}$ on $[\log2\pi,\infty)$ with a Dirichlet condition, whose eigenvalues are the ν_n^2; the trace $K^P(y)=\sum_n e^{-2\nu_n y}$	Section 8.7

symbol	meaning	defined in
δ	the displacement of an off line zero $\frac{1}{2} \pm \delta + i\gamma$, written d in Sections 5.2, 5.3, 5.4, 8.3, 8.4 and 12.2; in Chapter 2 and where its parameters are quoted, the tilt margin, with $0 < \delta < \pi/4$; in Section 4.3, the angle of a zero	Propositions 1.2.3, 2.1.1
η	the tilted height $\eta = \pi/4 - \delta$ (Dedekind's eta function in Sections 8.5, 4.3 and 11.3; Dirichlet's eta function $\eta(s) = (1 - 2^{1-s})\zeta(s)$ in Section 10.1 and, as $\eta(1 + v) = (1 - 2^{-v})\zeta(1 + v)$, in Chapters 6 and 7)	Proposition 2.1.1
$K_\eta(\varepsilon)$	the kernel constant $\int_{\mathbb{R}} \lvert\Phi(x \pm i\eta)\rvert\, e^{\varepsilon\lvert x\rvert}\, dx$ of the trapezoid bound, with $K_\eta = K_\eta(0)$	Section 2.1.1
ψ	the digamma function Γ'/Γ (Chebyshev's function $\sum_{n \le y} \Lambda(n)$ at real $y \ge 1$ in Sections 7.1, 9.1, 10.2 and 11.4, and $\psi(y, \chi_{12})$ in Section 11.3)	Section 1.2
γ_E	Euler's constant	Section 5.1
$\mathcal{N}$	the standard normal distribution function, with $\overline{\mathcal{N}} = 1 - \mathcal{N}$	Section 8.2
$\xi(s, \chi_{12})$, K	the completed L function $(12/\pi)^{s/2}\Gamma(s/2)L(s, \chi_{12}) = \int_0^\infty \eta(ix)x^{s/2}\, dx/x$; its kernel $K(u) = 2e^{u/2}\eta(ie^{2u})$, with $\xi(\frac{1}{2} + s, \chi_{12}) = \int_{\mathbb{R}} K(u)e^{su}\, du$ (a variable Y with density $K/\xi(\frac{1}{2}, \chi_{12})$ plays the role of X)	Section 8.5.1
$\xi(s, \chi_4)$, K_4, Y_4	the completed Dirichlet beta function $(4/\pi)^{(s+1)/2}\Gamma(\frac{s+1}{2})L(s, \chi_4)$, equal to $\int_0^\infty \eta(ix)^3 x^{(s+1)/2}\, dx/x$; its kernel $K_4(u) = 2e^{3u/2}\eta(ie^{2u})^3$, with $\xi(\frac{1}{2} + s, \chi_4) = \int_{\mathbb{R}} K_4(u)e^{su}\, du$; the variable with density $K_4/\xi(\frac{1}{2}, \chi_4)$	Section 8.5.7.1
$R(x)$, $R_h(x)$	Riesz's function $x\sum_{n \ge 1} \mu(n)n^{-2}e^{-x/n^2}$; its twist $x\sum_{n \ge 1} \mu(n)n^{-2}(x/n^2)^{ih}e^{-x/n^2}$, with $R_0 = R$ (in Chapters 6 and 7, R is the reciprocal Xi function)	Sections 10.1.8, 10.2.1
$\xi_K(s)$, C_K	the completed Dedekind zeta function $s(s-1)\lvert d_K\rvert^{s/2}\Gamma_{\mathbb{R}}(s)^{r_1}\Gamma_{\mathbb{C}}(s)^{r_2}\zeta_K(s)$ of a number field K, with $\xi_{\mathbb{Q}} = 2\xi$; the constant of $\sum_{N(\mathfrak{a}) \le x} \mu(\mathfrak{a})^2/N(\mathfrak{a})$, with $C = C_{\mathbb{Q}} = 1.7171377$	Sections 10.3.1, 10.3.3
$\lVert\cdot\rVert_{\mathrm{TV}}$	the total variation $\lvert\cdot\rvert(E)$ of a signed measure on the window E	Section 1.1
GGC, HCM, ID, SD, BO, EGGC	generalised gamma convolution; hyperbolically completely monotone; infinitely divisible; self decomposable; the Bondesson class; the extended class of generalised gamma convolutions, the weak limits of convolutions of gamma laws and their reflections	Section 1.3, Section 1.3.2

Sections 4.2 and 4.3 use the following notation.

symbol	meaning	defined in
s, $\varphi(s)$	the variable u of the clock; the clock $\varphi_{1/2}(s) = \xi(\frac{1}{2})/\xi(\frac{1}{2} + \sqrt{s})$	Section 4.2
W	the Mellin variable with $\mathbb{E}W^z = \Xi(0)/\Xi(z)$, not the heat trace	Section 4.2
T_α	Bosch's variable $(Z_\alpha/\tilde{Z}_\alpha)^\alpha$ with independent positive α stable $Z_\alpha, \tilde{Z}_\alpha$; here α is a stability index, not a basepoint	Section 4.2
X, p, ε_h, δ, θ, ϑ	under RH, the clock variable T; an exponent, not the Pólya density; an elasticity, not a resolution; the angle of a zero; angles, not the ordinate and the Riemann Siegel function	Section 4.3
ψ, S_{on}, R_{off}	$-\varphi'/\varphi = \sum_\gamma (s+\gamma^2)^{-1}$, the function of the Stieltjes form; its sums over the real and over the nonreal γ (in Section 11.4.7, the parts of G'/G in the variable u)	Theorems 4.2.4, 4.3.10
D_γ, $\mathcal{E}_{T_0}$, θ_0	for a nonreal zero $\gamma = a + ib$, the closed disc $\lvert s - (b^2 - a^2)\rvert \le 2ab$ with diameter from $-\gamma^2$ to $-\bar{\gamma}^2$; the parabolic sliver $\{x + iy :\ x < -T_0^2 + T_0 + \frac{1}{4},\ y^2 < \frac{1}{2} - x\}$ that contains every D_γ; its angle at the tip, seen from the origin	Theorem 4.3.10

The numerical checks behind the statements of Section 4.3 come from one script in the archive, whose blocks Appendix D relates to the statements they check, and, for Theorem 4.3.10, Remark 4.3.11 and Figure 4.3, from `pakes_disc.py` (Appendix A).

symbol	meaning	defined in

Chapters 6 and 7 use the following notation.

symbol	meaning	defined in
s, t, x	the variable u of the clock; the variable z of Thorin measures; the time of the heat trace $W(x)$	Section 6.1
τ, z	the ordinate; the argument of ζ and ξ	Section 6.1
G, R, φ	the gamma factor $\Gamma(\frac{1}{4}+\frac{\sqrt{s}}{2})\pi^{-\sqrt{s}/2}/\Gamma(\frac{1}{4})$; the reciprocal Xi function $\xi(\frac{1}{2})/\xi(\frac{1}{2}+\sqrt{s})$, the clock $1/G$ of Section 1.2; the normalised reciprocal zeta function $\varphi(s)=\zeta(\frac{1}{2})/\{(1-4s)\zeta(\frac{1}{2}+\sqrt{s})\}$, with $\varphi = RG$	(6.1.4)
R_σ, φ_σ, R_t	$\xi(\sigma)/\xi(\sigma+\sqrt{s})$, which the rest of the book writes φ_σ; the normalised reciprocal zeta function $\zeta(\sigma)/\{(1-s/(1-\sigma)^2)\zeta(\sigma+\sqrt{s})\}$ at σ, with $\varphi_{1/2}=\varphi$, not the φ_σ of the rest of the book; $H_t(0)/H_t(2i\sqrt{s})$, with $R_0 = R$ (also in Section 8.4, which writes u for s)	Sections 6.1, 7.2
θ, c, c_n	the Riemann Siegel theta function (ϑ elsewhere); $c=-\theta'(0)=\zeta'(\frac{1}{2})/\zeta(\frac{1}{2})=2.6860917$ (also quoted in Sections 5.1, 8.5 and Appendix A); $c_n = 2n+\frac{1}{2}$	Section 6.1, (6.1.6)
$W_0(x)$, $w(x)$, $M(t)$	the archimedean part $\frac{1}{\pi}\int_0^\infty e^{-x\tau^2}\theta'(\tau)\,d\tau$ of $W(x)$; the canonical function $w = W - W_0$ of φ; $N(\sqrt{t})-\theta(\sqrt{t})/\pi = 1+S(\sqrt{t})$	Section 6.1
X_φ, X_R	random variables with Laplace transforms φ and R, so that X_R has the law of the clock variable T	Section 6.2
κ, Y_κ	a Poisson width; the variable with $\mathbb{E}e^{-sY_\kappa}=\kappa/(\kappa+\sqrt{s})$, whose Thorin density is a Poisson kernel of width κ in $\sqrt{t}$	Section 6.2
$\eta(1+v)$, χ_4, $\mathcal{S}_n$	Dirichlet's eta function; the nontrivial character modulo 4; the truncated lattice variables $\mathcal{S}_n=\sum_{k\le n}k^{-2}(e_k+e'_k)$, with $\mathcal{S}=\frac{\pi^2}{2}S_2$ and tails $\mathcal{T}_n=\mathcal{S}-\mathcal{S}_n$ (also in Section 11.1)	Sections 6.1, 7.2
B_N, h_λ	the weights $B_N=\sum_{n^m=N}d_m a_n$ of the Vere-Jones representation, with $a_n=\Lambda(n)/(\sqrt{n}\log n)$ and $\sum_n d_n n^{-v}=1/\eta(1+v)$, which vanish unless N is a prime power p^n and are then sums of necklace polynomials at $p^{-1/2}$; the heat trace of the sawtooth measure U_λ, a Jacobi theta function	Section 6.1, Theorem 7.1.8

Appendix D

Verification code for the Bosch type theorem

The script behind the numerical checks of Section 4.3, `code_bosch/verify_bosch_xi_v7.py`, is in the book's archive (Appendix A) with its output, `verify_bosch_xi_v7_output.txt`, and the line numbers below are those of these two files; an identical copy of both is in `bosch_hcm_xi/`, beside the authors' notes. The script consists of thirteen blocks that run one after another in one process, each with its own imports; blocks 8 and 9 use `numpy` as well as `mpmath`, block 8 passes quadrature data to block 9, and block 13 calls the functions that block 12 defines. Run in an empty folder, the blocks write their results to nine text files, one per block except that block 8 writes none, blocks 5 to 7 share a file and so do blocks 9 and 10, and the output file joins these files in block order. Table D.1 gives, for each block, its lines in the script and in the output and the statements of Section 4.3 that it checks.

The script runs as it stands. In an empty folder it ends after about 400 seconds, and its nine files, joined in block order, give 95 lines, each of which occurs in the output file in the same order and agrees with it byte for byte, apart from the timings in seconds that some lines report. The output file is nevertheless not the record of a single run, and its other 14 lines come from code that is not in the script. Lines 37 to 39 (the zero sums with 100 and 200 zeros and the positivity of u_p at 148 points) and line 109 (the gaps among the first 1000 zeros) are reproduced exactly by reconstructions kept in the archive (Section A.5), although the grid of 148 points is a guess. Lines 93 to 102 come from an earlier version of block 13 with ten cases. Lines 103 to 107 run five of these cases again with the present block 13: for the removed frequencies $\{3\}$ they repeat line 102, and for $\{2,3\}$, $\{2,3,5,6\}$, $\{3,4\}$ and $\{4,5\}$ they give $\pi/h_* = 2.99997$, 3.0, 1.48313 and 1.31671 where lines 94, 97, 99 and 100 give 1.0, the value printed when the scan finds no failure below π. Lines 103 to 106 therefore supersede those four lines; the present block 13, run on all ten cases, reproduces lines 93, 95, 96, 98, 101 and 102, from which Table 4.5 takes its other rows. Line 108 adds the logistic model with no frequency removed.

Two results in the output need care in reading. For the toy with zeros $14e^{\pm 0.1i}$

block	script	output	what is checked	statement
1	1 to 50	1 to 10	HCM test by the signs of the Taylor coefficients of pair functions, for Bosch's family and for a toy with zeros $14e^{\pm 0.1i}$ and 21; minimum of $\operatorname{Im}\varepsilon_\varphi(re^{i\theta})/r$ over 885 points	Theorems 4.3.4, 4.3.6; Corollary 4.3.8
2	51 to 86	11 to 24	minimum of $\operatorname{Im}\varepsilon_h$ over 49 radii and 199 angles, for Bosch's family and for the toy	Theorems 4.3.2, 4.3.4, 4.3.6
3	87 to 116	25 to 36	u_p from (4.3.2) against (4.3.3) with 30 zeros; $\log\varphi(s^p)$ recovered from u_p	Theorem 4.3.13
none		37 to 39	(4.3.3) with 100 and 200 zeros; $u_p > 0$ at 148 points	Theorem 4.3.13; Corollary 4.3.15
4	117 to 148	40 to 49	density of $X_C = \sum_k e_k/(2k-1)^2$ against $(4/\pi)\eta(4it/\pi)^3$, and its Laplace transform at $s = 2$; a sufficient inequality in the prime region	Theorem 4.3.21; Proposition 4.3.19
5, 6	149 to 199	50 to 60	residue coefficients c_n from 200 zeros, their signs, sizes and gaps; the residue series against Talbot inversion	Theorem 4.3.23 and the numerical paragraph after it
7	200 to 219	61 to 66	the bound $m_p(t)$ of (4.3.4); $-\zeta'/\zeta(\sigma) < 1/(\sigma-1)$	Proposition 4.3.19; Table 4.4
8 to 10	220 to 318	67 to 80	g by quadrature and by series; Pick scan of $-g'/g$; $(\log g)''$ on $[0,1]$	Proposition 4.3.27; Theorem 4.3.28
11	319 to 343	81 to 86	the pair kernel; constants for $T_0 = 3\times 10^{12}$	Theorem 4.3.26
12, 13	344 to 419	87 to 108	Pick strip h_* and threshold $\beta_* = \pi/h_*$ in logistic models with frequencies removed	Table 4.5
none		109	gaps among the first 1000 zeros	Question 4.3.29

Table D.1: The blocks of the verification script: their lines in the script and in the output, what they check, and the statements of Section 4.3 they support. Output lines 37 to 39 and 109 come from code that is not in the script, and lines 93 to 102 from an earlier version of block 13.

and 21, neither test locates the threshold $p_* = 0.9363$. The sign test finds no violation at $p = 0.95$ (line 8), and the scan of block 2 misses the poles at $p = 0.94$ and $p = 0.97$, which lie between its grid points or beyond its last ray (lines 22 and 23, where the flag for poles is computed from their position, not detected); both tests see the failure only at $p = 1$ (lines 9 and 24). At $t = 0.003$ the terms of the residue series reach 2.7×10^{20} while the density is about 3.3×10^{-72}, so about 92 digits cancel, more than the 25 carried in line 56 and the 90 carried in line 60: the series values at $t = 0.003$ in these lines are rounding error, and the Talbot values there are negative. At 150 digits the residue series over the first 250 zeros and Talbot inversion agree on the density $3.32007507753 \times 10^{-72}$ at $t = 0.003$, with a largest term of 2.694×10^{20} and 91.9 digits cancelled; the reconstructions of Section A.5 contain this computation.

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